Yang–Mills Theory and the Mass Gap

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Four-dimensional lattice Yang–Mills theory with any compact simple gauge group $G \subset \mathrm{U}(N)$ has the mass gap $m \geq \operatorname{arccosh}\bigl(1+\frac{1}{12}\bigl(\frac{\kappa_G}{N\beta}-\frac{29}{2}\bigr)\bigr) > 0$ at strong coupling β, and one uniform cube estimate with constant ϖ at physical scale ℓ gives every continuum limit the mass gap $m \geq 16\varpi/(625e^2\ell)$.

I. INTRODUCTION

Wilson's lattice gauge theory [104] assigns an element Q_e of a compact group $G \subset \mathrm{U}(N)$ to every link e of $\mathbb{Z}^4$ and weighs a configuration by $\exp(N\beta\sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p)$, where Q_p is the ordered product of the four link variables around the plaquette p.[1] At small $|\beta|$ the theory is under control: it has a unique infinite-volume state, correlations decay exponentially, and the transfer matrix has a gap [90, 95].[2] The question is how far into the coupling range such statements can be proved with explicit constants, since the problem of the continuum limit [68] begins where strong coupling ends and Bałaban's weak-coupling analysis [6, 7, 13] takes over.

Two methods give explicit ranges. The Bakry–Émery method [4, 5] of Shen, Zhu and Zhu [96] bounds the Hessian of the action; in [19] we showed that the sharp bound is $4d = 16$ and that no curvature argument for the ungauged measure goes beyond $|\beta| < 1/32$. The Dobrushin method [41, 42][3] bounds the influence of each link on the conditional law of another; for $\mathrm{SU}(2)$ the single-link conditional law is a von Mises–Fisher law on the three-sphere, the sharp Kantorovich modulus[4] of that family is one quarter [19] (Theorem 364 in Appendix K), and every link has eighteen partners, which gives $\beta_W < 2/9$ and nothing beyond, because the coefficients are attained.

Both limits come from the same count. A link of $\mathbb{Z}^4$ lies in six plaquettes and shares each with three other links. Gauge invariance makes part of this redundant: in a box, the variables on a forest of links can be set to the identity without changing any gauge-invariant expectation [38].[5] The forest we use is a field of straight lines with one line through every site, chosen

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[1] This is the normalisation of Shen, Zhu and Zhu [96] and of our earlier paper [19]. The conventional inverse coupling is $\beta_W = N^2\beta$, so the weight is $\exp(\frac{\beta_W}{N}\sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p)$; for $\mathrm{SU}(2)$, identified with the unit quaternions, $\frac{1}{2}\mathrm{Tr}\,Q_p = \mathrm{Re}\,Q_p$ and $\beta_W = 4\beta$. Standard accounts of the lattice theory are the books of Montvay and Münster [85] and of Gattringer and Lang [52]; the Hamiltonian version is due to Kogut and Susskind [74].

[2] Strong-coupling results of this kind go back to Osterwalder and Seiler [90], who proved exponential clustering and a gap by a convergent expansion, and to Seiler's monograph [95]; the general framework for lattice gauge fields and their continuum limits is that of Brydges, Fröhlich and Seiler [27]. More recent work treats the strong-coupling regime by probabilistic methods: the survey of Chatterjee [35], his large-N solution of $\mathrm{SO}(N)$ lattice gauge theory at strong coupling [36] and its planar extension by Basu and Ganguly [16], the Langevin dynamics of Shen, Zhu and Zhu [96], and the area law in an enlarged region by Cao, Nissim and Sheffield [29]. For finite gauge groups Adhikari and Cao proved correlation decay at weak coupling [1]. In two dimensions the continuum Yang–Mills measure has been constructed by Gross, King and Sengupta [57], Driver [44] and Lévy [75]; in four dimensions Bałaban's renormalisation group, as expounded by Dimock [39], gives ultraviolet stability in finite volume, and Magnen, Rivasseau and Sénéor constructed the theory with an infrared cutoff [80]. None of these results gives a mass gap in physical units.

[3] Dobrushin's theorem: if the conditional law of each variable given all others moves, in Kantorovich distance, by at most C_{xy} times the distance by which the variable at y moves, and $\sup_x \sum_y C_{xy} < 1$, then the Gibbs measure is unique and correlations decay at the rate given by the Neumann series of the matrix C. Dobrushin and Shlosman [43] replaced single variables by finite boxes, which is the form used in Sec. IX.

[4] The Kantorovich distance between two probability measures on a metric space is the least expected distance over their couplings, equivalently the largest difference of their integrals over 1-Lipschitz functions [102]. The modulus of a family of laws indexed by a metric space is the best Lipschitz constant of the map from the index to the law in that distance.

[5] Creutz used a maximal tree to remove the gauge freedom completely; the temporal gauge, with all links in one direction fixed, is the special case of a family of straight lines. In a maximal tree almost every link is fixed in some region and none in others, so the number of partners of a free link is not uniform, and the temporal gauge leaves fourteen partners per link. The point here is a pattern that fixes exactly one link in every plaquette.

so that every plaquette contains exactly one fixed link. Then every free link has exactly twelve free partners, two in each of its plaquettes (Lemma 9), and both methods improve.

Theorem 1 (SU(2))**.** *Let $G = \mathrm{SU}(2)$ and $0 \le \beta_W < \frac{1}{3}$. For every gauge-invariant local observable O the expectations $W_\Lambda(O)$ in boxes with free boundary converge, along every sequence of boxes that exhausts $\mathbb{Z}^4$, to a limit $\omega(O)$ that does not depend on the sequence. The state ω is invariant under translations and the hypercubic group and reflection positive;*[6] *truncated correlations of Lipschitz observables decay at every rate below $-\log(3\beta_W)$; and the transfer matrix has spectral gap*

$$m \ge -\log(3\beta_W) > 0.$$

For $-\frac{1}{3} < \beta_W < 0$ all statements except reflection positivity and the gap hold.

The step behind Theorem 1 is short once the line field is in place. The number $\frac{1}{3}$ is also the end of the road for this kind of argument.

Theorem 2 (Ceiling)**.** *Let F' be any set of links of $\mathbb{Z}^4$ without cycles, and give two distinct links outside F' that share a plaquette the coefficient $|\beta|$, the largest single-link Kantorovich coefficient of* SU(2)*. Then the weighted Dobrushin condition with these coefficients fails for $\beta_W \ge \frac{1}{3}$. For the line field the coefficients are attained, so no single-link Kantorovich argument in that gauge reaches $\beta_W = \frac{1}{3}$.*

The ceiling concerns coefficients between single links. Take instead as a block the three free links of one plaquette. The free links of the line field split into such triples with period two (Lemma 12), the coupling inside the block plaquette is no longer paid for, and each link keeps ten outside partners. The price is the sensitivity of a law on three spheres instead of one, which we bound analytically by integrating out the middle link of the block.

Theorem 3 (SU(2) with plaquette blocks)**.** *Let $G = \mathrm{SU}(2)$ and $0 \le \beta_W \le 0.348$. The number $\alpha(\beta_W)$ of (4) is increasing, $\alpha(\frac{1}{3}) \le 0.9456$, $\alpha(0.345) \le 0.9837$ and $\alpha(0.348) \le 0.9936$. The conclusions of Theorem 1 hold with decay at every rate below $-\frac{1}{2}\log\alpha(\beta_W)$ and a transfer-matrix gap*

$$m \ge -\tfrac{1}{2}\log\alpha(\beta_W) > 0;$$

at $\beta_W = \frac{1}{3}$ this gives $m \ge 0.0279$. For $-0.348 \le \beta_W < 0$ all statements except reflection positivity and the gap hold.

At small coupling the rate of Theorem 1 is the better one, so both bounds stand side by side; Theorem 3 matters past $\beta_W = \frac{1}{3}$, where the gap of Theorem 1 has closed.

The curvature method gains as well, and for every N. Let $\mathrm{Hess}\, S_0$ be the Hessian of $\sum_p \mathrm{Re}\,\mathrm{Tr}\, Q_p$ on G^E and F the set of fixed links.

Theorem 4 (Curvature in the gauge)**.** *For every N, every field Q and every tangent vector $v = (X_e Q_e)$ with $X_e = 0$ for $e \in F$,*

$$|\,\mathrm{Hess}\, S_0(v,v)| \le \tfrac{29}{2}|v|^2,$$

against the sharp constant 16 *without gauge fixing. Consequently, for $G = \mathrm{SU}(N)$ and $|\beta| < \frac{1}{29}$, or $G = \mathrm{SO}(N)$, $N \ge 3$, and $|\beta| < \frac{N-2}{58N}$, the conclusions of Theorem 1 hold with the rate and gap of Theorem 5. For* SU(3) *this is $\beta_W < \frac{9}{29} = 0.310$.*

The constant $\frac{29}{2}$ is a statement about $\mathrm{U}(N)$, so it holds on every closed subgroup, and the curvature of the group does the rest. This gives the theorem for every compact simple gauge group, the exceptional groups included, with an explicit gap. Let $\rho : G \to \mathrm{U}(N)$ be a faithful unitary representation of a compact connected Lie group G, weigh a configuration by $\exp(N\beta\sum_p \mathrm{Re}\,\mathrm{Tr}\,\rho(Q_p))$, and let κ_G be the least Ricci[7] curvature of G for the bi-invariant metric $\langle X, Y\rangle = -\mathrm{Tr}(\rho(X)\rho(Y))$; we identify G with $\rho(G) \subset \mathrm{U}(N)$.

Theorem 5 (Every compact simple group)**.** *Suppose that the Lie algebra of G is semisimple, so that $\kappa_G > 0$, and that*

$$N|\beta| < \tfrac{2}{29}\,\kappa_G.$$

Then the conclusions of Theorem 1 hold, for $\beta < 0$ all of them except reflection positivity and the gap, with decay at every rate below

$$m_G(\beta) = \operatorname{arccosh}\Big(1 + \frac{1}{12}\Big(\frac{\kappa_G}{N|\beta|} - \frac{29}{2}\Big)\Big) > 0$$

and a transfer-matrix gap $m \ge m_G(\beta)$. For a simple G, $\kappa_G = h^\vee/(2\ell_\rho)$, where $h^\vee$ is the dual Coxeter number and ℓ_ρ the Dynkin index of ρ; Table I lists the values.

For $\mathrm{SU}(N)$, $\kappa_G = \frac{N}{2}$, and for $\mathrm{SO}(N)$, $\kappa_G = \frac{N-2}{4}$, so Theorem 5 contains the ranges of Theorem 4 and supplies their rate.[8] For small $|\beta|$ the existence of a gap is classical for every compact group, with no explicit range [90, Thm. 3.6]; what is new is an explicit range for every group, past the range $N|\beta| < \kappa_G/16$ that the Hessian bound 16 gives without gauge fixing, and an explicit gap.[9] As $\beta \to 0$ the gap grows like $\log\frac{\kappa_G}{6N|\beta|}$.

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[6] That is, $\omega(\vartheta(\bar F)F) \ge 0$ for every F depending on the links on one side of a lattice hyperplane, ϑ being the reflection in it. It gives the Hilbert space and the transfer matrix of Osterwalder and Schrader [88, 89] and of Osterwalder and Seiler [90], reconstructed in Appendix F.

[7] For a bi-invariant metric on a compact Lie group the Ricci curvature is $-\frac{1}{4}\mathrm{Tr}((\mathrm{ad}X)^2)$ in the direction of X. It is positive for every $X \ne 0$ exactly when the Lie algebra is semisimple, and κ_G is its minimum over unit vectors.

[8] The dual Coxeter number is $h^\vee = N$ for $\mathrm{SU}(N)$, $N-2$ for $\mathrm{SO}(N)$, $n+1$ for $\mathrm{Sp}(n)$, $4, 9, 12, 18, 30$ for G_2, F_4, E_6, E_7, E_8, and the Dynkin index of a representation is normalised to one on the defining representation of $\mathrm{SU}(N)$ [71, Ch. 6]. The identity $\kappa_G = h^\vee/(2\ell_\rho)$ is the statement that the Ricci tensor of a compact simple group with a bi-invariant metric is one quarter of the negative Killing form, $\mathrm{Ric}(X,X) = -\frac{1}{4}B(X,X)$ [84], together with $B(X,X) = 2h^\vee\,\mathrm{Tr}(\rho(X)^2)/\ell_\rho$. Lower Ricci bounds of the same kind, for special holonomy and for Riemannian submersions, are studied in [22, 53].

[9] The computation of Remark 377, from a certificate of T. Romanov, gives $h_F \le 14.420001$; with it in place of $\frac{29}{2}$, every range of Table I would grow by the factor $29/28.840002 = 1.0055$. No result of the paper uses it.

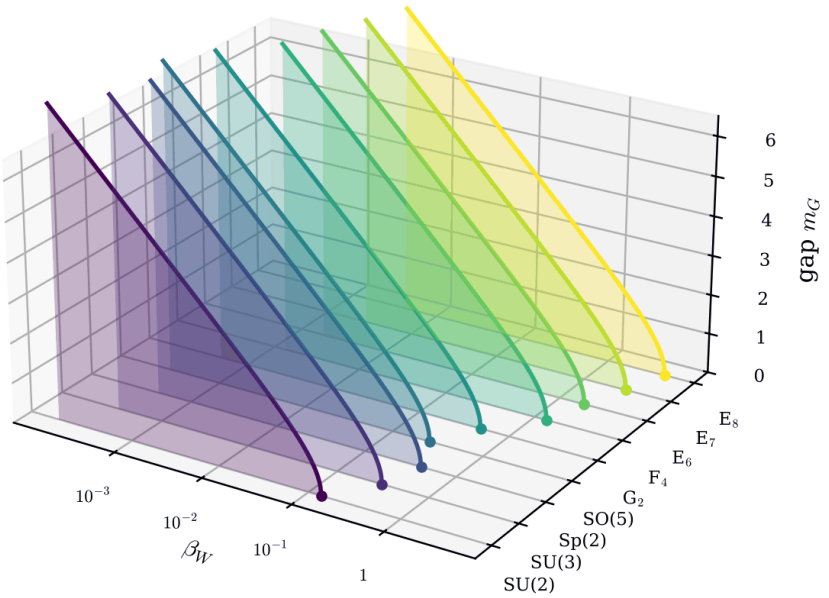


FIG. 1. The gap $m_G(\beta)$ of Theorem 5 as a function of $\beta_W = N^2\beta$ on a logarithmic scale, for nine groups in the representations of Table I. Each curve ends at the edge $\beta_W = 2N\kappa_G/29$ of the range, marked by a dot, where the gap tends to zero; as $\beta_W \to 0$ it grows like $\log(\kappa_G/(6N|\beta|))$. The curves differ only by the factor $N\kappa_G$, since m_G depends on $N|\beta|/\kappa_G$ alone.

TABLE I. Least Ricci curvature $\kappa_G = h^\vee/(2\ell_\rho)$ of the compact simple groups in a faithful representation ρ of least dimension N, and the range of Theorem 5 for the conventional inverse coupling $\beta_W = N^2\beta$.

Group	ρ	$h^\vee$	ℓ_ρ	κ_G	$\beta_W <$
$\mathrm{SU}(N)$, $N \geq 2$	$\mathbb{C}^N$	N	1	$\frac{N}{2}$	$\frac{N^2}{29}$
$\mathrm{SO}(N)$, $N \geq 5$	$\mathbb{R}^N$	$N-2$	2	$\frac{N-2}{4}$	$\frac{N(N-2)}{58}$
$\mathrm{SO}(3)$	$\mathbb{R}^3$	2	4	$\frac{1}{4}$	$\frac{3}{58}$
$\mathrm{Sp}(n)$, $n \geq 2$	$\mathbb{C}^{2n}$	$n+1$	1	$\frac{n+1}{2}$	$\frac{2n(n+1)}{29}$
G_2	7	4	2	1	$\frac{14}{29}$
F_4	26	9	6	$\frac{3}{4}$	$\frac{39}{29}$
E_6	27	12	6	1	$\frac{54}{29}$
E_7	56	18	12	$\frac{3}{4}$	$\frac{84}{29}$
E_8	248	30	60	$\frac{1}{4}$	$\frac{124}{29}$

No Bakry–Émery argument for the ungauged measure can pass $|\beta| = \frac{1}{32}$ [19, Thm. C]; the gauge fixing changes the measure, not the metric, which is why it can. For $\mathrm{SU}(2)$ the curvature range $\beta_W < \frac{4}{29}$ lies inside that of Theorem 1, and no curvature argument for the gauge-fixed $\mathrm{SU}(2)$ measure passes $\beta_W = \frac{1}{3}$: when one active link e is -1 and all others are 1, the six plaquettes through e have variable -1, and along a geodesic through Q_e the Hessian of $-\frac{\beta_W}{2}\sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p$ is $-6\beta_W$ times the metric of the unit sphere S^3, whose Ricci curvature is 2, so the Bakry–Émery tensor is negative there once $\beta_W > \frac{1}{3}$. For $\mathrm{SU}(3)$ the Dobrushin route goes further.

Theorem 6 ($\mathrm{SU}(3)$)**.** *Let $G = \mathrm{SU}(3)$ and $0 \leq \beta_W \leq 0.515$, put $r = 2\beta_W$ and*

$$\varepsilon(r) = 1 - \frac{5\,e^{-r/5}}{9\,Z_*(r)}, \qquad \alpha_1(\beta_W) = \frac{2r\sqrt{\varepsilon(r)}}{\gamma(r)},$$

where $\gamma(r) = \frac{283}{128} - \frac{r}{2}$ and $Z_*(r) = \int_{\mathrm{SU}(3)} e^{r\,\mathrm{Re}\,\mathrm{Tr}\,U}\,dU$. The number $\alpha_1(\beta_W)$ is increasing, $\alpha_1(\frac{1}{2}) \leq 0.9491$ and $\alpha_1(0.515) \leq 0.9924$. The conclusions of Theorem 1 hold with decay at every rate below $-\log\alpha_1(\beta_W)$ and a transfer-matrix gap

$$m \geq -\log\alpha_1(\beta_W) > 0;$$

at $\beta_W = \frac{1}{2}$ this gives $m \geq 0.052$.

The proof of Theorem 6 bounds the Kantorovich modulus of the one-link measure $\nu_b(dU) \propto e^{\mathrm{Re}\,\mathrm{Tr}(Ub)}dU$ on $\mathrm{SU}(3)$ uniformly over staple sums b. It uses a lower bound $\gamma(r)$ for the spectral gap of ν_b through Bochner's formula with a cross term (Lemma 29), the Poincaré inequality twice, and a bound for the mean of $|\nabla\,\mathrm{Re}\,\mathrm{Tr}(Uh)|^2$ under ν_b through Jensen's inequality and Haar integrals of degree three (Lemma 31); Z_* is a series in the moments of $\mathrm{Tr}\,U$, which are counts of standard tableaux (Lemma 30). A sharper bound for the same modulus, evaluated by a computer on boxes of staple sums, would carry the range to $\beta_W = \frac{3}{4}$; Remark 36 describes it, and no result uses it.

The plaquette blocks of Theorem 3 work for $\mathrm{SU}(3)$ as well. They use the same modulus of the one-link measure, but only through the three links of a block, so they need $\Lambda(2\beta_W)$ small and not $2r\Lambda(r) < 1$: near the end of Theorem 6 they give a larger gap, and they go past it.

TABLE II. Explicit ranges of uniqueness, decay and a transfer-matrix gap for four-dimensional lattice Yang–Mills theory. The first three rows hold for the ungauged measure; the last three are the results of this paper.

Method	SU(2), β_W	SU(3), β_W
Curvature, Shen–Zhu–Zhu [96]	$< \frac{1}{12}$	$< \frac{3}{16}$
Curvature, sharp [19]	$< \frac{1}{8}$	$< \frac{9}{32}$
Dobrushin, sharp quarter [19]	$< \frac{2}{9}$	
Curvature, forest gauge	$< \frac{4}{29}$	$< \frac{9}{29}$
Dobrushin, forest gauge	$< \frac{1}{3}$	≤ 0.515
Plaquette blocks, forest gauge	≤ 0.348	≤ 0.52

Theorem 7 (SU(3) with plaquette blocks)**.** *Let $G = \mathrm{SU}(3)$ and $0 \leq \beta_W \leq 0.52$, let r, $\varepsilon(r)$, $\gamma(r)$ be as in Theorem 6, and let $\alpha_3(\beta_W)$ be the number of (15) with $\Lambda(r)$, τ, λ_* and M replaced by $\sqrt{\varepsilon(r)}/\gamma(r)$, $\varepsilon(r)/\gamma(r)$, 0 and $2\sqrt{3}\,\varepsilon(r)/\gamma(r)$. Then $\alpha_3(\beta_W) < 1$, with $\alpha_3(0.515) \leq 0.9742$ and $\alpha_3(0.52) \leq 0.9927$. The conclusions of Theorem 1 hold with decay at every rate below $-\frac{1}{2}\log\alpha_3(\beta_W)$ and a transfer-matrix gap*

$$m \geq -\tfrac{1}{2}\log\alpha_3(\beta_W) > 0;$$

at $\beta_W = 0.515$ this gives $m \geq 0.013$, against 0.0076 from Theorem 6, and at $\beta_W = 0.52$, outside the range of Theorem 6, $m \geq 0.0036$.

The constants of Theorem 7 are the bounds of Lemma 31; Remark 39 records what computed constants would give.

All seven theorems are statements at fixed lattice spacing. They do not touch the continuum limit, the crossover to Bałaban's regime, or a gap in physical units, which together are the Clay problem. The obstruction proved in [20], that a lattice gap uniform in lattice units forces trivial continuum two-point functions, applies to all of them.

The continuum step. Sec. IX goes past that obstruction in one respect, through three implications, each proved.

(I1) A Dobrushin–Shlosman condition on one cube of fixed physical size, uniform along the continuum tuning, gives a mass gap in physical units for every continuum limit (Theorems 41 and 43), and so does exponential decay with constants of the renormalised correlations of a spanning set of observables (Theorem 44); this holds also for observables smoothed by the spatial gradient flow (Proposition 45). A spectral gap of the dynamics that resamples one such cube at a time, uniform along the tuning and required only on local gauge-invariant functions, gives the same conclusion (Theorems 64 and 82).

(I2) The flowed observables, renormalised by one condition at a single physical separation, have reflection positive continuum limits along subsequences, and such a limit is nontrivial exactly when one ratio of lattice two-point functions stays away from zero (Theorem 46); that ratio is bounded above and below by ratios of correlations of one slice field at fixed physical separations (Proposition 49).

(I3) Tree bounds on the truncated correlations give all the higher Schwinger functions as reflection positive distributions, imply the ratio bound, and carry the conditional mass gap to every polynomial in the smeared fields (Theorem 55); for a smooth packet they give the ratio with an explicit rate and a reference vector of finite energy (Proposition 59).

What is open. The inputs of (I1)–(I3), namely the cube condition along the tuning, the slice ratio in four dimensions and the tree bounds, are not proved, and so the nontrivial existence of the continuum limit and the gap in physical units remain open. Figure 3 shows how the proved statements, the implications and the open inputs fit together.

Sec. X measures the open inputs. The cube condition holds at strong coupling, fails at every fixed cube size at weak coupling, forces the gauge-fixed variables to be disordered on the scale of the cube, and cannot be reached from estimates on single lines of links; for SU(N) it needs cubes of side at least of order $(\beta_W/\log\beta_W)^{1/4}$ (Propositions 240 and 241). The spectral gap of the cube dynamics holds on every torus at strong coupling (Corollary 72) and fails on the torus of side $2n+4$ when n is small compared with $(\beta/\log\beta)^{1/4}$ (Theorem 77), but only through functions of nonzero centre charge, which its local form does not involve (Corollary 83). For a Gaussian field it holds at a fixed physical scale when the field has a mass and fails exactly, at every size, when it is massless (Proposition 93); it also fails for the quadratic approximation of the forest gauge itself (Proposition 94), and of every gauge fixed on an invariant forest with infinite trees (Proposition 95). The order-two cases of the tree bounds force the lattice gap below a fixed multiple of the lattice spacing, so no strong coupling expansion gives them (Proposition 58). The slice ratio holds in two dimensions, where it is the exact solution of lattice gauge theory (Remark 52), and fails for a field whose mass is fixed in lattice units; for the free field, conditioning an energy packet on averages over blocks leaves a fixed fraction of its ordinary variance at every lattice spacing, so a reduction to block averages has to be measured in the reflected norm (Proposition 53).

Across scales. Our Hexagon preprint [21] develops the multiscale side of the problem: how the averaged condition passes from the fine lattice to the law of the variables of a gauge-covariant block-spin map and back, what the condition at the coarsest scale requires of the single-link laws of the block variables, the cube condition for block averages and a cut-off of Bałaban's averaging operation that composes across scales at small fields, and the labelling of blocks into good and bad islands. None of these results gives (D) along the tuning, and they are not repeated here; this paper keeps the strong-coupling theorems, the continuum step and the estimates that bear directly on (D) and (E).

This paper substantially elaborates and revises our Hexagon preprint [21].

Plan. Sec. II constructs the line field. Sec. III shows that a Dobrushin bound for the gauge-fixed specification gives the full conclusion of Theorem 1. Secs. IV, V, VI, VII and VIII prove Theorems 1–7. Sec. IX proves (I1)–(I3), Sec. X states and measures their open inputs and proves the cube condition for free massive fields uniformly in the spacing (Sec. X B) and

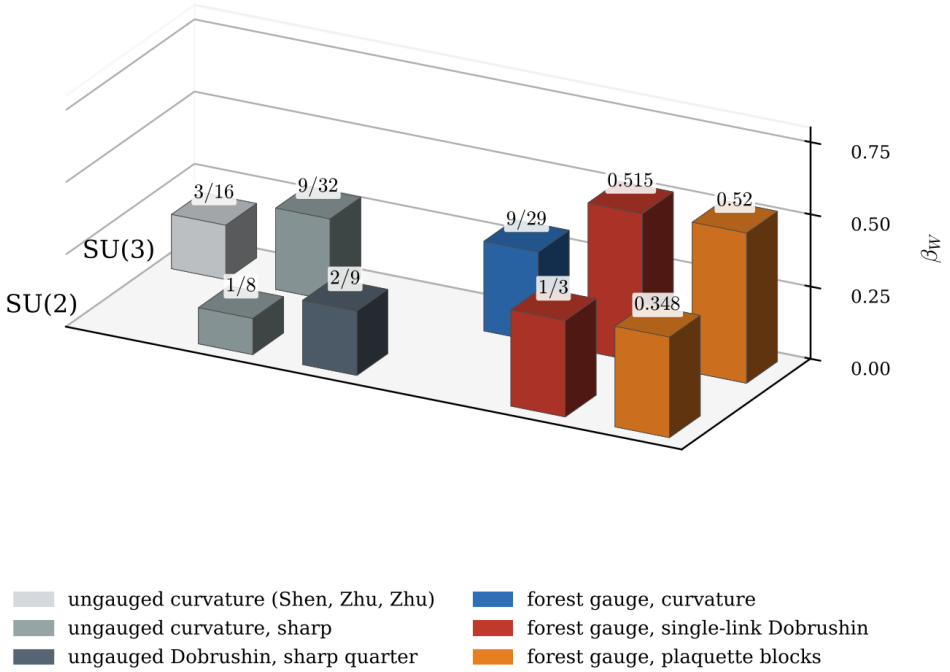


FIG. 2. Ranges of Table II. The height of each bar is the largest inverse coupling β_W covered by the method. Grey bars are methods for the ungauged measure; blue, red and orange bars use the forest gauge. For SU(2) the red bar ends at the ceiling of Theorem 2 and the orange bar, plaquette blocks, passes it (Theorem 3); for SU(3) the red bar is Theorem 6 and the orange bar, plaquette blocks, gives a larger gap there and passes it (Theorem 7).

its failure for the free photon together with its plaquette form at weak coupling (Sec. X D), Sec. X M builds the two-point continuum theory without the cube condition, and Sec. X O bounds plaquette correlations from below. Sec. XI sets out what is proved and what is open. The appendices contain the residue table of the line field, the comparison lemma and its version for blocks, the blocks of the local Hessian inequality, the details of the certified computation, the reconstruction of the transfer matrix with the equivalence of a gap and exponential clustering (Appendix F), the facts on spectral measures and on the character expansion that the continuum step uses (Appendices G and I), the Haar measure of SU(2) (Appendix J), and a complete proof of the sharp quarter for the single-link law of SU(2) (Appendices K–M), so that the paper does not depend on our earlier one for any step.

II. THE LINE FIELD

Write $e_0,\dots,e_3$ for the unit vectors of $\mathbb{Z}^4$ and (x,μ) for the link from x to $x+e_\mu$. A *direction field* is a map $\sigma:\mathbb{Z}^4\to\{0,1,2,3\}$; its links are $F=\{(x,\sigma(x)):x\in\mathbb{Z}^4\}$. At each site put $\pi_x(\mu)=\sigma(x+e_\mu)$. The plaquette at x in the plane $\{\mu,\nu\}$ has the links (x,μ), $(x+e_\mu,\nu)$, $(x+e_\nu,\mu)$, (x,ν), so it contains

$$[\sigma(x)\in\{\mu,\nu\}]+[\pi_x(\mu)=\nu]+[\pi_x(\nu)=\mu] \qquad (1)$$

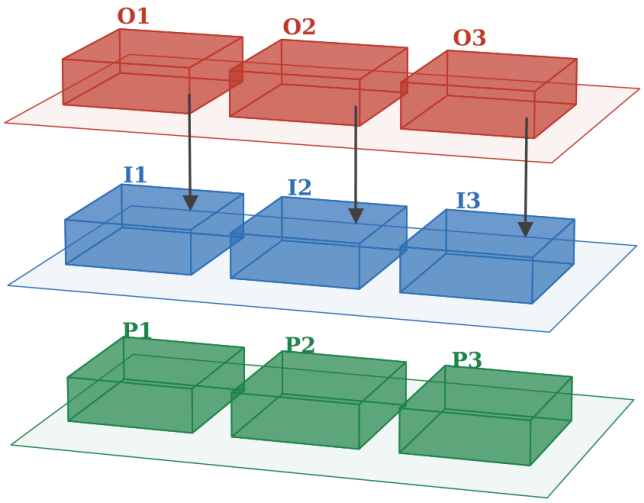
links of F.

Lemma 8. *Every plaquette of $\mathbb{Z}^4$ contains exactly one link of F if and only if, at every site x, π_x fixes $\sigma(x)$ and permutes the other three directions cyclically.*

Proof. Fix x, let $s=\sigma(x)$ and let R be the set of the other three directions. For a plane $\{s,\nu\}$ the count (1) is $1+[\pi_x(s)=\nu]+[\pi_x(\nu)=s]$; it equals one for all $\nu\in R$ exactly when $\pi_x(s)=s$ and $\pi_x(\nu)\neq s$ for $\nu\in R$, that is, when π_x fixes s and maps R into R. For a plane $\{\mu,\nu\}\subset R$ the count is $[\pi_x(\mu)=\nu]+[\pi_x(\nu)=\mu]$. Draw an arrow $\mu\to\pi_x(\mu)$ for every $\mu\in R$ with $\pi_x(\mu)\in R\setminus\{\mu\}$. Each vertex of R has at most one outgoing arrow, and the condition says that each of the three pairs of R carries exactly one arrow. Three arrows on three vertices with out-degree at most one give out-degree exactly one everywhere, and a tournament on three vertices with all out-degrees one is a directed three-cycle. □

From now on σ is the field of period two given in Table III of Appendix A, read from x modulo two. The table lists π_x at each of the sixteen residue classes; in each row π_x fixes $\sigma(x)$ and is a three-cycle on the rest, so every plaquette contains exactly one link of F by Lemma 8.[10]

[10] There are exactly eight period-two direction fields with this property, all equivalent to σ under the symmetries of the lattice. A field with one fixed link at every site and one in every plaquette exists only in four dimensions: a link lies in $2(d-1)$ plaquettes and a site carries $d(d-1)/2$ of them, and $2(d-1)=d(d-1)/2$ forces $d=4$. The statements of this section are also checked in the Lean certificate of the archive.



open inputs

O1 (D) or local BP along the tuning

O2 slice ratio in four dimensions

O3 tree bounds (T1)-(T3)

proved implications

I1 (D) or local BP: gap for every limit

I2 ratio iff nontrivial flowed limit

I3 tree bounds give the ratio

proved at fixed spacing

P1 gap for every simple group

P2 SU(2), SU(3) with blocks

P3 (D) and BP at strong coupling

arrows: an open input and the implication that consumes it

FIG. 3. The logical structure of the paper. Green: theorems at fixed lattice spacing (P1 the gap for every compact simple group, Theorem 5, P2 the plaquette blocks for SU(2) and SU(3), P3 the cube condition and the averaged condition BP at strong coupling, Corollary 72). Blue: the proved implications (I1)–(I3) of the continuum step, where (I1) takes either (D) or the local form of BP (Theorem 82). Red: their inputs, none of which is proved along the continuum tuning. Each arrow joins an open input to the implication that consumes it; no implication feeds its own input.

Lemma 9.

(i) *F is a disjoint union of bi-infinite straight lines, one through every site: every site has exactly one outgoing and one incoming link of F, and both point in the direction $\sigma(x)$.*

(ii) *For every box Λ (a product of four finite intervals, with the links and plaquettes it contains), $F \cap E(\Lambda)$ is a disjoint union of segments, hence a forest.*

(iii) *A link $e = (x,\mu) \notin F$ lies in six plaquettes. Four of them contain a link of F that shares an end point with e, and these lie in the planes $\{\mu,\sigma(x)\}$ and $\{\mu,\pi_x(\mu)\}$. In the other two, which lie in the remaining plane $\{\mu,\rho\}$, the link of F is opposite to e.*

(iv) *Every link outside F shares a plaquette with exactly twelve other links outside F, two in each of its plaquettes.*

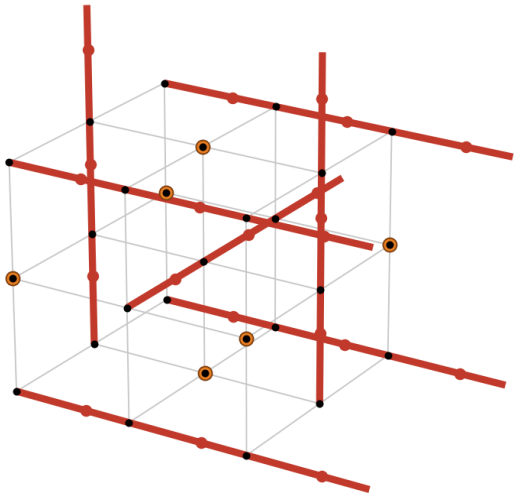
Proof. (i) Since π_x fixes $\sigma(x)$, $\sigma(x+e_{\sigma(x)}) = \sigma(x)$. Because σ has period two, $x - e_\mu \equiv x + e_\mu$, so also $\sigma(x - e_{\sigma(x)}) = \sigma(x)$, and the link $(x - e_{\sigma(x)}, \sigma(x))$ is in F. Conversely, if $(x - e_\nu, \nu) \in F$, then $\sigma(x - e_\nu) = \nu$, and straightness at $x - e_\nu$ gives $\sigma(x) = \nu$. So the incoming link is unique and continues the outgoing one.

(ii) A line meets a box in a segment, and distinct lines are disjoint.

(iii) The plaquettes of e are, for each $\nu \neq \mu$, the one at x and the one at $x - e_\nu$ in the plane $\{\mu,\nu\}$. A link of F sharing an end point with e is the outgoing or the incoming link of F at x or at $x + e_\mu$. At x both point in the direction $s = \sigma(x) \neq \mu$, and they lie in the plaquettes at x and at $x - e_s$ in the plane $\{\mu,s\}$. At $x + e_\mu$ both point in the direction $t = \pi_x(\mu)$, which lies in $R \setminus \{\mu\}$ because π_x is a three-cycle on R, and they lie in the plaquettes at x and at $x - e_t$ in the plane $\{\mu,t\}$. These four plaquettes are distinct because $s \neq t$. Each plaquette contains exactly one link of F, so in the two plaquettes of the third plane $\{\mu,\rho\}$ the link of F is a link of direction μ or ρ not touching e; the links of direction ρ in such a plaquette touch e, so it is the opposite link.

(iv) Each of the six plaquettes contains e, one link of F and two other links, and two distinct links share at most one plaquette. □

(a) the line field in the slice $x_3 = 0$



(b) one plaquette in the gauge

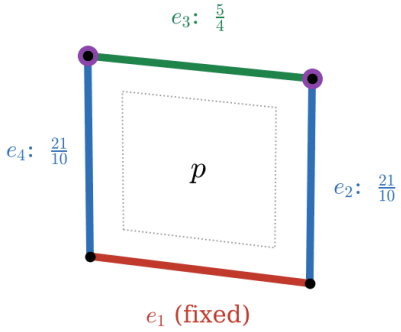


FIG. 4. (a) The line field in the slice $x_3 = 0$ of a box of side three. Thick red segments are the links of F in the directions $0, 1, 2$, with a dot marking the direction of travel; an orange ring marks a site whose line leaves the slice in direction 3. Every line is straight (Lemma 9(i)), and the segments that leave the box are its boundary stubs. (b) One plaquette with its fixed link e_1. The weights are those of the local inequality of Lemma 20 for the three free links: $\frac{21}{10}$ on the two links adjacent to e_1 and $\frac{5}{4}$ on the opposite one. The two purple corners are the ones that enter the corner form (7) once $Y_1 = 0$.

III. GAUGE FIXING AND THE FREE-BOUNDARY STATE

Let $G \subset \mathrm{U}(N)$ be compact with Haar measure dU, and let Λ be a box. The Wilson measure with free boundary is

$$W_\Lambda(dQ) = \frac{1}{Z_\Lambda} \exp\Big(N\beta \sum_{p \in P(\Lambda)} \mathrm{Re}\,\mathrm{Tr}\, Q_p\Big) \prod_{e \in E(\Lambda)} dQ_e .$$

The measure is invariant under the gauge transformations $Q_{(x,y)} \mapsto g_x Q_{(x,y)} g_y^{-1}$ with $g \in G^\Lambda$. Call the links outside F *active* and write A for their set. For a function O of the links, O^F is O with every link of F set to the identity.

Lemma 10 (Gauge fixing)**.** *Let μ_Λ^F be the probability measure on $G^{A \cap E(\Lambda)}$ whose density with respect to Haar measure is proportional to $\exp(N\beta \sum_{p \in P(\Lambda)} \mathrm{Re}\,\mathrm{Tr}\, Q_p^F)$. For every gauge-invariant O on $G^{E(\Lambda)}$,*

$$W_\Lambda(O) = \mu_\Lambda^F(O^F).$$

Proof. Each segment of $F \cap E(\Lambda)$ is a path $x_0, x_1, \dots, x_k$ along one direction. Given the variables on F, put $g_{x_0} = 1$ and $g_{x_{i+1}} = g_{x_i} Q_{(x_i, x_{i+1})}$ on every segment. The transformed field has $Q_e^g = 1$ on $F \cap E(\Lambda)$, and for fixed values on F the map $Q_A \mapsto Q_A^g$ is a product of left and right translations, which preserves Haar measure. Gauge invariance of the action and of O and the change of variables give the claim. □

No leftover holonomy appears because $F \cap E(\Lambda)$ is a forest. The measures μ_Λ^F are finite-volume versions of the specification γ^F on G^A: for $e \in A$, γ_e^F resamples Q_e from the law with density proportional to $\exp(N\beta \sum_{p \ni e} \mathrm{Re}\,\mathrm{Tr}\, Q_p^F)$ given all other active links. No infinite-volume gauge transformation is performed; γ^F is a specification in its own right. Fix a metric δ on G of diameter D that is bi-invariant (for $\mathrm{SU}(2)$ the chordal metric on S^3, $D = 2$; for $\mathrm{SU}(N)$ the Hilbert–Schmidt distance). The Kantorovich coefficient of e' on e is

$$C_{ee'} = \sup \frac{W_1\big(\gamma_e^F(\cdot \mid Q), \gamma_e^F(\cdot \mid Q')\big)}{\delta(Q_{e'}, Q'_{e'})},$$

the supremum over pairs of configurations that differ only at e'. It vanishes unless e and e' share a plaquette.

Proposition 11. *Suppose that every row and every column of $(C_{ee'})$ sums to at most $\alpha < 1$. Then:*

(i) *γ^F has exactly one Gibbs measure μ_F;*

(ii) *for every gauge-invariant local O the limit $\omega(O) = \lim W_\Lambda(O)$ exists along every sequence of boxes exhausting $\mathbb{Z}^4$ and equals $\mu_F(O^F)$;*

(iii) *ω is invariant under translations and the hypercubic group;*

(iv) *if $\beta \geq 0$, ω is reflection positive for reflections in hyperplanes through sites and in hyperplanes bisecting links;*

(v) *for gauge-invariant local O, O' with O Lipschitz in each link, $|\mathrm{Cov}_\omega(O, O')| \leq C_{O,O'} e^{-\lambda\, \mathrm{dist}(O,O')}$ for every $\lambda < -\log\alpha$;*

(vi) *if $\beta \geq 0$, the transfer matrix of ω has spectral gap at least $-\log\alpha$.*

Proof. (i) Gibbs measures exist by compactness, and the comparison lemma of Appendix B with $S=\emptyset$ shows that any two coincide.

(ii) Extend μ_Λ^F by Haar measure on the active links outside Λ, and let S be the set of active links with a plaquette outside $P(\Lambda)$; this includes every active link outside $E(\Lambda)$. For an active link $e\notin S$ all six plaquettes of e lie in $P(\Lambda)$, so the conditional law of Q_e under the extended measure is γ_e^F. Let T be the support of O^F and $\lambda<-\log\alpha$. The comparison lemma with weights $e^{-\lambda\,\mathrm{dist}(\cdot,T)}$ gives

$$|\mu_\Lambda^F(O^F)-\mu_F(O^F)|\le L\,\frac{\theta D}{1-\theta}\sum_{z\in S}e^{-\lambda\,\mathrm{dist}(z,T)},$$

with $\theta=\alpha e^{\lambda}<1$ and L a Lipschitz constant of O^F; the sum is at most $C_T R^3 e^{-\lambda R}$ when R is the distance from T to the complement of Λ. For O continuous but not Lipschitz, approximate uniformly. Lemma 10 gives $W_\Lambda(O)=\mu_\Lambda^F(O^F)$.

(iii) $W_\Lambda(\tau_v O)=W_{\Lambda-v}(O)$, and the boxes $\Lambda-v$ also exhaust $\mathbb{Z}^4$; by (ii) the limit does not depend on the sequence. The line field is a device for computing W_Λ and is never shifted. The same argument applies to the hypercubic group.

(iv) For $\beta\ge 0$ the measure W_Λ is reflection positive for every box symmetric under the reflection [90], and reflection positivity passes to the limit along such boxes, which is ω by (ii).

(v) $\mathrm{Cov}_\omega(O,O')=\mathrm{Cov}_{\mu_F}(O^F,O'^F)$, and setting the links of F to the identity keeps the Lipschitz property. The bound is Corollary 348.

(vi) Let ϑ be a time reflection and τ_t a time translation. For gauge-invariant local O that are Lipschitz in each link, (v) gives $|\omega(\vartheta(\bar O)\tau_t O)-|\omega(O)|^2|\le C_O e^{-\lambda t}$. On the support of O these functions are dense in the gauge-invariant part of $L^2(\omega)$: gauge-invariant functions are functions on the compact quotient of G^E by the gauge group, Lipschitz functions for the quotient metric are dense in L^2 of every finite measure on it, and they lift to gauge-invariant Lipschitz functions. By Proposition 351 the spectrum of the transfer matrix on the complement of the vacuum lies in $[0,e^{-\lambda}]$, for every $\lambda<-\log\alpha$. □

The proposition replaces the torus limits and the uniqueness of the ungauged Gibbs measure used in [19, Thm. F] by the free-boundary limit, which has all the properties the reconstruction of the transfer matrix needs.[11]

IV. SU(2) UP TO ONE THIRD

Proof of Theorem 1. Identify $\mathrm{SU}(2)$ with the unit quaternions and use the chordal metric. For an active link e and a plaquette $p\ni e$, the staple s_p is the product of the other three link variables in the order that makes $\mathrm{Re}(Q_e s_p)=\frac12\mathrm{Tr}\,Q_p$, with links of F equal to one. Since $N\beta\,\mathrm{Re}\,\mathrm{Tr}\,Q_p=4\beta\langle Q_e,\bar s_p\rangle$ in $\mathbb{R}^4$, the conditional law of Q_e is the von Mises–Fisher law $\nu_b(dx)\propto e^{\langle x,b\rangle}dx$ on S^3 with $b=4\beta\sum_{p\ni e}\bar s_p$. Changing a partner e' by δ in the chordal metric changes one staple by an isometric image of that displacement, so b moves by $4|\beta|\delta$, and the sharp quarter (Theorem 364), $W_1(\nu_b,\nu_{b'})\le\frac14|b-b'|$, gives $C_{ee'}\le|\beta|$. By Lemma 9(iv) every row and column has twelve nonzero entries, so $\alpha\le 12|\beta|=3|\beta_W|$. Proposition 11 applies for $|\beta_W|<\frac13$. □

Proof of Theorem 2. For the line field, every staple of an active link contains two active links, so the six staples can be chosen with sum zero while the staple through a given partner e' moves freely. At $b=0$ the derivative of ν_b in the direction h has Kantorovich norm $\frac14|h|$ by Lemma 363(i), so $C_{ee'}=|\beta|$ exactly, and every row sum is $12|\beta|=3|\beta_W|$.

Now let F' be any forest and C' the matrix of the statement. Suppose a positive weight w satisfies $C'w\le\alpha w$. For a box Λ_n of side n let C_n be the principal submatrix on the active links of $E(\Lambda_n)$; then $C_n w\le\alpha w$ on these links, and the Collatz–Wielandt inequality for nonnegative matrices gives $\rho(C_n)\le\alpha$. Since C_n is symmetric, $\rho(C_n)$ is at least the mean row sum. The box has $4n^4-O(n^3)$ links and n^4 sites, and a forest has fewer links than sites, so the number a of active links is at least $3n^4-O(n^3)$. Let a_p be the number of active links of the plaquette p; the box has $P=6n^4-O(n^3)$ plaquettes and $\sum_p a_p=6a-O(n^3)$. Two links share at most one plaquette, so the row sums add up to $|\beta|\sum_p a_p(a_p-1)$, and by convexity

$$\frac{|\beta|}{a}\sum_p a_p(a_p-1)\ge\frac{|\beta|P}{a}\,\bar a(\bar a-1)=6|\beta|(\bar a-1)-o(1),$$

with $\bar a=\sum_p a_p/P=6a/P-o(1)\ge 3-o(1)$. Hence $\alpha\ge 12|\beta|-o(1)$ for every n, and $\alpha\ge 12|\beta|=3|\beta_W|$. □

Without gauge fixing the same count gives eighteen partners and the ceiling $\beta_W=\frac29$ of [19, Prop. 13]; the temporal gauge, a family of lines in one direction, leaves fourteen. The line field is optimal among forests: twelve is the mean number of partners forced by the number of free links, and it is attained at every link.

V. SU(2) PAST ONE THIRD: PLAQUETTE BLOCKS

Throughout this section $G=\mathrm{SU}(2)$ is the group of unit quaternions with the chordal metric $|x-y|$, $\bar x$ is the conjugate, $\langle x,y\rangle=\mathrm{Re}(x\bar y)$ is the inner product of $\mathbb{R}^4$, and dx is the normalised Haar measure on S^3. Left and right multiplication by unit quaternions and conjugation are isometries, and $\mathrm{Re}(xyz)=\mathrm{Re}(yzx)=\langle y,\bar z\bar x\rangle$. Put

$$Z_0(c)=\int_{S^3}e^{\langle c,x\rangle}dx=\sum_{j\ge0}\frac{(|c|^2/4)^j}{j!\,(j+1)!},\qquad c\in\mathbb{R}^4,$$

let $\nu_c(dx)\propto e^{\langle c,x\rangle}dx$ be the von Mises–Fisher law, and write $g=\log Z_0$, so that $\nabla g(c)$ is the mean and $\mathrm{Hess}\,g(c)$ the covariance of ν_c.

[11] On a torus the line field closes up into cycles, and the holonomies along them cannot be removed by a gauge transformation; the argument never uses a torus. The state ω is defined only on gauge-invariant observables, which is all that the Osterwalder–Schrader reconstruction [88–90] uses.

A. The blocks

Let $\mathscr{B}$ be the set of plaquettes at x in the plane $\{0,1\}$ with $x_0 \equiv x_1$ and in the plane $\{2,3\}$ with $x_2 \equiv x_3$, congruences modulo two (Table IV of Appendix C lists them modulo $2\mathbb{Z}^4$, and Fig. 5 draws them). A *block plaquette* is a plaquette of $\mathscr{B}$, and its *block* is the set of its three active links.

Lemma 12. *The set $\mathscr{B}$ is invariant under translations by $2\mathbb{Z}^4$ and:*

(i) *the blocks partition the active links;*

(ii) *every plaquette $p \notin \mathscr{B}$ meets three distinct blocks, one in each of its active links;*

(iii) *every active link lies in exactly five plaquettes outside $\mathscr{B}$, so it shares a plaquette outside $\mathscr{B}$ with exactly ten active links, none of them in its own block;*

(iv) *if an active link j and a link of a block B lie in a common plaquette outside $\mathscr{B}$, then every link of B has its midpoint within distance 2 of the midpoint of j.*

Proof. Invariance under $2\mathbb{Z}^4$ is clear from the congruences. We use one fact throughout: two distinct links lie in at most one common plaquette, since they span it.

Every link lies in exactly one plaquette of $\mathscr{B}$. Indeed, a link (y,μ) with $\mu \in \{0,1\}$ lies in no plaquette of the plane $\{2,3\}$, and in exactly two of the plane $\{0,1\}$, those at y and at $y - e_\nu$, where ν is the other element of $\{0,1\}$. The base points y and $y - e_\nu$ differ only in the coordinate ν, so $y_0 - y_1$ has opposite parities at the two, and exactly one of them satisfies $x_0 \equiv x_1$. The directions $2,3$ are treated in the same way. Every plaquette of $\mathbb{Z}^4$ contains exactly one link of F (Lemma 8), so a block plaquette contains three active links, and every active link lies in exactly one block. This is (i).

(ii) Let $p \notin \mathscr{B}$ have active links e_1, e_2, e_3, and let q_i be the block plaquette of e_i. Then $q_i \neq p$, so $q_i \cap p = \{e_i\}$ by the fact above. If $q_i = q_k$ with $i \neq k$, then e_i, e_k lie in the two distinct plaquettes p and q_i, which is impossible. So the three blocks are distinct, and each meets p in one link.

(iii) An active link e lies in six plaquettes (Lemma 9), and by (i) exactly one of them is in $\mathscr{B}$. Each of the other five contains two further active links. These ten links are distinct, because a link other than e shares at most one plaquette with e, and none lies in the block of e, because a link f of that block sharing the plaquette $p \notin \mathscr{B}$ with e would put e, f in the two distinct plaquettes p and the block plaquette of e.

(iv) Let $f \in B$ share the plaquette $p \notin \mathscr{B}$ with j. Two links of one unit square have midpoints at distance $2^{-1/2}$ or 1, so the midpoint of j is within distance 1 of that of f, and the midpoint of f is within distance 1 of that of every link of its block plaquette. The triangle inequality gives the bound 2. □

Fix a block plaquette p with active links e_1, e_2, e_3. Reading the plaquette product cyclically and deleting its fixed link, $\frac{1}{2}\operatorname{Tr} Q_p = \operatorname{Re}(y_1 y_2 y_3)$, where y_i is Q_{e_i} or $Q_{e_i}^{-1}$. Every other plaquette containing e_i meets the block only in e_i by Lemma 12(ii) and contributes $4\beta\langle y_i, \bar{s}\rangle$ to the exponent, where s is the staple of y_i in it. Hence the conditional law of the block given all other active links is

$$\mu_b(dy) \propto \exp\Big(\sum_{i=1}^{3} \langle y_i, b_i\rangle + 4\beta \operatorname{Re}(y_1 y_2 y_3)\Big)\, dy_1\, dy_2\, dy_3, \quad (2)$$

with b_i the sum of $4\beta\bar{s}$ over the five staples, so that $|b_i| \le 20|\beta|$. Inversion is an isometry of S^3 and preserves Haar measure, so we may work with y instead of Q. On $(S^3)^3$ we use the metric $d(y,y') = \sum_i |y_i - y_i'|$.

B. The sensitivity of one block

Let λ_* and τ be a lower and an upper bound for the eigenvalues of $\operatorname{Hess} g(c)$ on the ball $|c| \le 24|\beta|$, and M an upper bound there for the norm of the third derivative $D^3 g(c)$ as a trilinear form. Put

$$\begin{aligned} w &= 8\beta^2(\tau - \lambda_*), \qquad s = \tfrac{1}{4} + (e^w - 1)e^w\sqrt{\tau/3}, \\ C &= 4|\beta| sA + \tfrac{320}{3}|\beta|^3 M e^w, \qquad K = \frac{(1+|\beta|)\,s}{1-C}, \end{aligned} \quad (3)$$

assuming $C < 1$, where A is an upper bound for $|\nabla g(c)|$ on the ball $|c| \le 20|\beta|$.

Lemma 13. *Let ν be a probability measure and $\lambda = e^R\nu/\nu(e^R)$ with $\operatorname{osc} R \le \omega$. For bounded f, h,*

$$\operatorname{Cov}_\lambda(f,h) \le \operatorname{Cov}_\nu(f,h) + (e^\omega - 1)e^\omega \operatorname{sd}_\nu(f)\operatorname{sd}_\nu(h).$$

Proof. We may assume $0 \le R \le \omega$. The density $\rho = e^R/\nu(e^R)$ lies in $[e^{-\omega}, e^\omega]$, so $\delta = \|\rho - 1\|_\infty \le e^\omega - 1$. With $\tilde{f} = f - \nu f$ and $\tilde{h} = h - \nu h$,

$$\begin{aligned} \operatorname{Cov}_\lambda(f,h) - \operatorname{Cov}_\nu(f,h) &= \nu\big((\rho-1)\tilde{f}\tilde{h}\big) \\ &\quad - \nu\big((\rho-1)\tilde{f}\big)\,\nu\big((\rho-1)\tilde{h}\big), \end{aligned}$$

and the right side is at most $(\delta + \delta^2)\operatorname{sd}_\nu(f)\operatorname{sd}_\nu(h)$ by the Cauchy–Schwarz inequality. □

Lemma 14. *Let $\beta \ge 0$, $a, c_0 \in \mathbb{R}^4$ with $|a|, |c_0| \le 20\beta$, and*

$$\kappa(dx) \propto \exp\big(\langle x, a\rangle + g(c_0 + 4\beta\bar{x})\big)\,dx \quad \text{on } S^3.$$

For every 1-Lipschitz f and every $h \in \mathbb{R}^4$:

(i) $\operatorname{Cov}_\kappa(f, \langle x, h\rangle) \le s|h|$;

(ii) *if κ' is the law with c_0' in place of c_0 and $|c_0'| = |c_0| = \rho$, then $|\kappa(f) - \kappa'(f)| \le (8\beta s\varphi'(\rho^2) + \frac{16}{3}\beta^2 M e^w)|c_0 - c_0'|$.*

Proof. (i) Taylor's formula at c_0 gives $g(c_0 + 4\beta u) = g(c_0) + 4\beta\langle \nabla g(c_0), u\rangle + R(u)$ with $R(u) = 8\beta^2 \operatorname{Hess} g(\xi)(u,u)$ for some ξ on the segment, $|\xi| \le 24\beta$. For $u = \bar{x}$ of norm one, $R(\bar{x}) \in [8\beta^2\lambda_*, 8\beta^2\tau]$, so $\operatorname{osc} R \le w$. Since $\langle \nabla g(c_0), \bar{x}\rangle = \langle x, \overline{\nabla g(c_0)}\rangle$, κ is the tilt $e^R\nu/\nu(e^R)$ of $\nu = \nu_c$ with $c = a + 4\beta\overline{\nabla g(c_0)}$. The sharp quarter, Theorem 364, gives $\operatorname{Cov}_\nu(f, \langle x, h\rangle) = \frac{d}{dt}\nu_{c+th}(f) \le \frac{1}{4}|h|$. A function that is 1-Lipschitz for the chordal metric is 1-Lipschitz for the geodesic

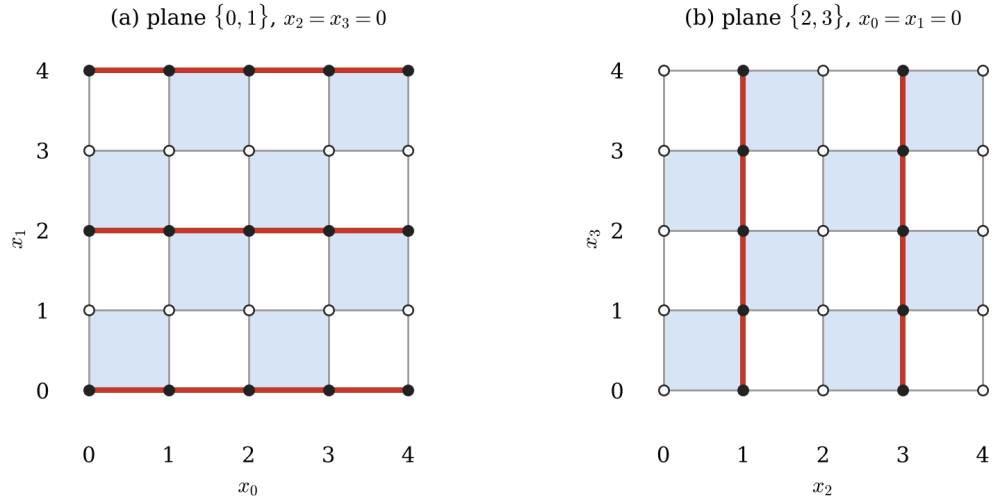


FIG. 5. Block plaquettes, shaded, in (a) a plane $\{0,1\}$ and (b) a plane $\{2,3\}$: the plaquettes at x with $x_0 \equiv x_1$, respectively $x_2 \equiv x_3$, modulo two. Red links belong to the line field F; a filled dot marks a site whose line runs in the plane, an open dot one whose line leaves it. The shaded squares form a checkerboard, so every link of the plane borders exactly one of them, and every square contains exactly one red link (Lemma 8); hence each block plaquette carries three active links and the blocks partition the active links (Lemma 12(i)).

one, so $|\nabla f| \le 1$ almost everywhere and the Poincaré inequality of Lemma 361 gives $\mathrm{sd}_\nu(f) \le 1/\sqrt{3}$; and $\mathrm{sd}_\nu\langle x,h\rangle \le \sqrt{\tau}|h|$ because $|c| \le |a| + 4\beta \le 24\beta$. Lemma 13 gives (i). The same argument applies to every law $e^{R'}\nu_{c'}/\nu_{c'}(e^{R'})$ with $|c'| \le 24\beta$ and $\operatorname{osc} R' \le w$.

(ii) We interpolate the exponents rather than the field. For $t \in [0,1]$ let $\kappa_t \propto \exp\big(\langle x,a\rangle + (1-t)g(c_0 + 4\beta\bar{x}) + tg(c_0' + 4\beta\bar{x})\big)dx$, so that $\kappa_0 = \kappa$, $\kappa_1 = \kappa'$ and $\frac{d}{dt}\kappa_t(f) = \mathrm{Cov}_{\kappa_t}(f,D)$ with $D(x) = g(c_0' + 4\beta\bar{x}) - g(c_0 + 4\beta\bar{x})$. Write $g(c + 4\beta u) = g(c) + 4\beta\langle \nabla g(c), u\rangle + R_c(u)$ as in (i), for $c \in \{c_0, c_0'\}$. Then κ_t is the tilt of ν_{c_t}, $c_t = a + 4\beta\,\overline{(1-t)\nabla g(c_0) + t\nabla g(c_0')}$, by $R_t = (1-t)R_{c_0} + tR_{c_0'}$, with $\operatorname{osc} R_t \le w$ and $|c_t| \le 24\beta$ because $|\nabla g| \le 1$; so (i) holds for κ_t, and the density of κ_t with respect to ν_{c_t} is at most e^w. Up to a constant, $D(x) = \langle x,h\rangle + N(\bar{x})$ with $h = 4\beta\,\overline{\nabla g(c_0') - \nabla g(c_0)}$ and $N = R_{c_0'} - R_{c_0}$. Since $\nabla g(c) = 2\varphi'(|c|^2)c$ and $|c_0| = |c_0'| = \rho$, $|h| = 8\beta\varphi'(\rho^2)|c_0 - c_0'|$, and by (i) the linear part contributes at most $s|h|$. For the rest, $\nabla N(u) = 16\beta^2 \int_0^1 \big(\mathrm{Hess}\, g(c_0' + 4\beta\sigma u) - \mathrm{Hess}\, g(c_0 + 4\beta\sigma u)\big)u\,d\sigma$, both arguments in the convex ball $|c| \le 24\beta$, so N is L-Lipschitz on the unit ball with $L = 16\beta^2 M|c_0 - c_0'|$. With $\nu = \nu_{c_t}$ and the Poincaré inequality of Lemma 361 again, $\mathrm{Var}_{\kappa_t}(N) \le \kappa_t((N - \nu N)^2) \le e^w \mathrm{Var}_\nu(N) \le e^w L^2/3$, and in the same way $\mathrm{Var}_{\kappa_t}(f) \le e^w/3$. So $\mathrm{Cov}_{\kappa_t}(f,N) \le e^w L/3$. Integrating over t gives (ii). □

Lemma 15. *Let $C < 1$. If b and b' satisfy $|b_i|, |b_i'| \le 20|\beta|$ and differ only in one coordinate b_i, then*

$$W_1(\mu_b, \mu_{b'}) \le K\,|b_i - b_i'|$$

in the metric d.

Proof. The map $y_1 \mapsto -y_1$ turns μ_b for β into the law for $-\beta$ with b_1 replaced by $-b_1$ and is an isometry, so we may take $\beta \ge 0$; the cyclic symmetry of $\mathrm{Re}(y_1y_2y_3)$ lets us take $i = 1$. Let f be 1-Lipschitz for d and $\varepsilon = |b_1 - b_1'|$.

The middle link. Given y_1, y_3, the law of y_2 is ν_c with $c = b_2 + 4\beta\,\overline{y_3y_1}$, which does not involve b_1. Changing y_1 by η moves c by $4\beta\eta$, and the sharp quarter moves ν_c by at most $\beta\eta$ in W_1. Hence $F(y_1,y_3) = \mathbb{E}[f \mid y_1, y_3]$ is $(1+\beta)$-Lipschitz in each variable, and $\mu_b(f) = \mu_b^{13}(F)$, where μ_b^{13} is the law of (y_1,y_3):

$$\mu_b^{13} \propto e^{\langle y_1,b_1\rangle + \langle y_3,b_3\rangle} Z_0\big(b_2 + 4\beta\,\overline{y_3y_1}\big).$$

Since $|b_2 + 4\beta\overline{y_3y_1}| = |b_2y_3 + 4\beta\bar{y}_1| = |y_1b_2 + 4\beta\bar{y}_3|$, its conditional laws are

$$\kappa_1(dy_1 \mid y_3) \propto e^{\langle y_1,b_1\rangle + g(b_2y_3 + 4\beta\bar{y}_1)}dy_1,$$
$$\kappa_3(dy_3 \mid y_1) \propto e^{\langle y_3,b_3\rangle + g(y_1b_2 + 4\beta\bar{y}_3)}dy_3,$$

both of the form of Lemma 14, with $|c_0| = |b_2|$.

Two sites. Integrating Lemma 14(i) along the segment from b_1 to b_1' gives $W_1(\kappa_1, \kappa_1') \le s\varepsilon$ for the same y_3. Moving y_3 to y_3' replaces $c_0 = b_2y_3$ by $c_0' = b_2y_3'$, of the same norm $\rho = |b_2| \le 20\beta$ and with $|c_0 - c_0'| = \rho|y_3 - y_3'|$. Since $\rho \cdot 2\varphi'(\rho^2) = |\nabla g(c_0)| \le A$, Lemma 14(ii) gives $W_1(\kappa_1(\cdot \mid y_3), \kappa_1(\cdot \mid y_3')) \le (4\beta sA + 20\beta \cdot \frac{16}{3}\beta^2 Me^w)|y_3 - y_3'| = C|y_3 - y_3'|$, and the same for κ_3, which does not depend on b_1. Let π be a coupling of μ_b^{13} and $\mu_{b'}^{13}$ minimising $d_1 + d_3$, $d_k = \mathbb{E}_\pi|y_k - y_k'|$. Replacing (y_1, y_1') by an optimal coupling of $\kappa_1(\cdot \mid y_3)$ and $\kappa_1'(\cdot \mid y_3')$ keeps a coupling, so minimality gives $d_1 \le s\varepsilon + Cd_3$; in the same way $d_3 \le Cd_1$. Hence $d_1 \le s\varepsilon/(1-C^2)$ and

$$|\mu_b(f) - \mu_{b'}(f)| \le (1+\beta)(d_1 + d_3) \le \frac{(1+\beta)s}{1-C}\,\varepsilon = K\varepsilon. \quad \square$$

C. The constants

Lemma 16. *Write $g(c)=\varphi(|c|^2)$ and $r=|c|$. For $c\neq 0$ the eigenvalues of* Hess $g(c)$ *are $2\varphi'(r^2)$, three times, and $2\varphi'(r^2)+4r^2\varphi''(r^2)$, and the norm of $D^3g(c)$ is*

$$\max\Big(|P+\Gamma|,\tfrac{2}{3}|P|\Big),\qquad P=12r\varphi''(r^2),\quad \Gamma=8r^3\varphi'''(r^2).$$

On the ball $r\le 2.088$ every eigenvalue of Hess g *lies between* 0.15435 *and $\frac14$, and the norm of D^3g is at most* 0.1071*; on the ball $r\le 1.74$, $|\nabla g|\le 0.38944$.*

Proof. Differentiating $\varphi(|c|^2)$ gives $D^2g(v,v)=2\varphi'|v|^2+4\varphi''\langle c,v\rangle^2$ and $D^3g(v,v,v)=12\varphi''\langle c,v\rangle|v|^2+8\varphi'''\langle c,v\rangle^3$. For $|v|=1$ and $t=\langle c/r,v\rangle\in[-1,1]$ the cubic form is $Pt+\Gamma t^3$, whose largest absolute value is attained at $t=\pm1$ or at an interior critical point, where it equals $\frac23|P||t|\le\frac23|P|$; by Banach's theorem [14] the supremum of a symmetric trilinear form on the diagonal of the unit sphere is its norm.

The numbers follow from the Bessel ratio $A=I_2/I_1$ of Lemma 362. Since $Z_0(c)=2I_1(r)/r$, the radial derivative of g is $A(r)$, so $2r\varphi'(r^2)=A(r)$ and $|\nabla g(c)|=A(r)$. Differentiating $2\varphi'(r^2)=A/r$ in r gives $4r\varphi''(r^2)=(rA'-A)/r^2$, so the eigenvalues are A/r, three times, and A', and $P=3(rA'-A)/r^2$; differentiating once more gives $4\varphi''+8r^2\varphi'''=A''/r-2(rA'-A)/r^3$, so $\Gamma=A''-P$ and $P+\Gamma=A''$. Write $\kappa=r$, $u=1+\sqrt{\kappa^2+9}$ and $v=\frac32+\sqrt{\kappa^2+25/4}$, so that $\kappa^2=(u-4)(u+2)=(v-4)(v+1)$. By Lemma 362(ii) and (iv), $\kappa/v\le A\le\kappa/u$, and $A'=1-A^2-3A/\kappa$.

Eigenvalues. By Lemma 362(iii), $A'\le A/\kappa\le\frac14$; at $c=0$, Hess $g=\frac14 I$. The function $1-a^2-3a/\kappa$ decreases in $a\ge0$, so $A'\ge1-\kappa^2/u^2-3/u=(8-u)/u^2$, and $(8-u)/u^2$ decreases in $u\in[4,8]$, hence in κ. At $\kappa=2.088$, $\kappa^2+9=13.359744<3.6551^2$, so $u<4.6551$ and $(8-u)/u^2>3.3449/21.67>0.15435$.

Gradient. $A(r)\le r/(1+\sqrt{r^2+9})$, which increases in r. At $r=1.74$, $r^2+9=12.0276>3.468^2$, so $A\le1.74/4.468<0.38944$.

Third derivative. First, $\frac23|P|=2|A-\kappa A'|/\kappa^2$ and $A-\kappa A'=4A-\kappa+\kappa A^2$, which increases in A. At $A=\kappa/v$ it equals $\kappa(v-4)/v^2\ge0$, and at $A=\kappa/u$ it equals $2\kappa(u-4)/u^2$; since $u-4=\kappa^2/(\sqrt{\kappa^2+9}+3)\le\kappa^2/6$ and $u\ge4$, $0\le\frac23|P|\le4(u-4)/(\kappa u^2)\le\kappa/24\le0.087$. Second, differentiating the Riccati equation gives $-A''=H(A,\kappa)$ with

$$H(a,\kappa)=\Big(1-a^2-\frac{3a}{\kappa}\Big)\Big(2a+\frac{3}{\kappa}\Big)-\frac{3a}{\kappa^2},$$

and $\partial_aH=-(2a+3/\kappa)^2+2(1-a^2-3a/\kappa)-3/\kappa^2\le2-12/\kappa^2<0$ for $a\ge0$ and $\kappa^2<6$. Hence $H(\kappa/u,\kappa)\le-A''\le H(\kappa/v,\kappa)$. Substituting $\kappa^2=(u-4)(u+2)$ and $\kappa^2=(v-4)(v+1)$,

$$H(\kappa/u,\kappa)=\frac{(u-4)(32+6u-2u^2)}{\kappa u^3}\ge0,$$
$$H(\kappa/v,\kappa)=\frac{(5v+8)(v-4)}{\kappa v^3}=\frac{\kappa(5v+8)}{(v+1)v^3},$$

the first because $4\le u<5.7$ for $\kappa\le2.088$, the last by $v-4=\kappa^2/(v+1)$. Call the last expression $G(\kappa)$. As a function of v, $\log G=\frac12\log(v-4)+\log(5v+8)-\frac12\log(v+1)-3\log v$, whose derivative is half of

$$\Big(\frac{1}{v-4}-\frac{6}{v}\Big)+\Big(\frac{10}{5v+8}-\frac{1}{v+1}\Big).$$

Both brackets decrease on $(4,6)$. At $\kappa=2.088$ one has $\kappa^2+\frac{25}{4}=10.609744$, which lies between 3.2572^2 and 3.2573^2, so $v\in[4.7572,4.7573]$ and the brackets are at least 0.059 and 0.140. So G increases on $(0,2.088]$, and

$$0\le-A''\le G(2.088)\le\frac{2.088\,(5\cdot4.7573+8)}{5.7572\cdot4.7572^3}\le\frac{66.371}{619.81}<0.1071.$$

The norm of D^3g is therefore at most $\max(0.1071,0.087)=0.1071$ on the ball $r\le2.088$. □

D. The proof of Theorem 3

Let $\gamma^{\mathscr B}$ be the block specification: for a block B, $\gamma^{\mathscr B}_B$ resamples the three links of B from (2). It comes from the same interaction as γ^F, so the two specifications have the same Gibbs measures. Define the block coefficients

$$C_{Bj}=\sup\frac{W_1\big(\gamma^{\mathscr B}_B(\cdot\mid Q),\gamma^{\mathscr B}_B(\cdot\mid Q')\big)}{|Q_j-Q'_j|}$$

for an active link $j\notin B$, the supremum over pairs of configurations that differ only at j, with W_1 for the metric d.

Proposition 17. *Suppose that $\sum_B C_{Bj}\le\alpha<1$ for every active link j. Then the conclusions (i)–(vi) of Proposition 11 hold with $-\frac12\log\alpha$ in place of $-\log\alpha$.*

Proof. The proof of Proposition 11 goes through with Lemma 349 of Appendix B in place of Lemma 347, and with the set S of part (ii), and the support of H in Corollary 348, enlarged to the union of the blocks that meet it. The enlargement multiplies the size of these sets by at most three and moves them by at most one, which changes only the constants. □

Proof of Theorem 3. Let $|\beta|\le0.348/4=0.087$ and take the constants of Lemma 16, whose balls $r\le2.088$ and $r\le1.74$ contain $|c|\le24|\beta|$ and $|c|\le20|\beta|$: $\lambda_*=0.15435$, $\tau=\frac14$, $M=0.1071$ and $A=0.38944$. With these fixed constants every quantity in (3) is increasing in $|\beta|$, and so is

$$\alpha(\beta_W)=40|\beta|K. \tag{4}$$

Changing an active link j by η changes one staple of each of its ten partners outside its block by an isometric image of the displacement, so the field b_i of each such partner i moves by at most $4|\beta|\eta$, and Lemma 15, applied once for each of these ten fields, gives $\sum_B C_{Bj}\le40|\beta|K$. At $|\beta|=0.087$, $8\beta^2=0.060552$ and $\tau-\lambda_*=0.09565$, so $w\le0.0057918$. Since

$e^w \le 1+w+w^2$ for $0 \le w \le 1$ and $\sqrt{1/12} \le 0.288676$, $e^w \le 1.0058254$ and $s \le \frac{1}{4} + 0.0058594 \cdot 0.288676 \le 0.2516915$. Then

$$\begin{aligned} C &\le 0.348 \cdot 0.2516915 \cdot 0.38944 \\ &\quad + \tfrac{320}{3}\, 0.087^3 \cdot 0.1071 \cdot 1.0058254 \\ &\le 0.0341107 + 0.0075670 \le 0.0416777, \end{aligned}$$

$K \le 1.087 \cdot 0.2516915/0.9583223 \le 0.285490$ and $\alpha \le 3.48 \cdot 0.285490 \le 0.99351$. The same evaluation at $|\beta| = 0.345/4$ gives $w \le 0.0056924$, $s \le 0.2516621$, $C \le 0.041185$, $K \le 0.285110$ and $\alpha \le 0.98363$, and at $|\beta| = \frac{1}{12}$ it gives $w \le 0.0053139$, $s \le 0.2515504$, $C \le 0.0393011$, $K \le 0.2836611$ and $\alpha \le 0.94554$. Proposition 17 applies. At $\beta_W = \frac{1}{3}$, $-\frac{1}{2}\log\alpha \ge \frac{1}{2}(y + \frac{y^2}{2})$ with $y = 1 - 0.9456 = 0.0544$, which is at least 0.0279. □

Remark 18. Most of the range is kept. A second bound for the same sensitivity, which keeps the law of the middle link instead of linearising it, evaluated in floating point on a quadrature grid and maximised over the fields b by a random search followed by the Nelder–Mead method, gives 0.2783 at $\beta_W = 0.34$, and $40|\beta|$ times it crosses one near $\beta_W = 0.357$; this evaluation is an illustration and no result uses it. No bound of this form passes $\beta_W = 0.3667$: at $b = 0$ the function $f = \langle y_1, e\rangle + \mathrm{Re}(y_2 y_3 e)$ is 1-Lipschitz for d and has $\mathrm{Cov}_{\mu_b}(f, \langle y_1, e\rangle) = \frac{1}{4}(1 + I_2(\beta_W)/I_1(\beta_W))$ exactly [21, Proposition 458], which is at least 0.2718 at $\beta_W = 0.351$, so the sensitivity is at least this number and $40|\beta|$ times it exceeds one for $\beta_W > 0.3667$. Past this point the gain from integrating out one link is spent, and larger blocks are needed, such as the free links of a cube of the line field, whose sensitivity no longer reduces to two sites.

VI. CURVATURE IN THE GAUGE

In this section $G \subset \mathrm{U}(N)$ is a closed connected subgroup with Lie algebra $\mathfrak{g}$ and the inner product $\langle X, Y\rangle = -\mathrm{Tr}(XY)$; the bound of Theorem 4 is proved first for $\mathrm{U}(N)$. A tangent vector at a field Q is $v = (X_e Q_e)$, $|v|^2 = \sum_e |X_e|^2$. For the plaquette at x in the plane $\mu < \nu$ number its links $e_1 = (x, \mu)$, $e_2 = (x + e_\mu, \nu)$, $e_3 = (x + e_\nu, \mu)$, $e_4 = (x, \nu)$, transport their vectors to x,

$$Y_1 = X_{e_1}, \quad Y_2 = Q_{e_1} X_{e_2} Q_{e_1}^*, \quad Y_3 = Q_{e_4} X_{e_3} Q_{e_4}^*, \quad Y_4 = X_{e_4}, \tag{5}$$

and write $W = Q_p$. The second derivative of $\mathrm{Re}\,\mathrm{Tr}\, Q_p$ along the geodesic $Q_e(t) = e^{tX_e} Q_e$ is

$$\begin{aligned} f_p = &\sum_{i=1}^{4} \mathrm{Re}\,\mathrm{Tr}(Y_i^2 W) + 2\,\mathrm{Re}\,\mathrm{Tr}(Y_1 Y_2 W) + 2\,\mathrm{Re}\,\mathrm{Tr}(Y_3 Y_4 W) \\ &- 2 \sum_{i \in \{1,2\},\, j \in \{3,4\}} \mathrm{Re}\,\mathrm{Tr}(Y_i W Y_j), \end{aligned} \tag{6}$$

so that $\mathrm{Hess}\, S_0(v,v) = \sum_p f_p$. To see this, write $Q_p = Q_{e_1} Q_{e_2} Q_{e_3}^* Q_{e_4}^*$. Since $Q_{e_1} e^{tX_{e_2}} = e^{tY_2} Q_{e_1}$ and $e^{-tX_{e_3}} Q_{e_4}^* = Q_{e_4}^* e^{-tY_3}$, the plaquette variable along the geodesic is $Q_p(t) = e^{tY_1} e^{tY_2} W e^{-tY_3} e^{-tY_4}$. Differentiating twice at $t = 0$ gives the four squares, the two products of neighbouring factors and the four products across W, and the squares and the product WY_3Y_4 are moved cyclically under the trace. The corner form of p is

$$c_p = 2\langle Y_1, Y_4\rangle - 2\langle Y_1, Y_2\rangle + 2\langle Y_2, W Y_3 W^*\rangle - 2\langle Y_3, Y_4\rangle. \tag{7}$$

It collects the inner products of the transported vectors of two links that meet at a right angle at a corner of p.

Lemma 19. *On $\mathbb{Z}^4$, or on a box with free boundary, for every field Q and every square-summable X,*

$$\begin{aligned} &\sum_y \Big[|D_y X|^2 + \sum_\mu \big| X_{y,\mu} + Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu} \big|^2 \Big] \\ &\qquad = 4|X|^2 + \sum_p c_p, \end{aligned}$$

where $D_y X = \sum_\mu [X_{y,\mu} - Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu}]$ is the covariant divergence and terms with a link outside the box are omitted.

Proof. At a site y write o_μ for the vector of the outgoing link in direction μ and i_μ for the transported vector of the incoming one, so that $D_y X = \sum_\mu (o_\mu - i_\mu)$. Every link is outgoing at one end and incoming at the other, so the squares $|o_\mu|^2 + |i_\mu|^2$, which appear once in $|D_y X|^2$ and once in $|o_\mu + i_\mu|^2$, add up to $4|X|^2$. The collinear cross term $-2\langle o_\mu, i_\mu\rangle$ of $|D_y X|^2$ cancels the term $+2\langle o_\mu, i_\mu\rangle$ of $|o_\mu + i_\mu|^2$. The remaining cross terms of $|D_y X|^2$ pair two perpendicular ends at y, and each such pair is a corner of exactly one plaquette, with the sign and the transport of (7) (this is the proof of [19, Lemma 1]). In a box two perpendicular ends at y span a plaquette of the box, because a box is a product of intervals. □

Lemma 20. *Let $k \in \{1,2,3,4\}$, let $k \pm 1$ (mod 4) be the positions adjacent to k and $k+2$ the opposite one. For every N, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$ with $Y_k = 0$,*

$$\pm f_p + \tfrac{9}{10} c_p \le \tfrac{21}{10}\big(|Y_{k-1}|^2 + |Y_{k+1}|^2\big) + \tfrac{5}{4}|Y_{k+2}|^2.$$

Proof. Replacing W by $-W$ changes the sign of f_p and fixes c_p, so the sign $+$ suffices. Conjugation by a unitary matrix preserves both sides, so we may take $W = \mathrm{diag}(w_a)$. The form $\mathscr{F} = \mathrm{RHS} - f_p - \frac{9}{10} c_p$ then splits into a block for each diagonal position a, in the real variables $y_i = -i(Y_i)_{aa}$, and a block for each pair $a \ne b$, in the complex variables $z_i = (Y_i)_{ab}$, since $(Y_i)_{ba} = -\bar{z}_i$. Appendix D shows that every block is positive definite. □

Proof of the bound in Theorem 4. Every plaquette has exactly one link of F, and X vanishes on it. By Lemma 9(iii) each active link is adjacent to the fixed link in four of its plaquettes and opposite to it in two. Summing Lemma 20 over the plaquettes,

$$\sum_p \big(\pm f_p + \tfrac{9}{10} c_p\big) \le \big(4 \cdot \tfrac{21}{10} + 2 \cdot \tfrac{5}{4}\big)|X|^2 = \tfrac{109}{10}|X|^2.$$

By Lemma 19, $\frac{9}{10}\sum_p c_p = \frac{9}{10}\Sigma - \frac{36}{10}|X|^2$, where $\Sigma \geq 0$ is the left side of the lemma. Hence

$$\pm \operatorname{Hess} S_0(v,v) + \tfrac{9}{10}\Sigma \leq \big(\tfrac{109}{10} + \tfrac{36}{10}\big)|X|^2 = \tfrac{29}{2}|X|^2.$$

In a box, links near the boundary lie in fewer plaquettes and receive less. □

The factor $\frac{9}{10}$ is chosen so that the corners are paid by the divergence identity at the price $\frac{36}{10}$ while the local blocks stay positive; Lemma 19 is the divergence identity of [19, Lemma 1] with the collinear term written as a square, so nothing is lost by Cauchy–Schwarz.

The two lemmas concern vectors in $\mathfrak{u}(N)$ and fields in $\mathrm{U}(N)$, so the bound holds without change on every closed subgroup $G \subset \mathrm{U}(N)$, the vectors X_e ranging over its Lie algebra $\mathfrak{g}$. It is also the Hessian of S_0 on G^E: the metric is bi-invariant, the geodesics of G through Q_e are the curves $e^{tX}Q_e$ with $X \in \mathfrak{g}$, which are geodesics of $\mathrm{U}(N)$, and the Hessian is read off from second derivatives along geodesics.

Lemma 21. *Let $G \subset \mathrm{U}(N)$ be closed and connected with simple Lie algebra $\mathfrak{g}$, let $\mathfrak{t} \subset \mathfrak{g}$ be a maximal abelian subalgebra, and write the eigenvalues of $\operatorname{ad} H$ on $\mathfrak{g}\otimes\mathbb{C}$ and of H on $\mathbb{C}^N$, for $H \in \mathfrak{t}$, as $i\alpha(H)$ and $i\mu(H)$, with α running over the roots and μ over the weights of $\mathbb{C}^N$ counted with multiplicity. Then $\mathrm{Ric}_G = \kappa_G\langle\cdot,\cdot\rangle$ with*

$$\kappa_G = \frac{\sum_\alpha \alpha(H)^2}{4\sum_\mu \mu(H)^2} \qquad (0 \neq H \in \mathfrak{t}).$$

Proof. For a bi-invariant metric the curvature is $R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$ on left-invariant fields [84, Sect. 7], so for an orthonormal basis (E_i) of $\mathfrak{g}$,

$$\begin{aligned}\mathrm{Ric}(X,X) &= \sum_i \langle R(E_i,X)X,E_i\rangle \\ &= \tfrac{1}{4}\sum_i |[X,E_i]|^2 = -\tfrac{1}{4}\operatorname{Tr}(\operatorname{ad}X)^2,\end{aligned}$$

since $\operatorname{ad}X$ is skew. The Killing form $B(X,Y) = \operatorname{Tr}(\operatorname{ad}X\operatorname{ad}Y)$ and $\langle\cdot,\cdot\rangle$ are invariant under the adjoint action, and $\mathfrak{g}$ is simple, so $B = -4\kappa\langle\cdot,\cdot\rangle$ for a constant κ and $\mathrm{Ric} = \kappa\langle\cdot,\cdot\rangle$. On $H \in \mathfrak{t}$ the two forms are $B(H,H) = -\sum_\alpha \alpha(H)^2$ and $\langle H,H\rangle = -\operatorname{Tr}H^2 = \sum_\mu \mu(H)^2$. □

Table I evaluates the ratio of Lemma 21 for every compact simple Lie algebra and a faithful representation of least dimension, on the root systems of [67, Sect. 12.1]; the weights of the representations of dimensions 27 and 56 of E_6 and E_7 are the roots of E_8 with prescribed inner products with a subsystem A_2, respectively A_1, projected to its orthogonal complement, by the decompositions $248 = (78,1)+(27,3)+(\overline{27},\bar{3})+(1,8)$ and $248 = (133,1)+(56,2)+(1,3)$ [97]. Taking for H the coroot $\theta^\vee$ of the highest root gives $\sum_\alpha \alpha(\theta^\vee)^2 = 4h^\vee$ and $\sum_\mu \mu(\theta^\vee)^2 = 2\ell_\rho$, so $\kappa_G = h^\vee/(2\ell_\rho)$, with the Dynkin index normalised to one on the defining representation of $\mathrm{SU}(N)$. The ratio depends on $\mathfrak{g}$ and ρ only, so it is the same for every compact group with Lie algebra $\mathfrak{g}$, simply connected or not.

Lemma 22. *Let $e \neq e'$ be two links of a plaquette p, and write*

$$\operatorname{Hess}\operatorname{Re}\operatorname{Tr} Q_p(v,v) = \sum_{e,e'\in p} H^p_{ee'}(X_e,X_{e'})$$

with bilinear forms satisfying $H^p_{ee'}(a,b) = H^p_{e'e}(b,a)$. Then $|H^p_{ee'}(a,b)| \leq |a|\,|b|$ for all $a,b \in \mathfrak{u}(N)$ and every field.

Proof. By (6) the terms of f_p that mix two positions $i \neq j$ are $2\operatorname{Re}\operatorname{Tr}(Y_1Y_2W)$, $2\operatorname{Re}\operatorname{Tr}(Y_3Y_4W)$ and $-2\operatorname{Re}\operatorname{Tr}(Y_iWY_j)$, so $H^p_{ee'}$ is $\pm\operatorname{Re}\operatorname{Tr}(abW)$ or $\pm\operatorname{Re}\operatorname{Tr}(aWb)$ after an isometric transport of a and b, and $|\operatorname{Tr}(abW)| \leq \|a\|_{\mathrm{HS}}\|bW\|_{\mathrm{HS}} = |a|\,|b|$. □

Corollary 23. *Let $G \subset \mathrm{U}(N)$ be closed and connected with semisimple Lie algebra and $N|\beta| < \frac{2}{29}\kappa_G$. For every box the measure μ^F_Λ satisfies the Bakry–Émery condition with $K = \kappa_G - \frac{29}{2}N|\beta| > 0$, uniformly in the box, and hence a logarithmic Sobolev inequality with constant $2/K$ and a Poincaré inequality with constant $1/K$.*

Proof. The set $\{Q_F = 1\}$ is a product factor, so its Ricci curvature is at least κ_G on every link, and the Hessian of the restricted action is the restriction of $N\beta\operatorname{Hess}S_0$ to $X_F = 0$. Apply the bound and [5, Cor. 5.7.2]. □

A. Decay with an explicit rate

Let A' be a finite set of active links and P' a finite set of plaquettes whose active links lie in A'. An action $S = \sum_{p\in P'} c_p \operatorname{Re}\operatorname{Tr} Q^F_p$ with $0 \leq c_p \leq 1$ is *admissible* if $|\operatorname{Hess}S(v,v)| \leq \frac{29}{2}|v|^2$ for all v tangent to $G^{A'}$. Two links lie in at most one common plaquette, so by Lemma 22 and Lemma 9(iv) the mixed blocks of its Hessian satisfy $|\operatorname{Hess}S_{ee'}(a,b)| \leq |a|\,|b|$, and each link has at most twelve partners. The action $S_\Lambda = \sum_{p\in P(\Lambda)} \operatorname{Re}\operatorname{Tr} Q^F_p$ of a box is admissible by the bound proved above, and so is every convex combination of two admissible actions. Let μ be the probability measure on $G^{A'}$ with density proportional to $e^{N\beta S}$ with respect to Haar measure, let $L = \Delta + N\beta\langle\nabla S,\nabla\cdot\rangle$ be its Langevin generator, which is self-adjoint in $L^2(\mu)$, and $P_t = e^{tL}$. Write ∇_e for the gradient in the factor of the link e, and for weights $w : A' \to (0,\infty)$ put $\Gamma^w(f) = \sum_e w_e|\nabla_e f|^2$; Γ is the case $w \equiv 1$.

Lemma 24. *Let $\lambda \geq 0$ and $w_e = e^{\lambda d_e}$, where $|d_e - d_{e'}| \leq 1$ whenever e and e' share a plaquette, and put*

$$K_\lambda = \kappa_G - N|\beta|\Big(\tfrac{29}{2} + 12\big(\cosh\tfrac{\lambda}{2} - 1\big)\Big).$$

Then $\Gamma^w(P_tf) \leq e^{-2K_\lambda t}P_t\Gamma^w(f)$ for every smooth f and $t \geq 0$.

Proof. Put $V = -N\beta S$. On the product manifold $M = G^{A'}$ the projection to the tangent space of one factor is parallel, the rough Laplacian commutes with it, and the Ricci tensor is block diagonal. Hence the Bochner–Weitzenböck formula $\nabla\Delta g = \bar{\Delta}\nabla g + \mathrm{Ric}(\nabla g)$, applied factor by factor, together with

the derivative of the drift, gives for smooth g

$$\begin{aligned}&\tfrac12 L\Gamma^w(g)-\Gamma^w(g,Lg)\\&\quad=\sum_e w_e\big(|\nabla\nabla_e g|^2+\mathrm{Ric}_G(\nabla_e g,\nabla_e g)\big)\\&\qquad+\sum_{e,e'} w_e\,\mathrm{Hess}\,V_{ee'}(\nabla_e g,\nabla_{e'}g),\end{aligned}$$

where $\Gamma^w(g,h)$ is the polarised form.[12] Put $u_e=\nabla_e g$ and $v_e=\sqrt{w_e}\,u_e$. By the symmetry of $\mathrm{Hess}\,V$,

$$\begin{aligned}\sum_{e,e'} w_e\,\mathrm{Hess}\,V_{ee'}(u_e,u_{e'})&=\mathrm{Hess}\,V(v,v)\\&+\sum_{e\neq e'}\Big(\cosh\tfrac{\lambda(d_e-d_{e'})}{2}-1\Big)\,\mathrm{Hess}\,V_{ee'}(v_e,v_{e'}),\end{aligned}$$

because $\frac{w_e+w_{e'}}{2\sqrt{w_e w_{e'}}}=\cosh\frac{\lambda(d_e-d_{e'})}{2}$. The first term is at least $-\frac{29}{2}N|\beta|\,|v|^2$. In the second only partners contribute, each with a coefficient at most $\cosh\frac{\lambda}{2}-1$ and a block of norm at most $N|\beta|$, and $2|v_e|\,|v_{e'}|\le|v_e|^2+|v_{e'}|^2$ with twelve partners per link bounds it by $12(\cosh\frac{\lambda}{2}-1)N|\beta|\,|v|^2$. Since $\mathrm{Ric}_G\ge\kappa_G$ and $|v|^2=\Gamma^w(g)$,

$$\tfrac12 L\Gamma^w(g)-\Gamma^w(g,Lg)\ge K_\lambda\Gamma^w(g).$$

For fixed t let $\varphi(s)=P_s\Gamma^w(P_{t-s}f)$, $0\le s\le t$. Then $\varphi'(s)=P_s\big(L\Gamma^w(g_s)-2\Gamma^w(g_s,Lg_s)\big)\ge 2K_\lambda\varphi(s)$ with $g_s=P_{t-s}f$, and $\varphi(0)\le e^{-2K_\lambda t}\varphi(t)$ is the claim. □

Lemma 25. *Suppose $K_\lambda>0$. Let f and g be smooth functions on $G^{A'}$ that depend only on the links of A_f and of B_g, and let D be the least number of steps from A_f to B_g in the graph on A' whose edges join links that share a plaquette. Then*

$$|\mathrm{Cov}_\mu(f,g)|\le\frac{e^{-\lambda D/2}}{K_\lambda}\,\|\nabla f\|_\infty\sum_{e\in B_g}\|\nabla_e g\|_\infty,$$

where $\|\nabla f\|_\infty^2=\sup\sum_e|\nabla_e f|^2$.

Proof. Since $K_0\ge K_\lambda>0$, Lemma 24 with $\lambda=0$ gives $\Gamma(P_t f)\le e^{-2K_0 t}\|\nabla f\|_\infty^2$, so $P_t f$ converges uniformly to a constant, which is $\mu(f)$ by invariance, M being connected. Since $\frac{d}{dt}\mu(gP_t f)=\mu(gLP_t f)=-\mu(\Gamma(g,P_t f))$,

$$\mathrm{Cov}_\mu(f,g)=\int_0^\infty\mu\big(\Gamma(P_t f,g)\big)\,dt.$$

Let d_e be the smaller of D and the graph distance from e to A_f, so that $|d_e-d_{e'}|\le 1$ for partners, $w_e=1$ on A_f, $w_e=e^{\lambda D}$ on B_g, and $\Gamma^w(f)\le\|\nabla f\|_\infty^2$. For $e\in B_g$, Lemma 24 gives $|\nabla_e P_t f|^2\le w_e^{-1}\Gamma^w(P_t f)\le e^{-\lambda D}e^{-2K_\lambda t}\|\nabla f\|_\infty^2$. Hence $|\Gamma(P_t f,g)|\le e^{-\lambda D/2}e^{-K_\lambda t}\|\nabla f\|_\infty\sum_{e\in B_g}\|\nabla_e g\|_\infty$, and integration in t gives the bound. □

[12] For $w\equiv 1$ this is Bochner's identity $\Gamma_2=|\nabla^2 g|^2+(\mathrm{Ric}+\mathrm{Hess}\,V)(\nabla g,\nabla g)$; the weights enter only through the factor index, which the connection, the Laplacian and the Ricci tensor preserve, so only the off-diagonal blocks of $\mathrm{Hess}\,V$ see them.

Proof of Theorem 5. Let $\lambda^*>0$ solve $K_{\lambda^*}=0$, so that $\frac{\lambda^*}{2}=m_G(\beta)$, and let $0<\lambda<\lambda^*$; then $K_\lambda>0$.

Decay in boxes. Lemma 25 with $S=S_\Lambda$ gives

$$|\mathrm{Cov}_{\mu_\Lambda^F}(f,g)|\le C_{f,g}e^{-\lambda D/2},$$

uniformly in the box.

Limit. For boxes $\Lambda\subset\Lambda'$ let $S_s=(1-s)S_\Lambda+sS_{\Lambda'}$ on the active links of Λ', $s\in[0,1]$, and μ_s the corresponding measure; S_s is admissible, μ_0 is μ_Λ^F times Haar measure on the active links outside Λ, and $\mu_1=\mu_{\Lambda'}^F$. For smooth f of the active links of Λ,

$$\frac{d}{ds}\mu_s(f)=N\beta\sum_{p\in P(\Lambda')\setminus P(\Lambda)}\mathrm{Cov}_{\mu_s}\big(f,\mathrm{Re}\,\mathrm{Tr}\,Q_p^F\big),$$

and $|\nabla_e\,\mathrm{Re}\,\mathrm{Tr}\,Q_p|\le\|Q_p\|_{\mathrm{HS}}=\sqrt{N}$. A step in the graph of Lemma 25 moves a link by at most one lattice unit, so a plaquette at graph distance k from the support T of f lies within distance $k+1$ of T, there are at most $C_T(k+1)^4$ of them, and every plaquette outside $P(\Lambda)$ is at graph distance at least $R-1$, R the distance from T to the complement of Λ. Lemma 25 gives

$$|\mu_{\Lambda'}^F(f)-\mu_\Lambda^F(f)|\le C_f R^4 e^{-\lambda R/2}.$$

The limit therefore exists along every increasing sequence of boxes, and two sequences are compared through their intersections, which are boxes. Continuous gauge-invariant O are uniform limits of smooth gauge-invariant ones, for instance of $e^{\tau\Delta}O$, since the heat semigroup of G^E commutes with left and right translations; with Lemma 10 the limit $\omega(O)$ of $W_\Lambda(O)$ exists for all of them.

Properties. Invariance and, for $\beta\ge 0$, reflection positivity follow as in Proposition 11(iii)–(iv). For Lipschitz O the functions $e^{\tau\Delta}O$ are smooth with the same Lipschitz constant, since $\mathrm{Ric}_G\ge 0$ [5, Sect. 3.3], so the bound of the first step passes to ω and to Lipschitz observables. Two links of one plaquette have time coordinates that differ by at most one, so observables at time separation t are at graph distance at least $t-c$, and truncated correlations decay at every rate below $m_G(\beta)$. The gap follows from Proposition 351 as in Proposition 11(vi). □

Proof of the rest of Theorem 4. By Lemma 21 and Table I, $\kappa_{\mathrm{SU}(N)}=\frac{N}{2}$ and $\kappa_{\mathrm{SO}(N)}=\frac{N-2}{4}$; for $\mathrm{SO}(4)$, whose Lie algebra is $\mathfrak{so}(3)\oplus\mathfrak{so}(3)$ acting on $\mathbb{R}^4\otimes\mathbb{C}=\mathbb{C}^2\otimes\mathbb{C}^2$, each ideal gives $\frac12$ by the same computation. The conditions of Theorem 4 are those of Theorem 5. □

Remark 26. Let h_F be the supremum of $\pm\mathrm{Hess}\,S_0$ on the gauge-fixed directions, so that uniform curvature reaches $|\beta|<1/(2h_F)$ and no further. At the identity field $-\mathrm{Hess}\,S_0$ restricted to $X_F=0$ has the eigenvalue $9+\sqrt{17}=13.123$ on the torus of side two, so $9+\sqrt{17}\le h_F\le\frac{29}{2}$. A gradient search over $\mathrm{SU}(2)$ fields of periods two, four and six finds nothing above 13.97401, the value at an abelian field; certificates made of plaquette forms and vertex forms on the period-two cell cannot go below 14.384. One of us (T. Romanov) [93] constructed such a certificate, with sixteen positive definite vertex

forms of size eight and nonnegative plaquette weights, rational with denominator 10^6. Its positivity is checked by computer, and with it the argument of Appendix N gives $h_F \le 14.420001$ (Remark 377). Like $\frac{29}{2}$ this would be a statement about $\mathrm{U}(N)$, so it could replace $\frac{29}{2}$ in Theorems 4 and 5, and the curvature method would then reach $|\beta| < 1/28.840002$, that is $\beta_W < 0.1387$ for SU(2) and $\beta_W < 0.3121$ for SU(3). This is a computation, used by no result of the paper, and concerns the lattice at fixed spacing, not the continuum limit. The theorems of this paper are stated with $\frac{29}{2}$, which is proved by hand. With this gauge the curvature method reaches $|\beta| < \frac{1}{29}$ and stops before $\frac{1}{26.25}$; the computation moves the first value to $1/28.840002$.

VII. SU(3) UP TO 0.515

In this section $G = \mathrm{SU}(3)$, $\beta \ge 0$ and $\beta_W = 9\beta$. The distance on SU(3) is $d(U,V) = \|U - V\|_{\mathrm{HS}}$, which is at most the geodesic distance for $\langle X, Y\rangle = -\mathrm{Tr}(XY)$; $K = \mathrm{conv}\,\mathrm{SU}(3)$; $A(M) = \frac{1}{2}(M - M^*)$ and P removes the trace. With this inner product $\mathrm{Ric} = \frac{3}{2}$, the Casimir of the fundamental representation is $\lambda_F = \frac{8}{3}$ and that of the adjoint representation is 6.

A. Reduction to one link

The conditional law of an active link is

$$\nu_b(dU) \propto e^{\mathrm{Re}\,\mathrm{Tr}(Ub)}\,dU, \qquad b = N\beta \sum_{p \ni e} S_p,$$

where S_p is the staple of e in p, a product of SU(3) matrices and identities, or its adjoint. Six staples give $b \in rK$ with $r = 6N\beta = 2\beta_W$. For a matrix h put $\varphi_h(U) = \mathrm{Re}\,\mathrm{Tr}(Uh)$ and

$$L_h(b) = \sup\{\mathrm{Cov}_{\nu_b}(g, \varphi_h) : g \text{ 1-Lipschitz for } d\},$$

$$\Lambda(r) = \sup_{b \in rK} \sup_{\|h\|_{\mathrm{HS}}=1} L_h(b).$$

Changing a partner e' changes one staple by a matrix of norm $d(Q_{e'}, Q'_{e'})$. Along the segment b_t from b to b', which stays in rK, $\frac{d}{dt}\nu_{b_t}(g) = \mathrm{Cov}_{\nu_{b_t}}(g, \mathrm{Re}\,\mathrm{Tr}(U(b' - b)))$, so $W_1(\nu_b, \nu_{b'}) \le N\beta\Lambda(r)\, d(Q_{e'}, Q'_{e'})$. With twelve partners per link,

$$\alpha \le 12N\beta\Lambda(r) = 2r\Lambda(r) = 4\beta_W\Lambda(2\beta_W). \tag{8}$$

Since K is convex and contains 0, the mean of $I, \omega I, \omega^2 I$ with $\omega^3 = 1$, $r'K \subset rK$ for $r' \le r$ and Λ is nondecreasing.

B. Gradients and one-forms

Lemma 27. *In the left trivialisation, $\nabla\varphi_c(U) = -PA(cU)$; if $V = \varphi_b$ then $\mathrm{Hess}\,V(X,X) = \mathrm{Re}\,\mathrm{Tr}(UX^2 b)$; and $\Delta\varphi_c = -\lambda_F\varphi_c$.*

Proof. $\frac{d}{dt}\mathrm{Re}\,\mathrm{Tr}(Ue^{tX}c)|_{t=0} = \mathrm{Re}\,\mathrm{Tr}(cUX) = -\langle PA(cU), X\rangle$ for $X \in \mathfrak{su}(3)$. On a group with a bi-invariant metric $\nabla_X X = 0$ for left-invariant X, so the Hessian is the second derivative along e^{tX}. The function φ_c is a matrix coefficient of the fundamental representation, on which the Casimir acts by λ_F. □

Lemma 28. *The bottom of the spectrum of the rough Laplacian $\nabla^*\nabla$ on L^2 one-forms of* SU(3) *is $\mu_0 = \frac{7}{6}$.*

Proof. Trivialise one-forms on the left, so that a one-form is a function $f : \mathrm{SU}(3) \to \mathfrak{su}(3)$. For an orthonormal basis X_a acting as left-invariant vector fields, $\nabla_{X_a} Y = \frac{1}{2}[X_a, Y]$ and $\nabla_{X_a} X_a = 0$, so $\nabla^*\nabla = -\sum_a (\rho(X_a) + \frac{1}{2}\,\mathrm{ad}\,X_a)^2$. On the isotypic part $V_\pi \otimes V_\pi^* \otimes \mathfrak{su}(3)$ of $L^2 \otimes \mathfrak{su}(3)$ put $A_a = \rho_\pi(X_a)$, $B_a = \mathrm{ad}\,X_a$ and $T_a = A_a + B_a$, the action on the tensor product. Since A_a and B_a commute, $A_a + \frac{1}{2}B_a = \frac{1}{2}(A_a + T_a)$ and $A_aT_a + T_aA_a = A_a^2 + T_a^2 - B_a^2$, so on a component $\sigma \subset \pi \otimes \mathrm{ad}$

$$\nabla^*\nabla = \tfrac{1}{2}(C_\pi + C_\sigma) - \tfrac{1}{4}C_{\mathrm{ad}} = \tfrac{1}{2}(C_\pi + C_\sigma) - \tfrac{3}{2},$$

with $C(p,q) = \frac{2}{3}(p^2 + q^2 + pq + 3p + 3q)$ for the representation of highest weight (p,q) [47]. The components of $(p,q) \otimes (1,1)$ are $(p+1, q+1)$, $(p+2, q-1)$, $(p-1, q+2)$, $(p+1, q-2)$, $(p-2, q+1)$, $(p-1, q-1)$ and (p,q), the last with multiplicity two, one or zero as both, one or none of p, q are positive, and only those with nonnegative labels occur. For $\pi = (0,0)$ the only component is $(1,1)$, with value $\frac{3}{2}$. For $\pi = (1,0)$ the components are $(2,1)$, $(0,2)$, $(1,0)$, with values $\frac{31}{6}$, $\frac{19}{6}$, $\frac{7}{6}$, and $(0,1)$ is conjugate. For $p + q \ge 2$ the smallest values occur at $(p-1, q-1)$, (p,q), $(p-2, q+1)$ or $(p+1, q-2)$, and each is at least $\frac{1}{3}(2(p^2 + q^2 + pq) + 3(p+q) - 3) - \frac{3}{2} \ge \frac{3}{2}$. □

C. The gap of the one-link measure

Lemma 29. *For $b \in rK$ the spectral gap of $L = \Delta + \langle \nabla V, \nabla \cdot\rangle$, $V = \varphi_b$, on $L^2(\nu_b)$ is at least*

$$\gamma(r) = \tfrac{39}{64}\mu_0 + \tfrac{3}{2} - \tfrac{r}{2} = \tfrac{283}{128} - \tfrac{r}{2}.$$

Proof. Let $-Lf = \lambda f$ with $\lambda > 0$, $\omega = df$ and $\eta = e^{V/2}\omega$; norms are in $L^2(dU)$. Bochner's formula integrated against ν_b gives $\lambda\nu_b|\omega|^2 = \nu_b|\nabla\omega|^2 + \nu_b((\mathrm{Ric} - \mathrm{Hess}\,V)(\omega,\omega))$, and since $e^{V/2}\nabla\omega = \nabla\eta - \frac{1}{2}dV \otimes \eta$,

$$\lambda\|\eta\|^2 = \|\nabla\eta - \tfrac{1}{2}dV \otimes \eta\|^2 + \int (\mathrm{Ric} - \mathrm{Hess}\,V)(\eta,\eta). \tag{9}$$

Put $\mathscr{A} = \nabla\eta$ and $\mathscr{B} = \frac{1}{2}dV \otimes \eta$. Then

$$\langle \mathscr{A}, \mathscr{B}\rangle = \tfrac{1}{4}\int \langle \nabla V, \nabla|\eta|^2\rangle = \tfrac{\lambda_F}{4}\int V|\eta|^2,$$

because $-\Delta V = \lambda_F V$. Writing $\|\mathscr{A} - \mathscr{B}\|^2 = \|\frac{5}{8}\mathscr{A} - \mathscr{B}\|^2 + \frac{39}{64}\|\mathscr{A}\|^2 - \frac{3}{4}\langle \mathscr{A}, \mathscr{B}\rangle$ and dropping the first term,

$$\|\mathscr{A} - \mathscr{B}\|^2 \ge \tfrac{39}{64}\mu_0\|\eta\|^2 - \tfrac{1}{2}\int V|\eta|^2$$

by Lemma 28. With $\mathrm{Ric}=\frac{3}{2}$, (9) gives $\lambda\|\eta\|^2 \ge (\frac{39}{64}\mu_0+\frac{3}{2})\|\eta\|^2 - \int(\mathrm{Hess}\,V(\eta,\eta)+\frac{1}{2}V|\eta|^2)$, and the claim follows from the pointwise bound

$$\mathrm{Re}\,\mathrm{Tr}(MX^2)+\tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,M \le \tfrac{r}{2}|X|^2, \qquad M=bU\in rK, \quad (10)$$

since $\mathrm{Hess}\,V(X,X)=\mathrm{Re}\,\mathrm{Tr}(MX^2)$ and $V=\mathrm{Re}\,\mathrm{Tr}\,M$. The maximum over $|X|=1$ of the left side is convex in M, so it suffices to take $M=rW$ with $W\in\mathrm{SU}(3)$ and, after a conjugation, $W=\mathrm{diag}(e^{i\theta_k})$ with $\sum_k\theta_k=0$; put $c_k=\cos\theta_k$. Write $X=iH$ with H Hermitian, traceless, $\mathrm{Tr}\,H^2=1$, with diagonal d_k and entries H_{jk}. Then

$$\mathrm{Re}\,\mathrm{Tr}(WX^2) = -\sum_k c_k d_k^2 - \sum_{j<k}(c_j+c_k)|H_{jk}|^2,$$

with $\sum_k d_k^2+2\sum_{j<k}|H_{jk}|^2=1$, so the maximum is the larger of $q_{\mathrm{off}}=\max_{j<k}-\frac{1}{2}(c_j+c_k)$ and $q_{\mathrm{diag}}=\max\{-\sum c_k d_k^2 : \sum d_k = 0,\ \sum d_k^2=1\}$. If l is the third index, $q_{\mathrm{off}}+\frac{1}{2}\sum_k c_k = \frac{1}{2}c_l \le \frac{1}{2}$. The value q_{diag} is minus the smaller eigenvalue of $\mathrm{diag}(c)$ compressed to the plane $\sum d_k=0$; the eigenvalues of the compression solve $\sum_k (c_k-\mu)^{-1}=0$, that is $3\mu^2-2e_1\mu+e_2=0$ with e_1,e_2 the elementary symmetric functions of the c_k, so $q_{\mathrm{diag}}=\frac{1}{3}(\sqrt{e_1^2-3e_2}-e_1)$. The inequality $q_{\mathrm{diag}}+\frac{1}{2}e_1\le\frac{1}{2}$ is $\sqrt{e_1^2-3e_2}\le\frac{1}{2}(3-e_1)$, that is $e_1^2-4e_2+2e_1-3\le 0$. When $\sum\theta_k=0$, $\sum c_k^2 = 1+2c_1c_2c_3$, and with this identity the left side equals $-2(1-c_1)(1-c_2)(1-c_3)\le 0$. □

The weight $\frac{5}{8}$ is the one that gives the coefficient $\frac{1}{2}$ of V, and (10) is an equality at $W=I$. At $r=\frac{3}{2}$, $\gamma=\frac{187}{128}$.[13]

D. The modulus by hand

For $h\in\mathbb{C}^{3\times 3}$ let $E_2(h)=\frac{1}{2}((\mathrm{Tr}\,h)^2-\mathrm{Tr}\,h^2)$, the sum of the principal 2×2 minors of h. All integrals are over $\mathrm{SU}(3)$ with Haar measure; put $t=\mathrm{Tr}\,U$,

$$Z_*(r)=\int e^{r\,\mathrm{Re}\,\mathrm{Tr}\,U}dU, \qquad m_n=\int(\mathrm{Re}\,\mathrm{Tr}\,U)^n dU.$$

Lemma 30.

(i) *The integral of a monomial of degree j in the entries of U and k in those of $\bar U$ vanishes unless $j\equiv k$ modulo three; $\int u_{ij}\bar u_{kl}dU=\frac{1}{3}\delta_{ik}\delta_{jl}$ and $\int u_{ab}u_{cd}u_{ef}dU=\frac{1}{6}\varepsilon_{ace}\varepsilon_{bdf}$.*

(ii) *For every h the integrals of $\mathrm{Tr}(hUhU)$ and $(\mathrm{Tr}\,hU)^2$ vanish, $\int|\mathrm{Tr}\,hU|^2dU=\frac{1}{3}\|h\|_{\mathrm{HS}}^2$,*

$$\int \mathrm{Tr}(hUhU)\,\mathrm{Tr}\,U\,dU = -\tfrac{1}{3}E_2(h),$$

$$\int(\mathrm{Tr}\,hU)^2\,\mathrm{Tr}\,U\,dU = \tfrac{1}{3}E_2(h),$$

and $|E_2(h)|\le\|h\|_{\mathrm{HS}}^2$.

(iii) *$\int t^a\bar t^b dU$ vanishes unless $a\equiv b$ modulo three, does not change when a and b are exchanged, and equals 1 at $(a,b)=(0,0)$, $(1,1)$, $(3,0)$, 2 at $(2,2)$, 3 at $(4,1)$, 5 at $(6,0)$, 6 at $(3,3)$, 11 at $(5,2)$, 21 at $(7,1)$ and 23 at $(4,4)$. Hence $m_0=1$, $m_1=0$,*

$$(m_2,\dots,m_8)=\left(\tfrac{1}{2},\tfrac{1}{4},\tfrac{3}{4},\tfrac{15}{16},\tfrac{65}{32},\tfrac{231}{64},\tfrac{973}{128}\right),$$

and $m_n\le 3^{n-8}m_8$ for $n\ge 8$.

Proof. (iii) By the proof of Lemma 34, $t^a=\sum_{\lambda\vdash a}f^\lambda s_\lambda(U)$, and on $\mathrm{SU}(3)$ the Schur polynomial s_λ vanishes when λ has more than three rows and is otherwise the irreducible character of highest weight $(\lambda_1-\lambda_3,\lambda_2-\lambda_3)$. By Schur orthogonality [28, Ch. 2], $\int t^a\bar t^b dU$ is the sum of $f^\lambda f^\mu$ over the pairs $\lambda\vdash a$, $\mu\vdash b$ with at most three rows and $\lambda-\mu$ a multiple of $(1,1,1)$; such pairs exist only when $a\equiv b$ modulo three, and the sum is symmetric in a,b. For $a\ge b$ and $a+b\le 8$ the pairs are $((1),(1))$; $((1,1,1),\varnothing)$; $\lambda=\mu\in\{(2),(1,1)\}$; $((2,1,1),(1))$; $((2,2,2),\varnothing)$; $\lambda=\mu\in\{(3),(2,1),(1,1,1)\}$; $((3,1,1),(2))$ and $((2,2,1),(1,1))$; $((3,2,2),(1))$; and $\lambda=\mu\in\{(4),(3,1),(2,2),(2,1,1)\}$. The hook length formula gives $f^{(2,1)}=2$, $f^{(3,1)}=f^{(1,1,1)}=3$, $f^{(2,2)}=2$, $f^{(2,2,2)}=5$, $f^{(3,1,1)}=6$, $f^{(2,2,1)}=5$ and $f^{(3,2,2)}=21$, hence the values, the last one being $1+9+4+9$. Since $(\mathrm{Re}\,\mathrm{Tr}\,U)^n = 2^{-n}\sum_a\binom{n}{a}t^a\bar t^{n-a}$,

$$m_2=\tfrac{2}{4},\quad m_3=\tfrac{2}{8},\quad m_4=\tfrac{6\cdot2}{16},\quad m_5=\tfrac{2\cdot5\cdot3}{32},$$
$$m_6=\tfrac{2\cdot5+20\cdot6}{64},\quad m_7=\tfrac{2\cdot21\cdot11}{128},\quad m_8=\tfrac{2\cdot8\cdot21+70\cdot23}{256}.$$

The eigenvalues of U have modulus one, so $|\mathrm{Re}\,\mathrm{Tr}\,U|\le 3$ and $|\mathrm{Re}\,\mathrm{Tr}\,U|^n\le 3^{n-8}(\mathrm{Re}\,\mathrm{Tr}\,U)^8$ for $n\ge 8$.

(i) The map $U\mapsto\omega U$, $\omega=e^{2\pi i/3}$, preserves Haar measure and multiplies the monomial by ω^{j-k}. For a unitary representation π of $\mathrm{SU}(3)$, $\int\pi(U)dU$ is the orthogonal projection onto the invariant vectors, whose dimension is the mean of the character of π. On $\mathbb{C}^3\otimes\overline{\mathbb{C}^3}$ the invariant vectors are the multiples of the identity, which gives the second formula. On $(\mathbb{C}^3)^{\otimes 3}$ the character is t^3, whose mean is 1 by (iii), and ε is invariant because $\det U=1$; the projection onto the line of ε, whose squared norm is 6, gives the third formula.

(ii) The first two integrals have degree two in U and none in $\bar U$, and $\int|\mathrm{Tr}\,hU|^2dU=\frac{1}{3}\sum|h_{ji}|^2$ by (i). By (i), $\int h_{pa}u_{aq}h_{qc}u_{cp}u_{ee}dU=\frac{1}{6}h_{pa}h_{qc}\varepsilon_{ace}\varepsilon_{qpe}$, summed over repeated indices, and $\sum_e\varepsilon_{ace}\varepsilon_{qpe}=\delta_{aq}\delta_{cp}-\delta_{ap}\delta_{cq}$, which gives $\frac{1}{6}(\mathrm{Tr}\,h^2-(\mathrm{Tr}\,h)^2)$; the second integral is the same with ε_{pqe} in place of ε_{qpe}, which changes the sign. Finally $E_2(h)$ is the trace of the second exterior power of h, whose singular values are the products $\sigma_i\sigma_j$, $i<j$, of those of h; so $|E_2(h)|\le\sum_{i<j}\sigma_i\sigma_j=\frac{1}{2}((\sum_i\sigma_i)^2-\sum_i\sigma_i^2)\le\sum_i\sigma_i^2$. □

Lemma 31. *Let $r\ge 0$ with $\gamma(r)>0$, put*

$$\varepsilon(r)=1-\frac{5\,e^{-r/5}}{9Z_*(r)},$$

and let $b\in rK$ and $\|h\|_{\mathrm{HS}}=1$. Then

$$\nu_b|\nabla\varphi_h|^2\le\varepsilon(r),\quad \mathrm{Var}_{\nu_b}(\varphi_h)\le\frac{\varepsilon(r)}{\gamma(r)},\quad L_h(b)\le\frac{\sqrt{\varepsilon(r)}}{\gamma(r)},$$

[13] The direct Bochner bound, with Minkowski's inequality in place of the cross term, gives $(\sqrt{\mu_0}-\frac{g}{2}r)^2+\frac{3}{2}-r$ with $g^2=\max_W\|PA(W)\|^2=2.0655$; it is exact at $b=0$, where the gap is $\frac{8}{3}$, but falls below γ for $r>0.24$ and vanishes before $r=1$.

so that $\Lambda(r) \le \sqrt{\varepsilon(r)}/\gamma(r)$. *The function* ε *is nondecreasing on* $[0,\infty)$.

Proof. By Lemma 27, $\nabla\varphi_h(U) = -PA(M)$ with $M = hU$, and $\|M\|_{\mathrm{HS}} = 1$. The anti-Hermitian part has $|A(M)|^2 = \frac{1}{2}(\|M\|^2 - \mathrm{Re}\,\mathrm{Tr}\,M^2)$, and P removes from it the multiple $\frac{1}{3}\mathrm{Tr}\,A(M) = \frac{i}{3}\,\mathrm{Im}\,\mathrm{Tr}\,M$ of the identity, whose squared norm is $\frac{1}{3}(\mathrm{Im}\,\mathrm{Tr}\,M)^2$. Hence

$$|\nabla\varphi_h|^2 = 1 - \tfrac{1}{2}F_h,$$
$$F_h(U) = 1 + \mathrm{Re}\,\mathrm{Tr}(hUhU) + \tfrac{2}{3}(\mathrm{Im}\,\mathrm{Tr}\,hU)^2,$$

and $F_h \ge 0$ because $|\mathrm{Re}\,\mathrm{Tr}\,M^2| \le \|M\|^2$. For complex numbers X, Y, $\mathrm{Re}\,X\,\mathrm{Re}\,Y = \frac{1}{2}\mathrm{Re}(XY) + \frac{1}{2}\mathrm{Re}(X\bar{Y})$ and $(\mathrm{Im}\,X)^2 = \frac{1}{2}|X|^2 - \frac{1}{2}\mathrm{Re}\,X^2$. With these identities, the vanishing of the integrals of degree two in U and one in $\bar{U}$, or the converse, and Lemma 30(ii),

$$\int F_h\,dU = 1 + \tfrac{2}{3}\cdot\tfrac{1}{6} = \tfrac{10}{9}, \qquad \int F_h\,\mathrm{Re}\,\mathrm{Tr}\,U\,dU = -\tfrac{2}{9}\,\mathrm{Re}\,E_2(h),$$

the second as the sum of $-\frac{1}{6}\,\mathrm{Re}\,E_2(h)$ and $\frac{2}{3}\cdot(-\frac{1}{12})\,\mathrm{Re}\,E_2(h)$. For $W \in \mathrm{SU}(3)$ the substitution $U \mapsto W^{-1}U$ and Lemma 30(ii) give

$$\int F_h\,\mathrm{Re}\,\mathrm{Tr}(WU)\,dU = \int F_{hW^{-1}}\,\mathrm{Re}\,\mathrm{Tr}\,U\,dU \ge -\tfrac{2}{9}.$$

Write $b = r\sum_i \lambda_i W_i$ as a convex combination of elements of $\mathrm{SU}(3)$ and $V = \mathrm{Re}\,\mathrm{Tr}(Ub)$. Under the probability measure $\pi = \frac{9}{10}F_h\,dU$ the mean of V is at least $-\frac{9}{10}\cdot\frac{2}{9}r = -\frac{r}{5}$, so Jensen's inequality gives $\int F_h e^V\,dU = \frac{10}{9}\pi(e^V) \ge \frac{10}{9}e^{-r/5}$. The function $c \mapsto \log\int e^{\mathrm{Re}\,\mathrm{Tr}(Uc)}dU$ is convex by Hölder's inequality and equals $\log Z_*(r)$ at every $c = rW$, $W \in \mathrm{SU}(3)$, by the invariance of Haar measure; hence $\int e^V dU \le Z_*(r)$ and $\nu_b(F_h) \ge \frac{10}{9}e^{-r/5}/Z_*(r)$, which is the first bound.

The spectral gap of ν_b is at least $\gamma(r)$ by Lemma 29. The Poincaré inequality gives $\mathrm{Var}_{\nu_b}(\varphi_h) \le \nu_b|\nabla\varphi_h|^2/\gamma(r)$, the second bound, and, for g 1-Lipschitz for d, hence for the geodesic distance, $\mathrm{Var}_{\nu_b}(g) \le 1/\gamma(r)$; so $\mathrm{Cov}_{\nu_b}(g,\varphi_h) \le \mathrm{sd}(g)\,\mathrm{sd}(\varphi_h) \le \sqrt{\varepsilon(r)}/\gamma(r)$. Finally $\log Z_*$ is convex in r with derivative $m_1 = 0$ at $r = 0$, so Z_* and ε are nondecreasing on $[0,\infty)$. □

Proof of Theorem 6. Let $r = 2\beta_W \le 1.03$, so that $\gamma(r) \ge \gamma(1.03) > 1.69$. By (8) and Lemma 31, the Dobrushin constant satisfies $\alpha \le 2r\Lambda(r) \le \alpha_1(\beta_W)$, and α_1 is increasing because ε is nondecreasing and γ is decreasing. By Lemma 30(iii), and since $\sum_{n\ge 9} x^n/n! \le \frac{x^9}{9!}(1-\frac{x}{10})^{-1}$ for $0 \le x < 10$,

$$Z_*(r) \le \sum_{n=0}^{8}\frac{m_n r^n}{n!} + \frac{m_8}{3^8}\,\frac{(3r)^9}{9!}\Big(1-\frac{3r}{10}\Big)^{-1}.$$

At $r = 1$ the terms of the sum, rounded up, are 1, 0, 0.25, 0.0416667, 0.03125, 0.0078125, 0.0028212, 0.0007162 and 0.0001886, with sum at most 1.3344552, and the last term is $\frac{973}{128\cdot 6561}\cdot\frac{19683}{362880}\cdot\frac{10}{7} < 9\cdot 10^{-5}$; so $Z_*(1) \le 1.334546$. Since $e^{-x} \ge 1 - x + \frac{x^2}{2} - \frac{x^3}{6}$ for $x \ge 0$, $e^{-1/5} \ge 0.818666$, so $\frac{5e^{-1/5}}{9Z_*(1)} \ge 0.340800$ and $\varepsilon(1) \le 0.659200 < 0.81192^2$. With $\gamma(1) = \frac{219}{128}$,

$$\alpha_1(\tfrac{1}{2}) \le \frac{2\cdot 0.81192\cdot 128}{219} \le 0.9491.$$

At $r = 1.03$ the terms are at most

$$1,\ 0,\ 0.265225,\ 0.0455303,\ 0.0351722,$$
$$0.0090569,\ 0.0033687,\ 0.0008808,\ 0.0002389,$$

with sum at most 1.3594728, and the last term is less than $1.2\cdot 10^{-4}$; so $Z_*(1.03) \le 1.359593$, $e^{-0.206} \ge 0.813760$, $\frac{5e^{-0.206}}{9Z_*(1.03)} \ge 0.332517$ and $\varepsilon(1.03) \le 0.667483 < 0.817^2$. With $\gamma(1.03) = \frac{5427}{3200}$, $\alpha_1(0.515) \le 2.06\cdot 0.817\cdot\frac{3200}{5427} \le 0.9924$. Proposition 11 applies with the Hilbert–Schmidt metric. With $y = 1-\alpha_1$, $-\log\alpha_1 \ge y + \frac{y^2}{2}$, which is at least 0.052 at $\beta_W = \frac{1}{2}$ and 0.0076 at 0.515. □

The losses in this bound are Jensen's inequality, the Poincaré inequality for φ_h, which replaces the true variance of φ_h by $\nu_b|\nabla\varphi_h|^2/\gamma(r)$, and the gap $\gamma(r)$ itself: at $b = 0$ the variance is $\frac{1}{6}$ and the gap $\frac{8}{3}$, so that $L_h(0) = \frac{1}{4}$, while the lemma gives $\frac{2}{3}/\gamma(0) = 0.302$. Evaluated in floating point, α_1 reaches one near $\beta_W = 0.5176$. The rest of this section describes a sharper bound for $\Lambda(r)$, with a corrector of degree two, the reduction to diagonal staple sums, exact moments and an interpolation over boxes of staple sums; its evaluation by a computer is the subject of Remark 36, and no result uses it.

E. A corrector of degree two

Lemma 32. *For* $b \in rK$, $\|h\| = 1$ *and every smooth real function* u *on* $\mathrm{SU}(3)$,

$$L_h(b) \le \sqrt{\nu_b|\nabla u|^2} + \sqrt{\mathrm{Var}_{\nu_b}(\varphi_h + Lu)/\gamma(r)}.$$

Proof. Let $-L\psi = \varphi_h - \nu_b\varphi_h$. For a 1-Lipschitz g, $|\nabla g| \le 1$ almost everywhere, because d is at most the geodesic distance, so $\mathrm{Cov}(g,\varphi_h) = \nu_b\langle\nabla g, \nabla\psi\rangle \le \nu_b|\nabla\psi|$. Write $\psi = u + \chi$. Since $\nu_b(Lu) = 0$, χ solves $-L\chi = X - \nu_b X$ with $X = \varphi_h + Lu$, and

$$\nu_b|\nabla\chi|^2 = \mathrm{Cov}(\chi, X) \le \sqrt{\mathrm{Var}\,\chi\ \mathrm{Var}\,X} \le \sqrt{\nu_b|\nabla\chi|^2\,\mathrm{Var}\,X/\gamma}$$

by Lemma 29. Hence $\nu_b|\nabla\chi|^2 \le \mathrm{Var}\,X/\gamma$, and $\nu_b|\nabla\psi| \le \sqrt{\nu_b|\nabla u|^2} + \sqrt{\nu_b|\nabla\chi|^2}$. □

The best u is ψ itself, and then the second term vanishes. With $u = s\varphi_h$, Lemma 27 gives $\varphi_h + Lu = (1 - s\lambda_F)\varphi_h + s\Gamma$ with $\Gamma = \langle\nabla V, \nabla\varphi_h\rangle$; the choice $s = 1/\lambda_F$ removes φ_h and leaves the drift. In floating point this blended corrector keeps $2r\sup_h L_h$ below one up to about $\beta_W = 0.715$. We take u in a space of polynomials of degree two.

Let $\mathscr{P}_2$ be the real vector space of real polynomials of degree one or two in the entries u_{ij} of U and $\bar{u}_{ij}$ of $\bar{U}$. It has dimension 189: a basis (B_k) consists of $\frac{1}{2}(m + \bar{m})$ and $\frac{1}{2i}(m - \bar{m})$ for each pair $\{m, \bar{m}\}$ of conjugate monomials with $m \ne \bar{m}$, and m itself when $m = \bar{m}$. Identify $h \in \mathbb{C}^{3\times 3}$ with $\mathbb{R}^{18}$ through the real and imaginary parts of its entries. For a real 189×18 matrix C put

$$q_h = \sum_k \sum_{m=1}^{18} C_{km} h_m B_k \in \mathscr{P}_2,$$

linear in h. For diagonal $b=\operatorname{diag}(a_1,a_2,x_3+iy_3)$ we have $V=\sum_k b_k\varphi_{F_k}$ with $(b_k)=(a_1,a_2,x_3,y_3)$ and $F=(e_{11},e_{22},e_{33},ie_{33})$, so

$$X_h=\varphi_h+Lq_h=\varphi_h+\Delta q_h+\sum_k b_k\langle\nabla\varphi_{F_k},\nabla q_h\rangle \tag{11}$$

is a polynomial of degree at most three, affine in b, while ∇q_h does not depend on b. The squares of these functions have degree at most six, so the quadratic forms $h\mapsto \nu_b|\nabla q_h|^2$ and $h\mapsto\nu_b[(X_h-\kappa)^2]$ are combinations of moments of ν_b of degree at most six.

To compute Δ and $\langle\nabla f,\nabla g\rangle$ exactly we use the basis of $\mathfrak{su}(3)$

$$e_{jk}-e_{kj},\quad i(e_{jk}+e_{kj})\quad(j<k),$$
$$i(e_{11}-e_{22}),\quad i(e_{11}+e_{22}-2e_{33}),$$

acting as left-invariant fields $X_af(U)=\frac{d}{dt}f(Ue^{tX_a})|_{t=0}$, with weights $w_a=\frac12$ for the first seven and $\frac16$ for the last. Then $\sum_a w_aX_a\otimes X_a$ is the Casimir tensor of $\langle X,Y\rangle=-\operatorname{Tr}(XY)$, so

$$\langle\nabla f,\nabla g\rangle=\sum_a w_aX_af\,X_ag,\qquad \Delta=\sum_a w_aX_a^2.$$

The fields map monomials to combinations of monomials of the same degree with coefficients in $\mathbb{Z}[i]$; for instance $\Delta\varphi_h=-\frac83\varphi_h$, as in Lemma 27.

F. Orbits of staple sums

Lemma 33. *Let $V,W\in\mathrm{SU}(3)$.*

(i) *$L_h(VbW)=L_{V^{-1}hW^{-1}}(b)$ and $L_h(\bar b)=L_{\bar h}(b)$.*

(ii) *Every $b\in\mathbb{C}^{3\times3}$ is in the orbit of some $\operatorname{diag}(a_1,a_2,z_3)$ with $a_1\ge a_2\ge|z_3|$ and $\operatorname{Im}z_3\ge0$.*

(iii) *A diagonal matrix lies in K if and only if it lies in the convex hull of the diagonal torus T of $\mathrm{SU}(3)$.*

Proof. (i) If U has law ν_{VbW} then WUV has law ν_b; the map preserves d and Haar measure, and $\varphi_h(U)=\varphi_{V^{-1}hW^{-1}}(WUV)$. For $\bar b$ use $U\mapsto\bar U$. (ii) Take a singular value decomposition $b=V_1\Sigma W_1$ and move the determinants of V_1,W_1 into the last entry; a complex conjugation makes $\operatorname{Im}z_3\ge0$. (iii) Let c be diagonal with nonzero entries of distinct moduli and let U maximise $\operatorname{Re}\operatorname{Tr}(cU)$ on $\mathrm{SU}(3)$. The derivative along $\mathfrak{su}(3)$ vanishes, so the anti-Hermitian part of cU is a multiple $i\lambda I$ of the identity and $cU=H+i\lambda I$ with H Hermitian. Then $H^2+\lambda^2I=cc^*$ is diagonal with distinct entries, H commutes with it and is diagonal, and $U\in T$. Hence $\max_{\mathrm{SU}(3)}\operatorname{Re}\operatorname{Tr}(cU)=\max_T\operatorname{Re}\operatorname{Tr}(ct)$ for such c, and by continuity for every diagonal c. A diagonal matrix outside $\operatorname{conv}T$ is separated from it by such a functional, which then separates it from K. □

Since every $b\in rK$ has operator norm at most r,

$$\Lambda(r)=\sup\Big\{\sup_{\|h\|=1}L_h(b):b\in\mathscr{R}_r\Big\},$$

where $\mathscr{R}_r$ is the set of $\operatorname{diag}(a_1,a_2,x_3+iy_3)\in r\operatorname{conv}T$ with $a_1\ge a_2\ge|x_3+iy_3|$ and $y_3\ge0$, a subset of the box $[0,r]^2\times[-r,r]\times[0,r]$ in the coordinates (a_1,a_2,x_3,y_3).

G. Exact moments and interpolation

Lemma 34. *For $D,D'\in\mathbb{C}^{3\times3}$ let $\mathbb{E}_H$ be the Haar mean on $\mathrm{SU}(3)$ and*

$$Z(D,D')=\mathbb{E}_H\exp\big(\tfrac12\operatorname{Tr}(UD)+\tfrac12\operatorname{Tr}(\bar UD')\big).$$

Then

$$Z(D,D')=\sum_{p\ge q\ge0}\frac{s_{(p,q)}(DD'^{\top})\,G_{p,q}(\det D)\,G_{p,q}(\det D')}{d_{p,q}},$$

with $G_{p,q}(z)=\sum_{k\ge0}c_{(p+k,q+k,k)}z^k$, $c_\lambda=2^{-|\lambda|}/\prod\mathrm{hooks}(\lambda)$, $d_{p,q}=\frac12(p-q+1)(q+1)(p+2)$ and $s_{(p,q)}$ the Schur polynomial. The moments of degree j in U and k in $\bar U$ are $\mathbb{E}_H[u^\alpha\bar u^\beta e^{\operatorname{Re}\operatorname{Tr}(Ub)}]=2^{|\alpha|+|\beta|}\partial_D^\alpha\partial_{D'}^\beta Z(b,\bar b)$, the entry u_{ij} pairing with D_{ji}.

Proof. We use $(\operatorname{Tr}X)^n=\sum_{\lambda\vdash n}f^\lambda s_\lambda(X)$, where f^λ is the number of standard tableaux of shape λ, given by the hook length formula [98, Cor. 7.21.6]. Expanding the exponential gives

$$e^{\frac12\operatorname{Tr}X}=\sum_{n\ge0}\frac{(\operatorname{Tr}X)^n}{2^nn!}=\sum_\lambda c_\lambda s_\lambda(X),$$

the sum over partitions with at most three parts. For $U\in\mathrm{SU}(3)$ one has $s_{\lambda+(k,k,k)}(UD)=(\det D)^k s_\lambda(UD)$, so

$$e^{\frac12\operatorname{Tr}(UD)}=\sum_{p\ge q}G_{p,q}(\det D)\,\chi_{p,q}(UD),$$

where $\chi_{p,q}$ is the irreducible character of highest weight $(p,q,0)$ and $d_{p,q}$ its dimension. Since $\bar U=(U^{-1})^\top$, we have $\chi_{p,q}(\bar UD')=\chi_{p,q}(U^{-1}D'^{\top})$, and Schur orthogonality [28, Ch. 2],

$$\mathbb{E}_H\big[\chi_\rho(UA)\chi_{\rho'}(U^{-1}B)\big]=\delta_{\rho\rho'}\frac{\chi_\rho(AB)}{d_\rho},$$

gives the expansion. The bound below gives absolute convergence. Finally $\operatorname{Re}\operatorname{Tr}(Ub)=\frac12\operatorname{Tr}(Ub)+\frac12\operatorname{Tr}(\bar U\bar b)$. □

The computation keeps $p\le34$ and $k\le11$. Every omitted partition has more than 34 boxes, and since $|s_\lambda(X)|\le\dim(\lambda)\|X\|^{|\lambda|}$ and $\sum_{\lambda\vdash n}f^\lambda\dim\lambda=3^n$ by Schur–Weyl duality, the omitted part of each factor is at most $\sum_{n>34}(\frac32\|D\|)^n/n!$; Cauchy's estimate on a polydisc around b turns this into a bound for every derivative (Appendix E).

Lemma 35 (Interpolation on a box)**.** *Let $Q=\prod_k[l_k,l_k+w_k]$ in the coordinates (a_1,a_2,x_3,y_3), with corners b_i and multilinear weights $\omega_i(b)\ge0$, $\sum_i\omega_i(b)=1$, and let $\varsigma=w_1^2+w_2^2+\max(w_3^2,w_4^2)$. For $b\in Q$:*

(i) *$Z(b)\ge e^{-\varsigma/8}\sum_i\omega_i(b)Z(b_i)$;*

(ii) *$\nu_b(f)\le e^{\varsigma/8}\max_i\nu_{b_i}(f)$ for every $f\ge0$ that does not depend on b;*

(iii) *if* $Y_b=(\mathscr{A}+\sum_k(b_k-\beta_k)\mathscr{B}_k)^2$ *with functions* $\mathscr{A},\mathscr{B}_k$ *of* U *and a fixed point* β*, then*

$$\nu_b(Y_b)\le e^{\varsigma/8}\Big(\max_i \nu_{b_i}(Y_{b_i})+\sum_k \frac{w_k^2}{4}\,\frac{\max_i Z(b_i)\nu_{b_i}(\mathscr{B}_k^2)}{\min_i Z(b_i)}\Big).$$

Proof. Write $E(b)=e^{\mathrm{Re}\,\mathrm{Tr}(Ub)}=\prod_k e^{c_k b_k}$, where

$$c=(\mathrm{Re}\,u_{11},\mathrm{Re}\,u_{22},\mathrm{Re}\,u_{33},-\mathrm{Im}\,u_{33}),$$

so that $|c_k|\le 1$ and $c_3^2+c_4^2\le 1$. In each coordinate the chord of $e^{c_k x}$ over $[l_k,l_k+w_k]$ lies above the function and, by Hoeffding's lemma [65] for a two-point variable, below $e^{c_k x}e^{c_k^2 w_k^2/8}$. Multiplying,

$$E(b)\le\sum_i \omega_i(b)E(b_i)\le E(b)\,e^{\sum_k c_k^2 w_k^2/8}\le E(b)\,e^{\varsigma/8}. \quad (12)$$

Integrating the right inequality gives (i), and (ii) follows from the left one and (i). For (iii) let $\mathscr{F}(b)=Y_bE(b)$ and I_k the linear interpolation in the k-th coordinate, so that $\mathscr{F}-I_1I_2I_3I_4\mathscr{F}=\sum_k I_1\cdots I_{k-1}(\mathscr{F}-I_k\mathscr{F})$. Along the k-th coordinate $\mathscr{F}$ is $\ell^2e^{c_kx}$ times a positive factor, with ℓ affine of slope $\mathscr{B}_k$, and $\partial_k^2\mathscr{F}=((c_k\ell+2\mathscr{B}_k)^2-2\mathscr{B}_k^2)E\ge -2\mathscr{B}_k^2E$. With the Green kernel of the interval, whose integral is at most $w_k^2/8$,

$$\mathscr{F}-I_k\mathscr{F}\le 2\mathscr{B}_k^2\int GE\le \tfrac{w_k^2}{4}\mathscr{B}_k^2\sum_i\omega_i^{(k)}E(b_i)$$

by the left inequality of (12), with weights $\omega_i^{(k)}\ge 0$ of sum one that do not depend on U. The operators I_j have nonnegative weights of sum one; integrate over U and divide by $Z(b)$, bounded below by (i). □

H. The cover

Fix a box Q and a matrix C. For any constant κ, $\mathrm{Var}(X_h)\le\nu_b[(X_h-\kappa)^2]$; take $\kappa=\langle\kappa(b),h\rangle$ with $\kappa(b)$ affine in b, fitted to the means at the corners. By (11), $X_h-\kappa=\mathscr{A}+\sum_k(b_k-\beta_k)\mathscr{B}_k$ with β the centre of Q and $\mathscr{B}_k=\langle\nabla\varphi_{F_k},\nabla q_h\rangle-\langle\partial_k\kappa,h\rangle$, and Lemmas 32 and 35 give, for all $b\in Q\cap\mathscr{R}_r$,

$$\sup_h L_h(b)\le\sqrt{e^{\varsigma/8}\max_i\lambda_A(b_i)}+\sqrt{\frac{e^{\varsigma/8}}{\gamma(r)}\Big(\max_i\lambda_V(b_i)+\frac{e_N}{\min_i Z(b_i)}\Big)}, \quad (13)$$

where $e_N=\sum_k\frac{w_k^2}{4}\max_i Z(b_i)\lambda_k(b_i)$ and λ_A, λ_V, λ_k are the largest eigenvalues of the quadratic forms $h\mapsto\nu_{b_i}|\nabla q_h|^2$, $h\mapsto\nu_{b_i}[(X_h-\kappa)^2]$ at $b=b_i$, and $h\mapsto\nu_{b_i}[\mathscr{B}_k^2]$ on $\mathbb{R}^{18}$. Their entries are moments of ν_{b_i} of degree at most six, computed with rigorous error bounds (Appendix E); each largest eigenvalue is bounded above by an $LDL^\top$ factorisation of tI minus the computed form with certified positive pivots, plus the norm of its error matrix by Weyl's inequality [62, 70].

The matrix C is chosen in each box in floating point. At the mean of the corner moments and at $b=\beta$, Lemma 32 for $u=q_h$ is a quadratic problem in C: for $\theta>0$ we minimise the trace of $\nu|\nabla q|^2+\theta\,\mathrm{Var}(X)/\gamma$ over the 18 coordinate directions h, a least-squares problem of size 189, keep the θ on a geometric grid of 48 values from 0.05 to 80 that gives the smallest bound, and round C to multiples of 2^{-30}. Any choice gives a valid bound.

A box is discarded when it contains no point with $a_1\ge a_2\ge|z_3|$, or when a linear functional $\ell(b)=\mathrm{Re}\,\mathrm{Tr}(c^*b)$ satisfies $\min_Q\ell>r\max_T\ell$, which separates it from $r\,\mathrm{conv}\,T$ and hence, by Lemma 33(iii), from $\mathscr{R}_r$.

Remark 36 (a computation, used by no theorem). At $r=\frac{3}{2}$, with $\gamma(r)=\frac{187}{128}$, the box $[0,r]^2\times[-r,r]\times[0,r]$ is cut into 8192 boxes of side $r/8$. Of these, 841 survive the two tests, and for 581 the right side B_Q of (13), evaluated with the error bounds of Appendix E, satisfies $2rB_Q<1$. Each of the other 260 is cut into sixteen boxes of side $r/16$; 1760 of these survive the tests, and all satisfy $2rB_Q<1$. The 2341 boxes cover $\mathscr{R}_r$; the largest bound is 0.999887, on the box $[1.3125,1.5]\times[0.9375,1.125]\times[0.5625,0.75]\times[0,0.1875]$, and the median is 0.983 (Fig. 6). This computation gives $2r\Lambda(r)\le 0.999887$ at $r=\frac{3}{2}$, and with it (8) and the monotonicity of Λ would give $\alpha\le 4\beta_W\Lambda(\frac{3}{2})<1.333183\,\beta_W<1$ for $\beta_W\le\frac{3}{4}$, and so the conclusions of Theorem 1, with $m\ge-\log(1.33319\,\beta_W)$, on that range. The same computation with $\gamma(r)=\frac{4643}{3200}$ and the threshold $2rB_Q<1.03$ at $r=\frac{38}{25}$ (of the 8192 boxes, 784 survive the tests and 690 satisfy the threshold; each of the other 94 is cut into sixteen, of which 373 survive and all satisfy it), and with $\gamma(r)=\frac{4611}{3200}$ and the threshold $2rB_Q<1.0275$ at $r=\frac{77}{50}$ (826 survive and 513 satisfy it; the other 313 are cut into sixteen, of which 2311 survive and 2281 satisfy it; the other 30 are cut into sixteen, of which 31 survive and all satisfy it), gives $2r\Lambda(r)\le 1.029967$ at $r=\frac{38}{25}$ and $2r\Lambda(r)\le 1.027468$ at $r=\frac{77}{50}$. No result of the paper uses these numbers.

VIII. SU(3) WITH PLAQUETTE BLOCKS

In this section $G=\mathrm{SU}(3)$, $\beta\ge 0$, $\beta_W=9\beta$, $N\beta=\beta_W/3$, and the notation is that of Sec. VII: d is the Hilbert–Schmidt distance, $K=\mathrm{conv}\,\mathrm{SU}(3)$, $\nu_c(dU)\propto e^{\mathrm{Re}\,\mathrm{Tr}(Uc)}dU$, $\varphi_h(U)=\mathrm{Re}\,\mathrm{Tr}(Uh)$ and $\Lambda(r)$ is the modulus of (8). Put $Z(c)=\int e^{\mathrm{Re}\,\mathrm{Tr}(Uc)}dU$ and $g=\log Z$ on $\mathbb{C}^{3\times3}$, viewed as $\mathbb{R}^{18}$ with the inner product $\langle x,y\rangle=\mathrm{Re}\,\mathrm{Tr}(xy^*)$. The gradient of g at c is the conjugate mean $m(c)^*$, $m(c)=\nu_c(U)\in K$, and $\mathrm{Hess}\,g(c)(h,h)=\mathrm{Var}_{\nu_c}(\varphi_h)$. Since $Z(VcW)=Z(c)$ for $V,W\in\mathrm{SU}(3)$, the same holds for g.

The blocks are those of Lemma 12, which concerns the line field only and holds for every group. For a block plaquette with active links e_1,e_2,e_3, $\mathrm{Re}\,\mathrm{Tr}\,Q_p=\mathrm{Re}\,\mathrm{Tr}(y_1y_2y_3)$ with $y_i=Q_{e_i}$ or $Q_{e_i}^{-1}$, and the conditional law of the block is

$$\mu_b(dy)\propto\exp\Big(\sum_i\mathrm{Re}\,\mathrm{Tr}(y_ib_i)+N\beta\,\mathrm{Re}\,\mathrm{Tr}(y_1y_2y_3)\Big)dy, \quad (14)$$

where b_i is $N\beta$ times the sum of the five staples of y_i outside

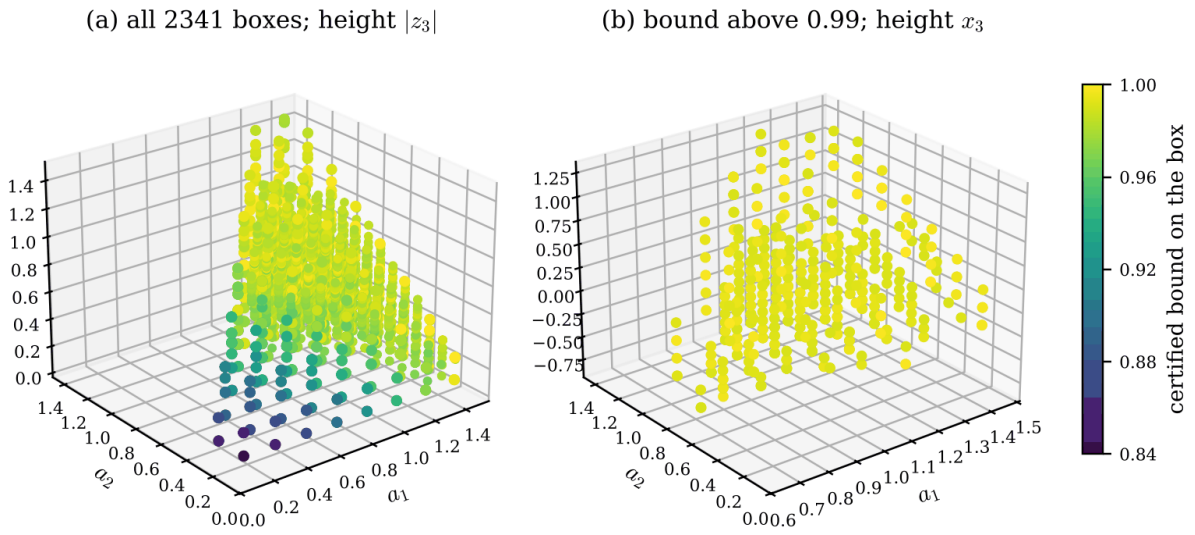


FIG. 6. The 2341 boxes of Remark 36 at $r=\frac{3}{2}$, plotted at their centres in the coordinates (a_1,a_2) of the staple sum $\operatorname{diag}(a_1,a_2,x_3+iy_3)$, coloured by the computed bound on $2r\sup_h L_h$ over the box; the larger points are the boxes of side $r/8$. (a) All boxes, with height $|x_3+iy_3|$. (b) The 419 boxes with bound above 0.99, with height x_3; the largest bounds sit in the interior of $\mathscr{R}_r$, not at its extreme points. No result uses this computation.

the block plaquette, so $b_i \in 5N\beta K$ and $\|b_i\|_{\mathrm{op}} \le 5N\beta$. On $\mathrm{SU}(3)^3$ we use $d(y,y') = \sum_i \|y_i - y_i'\|_{\mathrm{HS}}$.

A. The sensitivity of one block

Let $r = 2\beta_W = 6N\beta$. Let λ_* and τ be a lower and an upper bound for the eigenvalues of $\operatorname{Hess} g$ on rK, M an upper bound on rK for $|D^3 g(c)(u,v,v)|$ over unit vectors u,v, and $\gamma = \gamma(r)$ the gap of Lemma 29. Put

$$w = \tfrac{3}{2}(N\beta)^2(\tau-\lambda_*), \qquad s = \Lambda(r) + (e^w-1)e^w\sqrt{\tau/\gamma},$$
$$C = 5(N\beta)^2\Big(\tau s + \frac{\sqrt{3}N\beta M e^w}{\gamma}\Big),$$
$$K_3 = \frac{(1+N\beta\Lambda(r))\,s}{1-C}, \qquad \alpha_3 = 10N\beta K_3, \tag{15}$$

assuming $C<1$.

Lemma 37. *Let $a, c_0 \in 5N\beta K$ and*

$$\kappa(dx) \propto \exp\big(\mathrm{Re}\,\mathrm{Tr}(xa) + g(c_0+N\beta x)\big)\,dx \quad \text{on } \mathrm{SU}(3).$$

For every 1-Lipschitz f and every $h \in \mathbb{C}^{3\times 3}$:

(i) $\mathrm{Cov}_\kappa(f,\varphi_h) \le s\|h\|_{\mathrm{HS}}$;

(ii) *if c_0 moves in the direction k inside $5N\beta K$, then*

$$\big|\tfrac{d}{dt}\kappa(f)\big| \le N\beta\big(\tau s + \sqrt{3}N\beta M e^w/\gamma\big)\|k\|_{\mathrm{HS}}.$$

Proof. (i) The segment from c_0 to $c_0+N\beta x$ lies in $6N\beta K = rK$, and Taylor's formula gives

$$g(c_0+N\beta x) = g(c_0) + N\beta\,\mathrm{Re}\,\mathrm{Tr}(x\,m(c_0)) + R(x)$$

with $R(x) = \frac{1}{2}(N\beta)^2 \operatorname{Hess} g(\xi)(x,x)$. Since $\|x\|_{\mathrm{HS}}^2 = 3$, $R \in [\frac{3}{2}(N\beta)^2\lambda_*, \frac{3}{2}(N\beta)^2\tau]$ and $\operatorname{osc} R \le w$. Hence κ is the tilt $e^R\nu/\nu(e^R)$ of $\nu = \nu_{c'}$ with $c' = a + N\beta m(c_0) \in rK$. By definition of Λ, $\mathrm{Cov}_\nu(f,\varphi_h) \le \Lambda(r)\|h\|$. The function f is 1-Lipschitz for the geodesic distance, which dominates d, so $\mathrm{sd}_\nu f \le \gamma^{-1/2}$ by Lemma 29, and $\mathrm{sd}_\nu \varphi_h \le \sqrt{\tau}\|h\|$. Lemma 13 gives (i).

(ii) The derivative is $\mathrm{Cov}_\kappa(f, \langle \nabla g(c_0+N\beta x), k\rangle)$. Write $\nabla g(c_0+N\beta u) = \nabla g(c_0) + N\beta \operatorname{Hess} g(c_0)u + N(u)$. The linear term is φ_h with $\|h\| \le N\beta\tau\|k\|$ and contributes at most $N\beta\tau s\|k\|$ by (i). On the segment from c_0 the Hessian moves by at most $N\beta M\|u\| \le \sqrt{3}N\beta M$, so $x \mapsto \langle N(x),k\rangle$ is $\sqrt{3}(N\beta)^2 M\|k\|$-Lipschitz for d. The law κ is a tilt of ν by a function of oscillation at most w, so its spectral gap is at least $e^{-w}\gamma$ by the Holley–Stroock lemma [63], and the covariance of two Lipschitz functions under κ is at most e^w/γ times the product of their constants. □

Lemma 38. *Let $C<1$. If b, b' have all fields in $5N\beta K$ and differ only in b_i, then $W_1(\mu_b,\mu_{b'}) \le K_3\|b_i - b_i'\|_{\mathrm{HS}}$ for the metric d.*

Proof. By the cyclic symmetry of $\mathrm{Re}\,\mathrm{Tr}(y_1y_2y_3)$ we may take $i=1$. Given y_1, y_3, the law of y_2 is ν_c with $c = b_2 + N\beta y_3 y_1 \in rK$. Changing y_1 by η in d moves c by $N\beta\eta$ and ν_c by at most $N\beta\Lambda(r)\eta$ in W_1, as in (8); so $F(y_1,y_3) = \mathbb{E}[f \mid y_1, y_3]$ is

$(1+N\beta\Lambda(r))$-Lipschitz in each variable. The law of (y_1,y_3) is proportional to $e^{\mathrm{Re\,Tr}(y_1b_1)+\mathrm{Re\,Tr}(y_3b_3)}Z(b_2+N\beta y_3y_1)$, and $Z(b_2+N\beta y_3y_1)=Z(y_3^{-1}b_2+N\beta y_1)=Z(b_2y_1^{-1}+N\beta y_3)$, so its two conditional laws have the form of Lemma 37, with $c_0=y_3^{-1}b_2$ and $c_0=b_2y_1^{-1}$. Moving y_3 to y_3' moves $y_3^{-1}b_2$ along a segment in $5N\beta K$ of length at most $\|b_2\|_{\mathrm{op}}\|y_3-y_3'\|\le 5N\beta\|y_3-y_3'\|$. The coupling argument of Lemma 15 gives $d_1\le s\varepsilon+Cd_3$, $d_3\le Cd_1$ and $|\mu_b(f)-\mu_{b'}(f)|\le(1+N\beta\Lambda(r))(d_1+d_3)\le K_3\varepsilon$. □

B. The constants and the proof of Theorem 7

Proof of Theorem 7. Let $r=2\beta_W\le 1.04$ and $\gamma=\gamma(r)$. For $c\in rK$ and unit h, $\mathrm{Hess}\,g(c)(h,h)=\mathrm{Var}_{\nu_c}(\varphi_h)$ lies in $[0,\varepsilon(r)/\gamma]$ by Lemma 31, which also gives $\Lambda(r)\le\sqrt{\varepsilon(r)}/\gamma$; so $\lambda_*=0$ and $\tau=\varepsilon(r)/\gamma$ are admissible. The third derivative $D^3g(c)(u,v,v)$ is the third cumulant $\nu_c[(\varphi_u-\nu_c\varphi_u)(\varphi_v-\nu_c\varphi_v)^2]$. Since U and $m(c)$ lie in K, whose elements have Hilbert–Schmidt norm at most $\sqrt3$, $|\varphi_u(U)-\nu_c\varphi_u|=|\mathrm{Re\,Tr}((U-m(c))u)|\le 2\sqrt3$ for unit u, so $|D^3g(c)(u,v,v)|\le 2\sqrt3\,\mathrm{Var}_{\nu_c}(\varphi_v)\le 2\sqrt3\,\tau=M$. With these constants $\sqrt{\tau/\gamma}=\sqrt{\varepsilon(r)}/\gamma$, the bound used for $\Lambda(r)$, and $\sqrt3N\beta Me^w/\gamma=6N\beta\tau e^w/\gamma$. Every field met in Lemmas 37 and 38 lies in rK. Since ε is nondecreasing and γ decreasing in r, every quantity in (15) is nondecreasing in β_W while $C<1$, so it suffices to bound α_3 at $\beta_W=0.52$. Changing an active link j by η changes one staple of each of its ten partners outside its block by a matrix of norm η, so the field b_i of each such partner moves by $N\beta\eta$, and Lemma 38 gives $\sum_B C_{Bj}\le 10N\beta K_3=\alpha_3(\beta_W)$. Proposition 17, whose proof does not use the group, applies with the Hilbert–Schmidt metric.

For the numbers we use $e^w\le(1-w)^{-1}$. At $\beta_W=0.515$, $N\beta=\frac{103}{600}$, and the proof of Theorem 6 gives $\varepsilon(1.03)\le 0.667483<0.817^2$ and $\gamma=\frac{5427}{3200}$; hence $\Lambda(r)\le 0.48174$, $\tau\le 0.39358$, $w\le 0.017398$, $s\le 0.49043$, $C\le 0.064287$, $K_3\le 0.56747$ and $\alpha_3\le 0.9742$. At $\beta_W=0.52$, $N\beta=\frac{13}{75}$ and $\gamma=\frac{5411}{3200}$. The terms of the series for $Z_*(1.04)$ in the proof of Theorem 6 are at most 1, 0, 0.2704, 0.0468694, 0.0365581, 0.0095052, 0.0035697, 0.0009424 and 0.0002581, with sum at most 1.3681029, and the last term is less than $1.31\cdot10^{-4}$; so $Z_*(1.04)\le 1.368234$, $e^{-0.208}\ge 0.812132$, $\frac{5e^{-0.208}}{9Z_*(1.04)}\ge 0.329756$ and $\varepsilon(1.04)\le 0.670244<0.81869^2$. Hence $\Lambda(r)\le 0.48417$, $\tau\le 0.39638$, $w\le 0.017864$, $s\le 0.49314$, $C\le 0.066654$, $K_3\le 0.5727$ and $\alpha_3\le 0.9927$. With $y=1-\alpha_3$, $-\frac12\log\alpha_3\ge\frac12(y+\frac{y^2}{2})$, which is at least 0.013 at $\beta_W=0.515$ and 0.0036 at 0.52. □

Remark 39 (a computation, used by no theorem). The single-link criterion ends where $2r\Lambda(r)$ reaches one. Before the corrections for the three links the block sum is $\frac{10}{12}$ of the single-link sum $2r\Lambda(r)$, so it can stay below one while $2r\Lambda(r)$ is slightly above one. With the constants of Lemma 31 the corrections use most of that room: evaluated in floating point, α_3 of Theorem 7 reaches one near $\beta_W=0.522$, and the bound α_1 of Theorem 6 near 0.5176.

Computed constants go further. For $V,W\in\mathrm{SU}(3)$, $g(VcW)=g(c)$ and $x\mapsto VxW$ is an isometry, so the eigenvalues of $\mathrm{Hess}\,g$ and the norm of D^3g are constant on the orbits, and by Lemma 33 it suffices to bound them on $\mathscr{R}_r$. The derivatives of g are the cumulants of φ under ν_c. Cut the box $[0,r]^2\times[-r,r]\times[0,r]$ into boxes of side $r/8$ and keep the 841 (784 at $r=\frac{38}{25}$, 826 at $r=\frac{77}{50}$) that survive the two tests of Sec. VII H. On each box Q we compute, with rigorous error bounds, the moments of φ of degree at most three at the sixteen corners b_i and at the centre b_0, and put $k=m(b_0)$, the mean of φ at the centre.

Upper bound. For $b\in Q$ and every h, $\mathrm{Var}_{\nu_b}(\varphi_h)\le\nu_b[(\varphi_h-\langle k,h\rangle)^2]\le e^{\varsigma/8}\max_i\nu_{b_i}[(\varphi_h-\langle k,h\rangle)^2]$ by Lemma 35(ii).

Lower bound. Integrating the right inequality of (12) against $f\ge0$ and dividing by $Z(b)\le\sum_i\omega_iZ(b_i)$, the left one, gives $\nu_b(f)\ge e^{-\varsigma/8}\min_i\nu_{b_i}(f)$. With $f=(\varphi_h-\nu_b\varphi_h)^2$ this gives $\mathrm{Var}_{\nu_b}(\varphi_h)\ge e^{-\varsigma/8}\min_i\mathrm{Var}_{\nu_{b_i}}(\varphi_h)$.

Third cumulant. Put $\delta=m(b)-k$. The mean value theorem and the upper bound give $\|\delta\|\le\tau\rho+e$, with ρ the largest distance from b_0 in Q and e the error of the computed mean. Expanding the central moment about k,

$$D^3g(b)(u,v,v)=T_b(u,v,v)-\delta_u\,\mathrm{Var}(\varphi_v)\\-2\delta_v\,\mathrm{Cov}(\varphi_u,\varphi_v)-\delta_u\delta_v^2,$$

with $T_b=\nu_b[(\varphi-k)^{\otimes3}]$, $\delta_u=\langle\delta,u\rangle$ and the moments under ν_b. For a function G of either sign, writing $G=G^+-G^-$ and applying the two bounds above to $G^\pm$ gives $|\nu_b(G)|\le(\max_i|\nu_{b_i}(G)|+\varepsilon\max_i\nu_{b_i}|G|)/(1-\varepsilon)$ with $\varepsilon=1-e^{-\varsigma/8}$. For $G=(\varphi_u-k_u)(\varphi_v-k_v)^2$ we use $|\nu_{b_i}(G)|\le\|T_{b_i}\|$, the largest singular value of the 18×324 unfolding of T_{b_i}, and $\nu_{b_i}|G|\le2\sqrt3\,\nu_{b_i}[(\varphi_v-k_v)^2]$, since $\|U-k\|_{\mathrm{HS}}\le2\sqrt3$. The largest eigenvalues and singular values are bounded as in Sec. VII H, and the maxima over these boxes are given below (Appendix C). In this way every eigenvalue of $\mathrm{Hess}\,g$ on rK lies in $[0.056015,0.247316]$ at $r=\frac32$ and $|D^3g(c)(u,v,v)|\le0.182126$ for unit u,v; at $r=\frac{38}{25}$ the numbers are $[0.054978,0.248280]$ and 0.183546, and at $r=\frac{77}{50}$ they are $[0.053955,0.249240]$ and 0.184989. With these constants and the bounds for $2r\Lambda(r)$ of Remark 36, the quantities of (15) evaluated in ball arithmetic are, at $\beta_W=\frac34$, $w\le0.017935$, $s\le0.34088$, $C\le0.043520$, $K_3\le0.38609$ and $\alpha_3\le0.96521$; at $\beta_W=0.72$, $\alpha_3\le0.91815$; at $\beta_W=0.76$, $w\le0.018609$, $s\le0.34673$, $C\le0.045770$, $K_3\le0.39454$ and $\alpha_3\le0.99950$; at $\beta_W=0.77$, $w\le0.019298$, $s\le0.34186$, $C\le0.047232$, $K_3\le0.38953$ and $\alpha_3\le0.99978$. They would carry the block criterion to $\beta_W=0.77$, with a gap of 0.0177 at $\frac34$, where the cover of Remark 36 alone gives $1.1\cdot10^{-4}$. Evaluated in floating point with the constants of a scan of $\mathscr{R}_r$, the block sum stays below one up to $\beta_W\approx0.78$. No result of the paper uses these numbers.

IX. THE CONTINUUM STEP AT ONE PHYSICAL SCALE

The bounds of the preceding sections hold at fixed coupling and give a gap that is at least a fixed number in lattice units,

and [20] shows that no such bound can produce a continuum gap. This section replaces the single-link condition of Proposition 11 by a condition on one cube of n^4 sites, averaged over its translates in the manner of Dobrushin and Shlosman [43], and proves that the condition gives a gap of order $1/n$ in lattice units. If it holds with n proportional to $1/a$, that is, in a cube of fixed physical size, the gap survives in physical units and passes to every continuum limit (Theorem 43). The two statements that remain unproved, the cube estimate and the existence of the limit, are stated in Sec. X.

A. Averaged cubes

For a finite set $V \subset A$ let $\gamma_V(\cdot \mid Q)$ be the probability measure on G^V with density proportional to $\exp\bigl(N\beta \sum_{p \cap V \neq \emptyset} \operatorname{Re}\operatorname{Tr} Q_p^F\bigr)$ with respect to Haar measure, the links outside V being those of Q. Its conditional law of Q_e, $e \in V$, given the other links of V is γ_e^F. Write $d_V(Q_V, Q'_V) = \sum_{i \in V} \delta(Q_i, Q'_i)$ and, for $j \in A \setminus V$,

$$C_{V,j} = \sup \frac{W_1^{d_V}\bigl(\gamma_V(\cdot \mid Q), \gamma_V(\cdot \mid Q')\bigr)}{\delta(Q_j, Q'_j)},$$

the supremum over pairs that differ only at j; $W_1^{d_V}$ is the Kantorovich distance for d_V. The coefficient vanishes unless j shares a plaquette with a link of V.

Let $n \geq 2$ be even and let V be the set of active links (x,μ) with $x \in [0,n)^4$. The direction field has period two, so the translations by $2\mathbb{Z}^4$ preserve F, A and γ^F, and the translates $V_x = V + x$, $x \in 2\mathbb{Z}^4$, satisfy $C_{V_x, j+x} = C_{V,j}$. Every active link lies in exactly $N_n = (n/2)^4$ of them. A *class* is an orbit of $2\mathbb{Z}^4$ on A, that is, a residue of the base point modulo two together with a direction; there are at most 64 classes. For a class c put

$$K_c = \sum_{j \in c \setminus V} C_{V,j}.$$

Definition 40. For $0 < \alpha < 1$, the condition $D(n,\alpha)$ at the coupling β is $K_c \leq \alpha N_n$ for every class c.

Since $\sum_{x \in 2\mathbb{Z}^4,\, j \notin V_x} C_{V_x,j} = \sum_x C_{V, j-x}[j - x \notin V] = K_{c(j)}$ for the class $c(j)$ of j, the condition says that the total influence of a link on the translates that do not contain it is at most the fraction α of the number of translates that do. Figure 7 shows the cubes.

For $n = 2$ it is a bound on blocks of 48 links. Distances between links are those of their midpoints; a link of V and a link sharing a plaquette with V are at distance at most $\rho_n = 2n$, and c_n denotes the number of active links within distance ρ_n of a given one.

Theorem 41. *Let $\beta \geq 0$ and suppose that $D(n,\alpha)$ holds. Then statements* (i)–(iv) *of Proposition 11 hold,* (v) *holds for every $\lambda < \log(1/\alpha)/\rho_n$, and the transfer matrix of ω has spectral gap at least $\log(1/\alpha)/(2n)$.*

Proof. The proof of Proposition 11 uses the coefficients $C_{ee'}$ only through Lemma 347 and Corollary 348. Lemma 350 of Appendix B gives the same conclusions under $D(n,\alpha)$, for every $\lambda > 0$ with $\alpha e^{\lambda \rho_n} < 1$, with the constant $\theta D/(1-\theta)$ replaced by $D(1 + c_n e^{\lambda\rho_n})/(1-\theta)$ and $\theta = \alpha e^{\lambda\rho_n}$. The measures used there are invariant under every γ_{V_x} with V_x disjoint from the exceptional set: a Gibbs measure of γ^F and its conditional measures given the links of a finite set by Lemma 350(a), and the extended finite-volume measure of part (ii) because every plaquette that meets such a V_x lies in $P(\Lambda)$. Part (vi) then follows from Proposition 351 as before, with every $\lambda < \log(1/\alpha)/(2n)$. □

B. The continuum limit

Let $a_k = 2^{-k} a_0$ and let $\beta_k \geq 0$ be couplings, typically with $\beta_k \to \infty$. Suppose that the lattice states ω_k at β_k exist as in Proposition 11, with transfer matrices $\mathbb{T}_k$ on the reconstructed spaces $\mathscr{H}_k$ and vacua Ω_k (Appendix F). Let $\mathscr{T}$ be a complex vector space of test functions on $\mathbb{R}^4$ supported in $\{t > 0\}$ and, for each k, let $f \mapsto O_k(f)$ be a linear map from $\mathscr{T}$ to gauge-invariant local observables supported in positive lattice times; the example to keep in mind is $O_k(f) = Z_k \sum_x a_k^4 f(a_k x)\Phi(x)$ for a local gauge-invariant field Φ, a renormalisation constant Z_k and $\Phi(x)$ the translate of Φ to x. Write ϑ for the time reflection and τ_t for the time translation by t lattice steps.

Definition 42. A *continuum limit* of (ω_k, O_k) is a Hilbert space $\mathscr{H}$ with a unit vector Ω, a self-adjoint operator $H \geq 0$ with $H\Omega = 0$, and a linear map $f \mapsto \psi(f)$ from $\mathscr{T}$ to $\mathscr{H}$ with dense range, such that for all $f, g \in \mathscr{T}$ and all dyadic $\tau \geq 0$,

$$\langle \psi(f), e^{-\tau H} \psi(g) \rangle = \lim_{k \to \infty} \omega_k\bigl(\vartheta(\overline{O_k(f)})\, \tau_{\tau/a_k} O_k(g)\bigr),$$
$$\langle \Omega, \psi(f) \rangle = \lim_{k \to \infty} \omega_k\bigl(O_k(f)\bigr),$$

the limits taken over the k for which τ/a_k is an integer. It is *nontrivial* if $\mathscr{H} \neq \mathbb{C}\Omega$.

The Osterwalder–Schrader reconstruction [88, 89] produces such a limit from convergent renormalised Schwinger functions that satisfy the axioms; the definition records only what the next theorem uses.

Theorem 43. *Suppose that there are $\alpha < 1$, $\ell > 0$ and even n_k with $n_k a_k \leq \ell$ such that $D(n_k, \alpha)$ holds at β_k for every k. Then every continuum limit of (ω_k, O_k) satisfies*

$$\sigma(H) \subset \{0\} \cup [m_*, \infty), \qquad m_* = \frac{\log(1/\alpha)}{2\ell},$$

and 0 is a simple eigenvalue. If the limit is nontrivial, $\sigma(H) \cap [m_*, \infty) \neq \emptyset$, so H has a mass gap of at least m_*.

Proof. By Theorem 41 the state ω_k exists and $\mathbb{T}_k$ has gap at least $\log(1/\alpha)/(2n_k)$ on the complement of Ω_k. Let $f \in \mathscr{T}$, put $O = O_k(f)$ and $\varphi_k = \Psi_O - \omega_k(O)\Omega_k$. In the reconstruction of Appendix F, $\langle \Psi_F, \mathbb{T}_k^t \Psi_H \rangle = \omega_k(\vartheta(\bar{F})\tau_t H)$, so $\langle \Omega_k, \Psi_O \rangle = \omega_k(O)$, and $\mathbb{T}_k \Omega_k = \Omega_k$; hence, for integers $t \geq 0$,

$$\langle \varphi_k, \mathbb{T}_k^t \varphi_k \rangle = \omega_k\bigl(\vartheta(\bar{O})\, \tau_t O\bigr) - |\omega_k(O)|^2$$
$$\leq e^{-t \log(1/\alpha)/(2n_k)}\, \|\varphi_k\|^2.$$

(a) averaged cubes, $n = 4$ (b) one physical scale

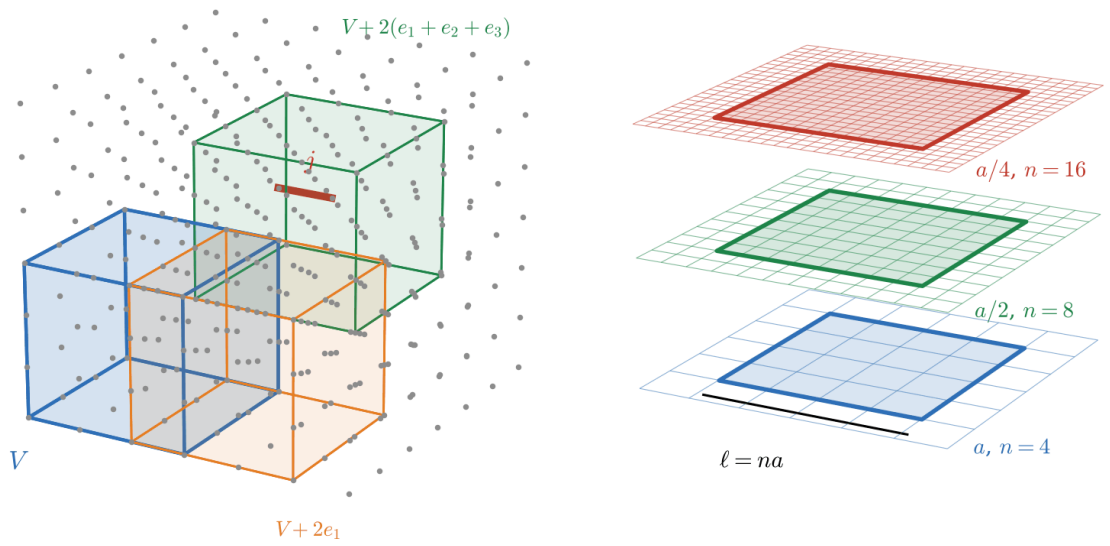


FIG. 7. (a) The cube V of side $n = 4$ and two of its translates by $2\mathbb{Z}^4$, drawn in three of the four dimensions. A link j outside V enters K_c through every translate whose boundary it touches, and $D(n,\alpha)$ compares that total with the $N_n = (n/2)^4$ translates that contain j. (b) The cube of Theorem 43: its physical side $\ell = na$ is held fixed while the spacing is halved, so n doubles at each step and the lattice gap $\log(1/\alpha)/(2n)$ is of order a/ℓ.

Take $\tau \geq 0$ dyadic and k large, so that $t = \tau/a_k$ is an integer; then $t/(2n_k) = \tau/(2n_k a_k) \geq \tau/(2\ell)$, and the right side is at most $e^{-m_*\tau}\|\varphi_k\|^2$. Both sides converge by Definition 42. With $\varphi = \psi(f) - \langle\Omega, \psi(f)\rangle\Omega$ and $e^{-\tau H}\Omega = \Omega$ the limit reads

$$\langle\varphi, e^{-\tau H}\varphi\rangle \leq e^{-m_*\tau}\|\varphi\|^2.$$

The maps $f \mapsto O_k(f)$ and $f \mapsto \psi(f)$ are linear, so the vectors φ form a linear subspace, and it is dense in $\Omega^\perp$ because the range of ψ is dense. The inequality extends to $\Omega^\perp$ by continuity, and from dyadic τ to all $\tau \geq 0$ by the strong continuity of $e^{-\tau H}$. For $\varphi \in \Omega^\perp$ with spectral measure μ_φ of H this says $\int e^{-\tau E} d\mu_\varphi(E) \leq e^{-m_*\tau}\mu_\varphi([0,\infty))$ for all $\tau \geq 0$; if $\mu_\varphi([0, m_* - \eta]) > 0$ for some $\eta > 0$, the left side is at least $\mu_\varphi([0, m_* - \eta])e^{-(m_*-\eta)\tau}$, which exceeds the right side for large τ. So μ_φ lives on $[m_*, \infty)$ for every $\varphi \in \Omega^\perp$, which gives both claims about 0, and the spectrum on $\Omega^\perp \neq \{0\}$ lies in $[m_*, \infty)$ and is not empty. □

The proof uses $D(n_k, \alpha)$ only through the decay of the two-point functions, and the same argument accepts any decay with constants that depend on the observable.

Theorem 44. *Let $m_0 > 0$ and let $\mathscr{T}_0$ span $\mathscr{T}$. Suppose that for each $f \in \mathscr{T}_0$ there are $C_f < \infty$, $\tau_f \geq 0$ and k_f such that*

$$\begin{aligned} C_k^f(\tau) &= \omega_k\big(\vartheta(\overline{O_k(f)})\,\tau_{\tau/a_k} O_k(f)\big) - |\omega_k(O_k(f))|^2 \\ &\leq C_f e^{-m_0\tau} \end{aligned}$$

for all $k \geq k_f$ and all dyadic $\tau \geq \tau_f$ with $\tau/a_k \in \mathbb{Z}$. Then every continuum limit of (ω_k, O_k) satisfies $\sigma(H) \subset \{0\} \cup [m_0, \infty)$, and 0 is a simple eigenvalue. If the limit is nontrivial, H has a mass gap of at least m_0.

Proof. For $f \in \mathscr{T}$ put $\varphi(f) = \psi(f) - \langle\Omega, \psi(f)\rangle\Omega$, which is linear in f. Since $e^{-\tau H}$ is self-adjoint and fixes Ω, Definition 42 gives $\langle\varphi(f), e^{-\tau H}\varphi(f)\rangle = \lim_k C_k^f(\tau)$. For $f \in \mathscr{T}_0$ let μ be the spectral measure of H at $\varphi(f)$. Then $\int e^{-\tau E} d\mu(E) \leq C_f e^{-m_0\tau}$ for dyadic $\tau \geq \tau_f$. If $\mu([0, m_0 - \eta]) > 0$ for some $\eta \in (0, m_0)$, the left side is at least $\mu([0, m_0 - \eta])e^{-(m_0-\eta)\tau}$, which is impossible for large τ. Hence the spectral projection $E_{[0,m_0)}$ of H vanishes on $\varphi(\mathscr{T}_0)$. By linearity it vanishes on $\varphi(\mathscr{T})$, and since its kernel is closed, it vanishes on the closure of $\varphi(\mathscr{T})$, which is $\Omega^\perp$ because $\psi(\mathscr{T})$ is dense. So $E_{[0,m_0)} = \langle\Omega, \cdot\rangle\Omega$, which gives both claims. □

Theorem 43 is the case $m_0 = m_*$, $\tau_f = 0$ and $C_f = \sup_{k \geq k_f}\|\varphi_k\|^2$, which is finite because $\|\varphi_k\|^2 = C_k^f(0)$ converges. Theorem 44 needs neither a lattice gap nor the gauge of this paper. It applies to the output of a cluster expansion of any coarse effective theory, provided the expansion bounds the correlations of a spanning set of renormalised observables in physical units.

Theorem 43 bypasses the obstruction of [20]: under the hypothesis the lattice gap is at least $\log(1/\alpha)/(2n_k)$, of order a_k/ℓ in lattice units, which tends to zero, and only its value in physical units is bounded below. In the language of [20] the hypothesis implies the gap part (R2) of Hypothesis R, with $m = m_*$, and the conclusion holds for every continuum limit without a separate assumption on the gap.

C. Smoothed observables

The observables $O_k(f)$ of Definition 42 need not be local. The renormalised fields that one expects to converge are

smoothed by the gradient flow of the Wilson action [78][14]; at positive flow time their correlation functions are finite to all orders of perturbation theory once the coupling is renormalised [79]. Flow in all four directions mixes the two sides of the reflection plane, but the spatial flow of [94, Sec. II.B] does not: it flows the links of each time slice under the Wilson action of that slice and leaves the temporal links alone.

Fix a bi-invariant Riemannian metric on G. For a configuration U of all links of $\mathbb{Z}^4$ (for the states of this paper, the links of F equal to 1), let $\Phi_s(U)$, $s\in\mathbb{R}$, solve

$$\frac{d}{ds}U_e = -\nabla_e \sum_{p\ni e,\ p \text{ spatial}} \big(N - \operatorname{Re}\operatorname{Tr} U_p\big)$$

for every spatial link e, the sum running over the four plaquettes of the time slice of e that contain it, with the temporal links held fixed and ∇_e the gradient in the variable U_e.

Proposition 45. (i) *Φ_s is defined on every configuration of the infinite lattice, $\Phi_s\circ\Phi_{s'}=\Phi_{s+s'}$, and Φ_s commutes with gauge transformations, with spatial and temporal translations and with the reflections in time. The link $\Phi_s(U)_e$ depends only on the spatial links of the time slice of e, and there are constants κ, c depending only on G such that, if Φ_s^R denotes the flow of the links of that slice within distance R of e with all other links held fixed,*

$$\delta\big(\Phi_s(U)_e, \Phi_s^R(U)_e\big) \le D\, e^{\kappa|s|}\, \frac{(c|s|)^{R+1}}{(R+1)!}$$

for every configuration U.

(ii) *Let ω be a state as in Appendix F and O a bounded gauge-invariant function of finitely many links at times in $[t_0,t_1]$, continuous in them. Then $O\circ\Phi_s$ is a bounded gauge-invariant function of the links at times in $[t_0,t_1]$ and a uniform limit of gauge-invariant local functions at those times.*

(iii) *Theorem 43 holds with $O_k(f)$ replaced by $O_k(f)\circ\Phi_{s_k}$ for any $s_k\in\mathbb{R}$, in particular for $s_k=\tau/a_k^2$ at a fixed flow time τ in physical units.*

(iv) *The image measure $(\Phi_s)_*\omega$ is reflection positive, and its transfer matrix is unitarily equivalent to that of ω, the unitary fixing the vacuum.*

Proof. (i) Embed G in a matrix algebra. The right side of the equation is a smooth function of the links of the four plaquettes containing e, tangent to G at U_e, so on the Banach space of bounded assignments of matrices to the spatial links it is a vector field that is bounded and Lipschitz on a neighbourhood of the configurations with values in G, with a Lipschitz constant that does not depend on the volume because every link meets finitely many others. The Picard–Lindelöf theorem gives a unique local solution; it stays on G because the field is tangent there, and since the field is bounded on G the solution exists for all s; uniqueness gives the group law. The action of a time slice and the metric are invariant under gauge transformations, translations and the reflections, so the flow commutes with them by uniqueness. Since the equation of a spatial link involves only links of its own slice, so does the solution. For the comparison, let $d_{e'}(s)$ be the distance between the two flows at a link e' of the slice. Smoothness on the compact group gives constants κ_0, c_1 with $d_{e'}(s)\le\int_0^{|s|}\big(\kappa_0 d_{e'}+c_1\sum_{e''\sim e'} d_{e''}\big)$ for e' within distance R of e, where $e''\sim e'$ means that the two links share a spatial plaquette (twelve neighbours), while $d_{e'}\le D$ beyond. Iterating the integral form of this inequality, a term that reaches the links beyond distance R needs at least $R+1$ steps of $\sim$, and $\sum_{k>R}x^k/k!\le e^x x^{R+1}/(R+1)!$; this gives the bound with $c=12c_1$ and $\kappa=\kappa_0+c$.

(ii) The truncated flow Φ_s^R commutes with gauge transformations for the same reason as Φ_s, so $O\circ\Phi_s^R$ is a gauge-invariant function of finitely many links at times in $[t_0,t_1]$. By (i) and the uniform continuity of O, it converges to $O\circ\Phi_s$ uniformly as $R\to\infty$.

(iii) The proof of Theorem 43 uses $O_k(f)$ only through the vectors $\Psi_{O_k(f)}$ and the identity $\langle\Psi_F,\mathbb{T}^t\Psi_H\rangle=\omega(\vartheta(\bar F)\tau_t H)$. By (ii) and the footnote in Appendix F, both extend to uniform limits of local functions at positive times, and $f\mapsto O_k(f)\circ\Phi_{s_k}$ is linear.

(iv) By (i), Φ_s and Φ_{-s} preserve the functions of the links at positive times and commute with ϑ and τ_t. Hence $(\Phi_s)_*\omega\big(\vartheta(\bar F)\tau_t H\big)=\omega\big(\vartheta(\overline{F\circ\Phi_s})\,\tau_t(H\circ\Phi_s)\big)$, which for $t=0$ and $H=F$ is nonnegative, and $F\mapsto F\circ\Phi_s$ induces an isometry between the reconstructed spaces that intertwines the transfer matrices and fixes the vacuum; Φ_{-s} gives its inverse. □

Part (iv) says that smoothing changes the variables and not the theory: it neither creates nor removes a gap. Its use is in the choice of the observables. With $O_k(f)$ the spatially flowed fields at a fixed physical flow time, for instance $Z_k\sum_x a_k^4 f(a_k x)\big(\mathscr{E}_\tau(x)-\omega_k(\mathscr{E}_\tau(x))\big)$ with $\mathscr{E}_\tau(x)$ the flowed action density of the slice, statement (E) asks for the convergence of correlation functions that are known to be finite in perturbation theory [79], and Theorem 43 applies to them. Theorem 46 below constructs their continuum limits along subsequences; their nontriviality is not proved here.

D. Renormalised flowed observables

The spatially flowed fields can be renormalised by one condition at a single physical separation, and their correlation functions then have continuum limits along subsequences without further assumptions. What remains open is whether such a limit is trivial, and that question reduces to one ratio of lattice two-point functions.

Fix a bounded gauge-invariant function Φ of finitely many spatial links of the slice $\{x_0=0\}$, continuous in them, a flow time $\tau>0$ in physical units, and a nonzero test function $f_0\in C_c(\mathbb{R}^4)$ supported in $\{t>0\}$. Put $s_k=\tau/a_k^2$, write Φ_x for the translate of Φ to $x\in\mathbb{Z}^4$, and let

$$A_k=\sum_{x\in\mathbb{Z}^4} a_k^4 f_0(a_k x)\big(\Phi_x\circ\Phi_{s_k}-\omega_k(\Phi_x\circ\Phi_{s_k})\big),$$

[14] The flow moves the gauge field along the negative gradient of the Wilson action. At positive flow time the flowed fields are smooth on the scale of the square root of the flow time, and in perturbation theory their gauge-invariant composite fields need no further renormalisation [78, 79].

$$C_k(\sigma) = \omega_k\big(\vartheta(\bar A_k)\,\tau_{\sigma/a_k} A_k\big)$$

for dyadic $\sigma \ge 0$ (the sum is finite, and σ/a_k is an integer for large k). By reflection positivity $C_k(0) \ge 0$; assume $C_k(0) > 0$ and set $Z_k = C_k(0)^{-1/2}$. Let $\mathscr{D}$ be the set of points $y = (t, y') \in \mathbb{R}^4$ with dyadic coordinates (in units of a_0) and $t > 0$, and for $y \in \mathscr{D}$ and k with $y/a_k \in \mathbb{Z}^4$ let A_k^y be the translate of A_k by y/a_k.

Theorem 46. (i) *For $y, y' \in \mathscr{D}$ and every k with $y/a_k, y'/a_k \in \mathbb{Z}^4$,*

$$Z_k^2 \left|\omega_k\big(\vartheta(\bar A_k^y) A_k^{y'}\big)\right| \le 1.$$

(ii) *Every sequence of k has a subsequence along which $K(y,y') = \lim Z_k^2\, \omega_k(\vartheta(\bar A_k^y) A_k^{y'})$ exists for all $y, y' \in \mathscr{D}$ and $\rho(\sigma) = \lim C_k(\sigma)/C_k(0)$ exists for all dyadic $\sigma > 0$. Along it there is a probability measure μ on $[0,\infty]$ with*

$$\rho(\sigma) = \int_{[0,\infty]} e^{-\sigma E}\, d\mu(E) \qquad (\sigma > 0),$$

the integrand being 0 at $E = \infty$.

(iii) *Along such a subsequence, let $\mathscr{T}$ be spanned by the translates $f_0(\cdot - y)$, $y \in \mathscr{D}$, and one more function $f_* \in C_c(\mathbb{R}^4)$ supported in $\{t > 0\}$ outside their span, and let O_k be the linear map with $O_k(f_0(\cdot - y)) = Z_k A_k^y$ and $O_k(f_*) = 1$. Then (ω_k, O_k) has a continuum limit in the sense of Definition 42, on which the dyadic spatial translations act by unitary operators commuting with H.*

(iv) *This limit is nontrivial if and only if $\mu([0,\infty)) > 0$, if and only if $\rho(\sigma) > 0$ for one dyadic $\sigma > 0$, and then $\rho(\sigma) > 0$ for all of them. Consequently a nontrivial limit exists along some subsequence if and only if $\limsup_k C_k(\sigma)/C_k(0) > 0$ for some dyadic $\sigma > 0$.*

(v) *If the hypothesis of Theorem 43 holds, then μ is carried by $[m_*, \infty]$ and $\rho(\sigma) \le e^{-m_*\sigma}$.*

Proof. By Proposition 45(ii) each $\Phi_x \circ \Phi_{s_k}$, and hence each A_k^y, is a uniform limit of gauge-invariant local functions at times ≥ 1, so the reconstruction of Appendix F applies to them (its footnote). Write $\Psi_k = Z_k \Psi_{A_k}$. Since $\langle \Omega_k, \Psi_{A_k}\rangle = \omega_k(A_k) = 0$ and $\|\Psi_k\|^2 = Z_k^2 C_k(0) = 1$, Ψ_k is a unit vector orthogonal to Ω_k. The relation $\langle \Psi_F, \mathbb{T}_k^t \Psi_H\rangle = \omega_k(\vartheta(\bar F)\tau_t H)$ for all F gives $\Psi_{\tau_t H} = \mathbb{T}_k^t \Psi_H$. Spatial translations commute with ϑ and with τ_t and leave ω_k invariant, so they act on the reconstructed space by unitaries $U_k(x)$ that commute with $\mathbb{T}_k$. Hence, for $y = (t, y')$,

$$Z_k \Psi_{A_k^y} = U_k(y'/a_k)\, \mathbb{T}_k^{t/a_k} \Psi_k.$$

(i) The left side of the inequality is $|\langle Z_k \Psi_{A_k^y}, Z_k \Psi_{A_k^{y'}}\rangle| \le 1$, because $\|U_k\| = 1$ and $0 \le \mathbb{T}_k \le 1$.

(ii) The set $\mathscr{D} \times \mathscr{D}$ is countable and the numbers are bounded by (i), so a diagonal subsequence gives all the limits K; the limits $\rho(\sigma)$ are among them, since $C_k(\sigma)/C_k(0) = \langle \Psi_k, \mathbb{T}_k^{\sigma/a_k} \Psi_k\rangle$ is the value of K at $y = y' = (\sigma/2, 0)$. Let ν_k be the spectral measure (Lemma 352) of $\mathbb{T}_k$ at Ψ_k, a probability measure on $[0,1]$, and μ_k its image under $\lambda \mapsto -a_k^{-1} \log \lambda \in [0,\infty]$. Then $C_k(\sigma)/C_k(0) = \int e^{-\sigma E} d\mu_k(E)$. The space $[0,\infty]$ is compact, so a further subsequence has $\mu_k \to \mu$ weakly (Lemma 353), and $E \mapsto e^{-\sigma E}$ is continuous on $[0,\infty]$ for $\sigma > 0$.

(iii) A finite combination $\sum_j c_j f_0(\cdot - y_j)$ with distinct y_j that vanishes has Fourier transform $\hat f_0(p) \sum_j c_j e^{-ip\cdot y_j} = 0$; since $\hat f_0$ is analytic and not zero, the exponential sum vanishes on an open set and every c_j is zero. So the translates are linearly independent and O_k is well defined. On the space V of finite sums $v = \sum_y c_y \delta_y$ over $\mathscr{D}$ put $B(v,w) = \sum \bar c_y c'_{y'} K(y,y')$. It is the limit of $\langle v_k, w_k\rangle$ with $v_k = \sum_y c_y Z_k \Psi_{A_k^y}$, hence positive semidefinite; let $\mathscr{H}_0$ be the completion of V modulo its null space. For dyadic $\sigma \ge 0$ and dyadic spatial x define $T(\sigma)\delta_y = \delta_{y+(\sigma,0)}$ and $U(x)\delta_y = \delta_{y+(0,x)}$. At level k these are $\mathbb{T}_k^{\sigma/a_k}$ and $U_k(x/a_k)$, so in the limit $B(T(\sigma)v, w) = B(v, T(\sigma)w)$, $B(T(\sigma)v, T(\sigma)v) \le B(v,v)$, $B(v, T(\sigma)v) \ge 0$, and $U(x)$ preserves B. Thus $T(\sigma)$ extends to commuting positive contractions on $\mathscr{H}_0$ with $T(\sigma)T(\sigma') = T(\sigma+\sigma')$, and $U(x)$ to unitaries commuting with them. For dyadic $\sigma > 0$ the kernel of $T(\sigma) = T(\sigma/2)^2$ equals that of $T(\sigma/2)$, so all these kernels coincide; and every δ_y with $y = (t, y')$ equals $T(t/2)\delta_{(t/2, y')}$, a vector in the range of a positive operator and so orthogonal to its kernel. Hence each $T(\sigma)$ is injective on $\mathscr{H}_0$, and $H_0 = -\log T(1)$ is a self-adjoint operator with $H_0 \ge 0$ and $e^{-\sigma H_0} = T(\sigma)$ for dyadic σ, by the uniqueness of positive roots. Put $\mathscr{H} = \mathbb{C}\Omega \oplus \mathscr{H}_0$, $H = 0 \oplus H_0$, $\psi(f_0(\cdot - y)) = \delta_y$ and $\psi(f_*) = \Omega$. The range of ψ is dense, and the identities of Definition 42 hold: for two translates they are the limits defining K, with $\tau_{\sigma/a_k} A_k^y = A_k^{y'+(\sigma,0)}$; and for f_* they read $\omega_k(1) = 1$ and $\omega_k(\vartheta(\bar A_k^y)) = \omega_k(\tau_{\sigma/a_k} A_k^y) = 0$.

(iv) The limit is nontrivial exactly when $\mathscr{H}_0 \ne 0$, that is when $\|\delta_y\|^2 = K(y,y) = \rho(2t)$ is positive for some $y = (t, y')$. By (ii) and Lemma 354(b), $\rho(\sigma) > 0$ for one $\sigma > 0$ if and only if $\mu([0,\infty)) > 0$, and then $\rho(\sigma) > 0$ for every $\sigma > 0$. For the last statement choose a subsequence along which $C_k(\sigma)/C_k(0)$ tends to its lim sup, and refine it by (ii).

(v) Under that hypothesis the proof of Theorem 43 gives $\nu_k([0, e^{-\log(1/\alpha)/(2n_k)}]) = 1$ because $\Psi_k \perp \Omega_k$, so μ_k is carried by $[\log(1/\alpha)/(2n_k a_k), \infty] \subset [m_*, \infty]$, and so is the weak limit μ. □

The measure μ is the spectral measure of the renormalised flowed field in the limit: its part on $[0,\infty)$ is what survives as a state of the continuum theory, and its atom at ∞ is the weight that the field puts on energies that diverge in physical units. The theorem constructs the renormalised continuum observables and their reconstruction for every tuning, and it reduces their nontriviality to the lattice inequality

$$\limsup_{k\to\infty} \frac{C_k(\sigma)}{C_k(0)} > 0$$

for one physical separation σ: the flowed two-point function of the smeared packet at a fixed physical distance must not become negligible against its reflected norm $C_k(0)$. If the lattice gap stayed above some $m > 0$ in lattice units, the ratio would be at most $e^{-m\sigma/a_k}$ and the limit would be trivial, so the inequality is a condition on the tuning $\beta_k \to \infty$; it is not proved here. Nor are the independence of the limit from the subsequence, Euclidean invariance, or bounds on products of

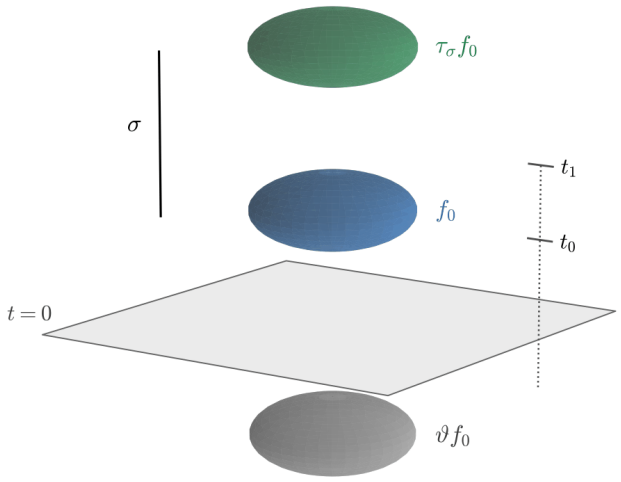


FIG. 8. The packet f_0 of Theorem 46, supported in the time window $[t_0,t_1]$ above the reflection plane $t=0$, its reflection ϑf_0 and its translate $\tau_\sigma f_0$ by σ in time. The correlation $C_k(\sigma)$ pairs the reflected packet with the translated one; $C_k(0)$ is the reflected norm that renormalises the packet.

more than two renormalised fields, for which (i) gives nothing because Z_k may diverge (Theorem 55 reduces them to tree bounds); (E) requires all of these.

The denominator $C_k(0)$ is the reflected norm of the packet smeared by f_0, not a coincidence limit of point fields, and the distinction matters. For the free massless scalar field in four dimensions with spatial momenta cut off at Λ, the covariance at spatial separation 0 is $(1-(1+\Lambda t)e^{-\Lambda t})/(4\pi^2t^2)$ at time $t>0$ and $\Lambda^2/(8\pi^2)$ at coincidence, so its ratio $2(1-(1+\Lambda t)e^{-\Lambda t})/(\Lambda t)^2$ tends to zero as $\Lambda\to\infty$, although the limiting field is the nontrivial Gaussian field; for packets smeared by f_0 the reflected norms of that field have finite positive limits and the ratio stays positive. The criterion of Theorem 46(iv) is about the smeared packets A_k only.

When the ratio is bounded below, renormalising at the separation σ instead of 0 removes the escape to infinite energy altogether, and one slab of the transfer matrix gives a sufficient condition for the bound. Write μ_k for the measures in the proof of Theorem 46.

Proposition 47. (i) *Suppose $C_k(\sigma)\geq cC_k(0)$ for a dyadic $\sigma>0$, a $c>0$ and every k of a sequence. Let μ_k' be the spectral measure, in energies $E=-a_k^{-1}\log\lambda$, of the unit vector $C_k(\sigma)^{-1/2}\Psi_{A_k'}$ with $A_k'=\tau_{\sigma/2a_k}A_k$. Then μ_k' is a probability measure on $[0,\infty)$,*

$$\mu_k'([\Lambda,\infty))\leq c^{-1}e^{-\sigma\Lambda},\qquad \int E^m\,d\mu_k'\leq c^{-1}\Big(\frac{m}{e\sigma}\Big)^m,$$

and $t\mapsto C_k(\sigma+t)/C_k(\sigma)$ is Lipschitz on $[0,\infty)$ with constant $(e\sigma c)^{-1}$. A subsequence of the μ_k' converges weakly to a probability measure on $[0,\infty)$ with the same bounds.

(ii) *Suppose $\langle\Psi_k,(1-\mathbb{T}_k)\Psi_k\rangle\leq Ca_k$ for a constant C and every k of a sequence. Then $C_k(\sigma)/C_k(0)\geq e^{-2C\sigma}$ whenever $Ca_k\leq\frac12$ and σ/a_k is an integer, and $\mu_k([\Lambda,\infty])\leq C(a_k+\Lambda^{-1})$ for every $\Lambda>0$.*

Proof. (i) We have $\Psi_{A_k'}=\mathbb{T}_k^{\sigma/2a_k}\Psi_{A_k}$, so the measure μ_k' is $e^{-\sigma E}d\mu_k(E)$ divided by its mass $C_k(\sigma)/C_k(0)\geq c$, and $e^{-\sigma E}$ vanishes at $E=\infty$. For $E\geq\Lambda$ we have $e^{-\sigma E}\leq e^{-\sigma\Lambda}$, and $E^me^{-\sigma E}\leq(m/e\sigma)^m$ for all $E\geq0$. The ratio in question is the integral of e^{-tE} against μ_k', and the inequality $|e^{-tE}-e^{-sE}|\leq|t-s|E$ with the moment bound for $m=1$ gives the Lipschitz bound. The tail bound makes the sequence tight on $[0,\infty)$, so Prokhorov's theorem gives the weak limit, and the bounds pass to it because $\{E>\Lambda\}$ is open and E^m is a monotone limit of bounded continuous functions.

(ii) Let ν_k be the spectral measure of $\mathbb{T}_k$ at Ψ_k on $[0,1]$, a probability measure, and $n=\sigma/a_k$. By Jensen's inequality for the convex function λ^n,

$$\frac{C_k(\sigma)}{C_k(0)}=\int\lambda^n d\nu_k\geq\Big(\int\lambda\,d\nu_k\Big)^n\geq(1-Ca_k)^n\geq e^{-2Ca_kn},$$

since $\log(1-x)\geq-2x$ on $[0,\frac12]$. Moreover $Ca_k\geq\int(1-e^{-a_kE})d\mu_k\geq(1-e^{-a_k\Lambda})\mu_k([\Lambda,\infty])$, and $1-e^{-x}\geq x/(1+x)$. □

Part (ii) asks for an upper bound on a one-slab Dirichlet form, the opposite direction to a mass gap, which bounds

such forms from below. In the box of Proposition 48, with g smooth, the time smearing supplies the factor a_k up to one ratio: in the notation of its proof, $\mathbb{T}_k = A$ and $\Psi_{A_k} = \sum_n w_n \mathbb{T}_k^n u$ with $w_n = a_k g(a_k n)$ and $u = a_k^3 F^c \psi_0$, and the identity $(1-\mathbb{T}_k)\Psi_{A_k} = \sum_n (w_n - w_{n-1})\mathbb{T}_k^n u$ gives $\langle \Psi_k, (1-\mathbb{T}_k)\Psi_k\rangle \le a_k \|g'\|_{L^1}\|u\|/\|\Psi_{A_k}\|$, so the hypothesis of (ii) holds when the vector u of a single slice stays comparable to the time-smeared packet. That comparison is not proved; Proposition 49 below bounds the ratio without it.

A one-slab bound of this kind also limits the mass that $D(n_k,\alpha)$ can supply. Let e^{-m_k} be the norm of $\mathbb{T}_k$ on $\Omega_k^\perp$. If a unit vector $\varphi_k \perp \Omega_k$ has $\langle \varphi_k, (1-\mathbb{T}_k)\varphi_k\rangle \le \kappa a_k < 1$, then the spectral measure of $\mathbb{T}_k$ at φ_k lives on $[0, e^{-m_k}]$, so $1-e^{-m_k} \le \kappa a_k$ and $m_k \le -\log(1-\kappa a_k)$. If in addition $D(n_k,\alpha)$ holds with $n_k a_k \le \ell$, Theorem 41 gives $m_k \ge \log(1/\alpha)/(2n_k) \ge a_k \log(1/\alpha)/(2\ell)$, and therefore

$$\frac{\log(1/\alpha)}{2\ell} \le \frac{-\log(1-\kappa a_k)}{a_k} = \kappa + O(a_k).$$

So the physical mass m_* of Theorem 43 is at most the constant κ of any one-slab bound. The two hypotheses are compatible because both are statements in physical units. The one-slab bound forces $m_k \to 0$, so it cannot come from an expansion whose convergence gives a lattice gap bounded below in lattice units.

In a periodic spatial box the assumption $C_k(0) > 0$ is a theorem. Let $G = \mathrm{SU}(N)$ with its defining representation, and replace $\mathbb{Z}^4$ by $\mathbb{Z}\times(\mathbb{Z}/L_k\mathbb{Z})^3$ with $L_k \ge 3$ and ω_k by the infinite-time Wilson state of the box. The proof of Theorem 46 uses only reflection positivity and the transfer matrix once the spatial translations are dropped, and parts (i)–(iv) hold in the box with $\mathscr{D}$ replaced by its points $(t,0)$, the time translates of f_0 being independent by the same Fourier argument in t.

Proposition 48. *In the box, let $f_0(t,x) = g(t)h(x)$ with $g,h \ge 0$, and let $\Phi = \sum_{i<j}(N - \mathrm{Re}\,\mathrm{Tr}\,Q_{ij})$ be the energy of the three spatial plaquettes at the origin. If $g(a_k n)h(a_k x) > 0$ for some $n \ge 1$ and some x, then $C_k(0) > 0$.*

Proof. Lüscher proved that the transfer matrix is strictly positive [77]; we recall the argument because the proof uses it in this form.

The one-link kernel. Write χ for the character of the defining representation and $b = \beta_W > 0$. Then

$$e^{\frac{b}{N}\mathrm{Re}\,\chi} = \sum_{m\ge 0} \frac{(b/2N)^m}{m!}(\chi+\bar\chi)^m,$$

and $(\chi+\bar\chi)^m$ is the character of $(\rho\oplus\bar\rho)^{\otimes m}$. Every irreducible representation σ occurs in one of these tensor powers: their matrix coefficients span the algebra of polynomials in the entries of q and $\bar q$, which contains the constants and separates points, so it is dense in $C(G)$ by the Stone–Weierstrass theorem, while the matrix coefficients of a σ that never occurred would be orthogonal to it by Schur orthogonality. So the coefficient of every irreducible character in $e^{\frac{b}{N}\mathrm{Re}\,\chi}$ is positive, and convolution with this function acts on each isotypic part of $L^2(G)$ by a positive scalar.

The transfer matrix. In the temporal gauge the transfer matrix of the box acts on the gauge-invariant part of $L^2(G^E)$, E the spatial links of one slice, as MKM, where K is the tensor product over E of the convolutions above and M is multiplication by the exponential of half the spatial plaquette weight [77, 90]. Both commute with gauge transformations, K is positive and injective by the first step, and M is positive and invertible, so $\langle v, MKMv\rangle = \langle Mv, KMv\rangle > 0$ for $v \ne 0$. Its kernel is a continuous positive function, and K is compact, so its largest eigenvalue λ_0 is simple with an eigenfunction $\psi_0 > 0$, and with $A = MKM/\lambda_0$, a function F of the slice at time $n \ge 0$ has $\Psi_F = A^n F\psi_0$ in the reconstruction, Ω being ψ_0.

Conclusion. Let $F = \sum_x h(a_k x)\Phi_x \circ \Phi_{s_k}$ and $F^c = F - \omega_k(F)$. Then $\Psi_{A_k} = a_k^4\, p(A) F^c\psi_0$ with $p(z) = \sum_{n\ge1} g(a_k n) z^n$, a polynomial with nonnegative coefficients, not all zero, so $p > 0$ on $(0,1]$ and $p(A)$ is injective. The function $\sum_x h(a_k x)\Phi_x$ vanishes at the configuration with every link 1 and is positive at one with a single nontrivial link at a site where $h(a_k x) > 0$; the flow Φ_{s_k} is a bijection, so F is not constant, and as F is continuous and $\psi_0 > 0$, $F^c\psi_0 \ne 0$. Hence $C_k(0) = \|\Psi_{A_k}\|^2 > 0$. □

The proposition removes the one assumption of Theorem 46 in a box; it says nothing about the ratio of part (iv), on which nontriviality depends.

The ratio of Theorem 46(iv) can be compared with correlations of a single slice field. Suppose that $f_0(t,x) = g(t)h(x)$ with $g \ge 0$ supported in $[t_0,t_1]$, $t_0 < t_1$ positive and dyadic, let

$$U_k = \sum_{x\in\mathbb{Z}^3} a_k^3 h(a_k x)\big(\Phi_{(0,x)}\circ\Phi_{s_k} - \omega_k(\Phi_{(0,x)}\circ\Phi_{s_k})\big)$$

be the centred field of the slice at time 0, smeared in space only, and put $u_k = \Psi_{U_k}$ and

$$D_k(r) = \langle u_k, \mathbb{T}_k^{r/a_k} u_k\rangle = \omega_k\big(\vartheta(\bar U_k)\,\tau_{r/a_k} U_k\big)$$

for $r \in a_k\mathbb{Z}$, $r \ge 0$. Let $n_0 = \lceil t_0/a_k\rceil$ and $n_1 = \lfloor t_1/a_k\rfloor$.

Proposition 49. *Assume $C_k(0) > 0$, and let $\sigma \ge 0$ with $\sigma/a_k \in \mathbb{Z}$.*

(i) *$D_k(r) > 0$ for every r, and*

$$\frac{D_k(2a_k n_1+\sigma)}{D_k(2a_k n_0)} \le \frac{C_k(\sigma)}{C_k(0)} \le \frac{D_k(2a_k n_0+\sigma)}{D_k(2a_k n_1)}.$$

(ii) *$\log D_k$ and $\log C_k$ are convex and nonincreasing on $a_k\mathbb{Z}_{\ge0}$. If $D_k(r') \ge c D_k(r)$ with $r < r'$ and $c \le 1$, then $D_k(\beta) \ge c^{\max(1,(\beta-\alpha)/(r'-r))} D_k(\alpha)$ whenever $r \le \alpha < \beta$; if $C_k(\sigma) \ge c C_k(0)$ with $\sigma > 0$, then $C_k(\sigma') \ge c^{\sigma'/\sigma} C_k(0)$ for $\sigma' \ge \sigma$.*

(iii) *Let k run through a sequence. If $D_k(r') \ge c D_k(r)$ for dyadic $0 < r < r'$ with $r \le 2t_0$, a $c \in (0,1]$ and every large k, then for every dyadic $\sigma \ge 0$ and every large k*

$$C_k(\sigma) \ge c^{\max(1,(2t_1-2t_0+\sigma)/(r'-r))}\, C_k(0).$$

Conversely, if $C_k(\sigma) \ge c C_k(0)$ for a dyadic $\sigma > 0$ and every k, then $D_k(2t_0+\sigma') \ge c^{\sigma'/\sigma} D_k(2t_1)$ for every dyadic $\sigma' \ge \sigma$ with $\sigma' > 2(t_1-t_0)$ and every large k.

Proof. For a function F of the slice at time 0 and an integer $n \geq 0$ we have $\Psi_{\tau_n F} = \mathbb{T}_k^n \Psi_F$, since both have inner product $\omega_k(\vartheta(\bar{G})\tau_n F)$ with every Ψ_G. By time translation invariance of ω_k, $A_k = \sum_n w_n \tau_n U_k$ with $w_n = a_k g(a_k n) \geq 0$, and $w_n = 0$ unless $n_0 \leq n \leq n_1$; so $\Psi_{A_k} = \sum_n w_n \mathbb{T}_k^n u_k$. Let ν be the spectral measure of $\mathbb{T}_k$ at u_k, on $[0,1]$, and $W(\lambda) = \sum_n w_n \lambda^n$. Then

$$C_k(\sigma) = \int W(\lambda)^2 \lambda^{\sigma/a_k}\, d\nu(\lambda), \qquad D_k(r) = \int \lambda^{r/a_k}\, d\nu(\lambda).$$

(i) Since $t_0 > 0$, $n_0 \geq 1$ and $W(0) = 0$; so $C_k(0) > 0$ forces $\nu((0,1]) > 0$, and then $D_k(r) > 0$. For $0 \leq \lambda \leq 1$, $\|w\|_1 \lambda^{n_1} \leq W(\lambda) \leq \|w\|_1 \lambda^{n_0}$; inserting these bounds in $C_k(\sigma)$ and $C_k(0)$ gives the two inequalities.

(ii) Convexity of $j \mapsto \log \int \lambda^j d\rho$ for a positive measure ρ on $[0,1]$ is Hölder's inequality, and the integrals decrease in j because $\lambda \leq 1$; take $\rho = \nu$ and $\rho = W^2\nu$. If $\beta \leq r'$, monotonicity gives $D_k(\beta) \geq D_k(r') \geq c D_k(r) \geq c D_k(\alpha)$. If $\beta > r'$, the difference quotients of the convex function $\log D_k$ increase in both end points, so $(\log D_k(\beta) - \log D_k(\alpha))/(\beta - \alpha) \geq (\log D_k(r') - \log D_k(r))/(r' - r) \geq \log c/(r' - r)$. The statement on C_k is the same comparison of quotients on $[0,\sigma]$ and $[0,\sigma']$.

(iii) For large k the times $2t_0$, $2t_1$ are multiples of a_k, and $2a_k n_0 \geq 2t_0$, $2a_k n_1 \leq 2t_1$. Since D_k is nonincreasing, (i) gives

$$\frac{D_k(2t_1+\sigma)}{D_k(2t_0)} \leq \frac{C_k(\sigma)}{C_k(0)} \leq \frac{D_k(2t_0+\sigma)}{D_k(2t_1)}.$$

The first claim is the left inequality with (ii) for $\alpha = 2t_0 \geq r$ and $\beta = 2t_1 + \sigma$. For the second, (ii) gives $C_k(\sigma') \geq c^{\sigma'/\sigma} C_k(0)$, and the right inequality at σ' gives the claim, the two times being ordered because $\sigma' > 2(t_1 - t_0)$. □

So for the packet $g \otimes h$ the open input of Theorem 46(iv) follows from an inequality between two correlations of one slice field,

$$\liminf_k D_k(r')/D_k(r) > 0 \quad \text{for one pair } 0 < r < r' \text{ with } r \leq 2t_0,$$

and by (iii) a uniform bound for one packet gives one for every packet $g' \otimes h$ with g' supported in $[t_1, \infty)$. The condition $r \leq 2t_0$ cannot be dropped: a bound at a later pair says nothing about an earlier packet, as $D_k(r) = (1-4^{-k})2^{-kr} + 4^{-k}2^{-r}$ shows, with $D_k(4)/D_k(3) \to \frac{1}{2}$ and $D_k(2)/D_k(1) \to 0$. A bound at a later pair does apply to the packet shifted forward in time, renormalised by its own reflected norm. The left inequality of (i) bounds the ratio without the comparison of u with the packet that was needed after Proposition 47(ii). Every bound on such ratios proved in this paper, at strong coupling or through Theorem 43, is an upper bound of the form $e^{-m(r'-r)/a_k}$; the inequality above is the opposite one and says that the slice field does not put its weight on energies that diverge in physical units. Reflection positivity and positivity of the transfer matrix do not give it, since a lattice theory with a fixed mass in lattice units has both and its ratio tends to zero. It holds for the free field.

Remark 50. Let D_k be defined as above for the massless free lattice field on $a_k\mathbb{Z}^4$ in place of Φ, with $h \in C_c^\infty(\mathbb{R}^3)$ and the lattice heat flow for a physical time $t_f > 0$ in place of the gradient flow. With $\hat{p}^2 = \sum_i (2a_k^{-1}\sin(a_k p_i/2))^2$, $\cosh(a_k\omega) = 1 + a_k^2\hat{p}^2/2$ and $\hat{h}_k(p) = \sum_x a_k^3 h(a_k x) e^{-ia_k p\cdot x}$,

$$D_k(r) = \int_{[-\pi/a_k, \pi/a_k]^3} |\hat{h}_k(p)|^2 e^{-2t_f\hat{p}^2} \frac{e^{-r\omega}}{2a_k^{-1}\sinh(a_k\omega)} \frac{d^3p}{(2\pi)^3}.$$

On the Brillouin zone $\hat{p} \geq 2|p|/\pi$ and $a_k^2\hat{p}^2/2 \leq 6$; as $\operatorname{arccosh}(1+y)/\sqrt{2y}$ decreases on $(0,6]$, $a_k\omega \geq \kappa a_k\hat{p}$ with $\kappa = \operatorname{arccosh} 7/\sqrt{12} > 0.76$, and $a_k^{-1}\sinh(a_k\omega) \geq \omega$. Since $|\hat{h}_k|$ is bounded uniformly in k, the integrands are dominated by a multiple of $e^{-8t_f|p|^2/\pi^2}/|p|$, which is integrable on $\mathbb{R}^3$, and they converge pointwise to $|\hat{h}(p)|^2 e^{-2t_f|p|^2} e^{-r|p|}/(2|p|)$. By dominated convergence $D_k(r')/D_k(r)$ tends to the corresponding positive ratio of continuum integrals; for $h = e^{-|x|^2}$, $t_f = 0.05$, $r = \frac{1}{2}$ and $r' = 1$ it is 0.6337.

Remark 51. The slice field of Proposition 49 is not linear: it is a gauge-invariant function of the spatial links of one slice. Its free analogue is the magnetic energy, and the conclusion of Remark 50 survives for it. Take the abelian free lattice field on $a_k\mathbb{Z} \times a_k\mathbb{Z}^3_{L_k}$ with $a_k L_k = \ell$ fixed, Feynman gauge, the lattice heat flow of the spatial links for a physical time $t_f \geq 0$, and $U_k = \sum_x a_k^3 \sum_{i<j} (F_{ij}(0,x)^2 - \omega_k(F_{ij}^2))$, the zero spatial momentum part of h (for SU(N) at tree level D_k is multiplied by N^2-1). Since F_{ij} is Gaussian, $\operatorname{Cov}(F^2, F'^2) = 2\operatorname{Cov}(F,F')^2$, and the mixed propagator of the magnetic field at spatial momentum p is $N_{ij,lm}(p)\, K_p(n)$, with a numerator that does not depend on the time momentum and $K_p(n) = e^{-\omega|n|a_k}/(2\sinh(a_k\omega))$. Hence

$$D_k(r) \propto \sum_{p \in \frac{2\pi}{\ell}\mathbb{Z}^3_{L_k} \setminus 0} S_k(p)\, e^{-4t_f\hat{p}^2} \frac{e^{-2r\omega}}{4a_k^{-2}\sinh^2(a_k\omega)},$$

with $S_k(p) = \sum_{i<j,l<m} |N_{ij,lm}(p)|^2 \leq 36\,\hat{p}^4$, and the bounds of Remark 50 dominate the summands by $(9/\kappa^2)\,\hat{p}^2 e^{-4t_f\hat{p}^2 - 2r\kappa\hat{p}}$, summable over $\frac{2\pi}{\ell}\mathbb{Z}^3$ for $r > 0$ uniformly in k, also for $t_f = 0$. The summands converge to the continuum ones, so for every $0 < r < r'$ the ratio $D_k(r')/D_k(r)$ converges to a positive limit, with or without the flow. For $\ell = 6$ the computed values of $D_k(\frac{1}{2})/D_k(\frac{1}{4})$ at $t_f = 0.05$ and $a_k^{-1} = 8, 12, 16, 24$ are 0.27341, 0.27239, 0.27201, 0.27174, against 0.27152 in the continuum; at $t_f = 0$ they are 0.0224, 0.0265, 0.0290, 0.0304, against $(r/r')^5 = \frac{1}{32}$ in infinite volume.

The behaviour at $r = 0$ separates the cases. With the flow, $D_k(0)$ stays bounded and the first-step energy $-a_k^{-1}\log(D_k(a_k)/D_k(0))$ is 6.37, 6.50, 6.56, 6.62 at the four spacings, bounded. Without the flow, $D_k(0) \sim a_k^{-5}$ and $D_k(a_k)/D_k(0) = 0.0198$ at every spacing, so the first-step energy grows like $3.92/a_k$ although every ratio at $r > 0$ converges: the condition of Proposition 49 at $r > 0$ is weaker than a bound on the first step. For the full energy density, which contains F_{0i} and is not a function of one slice, the electric numerator depends on the time momentum, and the second time difference of K_p is $-\delta_{n0}$ plus a multiple of K_p, a contact term: $D_k(0) \sim a_k^{-2}$ even with the flow, $D_k(a_k)/D_k(0) = 0.042$,

0.023, 0.0145 at $a_k^{-1} = 8, 12, 16$, while the ratios at $r > 0$ again converge to the same continuum values (0.27092, 0.27126, 0.27137). The values are numerical evaluations of the free-field formulas.

At next order the spatial flow, being nonlinear, fuses two hard modes into a soft one, and the state Ψ_{U_k} acquires a component of energy about $2q$ for each $q \lesssim a_k^{-1}$. Its norm is of order g^2 and ultraviolet finite, dominated by $q \approx \tau^{-1/2}$, but its mean energy is not: in the free vacuum the first-step energy is $E_{\mathrm{LO}}(1 + g^2(c + \frac{N}{3\pi^2}\log(\sqrt{\tau}/a_k)))$, and the logarithm has coefficient $\frac{8}{11}$ of $2b_0$, $b_0 = 11N/48\pi^2$. A first-step bound at fixed coupling therefore fails logarithmically, while the ratio at a fixed $r > 0$ only loses the norm of that component, damped by e^{-2qr}. This is why Proposition 49 is stated at $r > 0$; the decomposition $\|\mathbb{T}_k^{n/2} u\| \geq c^{n/2}\|v\| - \|w\|$ for $u = v + w$ with a first-step bound on the soft part v alone is the form a four-dimensional argument would use.

Remark 52. The inequality also holds along the right tuning in two-dimensional SU(N) lattice gauge theory, which is interacting and nonabelian but has no propagating gluons. Take the lattice $a_k\mathbb{Z} \times a_k\mathbb{Z}_{L_k}$ with $a_k L_k = \ell$ fixed, the weight $\prod_p \exp(\frac{\beta_k}{N}\operatorname{Re}\operatorname{Tr} U_p)$ with $\beta_k = 2N/(g^2 a_k^2)$, and the temporal gauge. A gauge-invariant function of the links of one slice depends only on the conjugacy class of their holonomy V around the circle; this includes every spatially flowed and spatially smeared slice field, since the flow is gauge covariant. Let w_β be the probability density proportional to $e^{\frac{\beta}{N}\operatorname{Re}\operatorname{Tr} U}$ with respect to Haar measure and, for an irreducible representation λ of dimension d_λ and character χ_λ, $\hat{c}_\lambda(\beta) = d_\lambda^{-1}\int \chi_\lambda w_\beta\, dU$. Between two slices put $W_x = V_x' V_x^{-1}$; the W_x are independent with density w_{β_k}, which is invariant under conjugation and inversion, and

$$V_1' \cdots V_L' = W_1 (V_1 W_2 V_1^{-1}) \cdots \\ \cdots (V_1 \cdots V_{L-1} W_L (V_1 \cdots V_{L-1})^{-1}) V_1 \cdots V_L,$$

so the new holonomy is QV with Q distributed as the L_k-fold convolution of w_{β_k}, independent of V. By Schur's lemma (Lemma 358) the transfer matrix acts on class functions by $\chi_\lambda \mapsto \hat{c}_\lambda(\beta_k)^{L_k}\chi_\lambda$. The coefficients are real, since w_β is invariant under $U \mapsto \bar{U}$, nonnegative, since the exponential of $\frac{\beta}{2N}(\chi_{\mathrm{F}} + \bar{\chi}_{\mathrm{F}})$ is a sum of characters with nonnegative coefficients, and below 1 for $\lambda \neq 0$. For a class function f with $\int f\, dU = 0$ and isotypic components of norms $|\hat{f}_\lambda|$, the slice correlation is therefore

$$D_k(r) = \sum_{\lambda \neq 0} |\hat{f}_\lambda|^2 e^{-r E_{\lambda,k}}, \qquad E_{\lambda,k} = -\frac{\ell}{a_k^2}\log \hat{c}_\lambda(\beta_k).$$

Laplace's method[15] at the unique maximum $U = 1$ of $\operatorname{Re}\operatorname{Tr} U$, with $U = e^X$, $\operatorname{Re}\operatorname{Tr} e^X = N + \frac{1}{2}\operatorname{Tr} X^2 + O(|X|^4)$, $\operatorname{Re}\operatorname{Tr} X^3 = 0$, $\operatorname{Tr}\rho_\lambda(X) = 0$ and $\sum_a \rho_\lambda(T_a)^2 = -2C_2(\lambda)$ for an orthonormal basis of $\mathfrak{su}(N)$ in the norm $-\operatorname{Tr} X^2$, gives $\hat{c}_\lambda(\beta) = 1 - NC_2(\lambda)/\beta + O(\beta^{-2})$, with the Casimir normalised by $C_2 = (N^2-1)/(2N)$ on the fundamental representation (Lemma 359). Hence $E_{\lambda,k} \to g^2 \ell C_2(\lambda)/2$, and since $D_k(r) \leq \|f\|^2$,

$$\liminf_k \frac{D_k(r')}{D_k(r)} \geq \frac{|\hat{f}_\mu|^2}{\|f\|^2} e^{-r' g^2 \ell C_2(\mu)/2} > 0$$

for every $\mu \neq 0$ with $\hat{f}_\mu \neq 0$; for $f = \operatorname{Re}\operatorname{Tr} V$ the ratio is exactly $e^{-(r'-r)E_{\mathrm{F},k}} \to e^{-(r'-r)g^2\ell(N^2-1)/(4N)}$. This is the exact solution of two-dimensional lattice gauge theory, whose continuum limit is the measure of [44, 57, 75], written as the inequality. In two dimensions g^2 has the dimension of a mass squared and there is no ultraviolet problem, so the argument has no four-dimensional analogue. Nor does the spatial flow give the inequality by itself: it damps the spectral weight at large spatial momentum and leaves the weight at zero spatial momentum unchanged. For the free lattice field of Remark 50 with a mass M_k and the same flow, every energy is at least $m_k = a_k^{-1}\operatorname{arccosh}(1 + a_k^2 M_k^2/2)$, so $D_k(r') \leq e^{-m_k(r'-r)} D_k(r)$, which tends to zero relative to $D_k(r)$ when $a_k M_k$ is fixed.

A route to the slice ratio in four dimensions replaces the packet by its conditional expectations given averages of the field over blocks of increasing size and controls the error of each step. The free field shows in which norm such errors can be small. In the ordinary L^2 norm the first step already fails: conditioning on the averages over blocks of side two leaves a fixed fraction of the variance of an energy packet, at every lattice spacing.

Proposition 53. *Let $d \geq 2$, let Λ be a periodic box in $\mathbb{Z}^d$ with even sides, let $\mu > 0$, and let ϕ be the Gaussian field on Λ with density proportional to $\exp\big(-\frac{1}{2}\sum_{x,\nu}(\nabla_\nu\phi(x))^2 - \frac{\mu^2}{2}\sum_x \phi(x)^2\big)$, where $\nabla_\nu\phi(x) = \phi(x+e_\nu) - \phi(x)$. Let $\mathscr{Z}$ be a set of blocks $B = 2z + \{0,1\}^d$, let $\mathfrak{B}$ be the σ-algebra generated by the averages of ϕ over the blocks in $\mathscr{Z}$, let $f \geq 0$ on Λ, and let*

$$F = \sum_{x\in\Lambda} f(x) \sum_{\nu=1}^{d} :(\nabla_\nu\phi(x))^2:,$$

where $:X^2: = X^2 - \mathbb{E}X^2$. For $B = 2z + \{0,1\}^d$ put $x_B = 2z$, $y_B = 2z + e_1 + e_2$ and

$$\alpha_B = \frac{1}{4}\sum_{w\in\{x_B, y_B\}} \Big(d\, f(w) + \sum_{\nu=1}^{d} f(w - e_\nu)\Big),$$

and $\sigma^2 = 2/(2d + \mu^2)$. (i) $\mathbb{E}\operatorname{Var}(F \mid \mathfrak{B}) \geq 2\sigma^4 \sum_{B\in\mathscr{Z}} \alpha_B^2$.

(ii) $\operatorname{Var} F \leq 2d^2 \sum_x f(x)^2$.

(iii) Let $g \geq 0$ be a Lipschitz function on $\mathbb{R}^d$ with compact support and $\int g^2 > 0$. For $n \geq 1$ let Λ_n contain $n\operatorname{supp} g$ in a fundamental domain, $f_n(x) = g(x/n)$, let $\mathscr{Z}_n$ contain every block that meets $n\operatorname{supp} g$, and let $\mu = \mu_n \to 0$. Then for the packets F_n of f_n,

$$\liminf_{n\to\infty} \frac{\mathbb{E}\operatorname{Var}(F_n \mid \mathfrak{B}_n)}{n^d \int g^2} \geq 2^{1-d}, \qquad \limsup_{n\to\infty} \frac{\operatorname{Var} F_n}{n^d \int g^2} \leq 2d^2,$$

[15] For a smooth function with a unique nondegenerate maximum, $\int e^{\beta\varphi} h$ has an expansion in powers of β^{-1} whose coefficients are Gaussian moments at the maximum; Lemma 359 carries it out here.

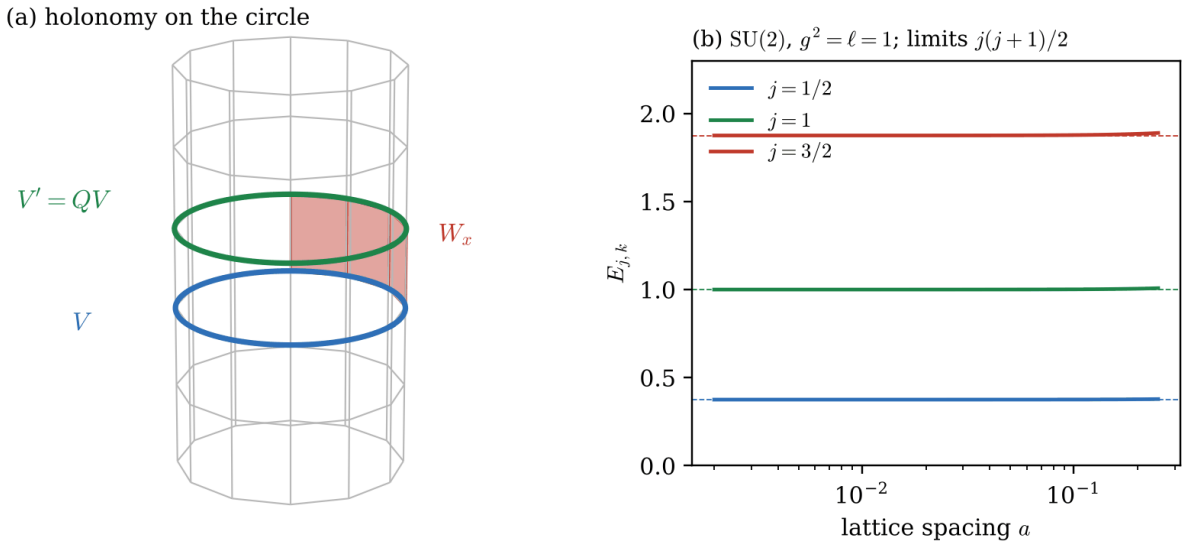


FIG. 9. Two-dimensional SU(N) (Remark 52). (a) The lattice $a_k\mathbb{Z} \times a_k\mathbb{Z}_{L_k}$ drawn as a cylinder: the holonomy V of one slice, the holonomy $V' = QV$ of the next, and the plaquette variables W_x between them. (b) The energies $E_{j,k} = -(\ell/a_k^2)\log \hat{c}_j(\beta_k)$ of SU(2) for $g^2 = \ell = 1$, with $\hat{c}_j(\beta) = I_{2j+1}(\beta)/I_1(\beta)$, against the lattice spacing; the dashed lines are the limits $j(j+1)/2$.

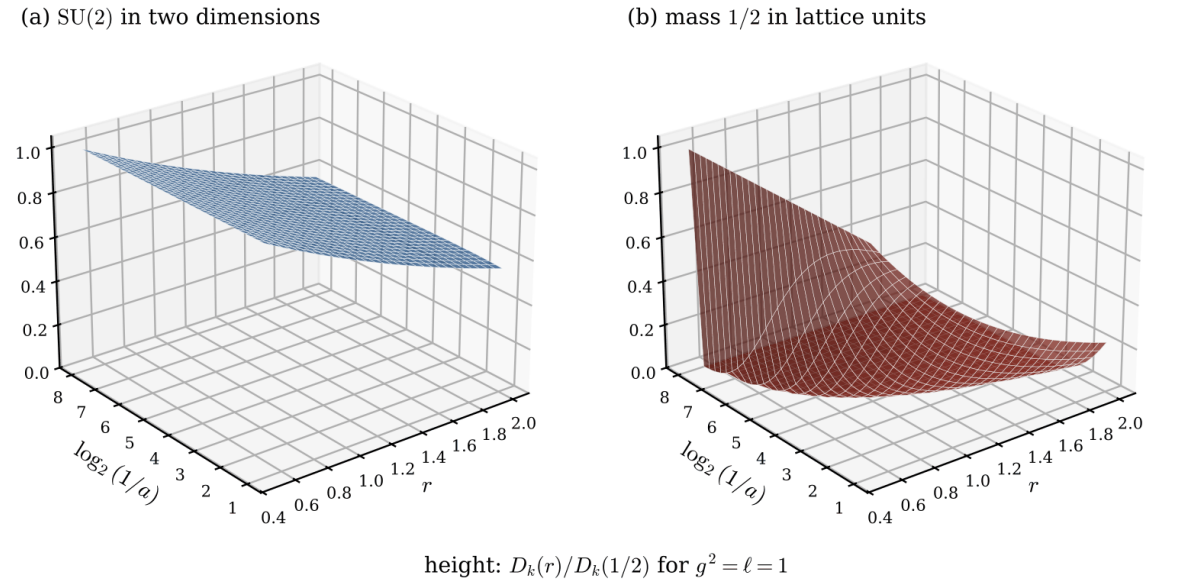


FIG. 10. The ratio $D_k(r)/D_k(\frac{1}{2})$ over the separation r and the lattice spacing a_k. (a) Two-dimensional SU(2) with $f = \mathrm{Re}\,\mathrm{Tr}\,V$ and $g^2 = \ell = 1$, where the ratio is $e^{-(r-1/2)E_{1/2,k}}$ and converges as $a_k \to 0$. (b) The term $e^{-(r-1/2)m_k}$ of the lowest energy m_k of a free field with mass $\frac{1}{2}$ in lattice units, which bounds the ratio from above and tends to zero at every $r > \frac{1}{2}$.

so $\mathbb{E}\,\mathrm{Var}(F_n \mid \mathfrak{B}_n) \geq (2^{-d}d^{-2} - o(1))\,\mathrm{Var}\,F_n$.

Proof. (i) No two of the sites x_B, y_B, $B \in \mathscr{Z}$, are neighbours: their coordinate sums are even, and a neighbour of a site has a coordinate sum of the other parity. Let $\mathfrak{H} \supseteq \mathfrak{B}$ be generated by $\mathfrak{B}$ and by $\phi(w)$ for every w outside $\{x_B, y_B : B \in \mathscr{Z}\}$. By the law of total variance, $\mathrm{Var}(F \mid \mathfrak{B}) = \mathbb{E}[\mathrm{Var}(F \mid \mathfrak{H}) \mid \mathfrak{B}] + \mathrm{Var}(\mathbb{E}[F \mid \mathfrak{H}] \mid \mathfrak{B})$, so $\mathbb{E}\,\mathrm{Var}(F \mid \mathfrak{B}) \geq \mathbb{E}\,\mathrm{Var}(F \mid \mathfrak{H})$. In the exponent of the density, a site w enters through $(d + \frac{\mu^2}{2})\phi(w)^2$ and through $-\phi(w)\phi(w')$ for each of its $2d$ neighbours w'. Given $\mathfrak{H}$, the block average fixes $s_B = \phi(x_B) + \phi(y_B)$, and with $u_B = \phi(x_B) - \phi(y_B)$ we have $\phi(x_B)^2 + \phi(y_B)^2 = \frac{1}{2}(s_B^2 + u_B^2)$. Since the free sites have only $\mathfrak{H}$-measurable neighbours, the conditional density of $(u_B)_{B \in \mathscr{Z}}$ is proportional to $\prod_B \exp(-\frac{1}{4}(2d + \mu^2)u_B^2 + c_B u_B)$ with $\mathfrak{H}$-measurable c_B: the u_B are conditionally independent Gaussians of variance σ^2. A term $(\nabla_v \phi(w))^2$ contains at most one free site, and if it con-

tains x_B, then $\phi(x_B)=\frac12(s_B+u_B)$ shows that it is $\frac14u_B^2$ plus a term linear in u_B plus an $\mathfrak{H}$-measurable term; the same holds for y_B with $-u_B$. The terms containing x_B are those with $w=x_B$ and those with $w=x_B-e_\nu$, d of each kind, so

$$F=\sum_{B\in\mathscr{Z}}\big(\alpha_Bu_B^2+\beta_Bu_B\big)+F_0$$

with $\mathfrak{H}$-measurable β_B and F_0. Writing $u_B=m_B+v_B$ with m_B the conditional mean, $\alpha_Bu_B^2+\beta_Bu_B=\alpha_Bv_B^2+(2\alpha_Bm_B+\beta_B)v_B+\text{const}$, and for a centred Gaussian v of variance σ^2, $\operatorname{Var}(\alpha v^2+\gamma v)=2\alpha^2\sigma^4+\gamma^2\sigma^2$. By conditional independence, $\operatorname{Var}(F\mid\mathfrak{H})\ge2\sigma^4\sum_B\alpha_B^2$.

(ii) For centred jointly Gaussian X,Y, $\operatorname{Cov}(:X^2:,:Y^2:)=2\operatorname{Cov}(X,Y)^2$. With $G_{\nu\rho}(x,y)=\mathbb{E}[\nabla_\nu\phi(x)\nabla_\rho\phi(y)]$,

$$\begin{aligned}\operatorname{Var}F&=2\sum_{x,y}f(x)f(y)\sum_{\nu,\rho}G_{\nu\rho}(x,y)^2\\&\le\sum_{x,y}\big(f(x)^2+f(y)^2\big)\sum_{\nu,\rho}G_{\nu\rho}(x,y)^2.\end{aligned}$$

On the dual torus Λ^*, with $\hat p^2=\sum_\nu|e^{ip_\nu}-1|^2$,

$$G_{\nu\rho}(x,y)=\frac{1}{|\Lambda|}\sum_{p\in\Lambda^*}\frac{(e^{ip_\nu}-1)(e^{-ip_\rho}-1)}{\hat p^2+\mu^2}e^{ip\cdot(x-y)},$$

and by Parseval $\sum_yG_{\nu\rho}(x,y)^2=|\Lambda|^{-1}\sum_p|e^{ip_\nu}-1|^2|e^{ip_\rho}-1|^2(\hat p^2+\mu^2)^{-2}\le1$, since each factor $|e^{ip_\nu}-1|^2$ is at most $\hat p^2$. The same holds for the sum over x, and (ii) follows.

(iii) If $\operatorname{Lip}g=K$, every site entering α_B lies within distance 3 of $2z$, so $|\alpha_B-d\,g(2z/n)|\le3dK/n$. Hence $\sum_{B\in\mathscr{Z}_n}\alpha_B^2\ge\sum_z(d\,g(2z/n)-3dK/n)_+^2$, the sum running over z with $2z\in n\operatorname{supp}g$, and this is $d^2(n/2)^d\int g^2\,(1+o(1))$ as a Riemann sum on the grid $(2/n)\mathbb{Z}^d$. Since $\sigma^4=4/(2d+\mu_n^2)^2\to d^{-2}$, (i) gives the first bound. By (ii) and $\sum_xg(x/n)^2=n^d\int g^2(1+o(1))$, the second follows. □

For $d=4$ and the energy density, $\operatorname{Var}F_n$ grows like $(\ell/a)^4$ at a fixed physical size ℓ of the packet, and no step of conditioning on block averages removes more than a fraction $1-2^{-d}d^{-2}+o(1)$ of it. The reflected norm $C_k(0)=\omega_k(\vartheta(\bar A_k)A_k)$, which renormalises the packet in Theorem 46, involves the covariance only between the support of ϑf_0 and that of f_0, at a fixed physical distance. The errors of a blocking route have therefore to be measured in that norm; Figure 11 shows the blocks between the reflection plane and the packet.

E. Higher correlations

Theorem 46 controls two renormalised fields at a time, and part (i) of it gives nothing for more: Z_k may diverge while the fields $\Phi_x\circ\Phi_{s_k}$ stay bounded, so a bound on the norm of the renormalised packet as an operator is not available. A reflection positive process with a fixed gap and a two-point function independent of the cutoff can have divergent four-point functions at separated times (Remark 57), so no statement about the two-point function alone can replace the following hypothesis on the truncated correlations. In this subsection Φ is real-valued, as the energy of the plaquettes is; for a complex field the conjugate field has to be adjoined and (T1) imposed on all mixed cumulants. Write $\varphi_k(x)=\Phi_x\circ\Phi_{s_k}$, ω_k^T for the truncated correlations (joint cumulants) under ω_k, and for $f\in C_c(\mathbb{R}^4)$

$$A_k(f)=\sum_{x\in\mathbb{Z}^4}a_k^4f(a_kx)\big(\varphi_k(x)-\omega_k(\varphi_k(x))\big),$$

so that $A_k=A_k(f_0)$ and $\bar A_k(f)=A_k(\bar f)$. Let $w_k:\mathbb{Z}^4\to[0,\infty)$ be even functions with $W_k=\sum_xw_k(x)<\infty$, let $r_k>0$ be amplitude scales, and consider:

(T1) *Tree bounds.* For every $n\ge2$ there is K_n, independent of k, with

$$\big|\omega_k^T\big(\varphi_k(x_1);\dots;\varphi_k(x_n)\big)\big|\le K_nr_k^n\sum_T\prod_{\{i,j\}\in T}w_k(x_i-x_j)$$

for all $x_1,\dots,x_n\in\mathbb{Z}^4$, the sum running over the trees T on $\{1,\dots,n\}$.

(T2) *Saturation.* $C_k(0)\ge c\,r_k^2a_k^4W_k$ for a $c>0$ and every k.

(T3) *Physical volume.* $B_*=\sup_ka_k^4W_k<\infty$.

The amplitude r_k separates the size of the field from the volume over which it is correlated; Remark 56 shows why it is needed. For a compact $Q\subset\mathbb{R}^4$ let $|Q|_k=a_k^4\#\{x\in\mathbb{Z}^4:a_kx\in Q\}$; for a cube of side L it is at most $(L+a_0)^4$.

Lemma 54. *Under* (T1)*, for $n\ge2$ and bounded $f_1,\dots,f_n$ with compact supports, f_1 supported in Q,*

$$\begin{aligned}&\big|\omega_k^T\big(A_k(f_1);\dots;A_k(f_n)\big)\big|\\&\qquad\le K_nn^{n-2}r_k^n\,|Q|_k\,(a_k^4W_k)^{n-1}\prod_{j=1}^n\|f_j\|_\infty.\end{aligned}$$

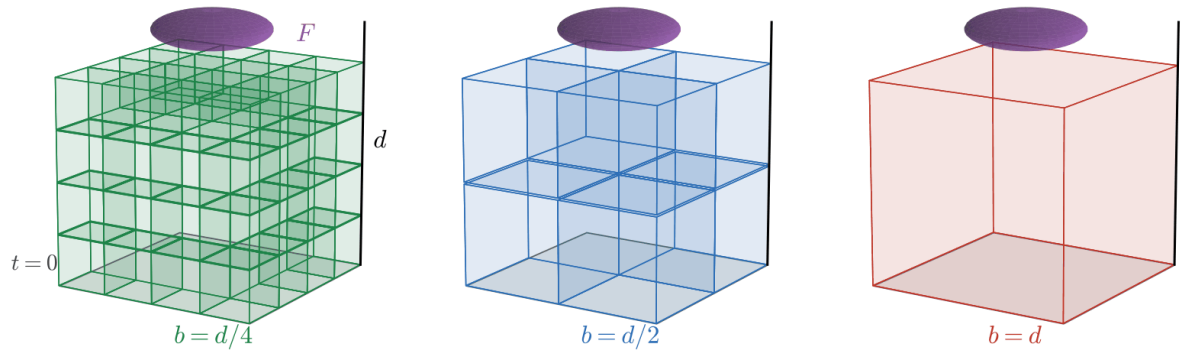
Proof. Joint cumulants of order $n\ge2$ are multilinear and do not change when constants are added to the variables, so the left side is at most $a_k^{4n}\sum_{x_1,\dots,x_n}\prod_j|f_j(a_kx_j)|$ times the absolute value of the truncated correlation, the sums being finite. Insert (T1) and fix a tree T. With x_1 fixed, sum over the other points one at a time, each time over a leaf of what remains of T other than 1; each such sum is at most W_k. The remaining sum over x_1 has at most $|Q|_ka_k^{-4}$ nonzero terms. There are n^{n-2} trees. □

Theorem 55. *Assume* (T1)–(T3).

(i) *For $n\ge2$, f_j as in the lemma,*

$$\begin{aligned}&\big|\omega_k^T\big(Z_kA_k(f_1);\dots;Z_kA_k(f_n)\big)\big|\\&\qquad\le K_nn^{n-2}c^{-n/2}\,B_*^{n/2-1}\,|Q|_k\prod_{j=1}^n\|f_j\|_\infty,\end{aligned}$$

and the moments $\omega_k\big(\prod_jZ_kA_k(f_j)\big)$ with all f_j supported in a cube Q are bounded by $M_n(Q)\prod_j\|f_j\|_\infty$, with $M_n(Q)$ independent of k.



block averages of side b between the plane $t=0$ and the packet F at height d

FIG. 11. Block averages between the reflection plane $t=0$ and a packet F at physical height d, for blocks of side $d/4$, $d/2$ and d. By Proposition 53 the first conditioning step already leaves a fixed fraction of the ordinary variance of the packet, so the errors of such steps have to be measured in the reflected norm.

(ii) *Every sequence of k has a subsequence along which*

$$S_n(f_1,\dots,f_n)=\lim_k \omega_k\Big(\prod_{j=1}^{n} Z_k A_k(f_j)\Big)$$

exists for all n and all $f_j\in C_c(\mathbb{R}^4)$. Each S_n is symmetric, multilinear, bounded by $M_n(Q)\prod_j\|f_j\|_\infty$ on functions supported in Q, hence a distribution on $(\mathbb{R}^4)^n$; it is invariant under all translations of $\mathbb{R}^4$ if the ω_k are translation invariant, and the kernel K of Theorem 46 along the subsequence is $K(y,y')=S_2\big(\overline{f_0(\cdot-y)}\circ\theta, f_0(\cdot-y')\big)$, with θ as in (iii).

(iii) *Put $S_0=1$, $S_1=0$, and let θ be the reflection $t\mapsto -t$ of $\mathbb{R}^4$. For every finite family of $c_\alpha\in\mathbb{C}$ and $f_{\alpha,j}\in C_c(\mathbb{R}^4)$ supported in $\{t>0\}$,*

$$\sum_{\alpha,\beta}\bar c_\alpha c_\beta\, S\Big(\prod_j \overline{f_{\alpha,j}}\circ\theta\cdot\prod_j f_{\beta,j}\Big)\ge 0,$$

where S of a product of n test functions means S_n. Hence the Osterwalder–Schrader construction gives a Hilbert space $\mathscr{H}$ with vacuum Ω and a dense set of vectors $\Psi(P)$ indexed by polynomials P in the smeared fields at positive times, with $\langle\Psi(P),\Psi(P')\rangle=S(\theta\bar P\cdot P')$; in particular the separated higher correlations of the renormalised fields are finite. The vector $\Psi(P_0)$ of the field smeared by f_0 is a unit vector orthogonal to Ω.

(iv) *Suppose the ω_k are translation invariant. Then the time translations by $t\ge 0$ define on $\mathscr{H}$ a strongly continuous semigroup e^{-tH} with $H\ge 0$ self-adjoint, and the spatial translations a strongly continuous unitary group commuting with H. Moreover there is a dyadic $\sigma>0$ with $C_k(\sigma)\ge\frac12 C_k(0)$ for all large k, so the limit of Theorem 46 is nontrivial.*

(v) *If $a_k^4W_k\to 0$ along the subsequence, every S_n is given by Wick's rule from S_2: the limit is the Gaussian field with covariance S_2.*

(vi) *If the ω_k are translation invariant and the hypothesis of Theorem 43 holds, then 0 is a simple eigenvalue of H and $\sigma(H)\subset\{0\}\cup[m_*,\infty)$ with $m_*=\log(1/\alpha)/(2\ell)$, and $\Omega^\perp\neq 0$.*

Proof. (i) By (T2), $Z_k^n=C_k(0)^{-n/2}$ is at most $(c\,r_k^2a_k^4W_k)^{-n/2}$; multiply the lemma by it, which cancels r_k^n, and use $(a_k^4W_k)^{n/2-1}\le B_*^{n/2-1}$ by (T3) for $n\ge 2$. A moment is the sum over partitions of $\{1,\dots,n\}$ of the products of the cumulants of the blocks; blocks of size one vanish because $A_k(f)$ is centred, and each block contributes the bound just proved with $|Q|_k\le(L+a_0)^4$ for a cube of side L. This gives $M_n(Q)$.

(ii) For each cube $Q_L=[-L,L]^4$, $L\in\mathbb{N}$, the continuous functions supported in Q_L form a separable space under the supremum norm. Choose a countable dense set $\mathscr{F}$ of functions; by (i) the countably many sequences $\omega_k(\prod Z_kA_k(f_j))$, $f_j\in\mathscr{F}$, are bounded, and a diagonal subsequence makes all of them converge. By multilinearity,

$$\Big|\omega_k\Big(\prod_j Z_kA_k(f_j)\Big)-\omega_k\Big(\prod_j Z_kA_k(g_j)\Big)\Big|$$
$$\le M_n(Q)\sum_i\|f_i-g_i\|_\infty\prod_{j\neq i}\max(\|f_j\|_\infty,\|g_j\|_\infty)$$

for functions supported in Q, so the limits exist for all $f_j\in C_c(\mathbb{R}^4)$ and obey the same bound. Symmetry and multilinearity pass to the limit. A multilinear form bounded in this way is separately continuous on $C_c^\infty(\mathbb{R}^4)$, so the Schwartz kernel theorem makes it a distribution. For dyadic y and large k, $y/a_k\in\mathbb{Z}^4$ and translation invariance of ω_k and of the flow (Proposition 45(i)) give $\omega_k(\prod Z_kA_k(f_j(\cdot-y)))=\omega_k(\prod Z_kA_k(f_j))$. For a real y take dyadic $y_m\to y$; the functions $f_j(\cdot-y_m)$ converge uniformly to $f_j(\cdot-y)$ with supports in one cube, so the bound above carries the invariance to y. The formula for K follows from $A_k^y=A_k(f_0(\cdot-y))$ and the argument of (iii) with $n=2$.

(iii) Let ϑ be the lattice reflection of Appendix F and θ_k the corresponding reflection of $a_k\mathbb{Z}^4$; it differs from θ by a translation of length at most a_k. Since Φ is real and ϑ commutes with the flow, $\vartheta(\bar{A}_k(f)) = A_k(\bar{f}\circ\theta_k)$. Let $P_k = \sum_\beta c_\beta \prod_j Z_k A_k(f_{\beta,j})$ and $R_k = \sum_\alpha \bar{c}_\alpha \prod_j Z_k A_k(\overline{f_{\alpha,j}}\circ\theta)$, so that $\vartheta(\bar{P}_k)$ is R_k with θ replaced by θ_k. Reflection positivity gives $\omega_k(\vartheta(\bar{P}_k)P_k)\geq 0$. By (i), applied in a cube containing all the supports and their reflections, and by multilinearity, $|\omega_k(\vartheta(\bar{P}_k)P_k) - \omega_k(R_kP_k)|$ is at most a constant independent of k times $\max_{\alpha,j}\|f_{\alpha,j}\circ\theta_k - f_{\alpha,j}\circ\theta\|_\infty$, which tends to zero by uniform continuity, while $\omega_k(R_kP_k)$ tends to the left side of (iii) by (ii). So the limit of the nonnegative numbers is the left side of (iii). The construction of $\mathscr{H}$ is the usual quotient of the polynomials at positive times by the null space of this form [88]. By the same approximation of θ by θ_k, $\|\Psi(P_0)\|^2 = \lim_k Z_k^2 C_k(0) = 1$, and $\langle\Omega,\Psi(P_0)\rangle = S_1(f_0) = 0$.

(iv) For a polynomial P at positive times let $\tau_t P$ be its time translate by $t\geq 0$, again at positive times. By translation invariance and $\theta\circ\tau_t = \tau_{-t}\circ\theta$, $\langle\Psi(\tau_tP),\Psi(P')\rangle = \langle\Psi(P),\Psi(\tau_tP')\rangle$, and $q_P(t) = \|\Psi(\tau_tP)\|^2 = \langle\Psi(P),\Psi(\tau_{2t}P)\rangle$. If P has degree at most d, every partition in (i) of the products in $q_P(t)$ has at most d blocks, each bounded by a multiple of $|Q|_k$ for a cube Q of side $O(1+t)$, so $q_P(t)\leq C_P(1+t)^{4d}$. The Cauchy–Schwarz inequality gives $q_P(t)\leq q_P(0)^{1/2}q_P(2t)^{1/2}$, hence by induction $q_P(t)\leq q_P(0)^{1-2^{-m}}q_P(2^mt)^{2^{-m}}$ for every m, and letting $m\to\infty$, $q_P(t)\leq q_P(0)$. So τ_t maps null vectors to null vectors and extends to a symmetric contraction $\mathsf{T}(t)$ of $\mathscr{H}$; $\mathsf{T}(s)\mathsf{T}(t) = \mathsf{T}(s+t)$, and $\mathsf{T}(t) = \mathsf{T}(t/2)^2\geq 0$. By the bound of (ii) and uniform continuity of the translated test functions, $\|\Psi(\tau_tP)-\Psi(P)\|\to 0$ as $t\to 0$, so the semigroup is strongly continuous on a dense set, hence everywhere, and $\mathsf{T}(t) = e^{-tH}$ with $H\geq 0$ self-adjoint [88]. Spatial translations preserve the form, have the opposite translations as inverses, commute with τ_t, and are strongly continuous by the same argument.

For the ratio let $f_{0,t} = f_0(\cdot-(t,0))$. For dyadic $t>0$ and large k, translation invariance gives $\tau_{t/a_k}A_k = A_k(f_{0,t})$, so by (iii)

$$C_k(t) - C_k(0) = \omega_k\big(A_k(\bar{f}_0\circ\theta_k)A_k(f_{0,t}-f_0)\big),$$

a covariance of two centred fields. The case $n=2$ of (i) bounds Z_k^2 times it by $L_0\|f_{0,t}-f_0\|_\infty$ with $L_0 = (K_2/c)\sup_k|Q|_k\|f_0\|_\infty$, Q a cube containing the supports of $f_0\circ\theta_k$ for all k. By uniform continuity of f_0 there is a dyadic $\sigma>0$ with $L_0\|f_{0,\sigma}-f_0\|_\infty\leq\frac{1}{2}$; then $C_k(\sigma)/C_k(0)\geq\frac{1}{2}$, and Theorem 46(iv) applies.

(v) For $n\geq 3$ the exponent $n/2-1$ in (i) is positive, so with $(a_k^4W_k)^{n/2-1}$ in place of $B_*^{n/2-1}$ the cumulants of order $n\geq 3$ of the renormalised fields tend to zero. The moments are the sums over partitions of products of cumulants, so in the limit only the partitions into pairs remain, which is Wick's rule.

(vi) As in the proof of Theorem 43, $\mathbb{T}_k$ has gap at least $\log(1/\alpha)/(2n_k)$ on the complement of Ω_k. For a polynomial P at positive times let $v_k = \Psi_{P_k} - \omega_k(P_k)\Omega_k$ with P_k its lattice version. For dyadic t and large k,

$$\omega_k\big(\vartheta(\bar{P}_k)\tau_{t/a_k}P_k\big) - |\omega_k(P_k)|^2 = \langle v_k,\mathbb{T}_k^{t/a_k}v_k\rangle\leq e^{-m_*t}\|v_k\|^2,$$

and both sides are finite combinations of moments, which converge by (ii) and the approximation of θ in (iii). The limit is $\langle v,e^{-tH}v\rangle\leq e^{-m_*t}\|v\|^2$ for $v = \Psi(P) - \langle\Omega,\Psi(P)\rangle\Omega$. These v are dense in $\Omega^\perp$, so the inequality holds on $\Omega^\perp$ for all $t\geq 0$, and the spectral theorem gives the spectral statement as in Theorem 43. By (iii), $\Psi(P_0)$ is a unit vector in $\Omega^\perp$. □

Part (iv) shows that the order-two cases of (T1) and (T2) already contain the lower bound on the ratio that Theorem 46(iv) needs; they are not harmless combinatorial assumptions, and proving them along the continuum tuning is at least as hard as that lower bound. Part (vi) passes the conditional mass gap of Theorem 43 to every polynomial in the smeared fields. Neither part proves Euclidean invariance, independence of the limit from the subsequence, or the reconstruction of the field operators, for which the growth of $M_n(Q)$ in n would also be needed.

Remark 56. The amplitude scale cannot be dropped for a bounded field. Suppose $\operatorname{Var}\varphi_k(0)\to 0$, as for the energy of the flowed plaquettes in lattice units when the flowed links approach the identity. Reflection invariance of ω_k, the Cauchy–Schwarz inequality and stationarity give

$$C_k(0)\leq\omega_k(|A_k|^2)\leq\Big(a_k^4\sum_x|f_0(a_kx)|\Big)^2\operatorname{Var}\varphi_k(0)\to 0.$$

With $r_k = 1$, (T2) then forces $a_k^4W_k\to 0$, and by part (v) the limit is Gaussian. With r_k^2 of the order of $\operatorname{Var}\varphi_k(0)$ this is no longer forced. Choosing r_k does not prove (T1) or (T2).

Remark 57. Let $0<\varepsilon<1$, $m>0$, let π be the probability measure on $\{-1,0,1\}$ with weights $\varepsilon/2$, $1-\varepsilon$, $\varepsilon/2$, and let $P_t = e^{-mt}I + (1-e^{-mt})\Pi$ on $L^2(\pi)$, where $\Pi f = \pi(f)$ is the projection onto the constants. This is a symmetric Markov semigroup with $P_t\geq 0$ and gap m, so its stationary process is reflection positive. For $X(s) = s/\sqrt{\varepsilon}$ we have $\pi(X) = \pi(X^3) = 0$, $\pi(X^2) = 1$, $\pi(X^4) = 1/\varepsilon$ and $P_tX = e^{-mt}X$, so the two-point function is e^{-mt} for every ε, while for $t_1\leq t_2\leq t_3\leq t_4$ and $q_i = e^{-m(t_{i+1}-t_i)}$ the four-point function is $q_3\,\pi\big(XP_{t_2-t_1}(XP_{t_3-t_2}X^2)\big) = q_1q_3(q_2/\varepsilon+1-q_2)$, with truncated part $q_1q_3(q_2/\varepsilon-q_2-2q_2^2)$; both diverge as $\varepsilon\to 0$. By part (i), no choice of amplitude scales makes (T1)–(T3) hold for this family with constants independent of ε.

The hypotheses (T1)–(T3) are uniform cluster estimates for the continuum tuning $\beta_k\to\infty$, and the next proposition shows that they are estimates at a critical point: with the order-two cases alone the lattice gap is at most a fixed multiple of a_k, and the tree majorant carries a fixed fraction of its weight at a fixed physical distance. Let $m_k\in[0,\infty]$ be the gap of $\mathbb{T}_k$ on the complement of Ω_k, so that $\|\mathbb{T}_kv\|\leq e^{-m_k}\|v\|$ for $v\perp\Omega_k$, and let $t_0>0$ be the least time in the support of f_0.

Proposition 58. (i) *For every dyadic $\sigma>0$ and every large k, $C_k(\sigma)\leq e^{-m_k\sigma/a_k}C_k(0)$. If $m_k/a_k\to\infty$, every limit of Theorem 46 is trivial; this is the case for $\beta_k=\beta$ fixed in the range of Theorem 5, where $m_k\geq m_G(\beta)$.*

(ii) *Suppose the ω_k are translation invariant and* (T1) *for $n=2$ and* (T2) *hold. Then $m_k\leq a_k\log 2/\sigma$ for every large k, where σ is the dyadic number of Theorem 55*(iv)*, which depends only on K_2/c and f_0. If also the hypothesis of Theorem 43 holds, then $m_*a_k\leq m_k\leq a_k\log 2/\sigma$, and in particular $m_*\leq\log 2/\sigma$.*

(iii) *Under* (T1) *for* $n=2$ *and* (T2)*, let* Q *be a cube containing the support of* f_0 *and* $\gamma = c/(K_2\|f_0\|_\infty^2 \sup_k |Q|_k)$*. Then for every* k

$$\sum_{z\in\mathbb{Z}^4,\ |z_0|\ge R_k} w_k(z) \ge \gamma W_k, \qquad R_k = \frac{2t_0}{a_k} - 1.$$

If $w_k(z) = e^{-\mu_k|z_0|}u_k(\bar z)$ *with* $\bar z$ *the spatial part of* z *and* $u_k \ge 0$ *summable on* $\mathbb{Z}^3$*, then* $\mu_k R_k \le \log(2/\gamma)$*, so the physical time decay rate of the majorant is at most* $\log(2/\gamma)/(2t_0 - a_k)$.

Proof. (i) In the proof of Theorem 46, $C_k(\sigma)/C_k(0) = \langle \Psi_k, \mathbb{T}_k^{\sigma/a_k}\Psi_k\rangle$ with Ψ_k a unit vector orthogonal to Ω_k. The operator $\mathbb{T}_k$ is self-adjoint and fixes Ω_k, so it preserves the complement, and the inequality follows. If $m_k/a_k \to \infty$ then $\rho(\sigma) = 0$ for every dyadic $\sigma > 0$ along every subsequence, and the limit is trivial by Theorem 46(iv). At fixed β in the range of Theorem 5 the gap is at least $m_G(\beta) > 0$ in lattice units, and $a_k \to 0$; the spatial flow does not change the transfer matrix (Proposition 45(iv)).

(ii) Theorem 55(iv) uses (T1) and (T2) only for $n = 2$, since the exponent $n/2 - 1$ in part (i) vanishes there, and gives $C_k(\sigma) \ge \frac12 C_k(0)$ for large k. With (i), $e^{-m_k\sigma/a_k} \ge \frac12$. Under the hypothesis of Theorem 43, the proof of that theorem gives $m_k \ge \log(1/\alpha)/(2n_k) \ge m_* a_k$.

(iii) $C_k(0) = \omega_k(A_k(\bar f_0 \circ \theta_k)A_k(f_0))$, a covariance of two centred fields, by the proof of Theorem 55(iv). A point x in the support of the first sum has $a_k x_0 \le -t_0 + a_k$, and a point y in the support of the second has $a_k y_0 \ge t_0$, so $|x_0 - y_0| \ge R_k$. By (T1) for $n = 2$,

$$\begin{aligned} C_k(0) &\le K_2 r_k^2 \|f_0\|_\infty^2 a_k^8 \sum_{a_k y\in Q} \sum_{|x_0-y_0|\ge R_k} w_k(x-y) \\ &\le K_2 r_k^2 \|f_0\|_\infty^2 |Q|_k\, a_k^4 \sum_{|z_0|\ge R_k} w_k(z), \end{aligned}$$

and (T2) bounds the left side below by $c\, r_k^2 a_k^4 W_k$. For the product form let $R' = \lceil R_k \rceil$; if $R' \le 0$ there is nothing to prove, and otherwise

$$\frac{\sum_{|j|\ge R'} e^{-\mu_k|j|}}{\sum_{j\in\mathbb{Z}} e^{-\mu_k|j|}} = \frac{2e^{-\mu_k R'}}{1+e^{-\mu_k}} \le 2e^{-\mu_k R_k},$$

while the factor $\sum_{\bar z} u_k(\bar z)$ cancels. So $2e^{-\mu_k R_k} \ge \gamma$. □

So reflection positivity and decay do not supply (T2): at fixed β in the range of Theorem 5 the measures are reflection positive with exponential clustering, and $C_k(\sigma)/C_k(0) \to 0$, so no choice of w_k, r_k and constants satisfies the order-two cases of (T1) and (T2) together there. For the same reason no estimate whose constants give a lattice gap bounded below, as a convergent strong coupling expansion does [90], can give the tree bounds along the tuning: (T1) and (T2) hold, if at all, only along couplings where the lattice gap closes at least as fast as a_k. A mass gap in physical units, which bounds the decay of the two-point function, does not give the constants K_n either. By Theorem 55(v) the limit can be non-Gaussian only if $\liminf_k a_k^4 W_k > 0$; for the product majorants of part (iii) the time factor of W_k is $\coth(\mu_k/2) \ge 2/\mu_k$, which is of order $1/a_k$ at least, so the time direction alone is compatible with this. At a fixed physical flow time the flowed fields are smooth on the scale $\sqrt{\tau}$; majorants of product form for them are not proved here.

The order-two cases of (T1) and (T2) say more about the reference vector than part (iv) of Theorem 55 records: for a smooth f_0 they bound every time difference of the packet, which makes its energy distribution tight with finite moments and gives the ratio with an explicit rate. Recall the spectral measures μ_k on $[0,\infty]$ of the proof of Theorem 46, so that $C_k(\sigma)/C_k(0) = \int e^{-\sigma E} d\mu_k(E)$.

Proposition 59. *Suppose the* ω_k *are translation invariant,* (T1) *holds for* $n = 2$ *and* (T2) *holds, and* f_0 *has* $m \ge 1$ *continuous time derivatives. Let* Q *be a cube containing the supports of* $f_0(\cdot - (s,0))$ *for* $0 \le s \le m a_0$*, and* $B_m = (K_2/c)\sup_k |Q|_k \|\partial_t^m f_0\|_\infty^2$.

(i) $\|(1-\mathbb{T}_k)^m \Psi_k\|^2 \le B_m a_k^{2m}$ *for every* k*, where* $\Psi_k = Z_k \Psi_{A_k}$.

(ii) $\int \big((1-e^{-a_k E})/a_k\big)^{2m} d\mu_k(E) \le B_m$ *and* $\mu_k([\Lambda,\infty]) \le B_m(a_k + \Lambda^{-1})^{2m}$ *for* $\Lambda > 0$*. Every weak limit* μ *of the* μ_k *satisfies* $\mu(\{\infty\}) = 0$ *and* $\int E^{2m} d\mu(E) \le B_m$*. If* (T1)–(T3) *hold, the unit vector* $\Psi(P_0)$ *of Theorem* 55(iii)–(iv) *lies in the domain of* H^m*, with* $\|H^m \Psi(P_0)\|^2 \le B_m$.

(iii) *If* $a_k\sqrt{B_1} \le \frac12$*, then* $C_k(\sigma) \ge e^{-2\sqrt{B_1}\sigma} C_k(0)$ *for every* $\sigma \in a_k\mathbb{Z}_{\ge 0}$*, and the gap* m_k *of Proposition 58 satisfies* $m_k \le -\log(1 - a_k\sqrt{B_1}) \le 2\sqrt{B_1}\, a_k$.

Proof. (i) Let $D_k f_0 = \sum_{j=0}^m (-1)^j \binom{m}{j} f_0(\cdot - (j a_k, 0))$. By translation invariance $\mathbb{T}_k^j \Psi_{A_k} = \Psi_{\tau_j A_k}$ and $\tau_j A_k = A_k(f_0(\cdot - (j a_k, 0)))$, so $(1-\mathbb{T}_k)^m \Psi_{A_k} = \Psi_{A_k(D_k f_0)}$, and as in the proof of Theorem 55(iv) its squared norm is the covariance $\omega_k(A_k(\overline{D_k f_0} \circ \theta_k) A_k(D_k f_0))$. The function $D_k f_0$ is supported in Q and is an m-th difference with step a_k, so $\|D_k f_0\|_\infty \le a_k^m \|\partial_t^m f_0\|_\infty$. Lemma 54 with $n = 2$ bounds the covariance by $K_2 r_k^2 |Q|_k a_k^4 W_k \|D_k f_0\|_\infty^2$, and (T2) bounds $C_k(0) = Z_k^{-2}$ below by $c\, r_k^2 a_k^4 W_k$.

(ii) By the spectral theorem the left side of (i) is $\int (1 - e^{-a_k E})^{2m} d\mu_k$, the integrand being 1 at $E = \infty$. For $E \ge \Lambda$, $(1-e^{-a_k E})/a_k \ge (1 - e^{-a_k\Lambda})/a_k \ge \Lambda/(1 + a_k\Lambda)$, because $1 - e^{-x} \ge x/(1+x)$; this gives the tail bound. The set $(\Lambda,\infty]$ is open in $[0,\infty]$, so $\mu((\Lambda,\infty]) \le \liminf_k \mu_k((\Lambda,\infty]) \le B_m \Lambda^{-2m}$, and $\mu(\{\infty\}) = 0$. For a continuous χ with $0 \le \chi \le 1$ and support in $[0, L]$, the same inequality gives $E \le (1 + a_k L)(1 - e^{-a_k E})/a_k$ on that support, so $\int \chi E^{2m} d\mu = \lim_k \int \chi E^{2m} d\mu_k \le B_m$; letting χ increase to 1 gives the moment bound. The vector $\Psi(P_0)$ has $\langle \Psi(P_0), e^{-\sigma H}\Psi(P_0)\rangle = \lim_k C_k(\sigma)/C_k(0) = \int e^{-\sigma E} d\mu$ for dyadic $\sigma > 0$ along the subsequence, so its spectral measure for H is μ.

(iii) By (i) with $m = 1$ and the Cauchy–Schwarz inequality, $\langle \Psi_k, (1-\mathbb{T}_k)\Psi_k\rangle \le a_k\sqrt{B_1}$, which is the hypothesis of Proposition 47(ii) with $C = \sqrt{B_1}$. Since Ψ_k is a unit vector orthogonal to Ω_k, the same quantity is at least $1 - e^{-m_k}$. □

Part (iii) makes Proposition 58(ii) explicit: the order-two tree and saturation bounds pin the lattice gap below $2\sqrt{B_1}\, a_k$,

a physical mass of at most $2\sqrt{B_1}$, and give the ratio of Theorem 46(iv) at every separation. The open input of Theorem 46 is therefore implied by these two bounds for one smooth f_0; nothing here proves them for the Wilson action.

F. An averaged cube condition

The condition $D(n,\alpha)$ enters the proof of Theorem 43 through the comparison lemma, which contracts Lipschitz seminorms and therefore needs the Kantorovich response of a cube for every exterior. It is posed in the forest gauge, and Propositions 94 and 95 and Theorem 236 show what that gauge costs. This subsection replaces it by a condition in the mean over the Wilson measure itself, on gauge-invariant functions and without gauge fixing: a Poincaré inequality for the dynamics that resamples cubes of side n. A finite-speed bound for that dynamics, which uses only the locality of its moves, turns the inequality into decay at a rate of order $1/n$ in lattice units, and the continuum step goes through as in Theorem 43.

Setting. Let $G \subset \mathrm{U}(N)$ be compact and nontrivial, δ a bi-invariant metric of diameter D on G, and φ a positive continuous class function on G, such as $e^{N\beta\,\mathrm{Re}\,\mathrm{Tr}(\cdot)}$ or the Villain weight of Proposition 243. The lattice Λ is either the torus $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$ with L even and $L \geq 2n+4$, or a box of $\mathbb{Z}^4$ with all sides at least $2n$; E is its set of links, P its set of plaquettes (on a box, those with four corners in it), and

$$\mu(dU) = \frac{1}{Z}\prod_{p\in P}\varphi(U_p)\prod_{e\in E}dU_e.$$

A link (y,ν) has base point y; on Λ_L differences of base points are taken in $(-L/2, L/2]^4$ and $|\cdot|_\infty$ is the corresponding distance. For $x \in \mathbb{Z}^4$ (or $x \in \Lambda_L$) let V_x be the set of links of E with base point in $x+[0,n)^4$, and let X be the set of x with $V_x \neq \emptyset$. Every link lies in exactly n^4 of the sets V_x, $x \in X$, and $|X| \geq 4n^4$. For $V \subset E$ let ∂V be the set of links of $E \setminus V$ that share a plaquette of P with a link of V; the conditional law $\gamma^0_V(\cdot \mid U)$ of U_V under μ given the other links has density proportional to $\prod_{p\cap V\neq\emptyset}\varphi(U_p)$ and depends only on $U_{\partial V}$. Write $\mathscr{P}_x$ for the conditional expectation given the links outside V_x, $\mathscr{P}_x f(U) = \int f\, d\gamma^0_{V_x}(\cdot \mid U)$, an orthogonal projection on $L^2(\mu)$, and $\mathscr{A}$ for the closed subspace of functions invariant under the gauge group G^Λ. By gauge covariance, $\gamma^0_{V_x}(\cdot \mid gU)$ is the image of $\gamma^0_{V_x}(\cdot \mid U)$ under the action of g on the links of V_x, so $\mathscr{P}_x$ maps $\mathscr{A}$ into itself.

Definition 60. For $n \geq 2$ and $\varpi > 0$, the condition $\mathrm{BP}(n,\varpi)$ for μ is

$$\varpi\,\mathrm{Var}_\mu(f) \leq \frac{1}{n^4}\sum_{x\in X}\mu\big((f-\mathscr{P}_x f)^2\big)$$

for every real $f \in \mathscr{A}$.

The right side is the Dirichlet form of the dynamics that resamples each cube at rate n^{-4}, so that every link is resampled at rate one, and $\mathrm{BP}(n,\varpi)$ says that this dynamics has spectral gap at least ϖ on gauge-invariant functions. It contains no supremum over exteriors and no gauge. Put $q_n = (2+1/n)^4$, so that $16 < q_n \leq \frac{625}{16}$ for $n \geq 2$, and let $\mathscr{K} = |X|^{-1}\sum_{x\in X}\mathscr{P}_x$ be the chain that resamples one cube chosen uniformly. For $Y \subset X$ let $\mathscr{K}_Y = |X|^{-1}\big(\sum_{x\in Y}\mathscr{P}_x + (|X|-|Y|)I\big)$, the same chain with the moves outside Y suppressed.

Write $y \rightsquigarrow z$ if $\partial V_z \cap V_y \neq \emptyset$, that is, if a move at z reads a link of V_y. A link of ∂V_z shares a plaquette with a link based in $z+[0,n)^4$, so its base point lies in $z+[-1,n]^4$; hence $y \rightsquigarrow z$ implies $|z-y|_\infty \leq n$, and for each z at most $(2n+1)^4 = q_n n^4$ points y have $y \rightsquigarrow z$. For a finite set $S \subset E$ put $Y_1(S) = \{y \in X : V_y \cap S \neq \emptyset\}$ and $Y_{j+1}(S) = Y_j(S) \cup \{y \in X : y \rightsquigarrow z \text{ for some } z \in Y_j(S)\}$.

Lemma 61 (finite speed of the cube chain)**.** *Let f be a bounded function of the links of a finite set $S \subset E$, let $m_0 \geq 0$ be an integer and $Y = Y_{m_0}(S)$, with $Y_0(S) = \emptyset$. Then $\mathscr{K}^t_Y f$ depends only on the links of $S \cup \bigcup_{y\in Y}\partial V_y$, and for every integer $t \geq 0$ and every configuration U*

$$\big|\mathscr{K}^t f(U) - \mathscr{K}^t_Y f(U)\big| \leq \frac{2|S|}{q_n}\,\|f\|_\infty \sum_{m>m_0}\frac{(q_n\tau)^m}{m!},$$

where $\tau = tn^4/|X|$.

Proof. If h depends only on the links of W, then $\mathscr{P}_x h = h$ when $V_x \cap W = \emptyset$, and $\mathscr{P}_x h$ depends only on the links of $(W \setminus V_x) \cup \partial V_x$ otherwise; induction on t gives the first claim.

Run two chains from U. At each step choose $x_s \in X$ uniformly, the choices independent. If $x_s \notin Y$, the first chain resamples V_{x_s} from $\gamma^0_{V_{x_s}}$ and the second does nothing. If $x_s \in Y$ and the two configurations agree on ∂V_{x_s}, both laws of V_{x_s} are the same, and both chains receive one common sample; otherwise they resample independently. The two marginal processes are the chains $\mathscr{K}$ and $\mathscr{K}_Y$, so $\mathscr{K}^t f(U) - \mathscr{K}^t_Y f(U)$ is the expectation of the difference of f at time t, which vanishes unless the set $\mathscr{D}_t$ of links where the two configurations differ meets S.

We claim that every link of $\mathscr{D}_t$ lies in $V_{x_{s_m}}$ for times $s_1 < \cdots < s_m \leq t$ with $x_{s_1} \notin Y$ and $x_{s_l} \rightsquigarrow x_{s_{l+1}}$ for $l < m$. For $t = 0$, $\mathscr{D}_0 = \emptyset$. At step t the links outside V_{x_t} keep their status. If $x_t \notin Y$, the links of V_{x_t} have the chain (t). If $x_t \in Y$, the links of V_{x_t} are in $\mathscr{D}_t$ only if a link $e' \in \partial V_{x_t}$ lies in $\mathscr{D}_{t-1}$; its chain ends at some $s_m < t$ with $e' \in V_{x_{s_m}}$, so $x_{s_m} \rightsquigarrow x_t$, and appending t gives a chain.

If $\mathscr{D}_t$ meets S, there is such a chain with $V_{x_{s_m}} \cap S \neq \emptyset$, so $x_{s_m} \in Y_1(S)$ and $x_{s_{m-l}} \in Y_{l+1}(S)$; since $x_{s_1} \notin Y$, $m > m_0$. For fixed m there are at most $\binom{t}{m} \leq t^m/m!$ choices of the times, at most $n^4|S|$ of x_{s_m} and at most $q_n n^4$ of each earlier cube given the next, and each prescribed sequence occurs with probability $|X|^{-m}$. So the probability that $\mathscr{D}_t$ meets S is at most $\frac{|S|}{q_n}\sum_{m>m_0}(q_n n^4 t/|X|)^m/m!$, and the difference of f is at most $2\|f\|_\infty$. □

For finite $S, S' \subset E$ let $d(S,S')$ be the least $|b-b'|_\infty$ over base points b of links of S and b' of links of S'.

Theorem 62 (decay from the averaged condition)**.** *Suppose that $\mathrm{BP}(n,\varpi)$ holds for μ. Then $\varpi \leq 4$, and for bounded*

$f,g\in\mathscr{A}$ that depend only on the links of finite sets S_f,S_g with $d=d(S_f,S_g)\geq 3n-2$,

$$|\mathrm{Cov}_\mu(f,g)|\leq\Big(e+\frac{k|S_f||S_g|}{64e^2}\Big)\|f\|_\infty\|g\|_\infty e^{-\varpi k/(e^2q_n)},$$
$$k=\Big\lfloor\frac{d+2}{n}\Big\rfloor-2.$$

In particular, for every d, $|\mathrm{Cov}_\mu(f,g)|\leq C(1+d)e^{-\varpi d/(e^2q_nn)}$ with C depending on $|S_f|$, $|S_g|$, $\|f\|_\infty$, $\|g\|_\infty$ only, and complex f,g satisfy both bounds with an extra factor four.

Proof. The bound $\varpi\leq 4$. The function $f=\mathrm{Re}\,\mathrm{Tr}\,U_p$ of one plaquette is gauge invariant and not constant on G^E, and μ has a positive density, so $\mathrm{Var}_\mu(f)>0$. Each $I-\mathscr{P}_x$ is an orthogonal projection that annihilates constants, so $\mu((f-\mathscr{P}_xf)^2)\leq\mathrm{Var}_\mu(f)$, and it vanishes unless V_x contains one of the four links of p, which happens for at most $4n^4$ points x.

The spectral part. Let $\mathscr{A}^0$ be the functions of $\mathscr{A}$ with mean zero. $\mathscr{K}$ is an average of orthogonal projections that preserve $\mathscr{A}^0$, so it is self-adjoint with $0\leq\mathscr{K}\leq I$ on $\mathscr{A}^0$, and $\mathrm{BP}(n,\varpi)$ reads $\langle h,(I-\mathscr{K})h\rangle\geq\varepsilon\|h\|^2$ on $\mathscr{A}^0$ with $\varepsilon=\varpi n^4/|X|$. Hence the spectrum of $\mathscr{K}$ on $\mathscr{A}^0$ lies in $[0,1-\varepsilon]$ and $\|\mathscr{K}^Th\|_2\leq e^{-\varepsilon T}\|h\|_2$ for every integer $T\geq 0$.

The decomposition. Since $(I-\mathscr{K})$ annihilates constants,

$$f-\mu(f)=\mathscr{K}^T\big(f-\mu(f)\big)+\sum_{t=0}^{T-1}\mathscr{K}^t(I-\mathscr{K})f,$$

and pairing with g in $L^2(\mu)$,

$$\mathrm{Cov}_\mu(f,g)=\langle\mathscr{K}^T(f-\mu f),g-\mu g\rangle+\sum_{t=0}^{T-1}\langle\mathscr{K}^tf,(I-\mathscr{K})g\rangle,$$

where the self-adjointness of $\mathscr{K}$ was used. The first term is at most $e^{-\varepsilon T}\|f\|_\infty\|g\|_\infty$.

The local part. Let $Z_g=Y_1(S_g)$, so $|Z_g|\leq n^4|S_g|$ and $(I-\mathscr{K})g=|X|^{-1}\sum_{x\in Z_g}(g-\mathscr{P}_xg)$. If h does not depend on the links of V_x, then $\mathscr{P}_xh=h$ and $\langle h,g-\mathscr{P}_xg\rangle=\langle h,g\rangle-\langle\mathscr{P}_xh,g\rangle=0$. Put $m_0=\lfloor(d+2)/n\rfloor-3\geq 0$, so $k=m_0+1$, and $Y=Y_{m_0}(S_f)$. Suppose a link w lies in V_x with $x\in Z_g$, and either $w\in S_f$ or $w\in\partial V_y$ with $y\in Y_j(S_f)$, $1\leq j\leq m_0$. In the second case there are $y=y_j\rightsquigarrow y_{j-1}\rightsquigarrow\cdots\rightsquigarrow y_1$ with $y_1\in Y_1(S_f)$, so $|y-y_1|_\infty\leq n(m_0-1)$; with a base point b of a link of S_f in $y_1+[0,n)^4$, the base point of w in $y+[-1,n]^4\cap(x+[0,n)^4)$, and a base point b' of a link of S_g in $x+[0,n)^4$,

$$|b-b'|_\infty\leq(n-1)+n(m_0-1)+n+2(n-1)=n(m_0+3)-3,$$

and the first case gives at most $2(n-1)$. Both are at most $n\lfloor(d+2)/n\rfloor-3<d$, which is impossible. So $\mathscr{K}_Y^tf$, which by Lemma 61 depends only on $S_f\cup\bigcup_{y\in Y}\partial V_y$, does not depend on the links of V_x for $x\in Z_g$, and

$$\big|\langle\mathscr{K}^tf,(I-\mathscr{K})g\rangle\big|=\big|\langle\mathscr{K}^tf-\mathscr{K}_Y^tf,(I-\mathscr{K})g\rangle\big|$$
$$\leq\|\mathscr{K}^tf-\mathscr{K}_Y^tf\|_\infty\frac{2|Z_g|}{|X|}\|g\|_\infty.$$

The bound of Lemma 61 increases with t; summing over $t<T$ and putting $\tau=Tn^4/|X|$, the second term is at most

$$\frac{4|S_f||S_g|\,\tau}{q_n}\|f\|_\infty\|g\|_\infty\sum_{m\geq k}\frac{(q_n\tau)^m}{m!}.$$

The choice of T. Take $T=\lfloor k|X|/(e^2q_nn^4)\rfloor$, so that $z=q_n\tau\leq k/e^2$ and $\varepsilon T=\varpi\tau\geq\varpi k/(e^2q_n)-\varpi n^4/|X|\geq\varpi k/(e^2q_n)-1$, because $\varpi\leq 4$ and $|X|\geq 4n^4$. Since $k!\geq(k/e)^k$,

$$\sum_{m\geq k}\frac{z^m}{m!}\leq\frac{z^k}{k!}e^z\leq\Big(\frac{ez}{k}\Big)^ke^z\leq e^{-k(1-e^{-2})},$$

and $\tau/q_n=z/q_n^2\leq k/(256e^2)$. As $\varpi/(e^2q_n)\leq 4/(16e^2)<1-e^{-2}$, both terms are at most the stated bound. For the second form, $k>d/n-3$, $k\leq d+1$ and $3\varpi/(e^2q_n)<1$; for $d<3n-2$ it holds with $C=e\|f\|_\infty\|g\|_\infty$, since $|\mathrm{Cov}_\mu(f,g)|\leq\|f\|_\infty\|g\|_\infty$ and $\varpi d/(e^2q_nn)<1$. Real and imaginary parts give the complex case. □

The proof uses the condition only on $\mathscr{A}$ and only in $L^2(\mu)$; the locality of the moves enters through the counting of chains, at speed e^2q_nn links per unit of the resampling time. The theorem gives the lattice gap and the continuum step. On the torus a *torus limit* at β is a state ω on the continuous gauge-invariant local functions of $\mathbb{Z}^4$ with $\omega(O)=\lim_j\mu_{L_j}(O)$ along a sequence $L_j\to\infty$, each O evaluated on Λ_{L_j} through a box of $\mathbb{Z}^4$ centred at the origin once L_j is large; by compactness of the marginals and a diagonal sequence every sequence $L_j\to\infty$ has a subsequence along which such a limit exists.

Corollary 63 (the lattice gap)**.** *Let $\mathscr{L}$ be an infinite set of even integers $L\geq 2n+4$ and suppose that $\mathrm{BP}(n,\varpi)$ holds for the Wilson measure of Λ_L at $\beta\geq 0$ for every $L\in\mathscr{L}$. Then every torus limit ω along $\mathscr{L}$ is invariant under translations and the hypercubic group and reflection positive, and its transfer matrix has spectral gap at least*

$$m_n=\frac{\varpi}{e^2q_nn}\geq\frac{16\varpi}{625e^2n}.$$

Proof. The measures μ_L are invariant under the translations and the hypercubic group of Λ_L, and for $\beta\geq 0$ and even L they are reflection positive for reflections in planes through sites and in planes bisecting links [90, 95]. Each of these properties is a set of identities or inequalities for finitely many local functions, so it passes to ω, and Appendix F applies. Let F be a continuous gauge-invariant function of finitely many links at positive times. The supports of $\vartheta(\bar F)$ and τ_tF are at distance d with $t-c_F\leq d\leq t+c_F$, c_F depending only on F, and Theorem 62 on Λ_L, $L\in\mathscr{L}$ large, bounds $|\mathrm{Cov}_{\mu_L}(\vartheta(\bar F),\tau_tF)|$ by $C_F(1+t)e^{-m_nt}$ with C_F independent of L. The limit gives $|\omega(\vartheta(\bar F)\tau_tF)-|\omega(F)|^2|\leq C_F(1+t)e^{-m_nt}$, using $\omega(\vartheta(\bar F))=\overline{\omega(F)}$ and translation invariance. These F are dense in the gauge-invariant part of $L^2(\omega)$ on each support, so Proposition 351 with every $m<m_n$ places the spectrum of the transfer matrix on $\Omega^\perp$ in $[0,e^{-m}]$, hence in $[0,e^{-m_n}]$. □

Theorem 64 (the continuum step from the averaged condition)**.** *Suppose that there are $\varpi > 0$, $\ell > 0$ and integers $n_k \geq 2$ with $n_k a_k \leq \ell$ such that, for every k, $\mathrm{BP}(n_k, \varpi)$ holds for the Wilson measure of Λ_L at β_k for all L in an infinite set $\mathscr{L}_k$ of even integers $L \geq 2n_k + 4$, and let ω_k be a torus limit along $\mathscr{L}_k$. Then every continuum limit of (ω_k, O_k) satisfies*

$$\sigma(H) \subset \{0\} \cup [m_*, \infty), \qquad m_* = \frac{16\varpi}{625 e^2 \ell},$$

and 0 is a simple eigenvalue. If the limit is nontrivial, H has a mass gap of at least m_*.

Proof. The proof of Theorem 43 uses $D(n_k, \alpha)$ only through the gap of $\mathbb{T}_k$ on $\Omega_k^{\perp}$. By Corollary 63 that gap is at least $16\varpi/(625e^2 n_k)$, and for $t = \tau/a_k$, $16\varpi t/(625e^2 n_k) \geq m_* \tau$; the rest of that proof is unchanged. □

Proposition 45(iii) holds with Theorem 64 in place of Theorem 43, by the same proof. The averaged condition follows from its supremum form, posed for the same measure without gauge fixing. On Λ_L let $C^0_{V,j}$ be defined as $C_{V,j}$ in Sec. IX with γ^0_V in place of γ_V and the links of E in place of the active links, put $K^0_\nu = \sum_j C^0_{V_0,j}$, the sum over the links $j \notin V_0$ of direction ν, and let $D^0_L(n, \alpha)$ be the condition $K^0_\nu \leq \alpha n^4$ for $\nu = 0, \dots, 3$. The translations of Λ_L preserve μ, so $C^0_{V_x, j+x} = C^0_{V_0, j}$.

Proposition 65. *On Λ_L:*

(i) *if $D^0_L(n, \alpha)$ holds with $\alpha < 1$, then $\mathrm{BP}(n, 1-\alpha)$ holds, for every real $f \in L^2(\mu)$;*

(ii) *for $\varphi \equiv 1$, that is $\beta = 0$, $\mathrm{BP}(n, 1)$ holds for every n and every real $f \in L^2(\mu)$.*

Proof. (i) Let $d_E(U, U') = \sum_{e \in E} \delta(U_e, U'_e)$ and $\kappa_0 = (1-\alpha) n^4 / L^4$. Let U, U' differ only at a link i. Couple $\mathscr{K}(U, \cdot)$ and $\mathscr{K}(U', \cdot)$ by choosing the same x. If $i \in V_x$, the two exteriors agree, the laws of the resampled cube coincide, and the identity coupling gives distance 0. If $i \notin V_x$, an optimal coupling of the two cube laws for d_{V_x} costs at most $C^0_{V_x, i} \delta(U_i, U'_i)$, and the link i adds $\delta(U_i, U'_i)$. The x with $i \notin V_x$ number $L^4 - n^4$, and $\sum_{x : i \notin V_x} C^0_{V_x, i} = K^0_\nu$ for the direction ν of i, so

$$W_1^{d_E}\big(\mathscr{K}(U, \cdot), \mathscr{K}(U', \cdot)\big) \leq \frac{L^4 - n^4 + \alpha n^4}{L^4} \delta(U_i, U'_i) = (1 - \kappa_0)\, \delta(U_i, U'_i).$$

Changing the links of two arbitrary configurations one at a time and using the triangle inequality for $W_1^{d_E}$ extends this to $(1-\kappa_0) d_E(U, U')$. Hence a function h with Lipschitz constant $\mathrm{Lip}(h)$ for d_E has $\mathrm{Lip}(\mathscr{K}^t h) \leq (1-\kappa_0)^t \mathrm{Lip}(h)$, and since $d_E \leq D|E|$,

$$\mathrm{Var}_\mu(\mathscr{K}^t h) = \tfrac{1}{2} \iint \big(\mathscr{K}^t h(U) - \mathscr{K}^t h(U')\big)^2 d\mu\, d\mu \leq \tfrac{1}{2} (1-\kappa_0)^{2t} \mathrm{Lip}(h)^2 D^2 |E|^2.$$

Let ν_h be the spectral measure of the self-adjoint operator $0 \leq \mathscr{K} \leq I$ at $h - \mu(h)$. Then $\mathrm{Var}_\mu(\mathscr{K}^t h) = \int \lambda^{2t} d\nu_h$, and if $\nu_h((1 - \kappa_0 + \eta, 1]) > 0$ for some $\eta > 0$ this exceeds the bound for large t; so ν_h lives on $[0, 1-\kappa_0]$. Lipschitz functions are uniformly dense in $C(G^E)$ by the Stone–Weierstrass theorem, hence dense in $L^2(\mu)$, so the spectral projection of $\mathscr{K}$ for $(1-\kappa_0, 1]$ vanishes on the functions of mean zero, and $\langle f, (I - \mathscr{K}) f \rangle \geq \kappa_0 \mathrm{Var}_\mu(f)$ for every f. Multiplying by L^4/n^4 gives $\mathrm{BP}(n, 1-\alpha)$.

(ii) For $\varphi \equiv 1$, μ is the product of Haar measures. Write $f = \sum_{S \subset E} f_S$ with f_S depending only on the links of S and $\int f_S\, dU_e = 0$ for $e \in S$; the f_S are orthogonal, $\mathrm{Var}_\mu(f) = \sum_{S \neq \emptyset} \|f_S\|^2$, and $\mathscr{P}_x f = \sum_{S \cap V_x = \emptyset} f_S$. Hence

$$\sum_x \mu\big((f - \mathscr{P}_x f)^2\big) = \sum_{S \neq \emptyset} \|f_S\|^2 \, \#\{x : S \cap V_x \neq \emptyset\} \geq n^4 \mathrm{Var}_\mu(f),$$

because a link of S lies in n^4 cubes. □

Whether $D(n, \alpha)$ in the forest gauge implies BP is not decided here: the forest gauge does not close on a torus, and on $\mathbb{Z}^4$ the two conditions concern different specifications. The averaged condition carries the full content of a gap, since by Theorem 62 it gives decay of every bounded gauge-invariant covariance at the rate $\varpi/(e^2 q_n n)$; like $D(n, \alpha)$, it fails in the Coulomb phase of the abelian theory at every cube size.

Corollary 66. *Let $G = \mathrm{U}(1)$ with the Villain weight, and let β_0 be as in Proposition 243. For $\beta > \beta_0$ there are no $n \geq 2$ and $\varpi > 0$ such that $\mathrm{BP}(n, \varpi)$ holds for the measures with free boundary of all boxes of side at least $2n$.*

Proof. The field strength Ψ_p of the proof of Proposition 243 is a bounded gauge-invariant function of the four links of p, and its mean vanishes under every box measure by the symmetry $U \mapsto \bar{U}$. If $\mathrm{BP}(n, \varpi)$ held for all boxes, Theorem 62 would give $|W_\Lambda(\Psi_p \Psi_{p'})| \leq C(1+d) e^{-\varpi d/(e^2 q_n n)}$ for parallel plaquettes at distance d, with C independent of Λ and p'. The state of [51] is the limit of these measures (proof of Proposition 243), so $\sum_{p'} |\omega(\Psi_p \Psi_{p'})| < \infty$, which contradicts [51, Sect. 2] as in that proof. □

Remark 67 (what the averaged condition settles and what it leaves). Proved: Theorem 64 gives a mass gap of at least $16\varpi/(625e^2\ell)$ in every nontrivial continuum limit of torus limits from the condition $\mathrm{BP}(n_k, \varpi)$ with $n_k a_k \leq \ell$, a spectral gap of the cube dynamics on gauge-invariant functions in L^2 of the Wilson measure, uniform in the torus; the condition follows from its supremum form $D^0_L(n, \alpha)$ with $\varpi = 1 - \alpha$, holds at $\beta = 0$ with $\varpi = 1$ (Proposition 65), holds on every torus at strong coupling for every cube size past an explicit $n_0(\beta)$, with $\varpi \to 1$ as $n \to \infty$ (Corollary 72), fails on the torus of side $2n+4$ once $\beta \gg n^4 \log \beta$, already on the functions of nonzero centre charge (Theorem 77, Corollary 83), and fails in the Coulomb phase of the Villain theory at every cube size (Corollary 66); the conclusion of Theorem 64 follows from the weaker local form $\mathrm{BP}_{\mathrm{loc}}$, which for $\mathrm{SU}(N)$ is implied by the condition on the neutral sector (Theorem 82). The obstructions of the forest gauge, Propositions 94, 95 and 238, Theorem 236 and Corollary 237, bound K_c from below through Lipschitz functions of the gauge-fixed links, transported by gauge transformations of $\mathbb{Z}^4$, which in the Gaussian models are the flat

fields dh; the averaged condition involves only gauge-invariant functions, on which every gauge transformation acts trivially, so these proofs do not apply to it. Nor do [21, Theorem 505] and [21, Proposition 455], which concern bounds uniform in the coarse field and arguments from rarity, since the condition contains no supremum and no labels. Not proved: the condition $\mathrm{BP}(n_k,\varpi)$ along the continuum tuning for any non-abelian group, or outside the strong-coupling ranges of Corollary 72, and its local form $\mathrm{BP}_{\mathrm{loc}}$ along the tuning; the relation between BP and $D(n,\alpha)$; and (E). Remark 79 lists what Sec. IX G proves about the dependence on n and β. Corollary 66 shows that a proof of BP has to use the non-abelian structure of G, as for (D). Neither the averaged condition nor the mass gap is proved here.

G. The averaged condition at strong coupling and at fixed cube size

This subsection proves $\mathrm{BP}(n,\varpi)$ on every torus, for every cube size, in explicit ranges of the coupling, with ϖ tending to its value at $\beta=0$ as $n\to\infty$, and proves that at weak coupling the condition fails on the torus of side $2n+4$ when n is small compared with $(\beta/\log\beta)^{1/4}$. The first statement comes from the supremum form $D_L^0(n,\alpha)$ of Proposition 65, which the single-link coefficients of the ungauged measure control through a Dobrushin–Shlosman estimate for one cube [43]. The second uses the center of $\mathrm{SU}(N)$ and the holonomies of the torus, which the cube moves can change only through the curvature of the configuration. Throughout, $\Lambda=\Lambda_L$ with L even, δ is a bi-invariant metric on G, and $C^0_{V,j}$, K^0_ν are as in Proposition 65. Two distinct links are *partners* if they lie in a common plaquette; for $L\ge 3$ every link has exactly eighteen partners, three in each of its six plaquettes, and two links share at most one plaquette. Write $C^0_{ee'}=C^0_{\{e\},e'}$ for the single-link coefficients; $C^0_{ee'}=0$ unless e and e' are partners.

Lemma 68. *Suppose that $C^0_{ee'}\le c$ for all partners e,e', with $18c<1$. Then for every finite $V\subset E$ and every link $j\notin V$,*

$$C^0_{V,j}\le\frac{1}{1-18c}\sum_{e\in V}C^0_{ej}.$$

Proof. Let U,U' differ only at j, and let $\gamma=\gamma^0_V(\cdot\mid U)$, $\gamma'=\gamma^0_V(\cdot\mid U')$. Under γ the conditional law of σ_e, $e\in V$, given the other links of V is $\gamma^0_e(\cdot\mid\sigma,U_{E\setminus V})$, and likewise for γ'. The couplings of γ and γ' form a weakly compact set on which $\Psi(\pi)=\sum_{e\in V}\mathbb{E}_\pi\delta(\sigma_e,\sigma'_e)$ is continuous; let π^* be a minimiser and $\rho(e)=\mathbb{E}_{\pi^*}\delta(\sigma_e,\sigma'_e)$. For $e\in V$ replace (σ_e,σ'_e) by a sample of an optimal coupling of the two single-link laws, chosen measurably [102, Cor. 5.22]; both marginals are invariant, so the result is a coupling of γ and γ'. Changing the partners of e one at a time bounds the new expected distance at e by $\sum_{k\in V}C^0_{ek}\rho(k)+C^0_{ej}\delta(U_j,U'_j)$, and minimality gives

$$\rho(e)\le\sum_{k\in V}C^0_{ek}\rho(k)+C^0_{ej}\delta(U_j,U'_j).$$

Sum over $e\in V$. Since k has eighteen partners, $\sum_{e\in V}C^0_{ek}\le 18c$, so $(1-18c)\sum_e\rho(e)\le\delta(U_j,U'_j)\sum_{e\in V}C^0_{ej}$, and $W_1^{d_V}(\gamma,\gamma')\le\sum_e\rho(e)$. □

Lemma 69. *Let $n\ge 2$, $L\ge 2n+4$ and let j be a link of direction ν.*

(i) $\sum_{x:\,j\notin V_x}\#\{e\in V_x : e \text{ a partner of } j\}=18n^3-3n^2$.

(ii) *The number B_n of x with $j\notin V_x$ such that V_x contains a partner of j is $7n^3+3n^2$.*

Proof. Let b be the base point of j. For each $\mu\neq\nu$ the plaquette at b in the plane $\{\mu,\nu\}$ gives the partners $(b+e_\nu,\mu)$, $(b+e_\mu,\nu)$, (b,μ), and the plaquette at $b-e_\mu$ gives $(b-e_\mu,\nu)$, $(b-e_\mu,\mu)$, $(b-e_\mu+e_\nu,\mu)$, so the base points of the partners are $b+\Delta$ with $\Delta\in\{e_\nu,e_\mu,0,-e_\mu,-e_\mu,e_\nu-e_\mu\}$. A link with base point $b+\Delta$ lies in V_x exactly when $z=b-x$ satisfies $z+\Delta\in Q=[0,n)^4$, and $j\in V_x$ exactly when $z\in Q$; since $L\ge 2n+4$ these conditions are read in $\mathbb{Z}^4$.

(i) Exchanging the sums, the left side is $\sum_\Delta\#\{z : z+\Delta\in Q,\ z\notin Q\}=\sum_\Delta(n^4-|Q\cap(Q-\Delta)|)$. This is 0 for $\Delta=0$, $n^4-n^3(n-1)=n^3$ for $\Delta=\pm e_\mu,e_\nu$, and $n^4-n^2(n-1)^2=2n^3-n^2$ for $\Delta=e_\nu-e_\mu$. Each μ contributes $6n^3-n^2$.

(ii) The set of $z\notin Q$ with $z+\Delta\in Q$ for some Δ consists of the layer $\{z_\nu=-1\}$ of $Q-e_\nu$, the layers $\{z_\mu=-1\}$ and $\{z_\mu=n\}$ of $Q\mp e_\mu$, each with the other coordinates in $[0,n)$, and the new points $\{z_\nu=-1,\ z_\mu=n\}$ of $Q+e_\mu-e_\nu$, with the two remaining coordinates in $[0,n)$. These sets are disjoint, with $n^3+3(2n^3+n^2)$ points. □

Theorem 70 (the averaged condition from single links)**.** *Let $n\ge 2$ and $L\ge 2n+4$.*

(i) *If $C^0_{ee'}\le c$ for all partners, with $18c<1$, then $K^0_\nu\le\alpha_n n^4$ for $\nu=0,\dots,3$, with*

$$\alpha_n=\frac{c\,(18n-3)}{(1-18c)\,n^2},$$

and if $\alpha_n<1$, $\mathrm{BP}(n,1-\alpha_n)$ holds for every real $f\in L^2(\mu)$.

(ii) *For every β,*

$$K^0_\nu\le(7n^3+3n^2)\,\Xi_n,\qquad \Xi_n=\max_j C^0_{V_0,j}.$$

The number Ξ_n and the condition $D^0_L(n,\alpha)$ do not depend on $L\ge 2n+4$, and $\mathrm{BP}(n,1-(7n^{-1}+3n^{-2})\Xi_n)$ holds whenever $(7n^3+3n^2)\Xi_n<n^4$.

The number α_n decreases in n and tends to 0.

Proof. Fix a link j of direction ν. By translation invariance, $K^0_\nu=\sum_{x:\,j\notin V_x}C^0_{V_x,j}$ (proof of Proposition 65).

(i) By Lemma 68 and Lemma 69(i),

$$K^0_\nu\le\frac{c}{1-18c}\sum_{x:\,j\notin V_x}\#\{e\in V_x \text{ partner of } j\}=\frac{c\,(18n^3-3n^2)}{1-18c}.$$

This is $D^0_L(n,\alpha_n)$, and Proposition 65(i) gives the condition $\mathrm{BP}(n,1-\alpha_n)$. Finally $(18n-3)/n^2 = 18/n - 3/n^2$ decreases for $n \geq 1$.

(ii) $C^0_{V_x,j}$ vanishes unless V_x contains a partner of j, which by Lemma 69(ii) happens for $B_n = 7n^3+3n^2$ points x with $j \notin V_x$. The law $\gamma^0_{V_0}(\cdot \mid U)$ depends on U only through the links of ∂V_0, whose base points lie in $[-1,n]^4$; for $L \geq 2n+4$ the links of $V_0 \cup \partial V_0$ and the plaquettes that meet V_0 are those of $\mathbb{Z}^4$, and the links of ∂V_0 take all values in $G^{\partial V_0}$. So every $C^0_{V_0,j}$, hence Ξ_n and K^0_ν, is the same for all $L \geq 2n+4$. The last claim is Proposition 65(i) with $\alpha = B_n \Xi_n / n^4$. □

Lemma 68 is Lemma 233 for the ungauged measure, where every link has eighteen partners instead of twelve. Part (ii) of the theorem is the exact form of the count of Remark 92 for that measure: if one link moves the law of a cube of side $n \geq 3$ by a total amount at most Ξ in Kantorovich distance, $D^0_L(n,\alpha)$ holds once $n \geq 8\Xi/\alpha$, since $7n^3+3n^2 \leq 8n^3$. The single-link coefficients are bounded as in Secs. IV–VII, and for a general group by curvature.

Lemma 71. *Let $G \subset \mathrm{U}(N)$ be closed and connected with semisimple Lie algebra, δ the geodesic distance of $\langle X,Y\rangle = -\mathrm{Tr}(XY)$, and $6N|\beta| < \kappa_G$. Then $C^0_{ee'} \leq N|\beta|/(\kappa_G - 6N|\beta|)$ for all partners.*

Proof. Given the other links, the law of U_e is $\nu \propto e^{-V} dU$ with $V(U) = -N\beta \sum_{p \ni e} \mathrm{Re}\,\mathrm{Tr}\, U_p$. Along $e^{tX}U$ the second derivative of $\mathrm{Re}\,\mathrm{Tr}\, U_p$ is $\mathrm{Re}\,\mathrm{Tr}(Y^2 W)$ with $|Y| = |X|$, by (6), and $|\mathrm{Tr}(Y^2W)| \leq |Y|^2$, so $\mathrm{Ric}_G + \mathrm{Hess}\, V \geq K_1 = \kappa_G - 6N|\beta| > 0$. The proof of Lemma 24 with one factor and $w \equiv 1$ gives $\Gamma(P_t h) \leq e^{-2K_1 t} P_t \Gamma(h)$ for the Langevin semigroup of ν, and as in the proof of Lemma 25,

$$|\mathrm{Cov}_\nu(g,h)| \leq \int_0^\infty |\nu(\Gamma(P_t h, g))|\, dt \leq \frac{\|\nabla g\|_\infty \|\nabla h\|_\infty}{K_1}.$$

Move the partner e' along a unit-speed geodesic $e^{sX}U_{e'}$, $0 \leq s \leq \delta(U_{e'}, U'_{e'})$, and let $\nu_s \propto e^{-V_s}$. Then $\frac{d}{ds}\nu_s(g) = -\mathrm{Cov}_{\nu_s}(g, \partial_s V_s)$; $\partial_s V_s$ comes from the one plaquette p that contains e and e', and its gradient in U_e is $-N\beta$ times the mixed block $H^p_{ee'}(\cdot, X)$, of norm at most one by Lemma 22. For smooth g that are 1-Lipschitz for δ this gives $|\nu_0(g) - \nu_\delta(g)| \leq N|\beta|\delta/K_1$. Convolution with a probability measure on G preserves the Lipschitz constant of a function for a bi-invariant metric, so smooth 1-Lipschitz functions approximate every 1-Lipschitz one uniformly, and the Kantorovich–Rubinstein duality gives the bound. □

Corollary 72 (explicit ranges)**.** *For every even $L \geq 2n+4$, $\mathrm{BP}(n,1-\alpha_n)$ holds for every real $f \in L^2(\mu)$ in the following cases.*

(i) *$G = \mathrm{SU}(2)$, $|\beta_W| < \frac{2}{9}$, with*

$$\alpha_n = \frac{|\beta_W|(18n-3)}{(4-18|\beta_W|)n^2},$$

whenever $|\beta_W| < 4n^2/(18n^2+18n-3)$; for $n=2$ this is $|\beta_W| < \frac{16}{105}$.[16] *At $|\beta_W| \leq \frac{1}{8}$, $\alpha_n \leq \alpha_2 \leq 0.59$ for every $n \geq 2$; at $|\beta_W| \leq \frac{1}{5}$, $\alpha_n \leq 0.982$ for every $n \geq 9$.*

(ii) *$G = \mathrm{SU}(3)$ and $\beta_W \geq 0$ with*

$$\zeta(\beta_W) = \frac{3r\sqrt{\varepsilon(r)}}{\gamma(r)} < 1, \qquad r = 2\beta_W,$$

where ε and γ are as in Theorem 6, with

$$\alpha_n = \frac{\zeta(\beta_W)\,(18n-3)}{18\,(1-\zeta(\beta_W))\,n^2}.$$

The function ζ is increasing, $\zeta(0.35) \leq 0.8584$, and for $\beta_W \leq 0.35$ and $n \geq 6$, $\alpha_n \leq 0.3368\,(18n-3)/n^2 \leq 0.9824$.

(iii) *$G \subset \mathrm{U}(N)$ closed and connected with semisimple Lie algebra, $24N|\beta| < \kappa_G$, with*

$$\alpha_n = \frac{N|\beta|(18n-3)}{(\kappa_G - 24N|\beta|)n^2},$$

whenever $N|\beta| < \kappa_G n^2/(24n^2+18n-3)$.

In each case, for fixed β in the range, $\mathrm{BP}(n,\varpi)$ holds with one $\varpi > 0$ for all n from some $n_0(\beta)$ on, uniformly in the torus, and the best constant tends to 1 as $n \to \infty$.

Proof. (i) As in the proof of Theorem 1, without the gauge: the law of U_e is the von Mises–Fisher law ν_b with $b = 4\beta \sum_{p \ni e} \bar{s}_p$, a partner moved by δ in the chordal metric moves b by $4|\beta|\delta$, and Theorem 364 gives $C^0_{ee'} \leq |\beta| = |\beta_W|/4$. Theorem 70(i) applies with $c/(1-18c) = |\beta_W|/(4-18|\beta_W|)$. At $|\beta_W| = \frac{1}{8}$, $\alpha_2 = \frac{33}{56}$; at $|\beta_W| = \frac{1}{5}$, $\alpha_n = (18n-3)/(2n^2)$ and $\alpha_9 = \frac{159}{162}$; α_n increases with $|\beta_W|$.

(ii) The six staples of a link are products of matrices of $\mathrm{SU}(3)$, so $b \in rK$, and as in (8) and Lemma 31, $C^0_{ee'} \leq N\beta\Lambda(r) \leq c$ with $18c = 3r\sqrt{\varepsilon(r)}/\gamma(r)$ in the Hilbert–Schmidt metric; ζ increases because ε is nondecreasing and γ decreasing. By Lemma 30(iii) and the tail bound in the proof of Theorem 6, at $r = 0.7$ the terms $m_n r^n/n!$, rounded up, are 1, 0, 0.1225, 0.0142917, 0.0075032, 0.0013131, 0.000332, 0.000059 and 0.0000109, with sum at most 1.1460099, and the tail is less than $4 \cdot 10^{-6}$; so $Z_*(0.7) \leq 1.146014$, $e^{-0.14} \geq 0.869342$, $\frac{5e^{-0.14}}{9Z_*(0.7)} \geq 0.421432$ and $\varepsilon(0.7) \leq 0.578568 < 0.76064^2$. With $\gamma(0.7) = \frac{1191}{640}$, $\zeta(0.35) \leq 2.1 \cdot 0.76064 \cdot \frac{640}{1191} \leq 0.8584$, and $\zeta/(18(1-\zeta)) \leq 0.3368$; at $n=6$, $(18n-3)/n^2 = \frac{35}{12}$.

(iii) Lemma 71 gives $c = N|\beta|/(\kappa_G - 6N|\beta|)$, so $18c < 1$ is $24N|\beta| < \kappa_G$ and $c/(1-18c) = N|\beta|/(\kappa_G - 24N|\beta|)$.

In each case $\alpha_n \to 0$; Proposition 73(ii) below bounds the best constant by $1 + 2/n$. □

[16] With $c = |\beta_W|/4$ the condition $18c < 1$ of Theorem 70 reads $|\beta_W| < \frac{2}{9}$, and $\alpha_n < 1$ reads $|\beta_W|(18n-3) < (4-18|\beta_W|)n^2$, that is, $|\beta_W|(18n^2+18n-3) < 4n^2$; at $n=2$ this is $105|\beta_W| < 16$. As $n \to \infty$ the threshold $4n^2/(18n^2+18n-3)$ increases to $\frac{2}{9}$.

For SU(3), the condition $\zeta < 1$ of (ii) holds up to $\beta_W \approx 0.391$ by a numerical evaluation that no statement uses, against $\frac{3}{16}$ from (iii); for SU(2), (i) reaches $\frac{2}{9}$ against $\frac{1}{12}$. These ranges lie inside those of Secs. IV–VIII, where the forest gauge gives a gap directly; what Corollary 72 adds is the averaged condition itself, on every torus, for every n past $n_0(\beta)$ and on all of $L^2(\mu)$. The gap it gives through Corollary 63 is of order $1/n$ and is not used. The dependence on n is constrained in general as follows. Write ϖ_n^* for the best constant in $\mathrm{BP}(n,\cdot)$ on Λ_L, and P_V for the conditional expectation given the links outside V.

Proposition 73. *Let $L \geq 2m+4$ and $2 \leq n \leq m$.*

(i) *If $\mathrm{BP}(n,\varpi)$ holds, then so does $\mathrm{BP}(m,(n/m)^4\varpi)$; that is, $n^4\varpi_n^* \leq m^4\varpi_m^*$.*

(ii) $\varpi_n^* \leq 1 + 2/n$.

(iii) *Let W_x be the cubes of side m. Suppose that $\mathrm{BP}(m,\varpi)$ holds and that for every x, every real $f \in \mathscr{A}$ and μ-almost every U,*

$$\varpi_{\mathrm{in}} P_{W_x}\big((f-P_{W_x}f)^2\big) \leq \frac{1}{n^4}\sum_{y:\,V_y\subset W_x} P_{W_x}\big((f-P_{V_y}f)^2\big).$$

Then $\mathrm{BP}(n,\varpi\varpi_{\mathrm{in}}(m/(m-n+1))^4)$ holds.

Proof. (i) If $V \subset W$, the σ-algebra of the links outside W is contained in that of the links outside V, so $P_W = P_W P_V$ and $\|f-P_W f\|^2 = \|f-P_V f\|^2 + \|P_V f - P_W f\|^2$. The cube V_y of side n lies in W_x exactly when $y-x \in [0,m-n]^4$, so for each x, $\mu((f-P_{W_x}f)^2)$ is at least the mean of $\mu((f-P_{V_y}f)^2)$ over these $(m-n+1)^4$ points y, and each y occurs for $(m-n+1)^4$ points x. Summing, $\sum_x \mu((f-P_{W_x}f)^2) \geq \sum_y \mu((f-P_{V_y}f)^2)$.

(ii) Let $f = \mathrm{Re}\,\mathrm{Tr}\,U_p$ for a plaquette p at y in the plane $\{\mu,\nu\}$; $\mathrm{Var}_\mu(f) > 0$. As in the proof of Theorem 62, $\mu((f-\mathscr{P}_x f)^2) \leq \mathrm{Var}_\mu(f)$, with equality to zero unless V_x contains a link of p, that is, unless $x \in b-[0,n)^4$ for a base point $b \in \{y, y+e_\mu, y+e_\nu\}$. By inclusion and exclusion these x number $3n^4-2n^3(n-1)-n^2(n-1)^2+n^2(n-1)^2 = n^4+2n^3$.

(iii) Integrating the hypothesis, $\varpi_{\mathrm{in}}\mu((f-P_{W_x}f)^2) \leq n^{-4}\sum_{y:\,V_y\subset W_x}\mu((f-P_{V_y}f)^2)$. Sum over x, use $\mathrm{BP}(m,\varpi)$ and count each y $(m-n+1)^4$ times: $\varpi\varpi_{\mathrm{in}}\mathrm{Var}_\mu(f) \leq (m-n+1)^4 m^{-4}n^{-4}\sum_y \mu((f-P_{V_y}f)^2)$. □

Part (iii) is the two-scale argument for block dynamics [82]; it transfers the condition from larger cubes to smaller ones, given a gap of the small-cube dynamics inside each large cube uniform in its exterior. Part (i) transfers it upwards only with the loss $(n/m)^4$, which, through Theorem 62, gives the rate $\varpi n^4/(e^2 q_m m^5)$ and nothing in physical units. So BP at one cube size implies no useful bound at larger sizes by itself.

The remaining results concern $G = \mathrm{SU}(N)$, $N \geq 2$, with center element $z = e^{2\pi i/N}I$, and use the holonomies of the torus. For a site y let P_y be the ordered product of the L links $(y+te_0,0)$, $0 \leq t < L$, the Polyakov loop of direction 0 through y, and $A = P_0$, B the analogous product of the links $(te_1,1)$; A and B are based at the site 0, so the traces $F_{ab} = \mathrm{Tr}(A^aB^b)$ are gauge invariant. For $c,d \in \mathbb{Z}$ the map $Z_{c,d}$ that multiplies the links $(y,0)$ with $y_0 = 0$ by z^c and the links $(y,1)$ with $y_1 = 0$ by z^d preserves μ: a plaquette that contains one of these links contains a second one of the same direction with the same base coordinate, entering with the opposite orientation, and z is central. It maps P_y to z^cP_y, A to z^cA, B to z^dB and F_{ab} to $z^{ac+bd}F_{ab}$.

Lemma 74. *Let $R = [0,a]\times[0,b]$ be a rectangle of $\mathbb{Z}^2$ and ϕ a map of its sites into Λ_L that sends unit steps in the two directions to links in two directions $\mu \neq \nu$ and is injective on plaquettes. The holonomy of the boundary of R, traversed counterclockwise from $(0,0)$, is an ordered product $\prod_k g_k U_{p_k}^{\pm1} g_k^{-1}$ over the ab plaquettes of $\phi(R)$, with $g_k \in G$. For every unitarily invariant norm it is within $\sum_{p\in\phi(R)}\|U_p - I\|$ of I.*

Proof. Holonomy is multiplicative under concatenation of paths and does not change when a step followed by its reverse is removed. Let ℓ_j be the boundary of $[0,a]\times[0,j]$ from $(0,0)$. After cancellation, $\ell_{j-1}^{-1}\ell_j$ is the left edge from $(0,0)$ to $(0,j-1)$, the boundary of the row $[0,a]\times[j-1,j]$ from $(0,j-1)$, and the left edge back; so $\mathrm{Hol}(\ell_b)$ is an ordered product of conjugates of the row holonomies. The same argument along a row writes the holonomy of its boundary as an ordered product of conjugates of the holonomies of the boundaries of its plaquettes, each a conjugate of U_p or U_p^*. For unitary X,Y, $\|XY-I\| \leq \|X-I\| + \|Y-I\|$ and $\|gXg^{-1}-I\| = \|X-I\|$. □

Lemma 75. *For $A,B \in \mathrm{SU}(N)$ put*

$$M(A,B) = \sum_{(a,b)\in I_N} |\mathrm{Tr}(A^aB^b)|^2,$$

where $I_N = \{0,\dots,N-1\}^2 \setminus \{(0,0)\}$. There are constants $c_N > 0$ and $\eta_N > 0$ such that $M(A,B) \geq c_N$ whenever $\|ABA^{-1}B^{-1}-I\|_{\mathrm{HS}} \leq \eta_N$. For $N = 2$ one may take $\eta_2 = 1$ and $c_2 = 0.64$.

Proof. Let A,B commute and suppose $M(A,B) = 0$. Diagonalise them together, with eigenvalues α_k, β_k. The power sums of the α_k of orders $1,\dots,N-1$ vanish, so by Newton's identities the characteristic polynomial of A is $x^N + (-1)^N \det A$, and $\alpha_k = \lambda\omega^k$ after reordering, with $\omega = e^{2\pi i/N}$; likewise $\beta_k = \lambda'\omega^{\tau(k)}$ for a permutation τ of $\mathbb{Z}/N\mathbb{Z}$. Then $\mathrm{Tr}(A^aB^b) = \lambda^a\lambda'^b\sum_k \omega^{ak+b\tau(k)}$ is a multiple of the Fourier coefficient at (a,b) of the indicator of the graph of τ in $(\mathbb{Z}/N\mathbb{Z})^2$. All coefficients except the one at 0 vanish, so the indicator is constant, which is impossible for a set of N points among N^2. Hence $M > 0$ on the compact set of commuting pairs. If no η_N, c_N existed, there would be pairs with $\|ABA^{-1}B^{-1}-I\| \to 0$ and $M \to 0$, and a limit point would be a commuting pair with $M = 0$.

For $N = 2$ put $x = \mathrm{Tr}\,A$, $y = \mathrm{Tr}\,B$, $w = \mathrm{Tr}\,AB$, real numbers in $[-2,2]$. By the Cayley–Hamilton theorem, $A^{-1} = x - A$, $B^{-1} = y - B$, $\mathrm{Tr}(AB^2) = yw - x$, $\mathrm{Tr}(A^2B) = xw - y$ and $\mathrm{Tr}(ABAB) = w^2 - 2$, so

$$\mathrm{Tr}(ABA^{-1}B^{-1}) = x^2 + y^2 + w^2 - xyw - 2.$$

Since $|\mathrm{Tr}(C-I)| \le \sqrt{2}\|C-I\|_{\mathrm{HS}}$, the commutator bound gives $x^2+y^2+w^2-xyw \ge 4-\sqrt{2}$, and with $m=\max(|x|,|y|,|w|)$, $3m^2+m^3 \ge 4-\sqrt{2} > 2.432 = 3(0.8)^2+0.8^3$. So $m>0.8$ and $M=x^2+y^2+w^2>0.64$. □

Lemma 76. *Let $G=\mathrm{SU}(N)$ and $\beta \ge 1$. There is A_N such that $\mu(N-\mathrm{Re}\,\mathrm{Tr}\,U_p) \le A_N(1+\log\beta)/\beta$ for every plaquette and every L; for $N=2$, $\mu(2-\mathrm{Tr}\,U_p) \le (\log\beta+6.11)/\beta$.*

Proof. Put

$$H=\sum_p (N-\mathrm{Re}\,\mathrm{Tr}\,U_p) \ge 0, \qquad \phi(t)=\log\int e^{-NtH}\,dU.$$

Then μ has density proportional to $e^{-N\beta H}$, $\phi'(\beta)=-N\mu(H)$, ϕ is convex and $\phi \le 0$. Convexity gives $\phi(\beta/2) \ge \phi(\beta)-\frac{\beta}{2}\phi'(\beta)$, hence $\mu(H) \le \frac{2}{N\beta}\log(1/e^{\phi(\beta)})$. Let δ be the geodesic distance of $-\mathrm{Tr}(XY)$. If every link lies within r of I, then $\delta(U_p,I) \le 4r$ by bi-invariance, and $N-\mathrm{Re}\,\mathrm{Tr}\,U \le \frac{1}{2}\delta(U,I)^2$, so $H \le 8r^2|P|$. With $v(r)$ the Haar measure of the ball of radius r about I, $|E|=4L^4$ and $|P|=6L^4$,

$$\mu(N-\mathrm{Re}\,\mathrm{Tr}\,U_p)=\frac{\mu(H)}{6L^4} \le \frac{2}{N\beta}\left(\tfrac{2}{3}\log\tfrac{1}{v(r)}+8N\beta r^2\right),$$

the plaquettes being equivalent under the symmetries of Λ_L. Haar measure has a smooth positive density in exponential coordinates, so $v(r) \ge v_N r^{N^2-1}$ for $r \le 1$; $r^2=1/(N\beta)$ gives the first claim. For $\mathrm{SU}(2)$, $U=\cos\theta+u\sin\theta$ in quaternions has $\delta(U,I)=\sqrt{2}\theta$ and the law of θ has density $\frac{2}{\pi}\sin^2\theta$, so $v(\sqrt{2}s) \ge \frac{2}{\pi}\cdot\frac{4}{\pi^2}\cdot\frac{s^3}{3}=\frac{8s^3}{3\pi^3}$ for $s \le \frac{\pi}{2}$. With $r=\sqrt{2}s$ and $s^2=1/(32\beta)$ the bracket times $\frac{1}{\beta}$ is at most $\frac{1}{6\beta}(6\log(32\beta)+6+4\log\frac{3\pi^3}{8}) \le (\log\beta+6.11)/\beta$. □

Theorem 77 (the averaged condition at fixed cube size)**.** *Let $G=\mathrm{SU}(N)$, $N \ge 2$, $\beta \ge 0$ and $n \ge 2$.*

(i) *If $\mathrm{BP}(n,\varpi)$ holds on Λ_L, then*

$$\frac{1}{L^3}\sum_{y:\,y_0=0}\mu\left(\mathrm{Tr}\,P_0\,\overline{\mathrm{Tr}\,P_y}\right) \le \frac{2N^2n^2}{\varpi L^2}.$$

(ii) *Let $N=2$ and $\beta \ge 1$. If $\mathrm{BP}(n,\varpi)$ holds on Λ_{2n+4}, then*

$$\varpi \le \frac{38400\,n^4(\log\beta+6.11)}{\beta}.$$

(iii) *There is C_N such that for $\beta \ge 1$, $\mathrm{BP}(n,\varpi)$ on Λ_{2n+4} implies $\varpi \le C_N n^4(1+\log\beta)/\beta$.*

In particular, along any sequence (β_k,n_k) with $n_k^4(1+\log\beta_k)/\beta_k \to 0$, no $\varpi>0$ satisfies $\mathrm{BP}(n_k,\varpi)$ uniformly in the torus.

Proof. (i) Let $g_y=\mathrm{Re}\,\mathrm{Tr}\,P_y$ and $f=L^{-3}\sum_{y:\,y_0=0}g_y$. By the symmetry $Z_{1,0}$, $\mu(f)=0$. The loop P_y meets V_x only if (y_1,y_2,y_3) lies in the projection of $x+[0,n)^4$, which holds for n^3 loops, and $\|g_y-\mathscr{P}_x g_y\|_2 \le \|g_y\|_2 \le N$. So $\|f-\mathscr{P}_x f\|_2 \le Nn^3/L^3$ for each of the L^4 points x, the right side of $\mathrm{BP}(n,\varpi)$ is at most N^2n^2/L^2, and $\mu(f^2) \le N^2n^2/(\varpi L^2)$. The same holds for the imaginary parts, and $\mu(|L^{-3}\sum_y \mathrm{Tr}\,P_y|^2)$ is the left side, by translation invariance.

(ii), (iii) Let $L=2n+4 \le 4n$, $\varepsilon=\mu(\|U_p-I\|_{\mathrm{HS}}^2)=2\mu(N-\mathrm{Re}\,\mathrm{Tr}\,U_p)$, the same for every plaquette, and let f be the real or the imaginary part of some F_{ab}, $(a,b)\in I_N$. Pick (c,d) with $ac+bd \not\equiv 0$ modulo N; the symmetry $Z_{c,d}$ gives $\mu(F_{ab})=0$.

The right side of BP. If V_x meets neither A nor B, $\mathscr{P}_x f=f$. Otherwise the coordinate x_2 satisfies $0 \in x_2+[0,n)$ modulo L, and one of $s=x_2-1$, $s=x_2+n$, read in $(-L/2,L/2]$, has $|s| \le n$ and $s \notin x_2+[0,n)$. The loops A', B' of directions 0, 1 through $y'=se_2$ avoid V_x, so $F'_{ab}=\mathrm{Tr}(A'^a B'^b)$ depends only on the links outside V_x, and $\|f-\mathscr{P}_x f\|_2 \le \|F_{ab}-F'_{ab}\|_2$. Let Γ be the holonomy of the straight path from 0 to y'. Lemma 74, for the rectangle $[0,L]\times[0,s]$ mapped onto the plane $\{0,2\}$ through 0, whose boundary is A, the path to y', A' reversed and the path back, gives $A=H_0\Gamma A'\Gamma^{-1}$ with $\|H_0-I\|_{\mathrm{op}} \le S_0$, the sum of $\|U_p-I\|_{\mathrm{op}}$ over the $L|s|$ plaquettes of the strip; likewise $B=H_1\Gamma B'\Gamma^{-1}$ with the strip in the plane $\{1,2\}$. Telescoping, $|F_{ab}-F'_{ab}| \le N(a\|H_0-I\|_{\mathrm{op}}+b\|H_1-I\|_{\mathrm{op}}) \le N(N-1)(S_0+S_1)$, and by the Cauchy–Schwarz inequality over the at most $2Ln$ plaquettes of the two strips, $\mu((S_0+S_1)^2) \le (2Ln)^2\varepsilon$. At most $2Ln^3$ points x have V_x meeting $A\cup B$, so

$$\frac{1}{n^4}\sum_x \mu\big((f-\mathscr{P}_x f)^2\big) \le 8N^2(N-1)^2L^3n\varepsilon.$$

The variance. Lemma 74 for the square $[0,L]^2$ mapped onto the plane $\{0,1\}$ through 0, whose boundary reads A, B, A^{-1}, B^{-1}, gives $\|ABA^{-1}B^{-1}-I\|_{\mathrm{HS}} \le \sum_p \|U_p-I\|_{\mathrm{HS}}$ over L^2 plaquettes, so this commutator exceeds η_N with probability at most $L^2\sqrt{\varepsilon}/\eta_N$. By Lemma 75, $\sum_{(a,b)\in I_N}\mu(|F_{ab}|^2) \ge c_N(1-L^2\sqrt{\varepsilon}/\eta_N)$. As $\mu(F_{ab})=0$, some f of the form above has $\mathrm{Var}_\mu(f) \ge c_N(1-L^2\sqrt{\varepsilon}/\eta_N)/(2(N^2-1))$. For $N=2$ the traces are real, and some F_{ab} has variance at least $0.64(1-L^2\sqrt{\varepsilon})/3$.

Conclusion. If $L^2\sqrt{\varepsilon} \le \eta_N/2$, $\mathrm{BP}(n,\varpi)$ gives $\varpi \le 2048(N^2-1)N^2(N-1)^2n^4\varepsilon/c_N$, and for $N=2$, $\varpi \le 300L^3n\varepsilon \le 19200n^4\varepsilon$. In the other case $\eta_N^2 < 4L^4\varepsilon \le 1024n^4\varepsilon$, and since $\varpi \le 2$ by Proposition 73(ii), $\varpi \le (2048/\eta_N^2)n^4\varepsilon$, which for $N=2$ is again at most $19200n^4\varepsilon$. Lemma 76 bounds ε by $2(\log\beta+6.11)/\beta$ for $N=2$ and by $2A_N(1+\log\beta)/\beta$ in general. □

Corollary 78 (the response of a cube at weak coupling)**.** *Let $G=\mathrm{SU}(2)$, $\beta \ge 1$ and $n \ge 2$, and put $X_n(\beta)=38400\,n^4(\log\beta+6.11)/\beta$. Then*

$$\Xi_n \ge \frac{n^4}{7n^3+3n^2}\big(1-X_n(\beta)\big),$$

and $D_L^0(n,\alpha)$ for one $L \ge 2n+4$, hence for all, implies $1-\alpha \le X_n(\beta)$.

Proof. By Theorem 70(ii), $D_L^0(n,\alpha)$ does not depend on L and holds with $\alpha=(7n^3+3n^2)\Xi_n/n^4$. If $\alpha<1$, Proposition 65(i) gives $\mathrm{BP}(n,1-\alpha)$ on Λ_{2n+4}, and Theorem 77(ii) gives $1-\alpha \le X_n(\beta)$; if $\alpha \ge 1$ the bound on Ξ_n holds trivially. □

So for $2 \le n \le (\beta/(76800(\log\beta+6.11)))^{1/4}$ one link outside the cube moves its law by at least $n/17$ in total, in the supremum over exteriors: the response grows linearly with the cube, the behaviour that Remark 92 assigns to cubes smaller than the correlation length.

Fig. 12 places the two results of this subsection in the plane of the coupling and the cube side.

Remark 79 (what is proved about the averaged condition). Proved: $\mathrm{BP}(n,1-\alpha_n)$ on every torus and on all of $L^2(\mu)$ for SU(2) with $|\beta_W| < \frac{2}{9}$, for SU(3) with $\zeta(\beta_W) < 1$ (in particular $\beta_W \le 0.35$, $n \ge 6$, $\varpi \ge 0.0176$), and for every compact connected group with semisimple Lie algebra with $24N|\beta| < \kappa_G$, for all n with $\alpha_n < 1$, where $\alpha_n = O(1/n)$ is explicit (Corollary 72); the supremum form D^0 is independent of the torus and follows from a bound $\Xi_n < n/(7+3/n)$ on the total response of a cube to one link (Theorem 70); the best constant satisfies $n^4\varpi_n^* \le m^4\varpi_m^*$ and $\varpi_n^* \le 1+2/n$, and it passes from large cubes to small ones given a uniform inner gap (Proposition 73). For SU(N) the condition forces the spatially averaged Polyakov correlator to be $O(n^2/L^2)$, and on the torus Λ_{2n+4} it fails once $\beta \gg n^4\log\beta$; for SU(2), $\varpi \le 38400\,n^4(\log\beta+6.11)/\beta$, and the cube response Ξ_n is at least of order n in that regime (Theorem 77, Corollary 78). Hence $\mathrm{BP}(n_k,\varpi)$ uniform in the torus requires $n_k^4 \ge \varpi\beta_k/(C_N(1+\log\beta_k))$ for $\beta_k \ge 1$; the continuum tuning, with n_k of order ℓ/a_k and β_k of order $\log(1/a_k)$, meets this requirement, and the obstruction comes from tori of side $2n+4$, which Theorem 64 does not need, since it uses an infinite set of L. Not proved: $\mathrm{BP}(n_k,\varpi)$ along the continuum tuning for any non-abelian group, or at any coupling outside the ranges above; a transfer from small cubes to large ones without the loss $(n/m)^4$; an obstruction on large tori at weak coupling. Sec. IX H shows that the obstruction on Λ_{2n+4} lies in the centre-charged part of $\mathscr{A}$, that the continuum step needs only the local form $\mathrm{BP}_{\mathrm{loc}}$, and that the comparison of cube sizes loses exactly $(n/m)^4$ (Remark 88); Sec. IX I reduces the failure of the neutral part on Λ_{2n+4} to a lower bound on the law of the Polyakov loop (Theorem 90).

H. The averaged condition at weak coupling: centre sectors and local slow modes

The condition $\mathrm{BP}(n_k,\varpi)$ along the continuum tuning is the spectral gap that Theorem 64 needs, and it is not proved here. This subsection separates the part of the condition that the continuum step uses from the part that Theorem 77 obstructs, and it records what each available route to the condition gives. Three facts organise it. For SU(N) the cube moves commute with the centre twists of the torus, so the condition splits into centre sectors; the test functions of Theorem 77 are charged, and every local gauge-invariant function is neutral. The proof of Theorem 62 uses the condition only through the spectral measures of local functions, so a local form of it suffices for the continuum step. Functions of a bounded set of links satisfy a Poincaré inequality for the cube chain uniformly in the torus, so the slow functions that the condition excludes are spread over growing sets of links.

The setting is that of Sec. IX F on the torus Λ_L, $L \ge 2n+4$ even, with $\mathscr{K} = L^{-4}\sum_x \mathscr{P}_x$. Put

$$\mathscr{E}_n(f) = \frac{1}{n^4}\sum_{x\in X}\mu\big(|f-\mathscr{P}_x f|^2\big), \qquad f \in L^2(\mu),$$

so that $\langle f,(I-\mathscr{K})f\rangle = n^4L^{-4}\mathscr{E}_n(f)$, and $\mathrm{BP}(n,\varpi)$ reads $\varpi\,\mathrm{Var}_\mu(f) \le \mathscr{E}_n(f)$ for real $f \in \mathscr{A}$; here $\mathrm{Var}_\mu(f) = \mu(|f-\mu(f)|^2)$. Let $\mathscr{A}_{\mathbb{C}}$ be the space of complex gauge-invariant functions in $L^2(\mu)$. For $G = \mathrm{SU}(N)$, $N \ge 2$, put $\omega = e^{2\pi i/N}$ and $z = \omega I$. For a direction ν and $t \in \mathbb{Z}/L\mathbb{Z}$ let Z_t^ν be the map that multiplies every link (y,ν) with $y_\nu = t$ by z and fixes the other links; the maps $Z_{c,d}$ of Sec. IX G are $(Z_0^0)^c(Z_0^1)^d$.

Proposition 80 (centre sectors)**.** *Let $G = \mathrm{SU}(N)$, $N \ge 2$.*

(i) *Each Z_t^ν preserves μ, maps $\mathscr{A}_{\mathbb{C}}$ onto itself, and satisfies $(\mathscr{P}_x f)\circ Z_t^\nu = \mathscr{P}_x(f\circ Z_t^\nu)$ for every x and every $f \in L^2(\mu)$.*

(ii) *For $f \in \mathscr{A}_{\mathbb{C}}$ the function $f\circ Z_t^\nu$ does not depend on t; write $f\circ Z^\nu$ for it.*

(iii) *For $r \in (\mathbb{Z}/N\mathbb{Z})^4$ let $\mathscr{A}_r = \{f \in \mathscr{A}_{\mathbb{C}} : f\circ Z^\nu = \omega^{r_\nu} f,\ \nu = 0,\dots,3\}$. Then $\mathscr{A}_{\mathbb{C}}$ is the orthogonal sum of the $\mathscr{A}_r$, each $\mathscr{A}_r$ is invariant under every $\mathscr{P}_x$, the constants lie in $\mathscr{A}_0$, and complex conjugation maps $\mathscr{A}_r$ onto $\mathscr{A}_{-r}$.*

(iv) *If $f \in \mathscr{A}_{\mathbb{C}}$ depends only on the links with base points in a box $b+[0,s)^4$ with $s \le L-1$, then $f \in \mathscr{A}_0$.*

(v) *$\mathrm{BP}(n,\varpi)$ holds if and only if $\varpi\,\mathrm{Var}_\mu(f) \le \mathscr{E}_n(f)$ for every r and every $f \in \mathscr{A}_r$; for $r \ne 0$ and $f \in \mathscr{A}_r$, $\mu(f) = 0$.*

Proof. (i) A plaquette that contains a link (y,ν) with $y_\nu = t$ lies in a plane $\{\nu,\nu'\}$ and contains exactly one other link of direction ν, with the same ν-coordinate and the opposite orientation; a plaquette in a plane without ν contains no link of direction ν. Since z is central, Z_t^ν leaves every U_p unchanged, and it preserves Haar measure, so it preserves μ. Since z is central, it also commutes with every gauge transformation, so it maps $\mathscr{A}_{\mathbb{C}}$ into itself, and onto it, as $(Z_t^\nu)^N$ is the identity. Write Z_t^ν as $Z_V \times Z'$ on $G^{V_x}\times G^{E\setminus V_x}$. The density of $\gamma^0_{V_x}(\cdot\,|\,U)$ at σ is proportional to the product of φ over the plaquettes meeting V_x of the configuration equal to σ on V_x and to U elsewhere, and Z_t^ν leaves these plaquettes unchanged; with the invariance of Haar measure under Z_V, the law $\gamma^0_{V_x}(\cdot\,|\,Z_t^\nu U)$ is the image of $\gamma^0_{V_x}(\cdot\,|\,U)$ under Z_V. Hence

$$\begin{aligned}\mathscr{P}_x(f\circ Z_t^\nu)(U) &= \int f(Z_V\sigma, Z'U_{E\setminus V_x})\,\gamma^0_{V_x}(d\sigma\,|\,U)\\ &= (\mathscr{P}_x f)(Z_t^\nu U).\end{aligned}$$

(ii) Let g be the gauge transformation equal to z^{-1} at the sites y with $y_\nu = t+1$ and to I elsewhere. It multiplies the links (y,ν) with $y_\nu = t$ by z and those with $y_\nu = t+1$ by z^{-1}, and it fixes every other link, because z is central and the two ends of a link of direction $\nu' \ne \nu$ have the same ν-coordinate. Hence Z_t^ν is Z_{t+1}^ν followed by g, and $f\circ Z_t^\nu = f\circ Z_{t+1}^\nu$ for gauge-invariant f.

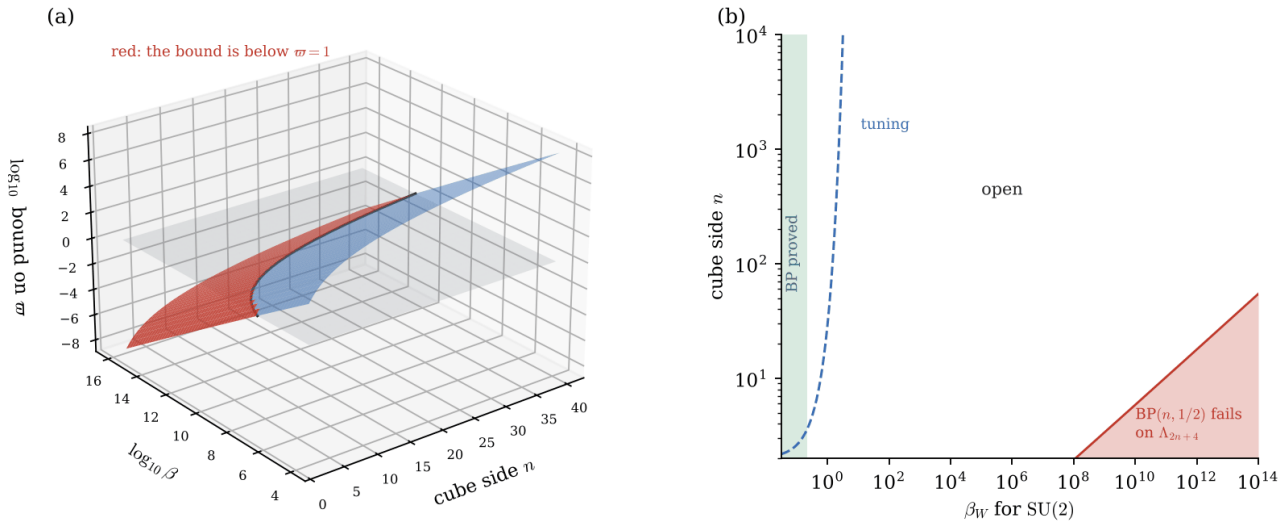


FIG. 12. The averaged condition at strong coupling and at fixed cube size. (a) The bound $\log_{10}\bigl(38400\,n^4(\log\beta+6.11)/\beta\bigr)$ of Theorem 77(ii) for SU(2) over the cube side n and $\log_{10}\beta$; where it lies below the plane $\varpi = 1$ (red), BP$(n,1)$ fails on Λ_{2n+4}, and the black curve is the crossing. (b) For SU(2) in the variables $\beta_W = 4\beta$ and n: the range $|\beta_W| < \frac{2}{9}$ of Corollary 72, where BP(n,ϖ) holds on every torus for all $n \geq n_0(\beta_W)$ (green); the region where Theorem 77(ii) excludes BP$(n,\frac{1}{2})$ on Λ_{2n+4} (red); and the continuum tuning $n \propto e^{3\pi^2\beta_W/11}$ at fixed physical cube size (dashed, schematic, the power prefactor omitted), which runs through the region where neither result applies.

(iii) By (i) and (ii) the maps $f \mapsto f\circ Z^\nu$ are unitary on $\mathscr{A}_\mathbb{C}$, commute with one another and have order N. So the operators $\Pi_r f = N^{-4}\sum_j \omega^{-r\cdot j} f\circ (Z^0)^{j_0}\cdots(Z^3)^{j_3}$, $j \in (\mathbb{Z}/N\mathbb{Z})^4$, are orthogonal projections onto the $\mathscr{A}_r$ with $\sum_r \Pi_r = I$ and $\Pi_r\Pi_{r'} = 0$ for $r \neq r'$, and by (i) they commute with every $\mathscr{P}_x$. Constants are fixed by every Z^ν, and $\bar f \circ Z^\nu = \overline{f\circ Z^\nu}$.

(iv) The residue $t = b_\nu - 1$ is not among $b_\nu,\dots,b_\nu+s-1$ modulo L, since $s \leq L-1$. So Z_t^ν fixes every link on which f depends, and $f\circ Z^\nu = f\circ Z_t^\nu = f$ for each ν.

(v) Let $f \in \mathscr{A}_\mathbb{C}$ and $f_r = \Pi_r f$. Since Π_r commutes with $\mathscr{P}_x$ and with the projection onto the constants, and the $\mathscr{A}_r$ are orthogonal, $\mathrm{Var}_\mu(f)$ and $\mathscr{E}_n(f)$ are the sums over r of $\mathrm{Var}_\mu(f_r)$ and $\mathscr{E}_n(f_r)$. Since $\mathscr{P}_x$ maps real functions to real functions, both are also the sums of their values at $\mathrm{Re}\,f$ and $\mathrm{Im}\,f$. So the inequality for real $f \in \mathscr{A}$, the inequality for all $f \in \mathscr{A}_\mathbb{C}$ and the inequality on every $\mathscr{A}_r$ are equivalent. For $r \neq 0$, $\mu(f) = \langle 1, f\rangle = 0$, since $1 \in \mathscr{A}_0$. □

A local gauge-invariant function, such as a Wilson loop of a contractible loop or a function of the plaquettes of a box, is therefore neutral, while the trace of a Polyakov loop of direction ν lies in $\mathscr{A}_r$ with $r = e_\nu$, since the loop crosses each slice $\{y_\nu = t\}$ once.[17] The continuum step needs the condition only on neutral functions, and only through local ones. Call $f \in L^2(\mu)$ *local* if it depends only on the links with base points in some box of side at most $L/2$.

Definition 81. Let G, φ and Λ_L be as in Sec. IX F, $n \geq 2$ and $\varpi > 0$, and put $\varepsilon = \varpi n^4/L^4$.

(a) BP$_{\mathrm{loc}}(n,\varpi)$: for every local real $f \in \mathscr{A}$ and every integer $t \geq 0$,

$$\bigl\|\mathscr{K}^t\bigl(f-\mu(f)\bigr)\bigr\|_2 \leq (1-\varepsilon)^t\bigl\|f-\mu(f)\bigr\|_2.$$

(b) For $G = \mathrm{SU}(N)$, BP$_0(n,\varpi)$: $\varpi\,\mathrm{Var}_\mu(f) \leq \mathscr{E}_n(f)$ for every real $f \in \mathscr{A}_0$.

For a self-adjoint $0 \leq \mathscr{K} \leq I$ and $g \in L^2(\mu)$ with spectral measure ν_g, $\|\mathscr{K}^t g\|_2^2 = \int \lambda^{2t} d\nu_g$; so the inequality of (a) for all t says that ν_g is carried by $[0, 1-\varepsilon]$ for $g = f - \mu(f)$, the spectral gap of the cube chain on the cyclic subspace of each local function. Condition (a) involves no function of links spread over the torus.[18]

Theorem 82 (the local form of the averaged condition)**.** *Let G, φ and Λ_L be as in Sec. IX F.*

(i) BP(n,ϖ) *implies* BP$_{\mathrm{loc}}(n,\varpi)$*; for* $G = \mathrm{SU}(N)$, BP(n,ϖ) *implies* BP$_0(n,\varpi)$*, which implies* BP$_{\mathrm{loc}}(n,\varpi)$.

(ii) *If* BP$_{\mathrm{loc}}(n,\varpi)$ *holds on* Λ_L*, then* $\varpi \leq 4$*, and both bounds of Theorem 62 hold for bounded* $f, g \in \mathscr{A}$ *with* f *local.*

[17] The twists Z_t^ν are the lattice form of the centre symmetry of the pure gauge theory on a torus. They change no plaquette, so they are invisible to the action, and the trace of a Polyakov loop, which they multiply by ω, is the order parameter of the deconfinement transition at finite temperature; by Proposition 80(v) its mean is zero on every torus.

[18] The distinction is the one between a spectral gap of the whole generator and a gap on the cyclic subspaces of the observables that are measured. Only the latter enters a bound on their correlations, and a lower bound on the Rayleigh quotient of a function does not give it; Appendix H proves both statements in the generality of a self-adjoint contraction.

(iii) *Corollary 63 and Theorem 64 hold with* $\mathrm{BP}_{\mathrm{loc}}(n,\varpi)$*, respectively* $\mathrm{BP}_{\mathrm{loc}}(n_k,\varpi)$*, in place of* BP*; so does the form of Proposition 45*(iii) *stated after Theorem 64.*

(iv) *If* $\mathrm{BP}_{\mathrm{loc}}(n,\varpi)$ *holds, then every local real* $f\in\mathscr{A}$ *and every real* $h\in\mathscr{A}$ *satisfy*

$$\mathrm{Cov}_\mu(f,h)^2\le \mathrm{Var}_\mu(f)\,\frac{\mathscr{E}_n(h)}{\varpi}.$$

If it fails, there are a local real $f\in\mathscr{A}$ *and a real* $h\in\mathscr{A}$ *with* $\mathscr{E}_n(h)<\varpi\,\mathrm{Var}_\mu(h)$ *and* $\mathrm{Cov}_\mu(f,h)=\mathrm{Var}_\mu(h)>0$*; for* $G=\mathrm{SU}(N)$*,* $h\in\mathscr{A}_0$*.*

Proof. (i) Let $\mathscr{B}$ be $\mathscr{A}$, or $\mathscr{A}_0$ for $G=\mathrm{SU}(N)$, and $\mathscr{B}^0$ its functions of mean zero. Every $\mathscr{P}_x$ maps $\mathscr{A}$ into itself (Sec. IX F) and $\mathscr{A}_0$ into itself (Proposition 80(iii)), and $\mu(\mathscr{P}_x g)=\mu(g)$; so $\mathscr{K}$ maps $\mathscr{B}^0$ into itself, and there it is self-adjoint with $0\le\mathscr{K}\le I$. If the inequality of $\mathrm{BP}(n,\varpi)$ holds on the real functions of $\mathscr{B}$, then $\langle g,(I-\mathscr{K})g\rangle\ge\varepsilon\|g\|_2^2$ on $\mathscr{B}^0$, the spectrum of $\mathscr{K}$ on $\mathscr{B}^0$ lies in $[0,1-\varepsilon]$, and $\|\mathscr{K}^t g\|_2\le(1-\varepsilon)^t\|g\|_2$ for $g\in\mathscr{B}^0$. A local real $f\in\mathscr{A}$ lies in $\mathscr{A}_0$ by Proposition 80(iv), since $L/2\le L-1$, so $g=f-\mu(f)\in\mathscr{B}^0$ in both cases.

(ii) Let $f=\mathrm{Re}\,\mathrm{Tr}\,U_p$ for a plaquette p, a local function with $\mathrm{Var}_\mu(f)>0$, and $g=f-\mu(f)$. By $\mathrm{BP}_{\mathrm{loc}}(n,\varpi)$ with $t=1$, $\langle g,\mathscr{K}g\rangle\le\|g\|_2\|\mathscr{K}g\|_2\le(1-\varepsilon)\|g\|_2^2$, so $\langle g,(I-\mathscr{K})g\rangle\ge\varepsilon\|g\|_2^2$. As in the proof of Theorem 62, $\langle g,(I-\mathscr{K})g\rangle=L^{-4}\sum_x\mu((g-\mathscr{P}_x g)^2)\le 4n^4L^{-4}\|g\|_2^2$. Hence $\varpi\le 4$. The proof of Theorem 62 uses $\mathrm{BP}(n,\varpi)$ in two places only: for the bound $\varpi\le 4$, just obtained, and in its spectral part, for $\|\mathscr{K}^T(f-\mu(f))\|_2\le e^{-\varepsilon T}\|f-\mu(f)\|_2$, which $\mathrm{BP}_{\mathrm{loc}}(n,\varpi)$ gives for local real f. The real and imaginary parts of a local function are local.

(iii) In the proof of Corollary 63, Theorem 62 is applied on Λ_L, $L\in\mathscr{L}$ large, to $\vartheta(\bar F)$ and $\tau_t F$, which depend on links with base points in a box whose side depends on F and t only, hence are local once L is large; (ii) applies. Theorem 64 and the form of Proposition 45(iii) use the condition only through Corollary 63.

(iv) Let E be the spectral projection of $\mathscr{K}$ on $L^2(\mu)$ for the interval $(1-\varepsilon,1]$, f local, real and in $\mathscr{A}$, and $g=f-\mu(f)$. Under $\mathrm{BP}_{\mathrm{loc}}(n,\varpi)$, $\int\lambda^{2t}d\nu_g\le(1-\varepsilon)^{2t}\nu_g([0,1])$ for all t, so $\nu_g((1-\varepsilon+\eta,1])=0$ for every $\eta>0$, and $Eg=0$. On the range of $I-E$, $I-\mathscr{K}\ge\varepsilon$, so for real $h\in\mathscr{A}$,

$$\begin{aligned}\mathrm{Cov}_\mu(f,h)^2&=\langle g,(I-E)h\rangle^2\\&\le\|g\|_2^2\,\frac{\langle(I-E)h,(I-\mathscr{K})(I-E)h\rangle}{\varepsilon}\\&\le\|g\|_2^2\,\frac{\langle h,(I-\mathscr{K})h\rangle}{\varepsilon}\\&=\mathrm{Var}_\mu(f)\,\frac{\mathscr{E}_n(h)}{\varpi}.\end{aligned}$$

If $\mathrm{BP}_{\mathrm{loc}}(n,\varpi)$ fails, some local real $f\in\mathscr{A}$ has $\nu_g((1-\varepsilon,1])>0$ for $g=f-\mu(f)$; put $h=Eg$. Spectral projections of $\mathscr{K}$ are strong limits of polynomials in $\mathscr{K}$, so h is real and lies in $\mathscr{A}$, and in $\mathscr{A}_0$ for $G=\mathrm{SU}(N)$, since g does. As $\mathscr{K}1=1$, $E1=1$ and $\mu(h)=\langle E1,g\rangle=0$; then $\mathrm{Var}_\mu(h)=\|Eg\|_2^2=\langle g,Eg\rangle=\mathrm{Cov}_\mu(f,h)>0$, and since ν_h is carried by $(1-\varepsilon,1]$, $\langle h,(I-\mathscr{K})h\rangle=\int(1-\lambda)\,d\nu_h<\varepsilon\|h\|_2^2$, that is, $\mathscr{E}_n(h)<\varpi\,\mathrm{Var}_\mu(h)$. □

Fig. 13 shows the sectors and the spectral picture behind Theorem 82.

By (iv), under $\mathrm{BP}_{\mathrm{loc}}$ the functions on which the cube chain is slow are nearly orthogonal to every local gauge-invariant function, with the defect of orthogonality bounded by their Dirichlet form, and a failure of $\mathrm{BP}_{\mathrm{loc}}$ produces a slow function correlated with a local one. Slow functions orthogonal to all local ones, and for $\mathrm{SU}(N)$ slow functions in the charged sectors, do not affect the continuum step. The Villain obstruction persists for the local form: the proof of Corollary 66 uses the condition only through Theorem 62 for the functions Ψ_p, and through the bound $\varpi\le 4$, which holds with Ψ_p in place of $\mathrm{Re}\,\mathrm{Tr}\,U_p$, a nonconstant function of the four links of p. So for the Villain theory at $\beta>\beta_0$ there are no n and $\varpi>0$ such that $\|\mathscr{K}^t(\Psi_p-\mu(\Psi_p))\|_2\le(1-\varpi n^4/|X|)^t\|\Psi_p-\mu(\Psi_p)\|_2$ for all t, all plaquettes and the measures with free boundary of all boxes of side at least $2n$. The local form, like the full one, has to use the non-abelian structure of G.

For $\mathrm{SU}(N)$ the obstruction of Sec. IX G lies in the charged sectors. Let $\mathscr{C}$ be the set of real $f\in\mathscr{A}$ with $\Pi_0 f=0$, in the notation of the proof of Proposition 80(iii); it is orthogonal to $\mathscr{A}_0$ and preserved by every $\mathscr{P}_x$.

Corollary 83 (the fixed-size obstruction is charged)**.** *Let* $G=\mathrm{SU}(N)$*,* $N\ge 2$*. Theorem 77 holds, with the same constants in* (i) *and* (ii) *and with another constant* C_N *in* (iii)*, when the inequality of* $\mathrm{BP}(n,\varpi)$ *is assumed only for* $f\in\mathscr{C}$*. Its hypotheses and conclusions concern only the restriction of the condition to* $\mathscr{C}$*; neither* $\mathrm{BP}_0(n,\varpi)$ *nor* $\mathrm{BP}_{\mathrm{loc}}(n,\varpi)$ *enters it.*

Proof. In the proof of Theorem 77(i) the condition is applied to $L^{-3}\sum_y\mathrm{Re}\,\mathrm{Tr}\,P_y$ and to the corresponding imaginary part. The trace of P_y lies in $\mathscr{A}_{e_0}$ and its conjugate in $\mathscr{A}_{-e_0}$, and $\pm e_0\ne 0$ in $(\mathbb{Z}/N\mathbb{Z})^4$, so both functions lie in $\mathscr{C}$. In the proof of (ii) and (iii), $F_{ab}=\mathrm{Tr}(A^aB^b)$ lies in $\mathscr{A}_r$ with $r=(a,b,0,0)\ne 0$ for $(a,b)\in I_N$, so the real and imaginary parts of F_{ab} lie in $\mathscr{C}$. The only other use of the condition is the bound $\varpi\le 2$ from Proposition 73(ii), whose test function is neutral. In its place take $f_*=\mathrm{Re}\,\mathrm{Tr}\,P_0\in\mathscr{C}$, a nonconstant continuous function, so $\mathrm{Var}_\mu(f_*)>0$. $\mathscr{P}_x f_*=f_*$ unless V_x meets the loop P_0, which happens for at most Ln^3 points x, and $\mu((f_*-\mathscr{P}_x f_*)^2)\le\mathrm{Var}_\mu(f_*)$; so $\varpi\le L/n\le 4$ on Λ_{2n+4}. In the case $\eta_N^2<4L^4\varepsilon\le 1024n^4\varepsilon$ of that proof this gives $\varpi\le(4096/\eta_N^2)n^4\varepsilon$, which for $N=2$ is at most $19200\,n^4\varepsilon$, so the constant 38400 of (ii) is unchanged. $\mathscr{C}$ is orthogonal to $\mathscr{A}_0$, which contains every local function of $\mathscr{A}$. □

Corollary 83 constrains every argument that proves the condition through a contraction of the cube chain on gauge orbits. On G^E let $d(U,U')=\min_g d_E(U,gU')$, the minimum over gauge transformations g, with d_E as in the proof of Proposition 65; since gauge transformations act by isometries of d_E, d is a pseudometric, constant on pairs of gauge orbits. Write W_1^d for the Kantorovich distance of d. By gauge covariance, $\mathscr{K}(gU',\cdot)$ is the image of $\mathscr{K}(U',\cdot)$ under g, so a contraction of $W_1^{d_E}$ by a factor $1-\kappa$, as in the proof of Proposition 65(i), gives one of W_1^d by the same factor.

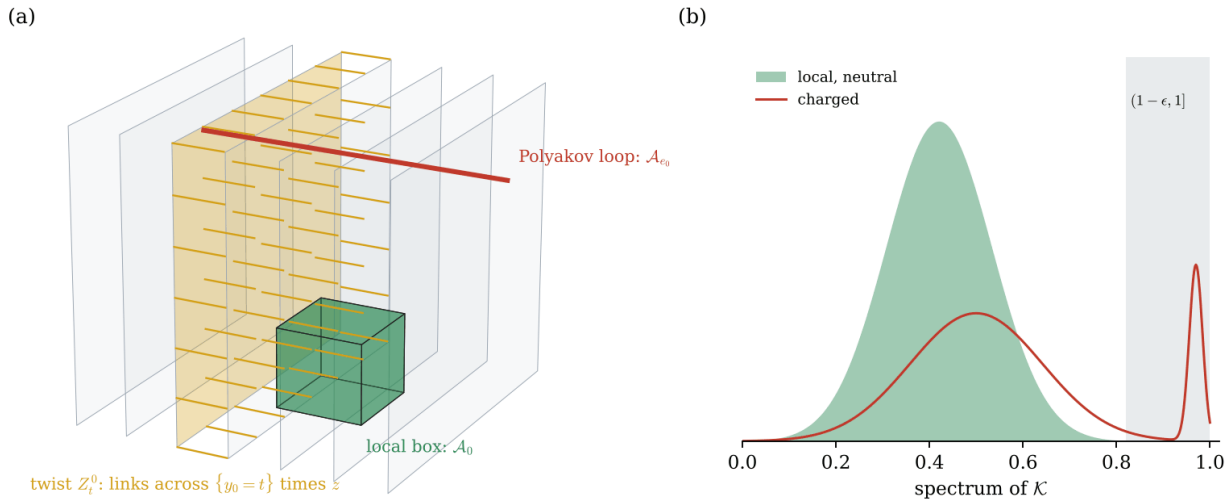


FIG. 13. Centre sectors and the local form of the averaged condition. (a) A three-dimensional section of the torus with the slices $\{y_0 = t\}$. The twist Z_t^0 multiplies the links that cross one slice by the central element z (gold) and leaves every plaquette unchanged; a Polyakov loop of direction 0 crosses every slice once and is charged, while a gauge-invariant function of a box that misses one slice is neutral (Proposition 80). (b) Spectral measures of the cube chain $\mathcal{K}$, schematic: under $\mathrm{BP}_{\mathrm{loc}}$ the measure of a local function puts no mass on $(1-\varepsilon, 1]$ (green), while a charged function may carry mass there (red), as the test functions of Theorem 77 do at weak coupling (Theorem 82, Corollary 83).

Corollary 84 (couplings on gauge orbits)**.** *Let $G = \mathrm{SU}(2)$, $\beta \geq 1$, $n \geq 2$ and $L = 2n+4$. If there is $\kappa > 0$ with*

$$W_1^d\big(\mathcal{K}(U,\cdot), \mathcal{K}(U',\cdot)\big) \leq (1-\kappa)\, d(U,U')$$

for all configurations U, U' of Λ_L, then

$$\kappa \leq \frac{38400\, n^8 (\log\beta + 6.11)}{L^4 \beta}.$$

Proof. Let h be gauge invariant and Lipschitz for d with constant $\mathrm{Lip}(h)$. For a coupling of $\mathcal{K}(U,\cdot)$ and $\mathcal{K}(U',\cdot)$ with expected distance W_1^d, $|\mathcal{K}h(U) - \mathcal{K}h(U')| \leq \mathrm{Lip}(h) W_1^d$, so $\mathcal{K}h$ is Lipschitz with constant $(1-\kappa)\mathrm{Lip}(h)$, and it is gauge invariant. As in the proof of Proposition 65(i), with $d \leq D|E|$, the spectral measure of $\mathcal{K}$ at $h - \mu(h)$ is carried by $[0, 1-\kappa]$. Such h are dense in $\mathscr{A}$: averaging over the compact gauge group is the orthogonal projection of $L^2(\mu)$ onto $\mathscr{A}$ and maps continuous functions to continuous ones, so the continuous functions of $\mathscr{A}$ are dense in it; a continuous gauge-invariant f is uniformly continuous for d, since $d(U,U')$ small gives g with $d_E(U, gU')$ small and $f(gU') = f(U')$; and $f_k(U) = \inf_V (f(V) + k\, d(U,V))$ is gauge invariant, Lipschitz for d with constant k, and converges to f uniformly. So the spectral projection of $\mathcal{K}$ for $(1-\kappa, 1]$ vanishes on the functions of $\mathscr{A}$ of mean zero, $\langle f, (I - \mathcal{K}) f\rangle \geq \kappa \mathrm{Var}_\mu(f)$ on $\mathscr{A}$, and $\mathrm{BP}(n, \kappa L^4/n^4)$ holds on Λ_{2n+4}. Theorem 77(ii) gives the bound. □

The proof uses only that d-Lipschitz functions are dense in the charged part $\mathscr{C}$. If d is replaced by its minimum over the centre twists as well, $d'(U,U') = \min d_E(U, gZU')$ over gauge transformations g and products Z of the maps Z_t^ν, the d'-Lipschitz functions are neutral, and the same argument, with the average over the gauge group and the group generated by the Z_t^ν, gives $\mathrm{BP}_0(n, \kappa L^4/n^4)$, which Theorem 77 does not constrain. A coupling argument for the cube chain at weak coupling and small n therefore has to contract a pseudometric that does not see the centre twists. Since $n^8/(2n+4)^4 \geq n^4/256$, the bound of Corollary 84 is at least $150\, n^4(\log\beta + 6.11)/\beta$, which exceeds one once $n^4 \geq \beta/150$; along the tuning, where n_k^4 grows faster than every power of β_k, the corollary does not exclude such couplings.

The remaining results hold for every G and φ of Sec. IX F, on tori and on boxes. Put $\omega_\varphi = \log(\max\varphi / \min\varphi)$, so that $\omega_\varphi \leq 2N^2|\beta|$ for the Wilson weight, and for a finite set S of links let P_S be the set of plaquettes of P that contain a link of S; $|P_S| \leq 6|S|$.

Proposition 85 (no local slow modes)**.** *Let $S \subset E$ be finite and let $f \in L^2(\mu)$ be real and depend only on the links of S. Then*

$$\mathrm{Var}_\mu(f) \leq e^{5\omega_\varphi |P_S|}\, \mathscr{E}_n(f).$$

In particular, if $\mathscr{E}_n(f) \leq \varpi\, \mathrm{Var}_\mu(f)$ with $0 < \varpi < 1$ and $\mathrm{Var}_\mu(f) > 0$, then $|P_S| \geq \log(1/\varpi)/(5\omega_\varphi)$; for the Wilson weight, $|S| \geq \log(1/\varpi)/(60N^2|\beta|)$.

Proof. For a finite $W \subset E$ write H_W for the normalised Haar measure on G^W and $\rho_W(\cdot \mid U)$ for the density of $\gamma_W^0(\cdot \mid U)$ with respect to H_W. It is proportional to a product of $|P_W|$ values of φ, so the ratio of two of its values is at most $e^{\omega_\varphi |P_W|}$, and as its integral is one, $e^{-\omega_\varphi|P_W|} \leq \rho_W \leq e^{\omega_\varphi|P_W|}$. Put $a = \omega_\varphi |P_S|$, and write P_W also for the conditional expectation given the links outside W; then $P_{V_x} = \mathscr{P}_x$.

Step 1. Let ν_S be the law of U_S under μ. Its density with respect to H_S is the mean of $\rho_S(\cdot \mid U)$ over U, so it lies in $[e^{-a}, e^a]$,

and $r_U = \rho_S(\cdot \mid U)/(d\nu_S/dH_S) \geq e^{-2a}$. Let $g = f - \mu(f)$, a function on G^S with $\nu_S(g) = 0$. Then $P_S g(U) = \int g\, r_U\, d\nu_S = \int g\,(r_U - e^{-2a})\, d\nu_S$, and the Cauchy–Schwarz inequality for the positive measure $(r_U - e^{-2a})\, d\nu_S$ of mass $1 - e^{-2a}$ gives

$$(P_S g)^2(U) \leq (1 - e^{-2a}) \int g^2 r_U \, d\nu_S = (1 - e^{-2a}) P_S(g^2)(U).$$

Integrating, $\mu((P_S g)^2) \leq (1 - e^{-2a})\mu(g^2)$, and since P_S is an orthogonal projection, $\mu((f - P_S f)^2) = \mu(g^2) - \mu((P_S g)^2) \geq e^{-2a} \operatorname{Var}_\mu(f)$.

Step 2. $\mu((f - P_S f)^2)$ is the mean over U of the variance of f under $\gamma_S^0(\cdot \mid U)$, which is at most $\int (f - H_S(f))^2 \rho_S\, dH_S \leq e^a \operatorname{Var}_{H_S}(f)$. So $\operatorname{Var}_\mu(f) \leq e^{3a} \operatorname{Var}_{H_S}(f)$.

Step 3. For $W \subset S$ let v_W be the variance of f in the links of W under H_W, the links of $S \setminus W$ fixed, a function of these links. With the decomposition $f = \sum_{T \subset S} f_T$ of the proof of Proposition 65(ii) for H_S, $H_S(v_W) = \sum_{T \cap W \neq \emptyset} \|f_T\|^2$, and since every link lies in exactly n^4 of the sets V_x,

$$\sum_{x \in X} H_S(v_{S \cap V_x}) = \sum_T \|f_T\|^2 \, \#\{x : T \cap V_x \neq \emptyset\} \geq n^4 \operatorname{Var}_{H_S}(f).$$

Step 4. Let $W = S \cap V_x \neq \emptyset$. Since $d\nu_S/dH_S \geq e^{-a}$ and $v_W \geq 0$, $H_S(v_W) \leq e^a \nu_S(v_W) = e^a \mu(v_W)$. For every configuration U, $v_W \leq \int (f - c)^2 dH_W$ with c the mean of f under $\gamma_W^0(\cdot \mid U)$, and $dH_W/d\gamma_W^0 \leq e^{\omega_\varphi |P_W|} \leq e^a$, so v_W is at most e^a times the variance of f under $\gamma_W^0(\cdot \mid U)$. Hence $\mu(v_W) \leq e^a \mu((f - P_W f)^2) \leq e^a \mu((f - \mathscr{P}_x f)^2)$, the last step because the links outside V_x are among those outside W.

Steps 3 and 4 give $\operatorname{Var}_{H_S}(f) \leq n^{-4} e^{2a} \sum_x \mu((f - \mathscr{P}_x f)^2) = e^{2a} \mathscr{E}_n(f)$, and with Step 2, $\operatorname{Var}_\mu(f) \leq e^{5a} \mathscr{E}_n(f)$. For the Wilson weight, $5\omega_\varphi |P_S| \leq 60 N^2 |\beta| |S|$. □

So a sequence of functions with $\mathscr{E}_n(f_j)/\operatorname{Var}_\mu(f_j) \to 0$, on tori of growing side at fixed n and β, depends on growing sets of links,[19] and no subsequence converges to a function of finitely many links. This does not give $\mathrm{BP}_{\mathrm{loc}}$: Proposition 85 bounds the Rayleigh quotient of a local function, which is the mean of $1 - \lambda$ under its spectral measure, while $\mathrm{BP}_{\mathrm{loc}}$ asks that this measure put no mass near $\lambda = 1$; by Theorem 82(iv) a failure of it produces a slow function correlated with a local one.

The transfer of the condition between cube sizes is limited in the same way. Write W_x for the cubes of side m and $\mathscr{E}_m$ for the corresponding form; the proof of Proposition 73(i) shows $\mathscr{E}_m \geq (n/m)^4 \mathscr{E}_n$ on $L^2(\mu)$.

Proposition 86. *Let $\varphi \equiv 1$, $2 \leq n \leq m$, $R \geq 2$, and let L be even with $L \geq \max(R+m, 2m+4)$. There is a real $f \in \mathscr{A}$ with $\operatorname{Var}_\mu(f) = 1$ and*

$$\mathscr{E}_m(f) \leq \left(\frac{n}{m}\right)^4 \left(\frac{R+m}{R-1}\right)^4 \mathscr{E}_n(f).$$

So no inequality $\mathscr{E}_m \geq c\,\mathscr{E}_n$ on $\mathscr{A}$ with $c > (n/m)^4$ holds on all tori, although $\mathrm{BP}(n,1)$ and $\mathrm{BP}(m,1)$ both hold for this measure.

Proof. Since G is compact and nontrivial, it has a nontrivial irreducible unitary representation, whose character has Haar mean zero; its real or imaginary part, normalised, is a real class function h with $\int h\, dU = 0$ and $\int h^2 dU = 1$. Let Q be the set of plaquettes of the plane $\{0,1\}$ with base point $y \in [0,R)^4$ and y_0, y_1 even. The plaquette at y has the links $(y,0)$, $(y+e_0,1)$, $(y+e_1,0)$, $(y,1)$. A link $(w,0)$ of such a plaquette determines y, which is w or $w - e_1$ according to the parity of w_1, and a link $(w,1)$ determines it through the parity of w_0; so two plaquettes of Q share no link. Put $f = \prod_{p \in Q} h(U_p)$, a gauge-invariant function. For one plaquette, integrating $h(U_p)$ over any one of its four links gives $\int h\, dU = 0$, by the invariance of Haar measure under translation and inversion; so in the decomposition of the proof of Proposition 65(ii), $h(U_p)$ has the single term $T = p$, and since the plaquettes of Q involve disjoint links, $f = f_T$ with T the union of their links, and $\|f\|_2 = 1 = \operatorname{Var}_\mu(f)$. That proof gives $k^4 \mathscr{E}_k(f) = N_k$, the number of x with $V_x \cap T \neq \emptyset$ for cubes of side k, $k = n, m$.

The base points of the links of T lie in $[0,R]^4$, so $N_k \leq (R+k)^4$, as $R + k \leq L$. If $x \in [0, R-2]^4$, the cube of side $k \geq 2$ at x contains the four points $x + ae_0 + be_1$, $a, b \in \{0,1\}$, of $[0,R)^4$, one of which has even coordinates 0 and 1 and is the base point of a plaquette of Q; so $N_k \geq (R-1)^4$. Hence $\mathscr{E}_m(f)/\mathscr{E}_n(f) = (n/m)^4 N_m/N_n \leq (n/m)^4 ((R+m)/(R-1))^4$. The last claim is Proposition 65(ii) and $R \to \infty$. □

At $\beta = 0$ the gaps at the two sizes are both at least one, while the Dirichlet forms differ by the factor $(n/m)^4$ on functions spread over large boxes. An upward transfer without that loss has to use more than the comparison of the forms, for instance an orthogonal decomposition such as that of Proposition 65(ii), which is specific to product measures.

Remark 87 (the routes to the averaged condition at weak coupling). The results above settle how far the following routes go.

Constructions. A test function bounds the best constant from above. A lower bound needs a Poincaré inequality, and Proposition 85 gives one for every function of a fixed finite set S of links, uniformly in the torus, with constant $e^{-5\omega_\varphi |P_S|}$. A Lyapunov argument derives a Poincaré inequality from a local one on a set of configurations and a drift towards that set; for the cube chain on Λ_L the set consists of configurations of all $4L^4$ links, a small-field set for instance, and the local inequality it needs is one for the restriction of μ to that set, for functions of all links, which Proposition 85 does not give. A coupling that contracts d_E or the pseudometric d on gauge orbits gives the condition on $\mathscr{A}$ on every torus to which it applies, and Corollaries 78 and 84 bound its rate on Λ_{2n+4} for $\mathrm{SU}(2)$; a coupling that contracts a pseudometric blind to the centre twists gives BP_0, which suffices by Theorem 82 and is open. The comparison with the cube chain of larger cubes reduces to the inner gap of Proposition 73(iii); that proof works sector by sector, so it gives $\mathrm{BP}_0(n,\cdot)$ from $\mathrm{BP}_0(m,\cdot)$ and the inner inequality for $f \in \mathscr{A}_0$. The comparison with the cube chain of the same size for another measure μ' through the

[19] The exponent $5\omega_\varphi |P_S|$ is the cost of comparing the conditional laws of the links of S with Haar measure, in the manner of the perturbation lemma of Holley and Stroock [63]; it depends on S through the plaquettes that touch it, and not on the torus. At fixed β it forces $|S| \to \infty$ as $\varpi \to 0$, linearly in $\log(1/\varpi)$.

density $d\mu/d\mu'$ costs the exponential of the oscillation of its logarithm, a sum of $6L^4$ plaquette terms; a bound uniform in the torus has to localise the comparison, which leads back to the inner gap. The comparison of Dirichlet forms between cube sizes loses exactly $(n/m)^4$ (Proposition 86).

Forced blow-up. If $\mathrm{BP}_{\mathrm{loc}}(n_k,\varpi)$ fails for every $\varpi>0$ along the tuning, Theorem 82(iv) gives local functions f_k and slow functions h_k, in $\mathscr{A}_0$ for $\mathrm{SU}(N)$, with $\mathrm{Cov}_\mu(f_k,h_k)=\mathrm{Var}_\mu(h_k)$ and $\mathscr{E}_{n_k}(h_k)/\mathrm{Var}_\mu(h_k)\to 0$, and Proposition 85 gives $60N^2\beta_k|S|\geq\log(\mathrm{Var}_\mu(h_k)/\mathscr{E}_{n_k}(h_k))$ for every set S of links on which h_k depends. A contradiction has to come from a property of these slow functions. The properties proved here for the tuned measures, tightness of the marginals, reflection positivity, the transfer matrix, gauge invariance and centre symmetry, are properties of the measures and of their torus limits, not of the cube chain, and the paper proves no implication from any of them to $\mathrm{BP}_{\mathrm{loc}}$. Such an implication is a converse of Theorem 62, from decay of correlations to relaxation of the cube chain. For local dynamics of spin systems the known converses assume a mixing condition uniform in the boundary condition [82], of the type of the supremum form D^0, which Corollary 78 excludes for n below a multiple of $(\beta/\log\beta)^{1/4}$. A limit object along the tuning would be a slow mode of a dynamics of continuum gauge fields, and no such dynamics is constructed here. Continuity in β does not connect the regimes: Corollary 72 holds at fixed β for $n\geq n_0(\beta)$, and on Λ_{2n+4} the best constant of the charged part passes from at least one at $\beta=0$ to $O(n^4\log\beta/\beta)$ (Theorem 77, Corollary 83); no monotonicity in β is known.

Residual sets. Theorem 82 and Corollary 83 identify the residual: the condition may fail on charged functions and on slow functions orthogonal to all local ones without affecting the continuum step, and what has to be excluded is a slow function correlated with a local neutral one. The Villain theory beyond β_0 has such slow functions, by the local form of Corollary 66. The failure of $\mathrm{BP}_{\mathrm{loc}}$ does not by itself give a trivial or Gaussian limit, since $\mathrm{BP}_{\mathrm{loc}}$ is sufficient for a gap and is not shown to be necessary; a trivialisation argument needs the converse of Theorem 62 again.

Perturbative and static estimates. By Theorem 82(ii), $\mathrm{BP}_{\mathrm{loc}}(n,\varpi)$ on infinitely many tori gives exponential decay of every local gauge-invariant covariance in the torus limit. The quadratic approximation of the Wilson measure at weak coupling consists of $\dim G$ copies of the free lattice Maxwell field, whose field strengths have correlations that decay as a power of the distance. An argument whose estimates apply equally to that Gaussian approximation, or order by order in an expansion around it, therefore cannot give $\mathrm{BP}_{\mathrm{loc}}$; the Villain theory, to which the setting applies, is a nonperturbative instance of the same failure. The renormalisation group bounds of [6, 7, 13] concern effective actions in a box of fixed physical size and do not address the cube chain. The Bakry–Émery bound of Lemma 71 requires $6N|\beta|<\kappa_G$. Reflection positivity gives the transfer matrix and the chessboard estimate bounds probabilities of local events uniformly in the torus; both are statements about the measure, not about the cube chain.

Remark 88 (what is proved about the averaged condition at weak coupling). Proved: for $\mathrm{SU}(N)$ the cube moves preserve the centre sectors of the torus, local gauge-invariant functions are neutral, and the condition splits into sectors (Proposition 80); the conclusions of Theorems 62 and 64 and of Corollary 63 follow from the local form $\mathrm{BP}_{\mathrm{loc}}$, which is implied by the neutral form BP_0; under $\mathrm{BP}_{\mathrm{loc}}$ slow functions are nearly orthogonal to local ones, and a failure of it produces a slow function correlated with a local one (Theorem 82); the obstruction of Theorem 77 holds for the charged part alone, and neither BP_0 nor $\mathrm{BP}_{\mathrm{loc}}$ enters it (Corollary 83); for $\mathrm{SU}(2)$ a contraction of the cube chain on the gauge orbits of Λ_{2n+4} has rate at most $38400\,n^8(\log\beta+6.11)/(L^4\beta)$ (Corollary 84); every function of a finite set S of links satisfies $\mathrm{Var}_\mu(f)\leq e^{5\omega_\varphi|P_S|}\mathscr{E}_n(f)$ on every torus and box (Proposition 85); and the upward comparison of the Dirichlet forms loses exactly $(n/m)^4$, already at $\beta=0$ (Proposition 86). Not proved: BP, BP_0 or $\mathrm{BP}_{\mathrm{loc}}$ along the continuum tuning for any non-abelian group; whether BP_0 or $\mathrm{BP}_{\mathrm{loc}}$ fails on Λ_{2n+4} at weak coupling, of which Theorem 90 reduces the first, for $\mathrm{SU}(2)$, to a lower bound on the law of the Polyakov loop; a converse of Theorem 62, from decay of correlations to the relaxation of the cube chain; a coupling that contracts a pseudometric blind to the centre twists; and the mass gap.

I. The neutral sector on small tori

Theorem 77 proves that the averaged condition fails on the torus Λ_{2n+4} at weak coupling, and Corollary 83 places that failure in the charged part of $\mathscr{A}$. This subsection treats the neutral part for $\mathrm{SU}(2)$. The neutral functions of a Polyakov loop change under the cube moves only through the curvature, as the charged ones do, so the right side of BP_0 is small for them; what is missing is a lower bound on their variance. A flow that turns every Polyakov loop of one direction inside its own one-parameter subgroup changes the Wilson action only through the curvature, and integration by parts along it bounds the concentration of the law of the Polyakov loop. The outcome is that BP_0 at weak coupling on Λ_{2n+4} forces the Polyakov loop towards the centre and towards the traceless elements of $\mathrm{SU}(2)$, and that the failure of BP_0 there reduces to one lower bound on the law of the Polyakov loop.

Throughout, $G=\mathrm{SU}(2)$ with the Wilson weight $\varphi=e^{2\beta\,\mathrm{Tr}(\cdot)}$, $\Lambda=\Lambda_L$ with L even, and $H=\sum_p(2-\mathrm{Tr}\,U_p)$, so that μ has density proportional to $e^{-2\beta H}$. Every $W\in\mathrm{SU}(2)$ can be written $W=\cos\theta\,I+\sin\theta\,u$ with $\theta\in[0,\pi]$ and u traceless anti-Hermitian with $u^2=-I$, and $\mathrm{Tr}\,W=2\cos\theta$; in these polar coordinates of the three-sphere the Haar measure is $\frac{2}{\pi}\sin^2\theta\,d\theta$ times the uniform measure of u. For $W\in\mathrm{SU}(2)$ put

$$J(W)=\mathrm{Tr}(W)\,\tfrac{1}{2}(W-W^*)=2\cos\theta\sin\theta\,u,$$

a traceless anti-Hermitian matrix that commutes with W and satisfies $J(gWg^{-1})=gJ(W)g^{-1}$ for $g\in\mathrm{SU}(2)$ and $J(-W)=J(W)$. As in Sec. IX G, P_b is the Polyakov loop of direction 0 through the site b, the ordered product of the L links $(b+te_0,0)$, $0\leq t<L$, and we put $T_b=\mathrm{Tr}\,P_b$ and $T=T_0$. By the symmetries of the torus every T_b, and the trace of every Polyakov loop of another direction, has the law of T under μ.

On the configuration space G^E let $\mathscr{V}$ be the vector field whose component at a link $(b,0)$ is $L^{-1}J(P_b)U_{(b,0)}$ and whose components at the links of the directions $1,2,3$ vanish; it is the velocity at $s=0$ of $U_{(b,0)}\mapsto e^{sJ(P_b)/L}U_{(b,0)}$. Its entries are polynomials in the entries of the links, so $\mathscr{V}$ is smooth, and for a function h of class C^1 we write $\mathscr{V}h$ for the derivative of h along $\mathscr{V}$. Put

$$\Theta=\mathscr{V}S-\operatorname{div}\mathscr{V},\qquad S=-2\beta\sum_p\operatorname{Tr}U_p,$$

where div is the divergence for the product of the Haar measures.

Lemma 89 (the holonomy flow)**.** *Let $L\ge 8$ and $\beta\ge 2$.*

(i) *For every site y, $\mathscr{V}P_y=J(P_y)P_y$; in particular $\mathscr{V}T=-\frac12 T(4-T^2)$.*

(ii) $\operatorname{div}\mathscr{V}=\sum_{y:\,y_0=0}(2T_y^2-2)$, *so* $|\operatorname{div}\mathscr{V}|\le 6L^3$.

(iii) $|\mathscr{V}S|\le 16\beta\sum_q(2-\operatorname{Tr}U_q)$, *the sum over the plaquettes q of the planes $\{0,\nu\}$, and* $\|\Theta\|_2\le 192L^4(\log\beta+5.44)$.

(iv) *For every real h of class C^1, $\mu(\mathscr{V}h)=\mu(h\Theta)$ and* $\mu(\mathscr{V}h)^2\le\operatorname{Var}_\mu(h)\|\Theta\|_2^2$.

Proof. (i) For $0\le t<L$ write U_t for the link $(y+te_0,0)$ and put $G_t=U_0\cdots U_{t-1}$, so that the loop based at $y+te_0$ is $P^{(t)}=U_t\cdots U_{L-1}U_0\cdots U_{t-1}=G_t^{-1}P_yG_t$. Then $J(P^{(t)})=G_t^{-1}J(P_y)G_t$, and differentiating the product,

$$\mathscr{V}P_y=\sum_{t=0}^{L-1}G_t\frac{J(P^{(t)})}{L}U_t\cdots U_{L-1}=\sum_{t=0}^{L-1}\frac{J(P_y)}{L}P_y=J(P_y)P_y.$$

With $P_y=\cos\theta+\sin\theta\,u$,

$$\begin{aligned}\operatorname{Tr}(J(P_y)P_y)&=2\cos\theta\sin\theta\operatorname{Tr}(u\cos\theta+u^2\sin\theta)\\&=-4\cos\theta\sin^2\theta=-\tfrac12T_y(4-T_y^2).\end{aligned}$$

(ii) The divergence of a vector field on G^E is the sum over the links e of the divergence in U_e of its component at e, the other links fixed. At $e=(y+te_0,0)$ the component is $L^{-1}J(U_eR)U_e$ with $R=U_{t+1}\cdots U_{L-1}U_0\cdots U_{t-1}$ independent of U_e. The right translation $U_e\mapsto W=U_eR$ preserves Haar measure and carries this field to $W\mapsto L^{-1}J(W)W$. Since $J(W)$ commutes with W, $e^{sJ(W)}W=\cos(\theta+s\sin2\theta)+\sin(\theta+s\sin2\theta)\,u$, so in polar coordinates the field $J(W)W$ is $\sin2\theta\,\partial_\theta$, with divergence

$$\frac{1}{\sin^2\theta}\,\partial_\theta\big(\sin^2\theta\sin2\theta\big)=8\cos^2\theta-2=2(\operatorname{Tr}W)^2-2$$

for $0<\theta<\pi$, and by continuity on all of $\mathrm{SU}(2)$. As $\operatorname{Tr}W=T_y$, the L links of the loop through y contribute $2T_y^2-2$, and there are L^3 such loops.

(iii) Only the plaquettes of the planes $\{0,\nu\}$ contain links of direction 0. The plaquette p at b in the plane $\{0,\nu\}$ is $U_p=U_{(b,0)}U_{(b+e_0,\nu)}U_{(b+e_\nu,0)}^{-1}U_{(b,\nu)}^{-1}$, and with $J_b=J(P_b)$ and $V=U_{(b,\nu)}$,

$$\mathscr{V}\operatorname{Tr}U_p=\frac1L\operatorname{Tr}\big(U_p\Delta_p\big),$$

where

$$\Delta_p=J_b-VJ_{b+e_\nu}V^{-1}=J(P_b)-J(VP_{b+e_\nu}V^{-1}).$$

The strip σ of the L plaquettes at $b+te_0$ in the plane $\{0,\nu\}$ has boundary holonomy $P_bVP_{b+e_\nu}^{-1}V^{-1}$, so by Lemma 74, $\|P_b-VP_{b+e_\nu}V^{-1}\|_{\mathrm{HS}}\le Q_\sigma=\sum_{q\in\sigma}\|U_q-I\|_{\mathrm{HS}}$. For $P,P'\in\mathrm{SU}(2)$, $|\operatorname{Tr}P-\operatorname{Tr}P'|\le\sqrt2\|P-P'\|_{\mathrm{HS}}$, $\|\frac12(P-P^*)\|_{\mathrm{HS}}\le\sqrt2$ and $\|\frac12(P-P^*)-\frac12(P'-P'^*)\|_{\mathrm{HS}}\le\|P-P'\|_{\mathrm{HS}}$, so J is 4-Lipschitz for the Hilbert–Schmidt norm and $\|\Delta_p\|_{\mathrm{HS}}\le 4Q_\sigma$. As Δ_p is anti-Hermitian, $\operatorname{Tr}\Delta_p$ is imaginary, and $\operatorname{Tr}(U_p\Delta_p)$ is real because $\mathscr{V}\operatorname{Tr}U_p$ is; hence $|\operatorname{Tr}(U_p\Delta_p)|=|\operatorname{Re}\operatorname{Tr}((U_p-I)\Delta_p)|\le 4\|U_p-I\|_{\mathrm{HS}}Q_\sigma$. Every plaquette p of the planes $\{0,\nu\}$ lies in exactly one strip σ, so

$$|\mathscr{V}S|\le\frac{8\beta}{L}\sum_\sigma Q_\sigma^2\le 8\beta\sum_\sigma\sum_{q\in\sigma}\|U_q-I\|_{\mathrm{HS}}^2,$$

by the Cauchy–Schwarz inequality over the L plaquettes of σ; with $\|U-I\|_{\mathrm{HS}}^2=2(2-\operatorname{Tr}U)$ this is the first claim, and $|\mathscr{V}S|\le 16\beta H$.

Put $\phi(t)=\log\int e^{-2tH}dU$, convex and decreasing, and $M=\log\mu(e^{\beta H})=\phi(\beta/2)-\phi(\beta)\ge 0$. Convexity gives $M\le-\frac{\beta}{2}\phi'(\beta/2)=\beta\mu_{\beta/2}(H)$, where $\mu_{\beta/2}$ is the Wilson measure at $\beta/2\ge1$, and Lemma 76 at $\beta/2$ bounds each of the $6L^4$ terms of $\mu_{\beta/2}(H)$ by $2(\log\beta+6.11-\log2)/\beta$; so $M\le 12L^4(\log\beta+5.42)$. For $y\ge0$ and $a\ge0$, $2ay+y^2\le(2a+2)(e^y-1)$, since $y\le e^y-1$ and $y^2\le2(e^y-1-y)$; so for every $X\ge0$, $X^2\le a^2+(2a+2)e^{X-a}$, and with $X=\beta H$ and $a=M$, $\beta^2\mu(H^2)\le M^2+2M+2\le(M+2)^2$. Hence $\|\mathscr{V}S\|_2\le 16(M+2)\le 192L^4(\log\beta+5.43)$ for $L\ge8$, and with (ii), $\|\Theta\|_2\le 192L^4(\log\beta+5.43)+6L^3\le 192L^4(\log\beta+5.44)$.

(iv) On the compact manifold G^E,

$$\int\mathscr{V}g\,dU=-\int g\operatorname{div}\mathscr{V}\,dU$$

for every g of class C^1. Applied to $g=he^{-S}$ this gives $\mu(\mathscr{V}h)=\mu(h(\mathscr{V}S-\operatorname{div}\mathscr{V}))=\mu(h\Theta)$. With $h=1$, $\mu(\Theta)=0$, so $\mu(\mathscr{V}h)=\mu((h-\mu(h))\Theta)$, and the Cauchy–Schwarz inequality gives the bound. □

Part (iv) with $h=\psi(T)$ and part (i) give, for every real ψ of class C^1 on $[-2,2]$,

$$\mu\big(T(4-T^2)\psi'(T)\big)^2\le 4\operatorname{Var}_\mu\big(\psi(T)\big)\|\Theta\|_2^2.\qquad(16)$$

The twist Z^0 preserves μ and reverses the sign of T, so the left side vanishes for odd ψ. With $\psi(t)=t^2/4$ it says that $\operatorname{Var}_\mu(T^2/4)\ge\mu(T^2(4-T^2))^2/(16\|\Theta\|_2^2)$: a law of the Polyakov loop with $\mu(T^2(4-T^2))$ bounded away from zero gives T^2 a variance at least of order $1/(L^8\log^2\beta)$.

Theorem 90 (the neutral sector on small tori)**.** *Let $G=\mathrm{SU}(2)$, $\beta\ge2$, $n\ge2$, $L\ge2n+4$ even, and $T=\operatorname{Tr}P_0$.*

(i) *If $\mathrm{BP}_0(n,\varpi)$ holds on Λ_L, then*

$$\mu\big(T^2(4-T^2)\big)^2\le\frac{9\cdot2^{19}L^{11}n(\log\beta+6.11)^3}{\varpi\beta}.$$

(ii) *On* Λ_{2n+4}, $\mathrm{BP}_0(n,\varpi)$ *implies*

$$\mu\big(T^2(4-T^2)\big) \le \frac{4.45\cdot 10^6 n^6(\log\beta+6.11)^{3/2}}{\sqrt{\varpi\beta}}.$$

(iii) *If* $\mu(T^2(4-T^2)) \ge c > 0$ *on* Λ_{2n+4}, *then* $\mathrm{BP}_0(n,\varpi)$ *on* Λ_{2n+4} *implies*

$$\varpi \le \frac{9\cdot 2^{41} n^{12}(\log\beta+6.11)^3}{c^2\beta}.$$

Proof. (i) Let $h = T^2/4$. It is real and gauge invariant; Z^0 maps P_0 to $-P_0$ and Z^ν, $\nu\neq 0$, fixes it, so $h\in\mathscr{A}_0$. If V_x meets no link of P_0, $\mathscr{P}_x h = h$, and V_x meets a link of P_0 for Ln^3 points x. For each of these choose s as in the proof of Theorem 77(ii), so that $|s|\le n$ and the loop $A' = P_{se_2}$ avoids V_x; that proof gives $P_0 = H_0\Gamma A'\Gamma^{-1}$ with $\|H_0 - I\|_{\mathrm{op}} \le S_0$, the sum of $\|U_p - I\|_{\mathrm{op}}$ over at most Ln plaquettes. Then $|T - \mathrm{Tr}A'| = |\mathrm{Tr}((H_0-I)\Gamma A'\Gamma^{-1})| \le 2S_0$, and as $|\frac{d}{dt}\frac{t^2}{4}| \le 1$ on $[-2,2]$, $h' = (\mathrm{Tr}A')^2/4$ satisfies $|h-h'|\le 2S_0$. Since h' depends only on the links outside V_x and $\mathscr{P}_x$ is the orthogonal projection onto such functions,

$$\mu\big((h-\mathscr{P}_x h)^2\big) \le \mu\big((h-h')^2\big) \le 4\mu(S_0^2) \le 4L^2n^2\varepsilon,$$

with $\varepsilon = \mu(\|U_p - I\|_{\mathrm{HS}}^2) \le 2(\log\beta+6.11)/\beta$ by Lemma 76. So $\mathscr{E}_n(h) \le n^{-4}Ln^3\cdot 4L^2n^2\varepsilon = 4L^3n\varepsilon$, and $\mathrm{BP}_0(n,\varpi)$ gives $\mathrm{Var}_\mu(h) \le 8L^3n(\log\beta+6.11)/(\varpi\beta)$. By (16) with $\psi(t) = t^2/4$, $\mu(T^2(4-T^2))^2 \le 16\,\mathrm{Var}_\mu(h)\|\Theta\|_2^2$, and Lemma 89(iii) gives $\|\Theta\|_2^2 \le 36864\,L^8(\log\beta+6.11)^2$. Since $16\cdot 8\cdot 36864 = 9\cdot 2^{19}$, (i) follows.

(ii), (iii) For $L = 2n+4 \le 4n$, $L^{11}n \le 2^{22}n^{12}$, and $3\cdot 2^{41/2} \le 4.45\cdot 10^6$. □

Under the Haar measure,

$$\mu\big(T^2(4-T^2)\big) = 16\cdot\frac{2}{\pi}\int_0^\pi \sin^4\theta\cos^2\theta\,d\theta = 2.$$

The quantity vanishes exactly when P_0 lies almost surely in $\{\pm I\}\cup\{W : \mathrm{Tr}W = 0\}$, and Theorem 90(ii) says that BP_0 with a fixed ϖ on Λ_{2n+4} at $\beta \gg n^{12}\log^3\beta$ forces the law of the Polyakov loop towards that set. A coupling of the cube chain that contracts a pseudometric blind to the centre twists at rate κ gives $\mathrm{BP}_0(n,\kappa L^4/n^4)$, so on Λ_{2n+4} it forces $\mu(T^2(4-T^2)) \le 4.45\cdot 10^6 n^8(\log\beta+6.11)^{3/2}/(L^2\sqrt{\kappa\beta})$.

The exceptional set is that of the method. Consider fields whose component at each link of direction 0 is $L^{-1}K(P)U$, with U the link and P the Polyakov loop based at its initial site. The cancellation in Lemma 89(iii) uses that K is equivariant under conjugation, and the part of the field that is odd under the centre contributes nothing, since for such a field $\mathscr{V}h$ is odd under Z^0 and has mean zero for every neutral h. An equivariant field $W\mapsto K(W)W$ on SU(2) has $K(W)$ fixed by conjugation with the stabiliser of W, so $K(\pm I) = 0$ and, for $W = \cos\theta + \sin\theta\,u$ with $0<\theta<\pi$, $K(W) = k(W)u$ with k real. If $\mathrm{Tr}W = 0$, then $-W = vWv^{-1}$ for every $v\in\mathrm{SU}(2)$ that anticommutes with u, so evenness and equivariance give $k(W)u = K(-W) = vK(W)v^{-1} = -k(W)u$ and $K(W) = 0$. So every even equivariant K vanishes on $\{\pm I\}\cup\{\mathrm{Tr}W = 0\}$, and $\mu(\mathscr{V}h)$ tends to zero for every neutral h of class C^1 when the law of the Polyakov loop concentrates there; integration by parts along such fields then gives no lower bound on the variance of a neutral function. The law of the Polyakov loop on Λ_{2n+4} at weak coupling is not determined here.

Remark 91 (what is proved about the neutral sector and what remains). Proved: for SU(2) on every torus with $L\ge 8$ and $\beta\ge 2$, the law of the Polyakov loop satisfies the anti-concentration bound (16) with $\|\Theta\|_2 \le 192L^4(\log\beta+5.44)$ (Lemma 89); $\mathrm{BP}_0(n,\varpi)$ on Λ_{2n+4} forces $\mu(T^2(4-T^2)) \le 4.45\cdot 10^6 n^6(\log\beta+6.11)^{3/2}(\varpi\beta)^{-1/2}$, so BP_0 fails there once $\beta \gg n^{12}\log^3\beta$ whenever $\mu(T^2(4-T^2))$ is bounded below (Theorem 90); and no vector field of the kind used can treat a Polyakov loop concentrated on the centre and the traceless elements. Not proved: a lower bound on $\mu(T^2(4-T^2))$ on Λ_{2n+4} at weak coupling, and so whether BP_0 fails there; whether $\mathrm{BP}_{\mathrm{loc}}$ fails there; BP_0 or $\mathrm{BP}_{\mathrm{loc}}$ along the continuum tuning for any non-abelian group; and the mass gap.

The routes to $\mathrm{BP}_{\mathrm{loc}}(n_k,\varpi)$ along the tuning stand as follows. A coupling blind to the centre twists is constrained on Λ_{2n+4} as above and not on the large tori that Theorem 64 uses. The comparison with larger cubes gives $\mathrm{BP}_0(n,\cdot)$ from $\mathrm{BP}_0(m,\cdot)$ and the inner inequality of Proposition 73(iii) for $f\in\mathscr{A}_0$, which must hold for almost every configuration outside the cube of side m, that is, a gap of the chain of cubes of side n for the measure of that cube under every boundary configuration; nothing is proved about it at weak coupling, and $\mathrm{BP}_0(m,\cdot)$ is the same problem at a larger cube size. Theorem 90 is an implication from a property of the measure alone, the spread of the Polyakov loop, to the failure of a gap of the cube chain; a converse of Theorem 62 needs an implication in the opposite direction, from properties of the measure such as reflection positivity and the transfer matrix to a gap of the cube chain, and none is proved. A comparison with a chain on block variables in the range of Corollary 72 needs two estimates on tori of every size: single-link bounds as in Theorem 70 for the effective measure of the block variables, and a Poincaré inequality for the fine links given the block variables, uniform in them, which is an inner inequality of the kind of Proposition 73(iii); neither is proved here, and the bounds of [6, 7, 13] concern effective actions in a box of fixed physical size. What remains for the continuum step is $\mathrm{BP}_{\mathrm{loc}}(n_k,\varpi)$ itself on infinitely many tori for each k, with one $\varpi>0$; by Theorem 82(iv) it bounds the correlation of every slow function with every local one, and its failure produces a slow neutral function correlated with a local one.

X. THE OPEN INPUTS

This section states the two unproved inputs of Sec. IX, the cube estimate and the limit, and measures the first: what can be controlled uniformly in the coupling, and what any proof of the cube estimate has to produce.

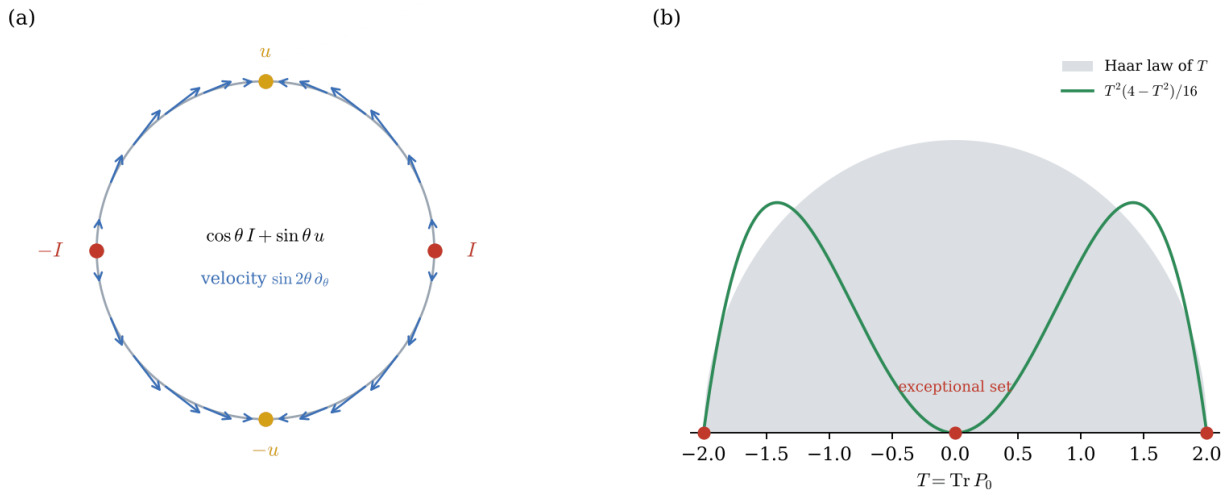


FIG. 14. The neutral sector on small tori. (a) A great circle of SU(2) through I, u, $-I$, $-u$, with the velocity $\sin 2\theta\,\partial_\theta$ by which the flow of Lemma 89 moves a Polyakov loop $P = \cos\theta\, I + \sin\theta\, u$. The flow fixes the centre $\pm I$ (red) and the traceless elements $\pm u$ (gold). (b) The Haar law of $T = \mathrm{Tr}\,P_0$, with density $\sqrt{4-T^2}/(2\pi)$, and the weight $T^2(4-T^2)$ of Theorem 90; the weight has Haar mean 2 and vanishes exactly at $T \in \{-2, 0, 2\}$, the set on which no flow of this kind gives a bound.

A. The two open statements

Theorem 43 reduces a continuum mass gap to two statements, neither of which is proved here or, to our knowledge, anywhere else.

(D) *The cube estimate.* There are $\alpha < 1$ and $\ell > 0$ such that $D(n, \alpha)$ holds with $na \le \ell$ along the continuum tuning $\beta = \beta(a) \to \infty$, uniformly in a.

(E) *The limit.* The renormalised Schwinger functions converge along $a_k = 2^{-k} a_0$ to Schwinger functions that satisfy the Osterwalder–Schrader axioms and are not those of the vacuum alone.

(D) and (E) together give the mass gap of the continuum theory; the remaining Clay requirements, Euclidean invariance and the local structure of the limit [68], belong to (E). (D) is a statement about a single cube of fixed physical size and a single coupling at each spacing; (E) is the construction problem, on which Bałaban's renormalisation group analysis [6, 7, 13] gives the ultraviolet part in finite volume. Theorem 64 draws the conclusion of (D) from an averaged condition, a spectral gap of the dynamics that resamples cubes of fixed physical size, on gauge-invariant functions of the Wilson measure on tori without gauge fixing; that condition is proved at strong coupling (Corollary 72) and is not proved along the tuning, nor is its local form $\mathrm{BP}_{\mathrm{loc}}$, which suffices (Sec. IX H).

For unsmoothed local fields at separated points, Sec. X L reduces (E), given the averaged condition along the tuning, to a moment bound on separated products whose constants grow at most like a power of the inverse separation and like $K^n (n!)^\beta$, one rotation of infinite order and uniqueness of the limit (Corollary 274), from which reflection positivity in every coordinate direction, clustering in every direction, the linear growth condition of [89] and, for constants fixed by one reflected correlation, nontriviality follow; without the rate in the separation the same inputs give these properties at separated points only (Corollary 249), and they do not imply temperedness (Proposition 276). For the two-point functions, with constants fixed by the largest reflected norm, that moment bound is equivalent to a comparison of reflected norms at short distance (Corollary 255), and its rate to a power rate in that comparison (Proposition 275); the comparison bounds the physical gap from above and fails in the strong-coupling range (Proposition 256). For $n \ge 3$ the moment bound does not follow from the two-point statements, reflection positivity and the gap, and its rate does not follow from bounds at each order (Propositions 284 and 286).

Remark 92. The following count relates (D) to a correlation length. Suppose that the gauge-fixed field responds to a change of one link on the boundary of a cube through a Poisson kernel with decay length ξ in lattice units. Then the response summed over the cube is of order ξ, so K_c is of order $n^3\xi$ and $D(n, \alpha)$ holds once $n \gtrsim \xi/\alpha$, that is, once $\ell \gtrsim \xi a/\alpha$, a fixed multiple of the physical correlation length. The hypothesis is that the gauge-fixed links decorrelate on the scale ξ, and it is a statement about the gauge, not about the gauge-invariant observables: the gauge here is built on straight lines, and in gauges of axial type the gauge-fixed variables can fluctuate on all scales while gauge-invariant observables cluster. Propositions 93 and 94 carry the count out exactly for Gaussian fields, and Proposition 94 shows that the quadratic approximation of the forest gauge fluctuates in this way. Where it does, (D) fails in this gauge and has to be posed for block-averaged variables, where it becomes part of the renormalisation group problem. [21, Sec. XII A] carries this out in the quadratic approximation.

The count of Remark 92 can be carried out exactly for a Gaussian field, the quadratic approximation of the gauge-fixed

theory at weak coupling, and shows what (D) asks of the gauge-fixed variables.

Proposition 93. *Let $d \geq 1$, $b > 0$, $t \geq 0$, let $n \geq 2$ be even and $V = [0,n)^d \cap \mathbb{Z}^d$. Given $\eta \in \mathbb{R}^{\mathbb{Z}^d \setminus V}$, let $\gamma_V(\cdot \mid \eta)$ be the Gaussian law on $\mathbb{R}^V$ with density proportional to*

$$\exp\Big(-\frac{b}{2}\Big[\sum_{\{x,y\}\cap V\neq\emptyset}(\phi_x-\phi_y)^2+t\sum_{x\in V}\phi_x^2\Big]\Big),$$

the first sum over nearest-neighbour pairs, with $\phi = \eta$ outside V. Define $C_{V,j}$, the classes c (residues modulo two) and K_c as above, with $d_V(\phi,\phi') = \sum_{x\in V}|\phi_x - \phi'_x|$ and $\delta(\eta_j,\eta'_j) = |\eta_j - \eta'_j|$, and put $N_n = (n/2)^d$. Then $K_c = \alpha_n(t)N_n$ for every class c, where

$$\alpha_n(t) = 1 - \frac{t}{n^d}\,\mathbf{1}^{\mathsf{T}}(t+L_n)^{-1}\mathbf{1}$$

and L_n is the Dirichlet Laplacian of V. In particular:

(i) *for $t = 0$, $K_c = N_n$ for every n and every b, so $D(n,\alpha)$ fails for every $\alpha < 1$;*

(ii) *for $t > 0$, $1 - \alpha_n(t) \geq (5/6)^d\, tn^2/(tn^2+10d)$; for $d = 4$, $t = m^2a^2$ and $\ell/2 \leq na \leq \ell$ this gives $D(n,\alpha)$ with $\alpha = 1-(5/6)^4(m\ell)^2/((m\ell)^2+160)$, uniformly in a.*

Proof. Let $A = t + L_n$ and $G = A^{-1}$, and let B be the matrix with $B_{xj} = 1$ when $x \in V$ and $j \notin V$ are neighbours. The law $\gamma_V(\cdot \mid \eta)$ has covariance $b^{-1}G$ and mean $GB\eta$. The matrix A has nonpositive off-diagonal entries and is positive definite, so $G \geq 0$ entrywise. Changing η_j by Δ translates the law by ΔGB_j, where B_j is the column of j. The coupling by translation gives $W_1^{d_V} \leq |\Delta|\sum_x (GB_j)_x$, and the 1-Lipschitz function $\sum_x \phi_x$ gives the reverse inequality, so $C_{V,j} = \mathbf{1}^{\mathsf{T}}GB_j$. Since $B\mathbf{1} = L_n\mathbf{1}$ (both count the exterior neighbours of each site),

$$\sum_{j\notin V} C_{V,j} = \mathbf{1}^{\mathsf{T}}GL_n\mathbf{1} = \mathbf{1}^{\mathsf{T}}\mathbf{1} - t\,\mathbf{1}^{\mathsf{T}}G\mathbf{1} = n^d\alpha_n(t).$$

The reflection $x_i \mapsto n-1-x_i$ preserves V and the coefficients and, as n is even, exchanges the two residues of the coordinate x_i; so the 2^d class sums are equal, and each is $n^d\alpha_n(t)/2^d = \alpha_n(t)N_n$. This proves the formula and (i).

For (ii) let $v(i) = (i+1)(n-i)$ for $0 \leq i < n$ and $\psi(x) = \prod_i v(x_i)$. The one-dimensional Dirichlet Laplacian maps v to the constant 2, so with $s_1 = \sum_i v(i) = n(n+1)(n+2)/6$ and $s_2 = \sum_i v(i)^2 = n(n+1)(n+2)(n^2+2n+2)/30$,

$$\mathbf{1}^{\mathsf{T}}\psi = s_1^d, \qquad \psi^{\mathsf{T}}A\psi = ts_2^d + 2ds_1s_2^{d-1}.$$

By the Cauchy–Schwarz inequality for the form of A^{-1}, $\mathbf{1}^{\mathsf{T}}G\mathbf{1} \geq (\mathbf{1}^{\mathsf{T}}\psi)^2/\psi^{\mathsf{T}}A\psi$, hence

$$1-\alpha_n(t) \geq \Big(\frac{s_1^2}{ns_2}\Big)^d \frac{t}{t+2ds_1/s_2} \geq \Big(\frac{5}{6}\Big)^d \frac{tn^2}{tn^2+10d},$$

since $s_1^2/(ns_2) = \frac{5}{6}(n^2+3n+2)/(n^2+2n+2) \geq \frac{5}{6}$ and $2ds_1/s_2 = 10d/(n^2+2n+2) \leq 10d/n^2$. With $d = 4$ and $tn^2 = m^2(na)^2 \geq (m\ell)^2/4$ the last fraction is at least $(m\ell)^2/((m\ell)^2+160)$. □

The Gaussian field is not Yang–Mills theory, and the proposition is a benchmark for the condition, not a statement about the Wilson measure. It shows that the averaged cube condition is met at a fixed physical scale by a field with a mass and fails exactly, at every size and every stiffness, for a massless one. Small local fluctuations alone therefore give nothing: in the massless Gaussian field on $\mathbb{Z}^d$, $d \geq 3$, the variance of $\phi_x - \phi_{x+e_1}$ is $1/(db)$, which tends to zero as $b \to \infty$, and still $K_c = N_n$. A proof of (D) along the tuning has to produce in the gauge-fixed variables the loss that the parameter t supplies here, from the Wilson action alone.

The quadratic approximation of the forest gauge itself, with the plaquette action in place of the Laplacian, is no better than the massless field of Proposition 93(i).

Proposition 94. *Let $b > 0$, let n be even and V the cube of Definition 40. For real functions φ on the links of $\mathbb{Z}^4$ that vanish on F, let $(d\varphi)_p$ be the sum of φ around the plaquette p with the signs of its orientation. Given $\eta \in \mathbb{R}^{A\setminus V}$, let $\gamma_V(\cdot \mid \eta)$ be the law on $\mathbb{R}^V$ with density proportional to*

$$\exp\Big(-\frac{b}{2}\sum_{p\cap V\neq\emptyset}(d\varphi)_p^2\Big),$$

with $\varphi = \eta$ on $A \setminus V$ and $\varphi = 0$ on F, and define $C_{V,j}$ and K_c as above with $d_V(\varphi,\varphi') = \sum_{i\in V}|\varphi_i - \varphi'_i|$ and $\delta(\eta_j,\eta'_j) = |\eta_j - \eta'_j|$. Then

$$\sum_c K_c \geq |V| = 48N_n,$$

so $K_c \geq N_n$ for some class c, and $D(n,\alpha)$ fails for every $\alpha < 1$, at every n and every b.

Proof. The laws. The quadratic form $\sum_{p\cap V\neq\emptyset}(d\varphi)_p^2$ with $\varphi = 0$ outside V vanishes only at $\varphi = 0$: the argument of Corollary 237, written additively, gives a function h on the sites with $\varphi_{(x,\mu)} = h(x) - h(x+e_\mu)$, constant outside the box and along the lines of F, hence constant. So $\gamma_V(\cdot \mid \eta)$ is a Gaussian law whose covariance does not depend on η and whose mean is $\sum_j \eta_j u_j$ for some vectors $u_j \in \mathbb{R}^V$. Two exteriors that differ in finitely many links give translates of one law by $t = \sum_j(\eta'_j - \eta_j)u_j$; the coupling by translation and the 1-Lipschitz function $\sum_i \operatorname{sign}(t_i)\varphi_i$ show that their distance is $\|t\|_1$. In particular $C_{V,j} = \|u_j\|_1$, and

$$W_1^{d_V}\big(\gamma_V(\cdot \mid \eta), \gamma_V(\cdot \mid \eta')\big) \leq \sum_j C_{V,j}\,|\eta'_j - \eta_j|.$$

Lines. By Lemma 9(i) the lines of F are straight, disjoint and cover the sites, and the two ends of an active link (x,μ) lie on different lines, since the line through x has direction $\sigma(x) \neq \mu$. For a line ℓ let h_ℓ be the indicator of its sites and $(dh_\ell)_{(x,\mu)} = h_\ell(x) - h_\ell(x+e_\mu)$. Then dh_ℓ vanishes on F and around every plaquette, so the density is invariant under $\varphi \mapsto \varphi + dh_\ell$, and $\gamma_V(\cdot \mid dh_\ell)$ is $\gamma_V(\cdot \mid 0)$ translated by the restriction of dh_ℓ to V. Hence

$$\sum_{i\in V}|(dh_\ell)_i| \leq \sum_{j\in A\setminus V} C_{V,j}\,|(dh_\ell)_j|.$$

Here $|(dh_\ell)_e| = 1$ when exactly one end of e lies on ℓ and 0 otherwise, so $\sum_\ell |(dh_\ell)_e| = 2$ for every active link e, the sum running over the finitely many lines that meet a link entering the inequality. Summing over these lines gives $2|V| \le 2\sum_j C_{V,j} = 2\sum_c K_c$. □

The proof uses only that gauge transformations constant on the lines of F move the whole cube while changing few boundary links, which is the behaviour of gauges of axial type anticipated in Remark 92. The class sums themselves, computed exactly for $n \le 8$, lie between 1.57 and 11.24 times N_n and increase with n. Below the correlation length the theory is weakly coupled, and the quadratic approximation is its leading order there. A proof of (D) along the tuning therefore has to show that the nonperturbative regime reduces the cube coefficients of the forest gauge below N_n, or (D) has to be posed for other variables.

The lines of F are not what makes the argument work. A constant field, corrected by a periodic gauge transformation so that it vanishes on the fixed links, costs no action and changes every exterior link by a bounded amount, and this happens in every gauge fixed on an invariant forest.

Proposition 95. *Let $b > 0$ and let F be a set of links of $\mathbb{Z}^4$, invariant under the translations by $2\mathbb{Z}^4$, that contains no closed path and whose trees are infinite and contain every site. Let A be the set of the other links, V the set of links $(x,\mu) \in A$ with $x \in [0,n)^4$, $n \ge 2$ even, and define the classes, $\gamma_V(\cdot \mid \eta)$, $C_{V,j}$ and K_c as in Proposition 94 with this F. Then there are numbers $a_c \ge 0$, not all zero, with*

$$\sum_c a_c K_c \ge N_n \sum_c a_c,$$

so $K_c \ge N_n$ for some class c, and $D(n,\alpha)$ fails for every $\alpha < 1$, at every n and every b.

Proof. The laws. Let φ vanish outside V and on F, with $(d\varphi)_p = 0$ for every plaquette p that meets V. The other plaquettes contain no link of V, so $d\varphi = 0$ on every plaquette, and $\varphi = dh$ for a function h on the sites. Since φ vanishes outside a box and the complement of a box is connected, h is constant there, and we take that constant to be 0. As φ vanishes on F, h is constant on every tree of F; each tree is infinite and meets the complement of the box, so $h = 0$ and $\varphi = 0$. Hence, as in the proof of Proposition 94, $\gamma_V(\cdot \mid \eta)$ is a nondegenerate Gaussian law with mean $\sum_j \eta_j u_j$, where only the links j that share a plaquette with V contribute, and $C_{V,j} = \|u_j\|_1$.

A flat field. For $c \in \mathbb{R}^4$ and a function h on $\mathbb{Z}^4$ of period two in each coordinate put $\psi_{(x,\mu)} = c_\mu + h(x) - h(x+e_\mu)$. Then $d\psi = 0$ on every plaquette and ψ is invariant under $2\mathbb{Z}^4$; if $\psi = 0$, adding ψ at (x,μ) and $(x+e_\mu,\mu)$ gives $2c_\mu = 0$, and then h is constant. These fields form a space of dimension $16 - 1 + 4 = 19$. Let f be the number of orbits of $2\mathbb{Z}^4$ in F. The box $[0,2m)^4$ has $16m^4$ sites, and the links of F with both ends in it form a forest, so there are at most $16m^4 - 1$ of them, while each orbit contributes at least $(m-1)^4$; letting $m \to \infty$ gives $f \le 16$. Vanishing on F is therefore at most 16 linear conditions, and some $\psi \ne 0$ of this form vanishes on F. It is nonzero on some link of A. Put $a_c = |\psi_j|$ for $j \in c$.

The inequality. Let η be the restriction of ψ to $A \setminus V$. The field equal to ψ on V and to η outside gives the quadratic form its least value 0, so by nondegeneracy the mean of $\gamma_V(\cdot \mid \eta)$ is the restriction of ψ to V, and $\psi_i = \sum_j \psi_j (u_j)_i$ for $i \in V$. Hence

$$\sum_{i \in V} |\psi_i| \le \sum_{j \in A \setminus V} C_{V,j} |\psi_j| = \sum_c a_c K_c.$$

A class consists of the active links with a given direction and a given residue of the base point modulo two, so it meets V in N_n links, and the left side is $N_n \sum_c a_c > 0$. So $K_c \ge N_n$ for some class with $a_c > 0$. □

For the direction field of Table III the flat fields vanishing on F form a space of dimension 7, and for the axial field $\sigma = 0$ one of dimension 10. The proposition is weaker than Proposition 94 for the forest gauge, which bounds the sum over all classes, and it says nothing about two kinds of condition. A mass gives the constant fields an action, which is why Proposition 93 distinguishes $t > 0$ from $t = 0$. The constrained condition D_c of [21, Theorem 334] holds the block averages fixed, and a constant field changes them. Within the quadratic approximation no gauge fixed on an invariant forest gives (D); the loss has to come from the Wilson action beyond its quadratic part, or (D) has to be posed for variables, such as the block averages, that a constant field moves.

Remark 96. Five other routes stop at (D) or (E), or earlier.

(a) *Disjoint blocks.* The block criterion of Lemma 349 with a partition into cubes of side n requires, for a link on a face, that its influence on the neighbouring cube be less than one. In the count of Remark 92 that influence is of order ξ, which grows like $1/a$. Averaging over the N_n translates divides it by the number of translates on whose boundary the link lies, a factor of order $1/n$, and this is why (D) is stated for averaged cubes.

(b) *Rescaling a fixed lattice gap.* Any bound uniform in lattice units, such as those of Theorems 1–7, gives trivial continuum two-point functions [20]; rescaling it produces no gap.

(c) *Assuming the gap.* Hypothesis R of [20] assumes the gap in physical units together with the limit. Theorem 43 replaces the gap part by the cube estimate (D), which concerns one box and one coupling at each spacing.

(d) *Computation.* At a fixed coupling and a fixed small n, $D(n,\alpha)$ is a finite set of Kantorovich bounds and could in principle be evaluated as in Remark 36; for $n = 2$ and SU(2) the block already has 48 links and 144 real variables. (D) asks for it at infinitely many couplings with $n \to \infty$, which no finite computation covers.

B. The cube condition for free massive fields

This subsection proves that the averaged cube condition of Definition 60 holds for the free massive scalar and Proca fields on every torus, with a constant that depends only on

the dimension and on the mass in units of the cube, hence uniformly in the lattice spacing at fixed physical size of the cube. The proof has three steps: a reduction of the inequality to linear functions of the field, which is exact for Gaussian measures; the identification of the Dirichlet form on linear functions with the additive Schwarz operator of the cubes; and an explicit stable decomposition of a field into fields supported in the cubes, the parabolic profile of Proposition 93. Upper bounds show that the constant vanishes with the mass, so that the mass term is what makes the condition hold.

Setting. In this subsection $\mu > 0$ is a lattice mass and $d \geq 1$, $n \geq 2$ and $L \geq n+1$ are integers; $\mathbb{T} = (\mathbb{Z}/L\mathbb{Z})^d$ is the torus, $e_1, \dots, e_d$ its unit vectors, $E = \mathbb{T} \times \{1, \dots, d\}$ its set of links, the link (y,i) running from y to $y+e_i$, and $P = \mathbb{T} \times \{(i,j) : 1 \leq i < j \leq d\}$ its set of plaquettes. For $f \in \mathbb{R}^{\mathbb{T}}$ let $\nabla_i f(y) = f(y+e_i) - f(y)$, let $d_0 f = (\nabla_i f(y))_{(y,i)} \in \mathbb{R}^E$, and for $A \in \mathbb{R}^E$ let $A_i(y) = A(y,i)$ and

$$(d_1 A)(y,i,j) = A(y,i) + A(y+e_i,j) - A(y+e_j,i) - A(y,j),$$

so that $d_1 d_0 = 0$; $\delta_0 = d_0^{\mathsf{T}}$ and $\delta_1 = d_1^{\mathsf{T}}$ are the adjoints for the standard inner products, and $-\Delta = \delta_0 d_0$ is the lattice Laplacian, $\langle f, -\Delta f \rangle = \sum_i \|\nabla_i f\|^2 =: \|\nabla f\|^2$. The scalar field has precision $Q_{\mathrm{s}} = -\Delta + \mu^2$ on $\mathbb{R}^{\mathbb{T}}$ and the Proca field has precision $Q_{\mathrm{P}} = \delta_1 d_1 + \mu^2$ on $\mathbb{R}^E$; both are positive definite. For $x \in \mathbb{T}$ let $V_x = x + \{0, \dots, n-1\}^d$, a set of n^d distinct sites since $L \geq n$, and let E_x be the set of links with base point in V_x. Every site lies in exactly n^d sets V_x and every link in exactly n^d sets E_x. The parameter that matters is $s = n\mu$: with $\mu = ma$ and $n = \ell/a$ it is $m\ell$, the mass times the physical size of the cube. The tori of Definition 60, $d = 4$ and L even with $L \geq 2n+4$, are included.

For a positive definite Q on $\mathbb{R}^N$ let γ_Q be the centred Gaussian law with covariance $C = Q^{-1}$, and for $V \subset \{1, \dots, N\}$ let $\mathscr{P}_V$ be the conditional expectation on $L^2(\gamma_Q)$ given the coordinates outside V; it is an orthogonal projection. For a family $(V_x)_{x \in X}$ of coordinate sets write $\mathscr{P}_x = \mathscr{P}_{V_x}$ and

$$\varpi(Q; (V_x)) = \inf_f \frac{n^{-d} \sum_{x \in X} \|f - \mathscr{P}_x f\|^2}{\mathrm{Var}_{\gamma_Q}(f)}, \tag{17}$$

the infimum over nonconstant $f \in L^2(\gamma_Q)$. Thus $\mathrm{BP}(n, \varpi)$ of Definition 60, with $L^2(\gamma_Q)$ in place of $\mathscr{A}$, holds exactly when $\varpi \leq \varpi(Q; (V_x))$. For the scalar field the family is $(V_x)_{x \in \mathbb{T}}$, for the Proca field it is $(E_x)_{x \in \mathbb{T}}$.

1. Reduction to linear functions

Let $H \subset L^2(\gamma_Q)$ be the space of linear functions $\ell_h(\phi) = \langle h, \phi \rangle$, $h \in \mathbb{R}^N$; $\mathbb{E}[\ell_h \ell_{h'}] = \langle h, Ch' \rangle$. For $m \geq 0$ let $\mathscr{Q}_m \subset L^2(\gamma_Q)$ be the space of polynomials of degree at most m in ϕ, and let $W_m = \mathscr{Q}_m \cap \mathscr{Q}_{m-1}^{\perp}$ ($\mathscr{Q}_{-1} = \{0\}$), the m-th Wiener chaos; W_0 is the constants and $W_1 = H$. Polynomials are dense in $L^2(\gamma_Q)$ [69, Chapter 2], so $L^2(\gamma_Q) = \bigoplus_{m \geq 0} W_m$, an orthogonal sum, and $\mathrm{Var}(f) = \sum_{m \geq 1} \|f_m\|^2$ for $f = \sum_m f_m$, $f_m \in W_m$. A subspace $G \subset \mathbb{R}^N$ is called compatible with Q if $QG = G$; then also $QG^{\perp} = G^{\perp}$, since Q is symmetric and invertible, so Q commutes with the orthogonal projection $P_{G^{\perp}}$ onto $G^{\perp}$. A function $f \in L^2(\gamma_Q)$ is G-invariant if it is a function of $P_{G^{\perp}}\phi$, that is, measurable with respect to $\sigma(P_{G^{\perp}}\phi)$; for a function defined at every ϕ this is the same as $f(\phi + w) = f(\phi)$ for all ϕ and all $w \in G$ (take $w = -P_G\phi$), which is gauge invariance when G is the space of gauge directions. The G-invariant functions form a closed subspace L^2_G, and ℓ_h is G-invariant exactly when $h \in G^{\perp}$.

Theorem 97 (the gap is attained on linear functions)**.** *Let Q be positive definite on $\mathbb{R}^N$, let $(V_x)_{x \in X}$ be a finite family of coordinate sets, and let G be a subspace compatible with Q (for instance $G = \{0\}$). For $\lambda \geq 0$ the following are equivalent:*

(i) $\sum_{x \in X} \|f - \mathscr{P}_x f\|^2 \geq \lambda \,\mathrm{Var}(f)$ *for every $f \in L^2_G$;*

(ii) $\sum_{x \in X} \|f - \mathscr{P}_x f\|^2 \geq \lambda \,\mathrm{Var}(f)$ *for every $f \in L^2_G \cap H$, that is, for $f = \ell_h$ with $h \in G^{\perp}$.*

In particular $\varpi(Q; (V_x))$ is the infimum in (17) over linear f, and the same holds for the infimum over L^2_G.

The proof uses the second quantization of the conditional expectations, which we set up in finite dimension. Let ξ be a standard Gaussian vector on $\mathbb{R}^N$ with law γ, and write $\ell_u(\xi) = \langle u, \xi \rangle$. The spaces $\mathscr{Q}_m$ and W_m are defined for γ as above; they are invariant under orthogonal changes of variables, since $\mathscr{Q}_m$ is. Let h_j be the Hermite polynomials, $h_0 = 1$, $h_1(t) = t$, $h_{j+1}(t) = t h_j(t) - j h_{j-1}(t)$; they are orthogonal in $L^2(\mathbb{R}, e^{-t^2/2} dt/\sqrt{2\pi})$ with $\|h_j\|^2 = j!$, and h_j has degree j and leading coefficient 1. For an orthonormal basis (e_i) of $\mathbb{R}^N$, the variables $\xi'_i = \langle e_i, \xi \rangle$ are independent standard Gaussians, and for a multi-index $\alpha \in \mathbb{N}^N$ we put $H_\alpha(\xi') = \prod_i h_{\alpha_i}(\xi'_i)$, $\alpha! = \prod_i \alpha_i!$ and $|\alpha| = \sum_i \alpha_i$. On $(\mathbb{R}^N)^{\otimes m}$ we use the inner product in which the tensors $e_{i_1} \otimes \cdots \otimes e_{i_m}$ are orthonormal, the symmetrization $\mathrm{Sym} = (m!)^{-1} \sum_\sigma \sigma$ over the permutations of the factors, and $\mathrm{Sym}^m(\mathbb{R}^N)$, its range.

Lemma 98 (the chaos as symmetric tensors)**.** (i) *For every orthonormal basis (e_i) and every $m \geq 0$ the polynomials $H_\alpha(\xi')$ with $|\alpha| = m$ form an orthogonal basis of W_m.*

(ii) *There is a linear map $J_m : (\mathbb{R}^N)^{\otimes m} \to W_m$ with $J_m(u_1 \otimes \cdots \otimes u_m) = \Pi_{W_m}(\ell_{u_1} \cdots \ell_{u_m})$, where Π_{W_m} is the orthogonal projection onto W_m. It satisfies $J_m = J_m \circ \mathrm{Sym}$, and $(m!)^{-1/2} J_m$ restricted to $\mathrm{Sym}^m(\mathbb{R}^N)$ is an isometry onto W_m.*

(iii) *Let $K \subset \mathbb{R}^N$ be a subspace, π the orthogonal projection onto K, $\mathscr{F}_K$ the σ-algebra generated by ℓ_u, $u \in K$, and $\mathbb{E}_K = \mathbb{E}[\,\cdot \mid \mathscr{F}_K]$. Then $\mathbb{E}_K J_m(t) = J_m(\pi^{\otimes m} t)$ for every $t \in (\mathbb{R}^N)^{\otimes m}$. In particular $\mathbb{E}_K$ maps W_m into itself and $\mathbb{E}_K = J_m \pi^{\otimes m} J_m^{-1}$ on W_m, where J_m^{-1} denotes the inverse of J_m restricted to $\mathrm{Sym}^m(\mathbb{R}^N)$.*

Proof. (i) $H_\alpha(\xi')$ is a polynomial of degree $|\alpha|$ in ξ', hence in ξ, and $\mathscr{Q}_m$ is also the span of the monomials ξ'^{β}, $|\beta| \leq m$. For $|\beta| < |\alpha|$ some $\beta_i < \alpha_i$, and by independence $\mathbb{E}[H_\alpha(\xi') \xi'^{\beta}] = \prod_i \mathbb{E}[h_{\alpha_i}(\xi'_i) \xi'^{\beta_i}_i] = 0$, since h_{α_i} is orthogonal to the polynomials of lower degree. So $H_\alpha(\xi') \in \mathscr{Q}_{|\alpha|} \cap \mathscr{Q}_{|\alpha|-1}^{\perp} = W_{|\alpha|}$,

and the H_α are mutually orthogonal, with $\|H_\alpha\|^2 = \alpha!$. Since $\xi'^\alpha - H_\alpha(\xi')$ has degree less than $|\alpha|$, induction on the degree gives $\mathscr{Q}_m = \operatorname{span}\{H_\beta : |\beta| \le m\}$; so $\{H_\beta : |\beta| \le m\}$ is an orthogonal basis of $\mathscr{Q}_m$, and $\{H_\alpha : |\alpha| = m\}$ one of $\mathscr{Q}_m \ominus \mathscr{Q}_{m-1} = W_m$.

(ii) The map $(u_1,\dots,u_m) \mapsto \Pi_{W_m}(\ell_{u_1}\cdots\ell_{u_m})$ is multilinear and symmetric, so it defines a linear J_m on $(\mathbb{R}^N)^{\otimes m}$ with $J_m = J_m \circ \mathrm{Sym}$. Fix an orthonormal basis (e_i). For $i_1,\dots,i_m$ with multiplicity vector α, $\ell_{e_{i_1}}\cdots\ell_{e_{i_m}} = \xi'^\alpha$ and $\xi'^\alpha - H_\alpha(\xi') \in \mathscr{Q}_{m-1}$, so $J_m(e_{i_1}\otimes\cdots\otimes e_{i_m}) = H_\alpha(\xi')$ by (i). The tensors $S_\alpha = (m!/\alpha!)^{1/2}\mathrm{Sym}(e_1^{\otimes\alpha_1}\otimes\cdots\otimes e_N^{\otimes\alpha_N})$, $|\alpha| = m$, form an orthonormal basis of $\mathrm{Sym}^m(\mathbb{R}^N)$: $\mathrm{Sym}(e^{\otimes\alpha})$ is $\alpha!/m!$ times the sum of the $m!/\alpha!$ distinct basis tensors with multiplicity vector α, so these symmetrizations are orthogonal for distinct α, have squared norm $\alpha!/m!$, and span $\mathrm{Sym}^m(\mathbb{R}^N)$. Then $J_m(S_\alpha) = (m!/\alpha!)^{1/2}H_\alpha(\xi')$, and these are orthogonal with squared norm $m!$ and span W_m by (i).

(iii) Both sides are linear in t, so it suffices to take $t = e'_{i_1}\otimes\cdots\otimes e'_{i_m}$ for an orthonormal basis (e'_i) with $e'_1,\dots,e'_k$ spanning K; such a basis exists, and (ii) holds for it because J_m was defined without reference to a basis. The σ-algebra $\mathscr{F}_K$ is generated by $\xi'_1,\dots,\xi'_k$, and the ξ'_i are independent, so for the multiplicity vector α of $(i_1,\dots,i_m)$

$$\mathbb{E}_K H_\alpha(\xi') = \prod_{i\le k} h_{\alpha_i}(\xi'_i)\prod_{i>k}\mathbb{E}[h_{\alpha_i}(\xi'_i)],$$

which is $H_\alpha(\xi')$ if $\alpha_i = 0$ for all $i > k$ and 0 otherwise, since $\mathbb{E}h_j(\xi'_i) = 0$ for $j \ge 1$. On the other side $\pi e'_i = e'_i$ for $i \le k$ and $\pi e'_i = 0$ for $i > k$, so $J_m(\pi e'_{i_1}\otimes\cdots\otimes\pi e'_{i_m})$ is $H_\alpha(\xi')$ if all $i_l \le k$ and 0 otherwise. The two agree. The last statement follows from (ii). □

Proof of Theorem 97. (i) implies (ii) since $L^2_G \cap H \subset L^2_G$. Assume (ii). Let $\xi = C^{-1/2}\phi$, a standard Gaussian vector, so that $\ell_h(\phi) = \langle C^{1/2}h, \xi\rangle$, and $f \mapsto f(C^{1/2}\cdot)$ is an isometry of $L^2(\gamma_Q)$ onto $L^2(\gamma)$ that maps $\mathscr{Q}_m$, W_m and H onto the corresponding spaces for ξ. For $x \in X$ let $W = V_x^c$. The σ-algebra generated by ϕ_y, $y \in W$, is generated by $\langle C^{1/2}\delta_y, \xi\rangle$, $y \in W$, that is, it is $\mathscr{F}_{K_x}$ with $K_x = C^{1/2}\mathbb{R}^W$, where $\mathbb{R}^W$ is the set of vectors supported in W. So $\mathscr{P}_x = \mathbb{E}_{K_x}$, and by Lemma 98(iii) it preserves each W_m and acts on it as $J_m\pi_x^{\otimes m}J_m^{-1}$, with π_x the orthogonal projection onto K_x.

Let $B = \sum_x(I-\pi_x)$ on $\mathbb{R}^N$. For $u \in \mathbb{R}^N$, $\ell_u(\xi) = J_1(u)$ and $\|\ell_u - \mathscr{P}_x\ell_u\|^2 = \|(I-\pi_x)u\|^2$, so (ii) for $f = \ell_h$, $h \in G^\perp$, reads $\langle u, Bu\rangle \ge \lambda|u|^2$ for $u \in T := C^{1/2}G^\perp$. As Q commutes with $P_{G^\perp}$, so does $C^{1/2}$ (a function of Q), and $T = G^\perp$.

Fix $m \ge 1$ and write $\pi_{(j)}$ for the operator on $(\mathbb{R}^N)^{\otimes m}$ that acts as π on the j-th factor and as the identity on the others. The $\pi_{(j)}$, $j = 1,\dots,m$, are commuting orthogonal projections and $\pi^{\otimes m} = \prod_j \pi_{(j)}$ is the orthogonal projection onto the intersection of their ranges, so $\pi^{\otimes m} \le \pi_{(j)}$ for each j, and averaging over j,

$$I - \pi^{\otimes m} \ge \frac{1}{m}\sum_{j=1}^m (I - \pi_{(j)}). \tag{18}$$

Applying this to each π_x and summing over x gives $\sum_x(I - \pi_x^{\otimes m}) \ge \frac{1}{m}\sum_j B_{(j)}$. For $t \in T^{\otimes m}$ and each j, $\langle t, B_{(j)}t\rangle \ge \lambda|t|^2$: writing $t = \sum_a t_a \otimes f_a$ with (f_a) an orthonormal basis of T in the j-th slot and t_a tensors in the other slots, $\langle t, B_{(j)}t\rangle = \sum_{a,b}\langle t_a, t_b\rangle\langle f_a, Bf_b\rangle = \mathrm{tr}(\Gamma B_T)$, where $\Gamma_{ab} = \langle t_a, t_b\rangle$ and $(B_T)_{ab} = \langle f_a, Bf_b\rangle$ are positive semidefinite and $B_T \ge \lambda$ by the hypothesis on T; so $\mathrm{tr}(\Gamma B_T) \ge \lambda\,\mathrm{tr}\,\Gamma = \lambda|t|^2$. Hence

$$\sum_x \langle t, (I - \pi_x^{\otimes m})t\rangle \ge \lambda|t|^2 \qquad (t \in T^{\otimes m}). \tag{19}$$

Now let $f \in L^2_G$ with chaos components $f_m \in W_m$. Choose an orthonormal basis (e'_i) of $\mathbb{R}^N$ with $e'_1,\dots,e'_k$ spanning T. Since f is a function of $P_{G^\perp}\phi$ and $P_{G^\perp}C^{1/2}\xi = C^{1/2}P_T\xi$, the function $f(C^{1/2}\xi)$ is a function of $P_T\xi$, that is, of $\xi'_1,\dots,\xi'_k$ alone. Its Hermite coefficients $\mathbb{E}[f(C^{1/2}\xi)H_\alpha(\xi')]$ vanish unless $\alpha_i = 0$ for all $i > k$, by independence and $\mathbb{E}h_j(\xi'_i) = 0$ for $j \ge 1$; since the normalized $H_\alpha(\xi')$ form an orthonormal basis of $L^2(\gamma)$ by Lemma 98(i) and the density of polynomials, $f(C^{1/2}\xi)$ is the sum of its expansion in them. So each f_m lies in the span of $H_\alpha(\xi')$ with $|\alpha| = m$ and α supported in $\{1,\dots,k\}$, which by Lemma 98(ii) is $J_m(\mathrm{Sym}^m(T))$. Write $f_m = (m!)^{-1/2}J_m(t_m)$ with $t_m \in \mathrm{Sym}^m(T) \subset T^{\otimes m}$. Since $\mathscr{P}_x$ preserves the orthogonal decomposition into chaoses, $\|f - \mathscr{P}_x f\|^2 = \sum_m \|f_m - \mathscr{P}_x f_m\|^2$, and $\|f_m - \mathscr{P}_x f_m\|^2 = |t_m - \pi_x^{\otimes m}t_m|^2 = \langle t_m, (I - \pi_x^{\otimes m})t_m\rangle$ by Lemma 98, using that $\pi_x^{\otimes m}$ preserves $\mathrm{Sym}^m(\mathbb{R}^N)$ and that $I - \pi_x^{\otimes m}$ is an orthogonal projection. By (19),

$$\begin{aligned}\sum_x \|f - \mathscr{P}_x f\|^2 &= \sum_{m\ge1}\sum_x \langle t_m, (I - \pi_x^{\otimes m})t_m\rangle \\ &\ge \lambda\sum_{m\ge1}|t_m|^2 = \lambda\sum_{m\ge1}\|f_m\|^2 = \lambda\,\mathrm{Var}(f),\end{aligned}$$

which is (i). The statements about ϖ follow with $\lambda = n^d\varpi$. □

Remark 99. The theorem is the analogue, for the sum of the projections $\mathscr{P}_x$, of a result of Amit [2] on their product, the Gibbs sampler with a Gaussian target, whose rate is likewise determined by the first chaos. Nothing beyond the finite torus is needed here: the field has finitely many coordinates and there are finitely many cubes. The conditional expectation $\mathscr{P}_x$ need not map L^2_G into itself; the proof only uses that it preserves the chaos decomposition.

2. *Linear functions and the additive Schwarz operator*

For $V \subset \{1,\dots,N\}$ let $R_V : \mathbb{R}^N \to \mathbb{R}^V$ be the restriction and $Q_{VV} = R_V Q R_V^{\mathsf{T}}$, a positive definite principal submatrix. In the Q-inner product $\langle v, w\rangle_Q = \langle v, Qw\rangle$ the Q-orthogonal projection onto $R_V^{\mathsf{T}}\mathbb{R}^V$, the fields supported in V, is $P_V^Q = R_V^{\mathsf{T}}Q_{VV}^{-1}R_V Q$. Put

$$S = \sum_{x\in X} R_{V_x}^{\mathsf{T}}Q_{V_xV_x}^{-1}R_{V_x}, \qquad P_{\mathrm{ad}} = \sum_{x\in X} P_{V_x}^Q = SQ, \tag{20}$$

the additive Schwarz operator of the family [76, 100].

Lemma 100 (linear functions)**.** *For $h \in \mathbb{R}^N$ and $V \subset \{1,\dots,N\}$, $\|\ell_h - \mathscr{P}_V\ell_h\|^2 = \langle h_V, Q_{VV}^{-1}h_V\rangle$ with $h_V = R_V h$. Hence $\sum_x\|\ell_h - \mathscr{P}_x\ell_h\|^2 = \langle h, Sh\rangle$, $\mathrm{Var}(\ell_h) = \langle h, Q^{-1}h\rangle$, and*

$n^d\varpi(Q;(V_x))$ is the smallest eigenvalue of $Q^{1/2}SQ^{1/2}$, which is unitarily equivalent to P_{ad} on $(\mathbb{R}^N,\langle\cdot,\cdot\rangle_Q)$. If G is compatible with Q, the infimum in (17) over L^2_G is n^{-d} times the smallest eigenvalue of the compression of $Q^{1/2}SQ^{1/2}$ to $G^\perp$.

Proof. In the notation of the proof of Theorem 97, with $W=V^c$, $\|\ell_h-\mathscr{P}_V\ell_h\|^2=|(I-\pi)u|^2$ with $u=C^{1/2}h$ and π the projection onto $K=C^{1/2}\mathbb{R}^W$. A vector v is orthogonal to K exactly when $C^{1/2}v\perp\mathbb{R}^W$, that is $C^{1/2}v\in R_V^{\mathsf T}\mathbb{R}^V$, so $K^\perp=Q^{1/2}R_V^{\mathsf T}\mathbb{R}^V$, and the orthogonal projection onto $K^\perp$ is $I-\pi=Q^{1/2}R_V^{\mathsf T}Q_{VV}^{-1}R_VQ^{1/2}$ (it is symmetric, idempotent, has range in $K^\perp$ and fixes $Q^{1/2}R_V^{\mathsf T}w$). So $|(I-\pi)u|^2=\langle u,(I-\pi)u\rangle=\langle Q^{1/2}u,R_V^{\mathsf T}Q_{VV}^{-1}R_VQ^{1/2}u\rangle=\langle h_V,Q_{VV}^{-1}h_V\rangle$, as $Q^{1/2}u=h$. Summing over x gives $\langle h,Sh\rangle$; the variance is $\langle h,Ch\rangle$. By Theorem 97 with $G=\{0\}$, $n^d\varpi$ is the infimum of $\langle h,Sh\rangle/\langle h,Q^{-1}h\rangle$, which is the smallest eigenvalue of $Q^{1/2}SQ^{1/2}$ (put $h=Q^{1/2}u$); and $Q^{1/2}$ is a unitary map from $(\mathbb{R}^N,\langle\cdot,\cdot\rangle_Q)$ to $(\mathbb{R}^N,\langle\cdot,\cdot\rangle)$ with $Q^{1/2}P_{\mathrm{ad}}Q^{-1/2}=Q^{1/2}SQ^{1/2}$. For G compatible with Q, $Q^{1/2}G^\perp=G^\perp$, and the infimum over $h\in G^\perp$ is the smallest eigenvalue of the compression to $G^\perp$. □

Lemma 101 (stable decompositions, P.-L. Lions)**.** *Let $T\subset\mathbb{R}^N$ be a subspace with $QT=T$ (for instance $T=\mathbb{R}^N$) and $C_0>0$. Suppose that every $v\in T$ can be written as $v=\sum_{x\in X}v_x$ with v_x supported in V_x and $\sum_x\langle v_x,Qv_x\rangle\le C_0\langle v,Qv\rangle$. Then $\langle h,Sh\rangle\ge C_0^{-1}\langle h,Q^{-1}h\rangle$ for every $h\in T$, and the infimum in (17) over $L^2_{T^\perp}$, in particular $\varpi(Q;(V_x))$ when $T=\mathbb{R}^N$, is at least $1/(C_0n^d)$.*

Proof. Let $h\in T$ and $v=Q^{-1}h\in T$. Since $v-P^Q_{V_x}v$ is Q-orthogonal to the fields supported in V_x, $\langle v_x,Qv\rangle=\langle v_x,QP^Q_{V_x}v\rangle$, and $\|P^Q_{V_x}v\|_Q^2=\langle v,QP^Q_{V_x}v\rangle=\langle h,R_{V_x}^{\mathsf T}Q_{V_xV_x}^{-1}R_{V_x}h\rangle$. By the Cauchy–Schwarz inequality

$$\begin{aligned}\langle v,Qv\rangle&=\sum_x\langle v_x,QP^Q_{V_x}v\rangle\le\Big(\sum_x\|v_x\|_Q^2\Big)^{1/2}\Big(\sum_x\|P^Q_{V_x}v\|_Q^2\Big)^{1/2}\\&\le C_0^{1/2}\langle v,Qv\rangle^{1/2}\langle h,Sh\rangle^{1/2},\end{aligned}$$

so $\langle h,Q^{-1}h\rangle=\langle v,Qv\rangle\le C_0\langle h,Sh\rangle$. The last claim follows from Lemma 100, since $T^\perp$ is compatible with Q and $(T^\perp)^\perp=T$. □

3. *The partition of unity*

A profile is a function $\tau:\mathbb{Z}\to[0,\infty)$ supported in $\{0,\dots,n-1\}$ with $\sum_s\tau(s)=1$; it is regarded as a function on $\mathbb{Z}/L\mathbb{Z}$, which is possible since $L\ge n$. Put

$$S_2=\sum_s\tau(s)^2,\qquad S_{11}=\sum_s\tau(s)\tau(s+1),$$
$$D_2=\sum_s(\tau(s+1)-\tau(s))^2=2(S_2-S_{11}),$$

the sums over $s\in\mathbb{Z}$, and $\alpha=nS_2$, $\beta=n^3D_2$. For $x\in\mathbb{T}$ let $\chi_x(y)=\prod_{i=1}^d\tau(y_i-x_i)$. Then χ_x is supported in V_x, $\sum_x\chi_x\equiv1$, and because $L\ge n+1$ the residues $-1,0,\dots,n-1$ are distinct in $\mathbb{Z}/L\mathbb{Z}$, so that for all y and $i\ne j$

$$\begin{aligned}\sum_x\chi_x(y)^2=S_2^d,\qquad\sum_x\chi_x(y)\chi_x(y+e_i)=S_2^{d-1}S_{11},\\\sum_x\chi_x(y+e_i)\chi_x(y+e_j)=S_2^{d-2}S_{11}^2,\end{aligned}\tag{21}$$

each sum over $x\in\mathbb{T}$ factorizing over the coordinates.

Lemma 102 (the scalar identity)**.** *For every profile and every $v\in\mathbb{R}^{\mathbb{T}}$,*

$$\begin{aligned}\sum_{x\in\mathbb{T}}\langle\chi_xv,Q_{\mathrm s}\chi_xv\rangle&=S_2^{d-1}\big[S_{11}\|\nabla v\|^2+(S_2\mu^2+dD_2)\|v\|^2\big]\\&\le S_2^{d-1}\Big(S_2+\frac{dD_2}{\mu^2}\Big)\langle v,Q_{\mathrm s}v\rangle.\end{aligned}$$

Proof. For a link (y,i), by (21),

$$\begin{aligned}\sum_x\big(\nabla_i(\chi_xv)(y)\big)^2&=\sum_x\big[\chi_x(y+e_i)^2v(y+e_i)^2+\chi_x(y)^2v(y)^2\\&\qquad-2\chi_x(y)\chi_x(y+e_i)v(y)v(y+e_i)\big]\\&=S_2^d\big(v(y+e_i)^2+v(y)^2\big)-2S_2^{d-1}S_{11}v(y)v(y+e_i).\end{aligned}$$

Summing over y, and using $2\sum_y v(y)v(y+e_i)=2\|v\|^2-\|\nabla_iv\|^2$,

$$\begin{aligned}\sum_x\|\nabla_i(\chi_xv)\|^2&=2S_2^d\|v\|^2-S_2^{d-1}S_{11}\big(2\|v\|^2-\|\nabla_iv\|^2\big)\\&=S_2^{d-1}\big[S_{11}\|\nabla_iv\|^2+D_2\|v\|^2\big].\end{aligned}$$

Summing over i and adding $\mu^2\sum_x\|\chi_xv\|^2=\mu^2S_2^d\|v\|^2$ gives the identity. The inequality follows from $S_{11}\le S_2$ (Cauchy–Schwarz) and $\langle v,Q_{\mathrm s}v\rangle=\|\nabla v\|^2+\mu^2\|v\|^2$. □

Lemma 103 (the parabolic profile)**.** *Let $\tau(s)=6(s+1)(n-s)/(n(n+1)(n+2))$ for $0\le s\le n-1$ and $\tau(s)=0$ otherwise. Then τ is a profile and*

$$\alpha_n:=nS_2=\frac{6(n^2+2n+2)}{5(n+1)(n+2)}\le\frac65,$$
$$\beta_n:=n^3D_2=\frac{12n^2}{(n+1)(n+2)}\le12,$$
$$\frac{\beta_n}{\alpha_n}=10\rho_n,\qquad\rho_n:=\frac{n^2}{n^2+2n+2}<1.$$

Proof. With $u=s+1$ and $m=n+1$, the formulas for $\sum_{u=1}^n u^k$, $k\le4$, give $\sum_{u=1}^n u(m-u)=n(n+1)(n+2)/6$ and $\sum_{u=1}^n u^2(m-u)^2=(m^5-m)/30=n(n+1)(n+2)(n^2+2n+2)/30$, and the increments of $u\mapsto u(m-u)$ on $0\le u\le n$ are $m-1-2u=n-2u$, with $\sum_{u=0}^n(n-2u)^2=n(n+1)(n+2)/3$. So $\sum\tau=1$, $S_2=6(n^2+2n+2)/(5n(n+1)(n+2))$ and $D_2=12/(n(n+1)(n+2))$, which give α_n, β_n and their ratio; $\alpha_n\le6/5$ since $n^2+2n+2\le(n+1)(n+2)$, and $\beta_n\le12$ since $n^2\le(n+1)(n+2)$. □

With a constant profile $\tau=1/n$ one has $\alpha=1$ but $\beta=2n$, and the bound below degenerates as $n\to\infty$; the smooth profile is what makes the constant uniform in n.

4. The scalar field

Theorem 104 (the cube condition for the free scalar field)**.** *Let $d \ge 1$, $n \ge 2$, $L \ge n+1$, $\mu > 0$ and $s = n\mu$. For the Gaussian field on $\mathbb{T}$ with precision $Q_{\mathrm{s}} = -\Delta + \mu^2$ and the cubes V_x, $x \in \mathbb{T}$,*

$$\varpi(Q_{\mathrm{s}}; (V_x)) \ge \frac{1}{\alpha_n^d + d\alpha_n^{d-1}\beta_n s^{-2}} = \alpha_n^{-d}\frac{s^2}{s^2 + 10d\rho_n} \ge \left(\frac{5}{6}\right)^d \frac{s^2}{s^2+10d} =: c_d(s),$$

with $\alpha_n, \beta_n, \rho_n$ as in Lemma 103. That is, $\mathrm{BP}(n, c_d(n\mu))$ *holds for every real $f \in L^2(\gamma_{Q_{\mathrm{s}}})$, and $c_d(s) \ge (5/6)^d(1+10d)^{-1}\min(1, s^2)$.*

Proof. Let τ be the parabolic profile and $v_x = \chi_x v$. Then v_x is supported in V_x, $\sum_x v_x = v$, and by Lemma 102 $\sum_x \langle v_x, Q_{\mathrm{s}} v_x\rangle \le C_0\langle v, Q_{\mathrm{s}} v\rangle$ with $C_0 = S_2^{d-1}(S_2 + dD_2/\mu^2)$. Lemma 101 with $T = \mathbb{R}^{\mathbb{T}}$ gives $\varpi \ge 1/(C_0 n^d)$, and $C_0 n^d = (nS_2)^d + d(nS_2)^{d-1} n^3 D_2/(n\mu)^2 = \alpha_n^d + d\alpha_n^{d-1}\beta_n s^{-2}$. The middle expression follows from $\beta_n/\alpha_n = 10\rho_n$, and the last inequality from $\alpha_n \le 6/5$ and $\rho_n < 1$. The claim about BP is the definition of ϖ, and the final bound is $s^2/(s^2+10d) \ge \min(1, s^2)/(1+10d)$. □

For $d = 4$ the bound is $\varpi \ge (5/6)^4 s^2/(s^2+40)$, which is 0.0118 at $s = 1$ and 0.0438 at $s = 2$. At $n = 2$ the bound of the theorem, with the n-dependent constants, is $s^2/(s^2+4d)$, and it coincides with the upper bound of Proposition 108 below, so it is exact; for every n the upper bound of Proposition 108(i) is the exact constant (Theorem 109).

5. The Proca field

The gauge transformations of the Proca field are $A \mapsto A + d_0\psi$, $\psi \in \mathbb{R}^{\mathbb{T}}$; let $G = d_0\mathbb{R}^{\mathbb{T}}$ be the space of gauge directions, so that $G^\perp = \ker\delta_0$, the transverse fields. A gauge-invariant function of A is a G-invariant one, and a linear function ℓ_h is gauge-invariant exactly when $\delta_0 h = 0$.

Lemma 105 (the Hodge identity on the torus)**.** *For every $A \in \mathbb{R}^E$, $\|d_1 A\|^2 + \|\delta_0 A\|^2 = \sum_{i=1}^d \|\nabla_i A\|^2$. Consequently Q_{P} commutes with the orthogonal projection onto G, so G is compatible with Q_{P}, and for transverse A, $\langle A, Q_{\mathrm{P}} A\rangle = \sum_i \|\nabla A_i\|^2 + \mu^2\|A\|^2$.*

Proof. Put $\nabla_i^- f(y) = f(y) - f(y - e_i)$, so that $(\delta_0 A)(y) = -\sum_i \nabla_i^- A_i(y)$, and $(\nabla_i^-)^{\mathsf{T}} = -\nabla_i$ on $\mathbb{R}^{\mathbb{T}}$; all these operators commute, being translations and their differences. Then

$$\begin{aligned}
\|d_1 A\|^2 &= \sum_{i<j}\|\nabla_i A_j - \nabla_j A_i\|^2 \\
&= \sum_{i\ne j}\|\nabla_j A_i\|^2 - 2\sum_{i<j}\langle \nabla_i A_j, \nabla_j A_i\rangle, \\
\|\delta_0 A\|^2 &= \sum_i \|\nabla_i^- A_i\|^2 + 2\sum_{i<j}\langle \nabla_i^- A_i, \nabla_j^- A_j\rangle \\
&= \sum_i \|\nabla_i A_i\|^2 + 2\sum_{i<j}\langle \nabla_j A_i, \nabla_i A_j\rangle,
\end{aligned}$$

since $\|\nabla_i^- f\| = \|\nabla_i f\|$ and $\langle \nabla_i^- A_i, \nabla_j^- A_j\rangle = -\langle A_i, \nabla_i \nabla_j^- A_j\rangle = -\langle A_i, \nabla_j^- \nabla_i A_j\rangle = \langle \nabla_j A_i, \nabla_i A_j\rangle$. Adding the two lines gives $\sum_{i,j}\|\nabla_j A_i\|^2$. Hence $\delta_1 d_1 = -\Delta\otimes I - d_0\delta_0$. Now $\delta_1 d_1$ vanishes on G (as $d_1 d_0 = 0$) and is symmetric, so it maps $G^\perp$ into $G^\perp$; thus $Q_{\mathrm{P}} = \delta_1 d_1 + \mu^2$ preserves G and $G^\perp$, which is the commutation. For transverse A the identity reads $\|d_1 A\|^2 = \sum_i \|\nabla A_i\|^2$. □

Lemma 106 (the decomposition of a Proca field)**.** *Let $d \ge 2$, let τ be a profile and $A_x(y,i) = \chi_x(y)A(y,i)$, so that A_x is supported in E_x and $\sum_x A_x = A$. Then, with $\kappa_d = (5d-4)/2$,*

$$\sum_{x\in\mathbb{T}}\langle A_x, Q_{\mathrm{P}} A_x\rangle \le S_2^{d-1}\left[S_2\|d_1 A\|^2 + (S_2\mu^2 + \kappa_d D_2)\|A\|^2\right] \le S_2^{d-1}\left(S_2 + \frac{\kappa_d D_2}{\mu^2}\right)\langle A, Q_{\mathrm{P}} A\rangle,$$

and if $\delta_0 A = 0$,

$$\sum_{x\in\mathbb{T}}\langle A_x, Q_{\mathrm{P}} A_x\rangle \le S_2^{d-1}\left[S_{11}\|d_1 A\|^2 + (S_2\mu^2 + dD_2)\|A\|^2\right] \le S_2^{d-1}\left(S_2 + \frac{dD_2}{\mu^2}\right)\langle A, Q_{\mathrm{P}} A\rangle.$$

Proof. Fix a plaquette $p = (y,i,j)$ and put $a = A(y,i) - A(y,j)$, $b = A(y+e_i, j)$, $c = A(y+e_j, i)$, so that $(d_1 A)(p) = a + b - c$ and $(d_1 A_x)(p) = \chi_x(y)a + \chi_x(y+e_i)b - \chi_x(y+e_j)c$. With $M = S_2^d$, $M' = S_2^{d-1}S_{11}$ and $M'' = S_2^{d-2}S_{11}^2$, (21) gives

$$\begin{aligned}
\sum_x (d_1 A_x)(p)^2 &= M(a^2+b^2+c^2) + 2M'(ab-ac) - 2M''bc \\
&= M(d_1 A)(p)^2 - 2(M-M')a(b-c) \\
&\quad + 2(M-M'')bc.
\end{aligned}$$

Here $M - M' = S_2^{d-1}(S_2 - S_{11}) = \frac{1}{2}S_2^{d-1}D_2 \ge 0$ and $M - M'' = S_2^{d-2}(S_2 - S_{11})(S_2 + S_{11}) \le S_2^{d-1}D_2$, using $S_{11} \le S_2$. Bound $2|a(b-c)| \le a^2 + (b-c)^2 \le a^2 + 2b^2 + 2c^2$ and $2|bc| \le b^2 + c^2$, and sum over the plaquettes. For fixed y, $\sum_{i<j}(A_i(y) - A_j(y))^2 = d\sum_i A_i(y)^2 - (\sum_i A_i(y))^2 \le d\sum_i A_i(y)^2$, so $\sum_p a^2 \le d\|A\|^2$; and each link occurs as b in $j-1$ plaquettes and as c in $d-i$ plaquettes, so $\sum_p (b^2+c^2) = (d-1)\|A\|^2$. Therefore

$$\begin{aligned}
\sum_x \|d_1 A_x\|^2 &\le M\|d_1 A\|^2 \\
&\quad + \left[(M-M')(3d-2) + (M-M'')(d-1)\right]\|A\|^2 \\
&\le S_2^d\|d_1 A\|^2 + S_2^{d-1}D_2\left[\frac{3d-2}{2} + d - 1\right]\|A\|^2,
\end{aligned}$$

and $\frac{3d-2}{2} + d - 1 = \kappa_d$. Adding $\mu^2\sum_x\|A_x\|^2 = \mu^2 S_2^d\|A\|^2$ gives the first bound, and $\langle A, Q_{\mathrm{P}} A\rangle = \|d_1 A\|^2 + \mu^2\|A\|^2$ the second.

If $\delta_0 A = 0$, Lemma 105 applied to A_x and to A gives $\|d_1 A_x\|^2 \le \sum_i \|\nabla(\chi_x A_i)\|^2$ and $\sum_i\|\nabla A_i\|^2 = \|d_1 A\|^2$. By the proof of Lemma 102,

$$\sum_x\sum_i \|\nabla(\chi_x A_i)\|^2 = S_2^{d-1}\sum_i\left[S_{11}\|\nabla A_i\|^2 + dD_2\|A_i\|^2\right] = S_2^{d-1}\left[S_{11}\|d_1 A\|^2 + dD_2\|A\|^2\right],$$

and the mass term is as before. □

Theorem 107 (the cube condition for the free Proca field)**.** *Let $d \geq 2$, $n \geq 2$, $L \geq n+1$, $\mu > 0$ and $s = n\mu$. For the Gaussian field on the links of $\mathbb{T}$ with precision $Q_{\mathrm{P}} = \delta_1 d_1 + \mu^2$ and the link cubes E_x, $x \in \mathbb{T}$:*

(i) *for every real $f \in L^2(\gamma_{Q_{\mathrm{P}}})$, $\mathrm{BP}(n, \varpi)$ holds with*

$$\varpi = \alpha_n^{-d} \frac{s^2}{s^2 + 10\kappa_d \rho_n} \geq \left(\frac{5}{6}\right)^d \frac{s^2}{s^2 + 5(5d-4)},$$

where $\kappa_d = (5d-4)/2$;

(ii) *for every real gauge-invariant $f \in L^2(\gamma_{Q_{\mathrm{P}}})$, $\mathrm{BP}(n, \varpi)$ holds with the constant of Theorem 104,*

$$\varpi = \alpha_n^{-d} \frac{s^2}{s^2 + 10d\rho_n} \geq c_d(s) = \left(\frac{5}{6}\right)^d \frac{s^2}{s^2 + 10d}.$$

In both cases the optimal constant is the infimum over linear functions of the respective class, and the inequality in (ii) does not require $\mathscr{P}_x$ to preserve gauge invariance.

Proof. (i) Lemma 106 with the parabolic profile gives a decomposition of every A with $C_0 = S_2^{d-1}(S_2 + \kappa_d D_2/\mu^2)$, and Lemma 101 with $T = \mathbb{R}^E$ gives $\varpi \geq 1/(C_0 n^d)$, $C_0 n^d = \alpha_n^d + \kappa_d \alpha_n^{d-1} \beta_n s^{-2} = \alpha_n^d (1 + 10\kappa_d \rho_n s^{-2})$. (ii) $T = G^{\perp}$ satisfies $Q_{\mathrm{P}} T = T$ by Lemma 105, and for $A \in T$ the second bound of Lemma 106 gives $C_0 = S_2^{d-1}(S_2 + dD_2/\mu^2)$; Lemma 101 bounds the infimum over L^2_G, the gauge-invariant functions, by $1/(C_0 n^d)$, which is the constant of Theorem 104. The statements about linear functions are Theorem 97 and Lemma 100; the last remark was made after Theorem 97. □

6. *Upper bounds*

Proposition 108 (the mass is needed)**.** *Let $n \geq 2$, $L \geq n+1$ and $s = n\mu$.*

(i) *For the scalar field,*

$$\varpi(Q_{\mathrm{s}}; (V_x)) \leq \frac{\mu^2}{n^d} \langle \mathbf{1}_V, (-\Delta_D + \mu^2)^{-1} \mathbf{1}_V \rangle \leq \frac{s^2}{s^2 + n^2 \lambda_1(n)} \leq \frac{s^2}{s^2 + 16d/9},$$

where $-\Delta_D$ is the Dirichlet Laplacian of the cube $V = \{0, \dots, n-1\}^d$ and $\lambda_1(n) = 4d \sin^2(\pi/(2n+2))$ is its smallest eigenvalue.

(ii) *For the Proca field with $d \geq 2$, the infimum in (17) over gauge-invariant functions, hence also $\varpi(Q_{\mathrm{P}}; (E_x))$, is at most*

$$\frac{\mu^2}{n^d} \langle h_V, (Q_{\mathrm{P}})^{-1}_{VV} h_V \rangle \leq \frac{s^2}{1 + s^2},$$

where $V = E_0$ and h_V is the indicator of the links of V in the direction e_1.

In particular both constants are at most s^2, and tend to 0 as $n\mu \to 0$.

Proof. (i) Take $h = \mathbf{1}$, so that $Q_{\mathrm{s}} h = \mu^2 h$ and $\mathrm{Var}(\ell_h) = \langle h, Q_{\mathrm{s}}^{-1} h \rangle = L^d/\mu^2$. For each x, Lemma 100 gives

$$\|\ell_h - \mathscr{P}_x \ell_h\|^2 = \langle \mathbf{1}_{V_x}, (Q_{\mathrm{s}})^{-1}_{V_x V_x} \mathbf{1}_{V_x} \rangle,$$

and $(Q_{\mathrm{s}})_{V_x V_x} = -\Delta_D + \mu^2$: two sites of V_x are neighbours on the torus exactly when they are neighbours in $\mathbb{Z}^d$, since their coordinates differ by at most $n - 1 < L - 1$. All x give the same value, so $\varpi \leq n^{-d} L^d \langle \mathbf{1}_V, (-\Delta_D + \mu^2)^{-1} \mathbf{1}_V \rangle \mu^2 / L^d$, the first bound. The eigenvalues of the Dirichlet Laplacian of the path on n sites are $2 - 2\cos(\pi k/(n+1))$, $1 \leq k \leq n$, and $-\Delta_D$ is the sum of d commuting copies, so its smallest eigenvalue is $\lambda_1(n) = d(2 - 2\cos(\pi/(n+1))) = 4d \sin^2(\pi/(2n+2))$ and $(-\Delta_D + \mu^2)^{-1} \leq (\lambda_1 + \mu^2)^{-1}$; with $\|\mathbf{1}_V\|^2 = n^d$ this gives $\varpi \leq \mu^2/(\lambda_1 + \mu^2) = s^2/(s^2 + n^2 \lambda_1)$. Finally $\sin t \geq 2t/\pi$ on $[0, \pi/2]$ gives $n^2 \lambda_1 \geq 4dn^2/(n+1)^2 \geq 16d/9$ for $n \geq 2$.

(ii) Let $h(y,i) = \delta_{i1}$, the constant field in the direction e_1. Then $d_1 h = 0$ and $\delta_0 h = 0$, so ℓ_h is gauge-invariant, $Q_{\mathrm{P}} h = \mu^2 h$ and $\mathrm{Var}(\ell_h) = L^d/\mu^2$. Fix x, let $V = E_x$ and $h_V = R_V h$, with $\|h_V\|^2 = n^d$, and write $I_i = \{x_i, \dots, x_i + n - 1\} \subset \mathbb{Z}/L\mathbb{Z}$. We claim that for every $w \in \mathbb{R}^E$ supported in E_x,

$$\langle h_V, w \rangle^2 \leq n^{d+2} \|d_1 w\|^2. \tag{22}$$

For $y_2 \in \mathbb{Z}/L\mathbb{Z}$ and $\tilde{y} = (y_3, \dots, y_d)$ let

$$W(y_2, \tilde{y}) = \sum_{y_1 \in \mathbb{Z}/L\mathbb{Z}} w(y_1, y_2, \tilde{y}; 1),$$
$$D(y_2, \tilde{y}) = \sum_{y_1 \in \mathbb{Z}/L\mathbb{Z}} (d_1 w)(y_1, y_2, \tilde{y}; 1, 2).$$

In the sum defining D the terms $w(y + e_1, 2) - w(y, 2)$ telescope over the period, so $D(y_2, \tilde{y}) = W(y_2, \tilde{y}) - W(y_2 + 1, \tilde{y})$. As w is supported in E_x, $W(y_2, \tilde{y}) = 0$ unless $y_2 \in I_2$ and $\tilde{y} \in I_3 \times \cdots \times I_d$; and $x_2 - 1 \notin I_2$ since $L \geq n+1$. Hence $W(x_2 + k, \tilde{y}) = -\sum_{j=0}^{k} D(x_2 - 1 + j, \tilde{y})$ for $0 \leq k \leq n-1$, and

$$\langle h_V, w \rangle = \sum_{\tilde{y}} \sum_{k=0}^{n-1} W(x_2 + k, \tilde{y}) = -\sum_{\tilde{y}} \sum_{j=0}^{n-1} (n - j) D_j(\tilde{y}),$$

with $D_j(\tilde{y}) = D(x_2 - 1 + j, \tilde{y})$ and the sums over $\tilde{y} \in I_3 \times \cdots \times I_d$. The value $(d_1 w)(y; 1, 2)$ is nonzero only if one of the links $(y, 1)$, $(y, 2)$, $(y + e_1, 2)$, $(y + e_2, 1)$ lies in E_x, which forces $y_1 \in I_1 \cup \{x_1 - 1\}$, a set of $n+1$ residues; so $D(t, \tilde{y})^2 \leq (n+1) \sum_{y_1} (d_1 w)(y_1, t, \tilde{y}; 1, 2)^2$ by the Cauchy–Schwarz inequality, and the plaquettes $(y; 1, 2)$ with $y_2 = x_2 - 1 + j$, $0 \leq j \leq n-1$, are distinct. By the Cauchy–Schwarz inequality again,

$$\langle h_V, w \rangle^2 \leq n^{d-2} \sum_{j=0}^{n-1} (n - j)^2 \cdot (n+1) \|d_1 w\|^2 = \frac{n^{d-1}(n+1)^2(2n+1)}{6} \|d_1 w\|^2 \leq n^{d+2} \|d_1 w\|^2,$$

since $6n^3 - (n+1)^2(2n+1) = (n-2)(4n^2 + 3n + 2) + 3 > 0$ for $n \geq 2$. This is (22). Now, with w ranging over the fields

supported in E_x and $t = \langle h_V, w\rangle$, so that $\|w\|^2 \geq t^2/n^d$,

$$\begin{aligned}\langle h_V, (Q_{\mathrm{P}})^{-1}_{VV} h_V\rangle &= \sup_w \big[2\langle h_V, w\rangle - \|d_1 w\|^2 - \mu^2\|w\|^2\big] \\ &\leq \sup_{t\in\mathbb{R}} \big[2t - t^2(n^{-d-2} + \mu^2 n^{-d})\big] \\ &= \frac{n^d}{n^{-2}+\mu^2}.\end{aligned}$$

By Lemma 100, $\langle h_V, (Q_{\mathrm{P}})^{-1}_{VV} h_V\rangle$ is $\|\ell_h - \mathscr{P}_x \ell_h\|^2$, the same for every x by translation invariance, and summing over the L^d cubes,

$$\begin{aligned}\frac{n^{-d}\sum_x \|\ell_h - \mathscr{P}_x\ell_h\|^2}{\mathrm{Var}(\ell_h)} &= \frac{\mu^2}{n^d}\langle h_V, (Q_{\mathrm{P}})^{-1}_{VV} h_V\rangle \\ &\leq \frac{\mu^2}{n^{-2}+\mu^2} = \frac{s^2}{1+s^2}.\end{aligned}$$

Since ℓ_h is gauge-invariant, this bounds the infimum over gauge-invariant functions, and the infimum over all functions is smaller. □

7. *The exact constant for the scalar field*

The bounds of Theorem 104 and Proposition 108(i) enclose the scalar constant, and the upper bound is attained: the infimum in (17) is the value of the quotient at the constant linear function. The proof diagonalizes the additive Schwarz operator of the cubes by Fourier series; what makes the zero mode minimal is the positivity of the Dirichlet resolvent of the cube (Fig. 15). Throughout this part $Q = Q_{\mathrm{s}}$, $V = V_0 = \{0,\dots,n-1\}^d$, $G = (Q_{VV})^{-1} = (-\Delta_D + \mu^2)^{-1}$ as in the proof of Proposition 108, and the spaces $\mathbb{R}^{\mathbb{T}}$, $\mathbb{R}^V$ are complexified where plane waves occur; inner products are conjugate linear in the first argument. Let $\widehat{\mathbb{T}} = (2\pi/L)\mathbb{Z}^d/2\pi\mathbb{Z}^d$ be the dual torus, and for $k \in \widehat{\mathbb{T}}$ let $e_k(y) = e^{ik\cdot y}$, $e_k^V = R_V e_k$, and

$$\hat{q}(k) = \mu^2 + \sum_{i=1}^{d}(2 - 2\cos k_i),$$

so that $Qe_k = \hat{q}(k)e_k$. For $u \in V$ write $w \sim u$, $w \notin V$ for the torus neighbours of u outside V; since $L \geq n+1$ these are the sites $u \pm e_i \notin V$ of $\mathbb{Z}^d$, read modulo L, and $k\cdot(w-u)$ is well defined modulo 2π.

Theorem 109 (the exact constant for the scalar field)**.** *Let $d \geq 1$, $n \geq 2$, $L \geq n+1$ and $\mu > 0$.*

(i) *The operator $Q^{1/2}SQ^{1/2}$ of Lemma 100 is diagonalized by the plane waves e_k, $k \in \widehat{\mathbb{T}}$, with eigenvalues*

$$\begin{aligned}\lambda_k &= \hat{q}(k)\langle e_k^V, Ge_k^V\rangle \\ &= n^d - \sum_{u,u'\in V}\sum_{w\sim u', w\notin V} G(u,u')\cos\big(k\cdot(w-u)\big).\end{aligned} \tag{23}$$

(ii) *G has positive entries, $\lambda_k \geq \lambda_0$ for every k, with equality only for $k = 0$, and $\lambda_0 = \mu^2\langle \mathbf{1}_V, G\mathbf{1}_V\rangle$.*

(iii) *Consequently*

$$\varpi(Q_{\mathrm{s}}; (V_x)) = \frac{\mu^2}{n^d}\langle \mathbf{1}_V, (-\Delta_D + \mu^2)^{-1}\mathbf{1}_V\rangle = 1 - \alpha_n(\mu^2),$$

with α_n as in Proposition 93; the infimum in (17) is attained at the constant linear function $\ell_{\mathbf{1}}$, and among linear functions only at its nonzero multiples. The first inequality of Proposition 108(i) is an equality.

Proof. (i) The translation $\tau_a f(y) = f(y-a)$ of the torus maps V_x to V_{x+a} and commutes with Q, so $\tau_a R_{V_x}^{\mathsf{T}} Q^{-1}_{V_xV_x} R_{V_x}\tau_a^{-1} = R^{\mathsf{T}}_{V_{x+a}} Q^{-1}_{V_{x+a}V_{x+a}} R_{V_{x+a}}$, and summing over x shows that S commutes with every τ_a. The plane waves are the common eigenvectors of the translations, with distinct characters for distinct k, so $Se_k = \hat{\sigma}(k)e_k$ with $\hat{\sigma}(k) = L^{-d}\langle e_k, Se_k\rangle$. In $\langle e_k, Se_k\rangle = \sum_x \langle R_{V_x}e_k, Q^{-1}_{V_xV_x} R_{V_x}e_k\rangle$ the translation τ_x identifies V_x with V and $Q_{V_xV_x}$ with $Q_{VV} = -\Delta_D + \mu^2$ (the proof of Proposition 108(i)), and carries $R_{V_x}e_k$ to $e^{ik\cdot x}e_k^V$; the phase drops out of the Hermitian form, so every term equals $\langle e_k^V, Ge_k^V\rangle$ and $\hat{\sigma}(k) = \langle e_k^V, Ge_k^V\rangle$. As $Q^{1/2}e_k = \hat{q}(k)^{1/2}e_k$, the e_k are an orthogonal eigenbasis of $Q^{1/2}SQ^{1/2}$ with eigenvalues $\lambda_k = \hat{q}(k)\langle e_k^V, Ge_k^V\rangle$.

For the second expression let $B : \mathbb{C}^{\mathbb{T}} \to \mathbb{C}^V$, $(Bf)(u) = \sum_{w\sim u, w\notin V} f(w)$. For every f and $u \in V$,

$$\begin{aligned}(Q_{VV}R_V f)(u) &= (2d+\mu^2)f(u) - \sum_{w\sim u, w\in V} f(w) \\ &= (R_V Qf)(u) + (Bf)(u),\end{aligned}$$

that is, $Q_{VV}R_V f = R_V Qf + Bf$. With $f = e_k$ this reads $Q_{VV}e_k^V = \hat{q}(k)e_k^V + Be_k$; applying G and pairing with e_k^V,

$$n^d = \langle e_k^V, e_k^V\rangle = \hat{q}(k)\langle e_k^V, Ge_k^V\rangle + \langle e_k^V, GBe_k\rangle,$$

so $\lambda_k = n^d - \langle e_k^V, GBe_k\rangle = n^d - \sum_{u,u',w} G(u,u')e^{ik\cdot(w-u)}$, the sum over $u,u' \in V$ and $w \sim u'$, $w \notin V$. Since λ_k is real, the exponential can be replaced by its real part, which is (23).

(ii) Write $Q_{VV} = (2d+\mu^2)(I - K)$, where $K = (2d+\mu^2)^{-1}A_V$ and A_V is the adjacency matrix of the cube, which is entrywise nonnegative and has spectral norm at most its maximal degree $2d$. So $\|K\| \leq 2d/(2d+\mu^2) < 1$ and $G = (2d+\mu^2)^{-1}\sum_{m\geq 0} K^m$ has nonnegative entries; the entry (u,u') of K^m is positive for m the graph distance of u and u', so $G(u,u') > 0$ for all u,u'. By (23),

$$\lambda_k - \lambda_0 = \sum_{u,u'\in V}\sum_{w\sim u', w\notin V} G(u,u')\big(1 - \cos(k\cdot(w-u))\big) \geq 0.$$

Equality forces $\cos(k\cdot(w-u)) = 1$ for every triple; the triple $u = u'$ with $u_i = n-1$ and $w = u + e_i$ gives $\cos k_i = 1$, so $k_i \in 2\pi\mathbb{Z}$ for every i, that is, $k = 0$ in $\widehat{\mathbb{T}}$. Finally $B\mathbf{1} = Q_{VV}\mathbf{1}_V - R_V Q\mathbf{1} = Q_{VV}\mathbf{1}_V - \mu^2\mathbf{1}_V$, so $\langle \mathbf{1}_V, GB\mathbf{1}\rangle = n^d - \mu^2\langle \mathbf{1}_V, G\mathbf{1}_V\rangle$ and $\lambda_0 = \mu^2\langle \mathbf{1}_V, G\mathbf{1}_V\rangle$.

(iii) By Lemma 100, $n^d\varpi(Q_{\mathrm{s}}; (V_x))$ is the smallest eigenvalue of $Q^{1/2}SQ^{1/2}$, which is λ_0 by (ii); the formula for α_n is that of Proposition 93 with $t = \mu^2$. The quotient in (17) at $\ell_{\mathbf{1}}$ is $n^{-d}\lambda_0$ by the proof of Proposition 108(i), and a linear function ℓ_h attains the infimum exactly when $Q^{-1/2}h$ lies in the λ_0-eigenspace, which is the span of $e_0 = \mathbf{1}$; since $Q^{1/2}\mathbf{1} = \mu\mathbf{1}$, these are the multiples of $\ell_{\mathbf{1}}$. □

Remark 110 (a second proof, and the structure of the spectrum). (a) The inequality $\lambda_k \geq \lambda_0$ also follows from the variational characterization $\langle b, Gb\rangle = \sup_{w\in\mathbb{C}^V}\big[2\,\mathrm{Re}\langle b,w\rangle - \langle w, Q_{VV}w\rangle\big] \geq |\langle b,w\rangle|^2/\langle w,Q_{VV}w\rangle$ with the trial function $w = ge_k^V$, $g = G\mathbf{1}_V$, where $(ge_k^V)(u) = g(u)e^{ik\cdot u}$. For real g on V, extended by zero, $|g(u+e_i)e^{ik_i} - g(u)|^2 = (g(u+e_i)-g(u))^2 + 2(1-\cos k_i)g(u)g(u+e_i)$, and $\sum_u g(u)g(u+e_i) \leq \|g\|^2$, so

$$\langle ge_k^V, Q_{VV}\, ge_k^V\rangle \leq \langle g, Q_{VV}g\rangle + (\hat{q}(k)-\mu^2)\|g\|^2.$$

With $\gamma = \langle \mathbf{1}_V, G\mathbf{1}_V\rangle = \langle g, Q_{VV}g\rangle \geq \mu^2\|g\|^2$ and $\langle e_k^V, ge_k^V\rangle = \gamma$ this gives $\langle e_k^V, Ge_k^V\rangle \geq \gamma^2/(\gamma\hat{q}(k)/\mu^2)$, that is $\lambda_k \geq \mu^2\gamma = \lambda_0$. This proof uses only that g is real, not the positivity of G; it is the route that Proposition 113 below follows for the Proca field, where positivity is not available.

(b) The sum in (23) is $\mathrm{Re}\langle e_k^V, f_k\rangle$, where $f_k = GBe_k$ is the function on V with $Q\tilde{f}_k = 0$ on V for the extension $\tilde{f}_k$ of f_k by e_k outside V: the massive harmonic extension of the boundary values of the plane wave. Its kernel $GB \geq 0$ is the Poisson kernel of the cube, whose row sums are $1-\mu^2(G\mathbf{1}_V)(u) < 1$, and (23) says that λ_k is n^d minus the Fourier transform of the nonnegative measure $\sum_{u,u',w} G(u,u')\delta_{w-u}$ on the torus, which is largest at $k=0$ (Bochner). In particular $\lambda_k \leq 2n^d - \lambda_0 < 2n^d$ for every k, so the spectrum of the additive Schwarz operator P_{ad} lies in $[\lambda_0, 2n^d - \lambda_0]$.

We write $\varpi_n^{(d)}(s)$ for the exact constant of Theorem 109 as a function of $s = n\mu$ and d, and $\varpi_n(s) = \varpi_n^{(d)}(s)$ when d is fixed. The Dirichlet Laplacian of the cube is the sum of d commuting copies of the Dirichlet Laplacian of the path on n sites, whose eigenvectors are $\phi_j(u) = \sin(\pi j(u+1)/(n+1))$, $1 \leq j \leq n$, with eigenvalues $4\sin^2\theta_j$, $\theta_j = \pi j/(2n+2)$, $\|\phi_j\|^2 = (n+1)/2$ and $\langle \mathbf{1}, \phi_j\rangle = \cot\theta_j$ for j odd and 0 for j even (the sum $\sum_{u=1}^n e^{i\pi ju/(n+1)}$ equals $2/(1-e^{i\pi j/(n+1)})$ for j odd and 0 for j even, and the imaginary part of the former is $\cot\theta_j$). The operator $-\Delta_D + \mu^2$ is not a tensor product, but the semigroup $e^{t\Delta_D}$ is, which gives the second formula below.

Corollary 111 (closed forms)**.** *Let $n \geq 2$, $s > 0$ and $\mu = s/n$.*

(i) (Fourier series) *With $\theta_j = \pi j/(2n+2)$ and the sum over $j \in \{1,3,5,\dots\}^d$, $j_i \leq n$,*

$$\varpi_n^{(d)}(s) = s^2\Big(\frac{2}{n(n+1)}\Big)^d \sum_j \frac{\prod_{i=1}^d \cot^2\theta_{j_i}}{s^2 + 4n^2\sum_{i=1}^d \sin^2\theta_{j_i}}.$$

(ii) (Product formula) *With*

$$H_n(t) = \langle \mathbf{1}, e^{t\Delta_D^{(1)}}\mathbf{1}\rangle = \frac{2}{n+1}\sum_{j\ \mathrm{odd}} \cot^2\theta_j\, e^{-4t\sin^2\theta_j},$$

the heat content of the path,

$$\varpi_n^{(d)}(s) = \frac{s^2}{n^{d+2}}\int_0^\infty e^{-s^2t/n^2} H_n(t)^d\, dt.$$

(iii) ($d=1$) *With $\kappa > 0$ defined by $2\sinh(\kappa/2) = \mu = s/n$,*

$$\varpi_n^{(1)}(s) = 1 - \frac{2\sinh(n\kappa/2)}{s\cosh((n+1)\kappa/2)}.$$

(iv) (The dimension)

$$1 - d\big(1-\varpi_n^{(1)}(s)\big) \leq \varpi_n^{(d)}(s) \leq \varpi_n^{(1)}(s),$$

and $\varpi_2^{(d)}(s) = s^2/(s^2+4d)$.

Proof. (i) Expand $\mathbf{1}_V$ in the orthogonal eigenbasis $\phi_{j_1}\otimes\cdots\otimes\phi_{j_d}$ of $-\Delta_D$:

$$\langle \mathbf{1}_V, G\mathbf{1}_V\rangle = \sum_j \prod_i \cot^2\theta_{j_i}/\big[((n+1)/2)^d(\mu^2 + 4\sum_i \sin^2\theta_{j_i})\big],$$

the sum over odd j_i, and multiply by $\mu^2 n^{-d} = s^2 n^{-d-2}$. (ii) $G = \int_0^\infty e^{-\mu^2 t}e^{t\Delta_D}dt$ and $e^{t\Delta_D} = \bigotimes_i e^{t\Delta_D^{(1)}}$, so $\langle \mathbf{1}_V, G\mathbf{1}_V\rangle = \int_0^\infty e^{-\mu^2 t}H_n(t)^d dt$. (iii) Let $g = G\mathbf{1}$ on the path. The solutions of $-(g(u+1)-2g(u)+g(u-1)) + \mu^2 g(u) = 0$ are combinations of $e^{\pm\kappa u}$ with $2\cosh\kappa = 2+\mu^2$, that is $4\sinh^2(\kappa/2) = \mu^2$, and

$$g(u) = \mu^{-2}\big[1 - \cosh(\kappa(u - \tfrac{n-1}{2}))/\cosh(\kappa\tfrac{n+1}{2})\big]$$

solves $(-\Delta_D + \mu^2)g = \mathbf{1}$ with $g(-1) = g(n) = 0$. Summing the geometric series,

$$\sum_{u=0}^{n-1}\cosh(\kappa(u-\tfrac{n-1}{2})) = \sinh(n\kappa/2)/\sinh(\kappa/2),$$

so

$$\mu^2 n^{-1}\langle \mathbf{1}, g\rangle = 1 - \sinh(n\kappa/2)/(n\sinh(\kappa/2)\cosh((n+1)\kappa/2)),$$

and $n\sinh(\kappa/2) = s/2$. (iv) Let $v = \mathbf{1}_V - \mu^2 G\mathbf{1}_V$, so that $(-\Delta_D+\mu^2)v = -\Delta_D\mathbf{1}_V =: b$, where $b(u) = \sum_i b_i(u)$ and $b_i(u) \in \{0,1,2\}$ counts the neighbours $u \pm e_i$ of u outside V; and let $v_i(u) = v^{(1)}(u_i)$, where $v^{(1)} = \mathbf{1} - \mu^2 G^{(1)}\mathbf{1}$ is the same function for the path, which satisfies $(-\Delta_D^{(1)} + \mu^2)v^{(1)} = b^{(1)}$ and $0 \leq v^{(1)} \leq 1$ (as $G^{(1)} \geq 0$ and $G^{(1)}\mathbf{1} \leq \mu^{-2}\mathbf{1}$). Writing $-\Delta_D = \sum_j(-\Delta_j)$ with $-\Delta_j$ the Dirichlet Laplacian in the coordinate j, one has $(-\Delta_i+\mu^2)v_i = b_i$ and, since v_i does not depend on u_j, $(-\Delta_j v_i)(u) = b_j(u)v_i(u)$ for $j \neq i$. Hence

$$(-\Delta_D+\mu^2)\Big(\sum_i v_i - v\Big) = \sum_i\sum_{j\neq i} b_j v_i \geq 0,$$

$$(-\Delta_D+\mu^2)(v - v_i) = \sum_{j\neq i} b_j(1-v_i) \geq 0,$$

and the positivity of G gives $v_i \leq v \leq \sum_j v_j$. Pairing with $\mathbf{1}_V$, and using $\langle \mathbf{1}_V, v\rangle = n^d(1-\varpi_n^{(d)}(s))$ and $\langle \mathbf{1}_V, v_i\rangle = n^{d-1}\langle \mathbf{1}, v^{(1)}\rangle = n^d(1-\varpi_n^{(1)}(s))$, gives the two inequalities. For $n=2$ every site of V has d neighbours in V, so $-\Delta_D\mathbf{1}_V = d\mathbf{1}_V$ and $\varpi_2^{(d)}(s) = \mu^2/(d+\mu^2) = s^2/(s^2+4d)$. □

Theorem 112 (the limit $n\to\infty$ at fixed $s = n\mu$)**.** *Let $d \geq 1$ and $s > 0$. Then*

$$\lim_{n\to\infty}\varpi_n^{(d)}(s) = \varpi_\infty^{(d)}(s) := s^2\langle \mathbf{1}, (-\Delta+s^2)^{-1}\mathbf{1}\rangle_{L^2([0,1]^d)}$$

$$= s^2\Big(\frac{8}{\pi^2}\Big)^d \sum_{k\in\{1,3,5,\dots\}^d} \frac{\prod_{i=1}^d k_i^{-2}}{s^2 + \pi^2|k|^2}$$

$$= s^2\int_0^\infty e^{-s^2t}H(t)^d\, dt,$$

where $-\Delta$ is the Dirichlet Laplacian of the unit cube and $H(t)=\frac{8}{\pi^2}\sum_{k\ \mathrm{odd}}k^{-2}e^{-\pi^2k^2t}$ is the heat content of the unit interval. In $d=1$, $\varpi_\infty^{(1)}(s)=1-\frac{2}{s}\tanh\frac{s}{2}$; in general

$$1-\frac{2d}{s}\tanh\frac{s}{2}\le\varpi_\infty^{(d)}(s)\le1-\frac{2}{s}\tanh\frac{s}{2},$$
$$\varpi_\infty^{(d)}(s)=T_ds^2+O(s^4)\quad(s\to0),$$

with $T_d=\langle\mathbf{1},(-\Delta)^{-1}\mathbf{1}\rangle_{L^2([0,1]^d)}$ the torsional rigidity of the unit cube, $T_1=1/12$. For $d=4$,

$$\varpi_\infty^{(4)}(1)=0.012954\ldots,\qquad\varpi_\infty^{(4)}(2)=0.048687\ldots,$$

and for $d=3$, $\varpi_\infty^{(3)}(1)=0.019564\ldots$, $\varpi_\infty^{(3)}(2)=0.071850\ldots$.

Proof. In the Fourier series of Corollary 111(i) fix j. As $n\to\infty$, $\theta_{j_i}=\pi j_i/(2n+2)\to0$,

$$\cot^2\theta_{j_i}\sim(2n+2)^2/(\pi^2j_i^2)\quad\text{and}\quad4n^2\sin^2\theta_{j_i}\to\pi^2j_i^2,$$

so the j-th term converges to

$$s^2(8/\pi^2)^d\prod_i j_i^{-2}/(s^2+\pi^2|j|^2).$$

Since $\cot^2\theta\le\theta^{-2}$, the j-th term is at most

$$(2/(n(n+1)))^d(4(n+1)^2/\pi^2)^d\prod_i j_i^{-2}$$
$$=(8(n+1)/(n\pi^2))^d\prod_i j_i^{-2}\le(12/\pi^2)^d\prod_i j_i^{-2},$$

which is summable over j; dominated convergence gives the series. The eigenfunctions of the Dirichlet Laplacian of $[0,1]^d$ are $\prod_i\sin(\pi k_ix_i)$, $k\in\mathbb{N}^d$, with eigenvalues $\pi^2|k|^2$, squared norms 2^{-d} and $\langle\mathbf{1},\prod_i\sin(\pi k_ix_i)\rangle=\prod_i2/(\pi k_i)$ for all k_i odd and 0 otherwise; so the series is $s^2\langle\mathbf{1},(-\Delta+s^2)^{-1}\mathbf{1}\rangle$. The integral form follows from $(s^2+\pi^2|k|^2)^{-1}=\int_0^\infty e^{-(s^2+\pi^2|k|^2)t}dt$ and Tonelli, since $H(t)^d=\sum_k(8/\pi^2)^d\prod_i k_i^{-2}e^{-\pi^2|k|^2t}$.

For $d=1$ the function $w=1-\cosh(s(x-\frac12))/\cosh(s/2)$ solves $-w''+s^2w=s^2$ with $w(0)=w(1)=0$, and $\int_0^1w=1-\frac{2}{s}\tanh\frac{s}{2}$. The two-sided bound is the limit of Corollary 111(iv), using Corollary 111(iii) and $\kappa\sim s/n$ (so $\sinh(n\kappa/2)/\cosh((n+1)\kappa/2)\to\tanh(s/2)$). The expansion at $s\to0$ is the series with $(s^2+\pi^2|k|^2)^{-1}=\pi^{-2}|k|^{-2}-s^2\pi^{-2}|k|^{-2}(s^2+\pi^2|k|^2)^{-1}$, both parts summable; $T_1=1/12$ is the coefficient of s^2 in $1-\frac{2}{s}\tanh\frac{s}{2}$. The numerical values are the integral evaluated by quadrature, with $H(t)=1-4\sqrt{t/\pi}+O(e^{-1/(16t)})$ for small t and the series otherwise; they agree with the lattice values of Corollary 111(ii) at $n=2048,4096$ extrapolated in $1/n$ (0.0129536 and 0.0486873) to five digits, and with a cell-centred finite difference solution of the continuum problem extrapolated in the mesh size (0.01296, 0.04871) to four. □

Numerically $T_2=0.03514\ldots$, $T_3=0.02017\ldots$, $T_4=0.01324\ldots$, and the lattice values approach the limit from above at the rate $1/n$: for $d=4$, $s=1$, $\varpi_n^{(4)}(1)$ is 0.0588, 0.0347, 0.0234, 0.0180, 0.0154, 0.0142, 0.0136 at $n=2,4,8,16,32,64,128$, and for $s=2$ it is 0.2000, 0.1250, 0.0862, 0.0671, 0.0578, 0.0532, 0.0509. In every case computed $\varpi_n^{(d)}(s)$ decreases in n; this monotonicity is not proved here. The lower bound $c_d(s)$ of Theorem 104 is uniform in n, so $\varpi_\infty^{(d)}(s)\ge c_d(s)$: for $d=4$, $0.0130\ge0.0118$ at $s=1$ and $0.0487\ge0.0438$ at $s=2$. Over the full range the ratio is at most $\alpha_n^d\le(6/5)^d$ times the factor $(s^2+10d\rho_n)/(s^2+n^2\lambda_1(n))$, and the Dirichlet resolvent is what the parabolic trial function approximates.

8. *The Proca field at zero momentum*

For the Proca field on gauge-invariant functions the same reduction to momentum space applies, but the proof of Theorem 109 does not: the precision of the link cube is not an M-matrix. What is proved is the exact scalar constant as a lower bound, and the exact value of the zero-momentum sector; that the latter is the infimum is observed numerically in every case computed. Let $d\ge2$, $V=V_0$, $E_V=E_0$, $G_{\mathrm P}=((Q_{\mathrm P})_{E_VE_V})^{-1}$, and for $1\le i\le d$ let $\varepsilon_i\in\mathbb{R}^{E_V}$ be the indicator of the links of E_V in the direction e_i, so that the test field of Proposition 108(ii) is $h_V=\varepsilon_1$. Put

$$M_0=\big(\langle\varepsilon_i,G_{\mathrm P}\varepsilon_j\rangle\big)_{i,j=1}^d,$$

and write $\varpi_{\mathrm P}^{\mathrm{inv}}$ for the infimum in (17) over gauge-invariant functions, the constant of Theorem 107(ii).

Proposition 113 (the Proca field: what is proved)**.** *Let $d\ge2$, $n\ge2$, $L\ge n+1$, $\mu>0$ and $s=n\mu$.*

(i) *$M_0=\alpha I+\beta(J-I)$ with $\alpha=\langle\varepsilon_1,G_{\mathrm P}\varepsilon_1\rangle$, $\beta=\langle\varepsilon_1,G_{\mathrm P}\varepsilon_2\rangle$ and J the all-ones matrix; its eigenvalues are $\alpha+(d-1)\beta$, on $\mathbf{1}$, and $\alpha-\beta$, on the vectors of sum zero.*

(ii) $\varpi_n^{(d)}(s)\le\varpi_{\mathrm P}^{\mathrm{inv}}\le\dfrac{\mu^2}{n^d}\min\big(\alpha-\beta,\alpha+(d-1)\beta\big)\le\dfrac{\mu^2}{n^d}\alpha$, *the last expression being the bound of Proposition 108(ii). The lower bound is the exact scalar constant of Theorem 109, which improves the constant of Theorem 107(ii). More precisely, for every $k\in\widehat{\mathbb{T}}$ and every transverse $a\in\mathbb{C}^d$ at k (that is, $\delta_0(ae_k)=0$), the Rayleigh quotient of the gauge-invariant linear function ℓ_{ae_k} is at least $n^{-d}\lambda_k\ge n^{-d}\lambda_0$, with λ_k as in (23).*

(iii) $\mu^2\alpha/n^d\ge\varpi_n^{(d-1)}(s)$, *the exact scalar constant in dimension $d-1$ with the same n and s.*

Proof. (i) A permutation π of $\{1,\ldots,d\}$ acts on sites by $(\pi y)_{\pi(i)}=y_i$ and on fields by $(U_\pi A)(\pi y,\pi(i))=A(y,i)$. U_π is orthogonal, preserves V, hence E_V, maps ε_i to $\varepsilon_{\pi(i)}$, and $(d_1U_\pi A)(\pi y;\pi(i),\pi(j))=\pm(d_1A)(y;i,j)$, so $\|d_1U_\pi A\|=\|d_1A\|$ and U_π commutes with $Q_{\mathrm P}$ and with the restriction to E_V, hence with $G_{\mathrm P}$. So $M_0(\pi(i),\pi(j))=M_0(i,j)$ for all π, and a symmetric matrix invariant under conjugation by all permutations is of the stated form.

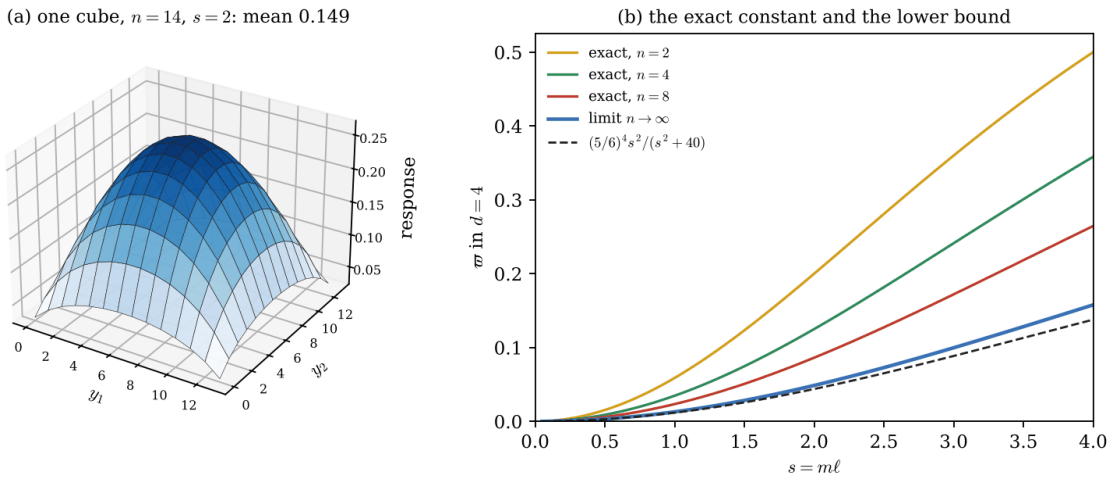


FIG. 15. The cube condition for the free massive scalar field. (a) The response $\mu^2(-\Delta_D+\mu^2)^{-1}\mathbf{1}$ of one square cube of side $n=14$ to a constant shift of its boundary data, at $s=n\mu=2$; its mean over the cube is the exact constant of Theorem 109 in $d=2$. (b) The exact constant $\varpi_n^{(4)}(s)$ for $n=2,4,8$, its limit $s^2\langle 1,(-\Delta+s^2)^{-1}1\rangle$ on the unit cube (Theorem 112), equal to 0.0130 at $s=1$ and 0.0487 at $s=2$, and the lower bound $(5/6)^4 s^2/(s^2+40)$ of Theorem 104, all against $s=m\ell$.

(ii) By Lemma 100, $n^d\varpi_{\mathrm{P}}^{\mathrm{inv}}$ is the smallest eigenvalue of the compression of $Q_{\mathrm{P}}^{1/2}S_{\mathrm{P}}Q_{\mathrm{P}}^{1/2}$ to $G^{\perp}=\ker\delta_0$, with S_{P} the Schwarz operator of the link cubes. As in the proof of Theorem 109, S_{P} and Q_{P} commute with translations, so they map each space $\mathbb{C}^d\otimes e_k=\{ae_k:a\in\mathbb{C}^d\}$ into itself, S_{P} acting by the matrix $\hat{\sigma}_{\mathrm{P}}(k)$ with $a^*\hat{\sigma}_{\mathrm{P}}(k)a=L^{-d}\langle ae_k,S_{\mathrm{P}}ae_k\rangle=\langle h_V,G_{\mathrm{P}}h_V\rangle$, $h_V=R_{E_V}(ae_k)$ (the translation τ_x identifies E_x with E_V and $(Q_{\mathrm{P}})_{E_xE_x}$ with $(Q_{\mathrm{P}})_{E_VE_V}$). The space $\ker\delta_0$ is the orthogonal sum of its intersections $T_k\otimes e_k$ with these spaces, where $T_k=\{a:\sum_i a_i(1-e^{-ik_i})=0\}$ for $k\neq 0$ and $T_0=\mathbb{C}^d$, and on $T_k\otimes e_k$ the operator $Q_{\mathrm{P}}=-\Delta\otimes I-d_0\delta_0+\mu^2$ (Lemma 105) acts as $\hat{q}(k)$. So the compression is block diagonal with blocks $\hat{q}(k)P_k\hat{\sigma}_{\mathrm{P}}(k)P_k$ on T_k, P_k the orthogonal projection onto T_k, and

$$n^d\varpi_{\mathrm{P}}^{\mathrm{inv}}=\min_{k\in\widehat{\mathbb{T}}}\ \min_{a\in T_k,|a|=1}\hat{q}(k)\langle h_V,G_{\mathrm{P}}h_V\rangle,\qquad h_V=R_{E_V}(ae_k). \tag{24}$$

At $k=0$, $\hat{q}(0)=\mu^2$, $T_0=\mathbb{C}^d$ and $\langle h_V,G_{\mathrm{P}}h_V\rangle=a^*M_0a$, so the inner minimum is $\mu^2\min(\alpha-\beta,\alpha+(d-1)\beta)$ by (i), which is at most $\mu^2\alpha$ (take $a=e_1$). This is the upper bound.

For the lower bound let $w\in\mathbb{C}^{E_V}$, extended by zero to E, and let $w_i=w(\cdot,i)\in\mathbb{C}^V$, extended by zero to $\mathbb{T}$. Lemma 105, applied to the real and imaginary parts of w, gives $\|d_1w\|^2=\sum_i\|\nabla w_i\|^2-\|\delta_0w\|^2\le\sum_i\|\nabla w_i\|^2$, and $\|\nabla w_i\|^2+\mu^2\|w_i\|^2=\langle w_i,(Q_{\mathrm{s}})_{VV}w_i\rangle$ since w_i is supported in V. Hence, identifying $\mathbb{C}^{E_V}$ with $\mathbb{C}^V\otimes\mathbb{C}^d$,

$$(Q_{\mathrm{P}})_{E_VE_V}\le(Q_{\mathrm{s}})_{VV}\otimes I_d,\qquad G_{\mathrm{P}}\ge G\otimes I_d$$

in the operator order. For $h_V=e_k^V\otimes a$ this gives $\langle h_V,G_{\mathrm{P}}h_V\rangle\ge|a|^2\langle e_k^V,Ge_k^V\rangle$, so each term in (24) is at least $\hat{q}(k)\langle e_k^V,Ge_k^V\rangle=\lambda_k\ge\lambda_0=n^d\varpi_n^{(d)}(s)$ by Theorem 109. The statement about the Rayleigh quotient of ℓ_{ae_k} is the same computation, since $\mathrm{Var}(\ell_{ae_k})=L^d|a|^2/\hat{q}(k)$ for transverse a.

(iii) Write $y=(y_1,y')$ with $y'\in V'=\{0,\dots,n-1\}^{d-1}$, let $\varphi\in\mathbb{R}^{V'}$, extended by zero, and let $w(y,1)=\varphi(y')$ for $y\in V$, $w(y,i)=0$ for $i\ge 2$; then w is supported in E_V. A plaquette $(y;i,j)$ with $i,j\ge 2$ carries no component of w, and for $(y;1,j)$, $j\ge 2$, $(d_1w)(y;1,j)=w(y,1)-w(y+e_j,1)$, which is $\varphi(y')-\varphi(y'+e_j)$ if $0\le y_1\le n-1$ and 0 otherwise. Hence $\|d_1w\|^2=n\|\nabla'\varphi\|^2=n\langle\varphi,-\Delta_D^{(d-1)}\varphi\rangle$, $\|w\|^2=n\|\varphi\|^2$ and $\langle\varepsilon_1,w\rangle=n\langle\mathbf{1},\varphi\rangle$, so that

$$\begin{aligned}\alpha&=\sup_{w\in\mathbb{R}^{E_V}}\big[2\langle\varepsilon_1,w\rangle-\langle w,Q_{\mathrm{P}}w\rangle\big]\\&\ge n\sup_{\varphi}\big[2\langle\mathbf{1},\varphi\rangle-\langle\varphi,(-\Delta_D^{(d-1)}+\mu^2)\varphi\rangle\big]\\&=n\langle\mathbf{1},(-\Delta_D^{(d-1)}+\mu^2)^{-1}\mathbf{1}\rangle,\end{aligned}$$

and

$$\mu^2\alpha/n^d\ge\mu^2n^{-(d-1)}\langle\mathbf{1},(-\Delta_D^{(d-1)}+\mu^2)^{-1}\mathbf{1}\rangle=\varpi_n^{(d-1)}(s).$$

□

Remark 114 (where the scalar argument fails, and what is observed). (a) *Positivity.* The entries of $(Q_{\mathrm{P}})_{E_VE_V}=(\delta_1d_1+\mu^2)_{E_VE_V}$ at the pairs $\{(y,i),(y+e_i,j)\}$ and $\{(y,i),(y-e_j,j)\}$, $i<j$, are $+1$, so the matrix is not an M-matrix, and G_{P} has negative entries (for $d=2$, $n=2$, $\mu=1/2$ its smallest entry is -0.88). The harmonic-extension kernel of Remark 110(b) is therefore not a positive kernel, and no analogue of (23) with a nonnegative measure is available. The variational route of Remark 110(a) with the trial field $(G_{\mathrm{P}}h_V^0)e_k$, $h_V^0=R_{E_V}a$, reduces the equality $n^d\varpi_{\mathrm{P}}^{\mathrm{inv}}=\mu^2(\alpha-\beta)$ to the inequality

$$\|d_1(ge_k)\|^2\le\|d_1g\|^2+(\hat{q}(k)-\mu^2)\|g\|^2\qquad\text{for } g=G_{\mathrm{P}}R_{E_V}a,\ a\in T_k,$$

where $(ge_k)(y,i) = g(y,i)e^{ik\cdot y}$: given it,

$$\hat q(k)\langle h_V, G_{\mathrm P} h_V\rangle \ge \hat q(k)|a^* M_0 a|^2/\langle ge_k, Q_{\mathrm P}\, ge_k\rangle \ge \mu^2 a^* M_0 a.$$

For the scalar field the corresponding inequality is pointwise in the links and holds for every real g; for the curl the four terms of $(d_1(ge_k))(p)$ carry different phases and signs, and

$$|\sum_m c_m e^{i\theta_m}|^2 - (\sum_m c_m)^2 = -\sum_{m<m'} c_m c_{m'}|e^{i\theta_m} - e^{i\theta_{m'}}|^2$$

has no sign. The inequality holds, with relative margin at least 0.14, in every case computed; it is not proved.

(b) *Observed.* In every case computed ($d = 2,3,4$, $n \le 4$, $L \le 6$, $0.6 \le s \le 9$), the minimum in (24) is attained at $k = 0$ only, the minimum over $k \ne 0$ exceeding it by a factor at least 1.07 (at $s = 9$) and typically between 2 and 25; $\beta > 0$; and so

$$n^d \varpi_{\mathrm P}^{\mathrm{inv}} = \mu^2(\alpha - \beta),$$

attained at the constant transverse field $h = e_1 - e_2$, which has $d_1 h = 0$ and $\delta_0 h = 0$. The test field ε_1 of Proposition 108(ii) is therefore not optimal: its quotient $\mu^2\alpha/n^d$ exceeds the constant by the factor $(1-\beta/\alpha)^{-1}$, which is 1.11 for $d = 4$, $n = 2$, $s = 1$ and tends to 1 as $n \to \infty$ (β/α is 0.10, 0.041, 0.015, 0.0047 at $n = 2,4,8,16$ for $d = 4$, $s = 1$). For $d = 4$, $n = 2$ the values $\mu^2(\alpha-\beta)/n^d$ are 0.0759 at $s = 1$ and 0.2436 at $s = 2$, the exact constants quoted in Remark 116(a). The lower bound of Proposition 113(ii) is not sharp: $\varpi_{\mathrm P}^{\mathrm{inv}}/\varpi_n^{(d)}(s)$ is 1.29 for $d = 4$, $n = 2$, $s = 1$.

(c) *The limit.* In every case computed, $\mu^2(\alpha-\beta)/n^d$ decreases in n at the rate $1/n$ towards the scalar constant of Theorem 112 *in one dimension less*: for $d = 3$, $s = 1$, extrapolation in $1/n$ from $n = 24, 32$ gives 0.03353 against $\varpi_\infty^{(2)}(1) = 0.03352$; for $d = 4$, $s = 1$, from $n = 16, 20$ it gives 0.0195 against $\varpi_\infty^{(3)}(1) = 0.01956$, and for $s = 2$ it gives 0.0721 against $\varpi_\infty^{(3)}(2) = 0.07185$ (the values at $n = 2,4,8,16,20$ are 0.0759, 0.0489, 0.0343, 0.0269, 0.0254 for $s = 1$ and 0.2436, 0.1669, 0.1216, 0.0971, 0.0921 for $s = 2$). This is what Proposition 113(iii) suggests: the field $\varphi(y')e_1$ with φ constant along e_1 is supported in E_V at no cost in the direction e_1, because the link cube E_V only constrains the tangential components of a field on its boundary (the normal component of a field on E_V may jump to zero across a face without producing a curl), and in the continuum $w = \varphi(x')e_1$, $\varphi = (-\Delta_{[0,1]^{d-1}} + s^2)^{-1}\mathbf{1}$, solves $(\delta d + s^2)w = e_1$ in the unit cube with vanishing tangential trace. On the lattice φe_1 fails to solve $(Q_{\mathrm P})_{E_V E_V} w = \varepsilon_1$ only by a residual on the face $y_1 = 0$, which accounts for the correction of order $1/n$. The convergence of $\mu^2(\alpha-\beta)/n^d$ to $\varpi_\infty^{(d-1)}(s)$ is not proved here. So, if the minimum stays at $k = 0$, the gauge-invariant Proca constant in $d = 4$ tends, at fixed $m\ell$, to the three-dimensional scalar constant, 0.0196 at $m\ell = 1$ and 0.0719 at $m\ell = 2$; the four-dimensional scalar constant tends to 0.0130 and 0.0487.

9. Uniformity in the lattice spacing, and the size of the constants

Corollary 115 (the hypothesis of Theorem 64 for free fields)**.** *Let $m > 0$ and $\ell > 0$, and for a sequence of spacings $a_k \to 0$ let $\mu_k = m a_k$ and $n_k = \ell/a_k$ (or any integers $n_k \ge 2$ with $\ell/2 \le n_k a_k \le \ell$). Then for every k and every torus side $L \ge n_k + 1$, the free scalar field with precision $-\Delta + \mu_k^2$, and the free Proca field with precision $\delta_1 d_1 + \mu_k^2$ on gauge-invariant functions, satisfy* $\mathrm{BP}(n_k, \varpi)$ *with*

$$\varpi = \Big(\frac{5}{6}\Big)^d \frac{(m\ell)^2}{(m\ell)^2 + 10d} \quad \text{if } n_k a_k = \ell \text{ for all } k,$$

and with

$$\varpi = \Big(\frac{5}{6}\Big)^d \frac{(m\ell)^2}{(m\ell)^2 + 40d} \quad \text{if only } \ell/2 \le n_k a_k \le \ell,$$

independent of k and L; the Proca field on all functions satisfies it with $10d$ replaced by $5(5d-4)$ and $40d$ by $20(5d-4)$. This is the hypothesis of Theorem 64, with $\mathscr{L}_k$ the even $L \ge 2n_k + 4$, for these Gaussian stand-ins of the Wilson measure. For $d = 4$ and $n_k a_k = \ell$ the gauge-invariant constant is $(5/6)^4(m\ell)^2/((m\ell)^2+40)$; by Proposition 108 no constant larger than $(m\ell)^2/(1+(m\ell)^2)$ is possible for the Proca field, nor larger than $(m\ell)^2/((m\ell)^2+16d/9)$ for the scalar field.

Proof. With $s_k = n_k\mu_k = m n_k a_k$, Theorems 104 and 107 give $c_d(s_k)$, which is increasing in s_k; for $n_k a_k = \ell$ it is $c_d(m\ell)$, and for $s_k \ge m\ell/2$ it is at least $(5/6)^d(m\ell)^2/((m\ell)^2+40d)$. The upper bound is Proposition 108(ii). □

Remark 116 (the size of the constants). (a) In the Gaussian setting the constant of Theorem 104 is close to optimal, and the exact constant is known: by Theorem 109 it is $\varpi_n^{(d)}(s) = \mu^2 n^{-d}\langle \mathbf{1}_V, (-\Delta_D + \mu^2)^{-1}\mathbf{1}_V\rangle$, with the closed forms of Corollary 111. The n-dependent bound $\alpha_n^{-d} s^2/(s^2 + 10d\rho_n)$ equals it at $n = 2$ for every d and s, and lies within 2.5% of it for $n = 3$, $d \le 4$ (and $n = 4$, $d = 1$); the ratio of the exact constant to the clean bound $c_d(s)$ tends to $\varpi_\infty^{(d)}(s)/c_d(s)$ as $n \to \infty$, which for $d = 4$ is 1.10 at $s = 1$ and 1.11 at $s = 2$. The gauge-invariant Proca constant exceeds the exact scalar constant, which is a lower bound for it by Proposition 113(ii), by a factor between 1.06 (at $s = 9$) and 2.0 (at $s = 0.6$) in the cases computed in $d = 2,3,4$, $n \le 4$, the factor decreasing in s, and the full Proca constant exceeds the bound of Theorem 107(i) by a factor between 1.3 and 3.0; the two Proca constants coincide except at large s (for $d = 4$, $n = 2$, $s = 4$ they are 0.5534 and 0.5571), the infimum being attained at zero momentum. For the gauge-invariant Proca field in $d = 4$ the exact constant on the tori $L = 4, 5$ with $n = 2$ is 0.0759 at $s = 1$ and 0.2436 at $s = 2$, which is $\mu^2(\alpha-\beta)/n^d$ of Proposition 113; the bounds are 0.0588 and 0.2 (the exact scalar constants at $n = 2$), the clean bounds $c_4(1) = 0.0118$ and $c_4(2) = 0.0438$.

(b) The parabolic profile is the trial function of the proof of Proposition 93(ii), and the bound of Theorem 104 is, term by term, the bound $(s_1^2/(ns_2))^d\, t/(t + 2ds_1/s_2)$ proved there for a single cube: $s_1^2/(ns_2) = 1/\alpha_n$ and $2ds_1 n^2/s_2 = 10d\rho_n$. So the dynamics that resamples all the cubes has, on the whole of

L^2, the gap that Proposition 93(ii) gives for the Kantorovich response of one cube; Proposition 65(i) is the statement, for the Wilson measure, that a one-cube bound of this kind implies BP, and for the Gaussian field Theorem 97 gives the conclusion directly. The quantity $1-\alpha_n(\mu^2)$ of Proposition 93 is $\mu^2 n^{-d}\langle \mathbf{1}_V, (-\Delta_D+\mu^2)^{-1}\mathbf{1}_V\rangle$, the value of the quotient in (17) at the constant linear function $\ell_{\mathbf{1}}$, and by Theorem 109 it is the exact constant: the gap of the dynamics on L^2 is exactly the Kantorovich response of one cube to a constant shift of its boundary data. The gap between the parabolic trial function and the exact Dirichlet resolvent is about 4% for $n \le 4$ and at most 11% in the limit $n \to \infty$ for $d = 4$, $s \le 2$.

(c) Corollary 115 shows that the hypothesis of Theorem 64, the uniform condition $\mathrm{BP}(\ell/a_k, \varpi)$ along a continuum tuning, is satisfied by the free massive fields with ϖ of order $(m\ell)^2$ for $m\ell \le 1$; a mass gap therefore implies the cube condition for these fields, with an explicit constant, and the constant tends to 0 with $m\ell$ by Proposition 108; the test function there is the zero mode, which has no Gaussian measure in the massless case. Inserted into the conclusion $m_* = 16\varpi/(625e^2\ell)$ of Theorem 64, the value $\varpi = c_4(1) = 0.0118$ at $m\ell = 1$ gives $m_* \approx 4.1\times 10^{-5}\, m$: about two orders of magnitude are the size of ϖ itself, and the factor $16/(625e^2) \approx 3.5\times 10^{-3}$ comes from the finite-speed argument of Lemma 61.

C. Stability of the cube condition under perturbations

Subsection X B proves the averaged cube condition of Definition 60 for the free massive fields, with a constant of order $(m\ell)^2$ that depends on the lattice spacing only through the mass in units of the cube. This subsection asks which perturbations of the Gaussian preserve it. The answer has a negative and a positive half. A lower bound on the Hessian of the Hamiltonian does not preserve it: a convex perturbation of the free field can drive the constant to zero, because the Dirichlet form of the cube dynamics, unlike the carré du champ of the Langevin dynamics, depends on the Hessian and is not monotone in it. What preserves it is a two-sided control of the Hessian, or more precisely a pointwise stable decomposition of the Hessian along the cubes; under that hypothesis the condition holds for every strictly log-concave measure that satisfies it, with a constant that is uniform in the volume and in the cube side, and the proof is Lions' lemma lifted to the Helffer–Sjöstrand representation. Bounded perturbations are not controlled by their oscillation: the Ising model on the torus is a perturbation of a product measure with bounded oscillation on every cube and violates the condition with a constant that vanishes with the volume. For the lattice gauge field in a fixed gauge the hypotheses of the positive theorems fail at every coupling, and the subsection records exactly where: the Hessian of the Wilson action has the flat transverse fields in its kernel at every configuration, and is not convex on a set of positive measure. The quadratic approximation of the Wilson action is massless, its massive regularisation has a cube constant that vanishes with the mass at every coupling, and the missing estimate is a mass that the generator part of the Witten Laplacian, not the potential, would have to supply.

Setting. We keep the notation of Sec. X B: $d \ge 1$, $n \ge 2$, $L \ge n+1$, $\mathbb{T} = (\mathbb{Z}/L\mathbb{Z})^d$, the coordinate space $\mathbb{R}^N$ (sites or links), the sets V_x, $x \in \mathbb{T}$ (for links, V_x stands for the set E_x of links with base point in $x + \{0,\dots,n-1\}^d$), each coordinate lying in exactly n^d of them, the profile τ with the constants S_2, S_{11}, D_2, the parabolic constants $\alpha_n, \beta_n, \rho_n$ of Lemma 103, and the functions $\chi_x(y) = \prod_i \tau(y_i - x_i)$, which we regard as diagonal operators on $\mathbb{R}^N$ (for links, $(\chi_x A)(y,i) = \chi_x(y)A(y,i)$). A *partition of unity* is a family $(\chi_x)_{x\in X}$ of diagonal operators with nonnegative entries, χ_x supported in V_x and $\sum_x \chi_x = I$; the family above is one, and for it $\sum_x \chi_x^2 = S_2^d I$ by (21). For a positive definite Q_0 we write

$$C_0(Q_0) = \inf\Big\{ C : \sum_x \langle \chi_x v, Q_0 \chi_x v\rangle \le C\langle v, Q_0 v\rangle \text{ for all } v \in \mathbb{R}^N \Big\}, \tag{25}$$

the stable decomposition constant of Q_0 for the parabolic partition of unity, and $C_0(Q_0; T)$ for the same infimum over v in a subspace T. Lemma 102 gives $C_0(Q_{\mathrm{s}}) \le S_2^{d-1}(S_2 + dD_2/\mu^2)$, so that $n^d C_0(Q_{\mathrm{s}}) \le \alpha_n^d(1 + 10d\rho_n s^{-2})$, $s = n\mu$, and Lemma 106 gives the same bound for $C_0(Q_{\mathrm{P}}; \ker \delta_0)$ and $S_2^{d-1}(S_2 + \kappa_d D_2/\mu^2)$ for $C_0(Q_{\mathrm{P}})$.

A *measure* in this subsection is a probability measure π on $\mathbb{R}^N$ with a density $Z^{-1}e^{-H}$, $H \in C^\infty(\mathbb{R}^N)$. For $x \in X$ let $\mathrm{Var}_x^\pi(f)$ be the conditional variance of f given the coordinates outside V_x, so that $\pi(\mathrm{Var}_x^\pi f) = \|f - \mathscr{P}_x f\|^2_{L^2(\pi)}$, and say that π satisfies $\mathrm{BP}(n, \varpi)$ on a class $\mathscr{C} \subset L^2(\pi)$ if

$$\varpi \,\mathrm{Var}_\pi(f) \le \frac{1}{n^d}\sum_{x\in X} \pi\big(\mathrm{Var}_x^\pi f\big) \qquad (f \in \mathscr{C}), \tag{26}$$

which for the Wilson measure and $\mathscr{C} = \mathscr{A}$ is Definition 60. Both sides of (26) make sense for any probability measure on a product of measurable spaces indexed by the coordinates, and Propositions 124, 125 and 126 use them in that generality. We write $\varpi(\pi)$ for the largest such ϖ on $\mathscr{C} = L^2(\pi)$ and $\varpi(Q)$ for $\varpi(\gamma_Q)$. The conditional law of π given the coordinates outside V_x has the Hamiltonian $\phi_{V_x} \mapsto H(\phi_{V_x}, \phi_{V_x^c})$ on $\mathbb{R}^{V_x}$, whose Hessian is the principal submatrix $(\mathrm{Hess}\, H)_{V_x V_x}$.

1. A Hessian lower bound does not suffice

Theorem 117 (convex perturbations can destroy the condition)**.** *Let $\mu_0 = \gamma_{Q_{\mathrm{s}}}$ be the free scalar field of Theorem 104 and, for $t \ge 0$, $\mu_t = Z_t^{-1}e^{-V_t}\mu_0$ with the convex function $V_t(\phi) = \frac{t}{2}\|\nabla\phi\|^2$. Then $\mathrm{Hess}(-\log\mu_t) = Q_{\mathrm{s}} + t(-\Delta) \ge Q_{\mathrm{s}}$ for every t, while*

$$\varpi(\mu_t) \le \frac{s^2}{s^2 + (1+t)\,16d/9} \xrightarrow[t\to\infty]{} 0.$$

The same holds for the Proca field with $V_t(A) = \frac{t}{2}\|d_1 A\|^2$, on gauge-invariant functions, with the bound $s^2/(s^2+1+t)$. Consequently there is no $c > 0$ such that, for all measures $\mu = e^{-V}\mu_0$ with V convex, $\mathrm{BP}(n, \varpi_0)$ for μ_0 implies $\mathrm{BP}(n, c\varpi_0)$ for μ; and the same holds with "$\mathrm{Hess}(-\log\mu) \ge \mathrm{Hess}(-\log\mu_0)$" in place of "$V$ convex".

Proof. μ_t is the Gaussian with precision $Q_t=(1+t)(-\Delta)+\mu^2=(1+t)(-\Delta+\mu_t^2)$, $\mu_t^2=\mu^2/(1+t)$. For $c>0$ the law γ_{cQ} is the image of γ_Q under $\phi\mapsto c^{-1/2}\phi$, which commutes with the conditional expectations $\mathscr{P}_x$, so $\varpi(cQ)=\varpi(Q)$ and $\varpi(Q_t)=\varpi(-\Delta+\mu_t^2)$. Proposition 108(i) with $s_t=n\mu_t$ gives $\varpi(Q_t)\le s_t^2/(s_t^2+16d/9)=s^2/(s^2+(1+t)16d/9)$. The Proca case is Proposition 108(ii) in the same way. The last claim follows because μ_0 itself satisfies $\mathrm{BP}(n,\varpi_0)$ with $\varpi_0=c_d(s)>0$ by Theorem 104, while $\inf_t\varpi(\mu_t)=0$. □

The mechanism is visible in Lemma 100: the Dirichlet form of the cube dynamics on linear functions is $\langle h,Sh\rangle$ with $S=\sum_x R_x^{\mathsf T}Q_{V_xV_x}^{-1}R_x$, which *decreases* when Q increases, while the variance $\langle h,Q^{-1}h\rangle$ also decreases; their ratio has no monotonicity. For the Langevin dynamics the Dirichlet form is $\pi(|\nabla f|^2)$, whose integrand does not depend on π, which is why the Brascamp–Lieb inequality [23] and the Bakry–Émery criterion [4] are monotone in the Hessian and the cube condition is not. The perturbation V_t is gradient-like: it stiffens the field across the boundaries of the cubes and so transfers variance to wavelengths longer than the cube, where a cube resampling cannot reach it. Zeroth-order (mass-like) perturbations behave in the opposite way, which is the content of the next results.

2. *Two-sided comparison for Gaussians*

Theorem 118 (two-sided Gaussian comparison)**.** *Let Q_0 and Q be positive definite with $Q_0\le Q\le KQ_0$ for some $K\ge1$. Then $\varpi(Q)\ge\varpi(Q_0)/K$. If G is a subspace compatible with both Q_0 and Q, the same holds for the infima in (17) over L^2_G.*

Proof. By Theorem 97 and Lemma 100, $n^d\varpi(Q)$ is the infimum over $h\neq0$ of $\sum_x\langle h_V,Q_{VV}^{-1}h_V\rangle/\langle h,Q^{-1}h\rangle$, with $V=V_x$ and $h_V=R_Vh$, and the infimum over L^2_G is the same infimum over $h\in G^\perp$. Principal submatrices preserve the order, so $Q_{VV}\le KQ_{0,VV}$ and $Q_{VV}^{-1}\ge K^{-1}Q_{0,VV}^{-1}$; and $Q^{-1}\le Q_0^{-1}$. Hence each quotient for Q is at least K^{-1} times the quotient for Q_0 at the same h. □

3. *Lions' lemma for the Witten Laplacian*

The Helffer–Sjöstrand representation [60, 64] expresses the variance of a strictly log-concave measure through the Witten Laplacian on one-forms. We use it in the following form. Let $\pi=Z^{-1}e^{-H}d\phi$ with $H\in C^\infty(\mathbb{R}^N)$ and $\mathrm{Hess}\,H(\phi)\ge cI$ for all ϕ and some $c>0$. Let $\mathscr{L}=-\Delta+\langle\nabla H,\nabla\rangle$ be the generator of the Langevin dynamics, the self-adjoint operator on $L^2(\pi)$ associated with the Dirichlet form $\pi(|\nabla f|^2)$, and on the space $L^2(\pi;\mathbb{R}^N)$ of vector fields let

$$\mathscr{A}=\mathscr{L}\otimes I+\mathrm{Hess}\,H,\tag{27}$$

that is, $\mathscr{L}$ acting on each component and the multiplication operator by the matrix $\mathrm{Hess}\,H(\phi)$; its quadratic form is

$$a(G,G)=\sum_{i=1}^N\pi\big(|\nabla G_i|^2\big)+\pi\big(\langle G,\mathrm{Hess}\,H\,G\rangle\big)\ge c\,\pi(|G|^2),$$

on the form domain $\mathscr{D}$ of fields whose components have square integrable weak gradients. Then for every $f\in C^\infty(\mathbb{R}^N)$ with bounded derivatives of all orders, the class to which the theorem below is first reduced,

$$\mathrm{Var}_\pi f=\pi\big(\langle\nabla f,\mathscr{A}^{-1}\nabla f\rangle\big)=\sup_{G\in\mathscr{D}}\big[2\pi(\langle\nabla f,G\rangle)-a(G,G)\big],\tag{28}$$

the supremum being attained at $G=\mathscr{A}^{-1}\nabla f$. The identity is the intertwining relation $\nabla\mathscr{L}u=\mathscr{A}\nabla u$ applied to $u=\mathscr{L}^{-1}(f-\pi(f))$, which exists because $\mathscr{L}$ has spectral gap at least c by the Bakry–Émery criterion; the second equality is the variational characterization of the inverse of a positive operator. Only two consequences of (28) are used below: the identity $\mathrm{Var}_\pi f=2\pi(\langle\nabla f,G^*\rangle)-a(G^*,G^*)$ at $G^*=\mathscr{A}^{-1}\nabla f=\nabla u$, and the inequality $\mathrm{Var}_\pi f\ge2\pi(\langle\nabla f,G\rangle)-a(G,G)$ for every $G\in\mathscr{D}$, which follows from $a(\nabla u,G)=\pi(\langle\nabla\mathscr{L}u,G\rangle)=\pi(\langle\nabla f,G\rangle)$ and $a\ge0$ alone, by expanding $a(\nabla u-G,\nabla u-G)\ge0$. We shall apply (28) to π and to its conditional laws given the coordinates outside a set V, whose Hamiltonians $\phi_V\mapsto H(\phi_V,\phi_{V^c})$ satisfy the same hypotheses with the same c; for these we write $\mathscr{A}_V$ and a_V, so that for a field g on $\mathbb{R}^V$,

$$a_V(g,g)=\sum_{i\in V}\pi_V\big(|\nabla_V g_i|^2\big)+\pi_V\big(\langle g,(\mathrm{Hess}\,H)_{VV}\,g\rangle\big),$$

with π_V the conditional law and ∇_V the gradient in the coordinates of V. Two features of these forms carry the whole argument: the conditional form differentiates in fewer variables than the global one, and its Hessian is a principal submatrix of the global Hessian.

Theorem 119 (the cube condition for log-concave measures)**.** *Let $\pi=Z^{-1}e^{-H}d\phi$ with $H\in C^\infty(\mathbb{R}^N)$ and $\mathrm{Hess}\,H\ge cI$, $c>0$. Let $(\chi_x)_{x\in X}$ be a partition of unity for the sets (V_x) and $m_2=\max_i\sum_x\chi_x(i)^2$. Let $T\subset\mathbb{R}^N$ be a subspace such that $\mathrm{Hess}\,H(\phi)$ commutes with the orthogonal projection onto T for every ϕ (for instance $T=\mathbb{R}^N$), and suppose that there is $C_H<\infty$ with*

$$\sum_{x\in X}\langle\chi_x g,\mathrm{Hess}\,H(\phi)\,\chi_x g\rangle\le C_H\langle g,\mathrm{Hess}\,H(\phi)\,g\rangle\tag{29}$$

for all $g\in T$ and $\phi\in\mathbb{R}^N$. Then for every real $f\in L^2(\pi)$ that is $T^\perp$-invariant, that is, a function of the orthogonal projection of ϕ onto T,

$$\mathrm{Var}_\pi(f)\le\max(m_2,C_H)\sum_{x\in X}\pi\big(\mathrm{Var}_x^\pi f\big).$$

In particular, with $\varpi=1/(n^d\max(m_2,C_H))$ and the constant C_H of (29) for the given T, the measure π satisfies $\mathrm{BP}(n,\varpi)$ on $L^2(\pi)$ when $T=\mathbb{R}^N$, and on the G-invariant functions L^2_G when $T=G^\perp$. For the partition of unity of Sec. X B, $m_2=S_2^d$, and when $T=\mathbb{R}^N$ every admissible C_H satisfies $C_H\ge S_2^d$, so that $\varpi=1/(n^dC_H)$.

Proof. Let first $f \in C^\infty(\mathbb{R}^N)$ be $T^\perp$-invariant with bounded derivatives of all orders. Then $f(\phi+w)=f(\phi)$ for $w \in T^\perp$, so $\nabla f(\phi) \in T$ for every ϕ. Let Π be the orthogonal projection of $L^2(\pi;\mathbb{R}^N)$ onto the fields with values in T, $(\Pi G)(\phi) = P_T G(\phi)$. It commutes with $\mathscr{L} \otimes I$, which acts in the same way on every component, and with the multiplication by $\operatorname{Hess} H(\phi)$ by hypothesis; so it commutes with $\mathscr{A}$ and with $\mathscr{A}^{-1}$, and the maximizer $G^* = \mathscr{A}^{-1}\nabla f = \mathscr{A}^{-1}\Pi\nabla f = \Pi G^*$ in (28) takes values in T. Thus

$$\operatorname{Var}_\pi(f) = 2\pi(\langle \nabla f, G^* \rangle) - a(G^*, G^*). \tag{30}$$

Put $G^{(x)} = \chi_x G^*$, a field supported in V_x (its components outside V_x vanish) with values in $\mathbb{R}^N$; since χ_x is a constant diagonal matrix, $\nabla G_i^{(x)} = \chi_x(i)\nabla G_i^*$, and $G^{(x)} \in \mathscr{D}$. As $\sum_x \chi_x = I$, $\sum_x G^{(x)} = G^*$ and

$$2\pi(\langle \nabla f, G^* \rangle) = \sum_x 2\pi\big(\langle \nabla_{V_x} f, G^{(x)} \rangle\big), \tag{31}$$

where $\nabla_{V_x} f$ is the restriction of ∇f to the coordinates of V_x. Fix x and the coordinates outside $V = V_x$. The restriction $g = G^{(x)}(\cdot, \phi_{V^c})$ is, for π-almost every exterior, a field on $\mathbb{R}^V$ in the form domain of $\mathscr{A}_V$, and the inequality in (28) for the conditional law gives

$$\operatorname{Var}_x^\pi f \geq 2\pi_V\big(\langle \nabla_V f, g \rangle\big) - a_V(g,g).$$

Integrating over the exterior,

$$\pi\big(\operatorname{Var}_x^\pi f\big) \geq 2\pi\big(\langle \nabla_{V_x} f, G^{(x)} \rangle\big) - \pi\big(a_{V_x}(G^{(x)}, G^{(x)})\big), \tag{32}$$

where

$$\pi\big(a_V(G^{(x)}, G^{(x)})\big) = \sum_{i \in V} \pi\big(|\nabla_V G_i^{(x)}|^2\big) + \pi\big(\langle G^{(x)}, \operatorname{Hess} H\, G^{(x)} \rangle\big);$$

in the last term we used that $G^{(x)}$ is supported in V, so that the principal submatrix $(\operatorname{Hess} H)_{VV}$ may be replaced by $\operatorname{Hess} H$.

Now $|\nabla_V G_i^{(x)}|^2 = \chi_x(i)^2 |\nabla_V G_i^*|^2 \leq \chi_x(i)^2 |\nabla G_i^*|^2$, since the gradient in the coordinates of V is part of the full gradient. Summing over x and using $\sum_x \chi_x(i)^2 \leq m_2$ for the first term and (29) at each ϕ, with $g = G^*(\phi) \in T$, for the second,

$$\begin{aligned}\sum_x \pi\big(a_{V_x}(G^{(x)}, G^{(x)})\big) &\leq m_2 \sum_i \pi\big(|\nabla G_i^*|^2\big) \\ &\quad + C_H\, \pi\big(\langle G^*, \operatorname{Hess} H\, G^* \rangle\big) \\ &\leq \max(m_2, C_H)\, a(G^*, G^*).\end{aligned}$$

Put $\lambda = 1/\max(m_2, C_H)$, so that $\lambda \sum_x \pi(a_{V_x}(G^{(x)}, G^{(x)})) \leq a(G^*, G^*)$. By (30) and (31),

$$\begin{aligned}\operatorname{Var}_\pi(f) &\leq \sum_x \Big[2\pi\big(\langle \nabla_{V_x} f, G^{(x)} \rangle\big) - \lambda\, \pi\big(a_{V_x}(G^{(x)}, G^{(x)})\big)\Big] \\ &= \lambda^{-1} \sum_x \Big[2\pi\big(\langle \nabla_{V_x} f, \lambda G^{(x)} \rangle\big) \\ &\qquad - \pi\big(a_{V_x}(\lambda G^{(x)}, \lambda G^{(x)})\big)\Big] \\ &\leq \lambda^{-1} \sum_x \pi\big(\operatorname{Var}_x^\pi f\big),\end{aligned}$$

by (32) applied to the fields $\lambda G^{(x)}$, which are again supported in V_x and in the form domains. This is the claim for smooth f with bounded derivatives.

For general $T^\perp$-invariant $f \in L^2(\pi)$, both sides of the claim are continuous in $L^2(\pi)$, so it suffices to approximate f in $L^2(\pi)$ by $T^\perp$-invariant functions of the kind treated. Truncation $f \mapsto \max(-M, \min(f, M))$ preserves $T^\perp$-invariance and converges in $L^2(\pi)$ as $M \to \infty$. For bounded f, convolution with a Gaussian kernel ρ_ε preserves invariance, since $(f * \rho_\varepsilon)(\phi + w) = \int f(\phi + w - y)\rho_\varepsilon(y)\,dy = (f * \rho_\varepsilon)(\phi)$ for $w \in T^\perp$, gives smooth functions with bounded derivatives, and converges to f in $L^2(\pi)$ as $\varepsilon \to 0$ because π has a density and $f * \rho_\varepsilon \to f$ in $L^2_{\mathrm{loc}}(d\phi)$ with $|f * \rho_\varepsilon| \leq \|f\|_\infty$. The statement about BP is (26). For the partition of unity of Sec. X B, $\sum_x \chi_x(i)^2 = S_2^d$ for every i by (21); and when $T = \mathbb{R}^N$, (29) is the operator inequality $\sum_x \chi_x \operatorname{Hess} H(\phi) \chi_x \leq C_H \operatorname{Hess} H(\phi)$, whose trace reads $S_2^d \operatorname{tr} \operatorname{Hess} H(\phi) \leq C_H \operatorname{tr} \operatorname{Hess} H(\phi)$, so $C_H \geq S_2^d$ because $\operatorname{tr} \operatorname{Hess} H(\phi) \geq Nc > 0$. (For $T = G^\perp$ the trace argument is not available; in the applications below C_H is an explicit bound of the form $S_2^{d-1}(S_2 + \cdots) \geq S_2^d$, and any number at least the best constant in (29) is an admissible C_H.) □

Remark 120. (a) For a Gaussian, $\operatorname{Hess} H = Q$ is constant and (29) is the stable decomposition hypothesis of Lemma 101 for the specific decomposition $v_x = \chi_x v$ (with $C_H = C_0(Q;T)$), so the theorem recovers Theorems 104 and 107 with the same constants. The chaos decomposition of Theorem 97 is replaced by the Helffer–Sjöstrand representation, which plays the same role: it turns a statement about all functions into a statement about vector fields, on which the cubes act by multiplication.

(b) When the V_x form a partition of the coordinates into M blocks and $\chi_x = \mathbf{1}_{V_x}$, $m_2 = 1$ and (29) says that the block-diagonal part $D(\phi)$ of $\operatorname{Hess} H(\phi)$ satisfies $D(\phi) \leq C_H \operatorname{Hess} H(\phi)$; the trace argument gives $C_H \geq 1$. The theorem then states that the random-scan block Gibbs sampler has spectral gap at least $1/(C_H M)$. Ascolani, Lavenant and Zanella [3, Theorem 3.2 and Corollary 3.7] prove, for strongly log-concave targets, entropy contraction at the rate $1/(\kappa^* M)$ and the same Poincaré bound with their coordinate-wise condition number κ^*, defined through block smoothness constants L_m with $(\operatorname{Hess} H)_{mm} \leq L_m I$ and $\operatorname{Hess} H \geq (\kappa^*)^{-1} \operatorname{diag}(L_m)$; since $D(\phi) \leq \operatorname{diag}(L_m)$, one has $C_H \leq \kappa^*$, and C_H may be taken pointwise in ϕ. For Gaussians the bound $1/(C_H M)$ with the best C_H is exact (Lemma 100). The overlapping cubes with the smooth partition of unity are what makes the constant uniform in n: for a partition into disjoint cubes of side n the constant of the free field is at most $s^2/(s^2 + 2dn)$, as the vector $v = \mathbf{1}$ in the best constant shows, because cutting a constant field along hard boundaries costs $2d/n$ per site against μ^2.

(c) The theorem needs the hypotheses only for the Hessian at every point; the exterior configuration enters nowhere, so the constant is uniform in the volume, and in the cube side as soon as C_H is.

Corollary 121 (zeroth-order convex perturbations)**.** *Let $\mu_0 = \gamma_{Q_s}$ be the free scalar field and $\mu = Z^{-1} e^{-V} \mu_0$ with $V(\phi) = \sum_{y \in \mathbb{T}} v_y(\phi_y)$, each $v_y \in C^\infty(\mathbb{R})$ convex (for instance $v_y(t) = \lambda t^4 + \kappa t^2$ with $\lambda, \kappa \geq 0$, or any convex polynomial). Then μ*

satisfies $\mathrm{BP}(n,\varpi)$ *on* $L^2(\mu)$ *with the constant of Theorem 104,*

$$\varpi = \alpha_n^{-d}\,\frac{s^2}{s^2+10d\rho_n} \geq c_d(s),$$

uniformly in L, in n and in the potentials v_y*. More generally, if* $R(\phi)\geq 0$ *is a diagonal matrix for every* ϕ *and* $\operatorname{Hess} H = Q_0 + R$*, then* $C_H \leq \max(C_0(Q_0), S_2^d) = C_0(Q_0)$ *for the parabolic partition of unity.*

Proof. $H = \frac{1}{2}\langle\phi, Q_{\mathrm{s}}\phi\rangle + V$ has $\operatorname{Hess} H = Q_{\mathrm{s}} + R$ with $R = \operatorname{diag}(v_y''(\phi_y)) \geq 0$, so $\operatorname{Hess} H \geq \mu^2$. For diagonal $R \geq 0$, $\sum_x \chi_x R \chi_x = R\sum_x \chi_x^2 = S_2^d R$, and Lemma 102 gives, for all g,

$$\sum_x \langle \chi_x g, (Q_{\mathrm{s}}+R)\chi_x g\rangle \leq C_0\langle g, Q_{\mathrm{s}} g\rangle + S_2^d\langle g, Rg\rangle \leq C_0\langle g, (Q_{\mathrm{s}}+R)g\rangle,$$

with $C_0 = S_2^{d-1}(S_2 + dD_2/\mu^2) \geq S_2^d$. Theorem 119 with $T = \mathbb{R}^N$ and $C_H = C_0$ gives $\varpi \geq 1/(n^d C_0)$, which is the constant of Theorem 104. The integrability of e^{-H} and the moments needed for (28) hold because a convex V is bounded below by an affine function. □

Corollary 122 (two-sided Hessian bounds)**.** *Let* Q_0 *be positive definite,* $K \geq 1$*, and let* $\pi = Z^{-1}e^{-H}d\phi$ *with* $H \in C^\infty$ *and*

$$Q_0 \leq \operatorname{Hess} H(\phi) \leq K Q_0 \qquad (\phi \in \mathbb{R}^N).$$

Then π *satisfies* $\mathrm{BP}(n,\varpi)$ *on* $L^2(\pi)$ *with* $\varpi = 1/(n^d K C_0(Q_0))$*. If moreover* $\operatorname{Hess} H(\phi)$ *commutes with the projection onto a subspace G for every* ϕ*, then* $\mathrm{BP}(n,\varpi_G)$ *holds on* L^2_G *with*

$$\varpi_G = \frac{1}{n^d \max\bigl(S_2^d,\, K C_0(Q_0; G^\perp)\bigr)}.$$

For the free scalar and Proca fields, with the explicit bounds of Lemmas 102 and 106 in place of C_0*, this gives the constants of Theorems 104 and 107 divided by K.*

Proof. For $g \in T$ ($T = \mathbb{R}^N$ or $G^\perp$), the upper bound at the vectors $\chi_x g$, which need not lie in T, and the lower bound at g give

$$\sum_x \langle \chi_x g, \operatorname{Hess} H \chi_x g\rangle \leq K\sum_x \langle \chi_x g, Q_0 \chi_x g\rangle \leq K C_0(Q_0;T)\langle g, Q_0 g\rangle \leq K C_0(Q_0;T)\langle g, \operatorname{Hess} H g\rangle,$$

and $\operatorname{Hess} H \geq Q_0 \geq \lambda_{\min}(Q_0) > 0$. Apply Theorem 119 with $C_H = K C_0(Q_0;T)$, or with $C_H = K\bar{C}$ for any upper bound $\bar{C}$ of $C_0(Q_0;T)$; for $T = \mathbb{R}^N$, $C_0(Q_0) \geq S_2^d$ by the trace argument in that proof, and the explicit bounds of the two lemmas are of the form $S_2^{d-1}(S_2 + \cdots) \geq S_2^d$. □

Corollary 123 (convex plaquette perturbations of the Proca field)**.** *Let* $d \geq 2$*,* $\mu_0 = \gamma_{Q_{\mathrm{P}}}$ *the free Proca field, and* $\mu = Z^{-1}e^{-V}\mu_0$ *with* $V(A) = \sum_{p\in P} v_p\bigl((d_1 A)(p)\bigr)$*, each* $v_p \in C^\infty(\mathbb{R})$ *convex with* $v_p'' \leq K$*. Then V is gauge-invariant and* μ *satisfies* $\mathrm{BP}(n,\varpi)$ *on the gauge-invariant functions with*

$$\varpi = \frac{1}{1+K}\,\alpha_n^{-d}\,\frac{s^2}{s^2+10d\rho_n} \geq \frac{c_d(s)}{1+K},$$

uniformly in L and n.

Proof. $\operatorname{Hess} V(A) = d_1^{\mathsf{T}} D(A) d_1$ with $D = \operatorname{diag}(v_p''((d_1A)(p)))$, $0 \leq D \leq K$; it vanishes on $G = d_0\mathbb{R}^{\mathbb{T}}$ and maps into $G^\perp$, so it commutes with the projection onto G, as does Q_{P} by Lemma 105; and $\operatorname{Hess} H = Q_{\mathrm{P}} + \operatorname{Hess} V \geq \mu^2$. For transverse g,

$$\sum_x \langle \chi_x g, \operatorname{Hess} V \chi_x g\rangle = \sum_x \|D^{1/2} d_1(\chi_x g)\|^2 \leq K\sum_x \|d_1(\chi_x g)\|^2 \leq K\sum_x \langle \chi_x g, Q_{\mathrm{P}}\chi_x g\rangle,$$

so by Lemma 106 (second bound)

$$\sum_x \langle \chi_x g, \operatorname{Hess} H \chi_x g\rangle \leq (1+K)C_0\langle g, Q_{\mathrm{P}} g\rangle \leq (1+K)C_0\langle g, \operatorname{Hess} H g\rangle,$$

$C_0 = S_2^{d-1}(S_2 + dD_2/\mu^2) \geq S_2^d = m_2$. Theorem 119 with $T = G^\perp$ and $C_H = (1+K)C_0$ gives the claim, since $n^d C_0 = \alpha_n^d(1 + 10d\rho_n s^{-2})$. □

The plaquette weight of the Wilson action, $v(\theta) = \beta(1-\cos\theta)$, has $v'' = \beta\cos\theta$ and is excluded by the corollary twice: it is not convex, and the Proca mass is absent. Both failures are structural, and Sec. X C 5 makes them precise.

4. *Bounded perturbations*

Proposition 124 (global Holley–Stroock bound)**.** *Let* μ_0 *satisfy* $\mathrm{BP}(n,\varpi_0)$ *on* $L^2(\mu_0)$ *and let* $\mu = Z^{-1}e^{-W}\mu_0$ *with W bounded. Let* $\omega = \sup W - \inf W$ *and*

$$\omega_{\mathrm{loc}} = \max_x \sup_{\phi_{V_x^c}} \operatorname{osc}_{\phi_{V_x}} W \leq \omega,$$

the largest oscillation of W over a cube at fixed exterior. Then μ *satisfies* $\mathrm{BP}(n, e^{-2\omega-\omega_{\mathrm{loc}}}\varpi_0)$ *on* $L^2(\mu)$*.*

Proof. The density $d\mu/d\mu_0 = e^{-W}/\mu_0(e^{-W})$ lies in $[e^{-\omega}, e^{\omega}]$, and the conditional densities given the exterior of V_x are $e^{-W}/\mu_{0,x}(e^{-W})$, in $[e^{-\omega_{\mathrm{loc}}}, e^{\omega_{\mathrm{loc}}}]$. Since $\operatorname{Var}_\nu(f) = \inf_c \nu((f-c)^2)$, a density in $[e^{-\omega}, e^{\omega}]$ changes variances by at most the factor $e^{\pm\omega}$. Hence

$$\operatorname{Var}_\mu f \leq e^{\omega}\operatorname{Var}_{\mu_0} f \leq \frac{e^{\omega}}{\varpi_0 n^d}\sum_x \mu_0\bigl(\operatorname{Var}_x^{\mu_0} f\bigr) \leq \frac{e^{2\omega}}{\varpi_0 n^d}\sum_x \mu\bigl(\operatorname{Var}_x^{\mu_0} f\bigr) \leq \frac{e^{2\omega+\omega_{\mathrm{loc}}}}{\varpi_0 n^d}\sum_x \mu\bigl(\operatorname{Var}_x^{\mu} f\bigr).$$

□

The global oscillation ω grows with the volume for a local perturbation $W = \sum_p W_p$, and the proposition is useless on large tori. Only the local oscillation ω_{loc}, of order $n^d \max_p \operatorname{osc} W_p$, is uniform in the volume, and one may ask for a bound $\varpi(\mu) \geq F(d,n,\omega_{\mathrm{loc}})\varpi(\mu_0)$. No such bound exists.

Proposition 125 (product measures)**.** *Every product probability measure on a finite product of measurable spaces indexed by the coordinates satisfies* $\mathrm{BP}(n,1)$ *on* L^2*, for every family* (V_x) *in which every coordinate lies in at least* n^d *sets.*

Proof. Let $\mathscr{E}_i$ be the expectation in the i-th coordinate. For $f \in L^2$ and $S \subset \{1,\dots,N\}$ put $f_S = \prod_{i\in S}(I-\mathscr{E}_i)\prod_{i\notin S}\mathscr{E}_i f$; the operators commute, $f = \sum_S f_S$, the f_S are mutually orthogonal, $f_\emptyset = \mathbb{E} f$ and $\mathscr{P}_x f_S = f_S$ if $S \cap V_x = \emptyset$ while $\mathscr{P}_x f_S = 0$ otherwise (it contains a factor $\mathscr{E}_i(I-\mathscr{E}_i)=0$). So $\|f-\mathscr{P}_x f\|^2 = \sum_{S\cap V_x\neq\emptyset}\|f_S\|^2$ and $\sum_x\|f-\mathscr{P}_x f\|^2 = \sum_{S\neq\emptyset}\|f_S\|^2 \#\{x : V_x\cap S\neq\emptyset\} \geq n^d \operatorname{Var} f$. □

Proposition 126 (no local Holley–Stroock bound)**.** *Let* $d \geq 3$*,* $L \geq n+1$ *even, and let* μ_β *be the Ising measure on* $\{\pm1\}^{\mathbb{T}}$*,* $\mu_\beta(\sigma) \propto \exp(\beta\sum_{\langle yz\rangle}\sigma_y\sigma_z)$*, the sum over the nearest-neighbour pairs of the torus. Then* $\mu_\beta = Z^{-1}e^{-W}\mu_0$ *with* μ_0 *the uniform measure, which satisfies* $\mathrm{BP}(n,1)$*, and* $W = -\beta\sum_{\langle yz\rangle}\sigma_y\sigma_z$*, whose local oscillation is at most* $2\beta d(n^d + n^{d-1})$*. There is* $\beta_1(d) < \infty$ *such that for* $\beta \geq \beta_1(d)$ *and every even* L*,*

$$\varpi(\mu_\beta) \leq \frac{2n^d}{L^d};$$

one may take $\beta_1(3) = 6.7$ *and* $\beta_1(4) = 8.6$*. Consequently there is no function* F *with* $\varpi(e^{-W}\mu_0) \geq F(d,n,\omega_{\mathrm{loc}}(W))\,\varpi(\mu_0)$ *on all tori: the cube condition is not stable under bounded local perturbations with constants uniform in the volume.*

Proof. The uniform measure is a product measure, so Proposition 125 gives $\mathrm{BP}(n,1)$; a cube meets at most $d(n^d+n^{d-1})$ bonds, each contributing oscillation 2β. Let $M = \sum_y \sigma_y$ and take $f = M$ in (26). Given the exterior of V_x, M varies only through $\sum_{y\in V_x}\sigma_y \in [-n^d, n^d]$, so $\operatorname{Var}_x^{\mu_\beta} M \leq n^{2d}$ and $n^{-d}\sum_x \mu_\beta(\operatorname{Var}_x M) \leq L^d n^d$. For the variance we use the infrared bound of Fröhlich, Simon and Spencer [50], in the torus form of [26, Theorem 3.3 and Corollary 3.4]: for L even, $\hat\sigma(k) = \sum_y \sigma_y e^{ik\cdot y}$ and $k \in \mathbb{T}^* \setminus \{0\}$, $\mathbb{T}^* = \frac{2\pi}{L}\{0,\dots,L-1\}^d$,

$$\mu_\beta\big(|\hat\sigma(k)|^2\big) \leq \frac{L^d}{2\beta D(k)}, \qquad D(k) = \sum_{j=1}^{d}(1-\cos k_j). \tag{33}$$

The constant depends on the normalisation of the Hamiltonian: [26] writes the bound for $-\frac{1}{2d}\sum_{\langle yz\rangle}\sigma_y\sigma_z$ at inverse temperature $2d\beta$, with $1-\hat J(k) = D(k)/d$, and in the present normalisation (33) is the second-order expansion at $h = \cos(k\cdot y)$ and $h = \sin(k\cdot y)$ of Gaussian domination, $\mu_\beta\big(e^{\beta\langle\sigma,-\Delta h\rangle}\big) \leq e^{\frac{\beta}{2}\langle h,-\Delta h\rangle}$, for which $\mu_\beta(\langle\sigma,h\rangle^2) \leq \|h\|^2/(2\beta D(k))$ when h is an eigenvector of $-\Delta$ with eigenvalue $2D(k)$. By Parseval, $\sum_{k\in\mathbb{T}^*}|\hat\sigma(k)|^2 = L^d\sum_y\sigma_y^2 = L^{2d}$, and $\hat\sigma(0) = M$, so

$$\mu_\beta(M^2) \geq L^{2d}\Big(1-\frac{G_L}{2\beta}\Big), \qquad G_L = \frac{1}{L^d}\sum_{k\in\mathbb{T}^*\setminus\{0\}}\frac{1}{D(k)}.$$

We bound G_L uniformly in L. For $k \in \mathbb{T}^*$ identify k with its representative in $(-\pi,\pi]^d$; then $1-\cos k_j \geq 2k_j^2/\pi^2$, so $D(k) \geq 2|k|^2/\pi^2$. For $k \neq 0$ let B_k be the cube of side $2\pi/L$ centred at k; the B_k are disjoint, contained in $[-2\pi,2\pi]^d$, and for $k' \in B_k$, $|k'| \leq |k| + \sqrt{d}\,\pi/L \leq (1+\sqrt{d}/2)|k|$ because $|k| \geq 2\pi/L$. Hence

$$G_L \leq \frac{\pi^2}{2}\Big(1+\frac{\sqrt{d}}{2}\Big)^2\frac{1}{(2\pi)^d}\int_{[-2\pi,2\pi]^d}\frac{dk'}{|k'|^2} =: I_d,$$

which is finite for $d \geq 3$; with $k' = 2\pi u$, $I_d = \frac{1}{8}(1+\frac{\sqrt{d}}{2})^2\int_{[-1,1]^d}|u|^{-2}du$, and the integral is 15.35 for $d=3$ and 17.15 for $d=4$, so $I_3 < 6.7$ and $I_4 < 8.6$. With $\beta_1(d) = I_d$, for $\beta \geq \beta_1$ we get $G_L/(2\beta) \leq \frac12$ and $\operatorname{Var}_{\mu_\beta}(M) = \mu_\beta(M^2) \geq L^{2d}/2$, using $\mu_\beta(M) = 0$ by the spin-flip symmetry. Then (26) with $f = M$ forces $\varpi \leq L^d n^d/(L^{2d}/2) = 2n^d/L^d$. The last claim follows by letting $L \to \infty$ at fixed n and β, since $\omega_{\mathrm{loc}}(W)$ and $\varpi(\mu_0) = 1$ do not depend on L. □

What survives for bounded perturbations is smallness measured against the Hessian, not against the oscillation: if $W \in C^\infty$ satisfies $-(1-\varepsilon)Q_0 \leq \operatorname{Hess} W \leq KQ_0$, then $\varepsilon Q_0 \leq \operatorname{Hess}(\frac12\langle\phi,Q_0\phi\rangle + W) \leq (1+K)Q_0$ and Corollary 122 gives $\mathrm{BP}(n,\varepsilon/((1+K)n^d C_0(Q_0)))$. For the massive field $Q_0 \geq \mu^2 = (ma)^2$, so a bounded perturbation with $\|\operatorname{Hess} W\|_{\mathrm{op}} \leq \delta$ is admissible by this route only when $\delta < (ma)^2$: in the continuum tuning the perturbation must be of order a^2 in lattice units, which is the lattice form of the statement that such a perturbation is irrelevant. Perturbations of order one in lattice units with small oscillation per term are the domain of the Dobrushin–Shlosman condition [41, 43], which the paper uses in Proposition 65; Proposition 126 shows that smallness of the oscillation per cube cannot replace it.

5. *The lattice gauge field*

We now place the Wilson measure against the hypotheses of Theorem 119. For definiteness we take the group $\mathrm{U}(1)$, with link variables $\theta \in (\mathbb{R}/2\pi\mathbb{Z})^E$, plaquette angles $\theta_p = (d_1\theta)(p)$ and action

$$H_\beta(\theta) = \beta\sum_{p\in P}(1-\cos\theta_p), \qquad \mu_\beta \propto e^{-H_\beta}\prod_e d\theta_e,$$

on the torus of Sec. X B with $d \geq 2$; the remarks on $\mathrm{SU}(N)$ follow the statements. A *linear gauge slice* is a subspace $S \subset \mathbb{R}^E$ with $S \oplus G = \mathbb{R}^E$, $G = d_0\mathbb{R}^{\mathbb{T}}$ the pure gauge directions: the Landau slice $S = G^\perp = \ker\delta_0$ and the axial slices, on which the links of a maximal tree are set to zero (the tree links are in bijection with G, since gauge transformations modulo constants are determined by their values on a spanning tree), are examples. Let $F = \ker d_1 \cap \ker\delta_0$ be the flat transverse fields.

The two consequences of the Helffer–Sjöstrand representation isolated after (28) survive the loss of convexity on the torus. The generator $\mathscr{L} = -\Delta + \langle\nabla H_\beta,\nabla\rangle$ on $L^2(\mu_\beta)$ has compact resolvent and kernel the constants, since the configuration space is compact and the density is smooth and positive, so $u_f = \mathscr{L}^{-1}(f-\mu_\beta(f))$ exists for every smooth f; the form

$$a(G,G) = \sum_{e\in E}\mu_\beta\big(|\nabla G_e|^2\big) + \mu_\beta\big(\langle G, \operatorname{Hess} H_\beta\, G\rangle\big)$$

is the quadratic form of the Witten Laplacian on one-forms of the flat torus with the weight e^{-H_β}, which is $d_H^* d_H + d_H d_H^*$ by the Bochner formula and hence nonnegative for every H_β; and the intertwining relation gives $a(\nabla u_f, G) = \mu_\beta(\langle \nabla f, G\rangle)$ for every G in the form domain. Hence, with $G_f := \nabla u_f$,

$$\begin{aligned}\operatorname{Var}_{\mu_\beta} f &= 2\mu_\beta(\langle \nabla f, G_f\rangle) - a(G_f, G_f)\\ &\geq 2\mu_\beta(\langle \nabla f, G\rangle) - a(G,G) \end{aligned} \tag{34}$$

for every such G, and the same holds for the conditional laws on the cubes, whose Hamiltonians are the restrictions of H_β, with the forms a_{V_x} defined as before. What is lost is only the lower bound $a \geq c$ and the invertibility of $\mathscr{A}$: by Proposition 127(iv) below the constant flat fields lie in its kernel.

Proposition 127 (the Hessian of the Wilson action)**.** *Let $d \geq 2$, $L \geq 3$ and $\beta > 0$.*

(i) *The Hessian is $\operatorname{Hess} H_\beta(\theta) = \beta\, d_1^{\mathsf{T}} D_\theta d_1$ with $D_\theta = \operatorname{diag}(\cos\theta_p)$; it vanishes on G and on F at every θ, and F is the d-dimensional space of the constant fields $A(y,i) = c_i$.*

(ii) *$\operatorname{Hess} H_\beta(0) = \beta\delta_1 d_1$, and $\ker \operatorname{Hess} H_\beta(0) = G \oplus F$. For every linear gauge slice S the restriction of $\operatorname{Hess} H_\beta(0)$ to S has a kernel of dimension at least d. Hence there is no positive definite Q_0 on S, nor on $G^\perp$, with $\operatorname{Hess} H_\beta \geq Q_0$ on a neighbourhood of 0, for any β.*

(iii) *For every $\beta > 0$ the set of θ at which $\operatorname{Hess} H_\beta(\theta)$ has a negative eigenvalue on $G^\perp$ is open and has positive μ_β measure; it contains every θ at which the $2(d-1)$ plaquettes through some link e have $\sum_{p\ni e}\cos\theta_p < 0$.*

(iv) *For every $\beta > 0$, $c_\beta = \mu_\beta(\cos\theta_p) > 0$, and for the constant field h in direction 1, $h(y,i) = \delta_{i1}$, the Witten form of μ_β and its cube restrictions satisfy*

$$a(h,h) = \mu_\beta\big(\langle h, \operatorname{Hess} H_\beta h\rangle\big) = 0,$$
$$\sum_x \mu_\beta\big(a_{V_x}(\chi_x h, \chi_x h)\big) = \beta c_\beta (d-1) S_2^{d-1} D_2 L^d > 0.$$

Proof. (i) H_β is a function of $d_1\theta$, so $\nabla H_\beta = \beta d_1^{\mathsf{T}} \sin(d_1\theta)$ and $\operatorname{Hess} H_\beta = \beta d_1^{\mathsf{T}} \operatorname{diag}(\cos\theta_p) d_1$, which vanishes on $\ker d_1 \supset G \cup F$. By the Hodge identity of Lemma 105, $A \in F$ has $\sum_i \|\nabla A_i\|^2 = 0$, so each A_i is constant on the connected torus; conversely constant fields have $d_1 A = 0 = \delta_0 A$. So F is the space of constants, of dimension d.

(ii) At $\theta = 0$ all cosines are 1. $\ker \delta_1 d_1 = \ker d_1$ and $\ker d_1 = G \oplus F$: indeed $G \subset \ker d_1$, $F \subset \ker d_1$, $G \perp F$, and for $A \in \ker d_1$ the Hodge decomposition $A = P_G A + P_{G^\perp} A$ has $d_1 P_{G^\perp} A = d_1 A - d_1 P_G A = 0$ and $\delta_0 P_{G^\perp} A = 0$, so $P_{G^\perp} A \in F$. For a slice S, $\dim(S \cap \ker d_1) = \dim S + \dim\ker d_1 - \dim(S + \ker d_1) \geq (N - \dim G) + (\dim G + d) - N = d$, as $S + \ker d_1 \supset S + G = \mathbb{R}^E$; and $S \cap \ker d_1 \subset \ker \operatorname{Hess} H_\beta(0)|_S$. A bound $\operatorname{Hess} H_\beta \geq Q_0$ near 0 would give $Q_0 \leq 0$ on this kernel; on $G^\perp$ the kernel contains $F \neq \{0\}$.

(iii) For a link e and θ as stated, $\langle \delta_e, \operatorname{Hess} H_\beta \delta_e\rangle = \beta \sum_{p\ni e}\cos\theta_p < 0$, each plaquette through e containing e with coefficient ± 1. Since $\operatorname{Hess} H_\beta$ is symmetric and vanishes on G, $\langle P_{G^\perp}\delta_e, \operatorname{Hess} H_\beta P_{G^\perp}\delta_e\rangle = \langle \delta_e, \operatorname{Hess} H_\beta \delta_e\rangle < 0$, and $P_{G^\perp}\delta_e \neq 0$ because $\delta_e \notin G$ ($d_1\delta_e \neq 0$). The set is open as $\operatorname{Hess} H_\beta$ is continuous in θ, nonempty (take $\theta_e = \pi$ and all other links zero, so that every plaquette through e has angle $\pm\pi$), and the density of μ_β is positive everywhere.

(iv) $\log Z(\beta) = \log \int e^{\beta \sum_p \cos\theta_p} d\theta$ is convex in β with derivative $|P|\,\mu_\beta(\cos\theta_p)$ (all plaquettes are equivalent under the translations and the hypercubic group, and μ_β is invariant), and its derivative at 0 is $|P| \int \cos\theta_p\, d\theta = 0$; the second derivative is $\operatorname{Var}_{\mu_\beta}(\sum_p \cos\theta_p) > 0$ since the plaquette angles are not constant, so the derivative is strictly increasing and $c_\beta > 0$ for $\beta > 0$. The constant vector field h has $\nabla h = 0$, so $a(h,h) = \mu_\beta(\langle h, \operatorname{Hess} H_\beta h\rangle) = \beta\mu_\beta(\sum_p \cos\theta_p (d_1 h)_p^2) = 0$ by (i), and likewise $a_{V_x}(\chi_x h, \chi_x h) = \beta \sum_p \cos\theta_p \big(d_1(\chi_x h)\big)_p^2$, whose expectation is $\beta c_\beta \|d_1(\chi_x h)\|^2$. Now $(\chi_x h)(y,i) = \chi_x(y)\delta_{i1}$, so $d_1(\chi_x h)(y,1,j) = \chi_x(y) - \chi_x(y+e_j) = -\nabla_j \chi_x(y)$ for $j \neq 1$ and $d_1(\chi_x h)(y,i,j) = 0$ for $i, j \neq 1$. By (21), $\sum_x (\nabla_j \chi_x(y))^2 = 2S_2^d - 2S_2^{d-1} S_{11} = S_2^{d-1} D_2$, and summing over the $(d-1)L^d$ pairs (y,j) gives the claim. □

For $\mathrm{SU}(N)$ the link variables are $U_e = e^{X_e}$, $X_e \in \mathfrak{su}(N)$, and the Hessian of $\beta \sum_p (1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p)$ at the trivial configuration is $\frac{\beta}{N}\delta_1 d_1$ on each of the $N^2 - 1$ components with respect to a basis of $\mathfrak{su}(N)$ orthonormal for $-\operatorname{Tr}(XY)$, so (ii) holds verbatim with $\dim F = d(N^2-1)$; the statement at the trivial configuration is a statement about the Maxwell form and does not see the group. The paper's Sec. VI records the analogue of (iii) for $\mathrm{SU}(2)$: the Bakry–Émery tensor is negative at large fields once $\beta_W > \frac{1}{3}$.

Corollary 128 (the positive theorems do not apply to the Wilson measure)**.** *For every $\beta > 0$, every $d \geq 2$ and every linear gauge slice S, the Wilson action restricted to S violates each hypothesis of Theorem 119 and of Corollaries 122 and 123: it is not convex on an open set of positive measure, and even where it is convex its Hessian has no positive lower bound, since $S \cap \ker d_1$, of dimension at least d, lies in the kernel of $\operatorname{Hess} H_\beta(\theta)|_S$ at every θ. Moreover, for the configuration-independent test field h of Proposition 127(iv) the inequality*

$$a(G,G) \geq \lambda \sum_x \mu_\beta\big(a_{V_x}(\chi_x G, \chi_x G)\big) \tag{35}$$

fails for every $\lambda > 0$, so that (35) cannot hold for all fields G, for any partition of unity and any β.

Proof. Proposition 127(ii), (iii) and (iv); for the kernel at every θ use that $\ker d_1 \subset \ker \operatorname{Hess} H_\beta(\theta)$ by (i) and the dimension count of (ii). For the last claim note that (35) at $G = h$ reads $0 \geq \lambda\beta c_\beta (d-1) S_2^{d-1} D_2 L^d > 0$; any other partition of unity with $\sum_x \chi_x = I$ and χ_x supported in V_x cuts the constant field h along the boundary of every cube, so that $\sum_x \|d_1(\chi_x h)\|^2 > 0$ as well. □

Inequality (35) for all G is sufficient for the cube condition, by the proof of Theorem 119 with (34) in place of (28), and it is not necessary: the field h is the gradient of no function on the torus, since $\nabla u = h$ would force $u(\theta + th) = u(\theta) + t|h|^2$, and

the generator part of the Witten form, $\sum_e \mu_\beta(|\nabla G_e|^2)$, vanishes on configuration-independent fields. So the corollary does not refute the cube condition for the Wilson measure. It locates the estimate that any proof along these lines must supply.

Proposition 129 (the quadratic approximation has no mass)**.** *The quadratic approximation of the Wilson action at the trivial configuration is the quadratic form $\beta\delta_1 d_1$, the free Maxwell action; it is degenerate on F, so it defines no Gaussian measure, even on a gauge slice. For its massive regularization, the Gaussian with precision $\beta(\delta_1 d_1+\mu^2)$, Proposition 108(ii) gives, on gauge-invariant functions and for every β,*

$$\varpi \le \frac{(n\mu)^2}{1+(n\mu)^2},$$

which tends to 0 as $\mu\to 0$. Consequently, along a continuum tuning with $n_k a_k=\ell$, a Gaussian approximation of this form satisfies $\mathrm{BP}(n_k,\varpi)$ uniformly in k only if its mass μ_k satisfies $\mu_k \ge \sqrt{\varpi/(1-\varpi)}\,a_k/\ell$, that is, only if it carries a physical mass $m\ge\sqrt{\varpi/(1-\varpi)}/\ell$; and the Wilson action has no bare mass term.

Proof. The precision $\beta(\delta_1 d_1+\mu^2)$ is a multiple of Q_{P}, and ϖ is homogeneous of degree zero in the precision (proof of Theorem 117), so Proposition 108(ii) applies. Solving $(n\mu)^2/(1+(n\mu)^2)\ge\varpi$ gives $n\mu\ge\sqrt{\varpi/(1-\varpi)}$, and $n_k\mu_k=\ell\mu_k/a_k$. □

Remark 130 (where a dynamical mass would have to enter). Theorem 64 shows that $\mathrm{BP}(n_k,\varpi)$ along the tuning gives a mass gap $m_*=16\varpi/(625e^2\ell)$, and Proposition 129 with Theorem 107 shows that in the Gaussian class with precision $\beta(\delta_1 d_1+\mu^2)$ the constant ϖ is of the order of the squared mass in units of the cube when that mass is at most one. For the Wilson measure the constant therefore has to be produced by what is absent from the quadratic approximation. Proposition 127 says where it cannot come from: not from a Hessian lower bound (there is none, in any gauge, at any β), and not from the potential term of the Witten form on flat directions, which vanishes identically. In the Helffer–Sjöstrand representation (34) the cube condition on gauge-invariant functions is implied by the following statement, which is the proof of Theorem 119 with configuration-dependent decompositions: there is $\lambda>0$ such that for every smooth gauge-invariant f the field $G_f=\nabla\mathscr{L}^{-1}(f-\mu_\beta(f))$ admits a decomposition $G_f=\sum_x G_f^{(x)}$ into fields in the form domain with $G_f^{(x)}(\theta)$ supported in V_x for every θ, such that

$$\sum_x \mu_\beta\big(a_{V_x}(G_f^{(x)},G_f^{(x)})\big)\le\lambda^{-1}a(G_f,G_f); \tag{36}$$

then $\mathrm{BP}(n,\lambda/n^d)$ holds, since that proof used (29) and m_2 only to produce such a decomposition. Corollary 128 shows that the decomposition $G^{(x)}=\chi_x G$, which under the hypotheses of Theorem 119 works for every field at once, cannot be controlled for the Wilson measure by a bound of the form (35) valid for all fields: a field that is constant in θ and flat in space has $a(G,G)=0$ and positive cube energy. Whether (36) holds with $G_f^{(x)}=\chi_x G_f$ for the fields G_f that actually occur is not decided here; these fields are orthogonal in $L^2(\mu_\beta;\mathbb{R}^E)$ to the constant flat fields, since $\mu_\beta(\langle\nabla u,h\rangle)=\mu_\beta(u\,\partial_h H_\beta)=0$ for flat h, but not pointwise in θ. If the flat part of G_f is what the cut costs, the term $\sum_e\mu_\beta(|\nabla_\theta G_e|^2)$ of the Witten form, the fluctuation of the field as a function of the configuration, has to pay for the curvature created at the cube boundaries. By Proposition 127(iv) that curvature costs $\beta c_\beta(d-1)S_2^{d-1}D_2$ per unit of $|h|^2$ for the flat field h. In the Gaussian class the same cut is paid by the mass term, which contributes $\beta\mu^2 S_2^d$ per unit of $|h|^2$ to the left side of (29) for the precision $\beta(\delta_1 d_1+\mu^2)$, and the ratio of the two rates is $(d-1)c_\beta D_2/(\mu^2 S_2)=(d-1)c_\beta\beta_n/(\alpha_n s^2)$, with $\beta_n/\alpha_n\to 10$ and $s=n\mu=m\ell$ along the tuning; so the mass term dominates the cut when $s^2\ge 10\rho_n(d-1)c_\beta$, that is $m\ell\gtrsim\sqrt{10(d-1)c_\beta}$, a requirement of the same form as that of Proposition 129. This is a form, in the present framework, of the heuristic statement that the mass must be dynamical: what is missing is a lower bound on the Dirichlet form of the Langevin generator of μ_β on the components of the one-forms G_f, uniform in the volume and in the coupling along the tuning, which on the flat directions plays the role of the mass term $(ma)^2$ of the Proca field. No such bound is known; the published control of the lattice measure is either at strong coupling, where the Bakry–Émery tensor is positive and the mass is of order $1/a$ (Sec. VI, [96]), or in the small-field region after a large-field split [13], where the quadratic approximation is massless. Both are consistent with Proposition 129: a lattice-unit mass gives $(n\mu)^2\to\infty$, a massless approximation gives 0.

D. The averaged cube estimate at weak coupling: necessary conditions, the free field, and the plaquette

Throughout, G, φ, $\Lambda=\Lambda_L$, E, P, μ, V_x, X, $\mathscr{P}_x$ and $\mathscr{A}$ are as in Sec. IX F, $n\ge 2$, so that $X=\Lambda_L$ on the torus, and

$$\mathscr{E}_n(f)=\frac{1}{n^4}\sum_{x\in X}\mu\big((f-\mathscr{P}_x f)^2\big)$$

is the Dirichlet form of Definition 60, as in Sec. IX H, so that $\mathrm{BP}(n,\varpi)$ reads $\varpi\,\mathrm{Var}_\mu(f)\le\mathscr{E}_n(f)$ for real $f\in\mathscr{A}$. For $V\subset E$ let $P(V)$ be the set of plaquettes of P that contain a link of V, and $\mathscr{F}_V$ the σ-algebra generated by the links outside V; thus $\mathscr{P}_x$ is the conditional expectation given $\mathscr{F}_{V_x}$. In this subsection ∂ denotes the boundary operator on chains, the transpose of the coboundary d (the set ∂V of Sec. IX F is not used). The lattice is the torus $\Lambda_L=(\mathbb{Z}/L\mathbb{Z})^4$; coordinates are taken in $\mathbb{Z}/L\mathbb{Z}$, and when $L\ge n+2$ the residues $x_i-1,x_i,\dots,x_i+n$ are distinct, so that x_i-1 and x_i+n lie outside $x_i+[0,n)$. A plaquette $(y,\mu\nu)$, $\mu<\nu$, has the links (y,μ), $(y+e_\mu,\nu)$, $(y+e_\nu,\mu)$, (y,ν), with base points $y,y+e_\mu,y+e_\nu,y$, and $U_p=U_{(y,\mu)}U_{(y+e_\mu,\nu)}U^{-1}_{(y+e_\nu,\mu)}U^{-1}_{(y,\nu)}$.

1. The plaquettes read by one cube

Lemma 131 (layers). *Let $L \geq n+2$, $x \in \Lambda_L$, let $\{\mu,\nu\}$ be a plane and ρ,σ the two other directions. For $t=(t_\rho,t_\sigma) \in (x_\rho+[0,n)) \times (x_\sigma+[0,n))$ let Q_t be the set of the $(n+1)^2$ plaquettes $(y,\mu\nu)$ with $y_\rho = t_\rho$, $y_\sigma = t_\sigma$, $y_\mu \in x_\mu+[-1,n-1]$, $y_\nu \in x_\nu+[-1,n-1]$; let $q_t \in Q_t$ be the plaquette with $y_\mu = x_\mu-1$, $y_\nu = x_\nu-1$, and $R_t = Q_t \setminus \{q_t\}$.*

(i) *The plaquettes of $P(V_x)$ in the plane $\{\mu,\nu\}$ form the disjoint union of the n^2 sets R_t; each has $(n+1)^2-1$ elements. In particular $|P(V_x)| = 6n^2\big((n+1)^2-1\big)$, and for a fixed plaquette p the number of x with $p \in P(V_x)$ is $n^2((n+1)^2-1)$, while the number of x such that all four links of p lie in V_x is $n^2(n-1)^2$.*

(ii) *No link of q_t lies in V_x, so $q_t \notin P(V_x)$; every link (y,μ) with $y_\nu \in \{x_\nu-1, x_\nu+n\}$ and every link (y,ν) with $y_\mu \in \{x_\mu-1, x_\mu+n\}$ lies outside V_x. The boundary 1-chain $\partial R_t = \partial Q_t - \partial q_t$ is supported on such links and on the links of q_t, hence on links outside V_x.*

(iii) *If G is abelian, $\prod_{p\in R_t} U_p = \prod_e U_e^{(\partial R_t)_e}$ is a function of the links outside V_x, hence $\mathscr{F}_{V_x}$-measurable.*

Proof. (i) A plaquette $(y,\mu\nu)$ lies in $P(V_x)$ if and only if one of the base points $y, y+e_\mu, y+e_\nu$ lies in $x+[0,n)^4$. All three have the transverse coordinates y_ρ, y_σ, so $y_\rho \in x_\rho+[0,n)$ and $y_\sigma \in x_\sigma+[0,n)$ are necessary, which gives the n^2 values of t. In the plane, with $A = (x_\mu+[0,n)) \times (x_\nu+[0,n))$, the condition is $(y_\mu,y_\nu) \in A \cup (A-e_\mu) \cup (A-e_\nu)$. A point of $(x_\mu+[-1,n-1]) \times (x_\nu+[-1,n-1])$ lies in A if both coordinates are at least x_μ, resp. x_ν; in $A-e_\mu$ if $y_\mu = x_\mu-1$ and $y_\nu \geq x_\nu$; in $A-e_\nu$ if $y_\nu = x_\nu-1$ and $y_\mu \geq x_\mu$; and the point $(x_\mu-1, x_\nu-1)$ lies in none of the three. Conversely the union lies in that square. So the plaquettes of $P(V_x)$ in the plane with transverse position t are exactly R_t, of cardinality $(n+1)^2-1$. Summing over the six planes gives $|P(V_x)|$. For fixed p, the number of x with $p \in P(V_x)$ equals, by translation, the number of plaquettes of $P(V_0)$ in the plane of p, that is $n^2((n+1)^2-1)$. All four links of $p=(y,\mu\nu)$ lie in V_x if and only if $y, y+e_\mu, y+e_\nu \in x+[0,n)^4$, that is $x_\rho, x_\sigma \in y_\rho+(-n,0], y_\sigma+(-n,0]$ (n values each) and $x_\mu \in [y_\mu+2-n, y_\mu]$, $x_\nu \in [y_\nu+2-n, y_\nu]$ ($n-1$ values each), which gives $n^2(n-1)^2$.

(ii) The base point of a link (y,μ) is y; if $y_\nu \in \{x_\nu-1, x_\nu+n\}$ then $y \notin x+[0,n)^4$ since $L \geq n+2$. The same holds for (y,ν) with $y_\mu \in \{x_\mu-1, x_\mu+n\}$. The links of q_t are (q_t,μ), (q_t,ν) with base point $(x_\mu-1, x_\nu-1)$ in the plane, (q_t+e_μ,ν) with base point $(x_\mu, x_\nu-1)$, and (q_t+e_ν,μ) with base point $(x_\mu-1, x_\nu)$; each has a coordinate $x_\mu-1$ or $x_\nu-1$, so none lies in V_x. Since ∂ is linear and each link interior to the square Q_t lies in two of its plaquettes with opposite orientations, ∂Q_t is supported on the links (y,μ) with $y_\nu \in \{x_\nu-1, x_\nu+n\}$, $y_\mu \in x_\mu+[-1,n-1]$, and (y,ν) with $y_\mu \in \{x_\mu-1, x_\mu+n\}$, $y_\nu \in x_\nu+[-1,n-1]$; and $\partial R_t = \partial Q_t - \partial q_t$ is supported on these links and those of q_t.

(iii) For abelian G, $U_p = \prod_e U_e^{(\partial p)_e}$, so $\prod_{p\in R_t} U_p = \prod_e U_e^{(\partial R_t)_e}$, and by (ii) only links outside V_x occur. □

Lemma 132 (walls). *In the setting of Lemma 131, put $s = x_\rho+n-t_\rho \in [1,n]$ and $Q'_t = Q_t + s e_\rho$, the translate of Q_t to the transverse position $(x_\rho+n, t_\sigma)$. There is an integer 2-chain W_t, supported on plaquettes of the planes $\{\mu,\rho\}$ and $\{\nu,\rho\}$ that do not lie in $P(V_x)$, such that $Q'_t \cap P(V_x) = \emptyset$ and such that every real 2-cochain F on Λ_L with $dF=0$ satisfies*

$$\sum_{p\in R_t} F_p = \sum_{p\in Q'_t} F_p + \langle F, W_t\rangle - F_{q_t}.$$

If G is abelian and U_p are the plaquette variables of a configuration U, then

$$\prod_{p\in R_t} U_p = \prod_{p\in Q'_t} U_p \prod_p U_p^{(W_t)_p} U_{q_t}^{-1}.$$

In both cases the left side is a function of the plaquette variables of plaquettes outside $P(V_x)$.

Proof. For $z \in \Lambda_L$ let $c(z)$ be the 3-cell with corner z spanned by e_μ, e_ν, e_ρ, oriented so that

$$\partial c(z) = (z+e_\rho,\mu\nu) - (z,\mu\nu) + \big[(z,\mu\rho) - (z+e_\nu,\mu\rho)\big] + \big[(z+e_\mu,\nu\rho) - (z,\nu\rho)\big]$$

up to a common sign of the two brackets, which plays no role below. Put $T = \sum_{j=0}^{s-1} \sum_{z\in Q_t} c(z+je_\rho)$, where $z \in Q_t$ stands for the corner of the plaquette. In ∂T the $\mu\nu$-faces telescope in j and give $\sum_{z\in Q_t}\big[(z+se_\rho,\mu\nu)-(z,\mu\nu)\big] = 1_{Q'_t} - 1_{Q_t}$. For fixed j the $\mu\rho$-faces of the cells $c(z+je_\rho)$, $z \in Q_t$, are $(z+je_\rho,\mu\rho)$ and $(z+je_\rho+e_\nu,\mu\rho)$ with opposite signs; as z_ν runs over $x_\nu+[-1,n-1]$ the faces with $z_\nu \in x_\nu+[0,n-1]$ cancel against those of the neighbour, and the survivors have ν-coordinate $x_\nu-1$ or $x_\nu+n$. In the same way the surviving $\nu\rho$-faces have μ-coordinate $x_\mu-1$ or $x_\mu+n$. Let $W_t = \partial T - (1_{Q'_t} - 1_{Q_t})$ be the chain of these survivors. A plaquette $(y,\mu\rho)$ has links with base points $y, y+e_\mu, y+e_\rho$, all with ν-coordinate y_ν; if $y_\nu \in \{x_\nu-1, x_\nu+n\}$ none of them lies in $x+[0,n)^4$, so the plaquette is not in $P(V_x)$. The same holds for the $\nu\rho$-faces. A plaquette of Q'_t has all base points with ρ-coordinate $x_\rho+n$, so it is not in $P(V_x)$ either. Since $L \geq n+2$, the coordinates $x_i-1,\dots,x_i+n$ occurring in T are distinct residues, so no two cells of T coincide and the cancellations are as described.

If $dF=0$ then $\langle F,\partial T\rangle = \langle dF, T\rangle = 0$, that is $\sum_{Q_t} F_p = \sum_{Q'_t} F_p + \langle F, W_t\rangle$, and subtracting F_{q_t} gives the first identity. For abelian G and every 3-cell c, $\prod_p U_p^{(\partial c)_p} = 1$, since each link of c lies in two of its faces with opposite orientations; hence $\prod_p U_p^{(\partial T)_p} = 1$, which is the second identity after multiplication by $U_{q_t}^{-1}$. Lemma 131(ii) shows $q_t \notin P(V_x)$. □

2. The free Maxwell field: the averaged condition fails at the diffusive rate

Let C^k be the real k-cochains on Λ_L with the ℓ^2 pairing $\langle\cdot,\cdot\rangle$, d the coboundary and $\partial = d^{\mathrm{T}}$, so that $d\circ d = 0$ and $\langle dA, S\rangle = \langle A, \partial S\rangle$. Let $B = dC^1 \subset C^2$ be the exact 2-cochains and Π the orthogonal projection of C^2 onto B. For $\beta > 0$

let γ_β be the centred Gaussian measure on B with density proportional to $e^{-\beta\|F\|^2/2}$ with respect to Lebesgue measure on B; its covariance is

$$\mathbb{E}_{\gamma_\beta}\big(\langle c,F\rangle\langle c',F\rangle\big)=\beta^{-1}\langle \Pi c,\Pi c'\rangle,\qquad c,c'\in C^2.$$

This is the lattice Maxwell field: $F=dA$ for a 1-cochain A determined up to dC^0, and $\langle S,F\rangle=\langle \partial S,A\rangle$ depends on A through the links of ∂S only. For $V\subset E$ let $\mathscr{G}_V$ be the σ-algebra generated by the variables $\langle S,F\rangle$ with ∂S supported on $E\setminus V$: since $\ker d=(\operatorname{ran}\partial)^\perp$ on C^1, these are exactly the linear functionals of A that are functions of F and depend on the links outside V only, so $\mathscr{G}_{V_x}$ is the analogue for γ_β of the restriction of $\mathscr{F}_{V_x}$ to $\mathscr{A}$. Every function of F is gauge invariant, and for σ-algebras $\mathscr{G}_x$, $x\in\Lambda_L$, with conditional expectations P_x we say that $(\gamma_\beta,(\mathscr{G}_x))$ satisfies $\mathrm{BP}(n,\varpi)$ if $\varpi\operatorname{Var}(f)\le n^{-4}\sum_x\mathbb{E}|f-P_xf|^2$ for every $f\in L^2(\gamma_\beta)$.

Lemma 133 (the response of a cube)**.** *Let $V\subset E$, C_V^1 the 1-cochains supported on V, $D_V=dC_V^1\subset B$ and P_{D_V} the orthogonal projection onto D_V. Then for every $c\in C^2$,*

$$\mathbb{E}_{\gamma_\beta}\big|\langle c,F\rangle-\mathbb{E}_{\gamma_\beta}(\langle c,F\rangle\mid\mathscr{G}_V)\big|^2=\beta^{-1}\|P_{D_V}c\|^2.$$

Proof. Let $\mathscr{S}_V=\{S\in C^2:(\partial S)_e=0\text{ for }e\in V\}$ and $H_V=\{\langle S,F\rangle:S\in\mathscr{S}_V\}\subset L^2(\gamma_\beta)$. The variables $\langle S,F\rangle$, $S\in C^2$, are jointly Gaussian and centred, so the conditional expectation of $\langle c,F\rangle$ given the σ-algebra they generate for $S\in\mathscr{S}_V$ is its orthogonal projection onto the (finite-dimensional, hence closed) linear span H_V. By the covariance formula, $\langle b,F\rangle\perp H_V$ for $b\in B$ if and only if $\langle b,\Pi S\rangle=\langle b,S\rangle=0$ for all $S\in\mathscr{S}_V$, that is $b\in\mathscr{S}_V^\perp$. The map $S\mapsto(\partial S)|_V$ from C^2 to $\mathbb{R}^V$ has the adjoint $a\mapsto da$ on C_V^1, since $\langle\partial S,a\rangle=\langle S,da\rangle$; hence $\mathscr{S}_V^\perp=dC_V^1=D_V$, and $D_V\subset dC^1=B$. Write $\Pi c=b_1+b_2$ with $b_2=P_{D_V}\Pi c=P_{D_V}c$ (as $D_V\subset B$) and $b_1=\Pi c-b_2\in B\cap D_V^\perp$. Then $\langle b_1,F\rangle\in H_V$: indeed $b_1\perp D_V=\mathscr{S}_V^\perp$ gives $b_1\in\mathscr{S}_V$ (a finite-dimensional space equals its double orthogonal complement), and $\langle b_1,F\rangle$ is one of the generators. And $\langle b_2,F\rangle\perp H_V$, so its conditional expectation vanishes. Since $\langle c,F\rangle=\langle\Pi c,F\rangle$ almost surely, $\langle c,F\rangle-\mathbb{E}(\langle c,F\rangle\mid\mathscr{G}_V)=\langle b_2,F\rangle$, whose second moment is $\beta^{-1}\|\Pi b_2\|^2=\beta^{-1}\|b_2\|^2$. □

Theorem 134 (no uniform averaged condition for the Maxwell field)**.** *Let $n\ge2$, $L\ge n+2$ and $\beta>0$. Let $\mathscr{G}_x$, $x\in\Lambda_L$, be σ-algebras on B such that F_p is $\mathscr{G}_x$-measurable for every plaquette $p\notin P(V_x)$; for instance $\mathscr{G}_x=\sigma(F_p:p\notin P(V_x))$ or $\mathscr{G}_x=\mathscr{G}_{V_x}$. Put*

$$f=\sum_{y\in\Lambda_L}\cos\Big(\frac{2\pi y_1}{L}\Big)F_{(y,12)}.$$

Then $\operatorname{Var}_{\gamma_\beta}(f)=L^4/(2\beta)$, and for every x,

$$\mathbb{E}_{\gamma_\beta}|f-P_xf|^2\le\frac{4\pi^2n^4(n+1)^2}{\beta L^2},$$
$$\frac{1}{n^4}\sum_x\mathbb{E}_{\gamma_\beta}|f-P_xf|^2\le\frac{4\pi^2(n+1)^2L^2}{\beta}.$$

Consequently, if $(\gamma_\beta,(\mathscr{G}_x))$ satisfies $\mathrm{BP}(n,\varpi)$, then

$$\varpi\le\frac{8\pi^2(n+1)^2}{L^2},$$

and there is no $\varpi>0$ such that $\mathrm{BP}(n,\varpi)$ holds for all L.

Proof. Let $c\in C^2$ be the cochain with $c_{(y,12)}=\cos(2\pi y_1/L)$ and all other components zero, so that $f=\langle c,F\rangle$. The cochain c is exact: with $g(m)=\sum_{0\le j<m}\cos(2\pi j/L)$, a function on $\mathbb{Z}/L\mathbb{Z}$ because $\sum_{j=0}^{L-1}\cos(2\pi j/L)=0$, let $A_{(y,2)}=g(y_1)$ and $A_{(y,a)}=0$ for $a\ne2$. Then $(dA)_{(y,12)}=A_{(y,1)}+A_{(y+e_1,2)}-A_{(y+e_2,1)}-A_{(y,2)}=g(y_1+1)-g(y_1)=c_{(y,12)}$, while $(dA)_{(y,02)}=A_{(y+e_0,2)}-A_{(y,2)}=0$, $(dA)_{(y,23)}=A_{(y,2)}-A_{(y+e_3,2)}=0$, and the components in the planes not containing the direction 2 vanish. Hence $\Pi c=c$ and $\operatorname{Var}(f)=\beta^{-1}\|c\|^2=\beta^{-1}L^3\sum_{y_1\in\mathbb{Z}/L\mathbb{Z}}\cos^2(2\pi y_1/L)=L^4/(2\beta)$, as $L\ge4$.

Fix x. Write $c=c^{\mathrm{ext}}+c^{\mathrm{in}}$ with $c^{\mathrm{in}}=c\,1_{P(V_x)}$. By Lemma 131(i) the support of c^{in} is the disjoint union of the layers R_t of the plane $\{1,2\}$, t ranging over the n^2 transverse positions. Put $\bar c=\cos(2\pi(x_1-1)/L)$, $\Phi_t=\sum_{p\in R_t}F_p$ and

$$g=\langle c^{\mathrm{ext}},F\rangle+\bar c\sum_t\Phi_t.$$

The variables F_p, $p\notin P(V_x)$, are $\mathscr{G}_x$-measurable by hypothesis; so is each Φ_t, because $F\in B$ satisfies $dF=0$ and Lemma 132 writes Φ_t as a linear combination of F_p with $p\notin P(V_x)$. Hence g is $\mathscr{G}_x$-measurable, $P_xg=g$, and since $I-P_x$ is an orthogonal projection,

$$\begin{aligned}\mathbb{E}|f-P_xf|^2&=\mathbb{E}|(I-P_x)(f-g)|^2\le\mathbb{E}|f-g|^2\\&=\beta^{-1}\|\Pi h\|^2\le\beta^{-1}\|h\|^2,\qquad h=c^{\mathrm{in}}-\bar c\sum_t1_{R_t}.\end{aligned}$$

The cochain h is supported on the $n^2((n+1)^2-1)$ plaquettes of the layers, and on $(y,12)\in R_t$ its value is $\cos(2\pi y_1/L)-\cos(2\pi(x_1-1)/L)$ with $y_1=x_1-1+j$, $j\in[0,n]$; since $\cos(2\pi\cdot/L)$ is well defined on $\mathbb{Z}/L\mathbb{Z}$ and $|\cos a-\cos b|\le|a-b|$, $|h_p|\le2\pi j/L\le2\pi n/L$. Therefore $\|h\|^2\le n^2((n+1)^2-1)(2\pi n/L)^2\le4\pi^2n^4(n+1)^2/L^2$, which is the bound for one cube; summing over the L^4 cubes and dividing by n^4 gives the Dirichlet form bound. If $\mathrm{BP}(n,\varpi)$ holds, then $\varpi L^4/(2\beta)\le4\pi^2(n+1)^2L^2/\beta$, that is $\varpi\le8\pi^2(n+1)^2/L^2$. □

Remark 135 (what the proof uses, and the exact first-chaos constant). The proof uses two facts only: that f is a slowly varying superposition of the field strength over one plane, and that the total flux of F through each layer read by a cube is conserved by the move at that cube up to exterior terms (Lemmas 131, 132). It does not use the Gaussian law beyond the identity $\mathbb{E}\langle h,F\rangle^2=\beta^{-1}\|\Pi h\|^2\le\beta^{-1}\|h\|^2$, and it does not depend on which of the two natural exterior σ-algebras is used. For $\mathscr{G}_x=\mathscr{G}_{V_x}$, Lemma 133 gives the exact Dirichlet form of every linear functional, $n^{-4}\sum_x\beta^{-1}\|P_{D_{V_x}}c\|^2=\beta^{-1}\langle c,Mc\rangle$ with $M=n^{-4}\sum_xP_{D_{V_x}}$, so that $\mathrm{BP}(n,\varpi)$ for γ_β implies $\varpi\le\varpi_1(n,L)$, the least eigenvalue of M on B. The operator M commutes with the translations of Λ_L and is block diagonal in Fourier space

with 6×6 blocks, one for each momentum, and B has dimension three in each block of nonzero momentum. A finite computation of these blocks gives $\varpi_1(n,L)L^2/(2\pi)^2 = 0.276$, 0.299, 0.317, 0.324, 0.329 for $n=2$ and $L=6,8,12,16,24$, and 0.469, 0.518, 0.537, 0.551 for $n=3$ and $L=8,12,16,24$, the minimum being attained at the lowest nonzero momentum (for $L\le 8$ all momenta were examined, for $L\ge 12$ those with $|k|_1\le 2$); the constant $8\pi^2(n+1)^2$ of the theorem is 55 to 70 times larger than the computed $\varpi_1 L^2$. These numbers are a finite computation and not a proof of any statement about ϖ_1 beyond the listed values; the theorem is the proved bound.

3. *Two necessary conditions for the averaged estimate*

For a bounded real $O\in\mathscr{A}$ depending on the links with base points in $b+[0,s)^4$, and $y\in\Lambda_L$, let $O_y = O\circ\tau_{-y}$ be its translate, depending on the links with base points in $b+y+[0,s)^4$, and let $\chi_L(O) = L^{-4}\operatorname{Var}_\mu\big(\sum_y O_y\big) = \sum_y \operatorname{Cov}_\mu(O_0,O_y)$ be its susceptibility; the last equality uses the translation invariance of μ on Λ_L.

Proposition 136 (susceptibilities)**.** *Let* $\mathrm{BP}(n,\varpi)$ *hold for* μ *on* Λ_L*, and let* O *be as above with* $1\le s$ *and* $n+s\le L$*. Then*

$$\chi_L(O)\le \frac{(n+s-1)^8}{n^4\varpi}\operatorname{Var}_\mu(O).$$

Proof. Put $f=\sum_y O_y$, a real element of $\mathscr{A}$ with $\operatorname{Var}_\mu(f) = L^4\chi_L(O)$. For fixed x, $(I-\mathscr{P}_x)O_y = 0$ unless a link of the support of O_y lies in V_x, that is unless $x-y-b\in u-v$ for some $u\in[0,s)^4$, $v\in[0,n)^4$; the set $(-n,s)^4$ of such differences has $(n+s-1)^4$ elements, which are distinct residues modulo L because $n+s-1<L$. For each such y, $\|(I-\mathscr{P}_x)O_y\|_2\le\|O_y-\mu(O_y)\|_2 = \operatorname{Var}_\mu(O)^{1/2}$, because $I-\mathscr{P}_x$ is an orthogonal projection annihilating constants. Hence $\|(I-\mathscr{P}_x)f\|_2\le (n+s-1)^4\operatorname{Var}_\mu(O)^{1/2}$ and $\mathscr{E}_n(f)\le n^{-4}L^4(n+s-1)^8\operatorname{Var}_\mu(O)$. The condition $\mathrm{BP}(n,\varpi)$ gives $\varpi L^4\chi_L(O)\le\mathscr{E}_n(f)$. □

Corollary 137 (correlation lengths)**.** *Let* $\mathscr{L}$ *be an infinite set of even integers* $L\ge 2n+4$ *such that* $\mathrm{BP}(n,\varpi)$ *holds for* μ_L *on* Λ_L *for every* $L\in\mathscr{L}$*, and let* ω *be a torus limit along* $\mathscr{L}$*. Let* f,g *be bounded local gauge-invariant functions and suppose that there are* $c_0>0$*,* $\xi>0$ *and infinitely many integers* d *with*

$$\big|\omega(f\,\tau_d g)-\omega(f)\omega(g)\big|\ge c_0 e^{-d/\xi},$$

where τ_d *is the translation by* d *lattice units in a fixed direction. Then*

$$\varpi\le\frac{e^2 q_n n}{\xi},\qquad \textit{equivalently}\qquad \xi\le\frac{e^2 q_n n}{\varpi}.$$

In the normalisation of Theorem 64, $na\le\ell$ *and* $\xi = 1/(ma)$ *with* m *the physical inverse correlation length, this reads* $\varpi\le e^2 q_n\, m\ell$*.*

Proof. For fixed d and all $L\in\mathscr{L}$ large enough, f and $\tau_d g$ depend on links of sets $S_f,S_g\subset\Lambda_L$ with $|S_f|,|S_g|$ independent of L and d, at torus distance $d(S_f,S_g)\ge d-c$ with c depending on f,g only. The second bound of Theorem 62, which holds for every distance, gives $|\operatorname{Cov}_{\mu_L}(f,\tau_d g)|\le C(1+d)e^{-\varpi d/(e^2q_n n)}$ with C independent of L and d (the factor $e^{\varpi c/(e^2q_n n)}\le e^c$ is absorbed in C, as $\varpi\le 4$ and $q_n>16$). The torus limit is the limit of $\mu_L(f\,\tau_d g)$, $\mu_L(f)$, $\mu_L(g)$ along $\mathscr{L}$, so $c_0e^{-d/\xi}\le C(1+d)e^{-\varpi d/(e^2q_n n)}$ for infinitely many d. Taking logarithms, dividing by d and letting $d\to\infty$ along these values gives $-1/\xi\le -\varpi/(e^2q_n n)$. □

4. *Local functions: the Dirichlet form between two conditional variances*

Proposition 138 (local functions)**.** *Let* $L\ge 2n+4$*, let* $f\in\mathscr{A}$ *be real and depend only on the links of a set* S *whose base points lie in a box* $b+[0,s)^4$ *with* $1\le s\le n$*. Then*

$$\Big(\frac{n-s+1}{n}\Big)^4\mu\big(\operatorname{Var}_\mu(f\mid\mathscr{F}_S)\big)\le\mathscr{E}_n(f)\le\Big(\frac{n+s-1}{n}\Big)^4\operatorname{Var}_\mu(f).$$

If f *depends only on the four links of a plaquette* p*, then*

$$\Big(\frac{n-1}{n}\Big)^2\mu\big(\operatorname{Var}_\mu(f\mid\mathscr{F}_p)\big)\le\mathscr{E}_n(f)\le\Big(1+\frac{2}{n}\Big)\operatorname{Var}_\mu(f),$$

where $\mathscr{F}_p$ *is the* σ*-algebra of the links not in* p*.*

Proof. If $\mathscr{G}\subset\mathscr{F}$ are σ-algebras and $E_\mathscr{G}$, $E_\mathscr{F}$ the conditional expectations, then $E_\mathscr{G}f = E_\mathscr{G}E_\mathscr{F}f$ and, by Pythagoras' theorem, $\mu((f-E_\mathscr{G}f)^2) = \mu((f-E_\mathscr{F}f)^2)+\mu((E_\mathscr{F}f-E_\mathscr{G}f)^2)\ge\mu((f-E_\mathscr{F}f)^2) = \mu(\operatorname{Var}_\mu(f\mid\mathscr{F}))$. If $S\subset V_x$ then $\mathscr{F}_{V_x}\subset\mathscr{F}_S$, so $\mu((f-\mathscr{P}_x f)^2)\ge\mu(\operatorname{Var}_\mu(f\mid\mathscr{F}_S))$. The box $b+[0,s)^4$ lies in $x+[0,n)^4$ for exactly $(n-s+1)^4$ values of x, which gives the lower bound. For the upper bound, $(I-\mathscr{P}_x)f=0$ unless a base point of S lies in $x+[0,n)^4$, which happens for at most $(n+s-1)^4$ values of x as in the proof of Proposition 136 ($n+s-1<L$), and each term is at most $\operatorname{Var}_\mu(f)$. For a plaquette, Lemma 131(i) gives $n^2(n-1)^2$ cubes containing all four links and $n^2((n+1)^2-1) = n^4+2n^3$ cubes reading at least one of them. □

5. *The single-link law of* SU(2) *and the plaquette at weak coupling*

We identify SU(2) with the unit quaternions $S^3\subset\mathbb{R}^4$, $U = u_0+u_1\mathbf{i}+u_2\mathbf{j}+u_3\mathbf{k}$, so that $\operatorname{Tr}U = 2u_0 = 2\langle U,1\rangle$ and $\operatorname{Tr}(US) = 2\operatorname{Re}(US) = 2\langle U,\bar S\rangle$, where $\bar S$ is the conjugate quaternion S^{-1} and $\langle\cdot,\cdot\rangle$ the Euclidean inner product of $\mathbb{R}^4$; Haar measure is the normalised surface measure. For a unit vector $\hat w\in S^3$ and $a>0$ let $\nu_{a,\hat w}$ be the law with density proportional to $e^{a\langle U,\hat w\rangle}$ with respect to Haar measure. Writing $U = \cos\theta\,\hat w+\sin\theta\,u$ with $\theta\in[0,\pi]$ and u in the unit sphere S^2 of $\hat w^\perp$, Haar measure is the product of the density $\frac{2}{\pi}\sin^2\theta\,d\theta$ and the uniform law of u; hence under $\nu_{a,\hat w}$ the angle θ has density proportional to $e^{a\cos\theta}\sin^2\theta$ on $[0,\pi]$ and u is uniform on S^2 and independent of θ. Expectations under this law are written $\mathbb{E}$.

Lemma 139 (the single-link law)**.** *Let $a>0$ and $\hat w\in S^3$.*

(i) *For every unit vector $v\in\mathbb{R}^4$ with $\langle v,\hat w\rangle=\cos\alpha$,*

$$\operatorname{Var}_{\nu_{a,\hat w}}\langle U,v\rangle=\cos^2\alpha\,\operatorname{Var}(\cos\theta)+\frac{\sin^2\alpha}{3}\,\mathbb{E}\sin^2\theta.$$

(ii) *$a\mathbb{E}\sin^2\theta=3\mathbb{E}\cos\theta$, so that $\mathbb{E}\cos^2\theta=1-3\mathbb{E}\cos\theta/a$; and $a\mathbb{E}\sin^4\theta=5\mathbb{E}(\sin^2\theta\cos\theta)\ge 0$.*

(iii) $\operatorname{Var}(\cos\theta)\ge(\mathbb{E}\cos\theta)^2/a^2$.

(iv) $\nu_{a,\hat w}(\cos\theta<0)\le 3\pi/(2a^2)$, *and*

$$\mathbb{E}\cos\theta\ge\frac{1-3\pi/a^2}{1+3/a}.$$

(v) *If $a\ge 6$, then $\mathbb{E}\cos\theta\ge 0.4921$ and $\operatorname{Var}_{\nu_{a,\hat w}}\langle U,v\rangle\ge 0.24/a^2$ for every unit vector v.*

Proof. (i) Write $v=\cos\alpha\,\hat w+\sin\alpha\,v'$ with v' a unit vector of $\hat w^{\perp}$ (any one if $\sin\alpha=0$). Then $\langle U,v\rangle=\cos\alpha\cos\theta+\sin\alpha\sin\theta\langle u,v'\rangle$. Since u is uniform on S^2 and independent of θ, $\mathbb{E}\langle u,v'\rangle=0$ and $\mathbb{E}\langle u,v'\rangle^2=\frac13$, so $\mathbb{E}\langle U,v\rangle=\cos\alpha\,\mathbb{E}\cos\theta$ and $\mathbb{E}\langle U,v\rangle^2=\cos^2\alpha\,\mathbb{E}\cos^2\theta+\frac13\sin^2\alpha\,\mathbb{E}\sin^2\theta$, which gives the formula.

(ii) For $k\ge 0$,

$$\frac{d}{d\theta}\big(e^{a\cos\theta}\sin^{2k+3}\theta\big)=e^{a\cos\theta}\sin^{2k+2}\theta\times\big((2k+3)\cos\theta-a\sin^2\theta\big),$$

and the left side integrates to zero over $[0,\pi]$. Dividing by $Z=\int_0^\pi e^{a\cos\theta}\sin^2\theta\,d\theta$ gives $a\mathbb{E}\sin^{2k+2}\theta=(2k+3)\mathbb{E}(\sin^{2k}\theta\cos\theta)$; $k=0$ and $k=1$ are the first and third identities, the second is $\mathbb{E}\cos^2\theta=1-\mathbb{E}\sin^2\theta$, and $\mathbb{E}\sin^4\theta\ge 0$ gives the sign.

(iii) Put $S=3\cos\theta-a\sin^2\theta$. For a smooth g on $[0,\pi]$,

$$\frac{d}{d\theta}\big(g(\theta)e^{a\cos\theta}\sin^3\theta\big)=e^{a\cos\theta}\sin^2\theta\times\big(g'(\theta)\sin\theta+g(\theta)S(\theta)\big),$$

so $\mathbb{E}(g'\sin\theta)=-\mathbb{E}(gS)$. With $g=1$, $\mathbb{E}S=0$; with $g=\cos\theta$, $\mathbb{E}\sin^2\theta=\mathbb{E}(S\cos\theta)=\operatorname{Cov}(\cos\theta,S)$. By the Cauchy–Schwarz inequality and (ii),

$$(\mathbb{E}\sin^2\theta)^2\le\operatorname{Var}(\cos\theta)\,\mathbb{E}S^2,$$
$$\mathbb{E}S^2=9\mathbb{E}\cos^2\theta-6a\mathbb{E}(\sin^2\theta\cos\theta)+a^2\mathbb{E}\sin^4\theta=9\mathbb{E}\cos^2\theta-a\mathbb{E}(\sin^2\theta\cos\theta)\le 9.$$

By (ii), $\mathbb{E}\sin^2\theta=3\mathbb{E}\cos\theta/a$, so the left side equals $9(\mathbb{E}\cos\theta)^2/a^2$; dividing by 9 gives the claim.

(iv) On $[0,\pi/2]$, $e^{a\cos\theta}\ge 1+a\cos\theta$, so $Z\ge\int_0^{\pi/2}(1+a\cos\theta)\sin^2\theta\,d\theta=\frac{\pi}{4}+\frac{a}{3}\ge\frac{a}{3}$. With $\theta=\pi/2+\phi$ and $\sin\phi\ge 2\phi/\pi$ on $[0,\pi/2]$,

$$\int_{\pi/2}^{\pi}e^{a\cos\theta}\sin^2\theta\,d\theta=\int_0^{\pi/2}e^{-a\sin\phi}\cos^2\phi\,d\phi\le\int_0^{\pi/2}e^{-2a\phi/\pi}d\phi\le\frac{\pi}{2a},$$

so $\nu_{a,\hat w}(\cos\theta<0)\le\frac{\pi}{2a}\cdot\frac{3}{a}$. Pointwise $\cos\theta\ge\cos^2\theta-2\cdot 1_{\{\cos\theta<0\}}$, since $\cos\theta\ge\cos^2\theta$ when $\cos\theta\ge 0$ and $\cos\theta\ge-1\ge\cos^2\theta-2$ always. Taking expectations and using (ii), $\mathbb{E}\cos\theta\ge 1-3\mathbb{E}\cos\theta/a-3\pi/a^2$, which is the second claim.

(v) The bound of (iv) increases with a, and at $a=6$ it equals $(1-\pi/12)/(3/2)\ge 0.4921$. By (iii), $\operatorname{Var}(\cos\theta)\ge 0.4921^2/a^2\ge 0.24/a^2$; by (ii), $\frac13\mathbb{E}\sin^2\theta=\mathbb{E}\cos\theta/a\ge 0.49/a\ge 0.24/a^2$ as $a\ge 6$. By (i), $\operatorname{Var}\langle U,v\rangle\ge(\cos^2\alpha+\sin^2\alpha)\,0.24/a^2$. □

Proposition 140 (the plaquette at weak coupling)**.** *Let $G=\mathrm{SU}(2)$ with $\varphi(U)=e^{-2\beta(2-\operatorname{Tr}U)}$, $\beta\ge 100$, $n\ge 2$ and $L\ge 2n+4$. Let p be a plaquette, e one of its links, and $\mathscr{F}_e$ the σ-algebra of all links other than e. Then*

(i) $\mu\big(\operatorname{Var}_\mu(\operatorname{Tr}U_p\mid\mathscr{F}_e)\big)\ge 1/(1200\beta^2)$;

(ii) $\operatorname{Var}_\mu(\operatorname{Tr}U_p)\le 4(\log\beta+6.11)/\beta$;

(iii) *the Dirichlet form satisfies*

$$\mathscr{E}_n(\operatorname{Tr}U_p)\ge\Big(\frac{n-1}{n}\Big)^2\frac{1}{1200\beta^2}\ge\Big(\frac{n-1}{n}\Big)^2\frac{\operatorname{Var}_\mu(\operatorname{Tr}U_p)}{4800\,\beta(\log\beta+6.11)}.$$

In particular the inequality of $\mathrm{BP}(n,\varpi)$ holds for the functions $\operatorname{Tr}U_p$ with $\varpi=\varpi_p(n,\beta)=((n-1)/n)^2/(4800\beta(\log\beta+6.11))$, uniformly in $L\ge 2n+4$ and in p.

Proof. (i) Under μ the conditional density of U_e given $\mathscr{F}_e$ is proportional to $\prod_{p'\ni e}\varphi(U_{p'})$, hence to $\exp(2\beta\sum_{p'\ni e}\operatorname{Tr}U_{p'})$, the product running over the six plaquettes containing e. For each of them, $U_{p'}$ is a product of four link variables or their inverses in which $U_e^{\pm1}$ occurs once; by cyclicity $\operatorname{Tr}U_{p'}=\operatorname{Tr}(U_e^{\pm1}S_{p'})$ with $S_{p'}\in\mathrm{SU}(2)$ the product of the other three factors in cyclic order, an $\mathscr{F}_e$-measurable variable, and since traces on $\mathrm{SU}(2)$ are real and $\operatorname{Tr}(M^{-1})=\overline{\operatorname{Tr}M}$, $\operatorname{Tr}(U_e^{-1}S)=\operatorname{Tr}(S^{-1}U_e)=\operatorname{Tr}(U_eS^{-1})$. Thus $\operatorname{Tr}U_{p'}=\operatorname{Tr}(U_eS'_{p'})=2\langle U_e,\bar S'_{p'}\rangle$ with $S'_{p'}\in\{S_{p'},S_{p'}^{-1}\}$, and the conditional density is proportional to $\exp(4\beta\langle U_e,w\rangle)$, $w=\sum_{p'\ni e}\bar S'_{p'}\in\mathbb{R}^4$, $|w|\le 6$. In the same way $\operatorname{Tr}U_p=2\langle U_e,v\rangle$ with $v=\bar S'_p$ a unit vector measurable with respect to $\mathscr{F}_e$. If $w\ne 0$ the conditional law is $\nu_{a,\hat w}$ with $a=4\beta|w|\le 24\beta$ and $\hat w=w/|w|$, so by Lemma 139(v), on the event $\{a\ge 6\}$,

$$\operatorname{Var}_\mu(\operatorname{Tr}U_p\mid\mathscr{F}_e)=4\operatorname{Var}_{\nu_{a,\hat w}}\langle U,v\rangle\ge\frac{0.96}{a^2}\ge\frac{0.96}{576\beta^2}=\frac{1}{600\beta^2}.$$

On the event $\{a<6\}$ (which contains $\{w=0\}$), $|w|<3/(2\beta)$ and

$$\sum_{p'\ni e}\operatorname{Tr}U_{p'}=2\langle U_e,w\rangle\le 2|w|<\frac{3}{\beta},$$
$$\text{so}\qquad\sum_{p'\ni e}(2-\operatorname{Tr}U_{p'})>12-\frac{3}{\beta}\ge 11.97,$$

and one of the six terms exceeds 1.99. By Markov's inequality and Lemma 76, which applies as $\beta \geq 1$, and by the equivalence of all plaquettes under the symmetries of Λ_L,

$$\mu(a<6) \leq \sum_{p' \ni e} \mu(2-\operatorname{Tr}U_{p'} > 1.99) \leq \frac{6}{1.99}\cdot\frac{\log\beta+6.11}{\beta} \leq 3.02\,\frac{\log\beta+6.11}{\beta} \leq \frac{1}{2},$$

the last step because $(\log\beta+6.11)/\beta$ decreases on $[1,\infty)$ and equals 0.1072 at $\beta=100$, where $3.02\cdot 0.1072<0.33$. Hence $\mu(\operatorname{Var}_\mu(\operatorname{Tr}U_p \mid \mathscr{F}_e)) \geq \mu(a\geq 6)/(600\beta^2) \geq 1/(1200\beta^2)$.

(ii) Since $0 \leq 2-\operatorname{Tr}U_p \leq 4$, $\operatorname{Var}_\mu(\operatorname{Tr}U_p) \leq \mu((2-\operatorname{Tr}U_p)^2) \leq 4\mu(2-\operatorname{Tr}U_p) \leq 4(\log\beta+6.11)/\beta$ by Lemma 76.

(iii) The σ-algebra $\mathscr{F}_p$ of the links not in p is contained in $\mathscr{F}_e$, so $\mu(\operatorname{Var}_\mu(\operatorname{Tr}U_p \mid \mathscr{F}_p)) \geq \mu(\operatorname{Var}_\mu(\operatorname{Tr}U_p \mid \mathscr{F}_e))$ as in the proof of Proposition 138, and that proposition gives $\mathscr{E}_n(\operatorname{Tr}U_p) \geq ((n-1)/n)^2\mu(\operatorname{Var}_\mu(\operatorname{Tr}U_p \mid \mathscr{F}_p))$. Combine with (i) and (ii). □

Remark 141 (where the factor $\beta\log\beta$ is lost). The two sides of Proposition 140(iii) are of different orders in β: the lower bound $((n-1)/n)^2/(1200\beta^2)$ on the Dirichlet form of $\operatorname{Tr}U_p$ comes from the single-link law, whose variance in the direction of v is of order a^{-2} with $a \asymp \beta$ when $\cos\alpha \neq 0$, while the variance bound (ii) comes from the first moment of $2-\operatorname{Tr}U_p$ through the crude inequality $(2-\operatorname{Tr}U_p)^2 \leq 4(2-\operatorname{Tr}U_p)$. A bound $\mu((2-\operatorname{Tr}U_p)^2) \leq C/\beta^2$ for $\beta \geq \beta_0$ would make $\varpi_p(n,\beta) \geq ((n-1)/n)^2/(1200C)$ independent of β. Such a bound, with $C = C_L(1+\ell)^2$, is Theorem 155, and Corollary 157 gives $\varpi = 1/(1200C_L(1+\ell)^2)$ for $\beta \geq 11$, with the factor $((n-1)/n)^2$ removed. For comparison, Theorem 77(ii) shows that $\mathrm{BP}(n,\varpi)$ itself fails on Λ_{2n+4} unless $\varpi \leq 38400n^4(\log\beta+6.11)/\beta$, through test functions built from Polyakov loops; for $\varpi \leq \varpi_p(n,\beta)$ the plaquette traces are not among the functions that violate it, and (iii) is a statement about those functions only.

6. *The compact* U(1) *models: what is proved and what is not*

Remark 142 (status of the abelian consistency check). The uniform averaged estimate must fail for a theory without a mass gap, and the paper records this for the Villain U(1) model: Corollary 66 shows that for $\beta > \beta_0$ no $\mathrm{BP}(n,\varpi)$ holds for the free-boundary measures of all boxes, by Theorem 62 and the non-summable two-point function of the field strength of [51, Sect. 2]. Theorem 134 adds the quantitative mechanism in the Gaussian limit of that phase: the flux of F through the layers read by a cube is conserved by the move up to exterior terms, so a superposition of the field strength with wavelength L relaxes at rate at most $8\pi^2(n+1)^2/L^2$, the diffusive rate. Two parts of the abelian check remain open and are stated here precisely.

(a) *The torus.* Theorem 64 and Corollary 63 are formulated for the tori Λ_L, while Corollary 66 concerns free-boundary boxes, because the state of [51] is the limit of those. The statement that would close the check is: for the Villain weight and $\beta > \beta_0$, $\mathrm{BP}(n,\varpi)$ on Λ_L implies $\varpi \leq C(n,\beta)/L^2$. The proof of Theorem 134 does not transfer as it stands: the field strength $\Psi_p = \varphi_\beta'(\theta_p)/\varphi_\beta(\theta_p)$ of Proposition 243 is a nonlinear function of the plaquette angle, the flux conservation of Lemma 132 holds for the compact variables $U_p = e^{i\theta_p}$, that is for the layer flux modulo 2π, and the step $\mathbb{E}|f-P_xf|^2 \leq \mathbb{E}|f-g|^2$ needs an exterior-measurable g within $O(n/L)$ of f on every layer, which requires control of the conditional law of the integer part of the layer flux given the exterior links. This is a statement about the Villain measure on the torus that is not available in [51] or [59].

(b) *Wilson's action.* [51, Sect. 2.11] prove the masslessness of the photon for the Villain form of the action, define the field strength for Wilson's form as well, state that most of their arguments extend to a general class of actions, and mark one step of the proof (the lower bound in their finite-volume estimate) as one that "does not obviously extend to Wilson's form" and would have to be replaced by a direct argument, which is not given there; the existence proof of the Coulomb phase in [59] is for the Villain form as well. No uniform-in-volume failure of $\mathrm{BP}(n,\varpi)$ is proved here for the Wilson U(1) action, at any β.

7. *The missing estimate*

Definition 143 (the uniform averaged estimate along the tuning)**.** Let $G = \mathrm{SU}(2)$ with the Wilson weight, let (β_k, a_k) be the couplings and lattice spacings of the continuum tuning of Theorem 64, with $\beta_k \to \infty$ and $a_k \to 0$, and let $\ell > 0$. The estimate (U) at the scale ℓ is: there are $\varpi > 0$ and integers $n_k \geq 2$ with $n_k a_k \leq \ell$ such that for every k and every even $L \geq 2n_k+4$ the Wilson measure of Λ_L at β_k satisfies $\mathrm{BP}(n_k,\varpi)$. Its local form $(\mathrm{U}_{\mathrm{loc}})$ is the same statement with $\mathrm{BP}_{\mathrm{loc}}$ of Definition 81 in place of BP.

Proposition 144 (what (U) gives and what it costs)**.** *Suppose* $(\mathrm{U}_{\mathrm{loc}})$ *holds at the scale* ℓ *with constants* ϖ, n_k*. Then:*

(i) *for every* k*, every torus limit* ω_k *along the even* $L \geq 2n_k+4$ *has a transfer matrix with spectral gap at least* $16\varpi/(625e^2n_k)$ *in lattice units, hence at least* $16\varpi a_k/(625e^2\ell)$ *in the units of* a_k*; and every continuum limit of the* (ω_k, O_k) *of Theorem 64 has a mass gap of at least* $16\varpi/(625e^2\ell)$ *if it is nontrivial;*

(ii) *for every* k *and every pair of bounded local gauge-invariant* f, g*: if* $|\omega_k(f\tau_d g) - \omega_k(f)\omega_k(g)| \geq c_0 e^{-d/\xi}$ *for some* $c_0 > 0$ *and infinitely many* d*, then* $\xi \leq e^2 q_{n_k} n_k/\varpi$*;*

(iii) *if* (U) *holds, then in addition* $n_k^4 \geq \varpi\beta_k/(38400(\log\beta_k + 6.11))$ *whenever* $\beta_k \geq 1$*, so* $n_k \to \infty$*, and the susceptibility bound of Proposition 136 holds with* (n_k,ϖ) *on every even* $L \geq 2n_k+4$ *for every* O *of side* s *with* $n_k + s \leq L$*, in particular for every local* O *in the sense of Sec. IX H.*

Proof. (i) is Corollary 63 and Theorem 64 in the form of Theorem 82(iii). (ii) is Corollary 137, whose proof uses only Theorem 62 for local f, which Theorem 82(ii) provides under $\mathrm{BP}_{\mathrm{loc}}$. (iii) Theorem 77(ii) applies on Λ_{2n_k+4}, an admissible torus, and gives $\varpi \le 38400 n_k^4(\log\beta_k + 6.11)/\beta_k$; Proposition 136 applies to every admissible torus, and a local O has side $s \le L/2 \le L - n_k$. □

Remark 145 (the exact obstruction). Proposition 144(i) shows that $(\mathrm{U}_{\mathrm{loc}})$, hence (U), implies the uniform lattice mass gap $m_k \ge 16\varpi a_k/(625e^2\ell)$ along the whole weak-coupling tuning, which is the open statement that the continuum step is meant to produce. So (U) is at least as strong as its intended consequence, and no proof of (U) can bypass the mass gap. The functions that Theorem 134 and Proposition 136 examine are the slowly varying superpositions of local observables over the whole torus: for the free Maxwell field their Dirichlet form is at most of order $(n/L)^2$ times their variance (Theorem 134; this is proved for that field only), while (U) requires their variance to be bounded by their Dirichlet form uniformly in L. So (U) implies that the susceptibility of every local observable is bounded uniformly in L at the scale n (Proposition 136) and that the correlation length of every torus limit is at most $e^2 q_n n/\varpi$ lattice units (Corollary 137); the converse implications are not proved.

What is available on the other side is of two kinds. First, the sufficient condition of the paper is the supremum condition $D_L^0(n,\alpha)$ with $\alpha < 1$, which gives $\mathrm{BP}(n, 1-\alpha)$ for all $L \ge 2n+4$ at once (Proposition 65, Theorem 70(ii)); it does not depend on L, and Corollary 78 excludes $D_L^0(n,\alpha)$ only for $n^4 < (1-\alpha)\beta/(38400(\log\beta + 6.11))$, for $\alpha \le \frac12$ only for $n^4 < \beta/(76800(\log\beta + 6.11))$; the choice $n_k = \lfloor \ell/a_k \rfloor$ leaves this range as soon as $a_k \le \ell\big(76800(\log\beta_k + 6.11)/\beta_k\big)^{1/4}/2$, that is for every tuning in which a_k tends to zero faster than $\beta_k^{-1/4}$. The first concrete test of (U) is therefore whether $D_L^0(n_k,\alpha)$ holds with $\alpha < 1$ uniformly in k; this is a statement about one cube of side $n(\beta)$ with arbitrary exterior, with no averaging, and its analogue for the massless Gaussian field fails (Proposition 93(i)). Second, the single-link bounds of Proposition 140 give the inequality of BP for the plaquette traces uniformly in L, but with a constant independent of β (Corollary 157), and only for these functions; the block-spin results of [21, Sec. X Q] give $\mathrm{BP}(n, \frac12)$ for the law μ^c of the block variables of blocks of side 2^k at $\beta_W \le k+2$ on tori of side at most $e^{2^k/(17k)}$ [21, Corollary 166], and their transfer to the Wilson measure is [21, Theorem 114], whose fibre hypothesis (F), $\mu'(\mathrm{Var}(f \mid c)) \le C_F \Phi(f)$ for all real $f \in \mathscr{A}'$, is the estimate on the conditional law of the fine field given the block field that is not proved; with it, (X) and the almost-orthogonality hypothesis of that theorem (its hypothesis (D), not the cube estimate (D) of Sec. X), the split inequality would give BP for the Wilson measure on the tori covered by the block-spin range, with a constant depending on C_F, ρ and R. None of these yields $\mathrm{BP}(n(\beta), \varpi)$ for a class of f containing the long-wavelength superpositions of Proposition 136.

The precise missing lemma is therefore: *for every k, $n = n_k \le \ell/a_k$ and every even $L \ge 2n+4$, every local real $f \in \mathscr{A}$ satisfies $\|\mathscr{K}^t(f - \mu(f))\|_2 \le (1 - \varpi n^4/L^4)^t \|f - \mu(f)\|_2$ for all $t \ge 0$*, that is $(\mathrm{U}_{\mathrm{loc}})$, which by Theorem 82(iii) suffices for the continuum step; by Theorem 82(iv) its failure produces a real $h \in \mathscr{A}_0$ with $\mathscr{E}_n(h) < \varpi\,\mathrm{Var}_\mu(h)$ correlated with a local function, and Theorem 134 exhibits the shape of such h in the massless case: a plane wave of the field strength of the longest wavelength. The results proved here bound the constant of (U) from above: by the inverse correlation length of the torus limits and by the susceptibilities (Corollary 137, Proposition 136), and, for the free Maxwell field, by $8\pi^2(n+1)^2/L^2$ (Theorem 134); nothing proved here bounds it from below except for the plaquette traces, and nothing here proves (U), $(\mathrm{U}_{\mathrm{loc}})$, the averaged condition along the tuning or a mass gap.

E. The plaquette at weak coupling: second moments without the logarithm, Wilson loops, and the Villain torus

Throughout this subsection $G = \mathrm{SU}(2)$ unless said otherwise, $\Lambda = \Lambda_L$ is the torus, E and P are its $4L^4$ links and $6L^4$ plaquettes, dU is normalised Haar measure, and for a plaquette p

$$s_p = 1 - \tfrac12 \operatorname{Tr} U_p \in [0,2], \qquad h_p = 2 - \operatorname{Tr} U_p = 2s_p.$$

The Wilson weight of Sec. IX F is $\varphi(U) = e^{-2\beta(2-\operatorname{Tr} U)} = e^{-bs_p}$ with $b = 4\beta$, the conventional coupling β_W of the introduction. For couplings $c = (c_p)_{p\in P}$, $c_p \ge 0$, put $Z(c) = \int_{G^E} \prod_p e^{-c_p s_p}\, dU$ and $Z_L(b) = Z(c)$ when all $c_p = b$, so that μ has density $e^{-bS}/Z_L(b)$ with $S = \sum_p s_p$. For $a > 0$ let

$$I(a) = \int_G e^{-a(1-\frac12 \operatorname{Tr} U)}\, dU = \frac{2}{\pi}\int_0^\pi e^{-a(1-\cos t)} \sin^2 t\, dt$$

be the single-plaquette integral (Weyl's formula, $\operatorname{Tr} U = 2\cos t$). All results of this subsection are statements about finite tori; none of them asserts $\mathrm{BP}(n, \varpi)$ for the Wilson measure, which is a statement about every real $f \in \mathscr{A}$ (Definition 60), and none bears on condition (D), on (U) of Definition 143 or on a mass gap.

1. *Two-sided bounds on the free energy with the same leading coefficient*

Lemma 146 (the single-plaquette integral)**.** *For every $a > 0$, $\sqrt{2/\pi}\, a^{-3/2}(1 - \frac{3}{4a}) \le I(a) \le \sqrt{2/\pi}\, a^{-3/2}$.*

Proof. With $u = 1 - \cos t$, $\sin^2 t\, dt = \sqrt{u(2-u)}\, du$ and $I(a) = \frac{2}{\pi}\int_0^2 e^{-au}\sqrt{u(2-u)}\, du$. Since $\sqrt{2-u} \le \sqrt{2}$ and $\int_0^\infty e^{-au}\sqrt{u}\, du = \frac{\sqrt{\pi}}{2} a^{-3/2}$, the upper bound follows. Since $\sqrt{2-u} \ge \sqrt{2}(1 - u/2)$ on $[0,2]$ and $e^{-au}\sqrt{u}(1-u/2) \le 0$ on $[2,\infty)$,

$$\begin{aligned} I(a) &\ge \frac{2\sqrt{2}}{\pi}\int_0^\infty e^{-au}\sqrt{u}\Big(1 - \frac{u}{2}\Big)\, du \\ &= \frac{2\sqrt{2}}{\pi}\Big(\frac{\sqrt{\pi}}{2} a^{-3/2} - \frac{3\sqrt{\pi}}{8} a^{-5/2}\Big). \end{aligned}$$

□

Lemma 147 (gauge fixing on the torus)**.** *Let G be a compact group, $T \subset E$ a maximal tree of Λ_L ($|T| = L^4 - 1$) and f an integrable gauge-invariant function on G^E. Then $\int_{G^E} f\,dU = \int_{G^{E\setminus T}} f(V, 1_T)\,dV$, where $(V, 1_T)$ equals V on $E \setminus T$ and the identity on T.*

Proof. As in Lemma 10, with the tree in place of the forest: fix a root x_0; the map $(g, V) \mapsto g \cdot (V, 1_T)$ from $\{g \in G^\Lambda : g_{x_0} = 1\} \times G^{E\setminus T}$ to G^E, $(g \cdot U)_{(x,\mu)} = g_x U_{(x,\mu)} g^{-1}_{x+e_\mu}$, is a bijection, since the conditions $(g^{-1} \cdot U)_e = 1$, $e \in T$, determine g uniquely along the tree from the root and then $V = (g^{-1} \cdot U)|_{E\setminus T}$. Ordering the links of T from the root, each U_e is a left and right translate of one new variable (g_x for $e \in T$, V_e for $e \notin T$) by previously determined ones, so the map preserves Haar measure; the numbers of factors agree, $(L^4 - 1) + (3L^4 + 1) = 4L^4$. Gauge invariance gives $f(g \cdot (V, 1_T)) = f(V, 1_T)$, and integrating out g gives the claim. □

We use the maximal tree T of the links $(y, 0)$ with $y_1 = y_2 = y_3 = 0$, $y_0 \le L - 2$; $(y, 1)$ with $y_2 = y_3 = 0$, $y_1 \le L - 2$; $(y, 2)$ with $y_3 = 0$, $y_2 \le L - 2$; and $(y, 3)$ with $y_3 \le L - 2$ (coordinates in $\{0, \dots, L-1\}$), and the four closing links $u_\mu = ((L-1)e_\mu, \mu)$. Put $F = T \cup \{u_0, \dots, u_3\}$, $|F| = L^4 + 3$.

Lemma 148 (the lower bound)**.** *Let $b > 0$ and $c_p \in [0, b]$ for all p. Then $Z(c) \ge I(6b)^{3L^4+1}$.*

Proof. By Lemma 147, $Z(c) = \int_{G^N} \prod_p e^{-c_p s_p(V,1_T)}\,dV$, $N = 3L^4 + 1$. Let ν_a have density $\rho_a(W) = e^{-a(1-\frac{1}{2}\operatorname{Tr} W)}/I(a)$, a class function, and write $dV = \prod_e \rho_a(V_e)^{-1} d\nu_a(V_e)$. By Jensen's inequality for the concave logarithm,

$$\log Z(c) \ge \mathbb{E}_{\nu_a^{\otimes N}}\Big[-\sum_p c_p s_p - \sum_{e\notin T} \log \rho_a(V_e)\Big].$$

Put $g(a) = \mathbb{E}_{\nu_a}(1 - \frac{1}{2}\operatorname{Tr} W)$, so $-\mathbb{E}\log\rho_a(V_e) = ag(a) + \log I(a)$. Let $k_p \le 4$ be the number of links of p outside T; in $(V, 1_T)$ the holonomy U_p is a product of k_p independent factors $V_e^{\pm 1}$ with conjugation-invariant laws, so by Schur's lemma each factor has matrix expectation $(1 - g(a))\mathbf{1}$ (also for V_e^{-1}, as $\operatorname{Tr} V^{-1} = \operatorname{Tr} V$ on SU(2)), hence $\mathbb{E} s_p = 1 - (1 - g(a))^{k_p} \le k_p g(a)$ (Bernoulli if $g \le 1$, trivially if $g > 1$). Every link lies in six plaquettes, so $\sum_p k_p = 6N$, and with $c_p \le b$,

$$\log Z(c) \ge N \log I(a) + N g(a)(a - 6b).$$

Take $a = 6b$. □

Lemma 149 (shelling)**.** *Let $(e_1, p_1), \dots, (e_M, p_M)$ be link–plaquette pairs with $e_j \in p_j$ and $e_k \notin p_j$ whenever $k > j$. Then $Z(c) \le \prod_{j=1}^M I(c_{p_j})$ for all couplings $c_p \ge 0$, and necessarily $M \le 3L^4 - 3$.*

Proof. Integrate U_{e_M} first. Among $p_1, \dots, p_M$ only p_M contains e_M; drop the weights $e^{-c_p s_p} \le 1$ of the other plaquettes through e_M. By cyclicity $s_{p_M} = 1 - \frac{1}{2}\operatorname{Tr}(U_{e_M}^{\pm 1} W)$ with W the product of the other three links of p_M, and $\int_G e^{-c(1-\frac{1}{2}\operatorname{Tr}(U^{\pm 1}W))} dU = I(c)$ by the invariance of Haar measure under inversion and translation. The remaining integrand involves $p_1, \dots, p_{M-1}$ and plaquettes not containing e_M; repeat with e_{M-1}, and so on; after M steps all remaining weights are at most 1. For the bound on M: the 1-chains $\partial p_1, \dots, \partial p_M$ are linearly independent, since in a vanishing combination the largest index j with nonzero coefficient has $e_j \in p_j$ and $e_j \notin p_i$ for $i < j$, so the coefficient of e_j is $\pm$ that of ∂p_j. Hence $M \le \operatorname{rank}\partial_2 = \dim\ker\partial_1 - \dim H_1(\Lambda_L; \mathbb{R}) = (4L^4 - (L^4 - 1)) - 4 = 3L^4 - 3$, as $\operatorname{rank}\partial_1 = L^4 - 1$. □

Lemma 150 (an explicit shelling)**.** *For every $L \ge 2$ there is a sequence as in Lemma 149 with $M = 3L^4 - 3$, whose links are exactly those of $E \setminus F$, and whose plaquettes in the plane $\{1, 2\}$ are the $L^3 - L$ plaquettes $(y, 12)$ with $y_3 = 0$ and $(y_1, y_2) \ne (0, L-1)$. When L is even, exactly $m_O = \frac{L}{2}(L^2 - 1) < L^3/2$ of them have y_0 and y_3 even.*

Proof. Read the condition in the order $j = 1, 2, \dots$: p_j contains e_j and otherwise only links of $F \cup \{e_1, \dots, e_{j-1}\}$; so the sequence is built by adding links, each new link completing a plaquette whose other three links are present, starting from F. Write $(y, \mu\nu)$ for plaquettes and (y, μ) for links.

Stage 1, the 2-torus $\{(x_0, x_1, *, *)\}$, given that it contains $(y, 2)$ with $y_3 = 0$ (all y_2), $(y, 3)$ with $y_3 \le L - 2$ (all y_2), and $(x_0, x_1, 0, L-1, 3)$. For $x_3 = 0, \dots, L-2$ and all x_2 the plaquette $(y, 23)$, $y = (x_0, x_1, x_2, x_3)$, has $(y, 2)$ (by induction on x_3), $(y + e_2, 3)$, $(y, 3)$ present and adds $(y + e_3, 2)$. Then for $x_2 = 0, \dots, L-2$ the plaquette $(y, 23)$ with $y = (x_0, x_1, x_2, L-1)$ has $(y, 2)$, $(y, 3)$ (by induction on x_2 from the closing link) and $(y + e_3, 2) = (x_0, x_1, x_2, 0, 2)$ present and adds $(y + e_2, 3)$. This adds $L^2 - 1$ links, all links of the 2-torus, with $L^2 - 1$ distinct plaquettes of the plane $\{2, 3\}$.

Stage 2, the 3-torus $\{(x_0, *, *, *)\}$, given that it contains $(y, 1)$ with $y_2 = y_3 = 0$ (all y_1), $(y, 2)$ with $y_3 = 0$, $y_2 \le L - 2$, $(y, 3)$ with $y_3 \le L - 2$, and $(x_0, 0, 0, L-1, 3)$, $(x_0, 0, L-1, 0, 2)$. Run Stage 1 in the slice $x_1 = 0$. For all x_1 and $x_2 \le L - 2$ the plaquette $(y, 12)$, $y = (x_0, x_1, x_2, 0)$, adds $(y + e_2, 1)$; for $x_3 \le L - 2$ and all x_1, x_2 the plaquette $(y, 13)$ adds $(y + e_3, 1)$; so all $(y, 1)$ are present. For $x_1 = L-1, L-2, \dots, 1$ in this order, $(y, 12)$ with $y = (x_0, x_1, L-1, 0)$ adds $(y, 2)$ and $(y, 13)$ with $y = (x_0, x_1, 0, L-1)$ adds $(y, 3)$, using the slice $x_1 + 1$ (the slice 0 for $x_1 = L-1$). Each slice $x_1 \ne 0$ now satisfies the hypothesis of Stage 1, which is run in each. Stage 2 adds $(L^2 - 1) + (L^3 - L) + 2(L-1) + (L-1)(L^2 - 1) = 2L^3 - 2$ links, all links of the 3-torus, and uses $L(L-1) + (L-1) = L^2 - 1$ plaquettes of the plane $\{1, 2\}$, all with $y_3 = 0$ and $(y_1, y_2) \ne (0, L-1)$.

Stage 3, the torus. F satisfies the hypothesis of Stage 2 in the slice $x_0 = 0$; run Stage 2 there. For $x_1 \le L - 2$ and all x_0 the plaquette $(y, 01)$, $y = (x_0, x_1, 0, 0)$, adds $(y + e_1, 0)$; for $x_2 \le L - 2$ the plaquette $(y, 02)$, $y = (x_0, x_1, x_2, 0)$, adds $(y + e_2, 0)$; for $x_3 \le L - 2$ the plaquette $(y, 03)$ adds $(y + e_3, 0)$; so all $(y, 0)$ are present. For $x_0 = L - 1, \dots, 1$ the plaquettes $(y, 01)$, $(y, 02)$, $(y, 03)$ with $y = (x_0, L-1, 0, 0)$, $(x_0, 0, L-1, 0)$, $(x_0, 0, 0, L-1)$ add $(y, 1)$, $(y, 2)$, $(y, 3)$, using the slice $x_0 + 1$. Each slice $x_0 \ne 0$ then satisfies the hypothesis of Stage 2, which is run in each. In all $(2L^3 - 2) + (L^4 - L) + 3(L-1) + (L-1)(2L^3 - 2) = 3L^4 - 3$ links are added, the links of $E \setminus F$, and the plaquettes of the plane $\{1, 2\}$ used are $L(L^2 - 1) = L^3 - L$, one in each slice x_0 for each $(y_1, y_2) \ne (0, L-1)$, all with $y_3 = 0$; for even L those with y_0 even number $\frac{L}{2}(L^2 - 1)$. At each step the other

three links of the plaquette were present, which is the condition of Lemma 149. □

Proposition 151 (two-sided bounds)**.** *Let $b > 1/8$, $c_p \in (0,b]$ for all p, and $(e_j, p_j)_{j \le 3L^4-3}$ the sequence of Lemma 150. Then*

$$\log Z(c) \ge (3L^4+1)\Big[\tfrac{1}{2}\log\tfrac{2}{\pi} - \tfrac{3}{2}\log(6b) + \log\Big(1-\frac{1}{8b}\Big)\Big],$$

$$\log Z(c) \le \sum_{j=1}^{3L^4-3}\Big[\tfrac{1}{2}\log\tfrac{2}{\pi} - \tfrac{3}{2}\log c_{p_j}\Big].$$

With all $c_p = b$ the two bounds on $\log Z_L(b)$ differ by

$$2\log\tfrac{\pi}{2} + \tfrac{9}{2}L^4\log 6 + \tfrac{3}{2}\log 6 + 6\log b + (3L^4+1)\log\frac{8b}{8b-1}:$$

the coefficient of $L^4\log b$ is $-\frac{9}{2}$ in both, and the residual $6\log b$ comes from the four links of $F\setminus T$, which no shelling can use (Lemma 149: $\operatorname{rank}\partial_2 = 3L^4-3$ *because $H_1(\Lambda_L)=\mathbb{R}^4$) and which the lower bound treats as massive.*

Proof. Lemmas 148, 149, 150 and 146; in the lower bound $3/(4\cdot 6b) = 1/(8b)$. □

The upper bound improves [21, Lemma 475], whose axial gauge uses $3L^3(L-1)$ plaquettes and the constant $(\pi/2)^{5/2}$, to $3L^4-3$ plaquettes and the constant $\sqrt{2/\pi}$.

2. The first moment of the plaquette without the logarithm

Theorem 152 (the plaquette energy)**.** *Let $\beta \ge 1$ and $L \ge 2$. For every plaquette p,*

$$\mu(2-\operatorname{Tr}U_p) \le \frac{1}{\beta}\Big(1.6 + \frac{2\log(4\beta)}{3L^4}\Big).$$

Proof. Put $\Phi(b) = \log Z_L(b)$; then $\Phi'(b) = -\mathbb{E}_b S$, $\Phi''(b) = \operatorname{Var}_b(S) \ge 0$, and convexity gives, for $\theta \in (0,1)$, $\mathbb{E}_b S \le (\Phi(\theta b)-\Phi(b))/((1-\theta)b)$. By Proposition 151 with couplings θb in the upper bound,

$$\Phi(\theta b)-\Phi(b) \le 2\log\tfrac{\pi}{2} + \tfrac{9}{2}L^4\log\tfrac{6}{\theta} + \tfrac{3}{2}(\log 6 + 3\log\theta) + 6\log b + (3L^4+1)\log\frac{8b}{8b-1}.$$

Take $\theta = 1/4$, so that $\frac{3}{2}(\log 6 + 3\log\theta) + 2\log\frac{\pi}{2} < 0$; with $\mathbb{E}_b s_p = \mathbb{E}_b S/(6L^4)$ by translation invariance,

$$\mathbb{E}_b s_p \le \frac{1}{b}\Big[\log 24 + \frac{2}{3}\Big(1+\frac{1}{3L^4}\Big)\log\frac{8b}{8b-1} + \frac{4\log b}{3L^4}\Big].$$

For $b = 4\beta \ge 4$ and $L \ge 2$ the middle term is at most $\frac{2}{3}\cdot\frac{49}{48}\log\frac{32}{31} < 0.0217$. Since $\mu(2-\operatorname{Tr}U_p) = 2\mathbb{E}_b s_p$ and $2/b = 1/(2\beta)$, the bracket is halved: $\frac{1}{2}\log 24 < 1.5891$ and $\frac{1}{2}\cdot 0.0217 < 0.0109$ give the constant 1.6, and $\frac{2}{b}\cdot\frac{4\log b}{3L^4} = \frac{2\log(4\beta)}{3L^4\beta}$. □

Theorem 152 replaces the bound $(\log\beta + 6.11)/\beta$ of Lemma 76: the logarithm there came from bounding the free energy at coupling $\beta/2$ by $\log Z \le 0$, while here both free energies are bounded with the same coefficient of $L^4\log b$, and the only logarithm left is the global term $6\log b$ divided by the number $6L^4$ of plaquettes. The weak-coupling value of $\mathbb{E}_b s_p$ is $3/(4b) + O(b^{-2})$, so the constant $\log 24$ is off by a factor of about 4.2.

3. The tail of the plaquette energy and the second moment

Let L be even. Put $Y = \{y \in \Lambda_L : y_0, y_3 \text{ even}\}$, $|Y| = L^4/4$, and $O = \{(y,12) : y \in Y\}$, a set of $L^4/4$ plaquettes. For a site reflection ϑ (in a hyperplane $x_\mu = k$, $k \in \mathbb{Z}$) put $(\vartheta U)_e = U_{\vartheta e}^{\pm 1}$, the inverse for the links perpendicular to the hyperplane, whose orientation is reversed. It maps plaquettes to plaquettes, and the holonomy of ϑp in ϑU is a conjugate of $U_p^{\pm 1}$.

Lemma 153 (chessboard estimate)**.** *Let G be a compact group, $\psi \ge 0$ a class function on G with $\psi(U^{-1}) = \psi(U)$, and μ the measure with density proportional to $\prod_p \psi(U_p)$ on G^E. Let $p = (0,12)$ and $\varepsilon \ge 0$.*

(i) *If $4 \mid L$, then $\mu(s_p > \varepsilon) \le \mu\big(\bigcap_{q\in O}\{s_q > \varepsilon\}\big)^{4/L^4}$.*

(ii) *If L is even, then $\mu(s_p > \varepsilon) \le \mu\big(\bigcap_{q\in O}\{s_q > \varepsilon\}\big)^{1/L^4}$.*

The same holds for $\{|\theta_p| > \delta\}$ in place of $\{s_p > \varepsilon\}$ when $G = \mathrm{U}(1)$, $U_p = e^{i\theta_p}$.

Proof. The measure is reflection positive for every reflection in a hyperplane through sites [90, 95]: every plaquette lies in the hyperplane or on one side of it, the weight factorises as $W_+ W_0 W_-$ with $W_0 \ge 0$ a function of the links in the hyperplane and $W_- = \vartheta W_+$ (here $\psi(U^{-1}) = \psi(U)$ and the class property are used), so that for F measurable with respect to the links on the positive side, $\int F\,\overline{\vartheta F}\prod_p \psi(U_p)\,dU = \int W_0 \big|\int F W_+\big|^2 \ge 0$, as ϑ preserves Haar measure. (i) Tile the torus by the $L^4/4$ blocks $y + [-1,1]\times[0,1]\times[0,1]\times[-1,1]$, $y \in Y$; the separating hyperplanes $x_0 \in 2\mathbb{Z}+1$, $x_3 \in 2\mathbb{Z}+1$, $x_1 \in \mathbb{Z}$, $x_2 \in \mathbb{Z}$ pass through sites, and the number of blocks in each direction, $L/2$ or L, is even. The chessboard estimate [49, Sect. 2], [26, Sect. 5.3] then gives $\mu(f) \le \mu(\prod_{y\in Y} f^{(y)})^{4/L^4}$ for every bounded $f \ge 0$ of the links with both endpoints in the block of 0, where $f^{(y)}$ is the image of f under the element of the group generated by the reflections that carries the block of 0 to that of y (for an even number of blocks in each direction this element is unique on blocks: a translation by y_μ when $y_\mu \equiv 0 \bmod 4$ in the directions $0,3$, resp. y_μ even in the directions $1,2$, otherwise the reflection in $x_\mu = y_\mu/2$, resp. $x_\mu = (y_\mu+1)/2$). For $f = 1_{\{s_p>\varepsilon\}}$ this element carries the plaquette $(0,12)$ to $(y,12)$, up to orientation, so $f^{(y)} = 1_{\{s_{(y,12)}>\varepsilon\}}$ by the invariance of the trace under conjugation and inversion. (ii) Use the unit blocks $y + [0,1]^4$, $y \in \Lambda_L$, L even; the chessboard estimate gives $\mu(f) \le \mu(\prod_{y\in\Lambda_L} f^{(y)})^{1/L^4}$, and the element carrying the block of 0 to that of y carries $(0,12)$ to $(y',12)$ with $y'_\mu = y_\mu$ for y_μ even and $y'_\mu = y_\mu + 1$ for y_μ odd in the directions $0,3$, and to

$(y,12)$ up to orientation in the directions $1,2$; so each indicator $1_{\{s_q>\varepsilon\}}$, $q\in O$, appears four times in the product, which is therefore $1_{\bigcap_{q\in O}\{s_q>\varepsilon\}}$. □

The $2\times1\times1\times2$ blocks of (i) are needed because a block of side 2 in the directions $0,3$ keeps the orbit of $(0,12)$ at one plaquette in four; for $L\equiv2 \bmod 4$ the number $L/2$ of such blocks is odd and the chessboard estimate does not apply to them (the reflection in $x_0=L/2$ maps a block onto itself), so (ii) is used instead.

Theorem 154 (the tail of the plaquette energy)**.** *Let $\beta\ge1$, $L\ge8$ even, $b=4\beta$, p a plaquette, $\varepsilon\ge0$ and $\lambda\in(0,1)$. Put*

$$A_\lambda = 18\log6+12\log\frac{8b}{8b-1}+\frac{3}{L}\log\frac{1}{1-\lambda} + \frac{1}{L^4}\Big[24\log b+6\log6+8\log\frac{\pi}{2}+4\log\frac{8b}{8b-1}\Big],$$

and $\kappa_L=1$ if $4\mid L$, $\kappa_L=4$ if $L\equiv2 \bmod 4$. Then

$$\mu(s_p>\varepsilon)\le\exp\big((-\lambda b\varepsilon+A_\lambda)/\kappa_L\big),$$

and $A_{9/10}\le33.5+24L^{-4}\log(4\beta)$.

Proof. By symmetry $p=(0,12)$. On $\{s_q>\varepsilon\}$, $1\le e^{\lambda b(s_q-\varepsilon)}$, so by Lemma 153

$$\mu(s_p>\varepsilon)^{\kappa_L L^4/4}\le e^{-\lambda b\varepsilon|O|}\,\frac{Z(c)}{Z_L(b)},$$

where $c_q=(1-\lambda)b$ for $q\in O$ and $c_q=b$ for $q\notin O$. In the shelling of Lemma 150, $m_O=\frac{L}{2}(L^2-1)$ plaquettes lie in O and the other $3L^4-3-m_O$ have coupling b; with the lower bound of Lemma 148 for $Z_L(b)$, Proposition 151 gives

$$\log\frac{Z(c)}{Z_L(b)}\le2\log\tfrac{\pi}{2}+\tfrac{9}{2}L^4\log6+\tfrac{3}{2}\log6+6\log b + \tfrac{3}{2}m_O\log\frac{1}{1-\lambda}+(3L^4+1)\log\frac{8b}{8b-1}.$$

Multiplying by $4/(\kappa_L L^4)$ and using $|O|=L^4/4$ and $6m_O/L^4<3/L$ gives the bound. For $\lambda=9/10$, $b\ge4$ and $L\ge8$: $18\log6<32.2517$, $12\log\frac{32}{31}<0.3810$, $\frac{3}{8}\log10<0.8635$ and $8^{-4}(6\log6+8\log\frac{\pi}{2}+4\log\frac{32}{31})<0.0036$, in all less than 33.5. □

Theorem 155 (the second moment of the plaquette)**.** *Let $\beta\ge1$ and $L\ge8$ even, $\ell=L^{-4}\log(4\beta)$, and $C_L=368$ if $4\mid L$, $C_L=439$ if $L\equiv2 \bmod 4$. For every plaquette p,*

$$\mathrm{Var}_\mu(\mathrm{Tr}\,U_p)\le\mu\big((2-\mathrm{Tr}\,U_p)^2\big)\le\frac{C_L(1+\ell)^2}{\beta^2}.$$

Proof. Put $\lambda=9/10$, $A=33.5+24\ell$, $\kappa=\kappa_L$, so that $\mu(s_p>\varepsilon)\le\min(1,e^{(A-\lambda b\varepsilon)/\kappa})$. With $\varepsilon_0=A/(\lambda b)$,

$$\begin{aligned}\mu(s_p^2)&=\int_0^2 2\varepsilon\,\mu(s_p>\varepsilon)\,d\varepsilon\\&\le\varepsilon_0^2+\int_{\varepsilon_0}^\infty 2\varepsilon e^{-\lambda b(\varepsilon-\varepsilon_0)/\kappa}d\varepsilon\\&=\frac{A^2+2\kappa A+2\kappa^2}{\lambda^2b^2}.\end{aligned}$$

For $x\ge0$, $(A_1+x)^2+2\kappa(A_1+x)+2\kappa^2\le(A_1^2+2\kappa A_1+2\kappa^2)(1+x/A_1)^2$ (expand both sides), and $24\ell/33.5\le\ell$; so with $A_1=33.5$, $A^2+2\kappa A+2\kappa^2\le(A_1^2+2\kappa A_1+2\kappa^2)(1+\ell)^2$. Finally $\mu((2-\mathrm{Tr}\,U_p)^2)=4\mu(s_p^2)$, $b=4\beta$, and $(A_1^2+2A_1+2)/(4\lambda^2)<367.7$, $(A_1^2+8A_1+32)/(4\lambda^2)<439$. □

Remark 156 (the residual ℓ). The factor $(1+\ell)^2$, $\ell=L^{-4}\log(4\beta)$, is the only trace of β beyond β^{-2}. It comes from the term $6\log b$ of Proposition 151, the mismatch between the $3L^4-3$ plaquettes a shelling can use and the $3L^4+1$ links the lower bound treats as massive, that is from $H_1(\Lambda_L)=\mathbb{R}^4$: the flat connections of the torus (commuting holonomies around the four cycles) are zero modes of the action, and neither bound integrates over them. The chessboard estimate raises the ratio of partition functions to the power $4/L^4$, so this global term enters the tail as b^{24/L^4}. For $\log(4\beta)\le L^4$ the bound reads $\mu((2-\mathrm{Tr}\,U_p)^2)\le4C_L/\beta^2$, and as $L\to\infty$ at fixed β it tends to C_L/β^2. Removing ℓ would require a lower bound on $Z_L(b)$ with the exact power of b, which is not determined here.

Corollary 157 (a β-independent constant for the plaquette traces)**.** *Let $\beta\ge11$, $n\ge2$, $L\ge2n+4$ even, $\ell=L^{-4}\log(4\beta)$, C_L as in Theorem 155. For every plaquette p and every link $e\in p$,*

(i) $\mu(\mathrm{Var}_\mu(\mathrm{Tr}\,U_p\mid\mathscr{F}_e))\ge1/(1200\beta^2)$*;*

(ii) $\mathscr{E}_n(\mathrm{Tr}\,U_p)\ge\mu(\mathrm{Var}_\mu(\mathrm{Tr}\,U_p\mid\mathscr{F}_e))$*, hence*

$$\mathscr{E}_n(\mathrm{Tr}\,U_p)\ge\frac{1}{1200\beta^2}\ge\frac{\mathrm{Var}_\mu(\mathrm{Tr}\,U_p)}{1200C_L(1+\ell)^2}.$$

In particular the inequality of $\mathrm{BP}(n,\varpi)$ *holds for the functions* $\mathrm{Tr}\,U_p$ *with* $\varpi=1/(1200C_L(1+\ell)^2)$*, and with* $\varpi=1/(1.77\cdot10^6)$ *for* $4\mid L$*,* $\varpi=1/(2.11\cdot10^6)$ *for* $L\equiv2 \bmod 4$*, whenever* $\log(4\beta)\le L^4$*, uniformly in β, L, n and p. This is a statement about these functions only: by Theorem 77(ii),* $\mathrm{BP}(n,\varpi)$ *itself fails on* Λ_{2n+4} *unless* $\varpi\le38400n^4(\log\beta+6.11)/\beta$*, so no β-independent constant holds for all of $\mathscr{A}$ at fixed n.*

Proof. (i) The proof of Proposition 140(i) gives $\mathrm{Var}_\mu(\mathrm{Tr}\,U_p\mid\mathscr{F}_e)\ge1/(600\beta^2)$ on $\{a\ge6\}$, $a=4\beta|w|$, and on $\{a<6\}$ it gives $\sum_{p'\ni e}s_{p'}=\frac12\sum_{p'\ni e}(2-\mathrm{Tr}\,U_{p'})>6-\frac{3}{2\beta}$, so that $s_{p'}>1-\frac{1}{4\beta}$ for one of the six plaquettes $p'\ni e$ (this step did not use $\beta\ge100$). By Markov's inequality, Theorem 152 and the equivalence of all plaquettes,

$$\mu(a<6)\le\frac{6\,\mu(s_p)}{1-1/(4\beta)}\le\frac{6\big(0.8+\frac{\log(4\beta)}{3L^4}\big)}{\beta-\frac14}\le\frac12$$

for $\beta\ge11$ and $L\ge8$: the middle expression is 0.447 at $\beta=11$, $L=8$, decreases in L, and decreases in β since its logarithmic derivative is negative. Hence $\mu(\mathrm{Var}_\mu(\mathrm{Tr}\,U_p\mid\mathscr{F}_e))\ge\frac12\cdot\frac{1}{600\beta^2}$.

(ii) In Sec. IX F, V_x is the set of links with base point in $x+[0,n)^4$, so a link $e=(y,\mu)$ lies in V_x for exactly the n^4 values $x\in y-[0,n)^4$, distinct residues as $L\ge n+2$. For each of them $\mathscr{F}_{V_x}\subset\mathscr{F}_e$, hence $\mu((f-\mathscr{P}_xf)^2)\ge\mu(\mathrm{Var}_\mu(f\mid\mathscr{F}_e))$ for every f, as in the proof of Proposition 138; dropping the other x

gives $\mathscr{E}_n(f) \ge n^{-4}\cdot n^4\,\mu(\mathrm{Var}_\mu(f \mid \mathscr{F}_e))$. Combine with (i) and Theorem 155; $1200\cdot 4\cdot 368 < 1.77\cdot 10^6$ and $1200\cdot 4\cdot 439 < 2.11\cdot 10^6$. □

This is the bound asked for in Remark 141, with $C = C_L(1+\ell)^2$. The argument of (ii) is general: $\mathscr{E}_n(f) \ge \max_e \mu(\mathrm{Var}_\mu(f \mid \mathscr{F}_e))$ for every $f \in L^2(\mu)$, which removes the factor $((n-1)/n)^2$ of Proposition 140(iii) (that factor counted the cubes containing all four links of p); the threshold $\beta \ge 100$ there becomes $\beta \ge 11$. Proposition 140(ii), (iii) are superseded for $\beta \ge 11$ and even $L \ge 8$.

4. *Wilson loops of fixed size*

For a closed lattice path $\gamma = (x_0, x_1, \dots, x_m = x_0)$ let $W_\gamma = U_{(x_0,x_1)}\cdots U_{(x_{m-1},x_m)}$, with $U_{(x,x+e_\mu)} = U_{(x,\mu)}$ and $U_{(x+e_\mu,x)} = U_{(x,\mu)}^{-1}$; W is multiplicative under concatenation, $W_{\gamma^{-1}} = W_\gamma^{-1}$, and is unchanged by inserting or removing a backtracking (x,y,x). Let $\theta(U) = \arccos(\frac12 \mathrm{Tr}\,U) \in [0,\pi]$, the geodesic distance from $\mathbf{1}$ in the bi-invariant metric of S^3, so that $\theta(gUg^{-1}) = \theta(U) = \theta(U^{-1})$ and $\theta(UV) \le \theta(U)+\theta(V)$.

Lemma 158 (lassos)**.** *Let $1 \le r \le L-1$, $\{\mu,\nu\}$ a plane, R the set of the r^2 plaquettes $(y+ie_\mu+je_\nu, \mu\nu)$, $0 \le i,j < r$, and γ the boundary loop of their union, based at y. Then $W_\gamma = \prod_{p\in R} g_p U_p^{\pm1} g_p^{-1}$ in a suitable order, with g_p the holonomy of a path from y to a corner of p; consequently $\theta(W_\gamma) \le \sum_{p\in R}\theta(U_p)$.*

Proof. Add the plaquettes row by row (j outer, i inner). After each step the union R_k is a staircase region (full rows below a partial row) with boundary a simple closed curve γ_k through y; the next plaquette p meets R_k along one or two consecutive edges, forming a path $\sigma \subset \gamma_k$, and γ_{k+1} replaces σ by the complementary path τ of ∂p. Writing $\gamma_k = \alpha\sigma\alpha'$ from y, $\gamma_{k+1} = \alpha\tau\alpha' \sim (\alpha\tau\sigma^{-1}\alpha^{-1})(\alpha\sigma\alpha')$ after inserting the backtrack $\alpha^{-1}\alpha$, so $W_{\gamma_{k+1}} = gU_p^{\pm1}g^{-1}W_{\gamma_k}$ with $g = W_\alpha$ times the holonomy along ∂p from the start of σ to the corner of p. Induction gives the product formula; the triangle inequality and the invariance of θ give the angle bound. For $r \le L-1$ the r^2 plaquettes are distinct and the region embeds in the torus. □

Theorem 159 (Wilson loops of size r)**.** *Let $\beta \ge 11$, $n \ge 2$, $L \ge 2n+4$ even, ℓ, C_L as above, γ the boundary of an $r\times r$ square of plaquettes, $1 \le r \le L-1$, e a link of γ, $W = W_\gamma$.*

(i) $\mu(\mathrm{Var}_\mu(\mathrm{Tr}\,W \mid \mathscr{F}_e)) \ge 1/(1200\beta^2)$*; this holds for every closed path through e exactly once, of any shape.*

(ii) $\mathrm{Var}_\mu(\mathrm{Tr}\,W) \le \mu\big((2-\mathrm{Tr}\,W)^2\big) \le \frac{\pi^4}{16} r^8 C_L(1+\ell)^2/\beta^2$*, which is at most $2241\,r^8(1+\ell)^2/\beta^2$ for $4 \mid L$ and $2673\,r^8(1+\ell)^2/\beta^2$ for $L \equiv 2 \bmod 4$.*

(iii) $\mathscr{E}_n(\mathrm{Tr}\,W) \ge \mu(\mathrm{Var}_\mu(\mathrm{Tr}\,W \mid \mathscr{F}_e))$*, hence*

$$\mathscr{E}_n(\mathrm{Tr}\,W) \ge \frac{1}{1200\beta^2} \ge \frac{\mathrm{Var}_\mu(\mathrm{Tr}\,W)}{1200\cdot\frac{\pi^4}{16}C_L\,r^8(1+\ell)^2};$$

the denominator is at most $2.69\cdot 10^6\,r^8(1+\ell)^2$ for $4 \mid L$ and $3.21\cdot 10^6\,r^8(1+\ell)^2$ for $L \equiv 2 \bmod 4$.

Proof. (i) As in the proof of Proposition 140(i), the conditional law of U_e given $\mathscr{F}_e$ is $\nu_{a,\hat w}$ with $a = 4\beta|w|$, and since e occurs once in γ, $\mathrm{Tr}\,W = \mathrm{Tr}(U_e^{\pm1}X) = 2\langle U_e, v\rangle$ with X the product of the other link variables and v an $\mathscr{F}_e$-measurable unit vector. Lemma 139(v) gives $\mathrm{Var}(\mathrm{Tr}\,W \mid \mathscr{F}_e) \ge 0.96/a^2 \ge 1/(600\beta^2)$ on $\{a \ge 6\}$, an event of probability at least $\frac12$ by the proof of Corollary 157(i); nothing depends on the shape of γ.

(ii) $h_W = 2 - \mathrm{Tr}\,W = 2(1-\cos\theta(W)) \le \theta(W)^2$. By Lemma 158 and the Cauchy–Schwarz inequality, $\theta(W)^2 \le r^2\sum_{p\in R}\theta(U_p)^2$, and $\theta(U_p)^2 \le \frac{\pi^2}{4}h_p$ since $1-\cos t \ge 2t^2/\pi^2$ on $[0,\pi]$. Hence $h_W \le \frac{\pi^2 r^2}{4}\sum_{p\in R}h_p$, and Minkowski's inequality with the equivalence of all plaquettes gives $\mu(h_W^2) \le \frac{\pi^4 r^4}{16}\big(\sum_{p\in R}\mu(h_p^2)^{1/2}\big)^2 = \frac{\pi^4 r^8}{16}\mu(h_p^2)$; apply Theorem 155; $\pi^4\cdot 368/16 < 2240.5$, $\pi^4\cdot 439/16 < 2673$.

(iii) As in Corollary 157(ii), $\mathscr{E}_n(\mathrm{Tr}\,W) \ge \mu(\mathrm{Var}_\mu(\mathrm{Tr}\,W \mid \mathscr{F}_e))$; combine with (i) and (ii); $1200\cdot 2241 < 2.69\cdot 10^6$, $1200\cdot 2673 < 3.21\cdot 10^6$. □

Remark 160 (where the dependence on r enters). The constant of Theorem 159(iii) is independent of β and L and depends on r only through the factor r^{-8}, which has two deterministic sources: the triangle inequality $\theta(W) \le \sum_p \theta(U_p)$ replaces the holonomy angle, in which the r^2 plaquette angles enter with the signs and rotations of the lassos, by the sum of their absolute values, and with the Cauchy–Schwarz inequality costs r^4 in h_W^2; the Minkowski step bounds the second moment of $\sum_p h_p$ by $r^4\mu(h_p^2)$, ignoring any decorrelation between distant plaquettes, and costs another r^4. The lower bound (i) uses one link of the loop and is independent of r; using all cubes that meet the loop would gain at most a factor of order r and would not change the order in β. No cancellation between plaquettes is proved; a bound on $\mathrm{Cov}_\mu(h_p,h_q)$ for distant p,q would reduce r^8 to r^4 in (ii) and is the natural next step. The true order in r is not claimed.

5. *The Villain* U(1) *model on the torus*

Let $G = \mathrm{U}(1)$ with link angles $\theta_e \in (-\pi,\pi]$, Haar measure $d\theta_e/(2\pi)$, and for a plaquette p let $\theta_p \in (-\pi,\pi]$ be the principal value of $(d\theta)_p$. The Villain measure μ has density proportional to $\prod_p \varphi_\beta(\theta_p)$, $\varphi_\beta(\theta) = \sum_{k\in\mathbb{Z}} e^{-\beta(\theta+2\pi k)^2/2}$, and the field strength of Proposition 243 is $\Psi_p = \varphi_\beta'(\theta_p)/\varphi_\beta(\theta_p)$. For $\theta \in (-\pi,\pi]$ let κ_θ be the law on $\mathbb{Z}$ with $\kappa_\theta(k) \propto e^{-\beta(\theta+2\pi k)^2/2}$ and put

$$\psi(\theta) = \sum_k k\,\kappa_\theta(k), \qquad \kappa(\theta) = \sum_k k^2\kappa_\theta(k),$$
$$\tilde F(\theta) = \sum_k (\theta+2\pi k)^2\kappa_\theta(k), \qquad \bar F(\theta) = \theta + 2\pi\psi(\theta),$$

so that $\Psi_p = -\beta\bar F(\theta_p)$. Put $\eta_\beta = 2.001\,e^{-2\pi^2\beta}$ and $\varepsilon_\beta = e^{-3\pi^2\beta}/(\pi\sqrt{3\pi\beta})$. Throughout this part $4 \mid L$, because Lemma 153(i) is used.

Lemma 161 (the Villain weight)**.** *Let $\beta \ge 1/4$ and $|\theta| \le \pi$.*

(i) $e^{-\beta\theta^2/2} \le \varphi_\beta(\theta) \le 2.001\, e^{-\beta\theta^2/2}$*; moreover* $\varphi_\beta(\theta) \le \varphi_\beta(0) \le 1+\eta_\beta$.

(ii) $\int_{-\pi}^{\pi} \varphi_\beta(\theta)\, d\theta = \sqrt{2\pi/\beta}$.

(iii) $|\psi(\theta)| \le \kappa(\theta) \le 2.001\, e^{-2\pi\beta(\pi-|\theta|)}$.

(iv) $\max(\theta^2, \bar F(\theta)^2) \le \tilde F(\theta)$, $-\partial_\beta \log\varphi_\beta(\theta) = \tilde F(\theta)/2$ *and* $\partial_\beta^2 \log\varphi_\beta(\theta) \ge 0$.

Proof. For $k \ne 0$, $(\theta+2\pi k)^2 - \theta^2 = 4\pi|k|(\pi|k| + \theta\operatorname{sgn} k) \ge 4\pi|k|(\pi - |\theta|) + 4\pi^2|k|(|k|-1)$. (i) The lower bound is the term $k=0$. Each of the terms $k=\pm1$ is at most $e^{-\beta\theta^2/2}$, and the sum of the terms $|k|\ge 2$ is at most $e^{-\beta\theta^2/2}\cdot 2e^{-4\pi^2\beta}/(1-e^{-8\pi^2\beta}) \le 0.001\, e^{-\beta\theta^2/2}$ for $\beta \ge 1/4$. By Poisson summation $\varphi_\beta(\theta) = (2\pi\beta)^{-1/2}\sum_n e^{-n^2/(2\beta)}\cos(n\theta)$ with positive coefficients, so

$$\varphi_\beta(\theta) \le \varphi_\beta(0) = \sum_k e^{-2\pi^2\beta k^2} \le 1 + \frac{2e^{-2\pi^2\beta}}{1-e^{-6\pi^2\beta}} \le 1+\eta_\beta.$$

(ii) Unfold the sum. (iii) $\kappa_\theta(k) \le e^{-\beta[(\theta+2\pi k)^2-\theta^2]/2}$, so

$$\kappa(\theta) \le e^{-2\pi\beta(\pi-|\theta|)} \sum_{k\ne0} k^2 e^{-2\pi^2\beta|k|(|k|-1)} \le 2.001\, e^{-2\pi\beta(\pi-|\theta|)}$$

for $\beta \ge 1/4$, and $|\psi| \le \sum |k|\kappa_\theta(k) \le \kappa$. (iv) $|\theta+2\pi k| \ge |\theta|$ for $|\theta| \le \pi$ gives $\theta^2 \le \tilde F$, and $\bar F^2 \le \tilde F$ is Jensen's inequality for κ_θ (note that $\bar F(\pm\pi) = 0$, so $\theta^2 \le \bar F^2$ is false in general). Differentiating $\log\varphi_\beta$ in β gives $-\tilde F/2$, and once more the variance of $(\theta+2\pi k)^2/2$ under κ_θ. □

Proposition 162 (two-sided bounds for Villain weights)**.** *Let $\beta \ge 1/4$, $N = 3L^4+1$, $Z_L^V(\beta) = \int \prod_p \varphi_\beta(\theta_p) \prod_e d\theta_e/(2\pi)$.*

(i) $\log Z_L^V(\beta) \ge N\big[-\log(2\sqrt{3\pi\beta}) + \log(1-\varepsilon_\beta)\big]$.

(ii) *Let $w_p \ge 0$ be even functions on $(-\pi,\pi]$ with $w_p \le M$, and $(e_j, p_j)_{j\le 3L^4-3}$ the sequence of Lemma 150. Then*

$$\int \prod_p w_p(\theta_p) \prod_e \frac{d\theta_e}{2\pi} \le M^{3L^4+3} \prod_{j=1}^{3L^4-3} \frac{1}{2\pi}\int_{-\pi}^{\pi} w_{p_j}(t)\, dt.$$

Proof. (i) By Lemma 147 with $G = \mathrm{U}(1)$,

$$Z_L^V(\beta) = \int \prod_p \varphi_\beta(\theta_p) \prod_{e\notin T} \frac{d\theta_e}{2\pi}, \qquad \theta_e = 0 \text{ on } T.$$

There θ_p is the principal value of $\sigma_p = \sum_{e\in p\setminus T} \pm\theta_e$, so $\theta_p^2 \le \sigma_p^2$ and $\varphi_\beta(\theta_p) \ge e^{-\beta\sigma_p^2/2}$. Let ν have density $e^{-3\beta t^2}/Z_6$ on $(-\pi,\pi]$, $Z_6 = \int_{-\pi}^{\pi} e^{-3\beta t^2} dt \ge \sqrt{\pi/(3\beta)} - e^{-3\pi^2\beta}/(3\pi\beta) = \sqrt{\pi/(3\beta)}(1-\varepsilon_\beta)$, by $\int_x^\infty e^{-at^2} dt \le e^{-ax^2}/(2ax)$. Jensen's inequality for $\nu^{\otimes N}$, with $\sigma^2 = \mathbb{E}_\nu t^2$ and $\mathbb{E}_\nu \sigma_p^2 = k_p\sigma^2$ (ν is symmetric), gives

$$\log Z_L^V(\beta) \ge -\frac{\beta}{2}\sum_p k_p\sigma^2 + N\Big(3\beta\sigma^2 + \log\frac{Z_6}{2\pi}\Big) = N\log\frac{Z_6}{2\pi},$$

as $\sum_p k_p = 6N$, and $Z_6/(2\pi) = (1-\varepsilon_\beta)/(2\sqrt{3\pi\beta})$. (ii) As in Lemma 149: integrating θ_{e_j} over the circle against $w_{p_j}(\theta_{p_j})$, where θ_{p_j} is the principal value of $\pm\theta_{e_j} + s$ with s independent of θ_{e_j}, gives $\frac{1}{2\pi}\int w_{p_j}$ by periodicity; the plaquettes dropped are at most $6L^4 - (3L^4-3)$ in number, each of weight at most M. □

Proposition 163 (moments of the plaquette angle)**.** *Let $L \ge 8$, $4 \mid L$, p a plaquette, $C_\theta = 2.12 + L^{-4}(\log\beta + 1.3)$ and $K_\beta = e^{11.54}\beta^{1/512}$.*

(i) *For $\beta \ge 2$:* $\max\big(\mu(\theta_p^2), \mu(\bar F(\theta_p)^2)\big) \le \mu(\tilde F(\theta_p)) \le C_\theta/\beta$.

(ii) *For $\beta \ge 1$: $\mu(|\theta_p| > \delta) \le K_\beta e^{-\beta\delta^2/4}$ for every $\delta \ge 0$.*

Proof. (i) Put $\Phi(\beta) = \log Z_L^V(\beta)$. By Lemma 161 (iv), $\Phi'(\beta) = -\frac12\sum_p \mu(\tilde F(\theta_p))$ and $\Phi'' \ge 0$, so $\sum_p \mu(\tilde F(\theta_p)) \le 2(\Phi(\beta/4) - \Phi(\beta))/(\frac34\beta)$. Proposition 162(ii) with $w_p = \varphi_{\beta/4} \le 1+\eta_{\beta/4}$ and Lemma 161(i), (ii) give $\Phi(\beta/4) \le (3L^4+3)\eta_{\beta/4} - \frac12(3L^4-3)\log(\pi\beta/2)$, and (i) of that proposition bounds $\Phi(\beta)$ below; hence

$$\begin{aligned}\Phi(\beta/4) - \Phi(\beta) \le 3L^4\Big[\log(2\sqrt{3\pi}) - \tfrac12\log\tfrac{\pi}{2}\Big] + 2\log\beta \\ + \tfrac32\log\tfrac{\pi}{2} + \log(2\sqrt{3\pi}) \\ + (3L^4+1)|\log(1-\varepsilon_\beta)| \\ + (3L^4+3)\eta_{\beta/4}.\end{aligned}$$

The bracket is less than 1.5891 and the constant $\frac32\log\frac{\pi}{2} + \log(2\sqrt{3\pi})$ less than 2.493. Dividing by $6L^4\cdot\frac34\beta/2 = \frac94 L^4\beta$ and using translation invariance,

$$\mu(\tilde F(\theta_p)) \le \frac{1}{\beta}\Big[\tfrac43\cdot 1.5891 + \frac{\frac89\log\beta + 1.109}{L^4} + \tfrac43\Big(1+\frac{1}{L^4}\Big)\big(|\log(1-\varepsilon_\beta)| + \eta_{\beta/4}\big)\Big];$$

for $\beta \ge 2$ and $L \ge 8$ the last term is below $1.4\cdot10^{-4}$ (it is $\eta_{\beta/4} \le 2.001 e^{-\pi^2}$ that forces $\beta \ge 2$: at $\beta = 1$ it would exceed 0.019), and $\frac43\cdot 1.5891 < 2.1189$, which gives C_θ. The first inequality is Lemma 161(iv).

(ii) Lemma 153(i) applies to $\psi(e^{i\theta}) = \varphi_\beta(\theta)$, nonnegative and even. With O the orbit of $p = (0,12)$, $\lambda = \frac12$ and $1_{\{|\theta_q|>\delta\}} \le e^{\lambda\beta(\theta_q^2-\delta^2)/2}$, Lemma 161(i) gives on O the weights $1_{\{|\theta_q|>\delta\}}\varphi_\beta(\theta_q) \le 2.001\, e^{-\beta\delta^2/4} w(\theta_q)$, $w(t) = e^{-\beta t^2/4} \le 1$, so that

$$\mu(|\theta_p| > \delta)^{L^4/4} \le \big(2.001\, e^{-\beta\delta^2/4}\big)^{L^4/4} \frac{\tilde Z}{Z_L^V(\beta)},$$

$\tilde Z$ the integral of Proposition 162(ii) for the weights w on O and φ_β elsewhere, all at most $1+\eta_\beta$. Of the shelled plaquettes $m_O = \frac{L}{2}(L^2-1)$ lie in O, with $\frac{1}{2\pi}\int w \le (\pi\beta)^{-1/2}$, and the

others give $(2\pi\beta)^{-1/2}$. Raising to the power $4/L^4$,

$$\begin{aligned}\log\mu(|\theta_p|>\delta) \le -\frac{\beta\delta^2}{4} &+ \log 2.001 \\ &+ 12\Big[\log(2\sqrt{3\pi}) - \tfrac{1}{2}\log(2\pi)\Big] \\ &+ \frac{8\log\beta + 6\log(2\pi) + 4\log(2\sqrt{3\pi})}{L^4} \\ &+ \frac{2m_O}{L^4}\log 2 + \Big(12 + \frac{4}{L^4}\Big)|\log(1-\varepsilon_\beta)| \\ &+ \frac{4(3L^4+3)}{L^4}\eta_\beta,\end{aligned}$$

the terms in $\log\beta$ before the power $4/L^4$ being $-\frac{1}{2}(3L^4-3)\log\beta$ from $\tilde{Z}$ and $\frac{1}{2}(3L^4+1)\log\beta$ from the lower bound for $Z_L^V(\beta)$, together $2\log\beta$. For $L\ge 8$, $\beta\ge 1$ the right side without $-\beta\delta^2/4$ is at most $0.6937+10.7506+0.0867+0.0045+10^{-7}+8L^{-4}\log\beta < 11.54+\frac{1}{512}\log\beta$. □

Lemma 164 (the Gaussian and the defects)**.** *Let $\tilde{P}$ be the probability measure on $(-\pi,\pi]^E\times\mathbb{Z}^P$ with density proportional to $\prod_p e^{-\beta((d\theta)_p+2\pi m_p)^2/2}$, and put $F = d\theta + 2\pi m \in C^2$ (real 2-cochains, θ read as a real 1-cochain). Then:*

(i) *the marginal of θ under $\tilde{P}$ is μ;*

(ii) *given θ, the m_p are independent, $m_p + j_p$ has the law κ_{θ_p} where $(d\theta)_p = \theta_p + 2\pi j_p$, $\mathbb{E}_{\tilde{P}}(F_p \mid \theta) = \bar{F}(\theta_p) = -\Psi_p/\beta$ and $\operatorname{Var}_{\tilde{P}}(F_p \mid \theta) = 4\pi^2 \operatorname{Var}_{\kappa_{\theta_p}}(k) \le 4\pi^2\kappa(\theta_p)$;*

(iii) *$F = G + 2\pi D$ with G distributed according to the Gaussian measure γ_β on $B = dC^1$ of Theorem 134, $D \in \ker\partial = B^\perp$ (with values in the discrete set $P_{B^\perp}\mathbb{Z}^P$), and G, D independent. In particular, for every exact $c\in B$, $\langle c,F\rangle = \langle c,G\rangle$ has the law $N(0,\|c\|^2/\beta)$ under $\tilde{P}$.*

Proof. (i), (ii): summing the density over m gives $\prod_p \varphi_\beta(\theta_p)$, since $(d\theta)_p + 2\pi m_p = \theta_p + 2\pi(m_p + j_p)$ and $m_p \mapsto m_p + j_p$ is a bijection of $\mathbb{Z}$; the conditional density of m given θ factorises over p, and $F_p = \theta_p + 2\pi k$ with $k = m_p + j_p \sim \kappa_{\theta_p}$. (iii) Let M be the measure on $\mathbb{R}^E\times\mathbb{Z}^P$ with density $e^{-\beta\|d\theta+2\pi m\|^2/2}$ with respect to Lebesgue times counting measure. The group $\mathbb{Z}^E$ acts on it by $(\theta,m)\mapsto(\theta+2\pi k, m-dk)$, preserving the density and F; $(-\pi,\pi]^E\times\mathbb{Z}^P$ is a fundamental domain, so $\tilde{P}$ is the normalised restriction of M to it, and the law of F is the same for every fundamental domain. Let $Z_1 = \ker d \subset \mathbb{R}^E$, $Z_1^{\mathbb{Z}} = Z_1\cap\mathbb{Z}^E$, a lattice of full rank in Z_1 (the kernel of an integer matrix), and $R\subset\mathbb{Z}^P$ a set of representatives of $\mathbb{Z}^P/d\mathbb{Z}^E$. A second fundamental domain is $\Omega\times R$ with $\Omega = \Omega_1\times Z_1^\perp$, Ω_1 a fundamental domain of $2\pi Z_1^{\mathbb{Z}}$ in Z_1: given (θ,m) there is $k_1\in\mathbb{Z}^E$, unique modulo $Z_1^{\mathbb{Z}}$, with $m - dk_1 \in R$, and then a unique $k_2\in Z_1^{\mathbb{Z}}$ with $\theta + 2\pi(k_1+k_2)\in\Omega$. On $\Omega\times R$ write $\theta = \theta_1+\theta_2$, $\theta_1\in\Omega_1$, $\theta_2\in Z_1^\perp$, and for $r\in R$ let $2\pi P_B r = dA_r$ with $A_r\in Z_1^\perp$, $D_r = P_{B^\perp}r$. Then $d\theta + 2\pi r = d(\theta_2+A_r) + 2\pi D_r$ and $\|d\theta+2\pi r\|^2 = \|d(\theta_2+A_r)\|^2 + 4\pi^2\|D_r\|^2$. The translation $\theta_2\mapsto\theta_2+A_r$ preserves Lebesgue measure on $Z_1^\perp$ and d is a linear bijection $Z_1^\perp\to B$; so under $\tilde{P}$, $G = d(\theta_2+A_r)$ has density proportional to $e^{-\beta\|G\|^2/2}$ on B, independent of θ_1 and of r, while $D = D_r$ has the law proportional to $e^{-2\pi^2\beta\|D_r\|^2}$ on R. This is the decomposition into Gaussian field and defect part behind the duality of [51, Sect. 2]; for $c\in B$, $\langle c,D\rangle = 0$ and $\langle c,G\rangle$ is centred Gaussian with variance $\beta^{-1}\|c\|^2$ (Theorem 134, $\Pi c = c$). □

Theorem 165 (the Villain torus: the constant is at most $C_1(n)/L^2 + R_n(\beta)$)**.** *Let $\beta\ge 20$, $n\ge 2$, $L\ge\max(8,n+2)$ with $4\mid L$, $q = (n+1)^2-1$, and C_θ, K_β as in Proposition 163. Put*

$$\begin{aligned}\varepsilon_1 &= 2.001\big(e^{-2\pi\beta} + K_\beta e^{-\beta(\pi-1)^2/4}\big),\\ \varepsilon_2 &= 4.004\big(e^{-4\pi\beta} + K_\beta e^{-\beta(\pi-1)^2/4}\big),\end{aligned}$$

and $f = \sum_{y\in\Lambda_L}\cos(2\pi y_1/L)\Psi_{(y,12)} \in \mathscr{A}$.

(i) *The variance of f is*

$$\operatorname{Var}_\mu(f) = \beta^2\Big(\frac{L^4}{2\beta} - \sum_p c_p^2\,\mu\operatorname{Var}_{\tilde{P}}(F_p\mid\theta)\Big) \ge \frac{\beta L^4}{4},$$

where $c_{(y,12)} = \cos(2\pi y_1/L)$ and $c_p = 0$ otherwise.

(ii) *For every $x\in\Lambda_L$,*

$$\begin{aligned}\mu\big((f-\mathscr{P}_x f)^2\big) \le 3\beta^2\Big[&\frac{4\pi^2 n^6 q^2 C_\theta}{\beta L^2} + 4\pi^2 n^4 q^2\varepsilon_2 \\ &+ \pi^2 n^4 (n+1)^4 q K_\beta e^{-\beta\pi^2/(4q^2)}\Big].\end{aligned}$$

(iii) *If $\mathrm{BP}(n,\varpi)$ holds for μ on Λ_L, then*

$$\begin{aligned}\varpi &\le \frac{C_1(n)}{L^2} + R_n(\beta), \qquad C_1(n) = 48\pi^2 C_\theta n^2 q^2,\\ R_n(\beta) &= 12\pi^2\beta\big[(n+1)^4 q K_\beta e^{-\beta\pi^2/(4q^2)} + 4q^2\varepsilon_2\big].\end{aligned}$$

Consequently $\limsup_{L\to\infty,\,4\mid L}\varpi_L \le R_n(\beta)$ for the best constant ϖ_L of the torus Λ_L, and $\varpi \le 2C_1(n)/L^2$ whenever $L^2 \le C_1(n)/R_n(\beta)$. For $n=2$: $C_1(2) < 2.6\cdot 10^5$ for $\log\beta + 1.3 \le L^4/100$, and $R_2(\beta) < 1.5\cdot 10^{-4}$ at $\beta = 1000$, $< 10^{-20}$ at $\beta = 2000$.

Proof. Let c be the exact cochain of the proof of Theorem 134, $\|c\|^2 = L^4/2$, and $X = \langle c,F\rangle$ under $\tilde{P}$. By Lemma 164, $f = -\beta\sum_p c_p\bar{F}(\theta_p) = -\beta\,\mathbb{E}_{\tilde{P}}(X\mid\theta)$ and $X\sim N(0,L^4/(2\beta))$.

(i) By the law of total variance and the conditional independence in Lemma 164(ii),

$$\operatorname{Var}(\mathbb{E}(X\mid\theta)) = \frac{L^4}{2\beta} - \sum_p c_p^2\,\mu\operatorname{Var}(F_p\mid\theta),$$

and $\mu\operatorname{Var}(F_p\mid\theta) \le 4\pi^2\mu(\kappa(\theta_p))$. By Lemma 161(iii) and Proposition 163(ii), $\mu(\kappa(\theta_p)) \le 2.001[e^{-2\pi\beta} + \mu(|\theta_p| > \pi - 1)] \le \varepsilon_1$. Since $\sum_p c_p^2 \le L^4$, $\operatorname{Var}_\mu(f) \ge \beta^2 L^4[1/(2\beta) - 4\pi^2\varepsilon_1]$, and $4\pi^2\beta\varepsilon_1 < 0.018$ at $\beta = 20$ and decreases in β (as $\beta^{1+1/512}e^{-\beta(\pi-1)^2/4}$ does for $\beta\ge 1$).

(ii) Fix x and use the layers R_t, t ranging over n^2 transverse positions, of Lemma 131 for the plane $\{1,2\}$, $|R_t| = q$, $\bar{c} =$

$\cos(2\pi(x_1-1)/L)$ and $h = c1_{P(V_x)} - \bar{c}\sum_t 1_{R_t}$ as in the proof of Theorem 134, so that $|h_p| \le 2\pi n/L$ and $\|h\|_1 \le 2\pi n^3 q/L$. Put $\Theta_t = \sum_{p\in R_t}\theta_p$ (a real sum) and let $\hat{\Theta}_t \in (-\pi,\pi]$ be the principal value of $\langle \partial R_t, \theta\rangle$. Since $\theta_p \equiv (d\theta)_p$ modulo 2π and $\sum_{R_t}(d\theta)_p = \langle \partial R_t,\theta\rangle$, $\Theta_t \equiv \hat{\Theta}_t$ modulo 2π; by Lemma 131(ii) the links of ∂R_t lie outside V_x, so $\hat{\Theta}_t$ is $\mathscr{F}_{V_x}$-measurable, and so is $\bar{F}(\theta_p)$ for $p \notin P(V_x)$. Hence

$$Y = -\beta\Big[\sum_{p\notin P(V_x)} c_p \bar{F}(\theta_p) + \bar{c}\sum_t \hat{\Theta}_t\Big]$$

is $\mathscr{F}_{V_x}$-measurable, $\mu((f-\mathscr{P}_x f)^2) \le \mu((f-Y)^2)$ as $I-\mathscr{P}_x$ is an orthogonal projection, and, using $\sum_{R_t}\bar{F}(\theta_p) = \Theta_t + 2\pi\sum_{R_t}\psi(\theta_p)$,

$$-\frac{f-Y}{\beta} = \langle h, \bar{F}(\theta)\rangle + \bar{c}\sum_t(\Theta_t - \hat{\Theta}_t) + 2\pi\bar{c}\sum_t\sum_{p\in R_t}\psi(\theta_p).$$

First, by Minkowski's inequality and Proposition 163 (i), $\mu(\langle h,\bar{F}(\theta)\rangle^2) \le \|h\|_1^2 C_\theta/\beta$. Second, $\Theta_t - \hat{\Theta}_t \in 2\pi\mathbb{Z}$ vanishes when $|\Theta_t| < \pi$, $|\Theta_t - \hat{\Theta}_t| \le |\Theta_t| + \pi \le (n+1)^2\pi$, and $|\Theta_t| \ge \pi$ forces $|\theta_p| \ge \pi/q$ for some $p \in R_t$, an event of probability at most $qK_\beta e^{-\beta\pi^2/(4q^2)}$ by Proposition 163(ii); hence

$$\mu\Big(\Big(\sum_t(\Theta_t-\hat{\Theta}_t)\Big)^2\Big) \le n^2\sum_t\mu\big((\Theta_t-\hat{\Theta}_t)^2\big) \le n^4(n+1)^4\pi^2 qK_\beta e^{-\beta\pi^2/(4q^2)}.$$

Third,

$$\mu(\psi(\theta_p)^2) \le \mu(\kappa(\theta_p)^2) \le 4.004\big[e^{-4\pi\beta} + \mu(|\theta_p| > \pi - 1)\big] \le \varepsilon_2,$$

so $\mu((\sum_t\sum_{R_t}\psi(\theta_p))^2) \le (n^2q)^2\varepsilon_2$. With $|\bar{c}| \le 1$ and $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ this is (ii).

(iii) $\mathscr{E}_n(f) \le n^{-4}L^4$ times the bound (ii), and $\varpi \le \mathscr{E}_n(f)/\operatorname{Var}_\mu(f) \le (4/(\beta L^4))\mathscr{E}_n(f)$ by (i). □

Remark 166 (what is proved for the torus, and what is not). Remark 142(a) asked for $\varpi \le C(n,\beta)/L^2$ on the Villain torus. Theorem 165 proves this for $L \le (C_1(n)/R_n(\beta))^{1/2}$, a range of size $e^{\beta\pi^2/(8q^2)}$ up to powers of β, and proves that the best constant of arbitrarily large tori is at most $R_n(\beta)$, of order $\beta e^{-\beta\pi^2/(4q^2)}$; it does not prove $\varpi_L \to 0$ as $L\to\infty$ at fixed β, and it needs $\beta \ge 20$ and $4 \mid L$. The remainder $R_n(\beta)$ is the second moment, summed over the n^2 layers read by a cube, of the two quantities the exterior principal value $\hat{\Theta}_t$ cannot see: the multiple of 2π by which the layer flux of the principal angles leaves $(-\pi,\pi]$, and the mean defect number $\sum_{R_t}\psi(\theta_p)$ of the field strength. Both are exponentially small in β and of order one in L, because $\mu((f-\mathscr{P}_x f)^2) \le \mu((f-Y)^2)$ replaces a conditional variance by an unconditional second moment. Closing the gap requires what Remark 142(a) named: a bound on $\operatorname{Var}_\mu(\sum_{p\in R_t}\bar{F}(\theta_p) \mid \mathscr{F}_{V_x})$, the conditional law of the defect part of the layer flux given the exterior links, decaying in L; by Lemma 164 that defect part is the charge of D in the stack of cubes of Lemma 132, since $\langle \partial T_t, F\rangle = 2\pi\langle T_t, dm\rangle$. Nothing is proved here for Wilson's U(1) action, as in Remark 142(b). What the theorem uses from the Villain form is the exact Gaussian law of $\langle c,F\rangle$ for exact c (the defects do not couple to exact test cochains) and the explicit conditional law κ_θ of the integer variables.

F. A renormalisation-group recursion for the cube constant

This subsection gives an exact form of the upward step of [21, Sec. X K]. There [21, Theorem 100] passes BP of Definition 60 from the law of the block variables to the fine measure under the split inequality S, a sufficient condition. Here the fine constant is characterised exactly, for every probability measure and every sub-σ-algebra of block variables with a positive fibre constant, by a one-parameter family of effective quadratic forms on the block variables, the Feshbach–Schur complements of the fine Dirichlet form, as in the spectral renormalisation group of [25]. Comparing these forms with the coarse Dirichlet form gives a two-sided recursion for the best constants along a tower. The comparison involves three quantities of one block-spin step: the fibre constant, the renormalisation ratio and the fluctuation response. For translation-invariant measures on scalar fields and block sums the recursion is an identity on the constant linear function, with the defect $\delta_j = 1 - \operatorname{Var}(X \mid \text{fine exterior})/\operatorname{Var}(X \mid \text{coarse exterior})$ of the mass X of one cube. For the free massive scalar field this defect is at most $\Gamma_d(s)/n_{j-1}$, so the defects are summable over the scales of a tower of fixed physical cube and their product lies between $(1+\Gamma_d(s)/n_0)^{-1}$ and 1; in this factorisation the factor s^2 of Theorem 109 enters through the coarsest scale and the susceptibility $1/\mu^2$. For SU(2) the subsection says which of Bałaban's estimates bear on each quantity and states, as the open lemma (L) of Definition 179, a condition across scales that, with (H1), would give BP along the tuning. That condition is not proved, and neither (H1) nor a mass gap is proved here.

Notation from [21]. A block-spin map of range r with blocks of side b is a measurable $c : G^{E'} \to G^E$ from the links E' of Λ_{bL} to the links E of Λ_L such that $c_{(y,v)}$ depends only on the links based in the blocks $B_w = bw + [0,b)^4$ with $|w-y|_\infty \le r$, and every gauge transformation g of Λ_{bL} satisfies $c\circ g = \hat{g}\circ c$ for a gauge transformation $\hat{g}$ of Λ_L, every one of which arises; the straight-path map is an example. For a gauge-invariant probability measure μ' on $G^{E'}$, $\mu^c = c_*\mu'$ is gauge invariant; $\mathscr{A}'$, $\mathscr{A}^c$ are the gauge-invariant functions in $L^2(\mu')$, $L^2(\mu^c)$, and $\mathscr{E}'_{n'}(f) = n'^{-4}\sum_{x'}\mu'((f-P'_{x'}f)^2)$, $\mathscr{E}^c_n(F) = n^{-4}\sum_y \mu^c((F-P^c_y F)^2)$ are the Dirichlet forms of the cube chains of Definition 60. For real $f \in \mathscr{A}'$ there is a real $F_f \in \mathscr{A}^c$ with $\mathbb{E}[f \mid c] = F_f \circ c$; then $\operatorname{Var}_{\mu'}(f) = \mu'(\operatorname{Var}(f \mid c)) + \operatorname{Var}_{\mu^c}(F_f)$ and $\mu^c((F_f - P^c_y F_f)^2) = \mu'(\operatorname{Var}(\mathbb{E}[f\mid c] \mid \mathscr{G}_y))$, where $\mathscr{G}_y \subset \sigma(c)$ is generated by the $c_{(z,v)}$ with $z \notin y + [0,n)^4$ [21, Lemma 98]; and $F\circ c \in \mathscr{A}'$ for real $F \in \mathscr{A}^c$. The *split inequality* $\mathrm{S}(n',n;C_A,C_B)$ for (μ',c) is

$$\frac{\mu'(\operatorname{Var}(f \mid c))}{C_A} + \frac{\mathscr{E}^c_n(F_f)}{C_B} \le \mathscr{E}'_{n'}(f) \quad \text{for every real } f \in \mathscr{A}',$$

and the *separate inequalities* are (A) $\mu'(\mathrm{Var}(f \mid c)) \le C_A \mathscr{E}'_{n'}(f)$ and (B) $\mathscr{E}^c_n(F_f) \le C_B \mathscr{E}'_{n'}(f)$ for every real $f \in \mathscr{A}'$ [21, Definition 99]. By [21, Theorem 100], S and $\mathrm{BP}_{\mu^c}(n, \varpi_c)$ give $\mathrm{BP}_{\mu'}(n', \min(1/C_A, \varpi_c/C_B))$, and (A), (B) give $\mathrm{BP}_{\mu'}(n', (C_A + C_B/\varpi_c)^{-1})$. A tower is a sequence of block-spin maps c_j from Λ_{L_j} to $\Lambda_{L_{j-1}}$, $L_j = b^j L_0$, $1 \le j \le K$, with measures $\mu_{j-1} = (c_j)_* \mu_j$ and cube sides n_j, $L_j \ge 2n_j + 4$; [21, Theorem 101] iterates S along it. Along the tuning [21, Corollary 102] takes $a_k = 2^{-k} a_0$, $n_k = 2^k n_0$, $\ell = n_k a_k$, tori Λ_L with $L = 2^k L_0$, L_0 even and $L_0 \ge 2n_0 + 4$, in an infinite set $\mathscr{L}_k$, the Wilson measure $\mu_{k,L}$ at β_k, maps $c_1, \dots, c_k$ with blocks of side two, and the tower $\mu^{(j)}_{k,L}$ with $\mu^{(k)}_{k,L} = \mu_{k,L}$; its hypotheses are (H1) $\mathrm{BP}(n_0, \varpi_0)$ for $\mu^{(0)}_{k,L}$, and (H2) $\mathrm{S}(n_j, n_{j-1}; C_{A,j}, C_{B,j})$ for $(\mu^{(j)}_{k,L}, c_j)$ with $C_{A,j} \le C_A$ and $\prod_{i=j}^k C_{B,i} \le \Gamma$, uniformly in k and $L \in \mathscr{L}_k$, and its conclusion is $\mathrm{BP}(n_k, \min(\varpi_0, 1/C_A)/\Gamma)$ for $\mu_{k,L}$. Finally [21, Theorem 94(i)]: if $n \ge 2r+2$, $L \ge 2n+4$ is even, $2 \le n' \le b(n-2r-1)+1$ and $bL \ge 2n'+4$, then $\mathscr{E}'_{n'}(F \circ c) \le (bn/n')^4 \mathscr{E}^c_n(F)$ for every real $F \in \mathscr{A}^c$.

1. The exact step

Setting. Let $\mathscr{H}$ be a real Hilbert space, A a bounded self-adjoint operator on $\mathscr{H}$ with $A \ge 0$, and Q an orthogonal projection on $\mathscr{H}$. Put $\mathscr{H}_c = Q\mathscr{H}$, $\mathscr{H}_f = (I-Q)\mathscr{H}$, and write $A_{cc} = QAQ$, $A_{fc} = (I-Q)AQ$, $A_{cf} = A_{fc}^{\mathsf{T}}$ and $A_{ff} = (I-Q)A(I-Q)$, regarded as operators between $\mathscr{H}_c$ and $\mathscr{H}_f$. Let $A^c \ge 0$ be a bounded self-adjoint operator on $\mathscr{H}_c$. Put

$$\varpi' = \inf_{f \ne 0} \frac{\langle f, Af\rangle}{\|f\|^2}, \qquad \varpi_c = \inf_{0 \ne g \in \mathscr{H}_c} \frac{\langle g, A^c g\rangle}{\|g\|^2},$$
$$\varpi_{\mathrm{fib}} = \inf_{0 \ne k \in \mathscr{H}_f} \frac{\langle k, Ak\rangle}{\|k\|^2},$$

with $\varpi_{\mathrm{fib}} = +\infty$ if $\mathscr{H}_f = \{0\}$. Restricting the infimum defining ϖ' to $\mathscr{H}_f$ gives $\varpi' \le \varpi_{\mathrm{fib}}$, so $\varpi' = 0$ if $\varpi_{\mathrm{fib}} = 0$; assume $\varpi_{\mathrm{fib}} > 0$. For $0 \le \lambda < \varpi_{\mathrm{fib}}$ and $g \in \mathscr{H}_c$ let

$$\tilde{\mathscr{E}}_\lambda(g) = \inf_{k \in \mathscr{H}_f} \big(\langle g+k, A(g+k)\rangle - \lambda \|k\|^2 \big),$$

the effective form at the spectral parameter λ. Let $k_0(g)$ be the minimiser at $\lambda = 0$ (Theorem 167(i)), the fluctuation response of g.

The measure-theoretic case. Let $(\Omega, \mathfrak{F}, \mu)$ be a probability space, $(\mathfrak{F}_x)_{x \in X'}$ a finite family of sub-σ-algebras with conditional expectations Π_x, $w' > 0$, and $\mathfrak{B} \subset \mathfrak{F}$ a sub-σ-algebra (the block variables) with conditional expectation Q. Let $(\mathfrak{G}_y)_{y \in X}$ be a finite family of sub-σ-algebras of $\mathfrak{B}$ with conditional expectations $\Pi_{\mathfrak{G}_y}$, and $w > 0$. Let $\mathscr{H}$ be a closed subspace of the real functions in $L^2(\mu)$ with mean zero that is mapped into itself by every Π_x and by Q, and suppose that every $\Pi_{\mathfrak{G}_y}$ maps $\mathscr{H}_c = Q\mathscr{H}$ into itself. Take

$$A = w' \sum_{x \in X'} (I - \Pi_x), \qquad A^c = w \sum_{y \in X} (Q - \Pi_{\mathfrak{G}_y})$$

on $\mathscr{H}$ and on $\mathscr{H}_c$. Since $\mathfrak{G}_y \subset \mathfrak{B}$, $\Pi_{\mathfrak{G}_y} Q = Q \Pi_{\mathfrak{G}_y} = \Pi_{\mathfrak{G}_y}$, so each $Q - \Pi_{\mathfrak{G}_y}$ is an orthogonal projection, A^c is w times a sum of orthogonal projections, and $\langle g, A^c g\rangle = w \sum_y \|g - \Pi_{\mathfrak{G}_y} g\|^2$ for $g \in \mathscr{H}_c$. This contains the setting of [21, Sec. X K]: take $\mu = \mu'$, $\mathfrak{F}_{x'}$ the links outside the fine cube $V'_{x'}$, $w' = n'^{-4}$, $\mathfrak{B} = \sigma(c)$, $\mathfrak{G}_y$ as in the notation above, $w = n^{-4}$ and $\mathscr{H}$ the real functions in $\mathscr{A}'$ of mean zero. Then $\langle f, Af\rangle = \mathscr{E}'_{n'}(f)$; by [21, Lemma 98], its proof and gauge covariance, $\mathscr{H}^c_c$ is the set of functions $F \circ c$ with real $F \in \mathscr{A}^c$ of mean zero, $\Pi_{\mathfrak{G}_y}(F \circ c) = (P^c_y F) \circ c$, $\langle F \circ c, A^c F \circ c\rangle = \mathscr{E}^c_n(F)$ and $\|k\|^2 = \mu'(\mathrm{Var}(f \mid c))$ for $f = F \circ c + k$, $k \in \mathscr{H}_f$; so ϖ' and ϖ_c are the best constants of $\mathrm{BP}_{\mu'}(n')$ and $\mathrm{BP}_{\mu^c}(n)$. Product spaces with a block-averaging map are the case $\Omega = S^{\Lambda'}$, $\mathfrak{B} = \sigma(c)$ for a measurable $c : S^{\Lambda'} \to T^{\Lambda}$ whose coordinate c_y depends on the spins of the block B_y, with $\mathfrak{F}_{x'}$ and $\mathfrak{G}_y$ the spins outside a fine cube and the block variables outside a coarse cube.

Theorem 167 (the exact step)**.** *In the setting above:*

(i) *For $0 \le \lambda < \varpi_{\mathrm{fib}}$ and $g \in \mathscr{H}_c$ the infimum defining $\tilde{\mathscr{E}}_\lambda(g)$ is attained exactly at $k_\lambda(g) = -(A_{ff} - \lambda)^{-1} A_{fc} g$, and*

$$\tilde{\mathscr{E}}_\lambda(g) = \langle g, A_{cc} g\rangle - \langle A_{fc} g, (A_{ff} - \lambda)^{-1} A_{fc} g\rangle.$$

The map $\lambda \mapsto \tilde{\mathscr{E}}_\lambda(g)$ is nonincreasing and concave, and $\tilde{\mathscr{E}}_0(g) = \min\{\langle f, Af\rangle : Qf = g\}$.

(ii) *For $0 \le \lambda < \varpi_{\mathrm{fib}}$, $\varpi' \ge \lambda$ holds if and only if $\tilde{\mathscr{E}}_\lambda(g) \ge \lambda \|g\|^2$ for every $g \in \mathscr{H}_c$. With $s = \sup\{\lambda \in [0, \varpi_{\mathrm{fib}}) : \tilde{\mathscr{E}}_\lambda \ge \lambda$ on $\mathscr{H}_c\}$,*

$$\varpi' = \min(s, \varpi_{\mathrm{fib}}).$$

(iii) *For $0 \le \lambda < \varpi_{\mathrm{fib}}$ and $g \in \mathscr{H}_c$,*

$$\tilde{\mathscr{E}}_0(g) - \frac{\lambda\, \varpi_{\mathrm{fib}}}{\varpi_{\mathrm{fib}} - \lambda} \|k_0(g)\|^2 \le \tilde{\mathscr{E}}_\lambda(g) \le \tilde{\mathscr{E}}_0(g) - \lambda \|k_0(g)\|^2.$$

Proof. (i) For $k \in \mathscr{H}_f$, $\langle g+k, A(g+k)\rangle - \lambda\|k\|^2 = \langle g, A_{cc} g\rangle + 2\langle k, A_{fc} g\rangle + \langle k, (A_{ff} - \lambda) k\rangle$. Since $A_{ff} \ge \varpi_{\mathrm{fib}}$ on $\mathscr{H}_f$, $R = A_{ff} - \lambda \ge \varpi_{\mathrm{fib}} - \lambda > 0$ is invertible on $\mathscr{H}_f$ with a bounded positive inverse, and

$$2\langle k, A_{fc} g\rangle + \langle k, Rk\rangle = \|R^{1/2} k + R^{-1/2} A_{fc} g\|^2 - \langle A_{fc} g, R^{-1} A_{fc} g\rangle,$$

which is minimal exactly when $R^{1/2} k = -R^{-1/2} A_{fc} g$, that is $k = k_\lambda(g)$. For fixed k the bracket is an affine nonincreasing function of λ; an infimum of such functions is nonincreasing and concave. At $\lambda = 0$ the functions f with $Qf = g$ are the $g + k$, $k \in \mathscr{H}_f$.

(ii) If $\varpi' \ge \lambda$, then for $k \in \mathscr{H}_f$, since $g \perp k$, $\langle g+k, A(g+k)\rangle - \lambda\|k\|^2 \ge \lambda\|g+k\|^2 - \lambda\|k\|^2 = \lambda\|g\|^2$, and the infimum over k gives $\tilde{\mathscr{E}}_\lambda(g) \ge \lambda\|g\|^2$. Conversely, for $f = g + k$ with $g = Qf$,

$$\langle f, Af\rangle - \lambda\|f\|^2 = \big(\langle f, Af\rangle - \lambda\|k\|^2\big) - \lambda\|g\|^2 \ge \tilde{\mathscr{E}}_\lambda(g) - \lambda\|g\|^2 \ge 0.$$

The set of admissible λ contains 0, since $\tilde{\mathscr{E}}_0 \ge 0$, and is an interval, since $\tilde{\mathscr{E}}_\lambda(g) - \lambda\|g\|^2$ is nonincreasing in λ. Restricting

the infimum defining ϖ' to $\mathscr{H}_f$ gives $\varpi' \le \varpi_{\mathrm{fib}}$. For every admissible λ, $\varpi' \ge \lambda$, so $\varpi' \ge s$. If $s < \varpi_{\mathrm{fib}}$, every $\lambda \in (s, \varpi_{\mathrm{fib}})$ is not admissible, so $\varpi' < \lambda$, and $\varpi' \le s$. If $s = \varpi_{\mathrm{fib}}$, then $\varpi' \ge s = \varpi_{\mathrm{fib}} \ge \varpi'$.

(iii) Put $a = A_{fc}g$, so that $k_0(g) = -A_{ff}^{-1}a$. By (i) and the resolvent identity $R^{-1} - A_{ff}^{-1} = \lambda A_{ff}^{-1}R^{-1}$, in which all operators are functions of A_{ff} and commute,

$$\tilde{\mathscr{E}}_0(g) - \tilde{\mathscr{E}}_\lambda(g) = \lambda\langle a, A_{ff}^{-1}R^{-1}a\rangle = \lambda\langle k_0(g), \phi(A_{ff})k_0(g)\rangle,$$

with $\phi(t) = t/(t-\lambda)$. On the spectrum of A_{ff}, contained in $[\varpi_{\mathrm{fib}}, \infty)$, $1 \le \phi(t) \le \varpi_{\mathrm{fib}}/(\varpi_{\mathrm{fib}} - \lambda)$, so $\|k_0\|^2 \le \langle k_0, \phi(A_{ff})k_0\rangle \le \varpi_{\mathrm{fib}}\|k_0\|^2/(\varpi_{\mathrm{fib}} - \lambda)$. □

In the measure-theoretic case $\tilde{\mathscr{E}}_0(g)$ is the least fine Dirichlet form of a function whose conditional mean given the block variables is g, and $k_0(g)$ is the part of that optimal function which has conditional mean zero. Part (ii) is an equivalence at each λ, so every sufficient condition for the upward step that gives $\varpi' \ge \lambda$ with $\lambda < \varpi_{\mathrm{fib}}$ amounts to the lower bound $\tilde{\mathscr{E}}_\lambda \ge \lambda$ on $\mathscr{H}_c$. The split inequality is one such condition.

Proposition 168 (the split inequality as a special case)**.** *If* $\mathrm{S}(n', n; C_A, C_B)$ *holds for* (μ', c)*, then* $\varpi_{\mathrm{fib}} \ge 1/C_A$ *and* $\tilde{\mathscr{E}}_\lambda(F \circ c) \ge \mathscr{E}_n^c(F)/C_B$ *for* $0 \le \lambda < 1/C_A$ *and every real* $F \in \mathscr{A}^c$ *of mean zero; so Theorem* 167(ii) *gives* $\varpi' \ge \min(1/C_A, \varpi_c/C_B)$*, which is [21, Theorem 100(i)].*

Proof. For $k \in \mathscr{H}_f$, $\mathbb{E}[k \mid c] = 0$, so F_k is constant and S with $f = k$ gives $\langle k, Ak\rangle \ge \|k\|^2/C_A$. For $f = F \circ c + k$ one may take $F_f = F$, and S gives $\langle f, Af\rangle - \lambda\|k\|^2 \ge (1/C_A - \lambda)\|k\|^2 + \mathscr{E}_n^c(F)/C_B \ge \mathscr{E}_n^c(F)/C_B$. If $\lambda < \min(1/C_A, \varpi_c/C_B)$, then $\tilde{\mathscr{E}}_\lambda(F \circ c) \ge \varpi_c\|F\|^2/C_B \ge \lambda\|F \circ c\|^2$, so $\varpi' \ge \lambda$; let λ increase to the minimum. □

2. *The recursion*

Three quantities of one step compare the effective forms with the coarse form. With the conventions $0/0 = 0$ and that the infimum and supremum run over $g \in \mathscr{H}_c$ with $\langle g, A^c g\rangle > 0$, put

$$\kappa = \inf_g \frac{\tilde{\mathscr{E}}_0(g)}{\langle g, A^c g\rangle}, \qquad \mathrm{K} = \sup_g \frac{\tilde{\mathscr{E}}_0(g)}{\langle g, A^c g\rangle},$$
$$\zeta = \sup_{0 \ne g \in \mathscr{H}_c} \frac{\|k_0(g)\|^2}{\|g\|^2},$$

the lower and upper renormalisation ratios and the fluctuation response of the step, and assume that $\tilde{\mathscr{E}}_0(g) = 0$ whenever $\langle g, A^c g\rangle = 0$ when K is used. In the measure-theoretic case of [21, Sec. X K], $\kappa \ge 1/C_B$ whenever the separate inequality (B) holds, and the best constant in (B) is $1/\kappa$, since (B) says $\mathscr{E}_n^c(F_f) \le C_B\mathscr{E}_{n'}'(f)$ for every f with $\mathbb{E}[f \mid c] = F_f \circ c$, that is $\mathscr{E}_n^c(F) \le C_B\tilde{\mathscr{E}}_0(F \circ c)$ for every real $F \in \mathscr{A}^c$ of mean zero.

Corollary 169 (the two-sided recursion)**.** *In the setting of Theorem* 167*,*

$$\min\Big(\frac{\varpi_{\mathrm{fib}}}{2}, \frac{\kappa\varpi_c}{1+2\zeta}\Big) \le \varpi' \le \min\Big(\varpi_{\mathrm{fib}}, \inf_{0 \ne g \in \mathscr{H}_c} \frac{\tilde{\mathscr{E}}_0(g)}{\|g\|^2 + \|k_0(g)\|^2}\Big) \le \min(\varpi_{\mathrm{fib}}, \mathrm{K}\varpi_c).$$

More precisely, $\varpi' \ge \lambda$ *for every* $\lambda \in [0, \varpi_{\mathrm{fib}})$ *with* $\kappa\varpi_c - \lambda\zeta\varpi_{\mathrm{fib}}/(\varpi_{\mathrm{fib}} - \lambda) \ge \lambda$*. If* $\zeta = 0$ *and* $\kappa = \mathrm{K}$*, then* $\varpi' = \min(\varpi_{\mathrm{fib}}, \kappa\varpi_c)$*.*

Proof. Let $\lambda \in [0, \varpi_{\mathrm{fib}})$ satisfy the displayed condition. By Theorem 167(iii), the definition of κ and $\langle g, A^c g\rangle \ge \varpi_c\|g\|^2$,

$$\tilde{\mathscr{E}}_\lambda(g) \ge \kappa\varpi_c\|g\|^2 - \frac{\lambda\varpi_{\mathrm{fib}}}{\varpi_{\mathrm{fib}} - \lambda}\zeta\|g\|^2 \ge \lambda\|g\|^2,$$

and Theorem 167(ii) gives $\varpi' \ge \lambda$. The number $\lambda = \min(\varpi_{\mathrm{fib}}/2, \kappa\varpi_c/(1+2\zeta))$ satisfies the condition, because $\varpi_{\mathrm{fib}}/(\varpi_{\mathrm{fib}} - \lambda) \le 2$ and $\lambda(1+2\zeta) \le \kappa\varpi_c$. For the upper bounds, $\varpi' \le \varpi_{\mathrm{fib}}$ by restriction to $\mathscr{H}_f$, and the function $f = g + k_0(g)$ has $\langle f, Af\rangle = \tilde{\mathscr{E}}_0(g)$ and $\|f\|^2 = \|g\|^2 + \|k_0(g)\|^2$. Dropping $\|k_0(g)\|^2$, $\tilde{\mathscr{E}}_0(g) \le \mathrm{K}\langle g, A^c g\rangle$ and the infimum over g of $\langle g, A^c g\rangle/\|g\|^2$ is ϖ_c. If $\zeta = 0$ and $\kappa = \mathrm{K}$, the lower bound holds for every $\lambda < \min(\varpi_{\mathrm{fib}}, \kappa\varpi_c)$ and the upper bound is $\min(\varpi_{\mathrm{fib}}, \kappa\varpi_c)$. □

In the measure-theoretic case $\mathrm{K} \le \sup_F \mathscr{E}_{n'}'(F \circ c)/\mathscr{E}_n^c(F)$, since $\tilde{\mathscr{E}}_0(g) \le \langle g, Ag\rangle$, and under the hypotheses of [21, Theorem 94(i)] recalled above this is at most $(bn/n')^4$ (the assumption $\tilde{\mathscr{E}}_0(g) = 0$ when $\langle g, A^c g\rangle = 0$ then holds); the upper bound of the corollary is then the descent $\varpi' \le (bn/n')^4\varpi_c$, sharpened by the fluctuation response. The lower bound is informative as soon as $\kappa > 0$ and $\zeta < \infty$; the three numbers ϖ_{fib}, κ, ζ are defined for every measure with $\varpi_{\mathrm{fib}} > 0$.

Corollary 170 (a tower)**.** *Let* $\mu_0, \dots, \mu_K$ *be the gauge-invariant measures of a tower as in [21, Theorem 101], with cube sides* n_j *and* $L_j \ge 2n_j + 4$*, let* ϖ_j *be the best constant of* $\mathrm{BP}_{\mu_j}(n_j)$*, and let* $\varpi_{\mathrm{fib},j}$*,* κ_j*,* ζ_j *be the quantities of the step* (μ_j, c_j) *in the measure-theoretic case,* $1 \le j \le K$*, and assume* $\varpi_{\mathrm{fib},j} > 0$*. Put* $r_j = \min(1, \kappa_j/(1+2\zeta_j))$ *and* $\varphi = \min_j \varpi_{\mathrm{fib},j}/2$*. Then*

$$\varpi_K \ge \min(\varpi_0, \varphi)\prod_{j=1}^{K} r_j.$$

In particular, if $\kappa_j \ge 1 - \varepsilon_j$ *and* $\zeta_j \le \varepsilon_j$ *with* $\varepsilon_j \in [0, \frac{1}{4}]$ *and* $\sum_j \varepsilon_j \le E$*, then* $\varpi_K \ge e^{-6E}\min(\varpi_0, \varphi)$*.*

Proof. The coarse constant of the step (μ_j, c_j) is ϖ_{j-1}, so Corollary 169 gives $\varpi_j \ge \min(\varphi, r_j\varpi_{j-1})$, as $r_j \le \kappa_j/(1 + 2\zeta_j)$. With $m_0 = \varpi_0$ and $m_j = \min(\varphi, r_j m_{j-1})$, induction gives $\varpi_j \ge m_j$, and since $r_j \le 1$, $m_K \ge \min(\varpi_0, \varphi)\prod_j r_j$. Under the last hypothesis $\kappa_j/(1+2\zeta_j) \ge (1-\varepsilon_j)/(1+2\varepsilon_j) \ge (1-\varepsilon_j)(1-2\varepsilon_j)$, and $\log(1-x) \ge -2x$ for $0 \le x \le \frac{1}{2}$, applied with $x = \varepsilon_j$ and $x = 2\varepsilon_j$, gives $r_j \ge e^{-6\varepsilon_j}$. □

The quantities κ_j and ζ_j enter only through the product of the r_j, so the constant at the finest level is bounded below uniformly in the number of steps when the defects $1-r_j$ are summable, given the condition at the coarsest level and a uniform fibre constant. In the other direction the upper bound of Corollary 169 gives $\varpi_j \le \mathrm{K}_j \varpi_{j-1}$ (under the assumption made for K), hence $\varpi_K \le \varpi_0 \prod_{j=1}^{K} \mathrm{K}_j$; so if, along a sequence of towers, the upper ratios satisfy $\mathrm{K}_j \le 1$ and $\sum_j (1-\mathrm{K}_j) = \infty$, the constants at the finest level tend to zero for bounded ϖ_0. The two products bracket the constant, and they agree, up to the cap by the fibre constants, when $\zeta_j = 0$ and $\kappa_j = \mathrm{K}_j$. For the free scalar field below the fluctuation response of the constant linear function vanishes and its ratio is $1-\delta_j$ with summable δ_j; this concerns one function, not the infimum κ_j or the supremum ζ_j.

3. *The constant function and the free field*

For measures on scalar fields with a linear block map the constant linear function gives an exact recursion with one defect, valid for every translation-invariant measure. Let $d \ge 1$, $b \ge 2$, $K \ge 1$, $n_0 \ge 2$, $L_0 \ge n_0 + 2$, and for $0 \le j \le K$ let $L_j = b^j L_0$, $n_j = b^j n_0$ and $m_j = b^{K-j}$. Let ν be a probability measure on $\mathbb{R}^{\mathbb{T}_K}$, $\mathbb{T}_K = (\mathbb{Z}/L_K\mathbb{Z})^d$, invariant under the translations of $\mathbb{T}_K$, with $\int \phi_x^2 \, d\nu < \infty$ and $\chi = L_K^{-d} \operatorname{Var}_\nu(\sum_x \phi_x) > 0$. The level-$j$ field is $\Phi_y^{(j)} = \sum_{x \in B_y^{(j)}} \phi_x$, $y \in \mathbb{T}_j = (\mathbb{Z}/L_j\mathbb{Z})^d$, with $B_y^{(j)} = m_j y + [0,m_j)^d$, and ν_j is its law; $\nu_K = \nu$. The cubes of $\mathbb{T}_j$ have side n_j, so that the cube $V_j = [0,n_j)^d$ of $\mathbb{T}_j$ is the union of the level-j blocks of the fine cube $V' = [0,n_K)^d$ for every j. Let $X = \sum_{x \in V'} \phi_x = \sum_{y \in V_j} \Phi_y^{(j)}$, $\mathfrak{G}_j = \sigma(\Phi_y^{(j)} : y \notin V_j)$ and

$$\sigma_j^2 = \nu\big(\operatorname{Var}(X \mid \mathfrak{G}_j)\big), \qquad \delta_j = 1 - \frac{\sigma_j^2}{\sigma_{j-1}^2} \quad (1 \le j \le K).$$

Assume $\sigma_K > 0$; then, by Proposition 171(ii) below, every $\sigma_j > 0$. For a function $f \in L^2(\nu_j)$ put $\mathscr{E}_{n_j}^{(j)}(f) = n_j^{-d} \sum_{y \in \mathbb{T}_j} \nu_j((f - P_y^{(j)} f)^2)$, with $P_y^{(j)}$ the conditional expectation given the $\Phi_z^{(j)}$, $z \notin V_j + y$, the Dirichlet form of (17) for ν_j. Let $\ell = \sum_x \phi_x - \nu(\sum_x \phi_x)$, which equals $\sum_y \Phi_y^{(j)}$ minus a constant for every j, let $\varpi_j^{(0)} = \mathscr{E}_{n_j}^{(j)}(\ell)/\operatorname{Var}(\ell)$ be its Rayleigh quotient, and let ϖ_j be the best constant of $\mathrm{BP}_{\nu_j}(n_j)$ on $L^2(\nu_j)$, the infimum of $\mathscr{E}_{n_j}^{(j)}(f)/\operatorname{Var}(f)$ over nonconstant f.

Proposition 171 (the recursion on the constant function)**.** *In this setting:*

(i) $\varpi_j \le \varpi_j^{(0)} = \sigma_j^2/(n_K^d \chi)$ *for* $0 \le j \le K$;

(ii) $\sigma_K^2 \le \sigma_{K-1}^2 \le \cdots \le \sigma_0^2$, *so* $\delta_j \in [0,1)$, *and*

$$\varpi_j^{(0)} = (1-\delta_j)\,\varpi_{j-1}^{(0)}, \qquad \prod_{j=1}^{K}(1-\delta_j) = \frac{\sigma_K^2}{\sigma_0^2} = \frac{\varpi_K^{(0)}}{\varpi_0^{(0)}};$$

(iii) *for the step from* ν_j *to* ν_{j-1}, $1 \le j \le K$, *in the measure-theoretic case of Theorem 167 with* $\mathscr{H}$ *the real functions in* $L^2(\nu_j)$ *of mean zero,* $\Pi_y = P_y^{(j)}$, $w' = n_j^{-d}$, $\mathfrak{B} = \sigma(\Phi^{(j-1)})$, $\mathfrak{G}_z = \sigma(\Phi_{z'}^{(j-1)} : z' \notin V_{j-1} + z)$ *and* $w = n_{j-1}^{-d}$ *(so that* $\langle F \circ \Phi^{(j-1)}, A^c F \circ \Phi^{(j-1)} \rangle = \mathscr{E}_{n_{j-1}}^{(j-1)}(F)$, ν_{j-1} *being the law of the block sums* $\Phi^{(j-1)}$ *of* $\Phi^{(j)}$*),* $\tilde{\mathscr{E}}_0(\ell) \le (1-\delta_j)\,\mathscr{E}_{n_{j-1}}^{(j-1)}(\ell)$, *so* $\kappa_j \le 1-\delta_j$ *when the fibre constant of the step is positive.*

Proof. (i) $\operatorname{Var}(\ell) = L_K^d \chi$. The translation of $\mathbb{T}_j$ by y is the translation of $\mathbb{T}_K$ by $m_j y$, which preserves ν, permutes the level-j blocks and fixes ℓ; it maps the cube V_j to $V_j + y$. Since $\ell - X$, which is $\sum_{y \notin V_j} \Phi_y^{(j)}$ minus a constant, is $\mathfrak{G}_j$-measurable, $\nu(\operatorname{Var}(\ell \mid \mathfrak{G}_j)) = \sigma_j^2$, and every cube of $\mathbb{T}_j$ contributes the same. So $\mathscr{E}_{n_j}^{(j)}(\ell) = n_j^{-d} L_j^d \sigma_j^2$, and with $L_K = m_j L_j$ and $n_K = m_j n_j$ the quotient is $\sigma_j^2/(n_K^d \chi)$. It bounds the best constant, ℓ being a test function.

(ii) For $z \notin V_{j-1}$ the block $B_z^{(j-1)}$ is disjoint from V', which is a union of level-$(j-1)$ blocks, and it is a union of level-j blocks $B_y^{(j)}$, all with $y \notin V_j$. So $\Phi_z^{(j-1)}$ is a sum of generators of $\mathfrak{G}_j$, and $\mathfrak{G}_{j-1} \subset \mathfrak{G}_j$. Hence $\nu(\operatorname{Var}(X \mid \mathfrak{G}_j)) = \|X - \mathbb{E}[X \mid \mathfrak{G}_j]\|^2 \le \|X - \mathbb{E}[X \mid \mathfrak{G}_{j-1}]\|^2$. The identity follows from (i), and the product telescopes.

(iii) ℓ is a function of $\Phi^{(j-1)}$ of mean zero, so $\ell \in \mathscr{H}_c$ and $\tilde{\mathscr{E}}_0(\ell) \le \langle \ell, A\ell \rangle = \mathscr{E}_{n_j}^{(j)}(\ell)$, and by (i) and (ii) the ratio of $\mathscr{E}_{n_j}^{(j)}(\ell)$ to $\mathscr{E}_{n_{j-1}}^{(j-1)}(\ell)$ is $\varpi_j^{(0)}/\varpi_{j-1}^{(0)} = 1-\delta_j$, as both have the denominator $\operatorname{Var}(\ell)$. As $\langle \ell, A^c \ell \rangle = \mathscr{E}_{n_{j-1}}^{(j-1)}(\ell) > 0$ by (i), $\kappa_j \le \tilde{\mathscr{E}}_0(\ell)/\langle \ell, A^c \ell \rangle$. □

So the Rayleigh quotient of the constant function obeys an exact recursion whose only defect is the information that the fine exterior of a cube carries on its mass beyond the block sums outside it. For the free field this quotient is the exact constant at the finest level, the fluctuation response of ℓ vanishes at every step, and the defects are bounded by the inverse cube size. Let γ be the centred Gaussian measure on $\mathbb{R}^{\mathbb{T}_K}$ with precision $-\Delta + \mu^2$, $\mu > 0$, as in Sec. X B, and $s = n_K \mu$; then $\chi = 1/\mu^2$.

Proposition 172 (the free field)**.** *Let* ν *in the setting above be* γ*, and put*

$$\Gamma_d(s) = \frac{7d}{4}\,(4d + s^2)\Big(\frac{6}{5}\Big)^{d-1}.$$

(i) $\varpi_K = \varpi_K^{(0)} = \varpi_{n_K}^{(d)}(s)$, *the exact constant of Theorem 109, and* $\varpi_j^{(0)} = \mu^2 \sigma_j^2 / n_K^d$ *for every* j.

(ii) *For every step the fluctuation response of* ℓ *vanishes:* $k_0(\ell) = 0$ *and* $\tilde{\mathscr{E}}_0(\ell) = \mathscr{E}_{n_j}^{(j)}(\ell) = (1-\delta_j)\mathscr{E}_{n_{j-1}}^{(j-1)}(\ell)$, *so* $\kappa_j \le 1 - \delta_j \le \mathrm{K}_j$.

(iii) *For* $0 \le j \le K$,

$$\sigma_j^2 - \sigma_K^2 \le \frac{7d}{12}\,\frac{n_K^{d+2}}{n_j}, \qquad \sigma_K^2 \ge \Big(\frac{5}{6}\Big)^{d-1} \frac{n_K^{d+2}}{12d + 3s^2};$$

hence, for $1 \le j \le K$,

$$\delta_j \le \frac{\Gamma_d(s)}{n_{j-1}}, \qquad \sum_{j=1}^{K} \delta_j \le \frac{b\Gamma_d(s)}{(b-1)n_0},$$

and

$$\Big(1+\frac{\Gamma_d(s)}{n_0}\Big)^{-1} \le \prod_{j=1}^{K}(1-\delta_j) \le 1.$$

(iv) $\varpi_0^{(0)} = s^2\sigma_0^2/n_K^{d+2}$, *and*

$$\varpi_{n_K}^{(d)}(s) \le \varpi_0^{(0)} \le \Big(1+\frac{\Gamma_d(s)}{n_0}\Big)\varpi_{n_K}^{(d)}(s),$$
$$\varpi_0^{(0)} \le \Big(1+\frac{\Gamma_d(s)}{n_0}\Big)\frac{s^2(n_K+1)(n_K+2)}{12n_K^2}.$$

Proof. (i) $\mathrm{Var}(\ell) = \langle \mathbf{1}, (-\Delta+\mu^2)^{-1}\mathbf{1}\rangle = L_K^d/\mu^2$. Given the field outside V', X has the conditional variance $\sigma_K^2 = \langle \mathbf{1}_{V'}, (-\Delta_D+\mu^2)^{-1}\mathbf{1}_{V'}\rangle$, with $-\Delta_D$ the Dirichlet Laplacian of V' (Lemma 100), so $\varpi_K^{(0)} = \mu^2\sigma_K^2/n_K^d$ is the constant of Theorem 109(iii), which applies since $L_K \ge n_K+2$. The formula for $\varpi_j^{(0)}$ is Proposition 171(i).

(ii) The law ν_j is Gaussian with a positive definite covariance, the conditional expectation given a set of coordinates or given the linear functions $\Phi^{(j-1)}$ maps linear functions to linear functions, and so A maps the space of linear functions of $\Phi^{(j)}$ into itself. It commutes with the translations of $\mathbb{T}_j$, which preserve ν_j and permute the cubes, and ℓ is translation invariant; so $A\ell$ is a translation-invariant linear function $\sum_y h_y \Phi_y^{(j)}$, and since the covariance is nondegenerate, h is constant and $A\ell = c\,\ell$. As $\ell = \sum_z \Phi_z^{(j-1)}$ is measurable for the block variables, $A_{fc}\ell = (I-Q)A\ell = 0$, so $k_0(\ell) = 0$ and $\tilde{\mathscr{E}}_0(\ell) = \langle \ell, A\ell\rangle$. The rest is Proposition 171(iii) and the definition of K_j (here $\varpi_{\mathrm{fib},j} \ge \varpi_j > 0$).

(iii) Let $\mathfrak{F} = \mathfrak{G}_K$, $j < K$, and $m = m_j \ge 2$, $n = n_K$. Since $\mathfrak{G}_j \subset \mathfrak{F}$, $\sigma_j^2 - \sigma_K^2 = \|\mathbb{E}[X \mid \mathfrak{F}] - \mathbb{E}[X \mid \mathfrak{G}_j]\|^2 \le \|\mathbb{E}[X \mid \mathfrak{F}] - Y\|^2$ for every $\mathfrak{G}_j$-measurable Y. Let $\tau = (-\Delta_D+\mu^2)^{-1}\mathbf{1}_{V'}$. The conditional mean of $\phi_{V'}$ given the exterior is $(-\Delta_D+\mu^2)^{-1}$ applied to the sum of the exterior neighbours, so $\mathbb{E}[X \mid \mathfrak{F}] = \sum_w p(w)\phi_w$, the sum over the sites $w \notin V'$ adjacent to V', with $p(w) = \tau(u_w)$ for the unique neighbour $u_w \in V'$ of w (unique since $L_K \ge n+2$). If $w = u \pm e_i$, then $u_i \in \{0, n-1\}$. The function $v(u) = \tau_1(u_i)$, $\tau_1(t) = (t+1)(n-t)/2$, satisfies $(-\Delta_D+\mu^2)v \ge \mathbf{1}_{V'}$, because τ_1 has second difference -1 and vanishes at -1 and n, while the Dirichlet differences in the other coordinates are $b_k(u)v(u) \ge 0$ as in the proof of Corollary 111(iv); the Green function of $-\Delta_D+\mu^2$ is positive (Theorem 109(ii)), so $\tau \le v$ and $p(w) \le \tau_1(0) = n/2$.

The sites w lie in the $2dn_j^{d-1}$ face blocks $B_y^{(j)}$ with one coordinate $y_i \in \{-1, n_j\}$ and the others in $[0, n_j)$; these are distinct and disjoint from V' because $L_j \ge n_j+2$, and in each the sites w fill the slice adjacent to V'. Put $Y = \sum_w p(w) m^{-d}\Phi_{y(w)}^{(j)}$, $y(w)$ the block of w, which is $\mathfrak{G}_j$-measurable. Then $\mathbb{E}[X \mid \mathfrak{F}] - Y = \langle h, \phi\rangle$ with $h = \sum_B h_B$, $h_B = \sum_{w\in B} p(w)(\delta_w - m^{-d}\mathbf{1}_B)$, and $\sum_x h(x) = 0$. In Fourier variables $(-\Delta+\mu^2)^{-1} \le (-\Delta)^+$ on functions of sum zero, and for every flow θ on the links with net outflow h, $\langle h, v\rangle = \sum_e \theta_e(v(e^-) - v(e^+)) \le \|\theta\|\|\nabla v\|$, so

$$\mathrm{Var}(\langle h, \phi\rangle) \le \langle h, (-\Delta)^+ h\rangle = \sup_v \frac{\langle h, v\rangle^2}{\|\nabla v\|^2} \le \|\theta\|^2.$$

In a face block write the sites as (z,t), $z \in F = [0,m)^{d-1}$ along the face and $t \in [0,m)$ the distance from the slice of the sources. The flow θ^1 that carries $p(z)(1-(t+1)/m)$ from (z,t) to $(z,t+1)$ has net outflow $\sum_z p(z)(\delta_{(z,0)} - m^{-1}\mathbf{1}_{\{z\}\times[0,m)})$ and energy $\sum_z p(z)^2 \sum_{r=1}^{m-1}(r/m)^2 \le m^d n^2/12$. The remainder $\psi(z,t) = (p(z)-\bar p)/m$, $\bar p$ the mean of p over F, is constant in t with mean zero on each slice; in each slice take the flow of least energy with net outflow $\psi(\cdot,t)$, of energy $\langle \psi, (-\Delta_F^N)^+\psi\rangle \le \|\psi(\cdot,t)\|^2/\lambda_1$ with $-\Delta_F^N$ the Neumann Laplacian of F and $\lambda_1 = 4\sin^2(\pi/(2m)) \ge 4/m^2$ its spectral gap. Summed over the m slices this flow θ^2 has energy at most $m \cdot \frac{m^2}{4}\cdot m^{-2}\sum_z (p(z)-\bar p)^2 \le m^d n^2/16$ (for $d=1$, $\psi = 0$). With $\theta_B = \theta^1 + \theta^2$, $\|\theta_B\|^2 \le 2(\frac{1}{12}+\frac{1}{16})m^d n^2 = \frac{7}{24}m^d n^2$ (the two flows use disjoint sets of links, so the factor 2 could be dropped), and the flows of distinct blocks use disjoint links. So $\sigma_j^2 - \sigma_K^2 \le 2dn_j^{d-1}\frac{7}{24}m^d n^2 = \frac{7d}{12}n^{d+2}/n_j$, using $n = mn_j$.

For the lower bound let $\varphi(t) = (t+1)(n-t)$ and $v = \varphi^{\otimes d}$ on V'; with $A = \sum_t \varphi(t) = n(n+1)(n+2)/6$, $B = \sum_t \varphi(t)^2 = n(n+1)(n+2)(n^2+2n+2)/30$ and the Dirichlet energy $2A$ of φ (Lemma 103), the Cauchy–Schwarz inequality gives

$$\sigma_K^2 \ge \frac{\langle \mathbf{1}, v\rangle^2}{\langle v, (-\Delta_D+\mu^2)v\rangle} = \frac{A^{2d}}{2dAB^{d-1}+\mu^2 B^d} \ge \frac{A(A^2/B)^{d-1}}{2d+s^2/2},$$

since $\mu^2 B/A = \mu^2(n^2+2n+2)/5 \le s^2/2$ for $n \ge 2$. With $A \ge n^3/6$ and $A^2/B = \frac{5}{6}n(n+1)(n+2)/(n^2+2n+2) \ge \frac{5}{6}n$ this is the bound. The two bounds give $(\sigma_{j-1}^2 - \sigma_K^2)/\sigma_K^2 \le \Gamma_d(s)/n_{j-1}$, and $\delta_j = (\sigma_{j-1}^2 - \sigma_j^2)/\sigma_{j-1}^2 \le (\sigma_{j-1}^2 - \sigma_K^2)/\sigma_K^2$. The sum is geometric, and the product is $\sigma_K^2/\sigma_0^2 \ge (1+\Gamma_d(s)/n_0)^{-1}$.

(iv) The formula is (i) with $\mu^2 = s^2/n_K^2$; the bounds follow from $\varpi_0^{(0)} = \varpi_K^{(0)}/\prod_j(1-\delta_j)$ and (iii). Finally $\sigma_K^2 = \langle \mathbf{1}, \tau\rangle \le \langle \mathbf{1}, v\rangle = n^{d-1}A/2$ by the comparison above, and $\sigma_0^2 \le (1+\Gamma_d(s)/n_0)\sigma_K^2$. □

Remark 173 (the massless limit and the photon). Let $\mu \to 0$ at fixed n_K, so that $s \to 0$. By Proposition 172(iii) the defects stay summable, with $\Gamma_d(0) = 7d^2(6/5)^{d-1}$, and their product stays in $[(1+\Gamma_d(s)/n_0)^{-1}, 1]$; on the constant function the steps of the tower lose a bounded factor uniformly in K. The constant nevertheless tends to zero: by (i) and the two bounds on σ_K^2 in the proof of (iii) and (iv),

$$\Big(\frac{5}{6}\Big)^{d-1}\frac{s^2}{12d+3s^2} \le \varpi_K = \frac{\mu^2\sigma_K^2}{n_K^d} \le \frac{s^2(n_K+1)(n_K+2)}{12n_K^2},$$

so $\varpi_K \asymp s^2$ uniformly in $n_K \ge 2$ for $0 < s \le 1$, in agreement with $\varpi_\infty^{(d)}(s) = T_d s^2 + O(s^4)$ of Theorem 112 ($T_1 = 1/12$ is

the limit of the upper bound for $d=1$). In the recursion $\varpi_K=\varpi_0^{(0)}\prod_j(1-\delta_j)$ the factor s^2 sits in $\varpi_0^{(0)}=\sigma_0^2/(n_K^d\chi)$: the conditional variance $\sigma_0^2\le(1+\Gamma_d(s)/n_0)\sigma_K^2$ of the mass of the cube given the block sums outside it at the coarsest scale stays bounded as $\mu\to0$, while the susceptibility $\chi=1/\mu^2$ diverges. In this factorisation the massless failure comes from the coarsest-scale factor, not from an accumulation of defects. The bound $\varpi\le 8\pi^2(n+1)^2/L^2$ of Theorem 134 for the photon is of the same order, $2(n+1)^2|p|^2$ with $|p|=2\pi/L$ the least nonzero momentum of the torus in place of μ, and its slow mode plays the part of ℓ. That mode is not translation invariant, so Proposition 171 does not apply to it as stated, and no gauge-invariant form of the recursion is claimed here.

Remark 174 (numerical checks). The following values were computed in double precision for $b=2$, $L_0=8$, $n_0=2$ and $d=1$; no statement rests on them. For $K=4$ and $s=1$ the best constants ϖ_j of $\mathrm{BP}_{v_j}(n_j)$, computed as the least eigenvalue of $n_j^{-1}Q_j^{1/2}S_jQ_j^{1/2}$ (Lemma 100), are 0.08253, 0.08525, 0.09196, 0.10609, 0.13327 from the finest level to the coarsest, and each equals $\varpi_j^{(0)}$ to six digits, although the precision matrices Q_j, $j<K$, have positive off-diagonal entries; that $\varpi_j=\varpi_j^{(0)}$ for $j<K$ is observed, not proved. For $K=6$ the defects from the finest step to the coarsest are 0.0088, 0.0215, 0.0442, 0.0819, 0.138, 0.206 at $s=1$, roughly halving toward the fine scales as Proposition 172(iii) predicts, and $\prod_j(1-\delta_j)=0.548$, 0.582, 0.781 at $s=10^{-4}$, 1, 4. The ratio of $\sigma_j^2-\sigma_K^2$ to the bound of (iii) was at most 0.25 in all cases checked ($d=1,2$), and the ratio of the lower bound for σ_K^2 to σ_K^2 lay between 0.47 ($d=2$, $s=3$) and 0.98 ($d=1$, $s=10^{-4}$). On linear functions of the block variables (the first chaos) the step quantities were $\varpi_{\mathrm{fib}}^{(1)}\in[1.03,1.16]$, $1-\kappa_j^{(1)}=0.052$, 0.116, 0.204, 0.292 for $j=4,\dots,1$ and $\zeta^{(1)}\le0.003$ for $d=1$, $s=1$, $K=4$; and $\varpi_{\mathrm{fib}}^{(1)}\in[0.87,0.98]$, $\kappa^{(1)}\ge0.66$, $\zeta^{(1)}\le0.01$ for $d=2$, $s=1$, $K=2$, $L_0=6$ (the measures are Gaussian, A and Q preserve the first chaos, and the minimiser of a linear block function is linear). These restrictions are one-sided: $\varpi_{\mathrm{fib}}^{(1)}\ge\varpi_{\mathrm{fib}}$, $\kappa^{(1)}\ge\kappa$, $\zeta^{(1)}\le\zeta$, so they are consistent with the open lemma (L) for the free field but do not test it.

Remark 175 (the convention of a fixed cube side). The cube sides $n_j=b^jn_0$ cover a fixed physical cube, as in [21, Corollary 102], where $n_ka_k=\ell$. If instead the same side n is used at every level, the cube of side n of $\mathbb{T}_j$ is the union of the level-j blocks of a cube of side $N_j=nm_j$ of $\mathbb{T}_K$. For the free field Proposition 171(i), with n in place of n_j, bounds the best constant of $\mathrm{BP}_{v_j}(n)$ by the quotient $\mu^2\gamma(\mathrm{Var}(X_{N_j}\mid\mathfrak{G}_j))/N_j^d$, with X_{N_j} the mass of the cube of side N_j and $\mathfrak{G}_j$ the block sums outside it, and, when $L_j\ge n+2$, Proposition 172(iii), (iv) applied to the tower with levels $j,\dots,K$ and cube sides nb^i place this quotient between $\varpi_{N_j}^{(d)}(N_j\mu)$ and $(1+\Gamma_d(N_j\mu)/n)\varpi_{N_j}^{(d)}(N_j\mu)$. By Theorem 112 this upper bound for the best constant is close to $T_d(nm_j\mu)^2$ when $nm_j\mu$ is small and nm_j large, so it decreases roughly by the factor b^{-2} at each step toward the fine scales, the dependence on s^2 of Sec. X B read along the tower. The two conventions are related at the level of BP constants by the analogue of Proposition 73: part (i) passes from a smaller to a larger cube with the factor of the fourth power of the ratio of sides (the d-th power for scalar fields), and part (iii) passes back under an interior condition.

4. *The angle between block and fibre functions*

The quantities κ and ζ are controlled by one constant, the cosine of the angle between $\mathscr{H}_c$ and $\mathscr{H}_f$ in the form A, together with the fine form of block functions. In the theory of multilevel methods this is the constant of the strengthened Cauchy–Schwarz inequality [45], which governs two-level iterations [106]. Put, with $0/0=0$,

$$\gamma=\sup_{g\in\mathscr{H}_c,k\in\mathscr{H}_f}\frac{|\langle g,Ak\rangle|}{\langle g,Ag\rangle^{1/2}\langle k,Ak\rangle^{1/2}}\in[0,1],$$

$$\theta=\inf_g\frac{\langle g,Ag\rangle}{\langle g,A^cg\rangle},$$

the infimum over $g\in\mathscr{H}_c$ with $\langle g,A^cg\rangle>0$. In the measure-theoretic case $\langle F\circ c,AF\circ c\rangle=\mathscr{E}'_{n'}(F\circ c)$ is the fine Dirichlet form of a function of the block variables, and $\gamma=0$ exactly when A maps $\mathscr{H}_c$ into itself.

Proposition 176 (the defect through the angle)**.** *In the setting of Theorem 167, for $g\in\mathscr{H}_c$,*

$$(1-\gamma^2)\langle g,Ag\rangle\le\tilde{\mathscr{E}}_0(g)\le\langle g,Ag\rangle,$$

$$\|k_0(g)\|^2\le\frac{\langle k_0(g),Ak_0(g)\rangle}{\varpi_{\mathrm{fib}}}\le\frac{\gamma^2\langle g,Ag\rangle}{\varpi_{\mathrm{fib}}}.$$

Hence $\kappa\ge(1-\gamma^2)\theta$ and $\zeta\le\gamma^2\|A_{cc}\|/\varpi_{\mathrm{fib}}$.

Proof. Let $a=\langle g,Ag\rangle$ and, for $k\in\mathscr{H}_f$, $t=\langle k,Ak\rangle$. Then $\langle g+k,A(g+k)\rangle=a+2\langle g,Ak\rangle+t\ge a-2\gamma\sqrt{at}+t\ge(1-\gamma^2)a$, since $t-2\gamma\sqrt{a}\sqrt{t}\ge-\gamma^2a$; the infimum over k gives the lower bound, and $k=0$ the upper one. The minimiser $k_0=k_0(g)$ satisfies $\langle k,A(g+k_0)\rangle=0$ for every $k\in\mathscr{H}_f$; with $k=k_0$, $\langle k_0,Ak_0\rangle=-\langle g,Ak_0\rangle\le\gamma\sqrt{a}\,\langle k_0,Ak_0\rangle^{1/2}$, so $\langle k_0,Ak_0\rangle\le\gamma^2a$, and $\langle k_0,Ak_0\rangle\ge\varpi_{\mathrm{fib}}\|k_0\|^2$ because $k_0\in\mathscr{H}_f$. The consequences follow from the definitions of κ, θ, ζ and $a\le\|A_{cc}\|\|g\|^2$. □

For the free field and $g=\ell$, $A\ell=c\ell\in\mathscr{H}_c$, so $\langle\ell,Ak\rangle=0$ for $k\in\mathscr{H}_f$: the angle vanishes on ℓ, which is the mechanism of Proposition 172(ii). The fluctuation response also turns the separate inequalities (A), (B) into a multiplicative step. [21, Theorem 100(ii)] gives $1/\varpi'\le C_A+C_B/\varpi_c$ under (A) and (B); along a tower with $C_{B,j}=1$ this yields only $1/\varpi_K\le 1/\varpi_0+\sum_jC_{A,j}$, which degenerates linearly in the number of steps, and the passage from (A), (B) to S with the constants $2C_A$, $2C_B$ costs a factor 2 per step.

Corollary 177 (separate inequalities with a summable response)**.** *In the setting of Corollary 170, suppose that for $1\le j\le K$ the separate inequalities* (A) *and* (B) *hold for (μ_j,c_j) with constants $C_{A,j}\le C_A$ and $C_{B,j}$, that $\prod_{i=j}^K C_{B,i}\le\Gamma$ for every j, with $\Gamma\ge1$, and that $\sum_j\zeta_j\le Z$. Then*

$$\varpi_K\ge\frac{\min(\varpi_0,1/(2C_A))}{\Gamma e^{2Z}}.$$

Proof. For $k \in \mathscr{H}_f$, $\mathbb{E}[k \mid c_j] = 0$, so $\|k\|^2 = \mu_j(\mathrm{Var}(k \mid c_j)) \le C_{A,j}\langle k, A\rangle k$ by (A), and $\varpi_{\mathrm{fib},j} \ge 1/C_A$. By (B), $\mathscr{E}^{(j-1)}_{n_{j-1}}(F) \le C_{B,j}\mathscr{E}^{(j)}_{n_j}(f)$ for every f with $\mathbb{E}[f \mid c_j] = F \circ c_j$, and the infimum over f gives $\kappa_j \ge 1/C_{B,j}$. Corollary 169 gives $\varpi_j \ge \min(\varphi, \rho_j \varpi_{j-1})$ with $\varphi = 1/(2C_A)$ and $\rho_j = 1/(C_{B,j}(1+2\zeta_j))$. With $m_0 = \varpi_0$ and $m_j = \min(\varphi, \rho_j m_{j-1})$, induction gives $\varpi_K \ge m_K$, and

$$m_K = \min\Big(\varpi_0 \prod_{l=1}^{K} \rho_l,\ \min_{1 \le i \le K} \varphi \prod_{l=i+1}^{K} \rho_l\Big).$$

Every product $\prod_{l=i+1}^{K} \rho_l$, $0 \le i \le K$, is at least $\Gamma^{-1} e^{-2Z}$, by $1 + 2\zeta \le e^{2\zeta}$ and $\Gamma \ge 1$ for the empty product. □

So the loss of the separate inequalities is the product of the constants $C_{B,j}$ and of the factors $1 + 2\zeta_j$; for the second to be bounded along a tower the fluctuation responses must be summable, and by Proposition 176 it suffices that the squared angles γ_j^2 are summable, with a bounded fine form of block functions and a fibre constant bounded below.

5. SU(2) *along the tuning: what is controlled and what is open*

Consider the tower of [21, Corollary 102]: $\mu^{(k)}_{k,L}$ is the Wilson measure at β_k, $\mu^{(j-1)}_{k,L} = (c_j)_* \mu^{(j)}_{k,L}$, $n_j = 2^j n_0$, and the quantities $\varpi_{\mathrm{fib},j}$, κ_j, K_j, ζ_j, γ_j are those of the step $(\mu^{(j)}_{k,L}, c_j)$ in the measure-theoretic case of Theorem 167. The lower bound of Corollary 170 is uniform in k and L when, at every step, the fibre constant is bounded below, the defect of the renormalisation ratio κ_j is summable and the fluctuation response is summable; these are sufficient, not shown to be necessary. In the other direction, by the upper bound of Corollary 169, if $\mathrm{K}_j \le 1$ for all j then the defects $1 - \mathrm{K}_j$ must be summable for a lower bound uniform in k with ϖ_0 bounded. Bałaban's renormalisation group bears on these quantities as follows.

The propagators. In [6] the renormalisation transformations and their propagators are studied for the vector field with Gaussian action, in [7] the averaging operations and their regularity classes, in [9] the existence and uniqueness up to gauge of the minimiser of the action at given averages (the background field), and in [8] regularity and decay of the propagators that depend on such a background field. The small-field fluctuation integrals of [10, 11] are expansions around the background field built on these propagators. The propagator bounds would be the natural starting point of a proof of a lower bound for $\varpi_{\mathrm{fib},j}$ on small fields, and their decay is the kind of input that would localise the fluctuation response k_0 near the boundary of the cube, as the flows of the proof of Proposition 172(iii) localise the free-field defect in a layer of blocks. They are statements about covariances and Green functions, not about the cube-chain form A, so they bound neither γ_j nor ζ_j as they stand.

The effective actions. In [10] a sequence of localised effective actions is constructed in the small-field approximation by cluster expansions in one-step transformations, with β-functions and a coupling constant renormalisation; in [11] the fluctuation integral is represented by an exponentiated cluster expansion, which completes the construction of the small-field effective actions. These describe, on small fields, the marginal law $\mu^{(j-1)}_{k,L}$ of the block variables, and with it the coarse form $\mathscr{E}^{(j-1)}_{n_{j-1}}$, together with the flow of the coupling that defines the tuning. The ratio κ_j is not a property of the marginal law: $\tilde{\mathscr{E}}_0(F \circ c_j)$ is the least fine Dirichlet form of a function with conditional mean $F \circ c_j$, and for the analogue of ℓ it is, by Proposition 171, the comparison of a conditional variance given the fine exterior of a cube with the conditional variance given the block variables outside it. That comparison concerns the joint law of fine variables near the cube and block variables away from it, and we are not aware of an estimate in [10, 11] stated for it.

The large fields. In [12, 13] the renormalisation operation $\mathbf{R}$ of the large-field regions is constructed, localised, exponentiated and bounded, which completes Bałaban's proof of ultraviolet stability of the four-dimensional pure gauge theory: bounds on the partition function in a box of fixed physical size, uniform in the spacing. These are upper bounds on integrals over the large-field configurations. A lower bound for $\varpi_{\mathrm{fib},j}$ is a Poincaré inequality for the conditional law given the block variables at every block configuration of positive probability, including the large ones, and such an inequality is not a consequence of bounds on partition functions.

The volume. The ultraviolet-stability bounds of [12, 13] concern a box of fixed physical size, while the quantities above are needed uniformly in $L \in \mathscr{L}_k$, an infinite family of tori of growing physical size.

So on the set of small fields the propagator bounds of [6, 8] are candidates for controlling the Gaussian part of the defect, through an argument of the kind of Proposition 172(iii) with the background propagator in place of $(-\Delta + \mu^2)^{-1}$; that argument is not carried out here. We know of no proved estimate that controls the non-Gaussian part of κ_j, the fluctuation response on large fields, or the fibre constant on large fields.

Remark 178 (why the defect should be suppressed by the cube size). Along the asymptotic-freedom tuning the effective coupling of the measure at level j is, by perturbative running, of order $g_j^2 \asymp 1/j$ for j large, so $\sum_{j \le k} g_j^2$ grows like $\log k$. If the upper ratios satisfied $\mathrm{K}_j \le 1 - c g_j^2$ at every step, the bound $\varpi_k \le \varpi_0 \prod_j \mathrm{K}_j$ that follows from Corollary 169 would decay like a negative power of k, that is of $\log(1/a_k)$, and the constant would tend to zero for bounded ϖ_0; a defect $1 - \kappa_j \asymp g_j^2$ would make the lower bound of Corollary 170 degenerate in the same way. In the convention of a fixed physical cube the weakly coupled steps $j \approx k$ have the largest cubes in lattice units, $n_{j-1} = 2^{j-1} n_0$, and for the free field the defect is at most $\Gamma_d(s)/n_{j-1}$ (Proposition 172(iii)). A uniform bound therefore needs the per-step defect of the interacting measure to be suppressed by the cube size, as a boundary-layer effect, and not merely by the coupling. This remark is heuristic and is used nowhere.

Definition 179 (the open lemma (L): summable renormalisation defect)**.** In the setting of [21, Corollary 102], the *open lemma* (L) is the statement that there are $\varphi > 0$ and $E < \infty$ such

that for every k, every $L \in \mathscr{L}_k$ and every $1 \le j \le k$ there is $\varepsilon_j = \varepsilon_j(k,L) \in [0, \frac{1}{4}]$ with $\sum_{j=1}^{k} \varepsilon_j \le E$ and the following hold for $\mu' = \mu_{k,L}^{(j)}$, $\mu^c = \mu_{k,L}^{(j-1)}$ and $c = c_j$, with $\mathscr{E}_{n_i}^{(i)}$ the Dirichlet form of $\mu_{k,L}^{(i)}$ for the cubes of side n_i:

(L1) (fibre) $2\varphi\,\mu'(\mathrm{Var}(f \mid c)) \le \mathscr{E}_{n_j}^{(j)}(f)$ for every real gauge-invariant $f \in L^2(\mu')$ with $\mathbb{E}[f \mid c] = 0$;

(L2) (renormalisation) $(1-\varepsilon_j)\,\mathscr{E}_{n_{j-1}}^{(j-1)}(F) \le \mathscr{E}_{n_j}^{(j)}(f)$ for every real gauge-invariant $F \in L^2(\mu^c)$ of mean zero and every real gauge-invariant $f \in L^2(\mu')$ with $\mathbb{E}[f \mid c] = F \circ c$;

(L3) (response) for every such F, the minimiser f_F of $\mathscr{E}_{n_j}^{(j)}$, which exists by (L1) and Theorem 167(i), under the constraint $\mathbb{E}[f \mid c] = F \circ c$ satisfies $\mu'(\mathrm{Var}(f_F \mid c)) \le \varepsilon_j \,\mathrm{Var}_{\mu^c}(F)$.

In the notation of Corollary 169: $\varpi_{\mathrm{fib},j} \ge 2\varphi$, $\kappa_j \ge 1-\varepsilon_j$ and $\zeta_j \le \varepsilon_j$.

The open lemma (L) is not proved, for SU(2) or for any interacting measure; for the free field, restricted to the constant function, (L2) holds with $\varepsilon_j = \delta_j \le \Gamma_d(s)/n_{j-1}$ (Proposition 172(iii)), summable, and within the cap $\varepsilon_j \le \frac{1}{4}$ for the steps with $n_{j-1} \ge 4\Gamma_d(s)$, and (L3) holds since the response vanishes; (L) is not proved on all functions. Part (L1) follows from (H2) through (A), with $2\varphi = 1/C_A$, and (L2) is (B) with $C_{B,j} = 1/(1-\varepsilon_j)$; (L) asks in addition that these constants have summable defects and that the responses are summable, and (H2) does not ask for the responses. Neither (L) nor (H2) is shown here to imply the other. The inequality $\inf\{\mathscr{E}_{n_j}^{(j)}(f) - \lambda\,\mu'(\mathrm{Var}(f \mid c)) : \mathbb{E}[f \mid c] = F \circ c\} \ge \rho\,\mathscr{E}_{n_{j-1}}^{(j-1)}(F)$ for all F, the form of (L2) at a spectral parameter $\lambda > 0$, is exactly $\mathrm{S}(n_j, n_{j-1}; 1/\lambda, 1/\rho)$, since $\mathscr{E}_{n_j}^{(j)}(f) - \lambda\,\mu'(\mathrm{Var}(f \mid c)) \ge \rho\,\mathscr{E}_{n_{j-1}}^{(j-1)}(F_f)$ for every f is the split inequality; (L2) is its form at $\lambda = 0$, whose analogue for scalar fields has on the constant function the defect δ_j of Proposition 171.

Proposition 180 (the conditional consequence)**.** *In the setting of [21, Corollary 102], suppose that* (H1) *and the open lemma* (L) *hold. Then* $\mathrm{BP}(n_k, \varpi)$ *holds for* $\mu_{k,L}$ *for all* k *and* $L \in \mathscr{L}_k$*, with*

$$\varpi = e^{-6E} \min(\varpi_0, \varphi),$$

and every continuum limit as in Theorem 64 satisfies $\sigma(H) \subset \{0\} \cup [16\varpi/(625e^2\ell), \infty)$.

Proof. Fix k and L. By (H1) the best constant of BP for $\mu_{k,L}^{(0)}$ at the side n_0 is at least ϖ_0. By (L1) the number $\min_j \varpi_{\mathrm{fib},j}/2$ of Corollary 170 is at least φ, and by (L2), (L3) $\kappa_j \ge 1-\varepsilon_j$ and $\zeta_j \le \varepsilon_j$ with $\varepsilon_j \in [0, \frac{1}{4}]$ and $\sum_j \varepsilon_j \le E$. Corollary 170, whose tower hypotheses hold since $L_j \ge 2n_j + 4$ as in the proof of [21, Corollary 102], gives that the best constant of $\mathrm{BP}(n_k)$ for $\mu_{k,L}$ is at least $e^{-6E}\min(\varpi_0, \varphi)$. Theorem 64 applies with $n_k a_k = \ell$. □

The proposition is conditional. By Theorem 64 its conclusion gives a mass gap only for a nontrivial continuum limit, and neither (H1), nor (L), nor the nontriviality of the limit is proved; (D) and (E) are not addressed.

Remark 181 (what is proved and what is open). Proved, for every probability measure and every sub-σ-algebra of block variables with a positive fibre constant: the exact characterisation of the fine constant by the effective forms (Theorem 167); the two-sided recursion in terms of ϖ_{fib}, κ, ζ and K (Corollary 169) and its tower form (Corollary 170); the control of κ and ζ by the angle γ (Proposition 176); the multiplicative step from the separate inequalities (Corollary 177); and, for translation-invariant measures with block sums, the exact recursion on the constant function (Proposition 171). For the massive free field: the recursion reproduces the exact constant of Theorem 109, the response of ℓ vanishes, and the defects are at most $\Gamma_d(s)/n_{j-1}$, so their product lies in $[(1+\Gamma_d(s)/n_0)^{-1}, 1]$ (Proposition 172); as $\mu \to 0$ the constant is of order s^2, and in the factorisation the factor s^2 sits in the coarsest-scale quotient (Remark 173). Open: whether $\varpi_j = \varpi_j^{(0)}$ for $j < K$ in the free field (observed numerically); the open lemma (L) on all functions, already for the free field; every quantity of the step for SU(2), in particular (L1) on large fields, the non-Gaussian part of (L2) and (L3); a gauge-invariant form of the recursion for the photon; and (H1).

G. The ratio, the strong-coupling range and the cube estimate

This subsection states the open input of Theorem 46 exactly, proves that it fails along every sequence of couplings in the range where the paper has a gap, shows that the hypothesis of Theorem 43 holds along some of these sequences and therefore cannot imply the input, and replaces the input by a condition that is necessary and sufficient and asks only for an upper bound on an energy and a lower bound on one overlap.

The notation is that of Theorem 46: translation invariant states ω_k at couplings β_k, with transfer matrices $\mathbb{T}_k$ and vacua Ω_k (Appendix F), the centred flowed packet A_k of a nonzero $f_0 \in C_c(\mathbb{R}^4)$ supported in $\{t > 0\}$, and $C_k(t) = \omega_k(\vartheta(\bar{A}_k)\tau_{t/a_k}A_k)$ for $t \in a_k\mathbb{Z}_{\ge 0}$, with $C_k(0) > 0$. For $u \ge 0$ and $\sigma > 0$ in $a_k\mathbb{Z}$ put

$$\rho_k(u,\sigma) = \frac{C_k(2u+\sigma)}{C_k(2u)},$$

so that $\rho(\sigma)$ of Theorem 46 is a limit of $\rho_k(0,\sigma)$. Let $f_{0,u} = f_0(\cdot - (u,0))$; for $u/a_k \in \mathbb{Z}$ its packet is $\tau_{u/a_k}A_k$, and $\Psi_{k,u} = C_k(2u)^{-1/2}\Psi_{\tau_{u/a_k}A_k}$ is the corresponding unit vector. For dyadic $u \ge 0$ consider

(N_u) $\liminf_{k\to\infty} \rho_k(u,\sigma) > 0$ for some dyadic $\sigma > 0$.

Proposition 182. (i) $C_k(t) > 0$ *for every* $t \in a_k\mathbb{Z}_{\ge 0}$*, and* $\log C_k$ *is convex and nonincreasing on* $a_k\mathbb{Z}_{\ge 0}$.

(ii) $0 < \rho_k(u,\sigma) \le 1$*;* $\rho_k(u,\sigma)$ *is nondecreasing in* u *and nonincreasing in* σ*, and* $\rho_k(u,\sigma') \ge \rho_k(u,\sigma)^{\sigma'/\sigma}$ *for* $\sigma' \ge \sigma$.

(iii) *For* $u \ge 0$ *dyadic, Theorem 46 applies to* $f_{0,u}$*, with reflected norm* $C_k(2u)$ *and ratio* $\rho_k(u,\sigma)$*. Every limit it produces*

along subsequences is nontrivial if and only if (N_u) *holds, and some limit is nontrivial if and only if* $\limsup_k \rho_k(u,\sigma) > 0$. *If* (N_u) *holds for one* σ, *it holds for every dyadic* $\sigma > 0$, *and* (N_u) *implies* $(\mathrm{N}_{u'})$ *for* $u' \geq u$.

(iv) *Let* $\mu_{k,u}$ *be the spectral measure, in energies* $E = -a_k^{-1}\log\lambda$, *of* $\mathbb{T}_k$ *at* $\Psi_{k,u}$. *If* $\rho_k(u,\sigma) \geq c > 0$, *then* $\mu_{k,u}([0,\Lambda]) \geq c/2$ *with* $\Lambda = \sigma^{-1}\log(2/c)$; *if* $\mu_{k,u}([0,\Lambda]) \geq v$, *then* $\rho_k(u,\sigma) \geq ve^{-\sigma\Lambda}$. *So* (N_u) *holds if and only if a fixed fraction of the spectral mass of* $\Psi_{k,u}$ *stays at bounded physical energy, uniformly in* k.

Proof. (i) The function f_0 is supported in $\{t > 0\}$, so the lattice points with $f_0(a_k x) \neq 0$ have $x_0 \geq 1$, and $A_k = \sum_{n\geq 1} \tau_n B_n$ is a finite sum in which B_n is a function of the spatial links of the slice at time 0, by Proposition 45(i). In the reconstruction [88–90], as in the proof of Theorem 46, $\Psi_{\tau_n B} = \mathbb{T}_k^n \Psi_B$, hence $\Psi_{A_k} = \mathbb{T}_k \xi$ with $\xi = \sum_n \mathbb{T}_k^{n-1}\Psi_{B_n}$. The transfer matrix is self-adjoint and positive [77, 90, 95], and its range is orthogonal to its kernel, so the spectral measure ν_k of $\mathbb{T}_k$ at $\Psi_k = C_k(0)^{-1/2}\Psi_{A_k}$ (Lemma 352) has no atom at 0. Therefore $C_k(t)/C_k(0) = \int \lambda^{t/a_k} d\nu_k > 0$. Convexity and monotonicity are Proposition 49(ii).

(ii) The first claims follow from (i). On a lattice, a convex function has chord slopes that increase in both end points; this gives monotonicity in u, and, comparing the chords $[2u, 2u+\sigma]$ and $[2u, 2u+\sigma']$, the power bound. Monotonicity in σ is that of C_k.

(iii) Translation invariance and $\vartheta\tau_u = \tau_{-u}\vartheta$ give, with times written in lattice steps,

$$\omega_k\big(\vartheta(\tau_u \bar{A}_k)\,\tau_{t+u}A_k\big) = \omega_k\big(\vartheta(\bar{A}_k)\,\tau_{t+2u}A_k\big),$$

which gives the reflected norm and the ratio. The function $f_{0,u}$ is nonzero and supported in $\{t > u\}$. If $\liminf_k \rho_k(u,\sigma) = 0$, a subsequence has $\rho_k(u,\sigma) \to 0$, and a refinement given by Theorem 46(ii) has a trivial limit by part (iv) of that theorem; if the liminf is at least $c > 0$, every limit has $\rho(\sigma) \geq c$ and is nontrivial. The statement with limsup is the last part of Theorem 46(iv). Independence of σ follows from (ii), by the power bound for $\sigma' \geq \sigma$ and by monotonicity for $\sigma' \leq \sigma$, and the last claim from monotonicity in u.

(iv) The vector $\Psi_{k,u}$ has no spectral weight at $\lambda = 0$, by the proof of (i), and $\rho_k(u,\sigma) = \int e^{-\sigma E} d\mu_{k,u}$. This integral is at most $\mu_{k,u}([0,\Lambda]) + e^{-\sigma\Lambda}$ and at least $e^{-\sigma\Lambda}\mu_{k,u}([0,\Lambda])$. □

The open input of Theorem 46 is therefore exactly (N_u) for one packet f_0 and one dyadic $u \geq 0$. It is a uniformity statement and nothing else: at each fixed k the ratio is positive by (i). The required uniformity is in k along the tuning, at fixed physical separations u and σ, so that $\sigma/a_k \to \infty$ lattice steps; the constant may depend on G, Φ, τ, f_0, u, σ and on the tuning, and nothing is required uniformly in f_0 or in τ. The ratio does not change when A_k is multiplied by a constant, so the renormalisation Z_k plays no role in it. The case $u = 0$ is the strongest and implies all the others. Proposition 47(i) assumes (N_0) and renormalises the packet shifted by $\sigma/2$. The hypothesis of [21, Lemma 462] is $(\mathrm{N}_{s/4})$ at $\sigma = s/2$; it is weaker than (N_0), as the example after Proposition 49 shows, and it suffices for a nontrivial limit of the packet shifted by $s/4$. Granted the convergence asked for in (E), the clause of (E) that the limit is not that of the vacuum alone is, for the flowed packets, (N_u) for one f_0 and one u; the convergence of the whole sequence, Euclidean invariance and the higher correlations remain [68]. No bound is required at one lattice step: the first-step energy $-a_k^{-1}\log\rho_k(u,a_k)$ may diverge while (N_u) holds, as Remark 51 shows for the slice field of the unflowed magnetic energy.

The strong-coupling range. In the range where the paper proves a gap, the ratio has two-sided bounds in lattice units, and the upper one rules out every continuum tuning there. Put

$$m_0 = -\tfrac{1}{2}\log 0.9936,$$

so that $m_0 \geq 0.0032$ by $-\log(1-x) \geq x$.

Theorem 183. *Let* $G = \mathrm{SU}(2)$, *let* $\beta_{W,k} \in [0, 0.348]$ *be any couplings, and let* ω_k *be the states of Theorems 1 and 3. Then for every* Φ, τ *and* f_0 *with* $C_k(0) > 0$, *every dyadic* $u \geq 0$ *and* $\sigma > 0$, *and every* k *with* $u/a_k, \sigma/a_k \in \mathbb{Z}$,

$$\rho_k(0,a_k)^{\sigma/a_k} \leq \rho_k(u,\sigma) \leq e^{-m_0\sigma/a_k}.$$

Consequently every limit of Theorem 46 along these couplings, for every packet $f_{0,u}$, *is trivial, and if* (N_u) *holds for some packet along a tuning* $(\beta_{W,k})$ *of* $\mathrm{SU}(2)$, *then* $\beta_{W,k} > 0.348$ *for all large* k.

Proof. By Theorem 3 the transfer matrix of ω_k has gap $m_k \geq -\frac{1}{2}\log\alpha(\beta_{W,k})$ on $\Omega_k^\perp$, and α is increasing with $\alpha(0.348) \leq 0.9936$, so $m_k \geq m_0$. The vector $\Psi_{k,u}$ is a unit vector orthogonal to Ω_k, because A_k is centred, and $\mathbb{T}_k$ preserves $\Omega_k^\perp$, so $\rho_k(u,\sigma) = \langle \Psi_{k,u}, \mathbb{T}_k^{\sigma/a_k}\Psi_{k,u}\rangle \leq e^{-m_0\sigma/a_k}$. For the lower bound, $\rho_k(u,\sigma) \geq \rho_k(0,\sigma)$ by Proposition 182(ii), and Jensen's inequality for the convex function λ^{σ/a_k} and the probability measure ν_k of the proof of Proposition 182(i) gives $\rho_k(0,\sigma) = \int \lambda^{\sigma/a_k} d\nu_k \geq (\int \lambda\, d\nu_k)^{\sigma/a_k} = \rho_k(0,a_k)^{\sigma/a_k}$, with $\rho_k(0,a_k) > 0$ by Proposition 182(i). The upper bound tends to zero as $a_k \to 0$, which gives triviality by Proposition 182(iii). If (N_u) held along a tuning with $\beta_{W,k} \leq 0.348$ for infinitely many k, the upper bound along those k would contradict it. □

Writing $\rho_k(u,\sigma) = e^{-\varepsilon_k\sigma/a_k}$, the theorem says $m_0 \leq \varepsilon_k \leq -\log\rho_k(0,a_k)$: both ends are rates in lattice units, and (N_u) asks for $\varepsilon_k = O(a_k)$. Reflection positivity gives the positivity of the lower end, through the range of $\mathbb{T}_k$, and the transfer-matrix gap gives the upper bound; neither bears on the size of ε_k/a_k except to exclude the range. The size of $\rho_k(0,a_k)$ is not estimated here; any estimate of it at strong coupling is in lattice units, and it is at most e^{-m_0}. The same holds for every compact simple G and couplings in $[0,\beta_1]$ with $N\beta_1 < \frac{2}{29}\kappa_G$, with $m_0 = m_G(\beta_1)$ of Theorem 5; for fixed β this is Proposition 58(i). So the strong-coupling range bears on the continuum tuning only negatively: a tuning with a nontrivial limit leaves it for good. That the tuning has to go to $\beta_W \to \infty$ is the expectation from asymptotic freedom and is not proved here.

The cube estimate in the same range. The hypothesis of Theorem 43 holds at fixed strong coupling with cubes of fixed physical size, and with every α.

Proposition 184. (i) *Suppose $D(n_k,\alpha_k)$ holds at β_k with $n_k a_k \le \ell$. Then $\rho_k(u,\sigma) \le \alpha_k^{\sigma/(2\ell)}$ for every u and σ. Hence (N_u) with constant c at σ forces $\alpha_k \ge c^{2\ell/\sigma}$ for large k, and the physical mass of Theorem 43 obeys $m_* \le \sigma^{-1}\log(1/c)$.*

(ii) *Let $G=\mathrm{SU}(2)$, $0 \le \beta_W < \frac{1}{3}$ fixed, $\ell > 0$, and $n_k = 2\lfloor \ell/(2a_k)\rfloor$. For every $\alpha \in (0,1)$, $D(n_k,\alpha)$ holds at β_W for all k with $n_k \ge 2$ and $5\beta_W \le \alpha(1-3\beta_W)n_k$, and $n_k a_k \le \ell$. So the hypothesis of Theorem 43 holds along the constant tuning, from some k on, with every $\alpha \in (0,1)$, while every limit of Theorem 46 is trivial.*

Proof. (i) By Theorem 41 the gap of $\mathbb{T}_k$ on $\Omega_k^\perp$ is at least $\log(1/\alpha_k)/(2n_k) \ge a_k \log(1/\alpha_k)/(2\ell)$, and $\rho_k(u,\sigma) \le e^{-m_k\sigma/a_k}$ as in the proof of Theorem 183. The bound on m_* is the case $\alpha_k = \alpha$, since then $\alpha^{\sigma/(2\ell)} = e^{-m_*\sigma}$.

(ii) Proposition 234 gives $D(n,\alpha)$ for every $\alpha \ge \hat\alpha_n$, and $\hat\alpha_n \le 5\beta_W/((1-3\beta_W)n)$. Triviality is Theorem 183. □

Part (i) shows that (D) is itself a triviality statement when its constant improves along the tuning: if $\alpha_k \to 0$ with $n_k a_k \le \ell$, every limit is trivial. Along the constant tuning of part (ii) with $\beta_W > 0$ this gives $\rho_k(u,\sigma) \le (5\beta_W/((1-3\beta_W)n_k))^{\sigma/(2\ell)}$, a power of a_k, from the cube estimate alone. Part (ii) shows that no argument that uses (D) only through the hypothesis of Theorem 43, or through Theorem 41, can give (N_u): that hypothesis is satisfied by Wilson measures whose continuum limits are trivial. Statement (D) of Sec. X adds that $\beta \to \infty$ along the tuning, and an implication that uses this and the weak-coupling structure besides (D) is not excluded; it would use information that (D) does not contain.

Trial states. The input is equivalent to the existence of states of bounded physical energy with a fixed overlap with the packet.

Lemma 185. *Let $\mathbb{T}$ be self-adjoint with $0 \le \mathbb{T} \le 1$, let $a > 0$ and let Ψ be a unit vector.*

(a) *Let χ be a unit vector with $\langle \chi, (1-\mathbb{T})\chi\rangle \le \kappa a$ and $|\langle \chi, \Psi\rangle| \ge v > 0$. Then for $\sigma \ge 2a$,*

$$\langle \Psi, \mathbb{T}^{\sigma/a}\Psi\rangle \ge \Big(v - \sqrt{\kappa\sigma/2}\Big)_+^2,$$

and if $a\Lambda \le 1$ with $\Lambda = 8\kappa/v^2$, the spectral projection $P = \mathbf{1}_{[e^{-a\Lambda},1]}(\mathbb{T})$ has $\|P\Psi\| \ge v/2$.

(b) *Conversely, if $\langle \Psi, \mathbb{T}^{\sigma/a}\Psi\rangle \ge c > 0$ with $\sigma/a \in \mathbb{Z}_{>0}$, put $\Lambda = \sigma^{-1}\log(2/c)$ and $P = \mathbf{1}_{[e^{-a\Lambda},1]}(\mathbb{T})$. Then $\chi = P\Psi/\|P\Psi\|$ is a unit vector with $\langle \chi, (1-\mathbb{T})\chi\rangle \le a\Lambda$ and $\langle \chi, \Psi\rangle = \|P\Psi\| \ge (c/2)^{1/2}$. If $\mathbb{T}\Omega = \Omega$ and $\Psi \perp \Omega$, then $\chi \perp \Omega$.*

Proof. (a) Let $m = \sigma/(2a) \ge 1$ and let ν_χ be the spectral measure at χ. For $\lambda \in [0,1]$, $0 \le 1-\lambda^m \le m(1-\lambda)$ by Bernoulli's inequality, so

$$\|(1-\mathbb{T}^m)\chi\|^2 = \int (1-\lambda^m)^2 d\nu_\chi \le m \int (1-\lambda)\, d\nu_\chi \le \frac{\kappa\sigma}{2}.$$

Hence $\|\mathbb{T}^m\Psi\| \ge |\langle \mathbb{T}^m\chi, \Psi\rangle| \ge |\langle \chi,\Psi\rangle| - \|(1-\mathbb{T}^m)\chi\| \ge v - \sqrt{\kappa\sigma/2}$, and $\langle \Psi, \mathbb{T}^{\sigma/a}\Psi\rangle = \|\mathbb{T}^m\Psi\|^2$. For the projection, $1-\lambda > 1-e^{-a\Lambda} \ge a\Lambda/(1+a\Lambda) \ge a\Lambda/2$ off $[e^{-a\Lambda},1]$, so $\|(1-P)\chi\|^2 \le 2\kappa/\Lambda = v^2/4$, and $\|P\Psi\| \ge |\langle \chi, P\Psi\rangle| \ge |\langle \chi,\Psi\rangle| - \|(1-P)\chi\| \ge v/2$.

(b) With ν the spectral measure at Ψ, $c \le \int \lambda^{\sigma/a} d\nu \le \|P\Psi\|^2 + e^{-\sigma\Lambda} = \|P\Psi\|^2 + c/2$. On the range of P, $1-\mathbb{T} \le 1-e^{-a\Lambda} \le a\Lambda$. Since $P\Omega = \Omega$, $\langle \Omega, P\Psi\rangle = \langle \Omega, \Psi\rangle = 0$. □

Corollary 186. (N_u) *holds if and only if there are $\kappa, v > 0$ and unit vectors χ_k in the reconstructed spaces of the ω_k such that, for all large k,*

$$\langle \chi_k, (1-\mathbb{T}_k)\chi_k\rangle \le \kappa a_k, \qquad |\langle \chi_k, \Psi_{k,u}\rangle| \ge v.$$

Then $\rho_k(u,\sigma) \ge (v - \sqrt{\kappa\sigma/2})_+^2$ for $\sigma \ge 2a_k$, and the χ_k may be taken of the form $\Psi_{X_k}/\|\Psi_{X_k}\|$ for gauge-invariant local functions X_k at positive times, at the cost of replacing (κ, v) by $(2\kappa, v/2)$. If moreover $D(n_k,\alpha)$ holds with $n_k a_k \le \ell$, then $m_* = \log(1/\alpha)/(2\ell) \le 8\kappa/v^2$.

Proof. Sufficiency is Lemma 185(a) with $\Psi = \Psi_{k,u}$, $a = a_k$, and Proposition 182(ii) extends the bound from small σ to every σ. Necessity is Lemma 185(b) with $\kappa = \sigma^{-1}\log(2/c)$ and $v = (c/2)^{1/2}$. The vectors Ψ_X with X local are dense (Appendix F), and both conditions are continuous in χ_k for fixed k, $1-\mathbb{T}_k$ being bounded. Under $D(n_k,\alpha)$, Theorem 41 puts the spectral measure of $\Psi_{k,u} \perp \Omega_k$ in $[0, e^{-a_k m_*}]$; if $m_* > 8\kappa/v^2$, the projection of Lemma 185(a) annihilates $\Psi_{k,u}$ for large k, which is impossible. □

For $\chi_k = \Psi_{X_k}/\|\Psi_{X_k}\|$ the two conditions are statements about lattice correlations:

$$\omega_k\big(\vartheta(\bar X_k)X_k\big) - \omega_k\big(\vartheta(\bar X_k)\tau_1 X_k\big) \le \kappa a_k\, \omega_k\big(\vartheta(\bar X_k)X_k\big),$$
$$\big|\omega_k\big(\vartheta(\bar X_k)\tau_{u/a_k}A_k\big)\big|^2 \ge v^2\, \omega_k\big(\vartheta(\bar X_k)X_k\big)\, C_k(2u).$$

The first is an upper bound on a one-slab Dirichlet form, which is the direction of variational and perturbative estimates and the opposite of a mass gap; the second is a lower bound on one correlation. The choice $\chi_k = \Psi_{k,u}$ is Proposition 47(ii), and the order-two tree and saturation bounds give it by Proposition 59(iii); the corollary shows that the trial vector need not be the packet itself.

A soft component of the slice field. For product packets the trial vector can be built from a part of the slice field of Proposition 49, and the conditions become one first-step bound for that part and one bound for the rest at the earliest time of the packet. Let $f_0 = g \otimes h$ with $g \ge 0$ supported in $[t_0,t_1]$, and let u_k, n_0, n_1 be as in Proposition 49.

Theorem 187. *Suppose that for all large k there is a decomposition $u_k = u_k^{\mathrm{s}} + u_k^{\mathrm{h}}$ with $\mathbb{T}_k^{n_0} u_k^{\mathrm{s}} \ne 0$ and, for constants $\kappa > 0$ and $\theta \in [0,1)$ with $\kappa a_k \le \frac{1}{2}$,*

(S1) $\langle u_k^{\mathrm{s}}, (1-\mathbb{T}_k)u_k^{\mathrm{s}}\rangle \le \kappa a_k \|u_k^{\mathrm{s}}\|^2$;

(S2) $\|\mathbb{T}_k^{n_0} u_k^{\mathrm{h}}\| \le \theta e^{-2\kappa(t_1-t_0)} \|\mathbb{T}_k^{n_0} u_k^{\mathrm{s}}\|$.

Then the hypothesis of Corollary 186 holds for $u = 0$ with $\kappa e^{4\kappa(t_1-t_0)}$ in place of κ and $v = (1-\theta)/(1+\theta)$; in particular (N_0) holds and

$$\frac{C_k(\sigma)}{C_k(0)} \ge \Big(\frac{1-\theta}{1+\theta} - \sqrt{\tfrac{1}{2}\kappa\sigma e^{4\kappa(t_1-t_0)}}\Big)_+^2$$

for $\sigma \geq 2a_k$.

Proof. By the proof of Proposition 49, $\Psi_{A_k} = W(\mathbb{T}_k)u_k$ with $W(\lambda) = \sum_n w_n \lambda^n$, $w_n \geq 0$, $w_n = 0$ unless $n_0 \leq n \leq n_1$, and $\|w\|_1 \lambda^{n_1} \leq W(\lambda) \leq \|w\|_1 \lambda^{n_0}$ on $[0,1]$, with $\|w\|_1 > 0$ because $C_k(0) > 0$. Write ν for the spectral measure at u_k^{s} and $y = \mathbb{T}_k^{n_0} u_k^{\mathrm{s}}$. For a probability measure, a nondecreasing and a nonincreasing function have $\int fg \leq \int f \int g$, because $\iint (f(x) - f(x'))(g(x) - g(x')) \leq 0$; with $f = \lambda^{2n_0}$, $g = 1 - \lambda$ and (S1),

$$\int \lambda^{2n_0}(1-\lambda)\, d\nu \leq \kappa a_k \|y\|^2.$$

So y satisfies (S1) as well, and Jensen's inequality for λ^{2j}, $j = n_1 - n_0$, gives $\|\mathbb{T}_k^j y\|^2 \geq \|y\|^2 (1 - \kappa a_k)^{2j} \geq \|y\|^2 e^{-4\kappa(t_1 - t_0)}$, since $\log(1-x) \geq -2x$ on $[0, \frac{1}{2}]$ and $a_k j \leq t_1 - t_0$. Hence

$$\|W(\mathbb{T}_k)u_k^{\mathrm{s}}\| \geq \|w\|_1 \|\mathbb{T}_k^{n_1} u_k^{\mathrm{s}}\| \geq \|w\|_1 e^{-2\kappa(t_1 - t_0)} \|y\|,$$

while $\|W(\mathbb{T}_k)u_k^{\mathrm{h}}\| \leq \|w\|_1 \|\mathbb{T}_k^{n_0} u_k^{\mathrm{h}}\| \leq \theta \|W(\mathbb{T}_k)u_k^{\mathrm{s}}\|$ by (S2). Let $\chi_k = W(\mathbb{T}_k)u_k^{\mathrm{s}} / \|W(\mathbb{T}_k)u_k^{\mathrm{s}}\|$. Its energy is

$$\frac{\int W^2(1-\lambda)\, d\nu}{\int W^2 d\nu} \leq \frac{\|w\|_1^2 \kappa a_k \|y\|^2}{\|w\|_1^2 e^{-4\kappa(t_1-t_0)} \|y\|^2} = \kappa a_k e^{4\kappa(t_1 - t_0)},$$

and with $r = \|W(\mathbb{T}_k)u_k^{\mathrm{h}}\| / \|W(\mathbb{T}_k)u_k^{\mathrm{s}}\| \leq \theta$ its overlap with $\Psi_k = \Psi_{A_k} / \|\Psi_{A_k}\|$ is at least $(1-r)/(1+r) \geq (1-\theta)/(1+\theta)$. Corollary 186 gives the rest. □

Condition (S1) asks for a soft component of the slice field whose first-step energy is bounded in physical units. Remark 51 shows why the whole slice field is not a candidate: at next order its first-step energy grows like $\log(\sqrt{\tau}/a_k)$, through a hard component of norm of order g^2. Condition (S2) asks that this hard component, propagated to the earliest time t_0 of the packet, be a fixed fraction of the soft part. For the free field of Remark 50, (S1) holds with $u_k^{\mathrm{h}} = 0$: since $(1 - e^{-a_k\omega})/a_k \leq \omega \leq a_k^{-1} \sinh(a_k\omega)$, the first-step energy $\langle u_k, (1 - \mathbb{T}_k)u_k \rangle / (a_k \|u_k\|^2)$ has a numerator whose integrand is at most a multiple of $e^{-8t_f|p|^2/\pi^2}$, and a denominator $D_k(0)$ with a positive limit, as in that remark. For the unflowed magnetic energy the whole slice field violates (S1) by a factor of order $1/a_k$, and the decomposition has to separate its hard modes. Both conditions concern energies of order $1/a_k$ at physical times below t_0; they are ultraviolet statements, the subject of a renormalisation group analysis between the scales a_k and t_0 [6, 7, 13], and they are not proved here. By Corollary 186 the trial-state condition is the least hypothesis that gives (N_u), since it is also necessary; by Proposition 184(ii) it is not a consequence of the cube estimate in the form used by Theorem 43; and (S1)–(S2) are its form for the slice field.

Remark 188. There is no circularity between (N_u) and (D). The proofs of Proposition 182, Lemma 185, Corollary 186 and Theorem 187 use reflection positivity, the transfer matrix and the spectral theorem only, and Theorem 46(i)–(iv), including the renormalisation $Z_k = C_k(0)^{-1/2}$, does not use (D). Conversely, Theorems 41 and 43 use no lower bound on any correlation. The two statements are opposite inequalities on the same spectral measures: (D) bounds ρ_k from above by $e^{-m_*\sigma}$, (N_u) from below. Their joint validity requires only the compatibility relations $\alpha \geq c^{2\ell/\sigma}$, $m_* \leq \sigma^{-1} \log(1/c)$ and $m_* \leq 8\kappa/\nu^2$ of Proposition 184 and Corollary 186, all of them statements in physical units. Neither implies the other. Strong-coupling Wilson measures satisfy the hypothesis of Theorem 43 and violate (N_u) (Proposition 184(ii)). In the free abelian theory the quadratic approximation of the forest gauge violates (D) at every size and stiffness (Proposition 94), while the zero-momentum packet of the magnetic energy has slice ratios with positive limits (Remark 51) and so satisfies (N_0) by Proposition 49(iii); (D) depends on the gauge and (N_u) does not. Theorem 55(iv) derives (N_0) from the order-two cases of (T1) and (T2), and Proposition 58 shows that those cases fail wherever the lattice gap is bounded below, in particular in the strong-coupling range, where the hypothesis of Theorem 43 holds.

H. Slab energies of local fields

This subsection tests the first condition of Corollary 186 against the most direct estimate for it: a pointwise bound on the change of a gauge-invariant function of one slice in one time step by the time-like plaquettes, combined with an upper bound on the plaquette energy. Both bounds are proved with explicit constants. The combination cannot give the condition along any tuning with $a_k \beta_k \to 0$, whatever the trial function, and the one-loop tuning is of this kind.

Let $G \subset \mathrm{U}(N)$ be a closed subgroup, let $|M|^2 = \mathrm{Tr}(M^*M)$ be the Frobenius norm, so that $|gMh| = |M|$ for $g, h \in G$, and put $s_p = 1 - N^{-1} \mathrm{Re}\,\mathrm{Tr}\, U_p$, so that $|U_p - I|^2 = 2N s_p$. Write $U_{(y,\mu)}$ for the link from y to $y + \hat{\mu}$, with $\mu = 0$ the time direction. Let ω be a state as in Appendix F, with transfer matrix $\mathbb{T}$, and let $\varepsilon_\omega = \sup_p \omega(s_p)$, which is $\omega(s_p)$ for every plaquette p when ω is invariant under translations and the hypercubic group. A *slice function* is a function X of the spatial links of the slice $x_0 = 0$ that depends only on the links of a finite set E, is invariant under the gauge transformations of that slice, and satisfies

$$|X(U) - X(V)| \leq \sum_{e \in E} L_e |U_e - V_e|$$

with constants $L_e \geq 0$; put $\Lambda(X) = \sum_{e \in E} L_e$. For a spatial link $e = ((0,x), i)$ let p_e be the time-like plaquette with lower edge e, with holonomy

$$U_{p_e} = U_{((0,x),i)} U_{((0,x+\hat{\imath}),0)} U^{-1}_{((1,x),i)} U^{-1}_{((0,x),0)}.$$

Lemma 189. *Let X be a bounded slice function.*

(i) $\langle \Psi_X, (1 - \mathbb{T})\Psi_X \rangle = \frac{1}{2}\omega(|\tau_1 X - X|^2)$*, and both sides are unchanged when a constant is added to X.*

(ii) *Pointwise,* $|\tau_1 X - X| \leq \sum_{e \in E} L_e |U_{p_e} - I|$.

(iii) $\langle \Psi_X, (1 - \mathbb{T})\Psi_X \rangle \leq N \Lambda(X)^2 \varepsilon_\omega$.

Proof. (i) The reflection ϑ in the plane $x_0 = 0$ fixes the spatial links of that slice, so $\vartheta(\bar{X}) = \bar{X}$ and $\|\Psi_X\|^2 = \omega(|X|^2)$. As in the proof of Proposition 49, $\Psi_{\tau_1 X} = \mathbb{T}\Psi_X$, so $\langle \Psi_X, \mathbb{T}\Psi_X \rangle =$

$\omega(\bar X\,\tau_1 X)$, a real number because $\mathbb{T}$ is self-adjoint. Translation invariance gives $\omega(|\tau_1 X|^2)=\omega(|X|^2)$, hence

$$\omega(|\tau_1 X - X|^2) = 2\omega(|X|^2) - 2\omega(\bar X\,\tau_1 X) = 2\langle \Psi_X, (1-\mathbb{T})\Psi_X\rangle .$$

The right side of the identity depends on X only through $\tau_1 X - X$.

(ii) For a site x of the slice put $g(x)=U_{((0,x),0)}$, and for $e=((0,x),i)$ put $V_e = g(x)U_{((1,x),i)}g(x+\hat\imath)^{-1}$. The function $\tau_1 X$ is X evaluated on the links $U_{((1,x),i)}$ in place of $U_{((0,x),i)}$, and gauge invariance under g gives $\tau_1 X(U) = X(V)$. Hence $|\tau_1 X - X| \le \sum_e L_e |V_e - U_e|$. Now $|V_e - U_e| = |U_e^{-1}V_e - I|$, and $U_e^{-1}V_e = U_e^{-1}U_{p_e}^{-1}U_e$, so $|V_e - U_e| = |U_{p_e}^{-1} - I| = |U_{p_e} - I|$.

(iii) By (i), (ii) and the Cauchy–Schwarz inequality,

$$\langle \Psi_X, (1-\mathbb{T})\Psi_X\rangle \le \frac{1}{2}\Lambda(X)\sum_e L_e\,\omega(|U_{p_e}-I|^2) \le N\Lambda(X)^2\varepsilon_\omega .$$

□

The plaquette energy of [21, Lemma 382] carries a factor $\log\beta_W$. It comes from the lower bound on the partition function alone; an upper bound in temporal gauge removes it.

Proposition 190. *Let $G=\mathrm{SU}(2)$ and let μ_L be the Wilson measure on the torus $\Lambda_L=(\mathbb{Z}/L\mathbb{Z})^4$, with $L\ge 2$ and $\beta_W\ge 1$. Then $\varepsilon_L = \mathbb{E}_{\mu_L} s_p$ satisfies*

$$\varepsilon_L \le \frac{1}{\beta_W}\Big(K_1 + \frac{K_2}{L^4} + \frac{15}{16}\Big(\frac{1}{L}+\frac{1}{3L^4}\Big)\log\beta_W + \frac{K_3}{\beta_W}\Big)$$

with $K_1 = 3.7148$, $K_2 = 0.4699$ and $K_3 = 0.149$. In particular every limit as $L\to\infty$ of the measures μ_L along a subsequence, on the local functions, has $\varepsilon \le (K_1 + K_3/\beta_W)/\beta_W$.

Proof. Write $\beta=\beta_W$, $Z(\beta)=\int e^{-\beta S}dU$ with $S=\sum_p s_p$, and $\theta\in[0,\pi]$ for the angle of an element $\cos\theta\, I + \sin\theta\, J$ of $\mathrm{SU}(2)$; it has Haar density $\frac{2}{\pi}\sin^2\theta$ (Lemma 360), it is a bi-invariant distance from I, and $\frac12\operatorname{Tr}U = \cos\theta$. The measure μ_L is invariant under the translations and the hypercubic group of the torus, so $\mathbb{E}_{\mu_L} S = 6L^4\varepsilon_L$.

Convexity. The function $\log Z$ is convex with derivative $-\mathbb{E}_{\mu_L}S$ at β, so for $0<\lambda<1$

$$6L^4(1-\lambda)\beta\,\varepsilon_L \le \log Z(\lambda\beta) - \log Z(\beta).$$

Upper bound. The integrand is gauge invariant, so the time links $U_{((t,x),0)}$ with $0\le t\le L-2$, which form a forest, may be set to I without changing Z; this is the substitution $U\mapsto U^g$ with g built along each tree from its root, which preserves the Haar measure. Drop the spatial plaquettes and the time-like plaquettes between the times $L-1$ and 0, whose weights are at most 1. For each spatial link (x,i) the remaining weights are $\prod_{t=0}^{L-2} f(V_t V_{t+1}^{-1})$ with $V_t = U_{((t,x),i)}$ and $f(U) = e^{-\beta(1-\frac12\operatorname{Tr}U)}$; integrating $V_0,\dots,V_{L-2}$ in turn, each integral is $z(\beta)=\int f\,dU$ by the invariance of the Haar measure. Hence $Z(\beta)\le z(\beta)^{3L^3(L-1)}$. With $y=\sin(\theta/2)$,

$$\begin{aligned} z(\beta) &= \frac{2}{\pi}\int_0^\pi e^{-\beta(1-\cos\theta)}\sin^2\theta\, d\theta \\ &= \frac{16}{\pi}\int_0^1 e^{-2\beta y^2} y^2\sqrt{1-y^2}\,dy \le \sqrt{\frac{2}{\pi}}\,\beta^{-3/2}, \end{aligned}$$

by $\sqrt{1-y^2}\le 1$ and $\int_0^\infty e^{-2\beta y^2}y^2dy = \frac{\sqrt\pi}{4}(2\beta)^{-3/2}$.

Lower bound. Fix a spanning tree of Λ_L and set its L^4-1 links to I, again without changing Z; $3L^4+1$ links remain. Let $0<\delta\le\pi/2$ and let ν be the Haar measure restricted to $\{\theta\le\delta\}$ and normalised. Since $\sin t/t$ decreases on $(0,\pi)$,

$$\frac{2}{\pi}\int_0^\delta \sin^2 t\,dt \ge \frac{2\delta^3}{3\pi}\Big(\frac{\sin\delta}{\delta}\Big)^2 .$$

By Jensen's inequality for the product of the measures ν on the remaining links, $\log Z(\beta) \ge (3L^4+1)\log\frac{2\delta^3}{3\pi}(\frac{\sin\delta}{\delta})^2 - \beta\sum_p \mathbb{E}_\nu s_p$. The set $\{\theta\le\delta\}$ is invariant under conjugation and inversion, so $\mathbb{E}_\nu U = mI$ and $\mathbb{E}_\nu U^{-1} = mI$ with $m = \mathbb{E}_\nu\cos\theta\in[0,1]$. A plaquette has at most four links off the tree, independent under ν, so $\mathbb{E}_\nu\frac12\operatorname{Tr}U_p = m^j$ with $j\le 4$ and $\mathbb{E}_\nu s_p = 1-m^j \le 4(1-m)\le 2\mathbb{E}_\nu\theta^2$. With $w(t)=\sin^2 t/t^2$ nonincreasing and t^2 increasing, Chebyshev's inequality for the measure proportional to t^2dt on $[0,\delta]$ gives $\mathbb{E}_\nu\theta^2 = \int_0^\delta t^2 w t^2 dt / \int_0^\delta w t^2 dt \le \frac35\delta^2$. Hence $\mathbb{E}_\nu s_p \le \frac65\delta^2$.

The constants. Take $\lambda=\frac15$ and $\delta^2 = c/\beta$ with $c=\frac58$, so that $\delta^2\le\frac58<\pi^2/4$. Dividing by $6L^4(1-\lambda)\beta$ and collecting the terms in $\log\beta$ gives the bound, with

$$K_1 = \tfrac54\Big(\tfrac14\log\tfrac{2}{\pi} - \tfrac34\log\lambda - \tfrac12\log\tfrac{2}{3\pi} - \tfrac34\log c + \tfrac65 c\Big),$$
$$K_2 = \tfrac{5}{24}\Big(-\log\tfrac{2}{3\pi} - \tfrac32\log c\Big),$$

after dropping the term $-\frac{5}{8L}(\frac12\log\frac{2}{\pi} - \frac32\log\lambda)$, which is negative. The factor $(\sin\delta/\delta)^2$ contributes at most $\frac54(1+\frac{1}{48})\frac{y}{1-y}$ with $y=\delta^2/6\le c/(6\beta)$, by $\sin\delta/\delta\ge 1-y$ and $-\log(1-y)\le y/(1-y)$, which is at most K_3/β inside the bracket. With $\lambda=\frac15$ and $c=\frac58$ the logarithms collect to

$$K_1 = \tfrac54\big(2\log 2 + \tfrac12\log 3 + \tfrac14\log\pi + \tfrac34\big),$$
$$K_2 = \tfrac{5}{24}\big(\tfrac72\log 2 + \log 3 + \log\pi - \tfrac32\log 5\big),$$

and $\frac54\cdot\frac{49}{48}\cdot\frac{c/6}{1-c/6} = \frac{1225}{8256} < 0.1484$. The series $2\operatorname{artanh}x = 2\sum_{k\ge0}x^{2k+1}/(2k+1)$, whose tail after any term is at most the next term divided by $1-x^2$, applied to $\log 2 = 2\operatorname{artanh}\frac13$, $\log\frac32 = 2\operatorname{artanh}\frac15$ and $\log\frac54 = 2\operatorname{artanh}\frac19$, gives $\log 2 < 0.6931472$, $\log 3 < 1.0986123$ and $\log 5 > 1.6094379$. With $\pi<3.1416$ and $\log(1+x)\le x - \frac{x^2}{2} + \frac{x^3}{3}$ at $x=0.0472$, $\log\pi < \log 3 + 0.046122 < 1.144735$. Hence $K_1 < 3.71474$ and $K_2 < 0.46984$. At fixed λ the value $c=\frac58$ minimises K_1. The last claim holds because s_p is a local function. □

For comparison, [21, Lemma 382] gives the bound $(\log\beta_W + 4.9082)/\beta_W$ for ε_L, which is larger at every $\beta_W\ge1$ in the limit $L\to\infty$. With Lemma 189, a slice function at a limit state of Proposition 190 has

$$\langle \Psi_X, (1-\mathbb{T})\Psi_X\rangle \le \frac{2\Lambda(X)^2}{\beta_W}\Big(K_1 + \frac{K_3}{\beta_W}\Big),$$

a bound of order one in lattice units at bounded $\Lambda(X)$. Whether it can serve as the first condition of Corollary 186 depends on the variance of X alone, and the next proposition shows that it cannot along the tunings of interest.

Proposition 191. *Let X be a real bounded slice function with $\mathrm{Var}_\omega X > 0$, and let $\chi = \Psi_{X-\omega(X)}/\|\Psi_{X-\omega(X)}\|$.*

(i) $\mathrm{Var}_\omega X \le N\Lambda(X)^2$. *Hence the bound of Lemma 189(iii) for $\langle \chi, (1-\mathbb{T})\chi\rangle$, which is $N\Lambda(X)^2\varepsilon_\omega/\mathrm{Var}_\omega X$, is at least ε_ω.*

(ii) *Let $G = \mathrm{SU}(2)$ and let ω_k be states as in Appendix F that are Wilson Gibbs states at β_k in the sense of Sec. X J. If that bound gives the first condition of Corollary 186 with a constant κ, then $(4\beta_k+1)a_k \ge 1/\kappa$. Along a tuning with $a_k\beta_k \to 0$, no choice of slice functions X_k gives that condition through Lemma 189(iii).*

(iii) *Let the Lie algebra $\mathfrak{g}$ of G have trivial centre. If X, as a function on G^E, is differentiable at the configuration with every link equal to I, its differential there vanishes.*

Proof. (i) For any two configurations U and V,

$$|X(U)-X(V)| \le \Lambda(X)\max_e |U_e - V_e| \le 2\sqrt{N}\Lambda(X),$$

since $|U_e| = |V_e| = \sqrt{N}$. A real random variable with values in an interval of length ℓ has variance at most $\ell^2/4$. By Lemma 189(i), $\langle \chi, (1-\mathbb{T})\chi\rangle = \langle \Psi_X, (1-\mathbb{T})\Psi_X\rangle/\mathrm{Var}_\omega X$, since $\|\Psi_{X-\omega(X)}\|^2 = \omega(|X-\omega(X)|^2)$.

(ii) By (i) the bound is at least ε_{ω_k}, which is at least $1/(4\beta_k+1)$ by Proposition 231. If it is at most κa_k, then $1 \le \kappa a_k(4\beta_k+1)$.

(iii) A constant gauge transformation h on the slice gives $X(hUh^{-1}) = X(U)$. For $A = (A_e) \in \mathfrak{g}^E$ let $\phi(A)$ be the derivative of $t \mapsto X((e^{tA_e})_e)$ at $t=0$; ϕ is linear, and $\phi(\mathrm{Ad}_h A) = \phi(A)$ for $h \in G$. Averaging over the Haar measure, $\phi(A) = \phi(\bar A)$ with $\bar A_e = \int \mathrm{Ad}_h A_e\, dh$. Each $\bar A_e$ is invariant under Ad_G, hence commutes with $\mathfrak{g}$, and so vanishes. The curves $t \mapsto e^{tA_e}$ span the tangent space at I. □

The one-loop tuning has $\beta_W = \frac{11}{3\pi^2}\log(1/a) + O(1)$ for $\mathrm{SU}(2)$, and the two-loop tuning adds a term of order $\log\log(1/a)$ [21, Remark 247], so $a_k\beta_k \to 0$ along either, and Proposition 191(ii) applies. Part (i) shows where the loss lies: the bound of Lemma 189(iii) is first order in the time-like plaquettes, and it gives an energy at most κa only if the variance of X is at least the fraction $\varepsilon_\omega/(\kappa a)$ of its largest possible value $N\Lambda(X)^2$. Part (iii) shows that a gauge-invariant function has no first-order term at the configuration with all links I. If X is continuously differentiable, its local Lipschitz constants at configurations within distance r of that configuration, or of a gauge transform of it, tend to 0 with r, while the bound of Lemma 189(ii) uses the global constants L_e.

The bound does give the trial-state condition in the opposite regime, at the cost of a lower bound on a variance and on one correlation, which are the infrared content of (N_u).

Corollary 192. *Let $G = \mathrm{SU}(2)$, let ω_k be states as in Appendix F that are limits as in Proposition 190 at $\beta_k \ge 1$, and let X_k be real slice functions with $X_k^c = X_k - \omega_k(X_k)$. Suppose that there are $\nu, v > 0$ such that for all large k*

$$\mathrm{Var}_{\omega_k} X_k \ge 2\nu\Lambda(X_k)^2,$$
$$\omega_k\big(X_k^c\,\tau_{u/a_k}A_k\big)^2 \ge v^2\,\mathrm{Var}_{\omega_k} X_k C_k(2u).$$

If moreover $a_k\beta_k \ge \kappa_0 > 0$ for all large k, then (N_u) holds, and $\rho_k(u,\sigma) \ge (v - \sqrt{\kappa\sigma/2})_+^2$ for $\sigma \ge 2a_k$, with $\kappa = (K_1+K_3)/(\nu\kappa_0)$.

Proof. The vector $\chi_k = \Psi_{X_k^c}/\|\Psi_{X_k^c}\|$ is a unit vector with $\|\Psi_{X_k^c}\|^2 = \mathrm{Var}_{\omega_k} X_k$. By Lemma 189 and Proposition 190, since $\beta_k \ge 1$ and $a_k\beta_k \ge \kappa_0$,

$$\langle \chi_k, (1-\mathbb{T}_k)\chi_k\rangle \le \frac{2\Lambda(X_k)^2(K_1+K_3)}{\beta_k\,\mathrm{Var}_{\omega_k} X_k} \le \frac{K_1+K_3}{\nu\beta_k} \le \kappa a_k.$$

Since X_k is a real function of the slice $x_0 = 0$, $\vartheta(X_k^c) = X_k^c$ and $\langle \chi_k, \Psi_{k,u}\rangle = \omega_k(X_k^c\tau_{u/a_k}A_k)/(\mathrm{Var}_{\omega_k} X_k C_k(2u))^{1/2}$, whose modulus is at least v. Corollary 186 gives the rest. □

The hypotheses of Corollary 192 are not proved here. The first asks that X_k take values spread over a fixed fraction of the range allowed by its Lipschitz constants, at couplings where the plaquettes are of size $\beta_k^{-1/2}$; the second is a lower bound on one correlation at physical distance. Together with Proposition 191, the corollary locates the estimate: the upper bound on the slab energy that reflection positivity and the plaquette energy provide is of order β_k^{-1} in lattice units, it meets the scale a_k only where $a_k\beta_k$ is bounded below, and even there it transfers the whole of (N_u) to lower bounds on a variance and on an overlap.

I. The axioms for the flowed fields

Statement (E) asks for more than the nontriviality of Sec. X G. This subsection determines, for the spatially flowed fields of Sec. IX E, which of the Osterwalder–Schrader axioms follow from the hypothesis of Theorem 43 once the higher renormalised correlations are bounded, and which further statements are needed. Three are needed; together they suffice, and each is necessary: a uniform moment bound (M), which contains (N_0); invariance under one spatial rotation of infinite order (W); and coincidence of the subsequential limits (U). The spatial flow leaves at most the symmetry $\mathrm{O}(3)\times\mathrm{O}(1)$ of space and time reflection, and Gaussian examples show that neither (W) nor (U) follows from the gap, reflection positivity, the lattice symmetries and the tree bounds.

The notation is that of Sec. IX E: Φ is a bounded real gauge-invariant function of finitely many spatial links of the slice $\{x_0 = 0\}$, continuous in them, $\varphi_k(x) = \Phi_x \circ \Phi_{s_k}$ with $s_k = \tau/a_k^2$, $A_k(f)$ is the centred packet of $f \in C_c(\mathbb{R}^4)$, f_0 is a fixed nonzero function supported in $\{t > 0\}$, $C_k(0) > 0$ and $Z_k = C_k(0)^{-1/2}$. Let W_3 be the group of the 48 signed permutations of the spatial coordinates, $\theta(t,\mathbf{x}) = (-t,\mathbf{x})$, and Γ the group of order 96 generated by W_3 and θ, the subgroup of the hyperoctahedral group B_4 that preserves the time axis. We assume that Φ is invariant under the action of W_3 on the links of the slice that fixes the site 0. Averaging Φ over W_3 achieves this; the sum of $N - \mathrm{Re}\,\mathrm{Tr}\,U_p$ over the twelve spatial plaquettes p with a corner at 0 is an example.

Throughout, the states ω_k at β_k satisfy the hypothesis of Theorem 43: $D(n_k,\alpha)$ holds with $n_k a_k \le \ell$. By Theorem 41

and Proposition 11, each ω_k is the restriction of the unique Gibbs measure of γ^F, is the free-boundary limit along every exhaustion of $\mathbb{Z}^4$, is invariant under translations and the hypercubic group and is reflection positive, and its transfer matrix has gap at least $\log(1/\alpha)/(2n_k) \geq m_* a_k$ on $\Omega_k^\perp$, with $m_* = \log(1/\alpha)/(2\ell)$. The infinite-volume state is therefore unique at each k, and the only non-uniqueness left is in the limit $k \to \infty$. The spatial flow commutes with Γ (Proposition 45(i) and the proof of Theorem 193(ii)) but not with the elements of B_4 that exchange time with a spatial axis, so Γ, not B_4, is the lattice symmetry of the flowed packets.

(M) *Moment bound.* For every $n \geq 1$ and every cube $Q \subset \mathbb{R}^4$ there are $M_n(Q) < \infty$ and k_0 such that

$$\Big|\omega_k\Big(\prod_{j=1}^n Z_k A_k(f_j)\Big)\Big| \leq M_n(Q) \prod_{j=1}^n \|f_j\|_\infty$$

for all $k \geq k_0$ and all $f_j \in C_c(\mathbb{R}^4)$ supported in Q.

By Theorem 55(i), (T1)–(T3) imply (M). The case $n = 2$ of (M) is not the bound of Theorem 46(i), which concerns reflected pairs of translates of f_0 and holds by reflection positivity alone; (M) concerns all test functions in Q, including those closer to the reflection plane than f_0, and it implies (N_0) (Theorem 193(v)). By Remark 57, no bound on the two-point functions gives (M) for $n = 4$.

A *polynomial* is a finite formal sum $P = \sum_\beta c_\beta \prod_{j=1}^{n_\beta} f_{\beta,j}$ with $c_\beta \in \mathbb{C}$ and $f_{\beta,j} \in C_c(\mathbb{R}^4)$; its lattice version is $P_k = \sum_\beta c_\beta \prod_j Z_k A_k(f_{\beta,j})$, and for a family $S = (S_n)$ with $S_0 = 1$ we put $S(P) = \sum_\beta c_\beta S_{n_\beta}(f_{\beta,1}, \dots, f_{\beta,n_\beta})$. The polynomials $\bar{P}$, θP, $\tau_y P$ and gP are obtained by replacing every $f_{\beta,j}$ by $\overline{f_{\beta,j}}$, $f_{\beta,j} \circ \theta$, $f_{\beta,j}(\cdot - y)$ and $f_{\beta,j} \circ g^{-1}$, products of polynomials are formal products, and P is *at positive times* if every $f_{\beta,j}$ is supported in $\{t > 0\}$. We write $\Psi(f_0)$ for $\Psi(P)$ with $P = f_0$.

Theorem 193. *Assume the hypothesis of Theorem 43 and* (M)*, with Φ real and W_3-invariant. Every sequence of k has a subsequence along which the following hold.*

(i) *For all n and all $f_j \in C_c(\mathbb{R}^4)$ the limit*

$$S_n(f_1, \dots, f_n) = \lim_k \omega_k\Big(\prod_{j=1}^n Z_k A_k(f_j)\Big)$$

exists. Each S_n is symmetric and multilinear, bounded by $M_n(Q)\prod_j \|f_j\|_\infty$ on functions supported in a cube Q, and a distribution on $(\mathbb{R}^4)^n$; $S_1 = 0$.

(ii) *Each S_n is invariant under the translations of $\mathbb{R}^4$ and under Γ: $S_n(f_1 \circ g^{-1}, \dots, f_n \circ g^{-1}) = S_n(f_1, \dots, f_n)$ for $g \in \Gamma$. Moreover $S(\bar{P}) = \overline{S(P)}$.*

(iii) *$S(\theta\bar{P} \cdot P) \geq 0$ for every polynomial P at positive times. The completion of the quotient of these polynomials by the null space of this form is a Hilbert space $\mathscr{H}$ with vectors $\Psi(P)$, $\langle \Psi(P), \Psi(P')\rangle = S(\theta\bar{P} \cdot P')$, and $\Omega = \Psi(1)$.*

(iv) *There are a self-adjoint $H \geq 0$ with $\Psi(\tau_{(t,0)}P) = e^{-tH}\Psi(P)$ for $t \geq 0$, a strongly continuous unitary representation U of $\mathbb{R}^3$ with $\Psi(\tau_{(0,x)}P) = U(x)\Psi(P)$, and a unitary representation V of W_3 with $\Psi(gP) = V(g)\Psi(P)$. They commute with H, $V(g)U(x)V(g)^{-1} = U(gx)$, $H\Omega = 0$, and U and V fix Ω.*

(v) *$\Psi(f_0)$ is a unit vector orthogonal to Ω, there is a dyadic $\sigma > 0$ with $C_k(\sigma) \geq \frac{1}{2}C_k(0)$ for all large k, and (N_0) holds.*

(vi) *$\sigma(H) \subset \{0\} \cup [m_*, \infty)$ and $\ker H = \mathbb{C}\Omega$. For polynomials P, P' at positive times and $t \geq 0$,*

$$\big|S(\theta\bar{P} \cdot \tau_{(t,0)}P') - \overline{S(P)}\,S(P')\big| \leq e^{-m_* t}\,\|\Psi(P) - S(P)\Omega\|\,\|\Psi(P') - S(P')\Omega\|.$$

The joint spectral measure of H and the generators of U is carried by $\{(0,0)\} \cup ([m_*, \infty) \times \mathbb{R}^3)$.

(vii) *If $M_n(Q) \leq C_n(1+L)^{p_n}$ for every cube Q of side L centred at 0, each S_n extends to a tempered distribution on $\mathbb{R}^{4n}$.*

Conversely, if for every n and all $f_j \in C_c(\mathbb{R}^4)$ the sequence $\omega_k(\prod_j Z_k A_k(f_j))$ is bounded, then (M) *holds.*

Proof. (i) The proof of Theorem 55(ii) uses part (i) of that theorem only through the bound $M_n(Q)$, and applies verbatim. $S_1 = 0$ because the packets are centred.

(ii) Translation invariance is proved as in Theorem 55(ii). Let $g \in \Gamma$; it maps $\mathbb{Z}^4$ to itself and fixes the site 0. It acts on configurations by moving the links, a link whose orientation is reversed carrying the inverse group element. Inversion is an isometry of the bi-invariant metric, and g preserves the Wilson action of every time slice, so this action maps solutions of the flow equation to solutions and commutes with Φ_s; for $g = \theta$ this is Proposition 45(i). Write gF for the function $U \mapsto F(g^{-1}U)$. Since Φ is W_3-invariant and θ fixes the spatial links of the slice $\{x_0 = 0\}$, $g\Phi_x = \Phi_{gx}$, hence $g\varphi_k(x) = \varphi_k(gx)$, and substituting $y = g^{-1}x$,

$$A_k(f \circ g^{-1}) = \sum_y a_k^4 f(a_k y)\big(\varphi_k(gy) - c_k\big) = gA_k(f),$$

where $c_k = \omega_k(\varphi_k(x))$ does not depend on x. The state ω_k is invariant under the hypercubic group, so $\omega_k(\prod_j A_k(f_j \circ g^{-1})) = \omega_k(\prod_j A_k(f_j))$, and the limits agree. Since Φ is real, $A_k(\bar{f}) = \overline{A_k(f)}$, which gives the last claim.

(iii) This is the proof of Theorem 55(iii), which uses only the bound of (i).

(iv) Let P be at positive times and $t \geq 0$ dyadic. For large k, $t/a_k \in \mathbb{Z}$, and translation invariance of ω_k and of the flow gives $(\tau_{(t,0)}P)_k = \tau_{t/a_k}P_k$. In the reconstruction of Appendix F, $\Psi_{\tau_n F} = \mathbb{T}_k^n \Psi_F$ (proof of Theorem 46), so

$$\omega_k\big(\vartheta(\bar{P}_k)\tau_{2t/a_k}P_k\big) = \big\|\mathbb{T}_k^{t/a_k}\Psi_{P_k}\big\|^2 \leq \omega_k\big(\vartheta(\bar{P}_k)P_k\big).$$

By (i) and the approximation of θ by the lattice reflection in the proof of Theorem 55(iii), the left side tends to $S(\theta\bar{P} \cdot \tau_{(2t,0)}P)$, which equals $\|\Psi(\tau_{(t,0)}P)\|^2$ by translation invariance and $\theta\tau_{(t,0)} = \tau_{(-t,0)}\theta$, and the right side tends to $\|\Psi(P)\|^2$. The bound of (i) makes $S(\theta\bar{P}' \cdot P')$ continuous under uniform convergence of the test functions in a fixed cube, so $\|\Psi(\tau_{(t,0)}P)\| \leq \|\Psi(P)\|$ for all $t \geq 0$, and $\|\Psi(\tau_{(t,0)}P) - \Psi(P)\| \to 0$ as $t \to 0$, each monomial of $\tau_{(t,0)}P - P$ being a telescoping sum of monomials with one factor $f(\cdot - (t,0)) - f$. Hence $\tau_{(t,0)}$ maps null

vectors to null vectors and induces contractions $\mathsf{T}(t)$ of $\mathscr{H}$ with $\mathsf{T}(s)\mathsf{T}(t)=\mathsf{T}(s+t)$, symmetric by translation invariance and $\theta\tau_{(t,0)}=\tau_{(-t,0)}\theta$, and strongly continuous on a dense set and hence everywhere. A strongly continuous semigroup of symmetric contractions is e^{-tH} with $H\geq 0$ self-adjoint [88]. The spatial translations and the elements of W_3 preserve the positive times, commute with θ and preserve the form by (ii); they have inverses of the same kind, and so induce unitaries, strongly continuous in x by the argument just given. The relations follow from those of the maps of $\mathbb{R}^4$, and $\tau_y 1=g1=1$.

(v) As in the proof of Theorem 55(iv), $C_k(t)-C_k(0)=\omega_k(A_k(\bar f_0\circ\theta_k)A_k(f_{0,t}-f_0))$ for dyadic t and large k, with $f_{0,t}=f_0(\cdot-(t,0))$ and θ_k the lattice reflection. With Q a cube containing the supports of $\bar f_0\circ\theta_k$ and of $f_{0,t}$ for all k and $t\leq 1$, (M) for $n=2$ gives $Z_k^2|C_k(t)-C_k(0)|\leq M_2(Q)\|f_0\|_\infty\|f_{0,t}-f_0\|_\infty$, which is at most $\frac12$ for a small dyadic t by uniform continuity. So $C_k(\sigma)/C_k(0)\geq\frac12$ with $\sigma=t$, which is (N_0). Further $\|\Psi(f_0)\|^2=S_2(\bar f_0\circ\theta,f_0)=\lim_k Z_k^2C_k(0)=1$ by the same approximation of θ, and $\langle\Omega,\Psi(f_0)\rangle=S_1(f_0)=0$.

(vi) The proof of Theorem 55(vi) uses part (ii) of that theorem only through the convergence of the moments, which (i) provides. It gives $\langle v,e^{-tH}v\rangle\leq e^{-m_*t}\|v\|^2$ for $v=\Psi(P)-\langle\Omega,\Psi(P)\rangle\Omega$, P at positive times. These vectors are dense in $\Omega^\perp$, which e^{-tH} leaves invariant, so the restriction of the positive operator e^{-tH} to $\Omega^\perp$ has norm at most e^{-m_*t}; the spectral theorem gives the claims on $\sigma(H)$ and $\ker H$. Here $\langle\Omega,\Psi(P)\rangle=S(P)$ and $\langle\Psi(P),\Omega\rangle=S(\theta\bar P)=\overline{S(P)}$ by (ii), and $\langle\Psi(P),e^{-tH}\Psi(P')\rangle-\langle\Psi(P),\Omega\rangle\langle\Omega,\Psi(P')\rangle=\langle v,e^{-tH}v'\rangle$ with v,v' the projections on $\Omega^\perp$, which gives the inequality. Since U commutes with H and fixes Ω, both reduce $\Omega^\perp$; the joint spectral measure at a vector of $\Omega^\perp$ is carried by $[m_*,\infty)\times\mathbb{R}^3$, and at Ω it is the point mass at $(0,0)$.

(vii) Let $\chi(x)=\prod_\mu(1-|x_\mu|)_+$, so that $\sum_{m\in\mathbb{Z}^4}\chi(x-m)=1$, and put $\chi_m=\chi(\cdot-m)$, $r=p_n+5$ and

$$N_r(f)=\sup_x(1+|x|_\infty)^r|f(x)|.$$

On the support of χ_m, $1+|x|_\infty\geq\max(1,|m|_\infty)\geq(2+|m|_\infty)/3$, so $\|\chi_m f\|_\infty\leq 3^r(2+|m|_\infty)^{-r}N_r(f)$. For $f_j\in C_c(\mathbb{R}^4)$, multilinearity gives the finite sum

$$S_n(f_1,\dots,f_n)=\sum_{m_1,\dots,m_n}S_n(\chi_{m_1}f_1,\dots,\chi_{m_n}f_n),$$

and the term of (m_j) has its supports in the cube of side $L=2+2\max_j|m_j|_\infty$, so it is at most

$$C_n\,2^{p_n}\prod_{j=1}^n(2+|m_j|_\infty)^{p_n}\,\|\chi_{m_j}f_j\|_\infty.$$

Summing, $|S_n(f_1,\dots,f_n)|\leq C_n2^{p_n}(3^rs)^n\prod_jN_r(f_j)$ with $s=\sum_{m\in\mathbb{Z}^4}(2+|m|_\infty)^{-5}<\infty$. Functions in $C_c(\mathbb{R}^4)$ are dense in $\mathscr{S}(\mathbb{R}^4)$ for N_r, so S_n extends to a jointly continuous multilinear form on $\mathscr{S}(\mathbb{R}^4)^n$, and the kernel theorem [92, Sect. V.3] gives a tempered distribution on $\mathbb{R}^{4n}$.

For the converse, let C_Q be the Banach space of continuous functions supported in Q with the supremum norm. For fixed k the form $B_k(f_1,\dots,f_n)=\omega_k(\prod_jZ_kA_k(f_j))$ is continuous on C_Q^n, since $|A_k(f)|\leq 2\|\Phi\|_\infty a_k^4\#\{x:a_kx\in Q\}\|f\|_\infty$. A family of continuous n-linear forms on a Banach space that is bounded at every point is bounded in norm: for $n=1$ this is the uniform boundedness principle [92], and for $n>1$, by induction, $\sup_k\|B_k(f_1,\cdot)\|<\infty$ for each f_1, so the linear maps $f_1\mapsto B_k(f_1,\cdot)$ into the Banach space of $(n-1)$-linear forms are bounded at every point, and hence in norm. The bound is $M_n(Q)$. □

Remark 194. If (M) is replaced by bounds on the dyadic packets only, namely $\sup_k Z_k^{n+n'}|\omega_k(\vartheta(\prod_{i\leq n}\bar A_k^{y_i})\prod_{j\leq n'}A_k^{y'_j})|<\infty$ for all $y_i,y'_j\in\mathscr{D}$, the construction of Theorem 46(iii) applies to polynomials in the vectors δ_y: every monomial is $\mathsf{T}(s)$ of a monomial at positive times for a small dyadic $s>0$, so the kernels of the positive contractions $\mathsf{T}(\sigma)$ are trivial, and $H=-\log\mathsf{T}(1)$ is self-adjoint and nonnegative, with unitary dyadic spatial translations; the proof of part (vi) gives the gap m_*. The limit is nontrivial when (N_u) holds for some u, but not only then: when $\rho\equiv 0$, the vectors δ_y vanish while products of them need not. Without continuity in the test functions, neither the distributions S_n nor continuous translations follow.

Under the tree bounds, the moment bound holds with polynomial constants, and the limit has two properties that (M) does not give: temperedness with an L^1 bound, and clustering in every direction. For $f\in\mathscr{S}(\mathbb{R}^4)$ put $|f|_{(1)}=\max(\|f\|_1,\|f\|_\infty)$; for a tree T on $\{1,\dots,n\}$, $\zeta\in(\mathbb{R}^4)^{E(T)}$ and $j\geq 2$, let ζ_j^T be the sum of ζ_e over the edges of the path from 1 to j in T.

Lemma 195. *Let S be a multilinear form on $\mathscr{S}(\mathbb{R}^4)^n$ with $|\mathsf{S}(f_1,\dots,f_n)|\leq\sigma\prod_j|f_j|_{(1)}$. Then S is given by a unique tempered distribution on $\mathbb{R}^{4n}$, and for $F\in\mathscr{S}(\mathbb{R}^{4n})$, with $L_i=-\partial_i^2+x_i^2+4$ in the $4n$ coordinates,*

$$|\mathsf{S}(F)|\leq\sigma\,\|L_1^3\cdots L_{4n}^3F\|_{L^2(\mathbb{R}^{4n})}\leq\sigma\,236^{4n}(\pi/2)^{2n}|F|_{32n},$$

where $|F|_m=\sup_{x,|\gamma|\leq m}(1+|x|^2)^{m/2}|\partial^\gamma F(x)|$ are the norms of the linear growth condition of [89].

Proof. The form is jointly continuous, so the kernel theorem [92, Sect. V.3] gives the distribution. Let h_k, $k\geq 0$, be the Hermite functions on $\mathbb{R}$, an orthonormal basis of $L^2(\mathbb{R})$ of real functions with $(-\partial^2+x^2)h_k=(2k+1)h_k$ and $xh_k=\sqrt{k/2}\,h_{k-1}+\sqrt{(k+1)/2}\,h_{k+1}$, whose tensor products h_κ, $\kappa\in\mathbb{N}^{4n}$, give an expansion of every $F\in\mathscr{S}(\mathbb{R}^{4n})$ that converges in $\mathscr{S}$ [92, Sect. V.3]. For real $g\in\mathscr{S}(\mathbb{R})$, $g(x)^2=\int_{-\infty}^x2gg'=-\int_x^\infty2gg'$, so $\|g\|_\infty^2\leq\|g\|_2\|g'\|_2$; with $\|h_k'\|_2^2\leq 2k+1$ this gives $\|h_k\|_\infty\leq(2k+1)^{1/4}$. Applying the recursion twice, $x^2h_k=\frac12\big(\sqrt{k(k-1)}\,h_{k-2}+(2k+1)h_k+\sqrt{(k+1)(k+2)}\,h_{k+2}\big)$, so $\|(1+x^2)h_k\|_\infty\leq(2k+\frac52)(2k+5)^{1/4}\leq(2k+5)^{5/4}$. Since $\int_{\mathbb{R}}(1+x^2)^{-1}dx=\pi$, a function f on $\mathbb{R}^4$ has $|f|_{(1)}\leq\pi^4\|\prod_{i=1}^4(1+x_i^2)f\|_\infty$, and for $\alpha\in\mathbb{N}^4$ therefore $|h_\alpha|_{(1)}\leq\pi^4\prod_i(2\alpha_i+5)^{5/4}$. The operator $L=\prod_iL_i^3$ is symmetric with $Lh_\kappa=\prod_i(2\kappa_i+5)^3h_\kappa$, so the coefficient $c_\kappa=\langle h_\kappa,F\rangle$ has $|c_\kappa|\leq\prod_i(2\kappa_i+5)^{-3}\|LF\|_2$. Hence

$$|\mathsf{S}(F)|\leq\sum_\kappa|c_\kappa||\mathsf{S}(h_\kappa)|\leq\sigma\|LF\|_2\Big(\pi\sum_{k\geq0}(2k+5)^{-7/4}\Big)^{4n},$$

and $\sum_k (2k+5)^{-7/4} \le 5^{-7/4} + \frac{2}{3}5^{-3/4} < 0.2592$, so the bracket is below 0.815. In one variable $L_i^3 = \sum c_{ab} x^a \partial^b$ with $a,b \le 6$ and integer coefficients, and expanding and normal ordering gives $\sum |c_{ab}| = 236$. Expanding the product, $\|LF\|_2 \le 236^{4n} \max \|x^a \partial^b F\|_2$ over multi-indices with entries at most 6, and since $\int_{\mathbb{R}^{4n}} \prod_i (1+x_i^2)^{-2} dx = (\pi/2)^{4n}$, $\|g\|_2 \le (\pi/2)^{2n} \|\prod_i (1+x_i^2) g\|_\infty$. Finally $\prod_i (1+x_i^2)|x_i|^{a_i} \le \prod_i (1+x_i^2)^4 \le (1+|x|^2)^{16n}$ and $|b| \le 24n$, so $\|\prod_i (1+x_i^2) x^a \partial^b F\|_\infty \le |F|_{32n}$. □

Theorem 196. *Let the ω_k be translation invariant and assume* (T1)–(T3). (i) (M) *holds, with $M_n(Q) \le C_n (1+L)^{2n}$ for cubes of side L centred at* 0.

(ii) *Along every subsequence on which the limits S_n of Theorem* 193(i) *exist there are a further subsequence, a number $b \in [0, B_*]$ and a positive measure ν on $\mathbb{R}^4$ with $\nu(\mathbb{R}^4) \le 1$ such that, for $n \ge 2$ and $f_j \in C_c(\mathbb{R}^4)$, the truncated functions S_n^T of the limit satisfy*

$$|S_n^T(f_1,\dots,f_n)| \le K_n c^{-n/2} b^{n/2-1} \sum_T \int |f_1(x)| \times \int \prod_{j=2}^n |f_j(x+\zeta_j^T)|\, d\nu^{\otimes E(T)}(\zeta)\, dx,$$

with $b^0 = 1$.

(iii) *For $n \ge 2$ and every i,*

$$|S_n^T(f_1,\dots,f_n)| \le K_n' \|f_i\|_1 \prod_{j\ne i} \|f_j\|_\infty$$

with $K_n' = K_n n^{n-2} c^{-n/2} B_*^{n/2-1}$. Hence S_n extends to $\mathscr{S}(\mathbb{R}^4)^n$ and

$$|S_n(f_1,\dots,f_n)| \le \sigma_n \prod_j |f_j|_{(1)},$$

where σ_n is the sum over the partitions of $\{1,\dots,n\}$ into blocks of size at least two of $\prod_B K'_{|B|}$, and Lemma 195 *applies. If $K_n \le K^n (n!)^\beta$ for all n, then $\sigma_n \le C^n (n!)^{\beta+2}$ with $C = Ke\, c^{-1/2} \max(1, B_*)^{1/2}$, and the Schwinger functions satisfy the linear growth condition of [89].*

(iv) *Let $\{1,\dots,n\} = I \cup J$ with I, J disjoint and nonempty, $y \in \mathbb{R}^4 \setminus \{0\}$, and $f_j^\lambda = f_j$ for $j \in I$, $f_j^\lambda = f_j(\cdot - \lambda y)$ for $j \in J$. Then*

$$\lim_{\lambda\to\infty} S_n(f_1^\lambda,\dots,f_n^\lambda) = S_{|I|}\big((f_j)_{j\in I}\big)\, S_{|J|}\big((f_j)_{j\in J}\big).$$

Under the hypotheses of Theorem 193, *$U(\lambda x)$ converges weakly to the projection on $\mathbb{C}\Omega$ as $\lambda \to \infty$ for every $x \in \mathbb{R}^3 \setminus \{0\}$, and Ω is the only vector invariant under U.*

Proof. (i) This is Theorem 55(i): a moment is a sum over partitions into blocks of size at least two, each contributing a factor $|Q|_k \le (L+a_0)^4$, and there are at most $n/2$ blocks.

(ii) By (T1) and (T2), as in the proof of Lemma 54, the truncated function of the renormalised packets at level k is at most $Z_k^n K_n r_k^n \sum_T a_k^{4n} \sum_x \prod_j |f_j(a_k x_j)| \prod_{\{i,j\}\in T} w_k(x_i - x_j)$. For fixed T change variables to x_1 and $z_e = x_{\text{child}} - x_{\text{parent}}$, the tree being rooted at 1; since w_k is even, the sum is $(a_k^4 W_k)^{n-1} \int h_k\, d\nu_k^{\otimes E(T)}$, where $\nu_k = W_k^{-1} \sum_z w_k(z) \delta_{a_k z}$ is a probability measure ($W_k > 0$, since $C_k(0) > 0$) and

$$h_k(\zeta) = a_k^4 \sum_{x\in\mathbb{Z}^4} |f_1(a_k x)| \prod_{j\ge 2} |f_j(a_k x + \zeta_j^T)|.$$

With $Z_k^n r_k^n \le (c\, a_k^4 W_k)^{-n/2}$ the bound becomes

$$K_n c^{-n/2} (a_k^4 W_k)^{n/2-1} \sum_T \int h_k\, d\nu_k^{\otimes E(T)}.$$

Choose a further subsequence along which $a_k^4 W_k \to b$ and $\nu_k \to \nu$ vaguely; then $\nu_k^{\otimes m} \to \nu^{\otimes m}$ vaguely, the masses being at most one. Let $h(\zeta) = \int |f_1(x)| \prod_{j\ge 2} |f_j(x+\zeta_j^T)|\, dx$. It is continuous, and it has compact support, because x and every $x + \zeta_j^T$ lie in compact sets when the integrand is nonzero, and ζ_e is the difference of two of these points. For ζ in $(a_k \mathbb{Z}^4)^{E(T)}$, $h_k(\zeta)$ is a Riemann sum for $h(\zeta)$ of a family of functions with supports in one compact set and a common modulus of continuity, so $\sup_\zeta |h_k(\zeta) - h(\zeta)| \to 0$. Hence $\int h_k d\nu_k^{\otimes E(T)} \to \int h\, d\nu^{\otimes E(T)}$. The truncated functions at level k are polynomials in finitely many moments, which converge, so the left side converges to $|S_n^T|$.

(iii) Relabel so that $i = 1$; the integral in (ii) is at most $\|f_1\|_1 \prod_{j\ge 2} \|f_j\|_\infty$, there are n^{n-2} trees, and $b^{n/2-1} \le B_*^{n/2-1}$. The moments are sums over partitions of products of truncated functions, which gives σ_n on $C_c(\mathbb{R}^4)$; the bound extends to $\mathscr{S}(\mathbb{R}^4)$, the closure of $C_c(\mathbb{R}^4)$ for $|\cdot|_{(1)}$ containing it. If $K_m \le K^m (m!)^\beta$, then, by $m^{m-2} \le e^m m!$, $K_m' \le C^m (m!)^{\beta+1}$; since $\prod_B |B|! \le n!$ and there are at most $n!$ partitions, $\sigma_n \le C^n (n!)^{\beta+2}$. With Lemma 195, $|S_n(F)| \le C^n (n!)^{\beta+2} 236^{4n} (\pi/2)^{2n} |F|_{32n}$, which is the condition of [89] with $s = 32$.

(iv) Assume $1 \in I$, by symmetry of S_n. In a partition of $\{1,\dots,n\}$ the blocks contained in I or in J do not change their truncated functions as λ varies, by translation invariance. For a block B meeting both, apply (ii) to $(f_j^\lambda)_{j\in B}$ with a root in $B \cap I$: the integrand is bounded by $|f_{\text{root}}(x)| \prod \|f_j\|_\infty$, integrable for $dx \otimes \nu^{\otimes E(T)}$, and tends to 0 pointwise as $\lambda \to \infty$, since for $j \in B \cap J$ the point $x + \zeta_j^T$ is fixed while the support of f_j^λ leaves every compact set. By dominated convergence the truncated function of B tends to 0, and the moment tends to the sum over the partitions into blocks in I and blocks in J, which is the product. For the last statement, let P, P' be at positive times; $\langle \Psi(P), U(\lambda x) \Psi(P') \rangle = S(\theta \bar{P} \cdot \tau_{(0,\lambda x)} P')$ tends to $S(\theta\bar{P}) S(P') = \langle \Psi(P), \Omega \rangle \langle \Omega, \Psi(P') \rangle$ by what was just proved, applied to each monomial. The operators are uniformly bounded and the vectors $\Psi(P)$ are dense, which gives weak convergence; a vector v with $U(x)v = v$ for all x then has $\|v\|^2 = |\langle \Omega, v \rangle|^2$, so $v \in \mathbb{C}\Omega$. □

Part (iv) is the cluster property of the axioms in every direction, for all n. The gap of Theorem 193(vi) gives it with the rate m_* in the time direction only. Spatial clustering does not follow from that gap and (M) without the tree bounds: the field of Proposition 199(b) below has the gap, satisfies (M), and does

not cluster in space. With the hypercubic group in place of Γ the exchange of time with a spatial axis would carry the time clustering to space; the spatial flow removes that exchange.

Rotations. For a family $S = (S_n)$ as in Theorem 193(i)–(ii) let

$$G_S = \big\{R \in \mathrm{O}(4) : S_n(f_1 \circ R^{-1}, \dots, f_n \circ R^{-1}) = S_n(f_1, \dots, f_n)\big\},$$

the condition holding for all n and $f_j \in C_c(\mathbb{R}^4)$. Let e_0 be the unit time vector, $\mathrm{SO}(3)_{\mathrm{sp}}$ the rotations that fix it and $\mathrm{O}(3) \times \mathrm{O}(1)$ the orthogonal maps that preserve the line $\mathbb{R}e_0$.

Proposition 197. (i) *G_S is a closed subgroup of* $\mathrm{O}(4)$ *that contains* Γ*, and its identity component is* $\{1\}$*,* $\mathrm{SO}(3)_{\mathrm{sp}}$ *or* $\mathrm{SO}(4)$.

(ii) *$G_S \supseteq \mathrm{O}(3) \times \mathrm{O}(1)$ if and only if G_S contains a rotation of the spatial coordinates of infinite order.*

(iii) *$G_S = \mathrm{O}(4)$ if and only if G_S contains a rotation R of infinite order with $Re_0 \notin \{\pm e_0\}$, and also if and only if G_S is infinite and contains some g with $ge_0 \notin \{\pm e_0\}$.*

(iv) *If $G_S \supseteq B_4$, then G_S is finite or equal to* $\mathrm{O}(4)$*, and it equals* $\mathrm{O}(4)$ *if and only if it contains an element of infinite order. One element of finite order outside B_4 does not suffice.*

Proof. (i) G_S is a group. If $R_m \to R$ with $R_m \in G_S$, the functions $f \circ R_m^{-1}$ converge uniformly to $f \circ R^{-1}$ with supports in one cube, and the bound of Theorem 193(i) gives $R \in G_S$. So G_S is a compact Lie group [28], and its Lie algebra $\mathfrak{g} \subset \mathfrak{so}(4)$ is invariant under $\mathrm{Ad}(g)$, $g \in \Gamma$. Let $E_{ij} = e_i e_j^{\mathsf{T}} - e_j e_i^{\mathsf{T}}$. The diagonal sign changes D_ε, $\varepsilon \in \{\pm 1\}^4$, lie in Γ, and $\mathrm{Ad}(D_\varepsilon)E_{ij} = \varepsilon_i\varepsilon_j E_{ij}$. The six characters $\varepsilon \mapsto \varepsilon_i\varepsilon_j$ are distinct, so the operators $2^{-4}\sum_\varepsilon \varepsilon_i\varepsilon_j \mathrm{Ad}(D_\varepsilon)$ project $\mathfrak{g}$ onto $\mathfrak{g} \cap \mathbb{R}E_{ij}$, and $\mathfrak{g}$ is spanned by the E_{ij} it contains. The permutations of the spatial axes act transitively on the pairs $\{i,j\} \subset \{1,2,3\}$ and on the pairs $\{0,i\}$. So $\mathfrak{g}$ is 0, $\mathfrak{so}(3)_{\mathrm{sp}} = \mathrm{span}\{E_{ij} : i,j \geq 1\}$, $\mathrm{span}\{E_{0i}\}$ or $\mathfrak{so}(4)$, and the third is not a Lie algebra, since $[E_{01}, E_{02}] = -E_{12}$. A connected Lie subgroup is determined by its Lie algebra.

(ii) If $R \in \mathrm{SO}(3)_{\mathrm{sp}}$ has infinite order, G_S is infinite, hence of positive dimension, and $G_S \cap (\mathrm{O}(3) \times \mathrm{O}(1))$ has the same property, so its identity component is a connected subgroup of $\mathrm{SO}(3)_{\mathrm{sp}}$ of positive dimension normalised by the signed permutations, and by the argument of (i), applied within $\mathfrak{so}(3)_{\mathrm{sp}}$, it is $\mathrm{SO}(3)_{\mathrm{sp}}$. With Γ this gives $\mathrm{O}(3) \times \mathrm{O}(1)$. The converse is clear.

(iii) If G_S is infinite, its identity component is $\mathrm{SO}(3)_{\mathrm{sp}}$ or $\mathrm{SO}(4)$ by (i). In the first case it is normal in G_S, and every $g \in G_S$ maps the vectors fixed by $\mathrm{SO}(3)_{\mathrm{sp}}$, the line $\mathbb{R}e_0$, to the vectors fixed by $g\mathrm{SO}(3)_{\mathrm{sp}}g^{-1} = \mathrm{SO}(3)_{\mathrm{sp}}$, so $ge_0 = \pm e_0$. Hence an infinite G_S with some $ge_0 \neq \pm e_0$ contains $\mathrm{SO}(4)$, and with θ it is $\mathrm{O}(4)$. A rotation R of infinite order with $Re_0 \neq \pm e_0$ makes G_S infinite and supplies g; the converses are clear.

(iv) The sign changes and all permutations of the four axes act on the pairs transitively, so $\mathfrak{g}$ is 0 or $\mathfrak{so}(4)$. The reflection s in the hyperplane orthogonal to $u = \frac{1}{2}(1,1,1,1)$, with B_4, generates a group that permutes the 24 vectors $\pm e_\mu$ and $\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1)$: s maps e_μ to $e_\mu - u$, and v to $v - 2\langle v,u\rangle u$ with $\langle v,u\rangle \in \{0, \pm\frac{1}{2}, \pm 1\}$ for the other vectors. These vectors span $\mathbb{R}^4$, so the group is finite. □

The spatial flow commutes with spatial rotations in the continuum, and in the free field it leaves exactly $\mathrm{O}(3) \times \mathrm{O}(1)$ (Proposition 199(c)). So for the spatially flowed fields at a positive flow time the symmetry to ask for is $\mathrm{O}(3) \times \mathrm{O}(1)$, and by part (ii) one rotation suffices:

(W) *One spatial rotation.* There is a rotation R of the spatial coordinates, of infinite order, with

$$\lim_{k\to\infty}\Big[\omega_k\Big(\prod_j Z_k A_k(f_j \circ R^{-1})\Big) - \omega_k\Big(\prod_j Z_k A_k(f_j)\Big)\Big] = 0$$

for all n and $f_j \in C_c(\mathbb{R}^4)$.

Under the hypotheses of Theorem 193, (W) holds if and only if every subsequential limit is invariant under $\mathrm{O}(3) \times \mathrm{O}(1)$: (W) puts R in each G_S; conversely, if the difference in (W) stayed above ε along a subsequence for some f_j, a further subsequence would give a limit S with $R \notin G_S$. Full Euclidean invariance, as (E) asks, is not available for these fields at a positive flow time; it has to be posed for observables whose regularisation is $\mathrm{O}(4)$-covariant, such as the four-dimensional flow, which at positive flow time mixes the two sides of the reflection plane (Sec. IX D), or the limit of vanishing flow time. For families invariant under B_4 part (iv) applies, and one rotation of infinite order gives $\mathrm{O}(4)$. In either case the rotation enters as a lattice statement: a Ward identity in which the breaking of the rotation is an insertion that vanishes in the limit, as the symmetry restoration of Symanzik's improvement programme predicts. It is not proved here.

Convergence of the whole sequence.

(U) *One limit.* Any two subsequential limits of Theorem 193(i) coincide.

Proposition 198. *Assume the hypotheses of Theorem 193.*

(i) *The moments $\omega_k(\prod_j Z_k A_k(f_j))$ converge for all n and $f_j \in C_c(\mathbb{R}^4)$ if and only if* (U) *holds.*

(ii) *Let (β_k) and (β'_k) satisfy the hypothesis of Theorem 43 with the same α, ℓ, and* (M) *and* (W) *with the same $M_n(Q)$ and R, and let β''_k be β_k for even k and β'_k for odd k. Then (β''_k) satisfies the same hypotheses, and if the moments along (β_k) and along (β'_k) converge to different limits, those along (β''_k) do not converge.*

(iii) *Suppose that every subsequential limit lies in a set $\mathfrak{F}$ of families on which $S \mapsto S_2(\bar{f}_0 \circ \theta, f_0(\cdot - (\sigma_0, 0)))$ is injective, for a dyadic $\sigma_0 > 0$, and that $\rho_k(0,\sigma_0) = C_k(\sigma_0)/C_k(0)$ converges. Then* (U) *holds.*

Proof. (i) If a moment does not converge, it has two limit points along two subsequences, and refining each by Theorem 193(i) gives two different limits. The converse is clear. (ii) The hypotheses are conditions at each k with constants uniform in k, and the even and the odd terms have different limits. (iii) Along a subsequence with limit S, $S_2(\bar{f}_0 \circ \theta, f_0(\cdot - (\sigma_0, 0))) = \lim Z_k^2 C_k(\sigma_0) = \lim \rho_k(0,\sigma_0)$ by the approximation of θ in the proof of Theorem 55(iii), the same number for every subsequence. □

Part (ii) shows that (U) is a property of the tuning that the other hypotheses do not contain. Part (iii) states it in the form

that Yang–Mills theory requires: if every limit is determined up to the choice of one physical scale, a single renormalisation condition fixes that scale. For Yang–Mills theory this is the statement that the continuum theory has no free dimensionless parameter; it is not proved here.

What these inputs cannot give. Gaussian fields show the limits of the inputs. They are not lattice gauge theories, and the hypothesis of Theorem 43 concerns the Wilson measure in the forest gauge; but the proofs of Theorems 193 and 196 use it only through the conclusions of Theorem 41, namely translation and lattice-symmetry invariance, reflection positivity and the transfer-matrix gap $m_* a_k$, and the fields below have these.

Proposition 199. *Let $m > 0$, $q_k = e^{-m a_k}$, and $f_0 \geq 0$.*

(a) *Let ϕ_k be the centred Gaussian field on $\mathbb{Z}^4$ with covariance $\prod_{\mu=0}^{3} q_k^{|x_\mu - y_\mu|}$, and let $A_k(f) = \sum_x a_k^4 f(a_k x)\phi_k(x)$, with C_k, Z_k formed from f_0 as before. Its law is invariant under translations and B_4 and reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links; its transfer matrix in each coordinate direction has spectrum $\{q_k^N : N \geq 0\}$ with 1 a simple eigenvalue, so its gap is $m a_k$. For large k,* (T1)–(T3) *hold with $r_k = 1$, $w_k(z) = \prod_\mu q_k^{|z_\mu|}$, $K_2 = 1$ and $K_n = 0$ for $n \geq 3$. The moments converge along the whole sequence to the Gaussian family with $S_2(f,g) = \kappa \iint f(x) g(y) e^{-m\|x-y\|_1} dx\, dy$, $\kappa > 0$. Its group G_S is exactly B_4, its Hamiltonian has $\sigma(H) \subset \{0\} \cup [m, \infty)$ with $m \in \sigma(H)$, and the joint spectrum of H and the spatial momentum contains $\{m\} \times \mathbb{R}^3$.*

(b) *Let ψ_k be the centred Gaussian field on $\mathbb{Z}^4$ with covariance $q_k^{|x_0 - y_0|}$, and form A_k, C_k, Z_k in the same way. Its law is invariant under translations and Γ, reflection positive in time, and its transfer matrix has gap $m a_k$.* (M)*, the bound of Theorem* 193(vii) *and* (N_0) *hold, and the moments converge along the whole sequence; the limit has $U(x) = 1$ for every $x \in \mathbb{R}^3$, so it does not cluster in space.* (T3) *fails.*

(c) *For $t_f > 0$ and $\mu > 0$, the centred Gaussian field on $\mathbb{R}^4$ with covariance $\widehat{C}(p) = e^{-2t_f|\mathbf{p}|^2}(|p|^2 + \mu^2)^{-1}$, the massive free field flowed by the spatial heat flow for the time t_f, is reflection positive and invariant under translations, and its group G_S is exactly $\mathrm{O}(3) \times \mathrm{O}(1)$.*

Proof. (a) The covariance is a product of the one-dimensional covariances $q^{|n|}$, which are positive definite, their Fourier transforms being $(1-q^2)/(1-2q\cos p + q^2) > 0$; invariance under translations and B_4 is clear. Fix a coordinate direction as time. The field at time 0 has a Gaussian law γ with covariance $\prod_{i \geq 1} q^{|x_i - y_i|}$, and the stationary chain $\phi_{t+1} = q\phi_t + (1-q^2)^{1/2}\xi_{t+1}$ with ξ_t independent of law γ has covariance $q^{|t-s|}$ times that of γ, so it is the field. It is a reversible Markov chain with transition operator $Pg(\phi) = \mathbb{E}\, g(q\phi + (1-q^2)^{1/2}\xi)$ on $L^2(\gamma)$. For h of finite support and $\mathscr{E}_h(\phi) = \exp(\phi(h) - \frac{1}{2}\mathbb{E}\phi(h)^2)$ one computes $P\mathscr{E}_h = \mathscr{E}_{qh}$; the part of $\mathscr{E}_h$ homogeneous of degree N in h spans the N-th Wiener chaos, so P is multiplication by q^N on it, and these chaoses span $L^2(\gamma)$ [55, Sect. 6.3]. So $0 \leq P \leq 1$ with spectrum $\{q^N\}$ and 1 simple. For F, F' functions of the field at times ≥ 0, the Markov property and reversibility give $\mathbb{E}[\vartheta(\bar{F})F'] = \langle g_F, g_{F'}\rangle_{L^2(\gamma)}$ with $g_F = \mathbb{E}[F \mid \phi_0 = \cdot]$, and for the reflection bisecting the links between times 0 and 1, with F, F' at times ≥ 1, $\langle g_F, P g_{F'}\rangle$ with $g_F = \mathbb{E}[F \mid \phi_1 = \cdot]$; both are positive for $F = F'$, and the transfer matrix is P. Gaussian fields have vanishing cumulants of order $n \geq 3$, and the covariance is $w_k(x-y)$, which is (T1). Further $a_k^4 W_k = a_k^4((1+q_k)/(1-q_k))^4 \to (2/m)^4$, which is (T3), and since $q_k^{|n|} = e^{-m a_k |n|}$, $C_k(0)$ is a Riemann sum converging to $\iint f_0(x) f_0(y) e^{-m\|\theta x - y\|_1} dx\, dy > 0$, which gives (T2) for large k and $Z_k \to \kappa^{1/2}$ with κ the inverse of this integral. The moments are sums of products of covariances, Riemann sums converging to the stated ones. Invariance of a Gaussian family under R is invariance of S_2, that is $e^{-m\|Rz\|_1} = e^{-m\|z\|_1}$ for all z; a linear isometry of the norm $\|\cdot\|_1$ maps the extreme points $\pm e_\mu$ of its unit ball to themselves, so $R \in B_4$. The spectral bound follows from the lattice gap $m a_k$ as in Theorem 193(vi). For $f = h \otimes g$ with h supported in $\{t > 0\}$ and $\int h e^{-mt} dt \neq 0$, $\langle \Psi(f), e^{-sH} U(x) \Psi(f)\rangle$ equals e^{-ms} times a function of x whose Fourier transform is a positive multiple of $|\hat{g}(\mathbf{p})|^2 \prod_{i=1}^3 2m/(p_i^2 + m^2)$. So the joint spectral measure at $\Psi(f)$ is carried by $\{m\} \times \mathbb{R}^3$, with support $\{m\} \times \mathbb{R}^3$ when $\hat{g}$ has no zeros.

(b) The field is $\psi_k(x) = X(x_0)$ for the chain of (a) with $\gamma = N(0,1)$, which gives the symmetry, reflection positivity and the gap. Write $F_f(t) = \int f(t, \mathbf{x})\, d\mathbf{x}$. The covariance of two packets converges to $\iint F_f(t) F_g(s) e^{-m|t-s|} dt\, ds$, which is positive for $f = \bar{f}_0 \circ \theta$, $g = f_0$, and bounded by $|Q|^2 \|f\|_\infty \|g\|_\infty$ for f, g supported in Q; with the Gaussian moment formula this gives (M) with $M_n(Q)$ a constant times $|Q|^n$, and convergence. The limiting covariance does not change when the test functions of one factor are translated in space, so neither does any moment, and $\langle \Psi(P), U(x)\Psi(P')\rangle = \langle \Psi(P), \Psi(P')\rangle$. The ratio $C_k(\sigma)/C_k(0)$ converges to a positive number, which is (N_0), and $\sum_z q_k^{|z_0|} = \infty$.

(c) For f supported in $\{t > 0\}$, the p_0 integral gives

$$\iint \bar{f}(\theta x) C(x-y) f(y)\, dx\, dy = \int e^{-2t_f|\mathbf{p}|^2} (2\omega)^{-1} \left| \int \hat{f}(t, \mathbf{p}) e^{-\omega t} dt \right|^2 d\mathbf{p} \geq 0,$$

with $\omega = (|\mathbf{p}|^2 + \mu^2)^{1/2}$ and $\hat{f}$ the spatial Fourier transform; reflection positivity of the Gaussian measure follows [55, Sect. 6.2]. An orthogonal R preserves $\widehat{C}$ if and only if $|(Rp)^{\mathrm{sp}}| = |p^{\mathrm{sp}}|$ for every p, the spatial parts, because $|Rp| = |p|$ and $s \mapsto e^{-2t_f s}$ is injective; that is, R preserves $|p_0|$, so $R^{\mathsf{T}} e_0 = \pm e_0$. □

Example (a) is reflection positive, has the gap in every direction and the lattice symmetry, satisfies the tree bounds with the best constants, and converges; its limit is invariant under no rotation outside B_4, so (W) does not follow from these properties. Its one-particle energy m does not depend on the momentum, so the spectral condition of the Wightman axioms, energy at least $|\mathbf{p}|$, fails; the gap transfers to $H \geq m_*$ on $\Omega^\perp$ and to the joint spectrum of Theorem 193(vi), and the spectral condition needs Lorentz covariance, which only $\mathrm{O}(4)$ invariance supplies. Example (b) shows that the time gap and (M) do not give spatial clustering. By Proposition 198(ii), example (a) at two

masses, interleaved, satisfies the hypotheses of Theorems 193 and 196 with uniform constants and has two limits.

Corollary 200. *Assume the hypothesis of Theorem 43, with Φ real and W_3-invariant. The moments $\omega_k(\prod_j Z_k A_k(f_j))$ converge for every n and all $f_j \in C_c(\mathbb{R}^4)$ to a family invariant under translations and $\mathrm{O}(3)\times\mathrm{O}(1)$ if and only if* (M), (W) *and* (U) *hold. The limit then satisfies* (i)–(vi) *of Theorem 193 with Γ replaced by $\mathrm{O}(3)\times\mathrm{O}(1)$: symmetric, reflection positive distributions, a Hamiltonian with $\sigma(H)\subset\{0\}\cup[m_*,\infty)$, $\ker H=\mathbb{C}\Omega$ and $\Omega^\perp\neq 0$, and unitary representations of the spatial translations and rotations commuting with H. If* (T1)–(T3) *hold as well, the limit is tempered and clusters in every direction, and it satisfies the linear growth condition of [89] when $K_n\le K^n(n!)^\beta$.*

Proof. Convergence gives bounded moments, hence (M) by Theorem 193; it gives (U), and with the invariance it gives (W) for every spatial rotation. Conversely, (M) and (U) give convergence by Theorem 193(i) and Proposition 198(i), and (W) gives the invariance by Proposition 197(ii). The representation of the spatial rotations is constructed as V in Theorem 193(iv), and its continuity follows as there. The rest is Theorems 193 and 196. □

So, given (D) along the tuning, the part of (E) that concerns the spatially flowed fields consists of three statements: (M), an ultraviolet bound on the products of renormalised packets that contains (N_0) and therefore fails along every tuning that stays in the strong-coupling range (Theorem 183); (W), the restoration of the spatial rotations; and (U), the uniqueness of the limit along the tuning. Each is necessary. Example (a) shows that (W) does not follow from (M), (U), the tree bounds and the conclusions of Theorem 41, and the same example at two interleaved masses shows that (U) does not follow from (M), the tree bounds and those conclusions. None of the three is proved. Euclidean invariance and the Wightman spectral condition are beyond these fields at a positive flow time.

Three further routes. The trial condition of Corollary 186 can be approached in three other ways: through upper bounds on correlations, through trial states smeared in time, and through a comparison with the Gaussian theory at weak coupling. The first never implies (N_u). The second removes the lattice scale from the first condition and leaves a lower bound on a ratio of correlations at physical times. The third gives (N_u) from a relative comparison at two physical separations; when that comparison is built scale by scale along the one-loop tuning, it must be exact to first order in the running coupling at every scale.

Proposition 201. *Let $G=\mathrm{SU}(2)$, let $\beta_{W,k}\in[0,0.348]$ be any couplings, let ω_k be the states of Theorems 1 and 3, and let m_0 be as in Theorem 183.*

(i) *Let F,H be bounded functions at positive times that are uniform limits of gauge-invariant local functions, with $\omega_k(F)=\omega_k(H)=0$. Then for every integer $n\ge 0$*

$$\big|\omega_k\big(\vartheta(\bar F)\tau_n H\big)\big|\le e^{-m_0 n}\|\Psi_F\|\,\|\Psi_H\|\le e^{-m_0 n}\|F\|_\infty\|H\|_\infty.$$

(ii) *Let* (H) *be a set of inequalities indexed by $i\in I$, each of the form*

$$\big|\omega_k\big(\vartheta(\bar F_{i,k})\tau_{n_{i,k}}H_{i,k}\big)\big|\le B_{i,k}\|\Psi_{F_{i,k}}\|\,\|\Psi_{H_{i,k}}\|\qquad(k\ge k_i),$$

with $F_{i,k},H_{i,k}$ as in (i), *integers $n_{i,k}\ge t_i/a_k$ for some $t_i>0$, and $B_{i,k}>0$ with $a_k\log B_{i,k}\to 0$ as $k\to\infty$. Then each inequality of* (H) *holds for the states ω_k for all large k, while (N_u) fails for every packet with $C_k(0)>0$ and every u, and every limit of Theorem 46 along these states is trivial. The same holds with $\|F_{i,k}\|_\infty\|H_{i,k}\|_\infty$ in place of $\|\Psi_{F_{i,k}}\|\|\Psi_{H_{i,k}}\|$.*

Proof. (i) Since $\langle\Omega,\Psi_F\rangle=\omega_k(F)=0$ and likewise for H, both vectors lie in $\Omega_k^\perp$, on which $\mathbb{T}_k$ has norm at most e^{-m_0}, as in the proof of Theorem 183. Hence $|\langle\Psi_F,\mathbb{T}_k^n\Psi_H\rangle|\le e^{-m_0 n}\|\Psi_F\|\|\Psi_H\|$, and $\|\Psi_F\|\le\|F\|_\infty$ by the footnote of Appendix F.

(ii) Since $a_k\log B_{i,k}\to 0$, for large k we have $a_k\log B_{i,k}\ge -m_0 t_i$, that is $B_{i,k}\ge e^{-m_0 t_i/a_k}\ge e^{-m_0 n_{i,k}}$, and (i) gives the inequality. The statements on (N_u) and on the limits are Theorem 183. □

The class (H) contains uniform bounds $B_{i,k}=B_i$, decay at a physical rate $B_{i,k}=B_i e^{-Mt_i}$ with any M, and bounds of order a_k^p, for connected correlations of any centred observables at physical separations, flowed or not. The same holds for every compact simple G at couplings in $[0,\beta_1]$ with $N\beta_1<\frac{2}{29}\kappa_G$, with $m_0=m_G(\beta_1)$ of Theorem 5. So a proof of (N_u) must contain a lower bound. In Corollary 186 it is $\langle\chi_k,\mathbb{T}_k\chi_k\rangle\ge 1-\kappa a_k$, a lower bound on a normalised correlation at one lattice step that is accurate to order a_k, together with a lower bound on one overlap. The spectral arguments of Lemma 185 and [21, Lemma 463] turn such a lower bound into a bound on the ratio, but they cannot produce one from upper bounds: the states above satisfy every inequality of (H), and the spectral measures of their centred vectors are carried by energies at least m_0/a_k.

Trial states smeared in time. A slice function enters Lemma 189 through its change in one lattice step, which Proposition 191 shows to be too large. Smearing the same function in time over a physical interval replaces that change by a telescoping sum, in which a lattice step costs a_k times the total variation of the time profile. Let ω be a state as in Appendix F, with transfer matrix $\mathbb{T}$ and vacuum Ω. Let X be a real bounded function of the spatial links of the slice $x_0=0$, invariant under the gauge transformations of that slice and a uniform limit of local such functions; by Proposition 45(ii) this includes $X=N^{-1}\operatorname{Re}\operatorname{Tr}W_\gamma\circ\Phi_s$ for the holonomy W_γ of a closed path γ in the slice and every flow time s. Put $X^c=X-\omega(X)$ and

$$D_X(n)=\omega\big(X^c\,\tau_n X^c\big)\qquad(n\in\mathbb{Z}_{\ge 0}).$$

Let $w=(w_n)_{n\in\mathbb{Z}}$ be nonnegative, with $w_n=0$ unless $n_0\le n\le n_1$, where $1\le n_0\le n_1$, and with $\|w\|_1=\sum_n w_n>0$; put $V(w)=\sum_{n\in\mathbb{Z}}|w_n-w_{n-1}|$ and

$$B=\sum_n w_n\,\tau_n X^c.$$

Proposition 202. (i) *D_X is nonnegative, nonincreasing and log-convex on $\mathbb{Z}_{\geq 0}$, and $\Psi_B \perp \Omega$. If $D_X(2n_1) > 0$, then $\|\Psi_B\|^2 \geq \|w\|_1^2 D_X(2n_1) > 0$, and the unit vector $\chi = \Psi_B/\|\Psi_B\|$ satisfies*

$$\langle \chi, (1-\mathbb{T})\chi \rangle \leq \frac{V(w)}{\|w\|_1} \frac{D_X(2n_0)}{D_X(2n_1)}.$$

(ii) *If $D_X(2n_1) > 0$ and $\langle \chi, (1-\mathbb{T})\chi \rangle \leq \eta \leq \frac{1}{2}$, then $D_X(2n_0+m) \geq e^{-2\eta m} D_X(2n_1)$ for every integer $m \geq 0$.*

(iii) *For a function H at positive times as in Appendix F, $\langle \Psi_B, \Psi_H \rangle = \omega(\vartheta(B)H)$.*

Proof. Let $x = \Psi_{X^c}$. As in the proof of Lemma 189(i), $\vartheta(X^c) = X^c$ and $\Psi_{\tau_n X^c} = \mathbb{T}^n x$, so $D_X(n) = \langle x, \mathbb{T}^n x \rangle = \int \lambda^n d\nu$ for the spectral measure ν of $\mathbb{T}$ at x, a finite positive measure on $[0,1]$. Hence $D_X \geq 0$, D_X is nonincreasing, and it is log-convex by Hölder's inequality. Since $\langle \Omega, x \rangle = \omega(X^c) = 0$ and $\mathbb{T}\Omega = \Omega$, $x \perp \Omega$ and so $\Psi_B = W(\mathbb{T})x \perp \Omega$, with $W(\lambda) = \sum_n w_n \lambda^n$.

(i) On $[0,1]$, $\|w\|_1 \lambda^{n_1} \leq W(\lambda) \leq \|w\|_1 \lambda^{n_0}$, so $\|\Psi_B\|^2 = \int W^2 d\nu \geq \|w\|_1^2 D_X(2n_1)$. Moreover

$$W(\lambda)(1-\lambda) = \sum_{n=n_0}^{n_1+1} (w_n - w_{n-1})\lambda^n,$$

which is nonnegative and at most $V(w)\lambda^{n_0}$ on $[0,1]$. Therefore

$$\begin{aligned} \langle \Psi_B, (1-\mathbb{T})\Psi_B \rangle &= \int W(\lambda) \cdot W(\lambda)(1-\lambda)\, d\nu \\ &\leq \|w\|_1 V(w) D_X(2n_0), \end{aligned}$$

and dividing by $\|\Psi_B\|^2$ gives the bound.

(ii) Let $\nu_\chi = W^2\nu / \int W^2 d\nu$ be the spectral measure at χ. By Jensen's inequality for λ^m, $\int \lambda^m d\nu_\chi \geq (1-\eta)^m \geq e^{-2\eta m}$, since $\log(1-y) \geq -2y$ on $[0, \frac{1}{2}]$. By the bounds on W,

$$\int \lambda^m d\nu_\chi = \frac{\int W^2 \lambda^m d\nu}{\int W^2 d\nu} \leq \frac{D_X(2n_0+m)}{D_X(2n_1)}.$$

(iii) This is the identity of Appendix F, with $\vartheta(\bar{B}) = \vartheta(B)$ because B is real. □

Along a tuning, take X_k as above for ω_k, a time profile $g \geq 0$ of bounded variation supported in $[t_0, t_1]$ with $0 < t_0 < t_1$ dyadic and $\int g > 0$, and $w_n = a_k g(a_k n)$, so that $B_k = \sum_n a_k g(a_k n)\, \tau_n X_k^c$ is the packet of X_k with time profile g; let χ_k be its unit vector, $\mathrm{Var}(g)$ the total variation of g on $\mathbb{R}$, and write $D_{X_k}(r)$ for $D_{X_k}(r/a_k)$ when $r \in a_k \mathbb{Z}_{\geq 0}$ is a physical time.

Corollary 203. (i) *If $D_{X_k}(2t_1) \geq c\, D_{X_k}(2t_0)$ with $c > 0$ for all large k, then $\langle \chi_k, (1-\mathbb{T}_k)\chi_k \rangle \leq \kappa a_k$ for all large k, with $\kappa = 2\,\mathrm{Var}(g)/(c \int g)$. The hypothesis holds with $c = c_1^{\max(1,(2t_1-2t_0)/(r'-r))}$ if $D_{X_k}(r') \geq c_1 D_{X_k}(r)$ for dyadic $0 < r < r'$ with $r \leq 2t_0$, a $c_1 \in (0,1]$ and all large k.*

(ii) *Conversely, if $\langle \chi_k, (1-\mathbb{T}_k)\chi_k \rangle \leq \kappa a_k$ for all large k, then $D_{X_k}(2t_0+s) \geq e^{-2\kappa s} D_{X_k}(2t_1)$ for all large k and every $s \in a_k \mathbb{Z}_{\geq 0}$; for $s > 2(t_1 - t_0)$ this is a lower bound on the ratio of two correlations of X_k at the physical times $2t_1 < 2t_0 + s$.*

(iii) *If the hypothesis of* (i) *holds and*

$$\left| \omega_k\big(\vartheta(B_k)\, \tau_{u/a_k} A_k\big) \right| \geq v \|\Psi_{B_k}\|\, C_k(2u)^{1/2}$$

with $v > 0$ for all large k, then (N_u) *holds and $\rho_k(u,\sigma) \geq (v - \sqrt{\kappa\sigma/2})_+^2$ for $\sigma \geq 2a_k$.*

Proof. For large k, t_0/a_k and t_1/a_k are integers, and Proposition 202 applies with $n_0 = t_0/a_k$ and $n_1 = t_1/a_k$; $n_0 \geq 1$. The points $a_k n$ partition $\mathbb{R}$, so $V(w) = a_k \sum_n |g(a_k n) - g(a_k(n-1))| \leq a_k\, \mathrm{Var}(g)$, while $\|w\|_1 = a_k \sum_n g(a_k n)$ is a Riemann sum of g, which is Riemann integrable, and so is at least $\frac{1}{2}\int g$ for large k. Then (i) follows from Proposition 202(i); the condition $D_{X_k}(2t_1) > 0$ holds because $D_{X_k}(2t_0) > 0$ is implicit in the hypothesis. Its second sentence is the log-convexity argument of Proposition 49(ii) with $\alpha = 2t_0$, $\beta = 2t_1$, which uses only that D_{X_k} is nonincreasing and log-convex. For (ii), $\kappa a_k \leq \frac{1}{2}$ for large k, and Proposition 202(ii) with $m = s/a_k$ gives the claim. For (iii), Proposition 202(iii) gives $|\langle \chi_k, \Psi_{k,u} \rangle| \geq v$, and Corollary 186 gives the rest. □

For trial states smeared in time, the first condition of Corollary 186 therefore holds along a tuning if and only if, up to the choice of the physical times, a ratio of two correlations of the smeared field at physical times is bounded below. No lattice-scale quantity enters: no Lipschitz constant, no plaquette energy, and no first step. Proposition 191 concerns single slices and does not apply to B_k. When Φ is real, X_k is the slice field U_k of a product packet $f_0 = g \otimes h$ and $u = 0$, $B_k = A_k$, the overlap is 1, and part (i) gives the hypothesis of Proposition 47(ii) from the slice ratio of Proposition 49(iii), without the comparison of u_k with the packet that was left open after Proposition 47. For a flowed loop $X_k = N^{-1} \mathrm{Re}\,\mathrm{Tr}\, W_{\gamma_k} \circ \Phi_{\tau/a_k^2}$, with γ_k the lattice approximation of a closed curve of physical size in the slice, the input that the route needs is the pair of lower bounds of Corollary 203(i) and (iii): a ratio of two flowed-loop correlations at physical times, and the overlap of the smeared loop with the flowed packet at physical distance. Both are lower bounds at physical separations, of the same kind as (N_u), and by Proposition 201 no upper bound supplies them. The overlap vanishes for every k if Ψ_{B_k} and Ψ_{A_k} lie in different eigenspaces of a unitary that commutes with $\mathbb{T}_k$; for instance, if $G = \mathrm{SU}(N)$ with $N \geq 3$ and ω_k is invariant under charge conjugation $U \mapsto \bar{U}$, then $X_k = N^{-1} \mathrm{Im}\,\mathrm{Tr}\, W_{\gamma_k} \circ \Phi_{\tau/a_k^2}$ is odd and the flowed energy packet is even, so X_k has to be chosen in the symmetry sector of the packet.

Comparison with the Gaussian theory. Let $C_k^{\mathrm{G}}(t) > 0$ be the correlations of the corresponding packet in a comparison theory, such as the Gaussian theory at tree level, with ratio $\rho_k^{\mathrm{G}} = C_k^{\mathrm{G}}(2u+\sigma)/C_k^{\mathrm{G}}(2u)$. If there are $Z_k > 0$ and $\delta_k \in [0,1)$ such that for $t = 2u$ and $t = 2u+\sigma$

$$(1-\delta_k) Z_k C_k^{\mathrm{G}}(t) \leq C_k(t) \leq (1+\delta_k) Z_k C_k^{\mathrm{G}}(t),$$

then dividing the lower bound at $2u+\sigma$ by the upper bound at $2u$ gives

$$\rho_k(u,\sigma) \geq \frac{1-\delta_k}{1+\delta_k} \rho_k^{\mathrm{G}}.$$

The normalisation Z_k cancels, and only the two physical separations $2u$ and $2u+\sigma$ enter. For the free fields of Remarks 50 and 51, the slice ratios have positive limits, so $\liminf_k \rho_k^{\mathrm{G}} > 0$

by the argument of Proposition 49(iii), which uses only reflection positivity and the transfer matrix, and (N_u) follows from $\limsup_k \delta_k < 1$. This relative comparison at two physical separations, uniform in k, is what a perturbative lower bound has to provide. It is not a statement of fixed order in the bare coupling: along the one-loop tuning $b\log(\sigma/a_k)/\beta_k \to 1$, so the terms $(\log(\sigma/a_k)/\beta_k)^n$ of bare perturbation theory are of order one at every n. The comparison has to be organised by scales, as in [21, Proposition 460], where repeating a comparison over scales multiplies the factors $1\pm\varepsilon$. The next proposition determines which losses per scale the one-loop tuning allows.

Proposition 204. *Let $b>0$, $h=b\log 2$, $a_k=2^{-k}a_0$, and let the couplings satisfy $|\beta_k - b\log(1/a_k)|\le M$ for all k. Let $\sigma>0$ be dyadic with $\beta_\sigma^- = b\log(1/\sigma)-M>0$, put $\beta_\sigma^+ = b\log(1/\sigma)+M$, and for k with $\sigma/a_k=2^{J_k}$, $J_k\in\mathbb{Z}_{\ge 0}$, put*

$$\beta_k^{(j)} = \beta_k - hj \qquad (0\le j\le J_k),$$

the one-loop coupling at the scale $2^j a_k$. Let $c_{k,j}\in[0,\frac12]$.

(i) $\beta_\sigma^- + h(J_k-j) \le \beta_k^{(j)} \le \beta_\sigma^+ + h(J_k-j)$.

(ii) *If $c_{k,j}\ge c/\beta_k^{(j)}$ for all j, with $c>0$, then*

$$\prod_{j=0}^{J_k}(1-c_{k,j}) \le \Big(\frac{\beta_\sigma^+}{\beta_\sigma^+ + b\log(2\sigma/a_k)}\Big)^{c/h},$$

which tends to 0 as $k\to\infty$.

(iii) *If $c_{k,j}\le c/(\beta_k^{(j)})^2$ for all j, then for every k*

$$\prod_{j=0}^{J_k}(1-c_{k,j}) \ge \exp\Big(-2c\Big(\frac{1}{(\beta_\sigma^-)^2}+\frac{1}{h\beta_\sigma^-}\Big)\Big).$$

More generally, $\prod_j(1-c_{k,j})\ge e^{-2\sum_j c_{k,j}}$, and the products are bounded below uniformly in k if and only if the sums $\sum_j c_{k,j}$ are bounded.

For SU(2) *and the one-loop tuning, $b=11/(3\pi^2)$ and $h\in[0.25751,0.25752]$.*

Proof. (i) Since $\log(1/a_k)=\log(1/\sigma)+J_k\log 2$, the hypothesis gives $\beta_\sigma^- + hJ_k \le \beta_k \le \beta_\sigma^+ + hJ_k$; subtract hj.

(ii) By $1-y\le e^{-y}$ and (i), with $i=J_k-j$,

$$\sum_{j=0}^{J_k} c_{k,j} \ge c\sum_{i=0}^{J_k}\frac{1}{\beta_\sigma^+ + hi} \ge c\int_0^{J_k+1}\frac{dt}{\beta_\sigma^+ + ht} = \frac{c}{h}\log\frac{\beta_\sigma^+ + h(J_k+1)}{\beta_\sigma^+},$$

since the integrand decreases, and $h(J_k+1)=b\log(2\sigma/a_k)\to\infty$.

(iii) Since $\log(1-y)\ge -2y$ on $[0,\frac12]$, $\prod_j(1-c_{k,j})\ge e^{-2\sum_j c_{k,j}}$, and $\prod_j(1-c_{k,j})\le e^{-\sum_j c_{k,j}}$ gives the converse of the last statement. By (i),

$$\sum_{j=0}^{J_k}\frac{1}{(\beta_k^{(j)})^2} \le \sum_{i\ge 0}\frac{1}{(\beta_\sigma^- + hi)^2} \le \frac{1}{(\beta_\sigma^-)^2} + \int_0^\infty \frac{dt}{(\beta_\sigma^- + ht)^2},$$

and the integral is $1/(h\beta_\sigma^-)$. For SU(2), $h = b\log 2 = 11\log 2/(3\pi^2)$, and $0.6931471805 < \log 2 < 0.6931471806$, from the series of $2\operatorname{artanh}\frac13$, with $3.1415926535 < \pi < 3.1415926536$ give $h\in[0.2575118, 0.2575119]$, so $4/h < 15.534$. □

With $g_j^2 = 4/\beta_k^{(j)}$, part (ii) says that a scale-by-scale comparison that loses a factor $1-cg_j^2$ at every scale between a_k and σ, however small $c>0$, gives a lower bound on $\rho_k(u,\sigma)$ that tends to zero, at the rate $(\log(1/a_k))^{-4c/h}$, with $4/h<15.534$;[20] such a bound carries no information in the limit. Part (iii) says that losses of second order in the running coupling give $\rho_k(u,\sigma)\ge \rho_k^{\mathrm{G}} e^{-2c(1/(\beta_\sigma^-)^2 + 1/(h\beta_\sigma^-))}$ uniformly in k, a bound that approaches ρ_k^{G} as σ decreases. So a perturbative lower bound on the ratio along the one-loop tuning has to account exactly for the first-order effect of every scale on the ratio, leaving errors whose sum over the scales is bounded. A first-order effect common to $C_k(2u)$ and $C_k(2u+\sigma)$ cancels in the ratio, and the proof has to exhibit that cancellation; bounding each correlation separately to first order is not enough. This is the summability of the one-step errors in [21, Proposition 460], now for the ratio, and the effective actions it requires are item (iv) of [21, Sec. XIII V], the renormalisation group analysis of Bałaban [6, 7, 13], which is not carried out here. None of the three routes gives (N_u) along the one-loop tuning without an input of this kind: a lower bound on correlations at physical separations, for the packet, for a smeared field that overlaps it, or for the Gaussian comparison.

Reductions of (M), (W) and (U). The rest of this subsection reduces each of the three statements to a lattice statement and determines what the bounds of this paper give for it. The constant Z_k is fixed by the reflected norm $C_k(0)$, while the moments in (M) are not reflected. Let

$$V_k = \omega_k\big(|A_k|^2\big), \qquad \Upsilon_k = Z_k^2 V_k = \frac{V_k}{C_k(0)},$$

the variance of the packet of f_0 and its ratio to the reflected norm, and consider:

$(\mathrm{M}_{\mathrm{loc}})$ For every $n\ge 1$ and every cube Q there are $M_n^{\mathrm{loc}}(Q)<\infty$ and k_0 with

$$\Big|\omega_k\Big(\prod_{j=1}^n A_k(f_j)\Big)\Big| \le M_n^{\mathrm{loc}}(Q)\, V_k^{n/2}\prod_{j=1}^n \|f_j\|_\infty$$

for $k\ge k_0$ and $f_j\in C_c(\mathbb{R}^4)$ supported in Q.

(M_Z) $\sup_k \Upsilon_k<\infty$.

Statement $(\mathrm{M}_{\mathrm{loc}})$ is (M) with Z_k replaced by the local normalisation $V_k^{-1/2}$; it does not change when Φ is multiplied by a constant, and it does not involve the reflection.

[20] Since $\beta_k = b\log(1/a_k)+O(1)$, this rate is a constant times $(g_0^2)^{4c/h}$ with $g_0^2 = 4/\beta_k$ the bare coupling: a power of the coupling at the lattice scale, which is not small at any fixed order of perturbation theory in g_0^2 but tends to zero along the tuning.

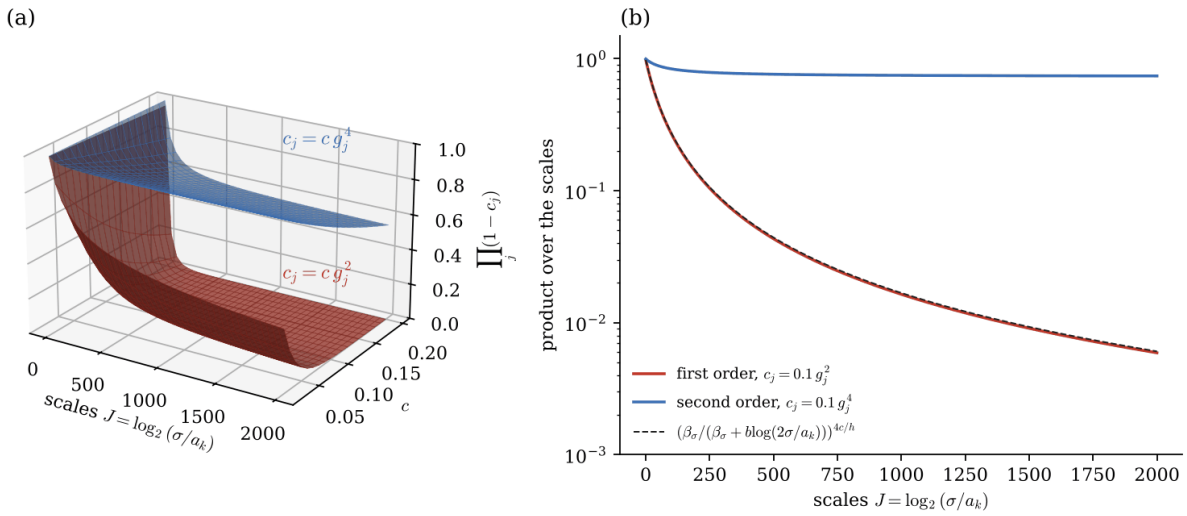


FIG. 16. Products of the factors $1-c_j$ over the $J+1$ scales between a_k and σ along the one-loop SU(2) tuning, with $\beta_W = 20$ at the scale σ, as in Proposition 204. (a) Losses of first order in the running coupling, $c_j = c\,g_j^2$, drive the product to zero for every $c > 0$; losses of second order, $c_j = c\,g_j^4$, keep it bounded below. (b) The section $c = 0.1$, with the bound of part (ii) dashed.

Proposition 205. *Let the states ω_k be invariant under translations and under ϑ and reflection positive, and let Φ be real.*

(i) $\Upsilon_k \geq 1$, *and for f_j supported in a cube Q*

$$\Big|\omega_k\Big(\prod_{j=1}^n Z_k A_k(f_j)\Big)\Big| \leq \big(2\|\Phi\|_\infty Z_k |Q|_k\big)^n \prod_{j=1}^n \|f_j\|_\infty.$$

In particular (M) *holds if* $\sup_k Z_k < \infty$.

(ii) (M) *holds if and only if* ($\mathrm{M}_{\mathrm{loc}}$) *and* ($\mathrm{M}_Z$) *hold. Precisely,* (M) *gives* ($\mathrm{M}_{\mathrm{loc}}$) *with* $M_n^{\mathrm{loc}} = M_n$ *and* $\Upsilon_k \leq M_2(Q_0)\|f_0\|_\infty^2$ *for a cube Q_0 containing the support of f_0, and* ($\mathrm{M}_{\mathrm{loc}}$) *and* ($\mathrm{M}_Z$) *give* (M) *with* $M_n = M_n^{\mathrm{loc}} \sup_k \Upsilon_k^{n/2}$.

(iii) *Assume* (T1). *Then* (T2) *holds if and only if* (M_Z) *holds and* $V_k \geq c' r_k^2 a_k^4 W_k$ *for a $c' > 0$ and all k. If* (T1), (T3) *and this lower bound on V_k hold, then* ($\mathrm{M}_{\mathrm{loc}}$) *holds.*

(iv) *Let $f_0 = g \otimes h$ be a product packet as in Proposition 49, choose an interval I of length $\ell_I > 0$ and $c_g > 0$ with $g \geq c_g$ on I, and put*

$$\kappa_g = \tfrac{1}{2} c_g^2 \ell_I (1 - e^{-1}), \quad \gamma_g = (t_1 - t_0 + \ell_I)\|g\|_\infty,$$
$$\Lambda_g = \max\Big(\frac{4}{\ell_I}, \frac{1}{2t_0}\Big), \quad \varepsilon_g(\Lambda) = \frac{\gamma_g^2 \Lambda e^{-2\Lambda t_0}}{\kappa_g}.$$

Assume that the states have the transfer matrices of Appendix F. If $a_k \leq \ell_I/4$, then for every $\Lambda \geq \Lambda_g$ and every $\sigma \in a_k \mathbb{Z}_{\geq 0}$

$$\rho_k(0,\sigma) \geq e^{-\Lambda\sigma}\big(1 - \varepsilon_g(\Lambda)\Upsilon_k\big).$$

Hence (M_Z) *implies* (N_u) *for every dyadic $u \geq 0$.*

(v) *Let $G = \mathrm{SU}(2)$, $\beta_{W,k} \in [0, 0.348]$ and ω_k the states of Theorem 183, and let f_0 be as in* (iv). *Then for $a_k \leq \min(\ell_I/4, m_0/(2\Lambda_g))$*

$$\Upsilon_k \geq \big(1 - e^{-m_0/2}\big)\frac{2 a_k \kappa_g}{m_0 \gamma_g^2} e^{m_0 t_0/a_k}.$$

(vi) *Let $m > 0$, $q_k = e^{-m a_k}$, let ψ_k be the field of Proposition 199*(a), *ξ_k a family of independent standard normal variables on $\mathbb{Z}^4$ independent of ψ_k, $\phi_k = \psi_k + a_k^{-5/2}\xi_k$, $A_k(f) = \sum_x a_k^4 f(a_k x)\phi_k(x)$, and $f_0 \geq 0$ nonzero. The law of ϕ_k is invariant under translations and B_4 and reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links, its transfer matrix in each coordinate direction has gap $m a_k$, $\rho_k(u,\sigma) = e^{-m\sigma}$ for all u, σ, and Z_k converges; but $\Upsilon_k \to \infty$, so* (M) *fails for $n = 2$.*

Proof. (i) Since ϑ is an automorphism with $\overline{\vartheta(F)} = \vartheta(\bar F)$ and ω_k is ϑ-invariant, $\omega_k(|\vartheta(\bar A_k)|^2) = \omega_k(\vartheta(|A_k|^2)) = V_k$, and the Cauchy–Schwarz inequality gives $C_k(0) \leq V_k$. Further $|A_k(f)| \leq 2\|\Phi\|_\infty a_k^4 \#\{x : a_k x \in Q\}\|f\|_\infty$.

(ii) Since Φ is real, $A_k(\bar f_0) = \overline{A_k(f_0)}$, and (M) with $n = 2$, $f_1 = \bar f_0$ and $f_2 = f_0$ is the bound on Υ_k. By (i), $Z_k^{-n} = C_k(0)^{n/2} \leq V_k^{n/2}$, which gives ($\mathrm{M}_{\mathrm{loc}}$). Conversely, since $Z_k^n V_k^{n/2} = \Upsilon_k^{n/2}$,

$$\Big|\omega_k\Big(\prod_j Z_k A_k(f_j)\Big)\Big| \leq M_n^{\mathrm{loc}} \Upsilon_k^{n/2} \prod_j \|f_j\|_\infty.$$

(iii) The packets are centred, so V_k is the joint cumulant of $A_k(\bar f_0)$ and $A_k(f_0)$, and Lemma 54 gives $V_k \leq K_2 r_k^2 |Q_0|_k a_k^4 W_k \|f_0\|_\infty^2$. If (T2) holds, then $V_k \geq C_k(0) \geq c\, r_k^2 a_k^4 W_k$ and $\Upsilon_k \leq K_2 |Q_0|_k \|f_0\|_\infty^2/c$, which is bounded since $|Q_0|_k \leq (L + a_0)^4$ for a cube of side L. Conversely $C_k(0) = V_k/\Upsilon_k \geq (c'/\sup_j \Upsilon_j) r_k^2 a_k^4 W_k$. For ($\mathrm{M}_{\mathrm{loc}}$), Lemma 54 and the lower bound give for a cumulant of order $p \geq 2$

$$\big|\omega_k^T\big(A_k(f_1); \ldots; A_k(f_p)\big)\big|$$
$$\leq K_p p^{p-2} c'^{-p/2} B_*^{p/2-1} |Q|_k V_k^{p/2} \prod_{j=1}^p \|f_j\|_\infty,$$

using (T3), and a moment is the sum over the partitions of $\{1,\dots,n\}$ without singletons of the products of the cumulants of the blocks, each partition contributing a factor $V_k^{n/2}$.

(iv) We use the notation of the proof of Proposition 49: $A_k=\sum_n w_n\tau_n U_k$ with $w_n=a_k g(a_k n)$, ν the spectral measure of $\mathbb{T}_k$ at u_k, $W(\lambda)=\sum_n w_n\lambda^n$, and $C_k(\sigma)=\int W^2\lambda^{\sigma/a_k}d\nu$. The function U_k depends only on the spatial links of the slice at time 0 (Proposition 45(i)), which ϑ fixes, so for $j\ge 0$, by translation invariance,

$$\omega_k\big(\bar U_k\,\tau_j U_k\big)=\omega_k\big(\vartheta(\bar U_k)\tau_j U_k\big)=\int\lambda^j d\nu,$$

and the same holds for $j<0$ with $|j|$ in place of j, the integral being real. Hence

$$V_k=\int\mathscr{Q}\,d\nu,\qquad \mathscr{Q}(\lambda)=\sum_{n,n'}w_n w_{n'}\lambda^{|n-n'|}.$$

Write $\lambda=e^{-a_k E}$ with $E\in[0,\infty]$. First, $W(\lambda)\le\|w\|_1\lambda^{n_0}\le\gamma_g e^{-Et_0}$, since $n_0\ge t_0/a_k$ and $\|w\|_1\le(t_1-t_0+a_k)\|g\|_\infty$. Second, let J be the set of n with $a_k n\in I$; it consists of $M\ge\ell_I/a_k-1\ge\frac34\ell_I/a_k$ consecutive integers, and $w_n\ge a_k c_g$ on J. Every $n\in J$ has on one side at least $p=\lceil M/2\rceil-1$ further elements of J, so, the terms of $\mathscr{Q}$ being nonnegative,

$$\mathscr{Q}(\lambda)\ge a_k^2c_g^2M\sum_{j=0}^{p}\lambda^j\ge a_k^2c_g^2M\,\frac{1-\lambda^{M/2}}{1-\lambda}.$$

For $E\ge 4/\ell_I$, $\lambda^{M/2}\le e^{-3E\ell_I/8}\le e^{-1}$ and $1-\lambda\le a_kE$, so $\mathscr{Q}(\lambda)\ge\frac34\ell_I c_g^2(1-e^{-1})/E\ge\kappa_g/E$. The function Ee^{-2Et_0} decreases for $E\ge 1/(2t_0)$, so for $\Lambda\ge\Lambda_g$, on the set $\{E>\Lambda\}$,

$$W^2\le\frac{\gamma_g^2}{\kappa_g}Ee^{-2Et_0}\mathscr{Q}\le\varepsilon_g(\Lambda)\mathscr{Q},$$

which also holds at $E=\infty$, where $W=0$. Therefore $\int_{\{E>\Lambda\}}W^2d\nu\le\varepsilon_g(\Lambda)V_k=\varepsilon_g(\Lambda)\Upsilon_k C_k(0)$, and

$$C_k(\sigma)\ge e^{-\Lambda\sigma}\int_{\{E\le\Lambda\}}W^2d\nu\ge e^{-\Lambda\sigma}\big(1-\varepsilon_g(\Lambda)\Upsilon_k\big)C_k(0).$$

If $\Upsilon_k\le\Upsilon$ for all k, choose $\Lambda\ge\Lambda_g$ with $\varepsilon_g(\Lambda)\Upsilon\le\frac12$; then $\rho_k(0,\sigma)\ge\frac12e^{-\Lambda\sigma}$, which is (N_0), and Proposition 182(iii) gives (N_u).

(v) By Theorem 183 with $\sigma=a_k$, $\rho_k(0,a_k)\le e^{-m_0}$. Part (iv) with $\sigma=a_k$ and $\Lambda=m_0/(2a_k)\ge\Lambda_g$ gives $e^{-m_0}\ge e^{-m_0/2}(1-\varepsilon_g(\Lambda)\Upsilon_k)$, so $\Upsilon_k\ge(1-e^{-m_0/2})/\varepsilon_g(m_0/(2a_k))$, which is the claim.

(vi) The covariance of ϕ_k is $\prod_\mu q_k^{|x_\mu-y_\mu|}+a_k^{-5}\delta_{xy}$, which is invariant under translations and B_4. Fix a coordinate direction as time. The process of the time slices of (ψ_k,ξ_k) is the stationary reversible Markov chain of the proof of Proposition 199(a) for ψ_k, together with independent slices of ξ_k; its transition operator is $P\otimes\Pi$, with P as there and Π the projection onto the functions that do not depend on the slice of ξ_k. It is positive with spectrum $\{q_k^N:N\ge0\}\cup\{0\}$ and 1 simple, and the argument given there gives reflection positivity and the gap. The packets in $C_k(\sigma)$, $\sigma\ge0$, are supported at times $x_0\le-1$ and $x_0\ge1$, where the variables ξ_k are independent and centred, so $C_k(\sigma)$ is that of example (a): with the time factor $q_k^{y_0-x_0}$, the substitution $y\mapsto y+(\sigma/a_k)e_0$ gives $C_k(2u+\sigma)=e^{-m\sigma}C_k(2u)$, and Z_k converges as in (a). The variances add, so $V_k\ge a_k^{-5}a_k^8\sum_x f_0(a_kx)^2$, which is a_k^{-1} times a Riemann sum converging to $\|f_0\|_2^2>0$. Hence $\Upsilon_k\to\infty$, and (M) for $n=2$ fails by (ii). □

By (ii), the renormalisation enters (M) only through (M_Z), a comparison of the variance of the packet with its reflected norm. By the proof of (iv), for a product packet V_k weighs the spectral measure of the slice field by $\mathscr{Q}\ge\kappa_g/E$ at energies $E\ge4/\ell_I$, while $C_k(0)$ weighs it by $W^2\le\gamma_g^2e^{-2Et_0}$: (M_Z) says that the part of the variance carried at high energy, which the reflection suppresses over the distance $2t_0$, is bounded by a multiple of the reflected norm, and it implies (N_0). In the strong-coupling range Υ_k grows exponentially in $1/a_k$ by (v). Example (vi) has the lattice symmetry, reflection positivity, the gap ma_k and ratios $\rho_k(u,\sigma)=e^{-m\sigma}$ independent of k, and (M_Z) fails for it; so (M_Z) does not follow from (N_u) and these properties. By Remark 57, $(\mathrm{M}_{\mathrm{loc}})$ does not follow from bounds on two-point functions. Every inequality that holds, with constants independent of k, for the states of Theorem 183 along all sequences of couplings in $[0,0.348]$ holds along sequences on which (M_Z) fails by (v), and (M) fails with it; so no such inequality implies (M_Z) or (M). This applies to every consequence of reflection positivity at all $\beta_W\ge0$, such as the chessboard estimates [49], and to the bounds of class (H) of Proposition 201; a proof of (M) has to use that the couplings leave the strong-coupling range. A sufficient condition without smearing follows from Hölder's and Minkowski's inequalities and translation invariance: if

$$\sup_k Z_k\big\|\varphi_k(0)-\omega_k(\varphi_k(0))\big\|_{L^n(\omega_k)}<\infty\qquad(n\ge1),$$

then $\|Z_kA_k(f)\|_{L^n(\omega_k)}$ is at most $|Q|_k\|f\|_\infty$ times this supremum, and (M) holds. This pointwise bound on the renormalised flowed field is not proved.

The flow does not bound the field. Part (i) of Proposition 205 is the bound that uses only $\|\Phi\|_\infty$. The spatial flow at a positive flow time does not improve it configuration by configuration.

Proposition 206. *Let $G=\mathrm{SU}(N)$, $N\ge2$, $\zeta=e^{2\pi i/N}$, and let U^* be the configuration with all temporal links equal to 1 and spatial links*

$$U^*_{(x,x+e_1)}=1,\quad U^*_{(x,x+e_2)}=\zeta^{x_1},\quad U^*_{(x,x+e_3)}=\zeta^{x_1+x_2},$$

multiples of the identity.

(i) *Every spatial plaquette of U^* is $\zeta^{\pm1}$, and $\Phi_s(gU^*)=gU^*$ and $\Phi_s(g1)=g1$ for every gauge transformation g and every $s\in\mathbb{R}$.*

(ii) *Let Φ be the sum of $N-\mathrm{Re}\,\mathrm{Tr}\,U_p$ over the twelve spatial plaquettes with a corner at 0. Then for all x and s*

$$\Phi_x\circ\Phi_s(U^*)=12N\big(1-\cos(2\pi/N)\big),\qquad\Phi_x\circ\Phi_s(1)=0.$$

(iii) *Let $\omega(O)=\mu(O^F)$ for gauge-invariant O, with μ a Gibbs measure of γ^F at $\beta\ge 0$, as for the states of Theorem 193, let Φ be as in* (ii) *and $f\ge 0$. Then the ω-essential supremum and infimum of $A_k(f)$ differ by at least $12N(1-\cos(2\pi/N))\,a_k^4\sum_x f(a_kx)$, at every flow time, and*

$$\|Z_kA_k(f)\|_{L^\infty(\omega)}\ge 6N\Big(1-\cos\frac{2\pi}{N}\Big)Z_k\,a_k^4\sum_x f(a_kx).$$

Proof. (i) Let $1\le i<j\le 3$. The plaquette at x in the plane of e_i and e_j is

$$U_{(x,x+e_i)}U_{(x+e_i,x+e_i+e_j)}U^{-1}_{(x+e_j,x+e_i+e_j)}U^{-1}_{(x,x+e_j)}.$$

For U^* and $(i,j)=(1,2),(1,3),(2,3)$ it is

$$\zeta^{x_1+1}\zeta^{-x_1},\quad \zeta^{x_1+x_2+1}\zeta^{-x_1-x_2},\quad \zeta^{x_1}\zeta^{x_1+x_2+1}\zeta^{-x_1}\zeta^{-x_1-x_2},$$

that is ζ; with the opposite orientation it is ζ^{-1}. Let e be a spatial link and X in the Lie algebra of $\mathrm{SU}(N)$. All links of U^* are central, so replacing U^*_e by $U^*_e e^{\varepsilon X}$ changes a plaquette $p\ni e$ into $c\,e^{\pm\varepsilon X}$ with $c=\zeta^{\pm1}$, and

$$\frac{d}{d\varepsilon}\operatorname{Re}\operatorname{Tr}\big(c\,e^{\pm\varepsilon X}\big)\Big|_{\varepsilon=0}=\pm\operatorname{Re}\big(c\operatorname{Tr}X\big)=0,$$

since $\operatorname{Tr}X=0$. So the gradient in the flow equation vanishes at U^*, and $\Phi_s(U^*)=U^*$ by the uniqueness in Proposition 45(i). The same computation with $c=1$ applies to the configuration 1, and the flow commutes with gauge transformations.

(ii) $N-\operatorname{Re}\operatorname{Tr}(\zeta^{\pm1}1)=N(1-\cos(2\pi/N))$ for each of the twelve plaquettes, and $N-\operatorname{Re}\operatorname{Tr}1=0$; by (i) the flow does not move either configuration.

(iii) As in the proof of Lemma 10, choosing a base point on each component of F gives a gauge transformation g with $(gU^*)_e=1$ on F; let Q^* be the restriction of gU^* to the active links, and $Q^1\equiv 1$. The functions $F_x=(\Phi_x\circ\Phi_{s_k})^F$ are uniform limits of continuous functions of finitely many active links (Proposition 45(ii)), and gauge invariance gives $F_x(Q^*)=\Phi_x\circ\Phi_{s_k}(U^*)$. Given $\eta>0$ and the finite set X_f of x with $f(a_kx)\ne 0$, there is a neighbourhood of Q^*, fixed by conditions on finitely many links B, on which $|F_x-F_x(Q^*)|<\eta$ for $x\in X_f$, and similarly for Q^1. Given the other links, the conditional law under μ of the links of B has density proportional to $\exp(N\beta\sum_p\operatorname{Re}\operatorname{Tr}Q_p^F)$ with respect to Haar measure, the sum running over the plaquettes meeting B, and this density is at least $e^{-2N^2\beta|P_B|}$, with $|P_B|$ the number of these plaquettes. So both neighbourhoods have positive probability. On them $A_k(f)$ is within $\eta a_k^4\sum_x f(a_kx)$ of its values at Q^* and Q^1, which differ by $12N(1-\cos(2\pi/N))a_k^4\sum_x f(a_kx)$ by (ii). Since η is arbitrary, this gives the first claim, and the L^∞ norm is at least half the difference. □

So the bound of Proposition 205(i) is attained up to a constant factor at every flow time, and it gives (M) only when Z_k is bounded. Away from the configurations of least action the flow has stationary points at which the flowed energy density keeps its lattice size; a proof of (M) by way of the flow has to bound the probability of the configurations near such points, which is a property of the measure and not of the flow.

Rotations as Ward identities. For $X\in\mathfrak{so}(4)$ and $f\in C^1_c(\mathbb{R}^4)$ let $L_Xf(x)=\nabla f(x)\cdot Xx$, the derivative of $f\circ e^{\varepsilon X}$ at $\varepsilon=0$, and $E_{\mu\nu}$ as in the proof of Proposition 197.

Lemma 207. *Let S be a family as in Theorem 193*(i)–(ii)*, $X\in\mathfrak{so}(4)$ and $n\ge 1$. Then $S_n(f_1\circ e^{-\varepsilon X},\dots,f_n\circ e^{-\varepsilon X})=S_n(f_1,\dots,f_n)$ for all $\varepsilon\in\mathbb{R}$ and $f_j\in C_c(\mathbb{R}^4)$ if and only if*

$$\sum_{j=1}^n S_n(f_1,\dots,L_Xf_j,\dots,f_n)=0$$

for all $f_j\in C_c^\infty(\mathbb{R}^4)$.

Proof. Let $R_\varepsilon=e^{\varepsilon X}$ and $F(\varepsilon)=S_n(f_1\circ R_\varepsilon,\dots,f_n\circ R_\varepsilon)$ for $f_j\in C_c^\infty(\mathbb{R}^4)$. Since R_ε commutes with X and $\nabla(f\circ R)(x)=R^{\mathsf T}\nabla f(Rx)$,

$$L_X(f\circ R_\varepsilon)(x)=\nabla f(R_\varepsilon x)\cdot R_\varepsilon Xx=(L_Xf)(R_\varepsilon x).$$

By Taylor's formula $h^{-1}(f\circ R_{\varepsilon+h}-f\circ R_\varepsilon)$ converges uniformly to $(L_Xf)\circ R_\varepsilon$ as $h\to 0$, with supports in a fixed cube for $|h|\le 1$, so the bound of Theorem 193(i) shows that F is differentiable with $F'(\varepsilon)$ equal to the sum of the lemma at the functions $f_j\circ R_\varepsilon$. If the sum vanishes, F is constant, which is the invariance for smooth f_j, and the bound and the density of C_c^∞ in C_c in the supremum norm on a cube give it for all f_j. Conversely invariance gives $F'(0)=0$. □

For $X\in\mathfrak{so}(4)$ and $f_j\in C_c^\infty(\mathbb{R}^4)$ put

$$\Delta_k^X(f_1,\dots,f_n)=\sum_{j=1}^n\omega_k\Big(Z_kA_k(L_Xf_j)\prod_{i\ne j}Z_kA_k(f_i)\Big),$$

the derivative at $\varepsilon=0$ of the lattice moment $\omega_k(\prod_j Z_kA_k(f_j\circ e^{\varepsilon X}))$, a finite sum. It measures the breaking of the rotation by the lattice and by the flow.

Proposition 208. *Assume the hypotheses of Theorem 193.*

(i) *$\Delta_k^X(f_1,\dots,f_n)\to 0$ for all n and $f_j\in C_c^\infty(\mathbb{R}^4)$ if and only if every subsequential limit is invariant under the group $\{e^{\varepsilon X}\}$.*

(ii) *Let $X\ne 0$ lie in $\mathfrak{so}(3)_{\mathrm{sp}}$, for instance $X=E_{12}$. Then* (W) *holds if and only if $\Delta_k^X(f_1,\dots,f_n)\to 0$ for all n and $f_j\in C_c^\infty(\mathbb{R}^4)$.*

(iii) *Let $X=E_{01}$. Then $\Delta_k^X(f_1,\dots,f_n)\to 0$ for all n and $f_j\in C_c^\infty(\mathbb{R}^4)$ if and only if every subsequential limit is invariant under* O(4)*. For the family of Proposition 199*(c) *the identity of Lemma 207 with $X=E_{01}$ fails.*

(iv) *If* (T1)–(T3) *hold and $a_k^4W_k\to 0$, the conditions in* (ii) *and* (iii) *need only be imposed for $n=2$.*

Proof. (i) Let S be the limit along a subsequence. By Theorem 193(i), applied to the functions $L_Xf_j\in C_c^\infty$, Δ_k^X converges along it to the sum of Lemma 207; if $\Delta_k^X\to 0$, the lemma gives the invariance. If $|\Delta_k^X(f_1,\dots,f_n)|\ge\eta>0$ along a subsequence, a refinement given by Theorem 193(i) has a limit for which the sum of Lemma 207 is at least η in absolute value, and that limit is not invariant.

(ii) $e^{\varepsilon X}$ is the rotation about the line $\ker X$ through an angle proportional to ε, of infinite order for all but countably many

ε. If (W) holds, every limit is invariant under $\mathrm{O}(3)\times\mathrm{O}(1)$, as stated after (W), and (i) gives $\Delta_k^X\to 0$. Conversely, by (i) each G_S contains a spatial rotation of infinite order, so $G_S\supseteq \mathrm{O}(3)\times\mathrm{O}(1)$ by Proposition 197(ii), and (W) follows.

(iii) $e^{\varepsilon E_{01}}$ rotates the $(0,1)$ plane, has infinite order for $\varepsilon\notin\pi\mathbb{Q}$ and moves e_0 off $\pm e_0$ for $\varepsilon\notin\pi\mathbb{Z}$; Proposition 197(iii) and (i) give the equivalence. The family of Proposition 199(c) has $G_S=\mathrm{O}(3)\times\mathrm{O}(1)$, which does not contain $e^{\varepsilon E_{01}}$ for $\varepsilon\notin\pi\mathbb{Z}$, so by Lemma 207 the identity fails for some f_j.

(iv) By Theorem 55(v) every limit is Gaussian, so it is invariant under a rotation if and only if S_2 is, and the proofs of (i) and of Lemma 207 apply to $n=2$ alone. □

So the lattice estimate that gives the spatial rotations is $\Delta_k^X\to 0$ for one $X\in\mathfrak{so}(3)_{\mathrm{sp}}$, with no rate; a bound $|\Delta_k^X(f_1,\dots,f_n)|\le C(f_1,\dots,f_n)a_k^p$ with $p>0$ would suffice and is not proved. For $X=E_{12}$ the rotation by $\pi/2$ lies in W_3, so $\varepsilon\mapsto\omega_k(\prod_j Z_kA_k(f_j\circ e^{\varepsilon X}))$ has period $\pi/2$, and the integral of $\Delta_k^X(f_1\circ e^{\varepsilon X},\dots,f_n\circ e^{\varepsilon X})$ over a period vanishes at every k: the lattice symmetry cancels the breaking on average over the angle, and (ii) asks for the cancellation at each angle in the limit. For the time–space generators the same identity is the condition for $\mathrm{O}(4)$, and part (iii) shows that it fails for the spatially flowed free field at a positive flow time.

The four-dimensional flow. The flow in all four directions preserves $\mathrm{O}(4)$, and it loses reflection positivity already for the free field.

Proposition 209. (i) *Let $t_f>0$ and $\mu>0$. The centred Gaussian field on $\mathbb{R}^4$ with covariance*

$$\widehat{C}(p)=e^{-2t_f|p|^2}(|p|^2+\mu^2)^{-1},$$

the massive free field flowed by the four-dimensional heat flow for the time t_f, is invariant under translations and $\mathrm{O}(4)$ and is not reflection positive: there is a real $f\in C_c^\infty(\mathbb{R}^4)$ supported in $\{t>0\}$ with $\iint f(\theta x)C(x-y)f(y)\,dx\,dy<0$.

(ii) *Let S^j, $j\ge 1$, be families as in Theorem 193(i)–(ii), each invariant under $\mathrm{O}(3)\times\mathrm{O}(1)$, bounded by $M_n(Q)\prod_j\|f_j\|_\infty$ uniformly in j, and with $S_n^j(f_1,\dots,f_n)\to S_n^0(f_1,\dots,f_n)$ for all n and $f_i\in C_c(\mathbb{R}^4)$. Then $G_{S^0}=\mathrm{O}(4)$ if and only if S^0 is invariant under the exchange of e_0 and e_1.*

Proof. (i) Invariance is clear. Put

$$H_{\mathbf{p}}(u)=\frac{1}{2\pi}\int e^{ip_0u}\widehat{C}(p_0,\mathbf{p})\,dp_0,$$

an even real function, smooth in u and continuous in $(\mathbf{p},u)$, with $|H_{\mathbf{p}}(u)|\le H_{\mathbf{p}}(0)\le H_0(0)$. For $f=g\otimes h$ with g supported in $(0,\infty)$ and g,h real, the spatial Fourier transform gives

$$\iint f(\theta x)C(x-y)f(y)\,dx\,dy=(2\pi)^{-3}\int|\hat h(\mathbf{p})|^2G_{\mathbf{p}}\,d\mathbf{p},$$
$$G_{\mathbf{p}}=\iint g(s)g(t)H_{\mathbf{p}}(s+t)\,ds\,dt.$$

Since $H_0(0)>0$ and $H_0''(0)=-(2\pi)^{-1}\int p_0^2\widehat{C}(p_0,0)\,dp_0<0$, $\log H_0$ is strictly concave on an interval $[0,2\delta]$. For $0<u_1<u_2<\delta$ this gives $H_0(u_1+u_2)^2>H_0(2u_1)H_0(2u_2)$, so the matrix $(H_0(u_i+u_j))_{i,j\le 2}$ has negative determinant and there is a real c with $\sum_{i,j}c_ic_jH_0(u_i+u_j)<0$. Let $j_\eta\ge 0$ be a mollifier with integral 1 supported in $[-\eta,\eta]$, $\eta<u_1$, and $g=c_1j_\eta(\cdot-u_1)+c_2j_\eta(\cdot-u_2)$; for small η, $G_0<0$ by continuity of H_0. Fix such g; then $\mathbf{p}\mapsto G_{\mathbf{p}}$ is continuous and bounded by $\|g\|_1^2H_0(0)$. For a real nonzero $h_1\in C_c^\infty(\mathbb{R}^3)$ and $h_L(\mathbf{x})=L^{-3/2}h_1(\mathbf{x}/L)$, the right side for $h=h_L$ is $(2\pi)^{-3}\int|\hat h_1(\mathbf{q})|^2G_{\mathbf{q}/L}\,d\mathbf{q}$, which tends to $\|h_1\|_2^2G_0<0$ as $L\to\infty$ by dominated convergence.

(ii) The bound makes each condition $S_n^0(f_1\circ R^{-1},\dots)=S_n^0(f_1,\dots)$ a limit of the same conditions for S^j, so $G_{S^0}\supseteq\mathrm{O}(3)\times\mathrm{O}(1)$, which is infinite. The exchange g has $ge_0=e_1$, and Proposition 197(iii) gives the claim; the converse is clear. □

At $t_f=0$ the function $H_{\mathbf{p}}(u)=e^{-\omega|u|}/(2\omega)$ is log-linear on $(0,\infty)$, and the field is reflection positive. The proof of (i) shows that reflection positivity needs the matrices $(H_{\mathbf{p}}(u_i+u_j))$ with $u_i>0$ to be positive semidefinite, which makes $H_{\mathbf{p}}$ log-convex on $(0,\infty)$; a smooth even function with $H''(0)<0$ is not log-convex near 0. So at a positive flow time the spatial flow keeps reflection positivity and loses $\mathrm{O}(4)$ (Proposition 199(c)), and the four-dimensional flow keeps $\mathrm{O}(4)$ and loses reflection positivity. By part (ii), at vanishing flow time, for limits S^0 of families that satisfy (W) at positive flow times with uniform bounds, $\mathrm{O}(4)$ reduces to one discrete symmetry, the exchange of time with a spatial axis, which the unflowed lattice theory has; the interchange of the limits $k\to\infty$ and $t_f\to 0$ that would carry it to S^0, and the renormalisation of the fields as $t_f\to 0$, are not treated here.

One limit under a renormalisation condition. Proposition 198(iii) uses the convergence of one ratio. Choosing the couplings so that it holds is a renormalisation condition:

(RC) There are a dyadic $\sigma_0>0$ and $\rho_*\in(0,1)$ with $\rho_k(0,\sigma_0)\to\rho_*$.

That a tuning with (RC) exists is not proved. Since (RC) contains (N_0), it is not weaker than the open input of Theorem 46.

Under (M), let $\mathfrak{X}$ be the set of families $S=(S_n)$ of multilinear forms on $C_c(\mathbb{R}^4)$ with $|S_n(f_1,\dots,f_n)|\le M_n(Q)\prod_j\|f_j\|_\infty$ for f_j supported in a cube Q. For $L\in\mathbb{N}$ let $\mathscr{F}_L$ be a countable $\mathbb{Q}(i)$-linear set of functions supported in $[-L,L]^4$, dense there in the supremum norm, let $P_1,P_2,\dots$ enumerate the tuples of elements of $\bigcup_L\mathscr{F}_L$, and put $d(S,S')=\sum_i 2^{-i}\min(1,|S(P_i)-S'(P_i)|)$. By the bounds and density, $d(S^m,S)\to 0$ if and only if $S_n^m(f_1,\dots,f_n)\to S_n(f_1,\dots,f_n)$ for all n and $f_j\in C_c(\mathbb{R}^4)$, and by the diagonal argument of Theorem 55(ii) $(\mathfrak{X},d)$ is a compact metric space. For $k\ge k_0$ the lattice moments $x_k=(\omega_k(\prod_jZ_kA_k(f_j)))_n$ lie in $\mathfrak{X}$; let $\mathfrak{L}$ be their set of limit points, the set of subsequential limits, and put

$$\delta_k(f_1,\dots,f_n)=\omega_{k+1}\Big(\prod_jZ_{k+1}A_{k+1}(f_j)\Big)-\omega_k\Big(\prod_jZ_kA_k(f_j)\Big).$$

Lemma 210. *If (x_k) is a sequence in a compact metric space with $d(x_k,x_{k+1})\to 0$, its set of limit points is nonempty and connected.*

Proof. The set L of limit points is nonempty and closed. If $L=A\cup B$ with A,B disjoint, closed and nonempty, let 3η be their distance and A_η,B_η their open η-neighbourhoods, at distance at least η. The sequence enters both infinitely often, and once $d(x_k,x_{k+1})<\eta$ it cannot pass from one to the other in one step, so infinitely many x_k lie in the compact complement of $A_\eta\cup B_\eta$, which then contains a limit point. That contradicts $L\subseteq A\cup B$. □

Proposition 211. (i) (RC) *implies* (N_u) *for every dyadic $u\ge 0$, and every limit of Theorem 46 along a subsequence is nontrivial and has $\rho(\sigma_0)=\rho_*$. Under the hypothesis of Theorem 43, $m_*\le\sigma_0^{-1}\log(1/\rho_*)$. For* SU(2), (RC) *implies $\beta_{W,k}>0.348$ for all large k. Under* (M), *every $S\in\mathfrak{L}$ has $S_2(\bar f_0\circ\theta,f_0(\cdot-(\sigma_0,0)))=\rho_*$.*

(ii) *Assume the hypotheses of Theorem 193. Then* (U) *holds if and only if the following hold:*

(U1) *$\delta_k(f_1,\dots,f_n)\to 0$ for all n and $f_j\in C_c(\mathbb{R}^4)$;*

(U2) *$\mathfrak{L}$ is totally disconnected.*

(iii) *Under the same hypotheses, if $\sum_k|\delta_k(f_1,\dots,f_n)|<\infty$ for all n and all f_j in a set whose linear span is dense in the continuous functions supported in each cube, then* (U) *holds.*

(iv) *Let $m>0$, $q_k=e^{-ma_k}$, $\varepsilon_k=2+\sin(\log k)$, $K_\varepsilon(\mathbf{z})=e^{-|\mathbf{z}|^2/\varepsilon}$ on $\mathbb{R}^3$, let ϕ_k be the centred Gaussian field on $\mathbb{Z}^4$ with covariance*

$$q_k^{|x_0-y_0|}K_{\varepsilon_k}\big(a_k(\mathbf{x}-\mathbf{y})\big),$$

$A_k(f)=\sum_x a_k^4 f(a_kx)\phi_k(x)$, and $f_0\ge 0$ nonzero. The law of ϕ_k is invariant under translations and Γ and reflection positive for the time reflections through sites and bisecting links, and its transfer matrix has gap ma_k. Moreover $\rho_k(u,\sigma)=e^{-m\sigma}$ for all u,σ, so (RC) *holds for every σ_0;* (T1)–(T3) *hold with $r_k=1$ for large k, and* (M), (W) *and* (U1) *hold, with $|\delta_k(f_1,\dots,f_n)|\le C/k$ for $f_j\in C_c^1(\mathbb{R}^4)$. But $\mathfrak{L}$ is homeomorphic to $[1,3]$, and* (U) *fails.*

Proof. (i) The first claims are Proposition 182(iii) and Theorem 46(ii). Under the hypothesis of Theorem 43 the gap is at least m_*a_k on $\Omega_k^\perp$, and as in the proof of Theorem 183 $\rho_k(0,\sigma_0)\le e^{-m_*\sigma_0}$, so $\rho_*\le e^{-m_*\sigma_0}$. The statement for SU(2) is Theorem 183. The last claim is the computation in the proof of Proposition 198(iii).

(ii) If (U) holds, x_k converges in $\mathfrak{X}$ by Proposition 198(i), so $d(x_k,x_{k+1})\to 0$, which is (U1), and $\mathfrak{L}$ is a point. Conversely (U1) is $d(x_k,x_{k+1})\to 0$, so $\mathfrak{L}$ is connected by Lemma 210; a nonempty, connected and totally disconnected set is a point, and (U) follows.

(iii) The moments of tuples from the set converge, being Cauchy; by multilinearity so do those of tuples from its span, and by (M) and density all moments converge. Proposition 198(i) gives (U).

(iv) K_ε is positive definite on $\mathbb{R}^3$, its Fourier transform being positive, hence on $a_k\mathbb{Z}^3$, and the field is the stationary Markov chain of the proof of Proposition 199(a) with γ the Gaussian law of covariance $K_{\varepsilon_k}(a_k(\mathbf{x}-\mathbf{y}))$; this gives reflection positivity, the gap and, since K_ε is radial, the invariance. As in Proposition 205(vi), the time factor of the covariance gives $C_k(2u+\sigma)=e^{-m\sigma}C_k(2u)$. The kernel $k_\varepsilon(z)=e^{-m|z_0|}K_\varepsilon(\mathbf{z})$ is bounded by 1 and Lipschitz, uniformly in $\varepsilon\in[1,3]$, and the covariance of $A_k(f)$ and $A_k(f')$ is a Riemann sum of mesh a_k for $\iint f(x)f'(y)k_{\varepsilon_k}(x-y)\,dx\,dy$, with error at most $C(f,f')a_k$ for $f,f'\in C_c^1$ and tending to 0 uniformly in ε for $f,f'\in C_c$. In particular $C_k(0)$ differs by $o(1)$ from $c(\varepsilon_k)$, where $c(\varepsilon)=\iint f_0(\theta x)f_0(y)k_\varepsilon(x-y)\,dx\,dy$ is positive and continuous on $[1,3]$. With $w_k(z)=q_k^{|z_0|}e^{-a_k^2|\mathbf{z}|^2/3}$, which bounds the covariance since $\varepsilon_k\le 3$, $a_k^4W_k\to(2/m)(3\pi)^{3/2}$, and (T1) holds with $K_2=1$, $K_n=0$ for $n\ge 3$, (T2) for large k, and (T3); Theorem 196(i) gives (M). Let S^ε be the Gaussian family with

$$S_2^\varepsilon(f,f')=c(\varepsilon)^{-1}\iint f(x)f'(y)k_\varepsilon(x-y)\,dx\,dy.$$

By Wick's rule, $|x_k(f_1,\dots,f_n)-S^{\varepsilon_k}(f_1,\dots,f_n)|$ tends to 0, and is at most Ca_k for $f_j\in C_c^1$. The derivative of k_ε in ε is bounded on compact sets uniformly in $\varepsilon\in[1,3]$ and $c>0$ there, so $|S^\varepsilon(f_1,\dots,f_n)-S^{\varepsilon'}(f_1,\dots,f_n)|\le C|\varepsilon-\varepsilon'|$, and $|\varepsilon_{k+1}-\varepsilon_k|\le\log(1+1/k)\le 1/k$; this gives the bound on δ_k, and (U1) follows by approximation and (M). Since $\log k\to\infty$ with increments tending to 0, the limit points of (ε_k) form $[1,3]$, so $\mathfrak{L}=\{S^\varepsilon:1\le\varepsilon\le 3\}$. Each S^ε is invariant under $\mathrm{O}(3)\times\mathrm{O}(1)$, since k_ε is, which gives (W). If $S_2^\varepsilon=S_2^{\varepsilon'}$, the continuous kernels $k_\varepsilon/c(\varepsilon)$ and $k_{\varepsilon'}/c(\varepsilon')$ coincide; at $z=0$ this gives $c(\varepsilon)=c(\varepsilon')$, and then $K_\varepsilon=K_{\varepsilon'}$, so $\varepsilon=\varepsilon'$. So $\varepsilon\mapsto S^\varepsilon$ is a continuous injection of $[1,3]$ onto $\mathfrak{L}$, a homeomorphism. □

So (RC) fixes the value of one functional on every limit and leaves (U1) and (U2) open: example (iv) satisfies (RC), (M), (W), the tree bounds and (U1), and its limits form an interval; the same field with ε_k alternating between 1 and 3 satisfies (RC), (M), (W), the tree bounds and (U2), and has two limits. Part (iii) is the form that a scale-by-scale analysis would supply. Along the one-loop tuning of Proposition 204, $\beta_k=hk+O(1)$, so a bound $|\delta_k(f_1,\dots,f_n)|\le C/\beta_k$, of first order in the running coupling, is of the size C/k of example (iv) and does not imply (U), while a bound $C\beta_k^{-p}$ with $p>1$ is summable and gives (U) by (iii). As for the ratio in Proposition 204, a bound on the change of the moments from one scale to the next gives (U) only if it is better than first order in the running coupling, or if the changes cancel between scales.

Remark 212. Let f_0 be invariant under the spatial rotations and $R\in\mathrm{SO}(3)_{\mathrm{sp}}$. The moments $x_k^R=(\omega_k(\prod_j Z_kA_k(f_j\circ R^{-1})))_n$ are those of the theory on the rotated lattice $R(a_k\mathbb{Z}^4)$, whose packet of f_0 is A_k; it has the same C_k, Z_k and ρ_k, and satisfies (RC) when the original theory does. The sequence $x_1,x_1^R,x_2,x_2^R,\dots$ converges if and only if (U) holds and the difference in (W) for R tends to 0: if it converges, both subsequences have the same limit, and conversely. For R of infinite order this is (U) and (W) together, the statement that the limit does not depend on which of the lattices $R(a_k\mathbb{Z}^4)$ is used. Example (a) of Proposition 199 satisfies (RC), since its time dependence factorises as in Proposition 205(vi), as well as (M), (U) and

the tree bounds, and not (W); example (iv) of Proposition 211 satisfies (RC), (M), (W) and the tree bounds, and not (U).

The time direction and the slice field. The two parts of (M) can be compared with correlations of slice fields, as the ratio was in Proposition 49. In the time direction the comparison is exact, because the transfer matrix is positive. In the spatial directions no such comparison is available: the spatial flow acts within each time slice and mixes the two sides of every spatial reflection plane. For $h' \in C_c(\mathbb{R}^3)$ put

$$U_k^{h'} = \sum_{x\in\mathbb{Z}^3} a_k^3 h'(a_k x)\big(\varphi_k(0,x) - \omega_k(\varphi_k(0,x))\big),$$

and, for $r \in a_k\mathbb{Z}_{\geq 0}$ and $\delta > 0$,

$$D_k^{h'}(r) = \omega_k\big(\bar U_k^{h'}\,\tau_{r/a_k} U_k^{h'}\big),$$
$$I_k^{h'}(\delta) = a_k \sum_{r\in a_k\mathbb{Z},\,0\leq r<\delta} D_k^{h'}(r).$$

For the spatial factor h of a product packet, $U_k^h = U_k$ and $D_k^h = D_k$ are those of Proposition 49; write $I_k = I_k^h$. As in the proof of Proposition 205(iv), $D_k^{h'}(r) = \langle u, \mathbb{T}_k^{r/a_k} u\rangle$ with $u = \Psi_{U_k^{h'}}$.

Lemma 213. *Let $\mathbb{T}$ be self-adjoint with $0 \leq \mathbb{T} \leq 1$ on a Hilbert space, let v_n, $n \in \mathbb{Z}$, be vectors, nonzero only for n in a set of N consecutive integers, put $D_n(j) = \langle v_n, \mathbb{T}^j v_n\rangle$ and*

$$B = \sum_{n,n'\in\mathbb{Z}} \langle v_n, \mathbb{T}^{|n-n'|} v_{n'}\rangle.$$

(i) $|B| \leq 2N \max_n \sum_{j=0}^{N-1} D_n(j)$.

(ii) *If $v_n = w_n v$ with $w_n \geq 0$, and $w_n \geq \eta$ on a set of $M \geq 1$ consecutive integers, then, with $D(j) = \langle v, \mathbb{T}^j v\rangle$,*

$$B \geq \eta^2 M \sum_{j=0}^{\lceil M/2\rceil - 1} D(j).$$

(iii) *$D(j)$ is nonnegative and nonincreasing in j, and $\sum_{j<pq} D(j) \leq q \sum_{j<p} D(j)$ for integers $p, q \geq 1$.*

Proof. (i) Since $\mathbb{T}^j \geq 0$, the Cauchy–Schwarz inequality for the form $\langle \cdot, \mathbb{T}^j \cdot\rangle$ gives $|\langle v_n, \mathbb{T}^j v_{n'}\rangle| \leq \frac12 (D_n(j) + D_{n'}(j))$. Summing over n, n' and using the symmetry, $|B| \leq \sum_n \sum_{n'} D_n(|n-n'|)$, and for fixed n the integers n' with $v_{n'} \neq 0$ give each value of $|n-n'|$ at most twice, all values below N.

(ii) All terms of B are nonnegative, since $D \geq 0$. Keep the pairs in the set J of M consecutive integers. An element of J has at least $\lceil M/2\rceil - 1$ further elements of J on one of its sides, so its row contributes at least $\eta^2 \sum_{j<\lceil M/2\rceil} D(j)$.

(iii) $D(j) = \int \lambda^j d\nu$ for the spectral measure ν of $\mathbb{T}$ at v, carried by $[0,1]$. Each of the q blocks of p consecutive terms is at most the first block. □

By (iii), $D_k^{h'}$ is nonincreasing and $I_k^{h'}(p\delta) \leq p I_k^{h'}(\delta)$ for integers $p \geq 1$ and $\delta \in a_k\mathbb{Z}$. By (i) and (ii), with $v_n = \Psi_{U_k^{h'}}$ for $0 \leq a_k n < \delta$, the number $\delta^{-1} I_k^{h'}(\delta)$ lies between one half and twice the variance of the average of $\tau_n U_k^{h'}$ over these n: $I_k^{h'}(\delta)$ is the variance of the slice field averaged over a physical time δ, times δ. In the rest of this paragraph the states are invariant under translations and ϑ and reflection positive, with the transfer matrices of Appendix F, Φ is real, and $f_0 = g \otimes h$ is a product packet as in Proposition 49, with an interval I of length $\ell_I > 0$ and $c_g > 0$ such that $g \geq c_g$ on I, as in Proposition 205(iv).

Proposition 214. *Let $\delta > 0$ be dyadic and let k be so large that $\delta/a_k \in \mathbb{Z}$ and $a_k \leq \min(\ell_I/4, \delta)$.*

(i) *Put $T_g = t_1 - t_0 + \delta$ and*

$$c_-(\delta) = \frac{3c_g^2 \ell_I}{4\lceil 8\delta/(3\ell_I)\rceil}, \qquad c_+(\delta) = 2\|g\|_\infty^2 T_g \Big\lceil \frac{T_g}{\delta}\Big\rceil.$$

Then $c_-(\delta) I_k(\delta) \leq V_k \leq c_+(\delta) I_k(\delta)$ and

$$\big(\tfrac34 c_g \ell_I\big)^2 D_k(2a_k n_1) \leq C_k(0) \leq \|g\|_\infty^2 T_g^2 D_k(2a_k n_0).$$

(ii) *Consider*

($\mathrm{Z_{sl}}$) *There are a dyadic $\delta > 0$ and $C < \infty$ with $I_k(\delta) \leq C D_k(\delta)$ for all large k.*

If ($\mathrm{Z_{sl}}$) *holds for one dyadic δ, it holds for every dyadic $\delta > 0$, and then $D_k(\delta) \geq (\delta/(2C)) D_k(\delta/2)$ and $a_k D_k(0) \leq C D_k(\delta)$ for all large k.*

(iii) ($\mathrm{M_Z}$) *holds if and only if* ($\mathrm{Z_{sl}}$) *holds. In particular* ($\mathrm{M_Z}$) *holds for one product packet with spatial factor h if and only if it holds for every product packet $g' \otimes h$ with g' as above, and it implies* ($\mathrm{N_u}$) *for every dyadic $u \geq 0$.*

Proof. (i) By time translation invariance, $A_k = \sum_n w_n \tau_n U_k$ with $w_n = a_k g(a_k n)$, and $V_k = \sum_{n,n'} w_n w_{n'} D_k(a_k|n-n'|)$ by the identity of the proof of Proposition 205(iv). The n with $w_n \neq 0$ lie in $[n_0, n_1]$, a set of $N \leq (t_1 - t_0)/a_k + 1$ integers, so $a_k N \leq T_g$. Lemma 213(i) with $v_n = w_n u_k$ and $|w_n| \leq a_k \|g\|_\infty$ gives

$$V_k \leq 2\|g\|_\infty^2 (a_k N) I_k(a_k N) \leq 2\|g\|_\infty^2 T_g \Big\lceil \frac{T_g}{\delta}\Big\rceil I_k(\delta),$$

by (iii) of the lemma. The set J of n with $a_k n \in I$ consists of $M \geq \ell_I/a_k - 1 \geq \frac34 \ell_I/a_k$ consecutive integers, and $w_n \geq a_k c_g$ there. Lemma 213(ii) gives, with $b = a_k \lceil M/2\rceil \geq \frac38 \ell_I$,

$$V_k \geq a_k^2 c_g^2 M \sum_{j<\lceil M/2\rceil} D_k(a_k j) = c_g^2 (a_k M) I_k(b) \geq \tfrac34 c_g^2 \ell_I I_k(b),$$

and $I_k(\delta) \leq \lceil \delta/b\rceil I_k(b) \leq \lceil 8\delta/(3\ell_I)\rceil I_k(b)$. The bounds on $C_k(0)$ are those of the proof of Proposition 49(i), $C_k(0) = \int W^2 d\nu$ with $\|w\|_1 \lambda^{n_1} \leq W(\lambda) \leq \|w\|_1 \lambda^{n_0}$, together with $\|w\|_1 \geq a_k c_g M \geq \frac34 c_g \ell_I$ and $\|w\|_1 \leq a_k N \|g\|_\infty \leq T_g \|g\|_\infty$.

(ii) Since D_k is nonincreasing, $I_k(\delta) \geq \frac{\delta}{2} D_k(\delta/2)$ and $I_k(\delta) \geq a_k D_k(0)$, which gives the two bounds; in particular $c = \delta/(2C) \leq 1$, because $D_k(\delta) \leq D_k(\delta/2)$ and $D_k(\delta/2) > 0$ by Proposition 49(i). Let δ' be dyadic. If $\delta' < \delta$, then $I_k(\delta') \leq I_k(\delta) \leq C D_k(\delta) \leq C D_k(\delta')$. If $\delta' > \delta$, Proposition 49(ii) with $r = \delta/2$, $r' = \delta$ gives $D_k(\delta') \geq c^e D_k(\delta)$ with $e = \max(1, 2(\delta' - \delta)/\delta)$, and

$$I_k(\delta') \leq I_k(\delta) + (\delta' - \delta) D_k(\delta) \leq (C + \delta')\, c^{-e} D_k(\delta').$$

(iii) Suppose $\Upsilon_k \le \Upsilon$ for all k, and take a dyadic $\delta \le 2t_0$. Then $2a_k n_0 \ge 2t_0 \ge \delta$, and by (i)

$$I_k(\delta) \le \frac{V_k}{c_-(\delta)} \le \frac{\Upsilon C_k(0)}{c_-(\delta)} \le \frac{\Upsilon \|g\|_\infty^2 T_g^2}{c_-(\delta)} D_k(\delta),$$

which is (Z_{sl}). Conversely, assume (Z_{sl}) at δ with the constant C, and take a dyadic $R \ge \max(2t_1, \delta)$. Then $D_k(2a_k n_1) \ge D_k(R) \ge c^e D_k(\delta)$ with $c = \delta/(2C)$ and $e = \max(1, 2(R-\delta)/\delta)$, as in (ii), and by (i)

$$\Upsilon_k = \frac{V_k}{C_k(0)} \le \frac{c_+(\delta) C c^{-e}}{(\frac{3}{4} c_g \ell_I)^2}$$

for all large k; the finitely many other Υ_k are finite, since $C_k(0) > 0$. Condition (Z_{sl}) involves h and not g, and (M_Z) implies (N_u) by Proposition 205(iv). □

So for product packets (M_Z) is a statement about the slice field alone: the variance of U_k averaged over a physical time δ is at most a constant times $D_k(\delta)/\delta$. Since $I_k(\delta) \ge a_k D_k(0)$, it contains the bound $a_k \operatorname{Var} U_k \le C D_k(\delta)$: the equal-time variance of the slice field may exceed its correlation at a physical separation by at most a factor of order $1/a_k$. In terms of the spectral measure μ_k of $\mathbb{T}_k$ at u_k, in energies $E = -a_k^{-1} \log \lambda$, $I_k(\delta)$ is the integral of $a_k(1-e^{-\delta E})/(1-e^{-a_k E})$, a weight between $(1-e^{-\delta E})/E$ and that plus a_k, and (Z_{sl}) bounds it by $\int e^{-\delta E} d\mu_k$; it bounds the spectral weight at high energy, counted with the factor $1/E$, by the weight at bounded physical energy. By (ii) it contains a ratio bound for D_k, hence (N_0), and it is strictly stronger than (N_u): for the field of Proposition 205(vi), $\rho_k(u,\sigma) = e^{-m\sigma}$ while $a_k D_k(0)$, which contains $a_k \cdot a_k^{-5} a_k^6 \sum_x h(a_k x)^2$, grows like a_k^{-1}, and $D_k(\delta)$ converges. Along the constant tunings of Proposition 184(ii) the hypothesis of Theorem 43 holds and (Z_{sl}) fails, since (N_u) fails there. So (M_Z) reduces to (N_u) in one direction only, and no argument that uses (D) only through the hypothesis of Theorem 43 gives it.

Proposition 215. *Let δ and k be as in Proposition 214.*

(i) *Let $g_1, g_2 \in C_c(\mathbb{R})$ be supported in an interval of length L, $y_1, y_2 \in a_k \mathbb{Z}^3$ and $f_i = g_i \otimes h(\cdot - y_i)$. Then*

$$\big|\omega_k\big(A_k(f_1)A_k(f_2)\big)\big| \le K_L V_k \|g_1\|_\infty \|g_2\|_\infty,$$

with $K_L = 2(L+\delta)\lceil (L+\delta)/\delta \rceil / c_-(\delta)$. So ($M_{loc}$) holds for $n = 2$ on these products, along every sequence of couplings.

(ii) (M_{loc}) *holds for $n = 2$ if and only if the following holds for one dyadic $\delta > 0$, and then it holds for every dyadic $\delta > 0$:*

(M_{sp}) *For every cube $Q_3 \subset \mathbb{R}^3$ there is $C < \infty$ with $I_k^{h'}(\delta) \le C I_k(\delta)$ for all large k and all $h' \in C_c(\mathbb{R}^3)$ supported in Q_3 with $\|h'\|_\infty \le 1$.*

(iii) (M_{loc}) *holds if and only if for every even $n \ge 2$ and every cube Q there are $C_n(Q)$ and k_0 with $\omega_k(|A_k(f)|^n) \le C_n(Q) V_k^{n/2} \|f\|_\infty^n$ for $k \ge k_0$ and $f \in C_c(\mathbb{R}^4)$ supported in Q. The case $n = 2$ is equivalent to (M_{sp}), and the case n implies the cases below it.*

(iv) *Let $m > 0$, $q_k = e^{-m a_k}$, $\kappa(\mathbf{z}) = e^{-|\mathbf{z}|^2}$ on $\mathbb{R}^3$, and let ψ_k be the centred Gaussian field on $\mathbb{Z}^4$ with covariance*

$$q_k^{|x_0 - y_0|} \Big(\kappa\big(a_k(\mathbf{x}-\mathbf{y})\big) + a_k^{-1} (-1)^{\Sigma_{i=1}^3 (x_i - y_i)} \Big),$$

$A_k(f) = \sum_x a_k^4 f(a_k x) \psi_k(x)$, and $f_0 = g \otimes h$ a product packet as in Proposition 49 with $g, h \ge 0$ nonzero and continuously differentiable. The law of ψ_k is invariant under translations and Γ and reflection positive for the time reflections through sites and bisecting links, its transfer matrix has gap $m a_k$, $\rho_k(u,\sigma) = e^{-m\sigma}$ for all u, σ, so (RC) *holds for every σ_0, and* (M_Z) *holds. But* (M_{sp}) *fails, and* (M_{loc}) *and* (M) *fail for $n = 2$.*

(v) *Let $m > 0$, $q_k = e^{-m a_k}$, $\varepsilon_k = a_k$, let $(\xi_n)_{n \in \mathbb{Z}}$ be the stationary Markov chain on $\{-1, 0, 1\}$ with the law π of Remark 57 at $\varepsilon = \varepsilon_k$ and transition operator $P = q_k I + (1 - q_k)\Pi$, and $X_n = \xi_n / \sqrt{\varepsilon_k}$. Let $\psi_k(x) = X_{x_0}$, and A_k, f_0 as in* (iv). *This field has the properties listed in* (iv), *and* (M_{sp}) *holds; but $\omega_k(A_k^4)/V_k^2 \to \infty$, so* ($M_{loc}$) *and* (M) *fail for $n = 4$.*

Proof. (i) Since Φ is real, $A_k(\bar f) = \overline{A_k(f)}$, so by the Cauchy–Schwarz inequality

$$\big|\omega_k\big(A_k(f_1)A_k(f_2)\big)\big| \le \prod_{i=1}^2 \omega_k\big(|A_k(f_i)|^2\big)^{1/2},$$

and it suffices to bound $\omega_k(|A_k(f)|^2)$ for $f = g \otimes h(\cdot - y)$ by $K_L V_k \|g\|_\infty^2$. By spatial translation invariance y may be taken to be 0. As in the proof of Proposition 214(i), $\omega_k(|A_k(f)|^2)$ is B of Lemma 213 with $v_n = a_k g(a_k n) u_k$, nonzero for n in a set of $N \le L/a_k + 1$ consecutive integers, so

$$\begin{aligned}\omega_k(|A_k(f)|^2) &\le 2\|g\|_\infty^2 (a_k N) I_k(a_k N) \\ &\le 2\|g\|_\infty^2 (L+\delta) \Big\lceil \frac{L+\delta}{\delta} \Big\rceil I_k(\delta),\end{aligned}$$

and Proposition 214(i) gives the claim.

(ii) Independence of δ: for dyadic $\delta' > \delta$, $I_k^{h'}(\delta') \le \lceil \delta'/\delta \rceil I_k^{h'}(\delta) \le \lceil \delta'/\delta \rceil C I_k(\delta')$, and for $\delta' < \delta$, $I_k^{h'}(\delta') \le I_k^{h'}(\delta) \le C I_k(\delta) \le C \lceil \delta/\delta' \rceil I_k(\delta')$. Assume ($M_{sp}$), and let $f \in C_c(\mathbb{R}^4)$ be supported in a cube $Q = [t_Q, t_Q + L] \times Q_3$. As in (i) it suffices to bound $\omega_k(|A_k(f)|^2)$. By time translation invariance $A_k(f) = \sum_n \tau_n(a_k U_k^{h_n})$ with $h_n = f(a_k n, \cdot)$, supported in Q_3 with $\|h_n\|_\infty \le \|f\|_\infty$, and nonzero for n in a set of $N \le L/a_k + 1$ consecutive integers. The identity of the proof of Proposition 205(iv) and translation invariance give $\omega_k(\bar U_k^{h_n} \tau_j U_k^{h_{n'}}) = \langle \Psi_{U_k^{h_n}}, \mathbb{T}_k^j \Psi_{U_k^{h_{n'}}} \rangle$ for $j \ge 0$, so $\omega_k(|A_k(f)|^2)$ is B of Lemma 213 with $v_n = a_k \Psi_{U_k^{h_n}}$, and

$$\begin{aligned}\omega_k(|A_k(f)|^2) &\le 2(a_k N) \max_n I_k^{h_n}(a_k N) \\ &\le \frac{2(L+\delta)C}{c_-(\delta)} \Big\lceil \frac{L+\delta}{\delta} \Big\rceil \|f\|_\infty^2 V_k,\end{aligned}$$

by (iii) of the lemma, (M_{sp}) and Proposition 214(i). Conversely, let $g_* \in C_c(\mathbb{R})$ satisfy $0 \le g_* \le 1$, $g_* = 0$ outside $[0,1]$ and $g_* = 1$ on $[\frac{1}{4}, \frac{3}{4}]$, and let h' be as in (M_{sp}). For $a_k \le \frac{1}{4}$ the n with $a_k n \in [\frac{1}{4}, \frac{3}{4}]$ are $M' \ge 1/(4a_k)$ consecutive integers, and

Lemma 213(ii) with $v_n = a_k g_*(a_k n)\Psi_{U_k^{h'}}$ and $\eta = a_k$ gives, with $b = a_k\lceil M'/2\rceil \geq \frac{1}{8}$,

$$\omega_k\big(|A_k(g_*\otimes h')|^2\big) \geq a_k M' I_k^{h'}(b) \geq \frac{I_k^{h'}(\delta)}{4\lceil 8\delta\rceil}.$$

Since $A_k(g_*\otimes \bar h') = \overline{A_k(g_*\otimes h')}$, $(\mathrm{M_{loc}})$ for $n=2$ on a cube $Q \supseteq [0,1]\times Q_3$ bounds the left side by $M_2^{\mathrm{loc}}(Q)V_k \leq M_2^{\mathrm{loc}}(Q)c_+(\delta)I_k(\delta)$, which is $(\mathrm{M_{sp}})$.

(iii) The bounds are $(\mathrm{M_{loc}})$ with f_j equal to $\bar f$ and f alternately, since $A_k(\bar f) = \overline{A_k(f)}$. Conversely, for $f_1,\dots,f_n$ supported in Q and an even $n' \geq n$, Hölder's inequality gives $|\omega_k(\prod_j A_k(f_j))| \leq \prod_j \omega_k(|A_k(f_j)|^{n'})^{1/n'}$. The case $n=2$ is $(\mathrm{M_{loc}})$ for $n=2$ by the Cauchy–Schwarz inequality, hence $(\mathrm{M_{sp}})$ by (ii), and the last claim is the monotonicity of L^n norms for the probability measure ω_k.

(iv) The spatial kernel is the sum of two positive definite functions on $\mathbb{Z}^3$, the first because the Fourier transform of κ is positive, the second because it is $\varepsilon_{\mathbf{x}}\varepsilon_{\mathbf{y}}$ with $\varepsilon_{\mathbf{x}} = (-1)^{x_1+x_2+x_3}$; both are invariant under the signed permutations of the spatial coordinates. The field is the stationary Markov chain of the proof of Proposition 199(a), with γ the Gaussian law of this spatial covariance, which gives the invariance, reflection positivity and the gap, and as in Proposition 205(vi) the time factor gives $\rho_k(u,\sigma) = e^{-m\sigma}$. Put $S_k = \sum_{\mathbf{x}} a_k^3 h(a_k\mathbf{x})\varepsilon_{\mathbf{x}}$. Grouping $\mathbf{x}$ with x_1 even with $\mathbf{x}+e_1$, $|S_k| \leq a_k\|\partial_1 h\|_\infty (L_h+2a_k)^3$, where L_h is the side of a cube containing the support of h. The part of V_k from the second term of the kernel is

$$a_k^{-1}\sum_{n,n'} a_k^2 g(a_k n)g(a_k n')q_k^{|n-n'|}S_k^2 \leq a_k^{-1}\Big(a_k\sum_n g(a_k n)\Big)^2 S_k^2,$$

which tends to 0, and the same holds for $C_k(0)$. The parts from κ are Riemann sums converging to

$$\iint f_0(x)f_0(y)e^{-m|x_0-y_0|}\kappa(\mathbf{x}-\mathbf{y})\,dx\,dy > 0,$$
$$\iint g(t)g(s)e^{-m(t+s)}dt\,ds \iint h(\mathbf{x})h(\mathbf{y})\kappa(\mathbf{x}-\mathbf{y})\,d\mathbf{x}\,d\mathbf{y} > 0.$$

So Υ_k converges, which is $(\mathrm{M_Z})$. Let

$$f_k(t,\mathbf{x}) = g(t)h(\mathbf{x})\cos\big(\pi(x_1+x_2+x_3)/a_k\big),$$

a continuous function supported in the support of f_0, with $\|f_k\|_\infty \leq \|g\|_\infty\|h\|_\infty$, equal to $g(a_k x_0)h(a_k\mathbf{x})\varepsilon_{\mathbf{x}}$ at the point $a_k x$. Both terms of the kernel are positive definite, so

$$\omega_k\big(A_k(f_k)^2\big) \geq a_k^{-1}\sum_{n,n'} a_k^2 g(a_k n)g(a_k n')q_k^{|n-n'|}\Big(\sum_{\mathbf{x}} a_k^3 h(a_k\mathbf{x})\Big)^2,$$

and the factor of a_k^{-1} on the right converges to

$$\iint g(t)g(s)e^{-m|t-s|}dt\,ds\Big(\int h\Big)^2 > 0,$$

while V_k converges. So $(\mathrm{M_{loc}})$ fails for $n=2$, and with it $(\mathrm{M_{sp}})$, by (ii), and (M), by Proposition 205(ii).

(v) $P = \Pi + q_k(I-\Pi)$ is symmetric on $L^2(\pi)$ with spectrum $\{1,q_k\}$ and 1 simple, and $P \geq 0$; the argument of the proof of Proposition 199(a) gives reflection positivity, and the gap is ma_k. The field does not depend on the spatial coordinates, which gives the invariance. Since $X \perp 1$ and $\pi(X^2)=1$, $P^jX = q_k^jX$ and the covariance is $q_k^{|x_0-y_0|}$, that of Proposition 199(b); so $\rho_k(u,\sigma) = e^{-m\sigma}$, and V_k and $C_k(0)$ converge to positive limits, which gives $(\mathrm{M_Z})$. Further $U_k^{h'} = (\sum_{\mathbf{x}} a_k^3 h'(a_k\mathbf{x}))X_0$, so $I_k^{h'}(\delta)/I_k(\delta) = |\sum a_k^3 h'(a_k\mathbf{x})|^2/(\sum a_k^3 h(a_k\mathbf{x}))^2$, which is bounded for h' as in $(\mathrm{M_{sp}})$, since the denominator converges to $(\int h)^2 > 0$. For times $s_1 \leq s_2 \leq s_3 \leq s_4$ with $j_i = s_{i+1}-s_i$, the Markov property and $P^j = \Pi + q_k^j(I-\Pi)$ give, as in Remark 57 and with $X^3 = X/\varepsilon$,

$$\mathbb{E}\Big[\prod_{i=1}^4 X_{s_i}\Big] = q_k^{j_1+j_3}\big(q_k^{j_2}/\varepsilon_k + 1 - q_k^{j_2}\big) \geq \varepsilon_k^{-1}q_k^{s_4-s_1}.$$

With $F_n = a_k^3\sum_{\mathbf{x}} f_0(a_k n, a_k\mathbf{x}) \geq 0$, $A_k = \sum_n a_k F_n X_n$, and $F_n = 0$ unless $a_k n \in [t_0,t_1]$, so

$$\omega_k(A_k^4) \geq \varepsilon_k^{-1}e^{-m(t_1-t_0)}\Big(a_k\sum_n F_n\Big)^4,$$

and $a_k\sum_n F_n \to \int f_0 > 0$. Since $\varepsilon_k = a_k \to 0$ and V_k converges, the ratio diverges. □

Part (i) is the part of $(\mathrm{M_{loc}})$ that the positivity of the transfer matrix gives: test functions in time, on both sides of the reflection plane and of either sign, cost no more than the packet f_0, for every state with a positive transfer matrix. What remains for $n=2$ is $(\mathrm{M_{sp}})$, a comparison between spatial profiles at one spacing, which involves neither the reflection nor the tuning. Example (iv) has a spatial mode at the lattice scale whose variance diverges, and the time structure of an exact Markov field; example (v) has no spatial structure and four-point functions of order $1/\varepsilon_k$, as in Remark 57. For a Gaussian field the linear heat flow in space at the flow time τ/a_k^2 multiplies the variance of the lattice-scale mode of (iv) by $e^{-c\tau/a_k^2}$, with $c>0$ fixed by the normalisation of the flow, and removes the divergence there. For the gauge field the flow does not do this configuration by configuration.

Proposition 216. *Let $G = \mathrm{SU}(N)$, $N \geq 2$, $\zeta = e^{2\pi i/N}$ and $c_N = N(1-\cos(2\pi/N))$. Let $m : \mathbb{Z}\to\mathbb{Z}$ satisfy $m(j+1)-m(j) = 1$ for $j \equiv 0,1$ and $m(j+1)-m(j) = 0$ for $j \equiv 2,3$ modulo 4, and let $U^\sharp$ be the configuration with all temporal links equal to 1 and spatial links*

$$U^\sharp_{(x,x+e_1)} = U^\sharp_{(x,x+e_3)} = 1, \qquad U^\sharp_{(x,x+e_2)} = \zeta^{m(x_1)}.$$

(i) *The spatial plaquettes of $U^\sharp$ in the plane of e_1,e_2 with base point x are $\zeta^{\pm1}$ for $x_1 \equiv 0,1$ and 1 for $x_1 \equiv 2,3$ modulo 4, all other spatial plaquettes are 1, and $\Phi_s(gU^\sharp) = gU^\sharp$ for every gauge transformation g and every $s\in\mathbb{R}$.*

(ii) *Let Φ be the sum of $N - \mathrm{Re}\,\mathrm{Tr}\,U_p$ over the twelve spatial plaquettes with a corner at 0. Then for all x and s*

$$\Phi_x\circ\Phi_s(U^\sharp) = 2c_N\Big(1+\cos\frac{\pi(x_1-1)}{2}\Big).$$

(iii) *Let $g \in C_c(\mathbb{R})$ and $h \in C_c(\mathbb{R}^3)$ be nonnegative and $h_k^\sharp(\mathbf{x}) = h(\mathbf{x})\cos(\pi(x_1/a_k - 1)/2)$, and let ω be as in Proposition 206(iii). At every flow time the ω-essential supremum and infimum of $A_k(g \otimes h_k^\sharp)$ differ by at least*

$$4c_N a_k^4 \sum_n g(a_k n) \sum_{\mathbf{x}: x_1 \equiv 1} h(a_k \mathbf{x}), \tag{4}$$

which converges to $c_N \int g \int h$. For $A_k(g \otimes h)$ the difference is at least

$$12 c_N a_k^4 \sum_n g(a_k n) \sum_{\mathbf{x}} h(a_k \mathbf{x}),$$

which converges to $12 c_N \int g \int h$.

Proof. (i) With the plaquette formula of the proof of Proposition 206(i), the plaquette in the plane of e_1, e_2 at x is $\zeta^{m(x_1+1)-m(x_1)}$; in the plane of e_1, e_3 all four links are 1, and in the plane of e_2, e_3 it is $\zeta^{m(x_1)}\zeta^{-m(x_1)} = 1$, because x and $x + e_3$ have the same first coordinate. All links are central, and the computation of the proof of Proposition 206(i) shows that the gradient in the flow equation vanishes; uniqueness and gauge covariance give the rest.

(ii) Only the four plaquettes of the plane of e_1, e_2 with a corner at x can be nontrivial; their base points have first coordinates x_1 and $x_1 - 1$, twice each. So $\Phi_x(U^\sharp) = 2c_N(d(x_1) + d(x_1 - 1))$ with $d(j) = m(j+1) - m(j)$, which is $2c_N$ times $1, 2, 1, 0$ for $x_1 \equiv 0, 1, 2, 3$, and the flow does not move $U^\sharp$.

(iii) With $c_x = \cos(\pi(x_1 - 1)/2) \in \{1, 0, -1, 0\}$ for $x_1 \equiv 1, 2, 3, 0$,

$$\sum_x a_k^4 g(a_k x_0) h_k^\sharp(a_k \mathbf{x})\, \Phi_x \circ \Phi_s(U^\sharp) = 2c_N \sum_x a_k^4 g(a_k x_0) h(a_k \mathbf{x})(c_x + c_x^2),$$

and $c_x + c_x^2$ is 2 for $x_1 \equiv 1$ and 0 otherwise; at the configuration 1 the sum is 0. The proof of Proposition 206(iii) applies with $U^\sharp$ in place of U^*; the centring constant cancels in the difference. The points with $x_1 \equiv 1$ modulo 4 form a sublattice of index four, which gives the limit. The second statement is Proposition 206(iii). □

At every flow time the packet with a spatial profile oscillating at the lattice scale has, at the stationary configurations, a range of values of the same order as the smooth packet, the ratio of the two lower bounds tending to $1/12$. A proof of ($\mathrm{M_{sp}}$) by way of the flow has therefore to bound the probability of the configurations near those with nontrivial central plaquettes, uniformly along the tuning; this is an estimate on large fields of the Wilson measure at the couplings β_k, of the kind of [21, Sec. XII E], compared with the size of $I_k(\delta)$, which is not known.

Remark 217. For the flowed fields along a tuning, with a product packet $f_0 = g \otimes h$, the five statements stand as follows.

($\mathrm{M_{loc}}$): the case $n = 1$ holds since the packets are centred, and the case $n = 2$ holds on the products of time profiles with the lattice translates of h (Proposition 215(i)), for every state with a positive transfer matrix. Not proved: ($\mathrm{M_{sp}}$), equivalent to the case $n = 2$, and the bounds $\omega_k(|A_k(f)|^n) \le C_n(Q) V_k^{n/2} \|f\|_\infty^n$ for even $n \ge 4$, which with it are equivalent to ($\mathrm{M_{loc}}$). Both concern one spacing at a time, without reflection; they are not consequences of the gap, (N_u), (RC), (M_Z), reflection positivity and the symmetry Γ, by Proposition 215(iv), (v), so no argument that uses (D) only through the transfer-matrix gap, together with (N_u), gives them.

(M_Z): equivalent to ($\mathrm{Z_{sl}}$), $I_k(\delta) \le C D_k(\delta)$ for the slice field, and independent of g (Proposition 214). Not proved. It implies (N_u) for every u; it is not implied by (N_u), the gap and the lattice symmetries (Proposition 205(vi)), nor by the hypothesis of Theorem 43 (Proposition 184(ii)). The missing estimate is a bound on the high-energy spectral weight of the slice field, with the factor $1/E$, and on $a_k \operatorname{Var} U_k$, by its correlation at a physical separation.

(W): equivalent to $\Delta_k^{E_{12}}(f_1, \dots, f_n) \to 0$ for all n and $f_j \in C_c^\infty(\mathbb{R}^4)$ (Proposition 208(ii)); a bound $C(f_1, \dots, f_n) a_k^p$ with $p > 0$ suffices. Not proved. Example (a) of Proposition 199 has the gap in every direction, (RC), (M), hence ($\mathrm{M_{sp}}$) and ($\mathrm{Z_{sl}}$), and (U), and not (W); so no argument that uses (D) only through the gap, together with (N_u) and (M), gives (W). The lattice symmetry gives the cancellation of $\Delta_k^{E_{12}}$ only on average over the angle.

(U1) and (U2), given (RC): (U1) is $\delta_k \to 0$, for which $\sum_k |\delta_k| < \infty$ on a set with dense span suffices (Proposition 211(iii)), and (U2) is the total disconnectedness of $\mathfrak{L}$. Not proved. Example (iv) of Proposition 211 and its alternating variant satisfy (RC), (M) and (W), each with one of (U1), (U2) and not the other, so neither follows from the gap, (N_u), (RC), (M) and (W); (U1) compares two spacings, which no statement at one spacing contains.

The examples are Gaussian and Markov fields, not lattice gauge theories. They show that a proof of any of the four open statements has to use properties of the Wilson measure along the tuning beyond translation and Γ invariance, reflection positivity, the transfer-matrix gap, the normalisation (RC) and the parts of (M) proved or assumed here.

The spatial spectral measure of the slice field. The comparison ($\mathrm{M_{sp}}$) is a statement about the distribution in spatial momentum of the time-integrated correlations of the slice field. In this paragraph the states are invariant under translations and ϑ and reflection positive, with the transfer matrices of Appendix F, Φ is real, $\delta > 0$ is dyadic and k is so large that $n_\delta = \delta/a_k$ is an even integer. Write $\mathring{\phi}_k(x) = \varphi_k(x) - \omega_k(\varphi_k(x))$,

$$K_k(z) = a_k \sum_{j=0}^{n_\delta - 1} \omega_k\big(\mathring{\phi}_k(0,0)\, \mathring{\phi}_k(j,z)\big),$$

$$K_k^V(z) = a_k^2 \sum_{n,n'=0}^{n_\delta - 1} \omega_k\big(\mathring{\phi}_k(n,0)\, \mathring{\phi}_k(n',z)\big)$$

for $z \in \mathbb{Z}^3$. For a finitely supported $c : \mathbb{Z}^3 \to \mathbb{C}$ put $\hat{c}(p) = \sum_x c_x e^{ip\cdot x}$ on $\mathbf{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, $U_k^c = \sum_x c_x \mathring{\phi}_k(0,x)$, and

$$I_k^c = a_k \sum_{j=0}^{n_\delta - 1} \omega_k\big(\bar{U}_k^c\, \tau_j U_k^c\big), \qquad V_k^c = \omega_k\Big(\Big| a_k \sum_{n=0}^{n_\delta - 1} \tau_n U_k^c \Big|^2\Big).$$

For $c_x = a_k^3 h'(a_k x)$, $U_k^c = U_k^{h'}$ and $I_k^c = I_k^{h'}(\delta)$. By translation invariance $I_k^c = \sum_{x,y} \bar{c}_x c_y K_k(y-x)$ and $V_k^c = \sum_{x,y} \bar{c}_x c_y K_k^V(y-x)$. For a closed box $\mathscr{B}_r(p_0) = \{p : |p_i - p_{0,i}| \le r,\ i = 1,2,3\}$ in $\mathbf{T}^3$, $r > 0$, and a finite positive measure ϖ on $\mathbf{T}^3$, put $\bar{\varpi}(r) = \sup_{p_0} \varpi(\mathscr{B}_r(p_0))$. A box of half-side r' is covered by $\lceil r'/r \rceil^3$ boxes of half-side r, so $\bar{\varpi}(r') \le \lceil r'/r \rceil^3 \bar{\varpi}(r)$.

Lemma 218. (i) *There are unique finite positive measures ϖ_k and ϖ_k^V on $\mathbf{T}^3$ with $K_k(z) = \int e^{ip\cdot z} d\varpi_k(p)$ and $K_k^V(z) = \int e^{ip\cdot z} d\varpi_k^V(p)$ for $z \in \mathbb{Z}^3$. For every finitely supported c*

$$I_k^c = \int |\hat{c}|^2 d\varpi_k, \qquad V_k^c = \int |\hat{c}|^2 d\varpi_k^V, \qquad \tfrac{\delta}{2} I_k^c \le V_k^c \le 2\delta I_k^c,$$

and $\varpi_k(\mathbf{T}^3) = K_k(0) \le \delta \|\Phi\|_\infty^2$.

(ii) *Let ϖ be a finite positive measure on $\mathbf{T}^3$, $L \ge 4a_k$, and $c_x = a_k^3 h'(a_k x)$ with h' supported in a cube of side L and $|h'| \le 1$. Then*

$$\int |\hat{c}|^2 d\varpi \le 7 \cdot 10^6 L^6 \bar{\varpi}(\pi a_k / L).$$

(iii) *Let $\chi \in C_c(\mathbb{R}^3)$ take values in $[0,1]$, be supported in a cube Q' of side $\ell \ge 4a_k$ and equal 1 on the cube of side $\ell/2$ with the same centre, and for $p_0 \in \mathbf{T}^3$ let $h'(\mathbf{y}) = \chi(\mathbf{y}) e^{-ip_0\cdot \mathbf{y}/a_k}$. Then for every finite positive measure ϖ*

$$\varpi\big(\mathscr{B}_{a_k/(3\ell)}(p_0)\big) \le 16384\, \ell^{-6} \int |\hat{c}|^2 d\varpi, \qquad c_x = a_k^3 h'(a_k x).$$

(iv) *Let $h \in C_c(\mathbb{R}^3)$ be real with $\int h \ne 0$, supported in a cube of side L_h, and $\rho_h = L_h^{-1}(|\int h|/(72 \int |h|))^{1/2}$. For all large k, with $c_x = a_k^3 h(a_k x)$, $|\hat{c}| \ge \frac{3}{4}|\int h|$ on $\mathscr{B}_{a_k\rho_h}(0)$, so $\int |\hat{c}|^2 d\varpi \ge \frac{1}{2}(\int h)^2 \varpi(\mathscr{B}_{a_k\rho_h}(0))$ for every finite positive measure ϖ.*

(v) *Let $\varkappa(p) = \sum_{i=1}^3 4\sin^2(p_i/2)$, $m \ge 1$, $h \in C_c^{2m}(\mathbb{R}^3)$ real, supported in a cube of side L_h, $\|h\|_{2m}$ the largest supremum of a derivative of order $2m$ of h, and $\Theta_m = 3^m \|h\|_{2m}(L_h + 2m + 1)^3$. For $a_k \le 1$ and $c_x = a_k^3 h(a_k x)$, $\varkappa^m |\hat{c}| \le \Theta_m a_k^{2m}$ on $\mathbf{T}^3$.*

Proof. (i) With $u_c = \Psi_{U_k^c}$, the identity of the proof of Proposition 205(iv) gives $\omega_k(\bar{U}_k^c \tau_j U_k^c) = \langle u_c, \mathbb{T}_k^j u_c \rangle \ge 0$ for $j \ge 0$, so $I_k^c \ge 0$ and K_k is positive definite; $V_k^c \ge 0$ shows the same for K_k^V. For a positive definite bounded K on $\mathbb{Z}^3$ and an integer $M \ge 1$, the function

$$f_M(p) = M^{-3} \sum_{x,y \in \{0,\dots,M-1\}^3} K(y-x) e^{-ip\cdot(y-x)}$$

is nonnegative, and $\int e^{ip\cdot z} f_M(p) \frac{dp}{(2\pi)^3} = K(z) \prod_i (1 - |z_i|/M)_+$. The measures $f_M\, dp/(2\pi)^3$ have mass $K(0)$, so by Lemma 353 a subsequence converges weakly to a measure with Fourier coefficients $K(z)$; trigonometric polynomials are dense in $C(\mathbf{T}^3)$, which gives uniqueness. Then $\sum_{x,y} \bar{c}_x c_y \int e^{ip\cdot(y-x)} d\varpi = \int |\hat{c}|^2 d\varpi$. The numbers $\langle u_c, \mathbb{T}_k^j u_c \rangle$ are real, so translation invariance and complex conjugation give $\omega_k(\overline{\tau_n U_k^c}\, \tau_{n'} U_k^c) = \langle u_c, \mathbb{T}_k^{|n-n'|} u_c \rangle$ for all n, n', and V_k^c is B of Lemma 213 with $v_n = a_k u_c$ for $0 \le n < n_\delta$. Part (i) of that lemma gives $V_k^c \le 2 n_\delta a_k I_k^c = 2\delta I_k^c$; part (ii) with $\eta = a_k$, $M = n_\delta$ and part (iii) give $V_k^c \ge \delta\, a_k \sum_{j < n_\delta/2} \langle u_c, \mathbb{T}_k^j u_c \rangle \ge \frac{\delta}{2} I_k^c$. Finally $|\omega_k(\mathring{\varphi}_k(0,0)\mathring{\varphi}_k(j,0))| \le \|\Phi\|_\infty^2$, since φ_k takes values in $[-\|\Phi\|_\infty, \|\Phi\|_\infty]$.

(ii) Translating c by a lattice vector multiplies $\hat{c}$ by a phase. With $n = \lceil L/(2a_k) \rceil$ the support of c may therefore be taken in $\{|x_i| \le n\}$; it has at most $(L/a_k + 1)^3$ points. Let $F_n(t) = \sum_{|j| \le n} (1 - \frac{|j|}{n+1}) e^{ijt} = \frac{1}{n+1} \big(\sin \frac{(n+1)t}{2} / \sin \frac{t}{2}\big)^2$ be the Fejér kernel and $W_n = 2F_{2n+1} - F_n$. Its coefficients equal 1 for $|j| \le n$, $\int |W_n| \frac{dt}{2\pi} \le 3$, and from $F_n \le n+1$ and $\sin \frac{t}{2} \ge \frac{t}{\pi}$ on $[0,\pi]$, $F_n(t) \le 2(n+1)/(1 + ((n+1)t/\pi)^2)$ for $|t| \le \pi$, so

$$|W_n(t)| \le \frac{10(n+1)}{1 + ((n+1)t/\pi)^2} \qquad (|t| \le \pi).$$

With $W(q) = \prod_i W_n(q_i)$, $\hat{c}(p) = \int \hat{c}(q) W(p-q) dq'$, where $dq' = dq/(2\pi)^3$, and the Cauchy–Schwarz inequality gives $|\hat{c}(p)|^2 \le 27 \int |\hat{c}(q)|^2 |W(p-q)| dq'$. Divide $\mathbf{T}^3$ into the $(2n+2)^3$ half-open boxes of side $\pi/(n+1)$ centred at the points $q + \pi j/(n+1)$, $j \in \{-n, \dots, n+1\}^3$; each lies in a closed box of half-side $\pi/(2(n+1)) \le \pi a_k/L$, and on the box with index j the i-th factor of $|W(p-q)|$ is at most $10(n+1)$ if $j_i = 0$ and at most $10(n+1)/(1 + (|j_i| - \frac{1}{2})^2)$ otherwise. Since $(1 + (x - \frac{1}{2})^2)^{-1}$ decreases on $[\frac{1}{2}, \infty)$, $\sum_{j \ge 2} (1 + (j - \frac{1}{2})^2)^{-1} \le \frac{\pi}{2} - \arctan \frac{1}{2} \le 1.1125$, using $\arctan \frac{1}{2} \ge \frac{1}{2} - \frac{1}{24}$. The bounds of the i-th factor therefore add up to at most $10(n+1)(1 + 2(0.8 + 1.1125)) \le 50(n+1)$, the term $j = 1$ being 0.8, and

$$\int |W(p-q)| d\varpi(p) \le \big(50(n+1)\big)^3 \bar{\varpi}(\pi a_k / L).$$

Integrating against $|\hat{c}(q)|^2 dq'$ and using $\int |\hat{c}|^2 dq' = \sum_x |c_x|^2 \le a_k^3 (L + a_k)^3$ and $n + 1 \le (L + 4a_k)/(2a_k)$,

$$\int |\hat{c}|^2 d\varpi \le \frac{27 \cdot 50^3}{8} (L + a_k)^3 (L + 4a_k)^3\, \bar{\varpi}(\pi a_k / L),$$

and $\frac{27 \cdot 50^3}{8} (L + a_k)^3 (L + 4a_k)^3 \le 421875 (\frac{5}{4})^3\, 8 L^6 < 7 \cdot 10^6 L^6$ for $a_k \le L/4$.

(iii) Here $\hat{c}(p) = \sum_x a_k^3 \chi(a_k x) e^{i(p - p_0)\cdot x}$. Let $\mathbf{y}_0$ be the centre of Q' and $x_0 \in \mathbb{Z}^3$ a nearest lattice point to $\mathbf{y}_0/a_k$. For $a_k x \in Q'$, $|x_i - x_{0,i}| \le \ell/(2a_k) + \frac{1}{2} \le \frac{5}{8} \ell/a_k$, so for $p \in \mathscr{B}_{a_k/(3\ell)}(p_0)$ the phase $(p - p_0)\cdot(x - x_0)$ is at most $\frac{5}{8}$ in absolute value and $|\hat{c}(p)| \ge \cos(1) \sum_x a_k^3 \chi(a_k x) \ge \frac{1}{2} a_k^3 \#\{x : a_k x \in Q''\}$, Q'' the middle cube. It has at least $\lfloor \ell/(2a_k) \rfloor \ge \ell/(4a_k)$ lattice points in each direction, so $|\hat{c}(p)| \ge \ell^3/128$ on the box, and $\varpi(\mathscr{B}_{a_k/(3\ell)}(p_0))(\ell^3/128)^2 \le \int |\hat{c}|^2 d\varpi$.

(iv) Let $\mathbf{x}_*$ be the centre of the cube divided by a_k. For $p \in \mathscr{B}_{a_k\rho_h}(0)$ and $a_k x$ in the cube, $\theta_x = p \cdot (x - \mathbf{x}_*)$ satisfies $|\theta_x| \le 3 a_k \rho_h L_h/(2a_k)$, so $\theta_x^2/2 \le \frac{9}{8} \rho_h^2 L_h^2 = |\int h|/(64 \int |h|)$, and

$$\begin{aligned} |\hat{c}(p)| &\ge \Big| \sum_x a_k^3 h(a_k x) \cos \theta_x \Big| \\ &\ge \Big| \sum_x a_k^3 h(a_k x) \Big| - \frac{|\int h|}{64 \int |h|} \sum_x a_k^3 |h(a_k x)|. \end{aligned}$$

For large k the Riemann sums are within $\frac{1}{8}|\int h|$ of $\int h$ and below $2 \int |h|$, which gives $|\hat{c}| \ge (\frac{7}{8} - \frac{1}{32})|\int h|$.

(v) With $(\Delta c)_x = \sum_i (c_{x+e_i} + c_{x-e_i} - 2c_x)$, $\widehat{\Delta c} = -\varkappa \hat{c}$, so $\varkappa^m |\hat{c}| \le \sum_x |(\Delta^m c)_x|$. The operator Δ^m is a sum of 3^m products of m second differences $D_i\phi(x) = \phi(x+e_i) + \phi(x-e_i) - 2\phi(x) = \int_{-1}^{1} (1-|t|)\partial_t^2 \phi(x+te_i)\,dt$, so $|(\Delta^m c)_x| \le 3^m a_k^{3+2m} \|h\|_{2m}$, and $(\Delta^m c)_x = 0$ unless x is within distance m of $a_k^{-1}\operatorname{supp} h$ in each coordinate, which leaves at most $(L_h/a_k + 2m + 1)^3$ points. □

Proposition 219. *Let $f_0 = g \otimes h$ be a product packet as in Proposition 49, so that $I_k = \int |\hat{c}_h|^2 d\varpi_k$ with $(c_h)_x = a_k^3 h(a_k x)$.*

(i) ($\mathrm{M_{sp}}$) *holds at δ if and only if there are $\rho > 0$ and $C < \infty$ with*

$$\bar{\varpi}_k(a_k\rho) \le C I_k(\delta) \quad \text{for all large } k;$$

if it holds for ρ' with C, it holds for every $\rho > 0$ with $\lceil \rho/\rho' \rceil^3 C$. If ($\mathrm{M_{sp}}$) *holds with the constant C for the unit cube, the bound holds with $\rho = \frac{1}{3}$ and $16384C$; conversely the bound gives* ($\mathrm{M_{sp}}$) *with the constant $7 \cdot 10^6 L^6 \lceil \pi/(L\rho) \rceil^3 C$ for cubes of side L.*

(ii) *If the bound of* (i) *holds with ρ and C, then $K_k(0) \le C \lceil \pi/(a_k\rho) \rceil^3 I_k(\delta)$ for all large k.*

(iii) *If $\int h \neq 0$,* ($\mathrm{M_{sp}}$) *holds as soon as $\bar{\varpi}_k(a_k\rho_h) \le C\varpi_k(\mathscr{B}_{a_k\rho_h}(0))$ for all large k, with ρ_h as in Lemma 218*(iv)*.*

Proof. (i) Since $\bar{\varpi}(r') \le \lceil r'/r \rceil^3 \bar{\varpi}(r)$ and $\bar{\varpi}$ is nondecreasing, the bound for ρ' gives it for every $\rho > 0$. If ($\mathrm{M_{sp}}$) holds for the unit cube, apply Lemma 218(iii) with $\ell = 1$ to a fixed χ and every p_0: $\varpi_k(\mathscr{B}_{a_k/3}(p_0)) \le 16384 I_k^{h'}(\delta) \le 16384 C I_k(\delta)$; if only real h' are admitted, the real and imaginary parts of the probe give the same with $4 \cdot 16384C$, since $|\hat{c}|^2 \le 2(|\widehat{\mathrm{Re}\, c}|^2 + |\widehat{\mathrm{Im}\, c}|^2)$. Conversely, for h' supported in a cube of side L with $|h'| \le 1$, Lemma 218(ii) gives $I_k^{h'}(\delta) \le 7 \cdot 10^6 L^6 \bar{\varpi}_k(\pi a_k/L) \le 7 \cdot 10^6 L^6 \lceil \pi/(L\rho) \rceil^3 \bar{\varpi}_k(a_k\rho)$ once $a_k \le L/4$.

(ii) $K_k(0) = \varpi_k(\mathbf{T}^3)$, and $\mathbf{T}^3$ is covered by $\lceil \pi/(a_k\rho) \rceil^3$ boxes of half-side $a_k\rho$.

(iii) Lemma 218(iv) gives $\varpi_k(\mathscr{B}_{a_k\rho_h}(0)) \le 2I_k(\delta)/(\int h)^2$, and (i) applies with $\rho = \rho_h$. □

So ($\mathrm{M_{sp}}$) says that no box of lattice momenta of side of order a_k carries more of the time-integrated spectral weight than a constant times $I_k(\delta)$; by (ii) it contains the bound $K_k(0) \le C' a_k^{-3} I_k(\delta)$ on the time-integrated autocorrelation at one site. Part (iii) reduces it, for $\int h \neq 0$, to a comparison of the weight of every box with that of the box at zero momentum. Reflection positivity in the spatial directions gives that comparison when it holds exactly. For $r \ge 0$, $i \in \{1,2,3\}$, let ι_i^{s} and ι_i^{b} be the reflections $x_i \mapsto -x_i$ and $x_i \mapsto -1 - x_i$ of $\mathbb{Z}^4$, and consider

(SR_r) Let $i \in \{1,2,3\}$ and $\iota \in \{\iota_i^{\mathrm{s}}, \iota_i^{\mathrm{b}}\}$. For every $c : \mathbb{Z}^4 \to \mathbb{C}$ of finite support in the half-space $\{x_i \ge r\}$,

$$\sum_{x,y} \bar{c}_x c_y\, \omega_k\big(\mathring{\phi}_k(\iota x)\, \mathring{\phi}_k(y)\big) \ge 0.$$

For a field $\varphi_k(x) = \Phi_x$ without flow, with Φ a function of the links within distance r of 0 in the slice, and a state reflection positive for the spatial reflections, (SR_r) holds, since Φ is W_3-invariant; for the flowed fields no r is known.

Theorem 220. *Assume* (SR_0) *for the slice fields at the spacing a_k.*

(i) *$\varpi_k^V(\mathscr{B}_r(p_0)) \le \varpi_k^V(\mathscr{B}_r(0))$ for every $p_0 \in \mathbf{T}^3$ and $r > 0$.*

(ii) *Let $h \in C_c(\mathbb{R}^3)$ be real with $\int h \neq 0$, and ρ_h as in Lemma 218*(iv)*. If* (SR_0) *holds for all large k, then for every cube of side L, all large k and all h' supported in it with $|h'| \le 1$,*

$$I_k^{h'}(\delta) \le \frac{6 \cdot 10^7 L^6}{(\int h)^2} \left\lceil \frac{\pi}{L\rho_h} \right\rceil^3 I_k^h(\delta),$$

so ($\mathrm{M_{sp}}$) *holds for the spatial factor h.*

Proof. (i) Fix i, write $x = (x_i, x_\perp)$, and let b be a finitely supported function of $x_\perp$. Put $k_b(n) = \sum_{x_\perp, y_\perp} \bar{b}_{x_\perp} b_{y_\perp} K_k^V(ne_i + y_\perp - x_\perp)$ for $n \in \mathbb{Z}$. For α finitely supported in $\mathbb{Z}_{\ge 0}$ let $c_{(t,x)} = a_k \mathbf{1}_{0 \le t < n_\delta} \alpha_{x_i} b_{x_\perp}$, supported in $\{x_i \ge 0\}$. By translation invariance the sum of (SR_0) for ι_i^{s} is $\sum_{j,l \ge 0} \bar{\alpha}_j \alpha_l k_b(j+l)$, and for ι_i^{b} it is $\sum_{j,l \ge 0} \bar{\alpha}_j \alpha_l k_b(j+l+1)$; both are nonnegative. On finitely supported sequences put $\langle \alpha, \beta \rangle_b = \sum_{j,l} \bar{\alpha}_j \beta_l k_b(j+l)$ and $(S\alpha)_j = \alpha_{j-1}$, with $\alpha_{-1} = 0$. Then $\langle \cdot, \cdot \rangle_b$ is positive semidefinite, S is symmetric with $\langle \alpha, S\alpha \rangle_b \ge 0$, and $\|S^m \alpha\|_b^2 \le (\sum_j |\alpha_j|)^2 \sup_n |k_b(n)|$ is bounded in m. Since S is symmetric, the Cauchy–Schwarz inequality gives $\|S^j \alpha\|_b^2 = \langle \alpha, S^{2j}\alpha \rangle_b \le \|\alpha\|_b \|S^{2j}\alpha\|_b$ for $j \ge 1$, and with $j = 1, 2, 4, \dots$, by induction

$$\|S\alpha\|_b \le \|\alpha\|_b^{1-2^{-q}} \|S^{2^q}\alpha\|_b^{2^{-q}} \qquad (q \ge 1),$$

so $\|S\alpha\|_b \le \|\alpha\|_b$. Hence S passes to the completion of the quotient by the null vectors as a self-adjoint operator with $0 \le S \le 1$, and $k_b(n) = \langle e_0, S^n e_0 \rangle_b$ for $n \ge 0$. By Lemma 352 there is a finite positive measure μ_b on $[0,1]$ with $k_b(n) = \int \lambda^n d\mu_b$, real, and $k_b(-n) = \overline{k_b(n)}$ because K_k^V is positive definite. So $k_b(n) = \int \lambda^{|n|} d\mu_b(\lambda)$ for all $n \in \mathbb{Z}$.

The measure ϖ_b on the circle with Fourier coefficients $k_b(n)$ is the image of $|\hat{b}(p_\perp)|^2 d\varpi_k^V(p)$ under $p \mapsto p_i$. Since $\int e^{int} P_\lambda(t) \frac{dt}{2\pi} = \lambda^{|n|}$ for the Poisson kernel $P_\lambda(t) = (1-\lambda^2)/(1 - 2\lambda\cos t + \lambda^2)$, $0 \le \lambda < 1$, uniqueness of Fourier coefficients gives $\varpi_b = \int P_\lambda \frac{dt}{2\pi} d\mu_b(\lambda)$, with $P_1 \frac{dt}{2\pi}$ read as the point mass at 0. Each P_λ is even and nonincreasing in $|t|$ on $[-\pi, \pi]$, so its superlevel sets are arcs centred at 0, and for an arc J of length $|J| < 2\pi$ and the arc J_0 of the same length centred at 0,

$$\begin{aligned} \int_J P_\lambda\, dt &= \int_0^\infty |J \cap \{P_\lambda > s\}|\, ds \\ &\le \int_0^\infty |J_0 \cap \{P_\lambda > s\}|\, ds = \int_{J_0} P_\lambda\, dt; \end{aligned}$$

the point mass also gives J_0 at least the weight of J. So $\varpi_b(J) \le \varpi_b(J_0)$, that is $\int (\mathbf{1}_{J_0} - \mathbf{1}_J)(p_i) |\hat{b}(p_\perp)|^2 d\varpi_k^V \ge 0$. Every continuous $g \ge 0$ on the transverse torus is a uniform limit of functions $|\hat{b}|^2$, since $\sqrt{g}$ is a uniform limit of trigonometric polynomials, so the signed measure $A \mapsto \varpi_k^V(\{p_i \in J_0,\ p_\perp \in A\}) - \varpi_k^V(\{p_i \in J,\ p_\perp \in A\})$ is nonnegative. Applying this with A a product of two closed arcs, successively for $i = 1, 2, 3$, moves each coordinate of the centre of $\mathscr{B}_r(p_0)$ to 0 without decreasing the weight; for $r \ge \pi$ there is nothing to prove.

(ii) Let k be so large that $a_k \le L/4$ and Lemma 218(iv) applies. Then Lemma 218(i), (ii), (iv) and part (i) give

$$I_k^{h'}(\delta) \le \tfrac{2}{\delta} V_k^{h'} \le \tfrac{2}{\delta} 7\cdot 10^6 L^6 \Big\lceil \frac{\pi}{L\rho_h}\Big\rceil^3 \varpi_k^V\big(\mathscr{B}_{a_k\rho_h}(0)\big)$$
$$\le \tfrac{2}{\delta} 7\cdot 10^6 L^6 \Big\lceil \frac{\pi}{L\rho_h}\Big\rceil^3 \frac{2V_k^h}{(\int h)^2}$$
$$\le \frac{56\cdot 10^6 L^6}{(\int h)^2}\Big\lceil \frac{\pi}{L\rho_h}\Big\rceil^3 I_k^h(\delta),$$

using $V_k^h \le 2\delta I_k^h(\delta)$. □

Proposition 221. *Let $m>0$ and $q_k = e^{-ma_k}$, let $\varkappa$ and Θ_2 be as in Lemma 218(v), and let Δ be the spatial lattice Laplacian, $(\Delta\phi)(x) = \sum_{i=1}^3 (\phi(x+e_i)+\phi(x-e_i)-2\phi(x))$.*

(a) *Let ψ_k be the field of Proposition 199(a) and $\varphi_k = \Delta^2\psi_k$. Its law is invariant under translations and Γ and reflection positive for the time reflections through sites and bisecting links, its transfer matrix has gap ma_k, $\rho_k(u,\sigma) = e^{-m\sigma}$, ($Z_{sl}$) and ($M_Z$) hold for every product packet, and (SR_2) holds. For every real $h \in C_c^4(\mathbb{R}^3)$ and every cube Q_3, (M_{sp}) fails.*

(b) *The field $\phi_k = \psi_k + a_k^{-5/2}\xi_k$ of Proposition 205(vi) satisfies (SR_0) and, for every nonzero real $h \in C_c(\mathbb{R}^3)$, (M_{sp}); and (Z_{sl}) fails.*

Proof. (a) The covariance of φ_k is $q_k^{|x_0-y_0|}$ times the spatial kernel with Fourier transform $\varkappa(p)^4\prod_i P_{q_k}(p_i) \ge 0$, $P_q(t) = (1-q^2)/(1-2q\cos t+q^2)$, as in the proof of Proposition 199(a). That proof, with γ the centred Gaussian law with this spatial covariance, gives reflection positivity for the time reflections and a transfer matrix with spectrum $\{q_k^N : N \ge 0\}$ and 1 simple; invariance under translations and Γ holds because Δ^2 commutes with translations and W_3. The slice field is linear, so $D_k^{h'}(r) = q_k^{r/a_k} D_k^{h'}(0)$, which gives the ratio and $I_k(\delta) = T_k D_k(0) \le \delta e^{m\delta} D_k(\delta)$ with $T_k = a_k\sum_{j<n_\delta} q_k^j$; ($M_Z$) follows from Proposition 214(iii). The field $\varphi_k(y)$ is a linear combination of ψ_k at points $y+z$ with $|z|_1 \le 2$ in the slice of y, with coefficients invariant under $z_i \mapsto -z_i$, so for c supported in $\{x_i \ge 2\}$ the sum of (SR_2) is $\omega_k(\overline{F\circ\iota}F)$ for a linear function F of ψ_k in $\{x_i \ge 0\}$, nonnegative by the reflection positivity of ψ_k in the spatial directions.

In the notation of Lemma 218, $d\varpi_k = T_k\varkappa^4\prod_i P_{q_k}(p_i)\frac{dp}{(2\pi)^3}$. For $c_x = a_k^3 h(a_k x)$, Lemma 218(v) gives $\varkappa^2|\hat c| \le \Theta_2 a_k^4$ for $a_k \le 1$. Since $\int P_q \frac{dt}{2\pi} = 1$,

$$I_k(\delta) \le T_k\Theta_2^2 a_k^8.$$

Let Q_3 have side ℓ, χ as in Lemma 218(iii) for $Q' = Q_3$, and $h_k'(\mathbf{y}) = \chi(\mathbf{y})\cos(\pi(y_1+y_2+y_3)/a_k)$, real, supported in Q_3, with $|h_k'| \le 1$; this is the h' of the lemma with $p_0 = (\pi,\pi,\pi)$. On $\mathscr{B}_{a_k/(3\ell)}(p_0)$, of half-side at most $\frac{1}{12}$, each $\sin^2(p_i/2) \ge \cos^2\frac{1}{24}$, so $\varkappa \ge 11$, and $P_{q_k}(p_i) \ge (1-q_k)/(1+q_k) \ge ma_k/4$ for $ma_k \le 1$. The lower bound of the proof of Lemma 218(iii), $|\hat c'| \ge \ell^3/128$ on that box, gives

$$I_k^{h_k'}(\delta) \ge T_k\, 11^4 \Big(\frac{\ell^3}{128}\Big)^2\Big(\frac{ma_k}{4}\Big)^3\Big(\frac{2a_k}{3\ell}\Big)^3\frac{1}{(2\pi)^3}$$
$$\ge 10^{-5}\, T_k\,\ell^3 m^3 a_k^6.$$

So $I_k^{h_k'}(\delta)/I_k(\delta) \ge 10^{-5}\ell^3 m^3\Theta_2^{-2}a_k^{-2} \to \infty$.

(b) The covariance is $\prod_\mu q_k^{|x_\mu-y_\mu|} + a_k^{-5}\mathbf{1}_{x=y}$. For c supported in $\{x_i \ge 0\}$ the second term contributes $a_k^{-5}\sum|c_x|^2$ over $x_i = 0$ for ι_i^{s} and 0 for ι_i^{b}, and the first is reflection positive by Proposition 199(a); this is (SR_0). The noise is independent in time, so $I_k^{h'}(\delta) = I_{\psi,k}^{h'}(\delta) + a_k^2\sum_x |h'(a_k x)|^2$, where the first term, for the field ψ_k, is at most $T_k(\sum_x a_k^3|h'(a_k x)|)^2 \le \delta(L+1)^6$ for h' supported in a cube of side L with $|h'| \le 1$ and $a_k \le 1$, and the second is at most $a_k^{-1}(L+1)^3$. For $h \ne 0$, $I_k(\delta) \ge a_k^2\sum_x h(a_k x)^2 \ge (2a_k)^{-1}\int h^2$ for large k, so $I_k^{h'}(\delta) \le 2(\delta+1)(L+1)^6 I_k(\delta)/\int h^2$, which is ($M_{sp}$). By Proposition 205(vi), $\Upsilon_k \to \infty$, so (M_Z) and, by Proposition 214(iii), (Z_{sl}) fail. □

So exact spatial reflection positivity gives (M_{sp}) for every spatial factor with $\int h \ne 0$, while positivity beyond a fixed lattice distance, as for local functions of a reflection positive field, does not; and neither gives (Z_{sl}). The flow enters through the size of the lattice-scale part of ϖ_k. For a function Φ as in Sec. XI put $\Delta\Phi = \sum_{i=1}^3(\Phi_{e_i}+\Phi_{-e_i}) - 6\Phi$. It is again a bounded real gauge-invariant W_3-invariant function of finitely many spatial links of the slice, since W_3 permutes the six translates, and since the flow commutes with translations, the field of $\Delta\Phi$ is $\Delta\varphi_k$.

Proposition 222. *Let Φ be as above, and m, h, Θ_m as in Lemma 218(v). Let $\varpi_k^{(m)}$ be the measure of Lemma 218 for $\Delta^m\Phi$.*

(i) *$\varpi_k^{(m)} = \varkappa^{2m}\varpi_k$, and for $a_k \le 1$ the integral $I_k(\delta)$ formed with $\Delta^m\Phi$ is at most $\Theta_m^2 a_k^{4m} K_k(0) \le \Theta_m^2\,\delta\,\|\Phi\|_\infty^2 a_k^{4m}$.*

(ii) *If (M_{sp}) holds for $\Delta^m\Phi$ and the spatial factor h, with the constant C for a cube of side ℓ, then for all large k*

$$\varpi_k\big(\{p : |p|_\infty \ge 2\}\big) \le 1.4\cdot 10^8\,\frac{C\Theta_m^2\delta\|\Phi\|_\infty^2}{\ell^3}\,a_k^{4m-3}.$$

This holds at every flow time τ, with constants that do not depend on τ.

Proof. (i) Summation by parts gives U_k^c for $\Delta^m\Phi$ equal to $U_k^{\Delta^m c}$ for Φ, and $\widehat{\Delta c} = -\varkappa\hat c$; so $\int|\hat c|^2 d\varpi_k^{(m)} = \int\varkappa^{2m}|\hat c|^2 d\varpi_k$ for all finitely supported c, which determines the measure. For $c_x = a_k^3 h(a_k x)$, Lemma 218(v) gives $\int\varkappa^{2m}|\hat c|^2 d\varpi_k \le \Theta_m^2 a_k^{4m}\varpi_k(\mathbf{T}^3)$, with $\varpi_k(\mathbf{T}^3) = K_k(0) \le \delta\|\Phi\|_\infty^2$ by Lemma 218(i).

(ii) Let $a_k \le \min(1,\ell/4)$ and $r = a_k/(3\ell) \le \frac{1}{12}$. Divide $\mathbf{T}^3$ into $\lceil 3\pi\ell/a_k\rceil^3 \le (4\pi\ell/a_k)^3$ half-open boxes of side at most $2r$. A box meeting $\{|p|_\infty \ge 2\}$ lies in $\mathscr{B}_r(p_0)$ for its centre p_0, which has a coordinate $|p_{0,i}| \ge 2-r \ge \frac{\pi}{2}$; on $\mathscr{B}_r(p_0)$ that coordinate lies at distance at least $\frac{\pi}{2}-\frac{1}{12}$ from 0 on the circle, and $4\sin^2(\frac{\pi}{4}-\varepsilon) = 2(\cos\varepsilon-\sin\varepsilon)^2 \ge 2(1-\varepsilon-\frac{\varepsilon^2}{2})^2 \ge 1$ for $\varepsilon = \frac{1}{24}$, so $\varkappa \ge 1$ there. By Lemma 218(iii) with Q' the cube of side ℓ, (M_{sp}) for $\Delta^m\Phi$ and (i),

$$\varpi_k(\mathscr{B}_r(p_0)) \le \varpi_k^{(m)}(\mathscr{B}_r(p_0)) \le 16384\,\ell^{-6}C\,\Theta_m^2\,\delta\,\|\Phi\|_\infty^2 a_k^{4m},$$

and summing over the boxes gives the bound, since $4\cdot 16384\cdot 64\pi^3 \le 1.4\cdot 10^8$; the factor 4 covers the case in which only

real h' are admitted, as in the proof of Proposition 219(i). The constants involve Φ only through $\|\Phi\|_\infty$, which the flow does not change. □

So a proof of (M_{sp}) for all the fields $\Delta^m\Phi$, which are admissible observables, must show that the time-integrated spectral weight of φ_k at lattice momenta tends to zero faster than every power of a_k. Such suppression is what the linear heat flow does to a Gaussian field at flow time τ/a_k^2; for the gauge field the flow leaves the central configurations unchanged, and with them their lattice-scale patterns (Proposition 216 and Proposition 223(iv) below).

Proposition 223. *Let $G=\mathrm{SU}(N)$, $\zeta=e^{2\pi i/N}$ and $c_N=N(1-\cos(2\pi/N))$. Let U° be a configuration of the spatial links of a slice with values in the centre $\{\zeta^j\}$, X a finitely supported assignment of elements of $\mathfrak{su}(N)$ to the spatial links, and $U^\varepsilon_e=U^\circ_e e^{\varepsilon X_e}$. For a plaquette p with links e_1,e_2 traversed forwards and e_3,e_4 backwards put $Y_p=X_{e_1}+X_{e_2}-X_{e_3}-X_{e_4}$, $|Y|^2=-\mathrm{Tr}\,Y^2$, and let $z_p\in\{\zeta^j\}$ be the plaquette of U°.*

(i) *The finite sum $S(\varepsilon)=\sum_p\big(\mathrm{Re\,Tr}\,U^\circ_p-\mathrm{Re\,Tr}\,U^\varepsilon_p\big)$ over the spatial plaquettes of the slice satisfies $S'(0)=0$ and $S''(0)=\sum_p\mathrm{Re}(z_p)\,|Y_p|^2$.*

(ii) *At $U^\circ=1$, $S''(0)=0$ if and only if there is a finitely supported $\Lambda:\mathbb{Z}^3\to\mathfrak{su}(N)$ with $X_{(x,y)}=\Lambda(x)-\Lambda(y)$ for every link, that is, X is an infinitesimal gauge transformation.*

(iii) *Let U^ζ have $U^\zeta_{e_*}=\zeta$ on one spatial link e_* and all other links* 1*, and let S_0'' be the form of* (ii)*. For $N\geq 5$, $S''(0)\geq\cos(2\pi/N)S_0''(0)\geq 0$, with equality to* 0 *only at infinitesimal gauge transformations; for $N=4$, $S''(0)\geq 0$; for $N=2,3$ and X supported on e_*, $S''(0)=4\cos(2\pi/N)|X_{e_*}|^2<0$.*

(iv) *For Φ the sum of $N-\mathrm{Re\,Tr}\,U_p$ over the twelve spatial plaquettes with a corner at* 0 *and every $s\in\mathbb{R}$, $\Phi_x\circ\Phi_s(U^\zeta)=c_Nn_x$, where $n_x=4$ at the two endpoints of e_*, $n_x=1$ at the eight other corners of the four plaquettes containing e_*, and $n_x=0$ elsewhere.*

Proof. (i) Central elements commute with everything, so $U^\varepsilon_p=z_pe^{\varepsilon X_{e_1}}e^{\varepsilon X_{e_2}}e^{-\varepsilon X_{e_3}}e^{-\varepsilon X_{e_4}}=z_pe^{Z_p(\varepsilon)}$ with $Z_p(\varepsilon)\in\mathfrak{su}(N)$ and, by the Baker–Campbell–Hausdorff formula, $Z_p(\varepsilon)=\varepsilon Y_p+O(\varepsilon^2)$. Since $\mathrm{Tr}\,Z_p=0$, $\mathrm{Tr}\,e^{Z_p}=N+\frac12\mathrm{Tr}\,Z_p^2+O(\varepsilon^3)=N-\frac{\varepsilon^2}{2}|Y_p|^2+O(\varepsilon^3)$, and only the finitely many plaquettes meeting the support of X contribute.

(ii) The vectors $\Lambda(x)-\Lambda(y)$ give $Y_p=0$ by telescoping. Conversely, $S''(0)=\sum_p|Y_p|^2=0$ means that X is a closed $\mathfrak{su}(N)$-valued cochain on the cubical complex of $\mathbb{Z}^3$. Choose a box B containing the endpoints of the support of X and a point $x_\infty\notin B$, and let $\Lambda(x)$ be minus the sum of $\pm X_e$ along a lattice path from x_∞ to x, the sign recording the orientation. Two paths with the same ends differ by a sum of boundaries of plaquettes, so Λ is well defined, and $X_{(x,y)}=\Lambda(x)-\Lambda(y)$. The complement of B in $\mathbb{Z}^3$ is connected and X vanishes on its links, so $\Lambda=0$ there.

(iii) The plaquettes of U^ζ are 1 except the four containing e_*, which are $\zeta^{\pm1}$, so $S''(0)=S_0''(0)-(1-\cos\frac{2\pi}{N})\sum_{p\ni e_*}|Y_p|^2$, and $\sum_{p\ni e_*}|Y_p|^2\leq S_0''(0)$. For $N\geq5$, $\cos\frac{2\pi}{N}>0$, which with (ii) gives the first claim, and $\cos\frac{\pi}{2}=0$ gives the case $N=4$. For X supported on e_*, $Y_p=\pm X_{e_*}$ for the four plaquettes containing e_* and $Y_p=0$ otherwise, and $\cos\pi=-1$, $\cos\frac{2\pi}{3}=-\frac12$.

(iv) By (i) the first variation of the action of the slice vanishes in every direction, so the right side of the flow equation vanishes at U^ζ and $\Phi_s(U^\zeta)=U^\zeta$ by uniqueness. Each of the four plaquettes containing e_* contributes $N-\mathrm{Re\,Tr}\,\zeta^{\pm1}=c_N$ to Φ_x for each of its four corners x; the two endpoints of e_* are corners of all four, and each of the other eight corners of exactly one. □

Spatial reflection positivity of a field and of its Laplacian. Theorem 220 turns (SR_0) for one field into (M_{sp}) for that field. The admissible fields are closed under $\Phi\mapsto\Delta\Phi$, and by Proposition 222 a proof of (M_{sp}) for all of them has to suppress the lattice-scale spectral weight. This paragraph shows that (SR_0) cannot hold for Φ and $\Delta\Phi$ together in a state that clusters in space, that the linear heat flow destroys (SR_0) at every positive flow time, and that (SR_0) for one field forces spectral weight at the lattice scale.

Write $\mathbf{T}=\mathbb{R}/2\pi\mathbb{Z}$, identified with $(-\pi,\pi]$. A finite positive measure ν on $\mathbf{T}$ is *of Poisson type* if there is a finite positive measure μ on $[0,1]$ with

$$\hat\nu(n)=\int e^{int}d\nu(t)=\int\lambda^{|n|}d\mu(\lambda)\qquad(n\in\mathbb{Z}),$$

with $0^0=1$. The moments determine μ, as in the proof of Lemma 352. Since $\int e^{int}P_\lambda(t)\frac{dt}{2\pi}=\lambda^{|n|}$ for $P_\lambda(t)=(1-\lambda^2)/(1+\lambda^2-2\lambda\cos t)$, $0\leq\lambda<1$, uniqueness of Fourier coefficients gives

$$\nu=\mu(\{1\})\,\delta_0+f\,\frac{dt}{2\pi},\qquad f=\int_{[0,1)}P_\lambda\,d\mu(\lambda).\tag{37}$$

For $i\in\{1,2,3\}$ write $p=(p_i,p_\perp)$ for $p\in\mathbf{T}^3$, with $p_\perp$ in the torus $\mathbf{T}^2$ of the other two coordinates, and put $\varkappa_\perp(p_\perp)=\sum_{j\neq i}4\sin^2(p_j/2)$, so that $\varkappa(p)=4\sin^2(p_i/2)+\varkappa_\perp(p_\perp)$. For a finite positive measure ϖ on $\mathbf{T}^3$ and a continuous $g\geq0$ on $\mathbf{T}^2$ let ϖ^i_g be the image of $g(p_\perp)\,d\varpi(p)$ under $p\mapsto p_i$. Call ϖ *of Poisson type in direction i* if every ϖ^i_g is of Poisson type.

Lemma 224. (i) *Let ν_r be measures of Poisson type on $\mathbf{T}$ with $\sup_r\nu_r(\mathbf{T})<\infty$, and let ν be a finite positive measure with $\hat\nu_r(n)\to\hat\nu(n)$ for every n. Then ν is of Poisson type.*

(ii) *If (SR_0) holds for the slice field at the spacing a_k, then ϖ^V_k is of Poisson type in every direction $i\in\{1,2,3\}$. The same holds for every real field on $\mathbb{Z}^4$ with translation invariant, bounded covariance that satisfies the inequalities of (SR_0), with K^V_k formed from its covariance.*

(iii) *If ν is of Poisson type, then for every integer $n\geq1$*

$$\hat\nu(0)-\hat\nu(n)\leq n\big(\hat\nu(0)-\hat\nu(1)\big),$$
$$\nu\big(\{t:2\leq|t|\leq\pi\}\big)\geq\frac{\pi-2}{2\pi}\big(\hat\nu(0)-\hat\nu(1)\big).$$

Proof. (i) Let μ_r be the measures of ν_r. Their masses $\mu_r([0,1])=\hat\nu_r(0)$ converge to $\hat\nu(0)$. If $\hat\nu(0)=0$, then $\nu=0$ and $\mu=0$ will do. Otherwise Lemma 353, applied to the probability measures $\mu_r/\hat\nu_r(0)$ for large r, gives a subsequence

along which μ_r converges weakly to a finite positive measure μ of mass $\hat\nu(0)$. Since $\lambda\mapsto\lambda^{|n|}$ is continuous on $[0,1]$, $\hat\nu(n)=\lim\int\lambda^{|n|}d\mu_r=\int\lambda^{|n|}d\mu$.

(ii) For a finitely supported b on the transverse lattice, the proof of Theorem 220(i) shows that $k_b(n)=\int e^{inp_i}|\hat b(p_\perp)|^2 d\varpi_k^V(p)$ equals $\int\lambda^{|n|}d\mu_b$ for all $n\in\mathbb{Z}$, so $\varpi^{V,i}_{k,|\hat b|^2}$ is of Poisson type. That proof uses only translation invariance, the boundedness and positive definiteness of K_k^V, and (SR_0). For a continuous $g\ge0$ on $\mathbf{T}^2$, $\sqrt g$ is a uniform limit of trigonometric polynomials $\hat b_r$, so $|\hat b_r|^2\to g$ uniformly, the Fourier coefficients converge, the masses are bounded, and (i) applies.

(iii) By the representation, $\hat\nu(0)-\hat\nu(n)=\int(1-\lambda^n)d\mu$ and $1-\lambda^n\le n(1-\lambda)$ on $[0,1]$. Further $P_\lambda(t)\ge(1-\lambda^2)/(1+\lambda)^2=(1-\lambda)/(1+\lambda)\ge(1-\lambda)/2$ for $0\le\lambda<1$, and the set $\{2\le|t|\le\pi\}$ has measure $2(\pi-2)$, so by (37) its ν-measure is at least $\frac{\pi-2}{2\pi}\int(1-\lambda)\,d\mu=\frac{\pi-2}{2\pi}(\hat\nu(0)-\hat\nu(1))$; the atom at $\lambda=1$ contributes 0 to both sides. □

The measures of $\Delta\Phi$ are those of Φ times $\varkappa^2$: the summation by parts of the proof of Proposition 222(i) gives V_k^c for $\Delta\Phi$ equal to $V_k^{\Delta c}$ for Φ, so the measure ϖ_k^V of $\Delta\Phi$ is $\varkappa^2\varpi_k^V$, and that of K_k is $\varkappa^2\varpi_k$.

Theorem 225. *Let Φ be as in Proposition 222, and assume (SR_0) at the spacing a_k for the slice fields of both Φ and $\Delta\Phi$.*

(i) *$\varpi_k^V=K_k^V(0)\,\delta_0$ and $\varpi_k=K_k(0)\,\delta_0$. So K_k^V and K_k are constant on $\mathbb{Z}^3$, and the time average $Y_k(\mathbf{x})=a_k\sum_{n=0}^{n_\delta-1}\mathring\phi_k(n,\mathbf{x})$ satisfies $Y_k(\mathbf{x})=Y_k(0)$ almost surely for every $\mathbf{x}$.*

(ii) *The slice field of $\Delta\Phi$ equals its mean almost surely at every point: $\Delta\mathring\phi_k(x)=0$ for every $x\in\mathbb{Z}^4$.*

(iii) *If $\omega_k(\mathring\phi_k(0,0)\,\mathring\phi_k(j,\mathbf{z}))\to0$ as $|\mathbf{z}|\to\infty$ for every integer $j\ge0$, then $\mathring\phi_k(x)=0$ almost surely for every x, and $C_k(0)=0$ for every packet f_0.*

Proof. (i) Fix i. By Lemma 224(ii), applied to Φ with $g=1,\varkappa_\perp,\varkappa_\perp^2$ and to $\Delta\Phi$ with $g=1$, the images ν_0,ν_1,ν_2 of ϖ_k^V, $\varkappa_\perp\varpi_k^V$, $\varkappa_\perp^2\varpi_k^V$ and the image ν' of $\varkappa^2\varpi_k^V$ under $p\mapsto p_i$ are of Poisson type, with measures μ_0,μ_1,μ_2,μ' on $[0,1]$. Put $k_l=\hat\nu_l$ and $F=\hat\nu'$. With $4\sin^2(t/2)=2-e^{it}-e^{-it}$,

$$\varkappa^2=(2-e^{it}-e^{-it})^2+2(2-e^{it}-e^{-it})\varkappa_\perp+\varkappa_\perp^2,\qquad t=p_i,$$

so ν' is ν_0, ν_1, ν_2 multiplied by these trigonometric polynomials in t and added, and

$$\begin{aligned}F(n)=\;&6k_0(n)-4k_0(n-1)-4k_0(n+1)\\&+k_0(n-2)+k_0(n+2)+4k_1(n)\\&-2k_1(n-1)-2k_1(n+1)+k_2(n).\end{aligned}$$

Insert $k_l(m)=\int\lambda^{|m|}d\mu_l$. For $n\ge2$ all the exponents $|n-j|$, $|j|\le2$, equal $n-j$, and the integrands combine to

$$\lambda^{n-2}(1-\lambda)^4,\qquad-2\lambda^{n-1}(1-\lambda)^2,\qquad\lambda^n$$

against μ_0, μ_1, μ_2; these are polynomial identities, valid at $\lambda=0$ with $0^0=1$. For $n=1$ the integrand against μ_0 is $6\lambda-4-4\lambda^2+\lambda+\lambda^3$, which for $\lambda>0$ equals $\lambda^{-1}(1-\lambda)^4-\lambda^{-1}+\lambda$, and the integrand against μ_1 is $4\lambda-2-2\lambda^2=-2(1-\lambda)^2$.

For $n\ge3$ the three integrands vanish at $\lambda=0$, and $F(n)=\int\lambda^n d\mu'$. So the finite signed measure

$$\lambda^3\mu'-\lambda(1-\lambda)^4\mu_0+2\lambda^2(1-\lambda)^2\mu_1-\lambda^3\mu_2$$

on $[0,1]$ annihilates every polynomial, hence every continuous function, and it is zero. Dividing by λ and by λ^2 on $(0,1]$,

$$\begin{aligned}\lambda^2\mu'&=(1-\lambda)^4\mu_0-2\lambda(1-\lambda)^2\mu_1+\lambda^2\mu_2,\\ \lambda\mu'&=\lambda^{-1}(1-\lambda)^4\mu_0-2(1-\lambda)^2\mu_1+\lambda\mu_2\end{aligned}\tag{38}$$

as measures on $(0,1]$. For $n=2$, $F(2)=\int\lambda^2d\mu'=\int_{(0,1]}\lambda^2d\mu'$ and $F(2)=\int_{[0,1]}((1-\lambda)^4d\mu_0-2\lambda(1-\lambda)^2d\mu_1+\lambda^2d\mu_2)$; integrating the first line of (38) over $(0,1]$ and subtracting leaves $\mu_0(\{0\})=0$. Integrating the second line over $[\varepsilon,1]$,

$$\int_{[\varepsilon,1]}\frac{(1-\lambda)^4}{\lambda}\,d\mu_0\le\nu'(\mathbf{T})+2\nu_1(\mathbf{T})$$

for every $\varepsilon>0$, so $\int_{(0,1]}\lambda^{-1}(1-\lambda)^4d\mu_0<\infty$, and since $(1-\lambda)^4\ge\frac{1}{16}$ on $(0,\frac12]$, $\int_{(0,1]}\lambda^{-1}d\mu_0<\infty$. Now, with $\mu_0(\{0\})=0$,

$$\begin{aligned}F(1)&=\int_{(0,1]}\Big(\frac{(1-\lambda)^4}{\lambda}-\frac{1}{\lambda}+\lambda\Big)d\mu_0\\&\quad-2\int_{[0,1]}(1-\lambda)^2d\mu_1+\int\lambda\,d\mu_2,\\F(1)&=\int\lambda\,d\mu'=\int_{(0,1]}\frac{(1-\lambda)^4}{\lambda}\,d\mu_0\\&\quad-2\int_{(0,1]}(1-\lambda)^2d\mu_1+\int\lambda\,d\mu_2,\end{aligned}$$

the second by the second line of (38). Subtracting,

$$\int_{(0,1]}\Big(\frac{1}{\lambda}-\lambda\Big)d\mu_0+2\mu_1(\{0\})=0.$$

Both terms are nonnegative and $1/\lambda-\lambda>0$ on $(0,1)$, so μ_0 is carried by $\{1\}$, k_0 is constant, and ν_0 is a point mass at 0. Thus $\varpi_k^V(\{p:p_i\ne0\})=0$ for $i=1,2,3$, and $\varpi_k^V=\varpi_k^V(\mathbf{T}^3)\delta_0=K_k^V(0)\delta_0$. By Lemma 218(i), $\frac{\delta}{2}\int|\hat c|^2d\varpi_k\le\int|\hat c|^2d\varpi_k^V\le2\delta\int|\hat c|^2d\varpi_k$ for every finitely supported c. As in the proof of Theorem 220(i) the functions $|\hat c|^2$ approximate every continuous $g\ge0$ on $\mathbf{T}^3$ uniformly, so $\frac{2}{\delta}\varpi_k^V-\varpi_k$ and $2\delta\varpi_k-\varpi_k^V$ are positive functionals on $C(\mathbf{T}^3)$, hence positive measures:

$$\tfrac{\delta}{2}\varpi_k\le\varpi_k^V\le2\delta\varpi_k.\tag{39}$$

So $\varpi_k(\{p\ne0\})=0$. Finally Y_k is real with $\omega_k(Y_k(0)Y_k(\mathbf{z}))=K_k^V(\mathbf{z})$, so $\omega_k((Y_k(\mathbf{x})-Y_k(0))^2)=2K_k^V(0)-2K_k^V(\mathbf{x})=0$.

(ii) The measure of K_k for $\Delta\Phi$ is $\varkappa^2\varpi_k=K_k(0)\varkappa(0)^2\delta_0=0$. So, with U_k^c formed from $\Delta\Phi$, $0=I_k^c=a_k\sum_{j<n_\delta}\langle u_c,\mathbb{T}_k^ju_c\rangle$ for every c, and each term is nonnegative because $\mathbb{T}_k\ge0$; the term

$j=0$ is $a_k\,\omega_k(|U_k^c|^2)$. With $c=\delta_{\mathbf{x}}$ this gives $\Delta\mathring\phi_k(0,\mathbf{x})=0$ almost surely, and time translation invariance gives every time.

(iii) $K_k^V(\mathbf{z})$ is a finite sum of the correlations

$$a_k^2\,\omega_k\big(\mathring\phi_k(n,0)\,\mathring\phi_k(n',\mathbf{z})\big),\qquad 0\le n,n'<n_\delta,$$

and by translation invariance each of them equals

$$a_k^2\,\omega_k\big(\mathring\phi_k(0,0)\,\mathring\phi_k(j,\pm\mathbf{z})\big),\qquad j=|n'-n|.$$

Under the hypothesis $K_k^V(\mathbf{z})\to 0$, and since it is constant by (i), $K_k^V=0$, so $\varpi_k^V=0$ and $\varpi_k=0$. As in (ii), now for Φ, $\omega_k(|\mathring\phi_k(0,\mathbf{x})|^2)=0$ for every $\mathbf{x}$, hence $\mathring\phi_k(x)=0$ almost surely for every $x\in\mathbb{Z}^4$. Every A_k is then 0 almost surely, and $C_k(0)=\omega_k(\vartheta(\bar A_k)A_k)=0$. □

Corollary 226. *Suppose that $D(n_k,\alpha)$ holds at β_k, as in Theorem 43, and that $C_k(0)>0$ for the packet f_0 of Φ. Then at every spacing a_k,* (SR$_0$) *fails for the slice field of Φ or for that of $\Delta\Phi$. So along such a sequence no class of admissible fields closed under $\Phi\mapsto\Delta\Phi$ satisfies* (SR$_0$)*, and Theorem 220 applies at the spacing a_k to at most one of Φ and $\Delta\Phi$.*

Proof. By Theorem 41, part (v) of Proposition 11 holds at β_k: covariances of gauge-invariant local functions, one of them Lipschitz in each link, decay exponentially in the distance. A continuous gauge-invariant function of finitely many links is a uniform limit of Lipschitz ones, which may be averaged over the gauge transformations at the sites they involve, since these act by isometries. By Proposition 45(ii), $\varphi_k(x)$ is a uniform limit of continuous gauge-invariant local functions at the time of x, and $|\mathrm{Cov}(F,F')-\mathrm{Cov}(G,G')|\le 2\|F-G\|_\infty\|F'\|_\infty+2\|F'-G'\|_\infty\|G\|_\infty$. So $\omega_k(\mathring\phi_k(0,0)\mathring\phi_k(j,\mathbf{z}))\to 0$ as $|\mathbf{z}|\to\infty$. If (SR$_0$) held for both fields, Theorem 225(iii) would give $C_k(0)=0$. □

The linear heat flow, the model of the smoothing that the flow is meant to provide, removes (SR$_0$) at every positive flow time. For $s>0$ let $p_s(\mathbf{z})$ be the probability that the continuous-time random walk on $\mathbb{Z}^3$ with generator Δ is at $\mathbf{z}$ at time s, starting from 0, so that $\sum_{\mathbf{z}}p_s(\mathbf{z})=1$ and $\sum_{\mathbf{z}}p_s(\mathbf{z})e^{ip\cdot\mathbf{z}}=e^{-s\varkappa(p)}$, and let $e^{s\Delta}$ act on the slices of a field on $\mathbb{Z}^4$ by $(e^{s\Delta}\psi)(t,\mathbf{x})=\sum_{\mathbf{y}}p_s(\mathbf{x}-\mathbf{y})\psi(t,\mathbf{y})$, a series that converges in L^2 for fields with bounded second moments.

Proposition 227. (i) *Let ϖ be a finite positive measure on $\mathbf{T}^3$, $s>0$ and $i\in\{1,2,3\}$. If ϖ and $e^{-2s\varkappa}\varpi$ are both of Poisson type in direction i, then $\varpi(\{p:p_i\ne 0\})=0$.*

(ii) *Let ψ_k be the field of Proposition 199*(a) *and $s>0$. Then* (SR$_0$) *holds for ψ_k and fails for $e^{s\Delta}\psi_k$, at every k.*

Proof. (i) Let $g\ge 0$ be continuous on $\mathbf{T}^2$ and $g'=ge^{-2s\varkappa_\perp}$. Since $e^{-2s\varkappa}=e^{-4s(1-\cos p_i)}e^{-2s\varkappa_\perp}$, the measure $\nu=\varpi^i_{g'}$ and the image $e^{-4s(1-\cos t)}\nu$ of $g\,e^{-2s\varkappa}\varpi$ are of Poisson type, with measures μ and μ^s. Write ν as in (37) and put

$$F(c)=\int_{[0,1)}\frac{1-\lambda^2}{1+\lambda^2-2\lambda c}\,d\mu(\lambda)$$

for c in $\Omega=\mathbb{C}\setminus[1,\infty)$, and F^s likewise with μ^s, so that $f(t)=F(\cos t)$ for $t\ne 0$. The denominator vanishes only at $c=(1+\lambda^2)/(2\lambda)\ge 1$. On a compact $K\subset\Omega$ with $|c|\le R$ and distance $d>0$ to $[1,\infty)$, it is at least $\frac12$ for $\lambda\le 1/(4R)$ and at least $2\lambda d\ge d/(2R)$ for larger λ, so the integrand is bounded on K uniformly in λ, and F is analytic on Ω. For $\mathrm{Im}\,c>0$,

$$\mathrm{Im}\,F(c)=\int_{[0,1)}\frac{2\lambda(1-\lambda^2)\,\mathrm{Im}\,c}{|1+\lambda^2-2\lambda c|^2}\,d\mu(\lambda)\ge 0,$$

and the same holds for F^s. The absolutely continuous parts of $e^{-4s(1-\cos t)}\nu$ in its two representations agree, so $F^s(\cos t)=e^{-4s(1-\cos t)}F(\cos t)$ for almost every t, hence for all $c\in(-1,1)$ by continuity, and on Ω, which is connected, by analyticity: $F^s(c)=e^{-4s(1-c)}F(c)$.

If $\mu((0,1))>0$, then $\mathrm{Im}\,F>0$ on the upper half plane, and $\theta(y)=\arg F(iy)\in(0,\pi)$ is continuous in $y>0$. Since $e^{-4s(1-iy)}=e^{-4s}e^{4isy}$,

$$\mathrm{Im}\,F^s(iy)=e^{-4s}|F(iy)|\sin\big(4sy+\theta(y)\big)\ge 0$$

and $F^s(iy)\ne 0$, so the continuous function $4sy+\theta(y)$ takes its values in the union of the disjoint closed intervals $[2\pi j,2\pi j+\pi]$, $j\in\mathbb{Z}$; on the connected set $(0,\infty)$ it stays in one of them, which is impossible since it exceeds $4sy$. So $\mu((0,1))=0$ and F is the constant $B=\mu(\{0\})\ge 0$. Then $\mathrm{Im}\,F^s(iy)=Be^{-4s}\sin(4sy)$, which is $-Be^{-4s}$ at $y=3\pi/(8s)$, so $B=0$. Hence $\nu=\mu(\{1\})\delta_0$. For $g=1$ this says $\int_{\{p_i\ne 0\}}e^{-2s\varkappa_\perp}d\varpi=0$, and $e^{-2s\varkappa_\perp}>0$.

(ii) The covariance of ψ_k is $q_k^{|x_0-y_0|}\prod_{i=1}^3 q_k^{|x_i-y_i|}$, so its measure ϖ_k^V is $T_k\prod_i P_{q_k}(p_i)\frac{dp}{(2\pi)^3}$ with $T_k=a_k^2\sum_{n,n'<n_\delta}q_k^{|n-n'|}>0$, and that of $e^{s\Delta}\psi_k$, whose spatial covariance kernel has Fourier transform $e^{-2s\varkappa}\prod_i P_{q_k}(p_i)$, is $e^{-2s\varkappa}\varpi_k^V$. Both covariances are bounded and translation invariant. The law of ψ_k is reflection positive for the spatial reflections through sites and bisecting links (Proposition 199(a)); for c supported in $\{x_i\ge 0\}$ the sum of (SR$_0$) is $\mathbb{E}[\overline{\psi_k(\imath c)}\,\psi_k(c)]\ge 0$ with $\psi_k(c)=\sum c_x\psi_k(x)$ and $\psi_k(\imath c)=\sum c_x\psi_k(\imath x)$, which is (SR$_0$). If (SR$_0$) held for $e^{s\Delta}\psi_k$, Lemma 224(ii) for both fields and (i) would give $\varpi_k^V(\{p_i\ne 0\})=0$, but ϖ_k^V has a positive density. □

The continuum analogue is the spatially flowed free field of Proposition 199(c). It is reflection positive in time, and the proof of Proposition 209(i), with x_1 in the role of time, shows that it is not reflection positive for the reflection of x_1: it uses only $H_0(0)>0$ and $H_0''(0)<0$, and here $H_0''(0)=-(2\pi)^{-1}\int p_1^2e^{-2t_fp_1^2}(p_1^2+\mu^2)^{-1}dp_1$. For one field, (SR$_0$) forces spectral weight at the lattice scale: by part (i) below with $n=L/a_k$, $\varpi_k(\{|p|_\infty\ge 2\})$ is at least $a_k/(12\delta L)$ times $K_k^V(0)-K_k^V(ne_i)$, the decorrelation of the time-averaged field over the physical distance L.

Proposition 228. *Assume* (SR$_0$) *at the spacing a_k for the slice field of Φ, and let $i\in\{1,2,3\}$ and $n\ge 1$.*

(i)

$$\begin{aligned}K_k^V(0)-K_k^V(ne_i)&\le n\big(K_k^V(0)-K_k^V(e_i)\big)\\&\le\frac{2\pi n}{\pi-2}\,\varpi_k^V\big(\{|p_i|\ge 2\}\big)\\&\le 12\,\delta\,n\,\varpi_k\big(\{|p|_\infty\ge 2\}\big).\end{aligned}$$

(ii) *For h' supported in a cube of side L' with $|h'|\le 1$, $a_k\le 1$ and $h'_n(\mathbf{y})=h'(\mathbf{y}-na_ke_i)$,*

$$I_k^{h'-h'_n}(\delta)\le\frac{4}{\delta}(L'+1)^6\big(K_k^V(0)-K_k^V(ne_i)\big).$$

(iii) *If in addition* $(\mathrm{M_{sp}})$ *holds for the slice field of $\Delta^m\Phi$, $m\ge 1$, and the spatial factor h, with the constant C for a cube of side ℓ, then for all large k*

$$K_k^V(0)-K_k^V(ne_i)\le 1.7\cdot 10^9\,\frac{C\Theta_m^2\delta^2\|\Phi\|_\infty^2}{\ell^3}\,n\,a_k^{4m-3}.$$

Proof. (i) By Lemma 224(ii) with $g=1$, $\nu=\varpi_{k,1}^{V,i}$ is of Poisson type, and $\hat\nu(n)=K_k^V(ne_i)$. Lemma 224 (iii) gives the first two inequalities. The third follows from (39), which uses only Lemma 218(i), and from $4\pi/(\pi-2)<12$, since $\pi-2>1.14$ and $4\pi<12.57<13.68$.

(ii) With $c_{\mathbf{x}}=a_k^3h'(a_k\mathbf{x})$ the coefficients of $h'-h'_n$ are $c-c(\cdot-ne_i)$, with Fourier transform $\hat c\,(1-e^{inp_i})$. So, by (39) and $|1-e^{inp_i}|^2=2(1-\cos np_i)$,

$$\begin{aligned}I_k^{h'-h'_n}(\delta)&\le\|\hat c\|_\infty^2\,\frac{2}{\delta}\int 2(1-\cos np_i)\,d\varpi_k^V\\&=\frac{4}{\delta}\|\hat c\|_\infty^2\big(K_k^V(0)-K_k^V(ne_i)\big),\end{aligned}$$

because K_k^V is real and even. Finally $\|\hat c\|_\infty\le\sum_{\mathbf{x}}a_k^3|h'(a_k\mathbf{x})|\le a_k^3(L'/a_k+1)^3\le(L'+1)^3$.

(iii) Proposition 222(ii) bounds $\varpi_k(\{|p|_\infty\ge 2\})$, and $12\cdot 1.4\cdot 10^8\le 1.7\cdot 10^9$. □

So $(\mathrm{SR_0})$ for Φ together with $(\mathrm{M_{sp}})$ for $\Delta^2\Phi$ gives, by (ii) and (iii) with $n=L/a_k$, $I_k^{h'-h'_n}(\delta)\le C'La_k^4$ for spatial translates by a physical distance L. Hence $I_k^{h'-h'_n}(\delta)/I_k(\delta)\to 0$ for every bounded L unless $\liminf_k a_k^{-4}I_k(\delta)<\infty$: the time-averaged slice field becomes invariant, relative to $I_k(\delta)$, under spatial translations by bounded physical distances. No lower bound on $I_k(\delta)$ along a tuning is proved here.

Remark 229. For the flowed fields of the Wilson measure along a tuning, the two statements stand as follows.

$(\mathrm{M_{sp}})$ is equivalent to the box bound of Proposition 219(i) and, for $\int h\ne 0$, follows from the comparison of Proposition 219(iii). Exact spatial reflection positivity $(\mathrm{SR_0})$ gives that comparison (Theorem 220). The Wilson action is reflection positive in every coordinate hyperplane [90], so for states whose restrictions to gauge-invariant functions are reflection positive for the spatial reflections, a field Φ_x without flow, depending on the links within distance r of x, satisfies (SR_r); the flowed field $\Phi_x\circ\Phi_{s_k}$ depends on all links of its slice, and Proposition 45(i) bounds the influence of links at distance R only for R beyond a multiple of $s_k=\tau/a_k^2$, so no (SR_r) is known for it. Positivity beyond a fixed distance does not suffice (Proposition 221(a)). Nor can $(\mathrm{SR_0})$ hold for a field and its Laplacian together: in a state that clusters in space, in particular under the hypothesis of Theorem 43, it forces $C_k(0)=0$ (Theorem 225, Corollary 226), and the linear heat flow destroys it at every positive flow time (Proposition 227). So Theorem 220 cannot give $(\mathrm{M_{sp}})$ for a class of fields closed under $\Phi\mapsto\Delta\Phi$, and for one field $(\mathrm{SR_0})$ forces spectral weight at the lattice scale (Proposition 228). Compactness of G gives $\bar\varpi_k\le\delta\|\Phi\|_\infty^2$, an upper bound of order one, and no lower bound on $I_k(\delta)$; by Proposition 222 a proof valid for the admissible fields $\Delta^m\Phi$ must suppress the lattice-scale spectral weight of φ_k faster than every power of a_k. For $G=\mathrm{SU}(2)$ the estimates of [21, Sec. XII E] give each plaquette a conditional probability at least $\nu_*e^{-3\beta_W}$ of being bad [21, Proposition 386], a power of a_k along couplings of order $\log(1/a_k)$, and the single-link defects of Proposition 223 are fixed by the flow at every flow time, with nonnegative second variation for $N\ge 4$ and a direction of negative second variation for $N=2,3$. Whether the flow over the time τ/a_k^2 moves the configurations near such defects to configurations with lattice-scale weight below every power of a_k is not known; this is the missing estimate.

$(\mathrm{Z_{sl}})$ is equivalent to (M_Z) and implies (N_u) (Proposition 214); for $G=\mathrm{SU}(2)$ it fails along every sequence with $\beta_{W,k}\le 0.348$ (Proposition 205(v)) and along the constant tunings (Proposition 184(ii)). Compactness bounds $D_k(0)$ above and gives no lower bound on $D_k(\delta)$, which is what $(\mathrm{Z_{sl}})$ requires; exact spatial reflection positivity together with $(\mathrm{M_{sp}})$ does not give it (Proposition 221(b)), and there the term that breaks it is constant in space, so that no spatial smoothing removes it. The missing estimate is that of Remark 217: a bound on the high-energy spectral weight of the slice field by its weight at bounded physical energy.

At a fixed k both inequalities hold with a constant depending on k whenever $C_k(0)>0$, since then $D_k(\delta)>0$ and $I_k(\delta)>0$ by Proposition 49(i); their content is the uniformity in k, and neither is proved along any sequence of couplings with $a_k\to 0$.

Status of (M), (W) and (U). None of the three statements is proved. Proposition 205(i)–(iii) uses translation and ϑ-invariance and reflection positivity of the states and the reality of Φ, part (iv) also the transfer matrix, and part (v) Theorem 183; none of them uses the hypothesis of Theorem 43 or (N_u). Proposition 206 is deterministic except for the positivity of the conditional densities in part (iii). Proposition 208 uses (M) and the invariance of the states under translations and Γ, through Theorem 193(i), (ii), and part (iv) the tree bounds. Proposition 209 concerns the free field and limits of families. Proposition 211(i) assumes (RC), whose existence along a tuning is not proved and which contains $(\mathrm{N_0})$, and the hypothesis of Theorem 43 for the bound on m_*; parts (ii) and (iii) assume (M). Lemma 213 and Propositions 214 and 215(i)–(iii) use translation and ϑ-invariance, reflection positivity and the positivity of the transfer matrix, and no hypothesis on the tuning; so do Lemma 218 and Propositions 219 and 222, while Theorem 220 assumes the exact spatial reflection positivity $(\mathrm{SR_0})$ of the slice field, which is not known for the flowed fields and, under the hypothesis of Theorem 43, fails for Φ or for $\Delta\Phi$ (Corollary 226). Proposition 223 is deterministic. What remains open is $(\mathrm{M_{loc}})$, whose time direction for $n=2$ is Proposition 215(i) and whose remainder is the spatial comparison $(\mathrm{M_{sp}})$ and the bounds of Proposition 215(iii) for even $n\ge 4$; (M_Z), which for product packets is the short-time bound $(\mathrm{Z_{sl}})$ on the slice field and is strictly stronger than $(\mathrm{N_0})$; the Ward identity $\Delta_k^X\to 0$ for one $X\in\mathfrak{so}(3)_{\mathrm{sp}}$; and, given (RC),

(U1) and (U2) (Remark 217).

J. Estimates uniform in the coupling

Some local quantities of the Wilson measure can be controlled uniformly in β_W, the volume and the boundary condition, by integrating by parts in a single link. Such estimates hold along every tuning, but they concern local fields in lattice units, and the three results below show how far they are from the open inputs: the force has bounded moments but its centred form is a null vector of the reflected form, the plaquette has fluctuations of order β_W^{-2} but this is a single-site variance, and bare links over a fixed physical distance become independent Haar variables.

Let $G=\mathrm{SU}(N)$, write $b=\beta_W$, so that the Wilson weight is e^{bS} with $S=\sum_p N^{-1}\operatorname{Re}\operatorname{Tr}Q_p$, and give $\mathfrak{g}$ the metric $\langle T,R\rangle=-\operatorname{Tr}(TR)$. For a link e let S_e be the sum of the six plaquette terms containing e, X_T the derivative along $U_e\mapsto e^{uT}U_e$ at $u=0$, ∇_e and Δ_e the gradient and Laplacian in U_e, $q=N^2-1$, $c_N=(N^2-1)/N$, and

$$E_e=b|\nabla_e S_e|^2,\qquad W_e=E_e-c_N S_e=b|\nabla_e S_e|^2+\Delta_e S_e.$$

Below, ω is a Wilson Gibbs state at $b>0$: a finite-volume measure in which e is not fixed by the boundary condition, or an infinite-volume state whose conditional law of U_e given the other links is proportional to $e^{bS_e}dU_e$ with dU_e the Haar measure.

Proposition 230. (i) *For every unit $T\in\mathfrak{g}$, $t\in\mathbb{R}$ and $0\le\eta<N/12$,*

$$\mathbb{E}_\omega\big[e^{t\sqrt{b}X_TS_e}\;\big|\;U_{e^c}\big]\le e^{3t^2/N},$$
$$\mathbb{E}_\omega\big[e^{\eta E_e}\;\big|\;U_{e^c}\big]\le(1-12\eta/N)^{-q/2},$$

so $\mathbb{E}_\omega E_e^m\le m!\,(24/N)^m 2^{q/2}$ for $m\ge1$, uniformly in b, the volume and the boundary condition. Moreover $\mathbb{E}_\omega E_e=c_N\mathbb{E}_\omega S_e$.

(ii) *For a bounded function F of finitely many links, continuously differentiable in U_e, $\mathbb{E}_\omega[W_eF]=-\mathbb{E}_\omega\langle\nabla_eS_e,\nabla_eF\rangle$. In particular $\mathbb{E}_\omega[W_eF]=0$ if F does not depend on U_e, and $\mathbb{E}_\omega[W_eW_f]=0$ if no plaquette contains both e and f.*

(iii) *Let ω be reflection positive as in Appendix F, and let $A=\sum_e h_eW_e$ be a finite sum over links whose endpoints have times at least 2. Then $\omega(\vartheta(\bar A)A)=0$, so $\Psi_A=0$: in the reconstructed Hilbert space the packets $\sum_e h_eE_e$ and $c_N\sum_e h_eS_e$ define the same vector.*

Proof. (i) Along $u\mapsto e^{uT}U_e$ a plaquette term containing e is $N^{-1}\operatorname{Re}\operatorname{Tr}(e^{uT}V)$ or $N^{-1}\operatorname{Re}\operatorname{Tr}(Ve^{-uT})$ with V unitary, whose second derivative is $N^{-1}\operatorname{Re}\operatorname{Tr}(T^2e^{\pm uT}V)$. Since $-T^2\ge0$ and $\operatorname{Tr}(-T^2)=1$, $|\operatorname{Tr}(T^2M)|\le1$ for unitary M, so the second derivative of S_e along the curve is at most $6/N$ in absolute value. Taylor's formula with $u=t/\sqrt{b}$ gives $b\big(S_e(e^{tT/\sqrt{b}}U_e)-S_e(U_e)\big)\ge t\sqrt{b}X_TS_e(U_e)-3t^2/N$. The conditional density is e^{bS_e}/Z with respect to Haar measure, and the left invariance of Haar measure gives $\mathbb{E}_\omega[e^{b(S_e(e^{tT/\sqrt{b}}U_e)-S_e(U_e))}\mid U_{e^c}]=1$, which with the Taylor bound proves the first inequality. For the second let G_0 be a standard Gaussian vector in $\mathfrak{g}$, independent of ω; then $e^{\eta|Y|^2}=\mathbb{E}e^{\sqrt{2\eta}\langle G_0,Y\rangle}$ for $Y=\sqrt{b}\nabla_eS_e$, and the first inequality in the direction of G_0 with $t=\sqrt{2\eta}|G_0|$ gives the bound $\mathbb{E}e^{6\eta|G_0|^2/N}=(1-12\eta/N)^{-q/2}$. With $\eta=N/24$ and $x^m\le m!\,\eta^{-m}e^{\eta x}$ the moment bound follows. For the mean, $\sum_A T_A^2=-c_NI$ in an orthonormal basis gives $\Delta_eS_e=-c_NS_e$, and the integral of the divergence of $e^{bS_e}\nabla_eS_e$ over G vanishes.

(ii) The integral over U_e of the divergence of $e^{bS_e}F\nabla_eS_e$ vanishes, and that divergence is $e^{bS_e}(W_eF+\langle\nabla_eS_e,\nabla_eF\rangle)$; integrate the conditional identity. The function W_f depends only on the links of the plaquettes containing f.

(iii) Each W_e in the sum depends on links with times at least 1, so $\vartheta(\bar A)$ depends only on links with times at most 0, none of which is a link e of the sum. By (ii), $\omega(\vartheta(\bar A)W_e)=0$ for each e. □

Proposition 231. *Let $G=\mathrm{SU}(2)$, $b\ge0$, ω a Wilson Gibbs state at b and p a plaquette with a link that is not fixed by the boundary condition. Then*

$$\operatorname{Var}_\omega\big(\tfrac12\operatorname{Tr}Q_p\big)\ge\frac{1}{2^{19}(1+6b)^2},\quad \mathbb{E}_\omega\big[1-\tfrac12\operatorname{Tr}Q_p\big]\ge\frac{1}{4b+1}.$$

Proof. Let e be the link and identify $\mathrm{SU}(2)$ with the unit quaternions. Given the other links, $U_e=x$ has density proportional to $e^{rx\cdot u}$ with $|u|=1$ and $0\le r\le6b$, since the six staples have norm one, and $\frac12\operatorname{Tr}Q_p=x\cdot v$ for a unit vector v fixed by the conditioning. Let $X=x\cdot u$ and $T=1-X$; the law of T has density proportional to $e^{-rt}\sqrt{t(2-t)}$ on $[0,2]$. Put $n=1+r$. Since $e^{-rt}\le e^2e^{-nt}$, the normalisation is at most $e^2\int_0^\infty\sqrt{2t}e^{-nt}dt=e^2\sqrt{\pi/2}\,n^{-3/2}<14n^{-3/2}$. On $I=[\frac{1}{4n},\frac{1}{2n}]$ the density is at least $\frac12n^{-1/2}e^{-1/2}$ and on $J=[\frac1n,\frac{3}{2n}]$ at least $\frac12n^{-1/2}e^{-3/2}$, so $P(T\in I)\ge\frac{1}{224}$ and $P(T\in J)\ge\frac{1}{336}$. With an independent copy T', $\operatorname{Var}T=\frac12\mathbb{E}(T-T')^2\ge P(T\in I)P(T\in J)(2n)^{-2}>2^{-19}n^{-2}$. By symmetry the conditional covariance of x is diagonal in u and $u^\perp$, with eigenvalues $\operatorname{Var}X$ and $\mathbb{E}(1-X^2)/3$; on I, $1-X^2=T(2-T)\ge\frac{3}{8n}$, so $\mathbb{E}(1-X^2)/3\ge\frac{1}{1792n}\ge2^{-19}n^{-2}$. Hence the conditional variance of $x\cdot v$ is at least $2^{-19}(1+6b)^{-2}$, and so is the variance, by the law of total variance. For the mean, integrating $\frac{d}{dX}\big(e^{rX}(1-X^2)^{3/2}\big)$ over $[-1,1]$ gives $3\mathbb{E}X=r\mathbb{E}(1-X^2)\le r(1-(\mathbb{E}X)^2)$, whence $\mathbb{E}X\le(\sqrt{9+4r^2}-3)/(2r)\le2r/(2r+3)$ for $r>0$, and $\mathbb{E}[1-x\cdot v]\ge1-\mathbb{E}X\ge3/(2r+3)\ge1/(4b+1)$. □

For the blocking statement let $G=\mathrm{SU}(2)$, $B\ge4$ an integer and, for a coarse edge (x,μ), $V_{(x,\mu)}=\prod_{r=0}^{B-1}U_{(Bx+r\hat\mu,\mu)}$ the ordered product of the fine links along it.

Proposition 232. *For every Wilson Gibbs state ω of $\mathrm{SU}(2)$ at $b>0$ and every set $\mathscr{E}$ of K coarse edges,*

$$\big\|\operatorname{Law}_\omega\big((V_e)_{e\in\mathscr{E}}\big)-m^{\otimes K}\big\|_{\mathrm{TV}}\le\frac{K}{2}e^{3b}\Big(\frac{4b}{4b+1}\Big)^{B-3}$$
$$\le\frac{K}{2}\exp\Big(3b-\frac{B-3}{4b+1}\Big),$$

where m is the Haar measure. If F is a function of the V_e of a set of coarse edges at positive times, $|F|\le1$, and ε is this bound

for those edges and their reflections, then $0 \le \omega(\vartheta(\bar F^c)F^c) \le 6\varepsilon$ *with* $F^c = F - \omega(F)$.

Proof. Select on each coarse edge the fine links with $1 \le r \le B-2$. No plaquette contains two selected links: collinear links do not share a plaquette, parallel coarse edges are B apart, and a plaquette containing a selected link of (x,μ) has its links in direction $\nu \neq \mu$ at μ-offsets $r, r+1 \in [1, B-1]$ from Bx, while coarse edges in direction ν lie at offsets in $B\mathbb{Z}$. Given the other links, the selected ones are therefore independent, each with density $e^{r_i x\cdot u_i}/Z(r_i)$, $0 \le r_i \le 6b$. Convolution with such a density acts on the representation of spin $l \ge 1$ by $D^{(l)}(u_i)$ times $I_{l+1}(r_i)/I_1(r_i)$, whose absolute value is at most $I_2(r_i)/I_1(r_i)$ because $I_\nu(r)$ decreases in $\nu \ge 0$, and $I_2(r_i)/I_1(r_i) = \mathbb{E}X \le 4b/(4b+1)$ by the proof of Proposition 231, so it contracts the mean-zero part of L^2 by that factor. The density is at most e^{6b} because $Z \ge 1$ by Jensen's inequality, so its distance to 1 in L^2 is at most e^{3b}. The conditional law of V_e is that of a product of $B-2$ such independent variables between two fixed ones, so its total variation distance to m is at most $\frac{1}{2}e^{3b}(4b/(4b+1))^{B-3}$, and the coarse edges are conditionally independent. Total variation distance from a product is at most the sum over the factors, and averaging over the conditioning does not increase it. For the last statement, under $m^{\otimes K}$ the covariance vanishes because F and its reflection depend on disjoint edges; replacing ω by $m^{\otimes K}$ changes $\omega(\vartheta(\bar F)F)$ by at most 2ε and $|\omega(F)|^2$ by at most 4ε. □

With $B = \ell/a_k$ for a fixed physical length ℓ and $b_k = O(\log(1/a_k))$ the bound tends to zero: the bare coarse links of a fixed coarse graph become independent Haar variables, their reflected form has rank one, and a renormalisation Z_k that keeps the reflected norm of $Z_k F^c$ at least $c > 0$ must grow at least like $(c/6\varepsilon)^{1/2}$. The renormalised flowed fields of Theorem 46 are not bounded functions of bare coarse links, and none of the three propositions bears on the slice ratio, the tree bounds or (D).

K. What the cube estimate requires

This subsection proves seven facts about $D(n,\alpha)$. It holds at small coupling for every even n above an explicit size (Proposition 234), and it survives small perturbations of the action (Proposition 235), although gauge-invariant perturbations still need strong coupling at the scale of the cube (Proposition 242). It forces the gauge-fixed variables of a cube with trivial exterior to stay a fixed distance from the identity on average (Theorem 236, and Proposition 238 for SU(2)), so that for every fixed n it fails once β is large (Corollary 237). And it cannot be obtained from bounds on the conditional law of a single line of links that are uniform in the configuration around the line (Proposition 246). For the compact abelian theory with the Villain action it fails at every size once β is large (Proposition 243). In [21, Sec. XII A] the estimate is posed for block averages, where its quadratic approximation holds along the tuning. None of these decides (D) along the continuum tuning, where n grows while β grows.[21]

For a finite $V \subset A$ and $j \in A \setminus V$ let $b(V,j)$ be the number of links of V that share a plaquette with j, and write $k \sim i$ when two active links share a plaquette.

Lemma 233. *Suppose that* $C_{ee'} \le c_0$ *for all active* e, e', *with* $12c_0 < 1$. *Then* $C_{V,j} \le c_0\, b(V,j)/(1-12c_0)$ *for every finite* $V \subset A$ *and every* $j \in A \setminus V$.

Proof. Fix Q, Q' that differ only at j, put $d = \delta(Q_j, Q'_j)$, $\mu = \gamma_V(\cdot \mid Q)$ and $\nu = \gamma_V(\cdot \mid Q')$. The couplings of μ and ν form a weakly compact set on which $\pi \mapsto \int d_V\, d\pi$ is continuous, so one of them, π, minimises it; let $\rho_i = \int \delta(X_i, Y_i)\, d\pi$. Fix $i \in V$. Under μ the conditional law of X_i given the other links of V is $\gamma_i^F(\cdot \mid X)$, and likewise for ν. Replace (X_i, Y_i) by a pair drawn from an optimal coupling of $\gamma_i^F(\cdot \mid X)$ and $\gamma_i^F(\cdot \mid Y)$, chosen measurably [102, Cor. 5.22], and keep the other coordinates. The new coupling still has marginals μ and ν, and changing the links in which X and Y differ one at a time gives

$$W_1\big(\gamma_i^F(\cdot \mid X), \gamma_i^F(\cdot \mid Y)\big) \le \sum_{k \in V, k \neq i} C_{ik}\, \delta(X_k, Y_k) + C_{ij}\, d.$$

Minimality of π therefore gives $\rho_i \le c_0 \sum_{k \in V, k \sim i} \rho_k + c_0 [i \sim j]\, d$. Sum over $i \in V$: every ρ_k occurs at most twelve times by Lemma 9(iv), so $(1-12c_0)\sum_i \rho_i \le c_0\, b(V,j)\, d$. Since $W_1^{d_V}(\mu,\nu) \le \sum_i \rho_i$, dividing by d proves the lemma. □

Proposition 234. *Let* $n = 2m$ *and let* V *be the cube of Definition 40. For every class* c,

$$\sum_{j \in c \setminus V} b(V,j) \le 10m^3 - 2m^2,$$

with equality for three classes. Consequently, if $C_{ee'} \le c_0$ *for all* e, e' *with* $12c_0 < 1$, *then* $D(n,\alpha)$ *holds for every* $\alpha \in (0,1)$ *with*

$$\alpha \ge \hat\alpha_n = \frac{c_0}{1-12c_0}\Big(\frac{20}{n} - \frac{8}{n^2}\Big).$$

For SU(2) *and* $|\beta_W| < \frac{1}{3}$ *this reads* $\hat\alpha_n = \frac{|\beta_W|}{1-3|\beta_W|}\big(\frac{5}{n} - \frac{2}{n^2}\big)$, *which is below one for every even* $n > 5|\beta_W|/(1-3|\beta_W|)$.

Proof. Let $j = (y,\mu)$ lie in the class c, with $y \equiv r$ modulo two. A link of V that shares a plaquette with j has base point $y+d$ with $d \in \{-1,0,1\}^4$ and at most two nonzero coordinates, and the twelve offsets d depend only on c. For a fixed offset we count the $y \equiv r$ with $y + d \in [0,n)^4$ and $y \notin [0,n)^4$. In the coordinate l there are m values of $y_l \equiv r_l$ with $y_l + d_l \in [0,n)$, and one of them lies outside $[0,n)$ exactly when $d_l = 1$ and $r_l = 1$ (the value $y_l = -1$) or $d_l = -1$ and $r_l = 0$ (the value $y_l = n$). If q is the number of such coordinates, the count is

[21] The four statements answer the question of which parts of (D) can be settled with the methods of this paper. The first places (D) in the range where the single-link method already works, the second and third show what any proof of (D) at weak coupling has to produce, and the fourth excludes the most direct one-dimensional route.

$m^4 - m^{4-q}(m-1)^q$, that is 0, m^3 or $2m^3 - m^2$ for $q = 0,1,2$. Summing over the twelve offsets, the class sum is $a_c m^3 - b_c m^2$, where b_c is the number of offsets with $q = 2$ and a_c is the number with $q = 1$ plus twice b_c. Reading the offsets from Table III, the pairs (a_c, b_c) over the 48 classes are $(2,0)$, $(7,0)$, $(7,1)$, $(9,1)$, $(9,2)$ and $(10,2)$ three times each and $(3,0)$, $(4,0)$, $(5,0)$, $(6,0)$ and $(8,1)$ six times each. In every case $a_c \le 10$ and $8 - a_c + b_c \ge 0$, so

$$(10m^3 - 2m^2) - (a_c m^3 - b_c m^2) = m^2\big((10 - a_c)m + b_c - 2\big) \ge m^2(8 - a_c + b_c) \ge 0,$$

with equality for the three classes with $(a_c, b_c) = (10,2)$. By Lemma 233, $K_c \le c_0(10m^3 - 2m^2)/(1 - 12c_0)$, and $N_n = m^4$ gives $K_c/N_n \le \hat{\alpha}_n$. For SU(2) the proof of Theorem 1 gives $c_0 = |\beta| = |\beta_W|/4$. □

At $\beta_W = \frac{3}{10}$ and $n = 16$, for instance, $\hat{\alpha}_{16} = \frac{117}{128}$, and Theorem 41 gives the gap $\log \frac{128}{117}/32$, much weaker than Theorem 1. The proposition shows that the cube condition is not empty; it does not extend the range, because the absorption $12c_0 < 1$ is the single-link condition itself.

The cube condition is also stable under small changes of the action, which is what a block renormalisation would need at its last scale, where the effective action is no longer of Wilson form. Let $r \ge 0$ and $\varepsilon \ge 0$, and let (w_X) be real functions on G^A indexed by finite sets $X \subset A$ of diameter at most r, with w_X depending only on the links of X, such that the family is invariant under the translations by $2\mathbb{Z}^4$, every active link lies in at most M of the sets X, every X has at most s links, and

$$|w_X(Q) - w_X(Q')| \le \varepsilon \sum_{i \in X} \delta(Q_i, Q_i')$$

for all Q, Q'. Let $\gamma_V^W(\cdot \mid Q)$ be the probability measure with density proportional to $\exp \sum_{X \cap V \neq \emptyset} w_X$ with respect to $\gamma_V(\cdot \mid Q)$, and define $C_{V,j}^W$, K_c^W and the condition $D^W(n,\alpha)$ as in Definition 40, with γ_V^W in place of γ_V.

Proposition 235. *Let n be even and suppose that $D(n,\alpha)$ holds at β. Then $D^W(n,\alpha')$ holds with*

$$\alpha' = \alpha\, e^{\Omega}(1 + 2\Omega) + 96\, s\, \Omega, \qquad \Omega = 48\, M\, D\, N_n\, \varepsilon.$$

In particular $\alpha' < 1$ for all ε below an explicit $\varepsilon_0(n, \alpha, M, s) > 0$.

Proof. The cube has $|V| = 48N_n$ links, so d_V has diameter $\Delta_V = 48N_n D$. For fixed exterior links put $u = \sum_{X \cap V \neq \emptyset} w_X$ as a function on G^V. Each link of V lies in at most M sets X, so u is $M\varepsilon$-Lipschitz for d_V and its oscillation is at most $M\varepsilon\Delta_V = \Omega$; we normalise u so that $\sup u = 0$, which does not change the tilted measures, and put $h = e^u$, so that $e^{-\Omega} \le h \le 1$ and h is $M\varepsilon$-Lipschitz.

A common tilt. Let μ, ν be probability measures on G^V and $\mu_h = h\mu/\mu(h)$, $\nu_h = h\nu/\nu(h)$. Let f be 1-Lipschitz for d_V; subtracting a constant we may take $|f| \le \Delta_V$. Then fh is $(1 + \Delta_V M\varepsilon)$-Lipschitz, $|\nu(fh)| \le \Delta_V \nu(h)$, $|\mu(h) - \nu(h)| \le M\varepsilon W_1^{d_V}(\mu,\nu)$ and $\mu(h) \ge e^{-\Omega}$, so

$$\mu_h(f) - \nu_h(f) = \frac{\mu(fh) - \nu(fh)}{\mu(h)} + \nu(fh)\,\frac{\nu(h) - \mu(h)}{\mu(h)\nu(h)} \le e^{\Omega}(1 + 2\Omega)\, W_1^{d_V}(\mu,\nu),$$

since $\Delta_V M\varepsilon = \Omega$. By Kantorovich duality $W_1^{d_V}(\mu_h, \nu_h) \le e^{\Omega}(1 + 2\Omega) W_1^{d_V}(\mu,\nu)$.

Two tilts. If u' is another weight with $|u - u'| \le \eta$, the density of $\nu_{h'}$ with respect to ν_h lies in $[e^{-2\eta}, e^{2\eta}]$, so the total variation distance of the two is at most $1 - e^{-2\eta} \le 2\eta$, and $W_1^{d_V}(\nu_h, \nu_{h'}) \le 2\eta\Delta_V$.

The coefficients. Let Q, Q' differ only at $j \notin V$, with $\delta = \delta(Q_j, Q_j')$, and let u, u' be the weights for the two exteriors. They differ only through the sets X that contain j and meet V; if there are m_j of them, $|u - u'| \le m_j \varepsilon \delta$. With $\mu = \gamma_V(\cdot \mid Q)$ and $\nu = \gamma_V(\cdot \mid Q')$, the triangle inequality and the two steps give

$$C_{V,j}^W \le e^{\Omega}(1 + 2\Omega)\, C_{V,j} + 2\Delta_V\, m_j\, \varepsilon.$$

The pairs (X, j) with $j \in X \setminus V$ and $X \cap V \neq \emptyset$ number at most s times the number of sets meeting V, hence at most $sM|V|$. Summing over $j \in c \setminus V$,

$$K_c^W \le e^{\Omega}(1 + 2\Omega)\, \alpha N_n + 2\Delta_V s M |V| \varepsilon = \big(\alpha e^{\Omega}(1 + 2\Omega) + 96\, s\, \Omega\big) N_n.$$

The family is invariant under $2\mathbb{Z}^4$, so the coefficients are, and this is $D^W(n,\alpha')$. □

The admissible perturbation is of order $1/N_n = (2/n)^4$, so the proposition is useful in a box of a fixed number of lattice spacings, which is how a renormalisation argument would use it: after enough block steps that the cube of physical size ℓ has become a cube of fixed side, the effective action has to lie within ε_0 of an action that satisfies the condition there. Whether the effective actions along the tuning do so is not decided here, and that is where the problem lies: at the scale where the blocked theory is strongly coupled enough for a condition of this kind, the running coupling is of order one, which is outside the range of the existing ultraviolet analysis. The proposition shows only that the condition itself is not the obstacle.

The next theorem uses the configuration in which every link outside V is the identity. Write $\gamma_V^1 = \gamma_V(\cdot \mid 1)$ and

$$\varepsilon_V(\beta) = \frac{1}{|V|} \sum_{i \in V} \int \delta(1, Q_i)\, d\gamma_V^1(Q)$$

for the mean distance of the gauge-fixed variables of the cube from the identity. Recall that δ is bi-invariant of diameter D.

Theorem 236. *Let n be even, V the cube of Definition 40, and suppose that $D(n,\alpha)$ holds at β. Then*

$$\varepsilon_V(\beta) \ge \frac{(1 - \alpha)\, D}{3}.$$

Proof. A flat configuration. By Lemma 9(i) and the period two of σ, the sixteen residues modulo two fall into eight pairs $\{r, r+e_{\sigma(r)}\}$, the residues met by one line of F. The eight residues 0010, 0011, 0100, 0110, 1001, 1011, 1100, 1101 form four of these pairs. Choose $z, w \in G$ with $\delta(z,w) = D$, put $g(x) = w$ if x is congruent to one of those eight residues and $g(x) = z$ otherwise, and let $Q^g_{(x,\mu)} = g(x)g(x+e_\mu)^{-1}$ for every link of $\mathbb{Z}^4$. Then g is constant on every line of F, so Q^g equals 1 on F, every plaquette variable of Q^g is 1, and Q^g has period two. By bi-invariance $\delta(1, Q^g_{(x,\mu)}) = \delta(g(x+e_\mu), g(x))$, which is D when x and $x+e_\mu$ receive different values and 0 otherwise; from Table III the first case occurs for exactly 32 of the 48 classes. Let δ_c be the common value of $\delta(1, Q^g_e)$ for e in the class c, so that $\sum_c \delta_c = 32D$.

Gauge covariance. Let $\Phi(Q)_{(x,\mu)} = g(x)Q_{(x,\mu)}g(x+e_\mu)^{-1}$ on G^V. The gauge transformation by g maps the configuration equal to Q_V on V and to 1 elsewhere to the one equal to $\Phi(Q_V)$ on V and to Q^g elsewhere; it leaves every plaquette trace unchanged and every link of F equal to 1, and Φ preserves Haar measure. Hence $\gamma_V(\cdot \mid Q^g) = \Phi_* \gamma^1_V$.

Upper bound. Only the finitely many links outside V that share a plaquette with V enter γ_V. Changing them one at a time from 1 to Q^g and applying the definition of $C_{V,j}$ at each step,

$$W_1^{d_V}\left(\gamma^1_V, \gamma_V(\cdot \mid Q^g)\right) \le \sum_{j \in A \setminus V} C_{V,j}\, \delta(1, Q^g_j) = \sum_c \delta_c K_c \le \alpha N_n \sum_c \delta_c = 32\,\alpha D N_n.$$

Lower bound. The function $f(Q) = -\sum_{i\in V} \delta(Q_i, Q^g_i)$ is 1-Lipschitz for d_V. By gauge covariance and bi-invariance, $\int f\, d\gamma_V(\cdot \mid Q^g) = -\int \sum_i \delta(Q_i, 1)\, d\gamma^1_V = -|V|\varepsilon_V$, and by the triangle inequality $\int f\, d\gamma^1_V \le -\sum_i \delta(1, Q^g_i) + |V|\varepsilon_V$. Every class has N_n links in V, so $\sum_i \delta(1, Q^g_i) = 32DN_n$ and

$$W_1^{d_V}\left(\gamma^1_V, \gamma_V(\cdot \mid Q^g)\right) \ge 32DN_n - 2|V|\varepsilon_V.$$

With $|V| = 48N_n$ the two bounds give $96N_n\varepsilon_V \ge 32(1-\alpha)DN_n$. □

Corollary 237. *For every even n and every $\alpha < 1$ there is $\beta_0(n,\alpha)$ such that $D(n,\alpha)$ fails at every $\beta \ge \beta_0(n,\alpha)$. In particular, if $D(n_k,\alpha)$ holds at couplings $\beta_k \to \infty$, then $n_k \to \infty$.*

Proof. For $\beta > 0$, γ^1_V has density proportional to $e^{-N\beta S}$ with respect to Haar measure, where $S = \sum_{p \cap V \neq \emptyset} (N - \operatorname{Re}\operatorname{Tr} Q^F_p)$ is continuous and nonnegative on G^V and vanishes exactly when $Q^F_p = 1$ for every plaquette p that meets V. We show that this forces $Q_V = 1$. The other plaquettes have all their links equal to 1, so every plaquette variable is 1; since the plaquettes of $\mathbb{Z}^4$ fill every closed lattice path, there is a function h on the sites with $Q_{(x,\mu)} = h(x)h(x+e_\mu)^{-1}$ on every link, and h takes equal values at the two ends of every link on which $Q = 1$. Every link with base point outside $[0,n)^4$ is such a link. From a site with a coordinate at least n one can move in every positive direction along links of this kind, and from a site with a negative coordinate x_ν one can move in the positive directions other than ν until another coordinate reaches n; so h is constant on the sites outside the box. Every line of F leaves the box, and h is constant along it, so h is constant and $Q_V = 1$.

Let U be a neighbourhood of 1 in G^V, $s = \min_{G^V \setminus U} S > 0$, and $U' \subset U$ a neighbourhood of 1 on which $S < s/2$. Then $\gamma^1_V(G^V \setminus U) \le e^{-N\beta s}/(|U'|e^{-N\beta s/2})$, which tends to zero, so $\varepsilon_V(\beta) \to 0$ as $\beta \to \infty$, and Theorem 236 excludes $D(n,\alpha)$ once $\varepsilon_V(\beta) < (1-\alpha)D/3$. □

For SU(2) the constant of Theorem 236 can be improved: a flat configuration with four values instead of two moves every class of active links, and the means of the gauge-fixed links are scalar.

Proposition 238. *Let $G = \mathrm{SU}(2)$ with the chordal metric, let n be even, V the cube of Definition 40, and suppose that $D(n,\alpha)$ holds at β. Then*

$$\frac{1}{|V|}\sum_{i\in V}\left|\int \operatorname{Re}\operatorname{Tr} Q_i\, d\gamma^1_V\right| \le 2\alpha,$$

$$\frac{1}{|V|}\sum_{i\in V}\int \delta(1,Q_i)^2\, d\gamma^1_V \ge 2(1-\alpha),$$

and $\varepsilon_V(\beta) \ge 1-\alpha$.

Proof. Write $\langle\cdot,\cdot\rangle$ for the Euclidean product of quaternions; then $\operatorname{Re}\operatorname{Tr} q = 2\langle q, 1\rangle$ and $\delta(p,q) = |p-q|$.

Four colours. Split the sixteen residues modulo two into the sets

$$X_1 = \{0000, 1000, 0111, 1111\},$$
$$X_2 = \{0001, 0101, 1010, 1110\},$$
$$X_3 = \{0010, 0011, 1100, 1101\},$$
$$X_4 = \{0100, 0110, 1001, 1011\}.$$

Each is the union of two of the eight pairs $\{r, r+e_{\sigma(r)}\}$, and Table III shows that r and $r+e_\mu$ lie in different sets for every residue r and every $\mu \neq \sigma(r)$. (The eight residues used in the proof of Theorem 236 form $X_3 \cup X_4$.) Let

$$z_1 = (\mathbf{i}+\mathbf{j}+\mathbf{k})/\sqrt{3}, \quad z_2 = (\mathbf{i}-\mathbf{j}-\mathbf{k})/\sqrt{3},$$
$$z_3 = (-\mathbf{i}+\mathbf{j}-\mathbf{k})/\sqrt{3}, \quad z_4 = (-\mathbf{i}-\mathbf{j}+\mathbf{k})/\sqrt{3},$$

unit quaternions with $\langle z_k, z_l\rangle = -\frac{1}{3}$ for $k \neq l$, so that $\delta(z_k, z_l) = d_* = (8/3)^{1/2}$. Put $g(x) = z_k$ when the residue of x lies in X_k, and $Q^g_{(x,\mu)} = g(x)g(x+e_\mu)^{-1}$. As in the proof of Theorem 236, Q^g equals 1 on F, every plaquette variable of Q^g is 1, and $\gamma_V(\cdot \mid Q^g) = \Phi_*\gamma^1_V$ with $\Phi(Q)_{(x,\mu)} = g(x)Q_{(x,\mu)}g(x+e_\mu)^{-1}$; now $\delta(1, Q^g_e) = d_*$ for every active link e.

Upper bound. Changing the exterior links one at a time,

$$W_1^{d_V}\left(\gamma^1_V, \gamma_V(\cdot \mid Q^g)\right) \le d_* \sum_c K_c \le 48\,\alpha\, d_* N_n = \alpha\, d_* |V|.$$

Lower bound. Conjugating every link of V by one unit quaternion leaves the exterior 1, the links of F, Haar measure and every plaquette trace unchanged, so it preserves γ^1_V. The

mean $m_i = \int Q_i \, d\gamma_V^1$, taken in the space of quaternions, therefore commutes with every unit quaternion, and $m_i = a_i 1$ with $a_i = \frac{1}{2}\int \mathrm{Re}\,\mathrm{Tr}\, Q_i \, d\gamma_V^1$ real. Since Φ acts on each link by a fixed left and right multiplication, the mean of Q_i under $\gamma_V(\cdot \mid Q^g)$ is $a_i Q_i^g$. Choose unit quaternions w_i with $\langle w_i, a_i Q_i^g - a_i 1\rangle = |a_i|\,|Q_i^g - 1|$. The function $f(Q) = \sum_{i\in V}\langle w_i, Q_i\rangle$ is 1-Lipschitz for d_V, and Kantorovich duality gives

$$W_1^{d_V}\big(\gamma_V^1, \gamma_V(\cdot \mid Q^g)\big) \geq \sum_{i\in V} |a_i|\,\delta(1, Q_i^g) = d_* \sum_{i\in V} |a_i|.$$

Conclusion. Comparing the two bounds, $\sum_i |a_i| \leq \alpha |V|$, which is the first inequality. Since $\delta(1,q)^2 = 2 - 2\langle q, 1\rangle$, $\int \delta(1,Q_i)^2 \, d\gamma_V^1 = 2(1-a_i) \geq 2(1-|a_i|)$, and averaging gives the second. As $\delta \leq 2$, $\delta \geq \delta^2/2$, and the bound on ε_V follows. □

The four colours need nothing from SU(2) except four elements whose pairwise quotients are conjugate, and every compact simple group contains such elements. For a metric that comes from a representation the bound then compares the cube with Haar measure.

Proposition 239. *Let G be a compact simple group, ρ a nontrivial irreducible unitary representation of G of dimension d, and $\delta(p,q) = \kappa^{-1/2}\|\rho(p) - \rho(q)\|_{\mathrm{HS}}$ for some $\kappa > 0$. Let n be even, V the cube of Definition 40, and suppose that $D(n,\alpha)$ holds at β. Then*

$$\frac{1}{|V|}\sum_{i\in V}\Big|\int \mathrm{Tr}\,\rho(Q_i)\, d\gamma_V^1\Big| \leq \alpha\, d,$$

$$\frac{1}{|V|}\sum_{i\in V}\int \delta(1,Q_i)^2\, d\gamma_V^1 \geq (1-\alpha)\int_G \delta(1,q)^2\, dq,$$

where dq is Haar measure, and $\varepsilon_V(\beta) \geq (1-\alpha)D/2$.

Proof. Four elements. Let $\varphi\colon \mathrm{SU}(2) \to G$ be the homomorphism of the SU(2) subgroup of a root, and $z_1, \dots, z_4$ the tetrahedral quaternions of Proposition 238. For $k \neq l$ the quaternion $z_k z_l^{-1}$ has real part $\langle z_k, z_l\rangle = -\frac{1}{3}$; unit quaternions with equal real parts are conjugate in SU(2), so the twelve elements $\varphi(z_k z_l^{-1})$ are conjugate in G, and by bi-invariance $\delta(\varphi(z_k), \varphi(z_l)) = d_*$ for all $k \neq l$, with one number d_*. If $d_* = 0$, then $\varphi(z_k z_l^{-1})$ lies in $\ker\rho$, a proper normal subgroup of the simple group G and therefore central; it then commutes with $\varphi(\mathrm{SU}(2))$, so it is central there, and $z_k z_l^{-1}$ lies in $\{\pm 1\}\ker\varphi \subset \{\pm 1\}$, which its real part $-\frac{1}{3}$ excludes. So $d_* > 0$.

The two bounds. Put $g(x) = \varphi(z_k)$ when the residue of x lies in X_k. The proof of Proposition 238 applies word for word: Q^g equals 1 on F and on every plaquette, $\delta(1, Q_e^g) = d_*$ on every active link, and $W_1^{d_V}(\gamma_V^1, \gamma_V(\cdot \mid Q^g)) \leq \alpha d_* |V|$. Simultaneous conjugation of the links of V preserves γ_V^1, so the mean $\int \rho(Q_i)\, d\gamma_V^1$ commutes with $\rho(G)$, and by Schur's lemma it is $c_i I$ with $c_i = d^{-1}\int \mathrm{Tr}\,\rho(Q_i)\, d\gamma_V^1$. Its mean under $\gamma_V(\cdot \mid Q^g)$ is $c_i \rho(Q_i^g)$. With matrices W_i of Hilbert–Schmidt norm $\kappa^{-1/2}$ chosen so that $\mathrm{Re}\,\mathrm{Tr}(W_i^* c_i(\rho(Q_i^g) - I)) = |c_i|\,\delta(1, Q_i^g)$, the function $f(Q) = \sum_i \mathrm{Re}\,\mathrm{Tr}(W_i^* \rho(Q_i))$ is 1-Lipschitz for d_V, and Kantorovich duality gives $W_1^{d_V} \geq d_* \sum_i |c_i|$. Hence $\sum_i |c_i| \leq \alpha |V|$, the first inequality.

Comparison with Haar measure. Since $\kappa\,\delta(1,q)^2 = 2d - 2\,\mathrm{Re}\,\mathrm{Tr}\,\rho(q)$ and $\int_G \mathrm{Tr}\,\rho(q)\, dq = 0$ for a nontrivial irreducible ρ, $\int_G \delta(1,q)^2 dq = 2d/\kappa$, while $\int \delta(1,Q_i)^2 d\gamma_V^1 = 2d(1 - \mathrm{Re}\, c_i)/\kappa \geq 2d(1-|c_i|)/\kappa$. Averaging gives the second inequality. Finally $\kappa D^2 \leq 4d$, because $\|\rho(p) - \rho(q)\|_{\mathrm{HS}} \leq 2\sqrt{d}$, so $\delta \geq \delta^2/D$ gives $\varepsilon_V \geq 2d(1-\alpha)/(\kappa D) \geq (1-\alpha)D/2$. □

For SU(N) with the Hilbert–Schmidt distance this improves the constant $\frac{1}{3}$ of Theorem 236 to $\frac{1}{2}$, and for SU(3), where $D^2 = 9$ and the Haar mean of δ^2 is 6, to $\frac{2}{3}$: in a cube with the identity outside, $D(n,\alpha)$ leaves the gauge-fixed links at least the fraction $1-\alpha$ of the disorder of independent Haar-distributed links.

The size $\beta_0(n,\alpha)$ in Corollary 237 comes from a compactness argument. For SU(2) it can be made explicit, and then reads as a lower bound on the side of any cube that satisfies the condition.

Proposition 240. *Let $G = \mathrm{SU}(2)$ with the chordal metric, so that $D = 2$, let $b = \beta_W \geq \frac{1}{9}$, let $n \geq 2$ be even and put $L(b) = 1 + \log 24 + \frac{3}{2}\log(9b)$. Then*

$$\frac{1}{|V|}\sum_{e\in V}\int \delta(1,Q_e)^2\, d\gamma_V^1 \leq \frac{n^2(n+2)(2n+1)\,L(b)}{b},$$

so $\varepsilon_V(\beta)^2$ is at most the same quantity. Consequently, if $D(n,\alpha)$ holds at b, then

$$n^4 \geq \frac{2(1-\alpha)\,b}{5\,L(b)}.$$

Proof. Put $m = n/2$. Under γ_V^1 every link outside V equals 1: the links of F always, the others by the choice of exterior. A plaquette that does not meet V therefore has plaquette variable 1.

Links as loops. Let $e = (x,\mu) \in V$, $y = x + e_\mu$, $\sigma = \sigma(x)$ and $\tau = \sigma(y)$. The directions σ, τ, μ are distinct: $\sigma \neq \mu$ because e is active; $\tau \neq \mu$ because otherwise the line of F through y would contain x, and the lines of F are disjoint; and $\tau \neq \sigma$ by Table III, in which $\sigma(x + e_\mu) \neq \sigma(x)$ whenever $\mu \neq \sigma(x)$. Since $0 \leq x_\sigma < n$, one of $n - x_\sigma$ and $x_\sigma + 1$ is at most m; call it s and let $\varepsilon = \pm 1$ be the sign for which $x' = x + s\varepsilon e_\sigma$ has σ-coordinate outside $[0,n)$. Choose $t \leq m$ and ε' in the same way for y and τ (note $y_\tau = x_\tau$). Let R_1 be the s plaquettes of the rectangle with corners x, x', $x' + e_\mu$, y in the plane $\{\sigma,\mu\}$, and R_2 the st plaquettes of the rectangle with corners y, $x' + e_\mu$, $x' + e_\mu + t\varepsilon' e_\tau$, $y + t\varepsilon' e_\tau$ in the plane $\{\sigma,\tau\}$. The union $R(e) = R_1 \cup R_2$ is a disc, and its boundary runs from x along the line of F through x to x', then to $x' + e_\mu$, along e_τ to $x' + e_\mu + t\varepsilon' e_\tau$, along e_σ to $y + t\varepsilon' e_\tau$, back along the line of F through y to y, and over e to x. The links of the first and fifth pieces lie in F. The links of the second and third pieces have base points with σ-coordinate $x'_\sigma \notin [0,n)$, and those of the fourth have base points with τ-coordinate outside $[0,n)$; none of them lies in V. So the holonomy of the boundary of

$R(e)$ is $Q_e^{\pm1}$. Adding the plaquettes of a disc one at a time along its boundary shows that this holonomy is a product of conjugates of the plaquette variables $(Q_p^F)^{\pm1}$, $p\in R(e)$, one for each plaquette. By bi-invariance and the triangle inequality,

$$\delta(1,Q_e)\le\sum_{p\in R(e)}\delta(1,Q_p^F),\qquad |R(e)|=s(t+1)\le m(m+1).$$

Multiplicity. If a plaquette p lies in $R_2(e)$, its plane is $\{\sigma,\tau\}$, μ is one of the two other directions, and the base point of p is $y+ae_\sigma+ce_\tau$ with $-m\le a,c\le m-1$. Given p, there are two ways to name the directions of its plane, two choices of μ and $(2m)^2$ of (a,c), and they determine y, hence e; so p lies in at most $16m^2$ of the sets $R_2(e)$. Likewise $p\in R_1(e)$ forces its plane to be $\{\sigma,\mu\}$ and its base point to be $x+ae_\sigma$ with $-m\le a\le m-1$, which leaves at most $4m$ links e. Every plaquette therefore lies in at most $16m^2+4m=4n^2+2n$ of the discs $R(e)$.

The energy. For the chordal metric $\delta(1,q)^2=2-\operatorname{Re}\operatorname{Tr}q$. Let $S=\sum_{p\cap V\neq\emptyset}\delta(1,Q_p^F)^2$; then γ_V^1 has density $e^{-bS/2}/Z$ with $Z=\int e^{-bS/2}$, the integral over $\mathrm{SU}(2)^V$ with Haar measure. The function $t\mapsto\log\int e^{-tS}$ is convex and vanishes at $t=0$, so its derivative at $t=b/2$, which is $-\int S\,d\gamma_V^1$, is at least $2\log Z/b$. A plaquette that meets V contains at most three links of V and the other links are 1, so $\delta(1,Q_p^F)^2\le3\sum_{e\in p\cap V}\delta(1,Q_e)^2$; every link lies in six plaquettes, so $S\le18\sum_{e\in V}\delta(1,Q_e)^2$ and $Z\ge I^{|V|}$ with $I=\int e^{-9b\,\delta(1,q)^2}dq$. The chordal distance is at most the geodesic one, and the geodesic ball of radius $r=(9b)^{-1/2}\le1$ has measure greater than $r^3/24$ (proof of Proposition 246), so $I\ge e^{-1}(9b)^{-3/2}/24$ and

$$\int S\,d\gamma_V^1\le\frac{2|V|L(b)}{b}.$$

Conclusion. The Cauchy–Schwarz inequality inside each disc squares the loop bound:

$$\begin{aligned}\delta(1,Q_e)^2&\le|R(e)|\sum_{p\in R(e)}\delta(1,Q_p^F)^2\\&\le m(m+1)\sum_{p\in R(e)}\delta(1,Q_p^F)^2.\end{aligned}$$

Summing over $e\in V$ and using the multiplicity,

$$\sum_{e\in V}\delta(1,Q_e)^2\le m(m+1)(4n^2+2n)S=\frac{n^2(n+2)(2n+1)}{2}S,$$

and integrating against γ_V^1 gives the first bound; the bound on ε_V follows by the Cauchy–Schwarz inequality. If $D(n,\alpha)$ holds, Proposition 238 gives $2(1-\alpha)\le n^2(n+2)(2n+1)L(b)/b$, and $(n+2)(2n+1)\le5n^2$ for $n\ge2$. □

The loops and their multiplicity use nothing of $\mathrm{SU}(2)$ but the bi-invariance of δ, and the energy uses only the form of the action and the measure of small balls. The bound therefore holds for every $\mathrm{SU}(N)$, with the same power of n.

Proposition 241. *Let $G=\mathrm{SU}(N)$, $N\ge2$, with the weight $\exp(\frac{b}{N}\sum_p\operatorname{Re}\operatorname{Tr}Q_p)$, $b=\beta_W\ge N/9$, and the Hilbert–Schmidt distance $\delta(p,q)=\|p-q\|_{\mathrm{HS}}$. Let $n\ge2$ be even and put*

$$L_N(b)=1+\frac{N^2-1}{2}\log(81\pi^2b)+N(N-1)\log\frac{12}{11}.$$

Then

$$\frac{1}{|V|}\sum_{e\in V}\int\delta(1,Q_e)^2\,d\gamma_V^1\le\frac{Nn^2(n+2)(2n+1)L_N(b)}{b}.$$

Consequently, if $D(n,\alpha)$ holds at b, then

$$n^4\ge\frac{2(1-\alpha)b}{5L_N(b)}.$$

Proof. Loops. The discs $R(e)$ of the proof of Proposition 240 and the bound $4n^2+2n$ on their multiplicity do not depend on the group, and the loop bound uses only bi-invariance and the triangle inequality. With $S=\sum_{p\cap V\neq\emptyset}\delta(1,Q_p^F)^2$ the same Cauchy–Schwarz step gives $\sum_{e\in V}\delta(1,Q_e)^2\le\frac12n^2(n+2)(2n+1)S$.

The energy. Since $\delta(1,q)^2=2N-2\operatorname{Re}\operatorname{Tr}q$, the measure γ_V^1 has density $e^{-bS/(2N)}/Z$ with respect to Haar measure. As in the proof of Proposition 240, the convexity of $t\mapsto\log\int e^{-tS}$ gives $\int S\,d\gamma_V^1\le2N\log(1/Z)/b$, and $S\le18\sum_{e\in V}\delta(1,Q_e)^2$ gives $Z\ge I^{|V|}$ with $I=\int e^{-9b\,\delta(1,q)^2/N}dq$.

Small balls. Give $\mathfrak{su}(N)$ the inner product $\operatorname{Tr}(X^*Y)$ and $\mathrm{SU}(N)$ the corresponding bi-invariant metric, of volume v_N. In exponential coordinates the Riemannian measure is $j(X)\,dX$, where dX is Lebesgue measure and

$$j(X)=\Big|\det\frac{1-e^{-\operatorname{ad}X}}{\operatorname{ad}X}\Big|=\prod_{k<l}\Big(\frac{\sin(\varphi_{kl}/2)}{\varphi_{kl}/2}\Big)^2$$

if X has eigenvalues $i\theta_1,\dots,i\theta_N$ and $\varphi_{kl}=\theta_k-\theta_l$ [61, Ch. II, Thm. 1.7]; in particular $j\le1$. Every $q\in\mathrm{SU}(N)$ is conjugate to a diagonal matrix with entries $e^{i\psi_k}$, $-\pi\le\psi_k<\pi$; then $\sum_k\psi_k=2\pi k_0$ with $|k_0|\le N/2$, and changing $|k_0|$ of the ψ_k by $\mp2\pi$ gives angles θ_k with sum zero and $|\theta_k|\le3\pi$. So $q=e^X$ with $X\in\mathfrak{su}(N)$ and $\|X\|_{\mathrm{HS}}\le3\pi\sqrt{N}$, and $v_N\le\omega_{N^2-1}(3\pi\sqrt{N})^{N^2-1}$, where ω_k is the volume of the unit ball of $\mathbb{R}^k$. Now let $\|X\|_{\mathrm{HS}}\le r\le1$. The eigenvalues have $|\theta_k|\le1<\pi$, so exp is injective on this ball; $\delta(1,e^X)^2=\sum_k|e^{i\theta_k}-1|^2\le\|X\|_{\mathrm{HS}}^2$; and $|\varphi_{kl}|/2\le(\theta_k^2+\theta_l^2)^{1/2}/\sqrt2\le2^{-1/2}$, so each factor of $j(X)$ is at least $(1-\frac{1}{12})^2$ because $\sin x/x\ge1-x^2/6$. Dividing by v_N, the Haar measure of $\{q:\delta(1,q)\le r\}$ is at least

$$\Big(\frac{11}{12}\Big)^{N(N-1)}\frac{\omega_{N^2-1}r^{N^2-1}}{v_N}\ge\Big(\frac{11}{12}\Big)^{N(N-1)}\Big(\frac{r}{3\pi\sqrt{N}}\Big)^{N^2-1}.$$

With $r=(N/9b)^{1/2}\le1$ this gives $I\ge e^{-1}$ times the right-hand side, and since $3\pi\sqrt{N}/r=9\pi\sqrt{b}$, $\log(1/I)\le L_N(b)$. Hence $\int S\,d\gamma_V^1\le2N|V|L_N(b)/b$, and the first bound follows from the loops.

Conclusion. Proposition 239 with the defining representation and $\kappa=1$, for which the Haar mean of $\delta(1,q)^2$ is $2N$, gives $2N(1-\alpha)\le Nn^2(n+2)(2n+1)L_N(b)/b$ when $D(n,\alpha)$ holds, and $(n+2)(2n+1)\le5n^2$. □

For every N, a cube on which the condition holds therefore has side at least of order $(\beta_W/\log\beta_W)^{1/4}$, with a constant that depends on N only through L_N, which grows like $\frac{N^2}{2}\log\beta_W$.

The cube condition is a statement about strong coupling at its own scale, and this persists under the perturbations of Proposition 235 when they respect the gauge. Call (w_X) *gauge invariant* if $w_X(Q^h) = w_X(Q)$ for every function h on the sites that is constant on every line of F, where $Q^h_{(x,\mu)} = h(x)Q_{(x,\mu)}h(x+e_\mu)^{-1}$. Such transformations keep the links of F equal to 1, so they act on G^A, and the restriction to the gauge of an action invariant under all gauge transformations is gauge invariant in this sense. Write ε^W_V for ε_V with $\gamma^W_V(\cdot \mid 1)$ in place of γ^1_V.

Proposition 242. *Let (w_X) be as in Proposition 235 and gauge invariant, let $n = 2m$, and suppose that $D^W(n,\alpha)$ holds.*

(i) $\varepsilon^W_V \geq (1-\alpha)D/3$.

(ii) *If $\eta \geq 0$, $0 \leq \kappa \leq 1$, and the event that $\delta(1, Q^F_p) \leq \eta$ for every plaquette p that meets V has probability at least $1-\kappa$ under $\gamma^W_V(\cdot \mid 1)$, then*

$$m(m+1)\,\eta + \kappa D \geq \frac{(1-\alpha)D}{3}.$$

Proof. (i) The proof of Theorem 236 uses the action only through the identity $\gamma_V(\cdot \mid Q^g) = \Phi_* \gamma^1_V$ and through the coefficients $C_{V,j}$. The function g of that proof is constant on every line of F, so the gauge transformation by g leaves every w_X unchanged, as it leaves every plaquette trace, and the same identity holds for γ^W_V. Only finitely many links outside V enter γ^W_V, because the sets X have bounded diameter, so the upper bound holds with $C^W_{V,j}$ and K^W_c in place of $C_{V,j}$ and K_c; the lower bound is unchanged.

(ii) Under $\gamma^W_V(\cdot \mid 1)$ every link outside V equals 1, so a plaquette that does not meet V has variable 1. The loop bound of the proof of Proposition 240, which uses only the bi-invariance of δ, gives $\delta(1,Q_e) \leq \sum_{p \in R(e)} \delta(1, Q^F_p) \leq m(m+1)\eta$ for every $e \in V$ on the event of the statement, and $\delta(1,Q_e) \leq D$ off it. Averaging over $e \in V$ and integrating, $\varepsilon^W_V \leq m(m+1)\eta + \kappa D$, and (i) completes the proof. □

In the small-field region of a gauge-covariant block renormalisation the blocked plaquette variables lie within a distance η that tends to zero with the running coupling, and large fields have small probability κ. If $\kappa < (1-\alpha)/3$, part (ii) allows the cube condition on the blocked lattice only when $m(m+1) \geq x$, where $x = ((1-\alpha)/3 - \kappa)D/\eta$, so the side must grow like $\eta^{-1/2}$. On such a cube $N_n = m^4 \geq x^2/4$, and since $\alpha' \geq \alpha + 96 s\Omega$, Proposition 235 gives $\alpha' < 1$ only for

$$\begin{aligned}\varepsilon &< \frac{1-\alpha}{4608\, sMDN_n} \\ &\leq \frac{(1-\alpha)\,\eta^2}{1152\, sMD^3\big((1-\alpha)/3-\kappa\big)^2},\end{aligned}$$

which is of order η^2. At a fixed side the condition is therefore a statement about strong coupling, and a renormalisation argument has to carry the blocked theory out of the small-field regime before the condition can be checked. That passage, at a running coupling of order one, is the step that remains open.

The next statement shows that the cube condition separates a theory with a massless photon from the expected behaviour of SU(N): for the compact abelian theory in its Coulomb phase it fails at every scale. Let $G = \mathrm{U}(1)$ with the chordal metric, $D = 2$, and replace the Wilson weight by the Villain weight

$$\varphi_\beta(\theta) = \sum_{k \in \mathbb{Z}} e^{-\beta(\theta + 2\pi k)^2/2},$$

so that the measure of a box Λ has density proportional to $\prod_{p \in P(\Lambda)} \varphi_\beta(\theta_p)$, where $e^{i\theta_p}$ is the plaquette variable. Define γ^F, $C_{V,j}$, K_c and $D(n,\alpha)$ as before with φ_β in place of the Wilson weight.

Proposition 243. *There is β_0 such that for the Villain theory at every $\beta > \beta_0$ the condition $D(n,\alpha)$ fails for every even $n \geq 2$ and every $\alpha < 1$. In particular, for couplings $\beta_k \to \infty$ and any even n_k, the hypothesis of Theorem 43 fails for this theory.*

Proof. Lemma 10, Proposition 11(i)–(iii) and (v), Lemma 347, Corollary 348 and Lemma 350 use the Wilson weight only as a positive continuous class function of the plaquette variable, with one exception: the Doeblin bound in Lemma 350(a), which for the Villain weight reads that each single-link density is at least $(\min\varphi_\beta/\max\varphi_\beta)^6 > 0$ times its maximum. Suppose that $D(n,\alpha)$ holds. As in the proof of Theorem 41, the limit $\omega(O) = \lim W_\Lambda(O)$ of the measures of boxes with free boundary conditions then exists along every sequence of boxes exhausting $\mathbb{Z}^4$, and for gauge-invariant local O, O' with O Lipschitz in each link,

$$|\mathrm{Cov}_\omega(O,O')| \leq \|O'\|_\infty L_O \frac{D(1 + c_n e^{\lambda\rho_n})}{1-\theta}\,|S_{O'}|\,e^{-\lambda\,\mathrm{dist}(O,O')}$$

for every $\lambda > 0$ with $\theta = \alpha e^{\lambda\rho_n} < 1$, where L_O is a Lipschitz constant of O and $S_{O'}$ the support of O'.

Put $\Psi_p = \varphi'_\beta(\theta_p)/\varphi_\beta(\theta_p)$; up to the factor $-\mathbf{i}$ this is the lattice field strength of Fröhlich and Spencer [51, Sect. 2] for the Villain action. It is real, odd and smooth in θ_p, so it is gauge invariant, local and Lipschitz in each of its four links, with constants that do not depend on p, and $\omega(\Psi_p) = 0$ by the symmetry $\theta \mapsto -\theta$. The bound above gives $\sum_{p'} |\omega(\Psi_p \Psi_{p'})| < \infty$, the sum over the plaquettes p' parallel to p. The state of [51] is the limit of the same free-boundary measures over boxes, which is ω. Fröhlich and Spencer prove, for β larger than some β_0, that the Fourier transform of the two-plaquette function of the field strength is discontinuous at zero momentum, so that this sum is infinite [51, Sect. 2]; see also [59]. This contradiction proves the first statement, and the second follows because $\beta_k > \beta_0$ for all large k. □

At large β the plaquette variables of the abelian theory are Gaussian to leading order, as are those of SU(N), and every property of the action used in this paper (compactness, locality, positivity and smoothness of the weight, reflection positivity, the forest gauge) is shared by the two theories. A proof of (D) along the continuum tuning therefore has to use the non-abelian structure of G; Proposition 243 holds at every cube size,

so letting the side grow with β, which Corollary 237 already requires, does not remove the need. Fröhlich and Spencer indicate how their result extends to the Wilson action without giving the details [51, Sect. 2], so the proposition is stated for the Villain action only.

Remark 244. The proof loses the area of a disc, of order n^2, in passing from plaquettes to links, and on loops that loss is of the right order. On the periodic lattice $(\mathbb{Z}/L\mathbb{Z})^4$ with L even and coordinates $0 \le x_j < L$, let $\phi = 4\pi/L^2$, $q(\theta) = \cos\theta + \mathbf{i}\sin\theta$, and put $U_{(x,2)} = q(\phi x_1)$, $U_{(x,1)} = q(-\phi L x_2)$ when $x_1 = L-1$, and $U = 1$ on every other link. Every plaquette of the plane $\{1,2\}$ equals $q(\phi)$, including those on the seam $x_1 = L-1$ and at the corner $x_1 = x_2 = L-1$, where the angle is $\phi - 4\pi$; all other plaquettes equal 1. So every plaquette variable lies within ϕ of the identity, while the boundary of the square $[0, L/2]^2$ of the plane $\{1,2\}$ has holonomy $q(\phi L^2/4) = -1$, at the largest possible distance $D = 2$, against the bound $\phi L^2/4 = \pi$ given by the disc. Every staple sum of this configuration is parallel to its link, so it is a stationary point of the Wilson action, and gradient flow [78] does not change it. Plaquette variables within $O(n^{-2})$ of the identity therefore do not force loops of length n near the identity, and an argument of the kind above that places gauge-fixed links near the identity needs plaquette bounds of order $o(n^{-2})$.

Up to the factor $L(b)^{1/2}$ the rate $b^{-1/2}$ of Proposition 240 is the right one, and the lower bound below holds for every n and every exterior.

Proposition 245. *Let $G = \mathrm{SU}(2)$ with the chordal metric and $b = \beta_W > 0$. For every Gibbs specification of the gauge-fixed theory, every active link e and every configuration of the other links, the conditional law of Q_e satisfies*

$$\int \delta(1, Q_e)\, d\mu_e \ge \frac{1}{9\max(6b,1)^{1/2}}.$$

In particular $\varepsilon_V(\beta) \ge \frac{1}{9}\max(6b,1)^{-1/2}$ for every n; at every fixed n, $\varepsilon_V(\beta)$ lies between constant multiples of $b^{-1/2}$ and $(L(b)/b)^{1/2}$.

Proof. Write $\langle\cdot,\cdot\rangle$ for the Euclidean product of quaternions, so that $\mathrm{Re}\,\mathrm{Tr}\, q = 2\langle q, 1\rangle$. The six plaquettes that contain e contribute $\exp(b\langle Q_e, W_p\rangle)$ to the conditional density, with $|W_p| = 1$; so μ_e has density $e^{\langle u,v\rangle}/Z(|v|)$ with respect to normalised Haar measure, for some v with $a = |v| \le 6b$. Put $a' = \max(a,1)$ and $\eta = a'^{-1/2} \le 1$. On the geodesic ball of radius η about $v/|v|$ the exponent is at least $a(1-\eta^2/2) \ge a - \frac{1}{2}$, and that ball has measure greater than $\eta^3/24$ (proof of Proposition 246); hence $Z(a) \ge e^{a-1/2}\eta^3/24$ and the density is at most $24e^{1/2}a'^{3/2}$ (for $a < 1$ directly, since $Z \ge 1$ by Jensen's inequality). A chordal ball of radius r is a geodesic ball of radius θ with $2\sin(\theta/2) = r$, so $\theta \le \pi r/2$, and its measure is $(\theta - \sin\theta\cos\theta)/\pi \le 2\theta^3/(3\pi) \le \pi^2 r^3/12$ (Lemma 360). With $r = \frac{2}{9}a'^{-1/2}$,

$$\mu_e\big(\delta(1,Q_e) < r\big) \le 24e^{1/2}a'^{3/2}\,\frac{\pi^2 r^3}{12} = 2e^{1/2}\pi^2\Big(\frac{2}{9}\Big)^3 < \frac{1}{2},$$

so the mean of $\delta(1,Q_e)$ is at least $r/2 = \frac{1}{9}a'^{-1/2}$. Averaging over $e \in V$ and over the configuration gives the bound on ε_V. □

Along the continuum tuning Theorem 236 says the following. If (D) holds, then in a cube of fixed physical size ℓ with the identity outside, the forest-gauge variables stay on average at distance at least $(1-\alpha)D/3$ from the identity, and at least $1-\alpha$ for SU(2), at every spacing. This is not a contradiction: $n = \ell/a$ grows like $1/a$, while asymptotic freedom makes $\beta(a)$ grow only like $\log(1/a)$, and the corollary concerns fixed n. The size bound of Proposition 240 is likewise met along the tuning, since its right side grows only like $\log(1/a)$. It turns the hypothesis of Remark 92, that the gauge-fixed links decorrelate on the scale of the cube, into a necessary condition: (D) requires the forest-gauge variables to be disordered at a fixed physical scale. For gauge-invariant observables of a massive theory that disorder is clustering; for the gauge-fixed variables it is not proved.

The last statement concerns the following strategy: control the conditional law of one line of links, given everything else, uniformly in the configuration around it, and assemble the cube from lines. Fix a site x, let $s = \sigma(x)$ and $\mu \ne s$, and let $c_k = (x + ke_s, \mu)$, $k \in \mathbb{Z}$. All c_k are active, since σ is constant along the line through x.

Proposition 246. *Let $G = \mathrm{SU}(2)$, identified with the unit quaternions, and $b = \beta_W \ge 1$. There is a configuration of the active links outside $\{c_k\}$, independent of b, for which the conditional law of $(Q_{c_k})_k$ has a ten-step transfer operator, similar to a positive self-adjoint operator K_b on $L^2(S^3)$ with simple top eigenvalue Λ_b, whose gap satisfies*

$$0 < 1 - \frac{\Lambda_{b,1}}{\Lambda_b} \le e^{20}\,24^{10}\,b^{15}\,e^{-b/1280},$$

where $\Lambda_{b,1}$ is the next point of the spectrum.

Proof. The interactions. By Lemma 9(iii), c_k lies in two plaquettes of the plane $\{\mu, s\}$, which contain c_{k-1} or c_{k+1}, the link $(x + ke_s, s)$ or $(x + (k-1)e_s, s)$ of F, and an active twist link of direction s; and in four plaquettes of the planes $\{\mu,t\}$ and $\{\mu,\rho\}$, each containing c_k, one link of F and two active links not on the line. The four plaquettes at c_k lie in the hyperplane of s-coordinate $x_s + k$, so these plaquettes at different k share no link; two plaquettes through c_k share no other link; and their links have directions other than s, so they are not twists. Set every twist to 1; then the plaquette at $x + ke_s$ in the plane $\{\mu, s\}$ contributes $b\langle q_k, q_{k+1}\rangle$ to the exponent, with $q_k = Q_{c_k}$. In each of the four other plaquettes at c_k set one of its two links to 1 and the other to X; after a cyclic rotation and possibly an inversion the plaquette variable is $q_k X^{\pm 1}$ or $q_k^{-1} X^{\pm 1}$, and since $\mathrm{Re}(q\bar{u}) = \mathrm{Re}(\bar{q}u) = \langle q, u\rangle$ for unit quaternions, X can be chosen so that the plaquette contributes $b\langle q_k, u\rangle$ for any prescribed unit vector u. Every other active link is set to 1. Four unit vectors add up to any vector of length at most 4 (two copies each of $t \pm (1-|t|^2)^{1/2}\nu$ with $t = v/4$ and $\nu \perp t$ a unit vector), so the conditional density of the line is proportional to $\exp\big(b\sum_k \langle q_k, q_{k+1}\rangle + \sum_k \langle h_k, q_k\rangle\big)$ for any fields h_k with $|h_k| \le 4b$.

The fields. Let $a=\frac12$, $c=\frac{\sqrt3}{2}$, let $e_1,\dots,e_4$ be the standard basis of $\mathbb{R}^4$, and let r_k have period ten with $(r_0,\dots,r_9)=a(e_1,-e_1,e_2,-e_2,e_3,-e_3,e_3,-e_2,e_2,-e_1)$. Put $h_k=b(2r_k-r_{k-1}-r_{k+1})$, so $|h_k|\le 4ab=2b$, and the configuration is independent of b because h_k/b is. With $v_k=q_k-r_k$ and $|q_k|=1$, $|r_k|=a$, summation by parts gives, over $0\le k\le 9$,

$$\frac12\sum_k|v_{k+1}-v_k|^2=\sum_k\bigl(1-\langle q_k,q_{k+1}\rangle\bigr)-\frac1b\sum_k\langle h_k,q_k\rangle + \frac12\sum_k|r_{k+1}-r_k|^2+B(q_0)-B(q_{10}),$$

with $B(q)=\langle q,r_0-r_9\rangle$. The boundary terms only conjugate the ten-step transfer kernel by a multiplication operator, so up to similarity it is

$$K_b(x,y)=\int_{(S^3)^9}\exp\Bigl(-\frac b2\sum_{k=0}^{9}|v_{k+1}-v_k|^2\Bigr)dq_1\cdots dq_9,$$

with $q_0=x$, $q_{10}=y$ and normalised Haar measure. The sequence $r_0,\dots,r_{10}$ is a palindrome, so $K_b=B_bB_b^*$ for the five-step kernel B_b; K_b is compact, positive and self-adjoint with a continuous positive kernel, its top eigenvalue Λ_b is simple, and its eigenfunction $\psi_b>0$ is even under the reflection $x_4\mapsto -x_4$, which commutes with K_b.

A barrier. Let $E=\frac12\sum_{k=0}^{9}|v_{k+1}-v_k|^2$ and suppose that $(q_0)_4\ge 0\ge(q_{10})_4$ and $E\le\frac{1}{1280}$. Put $\Delta=(20E)^{1/2}\le\frac18$ and $w=v_0$; by Cauchy–Schwarz $|w-v_k|\le\Delta$ for all k, and $0\le w_4\le\Delta$ because the r_k have no fourth coordinate. From $|v_k+r_k|=1$, $\bigl||w+r_k|^2-1\bigr|\le\Delta(2+\Delta)=:\eta$ for $0\le k\le 5$. The pairs $r_k=\pm ae_l$ give $|w_l|\le\eta/(2a)=\eta$ for $l=1,2,3$ and $|w|^2\ge\frac34-\eta$, so $\frac34\le 3\eta^2+\Delta^2+\eta\le 3(\frac{17}{64})^2+\frac{1}{64}+\frac{17}{64}=\frac{2019}{4096}$, a contradiction. Hence $K_b(x,y)\le e^{-b/1280}$ whenever $x_4\ge 0\ge y_4$.

The top eigenvalue. Let $\varepsilon=b^{-1/2}$. The geodesic ball of radius $\varepsilon\le 1$ in S^3 has normalised measure $\frac{2}{\pi}\int_0^{\varepsilon}\sin^2\theta\,d\theta\ge\frac{8\varepsilon^3}{3\pi^3}>\frac{\varepsilon^3}{24}$. If each q_k lies in the ball of radius ε about r_k+ce_4, then $|v_k-ce_4|\le\varepsilon$ and $E\le 20\varepsilon^2$, so the integrand is at least e^{-20}. Testing K_b on the normalised indicator of the ball about r_0+ce_4 gives $\Lambda_b\ge e^{-20}(\varepsilon^3/24)^{10}=e^{-20}24^{-10}b^{-15}$.

The gap. Let $\phi(x)=\operatorname{sign}(x_4)\psi_b(x)$, a unit vector orthogonal to ψ_b. Then

$$\Lambda_b-\langle\phi,K_b\phi\rangle=4\int_{x_4>0>y_4}\psi_b(x)K_b(x,y)\psi_b(y)\,dx\,dy\le e^{-b/1280},$$

because $\int_{x_4>0}\psi_b\le\frac12$ by Cauchy–Schwarz. By the variational principle $\Lambda_{b,1}\ge\langle\phi,K_b\phi\rangle$, and dividing by the lower bound on Λ_b proves the estimate; the gap is positive because Λ_b is simple. □

The exterior of Proposition 246 is legal at every coupling, so a bound on the conditional law of single lines that is uniform in the exterior cannot be better than $e^{-b/1280}$ up to powers of b, and an argument of this kind cannot give (D) along the tuning. A proof of (D) has to use what such a bound discards: the joint resampling of the transverse links inside the cube, which the averaged coefficients $C_{V,j}$ retain, or the smallness of the probability of such exteriors under the Wilson measure.

L. The limit at separated points

This subsection treats (E) for local fields without flow, under the averaged condition of Theorem 64. Two features separate it from Sec. XI. A local field without flow is covariant under the whole hypercubic group B_4, so reflection positivity and the gap pass from the time direction to every coordinate direction; the limit then clusters in every direction at the rate m_* without the tree bounds (Theorem 248(v)), and by Proposition 197(iv) one rotation of infinite order gives $\mathrm{O}(4)$. On the other hand, asymptotic freedom predicts for the action density a two-point function of order $|x-y|^{-8}$ at short distance, up to logarithms, and such a two-point function is incompatible with the moment bound (M) at coinciding supports (Proposition 252). The moment bound is therefore posed for separated supports, on which it determines distributions on the complement of the coinciding points (Theorem 248(i)). The Schwinger functions of [88] are defined on test functions that vanish, with all their derivatives, where two arguments coincide.

Throughout, the hypothesis of Theorem 64 holds, ω_k is the torus limit there and $m_*=16\varpi/(625e^2\ell)$. By Corollary 63, ω_k is invariant under translations and B_4, reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links, and its transfer matrix $\mathbb{T}_k$ has spectrum on $\Omega_k^{\perp}$ in $[0,e^{-m_*a_k}]$, since $m_{n_k}\ge 16\varpi/(625e^2n_k)\ge m_*a_k$. Let $r\ge 1$ be an integer and Φ a real continuous gauge-invariant function of the links with both endpoints in $[-r,r]^4$, invariant under the action of B_4 on these links described in the proof of Theorem 193(ii); the sum of $N-\operatorname{Re}\operatorname{Tr}U_p$ over the 24 plaquettes p with a corner at 0 is an example. Write Φ_x for its translate by $x\in\mathbb{Z}^4$, so that $g\Phi_x=\Phi_{gx}$ for $g\in B_4$, and $\tau_v\Phi_x=\Phi_{x+v}$ for the translation τ_v, $v\in\mathbb{Z}^4$. In this subsection

$$A_k(f)=\sum_{x\in\mathbb{Z}^4}a_k^4f(a_kx)\bigl(\Phi_x-c_k\bigr),\qquad c_k=\omega_k(\Phi_0),$$

for $f\in C_c(\mathbb{R}^4)$, and $Z_k>0$ are renormalisation constants.

Let e_μ be the unit vectors, θ_μ the reflection $x_\mu\mapsto -x_\mu$ of $\mathbb{R}^4$, $\pi_\mu\in B_4$ the transposition of the axes 0 and μ ($\pi_0=1$), ϑ the lattice reflection of Appendix F, $\vartheta_\mu=\pi_\mu\vartheta\pi_\mu$, and $\tau_t^\mu=\tau_{te_\mu}$. Let $\theta_{\mu,k}$ be the reflection of $\mathbb{R}^4$ with $\theta_{\mu,k}(a_kx)=a_k\vartheta_\mu(x)$ for $x\in\mathbb{Z}^4$; as in the proof of Theorem 55(iii) it is θ_μ followed by a translation along e_μ of length at most a_k, so the plane fixed by ϑ_μ is $\{x_\mu=c\}$ with $|c|\le\frac12$. For a cube Q and $\delta>0$ let $\mathscr{C}_n(Q,\delta)$ be the set of $(f_1,\dots,f_n)\in C_c(\mathbb{R}^4)^n$ with supports in Q and $\operatorname{dist}_\infty(\operatorname{supp}f_i,\operatorname{supp}f_j)\ge\delta$ for $i\ne j$, $\operatorname{dist}_\infty$ being the ℓ^∞ distance; a tuple is *separated* if it lies in some $\mathscr{C}_n(Q,\delta)$. A *separated polynomial* is a polynomial in the sense of Sec. XI each of whose monomials is a separated tuple; it is *at positive μ-times* if all its functions are supported in $\{x_\mu>0\}$. Put $\mathfrak{D}_n=\{(x_1,\dots,x_n)\in(\mathbb{R}^4)^n:x_i\ne x_j\text{ for }i\ne j\}$ and $\mathsf{K}_L=(2L+3)^4$.

($\mathrm{M_s}$) *Separated moment bound.* For every $n\ge 1$, every cube Q

and every $\delta > 0$ there are $M_n(Q,\delta) < \infty$ and k_0 such that

$$\Big|\omega_k\Big(\prod_{j=1}^{n} Z_k A_k(f_j)\Big)\Big| \le M_n(Q,\delta)\prod_{j=1}^{n}\|f_j\|_\infty$$

for all $k \ge k_0$ and $(f_1,\dots,f_n) \in \mathscr{C}_n(Q,\delta)$.

Posed for these packets, (M) contains the tuples with overlapping supports and implies ($\mathrm{M_s}$).

Lemma 247. (i) *For $f \in C_c(\mathbb{R}^4)$, $g \in B_4$ and $v \in \mathbb{Z}^4$,*

$$A_k(f\circ g^{-1}) = gA_k(f), \quad A_k(f(\cdot - a_k v)) = \tau_v A_k(f),$$
$$A_k(\bar f) = \overline{A_k(f)}, \quad \vartheta_\mu A_k(f) = A_k(f\circ\theta_{\mu,k}).$$

(ii) *If f is supported in $\{x_\mu \ge \varepsilon\}$ and $(r+1)a_k \le \varepsilon$, then $A_k(f)$ is a function of the links with both endpoints in $\{x_\mu \ge 1\}$, which lie on the positive side of the plane of ϑ_μ.*

(iii) *Let F, F' be continuous gauge-invariant functions of finitely many links with both endpoints in $\{x_\mu \ge 1\}$, and $t \ge 0$ an integer. Then $\omega_k(\vartheta_\mu(\bar F)F) \ge 0$,*

$$0 \le \omega_k\big(\vartheta_\mu(\bar F)\tau^\mu_{2t}F\big) \le \omega_k\big(\vartheta_\mu(\bar F)F\big),$$

and, with $N_\mu(F) = \omega_k(\vartheta_\mu(\bar F)F)^{1/2}$,

$$\big|\omega_k\big(\vartheta_\mu(\bar F)\tau^\mu_t F'\big) - \overline{\omega_k(F)}\,\omega_k(F')\big| \le e^{-m_* a_k t} N_\mu(F) N_\mu(F').$$

For $v \in \mathbb{Z}^4$ with $v_\mu = 0$, $N_\mu(\tau_v F) = N_\mu(F)$.

Proof. (i) Substitute $y = g^{-1}x$, $y = x - v$ and $y = \vartheta_\mu x$ in the sums, using $g\Phi_x = \Phi_{gx}$, $\tau_v\Phi_x = \Phi_{x+v}$ and $\vartheta_\mu\Phi_x = \Phi_{\vartheta_\mu x}$; the last holds because ϑ_μ is an element of B_4 followed by a lattice translation. The constant c_k does not depend on x by translation invariance, and it is real because Φ is real and ω_k is a state.

(ii) $f(a_k x) \ne 0$ forces $x_\mu \ge \varepsilon/a_k \ge r+1$, and Φ_x depends on the links with both endpoints in $x + [-r,r]^4$.

(iii) For $\mu = 0$ this is the reconstruction of Appendix F: with $\langle \Psi_F, \mathbb{T}'_k \Psi_{F'}\rangle = \omega_k(\vartheta(\bar F)\tau_t F')$ and the density of the vectors Ψ_G, $\Psi_{\tau_t F} = \mathbb{T}'_k\Psi_F$; then $\omega_k(\vartheta(\bar F)\tau_{2t}F) = \|\mathbb{T}'_k\Psi_F\|^2 \le \|\Psi_F\|^2$ by $0 \le \mathbb{T}_k \le 1$, and with $\Psi'_F = \Psi_F - \omega_k(F)\Omega_k$ the left side of the last inequality is $|\langle \Psi'_F, \mathbb{T}'_k\Psi'_{F'}\rangle| \le e^{-m_* a_k t}\|\Psi'_F\|\|\Psi'_{F'}\|$, where $\langle\Omega_k,\Psi_F\rangle = \omega_k(F)$ and $\|\Psi'_F\| \le \|\Psi_F\| = N_0(F)$. Since ϑ is θ_0 followed by a translation along e_0, it commutes with τ_v for $v_0 = 0$, and $N_0(\tau_v F)^2 = \omega_k(\tau_v(\vartheta(\bar F)F)) = N_0(F)^2$. For general μ apply this to $\pi_\mu F$ and $\pi_\mu F'$, using $\vartheta_\mu = \pi_\mu\vartheta\pi_\mu$, $\tau^\mu_t = \pi_\mu\tau_t\pi_\mu$ and the invariance of ω_k under π_μ. □

Theorem 248. *Assume the hypothesis of Theorem 64 and* ($\mathrm{M_s}$). *Every sequence of k has a subsequence along which the following hold.*

(i) *For every separated tuple the limit*

$$S_n(f_1,\dots,f_n) = \lim_k \omega_k\Big(\prod_{j=1}^{n} Z_k A_k(f_j)\Big)$$

exists. It is symmetric and multilinear, $S_1 = 0$, and on $\mathscr{C}_n(Q,\delta)$ it is bounded by $M_n(Q,\delta)\prod_j\|f_j\|_\infty$. There is a unique distribution on $\mathfrak{D}_n$ whose value at $f_1\otimes\cdots\otimes f_n$ is $S_n(f_1,\dots,f_n)$ for every separated tuple of smooth functions.

(ii) *Each S_n is invariant under the translations of $\mathbb{R}^4$ and under B_4, and $S(\bar P) = \overline{S(P)}$ for separated polynomials.*

(iii) *For every μ and every separated polynomial P at positive μ-times, $S(\theta_\mu\bar P\cdot P) \ge 0$. The completion of the quotient of these polynomials by the null space of this form is a Hilbert space $\mathscr{H}_\mu$ with vectors $\Psi_\mu(P)$, $\langle\Psi_\mu(P),\Psi_\mu(P')\rangle = S(\theta_\mu\bar P\cdot P')$, and $\Omega_\mu = \Psi_\mu(1)$; $P \mapsto \pi_\mu P$ induces a unitary map of $\mathscr{H}_0$ onto $\mathscr{H}_\mu$ that maps Ω_0 to Ω_μ.*

(iv) *There are a self-adjoint $H_\mu \ge 0$ with $\Psi_\mu(\tau_{te_\mu}P) = e^{-tH_\mu}\Psi_\mu(P)$ for $t \ge 0$ and a strongly continuous unitary representation U_μ of the hyperplane $e_\mu^\perp$ with $\Psi_\mu(\tau_y P) = U_\mu(y)\Psi_\mu(P)$, commuting with H_μ; $H_\mu\Omega_\mu = 0$ and U_μ fixes Ω_μ. Further $\sigma(H_\mu) \subset \{0\}\cup[m_*,\infty)$, $\ker H_\mu = \mathbb{C}\Omega_\mu$, and for P, P' at positive μ-times and $t \ge 0$*

$$\big|S(\theta_\mu\bar P\cdot\tau_{te_\mu}P') - \overline{S(P)}\,S(P')\big| \le e^{-m_* t}\,\|\Psi_\mu(P)\|\,\|\Psi_\mu(P')\|.$$

(v) *Let $(f_1,\dots,f_n) \in \mathscr{C}_n(Q,\delta)$, Q of side L, let $\{1,\dots,n\} = I\cup J$ with I, J disjoint and nonempty, $y \in \mathbb{R}^4\setminus\{0\}$, and $f_j^\lambda = f_j$ for $j \in I$, $f_j^\lambda = f_j(\cdot - \lambda y)$ for $j \in J$. There is $K < \infty$ independent of λ such that, for $\lambda|y|_\infty \ge L+2$,*

$$\Big|S_n(f_1^\lambda,\dots,f_n^\lambda) - S_{|I|}\big((f_j)_{j\in I}\big)S_{|J|}\big((f_j)_{j\in J}\big)\Big| \le Ke^{-m_*\lambda|y|_\infty}.$$

(vi) *$\mathscr{H}_0 \ne \mathbb{C}\Omega_0$ if and only if S_n does not vanish identically on the separated tuples for some $n \ge 2$. In that case $\sigma(H_\mu)\cap[m_*,\infty) \ne \emptyset$ for every μ: H_μ has a mass gap of at least m_*.*

(vii) *The set G_S of $R \in \mathrm{O}(4)$ with $S_n(f_1\circ R^{-1},\dots,f_n\circ R^{-1}) = S_n(f_1,\dots,f_n)$ for all n and all separated tuples is a closed subgroup of $\mathrm{O}(4)$ that contains B_4. It is finite or equal to $\mathrm{O}(4)$, and it equals $\mathrm{O}(4)$ if and only if it contains an element of infinite order.*

Conversely, if for every n and every separated tuple the sequence $\omega_k(\prod_j Z_k A_k(f_j))$ is bounded, then ($\mathrm{M_s}$) *holds.*

Proof. (i) Let $\chi_{m,p}(x) = \prod_\nu(1 - 2^m|x_\nu - p_\nu|)_+$ for $m \in \mathbb{N}$ and $p \in 2^{-m}\mathbb{Z}^4$, and let $\mathscr{G}$ be the countable set of the functions $\sum_{p\in F} c_p\chi_{m,p}$ with F finite and $c_p \in \mathbb{Q}+i\mathbb{Q}$. Since $\sum_p\chi_{m,p} = 1$ and $\chi_{m,p}(x) \ne 0$ only for $|x-p|_\infty < 2^{-m}$, for $f \in C_c(\mathbb{R}^4)$ and $\eta > 0$ there is $g \in \mathscr{G}$ with $\|f-g\|_\infty < \eta$ and support within ℓ^∞ distance η of $\operatorname{supp} f$: take $2^{-m} \le \eta$ with $|f(x)-f(p)| < \eta/2$ for $|x-p|_\infty \le 2^{-m}$, $F = \{p : f(p) \ne 0\}$ and $|c_p - f(p)| < \eta/2$. By ($\mathrm{M_s}$) the moment sequence of each of the countably many separated tuples of elements of $\mathscr{G}$ is bounded, and a diagonal subsequence makes all of them converge. Let $(f_j) \in \mathscr{C}_n(Q,\delta)$ and choose $g_j \in \mathscr{G}$ as above with $\eta \le \delta/4$. With Q' the cube Q enlarged by $\delta/4$ on every side, the tuples $(g_1,\dots,g_{i-1},f_i - g_i, f_{i+1},\dots,f_n)$ and (g_j) lie in $\mathscr{C}_n(Q',\delta/2)$, so for $k \ge k_0$ the moments B_k satisfy

$$|B_k(f_1,\dots,f_n) - B_k(g_1,\dots,g_n)| \le M_n(Q',\tfrac{\delta}{2})\sum_i\|f_i-g_i\|_\infty\prod_{j\ne i}\max(\|f_j\|_\infty,\|g_j\|_\infty).$$

Hence $B_k(f_1,\dots,f_n)$ is a Cauchy sequence. Symmetry and multilinearity pass to the limit, and so does the bound; $S_1 = 0$ because the packets are centred.

For the distribution, let $U_1,\dots,U_n$ be bounded open sets with pairwise disjoint closures, $U=U_1\times\cdots\times U_n$, and $M=M_n(Q,\delta)$ with $Q\supset\bigcup_j U_j$ and δ the least distance of the closures; $|S_n(f_1,\dots,f_n)|\le M\prod_j\|f_j\|_\infty$ for $f_j\in C_c(U_j)$. Fix a compact $K\subset U$, R with $U_j\subset(-R,R)^4$, and $\varphi_j\in C_c^\infty(U_j)$ equal to 1 near the projection of K. For $F\in C^\infty(\mathbb{R}^{4n})$ supported in K let c_κ, $\kappa\in\mathbb{Z}^{4n}$, be the Fourier coefficients of its $2R$-periodic extension, so that $F=\sum_\kappa c_\kappa e_\kappa$ on $(-R,R)^{4n}$ with $e_\kappa(x)=\prod_j e^{i\pi\kappa_j\cdot x_j/R}$. Integrating by parts against $1-(R/\pi)^2\Delta$, which multiplies e_κ by $1+|\kappa|^2$,

$$|c_\kappa|\le(2R)^{-4n}(1+|\kappa|^2)^{-2n-1}\big\|(1-(R/\pi)^2\Delta)^{2n+1}F\big\|_{L^1},$$

and $\sum_\kappa(1+|\kappa|^2)^{-2n-1}<\infty$. Put

$$T(F)=\sum_\kappa c_\kappa S_n\big(\varphi_1 e_{\kappa_1},\dots,\varphi_n e_{\kappa_n}\big).$$

The series converges absolutely, each term being at most $M\prod_j\|\varphi_j\|_\infty|c_\kappa|$, so T is continuous on the smooth functions supported in K. For $F=f_1\otimes\cdots\otimes f_n$ the expansions $f_j=\varphi_j\sum_{\kappa_j}c_{\kappa_j}^{(j)}e_{\kappa_j}$ converge uniformly with supports in $\operatorname{supp}\varphi_j$, so multilinearity and the bound give $T(F)=S_n(f_1,\dots,f_n)$. Conversely, for every F supported in K the partial sums $\sum_{|\kappa|_\infty\le N}c_\kappa\prod_j\varphi_j(x_j)e_{\kappa_j}(x_j)$ are finite sums of products and converge to $\prod_j\varphi_j(x_j)F=F$ in $C_c^\infty(U)$, since $\sum_\kappa|\kappa|^s|c_\kappa|<\infty$ for every s by the same bound with a higher power. So every distribution on U with the stated values on products agrees with T on the functions supported in K; in particular T does not depend on K, R and φ_j, and defines the unique such distribution T_U on U. Every point of $\mathfrak{D}_n$ has a neighbourhood of this form, namely a product of small open cubes around its n distinct entries, and the intersection of two such products is again one, on which T_U and $T_{U'}$ agree by uniqueness. By [66, Sect. 2.2] the T_U are the restrictions of one distribution on $\mathfrak{D}_n$, and it is unique by the same localisation.

(ii) For dyadic y and large k, $y/a_k\in\mathbb{Z}^4$, and Lemma 247(i) and translation invariance of ω_k give invariance of the moments under the translation by y. For real y take dyadic $y_m\to y$: the translated tuples lie in one $\mathscr{C}_n(Q,\delta)$ and converge uniformly, so the bound of (i) carries the invariance to y. For $g\in B_4$, $A_k(f\circ g^{-1})=gA_k(f)$, ω_k is B_4-invariant, and g preserves ℓ^∞ distances. The last claim follows from $A_k(\bar f)=\overline{A_k(f)}$ and $\omega_k(\bar G)=\overline{\omega_k(G)}$.

(iii) Let $P=\sum_\beta c_\beta\prod_j f_{\beta,j}$ have all its functions supported in $Q\cap\{x_\mu\ge\varepsilon\}$, $\varepsilon>0$, and its monomials in $\mathscr{C}(Q,\delta)$, and put $P_k=\sum_\beta c_\beta\prod_j Z_kA_k(f_{\beta,j})$. For $(r+1)a_k\le\varepsilon$, Lemma 247(ii),(iii) gives $\omega_k(\vartheta_\mu(\bar P_k)P_k)\ge0$, and by Lemma 247(i) $\vartheta_\mu(\bar P_k)$ is the lattice version of $\theta_\mu\bar P$ with θ_μ replaced by $\theta_{\mu,k}$. For $a_k\le\min(\varepsilon,\delta/4)$ the monomials of $\theta_\mu\bar P\cdot P$ and of the polynomials obtained by replacing θ_μ by $\theta_{\mu,k}$ in some factors lie in $\mathscr{C}(Q',\min(\varepsilon,\delta/2))$ for a cube Q' independent of k, because the reflected factors are supported in $\{x_\mu\le-\varepsilon+a_k\}$. By (i) and multilinearity the difference of the two moments is at most a constant times $\max_{\beta,j}\|f_{\beta,j}\circ\theta_{\mu,k}-f_{\beta,j}\circ\theta_\mu\|_\infty$, which tends to 0 by uniform continuity. So $S(\theta_\mu\bar P\cdot P)\ge0$. The form is Hermitian by (ii) and the symmetry of S_n, and the construction is that of Theorem 193(iii). Since $\pi_\mu\theta_0\pi_\mu=\theta_\mu$ and S is π_μ-invariant, $S(\theta_\mu\overline{\pi_\mu P}\cdot\pi_\mu P')=S(\theta_0\bar P\cdot P')$ for P,P' at positive 0-times, which gives the unitary map.

(iv) By the unitary map of (iii) it suffices to take $\mu=0$, with H_μ and U_μ the conjugates of H_0 and U_0. Let P be at positive 0-times and $t\ge0$ dyadic. For large k, $(\tau_{te_0}P)_k=\tau_{t/a_k}P_k$ by Lemma 247(i), and Lemma 247(iii) gives $\omega_k(\vartheta(\bar P_k)\tau_{2t/a_k}P_k)\le\omega_k(\vartheta(\bar P_k)P_k)$. By the approximation of $\theta_{0,k}$ by θ_0 in (iii), the left side tends to $S(\theta_0\bar P\cdot\tau_{2te_0}P)=\|\Psi_0(\tau_{te_0}P)\|^2$, by translation invariance and $\theta_0\tau_{te_0}=\tau_{-te_0}\theta_0$, and the right side tends to $\|\Psi_0(P)\|^2$. The translates $\tau_{te_0}P$, $0\le t\le\delta/2$, have their monomials in one $\mathscr{C}(Q',\delta/2)$, and each monomial of $\tau_{te_0}P-P$ is a telescoping sum of monomials with one factor $f(\cdot-te_0)-f$; so the bound of (i) gives $\|\Psi_0(\tau_{te_0}P)\|\le\|\Psi_0(P)\|$ for all $t\ge0$ and $\|\Psi_0(\tau_{te_0}P)-\Psi_0(P)\|\to0$ as $t\to0$. From here the proof of Theorem 193(iv) applies verbatim and gives H_0, and U_0 from the translations orthogonal to e_0, which commute with θ_0. For the gap, the proof of Lemma 247(iii) with $F=F'=P_k$ and t/a_k steps gives

$$\begin{aligned}&\omega_k\big(\vartheta(\bar P_k)\tau_{t/a_k}P_k\big)-|\omega_k(P_k)|^2\\&\quad\le e^{-m_*t}\big(\omega_k(\vartheta(\bar P_k)P_k)-|\omega_k(P_k)|^2\big),\end{aligned}$$

the right side being $e^{-m_*t}\|\Psi'_{P_k}\|^2$ in the notation of the proof of the lemma. In the limit, with $v=\Psi_0(P)-S(P)\Omega_0$ and $\langle\Omega_0,\Psi_0(P)\rangle=S(P)$, this reads $\langle v,e^{-tH_0}v\rangle\le e^{-m_*t}\|v\|^2$. As in the proofs of Theorems 43 and 193(vi), it extends to all $t\ge0$ and to the closure $\Omega_0^\perp$ of these vectors, and gives the claims on $\sigma(H_0)$, on $\ker H_0$ and the inequality, using $\langle\Psi_0(P),\Omega_0\rangle=\overline{S(P)}$ by (ii).

(v) Choose μ with $|y_\mu|=|y|_\infty=\eta$. Replacing every f_j by $f_j\circ\theta_\mu$ and y by $\theta_\mu y$, which changes no term by (ii), we may take $y_\mu=\eta>0$, and after a common translation of all f_j, $Q\subset\{-L-1\le x_\mu\le-1\}$. Let $P''=\prod_{j\in I}\bar f_j\circ\theta_\mu$ and $P_0'=\prod_{j\in J}f_j(\cdot-(L+2)e_\mu)$, both at positive μ-times, write $y=\eta e_\mu+y_\perp$ with $y_\perp\perp e_\mu$ and put $s=\lambda\eta-L-2\ge0$. Then $\prod_{j\in J}f_j(\cdot-\lambda y)=\tau_{se_\mu}\tau_{\lambda y_\perp}P_0'$ and $\theta_\mu\bar P''=\prod_{j\in I}f_j$, so by (iv)

$$S_n(f_1^\lambda,\dots,f_n^\lambda)=\langle\Psi_\mu(P''),e^{-sH_\mu}U_\mu(\lambda y_\perp)\Psi_\mu(P_0')\rangle,$$

while, by translation invariance,

$$\begin{aligned}S_{|I|}\big((f_j)_{j\in I}\big)&=\langle\Psi_\mu(P''),\Omega_\mu\rangle,\\S_{|J|}\big((f_j)_{j\in J}\big)&=\langle\Omega_\mu,\Psi_\mu(P_0')\rangle.\end{aligned}$$

The difference is the same matrix element between the projections of the two vectors on $\Omega_\mu^\perp$, which e^{-sH_μ} and U_μ preserve, so it is at most $e^{-m_*s}\|\Psi_\mu(P'')\|\|\Psi_\mu(P_0')\|$. This is the claim with $K=e^{m_*(L+2)}\|\Psi_\mu(P'')\|\|\Psi_\mu(P_0')\|$.

(vi) If all S_n with $n\ge1$ vanish, $\langle\Psi_0(P),\Psi_0(P')\rangle$ is the product of the constant terms, and $\mathscr{H}_0=\mathbb{C}\Omega_0$. Conversely, let $\mathscr{H}_0=\mathbb{C}\Omega_0$; by (iii) then $\mathscr{H}_\mu=\mathbb{C}\Omega_\mu$, so $S(\theta_\mu\bar P\cdot P')=\overline{S(P)}S(P')$ for P,P' at positive μ-times, and by (ii) $S(\theta_\mu\bar P)=\overline{S(P)}$. With translation invariance this says: if a separated tuple splits into $(f_j)_{j\in I}$ supported in $\{x_\mu<c\}$ and $(f_j)_{j\in J}$ supported in $\{x_\mu>c\}$, then $S_n=S_{|I|}S_{|J|}$. Suppose $S_m=0$ for $1\le m<n$, let $(f_j)\in\mathscr{C}_n(Q,\delta)$, $n\ge2$, and put $s=\delta/(2n)$ and $\chi_m^s(x)=$

$\prod_\nu (1-|2x_\nu/s-m_\nu|)_+$, $m\in\mathbb{Z}^4$, which have sum 1 and supports in closed cubes of side s. By multilinearity $S_n(f_1,\dots,f_n)$ is the finite sum of the terms $S_n(\chi^s_{m_1}f_1,\dots,\chi^s_{m_n}f_n)$. In a nonzero term pick $p\in\operatorname{supp}\chi^s_{m_1}f_1$ and $q\in\operatorname{supp}\chi^s_{m_2}f_2$, and μ with $|p_\mu-q_\mu|\ge\delta$. The projections of the n cubes on the μ axis have total length at most $ns<\delta$, so some c strictly between p_μ and q_μ lies in none of them; it splits the term into two nonempty groups as above, and the term is 0. Hence $S_n=0$. The last claim follows from (iv), since $\Omega^\perp_\mu\neq 0$ by (iii).

(vii) For $R\in\mathrm{O}(4)$, $|Rz|_\infty\ge\frac12|Rz|_2=\frac12|z|_2\ge\frac12|z|_\infty$, so R maps $\mathscr{C}_n(Q,\delta)$ into some $\mathscr{C}_n(Q',\delta/2)$ and G_S is well defined; it is a group, and it contains B_4 by (ii). If $R_m\to R$ with $R_m\in G_S$, the functions $f_j\circ R_m^{-1}$ converge uniformly to $f_j\circ R^{-1}$, and for large m the telescoping tuples lie in one $\mathscr{C}_n(Q',\delta/4)$, so the bound of (i) gives $R\in G_S$. The proof of Proposition 197(iv) uses only that the group is a closed subgroup of $\mathrm{O}(4)$ containing B_4.

For the converse, cover Q by the closed cubes of side $s=\delta/4$ that carry the functions χ^s_m, at most $(2L/s+3)^4$ of them. For $(f_j)\in\mathscr{C}_n(Q,\delta)$, every nonzero term $\omega_k(\prod_j Z_kA_k(\chi^s_{m_j}f_j))$ has the n cubes of $m_1,\dots,m_n$ at pairwise distance at least $\delta-2s=\delta/2$. For each such choice of cubes $C_1,\dots,C_n$ the forms $(h_j)\mapsto\omega_k(\prod_j Z_kA_k(h_j))$ on the Banach spaces of continuous functions supported in C_j are bounded at every point by hypothesis, hence bounded in norm by the uniform boundedness argument of the proof of Theorem 193. Summing over the finitely many choices gives $M_n(Q,\delta)$ with $k_0=1$. □

For the flowed fields of Sec. XI the gap gives clustering in the time direction only (Theorem 193(vi)), and spatial clustering needs the tree bounds; part (v) obtains it in every direction from the hypercubic covariance alone. The symmetry of the S_n is the Euclidean form of locality [88]. With

($\mathrm{W_s}$) there is $R\in\mathrm{O}(4)$ of infinite order with

$$\lim_k\Big[\omega_k\Big(\prod_j Z_kA_k(f_j\circ R^{-1})\Big)-\omega_k\Big(\prod_j Z_kA_k(f_j)\Big)\Big]=0$$

for every separated tuple,

($\mathrm{U_s}$) any two subsequential limits of Theorem 248(i) coincide,

the argument of Corollary 200 gives the following.

Corollary 249. *Assume the hypothesis of Theorem 64. The moments $\omega_k(\prod_j Z_kA_k(f_j))$ converge for every separated tuple to a family invariant under the translations and* $\mathrm{O}(4)$ *if and only if* ($\mathrm{M_s}$), ($\mathrm{W_s}$) *and* ($\mathrm{U_s}$) *hold. The limit then has properties* (i)–(vi) *of Theorem 248 with B_4 replaced by* $\mathrm{O}(4)$.

Proof. Convergence gives bounded moments, hence ($\mathrm{M_s}$) by the converse in Theorem 248, and it gives ($\mathrm{U_s}$) and ($\mathrm{W_s}$) for every R. Conversely, ($\mathrm{M_s}$) and ($\mathrm{U_s}$) give convergence of the whole sequence, since a nonconvergent moment would have two limit points and Theorem 248(i) would refine them to two different limits; ($\mathrm{W_s}$) puts R in G_S, and Theorem 248(vii) gives $G_S=\mathrm{O}(4)$. □

Nontriviality is not a further input once the constants are fixed by one reflected correlation. Fix $f_0\in C_c(\mathbb{R}^4)$, not zero, supported in $\{x_0\ge t_0\}$ with $t_0>0$, and for $(r+2)a_k\le t_0$ and $t\in a_k\mathbb{Z}_{\ge0}$ put

$$C_k(t)=\omega_k\big(\vartheta(\overline{A_k(f_0)})\,\tau_{t/a_k}A_k(f_0)\big).$$

Proposition 250. (i) *$C_k(0)\ge0$, and if $C_k(0)>0$, then $0<C_k(t)\le e^{-m_*t}C_k(0)$ for every $t\in a_k\mathbb{Z}_{\ge0}$.*

(ii) *Let $\sigma_0>0$ be dyadic, suppose $C_k(0)>0$ for large k, put $Z_k=C_k(\sigma_0)^{-1/2}$ and assume* ($\mathrm{M_s}$). *Every limit of Theorem 248 has $\langle\Psi_0(f_0),e^{-\sigma_0H_0}\Psi_0(f_0)\rangle=1$ and $\|\Psi_0(f_0)\|^2\ge e^{m_*\sigma_0}$, and is nontrivial. Moreover, with Q a cube that contains* $\operatorname{supp}f_0$ *and $\theta_{0,k}(\operatorname{supp}f_0)$ for all k,*

$$\frac{C_k(\sigma_0)}{C_k(0)}\ge\frac{1}{M_2(Q,t_0)\|f_0\|_\infty^2}$$

for all large k.

(iii) *Suppose that the transfer matrices $\mathbb{T}_k$ have spectrum on $\Omega_k^\perp$ in $[0,e^{-m_{\mathrm{lat}}}]$ for a fixed $m_{\mathrm{lat}}>0$ and all k. Then for every choice of Z_k satisfying* ($\mathrm{M_s}$), *every limit of Theorem 248 is trivial, and, when $C_k(0)>0$ for large k,* ($\mathrm{M_s}$) *fails for $Z_k=C_k(\sigma_0)^{-1/2}$.*

Proof. (i) Let $A=A_k(f_0)$. By Lemma 247(ii) and its proof, A and $A'=A_k(f_0(\cdot+a_ke_0))$ are at positive times, and $A=\tau_1A'$, so $\Psi_A=\mathbb{T}_k\Psi_{A'}$ lies in the range of $\mathbb{T}_k$, which is orthogonal to $\ker\mathbb{T}_k$. Also $\langle\Omega_k,\Psi_A\rangle=\omega_k(A)=0$. So the spectral measure ν of $\mathbb{T}_k$ at Ψ_A has no atom at 0 and is carried by $(0,e^{-m_*a_k}]$, and $C_k(t)=\int\lambda^{t/a_k}d\nu(\lambda)$. If $C_k(0)=\nu((0,1])>0$, the integrand is positive on a set of positive measure and at most e^{-m_*t}.

(ii) The pairs $(\bar f_0\circ\theta_0,f_0(\cdot-te_0))$, $t\ge0$, are separated by at least $2t_0$, and $\omega_k(\vartheta(\bar A)\tau_{t/a_k}A)=\omega_k(A_k(\bar f_0\circ\theta_{0,k})A_k(f_0(\cdot-te_0)))$ by Lemma 247(i). By the approximation of $\theta_{0,k}$ by θ_0 in the proof of Theorem 248(iii), $\langle\Psi_0(f_0),e^{-\sigma_0H_0}\Psi_0(f_0)\rangle=\lim_kZ_k^2C_k(\sigma_0)=1$ and $\|\Psi_0(f_0)\|^2=\lim_kC_k(0)/C_k(\sigma_0)\ge e^{m_*\sigma_0}$ by (i). Since $\langle\Omega_0,\Psi_0(f_0)\rangle=S_1(f_0)=0$, $\mathscr{H}_0\neq\mathbb{C}\Omega_0$. For $a_k\le t_0$ the pair $(\bar f_0\circ\theta_{0,k},f_0)$ lies in $\mathscr{C}_2(Q,t_0)$, so ($\mathrm{M_s}$) gives $Z_k^2C_k(0)\le M_2(Q,t_0)\|f_0\|_\infty^2$, which is the last claim.

(iii) For a separated polynomial P at positive 0-times and dyadic $t>0$, the proof of Theorem 248(iv) with $e^{-m_{\mathrm{lat}}t/a_k}$ in place of e^{-m_*t} gives $\langle v,e^{-tH_0}v\rangle\le\lim_ke^{-m_{\mathrm{lat}}t/a_k}\|\Psi'_{P_k}\|^2$, $v=\Psi_0(P)-S(P)\Omega_0$, where $\|\Psi'_{P_k}\|^2$ converges to $\|v\|^2$. So $\|e^{-tH_0/2}v\|^2=0$, and $v=0$ because $e^{-tH_0/2}$ is injective. Hence $\mathscr{H}_0=\mathbb{C}\Omega_0$. For $Z_k=C_k(\sigma_0)^{-1/2}$, $C_k(\sigma_0)/C_k(0)\le e^{-m_{\mathrm{lat}}\sigma_0/a_k}$ by the argument of (i), which contradicts (ii) for large k. □

The lower bound in (ii) is the analogue of ($\mathrm{N_0}$) for the packets without flow, and (iii) is the analogue of Theorem 183. The averaged condition itself does not exclude a lattice gap: it holds at strong coupling for every cube size.

Corollary 251. *Let $G=\mathrm{SU}(2)$ and $\beta_{W,k}\in[0,\frac18]$ for every k. Then the hypothesis of Theorem 64 holds with $\varpi=0.41$, $n_k=2$ and $\ell=2a_0$, every torus limit ω_k has transfer-matrix gap at least $m_{\mathrm{lat}}=3.28/(625e^2)>7\cdot10^{-4}$ in lattice units, and for every Φ and every choice of Z_k satisfying* ($\mathrm{M_s}$) *every limit of Theorem 248 is trivial.*

Proof. By Corollary 72(i), $\alpha_n \le \alpha_2 \le 0.59$ for $|\beta_W| \le \frac{1}{8}$ and every $n \ge 2$, so $\mathrm{BP}(2, 0.41)$ holds on Λ_L for every even $L \ge 8$, and $n_k a_k = 2a_k \le 2a_0$. By Corollary 63 with $n = 2$ the gap is at least $\varpi/(2e^2 q_2) \ge 16\varpi/(1250e^2) = 3.28/(625e^2)$, and $625e^2 < 4619$. Proposition 250(iii) applies. □

So the content of Theorem 248 beyond the averaged condition lies in $(\mathrm{M_s})$ for constants Z_k that keep the reflected correlation $C_k(\sigma_0)$ fixed. The bound (M) at coinciding supports cannot serve for the action density.

Proposition 252. *Let S_2 be a limit along a subsequence of $\omega_k(Z_k^2 A_k(f)A_k(g))$ for all separated pairs, and suppose that $S_2(f,g) = \iint f(x)g(y)K(x-y)\,dx\,dy$ for separated pairs, with $K \ge 0$ measurable on $\mathbb{R}^4 \setminus \{0\}$ and*

$$K(z) \ge c\,|z|^{-8}\Big(\log\frac{1}{|z|}\Big)^{-p} \qquad \text{for } 0 < |z| \le r_0,$$

where $c > 0$, $p \ge 0$ and $0 < r_0 < 1$. Then the moment bound (M) *of Sec. XI, posed for the packets $Z_k A_k(f)$, fails for $n = 2$ and $Q = [0,1]^4$.*

Proof. If (M) held for $n = 2$, the limit would satisfy $|S_2(f,g)| \le M_2(Q)\|f\|_\infty \|g\|_\infty$ for separated f, g supported in Q. Let $0 < \delta \le \min(\frac{1}{8}, \frac{r_0}{5})$, let f_δ be continuous with values in $[0,1]$, equal to 1 on $[\frac12 - 2\delta, \frac12 - \delta] \times [\frac14, \frac34]^3$ and supported in $[0, \frac12 - \frac{\delta}{2}] \times [0,1]^3$, and let g_δ be continuous with values in $[0,1]$, equal to 1 on $[\frac12 + \delta, \frac12 + 2\delta] \times [\frac18, \frac78]^3$ and supported in $[\frac12 + \frac{\delta}{2}, 1] \times [0,1]^3$. The supports are at distance at least δ. For x in the first box and y in the second with $|y_i - x_i| \le \delta$ for $i = 1,2,3$, $2\delta \le |x-y| \le \sqrt{19}\,\delta < 5\delta \le r_0$, so $K(x-y) \ge c(5\delta)^{-8}(\log\frac{1}{2\delta})^{-p}$; these pairs have measure $\frac{\delta}{8} \cdot 8\delta^4 = \delta^5$. Hence $S_2(f_\delta, g_\delta) \ge c\,5^{-8}\delta^{-3}(\log\frac{1}{2\delta})^{-p}$, which is unbounded as $\delta \to 0$, while $\|f_\delta\|_\infty \|g_\delta\|_\infty = 1$. □

The law of the proposition, with $p = 2$, is the one that asymptotic freedom predicts for the action density. It places no restriction on $(\mathrm{M_s})$ when K is bounded on $\{|z|_\infty \ge \delta\}$ for every $\delta > 0$.

The two-point functions. For $n = 2$ the separated moments reduce to one quantity of the lattice theory. For $\varepsilon > 0$ let $B_\varepsilon = [\varepsilon, \varepsilon + 1 + a_0] \times [0, 1 + a_0]^3$ and, for $(r+1)a_k \le \varepsilon$,

$$\mathscr{N}_k(\varepsilon) = \sup\big\{\omega_k\big(\vartheta(\overline{A_k(h)})A_k(h)\big) : \operatorname{supp} h \subset B_\varepsilon,\ \|h\|_\infty \le 1\big\},$$

which is finite because $|A_k(h)|$ is at most $2\|\Phi\|_\infty a_k^4 \|h\|_\infty$ times the number of sites x with $a_k x \in B_\varepsilon$. Let $\gamma_k \le e^{-m_* a_k}$ be the norm of $\mathbb{T}_k$ on $\Omega_k^\perp$.

Lemma 253. *Let $(r+1)a_k \le \varepsilon$ and let $g \in C_c(\mathbb{R}^4)$ be supported in a cube of side L and in $\{x_0 \ge \varepsilon\}$. Then*

$$\omega_k\big(\vartheta(\overline{A_k(g)})A_k(g)\big) \le \mathsf{K}_L^2\,\mathscr{N}_k(\varepsilon)\,\|g\|_\infty^2.$$

The same holds with ϑ_μ for g supported in $\{x_\mu \ge \varepsilon\}$.

Proof. Write $g = \sum_m h_m$ with $h_m = \chi_m^1 g$, χ_m^1 as in the proof of Theorem 248(vi); χ_m^1 is supported in the closed cube of side 1 centred at $m/2$, and $\chi_m^1 g \ne 0$ requires $m_\nu \in (2q_\nu - 1, 2q_\nu + 2L + 1)$ for the cube $\prod_\nu [q_\nu, q_\nu + L]$, so at most K_L pieces are nonzero, each with $\|h_m\|_\infty \le \|g\|_\infty$. Fix one, supported in $[s, s+1] \times \prod_i [u_i, u_i + 1]$ and in $\{x_0 \ge s'\}$, $s' = \max(s, \varepsilon)$. Let $n_0 = \lfloor (s' - \varepsilon)/a_k \rfloor \ge 0$, $v_i = \lfloor u_i/a_k \rfloor$, $v = (n_0, v_1, v_2, v_3)$ and $h' = h_m(\cdot + a_k v)$. Then h' is supported in $[s' - n_0 a_k, s + 1 - n_0 a_k] \times \prod_i [u_i - v_i a_k, u_i - v_i a_k + 1] \subset B_\varepsilon$, since $s' - n_0 a_k \in [\varepsilon, \varepsilon + a_k)$ and $u_i - v_i a_k \in [0, a_k)$. By Lemma 247(i), $A_k(h_m) = \tau_{n_0 e_0}\tau_{v_\perp} A_k(h')$ with $v_\perp = (0, v_1, v_2, v_3)$, and by Lemma 247(ii),(iii) and its proof, $\|\Psi_{A_k(h_m)}\| = \|\mathbb{T}_k^{n_0}\Psi_{\tau_{v_\perp}A_k(h')}\| \le \|\Psi_{A_k(h')}\| \le \mathscr{N}_k(\varepsilon)^{1/2}\|g\|_\infty$. The triangle inequality for the linear map $F \mapsto \Psi_F$ gives $\|\Psi_{A_k(g)}\| \le \mathsf{K}_L \mathscr{N}_k(\varepsilon)^{1/2}\|g\|_\infty$. For ϑ_μ apply this to $g \circ \pi_\mu$. □

Theorem 254. *Assume the hypothesis of Theorem 64, let $\varepsilon > 0$, suppose $\mathscr{N}_k(\varepsilon) > 0$ for large k, and put $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$.*

(i) *Let $c \in \mathbb{R}$, $d \ge 0$, f supported in a cube of side L in $\{x_\mu \le c - 2\varepsilon\}$ and g in a cube of side L' in $\{x_\mu \ge c + 2\varepsilon + d\}$. For $a_k \le \varepsilon$ and $(r+1)a_k \le \varepsilon$,*

$$Z_k^2\big|\omega_k\big(A_k(f)A_k(g)\big)\big| \le \mathsf{K}_L \mathsf{K}_{L'}\,\gamma_k^{\lfloor d/a_k \rfloor}\,\|f\|_\infty \|g\|_\infty.$$

(ii) *For $(f,g) \in \mathscr{C}_2(Q, \delta)$ with Q of side L and $\delta \ge 8\varepsilon$, and k as in* (i),

$$Z_k^2\big|\omega_k\big(A_k(f)A_k(g)\big)\big| \le \mathsf{K}_{4L/\delta}^2 \mathsf{K}_{\delta/4}^2\,\|f\|_\infty \|g\|_\infty.$$

(iii) *Every sequence of k has a subsequence along which $S_2(f,g) = \lim_k Z_k^2 \omega_k(A_k(f)A_k(g))$ exists for all f, g with $\operatorname{dist}_\infty(\operatorname{supp} f, \operatorname{supp} g) > 8\varepsilon$. It is symmetric, bilinear, bounded as in* (ii)*, invariant under the translations and B_4, $S_2(\bar f, \bar g) = \overline{S_2(f,g)}$, and $\sum_{i,j} \bar c_i c_j S_2(\bar g_i \circ \theta_\mu, g_j) \ge 0$ for all $c_i \in \mathbb{C}$ and g_i supported in $\{x_\mu > 4\varepsilon\}$.*

(iv) *For f, g supported in a cube of side L, $y \ne 0$ and $\lambda|y|_\infty > L + 8\varepsilon$,*

$$\big|S_2(f, g(\cdot - \lambda y))\big| \le \mathsf{K}_L^2 \|f\|_\infty \|g\|_\infty\, e^{-m_*(\lambda|y|_\infty - L - 4\varepsilon)}.$$

(v) *If $\gamma_k \le e^{-m_{\mathrm{lat}}}$ for a fixed $m_{\mathrm{lat}} > 0$ and all k, every S_2 of* (iii) *vanishes.*

Parts (i)–(iii) *use only the invariance of ω_k under translations and B_4 and its reflection positivity.*

Proof. (i) By the invariance of ω_k under π_μ and Lemma 247(i) we may take $\mu = 0$. Let $b_k \in a_k\mathbb{Z}$ with $c \le b_k < c + a_k$; by the invariance under the translation by $-b_k e_0/a_k$ we may take $b_k = 0$, so that f is supported in $\{x_0 \le -2\varepsilon\}$ and g in $\{x_0 \ge 2\varepsilon + d - a_k\}$. Let $h = \bar f \circ \theta_{0,k}$, supported in $\{x_0 \ge 2\varepsilon - a_k\} \subset \{x_0 \ge \varepsilon\}$; then $A_k(f) = \vartheta(\overline{A_k(h)})$. Let $n = \lfloor d/a_k \rfloor$ and $g' = g(\cdot + n a_k e_0)$, supported in $\{x_0 \ge 2\varepsilon - a_k\}$ because $n a_k \le d$. Then $A_k(g) = \tau_n A_k(g')$, and

$$\omega_k\big(A_k(f)A_k(g)\big) = \langle \Psi_{A_k(h)}, \mathbb{T}_k^n \Psi_{A_k(g')}\rangle.$$

Both vectors are orthogonal to Ω_k, the packets being centred, so the modulus is at most $\gamma_k^n \|\Psi_{A_k(h)}\|\,\|\Psi_{A_k(g')}\|$, and Lemma 253 bounds the norms by $\mathsf{K}_L \mathscr{N}_k(\varepsilon)^{1/2}\|f\|_\infty$ and $\mathsf{K}_{L'}\mathscr{N}_k(\varepsilon)^{1/2}\|g\|_\infty$.

(ii) Let $s = \delta/4$ and write $f = \sum_m \chi_m^s f$, $g = \sum_{m'} \chi_{m'}^s g$; at most $\mathsf{K}_{L/s} = \mathsf{K}_{4L/\delta}$ terms of each sum are nonzero. For a pair of nonzero pieces pick p, q in their supports, and μ with

$|p_\mu - q_\mu| \ge \delta$. The projections of the two cubes of side s on the μ axis contain p_μ and q_μ, so they are at distance at least $\delta - 2s = \delta/2 \ge 4\varepsilon$, and (i) applies to the pair, possibly with f and g exchanged, with $L = L' = \delta/4$ and $\gamma_k^{\lfloor d/a_k \rfloor} \le 1$. Summing over the pairs gives the bound.

(iii) The diagonal argument of the proof of Theorem 248(i), with (ii) in place of ($\mathrm{M_s}$) and approximants within $\eta < (\delta - 8\varepsilon)/4$, gives the limit for pairs at distance $\delta > 8\varepsilon$. Invariance and the conjugation identity follow as in Theorem 248(ii). For $g = \sum_i c_i g_i$ supported in $\{x_\mu \ge 4\varepsilon + \eta'\}$, $\eta' > 0$, $Z_k^2 \omega_k(\vartheta_\mu(\overline{A_k(g)})A_k(g)) \ge 0$ by Lemma 247, and it equals $Z_k^2 \omega_k(A_k(\bar g \circ \theta_{\mu,k})A_k(g))$; the pair $(\bar g \circ \theta_\mu, g)$ is at distance at least $8\varepsilon + 2\eta'$, and (ii) carries the approximation of $\theta_{\mu,k}$ by θ_μ as in Theorem 248(iii).

(iv) By B_4-invariance we may take $y_\mu = |y|_\infty = \eta > 0$, and by translation invariance the cube to be $\prod_\nu [q_\nu, q_\nu + L]$. Then f is supported in $\{x_\mu \le c - 2\varepsilon\}$ with $c = q_\mu + L + 2\varepsilon$, and $g(\cdot - \lambda y)$ in $\{x_\mu \ge c + 2\varepsilon + d\}$ with $d = \lambda\eta - L - 4\varepsilon > 0$. By (i) and $\gamma_k^{\lfloor d/a_k \rfloor} \le e^{-m_*(d-a_k)}$ the lattice quantities obey the bound with $e^{-m_*(d-a_k)}$, and the limit gives the claim.

(v) In the proof of (ii), now with $\delta > 8\varepsilon$, each pair of pieces satisfies (i) with $d = \delta/2 - 4\varepsilon > 0$, so the lattice quantity is at most $\mathsf{K}_{4L/\delta}^2 \mathsf{K}_{\delta/4}^2 \|f\|_\infty \|g\|_\infty e^{-m_{\mathrm{lat}}(d/a_k - 1)}$, which tends to 0. □

Two statements about the lattice two-point function control the small separations:

(SD) $\limsup_k \mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon) < \infty$ for every $\varepsilon' \in (0, \varepsilon)$;

(NE) there is $g_0 \in C_c(\mathbb{R}^4)$ supported in $\{x_0 > 0\}$ with

$$\liminf_k \frac{\omega_k\big(\vartheta(\overline{A_k(g_0)})A_k(g_0)\big)}{\mathscr{N}_k(\varepsilon)} > 0.$$

(SD) bounds the growth of the largest reflected norm of a packet of unit size as it approaches the reflection plane from the distance ε to any smaller distance, uniformly in the spacing; (NE) says that a fixed packet carries a fixed fraction of that largest norm.

Corollary 255. *Assume the hypothesis of Theorem 254.*

(a) (SD) *holds if and only if* ($\mathrm{M_s}$) *holds for* $n = 2$.

(b) *Under* (SD)*, every sequence of k has a subsequence along which $S_2(f,g)$ exists for every separated pair. It is given by a distribution on $\mathfrak{D}_2$, invariant under the translations and B_4, satisfies $S_2(\bar f, \bar g) = \overline{S_2(f,g)}$ and $S_2(\bar g \circ \theta_\mu, g) \ge 0$ for g supported in $\{x_\mu > 0\}$, and for f, g supported in a cube of side L, $y \ne 0$, $\varepsilon' \in (0, \varepsilon]$ and $\lambda|y|_\infty > L + 4\varepsilon'$,*

$$\big|S_2(f, g(\cdot - \lambda y))\big| \le C_{\varepsilon'} \mathsf{K}_L^2 \|f\|_\infty \|g\|_\infty e^{-m_*(\lambda|y|_\infty - L - 4\varepsilon')}$$

with $C_{\varepsilon'} = \limsup_k \mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon)$. The time translations define on the completion $\mathscr{H}^{(1)}$ of the functions supported in $\{x_0 > 0\}$ for the form $S_2(\bar g \circ \theta_0, g')$ a semigroup $e^{-tH^{(1)}}$ with $\sigma(H^{(1)}) \subset [m_*, \infty)$. Some limit S_2 is not identically zero if and only if $\limsup_k \omega_k(\vartheta(\overline{A_k(g)})A_k(g))/\mathscr{N}_k(\varepsilon) > 0$ for some g supported in $\{x_0 > 0\}$; under (NE) *every limit is not identically zero, and then $\sigma(H^{(1)})$ is a nonempty subset of $[m_*, \infty)$.*

Proof. (a) If ($\mathrm{M_s}$) holds for $n = 2$, let h be supported in $B_{\varepsilon'}$ with $\|h\|_\infty \le 1$. For $a_k \le \varepsilon'$ the pair $(\bar h \circ \theta_{0,k}, h)$ lies in $\mathscr{C}_2(Q, \varepsilon')$ for a cube Q independent of k and h, and $\omega_k(\vartheta(\overline{A_k(h)})A_k(h)) = \omega_k(A_k(\bar h \circ \theta_{0,k})A_k(h))$, so $\mathscr{N}_k(\varepsilon') \le M_2(Q, \varepsilon')\mathscr{N}_k(\varepsilon)$ for large k. Conversely, let $(f,g) \in \mathscr{C}_2(Q, \delta)$ and $\varepsilon' = \min(\varepsilon, \delta/8)$. The proof of Theorem 254(i), (ii) with ε' in place of ε gives $|\omega_k(A_k(f)A_k(g))| \le \mathsf{K}_{4L/\delta}^2 \mathsf{K}_{\delta/4}^2 \mathscr{N}_k(\varepsilon')\|f\|_\infty \|g\|_\infty$ for large k, and dividing by $\mathscr{N}_k(\varepsilon)$ and using (SD) gives ($\mathrm{M_s}$) for $n = 2$.

(b) By (a), the proof of Theorem 248(i) for $n = 2$ gives the limits and the distribution, and that of Theorem 248(ii), (iii) the invariance, the conjugation identity and the positivity. The decay follows from Theorem 254(i) with ε' in place of ε, multiplied by $\mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon)$, as in the proof of Theorem 254(iv). The semigroup comes from the proof of Theorem 248(iv) restricted to single functions. Since $\langle \Omega_k, \Psi_{A_k(g)} \rangle = 0$, Lemma 247(iii) gives $\omega_k(\vartheta(\overline{A_k(g)})\tau_{t/a_k}A_k(g)) \le e^{-m_* t}\omega_k(\vartheta(\overline{A_k(g)})A_k(g))$, hence $\langle w, e^{-tH^{(1)}} w \rangle \le e^{-m_* t}\|w\|^2$ on a dense set of w, and $\|e^{-tH^{(1)}}\| \le e^{-m_* t}$. If the lim sup is positive for g, a subsequence along which the ratio converges to it has a further subsequence with $S_2(\bar g \circ \theta_0, g)$ equal to that limit, by the approximation of $\theta_{0,k}$ by θ_0. Conversely, if $S_2(\bar g \circ \theta_\mu, g) = 0$ for all μ and all g supported in $\{x_\mu > 0\}$, the Cauchy–Schwarz inequality for these positive forms and translation invariance give $S_2(f,g) = 0$ whenever f and g lie on opposite sides of a coordinate hyperplane, and the decomposition in the proof of Theorem 254(ii) splits every separated pair into such pairs; by B_4-invariance $S_2(\bar g \circ \theta_\mu, g) = S_2(\overline{g \circ \pi_\mu} \circ \theta_0, g \circ \pi_\mu)$, so a nonzero S_2 has $S_2(\bar g \circ \theta_0, g) > 0$ for some g, and the ratio for g converges to it along the subsequence. Under (NE), $S_2(\bar g_0 \circ \theta_0, g_0)$ is at least the lim inf in (NE), so $\mathscr{H}^{(1)} \ne 0$. □

The short-distance comparison. The comparison (SD) is a statement about the spectrum of the transfer matrices. For $\Lambda > 0$ let $\mathbb{E}_k(\Lambda)$ be the spectral projection of $\mathbb{T}_k$ on $[e^{-\Lambda a_k}, 1]$, the states of energy at most Λ in physical units, put $\mathsf{K} = \mathsf{K}_{1+a_0} = (5 + 2a_0)^4$ and, for $(r+1)a_k \le \varepsilon$,

$$\mathscr{N}_k(\varepsilon; \Lambda) = \sup\big\{ \langle \Psi_{A_k(h)}, \mathbb{E}_k(\Lambda)\Psi_{A_k(h)} \rangle : \operatorname{supp} h \subset B_\varepsilon,\ \|h\|_\infty \le 1 \big\}.$$

(UV) For every $\varepsilon' \in (0, \varepsilon)$ there are $\Lambda < \infty$ and $c > 0$ with $\mathscr{N}_k(\varepsilon'; \Lambda) \ge c\,\mathscr{N}_k(\varepsilon')$ for all large k.

(UV) says that the largest reflected norm of a unit packet near the reflection plane is carried, up to a fixed fraction, by energies bounded in physical units.

Proposition 256. *Assume the hypothesis of Theorem 254.*

(i) *For $(r+1)a_k \le \varepsilon' < \varepsilon$,*

$$\mathscr{N}_k(\varepsilon) \le \mathsf{K}^2 \gamma_k^{2\lfloor (\varepsilon - \varepsilon')/a_k \rfloor} \mathscr{N}_k(\varepsilon') \le \mathsf{K}^2 e^{-2m_*(\varepsilon - \varepsilon' - a_k)} \mathscr{N}_k(\varepsilon').$$

(ii) (SD) *holds if and only if* (UV) *holds.*

(iii) *If* (SD) *holds, then for every $\varepsilon' \in (0, \varepsilon)$*

$$\limsup_k \frac{1}{a_k} \log \frac{1}{\gamma_k} \le \frac{\log(\mathsf{K}^2 C_{\varepsilon'})}{2(\varepsilon - \varepsilon')}, \qquad C_{\varepsilon'} = \limsup_k \frac{\mathscr{N}_k(\varepsilon')}{\mathscr{N}_k(\varepsilon)}.$$

In particular, if $\gamma_k \le e^{-m_{\mathrm{lat}}}$ for a fixed $m_{\mathrm{lat}} > 0$ and all k, then $\mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon) \to \infty$ for every $\varepsilon' \in (0,\varepsilon)$ and (SD) *fails; this is the case under the hypotheses of Corollary 251 whenever $\mathscr{N}_k(\varepsilon) > 0$ for large k.*

Proof. (i) Let h be supported in B_ε with $\|h\|_\infty \le 1$, put $n = \lfloor(\varepsilon-\varepsilon')/a_k\rfloor$ and $h' = h(\cdot + n a_k e_0)$, supported in a cube of side $1+a_0$ in $\{x_0 \ge \varepsilon'\}$. By Lemma 247(i) and the proof of Lemma 247(iii), $\Psi_{A_k(h)} = \mathbb{T}_k^n \Psi_{A_k(h')}$, and $\Psi_{A_k(h')} \perp \Omega_k$ because the packet is centred. So $\|\Psi_{A_k(h)}\| \le \gamma_k^n \|\Psi_{A_k(h')}\| \le \gamma_k^n \mathsf{K} \mathscr{N}_k(\varepsilon')^{1/2}$ by Lemma 253, and $\gamma_k^n \le e^{-m_* a_k n} \le e^{-m_*(\varepsilon-\varepsilon'-a_k)}$.

(ii) First a version of Lemma 253 for $\mathbb{E} = \mathbb{E}_k(\Lambda)$. For a function F of links in $\{x_0 \ge 1\}$ and $v \in \mathbb{Z}^4$ with $v_0 = 0$, the spectral measures of $\mathbb{T}_k$ at Ψ_F and at $\Psi_{\tau_v F}$ are measures on $[0,1]$ with the same moments $\omega_k(\vartheta(\bar F)\tau_j F) = \omega_k(\tau_v(\vartheta(\bar F)\tau_j F))$, $j \ge 0$, so they coincide and $\|\mathbb{E}\Psi_{\tau_v F}\| = \|\mathbb{E}\Psi_F\|$. If g is supported in a cube of side L in $\{x_0 \ge \varepsilon\}$, the proof of Lemma 253 writes $\Psi_{A_k(g)}$ as a sum of at most K_L vectors $\mathbb{T}_k^{n_m} \Psi_{\tau_{v_m} A_k(h'_m)}$ with $(v_m)_0 = 0$, h'_m supported in B_ε and $\|h'_m\|_\infty \le \|g\|_\infty$; since $\mathbb{E}$ commutes with $\mathbb{T}_k$ and $\|\mathbb{T}_k\| \le 1$,

$$\|\mathbb{E}\Psi_{A_k(g)}\| \le \mathsf{K}_L \mathscr{N}_k(\varepsilon;\Lambda)^{1/2} \|g\|_\infty.$$

Assume (UV) and let $\varepsilon' \in (0,\varepsilon)$, with Λ, c as in (UV). Take h supported in $B_{\varepsilon'}$ with $\|h\|_\infty \le 1$ and $\langle \Psi, \mathbb{E}\Psi\rangle \ge \frac{c}{2}\mathscr{N}_k(\varepsilon')$, $\Psi = \Psi_{A_k(h)}$, let $n = \lceil(\varepsilon-\varepsilon')/a_k\rceil$ and $h_n = h(\cdot - n a_k e_0)$, supported in a cube of side $1+a_0$ in $\{x_0 \ge \varepsilon\}$. With ν the spectral measure of $\mathbb{T}_k$ at Ψ,

$$\begin{aligned}\|\Psi_{A_k(h_n)}\|^2 &= \int \lambda^{2n} d\nu(\lambda) \ge e^{-2\Lambda a_k n} \nu\big([e^{-\Lambda a_k}, 1]\big) \\ &\ge e^{-2\Lambda(\varepsilon-\varepsilon'+a_k)} \tfrac{c}{2} \mathscr{N}_k(\varepsilon'),\end{aligned}$$

and Lemma 253 bounds the left side by $\mathsf{K}^2 \mathscr{N}_k(\varepsilon)$. So $\mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon) \le 2c^{-1}\mathsf{K}^2 e^{2\Lambda(\varepsilon-\varepsilon'+a_k)}$, which is (SD).

Conversely, let (SD) hold and $\varepsilon' \in (0,\varepsilon)$, and fix a number C greater than $\limsup_k \mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon)$, and put $\Lambda = \log(4C\mathsf{K}^2)/(\varepsilon-\varepsilon')$. For h, n and h' as in the proof of (i), $\|\mathbb{T}_k(1-\mathbb{E})\| \le e^{-\Lambda a_k}$, so

$$\|\Psi_{A_k(h)}\| = \|\mathbb{T}_k^n \Psi_{A_k(h')}\| \le \|\mathbb{E}\Psi_{A_k(h')}\| + e^{-\Lambda a_k n}\|\Psi_{A_k(h')}\|,$$

and the two forms of Lemma 253 give

$$\mathscr{N}_k(\varepsilon) \le 2\mathsf{K}^2\big(\mathscr{N}_k(\varepsilon';\Lambda) + e^{-2\Lambda a_k n}\mathscr{N}_k(\varepsilon')\big).$$

For large k, $a_k \le \frac12(\varepsilon-\varepsilon')$, so $a_k n \ge \frac12(\varepsilon-\varepsilon')$ and $2C\mathsf{K}^2 e^{-2\Lambda a_k n} \le \frac12$; with $\mathscr{N}_k(\varepsilon') \le C\mathscr{N}_k(\varepsilon)$ this gives $\mathscr{N}_k(\varepsilon') \le 4C\mathsf{K}^2\mathscr{N}_k(\varepsilon';\Lambda)$, which is (UV) at ε'.

(iii) By (i), $\mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon) \ge \mathsf{K}^{-2}\gamma_k^{-2n}$ with $n \ge (\varepsilon-\varepsilon')/a_k - 1$; in particular $\mathsf{K}^2 C_{\varepsilon'} \ge 1$. For $C > C_{\varepsilon'}$ and large k this gives $2n\log(1/\gamma_k) \le \log(\mathsf{K}^2 C)$, that is $a_k^{-1}\log(1/\gamma_k) \le \log(\mathsf{K}^2C)/(2(\varepsilon-\varepsilon'-a_k))$, and $C \downarrow C_{\varepsilon'}$ gives the bound. Under a lattice gap the same inequality gives $\mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon) \ge \mathsf{K}^{-2} e^{2m_{\mathrm{lat}}((\varepsilon-\varepsilon')/a_k - 1)} \to \infty$. Corollary 251 provides the lattice gap $m_{\mathrm{lat}} = 3.28/(625e^2)$. □

So (SD) bounds the physical gap from above, and it fails in the strong-coupling range of Corollary 251. It does not follow from invariance, reflection positivity and a gap bounded above and below in physical units: a heavy species with a growing weight removes it.

Proposition 257. *Let $m > 0$, $\rho \in (0,1)$, $q_k = e^{-m a_k}$, let $\phi_k^{(1)}$ and $\phi_k^{(2)}$ be independent centred Gaussian fields on $\mathbb{Z}^4$ with covariances $\prod_\mu q_k^{|x_\mu - y_\mu|}$ and $\prod_\mu \rho^{|x_\mu-y_\mu|}$, let $\lambda_k > 0$, $\phi_k = \phi_k^{(1)} + \lambda_k \phi_k^{(2)}$, $A_k(f) = \sum_x a_k^4 f(a_k x)\phi_k(x)$, and form $\mathscr{N}_k(\varepsilon)$ with these packets and $r = 0$.*

(a) *The law of ϕ_k is invariant under translations and B_4 and reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links. For $q_k \ge \rho$ the spectrum of its transfer matrix in each coordinate direction on $\Omega_k^\perp$ lies in $[0, q_k]$ and contains q_k: the gap is m in physical units.*

(b) *Given $\varepsilon > 0$, there are λ_k with $\mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon) \to \infty$ for every $\varepsilon' \in (0,\varepsilon)$, so that* (SD) *fails.*

(c) *For these λ_k and every Z_k for which* ($\mathrm{M_s}$) *holds at $n = 2$, $Z_k^2 \omega_k(A_k(f)A_k(g)) \to 0$ for every separated pair: every limit of the two-point functions vanishes.*

Proof. (a) Each $\phi_k^{(i)}$ is the field of Proposition 199(a), with q_k replaced by ρ for $i = 2$, which has the invariance and the reflection positivity, with the transfer operators P_1, P_2 of spectra $\{q_k^N\}$ and $\{\rho^N\}$, 1 simple, given in its proof. The pair $(\phi_k^{(1)}, \phi_k^{(2)})$ is then a reversible Markov chain in each coordinate direction with transfer operator $P_1 \otimes P_2$, of spectrum $\{q_k^N \rho^{N'}\}$ with 1 simple, and the argument of that proof gives $\mathbb{E}[\vartheta(\bar F)F'] = \langle g_F, g_{F'}\rangle$ for functions F, F' of the pair at positive times. The functions of ϕ_k at positive times are functions of the pair there, so ϕ_k is reflection positive, its transfer matrix is the restriction of $P_1 \otimes P_2$ to an invariant subspace, and its spectrum on $\Omega_k^\perp$ lies in $[0, \max(q_k,\rho)] = [0, q_k]$. For a site x at positive time, $g_{\phi_k(x)}$ is a sum of two orthogonal eigenvectors, of eigenvalues q_k and ρ, the first one nonzero, so q_k is in the spectrum.

(b) Let ϑ fix the plane $\{x_0 = c\}$, $|c| \le \frac12$, so that $\vartheta x = (2c - x_0, \mathbf{x})$. By independence and centring, $\omega_k(\vartheta(\overline{A_k(h)})A_k(h)) = \mathscr{N}^{(1)}(h) + \lambda_k^2 \mathscr{N}^{(2)}(h)$, with $\mathscr{N}^{(i)}(h) \ge 0$ formed with $\phi_k^{(i)}$. For h supported in $\{x_0 \ge a_k\}$,

$$\begin{aligned}\mathscr{N}^{(2)}(h) &= a_k^8 \sum_{\mathbf{x},\mathbf{y}} \overline{H(\mathbf{x})} H(\mathbf{y}) \prod_{i=1}^3 \rho^{|x_i - y_i|}, \\ H(\mathbf{x}) &= \sum_{x_0} h(a_k x)\rho^{x_0 - c}.\end{aligned}$$

If h is supported in B_ε with $\|h\|_\infty \le 1$ and $a_k \le 1$, then $|H| \le \rho^{\varepsilon/a_k - 1/2}/(1-\rho)$, at most $((2+a_0)/a_k)^3$ points $\mathbf{x}$ have $H(\mathbf{x}) \ne 0$, and $\sum_{\mathbf{y}} \prod_i \rho^{|x_i - y_i|} = ((1+\rho)/(1-\rho))^3$, so

$$\mathscr{N}^{(2)}(h) \le U_k(\varepsilon) = c_\rho a_k^5 \rho^{2\varepsilon/a_k}, \qquad c_\rho = \frac{(2+a_0)^3(1+\rho)^3}{\rho(1-\rho)^5}.$$

For $\varepsilon' > 0$ and $\tau \in (0, \frac12)$ let h_* be continuous with values in $[0,1]$, supported in $B_{\varepsilon'}$ and equal to 1 on $D = [\varepsilon'+\tau, \varepsilon'+1] \times$

$[\tau,1]^3$. All terms of $\mathscr{N}^{(2)}(h_*)$ are nonnegative; keeping $x=y$ with $x_0=\lceil(\varepsilon'+\tau)/a_k\rceil$ and $a_k\mathbf{x}\in[\tau,1]^3$, for $a_k\le\frac12(1-\tau)$,

$$\mathscr{N}^{(2)}(h_*)\ge L_k(\varepsilon')=\Big(\frac{1-\tau}{2}\Big)^3 a_k^5\rho^{2(\varepsilon'+\tau)/a_k+3}.$$

Let $\mathscr{N}_k^{(1)}(\varepsilon)$ be the supremum of $\mathscr{N}^{(1)}(h)$ over h supported in B_ε with $\|h\|_\infty\le1$. For $a_kx,a_ky\in D_\varepsilon=[\varepsilon+\frac14,\varepsilon+1]\times[\frac14,1]^3$ the covariance of $\phi_k^{(1)}(\vartheta x)$ and $\phi_k^{(1)}(y)$ is at least $e^{-m(2\varepsilon+6)}$; with the function h_* for ε and $\tau=\frac14$ this gives $\mathscr{N}_k^{(1)}(\varepsilon)\ge\ell_0=(\frac38)^8e^{-m(2\varepsilon+6)}>0$ for $a_k\le\frac38$, and it is finite. Put $\lambda_k^2=\mathscr{N}_k^{(1)}(\varepsilon)/U_k(\varepsilon)$. Then $\mathscr{N}_k(\varepsilon)\le\mathscr{N}_k^{(1)}(\varepsilon)+\lambda_k^2U_k(\varepsilon)=2\mathscr{N}_k^{(1)}(\varepsilon)$ and $\mathscr{N}_k(\varepsilon')\ge\lambda_k^2L_k(\varepsilon')$, so for $\tau<\varepsilon-\varepsilon'$

$$\begin{aligned}\frac{\mathscr{N}_k(\varepsilon')}{\mathscr{N}_k(\varepsilon)}&\ge\frac{L_k(\varepsilon')}{2U_k(\varepsilon)}\\&=\Big(\frac{1-\tau}{2}\Big)^3\frac{\rho^4(1-\rho)^5\rho^{-2(\varepsilon-\varepsilon'-\tau)/a_k}}{2(2+a_0)^3(1+\rho)^3}\to\infty.\end{aligned}$$

(c) Let $(f,g)\in\mathscr{C}_2(Q_0,\delta)$, Q_0 of side L_0, and fix $\varepsilon',\tau>0$ with $\varepsilon'+\tau<\min(\varepsilon,\delta/2)$ and $\tau<\frac12$. For $a_k\le\varepsilon'$ the pair $(\bar h_*\circ\theta_{0,k},h_*)$ lies in $\mathscr{C}_2(Q',\varepsilon')$ for a cube Q' independent of k, so $(\mathrm{M_s})$ gives $Z_k^2\lambda_k^2L_k(\varepsilon')\le Z_k^2\omega_k(\vartheta(A_k(h_*))A_k(h_*))\le M=M_2(Q',\varepsilon')$, that is $Z_k^2\mathscr{N}_k^{(1)}(\varepsilon)\le MU_k(\varepsilon)/L_k(\varepsilon')$. The covariance of $\phi_k^{(1)}$ is at most 1 and at most $(L_0/a_k+1)^4$ sites carry f or g, so $|\mathbb{E}[A_k^{(1)}(f)A_k^{(1)}(g)]|\le(L_0+1)^8\|f\|_\infty\|g\|_\infty$ for $a_k\le1$, and

$$Z_k^2\big|\mathbb{E}[A_k^{(1)}(f)A_k^{(1)}(g)]\big|\le\frac{M(L_0+1)^8}{\ell_0}\,\frac{U_k(\varepsilon)}{L_k(\varepsilon')}\|f\|_\infty\|g\|_\infty\to0.$$

For the second field the sites carrying f and g are at ℓ^∞ distance at least δ/a_k, and $\sum_{|z|_\infty\ge d}\prod_\mu\rho^{|z_\mu|}\le8\rho^d(1+\rho)^3/(1-\rho)^4$, so $\lambda_k^2|\mathbb{E}[A_k^{(2)}(f)A_k^{(2)}(g)]|\le\lambda_k^2C_1a_k^4\rho^{\delta/a_k}\|f\|_\infty\|g\|_\infty$ with $C_1=8(L_0+1)^4(1+\rho)^3/(1-\rho)^4$. Since $Z_k^2\lambda_k^2\le M/L_k(\varepsilon')$,

$$\begin{aligned}&Z_k^2\lambda_k^2\big|\mathbb{E}[A_k^{(2)}(f)A_k^{(2)}(g)]\big|\\&\quad\le MC_1\Big(\frac{2}{1-\tau}\Big)^3\frac{\rho^{(\delta-2\varepsilon'-2\tau)/a_k-3}}{a_k}\|f\|_\infty\|g\|_\infty\to0,\end{aligned}$$

because $\delta>2(\varepsilon'+\tau)$. The cross terms vanish. □

In the example the part $\lambda_k^2\mathscr{N}^{(2)}$ of the reflected norm, which produces the growth, is carried by energies at least $a_k^{-1}\log(1/\rho)\to\infty$, and (UV) fails, by the proof of Proposition 256(ii), which uses only the invariance, reflection positivity and the gap. For $\lambda_k=0$ it is the field of Proposition 199(a), for which (SD) and $(\mathrm{M_s})$ hold. Like that field, it is not a lattice gauge theory; it shows that (SD), equivalently (UV), is an input on the ultraviolet behaviour of the Wilson measure that reflection positivity and the gap do not supply.

The nonvanishing condition. Under (SD) the condition (NE) is exactly the nontriviality of the limits. Write $R_k(g)=\omega_k(\vartheta(\overline{A_k(g)})A_k(g))/\mathscr{N}_k(\varepsilon)$ for g supported in $\{x_0\ge(r+1)a_k\}$.

Proposition 258. *Assume the hypothesis of Theorem 254 and* (SD).

(i) *The following are equivalent:* (NE)*; every limit S_2 of Corollary 255*(b) *is not identically zero; every such limit has $\mathscr{H}^{(1)}\ne0$.*

(ii) *If $\omega_k(\Phi_0\Phi_z)\ge c_k^2$ for all large k and all $z\in\mathbb{Z}^4$ with $|z|_\infty\ge2r+1$, then $R_k(g_0)\ge1$ for large k and every continuous g_0 with values in $[0,1]$, equal to 1 on B_ε and supported in $\{x_0\ge\varepsilon/2\}$; in particular* (NE) *holds.*

Proof. (i) The first statement implies the second by Corollary 255(b). The second and the third are equivalent: $\mathscr{H}^{(1)}\ne0$ means $S_2(\bar g\circ\theta_0,g)>0$ for some g supported in $\{x_0>0\}$, and the proof of Corollary 255(b) shows that a limit with this property for no g vanishes identically.

Suppose (NE) fails, so that $\liminf_kR_k(g)=0$ for every $g\in C_c(\mathbb{R}^4)$ supported in $\{x_0>0\}$. Let $g_1,\dots,g_J$ be supported in a cube of side L and in $\{x_0\ge\eta\}$, $\eta>0$, let $D>0$ be dyadic and $G=\sum_jg_j(\cdot-jDe_1)$. For large k, $jD/a_k\in\mathbb{Z}$ and $R_k(g_j(\cdot-jDe_1))=R_k(g_j)$ by Lemma 247(i),(iii). For $i\ne j$ the cross term $\langle\Psi_{A_k(g_i(\cdot-iDe_1))},\Psi_{A_k(g_j(\cdot-jDe_1))}\rangle$ equals $\omega_k(A_k(\bar g_i(\cdot-iDe_1)\circ\theta_{0,k})A_k(g_j(\cdot-jDe_1)))$, two packets in cubes of side L whose projections on the e_1 axis are at distance at least $|i-j|D-L$; by Theorem 254(i) in the direction e_1 with $d=D-L-4\varepsilon$, its modulus is at most $\mathscr{N}_k(\varepsilon)\mathsf{K}_L^2e^{-m_*(D-L-4\varepsilon-1)}\max_j\|g_j\|_\infty^2$. Hence $\sum_jR_k(g_j)\le R_k(G)+J^2\mathsf{K}_L^2e^{-m_*(D-L-4\varepsilon-1)}\max_j\|g_j\|_\infty^2$ for large k, and since $\liminf_kR_k(G)=0$ and D is arbitrary, $\liminf_k\sum_jR_k(g_j)=0$.

Enumerate as $g_1,g_2,\dots$ the elements of the countable set $\mathscr{G}$ of the proof of Theorem 248(i) that are supported in $\{x_0>0\}$, and choose $k_1<k_2<\cdots$ with $\sum_{j\le J}R_{k_J}(g_j)<1/J$. Then $R_{k_J}(g_j)\to0$ for every j. Let g be supported in a cube of side L and in $\{x_0\ge\eta\}$, and let $\tau\in(0,\min(\eta,1)/2]$. That proof gives $g_j\in\mathscr{G}$ with $\|g-g_j\|_\infty<\tau$ and support within τ of $\operatorname{supp}g$, so $g-g_j$ is supported in a cube of side $L+1$ in $\{x_0\ge\eta/2\}$, and by Lemma 253

$$R_k(g)^{1/2}\le R_k(g_j)^{1/2}+\mathsf{K}_{L+1}\Big(\frac{\mathscr{N}_k(\eta/2)}{\mathscr{N}_k(\varepsilon)}\Big)^{1/2}\tau.$$

The ratio has finite lim sup, by (SD) if $\eta/2<\varepsilon$ and by Proposition 256(i) otherwise. As τ is arbitrary, $R_{k_J}(g)\to0$. By Corollary 255(b) a subsequence of (k_J) has a limit S_2, and by the approximation of $\theta_{0,k}$ by θ_0 in the proof of Theorem 248(iii), $S_2(\bar g\circ\theta_0,g)=\lim_JR_{k_J}(g)=0$ for every g supported in $\{x_0>0\}$. So this limit vanishes identically, and the second statement fails.

(ii) Let h be supported in B_ε with $\|h\|_\infty\le1$, and $(r+1)a_k\le\varepsilon/2$. By Lemma 247(i), $\omega_k(\vartheta(\overline{A_k(h)})A_k(h))=\sum_{x,y}a_k^8\overline{h(a_kx)}h(a_ky)K(x,y)$ with $K(x,y)=\omega_k(\Phi_{\vartheta x}\Phi_y)-c_k^2=\omega_k(\Phi_0\Phi_{y-\vartheta x})-c_k^2$. For $a_kx_0,a_ky_0\ge\varepsilon/2$, $x_0,y_0\ge r+1$ and the time component of $y-\vartheta x$ is $x_0+y_0-2c\ge2r+1$, so $K(x,y)\ge0$ and the sum is at most $\sum_{x,y}a_k^8g_0(a_kx)g_0(a_ky)K(x,y)=\omega_k(\vartheta(A_k(g_0))A_k(g_0))$, because $|h|\le g_0$. Taking the supremum over h gives $\mathscr{N}_k(\varepsilon)\le\omega_k(\vartheta(A_k(g_0))A_k(g_0))$. □

The condition of (ii) is a correlation inequality of Griffiths type for the connected two-point function of Φ; it is not proved

here for the Wilson measure.

Products of packets. For $n \ge 3$ the moment bound reduces to reflected norms of products. Consider

(RN) For every $m \ge 1$, every cube Q and every $\delta > 0$ there are $R_m(Q,\delta) < \infty$ and k_0 such that

$$Z_k^{2m}\omega_k\big(\vartheta(\bar P_k)P_k\big) \le R_m(Q,\delta)\prod_{j=1}^m \|f_j\|_\infty^2, \qquad P_k = \prod_{j=1}^m A_k(f_j),$$

for all $k \ge k_0$ and $(f_1,\dots,f_m) \in \mathscr{C}_m(Q,\delta)$ with supports in $\{x_0 \ge \delta\}$.

We write (RN_m) for the statement at a fixed m, and $(\mathrm{M}_{s,n})$ for (M_s) at a fixed n.

Proposition 259. *Assume the hypothesis of Theorem 64.*

(i) $(\mathrm{M}_{s,2m})$ *implies* (RN_m).

(ii) *Let $n \ge 2$. If (RN_i) holds for $1 \le i \le n-1$, then $(\mathrm{M}_{s,n})$ holds, with*

$$M_n(Q,\delta) = \mathsf{K}_{2nL/\delta}^n \max_{1\le i\le n-1}\big(R_i(Q^*,\delta_n)R_{n-i}(Q^*,\delta_n)\big)^{1/2}$$

for Q of side L, $Q^* = [-L-2, L+2]^4$ and $\delta_n = \delta/(16(n+1))$.

(iii) (M_s) *holds if and only if* (RN) *holds, and $(\mathrm{M}_{s,2})$ if and only if (RN_1). With $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$ as in Theorem 254, (RN_1) is equivalent to* (SD)*, and $(\mathrm{M}_{s,3})$ follows from* (SD) *and (RN_2).*

Proof. (i) Let $(f_j) \in \mathscr{C}_m(Q,\delta)$ be supported in $\{x_0 \ge \delta\}$ and $a_k \le \delta$. The functions $\bar f_j \circ \theta_{0,k}$ are supported in $\{x_0 \le a_k - \delta\}$, so the $2m$-tuple of these and the f_j lies in $\mathscr{C}_{2m}(Q',\delta)$ for a cube Q' containing Q and all $\theta_{0,k}(Q)$, and $\omega_k(\vartheta(\bar P_k)P_k) = \omega_k(\prod_j A_k(\bar f_j \circ \theta_{0,k})\prod_j A_k(f_j))$ by Lemma 247(i). $(\mathrm{M}_{s,2m})$ gives (RN_m) with $R_m(Q,\delta) = M_{2m}(Q',\delta)$.

(ii) Let $(f_j) \in \mathscr{C}_n(Q,\delta)$, $s = \delta/(2n)$, and decompose every f_j with the functions χ_κ^s of the proof of Theorem 248(vi); at most $\mathsf{K}_{L/s} = \mathsf{K}_{2nL/\delta}$ pieces of each are nonzero, each with norm at most $\|f_j\|_\infty$. Fix a nonzero term, with pieces in closed cubes $C_1,\dots,C_n$ of side s, points $p \in C_1$ and $q \in C_2$ in the supports and μ with $q_\mu - p_\mu \ge \delta$ (after exchanging f_1 and f_2). The projections of the C_j on the μ axis have total length $\delta/2$, so their complement in $[p_\mu, q_\mu]$ contains an interval of length at least $\delta/(2(n+1))$; let c be its midpoint. Every projection is at distance at least $\delta/(4(n+1)) = 4\delta_n$ from c; let I and J be the indices of the cubes below and above c, nonempty since $1 \in I$ and $2 \in J$. Translate all pieces by one vector of $a_k\mathbb{Z}^4$ so that c moves within $a_k/2$ of the plane $\{x_\mu = a_k c_\mu\}$ fixed by $\theta_{\mu,k}$, $|c_\mu| \le \frac12$, and so that the pieces lie in Q^*; the moment does not change. For $a_k \le 2\delta_n$ the pieces of I lie in $\{x_\mu \le a_k c_\mu - 3\delta_n\}$ and those of J in $\{x_\mu \ge a_k c_\mu + 3\delta_n\}$. With $F = \prod_{j\in I} A_k(\bar f_j' \circ \theta_{\mu,k})$ and $G = \prod_{j\in J} A_k(f_j')$, f_j' the translated pieces, Lemma 247(i) gives $\prod_{j\in I} A_k(f_j') = \vartheta_\mu(\bar F)$, and by the invariance under π_μ and the Cauchy–Schwarz inequality for the positive form of Lemma 247(iii),

$$\Big|\omega_k\Big(\prod_{j=1}^n A_k(f_j')\Big)\Big| = |\omega_k(\vartheta(\overline{\pi_\mu F})\pi_\mu G)| \le N_0(\pi_\mu F)N_0(\pi_\mu G).$$

The factors of $\pi_\mu F$ and $\pi_\mu G$ are supported in Q^* and in $\{x_0 \ge 3\delta_n - a_k\} \subset \{x_0 \ge \delta_n\}$, pairwise at distance at least δ, so $(\mathrm{RN}_{|I|})$ and $(\mathrm{RN}_{|J|})$ bound $Z_k^n N_0(\pi_\mu F)N_0(\pi_\mu G)$ by $(R_{|I|}R_{|J|})^{1/2}\prod_j \|f_j\|_\infty$ for large k. Summing over the at most $\mathsf{K}_{2nL/\delta}^n$ terms gives the claim.

(iii) The first two statements follow from (i) and (ii). For $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$, (RN_1) at $Q \supset B_{\varepsilon'}$ and $\delta = \varepsilon'$ gives $\mathscr{N}_k(\varepsilon') \le R_1\mathscr{N}_k(\varepsilon)$, which is (SD); conversely Lemma 253 bounds the left side of (RN_1) by $\mathsf{K}_L^2\|f\|_\infty^2$ times $\mathscr{N}_k(\delta)/\mathscr{N}_k(\varepsilon)$, which is bounded by (SD) for $\delta < \varepsilon$ and by Proposition 256(i) for $\delta \ge \varepsilon$. The last claim is (ii) for $n = 3$. □

The Cauchy–Schwarz inequality of reflection positivity bounds a moment by the reflected norms of the products of the two groups of packets on the two sides of a plane, and for $n \ge 3$ one group contains at least two packets; the reduction therefore stops at (RN_m) with $m \ge 2$, the reflected norms of products of separated packets. If the packets are jointly Gaussian, Wick's formula writes $\omega_k(\vartheta(\bar P_k)P_k)$ as a sum of $(2m-1)!!$ products of covariances of separated pairs, and (RN) follows from (RN_1) by (ii) with $n = 2$; for the Wilson measure (RN_m) with $m \ge 2$ is not proved.

Rotations at separated points. The hypercubic group fixes the lowest anisotropy that a separated two-point function can carry. Let $\mathscr{P}_l$ be the space of homogeneous polynomials of degree l on $\mathbb{R}^4$, $\mathscr{Y}_l \subset \mathscr{P}_l$ the harmonic ones, and $\mathscr{Y}_l^{B_4}$ the elements of $\mathscr{Y}_l$ invariant under B_4. Put

$$Y_4(z) = \sum_\mu z_\mu^4 - \tfrac12|z|^4,$$
$$Y_6(z) = \sum_\mu z_\mu^6 - \tfrac54|z|^2\sum_\mu z_\mu^4 + \tfrac{5}{16}|z|^6.$$

Lemma 260. *$\mathscr{Y}_l^{B_4} = 0$ for odd l, and $\dim\mathscr{Y}_{2j}^{B_4} = p(j) - p(j-1)$ for $j \ge 1$, where $p(j)$ is the number of partitions of j into at most four parts. In particular $\dim\mathscr{Y}_l^{B_4}$ is 0, 1, 1, 2 for $l = 2,4,6,8$, and $\mathscr{Y}_4^{B_4} = \mathbb{R}Y_4$, $\mathscr{Y}_6^{B_4} = \mathbb{R}Y_6$.*

Proof. A sign change $z_\mu \mapsto -z_\mu$ multiplies a monomial by -1 to the power of its degree in z_μ, so a polynomial invariant under the sign changes contains only monomials of even degree in every variable; it is a polynomial in $w_\mu = z_\mu^2$, and invariance under the permutations makes it symmetric in $w_0,\dots,w_3$. The monomial symmetric polynomials $m_\kappa(w)$, κ a partition of j into at most four parts, form a basis of the symmetric polynomials of degree j in four variables, so $\dim\mathscr{P}_{2j}^{B_4} = p(j)$ and $\mathscr{P}_l^{B_4} = 0$ for odd l. By [99, Ch. IV], $\mathscr{P}_l = \mathscr{Y}_l \oplus |z|^2\mathscr{P}_{l-2}$, and both summands are invariant under $\mathrm{O}(4)$, so the projections on them commute with B_4 and $\mathscr{P}_l^{B_4} = \mathscr{Y}_l^{B_4} \oplus |z|^2\mathscr{P}_{l-2}^{B_4}$; multiplication by $|z|^2$ is injective. This gives the dimension, and $p(0),\dots,p(4)$ are $1,1,2,3,5$. In $\mathbb{R}^4$, $\Delta|z|^{2i} = 2i(2i+2)|z|^{2i-2}$, $\Delta\sum_\mu z_\mu^{2i} = 2i(2i-1)\sum_\mu z_\mu^{2i-2}$ and $\Delta(|z|^2\sum_\mu z_\mu^4) = 24\sum_\mu z_\mu^4 + 12|z|^4$, so $\Delta Y_4 = 12|z|^2 - \frac12\cdot 24|z|^2 = 0$ and $\Delta Y_6 = 30\sum_\mu z_\mu^4 - \frac54(24\sum_\mu z_\mu^4 + 12|z|^4) + \frac{5}{16}\cdot 48|z|^4 = 0$. Both are invariant and nonzero, since $Y_4(e_0) = \frac12$ and $Y_6(e_0) = \frac{1}{16}$. □

Proposition 261. *Let S_2 be a bilinear form on the separated pairs, invariant under the translations and B_4, with $S_2(f,g) =$*

$\iint f(x)g(y)K(x-y)\,dx\,dy$ for separated pairs, where K is square integrable on every compact subset of $\mathbb{R}^4\setminus\{0\}$.

(i) *For $\varphi\in C_c(\mathbb{R}^4\setminus\{0\})$ and $g_\varepsilon=\varepsilon^{-4}\chi(\cdot/\varepsilon)$, $\chi\ge 0$ continuous with $\int\chi=1$ and support in $[-1,1]^4$,*

$$\int K\varphi\,dz=\lim_{\varepsilon\to 0}S_2(\varphi,g_\varepsilon),$$

so K is determined almost everywhere by S_2. For $R\in\mathrm{O}(4)$, $S_2(f\circ R^{-1},g\circ R^{-1})=S_2(f,g)$ for all separated pairs if and only if $K\circ R=K$ almost everywhere; in particular $K\circ g=K$ for $g\in B_4$.

(ii) *For almost every $\rho>0$, $K(\rho\,\cdot)\in L^2(S^3)$ and $K(\rho\omega)=\sum_{j\ge 0}K_{2j}(\rho,\omega)$ in $L^2(S^3)$, with $K_{2j}(\rho,\cdot)$ the restriction of an element of $\mathscr{Y}_{2j}^{B_4}$. Hence $K_2=0$, $K_0(\rho)$ is the mean of K on the sphere of radius ρ, and there are measurable k_4,k_6 with*

$$K_4(\rho,\omega)=k_4(\rho)Y_4(\omega),\qquad K_6(\rho,\omega)=k_6(\rho)Y_6(\omega).$$

(iii) *The following are equivalent:* (a) *S_2 is invariant under $\mathrm{O}(4)$ on the separated pairs;* (b) *it is invariant under one $R\in\mathrm{O}(4)$ of infinite order;* (c) *$K_{2j}(\rho,\cdot)=0$ for every $j\ge 2$ and almost every ρ;* (d) *$K(z)=K_0(|z|)$ for almost every z.*

Proof. (i) Let φ be supported in $\{|z|_\infty\ge 3\eta\}$ and $\varepsilon\le\eta$; the pair (φ,g_ε) is separated by at least 2η, and

$$S_2(\varphi,g_\varepsilon)=\int K(z)\varphi_\varepsilon(z)\,dz,\qquad \varphi_\varepsilon(z)=\int\varphi(z+y)g_\varepsilon(y)\,dy.$$

The functions φ_ε are supported in one compact subset of $\mathbb{R}^4\setminus\{0\}$ and converge uniformly to φ, and K is integrable there. For $R\in\mathrm{O}(4)$ the substitution $x=Ru$, $y=Rv$ gives $S_2(f\circ R^{-1},g\circ R^{-1})=\iint f(u)g(v)K(R(u-v))\,du\,dv$, so invariance holds if and only if the form with kernel $K\circ R-K$ vanishes on the separated pairs, and by the formula just proved this happens if and only if $K\circ R=K$ almost everywhere.

(ii) By Fubini's theorem in polar coordinates, $K(\rho\,\cdot)\in L^2(S^3)$ for almost every ρ, and for each of the finitely many $g\in B_4$, $K(\rho g\,\cdot)=K(\rho\,\cdot)$ for almost every ρ. The restrictions of the spaces $\mathscr{Y}_l$ to S^3 are orthogonal in $L^2(S^3)$ and span it [99, Ch. IV]; each is invariant under $\mathrm{O}(4)$, so the orthogonal projections on them commute with composition by elements of $\mathrm{O}(4)$. The component of degree l of $K(\rho\,\cdot)$ is therefore invariant under B_4, and Lemma 260 gives the claims.

(iii) Let G_K be the set of $R\in\mathrm{O}(4)$ with $K\circ R=K$ almost everywhere. On every closed annulus $A=\{r\le|z|\le r'\}$, $0<r<r'$, the map $R\mapsto K\circ R$ into $L^2(A)$ is continuous, as one sees by approximating K in $L^2(A)$ by continuous functions; so G_K is a closed subgroup of $\mathrm{O}(4)$, and it contains B_4 by (i). By the proof of Proposition 197(iv), which uses only these two properties, G_K is finite or equal to $\mathrm{O}(4)$, so (a) and (b) are equivalent by (i). If (a) holds, let $R_1,R_2,\dots$ be dense in $\mathrm{O}(4)$; for almost every ρ, $K(\rho R_i\,\cdot)=K(\rho\,\cdot)$ in $L^2(S^3)$ for every i, hence, by continuity of $R\mapsto u\circ R$ in $L^2(S^3)$, for every R, and each component $K_l(\rho,\cdot)$ is the restriction of an $\mathrm{O}(4)$-invariant element of $\mathscr{Y}_l$. Such an element is constant on spheres, hence equal to $c|z|^l$, and $\Delta|z|^l=l(l+2)|z|^{l-2}$ vanishes only for $l=0$; so (d) holds. (d) implies (c), (c) implies (d) because $K_2=0$, and (d) implies (a) by (i). □

So the part of a separated two-point kernel that is invariant under B_4 and not under $\mathrm{O}(4)$ has no components of degree below four, and its component of degree four has the fixed angular profile Y_4, with one radial coefficient k_4; by (i), $\int K(z)Y_4(z)\psi(|z|)\,dz=\lim_{\varepsilon\to 0}S_2(Y_4\psi(|\cdot|),g_\varepsilon)$ for $\psi\in C_c((0,\infty))$, a separated two-point function of packets. By (iii), $(\mathrm{W_s})$ for $n=2$ is the vanishing of all K_{2j} with $j\ge 2$. Reflection positivity and clustering do not remove the component of degree four, by Proposition 263(iv) below.

Two statements about the lattice moments

$$M_k(x)=\omega_k\Big(\prod_{j=1}^n(\Phi_{x_j}-c_k)\Big),\qquad x=(x_1,\dots,x_n)\in(\mathbb{Z}^4)^n,$$

give $(\mathrm{W_s})$. For $0<\delta<\Lambda$ let $\mathscr{W}_k(\delta,\Lambda)$ be the set of $x\in(\mathbb{Z}^4)^n$ with $\delta/a_k\le|x_i-x_j|\le\Lambda/a_k$ for $i\ne j$, $|\cdot|$ being the Euclidean norm. For $n\ge 2$:

(SB_n) for $0<\delta<\Lambda$,

$$b_n(\delta,\Lambda)=\limsup_k\sup_{x\in\mathscr{W}_k(\delta,\Lambda)}Z_k^n|M_k(x)|<\infty;$$

(RI_n) for $0<\delta<\Lambda$, $\limsup_k Z_k^n|M_k(x)-M_k(x')|=0$, the supremum over the pairs $x,x'\in\mathscr{W}_k(\delta,\Lambda)$ with $\big||x_i-x_j|-|x'_i-x'_j|\big|\le 4$ for all $i<j$.

(SB_n) is a pointwise bound of Symanzik type on the renormalised lattice moments at separated points. (RI_n) says that at separations of physical size, that is of order $1/a_k$ lattice units, the renormalised moments depend on the configuration only through its mutual Euclidean distances, up to an error that tends to zero, when these distances are changed by at most four lattice units. This is rotation invariance at large lattice distance, the form in which the restoration of rotation invariance predicted by Symanzik's improvement programme enters; neither statement is proved here for the Wilson measure.

Theorem 262. *Let the states ω_k be invariant under the translations of $\mathbb{Z}^4$, let Φ be as above and $Z_k>0$, and let $n\ge 2$.*

(i) (SB_n) *implies* $(\mathrm{M_s})$ *for this n.*

(ii) (SB_n) *and* (RI_n) *imply that for every $R\in\mathrm{O}(4)$ and every separated n-tuple*

$$\lim_k\Big[\omega_k\Big(\prod_j Z_kA_k(f_j\circ R^{-1})\Big)-\omega_k\Big(\prod_j Z_kA_k(f_j)\Big)\Big]=0.$$

(iii) *If* (SB_n) *and* (RI_n) *hold for every $n\ge 2$, then* $(\mathrm{M_s})$ *holds and* $(\mathrm{W_s})$ *holds for every R. Under the hypothesis of Theorem 64 every limit of Theorem 248 then has $G_S=\mathrm{O}(4)$, and with* $(\mathrm{U_s})$ *the conclusions of Corollary 249 hold.*

(iv) *Under* (SB_2) *and* (RI_2)*, every subsequence has a further subsequence along which $\lim_k Z_k^2\omega_k(A_k(f)A_k(g))=\iint f(x)g(y)K(x-y)\,dx\,dy$ for every separated pair, with $K(z)=K_0(|z|)$ and K_0 bounded on every compact subset of $(0,\infty)$.*

Proof. For $u\in\mathbb{R}^4$ let $\rho_k(u)$ be the point of $\mathbb{Z}^4$ nearest to u/a_k, ties broken by a fixed rule. Then $|u/a_k-\rho_k(u)|\le 1$, the half diagonal of the unit cube of $\mathbb{Z}^4$, and each set $\rho_k^{-1}(x)$ is, up to a

null set, the cube of side a_k centred at $a_k x$. Let $(f_j) \in \mathscr{C}_n(Q,\delta)$, Q of side L, so that all supports lie in a ball B of radius L, points of different supports are at Euclidean distance between δ and $2L$, and the same holds for the rotated functions $f_j \circ R^{-1}$ with the ball RB about the rotated centre. Let $a_k \le \min(\delta/4, L/2)$ and put $F(u) = \prod_j f_j(u_j)$, $\mathscr{B}_k(F) = \omega_k(\prod_j Z_k A_k(f_j))$ and

$$I_k(F) = \int F(u)\, Z_k^n M_k\big(\rho_k(u_1),\dots,\rho_k(u_n)\big)\, du.$$

(i) $\mathscr{B}_k(F) = Z_k^n a_k^{4n} \sum_x \prod_j f_j(a_k x_j) M_k(x)$, the nonzero terms have $x \in \mathscr{W}_k(\delta, 2L)$, and at most $(3L/a_k)^4$ sites x_j have $a_k x_j \in B$, because the cubes of side a_k about them are disjoint and lie in a ball of radius $L + a_k \le \frac{3}{2}L$ and volume at most $\frac{\pi^2}{2}(\frac{3}{2}L)^4 < (3L)^4$. So $|\mathscr{B}_k(F)| \le (b_n(\delta, 2L)+1)(3L)^{4n} \prod_j \|f_j\|_\infty$ for large k.

(ii) Since ρ_k is constant on the cubes,

$$I_k(F) - \mathscr{B}_k(F) = \sum_x Z_k^n M_k(x) \int_{\prod_j \rho_k^{-1}(x_j)} \big(F(u) - F(a_k x)\big) du.$$

A nonzero term has $|u_j - a_k x_j| \le a_k$ for some u_j in the support of f_j, hence $x \in \mathscr{W}_k(\delta/2, 3L)$, and the number of such x is at most $(3L/a_k)^{4n}$, the cubes about the x_j lying in a ball of radius $L + 2a_k \le 2L$ and volume $8\pi^2 L^4 < (3L)^4$. So $|I_k(F) - \mathscr{B}_k(F)| \le (b_n(\delta/2, 3L)+1)(3L)^{4n} \varpi_F(a_k)$ for large k, with $\varpi_F(s)$ the largest change of F when every u_j moves by at most s; it tends to 0, and the same holds for $F_R(u) = \prod_j f_j(R^{-1}u_j)$. By the substitution $u = Rv$,

$$I_k(F_R) - I_k(F) = \int F(v) Z_k^n \big[M_k(x'(v)) - M_k(x(v))\big] dv,$$

with $x_j(v) = \rho_k(v_j)$ and $x'_j(v) = \rho_k(Rv_j)$. For v in the support of F, $|x_i - x_j|$ and $|x'_i - x'_j|$ differ from $|v_i - v_j|/a_k = |Rv_i - Rv_j|/a_k$ by at most 2, hence from each other by at most 4, and both configurations lie in $\mathscr{W}_k(\delta/2, 3L)$. By (RI_n) the bracket is at most $\varepsilon_k \to 0$, and $|I_k(F_R) - I_k(F)| \le \varepsilon_k \prod_j \|f_j\|_{L^1}$.

(iii) The first claims are (i) and (ii) for every n, and the rest is Theorem 248(vii) and Corollary 249.

(iv) By translation invariance $M_k(x_1, x_2) = G_k(x_2 - x_1)$ with $G_k(y) = M_k(0, y)$; put $\kappa_k(z) = Z_k^2 G_k(\rho_k(z))$. For separated f, g as above, $y = \rho_k(v) - \rho_k(u)$ and $y' = \rho_k(v-u)$ have $|y|$ and $|y'|$ within 2 and 1 of $|v-u|/a_k$, so (RI_2) and the proof of (ii) give $Z_k^2 \omega_k(A_k(f)A_k(g)) - \iint f(u) g(v) \kappa_k(v-u)\, du\, dv \to 0$. For $r \le |z| \le r'$ and large k, $|\rho_k(z)| \in [r/(2a_k), 2r'/a_k]$, so $|\kappa_k| \le b_2(r/2, 2r') + 1$ there, and $|\kappa_k(Rz) - \kappa_k(z)| \to 0$ uniformly there for every $R \in \mathrm{O}(4)$ by (RI_2), since $|\rho_k(Rz)|$ and $|\rho_k(z)|$ differ by at most 2. A diagonal subsequence converges weakly-$*$ in L^∞ of every annulus to some K, and then $K \circ R = K$ almost everywhere for every R. For separated f, g, $\iint f(u) g(v) \kappa_k(v-u)\, du\, dv = \int \kappa_k(z) \Theta(z)\, dz$ with $\Theta(z) = \int f(u) g(u+z)\, du$ integrable and supported in an annulus, so the limit is $\iint f(u) g(v) K(v-u)\, du\, dv$, and $K(-z) = K(z)$. Proposition 261(iii) gives $K(z) = K_0(|z|)$ almost everywhere. □

Part (i) does not use the averaged condition, and parts (i), (ii), (iv) use of the states only their translation invariance. (SB_n) alone does not give $(\mathrm{W_s})$, and neither do reflection positivity, the gap in every coordinate direction and non-Gaussian moments, by the following family; it is not a lattice gauge theory, and, as in Remark 293, the proofs of this subsection use the states only through the properties in (i) below.

Proposition 263. *Let $m > 0$, $q_k = e^{-m a_k}$, $p \ge 1$, let $\phi^{(1)}, \dots, \phi^{(p)}$ be independent copies of the Gaussian field of Proposition 199(a), $\Phi_x = \prod_i \phi_x^{(i)}$, $A_k(f) = \sum_x a_k^4 f(a_k x) \Phi_x$ and $Z_k = 1$.*

(i) *The law of $(\phi^{(1)}, \dots, \phi^{(p)})$ is invariant under translations and B_4, reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links, and its transfer matrix in each coordinate direction has spectrum $\{q_k^N : N \ge 0\}$ with 1 a simple eigenvalue.*

(ii) *For all n and $f_j \in C_c(\mathbb{R}^4)$ the moments converge,*

$$\lim_k \omega_k\Big(\prod_j A_k(f_j)\Big) = \int \prod_j f_j(u_j) \Big(\sum_\pi \prod_{\{a,b\}\in\pi} e^{-m\|u_a - u_b\|_1}\Big)^p du,$$

the sum over the perfect matchings π of $\{1,\dots,n\}$; for f_j supported in a cube of side $L \ge a_k$ they are at most $(n!)^p (2L)^{4n} \prod_j \|f_j\|_\infty$. So (M), $(\mathrm{M_s})$, (SB_n) and $(\mathrm{U_s})$ hold.

(iii) *For $p \ge 2$ the limit is not Gaussian.*

(iv) *$G_S = B_4$, (RI_2) fails, and the kernel $K(z) = e^{-pm\|z\|_1}$ of S_2 has $k_4 \ne 0$ on a set of positive measure.*

Proof. (i) In each coordinate direction the field $(\phi^{(1)}, \dots, \phi^{(p)})$ is the stationary reversible Markov chain with transition operator $P^{\otimes p}$, P that of the proof of Proposition 199(a); the spectrum of $P^{\otimes p}$ consists of the products $q^{N_1 + \cdots + N_p}$, and 1 is simple. That proof uses only these properties.

(ii) By independence and Wick's formula, $\omega_k(\prod_j \Phi_{x_j}) = \big(\sum_\pi \prod_{\{a,b\}\in\pi} w_k(x_a - x_b)\big)^p$ with $w_k(z) = \prod_\mu q_k^{|z_\mu|} = e^{-m a_k \|z\|_1}$, so $c_k = 0$; this is at most the number of matchings to the power p when n is even and 0 otherwise, and the moments are Riemann sums of a bounded continuous function. At most $(L/a_k + 1)^4 \le (2L/a_k)^4$ sites x have $a_k x$ in a cube of side $L \ge a_k$, which gives the bound, hence (M), $(\mathrm{M_s})$ and (SB_n), and the convergence of the whole sequence gives $(\mathrm{U_s})$.

(iii) For $n = 4$ write $a = w_{12} w_{34}$, $b = w_{13} w_{24}$, $c = w_{14} w_{23}$ with $w_{ij} = e^{-m\|u_i - u_j\|_1}$. The Gaussian family with the same S_2 has the integrand $a^p + b^p + c^p$, and $(a+b+c)^p > a^p + b^p + c^p$ for $p \ge 2$ and $a, b, c > 0$; so S_4 differs from it for nonnegative nonzero f_j.

(iv) $B_4 \subset G_S \subset G_{S_2}$, and $G_{S_2} = B_4$ by the argument of Proposition 199(a) with pm in place of m. For $y = (2N, 0, 0, 0)$ and $y' = (N, N, N, N)$, $|y| = |y'| = 2N$ and $M_k(0, y) - M_k(0, y') = e^{-2pma_kN} - e^{-4pma_kN}$; with $N = \lceil \delta/a_k \rceil$ both lie in $\mathscr{W}_k(\delta, 3\delta)$ for large k, and the difference tends to $e^{-2pm\delta} - e^{-4pm\delta} > 0$, so (RI_2) fails. Finally $Y_4(z) = \frac{1}{2}\sum_\mu z_\mu^4 - \sum_{\mu<\nu} z_\mu^2 z_\nu^2$, and with $I_i = \int_{\mathbb{R}} |t|^i e^{-M|t|} dt = 2\, i!/M^{i+1}$, $M = pm$,

$$\int_{\mathbb{R}^4} K(z) Y_4(z)\, dz = 2 I_4 I_0^3 - 6 I_2^2 I_0^2 = \frac{768 - 384}{M^8} > 0.$$

In polar coordinates the left side is $\|Y_4\|^2_{L^2(S^3)}\int_0^\infty \rho^7 k_4(\rho)\,d\rho$, by the orthogonality of the components in Proposition 261(ii), so k_4 is not almost everywhere zero. □

For a product kernel $K(z)=\prod_\mu k(z_\mu)$ with $k\ge 0$ and moments $I_i=\int |t|^i k(t)\,dt$ the same computation gives $\int KY_4\,dz = 2I_0^2I_2^2(I_4I_0/I_2^2-3)$: the weighted coefficient $\int \rho^7 k_4\,d\rho$ has the sign of the excess kurtosis of k, which is 3 for $e^{-M|t|}$ and 0 for a Gaussian k. The family satisfies every hypothesis of Theorem 262 except (RI_2), so among the lattice statements used here it is (RI_n) that excludes it.

The scale. For a family $S=(S_n)$ on the separated tuples and $\lambda,c>0$ put

$$(D_{\lambda,c}S)_n(f_1,\dots,f_n)=c^nS_n\big(f_1(\lambda\,\cdot),\dots,f_n(\lambda\,\cdot)\big);$$

this is a family on the separated tuples, because the map taking f to $f(\lambda\,\cdot)$ takes $\mathscr{C}_n(Q,\delta)$ onto $\mathscr{C}_n(\lambda^{-1}Q,\delta/\lambda)$. Moreover $D_{\lambda,c}D_{\lambda',c'}=D_{\lambda\lambda',cc'}$. For a nontrivial limit S of Theorem 248 let $m(S)=\min(\sigma(H_0)\setminus\{0\})$, which exists and is at least m_* by Theorem 248(iv), (vi).

Lemma 264. *Let S be a limit of Theorem 248 and $\lambda,c>0$. For P at positive 0-times let $P^{\lambda,c}$ be obtained by replacing every $f_{\beta,j}$ by $f_{\beta,j}(\lambda\,\cdot)$ and every coefficient c_β by $c^{n_\beta}c_\beta$. Then $D_{\lambda,c}S(\theta_0\bar P\cdot P')=S(\theta_0\overline{P^{\lambda,c}}\cdot P'^{\lambda,c})$, and $\Psi'(P)\mapsto\Psi_0(P^{\lambda,c})$ extends to a unitary map V of the Hilbert space $\mathscr{H}'$ of $D_{\lambda,c}S$ onto $\mathscr{H}_0$ with $V\Omega'=\Omega_0$. With $H'=\lambda^{-1}V^{-1}H_0V$, $\Psi'(\tau_{te_0}P)=e^{-tH'}\Psi'(P)$ for $t\ge 0$, and $\sigma(H')=\lambda^{-1}\sigma(H_0)$; when $D_{\lambda,c}S$ is itself a limit of Theorem 248, its Hamiltonian in the time direction is H', and for nontrivial S, $m(D_{\lambda,c}S)=m(S)/\lambda$. If S is nontrivial and $D_{\lambda,c}S=S$, then $\lambda=c=1$.*

Proof. Since θ_0 is linear, $(f\circ\theta_0)(\lambda\,\cdot)=f(\lambda\,\cdot)\circ\theta_0$, which gives the identity; it shows that the form of $D_{\lambda,c}S$ is positive and that V is isometric, and V has dense range because $P=(P^{1/\lambda,1/c})^{\lambda,c}$. Further $(f(\cdot-te_0))(\lambda\,\cdot)=(f(\lambda\,\cdot))(\cdot-\frac{t}{\lambda}e_0)$, so $V\Psi'(\tau_{te_0}P)=\Psi_0(\tau_{(t/\lambda)e_0}P^{\lambda,c})=e^{-(t/\lambda)H_0}V\Psi'(P)$. A semigroup with this property is determined by its values on the dense set of the $\Psi'(P)$, which gives the claim on the Hamiltonian of a limit. If $D_{\lambda,c}S=S$, the two constructions coincide, $\sigma(H_0)=\lambda^{-1}\sigma(H_0)$, and the least point $m(S)$ of $\sigma(H_0)\setminus\{0\}$ satisfies $m(S)=m(S)/\lambda$; so $\lambda=1$, and $c^nS_n=S_n$ with $S_n\ne 0$ for some $n\ge 2$ by Theorem 248(vi) gives $c=1$. □

The normalisation of Proposition 250(ii) fixes the size of the field, not the scale: assigning the states to the spacings with a shift produces a dilated limit with the same normalisation.

Proposition 265. *Assume the hypothesis of Theorem 64, and suppose that the moments $\omega_k(\prod_j Z_kA_k(f_j))$ converge for every separated tuple to a nontrivial family S. Let $j\ge 1$, and let $\omega''_k=\omega_k$ for even k and $\omega''_k=\omega_{k+j}$ for odd k, with the packets A''_k formed from ω''_k at the spacing a_k.*

(i) *(ω''_k) satisfies the hypothesis of Theorem 64 with $2^j\ell$ in place of ℓ.*

(ii) *For every choice of $Z''_k>0$ for which* ($\mathrm{M_s}$) *holds along (ω''_k), the moments $\omega''_k(\prod_j Z''_kA''_k(f_j))$ either fail to converge for some separated tuple or converge to the trivial family.*

(iii) *If $C''_k(0)>0$ for large k and* ($\mathrm{M_s}$) *holds along (ω''_k) for $Z''_k=C''_k(\sigma_0)^{-1/2}$, then* ($\mathrm{U_s}$) *fails for it.*

Proof. (i) For odd k, ω_{k+j} satisfies $\mathrm{BP}(n_{k+j},\varpi)$ and $n_{k+j}a_k=2^jn_{k+j}a_{k+j}\le 2^j\ell$.

(ii) For odd k, $a_k=2^ja_{k+j}$ gives $A''_k(f)=2^{4j}A_{k+j}(f(2^j\cdot))$. Write $Z''_k=\zeta_kZ_k$ for even k and $Z''_k=\zeta_k2^{-4j}Z_{k+j}$ for odd k. The even moments are ζ_k^n times moments that converge to $S_n(f_1,\dots,f_n)$, and the odd ones ζ_k^n times moments that converge to $(D_{2^j,1}S)_n(f_1,\dots,f_n)$. Both families are nontrivial, so ($\mathrm{M_s}$) bounds ζ_k. Along a subsequence $\zeta_k\to\zeta_e$ on the even and $\zeta_k\to\zeta_o$ on the odd k, and the two limits are $D_{1,\zeta_e}S$ and $D_{2^j,\zeta_o}S$. If they coincide and $\zeta_e\zeta_o\ne 0$, then $D_{2^j,\zeta_o/\zeta_e}S=S$, contrary to Lemma 264; if exactly one of ζ_e,ζ_o vanishes, one limit is trivial and the other not. So a common limit has $\zeta_e=\zeta_o=0$ and is trivial.

(iii) By Proposition 250(ii) every limit is nontrivial, so by (ii) some separated moment does not converge, and by Corollary 249 ($\mathrm{U_s}$) fails. □

The scale is fixed by a mass. With f_0, σ_0, C_k as in Proposition 250 and γ_k the norm of $\mathbb{T}_k$ on $\Omega_k^\perp$, consider for $m_0>0$:

(MC_{m_0}) $\liminf_k a_k^{-1}\log(1/\gamma_k)\ge m_0$, and there are $c_0>0$ and k_1 with $C_k(t)\ge c_0e^{-m_0t}C_k(\sigma_0)$ for $k\ge k_1$ and $t\in a_k\mathbb{Z}$, $t\ge\sigma_0$;

and the statement that the limits differ at most by the scale and the size of the field:

($\mathrm{O_s}$) for any two limits S,S' of Theorem 248 along subsequences there are $\lambda,c>0$ with $S'=D_{\lambda,c}S$.

Theorem 266. *Assume the hypothesis of Theorem 64, $C_k(0)>0$ for large k, $Z_k=C_k(\sigma_0)^{-1/2}$ and* ($\mathrm{M_s}$).

(i) *Every limit S of Theorem 248 satisfies*

$$\langle\Psi_0(f_0),e^{-\sigma_0H_0}\Psi_0(f_0)\rangle=1,$$

and under (MC_{m_0}) *it has $m(S)=m_0$.*

(ii) *Under* (MC_{m_0}) *and* ($\mathrm{O_s}$), ($\mathrm{U_s}$) *holds and the moments converge for every separated tuple.*

(iii) *For the Gaussian field of Proposition 199*(a) *at mass m,* (MC_m) *holds; for the same field at two masses $m\ne m'$, interleaved,* ($\mathrm{O_s}$) *holds and* (MC_{m_0}) *fails for every m_0.*

Proof. (i) The first claim and nontriviality are Proposition 250(ii). The proof of Theorem 248(iv) with γ_k^{1/a_k} in place of e^{-m_*t} gives $\langle v,e^{-tH_0}v\rangle\le\limsup_k\gamma_k^{1/a_k}\|v\|^2\le e^{-m_0t}\|v\|^2$ on a dense set of $v\in\Omega_0^\perp$, so $m(S)\ge m_0$. By the approximation of $\theta_{0,k}$ by θ_0, as in the proof of Proposition 250(ii), $\langle\Psi_0(f_0),e^{-tH_0}\Psi_0(f_0)\rangle=\lim_kZ_k^2C_k(t)\ge c_0e^{-m_0t}$ for dyadic $t\ge\sigma_0$. Since $\Psi_0(f_0)\perp\Omega_0$, its spectral measure for H_0 is carried by $[m(S),\infty)$, and the left side is at most $e^{-m(S)t}\|\Psi_0(f_0)\|^2$; letting $t\to\infty$ gives $m(S)\le m_0$.

(ii) Let S,S' be two limits and $S'=D_{\lambda,c}S$. By (i) and Lemma 264, $m_0=m(S')=m(S)/\lambda=m_0/\lambda$, so $\lambda=1$; then $S'_2=c^2S_2$, and the normalisation in (i) gives $c=1$. Convergence follows as in the proof of Corollary 249.

(iii) The transfer matrix has $\gamma_k = q_k = e^{-ma_k}$. The packet $\vartheta(\overline{A_k(f_0)})$ involves sites at negative times and $\tau_{t/a_k}A_k(f_0)$ sites at positive times, so the time factor of the covariance is $q_k^{y_0-x_0}$ and $C_k(t) = e^{-mt}C_k(0)$; this is (MC_m). At two masses the limits are the two Gaussian families with kernels $\kappa e^{-m\|z\|_1}$ and $\kappa' e^{-m'\|z\|_1}$, and $D_{\lambda,c}$ with $\lambda = m/m'$ and $c^2 = \kappa'\lambda^8/\kappa$ maps the first to the second, Gaussian families being determined by S_2. The first part of (MC_{m_0}) needs $m_0 \le \min(m,m')$, and the second, since $C_k(t)/C_k(\sigma_0) = e^{-m_k(t-\sigma_0)}$ with $m_k \in \{m,m'\}$, needs $m_0 \ge \max(m,m')$. □

($\mathrm{O_s}$) is the form, for the separated moments, of the statement that the continuum theory has no free dimensionless parameter; it is not proved here, nor is (MC_{m_0}) for the Wilson measure along any tuning. The second part of (MC_{m_0}) asks that the reflected correlation of f_0 decay no faster than at the rate m_0, which holds when f_0 couples to the lowest state; together with the first part it fixes the mass of every limit. A condition fixing the ratio $C_k(\sigma_0)/C_k(0)$, the analogue of (RC), fixes the scale within a set of limits related by dilations only where the corresponding ratio of the limits is injective along that set, which is not proved.

The pointwise bound on an axis. Reflection positivity in the coordinate directions bounds the separated two-point moments at every point by their values on a coordinate axis, where they form a completely monotone function of the distance. For $n = 2$ write $g_k(s) = M_k(0, se_0)$ for $s \in \mathbb{Z}$, and $\lfloor t \rceil = \lfloor t + \frac{1}{2} \rfloor$. Let $h \in \{1,-1\}$ be such that $\vartheta x = (h - x_0, \mathbf{x})$ for $x = (x_0, \mathbf{x}) \in \mathbb{Z}^4$; the plane fixed by ϑ is $\{x_0 = h/2\}$.

Proposition 267. *Assume the hypothesis of Theorem 64, and let γ_k be the norm of $\mathbb{T}_k$ on $\Omega_k^\perp$.*

(i) *For integers $s \ge 2r+3$ and $j \ge 0$, $0 \le g_k(s+j) \le \gamma_k^j g_k(s)$, and for $s \ge 4r+6$*

$$0 \le g_k(s) - g_k(s+j) \le \frac{4j}{s}\, g_k\big(\lceil s/2 \rceil\big).$$

(ii) *For $y \in \mathbb{Z}^4$ with $|y|_\infty \ge 2r+4$, $|M_k(0,y)| \le g_k(|y|_\infty - 1)$.*

(iii) (SB_2) *holds if and only if* $\limsup_k Z_k^2 g_k(\lceil \delta/a_k \rceil) < \infty$ *for every $\delta > 0$. In that case, for every $\delta > 0$ and with $b(\delta) = \limsup_k Z_k^2 g_k(\lceil \delta/(4a_k) \rceil)$,*

$$Z_k^2 |M_k(0,y)| \le \big(b(\delta)+1\big) e^{-m_*(a_k|y|_\infty - \delta/2)}$$

for all large k and all $y \in \mathbb{Z}^4$ with $|y| \ge \delta/a_k$; in particular $b_2(\delta,\Lambda) \le b(\delta)+1$ for every $\Lambda > \delta$.

(iv) *Under* (SB_2), (RI_2) *holds if and only if for $0 < \delta < \Lambda$*

$$\limsup_k \sup_y Z_k^2 \big| M_k(0,y) - g_k(\lfloor |y| \rceil) \big| = 0,$$

the supremum over the $y \in \mathbb{Z}^4$ with $\delta/a_k \le |y| \le \Lambda/a_k$.

(v) *Under* (SB_2), *every limit of Theorem 248 has $S_2(f,g) = \iint f(x)g(y)K(x-y)\,dx\,dy$ for the separated pairs, with K measurable on $\mathbb{R}^4 \setminus \{0\}$, $K(-z) = K(z)$, $K \circ g = K$ almost everywhere for $g \in B_4$, and $|K(z)| \le (b(\delta)+1)e^{-m_*(|z|_\infty - \delta/2)}$ for almost every z with $|z| \ge 2\delta$, for every $\delta > 0$.*

Proof. For $j \ge r+1$ let $F_j = \Phi_{je_0} - c_k$, a real function of the links with both endpoints in $\{x_0 \ge 1\}$. Since $\vartheta\Phi_x = \Phi_{\vartheta x}$, $\vartheta F_j = \Phi_{(h-j)e_0} - c_k$, and by translation invariance

$$\omega_k\big(\vartheta(F_j)\tau_t F_{j'}\big) = M_k\big((h-j)e_0, (j'+t)e_0\big) = g_k(j+j'+t-h)$$

for integers $t \ge 0$. In particular $N_0(F_j)^2 = g_k(2j-h)$.

(i) By the proof of Lemma 247(iii), with $s_0 = 2r+2-h$ and $\Psi = \Psi_{F_{r+1}}$, $g_k(s_0+t) = \langle \Psi, \mathbb{T}_k^t \Psi \rangle$ for integers $t \ge 0$, and $\langle \Omega_k, \Psi \rangle = \omega_k(F_{r+1}) = 0$. Since $\mathbb{T}_k$ is self-adjoint, fixes Ω_k and has spectrum on $\Omega_k^\perp$ in $[0,\gamma_k]$, the spectral measure ν of $\mathbb{T}_k$ at Ψ is a positive measure on $[0,\gamma_k]$, and $g_k(s_0+t) = \int \lambda^t\, d\nu(\lambda)$, $0^0 = 1$. As $s_0 \le 2r+3$, this gives $g_k(s) \ge 0$ and $g_k(s+j) \le \gamma_k^j g_k(s)$ for $s \ge 2r+3$. Let $s \ge 4r+6$ and $u = s - s_0$. From $1-\lambda^j \le j(1-\lambda)$ and $\max_{0\le x\le 1} x^p(1-x) = p^p(p+1)^{-p-1} \le (p+1)^{-1}$, applied with $p = \lfloor u/2 \rfloor$ to $\lambda^u(1-\lambda) = \lambda^{\lceil u/2 \rceil}\lambda^p(1-\lambda)$,

$$\begin{aligned} g_k(s) - g_k(s+j) &= \int \lambda^u(1-\lambda^j)\,d\nu \le \frac{j}{p+1}\int \lambda^{\lceil u/2 \rceil}\,d\nu \\ &= \frac{j\, g_k(s_0 + \lceil u/2 \rceil)}{\lfloor u/2 \rfloor + 1}. \end{aligned}$$

Here $s_0 + \lceil u/2 \rceil = \lceil (s+s_0)/2 \rceil \ge \lceil s/2 \rceil \ge 2r+3$, so the numerator is at most $j\, g_k(\lceil s/2 \rceil)$, and $\lfloor u/2 \rfloor + 1 \ge (s - s_0 + 1)/2 \ge (s - 2r - 2)/2 \ge s/4$.

(ii) Let $s = |y|_\infty$. Choose $g \in B_4$, a permutation of the axes followed by a sign change, with $(gy)_0 = s$; then $M_k(0,y) = M_k(0,gy)$, since ω_k is invariant under B_4 and $g\Phi_x = \Phi_{gx}$. Write $gy = se_0 + v$ with $v_0 = 0$, and let $j = \lfloor (s+h)/2 \rfloor$ and $j' = s + h - j$, both at least $\lfloor (s-1)/2 \rfloor \ge r+1$. By translation invariance

$$M_k(0,gy) = M_k\big((h-j)e_0, j'e_0 + v\big) = \omega_k\big(\vartheta(F_j)\tau_v F_{j'}\big),$$

and $\tau_v F_{j'}$ is again a function of the links in $\{x_0 \ge 1\}$. The Cauchy–Schwarz inequality for the positive form of Lemma 247(iii) and $N_0(\tau_v F_{j'}) = N_0(F_{j'})$ give $|M_k(0,y)| \le (g_k(2j-h)g_k(2j'-h))^{1/2}$. If $s+h$ is even, $2j-h = 2j'-h = s$; otherwise the two arguments are $s-1$ and $s+1$. By (i) both products are at most $g_k(s-1)^2$.

(iii) Let $a_k \le \delta$. The pair $(0, \lceil \delta/a_k \rceil e_0)$ lies in $\mathscr{W}_k(\delta, 2\delta)$, and ($\mathrm{SB}_2$) gives

$$\limsup_k Z_k^2 g_k(\lceil \delta/a_k \rceil) \le b_2(\delta, 2\delta).$$

Conversely let $\delta > 0$, $s_1 = \lceil \delta/(4a_k) \rceil$ and $|y| \ge \delta/a_k$. Then $|y|_\infty \ge |y|/2 \ge \delta/(2a_k)$, which for large k is at least $2r+4$ and at least $s_1 + 1$. By (ii) and (i), and $\gamma_k \le e^{-m_* a_k}$,

$$|M_k(0,y)| \le g_k(|y|_\infty - 1) \le e^{-m_* a_k(|y|_\infty - 1 - s_1)} g_k(s_1),$$

and $a_k(1+s_1) \le 2a_k + \delta/4 \le \delta/2$ for $a_k \le \delta/8$. Since $Z_k^2 g_k(s_1) \le b(\delta)+1$ for large k, this is the bound, and $a_k|y|_\infty \ge \delta/2$ gives the last claim.

(iv) Suppose (RI_2), and let $\delta/a_k \le |y| \le \Lambda/a_k$. The point $y' = \lfloor |y| \rceil e_0$ has $||y| - |y'|| \le \frac{1}{2}$, both pairs $(0,y)$, $(0,y')$ lie in $\mathscr{W}_k(\delta/2, 2\Lambda)$ for large k, and $M_k(0,y') = g_k(\lfloor |y| \rceil)$; so ($\mathrm{RI}_2$)

with $(\delta/2, 2\Lambda)$ gives the condition. Conversely, let $x, x' \in \mathscr{W}_k(\delta,\Lambda)$ with $||x_1-x_2|-|x_1'-x_2'|| \le 4$, $y = x_2 - x_1$, $y' = x_2' - x_1'$, $s = \lfloor |y| \rceil$ and $s' = \lfloor |y'| \rceil$. By translation invariance $M_k(x) = M_k(0,y)$ and $M_k(x') = M_k(0,y')$, and

$$\begin{aligned} Z_k^2|M_k(x)-M_k(x')| &\le Z_k^2|M_k(0,y)-g_k(s)| \\ &\quad + Z_k^2|g_k(s)-g_k(s')| \\ &\quad + Z_k^2|M_k(0,y')-g_k(s')|. \end{aligned}$$

The first and last terms tend to 0 uniformly by the condition. Since $|s-s'| \le 5$ and $\min(s,s') \ge \delta/(2a_k)$ for large k, (i) bounds the middle term by $40 a_k \delta^{-1} Z_k^2 g_k(\lceil \delta/(4a_k) \rceil)$, which tends to 0 by (iii).

(v) Let S be a limit along a subsequence, f, g separated, supported in a cube Q of side L at ℓ^∞ distance $\eta > 0$, and $\Theta(z) = \int f(u) g(u+z)\, du$, continuous and supported in $\{|z|_\infty \ge \eta\} \cap [-L,L]^4$. With ρ_k as in the proof of Theorem 262 put $K_k(z) = Z_k^2 M_k(0, \rho_k(z))$. By translation invariance,

$$\begin{aligned} Z_k^2 \omega_k\big(A_k(f)A_k(g)\big) - \int K_k \Theta\, dz = \sum_w Z_k^2 M_k(0,w) \\ \times \Big(a_k^8 \sum_x f(a_k x) g(a_k x + a_k w) - \int_{\rho_k^{-1}(w)} \Theta\, dz \Big). \end{aligned}$$

The bracket vanishes unless $|w|_\infty \ge \eta/a_k - 2$ and $|w|_\infty \le L/a_k + 2$, and it is at most $a_k^4 \varepsilon_k$, where $\varepsilon_k \to 0$ by the uniform continuity of f, g and Θ, the inner sum being a Riemann sum for $\Theta(a_k w)$ with error uniform in w. At most $(2L/a_k + 5)^4$ vectors w contribute, and by (iii) with $\delta = \eta/2$ each has $Z_k^2|M_k(0,w)| \le b(\eta/2) + 1$ for large k. So the difference tends to 0. By (iii), K_k is bounded on every set $\{|z| \ge \delta\}$, $\delta > 0$, for large k, and a further subsequence converges weakly-$*$ in $L^\infty(\{|z| \ge \delta\})$ for every $\delta = 1/i$ to some K. Since Θ is integrable and supported in $\{|z| \ge \eta\}$, $\int K_k \Theta\, dz \to \int K \Theta\, dz$, which is the representation. For $|z| \ge 2\delta$ and large k, $|\rho_k(z)| \ge |z|/a_k - 1 \ge \delta/a_k$ and $a_k|\rho_k(z)|_\infty \ge |z|_\infty - a_k$, so (iii) gives $|K_k(z)| \le (b(\delta)+1) e^{-m_*(|z|_\infty - a_k - \delta/2)}$, a bound that passes to the weak-$*$ limit. Finally $M_k(0,-w) = M_k(0,w)$ and $M_k(0,gw) = M_k(0,w)$ for $g \in B_4$, while $\rho_k(-z) = -\rho_k(z)$ and $\rho_k(gz) = g\rho_k(z)$ off a null set; so K_k and K are even and B_4-invariant almost everywhere. □

So (SB_2) is a bound on one function of one variable, the renormalised two-point moment on a coordinate axis, and (RI_2) asks that the renormalised moment at a point be close to its value on the axis at the same Euclidean length. Under (SB_2) the kernel of (v) is square integrable on every compact subset of $\mathbb{R}^4 \setminus \{0\}$, so Proposition 261 applies to it, and (W_s) for $n = 2$ is the radial symmetry of K. Neither part follows from the other properties. For (RI_2) this is Proposition 263, whose fields satisfy (SB_n) for every n. For (SB_2) it is part (b) of the following proposition, whose fields satisfy the separated moment bound with constants independent of the separation, the mass condition and (U_s), and concentrate the two-point function on the coordinate axes, where the bound of (M_s) on averages over regions of physical size does not control the pointwise values.

Proposition 268. *Let $m > 0$ and $q_k = e^{-m a_k}$.*

(a) *Let J be a finite set, $q_{j,\mu} \in [0,1)$ for $j \in J$ and $\mu = 0,\dots,3$, and let $\psi = (\psi_j(x))_{j\in J, x\in\mathbb{Z}^4}$ be the centred Gaussian field with $\mathbb{E}\psi_j(x)\psi_{j'}(y) = \delta_{jj'} \prod_\mu q_{j,\mu}^{|x_\mu - y_\mu|}$, $0^0 = 1$. Its law is invariant under translations and reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links, and in the direction μ its transfer matrix $\mathbb{T}$ has $0 \le \mathbb{T} \le 1$, $\mathbb{T}\Omega = \Omega$ and norm at most $\max_j q_{j,\mu}$ on $\Omega^\perp$.*

(b) *Let $J = \{0,1,2,3\}$, $q_{j,\mu} = q_k$ for $j = \mu$ and 0 otherwise, $\Phi_x = \sum_j \psi_j(x)$, $A_k(f) = \sum_x a_k^4 f(a_k x)\Phi_x$, $f_0 \ge 0$ not zero, C_k formed from f_0, and $Z_k = C_k(\sigma_0)^{-1/2}$. The law is invariant under B_4, acting by $\psi_j(x) \mapsto \psi_{\bar g j}(gx)$ with $\bar g$ the permutation of the axes underlying g, so that $g\Phi_x = \Phi_{gx}$. Moreover $C_k(t) = e^{-mt} C_k(0)$ for $t \in a_k\mathbb{Z}_{\ge 0}$ and $a_k^{-3} C_k(0) \to \mathscr{J} = \int_{\mathbb{R}^3} (\int_{\mathbb{R}} f_0(s,\mathbf{x}) e^{-ms} ds)^2 d\mathbf{x} > 0$. There is $B < \infty$ such that for large k, every n and all $f_j \in C_c(\mathbb{R}^4)$ supported in a cube of side $L \ge a_k$,*

$$\Big| \omega_k\Big(\prod_{j=1}^n Z_k A_k(f_j) \Big) \Big| \le (n-1)!! \left(B(2L)^4 \right)^{n/2} \prod_{j=1}^n \|f_j\|_\infty$$

for even n, the moments vanishing for odd n; so (M_s) holds with constants independent of δ, and the growth condition (G_s) posed below holds with $p = 0$. The moments converge along the whole sequence to the Gaussian family with

$$S_2(f,g) = \frac{e^{m\sigma_0}}{\mathscr{J}} \sum_{\mu=0}^3 \int_{\mathbb{R}^4} \int_{\mathbb{R}} f(x) g(x + u e_\mu) e^{-m|u|}\, du\, dx,$$

so (U_s) holds, and (MC_m) holds. Further (SB_2) and (RI_2) fail, $G_S = B_4$, and there is no locally integrable K on $\mathbb{R}^4 \setminus \{0\}$ with $S_2(f,g) = \iint f(x) g(y) K(x-y)\, dx\, dy$ for all separated pairs.

(c) *Let $m' > m$, $q_k' = e^{-m' a_k}$, and let ω_k be the law of the field $\Phi = \psi_1$ of* (a) *with $J = \{1\}$, $q_{1,\mu} = q_k$, for even k, and of $\Phi = \psi_1 + \psi_2$ with $J = \{1,2\}$, $q_{1,\mu} = q_k$, $q_{2,\mu} = q_k'$, for odd k; let $f_0 \ge 0$ be not zero and $Z_k = C_k(\sigma_0)^{-1/2}$. The laws are invariant under translations and B_4, reflection positive as in* (a)*, with transfer matrices of norm at most q_k on $\Omega_k^\perp$. (M_s) and (MC_m) hold along the whole sequence. The moments converge along the even and along the odd k to Gaussian families S and S' with continuous kernels $\kappa e^{-m\|z\|_1}$ and $\kappa'(e^{-m\|z\|_1} + e^{-m'\|z\|_1})$, $\kappa, \kappa' > 0$, and $S' \ne D_{\lambda,c} S$ for all $\lambda, c > 0$. So (O_s) and (U_s) fail.*

Proof. (a) Fix $\mu = 0$ as time; the other directions are the same. The field at time 0 has a Gaussian law γ with covariance $C = \delta_{jj'} \prod_{i\ge 1} q_{j,i}^{|x_i - y_i|}$, positive definite as a product of the one-dimensional covariances $q^{|n|}$, $0 \le q < 1$, whose Fourier transforms $(1-q^2)/(1 - 2q\cos p + q^2)$ are positive. Let $(Qh)_j(\mathbf{x}) = q_{j,0} h_j(\mathbf{x})$ on the finitely supported h; Q commutes with C, so the stationary chain $\psi_{t+1} = Q\psi_t + (1-Q^2)^{1/2}\xi_{t+1}$, with ξ_t independent of law γ, has covariance $Q^{|t-s|}C$, which is that of the field. As in the proof of Proposition 199(a), its transition operator P satisfies $P\mathscr{E}_h = \mathscr{E}_{Qh}$, so on the N-th Wiener chaos, the closure of the parts of the $\mathscr{E}_h$ homogeneous of degree N in h, it acts as the N-th tensor power of Q on the symmetric tensors over the space of the h with the inner product

$\langle h, Ch'\rangle$. Q is self-adjoint there with $0 \le Q \le q_{\max} = \max_j q_{j,0}$, so $0 \le P \le 1$, P fixes the constants and has norm at most $q_{\max}^N \le q_{\max}$ on the chaos of order $N \ge 1$. The rest of that proof gives reflection positivity for both kinds of planes and identifies the transfer matrix with P.

(b) Here $\mathbb{E}\Phi_x\Phi_y = w_k(y-x)$ with $w_k(z) = q_k^{|z_\mu|}$ if $z \ne 0$ is a multiple of e_μ, $w_k(0) = 4$, and $w_k(z) = 0$ otherwise. Since $(gz)_{\bar g j} = \pm z_j$, the covariance of the transformed field is that of ψ, which gives the B_4 invariance; $\omega_k(\Phi_0) = 0$, so $c_k = 0$. For $f_0(a_k x) f_0(a_k y) \ne 0$ the time coordinate of $y + (t/a_k)e_0 - \vartheta x$ is $x_0 + y_0 - h + t/a_k \ge 1$, so only the term $\mu = 0$ contributes, with $\mathbf{x} = \mathbf{y}$, and

$$C_k(t) = e^{-mt} e^{mha_k} a_k^3 \cdot a_k^3 \sum_{\mathbf{x}} \Big(a_k \sum_{x_0} f_0(a_k x) e^{-m a_k x_0} \Big)^2 .$$

The last factor is a Riemann sum converging to $\mathscr{J}$, which is positive because $f_0 \ge 0$ is not zero. For the pairs,

$$\begin{aligned} &\omega_k\big(A_k(f)A_k(g)\big) \\ &\quad = a_k^3 \sum_\mu a_k^5 \sum_{x \in \mathbb{Z}^4, s \in \mathbb{Z}} f(a_k x) g(a_k x + a_k s e_\mu) e^{-m a_k |s|} . \end{aligned}$$

The inner sums are Riemann sums converging to $\int\int f(x) g(x + u e_\mu) e^{-m|u|} du\, dx$, and since at most $(L + a_k)^4/a_k^4 \le (2L)^4/a_k^4$ sites have $a_k x$ in the cube and $a_k \sum_s e^{-m a_k |s|} = a_k(1+q_k)/(1-q_k) \le 4/m$ for $m a_k \le 1$, they are at most $(4/m)(2L)^4 \|f\|_\infty \|g\|_\infty$. With $Z_k^2 a_k^3 = e^{m\sigma_0} a_k^3 / C_k(0) \to e^{m\sigma_0}/\mathscr{J}$ this gives the bound for $n = 2$ with $B = 32 e^{m\sigma_0}/(m\mathscr{J})$ and the limit of S_2, and Wick's formula, a sum over the $(n-1)!!$ pairings, gives the bound for all n and the convergence of all moments. Since $(n-1)!! \le (n!)^{1/2}$ and $(B(2L)^4)^{n/2} \le (4B^{1/2})^n (1+L)^{2n}$, the bound has the form of ($\mathrm{G_s}$) with $\beta = \frac12$, $q = 2$ and $p = 0$. By (a) the transfer matrix has norm at most q_k on $\Omega_k^\perp$, so $\liminf_k a_k^{-1} \log(1/\gamma_k) \ge m$, and $C_k(t) = e^{-m(t-\sigma_0)} C_k(\sigma_0)$; this is ($\mathrm{MC}_m$) with $c_0 = e^{m\sigma_0}$.

For $s \ge 1$, $g_k(s) = w_k(s e_0) = q_k^s$, so $Z_k^2 g_k(\lceil \delta/a_k \rceil) \ge (e^{m\sigma_0}/(2\mathscr{J})) a_k^{-3} e^{-m(\delta + a_k)} \to \infty$; as the pair $(0, \lceil \delta/a_k \rceil e_0)$ lies in $\mathscr{W}_k(\delta, 2\delta)$ for large k, (SB_2) fails. With $N = \lceil \delta/a_k \rceil + 1$, the points $y = N e_0$ and $y' = (N-1)e_0 + e_1$ have $N - 1 \le |y'| \le |y| = N$, both pairs $(0,y)$, $(0,y')$ lie in $\mathscr{W}_k(\delta, 2\delta)$ for large k, $M_k(0,y') = w_k(y') = 0$ and $Z_k^2 M_k(0,y) = Z_k^2 q_k^N \to \infty$; so ($\mathrm{RI}_2$) fails.

Let $T(\varphi) = \frac{e^{m\sigma_0}}{\mathscr{J}} \sum_\mu \int \varphi(u e_\mu) e^{-m|u|} du$ for $\varphi \in C_c(\mathbb{R}^4 \setminus \{0\})$; then $S_2(f,g) = T(\Theta)$ with $\Theta(z) = \int f(u) g(u+z)\, du$. If S_2 had a locally integrable kernel K, the argument of Proposition 261(i) would give $T(\varphi) = \int K \varphi\, dz$ for all φ, T being even. Take φ_ε with values in $[0,1]$, equal to 1 on the segment $\{u e_0 : 1 \le u \le 2\}$ and supported in its ε-neighbourhood N_ε; then $T(\varphi_\varepsilon) \ge \frac{e^{m\sigma_0}}{\mathscr{J}} \int_1^2 e^{-mu} du > 0$, while $\int |K| \varphi_\varepsilon \le \int_{N_\varepsilon} |K| \to 0$ as $\varepsilon \to 0$, since N_ε decreases to a null set. Let $R \in G_S$ and suppose $R^{-1} e_0$ is not one of the $\pm e_\nu$; then it has positive distance 3ε from the union A of the coordinate axes. Take $f, g \ge 0$ supported in the balls of radius ε about 0 and $R^{-1}e_0$, with $f(0) > 0$ and $g(R^{-1}e_0) > 0$. Then Θ is supported in the ball of radius 2ε about $R^{-1}e_0$, which misses A, so $S_2(f,g) = 0$, while $S_2(f \circ R^{-1}, g \circ R^{-1}) = T(\Theta \circ R^{-1}) \ge \frac{e^{m\sigma_0}}{\mathscr{J}} \int \Theta(R^{-1} u e_0) e^{-m|u|} du > 0$, because $\Theta(R^{-1}e_0) > 0$. So $R^{-1}e_0 \in \{\pm e_\nu\}$, and applying this to $\pi_\mu R \in G_S$ gives $R^{-1} e_\mu \in \{\pm e_\nu\}$ for every μ, that is, $R \in B_4$.

(c) The symmetries, positivity and the norm of the transfer matrices follow from (a), the covariances $\prod_\mu q^{|z_\mu|}$ being invariant under B_4. Let $C_k^{(i)}$ be formed with the covariance $\prod_\mu q_i^{|z_\mu|}$, $q_1 = q_k$, $q_2 = q_k'$. As in (b), the time factor of the covariance between the two packets of $C_k^{(i)}(t)$ is q_i^{t/a_k} times that at $t = 0$, so $C_k^{(1)}(t) = e^{-mt} C_k^{(1)}(0)$, $C_k^{(2)}(t) = e^{-m't} C_k^{(2)}(0)$, and $C_k^{(i)}(0)$ converges to $\mathscr{J}_1 = \iint f_0(x) f_0(y) e^{-m\|\theta_0 x - y\|_1} dx\, dy > 0$ and $\mathscr{J}_2$, defined with m' in place of m, by Riemann sums. For even k, $C_k = C_k^{(1)}$; for odd k, $C_k = C_k^{(1)} + C_k^{(2)}$, and for $t \ge \sigma_0$

$$\begin{aligned} \frac{C_k(t)}{C_k(\sigma_0)} &\ge \frac{e^{-mt} C_k^{(1)}(0)}{e^{-m\sigma_0}(C_k^{(1)}(0) + C_k^{(2)}(0))} \\ &\ge \frac{\mathscr{J}_1}{2(\mathscr{J}_1 + \mathscr{J}_2)} e^{-m(t-\sigma_0)} \end{aligned}$$

for large k. With the norm bound q_k this is (MC_m) with $c_0 = e^{m\sigma_0} \mathscr{J}_1/(2(\mathscr{J}_1 + \mathscr{J}_2))$. As in the proof of Proposition 199(a), the moments converge along the even and along the odd k to the Gaussian families with the stated kernels, $\kappa = e^{m\sigma_0}/\mathscr{J}_1$, $\kappa' = (e^{-m\sigma_0}\mathscr{J}_1 + e^{-m'\sigma_0}\mathscr{J}_2)^{-1}$, and the moments of packets supported in a cube of side L are bounded by $(n-1)!!$ times $((2L)^4 \sup_k Z_k^2 a_k^4 \sum_z (q_k^{|z|_1} + q_k'^{|z|_1}))^{n/2} \prod_j \|f_j\|_\infty$, where $a_k^4 \sum_z q_k^{|z|_1} = (a_k(1+q_k)/(1-q_k))^4$ is bounded; this gives ($\mathrm{M_s}$). The kernel of $D_{\lambda,c} S$ is $c^2 \lambda^{-8} \kappa e^{-(m/\lambda)\|z\|_1}$, by the substitution $u = \lambda x$, $v = \lambda y$. If $S' = D_{\lambda,c} S$, then the two continuous kernels coincide, since by Proposition 261(i) a continuous kernel is determined by S_2; at $z = t e_0$, after multiplication by e^{mt},

$$\kappa' \big(1 + e^{-(m'-m)t}\big) = c^2 \lambda^{-8} \kappa e^{(m - m/\lambda)t}$$

for all $t > 0$. The left side tends to $\kappa' > 0$ as $t \to \infty$, which forces $\lambda = 1$, and then the right side is constant while the left side is strictly decreasing. □

The mass. The gap of Theorem 64 gives the first half of (MC_{m_0}) with $m_0 = m_*$, and reflection positivity with ($\mathrm{M_s}$) gives the second half, but with a rate that is in general larger.

Proposition 269. *Assume the hypothesis of Theorem 64, $C_k(0) > 0$ for large k, $Z_k = C_k(\sigma_0)^{-1/2}$ and* ($\mathrm{M_s}$). *Let Q be a cube that contains* $\operatorname{supp} f_0$ *and $\theta_{0,k}(\operatorname{supp} f_0)$ for all k, and $m_1 = \sigma_0^{-1} \log(M_2(Q,t_0)\|f_0\|_\infty^2)$.*

(i) *$m_1 \ge m_*$; $\liminf_k a_k^{-1} \log(1/\gamma_k) \ge m_*$; and for large k and $t \in a_k\mathbb{Z}$ with $t \ge \sigma_0$,*

$$C_k(t) \ge e^{-m_1(t-\sigma_0)} C_k(\sigma_0).$$

(ii) *Every limit S of Theorem 248 has $m_* \le m(S) \le m_1$. If* (MC_{m_0}) *holds, then $m_* \le m_0 \le m_1$.*

(iii) *If S and S' are limits of Theorem 248 with $S' = D_{\lambda,c} S$, then $\lambda = m(S)/m(S')$ lies in $[m_*/m_1, m_1/m_*]$.*

Proof. (i) By Proposition 250(i),(ii), $(M_2(Q,t_0)\|f_0\|_\infty^2)^{-1} \le C_k(\sigma_0)/C_k(0) \le e^{-m_*\sigma_0}$ for large k, which gives $m_1 \ge m_*$ and $C_k(\sigma_0)/C_k(0) \ge e^{-m_1\sigma_0}$. The second claim is $\gamma_k \le e^{-m_* a_k}$. By the proof of Proposition 250(i), $C_k(t) = \int \lambda^{t/a_k} d\nu(\lambda)$ with a positive measure ν, so by Hölder's inequality, with $i = t/a_k$ and $p = \sigma_0/a_k$, $C_k(\sigma_0)^i \le C_k(0)^{i-p} C_k(t)^p$. Hence

$$C_k(t) \ge C_k(\sigma_0)\Big(\frac{C_k(\sigma_0)}{C_k(0)}\Big)^{(t-\sigma_0)/\sigma_0} \ge e^{-m_1(t-\sigma_0)} C_k(\sigma_0).$$

(ii) $m(S) \ge m_*$ is Theorem 248(iv). As in the proof of Theorem 266(i), (i) gives $\langle \Psi_0(f_0), e^{-tH_0}\Psi_0(f_0)\rangle = \lim_k Z_k^2 C_k(t) \ge e^{-m_1(t-\sigma_0)}$ for dyadic $t \ge \sigma_0$, while the left side is at most $e^{-m(S)t}\|\Psi_0(f_0)\|^2$; so $m(S) \le m_1$. Under (MC_{m_0}), $m(S) = m_0$ by Theorem 266(i).

(iii) By Lemma 264, $m(S') = m(S)/\lambda$, and (ii) applies to both. □

So the two halves of (MC_{m_0}) hold separately, with the rates m_* and m_1, and (MC_{m_0}) asks that they coincide. Since $C_k(t) \le \gamma_k^{(t-\sigma_0)/a_k} C_k(\sigma_0)$ for $t \ge \sigma_0$, by the spectral representation, its second half gives $a_k^{-1}\log(1/\gamma_k) \le m_0$ for $k \ge k_1$, so (MC_{m_0}) says that the lattice gaps in physical units converge to m_0 and that the reflected correlation of f_0 decays at this rate uniformly in k. This is not implied by the other properties: the interleaved two-mass field of Theorem 266(iii) has them and violates (MC_{m_0}) for every m_0. Nor is it necessary for ($\mathrm{U_s}$): for the products of Proposition 263 with $p \ge 2$, the transfer matrix has norm q_k on $\Omega_k^\perp$, attained on the first chaos of one factor, while the covariance of Φ is $e^{-pma_k\|z\|_1}$ and $C_k(t) = e^{-pmt} C_k(0)$ for $f_0 \ge 0$; so (MC_{m_0}) would need $m_0 \le m$ and $m_0 \ge pm$, while ($\mathrm{U_s}$) holds, also with $Z_k = C_k(\sigma_0)^{-1/2}$, which converges to a positive limit.

The dilations. ($\mathrm{O_s}$) is a statement about the set of limits. It is equivalent to a statement about the lattice moments alone: convergence after a change of the spacing by a bounded factor. Write $B_k(f_1,\dots,f_n) = \omega_k(\prod_j Z_k A_k(f_j))$ and $(D_{\lambda,c}B_k)(f_1,\dots,f_n) = c^n B_k(f_1(\lambda\cdot),\dots,f_n(\lambda\cdot))$. Since $A_k(f(\lambda\cdot)) = \lambda^{-4}\sum_x (\lambda a_k)^4 f(\lambda a_k x)(\Phi_x - c_k)$, $D_{\lambda,c}B_k$ is the moment family of the packets of the same states formed at the spacing λa_k, with the constants $c\lambda^{-4}Z_k$. Consider

(DS) there are $\Lambda \ge 1$, $\lambda_k \in [\Lambda^{-1},\Lambda]$, $c_k > 0$ and a nontrivial family T with $D_{\lambda_k,c_k}B_k \to T$ on every separated tuple.

A family is trivial if it vanishes on all separated tuples of length $n \ge 1$.

Proposition 270. *Assume the hypothesis of Theorem 64, $C_k(0) > 0$ for large k, $Z_k = C_k(\sigma_0)^{-1/2}$ and* ($\mathrm{M_s}$)*, and let m_1 be as in Proposition 269.*

(i) (DS) *implies* ($\mathrm{O_s}$)*: every limit S of Theorem 248 has $T = D_{\lambda,c}S$ for some $\lambda, c > 0$.*

(ii) ($\mathrm{O_s}$) *implies* (DS) *with $\Lambda = m_1/m_*$, and T can be taken to be any limit of Theorem 248.*

(iii) *Under* (DS), ($\mathrm{U_s}$) *holds if and only if (λ_k, c_k) converges in $(0,\infty)^2$.*

Proof. For $\lambda > 0$ the map $f \mapsto f(\lambda\cdot)$ is a bijection of the separated tuples. Let $(f_j) \in \mathscr{C}_n(Q,\delta)$, $\lambda_k \to \lambda \in [\Lambda^{-1},\Lambda]$. For large k the tuples $(f_j(\lambda_k\cdot))$, $(f_j(\lambda\cdot))$ and the telescoping tuples with one factor $f_i(\lambda_k\cdot) - f_i(\lambda\cdot)$ lie in one $\mathscr{C}_n(Q',\delta/(2\Lambda))$, and $f_i(\lambda_k\cdot) \to f_i(\lambda\cdot)$ uniformly; so ($\mathrm{M_s}$) and multilinearity give $B_k(f_1(\lambda_k\cdot),\dots) - B_k(f_1(\lambda\cdot),\dots) \to 0$. Hence, along a subsequence with $B_k \to S$ on every separated tuple,

$$B_k\big(f_1(\lambda_k\cdot),\dots,f_n(\lambda_k\cdot)\big) \to S_n\big(f_1(\lambda\cdot),\dots,f_n(\lambda\cdot)\big). \qquad (*)$$

By Proposition 250(ii) every limit is nontrivial.

(i) Let S be the limit along a subsequence, and pass to a further one with $\lambda_k \to \lambda$ and $c_k \to c \in [0,\infty]$. By $(*)$, $T_n(f) = \lim_k c_k^n B_k(f(\lambda_k\cdot))$ with $B_k(f(\lambda_k\cdot)) \to S_n(f(\lambda\cdot))$. If $c = \infty$, then $S_n(f(\lambda\cdot)) = 0$ for every separated tuple of length $n \ge 1$, and S is trivial; if $c = 0$, T is trivial. So $0 < c < \infty$ and $T = D_{\lambda,c}S$. For two limits, $D_{\lambda,c}S = D_{\lambda',c'}S'$ gives $S' = D_{\lambda/\lambda',c/c'}S$.

(ii) Let $\mathscr{G}$ be the countable set of the proof of Theorem 248(i), $F_1, F_2, \dots$ the separated tuples of its elements, and $d(B,B') = \sum_i 2^{-i}\min(1, |B(F_i) - B'(F_i)|)$ for families B, B'. By the approximation in that proof, a family bounded on every $\mathscr{C}_n(Q,\delta)$ by a multiple of $\prod_j \|f_j\|_\infty$ that vanishes at every F_i vanishes on all separated tuples; such bounds hold for every $D_{\lambda,c}S$ with S a limit. Fix a limit S, let $I = [m_*/m_1, m_1/m_*]$, $e_k(\lambda,c) = d(D_{\lambda,c}B_k, S)$, and choose $(\lambda_k,c_k) \in I \times (0,\infty)$ with $e_k(\lambda_k,c_k) \le \inf_{I\times(0,\infty)} e_k + 1/k$. Suppose $e_k(\lambda_k,c_k) \ge \varepsilon > 0$ along a subsequence. A further subsequence has $B_k \to S'$ on every separated tuple, and ($\mathrm{O_s}$) gives $S = D_{\lambda,c}S'$ with $\lambda \in I$ by Proposition 269(iii). Then $D_{\lambda,c}B_k \to S$ on every separated tuple by $(*)$ with constant $\lambda_k = \lambda$, so $e_k(\lambda,c) \to 0$ and $e_k(\lambda_k,c_k) \to 0$, a contradiction. Hence $e_k(\lambda_k,c_k) \to 0$. Now take any subsequence and a further one with $\lambda_k \to \lambda \in I$ and $B_k \to S'$. Some F_i, of length $n_i \ge 1$, has $S(F_i) \ne 0$, and $c_k^{n_i} B_k(F_i(\lambda_k\cdot)) \to S(F_i)$. If $c_k \to \infty$ along a further subsequence, then $B_k(F_l(\lambda_k\cdot)) \to 0$ for every l with $n_l \ge 1$, so by $(*)$ $D_{\lambda,1}S'$ vanishes at every F_l, and S' is trivial; if $c_k \to 0$, then $S(F_i) = 0$. So a further subsequence has $c_k \to c \in (0,\infty)$, and $D_{\lambda_k,c_k}B_k \to D_{\lambda,c}S'$ on every separated tuple by $(*)$. This family agrees with S at every F_l, hence everywhere. Every subsequence thus has a further subsequence along which $D_{\lambda_k,c_k}B_k \to S$, which is (DS) with $T = S$.

(iii) If $(\lambda_k,c_k) \to (\lambda,c)$, the proof of (i) gives $S = D_{1/\lambda,1/c}T$ for every limit S, which is ($\mathrm{U_s}$). Conversely, under ($\mathrm{U_s}$) let S be the limit. By the proof of (i), (c_k) has no subsequence tending to 0 or ∞, so (λ_k,c_k) lies in a compact subset of $(0,\infty)^2$, and every limit point (λ,c) has $D_{\lambda,c}S = T$. Two limit points (λ,c), (λ',c') give $D_{\lambda/\lambda',c/c'}S = S$, so $\lambda = \lambda'$ and $c = c'$ by Lemma 264. □

So ($\mathrm{O_s}$) says that the spacings a_k can be corrected by bounded factors λ_k so that the moments of the same states converge, and under it ($\mathrm{U_s}$) is the convergence of the corrections and of the constants, which (MC_{m_0}) gives by Theorem 266(ii). Neither ($\mathrm{O_s}$) nor ($\mathrm{U_s}$) follows from the properties proved here together with (MC_{m_0}): the fields of Proposition 268(c) are reflection positive, invariant under translations and B_4, have the gap ma_k in every coordinate direction and satisfy ($\mathrm{M_s}$) and (MC_m), and their two limits are not related by any $D_{\lambda,c}$. Conversely ($\mathrm{O_s}$) does not give (MC_{m_0}) (Theorem 266(iii)). The

proofs of Propositions 267, 269 and 270 use the averaged condition only through Lemma 247(iii) and the bound $\gamma_k \le e^{-m_* a_k}$, which the fields of Proposition 268 have with m in place of m_*.

Temperedness. The Schwinger functions of [88] are linear functionals on the space $\mathscr{S}_0(\mathbb{R}^{4n})$ of the $F \in \mathscr{S}(\mathbb{R}^{4n})$ that vanish, with all their derivatives, at every $x=(x_1,\dots,x_n)$ with $x_i = x_j$ for some $i \ne j$, and [89] adds the linear growth condition: $|S_n(F)| \le \sigma_n |F|_{sn}$ on $\mathscr{S}_0(\mathbb{R}^{4n})$ for a fixed integer s and $\sigma_n \le \alpha(n!)^{\beta'}$, with the norms $|F|_w = \sup_{x,|\gamma|\le w}(1+|x|^2)^{w/2}|\partial^\gamma F(x)|$ of Lemma 195. Theorem 248 gives the S_n only on $C_c^\infty(\mathfrak{D}_n)$. The test functions of $\mathscr{S}_0(\mathbb{R}^{4n})$ approach the coinciding points, where ($\mathrm{M_s}$) gives no control, since its constants $M_n(Q,\delta)$ may grow arbitrarily as $\delta \to 0$. The following rate closes this gap.

($\mathrm{G_s}$) *Growth at coinciding points.* There are $K,\beta,p,q \ge 0$ such that for every $n \ge 1$, every cube Q of side L and every $\delta \in (0,1]$ there is k_0 with

$$\Big|\omega_k\Big(\prod_{j=1}^n Z_k A_k(f_j)\Big)\Big| \le K^n (n!)^\beta (1+L)^{qn} \delta^{-pn} \prod_{j=1}^n \|f_j\|_\infty$$

for all $k \ge k_0$ and $(f_1,\dots,f_n) \in \mathscr{C}_n(Q,\delta)$.

Since $\mathscr{C}_n(Q,\delta) \subset \mathscr{C}_n(Q,1)$ for $\delta \ge 1$, ($\mathrm{G_s}$) implies ($\mathrm{M_s}$), and the limits of Theorem 248(i) obey the same bound.

For $x \in \mathbb{R}^{4n}$ write $x_{j\nu}$ for the coordinates of x_j, $|x|$ for the Euclidean norm, $|x|_\infty = \max_{j,\nu}|x_{j\nu}|$, and $d(x) = \min_{i\ne j}|x_i - x_j|_\infty$. For a coordinate direction μ let $\mathscr{S}_0^{\mu+}(\mathbb{R}^{4n})$ be the set of $F \in \mathscr{S}_0(\mathbb{R}^{4n})$ with $F(x)=0$ whenever $x_{j\mu} \le 0$ for some j, and $V_\mu = \{x \in \mathfrak{D}_n : x_{j\mu} > 0 \text{ for all } j\}$. Such an F vanishes with all its derivatives on $\{x_{j\mu} \le 0\}$, by continuity from the interior of this set. Fix $\eta_0 \in C_c^\infty((-\frac12,\frac12))$ with $\eta_0 \ge 0$ and $\int \eta_0 = 1$, let $\psi = \mathbf{1}_{[-1/2,1/2)} * \eta_0$, so that $0 \le \psi \le 1$, $\operatorname{supp}\psi \subset (-1,1)$ and $\sum_{m\in\mathbb{Z}} \psi(t-m) = 1$, and put $C_\psi = \max(1, \|\psi'\|_\infty, \|\psi''\|_\infty)$. For $h>0$ and $m=(m_1,\dots,m_n) \in (\mathbb{Z}^4)^n$ let $\psi_m^h(x) = \prod_{j,\nu}\psi(x_{j\nu}/h - m_{j\nu})$; call m *separated* if $|m_i - m_j|_\infty \ge 4$ for $i \ne j$, and let E_h be the sum of ψ_m^h over the separated m.

Lemma 271. (i) *E_h is smooth with values in $[0,1]$, $E_h = 0$ on $\{d < 2h\}$ and $E_h = 1$ on $\{d \ge 6h\}$. For every multi-index $\gamma \ne 0$, $|\partial^\gamma E_h| \le 2^{4n} h^{-|\gamma|} \prod_{j,\nu} \|\psi^{(\gamma_{j\nu})}\|_\infty$; in particular $|\partial^\gamma(1-E_h)| \le (2C_\psi)^{4n} h^{-|\gamma|}$ for all $\gamma \in \{0,1,2\}^{4n}$.*

(ii) *Let $F \in \mathscr{S}_0(\mathbb{R}^{4n})$, α a multi-index and w, N, w' integers with $w' \ge \max(w, |\alpha|+N)$. If $|x_i - x_j|_\infty = t \le 1$ for some $i \ne j$, then*

$$(1+|x|^2)^{w/2}|\partial^\alpha F(x)| \le 3^w \frac{(4t)^N}{N!}|F|_{w'}.$$

If $F \in \mathscr{S}_0^{\mu+}(\mathbb{R}^{4n})$ and $0 < x_{j\mu} = t \le 1$ for some j, the same holds with $2^w t^N/N!$ in place of $3^w(4t)^N/N!$.

(iii) *$C_c^\infty(\mathfrak{D}_n)$ is dense in $\mathscr{S}_0(\mathbb{R}^{4n})$ for every norm $|\cdot|_w$. Every $F \in \mathscr{S}_0^{\mu+}(\mathbb{R}^{4n})$ is, for every w, a $|\cdot|_w$-limit of finite sums of products $f_1 \otimes \cdots \otimes f_n$ with $f_j \in C_c^\infty(\mathbb{R}^4)$ supported in $\{x_\mu > 0\}$ and with pairwise disjoint supports.*

Proof. (i) The sum of all ψ_m^h is 1, and at each x at most two integers $m_{j\nu}$ have $\psi(x_{j\nu}/h - m_{j\nu}) \ne 0$, so at most 2^{4n} terms are nonzero near x; this gives smoothness and the derivative bounds, the same ones holding for $1-E_h$, the sum over the other m. If $\psi_m^h(x) \ne 0$, then $|x_j - hm_j|_\infty < h$ for all j. For separated m this gives $|x_i - x_j|_\infty > h|m_i - m_j|_\infty - 2h \ge 2h$, so $E_h = 0$ where $d < 2h$. If $d(x) \ge 6h$, it gives $h|m_i - m_j|_\infty > 4h$ for every m with $\psi_m^h(x) \ne 0$, so all these m are separated and $E_h(x) = 1$.

(ii) Let $u = x_i - x_j$, let $x(\tau)$ be x with x_i replaced by $x_j + \tau u$, and $g(\tau) = \partial^\alpha F(x(\tau))$. All derivatives of g vanish at $\tau = 0$, where two arguments coincide, so Taylor's formula gives $|g(1)| \le \sup_\tau |g^{(N)}(\tau)|/N!$, and $g^{(N)} = (u\cdot\nabla_{x_i})^N \partial^\alpha F$ is at most $|u|_1^N \le (4t)^N$ times $\max_{|\gamma|=N}|\partial^{\gamma+\alpha}F(x(\tau))| \le |F|_{w'}(1+|x(\tau)|^2)^{-w/2}$. Since $|x - x(\tau)| \le |u| \le 2t \le 2$, $(1+|x|^2)^{1/2} \le 3(1+|x(\tau)|^2)^{1/2}$. In the second case move $x_{j\mu}$ from 0 to t instead; the path has length $t \le 1$, which gives the factor 2^w.

(iii) Let $F \in \mathscr{S}_0(\mathbb{R}^{4n})$, fix w, let $\chi \in C_c^\infty(\mathbb{R}^{4n})$ with values in $[0,1]$ equal to 1 on $\{|x| \le 1\}$, $\chi_R = \chi(\cdot/R)$ with $R \ge 1$, and $0 < h \le \frac16$. Then $\chi_R E_h F \in C_c^\infty(\mathfrak{D}_n)$ by (i), and $F - \chi_R E_h F = (1-E_h)F + (1-\chi_R)E_h F$. By the Leibniz rule and (i), $(1+|x|^2)^{w/2}|\partial^\gamma((1-E_h)F)(x)|$ for $|\gamma| \le w$ is at most $c_w h^{-w} \max_{\alpha\le\gamma}(1+|x|^2)^{w/2}|\partial^\alpha F(x)|$ with c_w depending only on w, n and ψ, and it vanishes unless $d(x) < 6h$; there (ii) with $N = w+1$ bounds it by $c_w 3^w 24^{w+1} h|F|_{2w+1}$. The second term is supported in $\{|x| \ge R\}$ and its $|\cdot|_w$ norm is at most $c_{w,h}(1+R^2)^{-1/2}|F|_{w+1}$. Taking h small and then R large proves the first claim. For the second, let $\eta \in C^\infty(\mathbb{R})$ with values in $[0,1]$, $\eta = 0$ on $(-\infty,1]$ and $\eta = 1$ on $[2,\infty)$, and $G = \chi_R E_h \prod_j \eta(x_{j\mu}/h) F$, which is smooth with compact support in V_μ. The additional term $E_h(1 - \prod_j \eta(x_{j\mu}/h))F$ vanishes unless $0 < x_{j\mu} < 2h$ for some j, and the second case of (ii) bounds it in the same way. Finally, every point of V_μ has a neighbourhood $U_1 \times \cdots \times U_n$ of open cubes in $\{x_\mu > 0\}$ with pairwise disjoint closures; finitely many of them cover $\operatorname{supp} G$, a partition of unity subordinate to them [66, Sect. 1.4] writes G as a finite sum of smooth functions supported in such products, and for each the partial sums of the Fourier series in the proof of Theorem 248(i) are finite sums of products of the required form converging in C^∞ with supports in a fixed compact set, hence in every $|\cdot|_w$. □

Lemma 272. *Let $n \ge 2$, $C, p', q' \ge 0$, and let S be a multilinear form on the separated n-tuples with*

$$|S(f_1,\dots,f_n)| \le C(1+L)^{q'}\delta^{-p'}\prod_{j=1}^n \|f_j\|_\infty$$

for every cube Q of side L, every $\delta \in (0,1]$ and $(f_1,\dots,f_n) \in \mathscr{C}_n(Q,\delta)$. Let S also denote the distribution on $\mathfrak{D}_n$ with $S(f_1\otimes\cdots\otimes f_n) = S(f_1,\dots,f_n)$ for the separated tuples of smooth functions, which exists and is unique by the proof of Theorem 248(i). There is a number $c_\star$, independent of n, C, p', q' and S, such that, with $w = \lceil q'\rceil + \lceil p'\rceil + 12n + 1$,

$$|S(F)| \le c_\star^n C 6^{q'} 16^{p'} |F|_w \qquad \text{for } F \in C_c^\infty(\mathfrak{D}_n).$$

Hence S extends uniquely to a linear functional on $\mathscr{S}_0(\mathbb{R}^{4n})$ that is continuous for $|\cdot|_w$, with the same bound.

Proof. One piece. Let $h \in (0, \frac{1}{16}]$, m separated, $z_j = hm_j$, and $G \in C^\infty(\mathbb{R}^{4n})$ supported in $\prod_j (z_j + (-h,h)^4)$. The open cubes $U_j = z_j + (-\frac{3}{2}h, \frac{3}{2}h)^4$ are at mutual ℓ^∞ distance at least h, and lie in a cube Q of side $L \le 2\max_j |z_j|_\infty + 3h$. Let $B = \prod_j (z_j + [-2h,2h]^4)$, $e_\kappa(x) = \prod_j \exp(i\pi\kappa_j \cdot (x_j - z_j)/(2h))$ for $\kappa \in (\mathbb{Z}^4)^n$, $c_\kappa = (4h)^{-4n} \int_B G \bar{e}_\kappa$, and $\varphi_j \in C_c^\infty(U_j)$ with values in $[0,1]$, equal to 1 on $z_j + [-h,h]^4$. With $\mathsf{D} = \prod_{i=1}^{4n} (1 - (2h/\pi)^2 \partial_i^2)$ over the $4n$ coordinates, $\mathsf{D}\bar{e}_\kappa = \prod_i (1+\kappa_i^2)\bar{e}_\kappa$, so integration by parts gives $|c_\kappa| \le \prod_i (1+\kappa_i^2)^{-1} \|\mathsf{D}G\|_\infty$, and the same with powers of D shows that the partial sums of $\sum_\kappa c_\kappa \prod_j \varphi_j(x_j) e_{\kappa_j}(x_j)$ converge to G in C^∞ with supports in $\prod_j \operatorname{supp}\varphi_j$. Hence $S(G)$ is the sum of the values of S on the terms. Each term is c_κ times a product in $\mathscr{C}_n(Q,h)$ with factors of norm at most 1, and $\sum_{k\in\mathbb{Z}} (1+k^2)^{-1} \le 1 + 2\int_0^\infty \frac{dt}{1+t^2} < 5$, so

$$|S(G)| \le C(1+L)^{q'} h^{-p'} 5^{4n} \|\mathsf{D}G\|_\infty.$$

Decomposition. Let $F \in C_c^\infty(\mathfrak{D}_n)$, $h_l = 2^{-l-4}$ and $E_l = E_{h_l}$. Since d is bounded below on $\operatorname{supp} F$, Lemma 271(i) gives $E_l F = F$ for large l, and with $E_{l+1} - E_l = E_{l+1}(1-E_l) - E_l(1-E_{l+1})$,

$$F = E_0 F + \sum_{l\ge 0} \big(E_{l+1}(1-E_l)F - E_l(1-E_{l+1})F \big),$$

a finite sum. Each term has the form $\sum_m \psi_m^h H$ over separated m, with (h,H) equal to (h_0,F), $(h_{l+1},(1-E_l)F)$ or $(h_l,(1-E_{l+1})F)$; in the last two cases $H = (1-E_{h'})F$ with $h/h' \in \{\frac{1}{2},2\}$, and H vanishes where $d \ge 12h$.

Derivatives. Write $\mathsf{D} = \prod_i \mathsf{D}_i$ and $\psi_m^h = \prod_i \psi_i$. Since $\mathsf{D}_i(\psi_i G) = \psi_i G - (2h/\pi)^2(\psi_i'' G + 2\psi_i' \partial_i G + \psi_i \partial_i^2 G)$ and $|\psi_i^{(k)}| \le C_\psi h^{-k}$, induction over the coordinates gives

$$|\mathsf{D}(\psi_m^h H)| \le b_\psi^{4n} \max_{\gamma \in \{0,1,2\}^{4n}} h^{|\gamma|} |\partial^\gamma H|, \qquad b_\psi = 1 + 2C_\psi,$$

on the box $X_m = \prod_j (hm_j + [-h,h]^4)$, outside which $\psi_m^h = 0$, since $1 + 16C_\psi/\pi^2 \le b_\psi$. For $H = (1-E_{h'})F$, the Leibniz rule, Lemma 271(i), $h/h' \le 2$ and $h \le 1$ give $h^{|\gamma|}|\partial^\gamma H| \le 3^{|\gamma|}(2C_\psi)^{4n} \max_{\alpha\le\gamma} |\partial^\alpha F|$. Put $w_0 = \lceil q' \rceil + 8n$ and $N = \lceil p' \rceil + 4n + 1$, so that $w \ge \max(w_0, 8n+N)$. At a point of the support of H, $d \le 12h \le \frac{3}{4}$, and Lemma 271(ii) gives $(1+|x|^2)^{w_0/2} |\partial^\alpha F(x)| \le 3^{w_0}(48h)^N |F|_w / N!$ for $|\alpha| \le 8n$; for $H = F$ the same quantity is at most $|F|_w$.

Summation. For $x \in X_m$, $|z_j|_\infty \le |x|_\infty + h$, so the cube of the first step has $1 + L \le 2(1+|x|_\infty) \le 2(1+|x|^2)^{1/2}$. Evaluating at the point of the compact box X_m where $(1+|x|^2)^{-w_0/2}$ is largest, and using $q' - w_0 \le -8n$ and $(1+|x|^2)^{4n} \ge \prod_{j,\nu}(1 + x_{j\nu}^2)$, the piece $\psi_m^h H$ of a term of the second or third kind contributes at most

$$C 2^{q'} h^{-p'} \big(5 b_\psi \cdot 2C_\psi \cdot 9\big)^{4n} 3^{w_0} \frac{(48h)^N}{N!} |F|_w \sup_{x\in X_m} \prod_{j,\nu} (1+x_{j\nu}^2)^{-1}.$$

The sum over $m \in (\mathbb{Z}^4)^n$ of the supremum is the $4n$-th power of $\sum_{k\in\mathbb{Z}} \sup_{|t-hk|\le h} (1+t^2)^{-1} \le 3 + 2\int_0^\infty \frac{du}{1+h^2u^2} < \frac{7}{h}$. With $48^N/N! \le e^{48}$, $3^{w_0} \le 3^{q'+1} 3^{8n}$ and $h^{N-p'-4n} \le h$, the terms of the second and third kinds at the scale h contribute at most $C 6^{q'} c_1^n h |F|_w$ with $c_1 = 3^{10} e^{48} (5\cdot 7\cdot 9 \cdot 2C_\psi b_\psi)^4$, and each h_l occurs in at most two of them, with $\sum_l 2h_l = \frac{1}{4}$. The term $E_0 F$ contributes at most $C 2^{q'} 16^{p'} (5\cdot 7 \cdot 16 b_\psi)^{4n} |F|_w$ by the same count without the Taylor factor. This gives the bound with $c_\star = c_1 + (5\cdot 7\cdot 16 b_\psi)^4$.

The extension follows from the density of $C_c^\infty(\mathfrak{D}_n)$ in $\mathscr{S}_0(\mathbb{R}^{4n})$ for $|\cdot|_w$ (Lemma 271(iii)), and it is unique by the same density. □

For a function F of n points put

$$(\Theta_\mu F)(x) = F(\theta_\mu x_1, \dots, \theta_\mu x_n),$$
$$(\tau_v F)(x) = F(x_1 - v, \dots, x_n - v),$$
$$(F \otimes F')(x,x') = F(x) F'(x').$$

For $F_n \in \mathscr{S}_0^{\mu+}(\mathbb{R}^{4n})$ and $F'_{n'} \in \mathscr{S}_0^{\mu+}(\mathbb{R}^{4n'})$, $\Theta_\mu \bar{F}_n \otimes F'_{n'} \in \mathscr{S}_0(\mathbb{R}^{4(n+n')})$, because at a coincidence of a reflected point with an unreflected one some argument of F_n or of $F'_{n'}$ has μ-coordinate at most 0.

Theorem 273. *Assume the hypothesis of Theorem 64 and* ($\mathrm{M_s}$)*, and let S_n be limits as in Theorem 248.*

(i) *Let $n \ge 2$, and suppose that there are $C, p', q' \ge 0$ such that*

$$|S_n(f_1, \dots, f_n)| \le C(1+L)^{q'} \delta^{-p'} \prod_j \|f_j\|_\infty$$

on $\mathscr{C}_n(Q,\delta)$ for every cube Q of side L and $\delta \in (0,1]$. Then S_n extends uniquely to a linear functional on $\mathscr{S}_0(\mathbb{R}^{4n})$ continuous for the topology of $\mathscr{S}(\mathbb{R}^{4n})$, and $|S_n(F)| \le c_\star^n C 6^{q'} 16^{p'} |F|_w$ with w and $c_\star$ as in Lemma 272.

(ii) *Under* ($\mathrm{G_s}$)*, for every $n \ge 1$ and $F \in \mathscr{S}_0(\mathbb{R}^{4n})$,*

$$|S_n(F)| \le \sigma_n |F|_{sn}, \qquad \sigma_n = (6^q 16^p c_\star K)^n (n!)^\beta,$$

with $s = \lceil p \rceil + \lceil q \rceil + 13$, and $\sigma_n \le e^{6^q 16^p c_\star K} (n!)^{\beta+1}$, since $y^n \le e^y n!$ for $y \ge 0$; this is the linear growth condition of [89].

(iii) *Suppose that the hypothesis of* (i) *holds for every $n \ge 2$, with constants depending on n, and put $S_0 = 1$. Then the extended S_n are symmetric, invariant under the translations, B_4 and G_S, and satisfy $S_n(\bar{F}) = \overline{S_n(F)}$. For every μ and every finite family $F_n \in \mathscr{S}_0^{\mu+}(\mathbb{R}^{4n})$, $0 \le n \le n_0$ ($F_0 \in \mathbb{C}$),*

$$\sum_{n,n'=0}^{n_0} S_{n+n'}\big(\Theta_\mu \bar{F}_n \otimes F_{n'}\big) \ge 0,$$

and there are vectors $\Psi_\mu(F)$ of the space $\mathscr{H}_\mu$ of Theorem 248(iii) whose inner products are these sums, which agree with $\Psi_\mu(P)$ for separated polynomials P of smooth functions, are dense in $\mathscr{H}_\mu$, and satisfy $e^{-tH_\mu}\Psi_\mu(F) = \Psi_\mu(\tau_{te_\mu} F)$ for $t \ge 0$. For $y \perp e_\mu$, $y \ne 0$, and $F_n \in \mathscr{S}_0^{\mu+}(\mathbb{R}^{4n})$, $F'_{n'} \in \mathscr{S}_0^{\mu+}(\mathbb{R}^{4n'})$,

$$\lim_{\lambda\to\infty} S_{n+n'}\big(\Theta_\mu \bar{F}_n \otimes \tau_{\lambda y} F'_{n'}\big) = S_n(\Theta_\mu \bar{F}_n)\, S_{n'}(F'_{n'}).$$

Proof. (i) This is Lemma 272; continuity for $|\cdot|_w$ is continuity in $\mathscr{S}(\mathbb{R}^{4n})$.

(ii) For $n=1$, $S_1=0$. For $n\geq 2$ the limits satisfy the hypothesis of (i) with $C=K^n(n!)^{\beta}$, $q'=qn$ and $p'=pn$, and $\lceil qn\rceil+\lceil pn\rceil+12n+1\leq sn$, while $|F|_w\leq|F|_{sn}$ for $w\leq sn$.

(iii) Let g be a permutation of the arguments, or the diagonal action of a translation, of an element of B_4 or of an element of G_S. Then $F\mapsto F\circ g^{-1}$ maps $C_c^\infty(\mathfrak{D}_n)$ onto itself and is continuous for the topology of $\mathscr{S}(\mathbb{R}^{4n})$. The distributions $F\mapsto S_n(F\circ g^{-1})$ and S_n agree on the products of separated smooth functions by the symmetry of S_n, by Theorem 248(ii) and by the definition of G_S, hence on $C_c^\infty(\mathfrak{D}_n)$ by the uniqueness in Theorem 248(i), hence on $\mathscr{S}_0(\mathbb{R}^{4n})$ by density; the conjugation identity follows in the same way. Choose an integer s' such that the bounds of (i) for $n\leq 2n_0$ hold with $|\cdot|_{s'n}$, and by Lemma 271(iii) finite sums $P_n^{(i)}$ of products of smooth functions with disjoint supports in $\{x_\mu>0\}$ with $|P_n^{(i)}-F_n|_{2s'n_0}\to 0$, and $P_0^{(i)}=F_0$. Then $P^{(i)}=\sum_n P_n^{(i)}$ is a separated polynomial at positive μ-times, and $\sum_{n,n'}S_{n+n'}(\Theta_\mu\bar{P}_n^{(i)}\otimes P_{n'}^{(i)})=S(\theta_\mu\bar{P}^{(i)}\cdot P^{(i)})\geq 0$ by Theorem 248(iii), each $\Theta_\mu\bar{P}_n^{(i)}\otimes P_{n'}^{(i)}$ being a finite sum of products of separated smooth functions. Since $|G\otimes G'|_w\leq|G|_w|G'|_w$ and $|\Theta_\mu G|_w=|G|_w$, $\Theta_\mu\bar{P}_n^{(i)}\otimes P_{n'}^{(i)}\to\Theta_\mu\bar{F}_n\otimes F_{n'}$ in $|\cdot|_{s'(n+n')}$, and the positivity passes to the limit. The same convergence, applied to $P^{(i)}-P^{(i')}$, shows that $\Psi_\mu(P^{(i)})$ is a Cauchy sequence in $\mathscr{H}_\mu$; its limit $\Psi_\mu(F)$ does not depend on the approximants and has the stated inner products. Separated polynomials of smooth functions are among the families F, and their vectors are dense in $\mathscr{H}_\mu$: those of separated polynomials of continuous functions are dense by construction, and each of them is a limit of vectors of smooth ones, by uniform approximation with supports in slightly larger sets and the bound of Theorem 248(i). The relation with e^{-tH_μ} holds for the $P^{(i)}$ by Theorem 248(iv) and passes to the limit because τ_{te_μ} preserves $\mathscr{S}_0^{\mu+}$ and is continuous. For the last claim, by Theorem 248(v) applied to every pair of monomials, $\langle\Psi_\mu(P),U_\mu(\lambda y)\Psi_\mu(P')\rangle=S(\theta_\mu\bar{P}\cdot\tau_{\lambda y}P')$ tends to $S(\theta_\mu\bar{P})S(P')=\langle\Psi_\mu(P),\Omega_\mu\rangle\langle\Omega_\mu,\Psi_\mu(P')\rangle$ for separated polynomials P,P' at positive μ-times; the $U_\mu(\lambda y)$ are unitary and these vectors are dense, so $U_\mu(\lambda y)$ converges weakly to the projection on $\mathbb{C}\Omega_\mu$. With $U_\mu(\lambda y)\Psi_\mu(F')=\Psi_\mu(\tau_{\lambda y}F')$, obtained as for e^{-tH_μ}, and $\langle\Psi_\mu(F_n),\Omega_\mu\rangle=S_n(\Theta_\mu\bar{F}_n)$, $\langle\Omega_\mu,\Psi_\mu(F'_{n'})\rangle=S_{n'}(F'_{n'})$, this is the claim. □

With $(\mathrm{W_s})$ and $(\mathrm{U_s})$, Corollary 249 and Theorem 273 give the axioms of [88, 89]. The positivity of Theorem 273(iii) contains that of [88], whose test functions are supported where $0<x_{10}<\cdots<x_{n0}$, and the symmetry of the S_n makes the order of the arguments irrelevant.

Corollary 274. *Assume the hypothesis of Theorem 64,* $(\mathrm{G_s})$, $(\mathrm{W_s})$ *and* $(\mathrm{U_s})$. *Then the moments converge for every separated tuple, and their limits, extended to* $\mathscr{S}_0(\mathbb{R}^{4n})$ *by Theorem 273 and with* $S_0=1$, *satisfy the linear growth condition of [89] and the axioms of [88] of Euclidean invariance, reflection positivity, symmetry and clustering. The Hilbert space and the Hamiltonian of their reconstruction are* $\mathscr{H}_0$ *and* H_0, *with* $\sigma(H_0)\subset\{0\}\cup[m_*,\infty)$ *and* $\ker H_0=\mathbb{C}\Omega_0$. *If moreover* $C_k(0)>0$ *for large* k *and* $Z_k=C_k(\sigma_0)^{-1/2}$, *they are not the Schwinger functions of the vacuum alone.*

Proof. $(\mathrm{G_s})$ gives $(\mathrm{M_s})$, and Corollary 249 gives convergence, invariance under the translations and $\mathrm{O}(4)$, so $G_S=\mathrm{O}(4)$, and properties (i)–(vi) of Theorem 248. Theorem 273(ii) gives the growth condition, and part (iii) for $\mu=0$ gives the invariance on $\mathscr{S}_0$, positivity and clustering, and identifies the Hilbert space built from $\mathscr{S}_0^{0+}$ with $\mathscr{H}_0$ and its time translations with e^{-tH_0}; Theorem 248(iv) gives the spectrum. Proposition 250(ii) gives $\mathscr{H}_0\neq\mathbb{C}\Omega_0$, so some S_n with $n\geq 2$ is not zero, by Theorem 248(vi). □

By the reconstruction theorem of [89], Schwinger functions with these properties are those of a Wightman theory on $\mathscr{H}_0$ with Hamiltonian H_0. So the averaged condition along the tuning, $(\mathrm{G_s})$ with the constants of Proposition 250(ii), $(\mathrm{W_s})$ and $(\mathrm{U_s})$ together give (E) for the field Φ at separated points, with mass gap at least m_*. None of these four statements is proved. For $n=2$ the rate in $(\mathrm{G_s})$ is a rate in (SD):

$(\mathrm{SD_p})$ there are $C_0,p_0\geq 0$ such that for every $\varepsilon'\in(0,\varepsilon]$ there is k_0 with $\mathscr{N}_k(\varepsilon')\leq C_0(\varepsilon/\varepsilon')^{p_0}\mathscr{N}_k(\varepsilon)$ for $k\geq k_0$.

Proposition 275. *Assume the hypothesis of Theorem 254.*

(a) $(\mathrm{SD_p})$ *holds if and only if there are* $C,p',q'\geq 0$ *such that for every cube* Q *of side* L *and* $\delta\in(0,1]$ *there is* k_0 *with*

$$Z_k^2\big|\omega_k\big(A_k(f)A_k(g)\big)\big|\leq C(1+L)^{q'}\delta^{-p'}\|f\|_\infty\|g\|_\infty$$

for $k\geq k_0$ *and* $(f,g)\in\mathscr{C}_2(Q,\delta)$. *One can take* $q'=8$ *and* $p'=8+p_0$ *in one direction and* $p_0=p'$ *in the other.*

(b) *Under* $(\mathrm{SD_p})$ *every limit* S_2 *of Corollary 255(b) extends uniquely to a linear functional on* $\mathscr{S}_0(\mathbb{R}^8)$ *with* $|S_2(F)|\leq C'|F|_{\lceil p_0\rceil+41}$, *invariant under the translations and* B_4, *with* $S_2(\Theta_\mu\bar{f}\otimes f)\geq 0$ *for* $f\in\mathscr{S}_0^{\mu+}(\mathbb{R}^4)$.

Proof. (a) Let $(\mathrm{SD_p})$ hold, $(f,g)\in\mathscr{C}_2(Q,\delta)$ with $\delta\leq 1$, and $\varepsilon'=\min(\varepsilon,\delta/8)$. The proof of Corollary 255(a) gives, for large k depending only on δ, $Z_k^2|\omega_k(A_k(f)A_k(g))|\leq \mathsf{K}_{4L/\delta}^2\mathsf{K}_{\delta/4}^2\mathscr{N}_k(\varepsilon')\mathscr{N}_k(\varepsilon)^{-1}\|f\|_\infty\|g\|_\infty$, and $\mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon)\leq\max(1,C_0)\max(1,8\varepsilon)^{p_0}\delta^{-p_0}$. Since $\mathsf{K}_{4L/\delta}^2=(8L/\delta+3)^8\leq 11^8(1+L)^8\delta^{-8}$ and $\mathsf{K}_{\delta/4}^2\leq(\frac{7}{2})^8$, the bound holds with $q'=8$, $p'=8+p_0$. Conversely, the proof of Corollary 255(a) gives $\mathscr{N}_k(\varepsilon')\leq M\mathscr{N}_k(\varepsilon)$ for large k, where M is the constant of the bound for a cube Q_0 of side $L_0=2(\varepsilon+2+2a_0)$, which contains $B_{\varepsilon'}$ and $\theta_{0,k}(B_{\varepsilon'})$ for all k and $\varepsilon'\leq\varepsilon$, and for the separation $\min(\varepsilon',1)$. So $M\leq C(1+L_0)^{q'}\max(1,\varepsilon^{-p'})(\varepsilon/\varepsilon')^{p'}$, which is $(\mathrm{SD_p})$ with $p_0=p'$.

(b) By (a) the limits satisfy the hypothesis of Lemma 272 with $n=2$, $q'=8$, $p'=8+p_0$, and $\lceil q'\rceil+\lceil p'\rceil+25=\lceil p_0\rceil+41$. Invariance and positivity follow as in the proof of Theorem 273(iii), from those of Corollary 255(b) and Lemma 271(iii) with $n=1$. □

The rate cannot be dropped: the structure that Theorem 248 and Corollary 255(b) give at separated points does not imply temperedness.

Proposition 276. (i) *Let $K \geq 0$ be integrable on the compact subsets of $\mathbb{R}^4 \setminus \{0\}$ and $S_2(F) = \iint F(x,y)K(x-y)\,dx\,dy$ for $F \in C_c^\infty(\mathfrak{D}_2)$. If $|S_2(F)| \leq C|F|_w$ on $C_c^\infty(\mathfrak{D}_2)$, then $\int_{\delta\leq|z|_\infty\leq 1} K \leq c_w C\delta^{-w}$ for $\delta \in (0,1]$, with c_w depending only on w. Conversely, if $\int_{\delta\leq|z|_\infty\leq\rho} K \leq C(1+\rho)^{q'}\delta^{-p'}$ for $0<\delta\leq 1\leq\rho$, then S_2 extends to $\mathscr{S}_0(\mathbb{R}^8)$ as in Lemma 272.*

(ii) *Let $K(z) = \int_1^\infty e^{\sqrt{\lambda}-\lambda\|z\|_1}d\lambda$ for $z\neq 0$, $\|z\|_1 = \sum_\nu |z_\nu|$, and $S_2(f,g) = \iint f(x)g(y)K(x-y)\,dx\,dy$ for separated pairs. Then S_2 is given by a distribution on $\mathfrak{D}_2$, is symmetric, invariant under the translations and B_4, real on real functions, bounded on $\mathscr{C}_2(Q,\delta)$ by $L^8\mathsf{k}(\delta)\|f\|_\infty\|g\|_\infty$ with $\mathsf{k}(\delta) = \int_1^\infty e^{\sqrt{\lambda}-\lambda\delta}d\lambda$, and satisfies $|S_2(f,g(\cdot-\lambda y))| \leq e\mathsf{k}(1)L^8\|f\|_\infty\|g\|_\infty e^{-(\lambda|y|_\infty-L)}$ for f,g supported in a cube of side L and $\lambda|y|_\infty \geq L+1$. For g supported in $\{x_\mu>0\}$, $S_2(\bar g\circ\theta_\mu, g)\geq 0$; the time translations define on the completion $\mathscr{H}^{(1)}\neq 0$ of the functions supported in $\{x_0>0\}$ for the form $S_2(\bar g\circ\theta_0, g')$ a semigroup $e^{-tH^{(1)}}$ with $\sigma(H^{(1)})\subset[1,\infty)$. These are the properties of Corollary 255(b), with $m_*=1$. But $\int_{\delta\leq|z|_\infty\leq 2\delta} K \geq 240\,\delta^4 e^{1/(64\delta)}$ for $\delta\leq\frac{1}{32}$, so S_2 is bounded by no norm $|\cdot|_w$ on $C_c^\infty(\mathfrak{D}_2)$ and does not extend to a continuous functional on $\mathscr{S}_0(\mathbb{R}^8)$.*

Proof. (i) Let $\chi_0 \in C^\infty(\mathbb{R})$ be even, nonincreasing on $[0,\infty)$, equal to 1 on $[0,1]$ and to 0 on $[2,\infty)$, $\chi_1(z) = \prod_\nu \chi_0(z_\nu)$, and $\psi_\delta(z) = \chi_1(z) - \chi_1(2z/\delta)$. Then $0\leq\psi_\delta\leq 1$, $\psi_\delta = 1$ on $\{\delta\leq|z|_\infty\leq 1\}$, $\psi_\delta = 0$ on $\{|z|_\infty\leq\delta/2\}\cup\{|z|_\infty\geq 2\}$, and $|\partial^\gamma\psi_\delta| \leq c_\gamma\delta^{-|\gamma|}$. With $\phi\in C_c^\infty((-1,1)^4)$, $\phi\geq 0$, $\int\phi = 1$, the function $F_\delta(x,y) = \phi(x)\psi_\delta(x-y)$ lies in $C_c^\infty(\mathfrak{D}_2)$, has $|F_\delta|_w \leq c_w\delta^{-w}$, and $S_2(F_\delta) = \int\psi_\delta K \geq \int_{\delta\leq|z|_\infty\leq 1} K$. For the converse, $|S_2(f,g)| \leq \|f\|_\infty\|g\|_\infty\int_Q dx\int_{\delta\leq|z|_\infty\leq\max(L,1)} K \leq 2^{q'}C(1+L)^{q'+4}\delta^{-p'}\|f\|_\infty\|g\|_\infty$ on $\mathscr{C}_2(Q,\delta)$, and S_2 is the distribution of Lemma 272 by uniqueness.

(ii) For $|z|_\infty\geq\delta$, $\|z\|_1\geq\delta$ and $K(z)\leq\mathsf{k}(\delta)$; for $\|z\|_1 = u\geq 1$ and $\lambda\geq 1$, $\lambda u\geq\lambda+u-1$, so $K(z)\leq e\mathsf{k}(1)e^{-u}$. Hence K is continuous on $\mathbb{R}^4\setminus\{0\}$, which gives the distribution, and the two bounds follow, since $\|x-y\|_1\geq|x-y|_\infty$. Symmetry, reality and invariance hold because K is real, even and a function of $\|z\|_1$, which B_4 preserves. Let g,g' be supported in $\{x_0\geq t_0\}$, $t_0>0$. Substituting $x\mapsto\theta_0 x$ and $y\mapsto y+te_0$,

$$S_2\big(\bar g\circ\theta_0, g'(\cdot-te_0)\big) = \int_1^\infty e^{\sqrt{\lambda}-\lambda t}\,\mathsf{B}_\lambda(G_\lambda, G'_\lambda)\,d\lambda,$$

where G'_λ is formed from g' as G_λ from g, with

$$G_\lambda(\mathbf{x}) = \int g(x_0,\mathbf{x})e^{-\lambda x_0}dx_0,$$

$$\mathsf{B}_\lambda(u,v) = \iint \bar u(\mathbf{x})v(\mathbf{y})e^{-\lambda\|\mathbf{x}-\mathbf{y}\|_1}d\mathbf{x}\,d\mathbf{y}\qquad (u,v\colon\mathbb{R}^3\to\mathbb{C}).$$

The integrand is bounded by $e^{\sqrt{\lambda}-2\lambda t_0}\|g\|_1\|g'\|_1$. From $e^{-\lambda|\varsigma|} = \frac{1}{\pi}\int_{\mathbb{R}}\frac{\lambda}{\lambda^2+k^2}e^{ik\varsigma}dk$ one obtains

$$\mathsf{B}_\lambda(u,u) = \pi^{-3}\int|\hat u(k)|^2\prod_{i=1}^3\frac{\lambda}{\lambda^2+k_i^2}\,dk \geq 0.$$

So the form is positive, and

$$\|g(\cdot-te_0)\|^2 = \int_1^\infty e^{\sqrt{\lambda}-2\lambda t}\mathsf{B}_\lambda(G_\lambda,G_\lambda)\,d\lambda \leq e^{-2t}\|g\|^2.$$

The time translations therefore induce symmetric contractions $\mathsf{T}(t)$ of $\mathscr{H}^{(1)}$ with $\mathsf{T}(s)\mathsf{T}(t) = \mathsf{T}(s+t)$ and $\|\mathsf{T}(t)\|\leq e^{-t}$, strongly continuous by dominated convergence, so $\mathsf{T}(t) = e^{-tH^{(1)}}$ with $H^{(1)}\geq 1$, as in the proof of Theorem 193(iv). For $g\geq 0$ not zero, $G_\lambda\geq 0$ is not zero and $\mathsf{B}_\lambda(G_\lambda,G_\lambda)>0$, so $\mathscr{H}^{(1)}\neq 0$. Positivity for $\mu\neq 0$ follows from $\theta_\mu\pi_\mu = \pi_\mu\theta_0$ and the invariance under π_μ. Finally, let $0<|z|_\infty\leq 2\delta\leq\frac{1}{16}$, so that $u = \|z\|_1\leq 8\delta\leq\frac14$, and $\lambda_0 = (4u)^{-2}\geq 1$. For $\lambda\in[\lambda_0,2\lambda_0]$, $\sqrt{\lambda}-\lambda u\geq\frac{1}{4u}-\frac{1}{8u} = \frac{1}{8u}$, so $K(z)\geq\lambda_0 e^{1/(8u)}\geq e^{1/(64\delta)}$, and $\{\delta\leq|z|_\infty\leq 2\delta\}$ has volume $240\delta^4$. By (i), a bound $|S_2(F)|\leq C|F|_w$ would give $240\delta^4 e^{1/(64\delta)}\leq c_w C\delta^{-w}$ for all small δ, which is false; and a continuous functional on $\mathscr{S}_0(\mathbb{R}^8)$ is bounded by some $C|\cdot|_w$. □

The kernel of (ii) is the superposition, with weight $e^{\sqrt{\lambda}}d\lambda$, of the kernels $e^{-\lambda\|z\|_1}$ of the limits of Proposition 199(a). By (i), for a two-point function with a nonnegative kernel, a power rate as in Lemma 272 is also necessary for temperedness.

Remark 277. *Coinciding points.* The axioms of [88, 89] involve only $\mathscr{S}_0(\mathbb{R}^{4n})$, and positivity involves only test functions whose reflected and unreflected arguments are separated. Under ($\mathrm{G_s}$) the bound of Theorem 273(ii) extends by the Hahn–Banach theorem [92] to a functional on all of $\mathscr{S}(\mathbb{R}^{4n})$ with the same bound; averaging over the finite group generated by the permutations of the arguments and the diagonal action of B_4, which preserve every $|\cdot|_w$, makes it symmetric and B_4-invariant. Every such extension agrees with S_n on $\mathscr{S}_0(\mathbb{R}^{4n})$ and so satisfies the positivity of Theorem 273(iii); two of them differ by a distribution supported on the coinciding points. Translation invariance of the extension is not claimed here. For the action density the extension cannot be an integral: under the hypothesis of Proposition 252, the functions F_δ of the proof of Proposition 276(i) are supported in one compact set and bounded by $\|\phi\|_\infty$, while, since $|z|\leq 2|z|_\infty$, $S_2(F_\delta)\geq\int_{\delta\leq|z|_\infty\leq 2\delta}K \geq 240\cdot 4^{-8}c\,\delta^{-4}(\log\frac{1}{\delta})^{-p}$ for $2\delta\leq\min(1,r_0/2)$; so no extension of S_2 to $\mathbb{R}^8$ is a distribution of order zero, and its values at coinciding points are fixed only by a renormalisation, which the axioms do not see. The same law satisfies the hypothesis of Theorem 273(i) for $n=2$: if in addition $K(z)\leq C|z|^{-8}$ for $z\neq 0$, then $|S_2(f,g)|\leq\|f\|_\infty\|g\|_\infty L^4 C\int_{|z|\geq\delta}|z|^{-8}dz = \frac{\pi^2}{2}CL^4\delta^{-4}\|f\|_\infty\|g\|_\infty$ on $\mathscr{C}_2(Q,\delta)$, which is that hypothesis with $q'=p'=4$, while (M) fails by Proposition 252. At strong coupling (Corollary 251) every limit vanishes at separated points, and its extension is 0.

Asymptotic freedom at short distance. Renormalised perturbation theory predicts how the two-point function of Φ behaves at short distance, and this behaviour can be posed as a condition on the lattice states. Let $G_k(z) = \omega_k(\Phi_0\Phi_z) - c_k^2$ for $z\in\mathbb{Z}^4$, as in the proof of Theorem 262(iv), so that $\omega_k((\Phi_x-c_k)(\Phi_y-c_k)) = G_k(y-x)$; let $\log^+ t = \max(\log t, 0)$ and, for $\gamma\geq 0$ and $\rho>0$,

$$\ell(\rho) = 1+\log^+\frac{1}{\rho},\qquad w_\gamma(\rho) = \rho^{-8}\ell(\rho)^{-\gamma}.$$

(AF) There are $\gamma\geq 0$, $0<c_1\leq C_1$, $r_0\in(0,1]$ and k_0 such that

for all $k \geq k_0$ and $z \in \mathbb{Z}^4$ with $|z|_\infty \geq 2r+1$

$$Z_k^2 |G_k(z)| \leq C_1 w_\gamma(a_k|z|),$$

and $Z_k^2 G_k(z) \geq c_1 w_\gamma(a_k|z|)$ if moreover $a_k|z| \leq r_0$.

(AF_∞) There are $\gamma \geq 0$ and $C_{\mathrm{AF}} > 0$ such that every limit S_2 of $Z_k^2 \omega_k(A_k(f)A_k(g))$ on the separated pairs along a subsequence is $S_2(f,g) = \iint f(x)g(y)K(x-y)\,dx\,dy$ with K locally integrable on $\mathbb{R}^4 \setminus \{0\}$ and

$$\lim_{\rho \to 0} \operatorname*{ess\,sup}_{0<|z|\leq\rho} \Big| |z|^8 \Big(\log \frac{1}{|z|}\Big)^\gamma K(z) - C_{\mathrm{AF}} \Big| = 0.$$

(AF_β) For some $\Lambda > 0$, with $L_k = \log(1/(a_k\Lambda))$, the couplings are $\beta_{W,k} = 2N/g_k^2$ with $g_k^{-2} = 2b_0 L_k + (b_1/b_0)\log L_k + O(1)$.

(AF) is the two-sided form, at every lattice distance from $2r+1$ to r_0/a_k, of the law $|x-y|^{-8}(\log \frac{1}{|x-y|})^{-\gamma}$, with only a power bound at physical distances; (AF_∞) is the exact law of the limit at short distance; (AF_β) is the two-loop tuning of [21, Remark 247]. None of the three is proved here for the Wilson measure.

Remark 278. *The perturbative values.* Renormalised perturbation theory gives the following values; they are predictions and no proof below uses them. Let $F^a_{\mu\nu}$ be the field strength normalised by the action $\frac{1}{4}\int F^a_{\mu\nu}F^a_{\mu\nu}$, with generators $\mathrm{Tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, and let $\beta(g) = -b_0 g^3 + O(g^5)$, $b_0 = 11N/(48\pi^2)$ [58, 91]. The trace of the energy–momentum tensor, $\Theta = (\beta(g)/2g)F^a_{\mu\nu}F^a_{\mu\nu}$, is finite and invariant under the renormalisation group [30], so $\langle\Theta(x),\Theta(0)\rangle$, the connected two-point function, is $|x|^{-8}F(\bar g(|x|))$ with $\bar g(|x|)$ the running coupling at the scale $1/|x|$, and $F(g) = \frac{1}{4}b_0^2 g^4 T_0(1+O(g^2))$, where $T_0|x|^{-8}$ is the connected two-point function of $F^a_{\mu\nu}F^a_{\mu\nu}$ in the free theory. There, in the Feynman gauge, the propagator of the gauge field is $\delta^{ab}\delta_{\mu\nu}D(x)$ with $D(x) = 1/(4\pi^2|x|^2)$, and that of the field strength, which is gauge independent, is $\langle F^a_{\mu\nu}(x), F^b_{\rho\sigma}(0)\rangle = -\delta^{ab}P_{\mu\nu\rho\sigma}$ with

$$P_{\mu\nu\rho\sigma} = \delta_{\nu\sigma}D_{\mu\rho} - \delta_{\nu\rho}D_{\mu\sigma} - \delta_{\mu\sigma}D_{\nu\rho} + \delta_{\mu\rho}D_{\nu\sigma},$$

$D_{\mu\rho} = \partial_\mu\partial_\rho D = M_{\mu\rho}/(2\pi^2|x|^4)$, where $M = 4uu^{\mathsf{T}} - 1$, $u = x/|x|$. Since $\mathrm{tr}M = 0$ and $\mathrm{tr}M^2 = 12$, each of the four squared terms of the bracket, summed over μ,ν,ρ,σ, gives $4\cdot 12$, the two cross terms with a trace vanish and the other four give -12 each, so the sum of the squares is $(192-96)(2\pi^2|x|^4)^{-2} = 24\pi^{-4}|x|^{-8}$, and Wick's theorem gives $T_0 = 2(N^2-1)\cdot 24\pi^{-4} = 48(N^2-1)/\pi^4$. With $\bar g(|x|)^{-2} = 2b_0\log\frac{1}{|x|\Lambda}(1+o(1))$ this gives, as $|x| \to 0$,

$$\langle\Theta(x),\Theta(0)\rangle = \frac{3(N^2-1)}{\pi^4}\,|x|^{-8}\Big(\log\frac{1}{|x|}\Big)^{-2}(1+o(1)),$$

which is (AF_∞) with $\gamma = 2$ for every N and with $C_{\mathrm{AF}} = 3(N^2-1)/\pi^4$, a constant free of b_0 and Λ.[22] For the sum Φ of $N - \mathrm{Re}\,\mathrm{Tr}\,U_p$ over the 24 plaquettes with a corner at 0, $U_p = \exp(ia_k^2 g_k F^a_{\mu\nu}T^a + \cdots)$ gives $N - \mathrm{Re}\,\mathrm{Tr}\,U_p = \frac{1}{4}a_k^4 g_k^2 F^a_{\mu\nu}F^a_{\mu\nu} + \cdots$ (no sum) for p in the plane $\{\mu,\nu\}$, and each plane carries four of the 24 plaquettes, so $\Phi_x = \frac{1}{2}a_k^4 g_k^2 F^a_{\mu\nu}F^a_{\mu\nu} + \cdots = -(a_k^4/b_0)\Theta(1+O(g_k^2))$, up to a constant and fields of higher dimension. Thus $Z_k a_k^4 \to b_0$ makes $Z_k A_k(f)$ tend to $-\Theta(f)$; then (AF) holds with $\gamma = 2$ and Z_k of order a_k^{-4}, and $G_k(z)$ at a fixed lattice distance is of order g_k^4, that is of order $\beta_{W,k}^{-2}$ under (AF_β).

[22] For SU(2), $C_{\mathrm{AF}} = 9/\pi^4 \approx 0.092$; for SU(3), $C_{\mathrm{AF}} = 24/\pi^4 \approx 0.246$.

Theorem 279. *Assume the hypothesis of Theorem* 64 *and* (AF) *for the constants* Z_k.

(i) *Let* Q *be a cube of side* L *and* $\delta \in (0,1]$. *For* $k \geq k_0$ *with* $a_k \leq \min(1, \delta/(2r+1))$ *and for* $(f,g) \in \mathscr{C}_2(Q,\delta)$,

$$Z_k^2\big|\omega_k\big(A_k(f)A_k(g)\big)\big| \leq 100\,C_1(1+L)^4\delta^{-4}\|f\|_\infty\|g\|_\infty.$$

So ($\mathrm{M}_{\mathrm{s},2}$) *holds; every limit* S_2 *along a subsequence satisfies*

$$|S_2(f,g)| \leq C_1 \iint |f(x)||g(y)|w_\gamma(|x-y|)\,dx\,dy$$

on the separated pairs, and extends uniquely to a continuous linear functional on $\mathscr{S}_0(\mathbb{R}^8)$ *with* $|S_2(F)| \leq C|F|_{33}$.

(ii) *If* $f,g \geq 0$ *form a separated pair and* $|x-y| \leq r_0$ *for* $x \in \operatorname{supp} f$, $y \in \operatorname{supp} g$, *then* $S_2(f,g) \geq c_1 \iint f(x)g(y)w_\gamma(|x-y|)\,dx\,dy$. *In particular no limit* S_2 *vanishes identically, and the moment bound* (M) *of Sec.* X I, *posed for the packets* $Z_k A_k(f)$, *fails for* $n = 2$ *and* $Q = [0,1]^4$.

(iii) *For every* $\varepsilon > 0$, $\mathscr{N}_k(\varepsilon) > 0$ *for large* k *and*

$$0 < \liminf_k Z_k^2 \mathscr{N}_k(\varepsilon) \leq \limsup_k Z_k^2 \mathscr{N}_k(\varepsilon) < \infty.$$

(iv) (SD_p) *holds with* $p_0 = 4$, *hence* (SD) *and* (UV), *and* (NE) *holds. Every limit* S_2 *of Corollary* 255(b) *is not identically zero, extends to* $\mathscr{S}_0(\mathbb{R}^8)$ *as in Proposition* 275(b), *and its Hamiltonian* $H^{(1)}$ *has a nonempty spectrum in* $[m_*,\infty)$.

(v) *If the transfer matrices* $\mathbb{T}_k$ *have spectrum on* $\Omega_k^\perp$ *in* $[0, e^{-m_{\mathrm{lat}}}]$ *for a fixed* $m_{\mathrm{lat}} > 0$ *and all* k, *then* (AF) *fails for every choice of* Z_k; *this is the case under the hypotheses of Corollary* 251.

Proof. (i) By Lemma 247(i) and translation invariance,

$$\omega_k(A_k(f)A_k(g)) = \sum_{x,y} a_k^8 f(a_k x)g(a_k y)G_k(y-x).$$

A nonzero term has $|a_k x - a_k y|_\infty \geq \delta$, so $|y-x|_\infty \geq \delta/a_k \geq 2r+1$, and (AF) with $\ell \geq 1$ bounds $Z_k^2|G_k(y-x)|$ by $C_1(a_k|y-x|)^{-8} \leq C_1(a_k|y-x|_\infty)^{-8}$. There are $(2j+1)^4 - (2j-1)^4 = 64j^3 + 16j \leq 80j^3$ points $v \in \mathbb{Z}^4$ with $|v|_\infty = j \geq 1$, so with $J = \lceil \delta/a_k \rceil$

$$\sum_{|v|_\infty \geq J} a_k^4 (a_k|v|_\infty)^{-8} \leq 80 a_k^{-4} \sum_{j \geq J} j^{-5} \leq 80 a_k^{-4}\Big(J^{-5} + \frac{J^{-4}}{4}\Big) \leq 100\,\delta^{-4},$$

while at most $(L/a_k+1)^4$ sites x have $a_k x \in Q$, and $a_k^4(L/a_k+1)^4 \leq (1+L)^4$. This gives the bound, and ($\mathrm{M}_{\mathrm{s},2}$) follows since

$\mathscr{C}_2(Q,\delta)\subset\mathscr{C}_2(Q,1)$ for $\delta\ge 1$. In the same sum $Z_k^2|G_k(y-x)|\le C_1 w_\gamma(a_k|y-x|)$, and $(x,y)\mapsto|f(x)||g(y)|w_\gamma(|x-y|)$ is continuous with compact support in $\mathbb{R}^8$, since w_γ is continuous on $(0,\infty)$; its Riemann sums converge to its integral, which gives the bound on S_2. The extension is Lemma 272 with $n=2$, $C=100C_1$, $q'=p'=4$ and $w=4+4+24+1=33$.

(ii) For large k every nonzero term of the sum has $|y-x|_\infty\ge 2r+1$ and $a_k|y-x|\le r_0$, so the lower bound of (AF) gives $Z_k^2\omega_k(A_k(f)A_k(g))\ge c_1\sum_{x,y}a_k^8 f(a_kx)g(a_ky)w_\gamma(a_k|y-x|)$, and the Riemann sums converge as in (i). The integral is positive when f and g are not zero. For (M), let $s_0=r_0/2$ and $0<\delta\le r_0/16$; let f be continuous with values in $[0,1]$, equal to 1 on $[\frac12-2\delta,\frac12-\delta]\times[\frac{s_0}{4},\frac{3s_0}{4}]^3$ and supported in $[\frac12-3\delta,\frac12-\frac{\delta}{2}]\times[0,s_0]^3$, and g continuous with values in $[0,1]$, equal to 1 on $[\frac12+\delta,\frac12+2\delta]\times[\frac{s_0}{8},\frac{7s_0}{8}]^3$ and supported in $[\frac12+\frac{\delta}{2},\frac12+3\delta]\times[0,s_0]^3$. The supports lie in $[0,1]^4$, at distance δ, and all their points are within $(36\delta^2+3s_0^2)^{1/2}\le r_0$ of each other. The pairs with x in the first box, $x_0+\delta\le y_0\le x_0+4\delta$ with $y_0\in[\frac12+\delta,\frac12+2\delta]$, and $|y_i-x_i|\le\delta$ for $i=1,2,3$ have measure $\frac{\delta s_0^3}{8}\cdot\delta\cdot 8\delta^3=s_0^3\delta^5$, $f(x)g(y)=1$ and $2\delta\le|x-y|\le\sqrt{19}\,\delta<5\delta$, where $w_\gamma\ge(5\delta)^{-8}\ell(2\delta)^{-\gamma}$. Hence $S_2(f,g)\ge c_1 5^{-8}s_0^3\delta^{-3}(1+\log\frac{1}{2\delta})^{-\gamma}$, which is unbounded as $\delta\to0$, while $\|f\|_\infty\|g\|_\infty=1$; under (M) the limit would be bounded by $M_2(Q)$, as in the proof of Proposition 252.

(iii) Let Q_0 be the cube of side $L_0=2(\varepsilon+2+2a_0)$ of the proof of Proposition 275(a), which contains $B_{\varepsilon'}$ and $\theta_{0,k}(B_{\varepsilon'})$ for all k and $\varepsilon'\le\varepsilon$. For h supported in $B_{\varepsilon'}$ with $\|h\|_\infty\le1$ and $a_k\le\varepsilon'$, $\omega_k(\vartheta\overline{(A_k(h))}A_k(h))=\omega_k(A_k(\bar h\circ\theta_{0,k})A_k(h))$, and the pair lies in $\mathscr{C}_2(Q_0,\varepsilon')$; so by (i)

$$Z_k^2\mathscr{N}_k(\varepsilon')\le 100C_1(1+L_0)^4\max(1,\varepsilon'^{-4})$$

for large k, which with $\varepsilon'=\varepsilon$ is the upper bound. Suppose that $Z_k^2\mathscr{N}_k(\varepsilon)\to0$ along a subsequence. By ($M_{s,2}$) and the proof of Theorem 248(i) a further subsequence has a limit S_2 on all separated pairs. Let g be supported in a cube of side L in $\{x_0\ge\eta\}$, $\eta>0$, and $t\ge4\varepsilon+1$ dyadic. For large k, $\bar g\circ\theta_{0,k}$ is supported in $\{x_0\le-\eta+a_k\}$ and $g(\cdot-te_0)$ in $\{x_0\ge\eta+t\}$, so with $c=2\varepsilon-\eta+a_k$ and $d=0$ the inequality of Theorem 254(i), multiplied by $\mathscr{N}_k(\varepsilon)$, whose proof does not use $\mathscr{N}_k(\varepsilon)>0$, gives

$$Z_k^2\big|\omega_k\big(A_k(\bar g\circ\theta_{0,k})A_k(g(\cdot-te_0))\big)\big|\le \mathsf{K}_L^2Z_k^2\mathscr{N}_k(\varepsilon)\|g\|_\infty^2\to0.$$

By the approximation of $\theta_{0,k}$ by θ_0 in the proof of Theorem 248(iii), $S_2(\bar g\circ\theta_0,g(\cdot-te_0))=0$. The construction in the proof of Corollary 255(b), which uses only ($M_{s,2}$), gives a Hilbert space $\mathscr{H}^{(1)}$ and a self-adjoint $H^{(1)}\ge0$ with $S_2(\bar g\circ\theta_0,g(\cdot-te_0))=\|e^{-tH^{(1)}/2}w\|^2$, w the vector of g. Since $e^{-tH^{(1)}/2}$ is injective, $w=0$, that is $S_2(\bar g\circ\theta_0,g)=0$, for every g supported in $\{x_0>0\}$, and by the proof of Corollary 255(b) S_2 vanishes identically. This contradicts (ii). So $\liminf_k Z_k^2\mathscr{N}_k(\varepsilon)>0$, and in particular $\mathscr{N}_k(\varepsilon)>0$ for large k.

(iv) Let $c_\varepsilon=\frac12\liminf_k Z_k^2\mathscr{N}_k(\varepsilon)$ and $C_2=100C_1(1+L_0)^4$. By (iii), for $\varepsilon'\in(0,\varepsilon]$ and large k,

$$\begin{aligned}\mathscr{N}_k(\varepsilon')&\le\frac{C_2\max(1,\varepsilon'^{-4})}{c_\varepsilon}\,\mathscr{N}_k(\varepsilon)\\&\le C_0\Big(\frac{\varepsilon}{\varepsilon'}\Big)^4\mathscr{N}_k(\varepsilon),\end{aligned}$$

where $C_0=C_2\max(1,\varepsilon^{-4})/c_\varepsilon$; the second inequality holds because $\max(1,\varepsilon'^{-4})$ is at most $\max(1,\varepsilon^{-4})(\varepsilon/\varepsilon')^4$. This is ($SD_p$) with $p_0=4$; it gives (SD), and (UV) by Proposition 256(ii). Let $S_2^\sharp$ be a limit of Corollary 255(b), formed with $\mathscr{N}_k(\varepsilon)^{-1}$ in place of Z_k^2. By (i) and (iii) a further subsequence has $Z_k^2\mathscr{N}_k(\varepsilon)\to\lambda\in(0,\infty)$ and a limit S_2 with the constants Z_k, and then $S_2^\sharp=S_2/\lambda$, which is not identically zero by (ii). By Proposition 258(i) this is (NE), and Corollary 255(b) and Proposition 275(b) give the rest.

(v) Under (AF), (iii) and (iv) give $\mathscr{N}_k(\varepsilon)>0$ for large k and (SD), which Proposition 256(iii) excludes under a lattice gap; Corollary 251 provides the gap. □

So the short-distance law, in the two-sided form (AF), gives the two inputs (SD) and (NE) of Sec. X L for $n=2$, with the power rate (SD_p) that temperedness needs, and it excludes the trivial limits of the strong-coupling range. It says nothing about $n\ge3$: the bounds (RN_m) with $m\ge2$ of Proposition 259 are not consequences of it. It also fixes the constants Z_k up to bounded factors.

Proposition 280. *Assume* (AF) *for the constants Z_k, and let $z_0\in\mathbb{Z}^4$ with $|z_0|_\infty\ge2r+1$.*

(i) *For large k, $G_k(z_0)>0$ and*

$$c_1|z_0|^{-8}\le Z_k^2a_k^8\,\ell(a_k|z_0|)^\gamma G_k(z_0)\le C_1|z_0|^{-8}.$$

(ii) *$Z_k\ge c_1^{1/2}|z_0|^{-4}\|\Phi\|_\infty^{-1}a_k^{-4}\ell(a_k|z_0|)^{-\gamma/2}$ for large k; in particular $Z_k\to\infty$.*

(iii) *For large k and every $z\in\mathbb{Z}^4$ with $|z|_\infty\ge2r+1$ and $a_k|z|\le r_0$,*

$$\frac{c_1}{C_1}\,\frac{w_\gamma(a_k|z|)}{w_\gamma(a_k|z_0|)}\le\frac{G_k(z)}{G_k(z_0)}\le\frac{C_1}{c_1}\,\frac{w_\gamma(a_k|z|)}{w_\gamma(a_k|z_0|)}.$$

(iv) *Under the hypothesis of Theorem 64, if* (AF) *also holds for constants Z_k' with an exponent γ', then $\gamma'=\gamma$ and Z_k'/Z_k is bounded above and below by positive constants; both are comparable to $\mathscr{N}_k(\varepsilon)^{-1/2}$ for every $\varepsilon>0$.*

(v) *Under* (AF_β), *$Z_ka_k^4$ is bounded above and below by positive constants if and only if $\beta_{W,k}^\gamma G_k(z_0)$ is.*

Proof. (i) is (AF) at $z=z_0$, valid for large k because $a_k|z_0|\to0$. (ii) follows from (i), since $|G_k(z_0)|=|\operatorname{Cov}(\Phi_0,\Phi_{z_0})|\le\operatorname{Var}(\Phi_0)\le\|\Phi\|_\infty^2$ by the Cauchy–Schwarz inequality and translation invariance, and $\ell(a_k|z_0|)^\gamma a_k^8\to0$. (iii) is the quotient of the two bounds of (AF) at z and z_0. For (iv), Theorem 279(iii) applied to Z_k and to Z_k' makes $Z_k^2\mathscr{N}_k(\varepsilon)$ and $Z_k'^2\mathscr{N}_k(\varepsilon)$ bounded above and below, hence Z_k'/Z_k; and (i) for both gives $\ell(a_k|z_0|)^{\gamma-\gamma'}$ bounded above and below, which forces $\gamma'=\gamma$ since $\ell(a_k|z_0|)\to\infty$. For (v), (AF_β) gives

$\beta_{W,k}/\log(1/a_k) \to 4Nb_0 > 0$, and $\ell(a_k|z_0|)/\log(1/a_k) \to 1$, so by (i) $Z_k^2 a_k^8$ is bounded above and below exactly when $\beta_{W,k}^{\gamma} G_k(z_0)$ is. □

By Remark 278, perturbation theory predicts $\gamma = 2$ and $G_k(z_0)$ of order $\beta_{W,k}^{-2}$, hence Z_k of order a_k^{-4}, and for the plaquette sum $Z_k a_k^4 \to b_0$ in the normalisation of Θ. Part (iii) is the content of (AF) without reference to Z_k: the connected correlation of Φ decays from the lattice distance $|z_0|$ to every distance up to r_0/a_k like the eighth power of the distance, with the logarithmic correction, within fixed factors. Its lower half is a lower bound on correlations at physical separations, the kind of input that upper bounds do not supply (compare Proposition 201).

The two-sided law is consistent with the other properties of Sec. X L and implies neither ($\mathrm{W_s}$) nor the exact law (AF_∞).

Proposition 281. *Let $\gamma \geq 0$, $\mu_j = 2^{4(j-1)}(1+j)^{-\gamma/2}$ for $j \geq 0$, let $a_k \leq 1$ and $J_k = \lfloor \log_2(2/a_k) \rfloor$, let $\phi_k^{(j)}$, $0 \leq j \leq J_k$, be independent copies of the field of Proposition 199(a) with q_k replaced by $q_{k,j} = e^{-2^{j-1}a_k}$, and put $\phi_k = \sum_{j \leq J_k} \mu_j \phi_k^{(j)}$, $A_k(f) = \sum_x a_k^4 f(a_k x)\phi_k(x)$, $r = 0$ and $Z_k = 1$.*

(a) *The law of ϕ_k is invariant under translations and B_4 and reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links, and the spectrum of its transfer matrix in each coordinate direction on $\Omega_k^{\perp}$ lies in $[0, e^{-a_k/2}]$ and contains $e^{-a_k/2}$: the gap is $\frac{1}{2}$ in physical units.*

(b) (AF) *holds with this γ, $r_0 = \frac{1}{2}$, $c_1 = e^{-1}2^{-16-\gamma}$ and $C_1 = \max(2^{\gamma}(A_\gamma + B), B')$, where $A_\gamma = \sum_{i \geq 0} 2^{-8i}(1+i)^{\gamma}$, $B = \sum_{i \geq 1} 2^{8i} e^{-2^{2i-1}}$ and $B' = \sum_{j \geq 0} \sup_{t \geq 2^j} t^8 e^{-t}$.*

(c) *The moments converge along the whole sequence to the Gaussian family with $S_2(f,g) = \iint f(x)g(y)K(x-y)\,dx\,dy$, $K(v) = h(\|v\|_1)$, $h(u) = \sum_{j \geq 0} \mu_j^2 e^{-2^{j-1}u}$.* ($\mathrm{G_s}$) *holds with $\beta = \frac{1}{2}$ and $p = q = 2$, and* ($\mathrm{U_s}$) *holds; the conclusions of Theorem 279(i)–(iv) hold with $m_* = \frac{1}{2}$. The group G_S is B_4, so* ($\mathrm{W_s}$) *fails, and* (AF_∞) *fails.*

Proof. (a) As in the proof of Proposition 257(a), with $J_k + 1$ independent fields in place of two: the tuple is a reversible Markov chain in each coordinate direction with transfer operator $\bigotimes_j P_j$, of spectrum $\{\prod_j q_{k,j}^{N_j}\}$ with 1 simple, the functions of ϕ_k at positive times are functions of the tuple there, and the vector of $\phi_k(x)$ at positive time has a nonzero component of eigenvalue $q_{k,0} = e^{-a_k/2}$, the largest $q_{k,j}$.

(b) The covariance is $G_k(z) = \sum_{j \leq J_k} \mu_j^2 e^{-2^{j-1}u}$ with $u = a_k\|z\|_1$; for $z \neq 0$ let $\rho = a_k|z|$, so that $\rho \leq u \leq 2\rho$ and $u \geq a_k$. If $u \leq 2$, let $j_* = \lfloor \log_2(2/u) \rfloor \geq 0$, so that $\frac{1}{2} < 2^{j_*-1}u \leq 1$. The terms with $j = j_* - i$, $i \geq 0$, are at most $\mu_{j_*}^2 2^{-8i}(1+i)^{\gamma}$, because $(1+j_*)/(1+j) \leq 1+i$; those with $j = j_* + i$, $i \geq 1$, at most $\mu_{j_*}^2 2^{8i} e^{-2^{i-1}}$, because $2^{j-1}u > 2^{i-1}$. So $G_k(z) \leq (A_\gamma + B)(2^{j_*-1})^8(1+j_*)^{-\gamma} \leq (A_\gamma + B)\rho^{-8}(1+j_*)^{-\gamma}$. Here $1 + j_* \geq \max(1, \log_2(2/u)) \geq \max(1, \log_2(1/\rho))$, which is at least $\frac{1}{2}\ell(\rho)$, so $G_k(z) \leq 2^{\gamma}(A_\gamma + B)w_\gamma(\rho)$. If $u > 2$, then $\rho > 1$, $2^{j-1}u \geq 2^j$ and $\mu_j^2 \leq 2^{8(j-1)}$, so $G_k(z) \leq u^{-8}\sum_j (2^{j-1}u)^8 e^{-2^{j-1}u} \leq B' u^{-8} \leq B' w_\gamma(\rho)$. For the lower bound let $\rho \leq \frac{1}{2}$, so $u \leq 1$; then $j_* \leq J_k$ because $u \geq a_k$, and the term j_* alone is at least $(2u)^{-8}(1+j_*)^{-\gamma}e^{-1}$. Since $(2u)^{-8} \geq (4\rho)^{-8}$ and $1 + j_* \leq 1 + \log_2(2/\rho) = 2 + \log(1/\rho)/\log 2 \leq 2\ell(\rho)$, because $1/\log 2 < 1.5$, this is at least $e^{-1}2^{-16-\gamma}w_\gamma(\rho)$. The finiteness of A_γ, B, B' follows from the ratio test, since $t^8 e^{-t}$ decreases for $t \geq 8$.

(c) $\omega_k(A_k(f)A_k(g)) = \sum_{x,y} a_k^8 f(a_k x)g(a_k y)h_k(a_k\|y-x\|_1)$ with h_k the partial sum of h up to J_k. On a separated pair at distance δ only $u \geq \delta$ occurs, where $0 \leq h - h_k \leq \sum_{j > J_k} \mu_j^2 e^{-2^{j-1}\delta} \to 0$, and h is continuous on $(0,\infty)$; so the Riemann sums converge to $\iint f(x)g(y)K(x-y)\,dx\,dy$. The fields are centred Gaussian, so the moments of separated tuples are sums over the $(n-1)!!$ pairings of products of these two-point functions, and they converge as well; this gives ($\mathrm{U_s}$). Theorem 279(i) and $(n-1)!! \leq (n!)^{1/2}$ bound the moments of tuples in $\mathscr{C}_n(Q,\delta)$, $\delta \leq 1$, by $(100C_1)^{n/2}(n!)^{1/2}(1+L)^{2n}\delta^{-2n}\prod_j \|f_j\|_\infty$ for large k, which is ($\mathrm{G_s}$). The proofs of Theorem 279(i)–(iv) use the states only through their invariance, reflection positivity and gap, as in Remark 293. K is continuous on $\mathbb{R}^4 \setminus \{0\}$ and h is strictly decreasing, so by Proposition 261(i) an $R \in \mathrm{O}(4)$ lies in G_S only if $\|Rv\|_1 = \|v\|_1$ for all v; then R permutes the vertices $\pm e_\mu$ of the unit ball of $\|\cdot\|_1$, and $R \in B_4$. Finally, choose i_1 with $((1+i)/(2+i))^{\gamma} \geq \frac{1}{2}$ for $i \geq i_1$. Since $\mu_{i+1}^2 = 2^8((1+i)/(2+i))^{\gamma}\mu_i^2$,

$$h(\rho) \geq \sum_{i \geq i_1} \mu_{i+1}^2 e^{-2^i\rho} \geq 2^7\Big(h(2\rho) - \sum_{i < i_1} \mu_i^2\Big),$$

and $h(2\rho) \to \infty$ as $\rho \to 0$, so $\liminf_{\rho\to 0} h(\rho)/h(2\rho) \geq 2^7$. With $v = \rho e_0$ and $v' = \frac{\rho}{2}(1,1,1,1)$, $|v| = |v'| = \rho$, $\|v\|_1 = \rho$ and $\|v'\|_1 = 2\rho$, while (AF_∞), K being continuous, would give $K(v)/K(v') \to 1$. □

The example is not a lattice gauge theory. With (AF_∞) in place of (AF), the leading term at short distance is rotation invariant: if K is as in (AF_∞) and square integrable on compact subsets of $\mathbb{R}^4 \setminus \{0\}$, the components $K_{2j}(\rho,\cdot)$, $j \geq 1$, of Proposition 261(ii) satisfy $\rho^8(\log\frac{1}{\rho})^{\gamma}\|K_{2j}(\rho,\cdot)\|_{L^2(S^3)} \to 0$ as $\rho \to 0$ along almost every ρ, because the projection onto $\mathscr{Y}_{2j}$, $j \geq 1$, has norm one and annihilates the constant C_{AF}. This is not ($\mathrm{W_s}$), which asks that these components vanish.

Remark 282. *Status of (AF).* No part of (AF), (AF_∞) or (AF_β) is proved for the Wilson measure, in any regime of couplings that tends to the continuum, and (AF) fails in the strong-coupling range (Theorem 279(v)). Bałaban's renormalisation group [6, 7, 10, 11, 13] constructs effective actions in a small-field approximation with a coupling constant renormalisation by recursive renormalisation-group equations [10], and proves the ultraviolet stability of four-dimensional pure lattice gauge theories, that is, bounds on the partition function in a box of fixed physical size uniform in the spacing [13]; these bounds concern the partition function, not the two-point function of a local field, and the box is fixed. Magnen, Rivasseau and Sénéor [80] give the basis for a construction of the Schwinger functions of the SU(2) gauge field with a fixed infrared cut-off and no ultraviolet cut-off, in a regularised axial gauge; these are correlations of the gauge field in a gauge, with an infrared

cut-off, not of a gauge-invariant composite field such as Φ, and they are not states of the Wilson measure along a lattice tuning. An asymptotically free ultraviolet limit has been constructed by convergent expansions in the running coupling for the two-dimensional Gross–Neveu model [56], not for a gauge theory in four dimensions. What is missing is exactly the following inequality, for Φ the plaquette sum, the torus limits ω_k along the tuning (AF_β), $\gamma = 2$ and some $Z_k > 0$, $0 < c_1 \le C_1$, $r_0 > 0$: for large k and all $z \in \mathbb{Z}^4$ with $|z|_\infty \ge 2r+1$, where $G_k(z) = \omega_k(\Phi_0\Phi_z) - \omega_k(\Phi_0)^2$,

$$Z_k^2 |G_k(z)| \le C_1 w_2(a_k|z|),$$

and $Z_k^2 G_k(z) \ge c_1 w_2(a_k|z|)$ if moreover $a_k|z| \le r_0$; equivalently, by Proposition 280(iii), the Z_k-free two-sided comparison of $G_k(z)$ with $G_k(z_0)\, w_2(a_k|z|)/w_2(a_k|z_0|)$ for $2r+1 \le |z|_\infty$, $a_k|z| \le r_0$, together with the upper bound for $a_k|z| > r_0$ with $Z_k^2 = w_2(a_k|z_0|)/G_k(z_0)$. It is open, both halves, already in a box of fixed physical size. Given it, Theorem 279 supplies ($\mathrm{SD_p}$) and (NE) for $n = 2$; (RN_m) for $m \ge 2$, ($\mathrm{W_s}$), ($\mathrm{U_s}$) and the averaged condition along the tuning stay open.

Products and the two-point properties. After Proposition 259 the separated moments of order $n \ge 3$ rest on (RN_m), $m \ge 2$, and the rate ($\mathrm{G_s}$) on a rate in these bounds. Functions of two independent copies of the field of Proposition 199(a) show what the two-point properties do not give. Write γ_2 for the standard Gaussian measure on $\mathbb{R}^2$, $\|\cdot\|_p$ for the norm of $L^p(\gamma_2)$, and $T_M(u) = \max(-M, \min(M,u))$ for $M > 0$.

Lemma 283. *Let $m > 0$, $q_k = e^{-ma_k}$, let $\phi_k^{(1)}, \phi_k^{(2)}$ be independent copies of the field of Proposition 199(a), $\xi_x = (\phi_k^{(1)}(x), \phi_k^{(2)}(x))$, let $U : \mathbb{R}^2 \to \mathbb{R}$ be continuous, in $L^p(\gamma_2)$ for every $p < \infty$, with $\int U\, d\gamma_2 = 0$, and put $\Phi_x = U(\xi_x)$ and $A_k(f) = \sum_x a_k^4 f(a_k x)\Phi_x$, with $r = 0$.*

(i) *The law of Φ is invariant under translations and B_4 and reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links, and in each coordinate direction the spectrum of its transfer matrix on $\Omega_k^\perp$ lies in $[0, q_k]$. It contains q_k if $\int U(u) u_i\, d\gamma_2(u) \ne 0$ for $i = 1$ or $i = 2$.*

(ii) *Let $U_M = T_M \circ U - \int T_M \circ U\, d\gamma_2$, so that $|U_M| \le 2M$. For $n \ge 1$ and random vectors $\zeta_1, \dots, \zeta_n$ in $\mathbb{R}^2$, each of law γ_2,*

$$\Big|\mathbb{E}\prod_i U(\zeta_i)\Big| \le \|U\|_n^n,$$

$$\Big|\mathbb{E}\prod_i U_M(\zeta_i) - \mathbb{E}\prod_i U(\zeta_i)\Big| \le e_n(M),$$

with $e_n(M) = 2n(2\|U\|_n)^{n-1}\|U\mathbf{1}_{\{|U|>M\}}\|_n \to 0$ as $M \to \infty$. Hence, if f_j is supported in a cube of side L_j, the moments $\omega_k(\prod_j A_k(f_j))$ formed with U_M and with U differ by at most $e_n(M)\prod_j (L_j + a_k)^4 \|f_j\|_\infty$.

Proof. (i) As in the proof of Proposition 257(a), the pair $(\phi_k^{(1)}, \phi_k^{(2)})$ is invariant, reflection positive, and in each coordinate direction a reversible Markov chain with transfer operator $P \otimes P$, of spectrum $\{q_k^{N+N'}\}$ with 1 simple; the functions of Φ at positive times are functions of the pair there, so Φ has the stated invariance and positivity, and its transfer matrix is the restriction of $P \otimes P$ to an invariant subspace. For a site x at time $s \ge 1$, g_{Φ_x} is $(P\otimes P)^s$ applied to $U(\xi_x)$, viewed as a function of the field at one time, and $P \otimes P$ acts as q_k on the linear functions of that field. By Gaussian integration by parts the orthogonal projection of $U(\xi_x)$ on these linear functions has inner product $\int U(u) u_i\, d\gamma_2(u)$ with $\phi_k^{(i)}(x)$, so it is not zero, and g_{Φ_x} has a nonzero component of eigenvalue q_k.

(ii) The first bound is Hölder's inequality with n exponents n. Since $|T_M u| \le |u|$, $|T_M u - u| \le |u| \mathbf{1}_{\{|u|>M\}}$ and $\int U\, d\gamma_2 = 0$,

$$\|U_M\|_n \le 2\|U\|_n, \qquad \|U_M - U\|_n \le 2\|U\mathbf{1}_{\{|U|>M\}}\|_n.$$

The difference of the two products is the sum over j of the products $\prod_{i<j} U_M(\zeta_i)\,(U_M - U)(\zeta_j)\prod_{i>j} U(\zeta_i)$, and Hölder's inequality for each of them gives the second bound. By dominated convergence $e_n(M) \to 0$, as $|U|^n$ is integrable. The moments are $\sum_{x_1,\dots,x_n} a_k^{4n} \prod_j f_j(a_k x_j)\mathbb{E}\prod_j U(\xi_{x_j})$, each ξ_{x_j} has law γ_2, and $\sum_x a_k^4 |f_j(a_k x)| \le (L_j + a_k)^4 \|f_j\|_\infty$. □

The moments of exponentials are explicit. For a centred Gaussian vector $(X_1, \dots, X_n)$ with unit variances and covariances $\rho_{ii'}$, $t > 0$, and $u_{ii'} = e^{t^2\rho_{ii'}} - 1$,

$$\mathbb{E}\prod_{i=1}^n \big(e^{tX_i - t^2/2} - 1\big) = \sum_G \prod_{\{i,i'\}\in G} u_{ii'}, \tag{40}$$

the sum over the sets G of pairs whose union is $\{1, \dots, n\}$. Indeed $\mathbb{E}\prod_{i\in S} e^{tX_i - t^2/2} = \prod_{\{i,i'\}\subset S}(1 + u_{ii'})$ for $S \subset \{1,\dots,n\}$; expanding the product of the factors $e^{tX_i - t^2/2} - 1$ and these products, the coefficient of $\prod_{\{i,i'\}\in G} u_{ii'}$ is the sum of $(-1)^{n-|S|}$ over the S that contain every pair of G, which is 1 if G covers $\{1,\dots,n\}$ and 0 otherwise. When all $\rho_{ii'} \ge 0$, every term is nonnegative and nondecreasing in each $\rho_{ii'}$.

Proposition 284. *In the setting of Lemma 283 let $t_k > 0$ with $t_k \to \infty$, $\lambda_k = e^{-t_k^2/2}$ and*

$$U_k(u) = u_1 + \lambda_k\big(e^{t_k u_2 - t_k^2/2} - 1\big),$$

let $\varepsilon > 0$, form $\mathscr{N}_k(\varepsilon)$ with $\Phi_x = U_k(\xi_x)$, and put $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$ and $\ell_0 = (\frac{3}{8})^8 e^{-m(2\varepsilon+6)}$.

(a) *The spectrum of the transfer matrix on $\Omega_k^\perp$ lies in $[0, q_k]$ and contains q_k: the gap is m in physical units. For large k, $\ell_0 \le \mathscr{N}_k(\varepsilon) \le 2(1+2a_0)^8$. ($\mathrm{SD_p}$) holds with $p_0 = 0$; hence* (SD), (UV), (RN_1) *and* ($\mathrm{M_{s,2}}$) *hold.* (NE) *holds, and every limit S_2 of Corollary 255(b) is $S_2(f,g) = z\iint f(x)g(y)e^{-m\|x-y\|_1}dxdy$ with $z \in [\frac{1}{2}(1+2a_0)^{-8}, \ell_0^{-1}]$.*

(b) *Let $n \ge 3$, $\eta = 1/(4m(3n+5))$, $c_j = 3j\eta e_1$, and let $f_1, \dots, f_n$ be continuous with values in $[0,1]$, f_j equal to 1 on $c_j + [-\frac{\eta}{2}, \frac{\eta}{2}]^4$ and supported in $c_j + [-\eta, \eta]^4$. Then $(f_j) \in \mathscr{C}_n(Q, \eta)$ for a cube Q of side $3n\eta$, and for large k*

$$Z_k^n\, \omega_k\Big(\prod_{j=1}^n A_k(f_j)\Big) \ge \frac{(\eta/2)^{4n} e^{nt_k^2/4}}{2^{n(n-1)/2}(2(1+2a_0)^8)^{n/2}} \to \infty.$$

So ($\mathrm{M_{s,}}n$) *fails for every $n \ge 3$.*

(c) *Let $j \ge 2$, $\eta' = 1/(4m(3j+10))$, and let $f_1, \dots, f_j$ be as in* (b)*, with η' in place of η and centres $2\eta' e_0 + 3i\eta' e_1$,*

$i=1,\dots,j$. *They are supported in* $\{x_0\ge\eta'\}$, $(f_i)\in\mathscr{C}_j(Q',\eta')$ *for a cube* Q' *of side* $3j\eta'$, *and for large* k, *with* $P_k=\prod_i A_k(f_i)$,

$$Z_k^{2j}\,\omega_k\big(\vartheta(\bar P_k)P_k\big)\ge\frac{(\eta'/2)^{8j}e^{jt_k^2/2}}{2^{j(2j-1)}(2(1+2a_0)^8)^j}\to\infty.$$

So (RN_j) *fails for every* $j\ge2$.

(d) *There are* $M_k\to\infty$ *such that* (a)–(c) *hold for the fields* $\Phi_x=U_{k,M_k}(\xi_x)/(2M_k)$, *with* $U_{k,M}$ *formed from* U_k *as in Lemma* 283(ii), *which take values in* $[-1,1]$, *with* $\frac12\ell_0$ *and* $4(1+2a_0)^8$ *in place of* ℓ_0 *and* $2(1+2a_0)^8$ *and the right sides of* (b), (c) *halved. For these fields* $Z_k\to\infty$.

Proof. (a) The spectrum is given by Lemma 283(i), since $\int U_k u_1\,d\gamma_2=1$. Let $\rho(z)=\prod_\mu q_k^{|z_\mu|}=e^{-ma_k\|z\|_1}$ be the covariance of $\phi_k^{(i)}$ and $E_x=e^{t_k\phi_k^{(2)}(x)-t_k^2/2}$. By independence and centring, for h supported in $B_{\varepsilon'}$ with $\|h\|_\infty\le1$ and $(r+1)a_k\le\varepsilon'$,

$$\omega_k\big(\vartheta(\overline{A_k(h)})A_k(h)\big)=\sum_{x,y}a_k^8\overline{h(a_kx)}h(a_ky)\big(\rho+\lambda_k^2(e^{t_k^2\rho}-1)\big),$$

with $\rho=\rho(y-\vartheta x)$, since $\mathbb{E}[(E_x-1)(E_y-1)]=e^{t_k^2\rho(x-y)}-1$ by (40). For $a_kx_0,a_ky_0\ge\varepsilon'$ the time component of $y-\vartheta x$ is at least $2\varepsilon'/a_k-1$, so $\rho\le e^{-m(2\varepsilon'-a_k)}$, and $\lambda_k^2e^{t_k^2\rho}\le e^{-t_k^2(1-e^{-m(2\varepsilon'-a_k)})}\to0$. As $\sum_x a_k^4|h(a_kx)|\le(1+2a_0)^4$ and $\rho\le1$, $\mathscr{N}_k(\varepsilon')\le2(1+2a_0)^8$ for large k. For the function $h_*\ge0$ of the proof of Proposition 257(b), formed for ε with $\tau=\frac14$, both parts are nonnegative, and that proof gives $\sum_{x,y}a_k^8h_*(a_kx)h_*(a_ky)\rho(y-\vartheta x)\ge\ell_0$; so $\mathscr{N}_k(\varepsilon)\ge\ell_0$, and the ratio in (NE) with $g_0=h_*$ is at least $\ell_0/(2(1+2a_0)^8)$. Hence $\mathscr{N}_k(\varepsilon')\le C_0\mathscr{N}_k(\varepsilon)$ with $C_0=2(1+2a_0)^8/\ell_0$ for every $\varepsilon'\in(0,\varepsilon]$ and large k, which is (SD_p) with $p_0=0$. (UV) follows by Proposition 256(ii), (RN_1) by Proposition 259(iii) and $(\mathrm{M}_{\mathrm{s},2})$ by Corollary 255(a), whose proofs use the state only through the properties of Lemma 283(i). For $(f,g)\in\mathscr{C}_2(Q,\delta)$, Q of side L, the sites carrying f and g have $\|a_k(x-y)\|_1\ge\delta$, so the part of $Z_k^2\omega_k(A_k(f)A_k(g))$ with λ_k^2 is at most $Z_k^2(L+1)^8e^{-t_k^2(1-e^{-m\delta})}\|f\|_\infty\|g\|_\infty\to0$, and the other part is Z_k^2 times a Riemann sum of the continuous function $f(x)g(y)e^{-m\|x-y\|_1}$; along a subsequence with $Z_k^2\to z$ the limit is the stated one.

(b) Distinct c_j differ by 3η along e_1, so the supports are at ℓ^∞ distance at least η; they lie in $[2\eta,(3n+1)\eta]\times[-\eta,\eta]^3$, of ℓ^1 diameter at most $(3n-1)\eta+6\eta=1/(4m)$. Expanding $\prod_j U_k(\xi_{x_j})$ over the sets I of indices that take the first summand, independence gives

$$\mathbb{E}\prod_j U_k(\xi_{x_j})=\sum_I\mathbb{E}\Big[\prod_{j\in I}\phi_k^{(1)}(x_j)\Big]\lambda_k^{n-|I|}\mathbb{E}\Big[\prod_{j\notin I}(E_{x_j}-1)\Big].$$

The first factor is a sum of products of the covariances $\rho>0$ by Wick's formula, and the second is nonnegative by (40); the term $I=\emptyset$ and the term G of all pairs in (40) give

$$\mathbb{E}\prod_j U_k(\xi_{x_j})\ge\lambda_k^n\prod_{i<i'}\big(e^{t_k^2\rho(x_i-x_{i'})}-1\big).$$

For sites in the supports $\rho\ge e^{-1/4}\ge\frac34$; for $\frac34t_k^2\ge\log2$, each factor is at least $\frac12e^{t_k^2\rho}$, and the right side is at least $2^{-n(n-1)/2}\exp(t_k^2n(\frac38(n-1)-\frac12))\ge2^{-n(n-1)/2}e^{nt_k^2/4}$, since $n\ge3$. All terms of the moment are nonnegative; keeping the sites with $a_kx_j\in c_j+[-\frac{\eta}{2},\frac{\eta}{2}]^4$, at least $(\eta/(2a_k))^4$ for each j when $a_k\le\eta/2$, and using $Z_k^2\ge\frac12(1+2a_0)^{-8}$ from (a) gives the bound.

(c) The supports lie in $[\eta',3\eta']\times[2\eta',(3j+1)\eta']\times[-\eta',\eta']^2$, at ℓ^∞ distance at least η' from each other. By Lemma 247(i), $\vartheta(\bar P_k)=\prod_i A_k(f_i\circ\theta_{0,k})$, and the $2j$ functions $f_i\circ\theta_{0,k}$, f_i are nonnegative with supports in $\{x_0\le-\eta'+a_k\}$ and $\{x_0\ge\eta'\}$, whose union has ℓ^1 diameter at most $(6\eta'+a_k)+(3j-1)\eta'+4\eta'\le(3j+10)\eta'=1/(4m)$ for $a_k\le\eta'$. The proof of (b) with $n=2j$ applies, the map $x\mapsto\vartheta x$ being a bijection of $\mathbb{Z}^4$.

(d) Choose $M_k\ge k$ with $e_n(M_k)\le1/k$ for $n\le k$, e_n formed with U_k, and $\|U_{k,M_k}-U_k\|_2\le\frac12$. Then $\int U_{k,M_k}(u)u_1\,d\gamma_2\ge\frac12$, and Lemma 283 gives the spectrum and shows that, for each fixed n and fixed supports, the moments and the reflected norms formed with U_{k,M_k} differ from those of U_k by $O(1/k)$, uniformly in the functions of norm at most 1 with supports in the given cubes; in particular $\mathscr{N}_k(\varepsilon')$ changes by at most $(1+2a_0)^8/k$. The field $U_{k,M_k}(\xi_x)/(2M_k)$ has packets $A_k/(2M_k)$, reflected norms $\mathscr{N}_k/(4M_k^2)$ and Z_k multiplied by $2M_k$, so the moments of $Z_kA_k(f_j)$ are those formed with U_{k,M_k} and its own $\mathscr{N}_k(\varepsilon)$, and the claims follow from (a)–(c). □

The fields of Proposition 284 have every property that this subsection uses at separated points, the gap included, and (SD), (SD_p), (UV), (NE) and nontrivial two-point limits. The exponential part has variance $\lambda_k^2(e^{t_k^2}-1)=1-e^{-t_k^2}$ at each site, correlations that vanish at every fixed distance, and moments of every order $n\ge3$ that grow like $e^{nt_k^2/4}$ at points closer than $1/(4m)$; it does not enter the two-point functions. So (RN_m), $m\ge2$, does not follow from the invariance, the reflection positivity in the coordinate hyperplanes, the transfer-matrix gap and the two-point properties; any argument that uses the state only through these, such as a chessboard or multiple-reflection estimate or a bound by powers of the transfer matrix, gives neither (RN_2) nor $(\mathrm{M}_{\mathrm{s},3})$. The choice of Z_k does not matter: the unnormalised separated two-point moments are bounded in k, so every normalisation for which some separated two-point limit is nonzero has $\liminf_k Z_k>0$, and for it the moments in (b) and (c) still tend to infinity; this applies to $Z_k=C_k(\sigma_0)^{-1/2}$ of Proposition 250(ii) for $f_0\ge0$, $f_0\ne0$. Part (d) takes into account that Φ is bounded for the Wilson measure: for bounded fields (M_s) is automatic when Z_k is bounded, and in (d) $Z_k\to\infty$.

The rate (G_s) is a rate in (RN). Consider

(RN_p) there are $K_1,\beta_1,p_1,q_1\ge0$ such that for every $m\ge1$, every cube Q of side L and every $\delta\in(0,1]$ there is k_0 with

$$Z_k^{2m}\omega_k\big(\vartheta(\bar P_k)P_k\big)\le K_1^m(m!)^{\beta_1}(1+L)^{q_1m}\delta^{-p_1m}\prod_{j=1}^m\|f_j\|_\infty^2$$

for $P_k=\prod_{j=1}^m A_k(f_j)$, all $k\ge k_0$ and all $(f_1,\dots,f_m)\in\mathscr{C}_m(Q,\delta)$ with supports in $\{x_0\ge\delta\}$.

Proposition 285. *Assume the hypothesis of Theorem* 64.

(i) (G_s) *implies* (RN_p) *with* $K_1 = 4^\beta 7^{2q} K^2$, $\beta_1 = 2\beta$, $q_1 = 2q$, $p_1 = 2p$.

(ii) (RN_p) *implies* (G_s) *with* $K = 2^8 e^4 5^{q_1/2} (16e^2)^{p_1/2} K_1^{1/2}$, $\beta = 4 + \frac{1}{2}(\beta_1 + p_1)$, $q = 4 + \frac{1}{2}q_1$, $p = 4 + \frac{1}{2}p_1$.

(iii) *With* $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$ *as in Theorem 254, the case* $m = 1$ *of the bound in* (RN_p)*, with its own constants, holds if and only if* (SD_p) *holds. So under* (SD_p), (G_s) *holds if and only if the cases* $m \geq 2$ *hold with constants of the form in* (RN_p).

Proof. (i) Let $(f_j) \in \mathscr{C}_m(Q, \delta)$ have supports in $\{x_0 \geq \delta\}$, $Q = \prod_\nu [q_\nu, q_\nu + L]$, let $s \geq \delta$ be the least time in the supports, $n_0 = \lfloor (s - \delta)/a_k \rfloor$, $v_\nu = \lfloor q_\nu / a_k \rfloor$ for $\nu \neq 0$, $v_\perp = (0, v_1, v_2, v_3)$, and $f_j' = f_j(\cdot + a_k(n_0 e_0 + v_\perp))$, supported in $[\delta, L+2] \times [0, L+1]^3$ for $a_k \leq 1$. With $P_k' = \prod_j A_k(f_j')$, $P_k = \tau_{n_0 e_0} \tau_{v_\perp} P_k'$ by Lemma 247(i), and, as in the proof of Lemma 253, $\omega_k(\vartheta(\bar{P}_k) P_k) = \|\mathbb{T}_k^{n_0} \Psi_{\tau_{v_\perp} P_k'}\|^2 \leq \omega_k(\vartheta(\bar{P}_k') P_k')$. The last is the moment of the $2m$ functions $\bar{f}_j' \circ \theta_{0,k}$, f_j', which lie in $Q'' = [-L-3, L+3]^4$ and are pairwise at ℓ^∞ distance at least δ for $a_k \leq \delta$, the reflected ones being supported in $\{x_0 \leq -\delta + a_k\}$. ($G_s$) with $n = 2m$ and Q'', of side $2L+6$, gives the bound with $((2m)!)^\beta \leq 4^{\beta m} (m!)^{2\beta}$ and $7 + 2L \leq 7(1+L)$.

(ii) The moments of order one vanish. For $n \geq 2$, part (ii) of Proposition 259 gives ($M_{s,n}$) with $M_n(Q, \delta) = \mathsf{K}_{2nL/\delta}^n \max_i (R_i R_{n-i})^{1/2}$, R_i taken at Q^*, of side $2L+4$, and $\delta_n = \delta/(16(n+1)) \leq 1$. By ($RN_p$) and $i!(n-i)! \leq n!$,

$$(R_i R_{n-i})^{1/2} \leq K_1^{n/2} (n!)^{\beta_1/2} \big(5(1+L)\big)^{q_1 n/2} \times \big(16(n+1)\big)^{p_1 n/2} \delta^{-p_1 n/2}.$$

From $e^n \geq n^n/n!$, $(n+1)^n \leq e\, n^n \leq e^{n+1} n! \leq e^{2n} n!$. For $\delta \leq 1$, $4nL/\delta + 3 \leq 4n(1+L)/\delta$, so $\mathsf{K}_{2nL/\delta}^n \leq 4^{4n} n^{4n} (1+L)^{4n} \delta^{-4n} \leq 4^{4n} e^{4n} (n!)^4 (1+L)^{4n} \delta^{-4n}$. Multiplying gives the constants stated.

(iii) If (SD_p) holds, Lemma 253 gives, for f supported in a cube of side L in $\{x_0 \geq \delta\}$ and large k,

$$Z_k^2 \omega_k\big(\vartheta(\overline{A_k(f)}) A_k(f)\big) \leq \mathsf{K}_L^2 \|f\|_\infty^2 \frac{\mathscr{N}_k(\delta)}{\mathscr{N}_k(\varepsilon)},$$

with $\mathsf{K}_L^2 \leq 3^8 (1+L)^8$. The ratio is at most $C_0 \max(1, \varepsilon)^{p_0} \delta^{-p_0}$ for $\delta \leq \varepsilon$ and at most K^2 for $\delta > \varepsilon$, by Proposition 256(i). Conversely, if the case $m = 1$ holds, let $\varepsilon' \in (0, \varepsilon]$ and h be supported in $B_{\varepsilon'} \subset [0, L_0]^4$, $L_0 = \varepsilon + 1 + a_0$; the bound with $\delta = \min(\varepsilon', 1)$ gives $\mathscr{N}_k(\varepsilon') \leq C_0 (\varepsilon/\varepsilon')^{p_1} \mathscr{N}_k(\varepsilon)$ with $C_0 = K_1 (1 + L_0)^{q_1} \max(1, \varepsilon^{-p_1})$. The last claim follows from (i), (ii). □

Bounds on each moment do not give the factor $K^n (n!)^\beta$: in the next fields every moment is bounded uniformly in the separation, and (G_s) fails.

Proposition 286. *In the setting of Lemma 283 let* $t > 0$, $U(u) = e^{tu_2 - t^2/2} - 1$, $\varepsilon > 0$, *form* $\mathscr{N}_k(\varepsilon)$ *with* $\Phi_x = U(\xi_x)$, *and put* $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$, *with* ℓ_0 *as in Proposition 284.*

(a) *The gap is* m *in physical units. For large* k, $t^2 \ell_0 \leq \mathscr{N}_k(\varepsilon) \leq (1 + 2a_0)^8 (e^{t^2} - 1)$, ($SD_p$) *holds with* $p_0 = 0$, (NE) *holds, and for every* n *and all* f_j *supported in a cube of side* L, *separated or not,*

$$\Big| \omega_k\Big(\prod_{j=1}^n Z_k A_k(f_j) \Big) \Big| \leq \frac{2^n e^{t^2 n(n-1)/2} (L+1)^{4n}}{(t^2 \ell_0)^{n/2}} \prod_{j=1}^n \|f_j\|_\infty;$$

in particular (M_s) *and* (RN) *hold.*

(b) *Let* n *be even with* $n \geq e \log 2/t^2 + \frac{1}{2}$, $s = \lceil n^{1/4} \rceil$, $\eta = 1/(12(1+m)s)$, *let* $v_1, \dots, v_n$ *be distinct points of* $\{0, \dots, s-1\}^4$, *and let* f_j *be as in Proposition 284*(b) *with centres* $c_j = 3\eta v_j$. *Then* $(f_j) \in \mathscr{C}_n(Q, \eta)$ *for a cube* Q *of side at most* 1, *and for large* k

$$Z_k^n\, \omega_k\Big(\prod_{j=1}^n A_k(f_j) \Big) \geq \frac{(\eta/2)^{4n} e^{t^2 n(n-1)/(2e)}}{2^{n+1} \big((1+2a_0)^8 (e^{t^2} - 1)\big)^{n/2}}.$$

(G_s) *fails for all* K, β, p, q, *and no limit of Theorem 248 satisfies the bound of* (G_s).

(c) *For every sequence* $M_k \to \infty$, (a) *and* (b) *hold for the fields* $U_{M_k}(\xi_x)/(2M_k)$, *with values in* $[-1, 1]$, *with* 4^n *in place of* 2^n, $\frac{1}{2} t^2 \ell_0$ *in place of* $t^2 \ell_0$, *and the right side of* (b) *halved, for* k *large depending on* n*; for these fields* $Z_k \to \infty$.

Proof. (a) The gap is Lemma 283(i), with $\int U(u) u_2 \, d\gamma_2 = t$. As in the proof of Proposition 284(a), the reflected norm of $A_k(h)$ is $\sum_{x,y} a_k^8 \overline{h(a_k x)} h(a_k y)(e^{t^2 \rho} - 1)$, with $\rho = \rho(y - \vartheta x) \in (0, 1]$; the upper bound follows, and the lower one from $e^{t^2\rho} - 1 \geq t^2 \rho$ and the function h_*, which also bounds the ratio in (NE) with $g_0 = h_*$ below by $t^2 \ell_0/((1+2a_0)^8 (e^{t^2} - 1))$, and ($SD_p$) with $p_0 = 0$. By Lemma 283(ii) the moments are at most $\|U\|_n^n \prod_j (L+1)^4 \|f_j\|_\infty$, and $\|U\|_n \leq \|e^{tu_2 - t^2/2}\|_n + 1 = e^{t^2(n-1)/2} + 1 \leq 2e^{t^2(n-1)/2}$. This gives ($M_s$), and (RN) by Proposition 259(i).

(b) Distinct v_j differ by at least 1 in some coordinate, so the supports are at ℓ^∞ distance at least η; they lie in a cube of side $3\eta s \leq 1$, of ℓ^1 diameter at most $12\eta s \leq 1/m$, so $\rho \geq e^{-1}$ for all sites in them. By (40) and its monotonicity, $\mathbb{E} \prod_j U(\xi_{x_j})$ is at least its value for the Gaussian vector $X_i = e^{-1/2} Y + (1 - e^{-1})^{1/2} W_i$ with $Y, W_1, \dots, W_n$ independent standard normal, whose covariances are all e^{-1}. Conditionally on Y the factors are independent, and with $\sigma = te^{-1/2}$ this value is $\mathbb{E}[(e^{\sigma Y - \sigma^2/2} - 1)^n]$. For n even the integrand is nonnegative, and it is at least $2^{-n} e^{n\sigma Y - n\sigma^2/2}$ where $y_0 = (\log 2 + \sigma^2/2)/\sigma \leq Y$. Since $\mathbb{E}[e^{n\sigma Y} \mathbf{1}_{\{Y \geq y_0\}}] = e^{n^2\sigma^2/2} \mathbb{P}(Y \geq y_0 - n\sigma) \geq \frac{1}{2} e^{n^2\sigma^2/2}$ for $n\sigma \geq y_0$, that is $n \geq \log 2/\sigma^2 + \frac{1}{2}$, the value is at least $2^{-n-1} e^{\sigma^2 n(n-1)/2}$. Keeping the sites with $a_k x_j \in c_j + [-\frac{\eta}{2}, \frac{\eta}{2}]^4$, as in the proof of Proposition 284(b), gives the bound. If (G_s) held, these moments would be at most $K^n (n!)^\beta 2^{qn} \eta^{-pn}$, and $\eta^{-1} \leq 24(1+m) n^{1/4}$; the logarithm of this is $O(n \log n)$, while the logarithm of the lower bound is $t^2 n(n-1)/(2e) - O(n \log n)$, a contradiction for large even n. The limits of Theorem 248(i) satisfy the same lower bound.

(c) By Lemma 283(ii), $|\mathbb{E} \prod_j U_{M_k}(\zeta_j)| \leq (2\|U\|_n)^n$, the moments and reflected norms change by at most $e_n(M_k) \prod_j (L_j + 1)^4 \|f_j\|_\infty \to 0$ for each fixed n, and $\int U_{M_k}(u) u_2 \, d\gamma_2 \to t$; the rescaling by $2M_k$ is that of the proof of Proposition 284(d). □

One inequality between the moments and the separated two-point functions gives (M_s) and (G_s) from (SD) and (SD_p). For a cube Q and $\delta > 0$ put

$$\mu_k(Q,\delta) = \sup Z_k^2 \big|\omega_k\big(A_k(h)A_k(h')\big)\big|,$$

the supremum over the $(h,h') \in \mathscr{C}_2(Q,\delta)$ with $\|h\|_\infty, \|h'\|_\infty \le 1$.

(GD) There are $K_0, \beta_0 \ge 0$ such that for every $n \ge 3$, every cube Q and every $\delta > 0$ there is k_0 with

$$\Big|\omega_k\Big(\prod_{j=1}^n Z_k A_k(f_j)\Big)\Big| \le K_0^n (n!)^{\beta_0} \mu_k(Q,\delta)^{n/2} \prod_{j=1}^n \|f_j\|_\infty$$

for all $k \ge k_0$ and $(f_1,\dots,f_n) \in \mathscr{C}_n(Q,\delta)$.

Proposition 287. *Assume the hypothesis of Theorem 64.*

(i) *Under* (GD), ($M_{s,2}$) *implies* (M_s). *With* $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$ *as in Theorem 254,* (GD) *and* (SD) *imply* (M_s), *and* (GD) *and* (SD_p) *imply* (G_s) *with* $K = \max(1,K_0)C_2^{1/2}$, $\beta = \beta_0$, $q = 4$, $p = 4 + \frac{1}{2}p_0$, *where* $C_2 = (\frac{77}{2})^8 \max(1,C_0)\max(1,8\varepsilon)^{p_0}$.

(ii) *If for every k the packets* $A_k(f)$ *form a centred Gaussian family,* (GD) *holds with* $K_0 = 1$, $\beta_0 = \frac{1}{2}$.

(iii) (GD) *fails for* $n = 3$ *in the fields of Proposition 284.*

Proof. (i) ($M_{s,2}$) gives $\mu_k(Q,\delta) \le M_2(Q,\delta)$ for large k, and Corollary 255(a) gives ($M_{s,2}$) from (SD). Under (SD_p) the proof of Proposition 275(a) gives $\mu_k(Q,\delta) \le C_2(1+L)^8\delta^{-8-p_0}$ for Q of side L, $\delta \le 1$ and large k, which is the case $n = 2$; the moments of order one vanish.

(ii) By Wick's formula the moment is 0 for odd n and, for even n, the sum over the $(n-1)!!$ perfect matchings of the products of the $\omega_k(A_k(f_a)A_k(f_b))$; each pair lies in $\mathscr{C}_2(Q,\delta)$, and $((n-1)!!)^2 \le (n-1)!!\,n!! = n!$.

(iii) There ($M_{s,2}$) holds, so $\mu_k(Q,\eta)$ is bounded, while the left side of (GD) for the functions of Proposition 284(b) with $n = 3$ tends to infinity. □

So the case $m = 1$ of (RN) is (SD), and its rate is (SD_p) (Propositions 259(iii) and 285(iii)); the cases $m \ge 2$ are a further input on the Wilson measure, which the two-point properties, reflection positivity and the gap do not supply (Proposition 284); (G_s) is (RN) with constants $K_1^m(m!)^{\beta_1}(1+L)^{q_1 m}\delta^{-p_1 m}$ (Proposition 285), and bounds on each moment do not give the dependence on n (Proposition 286). The single inequality (GD), of Gaussian domination type, gives (M_s) from (SD) and (G_s) from (SD_p); it holds for Gaussian packets, fails in the fields of Proposition 284, and is not proved here for the Wilson measure.

The effective energy. Along each lattice of distances from the reflection plane the largest reflected norm is logarithmically convex, and (SD) becomes a bound, in physical units, on its rate of decay. For $(r+2)a_k \le s$ with $\mathscr{N}_k(s) > 0$ put

$$\mathfrak{E}_k(s) = \frac{1}{2a_k} \log \frac{\mathscr{N}_k(s)}{\mathscr{N}_k(s+a_k)}.$$

[23]

[23] For one correlator this is the effective mass of lattice spectroscopy; here it is taken of the largest reflected norm, a supremum over packets.

Lemma 288. *Assume the hypothesis of Theorem 64, let* $(r+2)a_k \le s$ *and* $\mathscr{N}(n) = \mathscr{N}_k(s+na_k)$ *for integers* $n \ge 0$.

(i) $\mathscr{N}(n)$ *is the supremum of* $\|\mathbb{T}_k^n \Psi_{A_k(h)}\|^2$ *over h supported in* B_s *with* $\|h\|_\infty \le 1$; $\mathscr{N}(n+1) \le \gamma_k^2 \mathscr{N}(n)$, *and* $\mathscr{N}(n)^2 \le \mathscr{N}(n-1)\mathscr{N}(n+1)$ *for* $n \ge 1$.

(ii) *If* $\mathscr{N}(0) > 0$, *then* $\mathscr{N}(n) > 0$ *for every n, and* $n \mapsto \mathfrak{E}_k(s+na_k)$ *is nonincreasing, with values in* $[a_k^{-1}\log(1/\gamma_k),\infty) \subset [m_*,\infty)$.

Proof. (i) Since $B_{s+na_k} = B_s + na_k e_0$, h is supported in B_{s+na_k} if and only if $h' = h(\cdot + na_k e_0)$ is supported in B_s, and then $A_k(h) = \tau_n A_k(h')$ by Lemma 247(i), so $\Psi_{A_k(h)} = \mathbb{T}_k^n \Psi_{A_k(h')}$ as in the proof of Lemma 247(iii). The packets are centred, so $\Psi_{A_k(h')} \perp \Omega_k$ and $\|\mathbb{T}_k^{n+1}\Psi_{A_k(h')}\| \le \gamma_k \|\mathbb{T}_k^n \Psi_{A_k(h')}\|$. For a vector Ψ and $f(n) = \|\mathbb{T}_k^n\Psi\|^2 = \langle \mathbb{T}_k^{n-1}\Psi, \mathbb{T}_k^{n+1}\Psi\rangle$, the Cauchy–Schwarz inequality gives $f(n)^2 \le f(n-1)f(n+1) \le \mathscr{N}(n-1)\mathscr{N}(n+1)$ when $\Psi = \Psi_{A_k(h')}$; the supremum over h' gives the last claim.

(ii) For h' supported in B_s, $h'' = h'(\cdot + a_k e_0)$ is supported in $\{x_0 \ge s - a_k\}$ and $(r+1)a_k \le s - a_k$, so $\Psi_{A_k(h')} = \mathbb{T}_k \Psi_{A_k(h'')}$ lies in the range of $\mathbb{T}_k$, which is orthogonal to $\ker \mathbb{T}_k = \ker \mathbb{T}_k^n$. So $\mathbb{T}_k^n \Psi_{A_k(h')} \ne 0$ when $\Psi_{A_k(h')} \ne 0$, and $\mathscr{N}(n) > 0$ by (i). The convexity in (i) says that $\log \mathscr{N}(n-1) - \log \mathscr{N}(n) \ge \log \mathscr{N}(n) - \log \mathscr{N}(n+1)$, and the bound $\mathscr{N}(n+1) \le \gamma_k^2 \mathscr{N}(n)$ gives the lower bound, with $\gamma_k \le e^{-m_* a_k}$. □

For a single packet, $(2a_k)^{-1}\log(\|\mathbb{T}_k^n\Psi\|^2/\|\mathbb{T}_k^{n+1}\Psi\|^2)$ decreases, as $n \to \infty$, to the least point of the support of its spectral measure, written in physical energy; so $\mathfrak{E}_k(s)$ is the energy, in physical units, at which the largest reflected norm decays at the distance s from the plane.

Proposition 289. *Assume the hypothesis of Theorem 254, and let* $C_{\varepsilon'}$ *be as in Proposition 256*(iii).

(i) (SD) *holds if and only if* $\limsup_k \mathfrak{E}_k(s) < \infty$ *for every* $s \in (0,\varepsilon)$. *Under* (SD), $\limsup_k \mathfrak{E}_k(s) \le \log(\mathsf{K}^4 C_{s/2})/s$; *conversely, for* $\varepsilon' \in (0,\varepsilon)$,

$$\limsup_k \frac{\mathscr{N}_k(\varepsilon')}{\mathscr{N}_k(\varepsilon)} \le \mathsf{K}^2 \exp\Big(2(\varepsilon-\varepsilon')\limsup_k \mathfrak{E}_k(\varepsilon')\Big).$$

(ii) *Under* (SD_p), *for every* $s \in (0,\varepsilon]$ *and all large k,*

$$\mathfrak{E}_k(s) \le \frac{\log(\mathsf{K}^4 C_0) + p_0 \log(2\varepsilon/s)}{s - 2a_k}.$$

Conversely, if there are $p, E_0 \ge 0$ *and, for every* $\varepsilon' \in (0,\varepsilon]$, *a* k_0 *with* $\mathfrak{E}_k(s) \le p/(2s) + E_0$ *for all* $s \in [\varepsilon',\varepsilon]$ *and* $k \ge k_0$, *then* (SD_p) *holds with* $p_0 = p$.

Proof. For $s \in (0,\varepsilon]$ and large k, $\mathscr{N}_k(s) \ge \mathsf{K}^{-2}\mathscr{N}_k(\varepsilon) > 0$ by Proposition 256(i), so $\mathfrak{E}_k(s)$ is defined. Let $n = \lfloor s/(2a_k) \rfloor$. By Lemma 288(ii), applied at $s - na_k$, $\mathfrak{E}_k(s) \le \mathfrak{E}_k(s - ia_k)$ for $1 \le i \le n$, so

$$\mathfrak{E}_k(s) \le \frac{1}{n}\sum_{i=1}^n \mathfrak{E}_k(s-ia_k) = \frac{1}{2a_k n}\log\frac{\mathscr{N}_k(s-na_k)}{\mathscr{N}_k(s)}.$$

Since $s-na_k \ge s/2$ and $s \le \varepsilon$, Proposition 256(i) gives $\mathscr{N}_k(s-na_k) \le \mathsf{K}^2\mathscr{N}_k(s/2)$ and $\mathscr{N}_k(s) \ge \mathsf{K}^{-2}\mathscr{N}_k(\varepsilon)$, and $2a_kn \ge s-2a_k$. Hence

$$\mathfrak{E}_k(s) \le \frac{1}{s-2a_k}\log\Big(\mathsf{K}^4\frac{\mathscr{N}_k(s/2)}{\mathscr{N}_k(\varepsilon)}\Big),$$

which gives the bound in (i) under (SD), and the bound in (ii) under (SD_p) at $\varepsilon' = s/2$.

Conversely, let $\varepsilon' \in (0,\varepsilon)$, $E' > \limsup_k \mathfrak{E}_k(\varepsilon')$ and $n = \lceil(\varepsilon-\varepsilon')/a_k\rceil$. For large k, Lemma 288(ii) gives

$$\log\frac{\mathscr{N}_k(\varepsilon')}{\mathscr{N}_k(\varepsilon'+na_k)} = 2a_k\sum_{j=0}^{n-1}\mathfrak{E}_k(\varepsilon'+ja_k) \le 2a_knE',$$

and $2a_kn \le 2(\varepsilon-\varepsilon'+a_k)$. Since $\varepsilon'+na_k \ge \varepsilon$, Proposition 256(i) gives $\mathscr{N}_k(\varepsilon'+na_k) \le \mathsf{K}^2\mathscr{N}_k(\varepsilon)$. Letting $k\to\infty$ and $E' \downarrow \limsup_k \mathfrak{E}_k(\varepsilon')$ gives the inequality of (i), hence (SD). Under the hypothesis of the converse in (ii) the same sum is at most $\sum_j(pa_k/(\varepsilon'+ja_k)+2a_kE_0) \le pa_k/\varepsilon' + p\log((\varepsilon+a_k)/\varepsilon')+2(\varepsilon+a_k)E_0$, so for $a_k \le \min(\varepsilon',\varepsilon,1)$

$$\frac{\mathscr{N}_k(\varepsilon')}{\mathscr{N}_k(\varepsilon)} \le \mathsf{K}^2 e^{p+2(\varepsilon+1)E_0}2^p\Big(\frac{\varepsilon}{\varepsilon'}\Big)^p,$$

which is (SD_p) for $\varepsilon' < \varepsilon$; for $\varepsilon' = \varepsilon$ it holds with $C_0 \ge 1$. □

So (SD) says that at every distance from the reflection plane the largest reflected norm decays at a rate bounded in physical units, and (SD_p) that this rate grows at most like $p_0/(2s)$ as $s\to 0$, up to the logarithm in (ii). Under a gap m_lat in lattice units the rate is at least m_lat/a_k at every distance, which is the failure of (SD) in Proposition 256(iii). By Proposition 256(ii) the bound on the rate is equivalent to the spectral condition (UV).

The abelian theory. For $G = \mathrm{U}(1)$ the positivity behind Proposition 258(ii) is a theorem.

Corollary 290. *Let $G = \mathrm{U}(1)$ with the Wilson weight and $\beta_k \ge 0$, assume the hypothesis of Theorem 254, and let $\Phi = b_0 - \sum_j b_j\cos\theta_j$ with $b_j \ge 0$ for $j \ge 1$ and each θ_j an integer combination of the angles of the links with both endpoints in $[-r,r]^4$; the sum of $1-\cos\theta_p$ over the 24 plaquettes with a corner at 0 is an example. Then $K_k(z) \ge 0$ for all $z \in \mathbb{Z}^4$. If* (SD) *holds, then* (NE) *holds, every limit S_2 of Corollary 255*(b) *is not identically zero, and $\mathscr{N}_k(\varepsilon) \le \omega_k(\vartheta(A_k(g_0))A_k(g_0))$ for large k and every g_0 as in Proposition 258*(ii).

Proof. Let μ_L be the Wilson measure on Λ_L. Its density with respect to the product of the normalised Lebesgue measures on the link angles is $Z^{-1}\exp(\beta_k\sum_p\cos\theta_p)$, where each θ_p is a signed sum of four link angles. By Ginibre's inequality [54], $\int\cos\theta\cos\theta' d\mu_L \ge \int\cos\theta\, d\mu_L\int\cos\theta' d\mu_L$ for any two integer combinations θ,θ' of link angles. So $\mathrm{Cov}_{\mu_L}(\Phi_0,\Phi_z) = \sum_{j,l\ge1} b_jb_l\,\mathrm{Cov}_{\mu_L}(\cos\theta_j,\cos\tau_z\theta_l) \ge 0$ once L is large, and the torus limit preserves this inequality for the local functions Φ_0, Φ_z. This is the hypothesis of Proposition 258(ii), whose proof gives the last claim, and Proposition 258(i) gives the others. □

For $\mathrm{U}(1)$ the averaged condition fails in the Coulomb phase of the Villain theory (Corollary 66), and (SD) fails under a gap in lattice units (Proposition 256(iii)); whether the hypotheses of the corollary hold together for $\mathrm{U}(1)$ along any tuning is not decided here. What the corollary excludes is the mechanism of the example below, whose two-point function changes sign.

Spatial momentum. The spatial translations act on the reconstructed space. For $v \in \mathbb{Z}^4$ with $v_0 = 0$, $\omega_k(\vartheta(\tau_v\bar F)\tau_vF') = \omega_k(\tau_v(\vartheta(\bar F)F')) = \omega_k(\vartheta(\bar F)F')$ for F,F' at positive times, so $\Psi_F \mapsto \Psi_{\tau_vF}$ extends to a unitary $U_k(v)$; U_k is a representation of $\mathbb{Z}^3$ that commutes with $\mathbb{T}_k$, since $\tau_v\tau_1 = \tau_1\tau_v$. By the spectral theorem for the commuting unitaries $U_k(e_i)$ there is a projection-valued measure P_k on $(-\pi,\pi]^3$ with $U_k(v) = \int e^{ip\cdot v}d\mathsf{P}_k(p)$. For $K>0$ let $\mathbb{P}_k(K) = \mathsf{P}_k(\{|p|_\infty \le Ka_k\})$, the states of spatial momentum at most K in physical units; it commutes with $\mathbb{T}_k$ and $\mathbb{E}_k(\Lambda)$. For $(r+1)a_k \le s$ let $\mathscr{N}_k^{\mathrm{sp}}(s;K)$ be the supremum of $\|\mathbb{P}_k(K)\Psi_{A_k(h)}\|^2$ over h supported in B_s with $\|h\|_\infty \le 1$, and consider

(LM) for some $K<\infty$, $c>0$ and $\varepsilon_1 \in (0,\varepsilon]$, $\mathscr{N}_k^{\mathrm{sp}}(\varepsilon_1;K) \ge c\,\mathscr{N}_k(\varepsilon)$ for all large k.

It says that, up to a fixed fraction, the largest reflected norm is carried by spatial momenta bounded in physical units, as (UV) says for the energy. The proofs below use the Fejér weights: for an integer $N \ge 1$,

$$F_N(q) = \sum_{|n|<N}\frac{1}{N}\Big(1-\frac{|n|}{N}\Big)e^{iqn} = \Big(\frac{\sin(Nq/2)}{N\sin(q/2)}\Big)^2$$

for $0<|q|\le\pi$, $F_N(0)=1$. Then $0\le F_N\le 1$, $F_N(q) \ge (2/\pi)^2$ for $N|q|\le\pi$, because $\sin x \ge 2x/\pi$ on $[0,\frac{\pi}{2}]$ and $N\sin(|q|/2) \le N|q|/2$, and $F_N(q) \le \pi^2/(Nq)^2$ for $0<|q|\le\pi$, because $\sin(|q|/2) \ge |q|/\pi$. For $v\in\mathbb{Z}^4$ with $v_0=0$ put $f_v^N = \prod_{i=1}^3 N^{-1}(1-|v_i|/N)_+$, so that $\sum_v f_v^N = 1$ and $\sum_v f_v^N e^{ip\cdot v} = F_N^{(3)}(p) = \prod_i F_N(p_i)$.

Proposition 291. *Assume the hypothesis of Theorem 254 and* (SD). *Then* (NE) *holds if and only if* (LM) *holds.*

Proof. Two facts are used. First, for g supported in a cube of side L in $\{x_0 \ge s\}$, $(r+1)a_k \le s$, the proof of Lemma 253 writes $\Psi_{A_k(g)}$ as a sum of at most K_L vectors $\mathbb{T}_k^nU_k(v)\Psi_{A_k(h')}$ with h' supported in B_s and $\|h'\|_\infty \le \|g\|_\infty$; since $\mathbb{P}_k(K)$ commutes with $\mathbb{T}_k$ and U_k,

$$\|\mathbb{P}_k(K)\Psi_{A_k(g)}\| \le \mathsf{K}_L\,\mathscr{N}_k^{\mathrm{sp}}(s;K)^{1/2}\|g\|_\infty.$$

Second, for g supported in $\{x_0\ge s\}$ and $g^N = \sum_v f_v^N g(\cdot - a_kv)$, Lemma 247(i) gives $A_k(g^N) = \sum_v f_v^N\tau_vA_k(g)$, so $\Psi_{A_k(g^N)} = F_N^{(3)}(\mathsf{P}_k)\Psi_{A_k(g)}$, where $F_N^{(3)}(\mathsf{P}_k) = \int F_N^{(3)}\,d\mathsf{P}_k$. By (SD) and Proposition 256(i), for every $s>0$ there is $C(s)$ with $\mathscr{N}_k(s) \le C(s)\mathscr{N}_k(\varepsilon)$ for large k.

Let (NE) hold with g_0 supported in $\{x_0\ge t_0\}$, $t_0>0$, and in a cube of side L, and $c_0>0$ with $4c_0 < \liminf_k R_k(g_0)$. Let $\delta\in(0,1]$, $a_k\le\delta$ and $N = \lfloor\delta/a_k\rfloor \ge \delta/(2a_k)$. The function $g_0^N - g_0$ is supported in a cube of side $L+2$ in $\{x_0\ge t_0\}$ and bounded by $\rho(\delta) = \sup\{|g_0(y)-g_0(y')| : |y-y'|_\infty \le \delta\}$, because $f_v^N \ne$

0 only for $|a_k v|_\infty < \delta$. For $|p|_\infty > Ka_k$ some $|p_i| > Ka_k$, so $F_N^{(3)}(p) \le (\pi/(NKa_k))^2 \le (2\pi/(\delta K))^2$. With Lemma 253,

$$\begin{aligned}&\|(1-\mathbb{P}_k(K))\Psi_{A_k(g_0)}\|\\ &\quad\le \|\Psi_{A_k(g_0-g_0^N)}\| + \|(1-\mathbb{P}_k(K))F_N^{(3)}(\mathsf{P}_k)\Psi_{A_k(g_0)}\|\\ &\quad\le \Big(\mathsf{K}_{L+2}\rho(\delta) + \frac{4\pi^2\mathsf{K}_L\|g_0\|_\infty}{\delta^2K^2}\Big)\mathscr{N}_k(t_0)^{1/2}.\end{aligned}$$

Fix δ, and then K, so that the bracket is at most $(c_0/C(t_0))^{1/2}$. For large k this gives

$$\|\mathbb{P}_k(K)\Psi_{A_k(g_0)}\| \ge (2c_0^{1/2} - c_0^{1/2})\mathscr{N}_k(\varepsilon)^{1/2},$$

and the first fact gives $\mathscr{N}_k^{\mathrm{sp}}(t_0;K) \ge c_0\mathsf{K}_L^{-2}\|g_0\|_\infty^{-2}\mathscr{N}_k(\varepsilon)$. If $t_0 \le \varepsilon$ this is (LM) with $\varepsilon_1 = t_0$. If $t_0 > \varepsilon$, every h supported in B_{t_0} is $h'(\cdot - na_k e_0)$ with $n \ge 0$ and h' supported in a cube of side $1+a_0$ in $\{x_0 \ge \varepsilon\}$, so $\|\mathbb{P}_k(K)\Psi_{A_k(h)}\| \le \|\mathbb{P}_k(K)\Psi_{A_k(h')}\|$ and $\mathscr{N}_k^{\mathrm{sp}}(t_0;K) \le \mathsf{K}^2\mathscr{N}_k^{\mathrm{sp}}(\varepsilon;K)$, which is (LM) with $\varepsilon_1 = \varepsilon$.

Conversely, let (LM) hold and, for large k, h_k be supported in B_{ε_1} with $\|h_k\|_\infty \le 1$ and $\|\mathbb{P}_k(K)\Psi_k\|^2 \ge \frac{c}{2}\mathscr{N}_k(\varepsilon)$, $\Psi_k = \Psi_{A_k(h_k)}$.

Energy. Let $n = \lfloor \varepsilon_1/(2a_k)\rfloor$ and $h_k' = h_k(\cdot + na_k e_0)$, supported in a cube of side $1+a_0$ in $\{x_0 \ge \varepsilon_1/2\}$. Then $\Psi_k = \mathbb{T}_k^n\Psi_{A_k(h_k')}$ and $\|\mathbb{T}_k^n(1-\mathbb{E}_k(\Lambda))\| \le e^{-\Lambda a_k n}$, so for $a_k \le \varepsilon_1/4$, by Lemma 253,

$$\|(1-\mathbb{E}_k(\Lambda))\Psi_k\| \le e^{-\Lambda\varepsilon_1/4}\mathsf{K}C(\tfrac{\varepsilon_1}{2})^{1/2}\mathscr{N}_k(\varepsilon)^{1/2}.$$

Fix Λ with

$$e^{-\Lambda\varepsilon_1/4}\mathsf{K}C(\tfrac{\varepsilon_1}{2})^{1/2} \le (c/8)^{1/2}.$$

Since $\mathbb{E}_k(\Lambda)$ and $\mathbb{P}_k(K)$ commute, $\|\mathbb{E}_k(\Lambda)\mathbb{P}_k(K)\Psi_k\| \ge (c/8)^{1/2}\mathscr{N}_k(\varepsilon)^{1/2}$.

Smoothing. Let $\sigma > 0$, $M = \lfloor \sigma/a_k\rfloor$, $N = \lfloor \pi/(Ka_k)\rfloor$, both at least 1, $w_n = M^{-1}(1-|n-M|/M)_+$ for $n \in \mathbb{Z}$, which vanish unless $1 \le n \le 2M-1$ and add up to 1, and

$$g_k = \sum_{n,v} w_n f_v^N h_k(\cdot - a_k(ne_0 + v)).$$

By Lemma 247(i), $\Psi_{A_k(g_k)} = W(\mathbb{T}_k)F_N^{(3)}(\mathsf{P}_k)\Psi_k$ with $W(\lambda) = \sum_n w_n\lambda^n \ge \lambda^{2M}$ on $[0,1]$. On the range of $\mathbb{E}_k(\Lambda)\mathbb{P}_k(K)$ the joint spectrum of $\mathbb{T}_k$ and U_k lies where $\lambda \ge e^{-\Lambda a_k}$, so $W \ge e^{-2\Lambda\sigma}$, and $|p|_\infty \le Ka_k$, so $N|p_i| \le \pi$ and $F_N^{(3)} \ge (2/\pi)^6$. Hence

$$\omega_k\big(\vartheta(\overline{A_k(g_k)})A_k(g_k)\big) \ge 4c'\mathscr{N}_k(\varepsilon),$$

with $4c' = \frac{c}{8}e^{-4\Lambda\sigma}(2/\pi)^{12}$.

Compactness. $A_k(g)$ depends only on the values of g on $a_k\mathbb{Z}^4$. Let $u(t) = M^{-1}(1-|t/a_k - M|/M)_+$, $\varsigma(t) = N^{-1}(1-|t/a_k|/N)_+$, $\kappa(y) = u(y_0)\prod_{i=1}^3\varsigma(y_i)$ and

$$G_k(y) = \sum_{z\in\mathbb{Z}^4}\kappa(y - a_k z)h_k(a_k z).$$

Then $G_k(a_k x) = g_k(a_k x)$, so $A_k(G_k) = A_k(g_k)$. The function G_k is supported in $S = B_{\varepsilon_1} + [0,2\sigma]\times[-\pi/K,\pi/K]^3$, a compact subset of $\{x_0 \ge \varepsilon_1\}$ independent of k. For $t,t' \in \mathbb{R}$, at most $2M$ terms of $\sum_{n\in\mathbb{Z}}u(t-a_kn)$ are nonzero, so this sum is at most 2, and $\sum_n|u(t-a_kn)-u(t'-a_kn)| \le 4|t-t'|/(Ma_k) \le 8|t-t'|/\sigma$; likewise $\sum_n\varsigma(t-a_kn) \le 2$ and $\sum_n|\varsigma(t-a_kn) - \varsigma(t'-a_kn)| \le 8K|t-t'|/\pi$, for $a_k \le \min(\sigma,\pi/K)$. Writing the difference of two products of four factors as a telescoping sum, $|G_k| \le 16$ and G_k is Lipschitz with constant $8(8/\sigma + 24K/\pi)$ for the ℓ^∞ norm. By the Arzelà–Ascoli theorem the G_k lie in a compact subset of $C(S)$.

One packet. Let L_S be the side of a cube containing S and $\eta > 0$ with $\mathsf{K}_{L_S}C(\varepsilon_1)^{1/2}\eta \le c'^{1/2}$. A finite set $G^1,\dots,G^J$ of the G_k is an η-net, so for each large k there is j with $\|G_k - G^j\|_\infty \le \eta$, and Lemma 253 gives $\|\Psi_{A_k(G_k)} - \Psi_{A_k(G^j)}\| \le c'^{1/2}\mathscr{N}_k(\varepsilon)^{1/2}$, hence $R_k(G^j) \ge c'$. Let $g_0 = \sum_j G^j(\cdot - jDe_1)$ with D dyadic. As in the proof of Proposition 258(i), for large k

$$R_k(g_0) \ge \sum_j R_k(G^j) - 16^2J^2\mathsf{K}_{L_S}^2 e^{-m_*(D-L_S-4\varepsilon-1)},$$

and $R_k \ge 0$. For D large the right side is at least $c'/2$ for all large k, and g_0 is continuous and supported in $\{x_0 \ge \varepsilon_1\}$: this is (NE). □

An example. (NE) is not implied by (SD) together with the properties used so far: in the following Gaussian field the largest reflected norms are carried by spatial momenta of order π/a_k at energies bounded in physical units, so that (UV) holds and (LM) fails.[24]

Proposition 292. *Let $m > 0$, $A_k = (128(\cosh(ma_k)-1))^{-1}$,*

$$D_k(p) = 1 + A_k\prod_{\mu=0}^3(2-2\cos p_\mu), \qquad p \in (-\pi,\pi]^4,$$

let ϕ_k be the centred Gaussian field on $\mathbb{Z}^4$ with covariance $\int e^{ip\cdot(x-y)}D_k(p)^{-1}dp/(2\pi)^4$, $A_k(f) = \sum_x a_k^4 f(a_kx)\phi_k(x)$, and form $\mathscr{N}_k(\varepsilon)$ and R_k with these packets and $r = 0$.

(a) *The law of ϕ_k is invariant under translations and B_4 and reflection positive for the reflections in coordinate hyperplanes through sites and bisecting links. Its transfer matrix in each coordinate direction has spectrum on $\Omega_k^\perp$ in $[0,e^{-ma_k}]$ containing e^{-ma_k}: the gap is m in physical units.*

(b) *For every $\varepsilon > 0$, $\varepsilon' \in (0,\varepsilon]$ and all large k,*

$$2^{-15}me^{-4m(\varepsilon+2)}a_k^4 \le \mathscr{N}_k(\varepsilon), \qquad \mathscr{N}_k(\varepsilon') \le (2+a_0)^4a_k^4;$$

so (SD$_{\mathrm{p}}$) *holds with $p_0 = 0$, and* (UV).

(c) *$R_k(g) \to 0$ for every $g \in C_c(\mathbb{R}^4)$ supported in $\{x_0 > 0\}$:* (NE) *fails, and every limit S_2 of Corollary 255*(b) *vanishes identically.*

(d) *With $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$,* (M$_{\mathrm{s}}$) *holds and every limit of Theorem 248 is trivial. With $Z_k = C_k(\sigma_0)^{-1/2}$, formed from any f_0 as before Proposition 250 with $C_k(0) > 0$,* (M$_{\mathrm{s},2}$) *fails.*

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[24] D_k depends on p only through the product of the four factors $2-2\cos p_\mu$, so the energy of a spatial momentum is small only where all three spatial factors are large, near the corner (π,π,π) of the Brillouin zone; the packets that carry the largest reflected norm live there, and smooth test functions do not see them.

Proof. (a) D_k is positive and invariant under permutations and sign changes of the coordinates, so the covariance is positive definite, real, and invariant under translations and B_4. Write $u_\mu = 2-2\cos p_\mu$, and for $p \in (-\pi,\pi]^3$ let $b(p) = A_k u_1 u_2 u_3$, $\lambda(p) \in [0,1)$ the root of $\lambda + \lambda^{-1} = 2 + 1/b(p)$ ($\lambda = 0$ if $b(p) = 0$), $E(p) = -\log \lambda(p)$ and $w(p) = (1-\lambda(p))/(1+\lambda(p)) = \tanh(E(p)/2)$. Then $1 + b u_0 = b\lambda^{-1}(1-\lambda e^{ip_0})(1-\lambda e^{-ip_0})$, and expanding $(1-\lambda^2)/((1-\lambda e^{ip_0})(1-\lambda e^{-ip_0})) = \sum_t \lambda^{|t|} e^{-ip_0 t}$ and using $b(\lambda^{-1}-\lambda) = 1/w$,

$$\frac{1}{1+b\,u_0} = w \sum_{t\in\mathbb{Z}} \lambda^{|t|} e^{-ip_0 t},$$

also for $b = 0$. So the covariance of $\phi_k(t,\mathbf{x})$ and $\phi_k(s,\mathbf{y})$ is $\int e^{ip\cdot(\mathbf{x}-\mathbf{y})} w(p) \lambda(p)^{|t-s|} dp/(2\pi)^3$. Let γ be the centred Gaussian law on $\mathbb{R}^{\mathbb{Z}^3}$ with covariance the multiplier w, Λ the convolution with multiplier λ, and ξ_t independent of law the centred Gaussian with multiplier $w(1-\lambda^2) \geq 0$. The stationary chain $\phi_{t+1} = \Lambda\phi_t + \xi_{t+1}$ has covariance with multiplier $w\lambda^{|t-s|}$, so it is the field; it is reversible because Λ and the covariance of γ commute and are symmetric. The proof of Proposition 199(a) applies with the operator Λ in place of q: the transition operator satisfies $P\mathscr{E}_h = \mathscr{E}_{\Lambda h}$, so it acts on the N-th Wiener chaos as the N-th tensor power of the restriction of Λ to the one-particle space, the completion of the finitely supported h for $\int |\hat h|^2 w\, dp/(2\pi)^3$, on which Λ is multiplication by λ. Since $b \leq 64A_k$ with equality only at $p = (\pi,\pi,\pi)$, $\cosh E = 1 + 1/(2b) \geq 1 + 1/(128A_k) = \cosh(ma_k)$, so $0 \leq \lambda \leq e^{-ma_k}$, with equality at that point. Hence $0 \leq P \leq e^{-ma_k N}$ on the N-th chaos for $N \geq 1$, and e^{-ma_k} lies in the spectrum on the first, λ being continuous and w positive. Reflection positivity and the identification of the transfer matrix with P follow as there, and B_4 gives the other directions.

For h supported in $\{x_0 \geq 1\}$ and ϑ fixing $\{x_0 = c\}$, $|c| \leq \frac12$, the time component of $y - \vartheta x$ is $x_0 + y_0 - 2c \geq 0$, and

$$\omega_k\big(\vartheta(\overline{A_k(h)})A_k(h)\big) = \int w|\hat H|^2 \frac{dp}{(2\pi)^3},$$

where

$$\hat H(p) = \sum_{x_0} a_k \lambda^{x_0 - c} \hat h_{x_0}(p), \qquad \hat h_{x_0}(p) = a_k^3 \sum_{\mathbf{x}} h(a_k x) e^{ip\cdot\mathbf{x}}.$$

Let h be supported in $\{x_0 \geq t\}$ with $t \geq a_k$. Since $\sum_{x_0 \geq t/a_k} \lambda^{x_0 - c} \leq \lambda^{t/a_k - 1/2}/(1-\lambda)$ and $w/(1-\lambda) \leq 1$, the Cauchy–Schwarz inequality with the weights λ^{x_0-c} gives

$$w|\hat H|^2 \leq a_k^2 \lambda^{2t/a_k - 1} \sum_{x_0} |\hat h_{x_0}|^2. \qquad (*)$$

(b) For h supported in $B_{\varepsilon'}$ with $\|h\|_\infty \leq 1$ and $a_k \leq \min(\varepsilon', 1)$, $(*)$ with $\lambda \leq 1$ and Parseval's identity give at most a_k^8 times the number of sites x with $a_k x \in B_{\varepsilon'}$, which is at most $(2+a_0)^4 a_k^{-4}$. For the lower bound let $a_k \leq \frac{1}{20}$, $ma_k \leq 2$, let b_0 be continuous with values in $[0,1]$, supported in $[\varepsilon, \varepsilon+1]$ and equal to 1 on $[\varepsilon + \frac14, \varepsilon + \frac34]$, let b_1 be the function equal to 1 on $[\frac14, \frac34]$, to 0 outside $[0,1]$ and linear in between, and $h_k(y) = b_0(y_0) \prod_{i=1}^3 \cos(\pi y_i/a_k) b_1(y_i)$, supported in B_ε with $\|h_k\|_\infty \leq 1$. Since $\cos(\pi x_i) = (-1)^{x_i}$, $\hat H(p) = T(\lambda)\prod_i \beta(p_i + \pi)$ with $T(\lambda) = \sum_{x_0} a_k b_0(a_k x_0) \lambda^{x_0 - c}$ and $\beta(q) = \sum_n a_k b_1(a_k n) e^{iqn}$. On $Q = \{p : |p_i - \pi| \leq \frac{\pi}{3} \bmod 2\pi\}$, $u_i \geq 3$, so $b \geq 27A_k$ and $\cosh E - 1 \leq 1/(54A_k) \leq 4(\cosh(ma_k) - 1) \leq \cosh(2ma_k) - 1$, since $\cosh 2x - 1 = 2(\cosh x - 1)(\cosh x + 1)$; so $ma_k \leq E \leq 2ma_k$ on Q. Then $w \geq \tanh(ma_k/2) \geq ma_k/4$, and $T(\lambda) \geq e^{-2ma_k((\varepsilon+1)/a_k + 1/2)} \sum_{x_0} a_k b_0(a_k x_0) \geq e^{-2m(\varepsilon+2)}/4$. By Parseval's identity $\int_{-\pi}^{\pi} |\beta|^2 dq/2\pi = a_k^2 \sum_n b_1(a_k n)^2 \geq a_k/4$, and since $|1 - e^{iq}| \geq 1$ for $|q| \geq \frac{\pi}{3}$ and b_1 is 4-Lipschitz,

$$\int_{\frac{\pi}{3} \leq |q| \leq \pi} |\beta|^2 \frac{dq}{2\pi} \leq a_k^2 \sum_n |b_1(a_k n) - b_1(a_k n - a_k)|^2 \leq 48 a_k^3,$$

so $\int_{|q|\leq \pi/3} |\beta|^2 dq/2\pi \geq a_k/8$. Restricting the integral to Q gives

$$\mathscr{N}_k(\varepsilon) \geq \frac{ma_k}{4} \cdot \frac{e^{-4m(\varepsilon+2)}}{16} \Big(\frac{a_k}{8}\Big)^3.$$

($\mathrm{SD_p}$) follows with $C_0 = 2^{15}(2+a_0)^4 e^{4m(\varepsilon+2)}/m$, and (UV) by Proposition 256(ii), whose proof uses only the invariance, the reflection positivity and the gap.

(c) Let g be supported in $\{x_0 \geq t_0\}$, $t_0 > 0$, and in a cube of side L, and $\delta \in (0,1]$. Let V be the set of p with $|p_i| < \delta$ for some i. There $u_i < \delta^2$ for that i, so $b \leq 16 A_k \delta^2$ and $\cosh E - 1 \geq 4(\cosh(ma_k) - 1)/\delta^2 \geq 2(ma_k/\delta)^2$; since $\cosh x - 1 \leq x^2$ for $0 \leq x \leq 1$, $E \geq ma_k/\delta$ for $ma_k \leq \delta$. By $(*)$ and Parseval's identity the part of the integral over V is at most $e^{-(m/\delta)(2t_0 - a_k)} a_k^8$ times $\sum_x |g(a_k x)|^2 \leq (L+1)^4 a_k^{-4} \|g\|_\infty^2$. On the complement, $|p_1| \geq \delta$ and $|1 - e^{ip_1}| \geq 2\delta/\pi$, and $(*)$ with $\lambda \leq 1$ bounds that part by

$$\Big(\frac{\pi}{2\delta}\Big)^2 a_k^8 \sum_x |g(a_k x) - g(a_k x - a_k e_1)|^2 \leq \Big(\frac{\pi}{2\delta}\Big)^2 (L+2)^4 a_k^4 \rho(a_k)^2,$$

with ρ the modulus of continuity of g for $|\cdot|_\infty$. With (b),

$$\limsup_k R_k(g) \leq \frac{2^{15}(L+2)^4 \|g\|_\infty^2}{m\, e^{-4m(\varepsilon+2)}} e^{-2mt_0/\delta},$$

and $\delta \to 0$ gives $R_k(g) \to 0$. The last claim is Corollary 255(b).

(d) By (b) and Corollary 255(a), whose proof uses only the invariance, the reflection positivity and the gap, ($\mathrm{M_{s,2}}$) holds for $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$. The moments of a centred Gaussian field are sums over pairings of products of covariances, and every pair of a separated tuple is separated; so ($\mathrm{M_s}$) holds with $M_n \leq (n-1)!! M_2^{n/2}$, and by (c) every limit vanishes for $n \geq 1$. For $Z_k = C_k(\sigma_0)^{-1/2}$, ($\mathrm{M_{s,2}}$) would give, as in the proof of Corollary 255(a), $\mathscr{N}_k(\varepsilon') \leq M_2(Q,\varepsilon') C_k(\sigma_0)$ for $\varepsilon' < \varepsilon$, while $C_k(\sigma_0) \leq C_k(0) = R_k(f_0)\mathscr{N}_k(\varepsilon)$ by Proposition 250(i), $R_k(f_0) \to 0$ by (c), and $\mathscr{N}_k(\varepsilon') \geq \mathsf{K}^{-2}\mathscr{N}_k(\varepsilon)$. □

So (NE) is an input of its own: reflection positivity, the invariance, a gap bounded above and below in physical units and ($\mathrm{SD_p}$) do not give it, and with the normalisation $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$ the separated moment bound holds for every n

while every limit is trivial. By Proposition 291, under (SD) it is exactly the statement that bounded spatial momenta carry a fixed fraction of the largest reflected norm, and by Corollary 290 it follows from (SD) for U(1). The example is not a lattice gauge theory. For the Wilson measure along the tuning the regimes are these: under a gap in lattice units, in particular in the strong-coupling range of Corollary 251, (SD) fails and the rate $\mathfrak{E}_k$ diverges like a_k^{-1}; under the short-distance law (AF) of Theorem 279, which asymptotic freedom predicts, ($\mathrm{SD_p}$) and (NE) hold; which of the two occurs, if either, is not decided here.

Remark 293. *What is proved.* Given the averaged condition along the tuning, Theorem 248 turns ($\mathrm{M_s}$) into Schwinger functions at separated points that are symmetric, invariant under the translations and B_4, reflection positive in every coordinate direction, with a Hamiltonian of gap m_* in every coordinate direction and exponential clustering in every direction; with the constants of Proposition 250(ii) they are nontrivial, and with ($\mathrm{W_s}$) and ($\mathrm{U_s}$) they converge and are O(4)-invariant (Corollary 249). The averaged condition enters only through Lemma 247(iii), that is, through Corollary 63. For the two-point functions, Theorem 254 and Corollary 255 reduce ($\mathrm{M_s}$) for $n=2$ and nontriviality to (SD) and (NE), statements about reflected norms of single packets under the Wilson measure. For the rotations, Proposition 261 shows that the anisotropy of a separated two-point kernel has no components of degree below four and the angular profile Y_4 in degree four, and Theorem 262 derives ($\mathrm{W_s}$) from (SB_n) and (RI_n); for the limit, Theorem 266 derives ($\mathrm{U_s}$) from (MC_{m_0}) and ($\mathrm{O_s}$). For the test functions of [88], Theorem 273 extends every S_n whose separated moments grow at most like a power of the inverse separation, and under ($\mathrm{G_s}$) gives the linear growth condition of [89], reflection positivity and clustering on $\mathscr{S}_0$ (Corollary 274); for $n=2$, ($\mathrm{G_s}$) is the rate ($\mathrm{SD_p}$) in (SD) (Proposition 275). For every n, ($\mathrm{G_s}$) is (RN) with constants $K_1^m (m!)^{\beta_1}(1+L)^{q_1 m}\delta^{-p_1 m}$ (Proposition 285), and the inequality (GD) gives ($\mathrm{M_s}$) from (SD) and ($\mathrm{G_s}$) from ($\mathrm{SD_p}$) (Proposition 287).

Which inputs are needed. ($\mathrm{M_s}$) with constants that keep a reflected correlation fixed is not implied by the averaged condition, by Corollary 251. ($\mathrm{W_s}$) is not implied by the other properties: the Gaussian field of Proposition 199(a) is invariant under translations and B_4, reflection positive in the coordinate hyperplanes, has gap ma_k in every coordinate direction, satisfies (M) and hence ($\mathrm{M_s}$) by Theorem 196(i), and the proofs above use the averaged condition only through these properties, with m in place of m_*. Its limit has the continuous kernel $\kappa e^{-m\|x-y\|_1}$, so invariance under R on the separated pairs forces $\|Rz\|_1 = \|z\|_1$ for $z \neq 0$, and $G_S = B_4$. The same example at two interleaved masses violates ($\mathrm{U_s}$), by the argument of Proposition 198(ii); it satisfies ($\mathrm{O_s}$) and violates (MC_{m_0}) for every m_0 (Theorem 266(iii)). Products of independent copies of the field are not Gaussian, have the same properties and $G_S = B_4$, and violate (RI_2) (Proposition 263). The normalisation by $C_k(\sigma_0)$ does not fix the scale (Proposition 265). The bounds (RN_m), $m \geq 2$, are not implied by the two-point properties: the fields of Proposition 284, bounded ones included, have the invariance, the reflection positivity, gap m, (SD), ($\mathrm{SD_p}$), (UV), (NE) and nonzero two-point limits, and violate ($\mathrm{M}_{s,n}$) for every $n \geq 3$ and (RN_m) for every $m \geq 2$. Bounds at each order do not give ($\mathrm{G_s}$): the fields of Proposition 286 satisfy ($\mathrm{M_s}$), (RN) and ($\mathrm{SD_p}$), with moment bounds uniform in the separation, and their moments grow faster than $K^n(n!)^\beta$ for all K, β.

Short-distance law. The two-sided condition (AF), the lattice form of the law $|x-y|^{-8}(\log \frac{1}{|x-y|})^{-\gamma}$ that asymptotic freedom predicts with $\gamma = 2$ (Remark 278), gives ($\mathrm{M}_{s,2}$), ($\mathrm{SD_p}$) with $p_0 = 4$, (SD), (UV) and (NE), excludes trivial limits and fails in the strong-coupling range (Theorem 279). It fixes Z_k up to bounded factors, with $Z_k \to \infty$ and Z_k comparable to $\mathscr{N}_k(\varepsilon)^{-1/2}$ (Proposition 280), it is consistent with a gap in physical units and with ($\mathrm{U_s}$), and it implies neither ($\mathrm{W_s}$) nor the exact law (AF_∞) (Proposition 281). It is not proved for the Wilson measure; the missing inequality is stated in Remark 282.

Relation to Theorem 225. Under the hypothesis of Theorem 43, Corollary 226 shows that, when $C_k(0) > 0$, (SR_0) fails at every spacing for the slice field of Φ or for that of $\Delta\Phi$. Nothing here uses (SR_0): the positivity used is that of the Wilson measure for products of packets with supports at distance at least ε from the reflection plane, that is, for functions of links at lattice distance at least $\varepsilon/a_k - r$ from it, and $r \geq 1$ for every nonconstant Φ. The nontriviality of Proposition 250(ii) comes from ($\mathrm{M_s}$) and the choice of Z_k, not from positivity at short distance.

What is not proved. ($\mathrm{M_s}$) for $n \geq 3$ with the constants of Proposition 250(ii), equivalently the bounds (RN_m), $m \geq 2$, on reflected norms of products of separated packets (Proposition 259), which do not follow from the two-point properties, reflection positivity and the gap (Proposition 284), and the inequality (GD) that gives them from (SD) (Proposition 287); (SD), equivalently the spectral condition (UV) (Proposition 256), which fails under a gap in lattice units, is not implied by reflection positivity and a gap bounded in physical units (Proposition 257), is a bound in physical units on the rate $\mathfrak{E}_k$ at which the largest reflected norm decays (Proposition 289), and follows from the short-distance law (AF) (Theorem 279); (NE), which under (SD) is the nontriviality of every limit (Proposition 258) and the condition (LM) that spatial momenta bounded in physical units carry a fixed fraction of the largest reflected norm (Proposition 291), which follows from (SD) for U(1) (Corollary 290) and is not implied by ($\mathrm{SD_p}$), reflection positivity, the invariance and a gap bounded above and below in physical units (Proposition 292); ($\mathrm{W_s}$), and (SB_n), (RI_n), of which (SB_2) is a bound on the axis and is not implied by the separated moment bounds (Propositions 267 and 268(b)); ($\mathrm{U_s}$), and (MC_{m_0}), of which the two halves hold separately with different rates (Proposition 269), and ($\mathrm{O_s}$), equivalently (DS) (Proposition 270); whether O(4) invariance of S_2 together with the other properties gives ($\mathrm{W_s}$); the rate ($\mathrm{G_s}$) at coinciding points, equivalently ($\mathrm{RN_p}$) (Proposition 285), and its form ($\mathrm{SD_p}$) for $n=2$, which with ($\mathrm{M_s}$) does not give ($\mathrm{G_s}$) (Proposition 286); and the averaged condition along the tuning. None of these is proved here, and the mass gap of the continuum theory is not proved. The extension of the S_n to the test functions of [88] with the linear growth condition of [89] is therefore proved only under ($\mathrm{G_s}$), and it does not follow from the properties at separated points: the two-point function of Proposition 276(ii) has those that Corollary 255(b) gives and

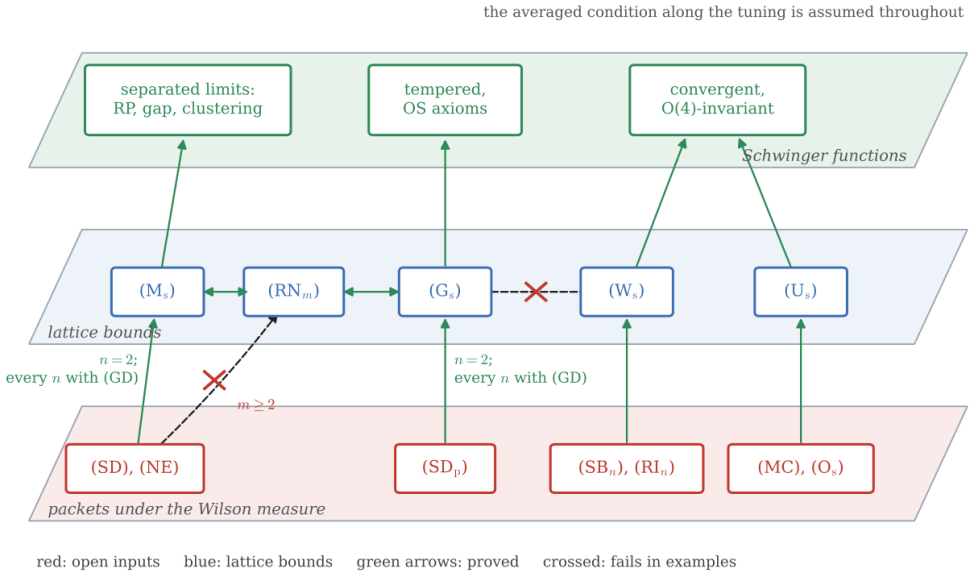


FIG. 17. The conditions of Sec. X L, given the averaged condition along the tuning. Bottom: inputs on reflected norms of packets under the Wilson measure, none of them proved. Middle: lattice bounds; (M_s) is (RN) (Proposition 259), and (G_s) is (RN) with factorial constants (Proposition 285). Top: what the bounds give in the limit (Theorems 248 and 273, Corollaries 274 and 249). From the bottom row: Theorem 254 and Proposition 275 for $n=2$, Proposition 287 for every n, Theorems 262 and 266. Crossed arrows fail in examples: the two-point properties do not give (RN_m), $m \geq 2$ (Proposition 284), and the bounds of the middle row do not give (W_s) (Proposition 199(a)).

is not tempered.

M. What the lattice gives without the averaged condition

The averaged condition BP(n_k, ϖ) of Theorem 64 enters Sec. X L at one place only, the decay $e^{-m_* a_k t}$ of Lemma 247(iii), that is, through the lattice gap of Corollary 63 (Remark 293). This subsection records what remains when that gap is removed: the torus limits exist for every coupling, they are invariant and reflection positive, their two-point functions have renormalised continuum limits along subsequences with a positive Hamiltonian on the one-field space, and the short-distance law (AF) alone supplies the two inputs (SD) and (NE). It then states the missing step as one lemma and proves that its hypotheses are necessary as well as sufficient.

Throughout, $\beta_k \geq 0$ is an arbitrary sequence of couplings; nothing is assumed about BP, about $D(n,\alpha)$ or about the tuning. For a state ω as in Appendix F with transfer matrix $\mathbb{T}$ and vacuum Ω put

$$\gamma(\omega) = \big\| \mathbb{T}|_{\Omega^\perp} \big\| \in [0,1],$$

the norm of $\mathbb{T}$ on the orthogonal complement of the vacuum; since $\mathbb{T}$ is self-adjoint and $\mathbb{T}\Omega = \Omega$, the subspace $\Omega^\perp$ is invariant under $\mathbb{T}$. We write $\gamma_k = \gamma(\omega_k)$. A *lattice gap* $m_{\mathrm{lat}} > 0$ means $\gamma_k \leq e^{-m_{\mathrm{lat}}}$ for all k; the *physical gap* at level k is $a_k^{-1} \log(1/\gamma_k) \in [0,\infty]$.

Lemma 294 (torus limits exist and are reflection positive)**.** *Let G be a compact group with a faithful unitary representation, $\beta \geq 0$, and let μ_L be the Wilson measure of $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$ at β.*

(i) *Every sequence of even integers $L \to \infty$ has a subsequence along which $\omega(O) = \lim_L \mu_L(O)$ exists for every continuous function O of finitely many links of $\mathbb{Z}^4$, each O being evaluated on Λ_L through a box centred at the origin once L is large. The functional ω is a state, given by a probability measure on the configurations of $\mathbb{Z}^4$.*

(ii) *ω is invariant under the translations of $\mathbb{Z}^4$ and under the hypercubic group B_4.*

(iii) *ω is reflection positive for the reflections in coordinate hyperplanes through sites and in coordinate hyperplanes bisecting links: $\omega(\vartheta_\mu(\bar F)F) \geq 0$ for every μ and every continuous gauge-invariant function F of finitely many links with both endpoints on the positive side of the plane of ϑ_μ.*

(iv) *The reconstruction of Appendix F applies to ω and gives $\mathscr{H}_\omega$, Ω, $F \mapsto \Psi_F$ and a self-adjoint $\mathbb{T}$ with $0 \leq \mathbb{T} \leq 1$, $\mathbb{T}\Omega = \Omega$ and $\langle \Psi_F, \mathbb{T}^t \Psi_H \rangle = \omega(\vartheta(\bar F)\tau_t H)$ for integers $t \geq 0$. For F with $\omega(F) = 0$, $\Psi_F \perp \Omega$ and $\|\mathbb{T}^t \Psi_F\| \leq \gamma(\omega)^t \|\Psi_F\|$.*

Proof. (i) Let E_m be the set of links of $\mathbb{Z}^4$ with both endpoints in $[-m,m]^4$. For $L > 2m+2$ the quotient map $\mathbb{Z}^4 \to \Lambda_L$ is

injective on $[-m,m]^4$, two of its points differing by less than L in every coordinate, so it identifies E_m with a set of links of Λ_L and a function of the links in E_m with a function on the configurations of Λ_L; this defines $\mu_L(O)$ for $O \in C(G^{E_m})$ and $L > 2m+2$. The image measure $\nu_{L,m}$ of μ_L on G^{E_m} is a Borel probability measure on a compact metrisable space, so by the Riesz representation theorem and the Banach–Alaoglu theorem every sequence of them has a weakly convergent subsequence, and a diagonal argument over $m \in \mathbb{N}$ gives a subsequence (L_j) with $\nu_{L_j,m} \to \nu_m$ weakly for every m. The projection $G^{E_{m+1}} \to G^{E_m}$ is continuous and carries $\nu_{L,m+1}$ to $\nu_{L,m}$, hence ν_{m+1} to ν_m. The Kolmogorov extension theorem on the countable product $G^{E(\mathbb{Z}^4)}$ gives a probability measure ν with marginals ν_m, and $\omega(O) = \int O\, d\nu = \lim_j \mu_{L_j}(O)$ for $O \in C(G^{E_m})$ and every m.

(ii) Let $v \in \mathbb{Z}^4$ and $O \in C(G^{E_m})$. The translate $\tau_v O$ lies in $C(G^{E_{m'}})$ with $m' = m + |v|_\infty$. For $L > 2m'+2$ the translation of Λ_L by the class of v is a bijection of its links that preserves the set of plaquettes and therefore μ_L, and it carries the embedding of E_m to that of $E_m + v$; hence $\mu_L(\tau_v O) = \mu_L(O)$, and the limit gives $\omega(\tau_v O) = \omega(O)$. The hypercubic group acts on Λ_L fixing the origin, permutes the plaquettes, and preserves $\operatorname{Re}\operatorname{Tr} Q_p$ under reversal of orientation because $\operatorname{Re}\operatorname{Tr} U^{-1} = \operatorname{Re}\operatorname{Tr} U$ for unitary U; it carries each link variable to the variable of the image link or to its inverse, and the product Haar measure is invariant under $U \mapsto U^{-1}$ on every link; so μ_L is B_4-invariant, and so is ω.

(iii) Let F be a continuous gauge-invariant function of the links in E_m lying on the positive side of the plane of ϑ_μ. For even $L > 2m+2$ the reflection ϑ_μ induces a reflection of Λ_L, in the plane $\{x_\mu = 0\}$ (and its antipode $\{x_\mu = L/2\}$) for the reflection through sites and in the plane $\{x_\mu = \frac{1}{2}\}$ for the reflection bisecting links, the embedded support of F lies strictly on one side of it, and Osterwalder and Seiler proved $\mu_L(\vartheta_\mu(\bar F)F) \geq 0$ for the Wilson measure of the torus with $\beta \geq 0$ for both kinds of reflection [90, 95]. This is one inequality for one local function, and it passes to the limit along (L_j).

(iv) By (ii) and (iii), ω is a state of the kind treated in Appendix F, which gives the reconstruction. If $\omega(F) = 0$ then $\langle \Omega, \Psi_F \rangle = \omega(F) = 0$, so $\Psi_F \in \Omega^\perp$, and $\|\mathbb{T}'\Psi_F\| \leq \gamma(\omega)^t \|\Psi_F\|$ because $\Omega^\perp$ is invariant under $\mathbb{T}$. □

The proof of Corollary 63 is this lemma followed by the decay of Theorem 62; the lemma is the part that needs no hypothesis on β.

The gap-free two-point theory. Let ω_k be torus limits at $\beta_k \geq 0$ as in Lemma 294, with $\mathbb{T}_k$, Ω_k and γ_k, and let Φ, r, $A_k(f)$, c_k, $\mathscr{N}_k(\varepsilon)$ be as in Sec. X L. For $\eta \geq 0$ and a direction μ let $\mathscr{V}_\mu^\eta$ be the set of $g \in C_c(\mathbb{R}^4)$ with $\operatorname{supp} g \subset \{x_\mu > \eta\}$, and for $g \in \mathscr{V}_0^0$ put

$$R_k(g) = \frac{\omega_k\big(\vartheta(\overline{A_k(g)})A_k(g)\big)}{\mathscr{N}_k(\varepsilon)}$$

when $\mathscr{N}_k(\varepsilon) > 0$. Let $\tau_y g = g(\cdot - y)$ for $y \in \mathbb{R}^4$.

Theorem 295 (the two-point function without the averaged condition)**.** *Let $\varepsilon > 0$ and let k be so large that $(r+1)a_k \leq \varepsilon$.*

(i) *If $\mathscr{N}_k(\varepsilon) = 0$, then $\omega_k(A_k(f)A_k(g)) = 0$ for every $(f,g) \in \mathscr{C}_2(Q,\delta)$ with $\delta \geq 8\varepsilon$. Consequently every sequence of k has a subsequence along which either $\mathscr{N}_k(\varepsilon) > 0$ for every k, or the connected two-point function G_k of Φ vanishes, for every k, at all $z \in \mathbb{Z}^4$ with $a_k|z|_\infty > 8\varepsilon$.*

In (ii)–(vi) *suppose $\mathscr{N}_k(\varepsilon) > 0$ for the k considered and put $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$.*

(ii) *Let $c \in \mathbb{R}$, $d \geq 0$, f supported in a cube of side L in $\{x_\mu \leq c - 2\varepsilon\}$ and g in a cube of side L' in $\{x_\mu \geq c + 2\varepsilon + d\}$. Then*

$$Z_k^2 \big|\omega_k\big(A_k(f)A_k(g)\big)\big| \leq \mathsf{K}_L \mathsf{K}_{L'}\, \gamma_k^{\lfloor d/a_k \rfloor}\, \|f\|_\infty \|g\|_\infty,$$

and for $(f,g) \in \mathscr{C}_2(Q,\delta)$ with Q of side L and $\delta \geq 8\varepsilon$,

$$Z_k^2 \big|\omega_k\big(A_k(f)A_k(g)\big)\big| \leq \mathsf{K}^2_{4L/\delta} \mathsf{K}^2_{\delta/4}\, \|f\|_\infty \|g\|_\infty.$$

(iii) *Every sequence of k has a subsequence along which $S_2(f,g) = \lim_k Z_k^2 \omega_k(A_k(f)A_k(g))$ exists for all $f,g \in C_c(\mathbb{R}^4)$ with $\operatorname{dist}_\infty(\operatorname{supp} f, \operatorname{supp} g) > 8\varepsilon$. It is symmetric and bilinear, bounded as in* (ii)*, invariant under the translations of $\mathbb{R}^4$ and under B_4, $S_2(\bar f, \bar g) = \overline{S_2(f,g)}$, and for every μ*

$$\sum_{i,j} \bar c_i c_j S_2(\bar g_i \circ \theta_\mu, g_j) \geq 0 \qquad (c_i \in \mathbb{C},\ g_i \in \mathscr{V}_\mu^{4\varepsilon}).$$

It is given by a unique distribution on the open set $\{(x,y) \in \mathbb{R}^8 : |x-y|_\infty > 8\varepsilon\}$.

(iv) *Let S_2 be a limit of* (iii) *and μ a direction. The form $\langle g, g' \rangle_\mu = S_2(\bar g \circ \theta_\mu, g')$ on $\mathscr{V}_\mu^{4\varepsilon}$ is Hermitian and positive semidefinite; let $\mathscr{H}_\mu^{(1)}$ be the completion of its quotient by its null space and $g \mapsto [g]$ the quotient map. There are a self-adjoint operator $H_\mu^{(1)} \geq 0$ on $\mathscr{H}_\mu^{(1)}$ and a strongly continuous unitary representation U_μ of the hyperplane $e_\mu^\perp$ commuting with it such that, for all $g, g' \in \mathscr{V}_\mu^{4\varepsilon}$, $t \geq 0$ and $y \perp e_\mu$,*

$$S_2\big(\bar g \circ \theta_\mu, \tau_{te_\mu + y} g'\big) = \big\langle [g], e^{-tH_\mu^{(1)}} U_\mu(y) [g'] \big\rangle.$$

The map $[g] \mapsto [g \circ \pi_\mu]$ is a unitary operator from $\mathscr{H}_0^{(1)}$ onto $\mathscr{H}_\mu^{(1)}$ that carries $H_0^{(1)}$ to $H_\mu^{(1)}$. If S_2 is defined on all separated pairs, the same holds with $\mathscr{V}_\mu^{4\varepsilon}$ replaced by $\mathscr{V}_\mu^0$.

(v) *For a limit S_2 of* (iii) *the following are equivalent: S_2 does not vanish identically; $S_2(\bar g \circ \theta_0, g) > 0$ for some $g \in \mathscr{V}_0^{4\varepsilon}$; $\mathscr{H}_0^{(1)} \neq 0$. Some limit of* (iii) *does not vanish identically if and only if $\limsup_k R_k(g) > 0$ for some $g \in \mathscr{V}_0^{4\varepsilon}$, and if $\liminf_k R_k(g_0) > 0$ for one $g_0 \in \mathscr{V}_0^{4\varepsilon}$, then no limit of* (iii) *vanishes identically. If S_2 is defined on all separated pairs, the equivalence holds with $\mathscr{V}_0^{4\varepsilon}$ replaced by $\mathscr{V}_0^0$.*

(vi) *If there is a lattice gap, $\gamma_k \leq e^{-m_{\mathrm{lat}}}$ for a fixed $m_{\mathrm{lat}} > 0$ and all k, then every limit of* (iii) *vanishes identically.*

The theorem is Theorem 254(i)–(iii),(v) and the constructions of Corollary 255(b) with the gap removed: γ_k is only the norm of $\mathbb{T}_k$ on $\Omega_k^\perp$, no bound on it is assumed, and the Hamiltonian $H_\mu^{(1)}$ is positive but no lower bound on its spectrum is claimed.

Proof. We use the following parts of Sec. X L, none of which uses the averaged condition: Lemma 247(i) and (ii), which are identities; the first two assertions of Lemma 247(iii), $\omega_k(\vartheta_\mu(\bar F)F) \geq 0$ and $0 \leq \omega_k(\vartheta_\mu(\bar F)\tau^\mu_{2t}F) \leq \omega_k(\vartheta_\mu(\bar F)F)$, whose proof uses only the reconstruction and $0 \leq \mathbb{T}_k \leq 1$; the identity $N_\mu(\tau_\nu F) = N_\mu(F)$ for $\nu_\mu = 0$ from the same proof; and Lemma 253, whose proof uses Lemma 247(i), (ii) and $\|\mathbb{T}_k^{n_0}\| \leq 1$. All of these hold for the states of Lemma 294.

(i) Suppose $\mathscr{N}_k(\varepsilon) = 0$. Then $\Psi_{A_k(h)} = 0$ for every h supported in B_ε, and Lemma 253 gives $\Psi_{A_k(g)} = 0$ for every g supported in a cube and in $\{x_0 \geq \varepsilon\}$, and, in the form with ϑ_μ, the same for g supported in $\{x_\mu \geq \varepsilon\}$ with the vectors of the reconstruction in the direction μ. Let f be supported in $\{x_\mu \leq c - 2\varepsilon\}$ and g in $\{x_\mu \geq c + 2\varepsilon\}$, both in cubes. The first three sentences of the proof of Theorem 254(i) use only the invariance of ω_k under π_μ and the translations and Lemma 247(i), and give $\omega_k(A_k(f)A_k(g)) = \langle \Psi_{A_k(h)}, \mathbb{T}_k^n \Psi_{A_k(g')} \rangle$ with h and g' supported in $\{x_0 \geq \varepsilon\}$; so $\omega_k(A_k(f)A_k(g)) = 0$. For $(f,g) \in \mathscr{C}_2(Q,\delta)$ with $\delta \geq 8\varepsilon$, the decomposition in the proof of Theorem 254(ii), with $s = \delta/4$, writes $\omega_k(A_k(f)A_k(g))$ as a finite sum of such terms, each with pieces in cubes of side s whose projections on some axis μ are at distance at least $\delta/2 \geq 4\varepsilon$; so the sum vanishes. Given $x, y \in \mathbb{Z}^4$ with $a_k|x-y|_\infty > 8\varepsilon$, choose $f, g \in C_c(\mathbb{R}^4)$ with values in $[0,1]$, supported in ℓ^∞-balls of radius $\rho < a_k/2$ about $a_k x$ and $a_k y$ with $2\rho \leq a_k|x-y|_\infty - 8\varepsilon$, and with $f(a_k x) = g(a_k y) = 1$; then $A_k(f) = a_k^4(\Phi_x - c_k)$, $A_k(g) = a_k^4(\Phi_y - c_k)$, the pair lies in $\mathscr{C}_2(Q, 8\varepsilon)$ for a cube Q containing both balls, and by translation invariance $a_k^8 G_k(y-x) = \omega_k(A_k(f)A_k(g)) = 0$. The dichotomy follows by passing to a subsequence on which the sign of $\mathscr{N}_k(\varepsilon)$ is constant.

(ii) The proof of Theorem 254(i) applies verbatim: the bound $|\langle \Psi_{A_k(h)}, \mathbb{T}_k^n \Psi_{A_k(g')} \rangle| \leq \gamma_k^n \|\Psi_{A_k(h)}\| \|\Psi_{A_k(g')}\|$ holds because both vectors are orthogonal to Ω_k, the packets being centred, and $\Omega_k^\perp$ is invariant under $\mathbb{T}_k$ (Lemma 294(iv)); the norms are bounded by Lemma 253. The second bound is the proof of Theorem 254(ii), which uses the first with $\gamma_k^{\lfloor d/a_k \rfloor} \leq 1$.

(iii) Let $\mathscr{G}$ be the countable set of the proof of Theorem 248(i). By (ii) the sequence $Z_k^2 \omega_k(A_k(g)A_k(g'))$ is bounded for each pair $g, g' \in \mathscr{G}$ at distance greater than 8ε, and a diagonal subsequence makes all of them converge. For f, g at distance $\delta > 8\varepsilon$, choose $g_1, g_2 \in \mathscr{G}$ as in that proof with $\|f - g_1\|_\infty, \|g - g_2\|_\infty < \eta$ and supports within η of those of f, g, where $\eta < (\delta - 8\varepsilon)/4$; the pairs $(f - g_1, g)$, $(g_1, g - g_2)$ and (g_1, g_2) lie in one $\mathscr{C}_2(Q', \delta')$ with $\delta' = \delta - 2\eta > 8\varepsilon$, and the second bound of (ii) shows that $Z_k^2 \omega_k(A_k(f)A_k(g))$ is a Cauchy sequence. Symmetry and bilinearity pass to the limit, and so does the bound. Invariance under dyadic translations and under B_4, and the conjugation identity, follow from Lemma 247(i) and the invariance of ω_k, and invariance under real translations from the bound and uniform continuity, exactly as in the proof of Theorem 248(ii), the translated pairs staying at distance greater than 8ε. For the positivity, let $g = \sum_i c_i g_i$ with $g_i \in \mathscr{V}_\mu^{4\varepsilon}$; then g is supported in $\{x_\mu \geq 4\varepsilon + \eta'\}$ for some $\eta' > 0$. For $(r+1)a_k \leq \varepsilon$ Lemma 247(ii),(iii) gives $\omega_k(\vartheta_\mu(\overline{A_k(g)})A_k(g)) \geq 0$, and by Lemma 247(i) this number equals $\omega_k(A_k(\bar g \circ \theta_{\mu,k})A_k(g))$. The function $\bar g \circ \theta_{\mu,k}$ is supported in $\{x_\mu \leq -4\varepsilon - \eta' + a_k\}$, so the pairs $(\bar g \circ \theta_{\mu,k}, g)$ and $(\bar g \circ \theta_\mu, g)$ lie in one $\mathscr{C}_2(Q', 8\varepsilon + \eta')$ for large k, and $\|\bar g \circ \theta_{\mu,k} - \bar g \circ \theta_\mu\|_\infty \to 0$ because $\theta_{\mu,k}$ is θ_μ followed by a translation of length at most a_k and g is uniformly continuous. By the bound of (ii) the difference of the two moments tends to 0, and $S_2(\bar g \circ \theta_\mu, g) = \lim_k Z_k^2 \omega_k(A_k(\bar g \circ \theta_{\mu,k})A_k(g)) \geq 0$; expanding g gives the stated inequality. The distribution is constructed from the bound exactly as in the proof of Theorem 248(i), with the products of open cubes replaced by products $U_1 \times U_2$ of bounded open sets whose closures are at ℓ^∞-distance greater than 8ε; these cover the open set in question, and the localisation argument given there applies.

(iv) Write $\langle \cdot, \cdot \rangle$ for $\langle \cdot, \cdot \rangle_\mu$ and τ_t for τ_{te_μ}. For $g, g' \in \mathscr{V}_\mu^{4\varepsilon}$ and $t \geq 0$ the pair $(\bar g \circ \theta_\mu, \tau_t g')$ has its supports in $\{x_\mu < -4\varepsilon\}$ and $\{x_\mu > 4\varepsilon + t\}$, at distance greater than 8ε, so S_2 is defined on it; and $t \mapsto S_2(\bar g \circ \theta_\mu, \tau_t g')$ is continuous on $[0,\infty)$ by the bound of (ii) and the uniform continuity of g', the pairs lying in one $\mathscr{C}_2(Q', \delta')$ with $\delta' > 8\varepsilon$ for t in a bounded set.

Hermitian symmetry. Since $\theta_\mu \in B_4$ and S_2 is symmetric, B_4-invariant and satisfies the conjugation identity, $\overline{\langle g, g' \rangle} = S_2(\bar g \circ \theta_\mu, g') = S_2(\bar g, g' \circ \theta_\mu) = S_2(g' \circ \theta_\mu, \bar g) = \overline{S_2(\bar g' \circ \theta_\mu, g)} = \overline{\langle g', g \rangle}$. Positivity is (iii). So the quotient by the null space is a pre-Hilbert space and $\mathscr{H}_\mu^{(1)}$ is its completion.

Time translations. We use $\theta_\mu e_\mu = -e_\mu$, hence $\overline{\tau_t g} \circ \theta_\mu = \tau_{-t}(\bar g \circ \theta_\mu)$, and the translation invariance of S_2. For $g, g' \in \mathscr{V}_\mu^{4\varepsilon}$ and $t \geq 0$,

$$\langle g, \tau_t g' \rangle = S_2\big(\tau_{-t}(\bar g \circ \theta_\mu), g'\big) = S_2\big(\overline{\tau_t g} \circ \theta_\mu, g'\big) = \langle \tau_t g, g' \rangle. \tag{a}$$

Next let $t \geq 0$ be dyadic and k so large that $n = t/a_k$ is an integer. By Lemma 247(ii) the function $F = A_k(g)$ depends on links in $\{x_\mu \geq 1\}$, and Lemma 247(iii) gives $0 \leq \omega_k(\vartheta_\mu(\bar F)\tau^\mu_{2n}F) \leq \omega_k(\vartheta_\mu(\bar F)F)$. By Lemma 247(i) the middle term is $\omega_k(A_k(\bar g \circ \theta_{\mu,k})A_k(\tau_{2t}g))$ and the right term is the same with $\tau_{2t}g$ replaced by g. Multiplying by Z_k^2 and passing to the limit, with the approximation of $\theta_{\mu,k}$ by θ_μ as in (iii), gives

$$0 \leq S_2(\bar g \circ \theta_\mu, \tau_{2t} g) \leq S_2(\bar g \circ \theta_\mu, g) \qquad (t \geq 0 \text{ dyadic}),$$

and by continuity in t for all $t \geq 0$. By (a) the middle term is $\langle \tau_t g, \tau_t g \rangle$. Hence $\|[\tau_t g]\| \leq \|[g]\|$, so $T(t)[g] = [\tau_t g]$ is a well defined linear contraction of the quotient, which extends to a contraction $T(t)$ of $\mathscr{H}_\mu^{(1)}$; by (a) it is symmetric, hence self-adjoint; $\langle [g], T(t)[g] \rangle = \|[\tau_{t/2} g]\|^2 \geq 0$ on a dense set, so $T(t) \geq 0$; and $T(t)T(s) = T(t+s)$, $T(0) = 1$. For strong continuity, let $g \in \mathscr{V}_\mu^{4\varepsilon}$ and $t \geq 0$; the function $h = \tau_t g - g$ lies in $\mathscr{V}_\mu^{4\varepsilon}$, and

$$\|T(t)[g] - [g]\|^2 = S_2(\bar h \circ \theta_\mu, h) \leq \mathsf{K}^2_{4L'/\delta'} \mathsf{K}^2_{\delta'/4} \|h\|^2_\infty$$

by the bound of (ii), for a cube of side L' and a separation $\delta' > 8\varepsilon$ that do not depend on $t \in [0,1]$. Since $\|h\|_\infty \to 0$ as $t \to 0$, $T(t) \to 1$ strongly on a dense set, hence on $\mathscr{H}_\mu^{(1)}$ because $\|T(t)\| \leq 1$, and $t \mapsto T(t)$ is strongly continuous on $[0,\infty)$ by the semigroup law. Now $T(1)$ is injective: $\ker T(2^{-m}) = \ker T(2^{-m-1})$ for every m, because $T(2^{-m}) = T(2^{-m-1})^2$ with $T(2^{-m-1}) \geq 0$; so a vector $v \in \ker T(1)$ lies in $\ker T(2^{-m})$ for all m, and $v = \lim_m T(2^{-m})v = 0$. Hence the spectral measure

of $T(1)$ has no mass at 0, and $H_\mu^{(1)} = -\log T(1)$, defined by the functional calculus on $(0,1]$, is self-adjoint with $H_\mu^{(1)} \geq 0$ and $e^{-H_\mu^{(1)}} = T(1)$. The positive square root of a positive operator is unique, so $T(2^{-m}) = T(1)^{2^{-m}}$ by induction on m, and $T(t) = e^{-tH_\mu^{(1)}}$ for every dyadic $t \geq 0$; both sides are strongly continuous in t, so the identity holds for all $t \geq 0$.

Spatial translations. For $y \perp e_\mu$, $\theta_\mu y = y$, so $\overline{\tau_y g} \circ \theta_\mu = \tau_y(\bar g \circ \theta_\mu)$, and the translation invariance of S_2 gives $\langle \tau_y g, \tau_y g' \rangle = \langle g, g' \rangle$; thus $U_\mu(y)[g] = [\tau_y g]$ defines an isometry of the quotient, with inverse $U_\mu(-y)$, which extends to a unitary operator; $U_\mu(y)U_\mu(y') = U_\mu(y+y')$, $U_\mu(y)$ commutes with $T(t)$ because $\tau_y \tau_t = \tau_t \tau_y$, and U_μ is strongly continuous by the same estimate as for T. Finally $\langle g, \tau_{te_\mu + y} g' \rangle = \langle [g], T(t)U_\mu(y)[g'] \rangle$ by definition.

The four directions. For $g \in \mathscr{V}_0^{4\varepsilon}$ the function $g \circ \pi_\mu$ vanishes unless $(\pi_\mu x)_0 = x_\mu > 4\varepsilon$, so it lies in $\mathscr{V}_\mu^{4\varepsilon}$, and the map is onto because π_μ is an involution. From $\pi_\mu \theta_\mu = \theta_0 \pi_\mu$ and the B_4-invariance of S_2,

$$\begin{aligned} S_2\big(\overline{g \circ \pi_\mu} \circ \theta_\mu, g' \circ \pi_\mu\big) &= S_2\big((\bar g \circ \theta_0) \circ \pi_\mu, g' \circ \pi_\mu\big) \\ &= S_2(\bar g \circ \theta_0, g'), \end{aligned}$$

so $[g] \mapsto [g \circ \pi_\mu]$ is a well defined isometry of the quotients, which extends to a unitary operator W_μ. Since $\pi_\mu e_0 = e_\mu$, $(\tau_{te_0} g) \circ \pi_\mu = \tau_{te_\mu}(g \circ \pi_\mu)$, so W_μ intertwines the two semigroups and therefore the two Hamiltonians.

If S_2 is defined on all separated pairs, every step above applies with 4ε replaced by 0: the pairs used are then separated, the positivity $\omega_k(\vartheta_\mu(\overline{A_k(g)})A_k(g)) \geq 0$ holds for g supported in $\{x_\mu \geq \eta\}$ once $(r+1)a_k \leq \eta$, and the approximation of $\theta_{\mu,k}$ by θ_μ keeps the pairs separated.

(v) If $S_2(\bar g \circ \theta_0, g) > 0$ for some $g \in \mathscr{V}_0^{4\varepsilon}$, then $S_2 \neq 0$ and $[g] \neq 0$. Conversely suppose $S_2(\bar g \circ \theta_0, g) = 0$ for all $g \in \mathscr{V}_0^{4\varepsilon}$. By (iv) with $\mu = 0$ and B_4-invariance, $S_2(\bar g \circ \theta_\mu, g) = S_2(\overline{g \circ \pi_\mu} \circ \theta_0, g \circ \pi_\mu) = 0$ for every μ and $g \in \mathscr{V}_\mu^{4\varepsilon}$, so the positive semidefinite form $\langle \cdot, \cdot \rangle_\mu$ vanishes on its diagonal and, by the Cauchy–Schwarz inequality, identically: $S_2(\bar g \circ \theta_\mu, g') = 0$ for $g, g' \in \mathscr{V}_\mu^{4\varepsilon}$. With translation invariance this says that $S_2(f,g) = 0$ whenever f is supported in $\{x_\mu < c - 4\varepsilon\}$ and g in $\{x_\mu > c + 4\varepsilon\}$ for some c and μ, as $f = \bar h \circ \theta_\mu$ with $h = \bar f \circ \theta_\mu$ after the translation by $-ce_\mu$. Let $(f,g) \in \mathscr{C}_2(Q,\delta)$ with $\delta > 8\varepsilon$, put $s = (\delta - 8\varepsilon)/4 > 0$, and write $f = \sum_m \chi_m^s f$, $g = \sum_{m'} \chi_{m'}^s g$ with the partition of unity of the proof of Theorem 248(vi), whose members are supported in closed cubes of side s; finitely many terms are nonzero. For a pair of nonzero pieces pick points p, q of their supports and μ with $|p_\mu - q_\mu| \geq \delta$; the projections of the two cubes on the μ axis are at distance at least $\delta - 2s = \delta/2 + 4\varepsilon > 8\varepsilon$, so the pieces are at ℓ^∞-distance greater than 8ε and lie on opposite sides of some c with margins greater than 4ε. By bilinearity $S_2(f,g)$ is the sum of the S_2 of the pairs of pieces, each of which is 0. So S_2 vanishes identically, and $\mathscr{H}_0^{(1)} = 0$ means exactly that $S_2(\bar g \circ \theta_0, g) = 0$ for all $g \in \mathscr{V}_0^{4\varepsilon}$. This proves the equivalence. If S_2 is defined on all separated pairs, the same argument with $\mathscr{V}_\mu^0$ in place of $\mathscr{V}_\mu^{4\varepsilon}$, pairs at a positive distance δ, $s = \delta/4$ and margins greater than 0 gives the equivalence on $\mathscr{V}_0^0$.

For $g \in \mathscr{V}_0^{4\varepsilon}$ and large k, Lemma 247(i) gives $R_k(g) = Z_k^2 \omega_k(A_k(\bar g \circ \theta_{0,k})A_k(g))$, and along any subsequence with a limit S_2, the approximation of $\theta_{0,k}$ by θ_0 in (iii) gives $R_k(g) \to S_2(\bar g \circ \theta_0, g)$. If some limit S_2 is not zero, pick g with $S_2(\bar g \circ \theta_0, g) > 0$; then $\limsup_k R_k(g) > 0$. If $\limsup_k R_k(g) > 0$, take a subsequence along which $R_k(g)$ tends to the $\limsup$ and refine it by (iii); the limit satisfies $S_2(\bar g \circ \theta_0, g) = \limsup_k R_k(g) > 0$. If $\liminf_k R_k(g_0) > 0$, every limit has $S_2(\bar g_0 \circ \theta_0, g_0) \geq \liminf_k R_k(g_0) > 0$.

(vi) Let $(f,g) \in \mathscr{C}_2(Q,\delta)$ with $\delta > 8\varepsilon$ and decompose as in the proof of the second bound of (ii), with pieces in cubes of side $s = \delta/4$ whose projections on some axis are at distance at least $\delta/2$. Each pair of pieces satisfies the hypothesis of the first bound of (ii) with $d = \delta/2 - 4\varepsilon > 0$, so the lattice quantity is at most $\mathsf{K}_{4L/\delta}^2 \mathsf{K}_{\delta/4}^2 \|f\|_\infty \|g\|_\infty e^{-m_{\mathrm{lat}}(d/a_k - 1)}$, which tends to 0. □

The structure that (iii) and (iv) give is, in the language of [88] restricted to the two-point function at separations greater than 8ε: symmetry, invariance under the translations and the hypercubic group, and reflection positivity in each of the four coordinate directions, with a positive Hamiltonian on the one-field space of each direction. It does not give invariance under $\mathrm{O}(4)$, the independence of the limit from the subsequence, a regularity or growth condition at coinciding points, clustering, or a lower bound on $\sigma(H_\mu^{(1)})$; and it does not decide whether $\mathscr{H}_\mu^{(1)} \neq 0$, which is the content of (v).

Corollary 296 (the physical gap under (SD))**.** *In the setting of Theorem 295, with $\mathscr{N}_k(\varepsilon) > 0$ for large k:*

(i) *for $(r+1)a_k \leq \varepsilon' < \varepsilon$, $\mathscr{N}_k(\varepsilon) \leq \mathsf{K}^2 \gamma_k^{2\lfloor(\varepsilon - \varepsilon')/a_k\rfloor} \mathscr{N}_k(\varepsilon')$;*

(ii) *if* (SD) *holds, then for every $\varepsilon' \in (0,\varepsilon)$, with $C_{\varepsilon'} = \limsup_k \mathscr{N}_k(\varepsilon')/\mathscr{N}_k(\varepsilon) < \infty$,*

$$\limsup_k \frac{1}{a_k} \log \frac{1}{\gamma_k} \leq \frac{\log(\mathsf{K}^2 C_{\varepsilon'})}{2(\varepsilon - \varepsilon')};$$

in particular a lattice gap is incompatible with (SD).

(iii) (SD) *holds if and only if* (UV) *holds, and if and only if* $(\mathrm{M_s})$ *holds for $n = 2$ with the constants $Z_k = \mathscr{N}_k(\varepsilon)^{-1/2}$.*

Proof. (i) is the first inequality of Proposition 256(i); its proof uses Lemma 247(i), the identity $\Psi_{A_k(h)} = \mathbb{T}_k^n \Psi_{A_k(h')}$ from the proof of Lemma 247(iii), $\Psi_{A_k(h')} \perp \Omega_k$ for the centred packet, $\|\mathbb{T}_k^n \Psi\| \leq \gamma_k^n \|\Psi\|$ on $\Omega_k^\perp$ (Lemma 294(iv)) and Lemma 253; the second inequality there, $\gamma_k \leq e^{-m_* a_k}$, is the only use of the averaged condition and is not needed. (ii) is the proof of Proposition 256(iii) from (i): with $n = \lfloor(\varepsilon - \varepsilon')/a_k\rfloor \geq (\varepsilon - \varepsilon')/a_k - 1$ and $C > C_{\varepsilon'}$, (i) gives $\gamma_k^{-2n} \leq \mathsf{K}^2 C$ for large k, that is $a_k^{-1} \log(1/\gamma_k) \leq \log(\mathsf{K}^2 C)/(2(\varepsilon - \varepsilon' - a_k))$, and $C \downarrow C_{\varepsilon'}$. Under a lattice gap the left side is at least $m_{\mathrm{lat}}/a_k \to \infty$. (iii) The equivalence with (UV) is Proposition 256(ii), whose proof uses only the commutation of the spectral projections with $\mathbb{T}_k$, $\|\mathbb{T}_k\| \leq 1$, Lemma 253 and spectral measures; the equivalence with $(\mathrm{M_{s,2}})$ is Corollary 255(a), whose proof uses Theorem 254(i),(ii) with ε' in place of ε, that is, Theorem 295(ii). □

The short-distance law without the averaged condition. The hypotheses of Theorem 279 are the averaged condition and (AF). The averaged condition can be dropped from parts (i)–(iv), with one change in the proof of (iv): (NE) is read off the lower bound of (AF) directly.

Theorem 297 ((AF) alone gives the two-point inputs)**.** *Let ω_k be torus limits at arbitrary $\beta_k \geq 0$ and assume* (AF) *for constants Z_k. Then:*

(i) *the bound of Theorem 279*(i) *holds, so* ($\mathrm{M_{s,2}}$) *holds for the constants Z_k, every limit S_2 of $Z_k^2\omega_k(A_k(f)A_k(g))$ along a subsequence satisfies $|S_2(f,g)| \leq C_1 \iint |f(x)||g(y)|w_\gamma(|x-y|)\,dx\,dy$ on the separated pairs, and extends uniquely to a continuous linear functional on $\mathscr{S}_0(\mathbb{R}^8)$;*

(ii) *for $f,g \geq 0$ forming a separated pair with $|x-y| \leq r_0$ for $x \in \operatorname{supp} f$, $y \in \operatorname{supp} g$, every such limit has $S_2(f,g) \geq c_1 \iint f(x)g(y)w_\gamma(|x-y|)\,dx\,dy$; in particular no limit vanishes identically;*

(iii) *for every $\varepsilon > 0$, $\mathscr{N}_k(\varepsilon) > 0$ for large k and $0 < \liminf_k Z_k^2\mathscr{N}_k(\varepsilon) \leq \limsup_k Z_k^2\mathscr{N}_k(\varepsilon) < \infty$;*

(iv) ($\mathrm{SD_p}$) *holds with $p_0 = 4$, hence* (SD) *and* (UV)*;* (NE) *holds; the physical gap $a_k^{-1}\log(1/\gamma_k)$ is bounded in k; and for the constants $Z_k^\sharp = \mathscr{N}_k(\varepsilon)^{-1/2}$ every limit $S_2^\sharp$ of $(Z_k^\sharp)^2\omega_k(A_k(f)A_k(g))$ along a subsequence exists on all separated pairs, is a positive multiple of a limit of* (i)*, does not vanish identically, extends to $\mathscr{S}_0(\mathbb{R}^8)$, and has $\mathscr{H}_\mu^{(1)} \neq 0$ for every μ;*

(v) (AF) *fails for every choice of Z_k if there is a lattice gap, in particular under the hypotheses of Corollary 251 or* [21, Corollary 322].

Proof. (i) and (ii) are proved exactly as Theorem 279(i),(ii): the proofs use Lemma 247(i), translation invariance, the two bounds of (AF), the counting of lattice points, the convergence of Riemann sums, and Lemma 272; the averaged condition is not used. In particular ($\mathrm{M_{s,2}}$) holds, so the diagonal argument of the proof of Theorem 248(i) gives limits on all separated pairs along subsequences, and the proofs of Theorem 248(ii) and (iii) for $n = 2$, which use only ($\mathrm{M_{s,2}}$), Lemma 247(i),(ii) and the positivity part of Lemma 247(iii), show that every such limit is symmetric, invariant under the translations and B_4, satisfies the conjugation identity, and has $S_2(\bar g \circ \theta_\mu, g) \geq 0$ for $g \in \mathscr{V}_\mu^0$. Theorem 295(iv), in its form for limits defined on all separated pairs, applies to these limits with the constants Z_k: its proof used of the constants only that S_2 is a limit of $Z_k^2\omega_k(A_k(\cdot)A_k(\cdot))$ with the stated invariance, positivity and boundedness, the bound of Theorem 295(ii) being replaced by ($\mathrm{M_{s,2}}$).

(iii) The upper bound is as in the proof of Theorem 279(iii): with Q_0 the cube of side $L_0 = 2(\varepsilon + 2 + 2a_0)$ there, for h supported in $B_{\varepsilon'}$ with $\|h\|_\infty \leq 1$, $\varepsilon' \leq \varepsilon$ and $a_k \leq \varepsilon'$, the pair $(\bar h \circ \theta_{0,k}, h)$ lies in $\mathscr{C}_2(Q_0, \min(\varepsilon', 1))$, so (i) gives

$$Z_k^2\mathscr{N}_k(\varepsilon') \leq C_2\max(1,\varepsilon'^{-4}), \qquad C_2 = 100C_1(1+L_0)^4, \quad \text{(b)}$$

for large k. For the lower bound suppose $Z_k^2\mathscr{N}_k(\varepsilon) \to 0$ along a subsequence, and refine it to a limit S_2 on all separated pairs. Let $g \in \mathscr{V}_0^0$ be supported in a cube of side L and in $\{x_0 \geq \eta\}$, $\eta > 0$, and let $t \geq 4\varepsilon + 1$ be dyadic. For large k the function $\bar g \circ \theta_{0,k}$ is supported in $\{x_0 \leq -\eta + a_k\}$ and $\tau_{te_0}g$ in $\{x_0 \geq \eta + t\}$; with $c = 2\varepsilon - \eta + a_k$ and $d = 0$ these are the supports of the first bound of Theorem 295(ii), whose proof does not use $\mathscr{N}_k(\varepsilon) > 0$ once it is multiplied by $\mathscr{N}_k(\varepsilon)$, and that bound gives

$$Z_k^2\big|\omega_k\big(A_k(\bar g \circ \theta_{0,k})A_k(\tau_{te_0}g)\big)\big| \leq \mathsf{K}_L^2 Z_k^2\mathscr{N}_k(\varepsilon)\,\|g\|_\infty^2 \to 0.$$

By the approximation of $\theta_{0,k}$ by θ_0, $S_2(\bar g \circ \theta_0, \tau_{te_0}g) = 0$. By Theorem 295(iv) for this S_2 on $\mathscr{V}_0^0$, this number is $\|e^{-tH_0^{(1)}/2}[g]\|^2$, and $e^{-tH_0^{(1)}/2}$ is injective, so $[g] = 0$, that is $S_2(\bar g \circ \theta_0, g) = 0$ for every $g \in \mathscr{V}_0^0$. By the argument of Theorem 295(v), applied with $\mathscr{V}_\mu^0$ and separated pairs, S_2 vanishes identically. This contradicts (ii). Hence $\liminf_k Z_k^2\mathscr{N}_k(\varepsilon) > 0$, and in particular $\mathscr{N}_k(\varepsilon) > 0$ for large k.

(iv) Let $c_\varepsilon = \frac{1}{2}\liminf_k Z_k^2\mathscr{N}_k(\varepsilon) > 0$. By (b) and (iii), for $\varepsilon' \in (0,\varepsilon]$ and large k,

$$\mathscr{N}_k(\varepsilon') \leq \frac{C_2\max(1,\varepsilon'^{-4})}{c_\varepsilon}\mathscr{N}_k(\varepsilon) \leq C_0\Big(\frac{\varepsilon}{\varepsilon'}\Big)^4\mathscr{N}_k(\varepsilon)$$

with $C_0 = C_2\max(1,\varepsilon^{-4})/c_\varepsilon$, which is ($\mathrm{SD_p}$) with $p_0 = 4$; it gives (SD), and (UV) and the bound on the physical gap by Corollary 296.

For (NE) let $g_0 \in C_c(\mathbb{R}^4)$ take values in $[0,1]$, be supported in $[\frac{r_0}{8},\frac{r_0}{4}] \times [0,\frac{r_0}{8}]^3$ and be positive on the interior of this box. Let $x,y \in \mathbb{Z}^4$ with $g_0(a_kx)g_0(a_ky) > 0$. The plane of ϑ is $\{x_0 = c\}$ with $|c| \leq \frac{1}{2}$, so $\vartheta x = (2c - x_0, x_1, x_2, x_3)$ and the time component of $y - \vartheta x$ is $x_0 + y_0 - 2c \geq r_0/(4a_k) - 1 \geq 2r + 1$ for large k, while $|y - \vartheta x| \leq |a_k^{-1}(a_ky - \theta_0(a_kx))| + 1 \leq a_k^{-1}r_0\sqrt{\frac{1}{4} + \frac{3}{64}} + 1 \leq r_0/a_k$ for large k. So both bounds of (AF) apply to every such pair, and with $p_k(x,y) = a_k^8 g_0(a_kx)g_0(a_ky)$ and $Y_k = Z_k^2\,\omega_k(\vartheta(\overline{A_k(g_0)})A_k(g_0))$,

$$\begin{aligned} Y_k &= \sum_{x,y} p_k(x,y)\,Z_k^2 G_k(y - \vartheta x) \\ &\geq c_1 \sum_{x,y} p_k(x,y)\,w_\gamma(a_k|y - \vartheta x|), \end{aligned}$$

the identity being Lemma 247(i) and translation invariance. On the support of $g_0 \otimes g_0$ the distance $|y - \theta_0 x|$ lies in $[\frac{r_0}{4}, r_0]$, where w_γ is Lipschitz, and $|a_k(y - \vartheta x)|$ differs from $|a_ky - \theta_0(a_kx)|$ by at most a_k; so the right side is a Riemann sum, up to an error $O(a_k)$, of the continuous compactly supported function $(x,y) \mapsto g_0(x)g_0(y)w_\gamma(|y - \theta_0 x|)$, and

$$\liminf_k Y_k \geq c_1 \iint g_0(x)g_0(y)\,w_\gamma(|y - \theta_0 x|)\,dx\,dy =: c_1 I > 0.$$

With (b) at $\varepsilon' = \varepsilon$ this gives

$$\liminf_k R_k(g_0) \geq \frac{c_1 I}{C_2\max(1,\varepsilon^{-4})} > 0,$$

which is (NE).

By (SD) and Corollary 296(iii), ($\mathrm{M_{s,2}}$) holds for $Z_k^\sharp$, so the limits $S_2^\sharp$ exist on all separated pairs along subsequences, and

by ($\mathrm{SD_p}$) and Proposition 275(a),(b), whose extension step uses only the proof of Corollary 255(a), that is Theorem 295(ii), and Lemma 272, they extend to $\mathscr{S}_0(\mathbb{R}^8)$; the invariance and positivity of the extensions, which that proposition takes from Theorem 273(iii) and Lemma 271(iii), are not needed here. Along a subsequence with a limit $S_2^\sharp$, refine so that $Z_k^2\mathscr{N}_k(\varepsilon)\to\lambda\in(0,\infty)$, by (iii), and so that the limit S_2 with the constants Z_k exists; then $S_2^\sharp = S_2/\lambda$, and $S_2^\sharp$ is not identically zero by (ii). By Theorem 295(v) in its form on $\mathscr{V}_\mu^0$, $\mathscr{H}_0^{(1)}\neq 0$, hence $\mathscr{H}_\mu^{(1)}\neq 0$ for every μ by the unitary equivalence of Theorem 295(iv).

(v) Under (AF), (iii) and (iv) give $\mathscr{N}_k(\varepsilon)>0$ for large k and (SD), and Corollary 296(ii) excludes a lattice gap. Corollary 251 provides the lattice gap $3.28/(625e^2)$ for SU(2) and $\beta_{W,k}\le\frac{1}{8}$, and [21, Corollary 322] the corresponding gap for $N\beta\le\kappa_G/48$. □

The theorem separates the two halves of the problem for $n=2$: the short-distance law (AF) is a statement about the Wilson measure at lattice distances between $2r+1$ and r_0/a_k and carries all of (SD), (UV) and (NE); the averaged condition is a statement about distances of order ℓ in physical units and carries the gap m_*. Neither implies the other, and (AF) forces the physical gap to stay bounded while the lattice gap tends to zero.

Flowed packets in a periodic spatial box. For the spatially flowed observables of Sec. IX D the analogous unconditional statement is already in the paper; we record it with the inventory of what it gives, since it is the only place where positivity of the renormalising norm is a theorem rather than a hypothesis.

Proposition 298 (flowed packets, unconditional part)**.** *Let $G=\mathrm{SU}(N)$, replace $\mathbb{Z}^4$ by $\mathbb{Z}\times(\mathbb{Z}/L_k\mathbb{Z})^3$ with $L_k\ge 3$, let ω_k be the infinite-time Wilson state of the box at an arbitrary $\beta_k>0$, and let Φ, τ, $f_0=g\otimes h$ with $g,h\ge 0$ and $g(a_k n)h(a_k x)>0$ for some $n\ge 1$ and some x, A_k, $C_k(\sigma)$ be as in Sec. IX D and Proposition 48. Then $C_k(0)>0$ for every k, and with $Z_k=C_k(0)^{-1/2}$ statements* (i)–(iv) *of Theorem 46 hold with $\mathscr{D}$ replaced by its points $(t,0)$: along a subsequence the renormalised flowed packets have a continuum limit in the sense of Definition 42, a Hilbert space with vacuum Ω, a self-adjoint $H\ge 0$ with $H\Omega=0$ and vectors $\psi(f)$ with dense range reproducing the limits of the reflected correlations at dyadic times, together with a spectral measure μ on $[0,\infty]$ with $\lim_k C_k(\sigma)/C_k(0)=\int e^{-\sigma E}d\mu(E)$; the limit is nontrivial if and only if $\mu([0,\infty))>0$, if and only if $\lim_k C_k(\sigma)/C_k(0)>0$ along the subsequence for one, equivalently for every, dyadic $\sigma>0$; and some subsequence has a nontrivial limit if and only if $\limsup_k C_k(\sigma)/C_k(0)>0$ for one dyadic $\sigma>0$. On the infinite lattice, with ω_k the torus limits of Lemma 294 and $C_k(0)>0$ assumed, the same holds with all of $\mathscr{D}$, and the dyadic spatial translations act by unitaries commuting with H.*

Proof. Proposition 48 gives $C_k(0)>0$ in the box, using Lüscher's strict positivity of the transfer matrix [77]. The proof of Theorem 46(i)–(iv) uses only the reflection positivity of ω_k, its transfer matrix with $0\le\mathbb{T}_k\le 1$, the spatial translations where $\mathscr{D}$ contains them, Proposition 45(ii) and Appendix G, as stated after Proposition 47; part (v) is the only part that uses a gap. On the infinite lattice the states of Lemma 294 have these properties. □

What Proposition 298 gives is the Hilbert space, the positive Hamiltonian, the time semigroup and the reflection-positive inner product of one renormalised flowed packet and its translates. It does not give Euclidean invariance, independence of the subsequence, correlations of more than two renormalised fields, clustering, a gap, or nontriviality; the last is the ratio condition, which Theorem 183 and Proposition 250(iii) show to fail wherever the lattice gap stays bounded below, so that it is a statement about the tuning $\beta_k\to\infty$ alone.

N. The missing step

We state the inputs of Sec. X L in one place, prove that they suffice for (E) at separated points, and prove that the two-point inputs are necessary. Let ω_k be torus limits along a tuning (a_k,β_k) as in Lemma 294 and Φ as in Sec. X L; the constants $Z_k>0$ are those of each input.

(I1) *The averaged condition along the tuning.* There are $\varpi>0$, $\ell>0$ and integers $n_k\ge 2$ with $n_k a_k\le\ell$ such that $\mathrm{BP}(n_k,\varpi)$ holds for the Wilson measure of Λ_L at β_k for all L in the infinite set of even integers along which ω_k is formed (the hypothesis of Theorem 64).

(I2) *The two-sided short-distance law.* (AF) holds for the constants Z_k: for $\gamma\ge 0$, $0<c_1\le C_1$, $r_0\in(0,1]$ and all large k, $Z_k^2|G_k(z)|\le C_1 w_\gamma(a_k|z|)$ for $|z|_\infty\ge 2r+1$, and $Z_k^2 G_k(z)\ge c_1 w_\gamma(a_k|z|)$ if moreover $a_k|z|\le r_0$.

(I3) *Reflected norms of products with a rate.* ($\mathrm{G_s}$) holds for the constants Z_k; by Proposition 259 this is equivalent, up to the rate, to (RN_m) for every $m\ge 1$.

(I4) *One rotation.* ($\mathrm{W_s}$): a rotation of infinite order is an asymptotic symmetry of the moments of the separated tuples.

(I5) *Uniqueness.* ($\mathrm{U_s}$): all subsequential limits of the moments of separated tuples coincide.

Theorem 299 (the inputs suffice)**.** (a) *Under (I1) and (I2), with $Z_k^\sharp=\mathscr{N}_k(\varepsilon)^{-1/2}$:* ($\mathrm{SD_p}$), (SD), (UV) *and* (NE) *hold; $Z_k/Z_k^\sharp$ is bounded above and below by positive constants; every limit S_2 of Corollary 255*(b) *is not identically zero, extends to $\mathscr{S}_0(\mathbb{R}^8)$, is invariant under the translations and B_4, reflection positive in every coordinate direction, and its one-field Hamiltonians $H_\mu^{(1)}$ have spectrum in $[m_*,\infty)$ and not empty, $m_*=16\varpi/(625e^2\ell)$.*

(b) *Under (I1)–(I5) with one sequence of constants Z_k, the moments $\omega_k(\prod_j Z_k A_k(f_j))$ converge for every separated tuple; their limits, extended to $\mathscr{S}_0(\mathbb{R}^{4n})$ by Theorem 273 and with $S_0=1$, satisfy the linear growth condition of [89] and the axioms of [88] of Euclidean invariance, reflection positivity, symmetry and clustering; the Hilbert space and Hamiltonian of their reconstruction are $\mathscr{H}_0$ and H_0 of Theorem 248, with $\sigma(H_0)\subset\{0\}\cup[m_*,\infty)$, $\ker H_0=\mathbb{C}\Omega_0$ and $\sigma(H_0)\cap[m_*,\infty)\neq$*

$\emptyset$*; and they are not the Schwinger functions of the vacuum alone. That is,* (E) *holds for the field* Φ *at separated points, with a mass gap of at least* m_*.

Proof. (a) Theorem 297(iii),(iv) gives ($\mathrm{SD_p}$), (SD), (UV), (NE), the comparability of the constants, the extension to $\mathscr{S}_0(\mathbb{R}^8)$ and the nonvanishing of every limit of Corollary 255(b); (I1) is not used for these. Under (I1), Corollary 63 gives $\gamma_k \le e^{-m_* a_k}$, and Corollary 255(b) gives $\sigma(H^{(1)}) \subset [m_*,\infty)$; the invariance, positivity and the unitary equivalence of the directions are Theorem 295(iii),(iv) on $\mathscr{V}_\mu^0$, and $\mathscr{H}_\mu^{(1)} \neq 0$ is Theorem 297(iv), so the spectrum is not empty.

(b) (I3) gives ($\mathrm{M_s}$). Corollary 274, whose hypotheses are (I1), ($\mathrm{G_s}$), ($\mathrm{W_s}$) and ($\mathrm{U_s}$), gives convergence, the growth condition, the four axioms, the identification of the reconstruction with $(\mathscr{H}_0, H_0)$, $\sigma(H_0) \subset \{0\} \cup [m_*,\infty)$ and $\ker H_0 = \mathbb{C}\Omega_0$. For nontriviality, Theorem 297(ii) shows that S_2 is not identically zero on the separated pairs, so $\mathscr{H}_0 \neq \mathbb{C}\Omega_0$ by Theorem 248(vi), and $\sigma(H_0) \cap [m_*,\infty) \neq \emptyset$ by Theorem 248(iv),(vi). The reconstruction theorem of [89] turns Schwinger functions with these properties into a Wightman theory on $\mathscr{H}_0$ with Hamiltonian H_0. □

Proposition 300 (the two-point inputs are necessary)**.** *Let* ω_k *be torus limits along any tuning,* $Z_k > 0$ *arbitrary constants, and suppose that for every* n *and every separated tuple the limit* $S_n(f_1,\dots,f_n) = \lim_k \omega_k(\prod_j Z_k A_k(f_j))$ *exists along the full sequence and that* S_2 *is not identically zero. Then:*

(i) ($\mathrm{M_s}$) *holds for the constants* Z_k*;*

(ii) *for every* $\varepsilon > 0$, $\mathscr{N}_k(\varepsilon) > 0$ *for large* k *and* $0 < \liminf_k Z_k^2 \mathscr{N}_k(\varepsilon) \le \limsup_k Z_k^2 \mathscr{N}_k(\varepsilon) < \infty$*; so* Z_k *is comparable to* $\mathscr{N}_k(\varepsilon)^{-1/2}$*;*

(iii) (SD), (UV) *and* (NE) *hold for the constants* $\mathscr{N}_k(\varepsilon)^{-1/2}$*;*

(iv) $\limsup_k a_k^{-1} \log(1/\gamma_k) < \infty$*: the physical gap of the torus limits stays bounded, and in particular the lattice gap tends to zero. Consequently no tuning along which the torus limits have a lattice gap, in particular none that stays in the range of Corollary 251 or* [21, Corollary 322]*, admits such limits, for any choice of* Z_k*.*

Proof. (i) Convergent sequences are bounded, so the hypothesis of the converse in Theorem 248 holds. We repeat its proof. Fix n, a cube Q and $\delta > 0$, and cover Q by the closed cubes of side $s = \delta/4$ that carry the functions χ_m^s of the proof of Theorem 248(vi); at most $(2L/s+3)^4$ of them meet Q. For $(f_j) \in \mathscr{C}_n(Q,\delta)$ every nonzero term $\omega_k(\prod_j Z_k A_k(\chi_{m_j}^s f_j))$ has its n cubes at pairwise distance at least $\delta - 2s = \delta/2$. Fix such cubes $C_1,\dots,C_n$, let X_j be the Banach space of continuous functions on $\mathbb{R}^4$ supported in C_j with the supremum norm, and let $B_k(h_1,\dots,h_n) = \omega_k(\prod_j Z_k A_k(h_j))$ on $X_1 \times \cdots \times X_n$. Each B_k is a bounded n-linear form, since $|A_k(h)| \le 2\|\Phi\|_\infty \|h\|_\infty \sum_x a_k^4 \mathbf{1}_{C_j}(a_k x)$, and for each point the sequence (B_k) is bounded because it converges. A pointwise bounded family of bounded n-linear forms on a product of Banach spaces is uniformly bounded: for $n = 1$ this is the uniform boundedness principle; for $n \ge 2$ and fixed h_1 the forms $B_k(h_1,\cdot)$ are bounded $(n-1)$-linear and pointwise bounded, so by induction $\sup_k \|B_k(h_1,\cdot)\| < \infty$, and the linear maps $h_1 \mapsto B_k(h_1,\cdot)$ into the Banach space of bounded $(n-1)$-linear forms are pointwise bounded, hence uniformly bounded. Summing the resulting bounds over the finitely many choices of cubes gives $M_n(Q,\delta)$.

(ii) Upper bound: for h supported in B_ε with $\|h\|_\infty \le 1$ and $a_k \le \varepsilon$, the pair $(\bar h \circ \theta_{0,k}, h)$ lies in $\mathscr{C}_2(Q_0,\varepsilon)$ for the cube Q_0 of the proof of Theorem 297(iii), and $\omega_k(\vartheta(\overline{A_k(h)})A_k(h)) = \omega_k(A_k(\bar h \circ \theta_{0,k})A_k(h))$; so (i) gives $Z_k^2 \mathscr{N}_k(\varepsilon) \le M_2(Q_0,\varepsilon)$. Lower bound: by (i) and the proofs of Theorem 248(ii),(iii) for $n = 2$, S_2 is symmetric, invariant, satisfies the conjugation identity and is reflection positive on $\mathscr{V}_\mu^0$, so Theorem 295(iv) on $\mathscr{V}_\mu^0$ applies, and by the argument of Theorem 295(v) with separated pairs, $S_2 \neq 0$ gives $g \in \mathscr{V}_0^0$ with $[g] \neq 0$. Since $e^{-\varepsilon H_0^{(1)}}$ is injective, $S_2(\overline{\tau_{\varepsilon e_0} g} \circ \theta_0, \tau_{\varepsilon e_0} g) = \|e^{-\varepsilon H_0^{(1)}}[g]\|^2 > 0$, and $g' = \tau_{\varepsilon e_0} g$ is supported in a cube of side L and in $\{x_0 \ge \varepsilon\}$. By Lemma 247(i) and Lemma 253,

$$Z_k^2\, \omega_k\big(A_k(\bar g' \circ \theta_{0,k}) A_k(g')\big) = Z_k^2\, \omega_k\big(\vartheta(\overline{A_k(g')}) A_k(g')\big) \le \mathsf{K}_L^2 Z_k^2 \mathscr{N}_k(\varepsilon)\, \|g\|_\infty^2,$$

and the left side tends to $S_2(\bar g' \circ \theta_0, g') > 0$ by the approximation of $\theta_{0,k}$ by θ_0. Hence $\liminf_k Z_k^2 \mathscr{N}_k(\varepsilon) > 0$.

(iii) For $\varepsilon' \in (0,\varepsilon)$, (ii) at ε' and at ε gives $\mathscr{N}_k(\varepsilon') \le M_2(Q_0,\varepsilon') Z_k^{-2} \le C\, \mathscr{N}_k(\varepsilon)$ for large k, which is (SD); (UV) follows by Corollary 296(iii). For (NE), the function g' of (ii) has $\liminf_k R_k(g') \ge S_2(\bar g' \circ \theta_0, g') / \limsup_k Z_k^2 \mathscr{N}_k(\varepsilon) > 0$.

(iv) is Corollary 296(ii) with (SD) from (iii); a lattice gap m_{lat} gives $a_k^{-1} \log(1/\gamma_k) \ge m_{\mathrm{lat}}/a_k \to \infty$. Corollary 251 and [21, Corollary 322] provide the lattice gaps in their ranges. □

Remark 301 (independence of the inputs). The examples of Sec. XL, states on $\mathbb{Z}^4$ invariant under translations and B_4 and reflection positive in the four coordinate directions, show that the consequences of the inputs do not imply one another. Proposition 257 gives such states with a physical gap bounded above and below for which (SD) fails, so the lattice gap $\gamma_k \le e^{-m_* a_k}$ that (I1) provides does not give (I2). Theorem 297(iv) shows that (I2) bounds the physical gap above and gives no lower bound on it, and Proposition 281 gives states satisfying (AF) with physical gap $\frac{1}{2}$; whether (AF) can hold with physical gap tending to zero is not decided by the examples. Proposition 284 gives states with a lattice gap, ($\mathrm{SD_p}$) and (NE) for which (RN_m), $m \ge 2$, fails, so a gap and the two-point conditions do not give (I3); these states do not satisfy (AF), and no example in Sec. XL decides whether (I1) and (I2) together give (I3). Proposition 286 gives states for which ($\mathrm{M_s}$) and (RN_m) hold for every m while the rate of ($\mathrm{G_s}$) fails, and Proposition 276 shows that the separated-point structure without a rate does not give temperedness, which is why (I3) carries the rate. Proposition 281 gives states satisfying (AF), ($\mathrm{G_s}$) and ($\mathrm{U_s}$) that are not invariant under any rotation of infinite order, so (I2), (I3) and (I5) do not give (I4).

Remark 302 (the exact obstruction). For the Wilson measure with $G = \mathrm{SU}(N)$ and the plaquette sum Φ the situation is the following. Everything in Sec. XM holds for every tuning. Of the inputs, (I1) is proved only in the strong-coupling ranges of Corollaries 72 and [21, Corollary 322], where Theorem 297(v)

and Proposition 300(iv) show that (I2) fails and no nontrivial limit exists; (I2) is proved in no range of couplings that tends to the continuum, and by Remark 282 it is equivalent to the Z_k-free two-sided comparison of Proposition 280(iii), open already in a box of fixed physical size; (I3), (I4) and (I5) are proved in no range. By Proposition 300, (SD) and (NE) with the resulting bound on the physical gap, which are consequences of (I2) by Theorem 297(iv), are also necessary for (E) with these packets, so that any proof of (E) along a tuning must establish them; (I2) itself is not shown to be necessary. The literature supplies none of the five for the four-dimensional Wilson measure: Bałaban's renormalisation group gives bounds on partition functions in a box of fixed physical size uniform in the spacing [6, 7, 10, 11, 13], not bounds on the connected two-point function of Φ; Magnen, Rivasseau and Sénéor give the basis for a construction of the Schwinger functions of the $\mathrm{SU}(2)$ gauge field in a regularised axial gauge with an infrared cut-off [80], not expectations of the Wilson measure along a lattice tuning; the stochastic quantisation of Yang–Mills in two and three dimensions [33, 34] and the state space of the three-dimensional theory [32] concern lower dimensions; and the weak-coupling correlation decay of Adhikari and Cao [1] concerns finite gauge groups. The smallest statement whose proof would change the picture is the two-sided law (I2) for the 24-plaquette sum Φ along the tuning (AF_β), whose finite-volume form in a box of fixed physical size is open as well: for the torus limits along that tuning, Theorem 297 would give the renormalisation, the short-distance condition and the nontriviality of the two-point limits, and with (I1) a two-point theory with mass gap m_*.

O. The connected plaquette correlation on a finite torus

This subsection records what the Wilson measure of a finite torus gives for the connected two-point function of the plaquette field Φ of Sec. XL at every coupling, and what it gives in the strong-coupling range. The results are unconditional. They do not prove either half of the two-sided law (AF) of Remark 282; the last part restates the lower half of (AF) along an axis in spectral terms and locates the step at which an argument for it leaves the range of the tools used here.

Conventions. Throughout, $G \subset \mathrm{U}(N)$ is a compact group with a faithful representation ρ, by which it is identified with a closed subgroup of $\mathrm{U}(N)$, so that G is a compact Lie group; $\chi = \mathrm{Tr}\,\rho$ is its character, $\bar\chi$ the character of the dual representation, and $\delta = 1$ if χ is real and $\delta = 0$ otherwise. From Lemma 309 on, ρ is irreducible, as it is for the defining representation of every compact simple group. The Wilson measure μ_L of $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$, $L \geq 6$ even, has density $\exp(N\beta\sum_p \mathrm{Re}\,\mathrm{Tr}\,U_p)$ with respect to the product Haar measure on the links, $\beta > 0$, as in Sec. I, and $\langle\cdot\rangle_L$ is its expectation. For a plaquette p put $F_p = N - \mathrm{Re}\,\mathrm{Tr}\,U_p \geq 0$, and for a site x let $\Phi_x = \sum_{p\ni x} F_p$ be the sum over the 24 plaquettes with a corner at x; this is the field Φ of Sec. XL with $r = 1$. Put

$$c_L = \langle\Phi_0\rangle_L, \qquad G_L(z) = \langle\Phi_0\Phi_z\rangle_L - c_L^2 \qquad (z \in \Lambda_L).$$

By the invariance of μ_L under translations and the hypercubic group (proof of Lemma 294(ii)), $G_L(z) = \langle\Phi_x\Phi_{x+z}\rangle_L - c_L^2$ for every x, $G_L(gz) = G_L(z)$ for $g \in B_4$, and $G_L(-z) = G_L(z)$. The torus limits ω of Lemma 294 are weak limits of μ_L along subsequences; for them $c = \omega(\Phi_0)$, $G(z) = \omega(\Phi_0\Phi_z) - c^2$, and $G(z) = \lim G_L(z)$ along the subsequence, since $\Phi_0\Phi_z$ is a continuous function of finitely many links.

1. *The transfer matrix of the torus*

For $\beta > 0$ and an irreducible representation R of G let

$$c_R(\beta) = \int_G e^{N\beta\,\mathrm{Re}\,\mathrm{Tr}\,g}\,\overline{\chi_R(g)}\,dg, \quad e^{N\beta\,\mathrm{Re}\,\mathrm{Tr}\,g} = \sum_R c_R(\beta)\chi_R(g),$$

the series converging absolutely and uniformly, this being the Peter–Weyl expansion of a smooth class function on a compact Lie group. Expanding the exponential, $c_R(\beta) = \sum_{n\geq 0}(N\beta/2)^n m_R^{(n)}/n!$, where $m_R^{(n)} \in \mathbb{Z}_{\geq 0}$ is the multiplicity of R in $(\rho\oplus\bar\rho)^{\otimes n}$; every R occurs in some tensor power because ρ is faithful (Stone–Weierstrass and Schur orthogonality, as in the proof of Proposition 48), so

$$c_R(\beta) > 0 \qquad \text{for every } R \text{ and every } \beta > 0. \tag{41}$$

Fix the time direction 0. Let $\Lambda^3 = (\mathbb{Z}/L\mathbb{Z})^3$ be the spatial torus, E^3 its links, P^3 its plaquettes, $\mathfrak{h} = L^2(G^{E^3})$ with respect to the product Haar measure, and let the gauge group $\mathscr{G} = G^{\Lambda^3}$ act on G^{E^3} by $(g\cdot U)_{xy} = g_x U_{xy} g_y^{-1}$ and on $\mathfrak{h}$ by the unitaries $(R_g\psi)(U) = \psi(g\cdot U)$. Let $P = \int_{\mathscr{G}} R_g\,dg$ be the orthogonal projection onto the gauge-invariant functions and $\mathfrak{h}_P = P\mathfrak{h}$. Define the kernel

$$\begin{aligned} T(U,U') &= D(U)\,K(U,U')\,D(U'),\\ D(U) &= \exp\Big(\frac{N\beta}{2}\sum_{p\in P^3}\mathrm{Re}\,\mathrm{Tr}\,U_p\Big),\\ K(U,U') &= \prod_{l\in E^3}\exp\big(N\beta\,\mathrm{Re}\,\mathrm{Tr}(U_l U_l'^{-1})\big), \end{aligned}$$

and the operator $(T\psi)(U) = \int T(U,U')\psi(U')\,dU'$ on $\mathfrak{h}$. This is the transfer matrix of Lüscher [77] in the temporal gauge, and of Osterwalder and Seiler [90]; its positivity is recalled in the proof of Proposition 48. What is needed here beyond that proof is the trace formula on a torus of finite time extent and the form of the plaquette insertions.

Lemma 303 (the transfer matrix)**.** (i) *T is a self-adjoint Hilbert–Schmidt operator on $\mathfrak{h}$ with $\langle\psi, T\psi\rangle \geq 0$ for all ψ, and T is injective. It commutes with every R_g, with P, and with the unitaries induced by the translations and the cubic symmetries of Λ^3. Consequently T maps $\mathfrak{h}_P$ into itself, and $\mathfrak{h}_P$ has an orthonormal basis $(e_j)_{j\geq 0}$ of eigenvectors of T with eigenvalues $\lambda_j > 0$, $\lambda_j \to 0$.*

(ii) $Z_L = \int \exp(N\beta\sum_p \mathrm{Re}\,\mathrm{Tr}\,U_p)\,dU = \mathrm{Tr}(PT^L)$.

(iii) *Let B be the operator on $\mathfrak{h}$ with kernel*

$$B(U,U') = \int \Phi(U,V,U')\,T(U,V)\,T(V,U')\,dV,$$
$$\Phi(U,V,U') = \sum_{p\in P^3,\,p\ni 0} (N - \mathrm{Re}\,\mathrm{Tr}\,V_p) + \sum_{l\in E^3,\,l\ni 0} \big[(N-\mathrm{Re}\,\mathrm{Tr}(U_l V_l^{-1})) + (N-\mathrm{Re}\,\mathrm{Tr}(V_l U_l'^{-1}))\big].$$

Then B is self-adjoint and Hilbert–Schmidt, commutes with every R_g and with P, and for $2\le t\le L-2$

$$c_L = \frac{\mathrm{Tr}(PT^{L-2}B)}{Z_L}, \quad \langle \Phi_0 \Phi_{te_0}\rangle_L = \frac{\mathrm{Tr}\big(PT^{L-t-2}BT^{t-2}B\big)}{Z_L}.$$

Proof. (i) The kernel is continuous and real, and symmetric because $K(U,U') = K(U',U)$, which follows from $\mathrm{Re}\,\mathrm{Tr}(U'U^{-1}) = \mathrm{Re}\,\mathrm{Tr}((UU'^{-1})^{-1}) = \mathrm{Re}\,\mathrm{Tr}(UU'^{-1})$ for unitary matrices; so T is self-adjoint and Hilbert–Schmidt. Write $T = \mathsf{D}K\mathsf{D}$ with D the multiplication by the bounded function D, which has values in a compact subinterval of $(0,\infty)$, so that D is invertible. The operator K is the tensor product over $l\in E^3$ of the convolution operators K_l on $L^2(G)$ with kernel $k_\beta(U_l U_l'^{-1})$, $k_\beta(g) = e^{N\beta\,\mathrm{Re}\,\mathrm{Tr}\,g} = \sum_R c_R(\beta)\chi_R(g)$. After the substitution $V = W^{-1}U$,

$$\int k_\beta(UV^{-1})R(V)_{ij}\,dV = \sum_m \Big(\int k_\beta(W)\,\overline{R(W)_{mi}}\,dW\Big) R(U)_{mj} = \frac{c_R(\beta)}{d_R} R(U)_{ij}$$

by the Schur orthogonality relations, so K_l acts on the matrix coefficients of each irreducible R, which form an orthogonal basis of $L^2(G)$, as multiplication by $c_R(\beta)/d_R > 0$ (41). Hence K is positive and injective, its eigenvalues on the product basis being the products $\prod_l c_{R_l}(\beta)/d_{R_l} > 0$, and $\langle\psi, T\psi\rangle = \langle \mathsf{D}\psi, K\mathsf{D}\psi\rangle \ge 0$; $T\psi = 0$ gives $K\mathsf{D}\psi = 0$, so $\psi = 0$. The kernel satisfies $T(g\cdot U, g\cdot U') = T(U,U')$ because every $\mathrm{Re}\,\mathrm{Tr}$ in it is the trace of a product around a closed loop of links, so T commutes with R_g, hence with P; the same holds for the lattice symmetries, which permute links and plaquettes and preserve $\mathrm{Re}\,\mathrm{Tr}\,U_p$ under reversal of orientation. A positive compact injective operator on the separable Hilbert space $\mathfrak{h}_P$ has an orthonormal eigenbasis with positive eigenvalues tending to 0.

(ii) Write the links of Λ_L as spatial links $U_{(x,s),i}$ and temporal links $U_{(x,s),0}$, $s\in\mathbb{Z}/L\mathbb{Z}$. Given a configuration, define $h_{(x,0)} = 1$ and $h_{(x,s)} = U_{(x,0),0}U_{(x,1),0}\cdots U_{(x,s-1),0}$ for $1\le s\le L-1$, and let $U' = h\cdot U$. Then $U'_{(x,s),0} = 1$ for $s\le L-2$ and $U'_{(x,L-1),0} = g_x := \prod_{s=0}^{L-1} U_{(x,s),0}$ (ordered product). The map $U\mapsto (U'_{(x,s),i}, U_{(x,s),0}\ (s\le L-2), g_x)$ preserves the product Haar measure: at each site the map $(U_0,\dots,U_{L-1})\mapsto (U_0,\dots,U_{L-2}, U_0\cdots U_{L-1})$ of G^L does, and for fixed temporal links the spatial links are transformed by a gauge transformation, which does. The Wilson density and every gauge-invariant observable depend only on $(U'_{(x,s),i}, g)$; the variables $U_{(x,s),0}$, $s\le L-2$, integrate to 1. With U_s the spatial configuration of slice s, the temporal plaquette at the link $l = \{x, x+e_i\}$ between the slices s and $s+1\le L-1$ is $U_{s,l}U_{s+1,l}^{-1}$, and between the slices $L-1$ and 0 it is $U_{L-1,l}g_{x+e_i}U_{0,l}^{-1}g_x^{-1} = U_{L-1,l}((g\cdot U_0)_l)^{-1}$. Distributing each spatial factor $D(U_s)^2$ over the two adjacent temporal steps gives

$$Z_L = \int_{\mathscr{G}} dg \int T(U_0,U_1)\cdots T(U_{L-2},U_{L-1}) \times T(U_{L-1}, g\cdot U_0)\prod_s dU_s.$$

The inner integral is $\int T^L(U_0, g\cdot U_0)\,dU_0 = \mathrm{Tr}(T^L R_g)$, since T^L has a continuous symmetric kernel and is trace class ($L\ge 2$), and $\int \mathrm{Tr}(T^L R_g)\,dg = \mathrm{Tr}(T^L P)$.

(iii) Place Φ_0 at the vertex $(0,1)$, so that it is a function of the slices $0,1,2$: its 12 spatial plaquettes lie in slice 1, and its 12 temporal plaquettes are $U_{0,l}U_{1,l}^{-1}$ and $U_{1,l}U_{2,l}^{-1}$ for the six spatial links l at 0; this is $\Phi(U_0,U_1,U_2)$. Inserting it into the integral of (ii), with the step carrying g chosen among the $L-2\ge 1$ steps not touched by the observable (the relabelling of the times is a symmetry of μ_L), gives $\langle\Phi_0\rangle_L Z_L = \mathrm{Tr}(PT^{L-2}B)$. For the two-point function place the second copy at $(0,t+1)$, a function of the slices $t, t+1, t+2$; for $2\le t\le L-2$ the two copies are separated by $t-2\ge 0$ and $L-t-2\ge 0$ transfer steps, $(t-2)+(L-t-2) = L-4\ge 2$, and the step carrying g is chosen in one of the two gaps. The kernel B is real and symmetric because $\Phi(U,V,U') = \Phi(U',V,U)$ and T is symmetric, and gauge covariant because Φ and T are. □

2. *Positivity, convexity and the dichotomy along an axis*

Let (λ_j, e_j) be as in Lemma 303(i), $B_{jk} = \langle e_j, Be_k\rangle$, $\pi_j = \lambda_j^L/Z_L$ and $b_j = B_{jj}/\lambda_j^2$. Then $\sum_j \pi_j = 1$, $\pi_j > 0$, and $Z_L c_L = \sum_j \lambda_j^{L-2} B_{jj}$ gives $c_L = \sum_j \pi_j b_j$. Put

$$V_L = \sum_j \pi_j b_j^2 - \Big(\sum_j \pi_j b_j\Big)^2 \ge 0.$$

Theorem 304 (the axial two-point function on a torus)**.** *Let $\beta > 0$ and $L\ge 6$ even. For $2\le t\le L-2$ and every direction μ,*

$$G_L(te_\mu) = G_L(te_0) = V_L + \frac{1}{Z_L}\sum_{j\ne k} \lambda_j^{L-t-2}\lambda_k^{t-2}\,|B_{jk}|^2,$$

the series converging absolutely. Consequently:

(i) *$G_L(te_0) \ge V_L \ge 0$ and $G_L(te_0) = G_L((L-t)e_0)$;*

(ii) *$t\mapsto G_L(te_0)$ is log-convex on $\{2,\dots,L-2\}$, that is $G_L(te_0)^2 \le G_L((t-1)e_0)\,G_L((t+1)e_0)$ for $3\le t\le L-3$, and nonincreasing on $\{2,\dots,L/2\}$;*

(iii) *either $G_L(te_0) > 0$ for every $t\in\{2,\dots,L-2\}$, or $G_L(te_0) = 0$ for every such t; the second alternative holds if and only if $B = c_L T^2$ on $\mathfrak{h}_P$.*

(iv) *In the second alternative of* (iii)*, $G_L(z) = 0$ for every $z\in\Lambda_L$ having a coordinate z_μ with $2\le |z_\mu|\le L-2$, and $\langle\Phi_0\rangle$ on $(\mathbb{Z}/L'\mathbb{Z})\times\Lambda^3$ equals c_L for every even $L'\ge 6$.*

Proof. The equality $G_L(te_\mu) = G_L(te_0)$ is the B_4-invariance of μ_L and Φ. By Lemma 303(iii) and the eigenbasis of $\mathfrak{h}_P$, using that B commutes with P so that $PT^{L-t-2}BT^{t-2}B = PT^{L-t-2}(PBP)T^{t-2}(PBP)$,

$$\begin{aligned} Z_L\langle\Phi_0\Phi_{te_0}\rangle_L &= \sum_{j,k}\lambda_j^{L-t-2}\lambda_k^{t-2}B_{jk}B_{kj} \\ &= \sum_{j,k}\lambda_j^{L-t-2}\lambda_k^{t-2}|B_{jk}|^2, \end{aligned}$$

since B is self-adjoint; the terms are nonnegative and their sum is the trace of $PT^{L-t-2}BT^{t-2}B$, which is trace class: T^aB is Hilbert–Schmidt for $a \geq 1$ and the product of two Hilbert–Schmidt operators is trace class, and if $t-2 = 0$ or $L-t-2 = 0$ the other exponent is $L-4 \geq 2$ and $BT^{L-4}B = (BT^{L/2-2})(T^{L/2-2}B)$. The diagonal terms give $\sum_j \lambda_j^{L-4}B_{jj}^2/Z_L = \sum_j \pi_j b_j^2$, and subtracting $c_L^2 = (\sum_j \pi_j b_j)^2$ yields the formula.

(i) Both parts are nonnegative, and $V_L \geq 0$ is the variance of b under π. Exchanging j and k in the sum shows the symmetry $t \mapsto L-t$.

(ii) Each term $\lambda_j^{L-t-2}\lambda_k^{t-2}|B_{jk}|^2 = C_{jk}(\lambda_k/\lambda_j)^t$ with $C_{jk} \geq 0$ is log-convex in t, a constant is log-convex, and sums of log-convex functions are log-convex (by the Cauchy–Schwarz inequality for the measure with weights C_{jk}). Grouping (j,k) with (k,j) gives $(\lambda_j\lambda_k)^{L/2-2}\big(x^{L/2-t}+x^{t-L/2}\big)|B_{jk}|^2$ with $x = \lambda_j/\lambda_k$, which is nonincreasing in t on $[2,L/2]$.

(iii) If $G_L(te_0) = 0$ for one t, then $V_L = 0$ and every term of the off-diagonal sum vanishes; as all $\lambda_j > 0$, $B_{jk} = 0$ for all $j \neq k$, so $Be_k = B_{kk}e_k = b_k\lambda_k^2 e_k$, and $V_L = 0$ with $\pi_k > 0$ for all k forces $b_k = \sum_j \pi_j b_j = c_L$ for all k. Hence $B = c_L T^2$ on $\mathfrak{h}_P$, and then the formula gives $G_L(te_0) = 0$ for every t. Conversely $B = c_L T^2$ on $\mathfrak{h}_P$ gives $B_{jk} = c_L\lambda_k^2\delta_{jk}$, $b_k = c_L$, $V_L = 0$.

(iv) Let $z = (t,\vec{z})$ with $2 \leq t \leq L-2$ and let $\tau_{\vec{z}}$ be the unitary of the spatial translation by $\vec{z}$; it commutes with T and P. The proof of Lemma 303(iii) gives $Z_L\langle\Phi_0\Phi_z\rangle_L = \mathrm{Tr}(PT^{L-t-2}BT^{t-2}\tau_{\vec{z}}B\tau_{\vec{z}}^{-1})$, which under $B = c_L T^2$ equals $c_L^2\,\mathrm{Tr}(PT^L) = c_L^2 Z_L$. The other directions follow from the B_4-invariance. For the time extent L', Lemma 303(ii),(iii) with L' in place of L give $\langle\Phi_0\rangle = \mathrm{Tr}(PT^{L'-2}B)/\mathrm{Tr}(PT^{L'}) = c_L$. □

Remark 305. Reflection positivity of μ_L in the plane $\{x_0 = 0\}$ through sites (with its antipode) and in the plane $\{x_0 = \frac{1}{2}\}$ bisecting links [90] gives $G_L(te_0) \geq 0$ for all $2 \leq t \leq L-2$: for even $t = 2s$ the function $\Phi_{se_0} - c_L$ depends on links at times in $[s-1, s+1] \subset [0, L/2]$ and is reflected onto $\Phi_{-se_0} - c_L$, and for odd $t = 2s+1$ the function $\Phi_{(s+1)e_0} - c_L$ depends on links at times in $[s, s+2] \subset [1, L/2]$ and is reflected onto $\Phi_{-se_0} - c_L$. Theorem 304 gives in addition the floor V_L, the log-convexity, the monotonicity and the exact form of the degenerate alternative. The quantity V_L is the variance, under the Gibbs weights π_j of the torus of time extent L, of the diagonal values b_j; it does not depend on t.

The torus limits inherit the structure. Let ω be a torus limit as in Lemma 294, and let ϑ' be the reflection in the plane $\{x_0 = \frac{1}{2}\}$ bisecting links, for which ω is reflection positive by Lemma 294(iii). The reconstruction of Appendix F applies to ϑ' as to ϑ: the positive-time functions are those of the links with both endpoints at times ≥ 1, the time translation τ_1 maps them into themselves, and $\vartheta'\tau_t = \tau_{-t}\vartheta'$; it gives $\mathbb{T}'$, Ω' and $F \mapsto \Psi'_F$ with $\langle\Psi'_F, \mathbb{T}'^n\Psi'_H\rangle = \omega(\vartheta'(\bar{F})\tau_n H)$ for $n \geq 0$. Let $\Psi' = \Psi'_{\Phi_{2e_0}-c}$ be the vector of the centred field at the site $2e_0$, whose links have both endpoints at times in $[1,3]$, and let σ be the spectral measure of $\mathbb{T}'$ at Ψ', a finite measure on $[0,1]$.

Corollary 306 (torus limits)**.** *For every torus limit ω at $\beta \geq 0$:*

(i) *$G(te_0) \geq 0$ for all $t \geq 2$, $t \mapsto G(te_0)$ is log-convex and nonincreasing on $\{2,3,\dots\}$, and for $n \geq 0$*

$$G((3+n)e_0) = \langle\Psi', \mathbb{T}'^n\Psi'\rangle = \int_{[0,1]} \lambda^n\, d\sigma(\lambda);$$

(ii) *either $G(te_0) > 0$ for all $t \geq 3$, or $G(te_0) = 0$ for all $t \geq 4$; the second alternative holds if and only if σ is concentrated at $\lambda = 0$;*

(iii) *in the first alternative the local rates*

$$m(t) = \log\big(G(te_0)/G((t+1)e_0)\big), \qquad t \geq 2,$$

are nonnegative, nonincreasing in t, and converge to $m_\Phi = -\log\max\mathrm{supp}\,\sigma \in [0,\infty)$, which satisfies $m_\Phi \geq \log(1/\gamma')$ with $\gamma' = \|\mathbb{T}'|_{\Omega'^\perp}\|$, the analogue of $\gamma(\omega)$ for the reflection ϑ'.

Proof. (i) For fixed t, $G(te_0) = \lim G_L(te_0)$ along the subsequence, and positivity, log-convexity on $\{2,\dots,L-2\}$ and monotonicity on $\{2,\dots,L/2\}$ pass to the limit, the ranges exhausting $\{2,3,\dots\}$. The identity for $n \geq 0$ is the reconstruction: $\vartheta'(\Phi_{2e_0} - c) = \Phi_{-e_0} - c$, $\tau_n(\Phi_{2e_0} - c) = \Phi_{(2+n)e_0} - c$, so $\omega(\vartheta'(\bar{F})\tau_n F) = G((3+n)e_0)$ with $F = \Phi_{2e_0} - c$, and this equals $\langle\Psi'_F, \mathbb{T}'^n\Psi'_F\rangle$.

(ii) If $G(t_0e_0) = 0$ for some $t_0 \geq 4$, then $\int\lambda^{t_0-3}d\sigma = 0$, so $\sigma((0,1]) = 0$ and $G(te_0) = \int\lambda^{t-3}d\sigma = 0$ for all $t \geq 4$; if $G(3e_0) = 0$ then $\sigma = 0$. If $\sigma((0,1]) > 0$ then $\int\lambda^n d\sigma > 0$ for all $n \geq 0$.

(iii) The ratios $\int\lambda^{n+1}d\sigma/\int\lambda^n d\sigma$ of the moments of a finite measure on $[0,1]$ with $\sigma((0,1]) > 0$ are at most 1, nondecreasing in n by the Cauchy–Schwarz inequality, and converge to $\max\mathrm{supp}\,\sigma$; so $m(t)$ is nonnegative, nonincreasing and converges to m_Φ for $t \geq 3$. For $t = 2$ use the log-convexity and monotonicity of (i) directly. The vector Ψ' is orthogonal to Ω' since $\omega(F) = 0$, so $\mathrm{supp}\,\sigma \subset [0, \|\mathbb{T}'|_{\Omega'^\perp}\|]$. □

3. *Strict positivity and the exceptional couplings*

The dichotomy of Theorem 304(iii) is decided for a single spatial plaquette at every coupling.

Theorem 307 (one plaquette)**.** *Let $\beta > 0$, $L \geq 6$ even, let p be a plaquette in a plane not containing the direction 0, and let $1 \leq t \leq L-1$. Then $\langle F_pF_{p+te_0}\rangle_L - \langle F_p\rangle_L^2 > 0$. The same holds for every direction μ not in the plane of p in place of e_0.*

Proof. Place p in the slice 0 and let M_p be the multiplication by F_p on $\mathfrak{h}$, a bounded self-adjoint operator commuting with every R_g. The proof of Lemma 303(iii), with M_p at the slices 0 and t and the step carrying g in one of the two gaps of lengths $t \geq 1$ and $L-t \geq 1$, gives $Z_L\langle F_pF_{p+te_0}\rangle_L = \mathrm{Tr}(PT^{L-t}M_pT^tM_p) =$

$\sum_{j,k}\lambda_j^{L-t}\lambda_k^t|M_{jk}|^2$ with $M_{jk}=\langle e_j,M_pe_k\rangle$, and $Z_L\langle F_p\rangle_L=\sum_j\lambda_j^L M_{jj}$. As in the proof of Theorem 304, the covariance is the sum of the variance of $(M_{jj})_j$ under $(\pi_j)_j$ and of $Z_L^{-1}\sum_{j\neq k}\lambda_j^{L-t}\lambda_k^t|M_{jk}|^2$, both nonnegative, and it vanishes only if $M_{jk}=0$ for $j\neq k$ and M_{jj} is constant, that is, only if $M_p\psi=c\psi$ for every $\psi\in\mathfrak{h}_P$. The constant function 1 lies in $\mathfrak{h}_P$, so F_p would be almost everywhere equal to c. The map $(U_l)_{l\in p}\mapsto U_p$ carries the product Haar measure on G^4 to the Haar measure of G, which has full support, and $g\mapsto\operatorname{Re}\operatorname{Tr}g$ is continuous; so $\operatorname{Re}\operatorname{Tr}g=c$ would hold on a dense set, hence everywhere, in particular $c=\operatorname{Re}\operatorname{Tr}1=N$, whereas $\operatorname{Re}\operatorname{Tr}g<N$ for $g\neq1$ and $G\neq\{1\}$. The statement for μ follows by a cubic symmetry. □

For the field Φ the first ingredient is analyticity in the coupling.

Lemma 308. *For fixed L and z, $\beta\mapsto G_L(z;\beta)$ is the restriction to $(0,\infty)$ of a function meromorphic on $\mathbb{C}$ and real-analytic on $\mathbb{R}$. If it does not vanish identically, its zero set in $(0,\infty)$ is closed and discrete.*

Proof. $Z_L(\beta)$, $\int\Phi_0e^{N\beta\sum\operatorname{Re}\operatorname{Tr}U_p}$ and $\int\Phi_0\Phi_ze^{N\beta\sum\operatorname{Re}\operatorname{Tr}U_p}$ are integrals over a compact space of functions entire in β with a uniform bound on compact sets of β, hence entire; $Z_L(\beta)>0$ for real β. The zeros of a real-analytic function on an interval that is not identically zero are isolated. □

The second ingredient is the strong-coupling expansion, which identifies the leading Taylor coefficient. We recall the polymer representation in the form of Osterwalder and Seiler [90] and Seiler [95]. From now on ρ is irreducible, so that χ and $\bar\chi$ are the only irreducible characters occurring in $\chi+\bar\chi$; $\delta=1$ if $\chi=\bar\chi$. For an irreducible $R\neq1$ let $n(R)$ be the least n with $m_R^{(n)}>0$; then $n(\rho)=n(\bar\rho)=1$, $n(R)\geq2$ otherwise, and $a_R(\beta)=c_R(\beta)/c_0(\beta)$ satisfies

$$\begin{aligned}&a_R(\beta)=O(\beta^{n(R)}),\qquad a_\rho(\beta)=\tfrac{1+\delta}{2}N\beta+O(\beta^2),\\&c_0(\beta)=1+O(\beta^2),\end{aligned}\tag{42}$$

the last because $\int\operatorname{Re}\operatorname{Tr}=0$ for a nontrivial irreducible ρ. For a plaquette p and $s\in\mathbb{C}$ let $c_R^{(p)}(\beta,s)$ be the character coefficients of $e^{sF_p}e^{N\beta\operatorname{Re}\operatorname{Tr}U_p}=e^{sN}e^{(N\beta-s)\operatorname{Re}\operatorname{Tr}U_p}$ and $a_R^{(p)}(s)=c_R^{(p)}(\beta,s)/c_0^{(p)}(\beta,s)$; then $a_R^{(p)}(0)=a_R(\beta)$ and, since $F_pe^{N\beta\operatorname{Re}\operatorname{Tr}}=\sum_n(N\beta)^n(N-\operatorname{Re}\operatorname{Tr})(\operatorname{Re}\operatorname{Tr})^n/n!$,

$$\partial_sa_R^{(p)}(0)=O(\beta^{n(R)-1}),\qquad\partial_sa_\rho^{(p)}(0)=-\tfrac{1+\delta}{2}+O(\beta).\tag{43}$$

For complex $|s|\leq r_N:=1/(2N)$ one has $|s\operatorname{Re}\operatorname{Tr}g|\leq\frac12$, so $\operatorname{Re}e^{-s\operatorname{Re}\operatorname{Tr}g}\geq e^{-1/2}\cos\frac12>\frac12$ and $|c_0^{(p)}(\beta,s)|\geq\frac12e^{-1/2}c_0(\beta)$, while $\sum_Rd_R|c_R^{(p)}(\beta,s)|\leq e^{N/2}\sum_n((N\beta+\frac12)/2)^n(2N)^n/n!=e^{N^2\beta+N}$, because $c_R^{(p)}(\beta,s)=e^{sN}\sum_n((N\beta-s)/2)^nm_R^{(n)}/n!$ and $\sum_Rd_Rm_R^{(n)}=(2N)^n$; hence

$$\sum_{R\neq1}d_R\,|a_R^{(p)}(s)|\leq C_N\quad(|s|\leq r_N,\ \beta\leq1/N^2),\tag{44}$$

and, by the same count restricted to $n\geq n(R)$, for $\beta\leq1/N^2$,

$$\sum_{R:n(R)\geq n}d_R\,a_R(\beta)\leq2e\,(N^2\beta)^n.$$

Lemma 309 (polymer expansion)**.** *A* polymer *is a pair $\gamma=(A,R)$ with A a nonempty finite set of plaquettes of Λ_L that is connected under the relation of sharing a link, and R an assignment of a nontrivial irreducible representation R_a to each $a\in A$, such that $I(\gamma)=\int\prod_{a\in A}\chi_{R_a}(U_a)\,dU\neq0$. Two polymers are compatible if their plaquette sets share no link. For sources s,s' at two plaquettes $p\neq q$ let $W(\gamma;s,s')=I(\gamma)\prod_{a\in A}a_{R_a}^{(a)}$ with $a_R^{(a)}=a_R^{(p)}(s)$, $a_R^{(q)}(s')$, $a_R(\beta)$ for $a=p$, $a=q$ and otherwise. Then:*

(i) *If a link lies in exactly one plaquette of A, then $I(A,R)=0$; in particular $|A|\geq6$ for every polymer. For every (A,R), $I(A,R)$ is the product of the I of the link-connected components of A.*

(ii) $\int e^{sF_p+s'F_q}e^{N\beta\sum\operatorname{Re}\operatorname{Tr}U_a}dU=\prod_ac_0^{(a)}\sum\prod_iW(\gamma_i;s,s')$, *the sum over finite families of pairwise compatible polymers, including the empty family.*

(iii) *There is $\beta_0=\beta_0(N)>0$ such that for $0\leq\beta\leq\beta_0$ and complex $|s|,|s'|\leq r_N$ the Mayer expansion*

$$\log\frac{\int e^{sF_p+s'F_q}e^{N\beta\sum\operatorname{Re}\operatorname{Tr}U_a}dU}{\prod_ac_0^{(a)}}=\sum_X\varphi(X)\prod_{\gamma\in X}W(\gamma;s,s')$$

over clusters X (finite multisets of polymers whose incompatibility graph is connected, φ the Ursell coefficients, $\varphi(X)=1$ for one polymer) converges absolutely, uniformly in L, s, s', and the sum of $|\varphi(X)\prod W|$ over the clusters X with $\sum_{\gamma\in X}\sum_{a\in A_\gamma\setminus\{p,q\}}n(R_a)\geq n$ and with some polymer containing p is at most $K_N(C_N\beta)^n$, with constants depending on N only.

Proof. (i) If the link e lies only in $a\in A$, write $U_a=XU_e^{\pm1}Y$ with X,Y independent of U_e; then

$$\chi_{R_a}(XU_e^{\pm1}Y)=\sum_{i,j,k}R_a(X)_{ij}R_a(U_e^{\pm1})_{jk}R_a(Y)_{ki},$$

$$\int_GR_a(U)_{jk}\,dU=0$$

for nontrivial irreducible R_a. A set of plaquettes in which every link lies in at least two plaquettes has at least six members, the faces of a cube being the least case. The factorisation is Fubini. (ii) Expand each plaquette factor as $c_0^{(a)}(1+\sum_{R\neq1}a_R^{(a)}\chi_R(U_a))$, multiply out, integrate, and group the plaquettes carrying a nontrivial representation into link-connected components; by (i) only the components with $I\neq0$ survive, and distinct components share no link. (iii) is the convergent strong-coupling cluster expansion of [90] (see also [95, Ch. 3]), with the sources carried along; we indicate the bounds. Since $|\chi_R|\leq d_R$, $|W(\gamma;s,s')|\leq\prod_{a\in A}d_{R_a}|a_{R_a}^{(a)}|$, and summing over the assignments R with (44) and the bound after it, the polymers with a given plaquette set A have total weight at most $C_N^2(2eN^2\beta)^{|A|-2}$; the number of link-connected sets of n plaquettes containing a given link is at most C^n with an absolute

C, and $|A| \geq 6$ by (i). Hence for every polymer γ_0 the polymers incompatible with it, which contain one of its at most $4|A_{\gamma_0}|$ links, satisfy $\sum_{\gamma \not\sim \gamma_0} |W(\gamma)| e^{|A_\gamma|} \leq |A_{\gamma_0}|$ once $\beta \leq \beta_0(N)$, which is the Kotecký–Preiss criterion [73] with $a(\gamma) = |A_\gamma|$, uniformly in L, s, s'; it gives the absolute convergence of the Mayer series, uniformly on the polydisc, and, applied with the activities of the plaquettes $a \neq p,q$ multiplied by $e^{n(R_a)}$ for $\beta \leq \beta_0(N)/e$, the bound $K_N(C_N\beta)^n$ on the clusters of order at least n through p. □

Lemma 310 (the minimal surfaces)**.** *Let $L \geq 6$ even, $2 \leq t < L/2$, let p be a plaquette with the corner 0 and q a plaquette with the corner te_0, and let Σ be a finite set of plaquettes of Λ_L containing p and q, connected under the relation of sharing a link, such that every link of every plaquette of Σ lies in at least two plaquettes of Σ. Then $|\Sigma \setminus \{p,q\}| \geq 4t$, with equality if and only if $\Sigma = \partial\mathscr{B}$ is the set of the $4t+2$ faces of a box $\mathscr{B} = [0,t] \times I \times J$, where $I = [a,a+1]$ and $J = [b,b+1]$ lie in two distinct spatial directions, $a,b \in \{-1,0\}$, and the fourth coordinate is 0. There are 12 such boxes; each has exactly three faces with the corner 0 and three with the corner te_0, so there are 108 triples $(\mathscr{B},p,q)$ with $p,q \in \partial\mathscr{B}$.*

Proof. Identify the times with $\{-L/2+1,\dots,L/2\}$; the plaquettes with the corner 0 have their vertices at times in $\{-1,0,1\}$, those with the corner te_0 at times in $\{t-1,t,t+1\}$. For a half-integer τ let E_τ be the set of temporal plaquettes of Σ with time extent $[\tau-\frac{1}{2},\tau+\frac{1}{2}]$; each is $[\tau-\frac{1}{2},\tau+\frac{1}{2}] \times e$ for a spatial link e, and we identify E_τ with a set of spatial links. If a spatial site y is an endpoint of a link of E_τ, the temporal link at y over $[\tau-\frac{1}{2},\tau+\frac{1}{2}]$ lies in a plaquette of Σ, hence in at least two, which are plaquettes of E_τ; so every vertex of the graph E_τ has degree 0 or at least 2, and E_τ, if nonempty, contains a cycle of the graph Λ^3, which has at least 4 links since $L \geq 6$: cycles of Λ^3 have even length, and a cycle of length 4 is the boundary of a unit square because a straight closed path needs length L.

Every plaquette has its vertices at one time or at two consecutive times, and two plaquettes sharing a link have a common time. Along a chain of plaquettes of Σ from p to q, consecutive members sharing a link, the union of the times of the members is therefore an arc of the time circle containing 0 and t, and the passage from a time s to $s+1$ inside the arc is made by a member with vertices at both times, that is, by a temporal plaquette over $[s,s+1]$. Let $s \in \{0,\dots,t-1\}$. If $E_{s+1/2} = \emptyset$, the arc does not contain the step from s to $s+1$, so it runs from 0 through $-1,-2,\dots$ to t, and $E_\tau \neq \emptyset$ for each of the $L-t$ half-integers τ between t and $L \equiv 0$; then $|\Sigma \setminus \{p,q\}| \geq 4(L-t)-2 > 4t$ because $L-t \geq t+1$. So assume $E_{s+1/2} \neq \emptyset$, hence $|E_{s+1/2}| \geq 4$, for all $s \in \{0,\dots,t-1\}$. The plaquettes p and q belong to at most one of these sets each, p to $E_{1/2}$ if p is temporal over $[0,1]$, q to $E_{t-1/2}$ if q is temporal over $[t-1,t]$. If p is temporal over $[0,1]$, its spatial link at time 0 lies in a second plaquette of Σ, which is a spatial plaquette at time 0 or a temporal plaquette over $[-1,0]$ (another temporal plaquette over $[0,1]$ through the same spatial link would be p itself); in the second case $|E_{-1/2}| \geq 4$. In every case $\Sigma \setminus \{p,q\}$ contains the $\sum_s |E_{s+1/2}| - [p \in E_{1/2}] - [q \in E_{t-1/2}] \geq 4t-2$ plaquettes of the $E_{s+1/2}$, $0 \leq s \leq t-1$, other than p,q, plus at least one further plaquette, at times in $\{-1,0\}$, for p if p is temporal, and one at times in $\{t,t+1\}$ for q if q is temporal; these are distinct since $t \geq 2$. If p is spatial, it is not counted in the E's and the bound $4t$ holds directly. The bound $4t$ follows.

Equality forces: $|E_{s+1/2}| = 4$ for $0 \leq s \leq t-1$, so $E_{s+1/2} = \partial Q_s$ for a unit square Q_s of Λ^3; $E_\tau = \emptyset$ for every other τ (a nonempty E_τ has at least four members, at most one of which is p or q, and would add at least three plaquettes to the count); and no spatial plaquettes besides p, q and one spatial plaquette at time 0 (resp. t) when p (resp. q) is temporal. For $0 \leq s \leq t-2$ the four spatial links ∂Q_s at time $s+1$ lie in the plaquettes of $E_{s+1/2}$ and need a second plaquette of Σ each; spatial plaquettes at time $s+1 \in [1,t-1]$ are excluded, so these links lie in $E_{s+3/2} = \partial Q_{s+1}$, and $Q_{s+1} = Q_s =: Q$. The links ∂Q at time 0 need a second plaquette, which is spatial at time 0 since $E_{-1/2} = \emptyset$: if p is spatial it is p, so $p = Q \times \{0\}$ and $0 \in Q$; if p is temporal, $p \in E_{1/2}$ and the single allowed spatial plaquette at time 0 must contain all four links, so it is $Q \times \{0\}$, and $0 \in p$ gives $0 \in Q$ again. Symmetrically at time t. Hence Σ consists of the tube $\bigcup_s \partial Q \times [s,s+1]$ and the two faces $Q \times \{0\}$, $Q \times \{t\}$, which is $\partial\mathscr{B}$ for $\mathscr{B} = [0,t] \times Q$ with $Q = I \times J \ni 0$ as stated; conversely each such $\partial\mathscr{B}$ satisfies the hypotheses with equality. The unit squares $Q \ni 0$ in the three spatial coordinate planes number $3 \cdot 4 = 12$. The faces of $\partial\mathscr{B}$ containing the vertex $(0,0)$ are the end face $Q \times \{0\}$ and the two side faces $e \times [0,1]$ for the two links e of ∂Q at 0; likewise at $(t,0)$. □

Lemma 311 (weight of a box)**.** *Let $\mathscr{B}$ be as in Lemma 310 and let R assign ρ or $\bar\rho$ to each face of $\partial\mathscr{B}$. Then $I(\partial\mathscr{B},R) = N^{-4t}$ if the assignment is* consistent*, that is, if for every link the two faces containing it carry mutually dual representations with respect to the orientations induced by the plaquette variables, and $I = 0$ otherwise. There are exactly $2-\delta$ consistent assignments. If some face carries a representation $R_a \notin \{\rho,\bar\rho\}$ while its neighbours carry ρ or $\bar\rho$, then $I = 0$.*

Proof. Each link e of $\partial\mathscr{B}$ lies in exactly two faces. Writing the two plaquette variables as $XU_e^\varepsilon Y$ and $X'U_e^{\varepsilon'}Y'$, the integral over U_e of the product of the two matrix coefficients of U_e vanishes unless the two representations of U_e appearing are dual to each other, that is, unless R' restricted along e is $\bar R$ with respect to the induced orientations; when it does not vanish it equals $\delta\delta/N$ by the Schur relations. This gives the last statement. Choosing the orientation of $\partial\mathscr{B}$ as the boundary of the box, each link is traversed in opposite directions by its two faces, so a consistent assignment is one in which every face carries ρ with respect to the boundary orientation, or every face carries $\bar\rho$; these two coincide when $\chi = \bar\chi$. For a consistent assignment the integral is the partition function of two-dimensional lattice gauge theory on the sphere $\partial\mathscr{B}$ with the face weight χ_ρ: after the link integrals, the remaining index contractions close up into one loop at each of the $4(t+1)$ vertices, giving N^V, against the factor N^{-1} of each of the $8t+4$ links, and $V-E = -4t$. □

Theorem 312 (leading strong-coupling term)**.** *Let ρ be irreducible, $L \geq 6$ even and $2 \leq t < L/2$. There are $\beta_0(N) > 0$ and*

$C(N,t)<\infty$, independent of L, such that for $0\le\beta\le\beta_0(N)$

$$\Big|G_L(te_0;\beta)-108\Big(\frac{1+\delta}{2}\Big)^{4t+1}\beta^{4t}\Big|\le C(N,t)\,\beta^{4t+1}.$$

The same holds for $G(te_0)$ in every torus limit.

Proof. Since the left side is the second mixed cumulant,

$$G_L(te_0)=\sum_{p\ni 0,\,q\ni te_0}\partial_s\partial_{s'}\log\int e^{sF_p+s'F_q}e^{N\beta\sum\mathrm{Re\,Tr}}\Big|_{s=s'=0},$$

a sum of 576 terms. Apply Lemma 309(iii) to each term: the series converges uniformly on the polydisc $|s|,|s'|\le r_N$, so it may be differentiated term by term. The volume factors $\prod_a c_0^{(a)}$ contribute $\log c_0^{(p)}(s)+\log c_0^{(q)}(s')$, whose mixed derivative is 0; a cluster contributes a nonzero mixed derivative only if some polymer of it contains p and some polymer contains q. For such a cluster X let Σ be the union of the plaquette sets of its polymers. It contains p and q, is connected under sharing a link (each polymer is, and incompatible polymers share a link), and every link of a plaquette of Σ lies in at least two plaquettes of the polymer containing that plaquette, by Lemma 309(i). By Lemma 310, $\sum_{\gamma\in X}\sum_{a\in A_\gamma\setminus\{p,q\}}n(R_a)\ge|\Sigma\setminus\{p,q\}|\ge 4t$, with equality only if $\Sigma=\partial\mathscr{B}$, every $n(R_a)=1$ for $a\ne p,q$, and no plaquette other than p,q belongs to two polymers of X. In the equality case X consists of a single polymer: a polymer whose plaquettes lie in $\{p,q\}$ does not exist (p and q share no link, and a polymer has at least six plaquettes), and if two polymers of X partitioned the faces of the sphere $\partial\mathscr{B}$, a link whose two faces lie in different polymers would lie in one plaquette of each, contradicting Lemma 309(i); moreover $R_p,R_q\in\{\rho,\bar\rho\}$ by the last statement of Lemma 311. By Lemma 309(iii) and the Cauchy estimates on the polydisc, the clusters of order at least $4t+1$ contribute at most $r_N^{-2}K_N(C_N\beta)^{4t+1}$ to the mixed derivative of each of the 576 terms. The order-$4t$ part is the sum over the 108 triples $(\mathscr{B},p,q)$ of $\partial_s\partial_{s'}W(\partial\mathscr{B},R;s,s')$ summed over the consistent R of Lemma 311, which by (42), (43) equals

$$\begin{aligned}&(2-\delta)N^{-4t}\Big(\frac{1+\delta}{2}N\beta\Big)^{4t}\Big(\frac{1+\delta}{2}\Big)^{2}\big(1+O(\beta)\big)\\&\quad=\Big(\frac{1+\delta}{2}\Big)^{4t+1}\beta^{4t}\big(1+O(\beta)\big),\end{aligned}$$

using $(2-\delta)(1+\delta)/2=1$ for $\delta\in\{0,1\}$. The $O(\beta)$ is uniform in L. The statement for torus limits follows since $G(te_0)=\lim G_L(te_0)$ along the subsequence. □

Corollary 313 (the exceptional set)**.** *Let ρ be irreducible and $L\ge 6$ even. The set $\mathscr{D}_L=\{\beta>0: G_L(2e_0;\beta)=0\}$ is closed and discrete in $(0,\infty)$, and there is $\beta_1(N)>0$ independent of L with $\mathscr{D}_L\cap(0,\beta_1(N)]=\emptyset$. For $\beta\notin\mathscr{D}_L$, $G_L(te_\mu;\beta)>0$ for all μ and all $2\le t\le L-2$; for $\beta\in\mathscr{D}_L$ the degenerate alternative of Theorem 304(iii),(iv) holds. For every torus limit at $\beta\in(0,\beta_1(N)]$, $G(te_0)>0$ for all $t\ge 2$.*

Proof. Put

$$\beta_1(N)=\min\Big(\beta_0(N),\ \min_{t=2,3,4}54\big(\tfrac{1+\delta}{2}\big)^{4t+1}/C(N,t)\Big).$$

By Theorem 312 with $t=2$, $G_L(2e_0;\beta)\ge\frac12\cdot108(\frac{1+\delta}{2})^9\beta^8>0$ for $0<\beta\le\beta_1(N)$, so $\beta\mapsto G_L(2e_0;\beta)$ does not vanish identically and Lemma 308 applies; the rest of the finite-torus statement is Theorem 304(iii),(iv). For a torus limit at $\beta\le\beta_1(N)$, Theorem 312 with $t=2,3,4$ (admissible since $L\to\infty$ along the subsequence) gives $G(te_0)>0$ for $t=2,3,4$, so the first alternative of Corollary 306(ii) holds and $G(te_0)>0$ for all $t\ge 2$. □

Proposition 314 (sum rule)**.** *With $S=\sum_p F_p$ the Wilson action of Λ_L, for every $\beta>0$*

$$\sum_{z\in\Lambda_L}G_L(z)=\frac{16}{L^4}\,\mathrm{Var}_L(S)=-\frac{16}{NL^4}\,\frac{d}{d\beta}\langle S\rangle_L>0.$$

Proof. $\sum_x\Phi_x=4S$ since every plaquette has four corners, so $\sum_z G_L(z)=L^{-4}\sum_{x,z}\mathrm{Cov}(\Phi_x,\Phi_{x+z})=L^{-4}\,\mathrm{Var}(4S)$. The density is $e^{-N\beta S}$ up to a constant, so $d\langle S\rangle_L/d\beta=-N\,\mathrm{Var}_L(S)$, and $\mathrm{Var}_L(S)>0$ because S is a nonconstant continuous function and the Wilson measure has a strictly positive density. □

Remark 315 (what is and is not proved about nonvanishing). For Φ on a finite torus, the connected correlation at an axial separation te_μ, $2\le t\le L-2$, is strictly positive for every coupling outside the closed discrete set $\mathscr{D}_L$, which is empty in $(0,\beta_1(N)]$, and its vanishing at one coupling is equivalent to the operator identity $B=c_LT^2$ on $\mathfrak{h}_P$, which forces the vanishing at every axial separation and at every z with a coordinate of modulus in $[2,L-2]$, and forces the one-point function to be independent of the time extent. Whether $\mathscr{D}_L$ is empty for every L is not decided here; it is decided for a single spatial plaquette (Theorem 307). For non-axial z only the real-analyticity of $\beta\mapsto G_L(z;\beta)$ (Lemma 308) and the vanishing in the degenerate alternative are obtained; neither the sign of $G_L(z;\beta)$ nor the sign of its first nonvanishing Taylor coefficient is asserted. The sum rule shows that G_L does not vanish identically in z at any coupling.

4. *Quantitative lower bounds along an axis*

The dichotomy of Theorem 304 comes with a rate. In the nondegenerate alternative put, for $\beta\notin\mathscr{D}_L$,

$$m_L(t;\beta)=\log\frac{G_L(te_0;\beta)}{G_L((t+1)e_0;\beta)}\qquad(2\le t\le L-3),$$

and for a torus limit in the first alternative of Corollary 306(ii) let $m(t)$ be as in Corollary 306(iii).

Theorem 316 (secant bound)**.** *Let $\beta>0$, $L\ge6$ even and $\beta\notin\mathscr{D}_L$. Then $t\mapsto m_L(t;\beta)$ is nonincreasing on $\{2,\dots,L-3\}$, $m_L(t;\beta)\ge0$ for $t\le L/2-1$, $m_L(L-1-t;\beta)=-m_L(t;\beta)$, and for $2\le t_0<t\le L-2$*

$$\begin{aligned}G_L(te_0;\beta)&\ge G_L(t_0e_0;\beta)\,e^{-(t-t_0)\,m_L(t_0;\beta)}\\&\ge G_L(t_0e_0;\beta)\,e^{-(t-t_0)\,m_L(2;\beta)}.\end{aligned}$$

For a torus limit ω at $\beta>0$ in the first alternative of Corollary 306(ii), and all $2\le t_0<t$,

$$G(te_0)\ge G(t_0e_0)\,e^{-(t-t_0)m(t_0)}\ge G(t_0e_0)\,e^{-(t-t_0)m(2)},$$

and $m(t_0)\downarrow m_\Phi$, so that the exponential rate of the bound improves to m_Φ as $t_0\to\infty$; moreover

$$G(te_0)\le G(t_0e_0)\,e^{-(t-t_0)m(t-1)},\quad \lim_{t\to\infty}\frac{\log G(te_0)}{t}=-m_\Phi,$$

so m_Φ is the exact exponential rate of G along the axis.

Proof. By Corollary 313, $G_L(te_0;\beta)>0$ for $2\le t\le L-2$, so $g(t)=\log G_L(te_0;\beta)$ is defined there, and Theorem 304(ii) says that g is convex on $\{2,\dots,L-2\}$: its increments $g(t+1)-g(t)=-m_L(t;\beta)$ are nondecreasing in t. The symmetry $g(t)=g(L-t)$ of Theorem 304(i) gives $m_L(L-1-t)=-m_L(t)$, and the monotonicity of Theorem 304(ii) gives $m_L(t)\ge 0$ for $t+1\le L/2$. For $t_0<t$,

$$\begin{aligned}g(t)-g(t_0)&=\sum_{s=t_0}^{t-1}\big(g(s+1)-g(s)\big)\\&\ge(t-t_0)\big(g(t_0+1)-g(t_0)\big),\end{aligned}$$

which is the first inequality; the second follows from $m_L(t_0)\le m_L(2)$. In a torus limit the same argument applies to $g(t)=\log G(te_0)$, $t\ge 2$, which is finite by the first alternative and the positivity $G(2e_0)\ge G(3e_0)>0$ of Corollary 306(i),(ii); the monotonicity and the limit m_Φ of the $m(t)$ are Corollary 306(iii). The upper bound is the convexity inequality $g(t)-g(t_0)\le(t-t_0)(g(t)-g(t-1))$. Finally $t^{-1}(g(t)-g(2))=-t^{-1}\sum_{s=2}^{t-1}m(s)\to-m_\Phi$ by Cesàro. □

At strong coupling the two numbers entering the bound are explicit.

Theorem 317 (explicit constants at strong coupling)**.** *Let ρ be irreducible. There are $\beta_2(N)\in(0,\beta_1(N)]$ and $C_2(N)<\infty$ such that for $0<\beta\le\beta_2(N)$ and every even $L\ge 8$ the numbers*

$$c(\beta)=108\Big(\frac{1+\delta}{2}\Big)^9\beta^8,\qquad m(\beta)=4\log\frac{2}{(1+\delta)\beta}$$

satisfy

$$\Big|\frac{G_L(2e_0;\beta)}{c(\beta)}-1\Big|\le C_2(N)\beta,\quad |m_L(2;\beta)-m(\beta)|\le C_2(N)\beta,$$

and consequently, for $2\le t\le L-2$,

$$\begin{aligned}G_L(te_0;\beta)&\ge\big(1-C_2(N)\beta\big)\,c(\beta)\,e^{-(t-2)\,(m(\beta)+C_2(N)\beta)}\\&=\big(1-C_2(N)\beta\big)\,108\Big(\frac{1+\delta}{2}\Big)^9\beta^8\\&\quad\times\Big(\frac{(1+\delta)\beta}{2}\Big)^{4(t-2)}e^{-C_2(N)\beta(t-2)}.\end{aligned}$$

The same inequalities hold for $G(te_0)$ and $m(2)$ in every torus limit at $\beta\in(0,\beta_2(N)]$, for all $t\ge 2$. The constants $\beta_2(N)$ and $C_2(N)$ do not depend on L.

Proof. Theorem 312 with $t=2$ and $t=3$ (admissible since $3<L/2$ for $L\ge 8$) gives, for $\beta\le\beta_0(N)$,

$$\begin{aligned}G_L(2e_0;\beta)&=108\Big(\frac{1+\delta}{2}\Big)^9\beta^8\big(1+\varepsilon_2(\beta)\big),\\G_L(3e_0;\beta)&=108\Big(\frac{1+\delta}{2}\Big)^{13}\beta^{12}\big(1+\varepsilon_3(\beta)\big),\end{aligned}$$

with $|\varepsilon_t(\beta)|\le C(N,t)\beta/(108((1+\delta)/2)^{4t+1})$. Choose $\beta_2(N)\le\beta_1(N)$ so small that $|\varepsilon_2|,|\varepsilon_3|\le 1/2$ on $(0,\beta_2(N)]$. Then both values are positive, and

$$m_L(2;\beta)=\log\frac{G_L(2e_0;\beta)}{G_L(3e_0;\beta)}=4\log\frac{2}{(1+\delta)\beta}+\log\frac{1+\varepsilon_2(\beta)}{1+\varepsilon_3(\beta)},$$

where

$$\big|\log\big((1+\varepsilon_2)/(1+\varepsilon_3)\big)\big|\le 2(|\varepsilon_2|+|\varepsilon_3|)$$

because $|\log(1+x)|\le 2|x|$ for $|x|\le 1/2$. This gives the two estimates with a $C_2(N)$ depending only on $C(N,2)$, $C(N,3)$ and δ. The lower bound on $G_L(te_0;\beta)$ is then Theorem 316 with $t_0=2$, and $e^{-(t-2)m(\beta)}=((1+\delta)\beta/2)^{4(t-2)}$. For torus limits, Theorem 312 holds for $G(te_0)$ as stated, the first alternative holds by Corollary 313, and the bound is Theorem 316. □

Remark 318. (a) The bound of Theorem 317 is sharp to leading order at $t=2,3$ and, by Theorem 312, sharp in the exponent at every fixed $t<L/2$: $G_L(te_0;\beta)=108((1+\delta)/2)^{4t+1}\beta^{4t}(1+O(\beta))$, while the bound gives $108((1+\delta)/2)^{4t+1}\beta^{4t}(1-O(t\beta))$. The factor $e^{-C_2(N)\beta(t-2)}$ is the price of using only the two shortest distances; it is removed, at any fixed t_0, by starting the secant at t_0 instead of 2, since $m_L(t_0;\beta)=4\log(2/((1+\delta)\beta))+O(\beta)$ for every fixed $t_0<L/2-1$ by the same computation.

(b) In the language of Sec. XL, the rate is a lattice mass. Since $m(t)\downarrow m_\Phi$, Theorem 317 gives $m_\Phi\le m(2)=4\log(2/((1+\delta)\beta))+O(\beta)$ in every torus limit at $\beta\le\beta_2(N)$, and by (a) the same upper bound holds with $m(t_0)$ for every fixed t_0. The equality $m_\Phi=4\log(2/((1+\delta)\beta))+O(\beta)$ would require the remainder $C(N,t)\beta^{4t+1}$ of Theorem 312 to be of the form $C(N)^t\beta^{4t+1}$ uniformly in t; this is not established here. The number $4\log(1/\beta)$ is the area-law cost of the four side plaquettes per unit length of the tube of Lemma 310; in the normalisation $\beta_W=N^2\beta$ of Sec. I the bound reads $m_\Phi\le 4\log(2N^2/((1+\delta)\beta_W))+O(\beta_W/N^2)$, the leading strong-coupling estimate of the lightest glueball mass in the plaquette channel, for which see [85].

a. The natural parameter and the degeneration of the bound. Let ν_β be the one-link measure $c_0(\beta)^{-1}e^{N\beta\,\mathrm{Re\,Tr}\,g}\,dg$ on G and

$$u(\beta)=\frac{a_\rho(\beta)}{N}=\frac{1}{N}\int_G\chi(g)\,d\nu_\beta(g)=\frac{1}{N}\int_G\mathrm{Re\,Tr}\,g\,d\nu_\beta(g),$$

the second equality because $g\mapsto g^{-1}$ preserves ν_β and $\chi(g^{-1})=\overline{\chi(g)}$. Then u is real-analytic on $[0,\infty)$, $u(\beta)=\frac{1+\delta}{2}\beta+O(\beta^2)$ by (42), $u'(\beta)=\mathrm{Var}_{\nu_\beta}(\mathrm{Re\,Tr}\,g)>0$ since $\mathrm{Re\,Tr}\,g$ is not constant on $G\neq\{1\}$, and $u(\beta)\uparrow 1$ as $\beta\to\infty$:

writing $\operatorname{Re}\operatorname{Tr} g = N - \frac{1}{2}\|g-1\|_{\mathrm{HS}}^2$, the set $\{\operatorname{Re}\operatorname{Tr} g < N-\varepsilon\}$ has ν_β-measure at most $e^{-N\beta\varepsilon/2}/\operatorname{Haar}\{\operatorname{Re}\operatorname{Tr} g \ge N-\varepsilon/2\} \to 0$. The leading strong-coupling term of Theorem 312 is $108\,\frac{1+\delta}{2}\,u(\beta)^{4t}(1+O(\beta))$, each of the $4t$ plaquettes of the box contributing one factor u (the activity a_ρ divided by the dimension N of Lemma 311), so that $m(\beta) = 4\log(1/u(\beta)) + O(\beta)$. As β grows, $u(\beta)$ increases to 1 and the formal rate $4\log(1/u(\beta))$ decreases to 0. What is proved is the following.

Proposition 319 (the bound at every coupling)**.** *Let ρ be irreducible and $L \ge 8$ even.*

(i) *For every $\beta \in (0,\infty)\setminus\mathscr{D}_L$ and $2 \le t \le L-2$,*

$$G_L(te_0;\beta) \ge G_L(2e_0;\beta)\, e^{-(t-2)m_L(2;\beta)},$$

where $\beta \mapsto G_L(2e_0;\beta) > 0$ and $\beta \mapsto m_L(2;\beta) \ge 0$ are real-analytic on $(0,\infty)\setminus\mathscr{D}_L$. For $\beta \in \mathscr{D}_L$ both sides vanish.

(ii) *On $(0,\beta_2(N)]$ the two functions are given by Theorem 317; in particular $m_L(2;\beta) \to \infty$ as $\beta \downarrow 0$: the exponential rate $4\log(2/((1+\delta)\beta))$ of the bound diverges while its prefactor $G_L(2e_0;\beta) \asymp \beta^8$ vanishes. For $\beta > \beta_2(N)$ no bound on $G_L(2e_0;\beta)$ from below or on $m_L(2;\beta)$ from above that is uniform in L is obtained here: Lemma 309(iii) controls the Taylor series only for $\beta \le \beta_0(N)$, and Lemma 308 gives analyticity in β for each L without bounds uniform in L.*

(iii) *If a sequence ω_k of torus limits at couplings β_k satisfies (AF) with constants $c_1 \le C_1$, γ, r_0, then for all $k \ge k_0$ with $4a_k \le r_0$*

$$\left| m_k(3) - 8\log\frac{4}{3} \right| \le \log\frac{C_1}{c_1} + \gamma\log\frac{\ell(3a_k)}{\ell(4a_k)},$$

where $m_k(3) = \log(G_k(3e_0)/G_k(4e_0))$ and the last term tends to 0 as $a_k \to 0$. In particular the lattice rates $m_k(3)$ stay in a bounded window, whereas the strong-coupling rate $4\log(2/((1+\delta)\beta_k))$ leaves every bounded window as $\beta_k \to 0$ and the formal rate $4\log(1/u(\beta_k))$ tends to 0 as $\beta_k \to \infty$.

Proof. (i) is Theorem 316 with $t_0 = 2$ together with Lemma 308 (the ratio of two positive real-analytic functions is real-analytic, and $m_L(2;\beta) \ge 0$ since $2 \le L/2-1$) and Theorem 304(iii). (ii) is Theorem 317 and a statement about the proofs given. For (iii), (AF) applied at $z = 3e_0$ and $z = 4e_0$, both with $|z|_\infty \ge 3$ and $a_k|z| \le r_0$, gives

$$\frac{c_1 w_\gamma(3a_k)}{C_1 w_\gamma(4a_k)} \le \frac{G_k(3e_0)}{G_k(4e_0)} \le \frac{C_1 w_\gamma(3a_k)}{c_1 w_\gamma(4a_k)},$$

and $w_\gamma(3a_k)/w_\gamma(4a_k) = (4/3)^8(\ell(4a_k)/\ell(3a_k))^\gamma$, with $1 \le \ell(3a)/\ell(4a) \to 1$ as $a \to 0$. □

b. What reflection positivity gives. The question whether reflection positivity alone yields a lower bound along a reflection axis has a precise answer.

Proposition 320 (reflection positivity and lower bounds)**.** *Let ω be a torus limit at $\beta \ge 0$, ϑ the reflection through the hyperplane $x_0 = 0$ and ϑ' the reflection through $x_0 = \frac{1}{2}$ of Lemma 294(iii).*

(i) *Reflection positivity gives $\omega((\Phi_{se_0} - c)\,\vartheta(\Phi_{se_0} - c)) = G(2se_0) \ge 0$ and $\omega((\Phi_{(s+1)e_0} - c)\,\vartheta'(\Phi_{(s+1)e_0} - c)) = G((2s+1)e_0) \ge 0$ for $s \ge 1$, hence $G(te_0) \ge 0$ for all $t \ge 2$; with the transfer-matrix reconstruction it gives the log-convexity and monotonicity of Corollary 306 and the secant bound of Theorem 316: a lower bound for $G(te_0)$ at every t in terms of the two values $G(t_0e_0)$, $G((t_0+1)e_0)$.*

(ii) *No positive lower bound on $G(te_0)$, $t \ge 2$, follows from reflection positivity, invariance and monotonicity alone: the function $G \equiv 0$ on $\{|z|_\infty \ge 2\}$ satisfies every inequality that these properties impose, and it is realised, with $c = \omega(\Phi_0) = 24N$, by the product Haar measure $\beta = 0$, which is a reflection-positive, translation- and hypercubically invariant torus limit. Within the Wilson measures at $\beta > 0$ the same conclusion holds at every $\beta \in \mathscr{D}_L$ on the torus Λ_L, by Theorem 304(iv).*

(iii) *Monotonicity gives $G(te_0) \ge \lim_{s\to\infty} G(se_0) = \sigma(\{1\})$ for all $t \ge 3$, the mass of the spectral measure of Corollary 306 at $\lambda = 1$; this is 0 whenever $\gamma' < 1$, in particular whenever ω has a mass gap for the reflection ϑ', so monotonicity yields no positive lower bound in the situations of interest.*

Proof. (i) For $F = \Phi_{se_0} - c$, which depends on the links in $\{x_0 \ge s-1\} \subset \{x_0 \ge 0\}$ when $s \ge 1$, reflection positivity for ϑ gives $\omega(\bar F\,\vartheta F) \ge 0$, and $\vartheta\Phi_{se_0} = \Phi_{-se_0}$ because ϑ maps the plaquettes with a corner at se_0 onto those with a corner at $-se_0$ and F_p is invariant under orientation reversal; so $\omega(\bar F\,\vartheta F) = \omega(\Phi_{se_0}\Phi_{-se_0}) - c^2 = G(2se_0)$ by invariance. For ϑ', $\Phi_{(s+1)e_0}$ depends on links in $\{x_0 \ge s\} \subset \{x_0 \ge 1\}$ for $s \ge 1$, and $\vartheta'\Phi_{(s+1)e_0} = \Phi_{-se_0}$, giving $G((2s+1)e_0) \ge 0$. The rest of (i) is Corollary 306 and Theorem 316, whose proofs use only the Osterwalder–Seiler reconstruction from reflection positivity.

(ii) At $\beta = 0$ the links are independent Haar variables. For $|z|_\infty \ge 2$ let $p \ni 0$ and $q \ni z$ be plaquettes; then $p \ne q$, since the corners of p lie in $\{|x|_\infty \le 1\}$, and two distinct plaquettes share at most one link. If they share no link, F_p and F_q are independent. If they share the link l, then conditionally on U_l the remaining three links of p are independent Haar variables, and, with the links of p taken in cyclic order from l and $P_0 = \int_G \rho(g)\,dg$ the orthogonal projection onto the G-invariant vectors of $\mathbb{C}^N$,

$$\begin{aligned} E(F_p \mid U_l) &= N - \operatorname{Re}\operatorname{Tr}\big(U_l^{\pm1} E(U_{l_1}^{\pm1} U_{l_2}^{\pm1} U_{l_3}^{\pm1})\big) \\ &= N - \operatorname{Re}\operatorname{Tr}(U_l^{\pm1} P_0^3) = N - \operatorname{Tr} P_0, \end{aligned}$$

because Haar measure is inversion invariant, $P_0^3 = P_0$ and $\rho(g)P_0 = P_0$; this is a constant, equal to $E(F_p)$. Hence $E(F_pF_q) = E(E(F_p \mid U_l)F_q) = E(F_p)E(F_q)$, since F_q depends on U_l and on links independent of those of p, and summing over $p \ni 0$, $q \ni z$ gives $G(z) = 0$. The product measure is reflection positive and invariant, and $c = 24(N - \operatorname{Tr} P_0) = 24N$ for irreducible ρ. The statement about $\mathscr{D}_L$ is Theorem 304(iv).

(iii) By Corollary 306(i), $G((3+n)e_0) = \int \lambda^n\, d\sigma \downarrow \sigma(\{1\})$ by dominated convergence, and σ is carried by $[0,\gamma']$ because $\Psi' \perp \Omega'$, as shown in the proof of Corollary 306(iii). □

Remark 321. Proposition 320 separates the two inputs of a lower bound: reflection positivity propagates positivity from two short distances to all distances along the axis, with the exact rate m_Φ in the limit, while the nonvanishing at short distance is a property of the specific measure, established here

by analyticity and the strong-coupling expansion outside the discrete set $\mathscr{D}_L$ (Corollary 313) and with explicit values for $\beta \le \beta_2(N)$ (Theorem 317). For a non-axial z there is no reflection in a coordinate hyperplane mapping 0 to z; the diagonal reflections are not among the reflection positivities proved in Lemma 294(iii), and the positivity $G(z) \ge 0$ for general z is not asserted here.

5. Comparison with the perturbative law and the missing step

Let ω_k be torus limits at couplings β_k with spacings $a_k \to 0$, and G_k, Z_k, $c_1 \le C_1$, γ, r_0, ℓ, w_γ as in (AF) of Sec. X L; with $r = 1$ the range of (AF) is $|z|_\infty \ge 3$, and for $z = te_0$ one has $|z| = t$. For $t \ge 2$ put $m_k(t) = \log(G_k(te_0)/G_k((t+1)e_0))$ when $G_k((t+1)e_0) > 0$. Theorem 316 gives, in the first alternative of Corollary 306(ii), the exponential lower bound

$$Z_k^2 G_k(te_0) \ge Z_k^2 G_k(t_0 e_0)\, e^{-(t-t_0)m_k(t_0)} \qquad (2 \le t_0 < t), \quad (45)$$

whereas the lower half of (AF) asks for $Z_k^2 G_k(te_0) \ge c_1 w_\gamma(a_k t)$, a power law, on $3 \le t \le r_0/a_k$. The two are compared first as functions of t and then in spectral terms.

Proposition 322 (exponential against power law)**.** (i) *If ω_k satisfies (AF), then for $3 \le t$ with $(t+1)a_k \le r_0$*

$$\Big| m_k(t) - 8\log\frac{t+1}{t} \Big| \le \log\frac{C_1}{c_1} + \gamma \log\frac{\ell(a_k t)}{\ell(a_k(t+1))},$$

and the lattice mass $m_{\Phi,k} = \lim_t m_k(t)$ of Corollary 306 satisfies $\limsup_k m_{\Phi,k} \le \log(C_1/c_1)$.

(ii) *Among all positive log-convex sequences with prescribed values at t_0 and t_0+1, the geometric sequence $G(t_0)e^{-(t-t_0)m(t_0)}$ is the smallest; so (45) is the best lower bound at $t > t_0+1$ that follows from the values at t_0, t_0+1 and log-convexity. If $G(t_0)$ and $G(t_0+1)$ are the values of the power law ct^{-8}, this bound is $ct_0^{-8}(1+1/t_0)^{-8(t-t_0)}$, equal to ct^{-8} at $t = t_0+1$ and strictly smaller for every $t \ge t_0+2$.*

(iii) *For every finite set $S \subset \{3,4,\dots\}$ and $c > 0$ there is a finite measure σ on $[0,1]$ such that $G(t) = \int \lambda^{t-3} d\sigma$ satisfies $\frac{1}{2}ct^{-8} \le G(t) \le ct^{-8}$ for $t \in S$ and $G(t) \le c\varepsilon^{t-3}$ for all $t \ge 3$, for some $\varepsilon \in (0,1)$. Hence the lower bounds $Z_k^2 G_k(te_0) \ge c_1 w_\gamma(a_k t)$ at the separations $t \in S$, together with every inequality that the spectral representation of Corollary 306 imposes, do not imply the lower bound at the separations $t \notin S$ with $t \le r_0/a_k$, once a_k is small.*

Proof. (i) is the computation of Proposition 319(iii) at the separations $t, t+1$, where $\ell(a_k t) \ge \ell(a_k(t+1))$ since ℓ is nonincreasing; the first alternative holds because the lower half of (AF) gives $G_k(te_0) > 0$ for $3 \le t \le r_0/a_k$. For the mass, $m_{\Phi,k} \le m_k(t)$ for every admissible t by Corollary 306(iii); taking $t+1 = \lfloor r_0/a_k \rfloor$ and letting $k \to \infty$, $8\log((t+1)/t) \to 0$ and $\log(\ell(a_k t)/\ell(a_k(t+1))) \le \log(\ell(r_0 - 2a_k)/\ell(r_0)) \to 0$.

(ii) A sequence $g(t) = \log G(t)$ is convex if and only if its increments are nondecreasing, so $g(t) \ge g(t_0) + (t-t_0)(g(t_0+1) - g(t_0))$, with equality for affine g, that is, geometric G. With $G(t_0) = ct_0^{-8}$ and $G(t_0+1) = c(t_0+1)^{-8}$ the geometric bound is $ct_0^{-8}(t_0/(t_0+1))^{8(t-t_0)}$, and $(1+1/t_0)^n > 1 + n/t_0$ for $n = t - t_0 \ge 2$ is Bernoulli's inequality, strict for $n \ge 2$.

(iii) The power law is a moment sequence: $c(3+n)^{-8} = \int_0^\infty e^{-(3+n)E}\, cE^7 dE/7!$, so $ct^{-8} = \int \lambda^{t-3} d\sigma_0$ with σ_0 the image of $cE^7 e^{-3E} dE/7!$ under $\lambda = e^{-E}$. Let $\sigma = \sigma_0|_{[\varepsilon',\varepsilon]}$ with $0 < \varepsilon' < \varepsilon < 1$. Then $G(t) \le ct^{-8}$ for all t, $G(t) \le \varepsilon^{t-3}\sigma_0([0,1]) = c\,3^{-8}\varepsilon^{t-3}$, and $G(t) \ge ct^{-8} - \sigma_0([0,\varepsilon')) - \sigma_0((\varepsilon,1])$ for every t. Since σ_0 has no atoms, $\sigma_0([0,\varepsilon')) \to 0$ as $\varepsilon' \to 0$ and $\sigma_0((\varepsilon,1]) \to 0$ as $\varepsilon \to 1$; choose ε close to 1 and then ε' close to 0 so that both are at most $\frac{1}{4}c(\max S)^{-8}$. Then $G(t) \ge \frac{1}{2}ct^{-8}$ on S. For the last statement take $c = 2c_1 a_k^{-8}$: then $G(t) \ge c_1 a_k^{-8} t^{-8} \ge c_1 w_\gamma(a_k t)$ on S, G has the spectral form of Corollary 306(i), and $G(t) \le 2c_1 a_k^{-8} 3^{-8}\varepsilon^{t-3} < c_1 w_\gamma(a_k t)$ for t large with $a_k t \le r_0$, since $\varepsilon^t t^8 \ell(a_k t)^\gamma \to 0$ and $\ell(a_k t) \le \ell(a_k)$. □

a. Spectral form of (AF). Let σ_k be the spectral measure of Corollary 306(i) for ω_k, and let τ_k be the image of $Z_k^2\sigma_k$ under $\lambda \mapsto E = -\log\lambda \in [0,\infty]$, the energy distribution, in lattice units of the time direction, of the state $Z_k(\Phi_{2e_0} - c_k)\Omega'$ for the transfer matrix of the link reflection. Then

$$Z_k^2 G_k((3+n)e_0) = \int_{[0,\infty)} e^{-nE}\, d\tau_k(E) \qquad (n \ge 1), \quad (46)$$

and $\tau_k([0,\infty)) = Z_k^2 G_k(3e_0)$.

Proposition 323 (spectral form of the two-point law)**.** *Let ω_k satisfy (AF), and put $A = \log(2C_1/c_1)$, $B = 8 + \gamma$.*

(i) (*Upper half.*) *For $0 < E \le 1$,*

$$\tau_k([0,E]) \le eC_1\, w_\gamma\big(a_k(3 + \lfloor 1/E \rfloor)\big) \le eC_1\, a_k^{-8} E^8.$$

(ii) (*Lower half.*) *For $4 \le t \le r_0/a_k$ the number $M_k(t) = \log\big(2C_1 w_\gamma(3a_k)/(c_1 w_\gamma(a_k t))\big)$ satisfies $\log 2 \le M_k(t) \le A + B\log(t/3)$ and*

$$\tau_k\Big(\Big[0, \frac{M_k(t)}{t-3}\Big]\Big) \ge \frac{c_1}{2}\, w_\gamma(a_k t).$$

Consequently, for $0 < E \le 1$ and $t_E = 3 + \lceil (2/E)(A + B\log(2B/E)) \rceil$, if $t_E \le r_0/a_k$ then

$$\begin{aligned}\tau_k([0,E]) &\ge \frac{c_1}{2}\, w_\gamma(a_k t_E) \\ &\ge \frac{c_1}{2}\, a_k^{-8} \Big(\frac{E}{C_3 \log(4/E)}\Big)^8 \ell(a_k/E)^{-\gamma},\end{aligned}$$

with $C_3 = 4 + 2A + 2B\log(2B) + 2B$; the condition $t_E \le r_0/a_k$ holds for $E \ge C_4(a_k/r_0)\log(r_0/a_k)$ when a_k/r_0 is small, for a C_4 depending on C_1/c_1 and γ.

(iii) (*Converse.*) *If*

$$\tau_k([0,E]) \ge C^{-1} a_k^{-8} \Big(\frac{E}{\log(4/E)}\Big)^8 \ell(a_k/E)^{-\gamma}$$

for $E_k \le E \le 1$, then for $11 \le t \le 3 + 8/E_k$

$$Z_k^2 G_k(te_0) \ge \frac{(8/e)^8}{C(1+\log 16)^\gamma}\, \frac{w_\gamma(a_k t)}{\big(\log((t-3)/2)\big)^8},$$

the lower half of (AF) up to the factor $(\log((t-3)/2))^{-8}$*; and if* $\tau_k([0,E]) \le C' a_k^{-8} E^8$ *for* $0 < E \le 1$ *and* $\tau_k([0,\infty)) \le C' a_k^{-8}$*, then* $Z_k^2 G_k(te_0) \le C'' a_k^{-8} t^{-8}$ *for* $t \ge 4$*, the upper half with* $\gamma = 0$*.*

Proof. (i) For $n \ge 1$ and $E > 0$,

$$\tau_k([0,E]) \le e^{nE} \int e^{-nE'} d\tau_k(E') = e^{nE} Z_k^2 G_k((3+n)e_0)$$
$$\le e^{nE} C_1 w_\gamma(a_k(3+n))$$

by the upper half of (AF), valid at every $|z|_\infty \ge 3$. With $n = \lfloor 1/E \rfloor \ge 1$ one has $nE \le 1$ and $3+n \ge 1/E$, and $w_\gamma(\rho) \le \rho^{-8}$.

(ii) By Corollary 306(i) and (AF),

$$c_1 w_\gamma(a_k t) \le Z_k^2 G_k(te_0) \le Z_k^2 G_k(3e_0) \le C_1 w_\gamma(3a_k),$$

so $M_k(t) \ge \log 2$; and

$$\frac{w_\gamma(3a_k)}{w_\gamma(a_k t)} = \Big(\frac{t}{3}\Big)^8 \Big(\frac{\ell(a_k t)}{\ell(3a_k)}\Big)^\gamma,$$
$$\frac{\ell(3a_k)}{\ell(a_k t)} = 1 + \frac{\log(t/3)}{\ell(a_k t)} \le 1 + \log(t/3),$$

so $M_k(t) \le A + 8\log(t/3) + \gamma\log(t/3)$. Split (46) at E_0 with $n = t-3 \ge 1$:

$$Z_k^2 G_k(te_0) \le \tau_k([0,E_0]) + e^{-(t-3)E_0}\, \tau_k((E_0,\infty))$$
$$\le \tau_k([0,E_0]) + e^{-(t-3)E_0} C_1 w_\gamma(3a_k).$$

With $E_0 = M_k(t)/(t-3)$ the last term is $\frac{1}{2} c_1 w_\gamma(a_k t)$ by the definition of $M_k(t)$, and the lower half of (AF) gives the claim. For the consequence: for $s \ge 0$ and $0 < \eta \le 1$, $\log(1+s) \le \eta s + \log(1/\eta)$, since the left side minus ηs is maximal at $s = 1/\eta - 1$ with value $\log(1/\eta) - 1 + \eta$. With $\eta = E/(2B)$, $s = t-3$ and $t/3 \le 1 + (t-3)$,

$$M_k(t) \le A + B\log(1+(t-3)) \le A + \tfrac{1}{2}E(t-3) + B\log(2B/E),$$

so $M_k(t)/(t-3) \le E$ as soon as $(t-3)E/2 \ge A + B\log(2B/E)$, which holds for $t = t_E$. If $t_E \le r_0/a_k$ the first bound applies at t_E and gives $\tau_k([0,E]) \ge \tau_k([0, M_k(t_E)/(t_E - 3)]) \ge \frac{1}{2} c_1 w_\gamma(a_k t_E)$. Finally $t_E \le 4 + (2/E)(A + B\log(2B) + B\log(1/E)) \le C_3 \log(4/E)/E$ for $E \le 1$, using $1 \le \log(4/E)$ and $\log(1/E) \le \log(4/E)$ for $E \le 1$, and $t_E \ge 1/E$ gives $\ell(a_k t_E) \le \ell(a_k/E)$, ℓ being nonincreasing. The condition $t_E \le r_0/a_k$ follows from $C_3 \log(4/E)/E \le r_0/a_k$; with $E = C_4 y \log(1/y)$, $y = a_k/r_0$, this reads $C_3 \log(4/(C_4 y \log(1/y))) \le C_4 \log(1/y)$ and holds for $C_4 = 2C_3$ once y is small, and $\log(4/E)/E$ is decreasing in E.

(iii) For $t \ge 11$ put $E = 8/(t-3) \le 1$; if $E \ge E_k$, that is $t \le 3 + 8/E_k$, then by (46)

$$Z_k^2 G_k(te_0) \ge e^{-(t-3)E} \tau_k([0,E])$$
$$\ge \frac{a_k^{-8}}{e^8 C} \Big(\frac{8}{(t-3)\log((t-3)/2)}\Big)^8 \ell\Big(\frac{a_k(t-3)}{8}\Big)^{-\gamma},$$

and $(t-3)/8 \ge t/16$ for $t \ge 6$ gives

$$\ell\big(a_k(t-3)/8\big) \le \ell(a_k t/16) \le \ell(a_k t) + \log 16$$
$$\le (1 + \log 16)\, \ell(a_k t),$$

while $(t-3)^{-8} \ge t^{-8}$. For the upper half, integrate by parts in (46) with $n = t-3 \ge 1$:

$$Z_k^2 G_k(te_0) = \int_0^\infty n e^{-nE} \tau_k([0,E])\, dE$$
$$\le C' a_k^{-8} \int_0^1 n e^{-nE} E^8\, dE + e^{-n} C' a_k^{-8}$$
$$\le C' a_k^{-8} (8!\, n^{-8} + e^{-n}),$$

$e^{-n} \le 8!\, n^{-8}$, and $(t-3)^{-8} \le 4^8 t^{-8}$ for $t \ge 4$. □

Remark 324. Proposition 323 says that (AF) along an axis is, up to powers of logarithms, the statement that the energy distribution of the local excitation $Z_k(\Phi_{2e_0} - c_k)\Omega'$ has weight of order $a_k^{-8} E^8$ below every lattice energy E between $(a_k/r_0)\log(r_0/a_k)$ and 1: in physical units $E_{\rm ph} = E/a_k$, weight of order $E_{\rm ph}^8$ below every energy between $\log(r_0/a_k)/r_0$ and the cutoff $1/a_k$, the distribution $E_{\rm ph}^7 dE_{\rm ph}$. The lower half concerns the low-energy part at every intermediate scale; the exponential bound (45) controls the Laplace transform (46) at a single n, hence, by Proposition 322(ii),(iii), nothing of this kind.

b. The two inputs of the lower half. This subsection provides, for the sequence ω_k, two kinds of information: the positivity, log-convexity and dichotomy of Theorem 304 and Corollary 306, valid at every k with no uniformity in k, and the explicit values of Theorem 317, valid for $\beta_k \le \beta_2(N)$ only. A continuum sequence has $\beta_k \to \infty$ by (AF_β) and leaves the second range; inside it, Theorem 312 shows that the two-point function is exponential in t, $G_k(te_0) \asymp \beta_k^{4t}$, in accordance with Theorem 279(v).

Corollary 325 (the two inputs)**.** *Let* ω_k *be torus limits at couplings* $\beta_k \to \infty$ *in the first alternative of Corollary* 306(ii)*, and consider*

(a) (*cutoff scale*) $\liminf_k Z_k^2 G_k(3e_0)/w_\gamma(3a_k) > 0$*, and*

(b) (*intermediate scales*) *for all* $k \ge k_0$ *and* $E_k \le E \le 1$*,*

$$\tau_k([0,E]) \ge C^{-1} a_k^{-8} \Big(\frac{E}{\log(4/E)}\Big)^8 \ell(a_k/E)^{-\gamma}.$$

The lower half of (AF) implies (a) *and* (b) *with* $E_k = C_4(a_k/r_0)\log(r_0/a_k)$*. Conversely* (a) *and* (b) *imply the lower half of (AF) up to the factor* $(\log((t-3)/2))^{-8}$ *on* $3 \le t \le 3 + 8/E_k$*; with the* E_k *above this is the range* $a_k t \le 8r_0/(C_4 \log(r_0/a_k))(1+o(1))$*, shorter than* $a_k t \le r_0$ *by a logarithm. Neither* (a) *nor* (b) *follows from the results of this subsection:* (a) *is a lower bound at three lattice units for the Wilson measure at* $\beta_k \to \infty$*, where the strong-coupling expansion does not reach, analyticity in* β *gives no bound uniform in k, and reflection positivity gives* $G_k(3e_0) \ge 0$ *only;* (b) *is a lower bound on the low-energy spectral weight of the local excitation at every scale between* r_0 *and the cutoff, which by Proposition 322(iii) is not implied by* (a)*, nor by the lower half of (AF) at any finite set of separations, together with every consequence of reflection positivity.*

Proof. If the lower half of (AF) holds, (a) is its case $t = 3$ and (b) is Proposition 323(ii) with $C = 2C_3^8/c_1$. Conversely (b) gives, by Proposition 323(iii), the lower half for $11 \le$

$t \le 3 + 8/E_k$ up to the factor $(\log((t-3)/2))^{-8}$; (a) gives it at $t = 3$ since $w_\gamma(3a_k) \ge 3^{-8}\ell(3a_k)^{-\gamma}a_k^{-8}$ is the right side of (AF) there; and Corollary 306(i) gives it at $4 \le t \le 10$ from the value at $t = 11$, since $G_k(te_0) \ge G_k(11e_0)$ there and $w_\gamma(a_k t) \le (11/t)^8 w_\gamma(11a_k) \le (11/4)^8 w_\gamma(11a_k)$ for $4 \le t \le 11$, ℓ being nonincreasing. The negative statements are Proposition 319(ii), Theorem 312, Proposition 320 and Proposition 322(iii). □

Remark 326. Statement (b) at the scale E concerns, in the renormalisation-group description of Bałaban [10, 11, 13], the effective theory after the block-spin steps that bring the lattice spacing from a_k to about a_k/E, with the two insertions of Φ made at the cutoff scale and carried through every step. As recalled in Remarks 282 and 302, the bounds of [13] concern the partition function of a box of fixed physical size, with the large-field regions localised and exponentiated, and give no lower bound on a connected correlation. What Corollary 325 adds to those remarks is the form of the missing bound: a lower bound on the low-energy weight of the two-source generating function at every intermediate scale, with the large-field contribution dominated by the small-field connected part of order $a_k^{-8}E^8$ at that scale, and the same at the cutoff scale for (a). Positivity, analyticity and the strong-coupling expansion give the lower bound at every scale from the values at two consecutive separations, but the values at two separations, at any fixed scale, do not determine the low-energy weight at the others.

Remark 327 (status). Proved here, for every $\beta > 0$, every even torus $L \ge 6$ and every compact $G \subset \mathrm{U}(N)$ with ρ irreducible: the axial connected correlation of Φ is nonnegative, log-convex and monotone, and either strictly positive at all axial separations or zero at all of them, the latter only on a closed discrete set $\mathscr{D}_L$ of couplings that misses $(0, \beta_1(N)]$; for a single spatial plaquette it is strictly positive at every β; the exponential lower bound (45) holds with the exact rate m_Φ in torus limits; its constants are $c(\beta) = 108((1+\delta)/2)^9\beta^8$ and $m(\beta) = 4\log(2/((1+\delta)\beta))$ to leading order at strong coupling; reflection positivity propagates positivity but does not create it. Not proved: that $\mathscr{D}_L$ is empty for every L; positivity at non-axial separations; any bound uniform in the lattice spacing along a continuum sequence; either half of (AF). Nothing here bears on the existence of a continuum limit or on a mass gap; Corollary 325 states the two inputs that an argument for the lower half of (AF) along an axis would have to supply.

P. Compactness of the smeared plaquette field from the free energy

This subsection proves an unconditional compactness statement for the renormalised, smeared plaquette field along an arbitrary sequence of couplings and spacings, and identifies exactly which property of the partition function controls it. The tool is the chessboard estimate of the Wilson measure on a torus, which reduces every exponential moment of a smeared plaquette field to the free energy of the same torus at a shifted coupling. Nothing below assumes the averaged condition or any decay, and everything is proved for every $\beta_k \ge 0$, except where the short-distance law (AF$^\square$) of Sec. X P 3 or the stability hypothesis (B) of Sec. X P 4 is stated as a hypothesis. Multiple reflections were used for the same purpose by O'Carroll and Faria da Veiga [87], who bound the generating functional of a fixed number of unsmeared gauge-invariant plaquette fields uniformly in the spacing and the volume; the bound below is for fields smeared over $\asymp a^{-4}$ plaquettes, and it is an identity for the constant smearing, which is what makes it a criterion rather than an estimate. The relation to the inputs of Sec. X N is in Sec. X P 3 and the relation to the renormalisation group bounds of Bałaban in Sec. X P 4.

Setting. Let $G \subset \mathrm{U}(N)$ be a closed subgroup, $G \ne \{1\}$, with the Wilson weight $\exp(N\beta\sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p)$ of Sec. I, $\beta \ge 0$. Fix $L > 0$ and let $\mathbb{T}_L = (\mathbb{R}/L\mathbb{Z})^4$. For an even integer $M \ge 4$ let $\Lambda_M = (\mathbb{Z}/M\mathbb{Z})^4$, $a = L/M$, $\mu_{M,\beta}$ the Wilson measure of Λ_M at β, $S = \sum_p \mathrm{Re}\,\mathrm{Tr}\,Q_p$ the sum over the $6M^4$ plaquettes, and

$$Z_M(\beta) = \int e^{N\beta S}\, dQ, \qquad p_M(\beta) = M^{-4}\log Z_M(\beta),$$

with normalised Haar measure dQ on $G^{E(\Lambda_M)}$; $Z_M(\beta)$ and $p_M(\beta)$ are defined by these formulas for every real β, and $\mu_{M,\beta}$ is the probability measure with density $e^{N\beta S}/Z_M(\beta)$. The function p_M is real analytic and convex on $\mathbb{R}$, with

$$p_M'(\beta) = \frac{N}{M^4}\,\mu_{M,\beta}(S) = 6N\,\mu_{M,\beta}(\mathrm{Re}\,\mathrm{Tr}\,Q_p),$$
$$p_M''(\beta) = \frac{N^2}{M^4}\,\mathrm{Var}_{\mu_{M,\beta}}(S) > 0,$$

(the second equality in the first line by the invariance of $\mu_{M,\beta}$ under translations and the hypercubic group, which make the six orientations equivalent), and $|p_M'| \le 6N^2$ because $|\mathrm{Re}\,\mathrm{Tr}\,Q_p| \le N$; the variance is positive because S is not constant on $G^{E(\Lambda_M)}$ ($\mathrm{Re}\,\mathrm{Tr}\,g < N$ for $g \in G \setminus \{1\}$, so changing one link of the identity configuration to g lowers S, and S is continuous while Haar measure charges every open set). Its second-order remainder

$$R_M(\beta;h) = p_M(\beta+h) - p_M(\beta) - h\,p_M'(\beta) = \int_0^h \big(p_M'(\beta+s) - p_M'(\beta)\big)\,ds \ \ge 0$$

is convex in h, vanishes at $h = 0$, is strictly increasing on $h \ge 0$ and strictly decreasing on $h \le 0$, and satisfies $R_M(\beta;sh) \le sR_M(\beta;h)$ for $s \in [0,1]$.

For a site $x \in \Lambda_M$ let $C_x = x + [0,1]^4$ be the closed unit cube with lower corner x, $m_x = x + \frac{1}{2}\mathbf{1}$ its centre, and

$$\psi_p = N - \mathrm{Re}\,\mathrm{Tr}\,Q_p \in [0, 2N], \qquad \Phi_x^\square = \sum_{p \subset C_x} \psi_p,$$

the sum over the 24 two-dimensional faces of C_x; $\Phi_x^\square$ is a gauge-invariant function of the links of C_x and is invariant under the hypercubic group of C_x. Every plaquette lies in exactly four unit cubes (two choices of the lower corner in each of its two transverse directions), so

$$\sum_{x \in \Lambda_M} \Phi_x^\square = 4\sum_p \psi_p = 24NM^4 - 4S. \tag{rU7.1}$$

For a bounded function f on $\mathbb{T}_L$, $Z>0$ and the measure $\mu=\mu_{M,\beta}$ with $c=\mu(\Phi_0^{\square})=24\mu(\psi_p)$, put

$$\mathscr{A}(f)=\mathscr{A}_{M,\beta,Z}(f)=Z\sum_{x\in\Lambda_M}a^4 f(am_x)\,(\Phi_x^{\square}-c),$$
$$\|f\|_{1,a}=\sum_x a^4|f(am_x)|,$$

so that $\|f\|_{1,a}\le L^4\|f\|_\infty$, and $\|f\|_{1,a}\to\|f\|_{L^1}$ as $a\to 0$ for Riemann integrable f. The 24-plaquette star field Φ_x (the sum of ψ_p over the plaquettes with a corner at x, the example of a local field in Sec. X L) and $\Phi^{\square}$ are two smearings of the same plaquette field, $\sum_x f(ax)\Phi_x=\sum_p\psi_p\sum_{\text{corners } y \text{ of } p}f(ay)$ and $\sum_x f(am_x)\Phi_x^{\square}=\sum_p\psi_p\sum_{\text{cubes } C_x\supset p}f(am_x)$, with weights that differ by at most $8\,\mathrm{w}_f(a)$, w_f the modulus of continuity of f (corners and centres of the four cubes are at distance $a/\sqrt{2}$ from the centre of p); the results below are stated for $\Phi^{\square}$ because its chessboard reflections are isotropic (Lemma 329).

Lemma 328 (the constant source is a shift of the coupling)**.** *For every $\lambda\in\mathbb{R}$,*

$$\mu_{M,\beta}\Big(\exp\Big(\lambda\sum_{x\in\Lambda_M}(\Phi_x^{\square}-c)\Big)\Big)=\exp\Big(M^4R_M\big(\beta;-\tfrac{4\lambda}{N}\big)\Big).$$

Proof. By (rU7.1) and $M^4c=\mu(\sum_x\Phi_x^{\square})=24NM^4-4\mu(S)$, $\sum_x(\Phi_x^{\square}-c)=-4(S-\mu(S))$. Since $e^{N\beta S}e^{-4\lambda S}=e^{N(\beta-4\lambda/N)S}$,

$$\begin{aligned}\mu\big(e^{-4\lambda(S-\mu(S))}\big)&=e^{4\lambda\mu(S)}\,\frac{Z_M(\beta-4\lambda/N)}{Z_M(\beta)}\\&=\exp\Big(M^4\Big[\tfrac{4\lambda}{N}\,p_M'(\beta)\\&\qquad+p_M\big(\beta-\tfrac{4\lambda}{N}\big)-p_M(\beta)\Big]\Big),\end{aligned}$$

using $\mu(S)=M^4p_M'(\beta)/N$; the bracket is $R_M(\beta;-4\lambda/N)$. □

Lemma 329 (chessboard bound for the plaquette field)**.** *Let $M\ge 4$ be even, $\beta\ge 0$, and $\lambda:\Lambda_M\to\mathbb{R}$. Then*

$$\mu_{M,\beta}\Big(\exp\Big(\sum_x\lambda_x(\Phi_x^{\square}-c)\Big)\Big)\le\exp\Big(\sum_{x\in\Lambda_M}R_M\big(\beta;-\tfrac{4\lambda_x}{N}\big)\Big).$$

Proof. Reflection positivity. Let ϑ be the reflection of Λ_M in the hyperplane $\{x_\mu=c_0\}$, $c_0\in\mathbb{Z}$, which also fixes the antipodal hyperplane $\{x_\mu=c_0+M/2\}$; it maps links to links, preserving orientation for links in the two hyperplanes and otherwise mapping a link to a link, and plaquettes to plaquettes. Let $\mathfrak{A}_+$ be the algebra of continuous functions of the links with both endpoints in the closed half $\{c_0\le x_\mu\le c_0+M/2\}$, $\mathfrak{A}_-=\mathfrak{A}_+\circ\vartheta$, and $(\vartheta F)(Q)=F(Q\circ\vartheta)$. Every plaquette lies in one closed half or in both, the latter exactly when it lies in one of the two hyperplanes, so $S=S_++\vartheta S_+$ with $S_+\in\mathfrak{A}_+$ the sum of $\mathrm{Re}\,\mathrm{Tr}\,Q_p$ over the plaquettes in the positive closed half, those in the hyperplanes counted with weight $\frac12$. Haar measure is invariant under $Q_e\mapsto Q_{\vartheta e}$ and under inversion, so, integrating first over the links not in the hyperplanes,

$$\int\overline{\vartheta F}\,F\,e^{N\beta S}\,dQ=\int dQ_0\Big|\int F\,e^{N\beta S_+}\,dQ_+\Big|^2\ge 0\qquad(F\in\mathfrak{A}_+),$$

where dQ_0 is Haar measure on the links in the two hyperplanes and dQ_+ on the other links of the positive closed half, which is the reflection positivity of $\mu_{M,\beta}$ in every hyperplane of sites, proved by Osterwalder and Seiler [90].

Chessboard estimate. The torus is tiled by the closed unit cubes C_x, neighbouring cubes share a face, and the reflections in the hyperplanes containing the faces are those of the preceding paragraph; a function of the links of C_x lies in $\mathfrak{A}_+$ for every such hyperplane with C_x on its positive side. For such block functions F_x the chessboard estimate of Fröhlich, Israel, Lieb and Simon [49], in the form of [26, Theorem 5.8] with blocks of side $B=1$ and $M/B=M$ even (reflections through planes of sites), gives

$$\mu\Big(\prod_x F_x\Big)\le\prod_x\mu\Big(\prod_y\vartheta_{y\leftarrow x}F_x\Big)^{1/M^4},$$

where $\vartheta_{y\leftarrow x}F_x$ is F_x carried into the cube C_y by a chain of reflections in faces (independent of the chain, since reflections in different directions commute and $\vartheta^2=1$). In [26] the variables sit on sites and a B-block function depends on the sites of $\{0,\dots,B\}^d$; here they sit on links and a block function depends on the links with both endpoints in a closed unit cube, and the proof of [26, Theorem 5.8] carries over word for word. It uses only (a) the Cauchy–Schwarz inequality for the positive semidefinite symmetric form $(F,F')\mapsto\mu(F\,\vartheta F')$ on real elements of $\mathfrak{A}_+$, which is the case F real of the display above together with the invariance of μ under ϑ; (b) the algebraic fact that ϑ maps $\mathfrak{A}_+$ onto $\mathfrak{A}_-$ as an algebra homomorphism and that products of block functions in one closed half lie in $\mathfrak{A}_+$, the links in the hyperplane belonging to both halves; and (c) the cyclic invariance of the multilinear functional in its proof, which follows from the invariance of μ under the reflections of the torus. All three hold for $\mu_{M,\beta}$, which is invariant under every reflection and translation of Λ_M because Haar measure is invariant under permutations and inversions of the link variables and S is invariant under these maps. Take $F_x=\exp(\lambda_x\Phi_x^{\square})>0$. A reflection in a face hyperplane maps the 24 faces of C_x onto the 24 faces of the image cube and preserves $\mathrm{Re}\,\mathrm{Tr}\,Q_p$, since $\mathrm{Re}\,\mathrm{Tr}\,Q^{-1}=\mathrm{Re}\,\mathrm{Tr}\,Q$ for unitary Q; so $\vartheta_{y\leftarrow x}F_x=\exp(\lambda_x\Phi_y^{\square})$ and $\prod_y\vartheta_{y\leftarrow x}F_x=\exp(\lambda_x\sum_y\Phi_y^{\square})$. Multiplying both sides of the chessboard estimate by the constant $\exp(-\sum_x\lambda_x c)=\prod_x\exp(-\lambda_xM^4c)^{1/M^4}$ and applying Lemma 328 to each factor gives the claim. □

For $Z>0$ and $T\ge 0$ put

$$\Xi_{M,\beta,Z}(T)=a^{-4}\max\Big\{R_M\big(\beta;\tfrac{4TZa^4}{N}\big),\,R_M\big(\beta;-\tfrac{4TZa^4}{N}\big)\Big\},$$

the larger second-order remainder of the free energy, per unit physical volume, at a coupling shift of size $4TZa^4/N$.

Theorem 330 (exponential moments and the free energy)**.** *Let $M\ge 4$ be even, $\beta\ge 0$, $Z>0$, and $\mu=\mu_{M,\beta}$.*

(i) *For every bounded $f\ne 0$ on $\mathbb{T}_L$ and $t\in\mathbb{R}$,*

$$\log\mu\big(e^{t\mathscr{A}(f)}\big)\le\frac{\|f\|_{1,a}}{\|f\|_\infty}\,\Xi_{M,\beta,Z}\big(|t|\,\|f\|_\infty\big).$$

(ii) *For $f\equiv 1$ there is equality up to the choice of sign:*

$$\log\mu\big(e^{\pm t\mathscr{A}(1)}\big)=L^4a^{-4}R_M\big(\beta;\mp\tfrac{4tZa^4}{N}\big),$$

so $\max_\pm\log\mu(e^{\pm t\mathscr{A}(1)})=L^4\,\Xi_{M,\beta,Z}(|t|)$.

(iii) *For every bounded f,*

$$\begin{aligned}\mu\big(\mathscr{A}(f)^2\big)&\le\frac{16Z^2a^4p_M''(\beta)}{N^2}\,\|f\|_{1,a}\|f\|_\infty\\&=\frac{\|f\|_{1,a}\|f\|_\infty}{L^4}\,\mu\big(\mathscr{A}(1)^2\big).\end{aligned}$$

(iv) *Consequently, for a sequence (M_k,β_k,Z_k) with M_k even, $a_k=L/M_k$ and $\mu_k=\mu_{M_k,\beta_k}$, $\mathscr{A}_k=\mathscr{A}_{M_k,\beta_k,Z_k}$, the following are equivalent:*

(a) $\sup_k\mu_k(e^{t\mathscr{A}_k(1)})<\infty$ *for every $t\in\mathbb{R}$;*

(b) $\sup_k\Xi_{M_k,\beta_k,Z_k}(T)<\infty$ *for every $T>0$;*

(c) $\sup_k\mu_k(e^{t\mathscr{A}_k(f)})<\infty$ *for every $t\in\mathbb{R}$ and every bounded f on $\mathbb{T}_L$.*

When they hold, for every $n\ge1$, $T>0$ and bounded $f\neq0$,

$$\mu_k\big(|\mathscr{A}_k(f)|^n\big)\le 2\,n!\,T^{-n}\exp\Big(\frac{\|f\|_{1,a_k}}{\|f\|_\infty}\sup_k\Xi_{M_k,\beta_k,Z_k}(T\|f\|_\infty)\Big).$$

Proof. (i) Apply Lemma 329 with $\lambda_x=tZa^4f(am_x)$ and put $h_x=-4\lambda_x/N$, $H=4|t|Za^4\|f\|_\infty/N$, so $|h_x|\le H$. By convexity and $R_M(\beta;0)=0$, $R_M(\beta;h_x)\le(|h_x|/H)R_M(\beta;\mathrm{sgn}(h_x)H)\le(|f(am_x)|/\|f\|_\infty)\max_\pm R_M(\beta;\pm H)$. Summing over x,

$$\begin{aligned}\sum_x R_M(\beta;h_x)&\le\frac{\sum_x|f(am_x)|}{\|f\|_\infty}\max_\pm R_M(\beta;\pm H)\\&=\frac{\|f\|_{1,a}}{a^4\|f\|_\infty}\max_\pm R_M(\beta;\pm H),\end{aligned}$$

which is the right side of (i), since $a^{-4}\max_\pm R_M(\beta;\pm H)=\Xi_{M,\beta,Z}(|t|\|f\|_\infty)$.

(ii) is Lemma 328 with $\lambda=\pm tZa^4$, as $M^4=L^4a^{-4}$.

(iii) Since $\mathscr{A}(f)$ is a bounded centred random variable, the left side of (i) is $\frac12t^2\mu(\mathscr{A}(f)^2)+O(|t|^3)$; since p_M is smooth, $R_M(\beta;\pm H)=\frac12H^2p_M''(\beta)+O(H^3)$, and $H=4|t|Za^4\|f\|_\infty/N$, so the right side is $\frac12(\|f\|_{1,a}/\|f\|_\infty)a^{-4}H^2p_M''(\beta)+O(|t|^3)$. Dividing (i) by $t^2/2$ and letting $t\to0$ gives $\mu(\mathscr{A}(f)^2)\le(\|f\|_{1,a}/\|f\|_\infty)\,a^{-4}(4Za^4\|f\|_\infty/N)^2p_M''(\beta)$, the first bound. By (rU7.1), $\mathscr{A}(1)=-4Za^4(S-\mu(S))$, so $\mu(\mathscr{A}(1)^2)=16Z^2a^8\,\mathrm{Var}(S)=16Z^2a^4L^4p_M''(\beta)/N^2$, which gives the second form.

(iv) (c)⇒(a) is trivial; (a)⇒(b) is (ii), since (a) for $\pm t$ bounds $L^4\Xi_k(|t|)$; (b)⇒(c) is (i) with $\|f\|_{1,a_k}/\|f\|_\infty\le L^4$. For the moment bound, $|x|^n/n!\le e^{T|x|}T^{-n}\le T^{-n}(e^{Tx}+e^{-Tx})$, and (i) bounds both exponential moments. □

Remark 331. Part (iv) is a sharp criterion: the whole family of smeared plaquette fields on the torus of side L, renormalised by Z_k, has uniformly bounded exponential moments if and only if the constant smearing does, if and only if the second-order Taylor remainder of the free energy per unit physical volume at coupling shifts of order $Z_ka_k^4$ stays bounded. Since $M^4[R_M(\beta;h)+R_M(\beta;-h)]=\log Z_M(\beta+h)-2\log Z_M(\beta)+\log Z_M(\beta-h)$ and both remainders are nonnegative, (b) is equivalent to

$$\sup_k L^{-4}\Big|\Delta^2_{h_k(T)}\log Z_{M_k}(\beta_k)\Big|<\infty\quad(T>0),$$

with $h_k(T)=4TZ_ka_k^4/N$, a bound on the second difference of the logarithm of the partition function at three couplings. This is the exact form in which a statement about partition functions controls correlations of the plaquette field (Sec. X P 4). Part (iii) shows that the plaquette susceptibility $p_M''(\beta)/N^2=M^{-4}\,\mathrm{Var}(S)=\sum_p\sum_{p'}\mathrm{Cov}(\mathrm{Re\,Tr}\,Q_p,\mathrm{Re\,Tr}\,Q_{p'})$, the inner sum over all plaquettes p' of the torus and the outer one over the six plaquettes p with lower corner 0 (one for each orientation), is, up to the factor $16Z^2a^4L^4$, the variance of the constant smearing and dominates every smeared variance: no smearing, however singular or however close to two coinciding points, has a larger variance per unit of $\|f\|_{1,a}\|f\|_\infty$ than the constant one. An upper bound on such a susceptibility by the variance of the sum of $\mathrm{Re\,Tr}\,Q_p$ over the six plaquettes with lower corner 0 is a consequence of the averaged condition by Proposition 136.

The free-energy normalisation. The function

$$H\mapsto a^{-4}\max_\pm R_M(\beta;\pm H)$$

is continuous, vanishes at 0 and is strictly increasing on $[0,\infty)$. Let

$$\begin{aligned}h_{M,\beta}&=\sup\big\{H>0:\ a^{-4}\max_\pm R_M(\beta;\pm H)\le1\big\}\in(0,\infty],\\Z^{\mathrm{FE}}_{M,\beta}&=\frac{N}{4a^4}\min(h_{M,\beta},1).\end{aligned}$$

On the torus $h_{M,\beta}<\infty$: as $H\to\infty$, $p_M'(\beta+H)\to6N^2$, because $\mu_{M,\beta+H}$ concentrates on the maximum $S=6NM^4$ of S, attained on the flat configurations, while $p_M'(\beta)<6N^2$; so $R_M(\beta;H)\ge\int_{H/2}^H(p_M'(\beta+s)-p_M'(\beta))\,ds\to\infty$. The minimum with 1 is only needed for the infinite-lattice version of Sec. X P 1. With this choice,

$$\begin{aligned}\Xi_{M,\beta,Z^{\mathrm{FE}}}(T)&=a^{-4}\max_\pm R_M\big(\beta;\pm T\min(h_{M,\beta},1)\big)\\&\le T\qquad(0\le T\le1)\end{aligned}\tag{rU7.2}$$

by $R_M(\beta;sh)\le sR_M(\beta;h)$, $s\in[0,1]$, and the definition of $h_{M,\beta}$.

Corollary 332 (unconditional bounds on the torus)**.** *Let $M_k\ge4$ be even, $a_k=L/M_k\to0$, $\beta_k\ge0$ arbitrary, $Z_k=Z^{\mathrm{FE}}_{M_k,\beta_k}$ and $\mathscr{A}_k$ the corresponding smeared field. Then for every bounded $f\neq0$ on $\mathbb{T}_L$, every k and every t with $|t|\,\|f\|_\infty\le1$,*

$$\log\mu_k\big(e^{t\mathscr{A}_k(f)}\big)\le|t|\,\|f\|_{1,a_k}\le|t|\,\|f\|_\infty L^4,$$

and for every $n \geq 1$

$$\begin{aligned}
\mu_k\big(\mathscr{A}_k(f)^2\big) &\leq 2e^{L^4}\|f\|_\infty\|f\|_{1,a_k},\\
\mu_k\big(|\mathscr{A}_k(f)|^n\big) &\leq 2\,n!\,e^{n}\big(\|f\|_\infty+\tfrac{1}{n}\|f\|_{1,a_k}\big)^n,\\
\mu_k\big(|\mathscr{A}_k(f)|^n\big) &\leq 2\,n!\,e^{L^4}\|f\|_\infty^n,\\
\Big|\mu_k\Big(\prod_{j=1}^n \mathscr{A}_k(f_j)\Big)\Big| &\leq 2\,n!\,e^{L^4}\prod_{j=1}^n\|f_j\|_\infty.
\end{aligned}$$

Proof. Theorem 330(i) and (rU7.2) give the first bound. For $0<t\leq 1/\|f\|_\infty$ and $y\in\mathbb{R}$, $|y|^n\leq n!\,t^{-n}(e^{ty}+e^{-ty})$ and $t^2y^2\leq e^{ty}+e^{-ty}-2$; with the first bound, $\mu_k(|\mathscr{A}_k(f)|^n)\leq 2n!\,t^{-n}e^{t\|f\|_{1,a_k}}$ and $\mu_k(\mathscr{A}_k(f)^2)\leq 2t^{-2}(e^{t\|f\|_{1,a_k}}-1)$. The choice $t=1/\|f\|_\infty$ with $e^u-1\leq ue^u$ and $\|f\|_{1,a_k}\leq L^4\|f\|_\infty$ gives the second-moment bound and the third bound; the choice $t=(\|f\|_\infty+\|f\|_{1,a_k}/n)^{-1}$, for which $t\|f\|_{1,a_k}\leq n$, gives the second. The product bound follows by Hölder's inequality, $|\mu(\prod_j\mathscr{A}(f_j))|\leq\prod_j\mu(|\mathscr{A}(f_j)|^n)^{1/n}$. □

Corollary 333 (the variance normalisation)**.** *With $Z^{\mathrm{V}}_{M,\beta}=N/(4a^2p_M''(\beta)^{1/2})$ and $\mathscr{A}=\mathscr{A}_{M,\beta,Z^{\mathrm{V}}_{M,\beta}}$,*

$$\mu_{M,\beta}\big(\mathscr{A}(1)^2\big)=L^4,\qquad \mu_{M,\beta}\big(\mathscr{A}(f)^2\big)\leq\|f\|_{1,a}\|f\|_\infty$$

for every even $M\geq 4$, every $\beta\geq 0$ and every bounded f on $\mathbb{T}_L$.

Proof. Theorem 330(iii) with $16Z^2a^4p_M''(\beta)/N^2=1$. □

The variance normalisation gives second moments only; by Theorem 330(iv) it gives exponential moments exactly when the remainders $a^{-4}R_M(\beta;\pm Ta^2p_M''(\beta)^{-1/2})$ stay bounded, which holds, for instance, if p_M'' varies by a bounded factor on the coupling window $|\beta'-\beta|\leq Ta^2p_M''(\beta)^{-1/2}$, since then $a^{-4}R_M\leq\frac{1}{2}T^2\sup p_M''/p_M''(\beta)$.

1. *The infinite lattice*

The torus bounds pass to the torus limits of Lemma 294 (there the torus has side L; here the side is M and L is the physical length). The free energy is replaced by a limit of free energies and its derivative at β by a subgradient.

Lemma 334 (free energy of a torus limit)**.** *Let $\beta\geq 0$ and let (M_j) be a sequence of even integers along which the torus measures $\mu_{M_j,\beta}$ converge to a state $\omega=\omega_\beta$ as in Lemma 294*(i)*. Then (M_j) has a subsequence, again denoted (M_j), along which $p_{M_j}(\beta')\to p^\infty(\beta')$ locally uniformly in $\beta'\in\mathbb{R}$ and $p'_{M_j}(\beta)\to D$, where p^∞ is convex with $|p^\infty(\beta')-p^\infty(\beta'')|\leq 6N^2|\beta'-\beta''|$, $D=6N\,\omega(\mathrm{Re\,Tr}\,Q_p)$, and D is a subgradient of p^∞ at β. Consequently*

$$R^\infty(h)=p^\infty(\beta+h)-p^\infty(\beta)-hD\ \geq 0\qquad(h\in\mathbb{R})$$

is convex, $R^\infty(0)=0$, $R^\infty(sh)\leq sR^\infty(h)$ for $s\in[0,1]$, and R^∞ is nondecreasing on $[0,\infty)$ and nonincreasing on $(-\infty,0]$. Moreover $c_{M_j}=\mu_{M_j,\beta}(\Phi_0^\square)\to c^\infty=\omega(\Phi_0^\square)=24(N-D/(6N))$, and for every finitely supported $\lambda:\mathbb{Z}^4\to\mathbb{R}$,

$$\omega\Big(\exp\sum_{x\in\mathbb{Z}^4}\lambda_x(\Phi_x^\square-c^\infty)\Big)\leq\exp\Big(\sum_{x\in\mathbb{Z}^4}R^\infty(-4\lambda_x/N)\Big).$$

Proof. Each p_M is convex with $p_M(0)=0$ and $|p'_M|\leq 6N^2$, so the family (p_M) is uniformly bounded and equi-Lipschitz on compact sets; by Arzelà–Ascoli and a diagonal argument a subsequence converges locally uniformly to a function p^∞, which is convex and $6N^2$-Lipschitz as a locally uniform limit of such functions. Passing to a further subsequence, $p'_{M_j}(\beta)=6N\mu_{M_j,\beta}(\mathrm{Re\,Tr}\,Q_p)$ converges, and its limit is $6N\omega(\mathrm{Re\,Tr}\,Q_p)=D$ because $\mathrm{Re\,Tr}\,Q_p$ is a continuous function of the four links of p; here $p'_M(\beta)=6N\mu_{M,\beta}(\mathrm{Re\,Tr}\,Q_p)$ holds for each of the six plaquettes p with lower corner 0, and so does the limit, by the B_4 invariance of ω (Lemma 294(ii)). For each j and h, $p_{M_j}(\beta+h)\geq p_{M_j}(\beta)+hp'_{M_j}(\beta)$ by convexity; letting $j\to\infty$ gives $p^\infty(\beta+h)\geq p^\infty(\beta)+hD$, that is, D is a subgradient and $R^\infty\geq 0$. The remaining properties of R^∞ follow from convexity and $R^\infty(0)=0$ exactly as for R_M. The convergence $c_{M_j}\to c^\infty$ holds because $\Phi_0^\square$ is a continuous function of finitely many links, and $c_{M_j}=24(N-p'_{M_j}(\beta)/(6N))$.

For the last bound fix λ with support in a box of side $2m$ centred at the origin. For $M_j>2m+4$ the box embeds in Λ_{M_j} and Lemma 329 gives $\mu_{M_j,\beta}(\exp\sum_x\lambda_x(\Phi_x^\square-c_{M_j}))\leq\exp(\sum_x R_{M_j}(\beta;-4\lambda_x/N))$. The function $\exp\sum_x\lambda_x(\Phi_x^\square-c)$ is continuous in finitely many links and jointly continuous in c, and $\exp\sum_x\lambda_x(\Phi_x^\square-c_{M_j})-\exp\sum_x\lambda_x(\Phi_x^\square-c^\infty)$ tends to 0 uniformly on the configuration space, so the left side converges to the left side of the claim. On the right, $R_{M_j}(\beta;h)\to R^\infty(h)$ for each h by the locally uniform convergence of p_{M_j} and the convergence of $p'_{M_j}(\beta)$. □

The sequence of states. Let $\beta_k\geq 0$ be arbitrary and $a_k\to 0$, $a_k\leq 1$. For each k fix a torus limit ω_k of μ_{M,β_k} along a sequence of even M for which also the conclusions of Lemma 334 hold, with p_k^∞, D_k, R_k^∞ and $c_k=\omega_k(\Phi_0^\square)$. For a bounded function f on $\mathbb{R}^4$ with compact support, $Z>0$ and $m_x=x+\frac{1}{2}\mathbf{1}$, put

$$A_{k,Z}(f)=Z\sum_{x\in\mathbb{Z}^4}a_k^4f(a_km_x)\big(\Phi_x^\square-c_k\big),$$

$$\|f\|_{1,k}=\sum_{x\in\mathbb{Z}^4}a_k^4|f(a_km_x)|,\qquad \mathfrak{m}_k(f)=\frac{\|f\|_{1,k}}{\|f\|_\infty}\quad(f\neq 0),$$

a finite sum. If f vanishes outside a closed cube Q of side ℓ, then at most $(\ell/a_k+1)^4$ points of the lattice $a_k(\mathbb{Z}^4+\frac{1}{2}\mathbf{1})$ lie in Q, so

$$\mathfrak{m}_k(f)\leq(\ell+a_k)^4\leq(\ell+1)^4.\qquad\text{(rU7.3)}$$

For f Riemann integrable, $\|f\|_{1,k}\to\|f\|_{L^1}$ as $k\to\infty$, since $\|f\|_{1,k}$ is a Riemann sum of $|f|$ over the cubes of side a_k centred at the points a_km_x.

Theorem 335 (chessboard bound and free-energy normalisation on $\mathbb{Z}^4$)**.** *In the setting above let*

$$\Xi_k^\infty(T)=a_k^{-4}\max_{\pm}R_k^\infty\big(\pm 4TZ_ka_k^4/N\big),$$

$$h_k^\infty=\sup\big\{H>0:\ a_k^{-4}\max_{\pm}R_k^\infty(\pm H)\leq 1\big\}\in(0,\infty],$$

and $Z_k^{\mathrm{FE}} = \frac{N}{4a_k^4}\min(h_k^\infty, 1)$.

(i) *For every* $Z_k > 0$*, every bounded* $f \neq 0$ *with compact support and every* $t \in \mathbb{R}$, $\log \omega_k(e^{tA_{k,Z_k}(f)}) \le \mathfrak{m}_k(f)\,\Xi_k^\infty(|t|\,\|f\|_\infty)$.

(ii) *With* $Z_k = Z_k^{\mathrm{FE}}$, $\Xi_k^\infty(T) \le T$ *for* $0 \le T \le 1$*; hence, writing* $A_k = A_{k,Z_k^{\mathrm{FE}}}$*, for* $|t|\,\|f\|_\infty \le 1$

$$\log \omega_k\big(e^{tA_k(f)}\big) \le |t|\,\|f\|_{1,k},$$

and for every $n \ge 1$ *and every* $f, f_1, \dots, f_n$ *vanishing outside a cube of side* ℓ*,*

$$\omega_k\big(|A_k(f)|^n\big) \le 2n!\, e^{(\ell+1)^4}\|f\|_\infty^n,$$
$$\omega_k\big(A_k(f)^2\big) \le 2e^{(\ell+1)^4}\,\|f\|_\infty\|f\|_{1,k},$$

$$\Big|\omega_k\Big(\prod_{j=1}^n A_k(f_j)\Big)\Big| \le 2n!\, e^{(\ell+1)^4}\prod_{j=1}^n \|f_j\|_\infty.$$

These bounds hold for every k*, with no condition on* β_k*.*

Proof. (i) is the last bound of Lemma 334 with $\lambda_x = tZ_k a_k^4 f(a_k m_x)$, followed by the convexity argument in the proof of Theorem 330(i), which used only $R(0) = 0$ and convexity.

(ii) If $h_k^\infty < \infty$ then $a_k^{-4}\max_\pm R_k^\infty(\pm h_k^\infty) \le 1$ by continuity of R_k^∞, and $4TZ_k^{\mathrm{FE}}a_k^4/N = T\min(h_k^\infty, 1) \le Th_k^\infty$, so $\Xi_k^\infty(T) \le T\,a_k^{-4}\max_\pm R_k^\infty(\pm h_k^\infty) \le T$ for $T \le 1$ by $R^\infty(sh) \le sR^\infty(h)$. If $h_k^\infty = \infty$ then $a_k^{-4}\max_\pm R_k^\infty(\pm H) \le 1$ for all H and the same argument with $H = 1$ applies. The exponential bound follows from (i). For the moments, $|y|^n/n! \le T^{-n}(e^{Ty} + e^{-Ty})$ with $T = 1/\|f\|_\infty$ and $\mathfrak{m}_k(f) \le (\ell+1)^4$ give the first bound. For the second, $t^2y^2 \le e^{ty} + e^{-ty} - 2$ with $t = 1/\|f\|_\infty$ gives $\omega_k(A_k(f)^2) \le \|f\|_\infty^2 \cdot 2(e^{\|f\|_{1,k}/\|f\|_\infty} - 1) \le 2\|f\|_\infty\|f\|_{1,k}e^{\mathfrak{m}_k(f)}$, using $e^u - 1 \le ue^u$. The product bound is Hölder's inequality. □

Remark 336. (a) Theorem 335(ii) is a compactness statement for the family of all smeared, centred plaquette fields along *any* sequence of couplings: the normalisation Z_k^{FE} is read off from the infinite-volume free energy alone, and with it all moments are bounded uniformly in k, including at coinciding points and in the ultraviolet limit, by constants depending only on the size of the support and on $\|f\|_\infty$. It assumes nothing about decay, about the tuning or about the averaged condition. Its price is paid in Sec. X P 3: when the short-distance law (AF$^\square$) holds for constants Z_k', $Z_k^{\mathrm{FE}}/Z_k' \to 0$, and the limits obtained with Z_k^{FE} have vanishing separated two-point functions (Theorem 342).

(b) The second-moment bound is of the form $\|f\|_\infty\|f\|_{1,k}$, not $\|f\|_\infty^2$; this two-norm form is what makes the limit field continuous under translations and reflections of step functions (Theorem 339).

(c) By Theorem 330(ii), for the torus states the choice Z^{FE} is sharp up to the factor $\min(h, 1)$: with any $Z_k \ge TZ_k^{\mathrm{FE}}$, $T > 1$, and $h_k \le 1$, the exponential moment of $\pm\mathscr{A}_k(1)$ at $t = 1$ is at least e^{L^4} in one of the signs, and with $Z_k/Z_k^{\mathrm{FE}} \to \infty$ it diverges.

Test functions. Let $\mathscr{R}$ be the space of bounded, compactly supported, Riemann integrable functions $f : \mathbb{R}^4 \to \mathbb{R}$ (bounded, compactly supported, continuous outside a Lebesgue null set); it is an algebra containing $C_c(\mathbb{R}^4)$ and the indicators of boxes, and it is stable under translations $\tau_v f = f(\cdot - v)$, under the hypercubic group B_4 acting by $f \mapsto f \circ R^{-1}$, and under the reflections $\theta_{\mu,c}$ in the planes $\{x_\mu = c\}$. Let $\mathscr{G} \subset \mathscr{R}$ be the countable $\mathbb{Q}$-vector space spanned by the indicators of boxes $\prod_\mu [a_\mu, b_\mu)$ with rational corners.

Lemma 337 (approximation)**.** *Let* $f \in \mathscr{R}$ *vanish outside a cube* Q *of side* ℓ *and* $\varepsilon \in (0, 1]$*. There is* $g \in \mathscr{G}$ *vanishing outside a cube of side* $\ell + 1$ *containing* Q *with* $\|g\|_\infty \le \|f\|_\infty + 1$ *and* $\|f - g\|_{L^1} < \varepsilon$*. For such* f, g *and all large* k*,*

$$\omega_k\big(A_k(f-g)^2\big) \le 4\varepsilon\,(2\|f\|_\infty + 1)\,e^{(\ell+2)^4}.$$

Moreover, for $g \in \mathscr{G}$ *and* $v, w \in \mathbb{R}^4$ *with* $|v - w|_\infty \le 2a_k$*,*

$$\omega_k\big(A_k(\tau_v g - \tau_w g)^2\big) \le C_g\, a_k,$$

with C_g *depending only on* g*.*

Proof. By the Darboux criterion there is a partition of a cube $Q' \supset Q$ of side $\ell + 1$ with rational corners into boxes B_i such that $\sum_i(\sup_{B_i} f - \inf_{B_i} f)|B_i| < \varepsilon/2$; put $g = \sum_i q_i \mathbf{1}_{B_i}$ with rational $q_i \in [\inf_{B_i} f, \inf_{B_i} f + \varepsilon/(2|Q'|)]$. Then $\|g\|_\infty \le \|f\|_\infty + 1$ (as $|Q'| \ge 1$) and $\|f - g\|_{L^1} \le \sum_i(\sup_{B_i} f - \inf_{B_i} f)|B_i| + \varepsilon/2 < \varepsilon$. (Such a partition exists: f is Riemann integrable on Q', so the upper and lower Darboux sums of every sufficiently fine grid partition differ by less than $\varepsilon/2$, and the grid points can be taken rational.) Since $f - g \in \mathscr{R}$, $\|f - g\|_{1,k} \to \|f - g\|_{L^1}$, so $\|f - g\|_{1,k} < 2\varepsilon$ for large k, and the second bound of Theorem 335(ii) with $\|f - g\|_\infty \le 2\|f\|_\infty + 1$ and the cube of side $\ell + 1$ gives the claim. For the last bound write $g = \sum_i q_i \mathbf{1}_{B_i}$ with boxes B_i contained in a cube of side ℓ_g, let n_g be the total number of 3-dimensional faces of the B_i, and put $r = |v - w|_\infty \le 2a_k \le 2$. The function $h = \tau_v g - \tau_w g$ is bounded by $2\|g\|_\infty$ and vanishes outside a cube of side $\ell_g + 2$. If $h(y) \neq 0$, then for some i exactly one of $y - v$, $y - w$ lies in B_i; the segment between them has ℓ^∞-length r, so $y - v$ lies within ℓ^∞-distance $2r$ of some face F of B_i, that is, $y \in v + F + [-2r, 2r]^4$. This set is a box with one side at most $4r \le 8a_k$ and three sides at most $\ell_g + 8a_k$, so it contains at most $9(\ell_g/a_k + 9)^3$ points of the lattice $a_k(\mathbb{Z}^4 + \frac{1}{2}\mathbf{1})$. Hence, with $a_k \le 1$, $\|h\|_{1,k} \le 2\|g\|_\infty \cdot 9n_g a_k^4(\ell_g/a_k + 9)^3 \le 18n_g\|g\|_\infty(\ell_g + 9)^3 a_k$, and the second-moment bound of Theorem 335(ii), with a cube of side $\ell_g + 2$, gives the claim with $C_g = 72n_g\|g\|_\infty^2(\ell_g + 9)^3 e^{(\ell_g+3)^4}$. □

2. *Subsequential continuum limits*

We treat the torus of side L and the infinite lattice together. In the *torus setting* $X = \mathbb{T}_L$, $\nu_k = \mu_{M_k,\beta_k}$ with $M_k \ge 4$ even, $a_k = L/M_k \to 0$, $A_k = \mathscr{A}_{M_k,\beta_k,Z^{\mathrm{FE}}_{M_k,\beta_k}}$, $\|f\|_{1,k} = \|f\|_{1,a_k}$, $\mathscr{R}(X)$ is the space of bounded functions on $\mathbb{T}_L$ continuous outside a null set, and for a direction μ and $c \in \mathbb{R}$ the *positive region* is $P_{\mu,c} = \{c < x_\mu < c + \frac{L}{2}\}$ (modulo L). In the *infinite-lattice setting* $X = \mathbb{R}^4$, $\nu_k = \omega_k$, $A_k = A_{k,Z_k^{\mathrm{FE}}}$ and $\|f\|_{1,k}$ are

those of Theorem 335, $\mathscr{R}(X)=\mathscr{R}$, and $P_{\mu,c}=\{x_\mu>c\}$. In both, $\beta_k\ge 0$ is arbitrary, $\theta_{\mu,c}$ is the reflection $x_\mu\mapsto 2c-x_\mu$, $\theta f=f\circ\theta$, $\tau_v f=f(\cdot-v)$, B_4 acts on X fixing 0, and $\Gamma(X)$ is the group generated by the translations of X and B_4. The bounds of Corollary 332, and in the infinite-lattice setting those of Theorem 335(ii) together with the bound $\nu_k(|A_k(f)|^n)\le 2n!\,e^n(\|f\|_\infty+\|f\|_{1,k}/n)^n$, which follows from Theorem 335(ii) exactly as in the proof of Corollary 332, hold in both settings. On $\mathbb{T}_L$ we let $\mathscr{G}$ be the $\mathbb{Q}$-span of the indicators of the images in $\mathbb{T}_L$ of boxes $\prod_\mu[a_\mu,b_\mu)$ with $0\le b_\mu-a_\mu\le L$ and corners in the countable dense set $\mathbb{Q}+L\mathbb{Q}$, which contains 0 and L; then Lemma 337 holds on $\mathbb{T}_L$ with the same proof, the cube Q' being a fundamental domain $[0,L)^4$ when $\ell+1>L$, and with $e^{(\ell+2)^4}$, $e^{(\ell_g+3)^4}$ replaced by e^{L^4} (Corollary 332).

Lemma 338 (exact lattice symmetries)**.** (a) *For $\gamma=\tau_v$ with $v\in a_k\mathbb{Z}^4$, or $\gamma\in B_4$, the law under ν_k of $(A_k(f_j\circ\gamma^{-1}))_{j\le n}$ equals that of $(A_k(f_j))_{j\le n}$, for all $f_j\in\mathscr{R}(X)$.*

(b) *Let $c\in a_k\mathbb{Z}$, let $f_1,\dots,f_n\in\mathscr{R}(X)$ vanish outside a compact subset of $P_{\mu,c}$, and let $F:\mathbb{R}^n\to\mathbb{C}$ be bounded and continuous. Then, with $\theta=\theta_{\mu,c}$,*

$$\nu_k\Big(\overline{F\big(A_k(\theta f_1),\dots,A_k(\theta f_n)\big)}\,F\big(A_k(f_1),\dots,A_k(f_n)\big)\Big)\ge 0.$$

Proof. (a) The translation of the lattice by $v/a_k\in\mathbb{Z}^4$ maps C_x onto C_{x+v/a_k} and m_x to m_x+v/a_k. An element $g\in B_4$ maps $\mathbb{Z}^4$ to itself and $\frac12\mathbf{1}$ to a vector with entries $\pm\frac12$, so it maps m_x to m_y with $y=gx+g(\frac12\mathbf{1})-\frac12\mathbf{1}\in\mathbb{Z}^4$, and the cube C_x onto C_y. In both cases, with the action on link variables of the proof of Lemma 294(ii), $\Phi^\square_x$ is carried to $\Phi^\square$ of the image cube, because $\operatorname{Re}\operatorname{Tr}Q^{-1}=\operatorname{Re}\operatorname{Tr}Q$. Hence $A_k(f\circ\gamma^{-1})$ is the image of $A_k(f)$ under the corresponding transformation of configurations, which preserves ν_k (on the torus directly, on $\mathbb{Z}^4$ by Lemma 294(ii)); the constant c_k does not depend on x.

(b) Let ϑ be the lattice reflection in $\{x_\mu=c/a_k\}$. If $f_j(a_km_x)\ne 0$ then $a_km_x\in P_{\mu,c}$; since $m_{x,\mu}-c/a_k$ is a half-integer, it is at least $\frac12$, so $x_\mu\ge c/a_k$, and on the torus also $x_\mu+1\le c/a_k+M_k/2$. So $A_k(f_j)$ is a function of the links of cubes in the closed half on the positive side of ϑ, and $\vartheta A_k(f_j)=A_k(\theta f_j)$, since ϑ maps C_x onto the cube with centre ϑm_x and $\theta(a_km_x)=a_k\vartheta(m_x)$. On the torus the claim is the reflection positivity proved in Lemma 329 for the continuous function $F(A_k(f_1),\dots)$ of the links of the closed half. On $\mathbb{Z}^4$ the variables $A_k(f_j)$ depend on finitely many links; for each large M in the sequence defining ω_k the same inequality holds for μ_{M,β_k} with the reflection of Λ_M in $\{x_\mu=c/a_k\}$, and it passes to the limit. □

Theorem 339 (subsequential limits of the smeared plaquette field)**.** *In either setting, every sequence of k has a subsequence along which the following holds: there are a probability space $(\Omega,\mathscr{F},\mathrm{P})$ and real random variables $A(f)$, $f\in\mathscr{R}(X)$, such that*

(i) *for all $f_1,\dots,f_n\in\mathscr{R}(X)$, $(A_k(f_j))_j\to(A(f_j))_j$ in law, $\nu_k(\prod_jA_k(f_j))\to\mathrm{E}\prod_jA(f_j)$, and $\nu_k(e^{tA_k(f)})\to\mathrm{E}e^{tA(f)}$ for $|t|\,\|f\|_\infty<1$;*

(ii) *$f\mapsto A(f)$ is linear almost surely, $\mathrm{E}A(f)=0$, and for $|t|\,\|f\|_\infty\le 1$ and $n\ge 1$*

$$\begin{aligned}\log\mathrm{E}e^{tA(f)}&\le|t|\,\|f\|_{L^1},\\ \mathrm{E}|A(f)|^n&\le 2n!\,e^n\big(\|f\|_\infty+\tfrac1n\|f\|_{L^1}\big)^n,\\ \mathrm{E}A(f)^2&\le 2\|f\|_\infty\|f\|_{L^1}e^{\|f\|_{L^1}/\|f\|_\infty},\\ \Big|\mathrm{E}\prod_{j=1}^nA(f_j)\Big|&\le 2n!\,e^n\prod_{j=1}^n\big(\|f_j\|_\infty+\|f_j\|_{L^1}\big);\end{aligned}$$

(iii) *for every $\gamma\in\Gamma(X)$ the laws of $(A(f_j\circ\gamma^{-1}))_j$ and $(A(f_j))_j$ coincide, and $v\mapsto A(\tau_vf)$ is continuous into $L^2(\mathrm{P})$;*

(iv) *for every direction μ, every $c\in\mathbb{R}$, all $f_1,\dots,f_n\in\mathscr{R}(X)$ vanishing outside a compact subset of $P_{\mu,c}$ and every bounded continuous F, with $\theta=\theta_{\mu,c}$,*

$$\mathrm{E}\Big[\overline{F\big(A(\theta f_1),\dots,A(\theta f_n)\big)}\,F\big(A(f_1),\dots,A(f_n)\big)\Big]\ge 0;$$

in particular $\sum_{i,j}\bar\alpha_i\alpha_j\mathrm{E}e^{i(A(f_j)-A(\theta f_i))}\ge 0$ for $\alpha_i\in\mathbb{C}$, and the same holds with F a polynomial.

(v) *In the torus setting, with $\eta_k=\min(h_{M_k,\beta_k},1)$, for $|t|<1$*

$$\begin{aligned}\log\mathrm{E}e^{\pm tA(1)}&=\lim_k L^4a_k^{-4}R_{M_k}(\beta_k;\mp t\eta_k),\\ \mathrm{E}A(1)^2&=L^4\lim_k\eta_k^2\,p''_{M_k}(\beta_k)/a_k^4,\end{aligned}$$

the limits existing along the subsequence; so $A\ne0$ if and only if this last limit is positive.

Proof. Let $f\in\mathscr{R}(X)$. The bounds quoted before Lemma 338 give

$$\sup_k\nu_k(|A_k(f)|^n)<\infty\quad(n\ge1),\qquad\sup_k\nu_k\big(e^{|A_k(f)|/\|f\|_\infty}\big)<\infty$$

(in the infinite-lattice setting with constants depending on a cube containing the support); hence, for each finite family, the products $\prod_jA_k(f_j)$ and the variables $e^{tA_k(f)}$, $|t|\,\|f\|_\infty<1$, are uniformly integrable, by Hölder's inequality and because $e^{|t||y|}$ is a power <1 of $e^{|y|/\|f\|_\infty}$.

Step 1 (a countable family). Let $\mathscr{G}$ be the countable space of rational step functions of Lemma 337. For each finite family in $\mathscr{G}$ the laws of $(A_k(g_j))_j$ on $\mathbb{R}^n$ are tight, by the second moments. A diagonal argument over the countably many finite families gives a subsequence along which all these laws converge; the limits are consistent under projections, and the Kolmogorov extension theorem gives random variables $(A(g))_{g\in\mathscr{G}}$ with these finite-dimensional laws. For rational q,q', $A_k(qg+q'g')=qA_k(g)+q'A_k(g')$ identically, so the law of $(A_k(qg+q'g'),A_k(g),A_k(g'))$ is carried by a closed hyperplane, and so is its limit: $A(qg+q'g')=qA(g)+q'A(g')$ almost surely.

Step 2 (extension). Let $f\in\mathscr{R}(X)$ and let $g_m\in\mathscr{G}$ be as in Lemma 337 with $\|f-g_m\|_{L^1}\to0$. By the second-moment bound, with a constant C_f depending only on $\|f\|_\infty$ and on a cube containing the supports,

$$\begin{aligned}\limsup_k\nu_k\big(A_k(f-g_m)^2\big)&\le C_f\|f-g_m\|_{L^1},\\ \mathrm{E}\big(A(g_m)-A(g_{m'})\big)^2&\le C_f\|g_m-g_{m'}\|_{L^1},\end{aligned}$$

the second because $\mathrm{E}(A(g)-A(g'))^2=\lim_k \nu_k(A_k(g-g')^2)$ by uniform integrability and $\|g-g'\|_{1,k}\to\|g-g'\|_{L^1}$. So $A(g_m)$ is Cauchy in $L^2(\mathrm{P})$; let $A(f)$ be its limit. The same two bounds show that the limit is the same for every sequence $g_m\in\mathscr{G}$ with $\sup_m\|g_m\|_\infty<\infty$, supports in a fixed cube and $\|f-g_m\|_{L^1}\to 0$, and that it agrees with Step 1 for $f\in\mathscr{G}$. For $f_1,\dots,f_n$ and $t\in\mathbb{R}^n$, using $|e^{iy}-e^{iy'}|\le|y-y'|$,

$$\begin{aligned}&\Big|\nu_k\big(e^{i\sum_j t_jA_k(f_j)}\big)-\mathrm{E}e^{i\sum_j t_jA(f_j)}\Big|\\&\quad\le\Big|\nu_k\big(e^{i\sum_j t_jA_k(g_{j,m})}\big)-\mathrm{E}e^{i\sum_j t_jA(g_{j,m})}\Big|\\&\qquad+\sum_j|t_j|\big(\delta_{j,m,k}+\delta_{j,m}\big),\end{aligned}$$

with $\delta_{j,m,k}=\nu_k(A_k(f_j-g_{j,m})^2)^{1/2}$ and $\delta_{j,m}=\mathrm{E}((A(f_j)-A(g_{j,m}))^2)^{1/2}$; letting $k\to\infty$ and then $m\to\infty$ gives convergence in law. Convergence of the moments and of the exponential moments follows by uniform integrability. For linearity let $f,f'\in\mathscr{R}(X)$, $\lambda,\lambda'\in\mathbb{R}$, let g_m,g'_m be sequences as above for f,f', and let $q_m,q'_m\in\mathbb{Q}$ tend to λ,λ'; then $q_mg_m+q'_mg'_m\in\mathscr{G}$ is such a sequence for $\lambda f+\lambda' f'$, and $A(q_mg_m+q'_mg'_m)=q_mA(g_m)+q'_mA(g'_m)$ almost surely by Step 1, so $A(\lambda f+\lambda' f')=\lambda A(f)+\lambda' A(f')$ almost surely. (Linearity is meant in this sense: almost surely for each finite linear relation.) Finally $\mathrm{E}A(f)=\lim_k\nu_k(A_k(f))=0$.

Step 3 (bounds). For continuous $\varphi\ge 0$ and convergence in law, $\mathrm{E}\varphi(A(f))\le\liminf_k\nu_k(\varphi(A_k(f)))$; applied to e^{ty}, $|y|^n$, y^2 with the bounds of Corollary 332 or Theorem 335(ii) and $\|f\|_{1,k}\to\|f\|_{L^1}$, this gives the first three bounds of (ii); the last is Hölder's inequality.

Step 4 (symmetries). For $\gamma\in B_4$, Lemma 338(a) holds for every k and passes to the limit in law. For $v\in X$ let $v_k\in a_k\mathbb{Z}^4$ with $|v-v_k|_\infty\le a_k$. For $g\in\mathscr{G}$, $\nu_k(A_k(\tau_vg-\tau_{v_k}g)^2)\le C_ga_k$ by Lemma 337, and the law of $(A_k(\tau_{v_k}g_j))_j$ is that of $(A_k(g_j))_j$; so the limit laws of $(A(\tau_vg_j))_j$ and $(A(g_j))_j$ coincide. Since τ_vg_m approximates τ_vf as g_m approximates f, this extends to $\mathscr{R}(X)$. Continuity: for $f\in\mathscr{R}(X)$ and $h=\tau_vf-f$, $|v|\le 1$, the third bound of (ii) (proved in Step 3, which does not use Step 4) and $\|h\|_{L^1}\le\|h\|_\infty|\{h\ne 0\}|$ give $\mathrm{E}(A(\tau_vf)-A(f))^2=\mathrm{E}A(h)^2\le 4\|f\|_\infty e^K\|\tau_vf-f\|_{L^1}\to 0$ as $v\to 0$, with K the volume of the 1-neighbourhood of the support of f (or $K=L^4$ on the torus).

Step 5 (positivity). Since $0\in a_k\mathbb{Z}$ for every k, Lemma 338(b) with $c=0$ holds for every k; the function $(y,y')\mapsto\overline{F(y')}F(y)$ is bounded and continuous on $\mathbb{R}^{2n}$, and the joint law of $(A_k(f_j),A_k(\theta_{\mu,0}f_j))_j$ converges, so the inequality passes to the limit. For general c put $g_j=f_j(\cdot+ce_\mu)$, which vanish outside a compact subset of $P_{\mu,0}$; since $\theta_{\mu,c}x-ce_\mu=\theta_{\mu,0}(x-ce_\mu)$, the family $(f_j,\theta_{\mu,c}f_j)_j$ is the translate by ce_μ of $(g_j,\theta_{\mu,0}g_j)_j$, and (iii) reduces the claim to $c=0$. For a polynomial F, truncate: $F_R=F\chi(|y|/R)$ with a continuous cut-off χ, and let $R\to\infty$ using the moment bounds.

Step 6 (the constant smearing). By Theorem 330(ii) and $4Z_k^{\mathrm{FE}}a_k^4/N=\eta_k$,

$$\log\mu_k\big(e^{\pm tA_k(1)}\big)=L^4a_k^{-4}R_{M_k}(\beta_k;\mp t\eta_k),$$

and by (i) the left side converges for $|t|<1$; $\mu_k(A_k(1)^2)=16(Z_k^{\mathrm{FE}})^2a_k^4L^4p''_{M_k}(\beta_k)/N^2$ by the proof of Theorem 330(iii), and it converges to $\mathrm{E}A(1)^2$ by uniform integrability. If $\mathrm{E}A(1)^2>0$ then $A\ne 0$; if $\mathrm{E}A(1)^2=0$, then $A(1)=0$ and $\mathrm{E}A(f)^2\le$ the limit of $\|f\|_{1,a_k}\|f\|_\infty L^{-4}\mu_k(A_k(1)^2)$ by Theorem 330(iii), so $A=0$. □

Remark 340 (what the theorem does and does not give). Theorem 339 is the compactness half of (E) for the smeared plaquette field, with no hypothesis on the couplings: a random field on $\mathbb{T}_L$ or $\mathbb{R}^4$, invariant under translations and B_4, reflection positive for all coordinate reflections, with moments of factorial growth bounded by the sup and L^1 norms of the test functions; on $\mathbb{R}^4$ these bounds grow polynomially in the volume of the support. It does not decide whether $A\ne 0$ (on the torus this is the free-energy condition (v)), whether the limit depends on the subsequence, whether it is $\mathrm{O}(4)$ invariant, or whether it is Gaussian. The next subsection shows that, whenever the short-distance law holds, the normalisation Z^{FE} is necessarily too small to see correlations at separated points.

3. *The free-energy normalisation against the short-distance law*

Let $G_k^\square(z)=\nu_k(\Phi_0^\square\Phi_z^\square)-c_k^2$, $z\in\mathbb{Z}^4$, in either setting of Sec. XP2 (on the torus $z\in\Lambda_{M_k}$ and $|z|$ is the torus distance), so that $\nu_k((\Phi_x^\square-c_k)(\Phi_y^\square-c_k))=G_k^\square(y-x)$ by translation invariance. With $w_\gamma(\rho)=\rho^{-8}(1+\log^+\frac{1}{\rho})^{-\gamma}$ as in Sec. XL, the short-distance law for the cube field reads:

($\mathrm{AF}^\square$) there are $\gamma\ge 0$, $0<c_1\le C_1$, $r_0\in(0,\frac14]$ and k_0 such that for $k\ge k_0$ and $|z|_\infty\ge 2$, $Z_k'^2|G_k^\square(z)|\le C_1w_\gamma(a_k|z|)$, and $Z_k'^2G_k^\square(z)\ge c_1w_\gamma(a_k|z|)$ if moreover $a_k|z|\le r_0$.

We call the two inequalities the upper and the lower half. For $|z|_\infty\ge 2$ the cubes C_0 and C_z have no common link, which is why 2 replaces the $2r+1$ of (AF). The law is the same statement as (AF) for the gauge-invariant field $\Phi^\square$, centred at the points m_x.

Lemma 341 (two-norm second moments exclude the short-distance law)**.** *Let $Z_k,Z'_k>0$ and suppose that, for some $C<\infty$, all large k and all $f\in\mathscr{R}(X)$ vanishing outside a fixed cube $Q_*\subset X$ of side $\ell_*\in(0,1]$,*

$$\nu_k\big(A_{k,Z_k}(f)^2\big)\le C\|f\|_\infty\|f\|_{1,k},$$

where $A_{k,Z}$ is the field with normalisation Z. If the lower half of ($\mathrm{AF}^\square$) holds for the constants Z'_k, then $Z_k/Z'_k\to 0$.

Proof. Suppose $Z_k/Z'_k\ge\vartheta>0$ along a subsequence. Let $0<s\le r_0/4$ (on $\mathbb{T}_L$ also $3s\le L/2$, so that the torus distance of two points of Q_* used below is their Euclidean distance) and let $B=b+[0,s)^4$ and $B'=2se_0+B$ with b chosen so that $B\cup B'\subset Q_*$ (possible once $3s\le\ell_*$), $f=\mathbf{1}_B$, $g=\mathbf{1}_{B'}$. If $a_km_x\in B$ and $a_km_y\in B'$ then $s<|a_km_y-a_km_x|<\sqrt{12}\,s<4s\le r_0$ and $|y-x|_\infty>s/a_k-1\ge 2$ for large k. On $[s,4s]$, $w_\gamma(\rho)\ge(4s)^{-8}(1+\log\frac{1}{s})^{-\gamma}$. So the lower half of ($\mathrm{AF}^\square$) gives, for

large k,

$$\begin{aligned}\nu_k\big(A_{k,Z_k}(f)A_{k,Z_k}(g)\big)&=Z_k^2\sum_{x,y}a_k^8 f(a_k m_x)g(a_k m_y)G_k^{\square}(y-x)\\&\geq \vartheta^2 c_1(4s)^{-8}\Big(1+\log\frac{1}{s}\Big)^{-\gamma}\|f\|_{1,k}\|g\|_{1,k},\end{aligned}$$

while the hypothesis and the Cauchy–Schwarz inequality bound the left side by

$$C(\|f\|_{1,k}\|g\|_{1,k})^{1/2}.$$

Since $\|f\|_{1,k},\|g\|_{1,k}\to s^4$,

$$\vartheta^2 c_1 4^{-8}\Big(1+\log\frac{1}{s}\Big)^{-\gamma}\leq Cs^4$$

for every small $s>0$, which fails as $s\to 0$. □

Theorem 342 (what the free energy can see)**.** *In either setting of Sec. XP 2, let $\beta_k\geq 0$ be arbitrary.*

(i) *$Z_k^{\mathrm{FE}}\leq N/(4a_k^4)$ for every k.*

(ii) *If the lower half of* (AF$^{\square}$) *holds for constants Z_k', then $Z_k'\geq c\,a_k^{-4}(1+\log\frac{1}{a_k})^{-\gamma/2}$ for large k, with $c=c_1^{1/2}4^{-4}(48N)^{-1}$, and $Z_k^{\mathrm{FE}}/Z_k'\to 0$. Moreover, for every $T>0$, no bound of Theorem 330*(iv)(b) *holds for the constants Z_k': in the torus setting*

$$\sup_k L^{-4}\big(\log Z_{M_k}(\beta_k+h_k)-2\log Z_{M_k}(\beta_k)+\log Z_{M_k}(\beta_k-h_k)\big)=\infty,\qquad h_k=\frac{4TZ_k'a_k^4}{N},$$

and in the infinite-lattice setting $\sup_k\Xi_k^{\infty}(T)=\infty$ for the constants Z_k'.

(iii) *If* (AF$^{\square}$) *holds for constants Z_k', then every limit A of Theorem 339 has $\mathrm{E}A(f)A(g)=0$ whenever the supports of $f,g\in\mathscr{R}(X)$ are at positive distance, and there is $\sigma^2\geq 0$ with*

$$\mathrm{E}A(f)A(g)=\sigma^2\int_X f(x)g(x)\,dx\qquad(f,g\in\mathscr{R}(X)).$$

(iv) *Under the hypothesis of* (iii)*, for $f,g\in\mathscr{R}(X)$ vanishing outside a compact subset of $P_{\mu,c}$, $\mathrm{E}A(\theta_{\mu,c}f)A(g)=0$. In the Hilbert space obtained from the positive form $(F,F')\mapsto\mathrm{E}[\overline{F(A(\theta_{\mu,c}f_j))}F'(A(g_j))]$ of Theorem 339*(iv) *(polynomials in fields smeared in $P_{\mu,c}$, modulo null vectors), every single smeared field $A(f)$ is a null vector, since its norm is $\mathrm{E}A(\theta_{\mu,c}f)A(f)=0$: no one-field vector survives.*

Proof. (i) $\min(h,1)\leq 1$.

(ii) For $|z|_\infty=2$ along a coordinate axis, $a_k|z|=2a_k\leq r_0$ for large k, and $|G_k^{\square}(z)|\leq\|\Phi^{\square}\|_\infty^2\leq(48N)^2$ because $0\leq\Phi^{\square}\leq 48N$; the lower half gives $Z_k'^2(48N)^2\geq c_1(2a_k)^{-8}(1+\log\frac{1}{2a_k})^{-\gamma}$, which gives the bound on Z_k' (with room to spare in the constant). By Corollary 332 or Theorem 335(ii), the hypothesis of Lemma 341 holds for $Z_k=Z_k^{\mathrm{FE}}$ and $Q_*=[0,1)^4$ with $C=2e^{16}$ (on $\mathbb{T}_L$, with the image of $[0,\min(1,L/2))^4$ and $C=2e^{L^4}$), so $Z_k^{\mathrm{FE}}/Z_k'\to 0$. If the displayed supremum were finite for some T, then by Remark 331 $K_T=\sup_k\Xi_{M_k,\beta_k,Z_k'}(T)<\infty$, and by Theorem 330(i) with $t=\pm T/\|f\|_\infty$ and $t^2y^2\leq e^{ty}+e^{-ty}-2$, for bounded $f\neq 0$,

$$\begin{aligned}\mu_k\big(A_{k,Z_k'}(f)^2\big)&\leq\frac{2\|f\|_\infty^2}{T^2}\big(e^{K_T\|f\|_{1,a_k}/\|f\|_\infty}-1\big)\\&\leq\frac{2K_Te^{K_TL^4}}{T^2}\|f\|_\infty\|f\|_{1,a_k};\end{aligned}$$

Lemma 341 with $Z_k=Z_k'$ would give $1\to 0$. The infinite-lattice case is the same with Theorem 335(i) and $\mathfrak{m}_k(f)\leq 2^4$ for f vanishing outside $[0,1)^4$, by (rU7.3).

(iii) Let the supports of f,g be at distance $\delta>0$. For large k, every pair with $f(a_km_x)g(a_km_y)\neq 0$ has $|y-x|_\infty\geq 2$ and $a_k|y-x|\geq\delta/2$, where $w_\gamma\leq(\delta/2)^{-8}$; the upper half gives

$$\begin{aligned}&\big|\nu_k\big(A_k(f)A_k(g)\big)\big|\\&=\Big(\frac{Z_k^{\mathrm{FE}}}{Z_k'}\Big)^2\Big|Z_k'^2\sum_{x,y}a_k^8 f(a_km_x)g(a_km_y)G_k^{\square}(y-x)\Big|\\&\leq\Big(\frac{Z_k^{\mathrm{FE}}}{Z_k'}\Big)^2\frac{2^8C_1}{\delta^8}\|f\|_{1,k}\|g\|_{1,k}\to 0\end{aligned}$$

by (ii), and the left side tends to $\mathrm{E}A(f)A(g)$ by Theorem 339(i). For boxes B,B' (products of half-open intervals) with $B\cap B'=\emptyset$ let $B_\varepsilon\subset B$ be the box of the points at ℓ^∞-distance at least ε from the complement of B; then $\mathrm{E}A(\mathbf{1}_{B_\varepsilon})A(\mathbf{1}_{B'})=0$, and by the second-moment bound of Theorem 339(ii) and Cauchy–Schwarz, $|\mathrm{E}A(\mathbf{1}_{B\setminus B_\varepsilon})A(\mathbf{1}_{B'})|\leq C(|B\setminus B_\varepsilon||B'|)^{1/2}\to 0$; so $\mathrm{E}A(\mathbf{1}_B)A(\mathbf{1}_{B'})=0$. Let $m(B)=\mathrm{E}A(\mathbf{1}_B)^2$. By linearity and these cross terms, m is additive over finite disjoint unions that are boxes; by Theorem 339(iii) it is translation invariant. A cube Q of side $2^{-n}\ell$ tiles the cube of side ℓ by 2^{4n} translates, so $m(Q)=|Q|\,\sigma^2$ with $\sigma^2=m(Q_0)/|Q_0|$ for a fixed cube Q_0 (on $\mathbb{T}_L$, $Q_0=\mathbb{T}_L$), and every box with corners in $b_0+2^{-n}\ell_0\mathbb{Z}^4$, b_0 a corner and ℓ_0 the side of Q_0, is a disjoint union of such cubes, so $m(B)=\sigma^2|B|$ for these boxes. For a general box B and such boxes B_j with $|B\triangle B_j|\to 0$, $|m(B)^{1/2}-m(B_j)^{1/2}|\leq\mathrm{E}(A(\mathbf{1}_B-\mathbf{1}_{B_j})^2)^{1/2}\leq(C|B\triangle B_j|)^{1/2}\to 0$. Hence for step functions on a common family of disjoint boxes, $\mathrm{E}A(f)A(g)=\sum_i f_ig_i\sigma^2|B_i|=\sigma^2\int fg$, and the identity extends to $\mathscr{R}(X)$ by the approximation of Lemma 337 and the L^2 bounds of Step 2 of the proof of Theorem 339, since $\int fg$ is continuous under L^1 convergence with bounded sup norms.

(iv) $\theta_{\mu,c}f$ vanishes outside a compact subset of $\theta_{\mu,c}(P_{\mu,c})$, which is at positive distance from that of g; apply (iii). □

Remark 343. Theorem 342 makes the price of Theorem 339 exact. Every normalisation that controls the coinciding-point moments of the smeared plaquette field with the two-norm bound $\|f\|_\infty\|f\|_{1,k}$ is negligible against any normalisation for which the short-distance law holds; under the full law the limit sees only a contact term, its covariance is that of white noise, and its single smeared fields carry no vector of the reflection-positive Hilbert space. This is the lattice counterpart of Proposition 252, now with no hypothesis on the limit: the renormalised field of (E) must be controlled at separated

points only, as in (M_s). Nothing in (iii) says that the higher correlations of the limit vanish at separated points, or that it is Gaussian; whether $\sigma^2 > 0$ is the free-energy condition of Theorem 339(v).

4. *Partition functions, sources, and what remains*

What is known. Bałaban's renormalisation group analysis of four-dimensional lattice gauge theories on a periodic lattice [6, 7, 9–11, 13] culminates, in the last of these papers, in the statement that if the sequence of effective coupling constants stays in an interval $(0,\gamma]$ with γ small, the effective densities produced by the renormalisation transformations have the form and satisfy the bounds of his inductive scheme, from which ultraviolet stability bounds of the same type as for superrenormalisable models follow (we rely here on the review of [13] in zbMATH, not on a reading of the original papers; no theorem of these papers is quoted below, and their bounds enter only through hypothesis (B), which is assumed). Jaffe and Witten, describing Bałaban's analysis of the three-dimensional lattice theory on a torus, call its estimates uniform in the lattice spacing and note that "these results need to be extended to the study of expectations of gauge-invariant functions of the fields"; they describe the four-dimensional extension [10] as including an analysis of asymptotic freedom and of large fields [68]. Magnen, Rivasseau and Sénéor [80] construct the Schwinger functions of the pure SU(2) gauge field in four dimensions with an infrared cut-off and no ultraviolet cut-off, in a regularised axial gauge whose momentum cut-off breaks gauge invariance and is compensated by counterterms, and verify the Slavnov identities non-perturbatively; the infrared cut-off is not removed and the Osterwalder–Schrader axioms are not established there. Gauge-invariant smoothed observables via the Yang–Mills heat flow, proposed by Charalambous and Gross (see [32]) and used in lattice gauge theory by Lüscher [78], are the basis of the state space of Cao and Chatterjee for three-dimensional theories [31, 32], which defines regularised Wilson loops under a Gaussian-free-field-like assumption on the theory; Chevyrev [37] and Chandra, Chevyrev, Hairer and Shen [33, 34] construct state spaces of distributional connections in two and three dimensions. None of these gives a bound on correlations of gauge-invariant observables of the four-dimensional Wilson measure along a lattice tuning. O'Carroll and Faria da Veiga [87] give, by multiple reflections, bounds on a normalised generating functional of a fixed number r of unsmeared gauge-invariant plaquette fields that are uniform in the volume, the spacing and the coupling in a bounded range; since r is fixed, so these bounds do not control fields smeared over $\asymp a^{-4}$ plaquettes.

From partition functions to correlations. Theorem 330(iv) and Remark 331 are the exact interface: uniform exponential moments of all smeared plaquette fields with normalisation Z_k on $\mathbb{T}_L$ are equivalent to the bound

$$\sup_k L^{-4}\left|\Delta^2_{h_k(T)}\log Z_{M_k}(\beta_k)\right| < \infty \quad (T>0), \qquad h_k(T) = \frac{4TZ_k a_k^4}{N}, \tag{rU7.4}$$

a statement about the partition function at three homogeneous couplings. A stability bound of Bałaban's type feeds into (rU7.4) as follows.

Corollary 344 (stability bounds at three couplings)**.** *Suppose that there are $K, E < \infty$ and functions F_k on $W_k = [\beta_k - a_k^2, \beta_k + a_k^2]$, $W_k \subset [0,\infty)$, such that*

$$\begin{aligned} &\left|\log Z_{M_k}(\beta') - F_k(\beta')\right| \le KL^4 \quad (\beta' \in W_k),\\ &\left|F_k(\beta_k+H) - 2F_k(\beta_k) + F_k(\beta_k - H)\right| \le EL^4\frac{H^2}{a_k^4} \qquad (|H| \le a_k^2). \end{aligned} \tag{B}$$

Then $a_k^{-4}\max_\pm R_{M_k}(\beta_k;\pm H) \le EH^2/a_k^4 + 4K$ for $|H| \le a_k^2$, $Z_k^{\mathrm{FE}} \ge N/(4(E+4K)a_k^2)$ when $a_k \le 1$ and $E+4K \ge 1$, and the fields with $Z_k = N/(4a_k^2)$ have uniformly bounded exponential moments: $\log \mu_k(e^{t\mathscr{A}_k(f)}) \le (E+4K)\,|t|\,\|f\|_{1,a_k}$ for $|t|\,\|f\|_\infty \le 1$.

Proof. Both remainders are nonnegative and $M^4(R_M(\beta;H) + R_M(\beta;-H)) = \Delta_H^2 \log Z_M(\beta)$, so, with $M_k^4 = L^4 a_k^{-4}$,

$$a_k^{-4}\max_\pm R_{M_k}(\beta_k;\pm H) \le L^{-4}\Delta_H^2 \log Z_{M_k}(\beta_k) \le \frac{EH^2}{a_k^4} + 4K.$$

For $|H| \le a_k^2$ the right side is at most $E+4K$, and by convexity $a_k^{-4}\max_\pm R(\pm s a_k^2) \le s(E+4K)$ for $s \in [0,1]$; so $h_{M_k,\beta_k} \ge a_k^2/(E+4K)$, which gives the bound on Z_k^{FE} as $a_k^2/(E+4K) \le 1$. With $Z_k = N/(4a_k^2)$, $4TZ_k a_k^4/N = Ta_k^2$ and $\Xi_{M_k,\beta_k,Z_k}(T) \le T(E+4K)$ for $T \le 1$; Theorem 330(i) gives the last claim. □

Ultraviolet stability bounds of the type described above would be expected to give, at each fixed coupling, an estimate of the first kind in (B) with F_k the extracted vacuum energy; we have not verified this against the original papers. The uniformity of such an estimate over a window of couplings of width a_k^2, and the second condition in (B), regularity of the extracted vacuum energy on that window, are further inputs that we have not found stated in the descriptions of these papers available to us. Corollary 344 is therefore a conditional statement with (B) as its hypothesis. Granted (B), the normalisation a_k^{-2} is reached, but no further:

Corollary 345 (no partition-function route to separated correlations)**.** *If the lower half of* ($\mathrm{AF}^{\square}$) *holds for the torus states with constants Z_k', then* (rU7.4) *fails for $Z_k = Z_k'$ and every $T > 0$, while $h_k(T) \ge 4cT(1+\log\frac{1}{a_k})^{-\gamma/2}/N$: no bound on homogeneous partition functions at couplings β_k and $\beta_k \pm h_k$, however precise, controls the renormalised field of the short-distance law through the criterion of Theorem 330(iv), although the coupling shifts involved are bounded below by a negative power of $1+\log\frac{1}{a_k}$ only.*

Proof. Theorem 342(ii). □

The additional input. What is needed is the partition function with an inhomogeneous coupling. For bounded f and $t \in \mathbb{R}$,

by the definition of $\mathscr{A}$ and $\sum_x f(am_x)\Phi_x^{\square} = \sum_p \psi_p w_p(f)$ with $w_p(f) = \sum_{x:C_x\supset p} f(am_x)$,

$$\mu_{M,\beta}\big(e^{t\mathscr{A}(f)}\big) = e^{-tZa^4 c\sum_x f(am_x)}\, e^{tZa^4 N\sum_p w_p(f)} \times \frac{Z_M[(\beta - tZa^4 w_p(f)/N)_p]}{Z_M(\beta)},$$

where

$$Z_M[(\beta_p)_p] = \int \exp\Big(N\sum_p \beta_p \operatorname{Re}\operatorname{Tr} Q_p\Big)\, dQ.$$

The mixed second derivative in (s,u) at 0 of

$$\log Z_M\big[\big(\beta - sZa^4 w_p(f)/N - uZa^4 w_p(g)/N\big)_p\big]$$

is the connected correlation $\mu(\mathscr{A}(f)\mathscr{A}(g))$; at the normalisation of (AF$^{\square}$), by Theorem 342(ii), the coupling profile is a bump of height at least of order $(1+\log\frac{1}{a_k})^{-\gamma/2}$ on the physical supports of f and g. The input that would pass from the renormalisation group to correlations of the smeared plaquette field is therefore a stability bound for $\log Z_M$ with a coupling varying on physical scales, with an error in the mixed second derivative that tends to 0 for separated f,g, rather than an error $O(L^4)$ in each partition function. By Lemma 341 no bound of the two-norm form $C\|f\|_\infty\|f\|_{1,k}$ on the variances holds at that normalisation, so such an input can only be posed at separated supports, as (M$_s$) is for $n=2$.

Reconstruction in one direction. The proved properties of the limits give, in the infinite-lattice setting, the Hilbert space and the transfer semigroup of one coordinate direction.

Proposition 346 (one-direction reconstruction)**.** *Let A be a limit of Theorem 339 in the infinite-lattice setting, μ a direction, $\theta = \theta_{\mu,0}$, $\tau_t = \tau_{te_\mu}$, and let $\mathfrak{P}_+$ be the set of random variables $F = \Phi(A(f_1),\dots,A(f_n))$ with $\Phi:\mathbb{R}^n\to\mathbb{C}$ bounded and continuous and $f_j \in \mathscr{R}$ vanishing outside a compact subset of $P_{\mu,0}$; put $\theta F = \Phi(A(\theta f_1),\dots,A(\theta f_n))$ and $\tau_t F = \Phi(A(\tau_t f_1),\dots,A(\tau_t f_n))$ for $t\ge 0$. Then $\langle F,F'\rangle = \mathrm{E}[\overline{\theta F}\,F']$ is a positive semidefinite Hermitian form on $\mathfrak{P}_+$; let $\mathscr{H}_\mu$ be the completion of $\mathfrak{P}_+$ modulo its null space, $[F]$ the class of F and $\Omega = [1]$. There is a self-adjoint operator $H_\mu \ge 0$ on $\mathscr{H}_\mu$ with $[\tau_t F] = e^{-tH_\mu}[F]$ for $t\ge 0$, $F\in\mathfrak{P}_+$, and $H_\mu\Omega = 0$.*

Proof. Positivity is Theorem 339(iv) with $c=0$. Since $\theta\in B_4$, Theorem 339(iii) shows that the families $(A(f_j),A(\theta f_i'))$ and $(A(\theta f_j),A(f_i'))$ have the same law, so

$$\langle F',F\rangle = \overline{\langle F,F'\rangle},$$

and the Cauchy–Schwarz inequality holds. For test functions

$$\theta(\tau_t f) = \tau_{-t}(\theta f);$$

translating the family $(A(\tau_{-t}\theta f_j),A(\tau_t f_j))_j$ by te_μ and using Theorem 339(iii),

$$\langle \tau_t F,\tau_t F\rangle = \langle F,\tau_{2t}F\rangle \le \|F\|\,\|\tau_{2t}F\|,$$

where $\|F\| = \langle F,F\rangle^{1/2}$. Applying this with t replaced by $2t,4t,\dots,$

$$\|\tau_t F\| \le \|F\|^{1-2^{-m}}\,\|\tau_{2^m t}F\|^{2^{-m}} \qquad (m\ge 1),$$

and $\|\tau_s F\|\le \sup|\Phi|$ for every s; letting $m\to\infty$ gives $\|\tau_t F\|\le\|F\|$. Hence $T_t[F] = [\tau_t F]$ is well defined (null vectors go to null vectors) and extends to a contraction of $\mathscr{H}_\mu$, with $T_0 = 1$, $T_sT_t = T_{s+t}$ and $T_t\Omega = \Omega$. Translating $(A(\theta g_i),A(\tau_t f_j))$ by $-te_\mu$ gives $\langle G,\tau_t F\rangle = \langle \tau_t G,F\rangle$, so T_t is self-adjoint, and $T_t = T_{t/2}^2 \ge 0$. For strong continuity, by Cauchy–Schwarz in $L^2(\mathrm{P})$ and the equality of the laws of $(A(\theta\tau_t f_j),A(\theta f_j))_j$ and $(A(\tau_t f_j),A(f_j))_j$,

$$\|T_t[F]-[F]\|^2 \le \mathrm{E}|\tau_t F - F|^2,$$

which tends to 0 as $t\to 0$ because $A(\tau_t f_j)\to A(f_j)$ in $L^2(\mathrm{P})$ (Theorem 339(iii)) and Φ is bounded and continuous. Let $T = T_1$; then $0\le T\le 1$. If $T\psi = 0$ then, by induction on m, $\|T_{2^{-m-1}}\psi\|^2 = \langle\psi,T_{2^{-m}}\psi\rangle = 0$, so $T_{2^{-m}}\psi = 0$ for all m, hence $\psi = \lim_m T_{2^{-m}}\psi = 0$. So T is injective and $H_\mu = -\log T$, defined by the spectral theorem [92, Sec. VII.2], is self-adjoint and $H_\mu\ge 0$. Since $T_{2^{-m}}\ge 0$ and $(T_{2^{-m}})^{2^m} = T$, $T_{2^{-m}}$ is the positive 2^m-th root $e^{-2^{-m}H_\mu}$ of T, so $T_t = e^{-tH_\mu}$ for dyadic $t\ge 0$, and for all $t\ge 0$ by the strong continuity of both sides; $T_t\Omega = \Omega$ gives $H_\mu\Omega = 0$. □

On the torus the same form gives a Hilbert space and the vacuum Ω, but the positive region $P_{\mu,c}$ is bounded and τ_t does not map it into itself for every $t>0$, so no semigroup is claimed there. By Theorem 342(iv), under (AF$^{\square}$) every single smeared field is a null vector of $\mathscr{H}_\mu$.

What remains for (E), reconstruction, and nontriviality. For the limits of Theorem 339 the following are proved: existence along subsequences for every coupling sequence, translation and B_4 invariance, reflection positivity in every coordinate direction, centring, moment bounds $|\mathrm{E}\prod_j A(f_j)|\le 2n!\,e^n\prod_j(\|f_j\|_\infty + \|f_j\|_{L^1})$, which are of the type of the linear growth condition of [89] for product test functions with a norm of order zero (we do not show that the moments extend to tempered distributions), and, on $\mathbb{R}^4$, the reconstruction of Proposition 346 in each coordinate direction. Not proved, and needed: (a) O(4) invariance; the set of $R\in\mathrm{O}(4)$ leaving the law of the limit invariant is a subgroup containing B_4, closed because $f\mapsto A(f)$ is continuous into $L^2(\mathrm{P})$ under L^1 convergence with bounded sup norms and supports in a fixed cube (Step 2 of the proof of Theorem 339), so by the group-theoretic argument of Theorem 248(vii) one rotation of infinite order would suffice; (b) independence of the subsequence; (c) clustering; (d) a normalisation that sees separated points: by Theorem 342, under (AF$^{\square}$) the limits of Theorem 339 have white-noise covariance and no one-field vectors, so a nontrivial two-point function at separated points requires $Z_k/Z_k^{\mathrm{FE}}\to\infty$ and bounds posed at separated supports only, as in the inputs (I2)–(I5) of Sec. X N; (e) for the mass gap, the averaged condition (I1) along the tuning; (f) for nontriviality in the sense of a non-free theory, beyond $S_2\neq 0$ at separated points, a non-vanishing connected four-point function, since a generalised free field satisfies all the other requirements. None of (a)–(f)

is proved here, and neither the mass gap, nor (D), nor (E) is claimed.

Bounded observables. Wilson loops of physical size, with or without gradient flow, are bounded by N in absolute value, so their joint laws along any tuning are tight; for them the content of (E) lies entirely in renormalisation and nontriviality, which for spatially flowed packets is the ratio condition of Proposition 298. Theorem 339 is the corresponding statement for an unbounded field, where tightness itself has content: it requires the free-energy normalisation, and Theorem 342 shows that under (AF$^{\square}$) this normalisation is the obstruction.

XI. CONCLUSION

For every compact simple gauge group, four-dimensional lattice Yang–Mills theory has a unique infinite-volume state, exponential clustering and a transfer-matrix gap whenever $N|\beta| < \frac{2}{29}\kappa_G$ (Theorem 5), and plaquette blocks extend the range for SU(2) and SU(3) (Theorems 3 and 7). One averaged estimate in a cube of fixed physical size, the bound $\mathrm{BP}(n_k, \varpi)$ with $n_k a_k \le \ell$, places the spectrum of every continuum limit in $\{0\} \cup [16\varpi/(625e^2\ell), \infty)$, so every nontrivial limit has a mass gap (Theorem 64). The averaged estimate holds on every torus at strong coupling (Corollary 72). The free massive scalar and Proca fields satisfy the averaged estimate itself on every torus, uniformly in the lattice spacing (Corollary 115). For the scalar field the best constant is the response of one cube to a constant shift of its boundary data, and it converges, as the spacing goes to zero, to $s^2\langle 1, (-\Delta + s^2)^{-1} 1\rangle$ on the unit cube with $s = m\ell$ (Theorems 109 and 112); no constant survives as $m\ell \to 0$ (Proposition 108), so for these fields the estimate holds exactly when a mass is present at the scale of the cube. The free photon violates it at the diffusive rate $\varpi \le 8\pi^2(n+1)^2/L^2$ (Theorem 134); every measure satisfying it has local observables of finite susceptibility at the scale of the cube (Proposition 136); and for SU(2) at weak coupling it holds on plaquette traces and on Wilson loops of fixed size with a constant independent of β (Corollary 157 and Theorem 159), while on all gauge-invariant functions it fails at fixed n as $\beta \to \infty$ (Theorem 77). The estimate is not monotone in the Hessian (Theorem 117), but it survives convex perturbations with bounded Hessian, in particular convex plaquette potentials added to the Proca field (Theorem 119 and Corollary 123); bounded perturbations can destroy it, as the Ising model shows (Proposition 126), and the Wilson action has flat directions in every gauge, so these arguments do not reach it (Corollary 128).

Two inputs remain: the averaged estimate along the tuning, which is the cube estimate (D) in averaged form, and a nontrivial continuum limit with the properties of Sec. XL, which is (E). Without any of them the torus limits exist and are reflection positive at every coupling, and their two-point functions have renormalised continuum limits with a positive Hamiltonian (Theorem 295); the short-distance law (AF) together with the averaged estimate gives a nontrivial two-point theory with a mass gap, and every nontrivial limit of these correlations along the whole tuning forces (SD), (UV), (NE) and a bounded physical gap (Theorem 299 and Proposition 300). Along a lattice axis the connected plaquette correlation of every torus is given by a transfer-matrix spectral formula at every coupling: it is nonnegative and log-convex, and it is either positive at all separations or zero at all of them; at strong coupling its leading term is $108((1+\delta)/2)^{4t+1}\beta^{4t}$ (Theorems 304 and 312). The lower half of (AF) along an axis is equivalent, up to logarithms, to a lower bound at the cutoff scale together with a lower bound on low-energy spectral weight at every intermediate scale (Corollary 325). Given both inputs, the results above give the continuum theory a mass gap; without them the mass gap is not proved here.

Lattice gauge fields also define four-dimensional gauge theories with chiral fermions in every anomaly-free representation, once the lattice field is interpolated to the continuum and coupled to continuum fermions [105]. The mass gap problem concerns the pure gauge theory, to which every result here refers.

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This paper rests on the work of many earlier scientists. Lattice gauge theory is due to Wilson [104] and its Hamiltonian form to Kogut and Susskind [74]; reflection positivity and the reconstruction of the transfer matrix to Osterwalder and Schrader [88, 89], and their lattice form, with the first proof of a gap at strong coupling, to Osterwalder and Seiler [90, 95]. The uniqueness method is Dobrushin's [41–43], the curvature method is that of Bakry and Émery [4, 5] in the form of Shen, Zhu and Zhu [96], the covariance estimate under Dobrushin's condition is Föllmer's [48], the chessboard estimate is that of Fröhlich, Israel, Lieb and Simon [49] and Fröhlich and Spencer [51], disagreement percolation is due to van den Berg and Maes [101], the Brascamp–Lieb inequality to Brascamp and Lieb [23], the correlation representation to Helffer and Sjöstrand [64] and Naddaf and Spencer [86], the perturbation lemma to Holley and Stroock [63], the derivative of a matrix function to Daleckii and Krein [40], and the strict positivity of the transfer matrix to Lüscher [77], the gauge fixing on a forest is Creutz's [38], and the curvature of Lie groups is Milnor's [84]. The analysis of the single-link law uses results of Barta [15], Biernacki and Krzyż [24], McShane [83], Stein and Weiss [99] and Watson [103], and the continuum problem is the one formulated by Jaffe and Witten [68] and studied by Bałaban [6, 7, 13] through the renormalisation group, in the account of Dimock [39]; gauge fields with cut-offs were constructed by Brydges, Fröhlich and Seiler [27] and by Magnen, Rivasseau and Sénéor [80]; recent probabilistic work on lattice Yang–Mills theory is due to Chatterjee [35, 36], Basu and Ganguly [16], Adhikari and Cao [1], and Cao, Nissim and Sheffield [29], and the two-dimensional theory was constructed by Driver [44], Gross, King and Sengupta [57] and Lévy [75].

DATA AND CODE AVAILABILITY

The current manuscript and complete TeX inputs are in the `papers/` directory of `https://github.com/creelie`

/4Y-M, research release v4.43.0. The permanent work DOI is 10.5281/zenodo.23444338. The earlier DOI links below identify historical computational records.

The proofs are analytic. Remarks 36 and 39 report evaluations for SU(3) in interval arithmetic with rigorous error bounds, and Remark 377 reports a computation with a certificate checked in exact arithmetic; no result uses them. Every number used in a proof is derived in the text. The programs of these steps, in Python with python-flint (FLINT and Arb) and a C kernel for products of Taylor polynomials, their outputs and the list of certified boxes are in the repository https://github.com/creelie/4Y-M, archived on Zenodo (doi:10.5281/zenodo.23444338; release v4.37.0, doi:10.5281/zenodo.23992883), together with a Lean 4 certificate of the line field, Lean 4 files for the comparison arithmetic and for the counting and trace identities and the constants of Secs. IX F–IX I, the Lean 4 files and checks of the results of [21] that are not repeated here, independent exact or numerical checks of the other finite statements, and our checks of the certificate of [93]. The code was written by Deep Bhattacharjee, James Montgomery Lockwood and Tristan Romanov.

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Use of artificial intelligence. The computations, the checks and parts of the drafting were done with AI assistance, through Claude Code with Claude Opus 5.5 and Claude Fable 5.1, and with GPT-6.1 Sol and GPT-6 Astra, under the direction of Deep Bhattacharjee, who checked every statement and is responsible for the text. The combination of the forest gauge with the sharp quarter behind Theorem 1 was first assembled by Anaxia, the reasoning engine of the BLLM system [17, 18]; the line field, the free-boundary limit, the ceiling, the curvature bound, the SU(3) part, the plaquette blocks and the certified constants are the authors' own.

Appendix A: The line field

Table III lists, for each residue class of x modulo two, the direction $\sigma(x)$ and the permutation $\pi_x(\mu) = \sigma(x+e_\mu)$. In every row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions, which is the condition of Lemma 8.

Appendix B: The comparison lemma

Let (G,δ) be a compact group with a bi-invariant metric of diameter D, $\Omega = G^A$, and $C_{ee'}$ the coefficients of Sec. III, with row and column sums at most $\alpha < 1$ and $C_{ee'} = 0$ unless e, e' share a plaquette, so that $|e - e'| \le 1$ for the distance of link midpoints. Fix $\lambda > 0$ with $\theta = \alpha e^{\lambda} < 1$.

TABLE III. The line field: x modulo two (digits $x_0x_1x_2x_3$), the direction $\sigma(x)$ and the values $\sigma(x+e_\mu)$ for $\mu = 0,1,2,3$.

x	$\sigma(x)$	$\mu=0$	$\mu=1$	$\mu=2$	$\mu=3$
0000	0	0	2	3	1
1000	0	0	3	1	2
0100	2	3	0	2	1
1100	3	2	0	1	3
0010	3	1	2	0	3
1010	1	3	1	0	2
0110	2	1	3	2	0
1110	1	2	1	3	0
0001	1	2	1	3	0
1001	2	1	3	2	0
0101	1	3	1	0	2
1101	3	1	2	0	3
0011	3	2	0	1	3
1011	2	3	0	2	1
0111	0	0	3	1	2
1111	0	0	2	3	1

Lemma 347. *Let $S \subset A$, possibly infinite, and let μ_1, μ_2 be probability measures on Ω invariant under γ_e^F for every $e \notin S$. Let $T \subset A \setminus S$ be finite, $w_b = e^{-\lambda \operatorname{dist}(b,T)}$, and suppose $\sum_{z\in S} w_z < \infty$. There is a coupling of μ_1, μ_2 whose link distances $\rho(b) = \mathbb{E}\,\delta(\sigma_b, \sigma'_b)$ satisfy*

$$\sum_{b\notin S} w_b \rho(b) \le \frac{\theta D}{1-\theta} \sum_{z\in S} w_z.$$

If $S = \emptyset$ then $\mu_1 = \mu_2$.

Proof. The couplings form a weakly compact set and $\Psi(\pi) = \sum_b w_b \mathbb{E}_\pi \delta(\sigma_b, \sigma'_b)$ is weakly continuous because $\sum_b w_b < \infty$; let π^* be a minimiser. For $e \notin S$ replace the pair (σ_e, σ'_e) by a sample of an optimal coupling of $\gamma_e^F(\cdot \mid \sigma)$ and $\gamma_e^F(\cdot \mid \sigma')$, chosen measurably [102, Cor. 5.22]. Both μ_i are invariant, so the result is again a coupling, and changing the finitely many links that enter γ_e^F one at a time gives $W_1 \le \sum_{e'} C_{ee'} \delta(\sigma_{e'}, \sigma'_{e'})$. Minimality gives $\rho(e) \le \sum_{e'} C_{ee'} \rho(e')$. Multiply by w_e and sum over $e \notin S$; where $C_{ee'} \ne 0$, $w_e \le e^{\lambda} w_{e'}$, so

$$\sum_{e\notin S} w_e \rho(e) \le \theta \Big[\sum_{e'\notin S} w_{e'} \rho(e') + \sum_{z\in S} w_z \rho(z) \Big],$$

all sums are finite, and $\rho \le D$ on S. If $S = \emptyset$, $\rho \equiv 0$. □

Corollary 348. *Let μ be the Gibbs measure of γ^F, F a local function with support S_F, L_F-Lipschitz in each link, and H bounded with finite support S_H disjoint from S_F. Then*

$$|\operatorname{Cov}_\mu(F,H)| \le \|H\|_\infty L_F \frac{\theta D}{1-\theta} |S_H|\, e^{-\lambda \operatorname{dist}(S_F, S_H)}.$$

Proof. Condition on the links of S_H [72, Ch. 3]; the conditional measures $\mu^\sigma = \mu(\cdot \mid \sigma_{S_H})$ are invariant under γ_e^F for $e \notin S_H$.

Lemma 347 with $S = S_H$ and $T = S_F$, for which $w_b = 1$ on S_F and $w_z \le e^{-\lambda \operatorname{dist}(S_F,S_H)}$ on S_H, gives $|\mu^\sigma(F) - \mu^{\sigma'}(F)| \le L_F \sum_{b\in S_F} \rho(b) \le L_F \frac{\theta D}{1-\theta} |S_H| e^{-\lambda \operatorname{dist}(S_F,S_H)}$ for any two boundary values. Since H depends only on σ_{S_H}, $\operatorname{Cov}_\mu(F,H) = \iint H(\sigma)\big(\mu^\sigma(F) - \mu^{\sigma'}(F)\big)\, d\mu(\sigma)\, d\mu(\sigma')$, and the bound follows. □

Lemma 349. *Let the blocks partition A, let C_{Bj} be the block coefficients of Sec. V, with $\sum_B C_{Bj} \le \alpha < 1$ for every j and $C_{Bj} = 0$ unless every link of B lies within distance 2 of j, and fix $\lambda > 0$ with $\theta = \alpha e^{2\lambda} < 1$. Let $S \subset A$ be a union of blocks and let μ_1, μ_2 be invariant under $\gamma_B^{\mathscr{B}}$ for every block $B \not\subset S$. With T, w_b as in Lemma 347 and $\sum_{z\in S} w_z < \infty$, there is a coupling with $\sum_{b\notin S} w_b \rho(b) \le \frac{\theta D}{1-\theta} \sum_{z\in S} w_z$, and $\mu_1 = \mu_2$ if $S = \emptyset$.*

Proof. Take a minimiser π^* of Ψ as before. For a block $B \not\subset S$ replace (σ_B, σ'_B) by an optimal coupling, for the metric d, of the two block laws; it is again a coupling, and changing the boundary links one at a time bounds the expected new d-distance by $\sum_j C_{Bj}\rho(j)$. Put $\bar w_B = \max_{i\in B} w_i$. Minimality gives

$$\sum_{i\in B} w_i \rho(i) \le \bar w_B \sum_j C_{Bj}\rho(j),$$

and $\bar w_B \le e^{2\lambda} w_j$ whenever $C_{Bj} \ne 0$. Summing over the blocks outside S and using the column sums,

$$\sum_{b\notin S} w_b \rho(b) \le \theta \Big[\sum_{j\notin S} w_j \rho(j) + \sum_{z\in S} w_z \rho(z) \Big],$$

which gives the claim as in Lemma 347. □

Lemma 350. *Let $n \ge 2$ be even, let V_x, $x \in 2\mathbb{Z}^4$, be the translates of Sec. IX, and suppose that $D(n,\alpha)$ holds.*

(a) *If a probability measure μ on G^A is invariant under γ_e^F for every e in a finite set V, it is invariant under γ_V.*

(b) *Fix $\lambda > 0$ with $\theta = \alpha e^{\lambda\rho_n} < 1$. Let $S \subset A$, and let μ_1, μ_2 be invariant under γ_{V_x} for every x with $V_x \cap S = \emptyset$. With T, w_b as in Lemma 347 and $\sum_{z\in S} w_z < \infty$, there is a coupling with*

$$\sum_{b\notin S} w_b \rho(b) \le \frac{D(1 + c_n e^{\lambda\rho_n})}{1-\theta} \sum_{z\in S} w_z,$$

and $\mu_1 = \mu_2$ if $S = \emptyset$.

Proof. (a) Let ν_η be the conditional law of Q_V given the links η outside V. For bounded f on G^V and bounded g of η, $\gamma_e^F(fg) = g\,\gamma_e^F f$ for $e \in V$, so invariance gives $\int g(\eta)\big(\nu_\eta(\gamma_e^F f) - \nu_\eta(f)\big)\, d\mu = 0$ for all g; taking f in a countable dense set of continuous functions, ν_η is invariant under every $\gamma_e^F(\cdot \mid \cdot, \eta)$, $e \in V$, for μ-almost every η. So is $\gamma_V(\cdot \mid \eta)$, whose conditional laws of single links are the γ_e^F. The composition P_η of the kernels γ_e^F, $e \in V$, in a fixed order has a density with respect to Haar measure on G^V bounded below by a positive constant, because each single-link density is at least $e^{-12N^2\beta}$ times its maximum over the group. A Markov kernel with such a lower bound has at most one invariant probability measure (Doeblin's condition), so $\nu_\eta = \gamma_V(\cdot \mid \eta)$.

(b) Let π^* minimise Ψ as in Lemma 347 and call x admissible if $V_x \cap S = \emptyset$. For admissible x replace $(\sigma_{V_x}, \sigma'_{V_x})$ by an optimal coupling, for d_{V_x}, of the two block laws; both μ_i are invariant, so the result is a coupling, and changing the boundary links one at a time bounds the new expected d_{V_x}-distance by $\sum_j C_{V_x,j}\rho(j)$. With $\bar w_x = \max_{i\in V_x} w_i$, minimality gives

$$\sum_{i\in V_x} w_i \rho(i) \le \bar w_x \sum_{j\notin V_x} C_{V_x,j}\rho(j),$$

and $\bar w_x \le e^{\lambda\rho_n} w_j$ whenever $C_{V_x,j} \ne 0$. Let S' be the set of links outside S that lie in a translate meeting S; each is within distance ρ_n of a link of S, so $\sum_{i\in S'} w_i \le c_n e^{\lambda\rho_n} \sum_{z\in S} w_z$. A link outside $S \cup S'$ lies in N_n translates, all admissible. Summing over admissible x and using $\sum_{x:\, j\notin V_x} C_{V_x,j} = K_{c(j)} \le \alpha N_n$,

$$N_n \sum_{i\notin S\cup S'} w_i\rho(i) \le \theta N_n \Big[\sum_{i\notin S\cup S'} w_i\rho(i) + \sum_{i\in S\cup S'} w_i\rho(i) \Big].$$

All sums are finite and $\rho \le D$, so $\sum_{i\notin S\cup S'} w_i\rho(i) \le \frac{\theta D}{1-\theta}(1 + c_n e^{\lambda\rho_n}) \sum_{z\in S} w_z$; adding $\sum_{i\in S'} w_i\rho(i) \le D c_n e^{\lambda\rho_n} \sum_{z\in S} w_z$ gives the bound. If $S = \emptyset$ then $S' = \emptyset$ and $\rho \equiv 0$. □

The covariance bound of Corollary 348 follows from part (b) in the same way, with $\theta D/(1-\theta)$ replaced by $D(1 + c_n e^{\lambda\rho_n})/(1-\theta)$, since the conditional measures μ^σ there are invariant under γ_{V_x} for $V_x \cap S_H = \emptyset$ by part (a).

Appendix C: The plaquette blocks

Table IV lists the block plaquettes of Sec. V modulo $2\mathbb{Z}^4$: the plaquette at x in the plane $\{\mu,\nu\}$ is a block plaquette exactly when x modulo two and $\{\mu,\nu\}$ appear in the table. Eight lie in the plane $\{0,1\}$ and eight in $\{2,3\}$; together they contain the 48 active links of a cell of side two, each once.

TABLE IV. The sixteen block plaquettes modulo two: the base point x (digits $x_0x_1x_2x_3$ modulo two) and the plane.

x	plane	x	plane
0000	$\{0,1\}$	0000	$\{2,3\}$
0001	$\{0,1\}$	0011	$\{2,3\}$
0010	$\{0,1\}$	0100	$\{2,3\}$
0011	$\{0,1\}$	0111	$\{2,3\}$
1100	$\{0,1\}$	1000	$\{2,3\}$
1101	$\{0,1\}$	1011	$\{2,3\}$
1110	$\{0,1\}$	1100	$\{2,3\}$
1111	$\{0,1\}$	1111	$\{2,3\}$

In closed form, the plaquette at x in $\{0,1\}$ is a block plaquette when $x_0 \equiv x_1$ and the plaquette at x in $\{2,3\}$ when $x_2 \equiv x_3$; the blocks do not depend on the line field beyond the fact that every plaquette contains one fixed link. The bounds of

Lemma 16 and the chain (3) are proved by hand in Sec. V. The archive also evaluates the functions of Lemma 16 in ball arithmetic at 128 bits, from the series of Z_0 and of its first three derivatives in $s = r^2$ summed to forty-eight terms with the tail bounded by twice the first omitted term; these sharper constants are not used by any result.

For SU(3) the constants of Remark 39 are evaluated on the 841 boxes of side $\frac{3}{16}$ at $r = \frac{3}{2}$ and the 784 boxes of side $\frac{19}{100}$ at $r = \frac{38}{25}$ and the 826 boxes of side $\frac{77}{400}$ at $r = \frac{77}{50}$ with the moments of Appendix E, and the constants of (15) and α_3 are evaluated from these bounds and the covers of Remark 36 in ball arithmetic at 120 bits. The covers at $r = \frac{38}{25}$ and $\frac{77}{50}$ are computed as in Appendix E with $r = \frac{38}{25}$ and the threshold 1.03, and with $r = \frac{77}{50}$ and the threshold 1.0275. No result of the paper uses these numbers; Theorem 7 uses the constants of Lemma 31.

Appendix D: The blocks of Lemma 20

Diagonal blocks. Let $w_a = e^{i\phi}$. On the diagonal part, $f_p = -\cos\phi\,(y_1+y_2-y_3-y_4)^2$ and $c_p = 2(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$, so

$$\mathscr{F} = \sum_i \alpha_i y_i^2 + \cos\phi\,(y_1+y_2-y_3-y_4)^2 - \tfrac{9}{5}(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$$

with $y_k = 0$ and weights $\alpha_{k\pm1} = \frac{21}{10}$, $\alpha_{k+2} = \frac{5}{4}$. It is affine in $\cos\phi$, so positivity at $\cos\phi = \pm1$ suffices. For $k = 1$ and $\cos\phi = 1$ the matrix in (y_2, y_3, y_4) is

$$\begin{pmatrix} \frac{31}{10} & -\frac{19}{10} & -1 \\ -\frac{19}{10} & \frac{9}{4} & \frac{19}{10} \\ -1 & \frac{19}{10} & \frac{31}{10} \end{pmatrix},$$

and Table V lists the leading principal minors for every position; all are positive.

TABLE V. Leading principal minors of the diagonal blocks of Lemma 20.

Fixed link	$\cos\phi = 1$	$\cos\phi = -1$
e_1, e_4	$\frac{31}{10}, \frac{673}{200}, \frac{8421}{2000}$	$\frac{11}{10}, \frac{53}{200}, \frac{21}{2000}$
e_2	$\frac{31}{10}, \frac{861}{100}, \frac{8421}{2000}$	$\frac{11}{10}, \frac{21}{100}, \frac{21}{2000}$
e_3	$\frac{9}{4}, \frac{673}{200}, \frac{8421}{2000}$	$\frac{1}{4}, \frac{53}{200}, \frac{21}{2000}$

Pair blocks. Let $w_a = e^{i(\chi+\vartheta)}$ and $w_b = e^{i(\chi-\vartheta)}$. The terms of f_p carry $w_a + \bar{w}_b = 2\cos\chi\, e^{i\vartheta}$ or its conjugate, and c_p carries $w_a\bar{w}_b = e^{2i\vartheta}$, so the Hermitian matrix $M(m,\vartheta)$ of $\mathscr{F}$, $m = \cos\chi$, is affine in m with $M(-1,\vartheta) = M(1,\vartheta+\pi)$, and $m = 1$ suffices. For the fixed link e_2, in the variables (z_1, z_3, z_4) and with $c = \cos\vartheta$,

$$M = \begin{pmatrix} \frac{21}{5} + 2c & -2e^{i\vartheta} & -\frac{9}{5} - 2e^{i\vartheta} \\ -2e^{-i\vartheta} & \frac{21}{5} + 2c & \frac{9}{5} + 2e^{-i\vartheta} \\ -\frac{9}{5} - 2e^{-i\vartheta} & \frac{9}{5} + 2e^{i\vartheta} & \frac{5}{2} + 2c \end{pmatrix}.$$

With $u = \cos^2(\vartheta/2) \in [0,1]$ its leading principal minors are

$$\frac{20u+11}{5}, \quad \frac{(20u+1)(20u+21)}{25}, \quad \frac{400u(40u^2-23u+4)+21}{250},$$

and $40u^2 - 23u + 4$ has negative discriminant $529 - 640$. The other positions give the same determinant and the leading minors $\frac{20u+11}{5}$ or $\frac{8u+1}{2}$ and $\frac{800u^2-180u+53}{50}$, whose discriminant $32400 - 169600$ is negative. Every block is positive definite.

The inequality is not tight. In floating point the least eigenvalue of the whole form over $W \in \mathrm{U}(N)$, $N \le 4$, is 0.0149, which is also the least eigenvalue of the exact blocks; this slack is what the choice of the weights $\frac{21}{10}$, $\frac{5}{4}$, $\frac{9}{10}$ trades against the constant $\frac{29}{2}$.

Appendix E: The certified computation

This appendix describes the computation of Remark 36, which no result of the paper uses.

Truncation. For a corner b of a box the moments of Lemma 34 are derivatives of Z at $(b, \bar{b})$. The truncated series is $\mathbb{E}_H[A_t(UD)B_t(\bar{U}D')]$ with A_t the part of $\sum_\lambda c_\lambda s_\lambda(UD)$ over partitions with $\lambda_1 - \lambda_3 \le 34$ and $\lambda_3 \le 11$, and B_t likewise. Every omitted partition has $|\lambda| > 34$, because $3 \cdot 12 > 34$, and the omitted part is at most $T(\|D\|) = \sum_{n>34}(\frac{3}{2}\|D\|)^n/n!$ in modulus. On the polydisc $|D_{ij} - b_{ij}| \le \rho$, $\|D\| \le \|b\| + 3\rho$, and Cauchy's estimate bounds the α-th derivative of the omitted part by $\alpha!\rho^{-|\alpha|}T(\|b\| + 3\rho)$. The derivatives of the full exponential are at most $2^{-|\alpha|}e^{\nu/2}$ with $\nu = \sum_k |b_k|$, since $|\mathrm{Tr}(Ub)| \le \sum_k |b_k|$ for diagonal b. With $\varepsilon_\alpha = 2^{|\alpha|}\alpha!\rho^{-|\alpha|}T(\|b\| + 3\rho)$ the truncated moment differs from the true one by at most $\varepsilon_\alpha e^{\nu/2} + (e^{\nu/2} + \varepsilon_\alpha)\varepsilon_\beta$, and ρ is chosen to minimise $\rho^{-6}T$. The largest error bound of an unnormalised moment at any corner is 1.7×10^{-11}.

Weight zero. Conjugation by the diagonal torus T preserves Haar measure and $\mathrm{Re}\,\mathrm{Tr}(Ub)$ for diagonal b, and multiplies u_{ij} by $t_i t_j^{-1}$; a moment of a monomial on which T acts by a nontrivial character is therefore 0, and only 8278 of the 134596 monomials of degree at most six have the trivial character. The functions $\mathrm{Tr}(X)$, $e_2(X)$, $\det X$ of $X = DD'^{\top}$ and $\det D$, $\det D'$ are invariant under the same action on the variables, so after these are formed in the full ring of Taylor polynomials of degree six in the 18 variables, the series is computed in the subring of weight-zero monomials.

Rounding. The truncated series is a Taylor polynomial of total degree six in the 36 real directions of $(D - b, D' - \bar{b})$, computed in double precision. Every operation is repeated on absolute values and carries an operation count n, so that the computed coefficient differs from the exact one by at most $\gamma_n = nu/(1-nu)$, $u = 2^{-53}$, times the absolute-value coefficient [62, Sec. 3.1]. With C in $2^{-30}\mathbb{Z}$, the polynomials $2^{31}q_h$, $X_a(2^{31}q_h)$, $12 \cdot 2^{31}(\varphi_h + \Delta q_h)$ and $12 \cdot 2^{31}\langle \nabla\varphi_{F_k}, \nabla q_h\rangle$ for the coordinate vectors h have coefficients in $\mathbb{Z}[i]$, and every partial sum in their computation is bounded, through the same operations on absolute values, by less than 2^{50}; they are therefore computed exactly. Normalisation by Z, the products with the corner coordinates, the quadratic forms and their error matrices are

formed with the same bookkeeping, and the largest eigenvalues are bounded in ball arithmetic [70].

Separation. The functional c of a discarded box is the dual solution of a linear program for the gauge of the centre of the box with respect to a 72×72 grid of T. Only the final inequality is used: $\max_T \ell$ is bounded by its maximum over a 512×512 grid plus $\frac{1}{2}H\delta^2/2$, with $\delta = 2\pi/512$ and $H = |c_1|+|c_2|+2|c_3|$ a bound for its Hessian on the torus, and the inequality is checked with relative margin 10^{-12}.

Appendix F: The transfer matrix

This appendix recalls the Osterwalder–Schrader reconstruction for the lattice [88–90] in the form used for the gaps, following our earlier paper [19].

Let ω be a state on the gauge-invariant local functions of $\mathbb{Z}^4$ that is invariant under translations and the hypercubic group and reflection positive for reflections in planes through sites and in planes bisecting links; by Proposition 11(iii)–(iv) and its analogues in the other theorems, the states of this paper are of this kind. The reconstruction gives a Hilbert space with a vector Ω, a map $F\mapsto \Psi_F$ with dense range, and a self-adjoint $\mathbb{T}$ with $0\le \mathbb{T}\le 1$, $\mathbb{T}\Omega=\Omega$ and $\langle \Psi_F, \mathbb{T}^t\Psi_H\rangle = \omega(\vartheta(\bar F)\tau_t H)$, where ϑ is the time reflection and τ_t the time translation.[25]

Proposition 351. *For $m>0$ the following are equivalent:*

(a) *the spectrum of $\mathbb{T}$ on $\Omega^\perp$ lies in $[0,e^{-m}]$;*

(b) *for every F in a class of gauge-invariant local functions supported at times in a bounded interval and dense in $L^2(\omega)$ on each such support, there is $C_F<\infty$ with $0\le \omega(\vartheta(\bar F)\tau_t F)-|\omega(F)|^2\le C_F e^{-mt}$ for all $t\ge 0$.*

Under (a) *one may take $C_F=\|F\|_\infty^2$ for every bounded F.*

Proof. If (a) holds, then

$$\omega(\vartheta(\bar F)\tau_t F)-|\omega(F)|^2 = \langle \Psi'_F, \mathbb{T}^t\Psi'_F\rangle$$

with $\Psi'_F=\Psi_F-\langle\Omega,\Psi_F\rangle\Omega\in\Omega^\perp$; this number lies in $[0,e^{-mt}\|\Psi_F\|^2]$, and $\|\Psi_F\|^2\le\|F\|_\infty^2$. Conversely, let σ be the spectral measure of $\mathbb{T}$ at Ψ'_F and let $\delta>0$. Then

$$\sigma\big((e^{-m+\delta},1]\big)e^{(-m+\delta)t}\le C_F e^{-mt}$$

for all t, so $\sigma((e^{-m+\delta},1])=0$. The spectral projection of $(e^{-m},1]$ restricted to $\Omega^\perp$ therefore annihilates the vectors Ψ'_F of the class, which are dense by the footnote above, and it vanishes. □

[25] Ψ_F is the class of F in the completion of the positive-time local functions for the form $\langle F,H\rangle=\omega(\vartheta(\bar F)H)$, which is positive semidefinite by reflection positivity. Since $|\omega(\vartheta(\bar F)F)|\le\omega(|F|^2)$ by the Cauchy–Schwarz inequality and the invariance of ω under ϑ, functions that are close in $L^2(\omega)$ give close vectors.

The class in (b) cannot be replaced by one observable. On $\mathbb{C}^3$ let $\mathbb{T}=\mathrm{diag}(1,e^{-m},e^{-m/2})$, $\Omega=e_1$ and $\Psi'_F=e_2$: the function in (b) is e^{-mt}, while the spectrum on $\Omega^\perp$ reaches $e^{-m/2}$. The decay of one two-point function places only the spectral measure of Ψ'_F in $[0,e^{-m}]$, and the gap follows when such vectors are dense in $\Omega^\perp$, as in (b) and in Theorem 44. Proposition 8.8 of [46] draws the gap on $\Omega^\perp$ from the two-point function of one fixed observable.

Appendix G: Spectral measures and Laplace transforms

This appendix collects the facts on spectral measures that the continuum step uses, with proofs.

Lemma 352. *Let T be a self-adjoint operator on a Hilbert space with $0\le T\le 1$ and let ψ be a unit vector. There is a unique probability measure ν on $[0,1]$ with $\langle\psi,h(T)\psi\rangle=\int h\,d\nu$ for every bounded Borel function h on $[0,1]$. If $T\psi'=\psi'$ implies $\langle\psi',\psi\rangle=0$, then $\nu(\{1\})=0$. If μ is the image of ν under $\lambda\mapsto -c^{-1}\log\lambda\in[0,\infty]$ for some $c>0$, then $\langle\psi,T^{\sigma/c}\psi\rangle=\int_{[0,\infty]}e^{-\sigma E}d\mu(E)$ for $\sigma>0$, with the integrand 0 at $E=\infty$.*

Proof. The spectral theorem [92, Sec. VII.2] gives the measure ν, carried by the spectrum of T, which lies in $[0,1]$; uniqueness holds because the moments $\int\lambda^n d\nu=\langle\psi,T^n\psi\rangle$ determine a measure on a compact interval, polynomials being dense in $C[0,1]$. The spectral projection $P=\mathbf{1}_{\{1\}}(T)$ projects onto the eigenvectors of T with eigenvalue 1, so $\nu(\{1\})=\|P\psi\|^2=\langle P\psi,\psi\rangle=0$ under the hypothesis. The last identity is the change of variables $\lambda^{\sigma/c}=e^{-\sigma E}$ for $\lambda=e^{-cE}$, with $\lambda=0$ corresponding to $E=\infty$. □

Lemma 353. *Every sequence of probability measures on a compact metric space K has a subsequence that converges weakly, that is, against every continuous function, to a probability measure on K.*

Proof. $C(K)$ is separable; let (h_j) be dense. The numbers $\int h_j\,d\mu_k$ are bounded by $\|h_j\|_\infty$, so a diagonal subsequence makes all of them converge. By density the limit $\Lambda(h)=\lim\int h\,d\mu_k$ exists for every $h\in C(K)$; Λ is linear, positive and $\Lambda(1)=1$, so by the Riesz representation theorem it is integration against a probability measure. □

Lemma 354. *Let μ be a probability measure on $[0,\infty]$ and $\rho(\sigma)=\int e^{-\sigma E}d\mu(E)$ for $\sigma>0$.*

(a) *ρ is nonincreasing and log-convex: $\rho(\sigma)^2\le\rho(\sigma-s)\rho(\sigma+s)$ for $0<s<\sigma$.*

(b) *$\rho(\sigma)>0$ for one $\sigma>0$ if and only if $\mu([0,\infty))>0$, and then $\rho(\sigma)>0$ for every $\sigma>0$.*

(c) *$\rho(\sigma)\to\mu([0,\infty))$ as $\sigma\downarrow 0$.*

(d) *If μ is carried by $[m,\infty]$, then $\rho(\sigma)\le e^{-m\sigma}$.*

(e) *For $M>0$, $\mu([0,M])\ge\rho(\sigma)-e^{-\sigma M}$.*

(f) *If $\mu_k \to \mu$ weakly on $[0,\infty]$, then $\int e^{-\sigma E} d\mu_k \to \rho(\sigma)$ for every $\sigma > 0$.*

Proof. (a) The integrand decreases in σ. By the Cauchy–Schwarz inequality,

$$\Big(\int e^{-\sigma E} d\mu\Big)^2 \le \int e^{-(\sigma-s)E} d\mu \int e^{-(\sigma+s)E} d\mu.$$

(b) The integrand is positive exactly on $[0,\infty)$. (c) The integrand increases to $\mathbf{1}_{[0,\infty)}$ as $\sigma \downarrow 0$, and monotone convergence applies. (d) The integrand is at most $e^{-m\sigma}$ on $[m,\infty]$. (e) Split the integral at M: it is at most $\mu([0,M]) + e^{-\sigma M}$. (f) For $\sigma > 0$ the function $E \mapsto e^{-\sigma E}$, extended by 0 at ∞, is continuous on $[0,\infty]$. □

By (b) and (e), a lower bound $\rho(\sigma) \ge c > 0$ at one σ places mass at least $c/2$ on the bounded interval $[0, \sigma^{-1}\log(2/c)]$; for the measures μ_k of Theorem 46 a bound uniform in k therefore makes them tight on $[0,\infty)$ with a fixed fraction of their mass.

Part (b) also excludes a positive Laplace representation for a spectral cutoff of compact support. If $\chi(\lambda) = \int_{[0,\infty)} e^{-s\lambda} d\nu(s)$ with ν a positive measure and $\chi(0) = \nu([0,\infty)) > 0$, then (b), applied to $\nu/\chi(0)$, gives $\chi(\lambda) > 0$ for every $\lambda > 0$, so χ vanishes nowhere. The cutoff χ_σ of [46, Sec. 9, Eqs. (9.11), (9.12)] equals 1 on $[0,\sigma]$ and 0 on $[2\sigma,\infty)$ and is asserted to have such a representation.

Appendix H: Spectral gaps on cyclic subspaces

The local form $\mathrm{BP}_{\mathrm{loc}}$ of Definition 81 asks for a spectral gap of the cube chain on the cyclic subspace of each local function. This appendix records the elementary facts about such gaps that Sec. IX H uses, for a self-adjoint operator K on a real or complex Hilbert space H with $0 \le K \le I$. For $g \in H$ let ν_g be its spectral measure, so that $\langle g, \phi(K) g\rangle = \int \phi \, d\nu_g$ for bounded Borel ϕ on $[0,1]$, and for a Borel set B let $E(B)$ be the spectral projection.

Lemma 355. *Let $g \in H$ and $0 < \varepsilon < 1$. The following are equivalent.*

(i) $\|K^t g\| \le (1-\varepsilon)^t \|g\|$ *for every integer* $t \ge 0$.

(ii) $\nu_g((1-\varepsilon, 1]) = 0$.

(iii) $E((1-\varepsilon,1])g = 0$.

Under these conditions $\langle g', (I-K)g'\rangle \ge \varepsilon \|g'\|^2$ for every g' in the closed cyclic subspace generated by g.

Proof. $\|E(B)g\|^2 = \nu_g(B)$, so (ii) and (iii) are equivalent. Under (ii), $\|K^t g\|^2 = \int \lambda^{2t} d\nu_g \le (1-\varepsilon)^{2t} \nu_g([0,1])$, which is (i). If (ii) fails, there is $\eta > 0$ with $m = \nu_g((1-\varepsilon+\eta, 1]) > 0$, since these sets increase to $(1-\varepsilon,1]$ as $\eta \downarrow 0$; then $\|K^t g\|^2 \ge m(1-\varepsilon+\eta)^{2t}$, which exceeds $(1-\varepsilon)^{2t}\|g\|^2$ for large t, and (i) fails. The closed cyclic subspace of g is the closure of $\{\phi(K)g\}$ over polynomials ϕ; it lies in the range of $E([0,1-\varepsilon])$, since $E((1-\varepsilon,1])$ commutes with K, and on that range $I - K \ge \varepsilon$. □

Lemma 356. *Let $F \subset H$ and $0 < \varepsilon < 1$. Every $g \in F$ satisfies the conditions of Lemma 355 if and only if the range of $E((1-\varepsilon,1])$ is orthogonal to F, and then every vector of the closed linear span of F satisfies them. In that case, for every g in that span and every $h \in H$,*

$$|\langle g, h\rangle|^2 \le \|g\|^2 \, \frac{\langle h, (I-K)h\rangle}{\varepsilon}.$$

Proof. $E = E((1-\varepsilon,1])$ is an orthogonal projection, so $Eg = 0$ if and only if g is orthogonal to the range of E; the set of such g is a closed subspace. For g in it, g and $h' = (I-E)h$ lie in the range R of $I-E$, which is invariant under K and on which $I - K \ge \varepsilon$. Let T be the restriction of $(I-K)^{1/2}$ to R; it is invertible with $\|T^{-1}\| \le \varepsilon^{-1/2}$. Then $\langle g, h\rangle = \langle g, h'\rangle = \langle T^{-1}g, Th'\rangle$, and

$$\begin{aligned} |\langle g, h\rangle|^2 &\le \frac{\|g\|^2}{\varepsilon} \, \langle h', (I-K)h'\rangle \\ &\le \frac{\|g\|^2}{\varepsilon} \, \langle h, (I-K)h\rangle, \end{aligned}$$

the last step because E commutes with K and $\langle Eh, (I-K)Eh\rangle \ge 0$. □

So $\mathrm{BP}_{\mathrm{loc}}(n, \varpi)$ on Λ_L says that the spectral projection of the cube chain for $(1 - \varpi n^4/L^4, 1]$ annihilates the closed span of the local gauge-invariant functions of mean zero, and the inequality of Theorem 82(iv) is Lemma 356 for that span. The span is not an algebra: a product of two local functions in distant boxes is not local, and $\mathrm{BP}_{\mathrm{loc}}$ says nothing about it. A Poincaré inequality for one function bounds the mean of $1-\lambda$ under its spectral measure and does not give a gap on its cyclic subspace.

Lemma 357. *For all $0 < \eta < \varepsilon < 1$ and $0 < \theta < \frac{\pi}{2}$ there are H, K and g with $\|g\| = 1$, $\langle g, (I-K)g\rangle \ge \cos^2\theta$, and $\|K^t g\| \ge (1-\eta)^t \sin\theta$ for every integer $t \ge 0$; so g fails the conditions of Lemma 355.*

Proof. Take $H = \mathbb{R}^2$, $K = \mathrm{diag}(0, 1-\eta)$ and $g = (\cos\theta, \sin\theta)$. Then $\langle g, (I-K)g\rangle = \cos^2\theta + \eta\sin^2\theta$ and $K^t g = (0, (1-\eta)^t \sin\theta)$ for $t \ge 1$, and $\nu_g(\{1-\eta\}) = \sin^2\theta > 0$ with $1-\eta \in (1-\varepsilon, 1]$. □

With θ close to 0 the Rayleigh quotient of g is close to one, the best possible value, while a fraction $\sin^2\theta$ of its spectral mass relaxes at the rate η only. This is why Proposition 85, a Poincaré inequality for each function of a finite set of links, does not give $\mathrm{BP}_{\mathrm{loc}}$.

Appendix I: The character expansion

Let G be a compact connected Lie group with normalised Haar measure dU. For an irreducible unitary representation ρ_λ of dimension d_λ write $\chi_\lambda = \mathrm{Tr}\,\rho_\lambda$. The characters form an orthonormal basis of the square-integrable class functions (Peter–Weyl and Schur orthogonality [28, Chs. 2–4]).

Lemma 358. *Let $w \in L^1(G)$ be a class function with $w(U^{-1}) = w(U)$, and put*

$$\hat{c}_\lambda = d_\lambda^{-1} \int w\, \chi_\lambda \, dU.$$

Then $\int w(U)\rho_\lambda(U)\,dU = \hat{c}_\lambda I$, and $\int w(U)\chi_\lambda(U^{-1}V)\,dU = \hat{c}_\lambda \chi_\lambda(V)$.

Proof. $M = \int w(U)\rho_\lambda(U)dU$ commutes with every $\rho_\lambda(V)$, since $w(VUV^{-1}) = w(U)$ and Haar measure is invariant; by Schur's lemma $M = cI$, and taking traces gives $c = \hat{c}_\lambda$. Then $\int w(U)\chi_\lambda(U^{-1}V)dU = \mathrm{Tr}(\int w(U)\rho_\lambda(U^{-1})dU\,\rho_\lambda(V))$, and the substitution $U \mapsto U^{-1}$ with $w(U^{-1}) = w(U)$ turns the integral into M. □

Lemma 359. *Let $G = \mathrm{SU}(N)$ and $\beta > 0$, and let*

$$w_\beta(U) = e^{\frac{\beta}{N}\mathrm{Re\,Tr}\,U} \Big/ \int e^{\frac{\beta}{N}\mathrm{Re\,Tr}\,V} dV.$$

Then $\hat{c}_\lambda(\beta) = d_\lambda^{-1}\int w_\beta \chi_\lambda\, dU$ is real, $0 \le \hat{c}_\lambda(\beta) \le 1$, and $\hat{c}_\lambda(\beta) < 1$ for $\lambda \neq 0$. Moreover

$$\hat{c}_\lambda(\beta) = 1 - \frac{NC_2(\lambda)}{\beta} + O(\beta^{-2}) \qquad (\beta \to \infty),$$

where $-\sum_a \rho_\lambda(T_a)^2 = 2C_2(\lambda)I$ for a basis (T_a) of $\mathfrak{su}(N)$ orthonormal for $|X|^2 = -\mathrm{Tr}X^2$, so that $C_2 = (N^2-1)/(2N)$ on the fundamental representation.

Proof. Since $w_\beta(\bar{U}) = w_\beta(U)$ and $\chi_\lambda(\bar{U}) = \overline{\chi_\lambda(U)}$, and complex conjugation preserves Haar measure, $\int w_\beta \chi_\lambda = \int w_\beta \bar{\chi}_\lambda$, which is the complex conjugate of the left side; so $\hat{c}_\lambda$ is real. Next, w_β is a multiple of $\exp(\frac{\beta}{2N}(\chi_F + \bar{\chi}_F)) = \sum_m \frac{1}{m!}(\frac{\beta}{2N})^m(\chi_F + \bar{\chi}_F)^m$, and each power is the character of a representation, hence a sum of irreducible characters with non-negative integer coefficients. By orthonormality $\int w_\beta \bar{\chi}_\lambda \ge 0$, so $\hat{c}_\lambda \ge 0$. As $|\chi_\lambda(U)| \le d_\lambda$, with $\mathrm{Re}\,\chi_\lambda(U) = d_\lambda$ only where $\rho_\lambda(U) = I$, a proper closed subgroup when $\lambda \neq 0$ and so a Haar null set, and $w_\beta > 0$ everywhere, $\hat{c}_\lambda = d_\lambda^{-1}\int w_\beta \,\mathrm{Re}\,\chi_\lambda < 1$ for $\lambda \neq 0$.

For the expansion, $\mathrm{Re\,Tr}\,U \le N$ with equality only at $U = I$, so outside a neighbourhood $\{U = e^X : |X| < \delta\}$ both integrals $\int e^{\frac{\beta}{N}\mathrm{Re\,Tr}\,U}\chi_\lambda$ and $\int e^{\frac{\beta}{N}\mathrm{Re\,Tr}\,U}$ are smaller than their values near I by a factor $e^{-c\beta}$. Near I write $U = e^X$; the Haar density in these coordinates is $j(X) = 1 + O(|X|^2)$, even in X. For antihermitian X, $\mathrm{Tr}X^3$ is imaginary, so

$$\tfrac{1}{N}\mathrm{Re\,Tr}\,e^X = 1 - \tfrac{1}{2N}|X|^2 + r(X), \qquad 0 \le r(X) \le C|X|^4,$$

and $d_\lambda^{-1}\chi_\lambda(e^X) = 1 + d_\lambda^{-1}\mathrm{Tr}\,\rho_\lambda(X) + \frac{1}{2d_\lambda}\mathrm{Tr}\,\rho_\lambda(X)^2 + c_3(X) + O(|X|^4)$, with c_3 cubic and $\mathrm{Tr}\,\rho_\lambda(X) = 0$, since $\mathfrak{su}(N)$ is perfect. Let $\langle\cdot\rangle$ denote the average against the Gaussian weight $e^{-\beta|X|^2/(2N)}$, under which the coordinates x_a of $X = \sum x_a T_a$ are independent with variance N/β. With $h = j\,e^{\beta r}$,

$$\begin{aligned}\hat{c}_\lambda(\beta) &= \frac{\langle d_\lambda^{-1}\chi_\lambda(e^X)\,h\rangle}{\langle h\rangle} + O(e^{-c\beta}) \\ &= 1 + \frac{\langle \frac{1}{2d_\lambda}\mathrm{Tr}\,\rho_\lambda(X)^2 h\rangle}{\langle h\rangle} + \frac{\langle (c_3(X) + O(|X|^4))h\rangle}{\langle h\rangle},\end{aligned}$$

the averages being taken over $|X| < \delta$. Under $\langle\cdot\rangle$, $|X|^{2m}$ has mean of order β^{-m} and $\beta|r(X)| \le C\beta|X|^4$ has mean of order β^{-1}, so $h = 1 + O(\beta^{-1})$ in every moment. The weight h is even in X, since $\mathrm{Tr}X^k$ is imaginary for odd k and j is even, so the odd term c_3 contributes nothing; the quartic remainder and the correlation of the quadratic term with $h-1$ contribute $O(\beta^{-2})$. Finally $\langle \mathrm{Tr}\,\rho_\lambda(X)^2\rangle = \frac{N}{\beta}\sum_a \mathrm{Tr}\,\rho_\lambda(T_a)^2 = -\frac{2NC_2(\lambda)d_\lambda}{\beta}$, which gives the expansion. For $\mathrm{SU}(2)$ and the representation of spin j the integrals are explicit, $\hat{c}_j(\beta) = I_{2j+1}(\beta)/I_1(\beta)$ with the modified Bessel functions I_n, in agreement with $C_2 = j(j+1)$. □

Appendix J: Haar measure on SU(2)

Write $\mathbf{i},\mathbf{j},\mathbf{k}$ for the matrices $i\sigma_3, i\sigma_2, i\sigma_1$. The map $(a,b) \mapsto aI + b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ identifies $\mathbb{R}^4$ with the real span of $\mathrm{SU}(2)$, and $\mathrm{SU}(2)$ with the unit sphere S^3; the Hilbert–Schmidt norm of $aI + b\cdot(\mathbf{i},\mathbf{j},\mathbf{k})$ is $\sqrt{2}\,(a^2 + |b|^2)^{1/2}$. Left and right multiplication by elements of $\mathrm{SU}(2)$ preserve this norm, so they act on S^3 by rotations, and the normalised Haar measure is the normalised surface measure of S^3. Every $U \in \mathrm{SU}(2)$ is conjugate to $\mathrm{diag}(e^{i\theta}, e^{-i\theta})$ for one $\theta \in [0,\pi]$, its eigenangle, and $\frac{1}{2}\mathrm{Tr}\,U = \cos\theta$ is the inner product of U and I in $\mathbb{R}^4$; so θ is the angle between U and I on S^3.

Lemma 360. (i) *The eigenangle θ has Haar density $\frac{2}{\pi}\sin^2\theta$ on $[0,\pi]$.*

(ii) $\mathrm{Haar}\{\theta \le t\} = \frac{1}{\pi}(t - \sin t\cos t) \le \frac{2t^3}{3\pi}$ *for $0 \le t \le \pi$.*

(iii) $|U - I| = 2\sqrt{2}\sin\frac{\theta}{2}$, *and the distance of the bi-invariant metric induced by the Hilbert–Schmidt inner product is $d(U,I) = \sqrt{2}\,\theta$. With this metric* $\mathrm{SU}(2)$ *is the round sphere of radius $\sqrt{2}$.*

Proof. (i) The unit vectors at angle θ from I form a two-sphere of radius $\sin\theta$ and area $4\pi\sin^2\theta$, and S^3 has area $2\pi^2$.

(ii) $\int_0^t \sin^2\theta\,d\theta = \frac{1}{2}(t - \sin t\cos t)$, and $\sin^2\theta \le \theta^2$.

(iii) The eigenvalues of $U - I$ are $e^{\pm i\theta} - 1$, so $|U-I|^2 = 2|e^{i\theta} - 1|^2 = 8\sin^2\frac{\theta}{2}$. By the first paragraph the metric is that of the sphere of radius $\sqrt{2}$, on which the points at angle $\theta \le \pi$ are at distance $\sqrt{2}\,\theta$. □

Appendix K: The single-link law of SU(2)

This appendix proves the sharp quarter, $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$ for the one-link law of $\mathrm{SU}(2)$, which Theorems 1 and 3 use, and the exact value at $b = 0$, which Theorem 2 uses. The argument is that of our earlier paper [19]; it is given in full so that the present paper is self-contained.

1. The von Mises–Fisher law

Identify $\mathrm{SU}(2)$ with the unit quaternions, that is with the unit sphere $S^3 \subset \mathbb{R}^4$, so that $\mathrm{Re\,Tr}\,U = 2x_0(U)$ and $\mathrm{Re\,Tr}(UV) = 2\langle u, \bar{v}\rangle$ for the Euclidean inner product of $\mathbb{R}^4$ and quaternion

conjugation $v \mapsto \bar v$. Fix a link e and the field off e. Every plaquette through e can be rotated cyclically so that its trace reads $\operatorname{Re}\operatorname{Tr}(Q_e A_p)$ with $A_p \in \mathrm{SU}(2)$, the staple of e in p. The action therefore depends on Q_e only through

$$N\beta \sum_{p \ni e} \operatorname{Re}\operatorname{Tr}(Q_e A_p) = \langle x, b\rangle, \qquad b = 4\beta \sum_{p \ni e} \bar A_p \in \mathbb{R}^4,$$

with $x = Q_e \in S^3$. Since there are six staples, each of norm one,

$$\kappa := |b| \le y := 24|\beta|. \tag{K1}$$

The conditional law of Q_e is the von Mises–Fisher law[26]

$$\nu_b(dx) = Z(\kappa)^{-1} e^{\langle x, b\rangle}\, \sigma(dx) \tag{K2}$$

on the unit sphere, with σ the normalised surface measure. We use coordinates in which $b = \kappa e_1$. The density of $t = x_1$ is proportional to $(1-t^2)^{1/2} e^{\kappa t}$ on $[-1,1]$, so $Z(\kappa) = 2I_1(\kappa)/\kappa$ with the modified Bessel function I_1, normalised so that $Z(0) = 1$.

Lemma 361. *For every $\kappa \ge 0$ the law ν_b on the unit sphere S^3 satisfies the Poincaré inequality $\operatorname{Var}_{\nu_b}(g) \le \frac{1}{3}\int |\nabla g|^2 d\nu_b$, and the constant $\frac{1}{3}$ is attained at $\kappa = 0$.*

Proof. In the polar angle $\theta \in (0,\pi)$ from e_1 the generator of the reversible diffusion of ν_b is $L = \partial_\theta^2 + b(\theta)\partial_\theta + \sin^{-2}\theta\, \Delta_{S^2}$ with $b(\theta) = 2\cot\theta - \kappa\sin\theta$. On spherical harmonics of degree ℓ on S^2 it acts as a Sturm–Liouville operator L_ℓ with potential $\ell(\ell+1)\sin^{-2}\theta$. A ground state comparison of Barta's type [15] with the trial function $\psi = \sin\theta\, e^{a\kappa\cos\theta}$ gives $-L_1\psi \ge 3\psi$ for $a = -\frac{1}{5}$, and the same trial function with $a = -\frac{2}{5}$ gives the bound 3 for the derivatives of zonal eigenfunctions; Appendix L carries out both computations. Hence every nonzero eigenvalue of $-L$ is at least 3. At $\kappa = 0$ the first eigenvalue of the Laplacian of the unit S^3 is 3. □

2. Moments

Let $A(\kappa) = \mathbb{E}_{\nu_b} x_1 = Z'(\kappa)/Z(\kappa) = I_2(\kappa)/I_1(\kappa)$.

Lemma 362. *For every $\kappa > 0$, in coordinates with $b = \kappa e_1$,*

(i) $\mathbb{E}x_2^2 = \mathbb{E}x_3^2 = \mathbb{E}x_4^2 = A/\kappa$, $\mathbb{E}(1-x_1^2) = 3A/\kappa$, *the covariance matrix of x is* $\operatorname{diag}(A', A/\kappa, A/\kappa, A/\kappa)$, *and* $A' = 1 - A^2 - 3A/\kappa$;

(ii) $$\frac{\kappa}{2+\sqrt{\kappa^2+4}} \le A(\kappa) \le \frac{\kappa}{1+\sqrt{\kappa^2+9}};$$

(iii) $A' \le A/\kappa \le \big(1+\sqrt{\kappa^2+9}\big)^{-1} \le \frac{1}{4}$;

(iv) $A(\kappa) \ge \kappa\big/\big(\frac{3}{2} + \sqrt{\kappa^2 + 25/4}\big)$.

Proof. (i) Integrating by parts against $\frac{d}{dt}(1-t^2)^{3/2} = -3t(1-t^2)^{1/2}$ gives $\kappa\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, and the three coordinates x_2, x_3, x_4 are exchangeable under ν_b, with $\sum_i x_i^2 = 1$. The off-diagonal covariances vanish by the reflections $x_i \mapsto -x_i$, $i \ge 2$, and $A' = \operatorname{Var} x_1$ because $A = (\log Z)'$. (ii) By (i), A solves the Riccati equation $A' = 1 - A^2 - 3A/\kappa$ with $A(\kappa) = \kappa/4 + O(\kappa^3)$. Appendix L shows that the left side of (ii) is a subsolution and the right side a supersolution of this equation, with the same behaviour at 0, and concludes by comparison. (iii) The middle inequality is (ii). For the first, put $s = \sqrt{\kappa^2+4}$. By (ii), $4A/\kappa \ge 4/(2+s)$ and $1 - A^2 \le 1 - \kappa^2/(2+s)^2$, and

$$1 - \frac{\kappa^2}{(2+s)^2} \le \frac{4}{2+s} \iff (2+s)^2 - \kappa^2 \le 4(2+s)$$
$$\iff s^2 - \kappa^2 \le 4,$$

which holds with equality. So $A' = 1 - A^2 - 3A/\kappa \le A/\kappa$. (iv) is the comparison of (ii) with a third barrier, which improves the lower bound of (ii); it is carried out in the same appendix. □

3. The Lipschitz rate of the family

Let δ be the chordal metric $\delta(x,x') = |x - x'|$ on S^3 and W_1 the Kantorovich distance for δ. For $c \in \mathbb{R}^4$ and a unit vector $h \in \mathbb{R}^4$ put

$$L(c,h) = \sup_g \operatorname{Cov}_{\nu_c}\big(g, \langle h, x\rangle\big), \tag{K3}$$

the supremum over 1-Lipschitz functions g for δ. Since $\frac{d}{dt}\nu_{c+th}(g) = \operatorname{Cov}_{\nu_{c+th}}(g, \langle h, x\rangle)$, the Kantorovich–Rubinstein duality gives, for all b, b',

$$W_1(\nu_b, \nu_{b'}) \le |b - b'| \sup_{c \in [b,b']} L\Big(c, \frac{b'-b}{|b'-b|}\Big). \tag{K4}$$

By rotation invariance $L(c,h)$ depends only on $\kappa = |c|$ and on the angle φ between h and $c^\perp$, and we write $L(\kappa, \varphi)$.

Lemma 363. *For all c and h, $L(c,h) \le 1/(2\sqrt{3})$. Moreover:*

(i) *at $c = 0$, $L(0,h) = \frac{1}{4}$ for every h;*

(ii) *if $h \perp c$, then $L(c,h) = \mathbb{E}_{\nu_c} x_2^2 = A(\kappa)/\kappa \le \frac{1}{4}$;*

(iii) *if $\kappa \ge \sqrt{88}/3 \approx 3.127$, then $L(c,h) \le \frac{1}{4}$ for every h.*

Proof. A 1-Lipschitz function for δ is 1-Lipschitz for the geodesic distance, which is larger, so $|\nabla g| \le 1$ almost everywhere, and Lemma 361 gives $\operatorname{Var} g \le \frac{1}{3}$. By Lemma 362(i), (iii) the largest eigenvalue of the covariance of x is $A/\kappa \le \frac{1}{4}$, so $\operatorname{Var}\langle h, x\rangle \le A/\kappa$. The Cauchy–Schwarz inequality gives

$$L(c,h) \le \big(\tfrac{1}{3}A/\kappa\big)^{1/2} \le \Big(3\big(1+\sqrt{\kappa^2+9}\big)\Big)^{-1/2} \le \frac{1}{2\sqrt{3}}. \tag{K5}$$

[26] On the sphere $S^{p-1} \subset \mathbb{R}^p$ the von Mises–Fisher law with mean direction μ and concentration κ has density proportional to $e^{\kappa\langle x,\mu\rangle}$ with respect to the surface measure, and its normalising constant is $(2\pi)^{p/2} I_{p/2-1}(\kappa)/\kappa^{p/2-1}$; it is the maximum entropy law with prescribed mean resultant, and it plays on the sphere the role of the Gaussian in $\mathbb{R}^p$. For $p = 4$ the Bessel index is 1. The mean resultant length $\mathbb{E}\langle x,\mu\rangle = I_{p/2}(\kappa)/I_{p/2-1}(\kappa)$ is the function A of Lemma 362. See Mardia and Jupp [81], Ch. 9, for these facts and for the tangent-normal decomposition used in Appendix L.

(i) and (ii) are reflection arguments. Let R be the reflection in the hyperplane $h^{\perp}$, and suppose ν_c is R-invariant, which is the case when $c=0$ or $h\perp c$. Then $\mathbb{E}\langle h,x\rangle=0$ and

$$\begin{aligned}\operatorname{Cov}(g,\langle h,x\rangle)&=\mathbb{E}\big[(g(x)-g(Rx))\langle h,x\rangle;\ \langle h,x\rangle>0\big]\\&\le\mathbb{E}\big[2\langle h,x\rangle^2;\ \langle h,x\rangle>0\big]=\mathbb{E}\langle h,x\rangle^2,\end{aligned}$$

since $\delta(x,Rx)=2|\langle h,x\rangle|$. Equality holds for $g=\langle h,x\rangle$, which is 1-Lipschitz. At $c=0$, $\mathbb{E}\langle h,x\rangle^2=\frac14$; for $h\perp c$, $\mathbb{E}\langle h,x\rangle^2=A/\kappa$. (iii) By (K5), $L\le\frac14$ as soon as $1+\sqrt{\kappa^2+9}\ge\frac{16}{3}$, that is $\kappa^2\ge\frac{88}{9}$. □

The reflection settles the directions $h\perp c$ and the point $c=0$; for an oblique h the reflection no longer preserves ν_c, and the Cauchy–Schwarz bound (K5) is too weak for $\kappa<\sqrt{88}/3$. The next theorem closes this case.

Theorem 364 (The sharp quarter)**.** *For every $c\in\mathbb{R}^4$ and every unit vector $h\in\mathbb{R}^4$,*

$$L(c,h)\le\tfrac14,$$

with equality at $c=0$. Equivalently, $W_1(\nu_b,\nu_{b'})\le\frac14|b-b'|$ for all $b,b'\in\mathbb{R}^4$, and the constant $\frac14$ is the best possible.

The equivalence is (K4) in one direction and the derivative formula above it in the other; the sharpness is Lemma 363(i). The metric matters.[27] The proof has four steps. A Lipschitz function on S^3 extends to the unit ball $B\subset\mathbb{R}^4$, and Green's formula in B bounds the covariance by a negative Sobolev norm of the density (Lemma 365). This norm splits into a zonal and a transverse part, both explicit series in modified Bessel functions (Lemma 366). The transverse part is at most $\frac14$ because a certain power series has nonnegative coefficients (Proposition 368), and the zonal part is at most $\frac14$ by a Riccati comparison (Proposition 369).

4. Harmonic extension

Let σ be the normalised surface measure of S^3 and H_k the space of spherical harmonics of degree k, the restrictions to S^3 of harmonic homogeneous polynomials of degree k on $\mathbb{R}^4$. The spaces H_k are orthogonal in $L^2(\sigma)$ and span it; we write u_k for the component of u in H_k, and for u with $\int u\,d\sigma=0$ we put

$$\mathscr{Q}(u)=\sum_{k\ge1}\frac{1}{k}\,\|u_k\|^2_{L^2(\sigma)}. \tag{K6}$$

This is $\langle u,\Lambda^{-1}u\rangle$ for the Dirichlet-to-Neumann operator Λ of B, which acts on H_k as multiplication by k.

Lemma 365. *Let u be a real-analytic function on S^3 with $\int u\,d\sigma=0$ and let g be 1-Lipschitz for the chordal metric. Then*

$$\int_{S^3} g\,u\,d\sigma\le\tfrac12\,\mathscr{Q}(u)^{1/2}.$$

Proof. The chordal metric is the Euclidean metric of $\mathbb{R}^4$ restricted to S^3, so McShane's extension $G(y)=\inf_{x\in S^3}\big(g(x)+|x-y|\big)$ is 1-Lipschitz on $\mathbb{R}^4$ and equals g on S^3 [83]; by Rademacher's theorem $|\nabla G|\le1$ almost everywhere. Let $\psi=\sum_{k\ge1}k^{-1}r^k u_k(\omega)$ in polar coordinates $y=r\omega$. Each term is a harmonic polynomial, the series converges with all derivatives on the closed ball because u is real-analytic,[28] and $\partial_r\psi=u$ on S^3. Green's formula for the Lipschitz function G and the smooth harmonic function ψ reads $\int_{S^3}G\,\partial_r\psi\,dS=\int_B\nabla G\cdot\nabla\psi\,dy$. With $|S^3|=2\pi^2$ and $|B|=\pi^2/2$ the Cauchy–Schwarz inequality gives

$$\int g\,u\,d\sigma=\frac{1}{2\pi^2}\int_B\nabla G\cdot\nabla\psi\,dy\le\frac{|B|^{1/2}}{2\pi^2}\Big(\int_B|\nabla\psi|^2dy\Big)^{1/2}.$$

Green's formula once more gives $\int_B|\nabla\psi|^2=\int_{S^3}\psi\,\partial_r\psi\,dS=2\pi^2\sum_k k^{-1}\|u_k\|^2=2\pi^2\mathscr{Q}(u)$, and the right side becomes $\frac12\mathscr{Q}(u)^{1/2}$. □

The lemma is a form of Thomson's principle. By Kantorovich–Rubinstein duality, $\sup_g\int g\,u\,d\sigma$ is the least cost of a flow in $\mathbb{R}^4$ that carries the positive part of $u\,d\sigma$ to its negative part, the cost being the total length travelled by the mass.[29] The harmonic field $\nabla\psi$ is one such flow, through the interior of the ball, and the lemma computes its cost. At $c=0$ the flow is optimal: for $u=\langle h,x\rangle$ one has $\psi=\langle h,y\rangle$, the field $\nabla\psi=h$ is constant, the Cauchy–Schwarz inequality is an equality, and $\mathscr{Q}(u)=\|u\|^2=\frac14$.

5. Spectral decomposition

Fix $c=\kappa e_1$ with $\kappa>0$ and write $h=\cos\varphi\,e_1+\sin\varphi\,e_2$, which is no loss of generality by rotation invariance. Let $\rho=d\nu_c/d\sigma=e^{\kappa x_1}/Z(\kappa)$, $Z(\kappa)=\int e^{\kappa x_1}d\sigma=2I_1(\kappa)/\kappa$, and

$$u=\big(\langle h,x\rangle-\mathbb{E}_{\nu_c}\langle h,x\rangle\big)\rho,$$

so that $\operatorname{Cov}_{\nu_c}(g,\langle h,x\rangle)=\int g\,u\,d\sigma$ and $\int u\,d\sigma=0$. Let U_n be the Chebyshev polynomials of the second kind and put

$$d_n(\kappa)=\frac{(n+1)\,I_{n+1}(\kappa)}{I_1(\kappa)},\qquad n\ge0. \tag{K7}$$

[27] In the geodesic metric of S^3 the statement is false already at $c=0$: the function $\arcsin x_2$ is geodesically 1-Lipschitz and its covariance with x_2 under the uniform law is $\frac{8}{9\pi}\approx0.283>\frac14$. A proof therefore has to use that the chordal distance between two points is shorter than their geodesic distance, that is, it has to see the sphere from the ball it bounds. The proof below does exactly this: it extends a chordally Lipschitz function to the ball and integrates by parts there.

[28] For real-analytic u the norms $\|u_k\|$ decay geometrically in k, and the supremum of a function in H_k is at most $(\dim H_k)^{1/2}=k+1$ times its $L^2(\sigma)$ norm, so the series for ψ and for each of its derivatives converge uniformly on $\bar B$.

[29] This is the Beckmann formulation of the transport problem with cost $|x-y|$: the optimal cost equals the least $\int|V|$ over vector measures V on $\mathbb{R}^4$ with divergence $u\,d\sigma$. Restricting the flow to the closed ball does not change the infimum, since projecting onto $\bar B$ shortens paths. The gradient $\nabla\psi$ on B is one admissible flow. It minimises the energy $\int|V|^2$ rather than the cost $\int|V|$, and the Cauchy–Schwarz inequality converts one into the other at the price of the factor $|B|^{1/2}$.

Lemma 366. *With this notation*

$$\mathscr{Q}(u) = \cos^2\varphi\,\mathscr{Q}_z(\kappa) + \sin^2\varphi\,\mathscr{Q}_x(\kappa),$$

where

$$\mathscr{Q}_z = \sum_{n\ge1} \frac{d_n'(\kappa)^2}{n}, \qquad \mathscr{Q}_x = \frac{1}{3\kappa^2}\sum_{n\ge1}(n+2)\,d_n(\kappa)^2.$$

Proof. The functions $U_n(x_1)$, $n\ge0$, are the zonal harmonics: $U_n(x_1)\in H_n$, and they are orthonormal in $L^2(\sigma)$ (Appendix M). The expansion $e^{\kappa t} = \frac{2}{\kappa}\sum_n (n+1)I_{n+1}(\kappa)U_n(t)$ of the same appendix gives $\rho = \sum_n d_n(\kappa)U_n(x_1)$, with $d_0 = 1$.

Write $u = \cos\varphi\, u_z + \sin\varphi\, u_x$ with $u_z = (x_1 - A)\rho$ and $u_x = x_2\rho$; here $\mathbb{E}_{\nu_c} x_1 = A$ and $\mathbb{E}_{\nu_c} x_2 = 0$. The reflection $x_2 \mapsto -x_2$ maps each H_k to itself, fixes u_z and reverses u_x, so the components of u_z and u_x in each H_k are orthogonal and the cross term in $\mathscr{Q}(u)$ vanishes.

Since $A = Z'/Z$, we have $u_z = \partial_\kappa\rho = \sum_n d_n'(\kappa)U_n(x_1)$, and the component of u_z in H_n is $d_n'U_n(x_1)$, of norm $|d_n'|$. This gives $\mathscr{Q}_z$.

For u_x, let $y(t) = \cos t\, e_1 + \sin t\, e_2$. The law $\nu_{\kappa y(t)}$ has density $\rho_t = e^{\kappa\langle y(t),x\rangle}/Z(\kappa) = \sum_n d_n(\kappa)U_n(\langle y(t),x\rangle)$, and $\partial_t\rho_t|_{t=0} = \kappa x_2\rho$. The function $U_n(\langle y(t),x\rangle)$ lies in H_n for every t, hence so does its derivative $x_2U_n'(x_1)$ at $t=0$, and the component of u_x in H_n is $\kappa^{-1}d_n x_2 U_n'(x_1)$. Its norm follows from the eigenvalue $n(n+2)$ of the Laplacian of S^3 on H_n. For $f = U_n(x_1)$,

$$\begin{aligned} n(n+2) = n(n+2)\|f\|^2 &= \int |\nabla_{S^3} f|^2 d\sigma \\ &= \int U_n'(x_1)^2(1-x_1^2)\,d\sigma, \end{aligned}$$

since the tangential gradient of x_1 has squared length $1-x_1^2$. Given x_1, the point (x_2,x_3,x_4) is uniform on the sphere of radius $(1-x_1^2)^{1/2}$, so $\mathbb{E}[x_2^2\,|\,x_1] = \frac13(1-x_1^2)$ and $\|x_2U_n'(x_1)\|^2 = \frac13 n(n+2)$. Summing $n^{-1}\kappa^{-2}d_n^2\cdot\frac13 n(n+2)$ over $n\ge1$ gives $\mathscr{Q}_x$. □

By Lemmas 365 and 366,

$$L(\kappa,\varphi) \le \tfrac12 \max\big(\mathscr{Q}_z(\kappa),\mathscr{Q}_x(\kappa)\big)^{1/2}, \tag{K8}$$

and Theorem 364 follows once both parts are at most $\frac14$. Both are $\frac14$ at $\kappa = 0$, by continuity from the values $d_1 \sim \kappa/2$, $d_1' \to \frac12$ and $d_n = O(\kappa^n)$.

6. Three Bessel sums

The series of Lemma 366 reduce to closed forms. Write I_j for $I_j(\kappa)$.

Lemma 367. *For every $\kappa > 0$,*

$$\sum_{j\ge1} j^2 I_j^2 = \frac{\kappa}{4} I_1(2\kappa), \qquad \sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}\big(I_0^2 + I_1^2\big),$$
$$I_1(2\kappa) = 2\sum_{j\ge1} I_{j-1}I_j.$$

The first and third identities follow from Parseval's formula for $e^{\kappa\cos\theta} = \sum_{j\in\mathbb{Z}} I_j e^{ij\theta}$, and the second from the recurrences of I_j by telescoping; Appendix M gives the proofs.

7. The transverse part

Proposition 368. $\mathscr{Q}_x(\kappa) \le \frac14$ *for every* $\kappa \ge 0$.

Proof. By (K7), $\mathscr{Q}_x = (3\kappa^2 I_1^2)^{-1}\sum_{j\ge2} j^2(j+1)I_j^2$, and the sum equals $\sum_{j\ge1}(j^3+j^2)I_j^2 - 2I_1^2$. By Lemma 367, $\mathscr{Q}_x \le \frac14$ is equivalent to

$$\mathscr{D}(\kappa) := (8+2\kappa^2)I_1^2 - \kappa^2 I_0^2 - \kappa I_1(2\kappa) \ge 0. \tag{K9}$$

Each term is an entire function of κ that is even after the factor κ^2 is removed, so $\mathscr{D}(\kappa) = \sum_{n\ge0}\mathscr{D}_n\kappa^{2n}$. Lemma 371 of Appendix M computes the coefficients: $\mathscr{D}_0 = \mathscr{D}_1 = 0$, and for $n = m+1 \ge 2$

$$\mathscr{D}_n = \frac{N_m}{4^m\, m!\,(m+1)!\,(m+1)(m+2)},$$

with

$$\begin{aligned} N_m = \binom{2m}{m}\big(4(2m+1) + (m-1)(m+1)(m+2)\big) \\ - 4^m(m+1)(m+2). \end{aligned}$$

The numerator N_m vanishes for $m = 0,1,2$ and is positive for $m \ge 3$ (Lemma 371). So every coefficient of $\mathscr{D}$ is nonnegative and $\mathscr{D} \ge 0$. □

The first nonzero coefficient is $\mathscr{D}_4 = \frac{1}{2304}$, so $\frac14 - \mathscr{Q}_x = \mathscr{D}/(12\kappa^2 I_1^2)$ is of order κ^4 at 0: the transverse part leaves the value $\frac14$ very slowly, which is why no estimate that loses a constant can prove the proposition.

8. The zonal part

Let $A = I_2/I_1$ as before, so that $d_1 = 2A$ and, by Lemma 362(i), $A' = 1 - A^2 - 3A/\kappa$.

Proposition 369. *For* $0 < \kappa \le \sqrt{88}/3$,

$$\mathscr{Q}_z(\kappa) \le \frac14 + 4A'^2 + \frac{A^2}{4} - \frac{A}{2\kappa} - \frac{2A^2}{\kappa^2} < \frac14.$$

Proof. The higher modes. With $r_\nu = I_{\nu+1}/I_\nu$ the recurrence $I_\nu' = I_{\nu+1} + \frac{\nu}{\kappa}I_\nu$ gives

$$\frac{d_n'}{d_n} = \frac{I_{n+1}'}{I_{n+1}} - \frac{I_1'}{I_1} = \frac{n}{\kappa} + r_{n+1} - r_1.$$

By Lemma 372 of Appendix M, $r_{n+1} \le r_1$, and $r_1 = A \le \kappa/4$ by Lemma 362(iii). Hence $-n/\kappa \le n/\kappa - \kappa/4 \le d_n'/d_n \le n/\kappa$ for $n \ge 2$ and $\kappa^2 \le 16$, so $d_n'^2 \le n^2 d_n^2/\kappa^2$. With $d_1' = 2A'$,

$$\mathscr{Q}_z \le 4A'^2 + \frac{1}{\kappa^2}\sum_{n\ge2} n\,d_n^2 = 4A'^2 + \frac{S_3 - S_2 - 4I_2^2}{\kappa^2 I_1^2},$$

where $S_2 = \sum_{j\ge1} j^2 I_j^2$ and $S_3 = \sum_{j\ge1} j^3 I_j^2$; the second form uses $n\,d_n^2 I_1^2 = (j-1)j^2 I_j^2$ with $j = n+1$.

The closed form. The recurrences $I_0 - I_2 = \frac{2}{\kappa} I_1$ and $I_1 - I_3 = \frac{4}{\kappa} I_2$ give $I_0/I_1 = 2/\kappa + A$ and $I_2 I_3 / I_1^2 = A - 4A^2/\kappa$. By Lemma 367, $S_3/(\kappa^2 I_1^2) = \frac{1}{4}(2/\kappa + A)^2 + \frac{1}{4}$, and, keeping three terms of the third identity,

$$\frac{S_2}{\kappa^2 I_1^2} \geq \frac{I_0 I_1 + I_1 I_2 + I_2 I_3}{2\kappa I_1^2} = \frac{1}{2\kappa}\Big(\frac{2}{\kappa} + 3A - \frac{4A^2}{\kappa}\Big).$$

Collecting terms, $\mathscr{Q}_z \leq \frac{1}{4} + 4A'^2 + T(A)$ with $T(a) = \frac{a^2}{4} - \frac{a}{2\kappa} - \frac{2a^2}{\kappa^2}$, which is the first inequality.

The barrier. Let $s = \sqrt{\kappa^2 + 25/4}$ and $A_\flat = \kappa/(\frac{3}{2} + s)$. By Lemma 362(iv), $A_\flat \leq A \leq \kappa/4$. The function $a \mapsto 1 - a^2 - 3a/\kappa$ decreases on $a \geq 0$, and $A' \geq 0$ because $A' = \operatorname{Var} x_1$, so $0 \leq A' \leq \alpha := 1 - A_\flat^2 - 3A_\flat/\kappa$. For $\kappa^2 \leq 12$ the quadratic T decreases on $[0, \kappa/4]$, since

$$T(a) - T(a') = (a' - a)\,\frac{2\kappa - (a + a')(\kappa^2 - 8)}{4\kappa^2}$$

and $(a + a')(\kappa^2 - 8) \leq \frac{\kappa}{2}(\kappa^2 - 8) \leq 2\kappa$ there. So $\mathscr{Q}_z \leq \frac{1}{4} + 4\alpha^2 + T(A_\flat)$. With $\kappa^2 = s^2 - \frac{25}{4}$ and $A_\flat/\kappa = 2/(2s+3)$ both terms are rational in s: $\alpha = 16/(2s+3)^2$, and

$$4\alpha^2 + T(A_\flat) = \frac{(2s-5)\,p(s)}{4(2s+3)^4}, \qquad p(s) = 8s^3 + 28s^2 - 98s - 695. \tag{K10}$$

For $0 < \kappa \leq \sqrt{88}/3$ we have $\frac{5}{2} < s \leq \sqrt{577}/6 < 4.004$. On $s \geq \frac{5}{2}$ the cubic p increases, since $p'(s) = 24s^2 + 56s - 98 > 0$, and $p(4.004) < -120$. So the right side of (K10) is negative. □

9. Proof of Theorem 364

At $c = 0$ the theorem is Lemma 363(i), and for $|c| \geq \sqrt{88}/3$ it is Lemma 363(iii). For $0 < |c| < \sqrt{88}/3$, Propositions 368 and 369 give $\mathscr{Q}_x, \mathscr{Q}_z \leq \frac{1}{4}$, and (K8) gives $L \leq \frac{1}{4}$. □

Appendix L: Computations for the single-link law

1. Barta's inequality

Let $\rho(\theta) = \sin^2\theta\, e^{\kappa\cos\theta}$ on $(0,\pi)$, let V be a continuous potential, and let

$$T = -\rho^{-1}\partial_\theta\,\rho\,\partial_\theta + V.$$

If $\psi > 0$ is smooth on $(0,\pi)$ with $\rho\psi^2 = O(\sin^4\theta)$ at the end points and $T\psi \geq \lambda_0\psi$, then

$$\int (|h'|^2 + Vh^2)\rho\, d\theta \geq \lambda_0 \int h^2 \rho\, d\theta$$

for every smooth h vanishing to first order at 0 and π. Indeed, writing $h = \psi u$ and integrating by parts, the boundary terms $\rho\psi\psi' u^2$ vanish and

$$\int (|h'|^2 + Vh^2)\rho = \int \rho\psi^2 |u'|^2 + \int \frac{T\psi}{\psi} h^2 \rho \;\geq\; \lambda_0 \int h^2 \rho.$$

This is the inequality of Barta [15] in the form of a ground state transformation.

2. The computations of Lemma 361

The generator of the reversible diffusion of ν_b on S^3 is $L = \Delta_{S^3} + \kappa \nabla x_1 \cdot \nabla$. In the polar angle θ from e_1 the round metric is $d\theta^2 + \sin^2\theta\, g_{S^2}$, so $\Delta_{S^3} = \partial_\theta^2 + 2\cot\theta\,\partial_\theta + \sin^{-2}\theta\,\Delta_{S^2}$, and $\nabla x_1 \cdot \nabla = -\sin\theta\,\partial_\theta$. On $h(\theta)Y_\ell$ with a spherical harmonic Y_ℓ of degree ℓ on S^2 the operator acts as $L_\ell = \partial_\theta^2 + b\,\partial_\theta - \ell(\ell+1)\sin^{-2}\theta$, $b = 2\cot\theta - \kappa\sin\theta = \rho'/\rho$, so $-L_\ell = T$ with $V = \ell(\ell+1)\sin^{-2}\theta$.

Degree $\ell \geq 1$. Take $\psi = \sin\theta\, e^{a\kappa\cos\theta}$. Then $\psi'/\psi = \cot\theta - a\kappa\sin\theta$ and $\psi''/\psi = (\psi'/\psi)^2 + (\psi'/\psi)'$, and a short computation gives

$$\frac{-L_1\psi}{\psi} = 3 + (1+5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta.$$

For $a = -\frac{1}{5}$ this is $3 + \frac{4}{25}\kappa^2\sin^2\theta \geq 3$. Since $V = \ell(\ell+1)\sin^{-2}\theta \geq 2\sin^{-2}\theta$ for $\ell \geq 1$, Barta's inequality gives $-L_\ell \geq 3$ on every sector with $\ell \geq 1$.

Degree $\ell = 0$. Let $g(\theta)$ be a zonal eigenfunction, $-L_0 g = \lambda g$, g not constant, so that $g'(0) = g'(\pi) = 0$. Differentiating $g'' + bg' = -\lambda g$ gives, for $h = g'$,

$$-\big(h'' + bh'\big) - b'h = \lambda h, \qquad -b' = \frac{2}{\sin^2\theta} + \kappa\cos\theta,$$

so h is an eigenfunction of T with $V = 2\sin^{-2}\theta + \kappa\cos\theta$, with eigenvalue λ, and $h \not\equiv 0$ vanishes at both ends. With the same ψ,

$$\frac{T\psi}{\psi} = 3 + (2+5a)\kappa\cos\theta - a(1+a)\kappa^2\sin^2\theta,$$

which for $a = -\frac{2}{5}$ is $3 + \frac{6}{25}\kappa^2\sin^2\theta \geq 3$. Barta's inequality applied to h gives $\lambda \geq 3$. Together with the case $\ell \geq 1$ every nonzero eigenvalue of $-L$ is at least 3, which is Lemma 361; the spectrum is discrete because S^3 is compact and the drift is smooth.

3. Moments

The density of $t = x_1$ under ν_b is proportional to $(1-t^2)^{1/2}e^{\kappa t}$. For every smooth ϕ on $[-1,1]$,

$$\frac{d}{dt}\Big[(1-t^2)^{3/2}e^{\kappa t}\Big] = (1-t^2)^{1/2}e^{\kappa t}\big(\kappa(1-t^2) - 3t\big),$$

and integration by parts gives $\mathbb{E}[\phi'(t)(1-t^2)] = \mathbb{E}\big[\phi(t)\big(3t - \kappa(1-t^2)\big)\big]$. With $\phi = 1$ this is $\kappa\mathbb{E}(1-x_1^2) = 3\mathbb{E}x_1$, the identity of Lemma 362(i). With $\phi = t^k$ it gives the recursion

$$m_{k+2} = m_k - \frac{k+3}{\kappa}\,m_{k+1} + \frac{k}{\kappa}\,m_{k-1}, \qquad m_k = \mathbb{E}x_1^k,$$

from $m_0=1$ and $m_1=A$; all polynomial moments of ν_b are rational in κ and $A(\kappa)$.[30]

4. The Riccati barriers

By Lemma 362(i), A solves $A'=\mathscr{F}(\kappa,A):=1-A^2-3A/\kappa$ on $(0,\infty)$, and $A(\kappa)=\frac{\kappa}{4}-\frac{\kappa^3}{96}+O(\kappa^5)$ from the series of I_1 and I_2. Let $s=\sqrt{\kappa^2+4}$, $u=\sqrt{\kappa^2+9}$, $\phi_-=\kappa/(2+s)$ and $\phi_+=\kappa/(1+u)$. Direct differentiation gives

$$\phi_-'-\mathscr{F}(\kappa,\phi_-)=-\frac{\kappa^2}{s(s+2)^2}\le 0,$$
$$\phi_+'-\mathscr{F}(\kappa,\phi_+)=\frac{(u-3)^2}{u(1+u)^2}\ge 0,$$

and $\phi_\pm=\frac{\kappa}{4}+O(\kappa^3)$.[31] The comparison is singular at 0 because of the term $3A/\kappa$, and we carry it out with an integrating factor. Let $D=A-\phi_-$. Then

$$\begin{aligned}D'&=\mathscr{F}(\kappa,A)-\mathscr{F}(\kappa,\phi_-)-\big(\phi_-'-\mathscr{F}(\kappa,\phi_-)\big)\\&=-\Big(A+\phi_-+\frac{3}{\kappa}\Big)D+R,\end{aligned}$$

with $R=\kappa^2/(s(s+2)^2)\ge 0$. Put $\mu(\kappa)=\kappa^3\exp\int_0^\kappa(A+\phi_-)$. Then $(\mu D)'=\mu R\ge 0$, and $\mu D=O(\kappa^6)\to 0$ as $\kappa\to 0$, so $\mu D\ge 0$ and $A\ge\phi_-$. The same argument with $D=\phi_+-A$ and $R=(u-3)^2/(u(1+u)^2)$ gives $A\le\phi_+$. This is Lemma 362(ii).

For (iv) let $s=\sqrt{\kappa^2+25/4}$ and $\phi_\flat=\kappa/(\frac{3}{2}+s)$. Since $s^2-\frac{9}{4}=\kappa^2+4$, the same differentiation gives

$$\phi_\flat'-\mathscr{F}(\kappa,\phi_\flat)=-\frac{5\,(s-\frac{5}{2})}{2s\,(\frac{3}{2}+s)^2}\le 0,$$

so $\phi_\flat$ is a subsolution, and $\phi_\flat=\frac{\kappa}{4}-\frac{\kappa^3}{80}+O(\kappa^5)$.[32] Hence $A-\phi_\flat=\frac{\kappa^3}{480}+O(\kappa^5)$, and the integrating-factor argument above, with $D=A-\phi_\flat$, $R=\frac{5(s-5/2)}{2s(3/2+s)^2}\ge 0$ and $\mu=\kappa^3\exp\int_0^\kappa(A+\phi_\flat)$, gives $\mu D=O(\kappa^6)$ at 0, $(\mu D)'=\mu R\ge 0$, and so $A\ge\phi_\flat$. Since $\frac{3}{2}+\sqrt{\kappa^2+25/4}<2+\sqrt{\kappa^2+4}$ for $\kappa>0$, this improves the lower bound of (ii).

[30] For instance $m_2=1-3A/\kappa$, $m_3=A-3/\kappa+12A/\kappa^2$ and $m_4=m_2-5m_3/\kappa+2A/\kappa$. The joint moments with x_2,x_3,x_4 follow from exchangeability: $\mathbb{E}[\phi(x_1)x_2^2]=\frac{1}{3}\mathbb{E}[\phi(x_1)(1-x_1^2)]$.

[31] For the first identity write $\phi_-=(s-2)/\kappa$, which follows from $s^2-4=\kappa^2$; then $\phi_-'=(\kappa^2/s-s+2)/\kappa^2$ and $1-\phi_-^2-3\phi_-/\kappa=(\kappa^2-(s-2)^2-3(s-2))/\kappa^2$, and the difference simplifies with $s^2=\kappa^2+4$. The second is the same computation with $\phi_+=\kappa(u-1)/(u^2-1)=\kappa(u-1)/(\kappa^2+8)$ in place of ϕ_- and $u^2=\kappa^2+9$ in place of $s^2=\kappa^2+4$. After clearing the denominator $\kappa^2(1+u)^2$ the numerator becomes $\kappa^2(u-3)^2/u$, which is the stated residual.

[32] With $\phi_\flat=\kappa(s-\frac{3}{2})/(\kappa^2+4)$ one has $\phi_\flat'=(\frac{3}{2}s+\frac{25}{4})/(s(\frac{3}{2}+s)^2)$, using $ds/d\kappa=\kappa/s$; and $\mathscr{F}(\kappa,\phi_\flat)=1-\phi_\flat^2-3/(\frac{3}{2}+s)$. Clearing the denominator $s(\frac{3}{2}+s)^2$ and replacing κ^2 by $s^2-\frac{25}{4}$ leaves $-\frac{5}{2}s+\frac{25}{4}$ in the numerator, which is the stated residual. The expansion follows from $s=\frac{5}{2}+\frac{\kappa^2}{5}+O(\kappa^4)$.

Appendix M: Bessel functions

This appendix proves the auxiliary statements used in Appendix K for Theorem 364. Throughout, $I_j=I_j(\kappa)$ with $\kappa>0$, and $I_{-j}=I_j$ for integers j. We use the series

$$I_\nu(\kappa)=\sum_{i\ge 0}\frac{(\kappa/2)^{2i+\nu}}{i!\,(i+\nu)!},$$

the integral $I_j(\kappa)=\pi^{-1}\int_0^\pi e^{\kappa\cos\theta}\cos j\theta\,d\theta$, the Fourier series $e^{\kappa\cos\theta}=\sum_{j\in\mathbb{Z}}I_j e^{ij\theta}$ and the recurrences

$$I_{j-1}-I_{j+1}=\frac{2j}{\kappa}I_j,\quad I_j'=I_{j+1}+\frac{j}{\kappa}I_j,\tag{M1}$$

all of which are in Watson's treatise [103], §§3.7, 3.71, 2.22. All series below converge absolutely, because $0<I_j\le(\kappa/2)^j e^\kappa/j!$.

1. Zonal harmonics

Under σ the coordinate x_1 has the density $\frac{2}{\pi}(1-t^2)^{1/2}$ on $[-1,1]$, the area of the two-sphere $\{x_1=t\}$ being proportional to $1-t^2$ and its normal speed to $(1-t^2)^{-1/2}$. The Chebyshev polynomials $U_n(\cos\theta)=\sin((n+1)\theta)/\sin\theta$ satisfy

$$\int U_nU_m\,d\sigma=\frac{2}{\pi}\int_0^\pi\sin((n+1)\theta)\sin((m+1)\theta)\,d\theta=\delta_{nm},$$

and $U_n=C_n^{(1)}$ is the Gegenbauer polynomial of index $\frac{4-2}{2}=1$, so $U_n(\langle y,x\rangle)$ is the zonal harmonic of degree n on S^3 with pole y: $|x|^nU_n(\langle y,x\rangle/|x|)$ is a harmonic polynomial on $\mathbb{R}^4$ [99], Ch. IV, §2.

Lemma 370. *$e^{\kappa t}=\frac{2}{\kappa}\sum_{n\ge 0}(n+1)I_{n+1}(\kappa)U_n(t)$ in $L^2\big(\frac{2}{\pi}(1-t^2)^{1/2}dt\big)$, and uniformly on $[-1,1]$.*

Proof. With $t=\cos\theta$ and $2\sin\theta\sin((n+1)\theta)=\cos n\theta-\cos(n+2)\theta$, the n-th coefficient is

$$\begin{aligned}\frac{2}{\pi}\int_0^\pi e^{\kappa\cos\theta}\sin((n+1)\theta)\sin\theta\,d\theta&=I_n-I_{n+2}\\&=\frac{2(n+1)}{\kappa}I_{n+1},\end{aligned}$$

by (M1). Since $|U_n|\le n+1$ and I_{n+1} decays factorially, the series converges uniformly. □

2. Proof of Lemma 367

Let $F(\theta)=e^{\kappa\cos\theta}=\sum_j I_j e^{ij\theta}$. Then

$$F'(\theta)=-\kappa\sin\theta\,F(\theta)=\sum_j ijI_je^{ij\theta},$$

and Parseval's formula gives

$$\sum_{j\in\mathbb{Z}} j^2 I_j^2 = \frac{\kappa^2}{2\pi}\int_0^{2\pi} \sin^2\theta\, e^{2\kappa\cos\theta}\,d\theta$$
$$= \frac{\kappa^2}{2}\big(I_0(2\kappa) - I_2(2\kappa)\big) = \frac{\kappa}{2} I_1(2\kappa),$$

the last step by (M1) at the argument 2κ. Halving gives the first identity. The coefficient of $e^{i\theta}$ in $F^2 = e^{2\kappa\cos\theta}$ is $I_1(2\kappa) = \sum_{j\in\mathbb{Z}} I_j I_{1-j}$, and the terms with $j \le 0$ repeat those with $j \ge 1$; this is the third identity.

For the second, $jI_j = \frac{\kappa}{2}(I_{j-1} - I_{j+1})$ by (M1), so

$$\sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}\sum_{j\ge1} j\big(I_{j-1}^2 + I_{j+1}^2 - 2I_{j-1}I_{j+1}\big).$$

Shifting indices, $\sum_{j\ge1} j(I_{j-1}^2 + I_{j+1}^2) = I_0^2 + \sum_{m\ge1} 2mI_m^2$, and $\sum_{m\ge1} 2mI_m^2 = \kappa\sum_{m\ge1}(I_{m-1}I_m - I_mI_{m+1}) = \kappa I_0 I_1$ by telescoping. For the mixed term write $j = \frac12((j-1)+(j+1))$ and apply (M1) to $(j-1)I_{j-1}$ and to $(j+1)I_{j+1}$, using $I_{-1} = I_1$ at $j = 1$:

$$jI_{j-1}I_{j+1} = \frac{\kappa}{4}\big[(I_{j-1}I_j - I_jI_{j+1})$$
$$+ (I_{j-2}I_{j+1} - I_{j-1}I_{j+2})\big].$$

Both brackets telescope, to I_0I_1 and to $I_{-1}I_2 = I_1I_2$, so $\sum_{j\ge1} jI_{j-1}I_{j+1} = \frac{\kappa}{4}(I_0I_1 + I_1I_2)$. Altogether

$$\sum_{j\ge1} j^3 I_j^2 = \frac{\kappa^2}{4}\Big(I_0^2 + \frac{\kappa}{2}I_1(I_0 - I_2)\Big) = \frac{\kappa^2}{4}\big(I_0^2 + I_1^2\big). \qquad \square$$

3. The coefficients of $\mathscr{D}$

Lemma 371. *Write $\mathscr{D}(\kappa) = \sum_{n\ge0}\mathscr{D}_n\kappa^{2n}$ for the function of (K9). Then $\mathscr{D}_0 = \mathscr{D}_1 = 0$, and for $m \ge 1$*

$$\mathscr{D}_{m+1} = \frac{N_m}{4^m\, m!\,(m+1)!\,(m+1)(m+2)},$$

with $N_m = \binom{2m}{m}\big(4(2m+1) + (m-1)(m+1)(m+2)\big) - 4^m(m+1)(m+2)$. Moreover $N_m = 0$ for $m \le 2$ and $N_m > 0$ for $m \ge 3$.

Proof. The coefficients. Multiplying the series of I_0 and I_1 and collecting powers,

$$I_0^2 = \sum_{m\ge0}\alpha_m\kappa^{2m}, \qquad I_1^2 = \sum_{m\ge0}\beta_m\kappa^{2m+2},$$
$$\kappa I_1(2\kappa) = \sum_{m\ge0}\frac{\kappa^{2m+2}}{m!\,(m+1)!},$$

where

$$\alpha_m = \frac{1}{4^m}\sum_{i=0}^{m}\frac{1}{i!^2(m-i)!^2},$$
$$\beta_m = \frac{1}{4^{m+1}}\sum_{i=0}^{m}\frac{1}{i!\,(i+1)!\,(m-i)!\,(m-i+1)!}.$$

By Vandermonde's identity, $\sum_i \binom{m}{i}^2 = \binom{2m}{m}$ and $\sum_i \binom{m}{i}\binom{m+2}{i+1} = \binom{2m+2}{m+1}$, so

$$\alpha_m = \frac{\binom{2m}{m}}{4^m\, m!^2}, \qquad \beta_m = \frac{\binom{2m+2}{m+1}}{4^{m+1} m!\,(m+2)!}.$$

The coefficient of κ^{2m+2} in $\mathscr{D}$ is $\mathscr{D}_{m+1} = 8\beta_m + 2\beta_{m-1} - \alpha_m - 1/(m!(m+1)!)$, with $\beta_{-1} = 0$; for $m = 0$ it is $\frac84 - 1 - 1 = 0$, and $\mathscr{D}_0 = 0$ because every term of $\mathscr{D}$ is $O(\kappa^2)$. For $m \ge 1$ put $\Delta = 4^m\, m!\,(m+1)!\,(m+1)(m+2)$ and $b = \binom{2m}{m}$. With $\binom{2m+2}{m+1} = \frac{2(2m+1)}{m+1}b$,

$$8\beta_m = \frac{4(2m+1)\,b}{\Delta}, \qquad \alpha_m = \frac{(m+1)^2(m+2)\,b}{\Delta},$$
$$2\beta_{m-1} = \frac{2m(m+1)(m+2)\,b}{\Delta},$$
$$\frac{1}{m!(m+1)!} = \frac{4^m(m+1)(m+2)}{\Delta},$$

and $2m - (m+1) = m-1$ gives $\mathscr{D}_{m+1} = N_m/\Delta$.

The sign. Directly, $N_0 = N_1 = N_2 = 0$ ($2-2$, $24-24$, $192-192$), $N_3 = 80$, $N_4 = 1140$ and $N_5 = 10416$. For $m \ge 6$ we first show $16^m \le 4m\binom{2m}{m}^2$ for $m \ge 1$ by induction: it is an equality at $m = 1$, and $(m+1)\binom{2m+2}{m+1} = 2(2m+1)\binom{2m}{m}$ gives

$$4(m+1)\binom{2m+2}{m+1}^2 = \frac{16(2m+1)^2}{m+1}\binom{2m}{m}^2$$
$$\ge \frac{16(2m+1)^2}{4m(m+1)}16^m \ge 16^{m+1}.$$

Since $(m-1)^2 \ge 4m$ for $m \ge 6$, it follows that $\binom{2m}{m}^2(m-1)^2 \ge 16^m$, that is $\binom{2m}{m}(m-1) \ge 4^m$, and

$$N_m = \left(\binom{2m}{m}(m-1) - 4^m\right)(m+1)(m+2)$$
$$+ 4(2m+1)\binom{2m}{m} > 0.$$

$\square$

4. Monotonicity of Bessel ratios

Lemma 372. *For $n \ge 1$ and $\kappa > 0$, $I_{n+2}(\kappa)/I_{n+1}(\kappa) \le I_2(\kappa)/I_1(\kappa)$.*

Proof. We use a lemma of Biernacki and Krzyż [24]: if $f = \sum a_j x^j$ and $g = \sum b_j x^j$ have positive coefficients and a_j/b_j is nonincreasing in j, then f/g is nonincreasing on $x > 0$. Indeed

$$f'g - fg' = \sum_{i,j}(i-j)a_ib_jx^{i+j-1}$$
$$= \frac12\sum_{i,j}(i-j)b_ib_j\Big(\frac{a_i}{b_i} - \frac{a_j}{b_j}\Big)x^{i+j-1} \le 0.$$

Take $x=\kappa^2$, $f=I_{n+1}(\kappa)/\kappa^{n+1}$ and $g=I_1(\kappa)/\kappa$. Then $a_j/b_j=2^{-n}(j+1)!/(j+n+1)!$ decreases in j, so $I_{n+1}/(\kappa^n I_1)$ is nonincreasing in κ. Its logarithmic derivative is, by the second recurrence of (M1),

$$\frac{n+1}{\kappa}+\frac{I_{n+2}}{I_{n+1}}-\frac{n}{\kappa}-\frac{1}{\kappa}-\frac{I_2}{I_1} = \frac{I_{n+2}}{I_{n+1}}-\frac{I_2}{I_1}\le 0. \qquad \square$$

Appendix N: The certificate bound

This appendix reduces a bound on h_F to finitely many positivity statements, by hand, and reports the computation of [93] that verifies them for $h_F\le 14.420001$ (Remark 26); no result of the paper uses that value. The argument is the proof of the bound in Theorem 4 with two changes. The divergence identity of Lemma 19, taken there with the factor $\frac{9}{10}$, is replaced by a positive semidefinite form at each site, and the weights $\frac{21}{10}$ and $\frac{5}{4}$ of Lemma 20 are replaced by weights chosen plaquette by plaquette. The reduction is proved by hand; the positivity of ninety-six matrices of trigonometric polynomials and the arithmetic of finitely many rational numbers are checked by computer.[33]

1. Ends and corners

At a site y the eight *ends* are

$$o_\mu(y)=X_{y,\mu}, \qquad i_\mu(y)=Q^*_{y-e_\mu,\mu}X_{y-e_\mu,\mu}Q_{y-e_\mu,\mu},$$

for $\mu=0,1,2,3$, as in the proof of Lemma 19; the end of a link in F or outside the box is 0. Each link $e=(y,\mu)$ has the end o_μ at its tail y and the end i_μ at its head $y+e_\mu$, both of norm $|X_e|$.

Lemma 373. *Let p be the plaquette at x in the plane $\mu<\nu$. Then*

$$\begin{aligned}\langle Y_1,Y_4\rangle&=\langle o_\mu,o_\nu\rangle(x),\\ \langle Y_1,Y_2\rangle&=\langle i_\mu,o_\nu\rangle(x+e_\mu),\\ \langle Y_3,Y_4\rangle&=\langle o_\mu,i_\nu\rangle(x+e_\nu),\\ \langle Y_2,WY_3W^*\rangle&=\langle i_\nu,i_\mu\rangle(x+e_\mu+e_\nu).\end{aligned}$$

Every pair of ends at one site in two different directions occurs in exactly one of these identities, for exactly one plaquette.

Proof. The first identity is the definition of the ends. The inner product is invariant under conjugation by a unitary matrix, so $\langle X_{e_1},Q_{e_1}X_{e_2}Q^*_{e_1}\rangle=\langle Q^*_{e_1}X_{e_1}Q_{e_1},X_{e_2}\rangle$, which is the second identity, and the third follows in the same way from $\langle Q_{e_4}X_{e_3}Q^*_{e_4},X_{e_4}\rangle=\langle X_{e_3},Q^*_{e_4}X_{e_4}Q_{e_4}\rangle$. For the fourth, $WQ_{e_4}=Q_{e_1}Q_{e_2}Q^*_{e_3}$, so that $WY_3W^*=RX_{e_3}R^*$ with $R=Q_{e_1}Q_{e_2}Q^*_{e_3}$, and conjugating both vectors by $(Q_{e_1}Q_{e_2})^*$ gives $\langle Q^*_{e_2}X_{e_2}Q_{e_2},Q^*_{e_3}X_{e_3}Q_{e_3}\rangle$. The links e_2 and e_3 both have their head at $x+e_\mu+e_\nu$. For the last statement, the pairs (o_μ,o_ν), (i_μ,o_ν), (o_μ,i_ν) and (i_μ,i_ν) at y come from the plaquettes in the plane $\mu\nu$ at y, $y-e_\mu$, $y-e_\nu$ and $y-e_\mu-e_\nu$ respectively. There are $4\binom{4}{2}=24$ such pairs at each site and $6\cdot4=24$ corners of plaquettes per site. $\square$

For each class of sites modulo 2 let G_y be a real symmetric 8×8 matrix indexed by the ends, with $G_y[o_\mu,i_\mu]=0$ for every μ. For the plaquette p at x in the plane $\mu<\nu$ put

$$\begin{aligned}c_p^G=\;&2G_x[o_\mu,o_\nu]\langle Y_1,Y_4\rangle+2G_{x+e_\mu}[i_\mu,o_\nu]\langle Y_1,Y_2\rangle\\ &+2G_{x+e_\nu}[o_\mu,i_\nu]\langle Y_3,Y_4\rangle\\ &+2G_{x+e_\mu+e_\nu}[i_\mu,i_\nu]\langle Y_2,WY_3W^*\rangle.\end{aligned}$$

The matrix G^0 with $G^0[a,a]=2$, $G^0[o_\mu,o_\nu]=G^0[i_\mu,i_\nu]=1$ and $G^0[o_\mu,i_\nu]=-1$ for $\mu\ne\nu$ is the matrix of the form $|\sum_\mu(o_\mu-i_\mu)|^2+\sum_\mu|o_\mu+i_\mu|^2$ of Lemma 19, and $c_p^{G^0}$ is the corner form c_p of (7).

Lemma 374. *On $\mathbb{Z}^4$, or on a box with free boundary, for every field Q and every square-summable X,*

$$\begin{aligned}&\sum_y\sum_{a,b}G_y[a,b]\,\langle \mathrm{end}_a,\mathrm{end}_b\rangle(y)\\ &\quad=\sum_p c_p^G+\sum_e\big(G_{t(e)}[o,o]+G_{h(e)}[i,i]\big)|X_e|^2,\end{aligned}$$

where $t(e)$ and $h(e)$ are the tail and the head of e, and o, i are its two ends. If every G_y is positive semidefinite, the left side is nonnegative.

Proof. Split the sum over a,b at each site into the diagonal terms, the pairs (o_μ,i_μ), whose coefficient is 0, and the pairs of ends in different directions, each counted twice. The diagonal terms give the last sum, since each link has one end at its tail and one at its head, and the pairs in different directions give $\sum_p c_p^G$ by Lemma 373. In a box the argument at the end of the proof of Lemma 19 applies. At each site the left side is the trace of G_y times the Gram matrix of the eight ends, and the trace of a product of two positive semidefinite matrices is nonnegative. $\square$

2. From plaquettes to the Hessian

Number the positions of a plaquette as in (5) and let k_p be the position of its link in F; by Lemma 9 there is exactly one.

Lemma 375. *Let every G_y be positive semidefinite with $G_y[o_\mu,i_\mu]=0$, and let $a_{p,r}\ge0$ for the three positions $r\ne k_p$ of each plaquette. Suppose that for every plaquette p, every N, every $W\in\mathrm{U}(N)$ and all $Y_1,\dots,Y_4\in\mathfrak{u}(N)$ with $Y_{k_p}=0$,*

$$f_p+c_p^G\le\sum_{r\ne k_p}a_{p,r}|Y_r|^2. \tag{N1}$$

[33] The constant 14.420001 is not the best possible for this kind of certificate: by the search reported in Remark 26 it cannot go below 14.384, while the true value of h_F is at least 13.123.

Then $|\operatorname{Hess} S_0(v,v)| \le h|v|^2$ *for every field* Q *and every* $v = (X_e Q_e)$ *with* $X_e = 0$ *for* $e \in F$, *on* $\mathbb{Z}^4$ *and on every box, where*

$$h = \max_{e \notin F} \Big(\sum_{p \ni e} a_{p,e} + G_{t(e)}[o,o] + G_{h(e)}[i,i] \Big)$$

and $a_{p,e}$ *is the weight of the position of* e *in* p.

Proof. Summing (N1) over the plaquettes and adding the identity of Lemma 374, whose left side is nonnegative,

$$\begin{aligned} \operatorname{Hess} S_0(v,v) = \sum_p f_p &\le \sum_p \Big(\sum_r a_{p,r} |Y_r|^2 - c_p^G \Big) \\ &\le \sum_{e \notin F} \Big(\sum_{p \ni e} a_{p,e} + G_{t(e)}[o,o] + G_{h(e)}[i,i] \Big) |X_e|^2 \\ &\le h|v|^2, \end{aligned}$$

since $|Y_r| = |X_e|$ for the link e at position r. In a box the links near the boundary lie in fewer plaquettes, and the weights are nonnegative. For the lower bound, apply (N1) at $-W$. By (6) this changes the sign of f_p, while c_p^G does not change, and the same chain gives $-\operatorname{Hess} S_0(v,v) \le h|v|^2$. □

The hypothesis is needed on $\mathrm{U}(N)$ and not only on $\mathrm{SU}(N)$, since $-W \notin \mathrm{SU}(N)$ when N is odd.

Lemma 376. *If* (N1) *holds for* $N = 2$ *and every* $W = \operatorname{diag}(e^{i\varphi_1}, e^{i\varphi_2})$, *it holds for every* N *and every* $W \in \mathrm{U}(N)$.

Proof. Both sides are invariant under conjugation of W and all Y_i by one unitary matrix, so we may take $W = \operatorname{diag}(w_1, \dots, w_N)$. Every term of f_p, of c_p^G and of $|Y_r|^2$ is a multiple of $\operatorname{Re}\operatorname{Tr}(Y_i D_1 Y_j D_2)$ with $D_1, D_2 \in \{I, W, W^*\}$, and

$$\operatorname{Re}\operatorname{Tr}(Y_i D_1 Y_j D_2) = \operatorname{Re} \sum_{a,b} (Y_i)_{ab} (D_1)_{bb} (Y_j)_{ba} (D_2)_{aa}.$$

The difference of the two sides of (N1) is therefore a sum of one term for each a, in the entries $(Y_i)_{aa}$, and one term for each pair $a < b$, in the entries $(Y_i)_{ab}$ and $(Y_i)_{ba} = -\overline{(Y_i)_{ab}}$, as in the proof of Lemma 20. The term of the pair $a < b$ is the same difference for $N = 2$ at $W' = \operatorname{diag}(w_a, w_b)$, evaluated on the matrices $Y_i' \in \mathfrak{u}(2)$ with off-diagonal entries $(Y_i)_{ab}$, $(Y_i)_{ba}$ and zero diagonal. The term of a is the difference at W', for any $b \ne a$, evaluated on $Y_i' = \operatorname{diag}((Y_i)_{aa}, 0)$. In both cases $Y'_{k_p} = 0$, so each term is nonnegative. □

In the orthonormal basis $iI/\sqrt{2}, i\sigma_1/\sqrt{2}, i\sigma_2/\sqrt{2}, i\sigma_3/\sqrt{2}$ of $\mathfrak{u}(2)$, with the Pauli matrices σ_j, the difference of the two sides of (N1) at $N = 2$ is a quadratic form on the three free positions, with a real symmetric 12×12 matrix $M_p(\varphi)$. Its entries are affine combinations of 1, $\cos\varphi_a$, $\sin\varphi_a$, $\cos(\varphi_1 - \varphi_2)$ and $\sin(\varphi_1 - \varphi_2)$: the form f_p is linear in W, and W enters c_p^G only through $\langle Y_2, W Y_3 W^* \rangle$. Since W commutes with I and σ_3, $M_p(\varphi)$ splits into two blocks of size six, one on the span of $iI, i\sigma_3$ and one on the span of $i\sigma_1, i\sigma_2$, at every position. By Lemmas 375 and 376, a certificate is a choice of G_y and $a_{p,r}$ such that every $M_p(\varphi)$ is positive semidefinite at every φ, and it gives $h_F \le h$.

3. The certificate

The fixed set F and the field of ends have period two, so it suffices to give G_y for the sixteen classes of sites and $a_{p,r}$ for the ninety-six classes of plaquettes, six planes at each of sixteen sites. The certificate of [93] consists of sixteen positive definite matrices G_y of size eight with $G_y[o_\mu, i_\mu] = 0$, and of nonnegative weights $a_{p,r}$, all rational with denominator 10^6. We checked that the set of fixed links used there is our F up to a symmetry of the lattice. Of the sixty-four links of a period cell, sixteen lie in F, one ending at each site. For each of the other forty-eight the bracket in the definition of h is computed exactly, and the largest is $14420001/10^6$.

Remark 377 (a computation, used by no theorem). The certificate of [93] comes with, for each of the 192 blocks of size six, an integer symmetric matrix P of size thirty, stated to be positive definite and to satisfy, with $z = (1, \cos\varphi_1, \sin\varphi_1, \cos\varphi_2, \sin\varphi_2)$,

$$2^{24} \cdot 4 \cdot 10^6 M(\varphi)_{jk} = \sum_{a,b=0}^{4} z_a z_b P_{6a+j,\,6b+k}$$

for all j, k, as polynomials modulo $\cos^2\varphi_a + \sin^2\varphi_a = 1$. Granted these finite statements, for $x \in \mathbb{R}^6$ and $w = z \otimes x$ one has $2^{24} \cdot 4 \cdot 10^6 x^{\mathsf{T}} M(\varphi) x = w^{\mathsf{T}} P w$, which is positive for $x \ne 0$, since $z_0 = 1$ gives $w \ne 0$; so every block is positive definite at every φ, and Lemmas 375 and 376 give, for every N, every field Q and every $v = (X_e Q_e)$ with $X_e = 0$ for $e \in F$, on $\mathbb{Z}^4$ and on every box,

$$|\operatorname{Hess} S_0(v,v)| \le 14.420001\, |v|^2.$$

The finite statements are checked by computer, not by hand, so this bound is a computation; no result of the paper uses it.

The bound of Remark 377 rests on computations of three kinds. The positivity of the G_y, the signs of the weights and the forty-eight totals are finite statements about rational numbers. The identities between $M(\varphi)$ and the matrices P, and the positive definiteness of the P, are finite statements about integers. All of these are checked by the Lean 4 kernel in [93], on no axioms beyond `propext` and `Quot.sound`; the program there also checks the positivity at every phase a second time, by interval arithmetic on a subdivision of the torus of phases. Lemmas 373 to 376 and the formula (6) for the Hessian are the written proofs above and are not formalised.[34] We checked the finite data in exact arithmetic and the positivity at every phase by eigenvalues on an adaptive grid with a Lipschitz bound, in floating point, and J. M. Lockwood checked it again in exact rational arithmetic with outward-rounded interval elimination, finding least eigenvalue above $559/716800$ for every plaquette.

With 14.420001 in place of $\frac{29}{2}$ the proofs of Theorems 4 and 5 would go through unchanged and give the ranges stated

[34] Formalising them needs matrices over $\mathbb{C}$, the group $\mathrm{U}(N)$ and second derivatives along geodesics, that is, a library such as Mathlib; the Lean files of [93] use only the core library.

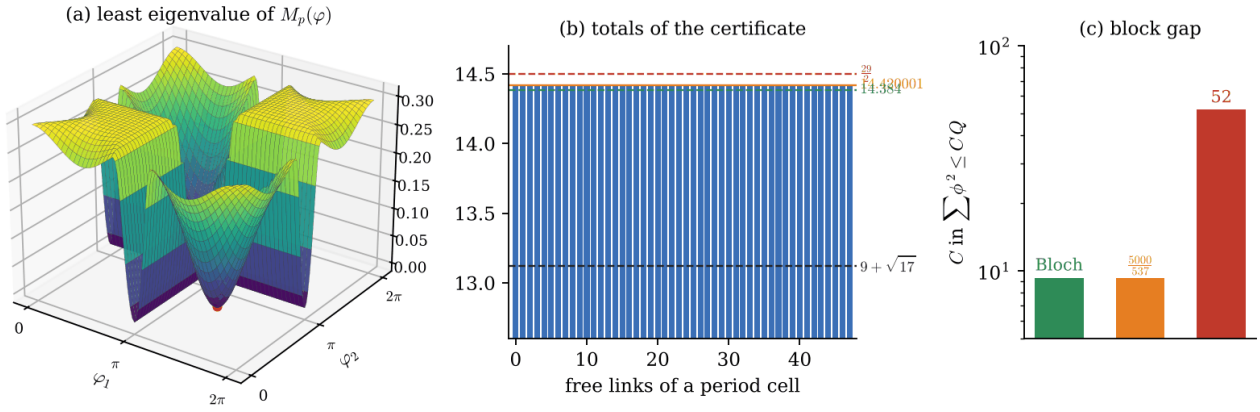


FIG. 18. The certificate of Appendix N. (a) The least eigenvalue of $M_p(\varphi)$ over the torus of phases for the plaquette class with the smallest minimum, about 0.0057 on a grid of 96^2 phases (red dot); the exact computation of Remark 377 finds it positive at every phase. (b) The forty-eight totals in the definition of h, in increasing order, against $\frac{29}{2}$, the value 14.420001 of the largest, the floor 14.384 for certificates on the period-two cell, and the lower bound $9+\sqrt{17}$ for h_F. (c) The constant C in $\sum\phi^2 \le CQ$ for the block gauge: 52 from [21, Theorem 330], $5000/537$ from the computation of [21, Remark 331], and the bulk Bloch value $1/0.10749 = 9.3034$, the best constant when W is the set of all blocks. The eigenvalues in (a) and the Bloch value are in floating point.

in Remark 26. Those theorems are stated with $\frac{29}{2}$, whose proof uses no computer.

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